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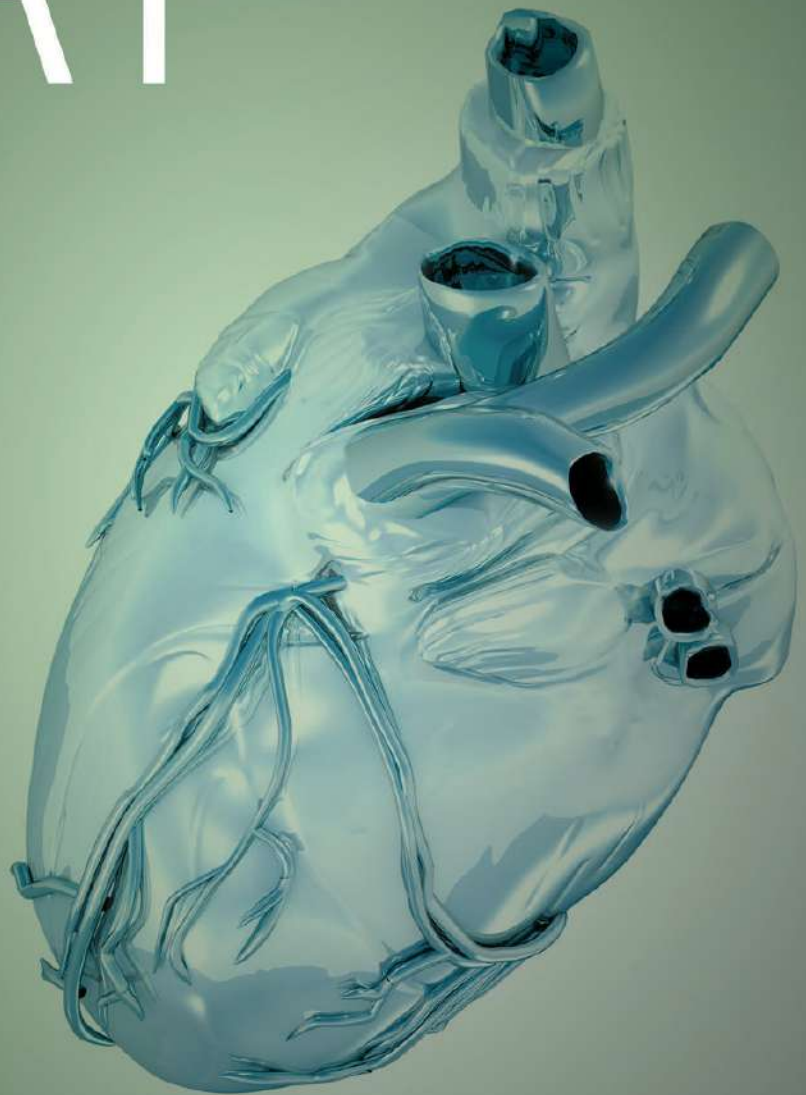
Biology Review

2021–2022

Online + Book

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MCAT® Biology Review

2021-2022

Edited by Alexander Stone Macnow, MD

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Getting Started Checklist

Getting Started Checklist



Register for your free online assets—including full-length practice, Science Review Videos, and additional practice materials—at **www.kaptest.com/booksonline**



Download the Kaplan app to access the digital materials on your mobile device.



Create a study calendar that ensures you complete content review and sufficient practice by Test Day!



As you finish a chapter (including the practice problems), check it off on the table of contents.



Register to take the MCAT at **www.aamc.org/mcat**



Set aside time during your prep to make sure the rest of your application—personal statement, recommendations, and other materials—is ready to go!



Take a moment to admire your completed checklist, then get back to the business of prepping for this exam!

Preface

And now it starts: your long, yet fruitful journey toward wearing a white coat. Proudly wearing that white coat, though, is hopefully only part of your motivation. You are reading this book because you want to be a healer.

If you're serious about going to medical school, then you are likely already familiar with the importance of the MCAT in medical school admissions. While the holistic review process puts additional weight on your experiences, extracurricular activities, and personal attributes, the fact remains: along with your GPA, your MCAT score remains one of the two most important components of your application portfolio—at least early in the admissions process. Each additional point you score on the MCAT pushes you in front of thousands of other students and makes you an even more attractive applicant. But the MCAT is not simply an obstacle to overcome; it is an opportunity to show schools that you will be a strong student and a future leader in medicine.

We at Kaplan take our jobs very seriously and aim to help students see success not only on the MCAT, but as future physicians. We work with our learning science experts to ensure that we're using the most up-to-date teaching techniques in our resources. Multiple members of our team hold advanced degrees in medicine or associated biomedical sciences, and are committed to the highest level of medical education. Kaplan has been working with the MCAT for over 50 years and our commitment to premed students is unflagging; in fact, Stanley Kaplan created this company when he

had difficulty being accepted to medical school due to unfair quota systems that existed at the time.

We stand now at the beginning of a new era in medical education. As citizens of this 21st-century world of healthcare, we are charged with creating a patient-oriented, culturally competent, cost-conscious, universally available, technically advanced, and research-focused healthcare system, run by compassionate providers. Suffice it to say, this is no easy task. Problem-based learning, integrated curricula, and classes in interpersonal skills are some of the responses to this demand for an excellent workforce—a workforce of which you'll soon be a part.

We're thrilled that you've chosen us to help you on this journey. Please reach out to us to share your challenges, concerns, and successes. Together, we will shape the future of medicine in the United States and abroad; we look forward to helping you become the doctor you deserve to be.

Good luck!

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About *Scientific American*

As the world's premier science and technology magazine, and the oldest continuously published magazine in the United States, *Scientific American* is committed to bringing the most important developments in modern science, medicine, and technology to our worldwide audience in an understandable, credible, and provocative format.

Founded in 1845 and on the “cutting edge” ever since, *Scientific American* boasts over 150 Nobel laureate authors including Albert Einstein, Francis Crick, Stanley Prusiner, and Richard Axel. *Scientific American* is a forum where scientific theories and discoveries are explained to a broader audience.

Scientific American published its first foreign edition in 1890, and in 1979 was the first Western magazine published in the People's Republic of China. Today, *Scientific American* is published in 14 foreign language editions. *Scientific American* is also a leading online destination (www.ScientificAmerican.com), providing the latest science news and exclusive features to millions of visitors each month.

The knowledge that fills our pages has the power to spark new ideas, paradigms, and visions for the future. As science races forward, *Scientific American* continues to cover the promising strides, inevitable setbacks and challenges, and new medical discoveries as they unfold.

About the MCAT

ANATOMY OF THE MCAT

Here is a general overview of the structure of Test Day:

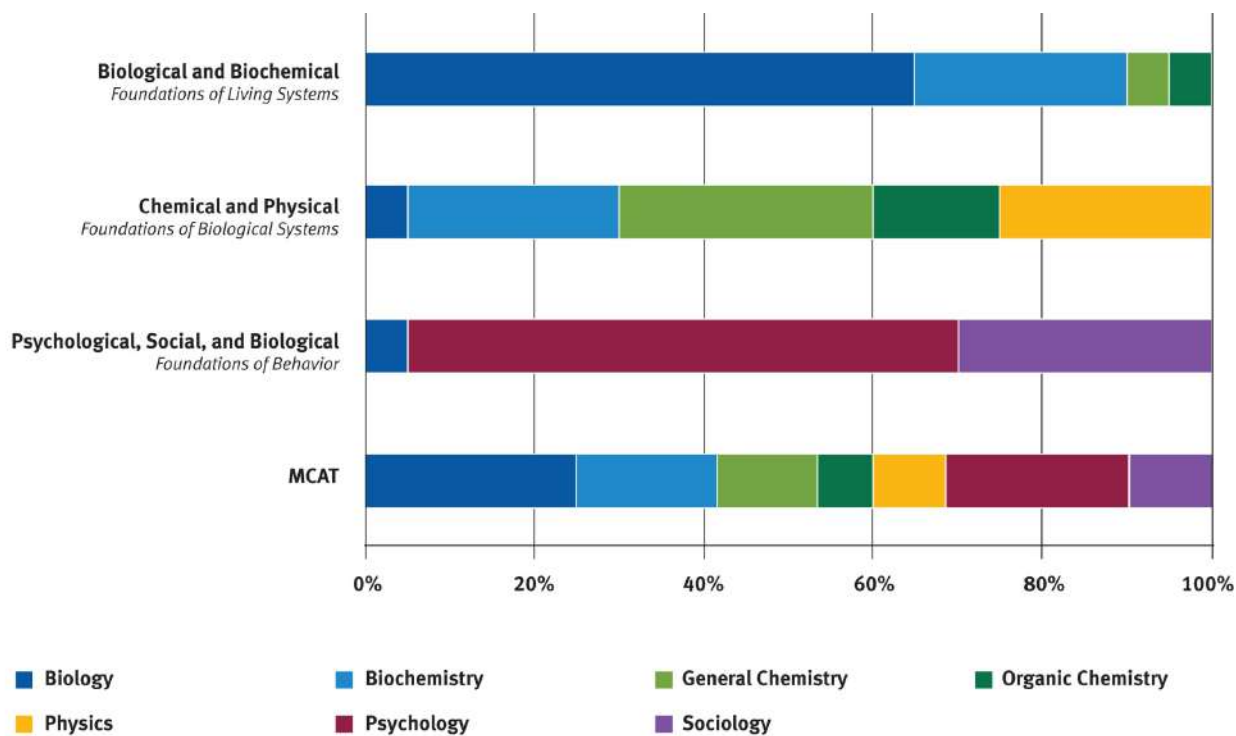
Section	Number of Questions	Time Allotted
Test-Day Certification		4 minutes
Tutorial (optional)		10 minutes
Chemical and Physical Foundations of Biological Systems	59	95 minutes
Break (optional)		10 minutes
Critical Analysis and Reasoning Skills (CARS)	53	90 minutes
Lunch Break (optional)		30 minutes
Biological and Biochemical Foundations of Living Systems	59	95 minutes
Break (optional)		10 minutes
Psychological, Social, and Biological Foundations of Behavior	59	95 minutes
Void Question		3 minutes
Satisfaction Survey (optional)		5 minutes

The structure of the four sections of the MCAT is shown below.

Chemical and Physical Foundations of Biological Systems

Time	95 minutes
Format	• 59 questions • 10 passages • 44 questions are passage-based, and 15 are discrete (stand-alone) questions. • Score between 118 and 132
What It Tests	• Biochemistry: 25% • Biology: 5% • General Chemistry: 30% • Organic Chemistry: 15% • Physics: 25%
Critical Analysis and Reasoning Skills (CARS)	
Time	90 minutes
Format	• 53 questions • 9 passages • All questions are passage-based. There are no discrete (stand-alone) questions. • Score between 118 and 132
What It Tests	Disciplines: • Humanities: 50% • Social Sciences: 50% Skills: • <i>Foundations of Comprehension</i>: 30% • <i>Reasoning Within the Text</i>: 30% • <i>Reasoning Beyond the Text</i>: 40%
Biological and Biochemical Foundations of Living Systems	
Time	95 minutes
Format	• 59 questions • 10 passages • 44 questions are passage-based, and 15 are discrete (stand-alone) questions.

	• Score between 118 and 132
What It Tests	• Biochemistry: 25% • Biology: 65% • General Chemistry: 5% • Organic Chemistry: 5%
Psychological, Social, and Biological Foundations of Behavior	
Time	95 minutes
Format	• 59 questions • 10 passages • 44 questions are passage-based, and 15 are discrete (stand-alone) questions. • Score between 118 and 132
What It Tests	• Biology: 5% • Psychology: 65% • Sociology: 30%
Total	
Testing Time	375 minutes (6 hours, 15 minutes)
Total Seat Time	447 minutes (7 hours, 27 minutes)
Questions	230
Score	472 to 528



SCIENTIFIC INQUIRY AND REASONING SKILLS (SIRS)

The AAMC has defined four *Scientific Inquiry and Reasoning Skills* (SIRS) that will be tested in the three science sections of the MCAT:

- 1. *Knowledge of Scientific Concepts and Principles* (35% of questions)
- 2. *Scientific Reasoning and Problem-Solving* (45% of questions)
- 3. *Reasoning About the Design and Execution of Research* (10% of questions)
- 4. *Data-Based and Statistical Reasoning* (10% of questions)

Let's see how each one breaks down into more specific Test Day behaviors. Note that the bullet points of specific objectives for each of the SIRS are taken directly from the *Official Guide to the MCAT Exam*; the descriptions of what

these behaviors mean and sample question stems, however, are written by Kaplan.

Skill 1: Knowledge of Scientific Concepts and Principles

This is probably the least surprising of the four SIRS; the testing of science knowledge is, after all, one of the signature qualities of the MCAT. Skill 1 questions will require you to do the following:

- Recognize correct scientific principles
- Identify the relationships among closely related concepts
- Identify the relationships between different representations of concepts (verbal, symbolic, graphic)
- Identify examples of observations that illustrate scientific principles
- Use mathematical equations to solve problems

At Kaplan, we simply call these Science Knowledge or Skill 1 questions. Another way to think of Skill 1 questions is as “one-step” problems. The single step is either to realize which scientific concept the question stem is suggesting or to take the concept stated in the question stem and identify which answer choice is an accurate application of it. Skill 1 questions are particularly prominent among discrete questions (those not associated with a passage). These questions are an opportunity to gain quick points on Test Day—if you know the science concept attached to the question, then that’s it! On Test Day, 35% of the questions in each science section will be Skill 1 questions.

Here are some sample Skill 1 question stems:

- How would a proponent of the James–Lange theory of emotion interpret

- the findings of the study cited in the passage?
- Which of the following most accurately describes the function of FSH in the human female menstrual cycle?
 - If the products of Reaction 1 and Reaction 2 were combined in solution, the resulting reaction would form:
 - Ionic bonds are maintained by which of the following forces?

Skill 2: Scientific Reasoning and Problem-Solving

The MCAT science sections do, of course, move beyond testing straightforward science knowledge; Skill 2 questions are the most common way in which it does so. At Kaplan, we also call these Critical Thinking questions. Skill 2 questions will require you to do the following:

- Reason about scientific principles, theories, and models
- Analyze and evaluate scientific explanations and predictions
- Evaluate arguments about causes and consequences
- Bring together theory, observations, and evidence to draw conclusions
- Recognize scientific findings that challenge or invalidate a scientific theory or model
- Determine and use scientific formulas to solve problems

Just as Skill 1 questions can be thought of as “one-step” problems, many Skill 2 questions are “two-step” problems, and more difficult Skill 2 questions may require three or more steps. These questions can require a wide spectrum of reasoning skills, including integration of multiple facts from a passage, combination of multiple science content areas, and prediction of an experiment’s results. Skill 2 questions also tend to ask about science content without actually mentioning it by name. For example, a question might describe the results of one experiment and ask you to predict the results of a

second experiment without actually telling you what underlying scientific principles are at work—part of the question’s difficulty will be figuring out which principles to apply in order to get the correct answer. On Test Day, 45% of the questions in each science section will be Skill 2 questions.

Here are some sample Skill 2 question stems:

- Which of the following experimental conditions would most likely yield results similar to those in Figure 2?
- All of the following conclusions are supported by the information in the passage EXCEPT:
- The most likely cause of the anomalous results found by the experimenter is:
- An impact to a man’s chest quickly reduces the volume of one of his lungs to 70% of its initial value while not allowing any air to escape from the man’s mouth. By what percentage is the force of outward air pressure increased on a 2 cm^2 portion of the inner surface of the compressed lung?

Skill 3: Reasoning About the Design and Execution of Research

The MCAT is interested in your ability to critically appraise and analyze research, as this is an important day-to-day task of a physician. We call these questions Skill 3 or Experimental and Research Design questions for short. Skill 3 questions will require you to do the following:

- Identify the role of theory, past findings, and observations in scientific questioning
- Identify testable research questions and hypotheses
- Distinguish between samples and populations and distinguish results that support generalizations about populations

- Identify independent and dependent variables
- Reason about the features of research studies that suggest associations between variables or causal relationships between them (such as temporality and random assignment)
- Identify conclusions that are supported by research results
- Determine the implications of results for real-world situations
- Reason about ethical issues in scientific research

Over the years, the AAMC has received input from medical schools to require more practical research skills of MCAT test takers, and Skill 3 questions are the response to these demands. This skill is unique in that the outside knowledge you need to answer Skill 3 questions is not taught in any one undergraduate course; instead, the research design principles needed to answer these questions are learned gradually throughout your science classes and especially through any laboratory work you have completed. It should be noted that Skill 3 comprises 10% of the questions in each science section on Test Day.

Here are some sample Skill 3 question stems:

- What is the dependent variable in the study described in the passage?
- The major flaw in the method used to measure disease susceptibility in Experiment 1 is:
- Which of the following procedures is most important for the experimenters to follow in order for their study to maintain a proper, randomized sample of research subjects?
- A researcher would like to test the hypothesis that individuals who move to an urban area during adulthood are more likely to own a car than are those who have lived in an urban area since birth. Which of the following studies would best test this hypothesis?

Skill 4: Data-Based and Statistical Reasoning

Lastly, the science sections of the MCAT test your ability to analyze the visual and numerical results of experiments and studies. We call these Data and Statistical Analysis questions. Skill 4 questions will require you to do the following:

- Use, analyze, and interpret data in figures, graphs, and tables
- Evaluate whether representations make sense for particular scientific observations and data
- Use measures of central tendency (mean, median, and mode) and measures of dispersion (range, interquartile range, and standard deviation) to describe data
- Reason about random and systematic error
- Reason about statistical significance and uncertainty (interpreting statistical significance levels and interpreting a confidence interval)
- Use data to explain relationships between variables or make predictions
- Use data to answer research questions and draw conclusions

Skill 4 is included in the MCAT because physicians and researchers spend much of their time examining the results of their own studies and the studies of others, and it's very important for them to make legitimate conclusions and sound judgments based on that data. The MCAT tests Skill 4 on all three science sections with graphical representations of data (charts and bar graphs), as well as numerical ones (tables, lists, and results summarized in sentence or paragraph form). On Test Day, 10% of the questions in each science section will be Skill 4 questions.

Here are some sample Skill 4 question stems:

- According to the information in the passage, there is an inverse correlation between:
- What conclusion is best supported by the findings displayed in Figure 2?
- A medical test for a rare type of heavy metal poisoning returns a positive result for 98% of affected individuals and 13% of unaffected individuals. Which of the following types of errors is most prevalent in this test?
- If a fourth trial of Experiment 1 was run and yielded a result of 54% compliance, which of the following would be true?

SIRS Summary

Discussing the SIRS tested on the MCAT is a daunting prospect given that the very nature of the skills tends to make the conversation rather abstract. Nevertheless, with enough practice, you'll be able to identify each of the four skills quickly, and you'll also be able to apply the proper strategies to solve those problems on Test Day. If you need a quick reference to remind you of the four SIRS, these guidelines may help:

Skill 1 (Science Knowledge) questions ask:

- Do you remember this science content?

Skill 2 (Critical Thinking) questions ask:

- Do you remember this science content? And if you do, could you please apply it to this novel situation?
- Could you answer this question that cleverly combines multiple content areas at the same time?

Skill 3 (Experimental and Research Design) questions ask:

- Let's forget about the science content for a while. Could you give some insight into the experimental or research methods involved in this situation?

Skill 4 (Data and Statistical Analysis) questions ask:

- Let's forget about the science content for a while. Could you accurately read some graphs and tables for a moment? Could you make some conclusions or extrapolations based on the information presented?

CRITICAL ANALYSIS AND REASONING SKILLS (CARS)

The *Critical Analysis and Reasoning Skills* (CARS) section of the MCAT tests three discrete families of textual reasoning skills; each of these families requires a higher level of reasoning than the last. Those three skills are as follows:

- .. *Foundations of Comprehension* (30% of questions)
- !. *Reasoning Within the Text* (30% of questions)
- }. *Reasoning Beyond the Text* (40% of questions)

These three skills are tested through nine humanities- and social sciences-themed passages, with approximately 5 to 7 questions per passage. Let's take a more in-depth look into these three skills. Again, the bullet points of specific objectives for each of the CARS are taken directly from the *Official Guide to the MCAT Exam*; the descriptions of what these behaviors mean and sample question stems, however, are written by Kaplan.

Foundations of Comprehension

Questions in this skill will ask for basic facts and simple inferences about the passage; the questions themselves will be similar to those seen on reading comprehension sections of other standardized exams like the SAT® and ACT®. *Foundations of Comprehension* questions will require you to do the following:

- Understand the basic components of the text
- Infer meaning from rhetorical devices, word choice, and text structure

This admittedly covers a wide range of potential question types including Main Idea, Detail, Inference, and Definition-in-Context questions, but finding the correct answer to all *Foundations of Comprehension* questions will follow from a basic understanding of the passage and the point of view of its author (and occasionally that of other voices in the passage).

Here are some sample *Foundations of Comprehension* question stems:

- **Main Idea**—The author’s primary purpose in this passage is:
- **Detail**—Based on the information in the second paragraph, which of the following is the most accurate summary of the opinion held by Schubert’s critics?
- **(Scattered) Detail**—According to the passage, which of the following is FALSE about literary reviews in the 1920s?
- **Inference (Implication)**—Which of the following phrases, as used in the passage, is most suggestive that the author has a personal bias toward narrative records of history?
- **Inference (Assumption)**—In putting together her argument in the passage, the author most likely assumes:
- **Definition-in-Context**—The word “obscure” (paragraph 3), when used in reference to the historian’s actions, most nearly means:

Reasoning Within the Text

While *Foundations of Comprehension* questions will usually depend on interpreting a single piece of information in the passage or understanding the passage as a whole, *Reasoning Within the Text* questions require more thought because they will ask you to identify the purpose of a particular piece of information in the context of the passage, or ask how one piece of information relates to another. *Reasoning Within the Text* questions will require you to:

- Integrate different components of the text to draw relevant conclusions

The CARS section will also ask you to judge certain parts of the passage or even judge the author. These questions, which fall under the *Reasoning Within the Text* skill, can ask you to identify authorial bias, evaluate the credibility of cited sources, determine the logical soundness of an argument, identify the importance of a particular fact or statement in the context of the passage, or search for relevant evidence in the passage to support a given conclusion. In all, this category includes Function and Strengthen–Weaken (Within the Passage) questions, as well as a smattering of related—but rare—question types.

Here are some sample *Reasoning Within the Text* question stems:

- **Function**—The author’s discussion of the effect of socioeconomic status on social mobility primarily serves which of the following functions?
- **Strengthen–Weaken (Within the Passage)**—Which of the following facts is used in the passage as the most prominent piece of evidence in favor of the author’s conclusions?
- **Strengthen–Weaken (Within the Passage)**—Based on the role it plays in

the author's argument, *The Possessed* can be considered:

Reasoning Beyond the Text

The distinguishing factor of *Reasoning Beyond the Text* questions is in the title of the skill: the word *Beyond*. Questions that test this skill, which make up a larger share of the CARS section than questions from either of the other two skills, will always introduce a completely new situation that was not present in the passage itself; these questions will ask you to determine how one influences the other. *Reasoning Beyond the Text* questions will require you to:

- Apply or extrapolate ideas from the passage to new contexts
- Assess the impact of introducing new factors, information, or conditions to ideas from the passage

The *Reasoning Beyond the Text* skill is further divided into Apply and Strengthen–Weaken (Beyond the Passage) questions, and a few other rarely appearing question types.

Here are some sample *Reasoning Beyond the Text* question stems:

- **Apply**—If a document were located that demonstrated Berlioz intended to include a chorus of at least 700 in his *Grande Messe des Morts*, how would the author likely respond?
- **Apply**—Which of the following is the best example of a “virtuous rebellion,” as it is defined in the passage?
- **Strengthen–Weaken (Beyond the Passage)**—Suppose Jane Austen had written in a letter to her sister, “My strongest characters were those forced by circumstance to confront basic questions about the society in which they lived.” What relevance would this have to the passage?

- **Strengthen–Weaken (Beyond the Passage)**—Which of the following sentences, if added to the end of the passage, would most WEAKEN the author’s conclusions in the last paragraph?

CARS Summary

Through the *Foundations of Comprehension* skill, the CARS section tests many of the reading skills you have been building on since grade school, albeit in the context of very challenging doctorate-level passages. But through the two other skills (*Reasoning Within the Text* and *Reasoning Beyond the Text*), the MCAT demands that you understand the deep structure of passages and the arguments within them at a very advanced level. And, of course, all of this is tested under very tight timing restrictions: only 102 seconds per question—and that doesn't even include the time spent reading the passages.

Here’s a quick reference guide to the three CARS skills:

Foundations of Comprehension questions ask:

- Did you understand the passage and its main ideas?
- What does the passage have to say about this particular detail?
- What must be true that the author did not say?

Reasoning Within the Text questions ask:

- What’s the logical relationship between these two ideas from the passage?
- How well argued is the author’s thesis?

Reasoning Beyond the Text questions ask:

- How does this principle from the passage apply to this new situation?
- How does this new piece of information influence the arguments in the passage?

Scoring

Each of the four sections of the MCAT is scored between 118 and 132, with the median at 125. This means the total score ranges from 472 to 528, with the median at 500. Why such peculiar numbers? The AAMC stresses that this scale emphasizes the importance of the central portion of the score distribution, where most students score (around 125 per section, or 500 total), rather than putting undue focus on the high end of the scale.

Note that there is no wrong answer penalty on the MCAT, so you should select an answer for every question—even if it is only a guess.

The AAMC has released the 2019–2020 correlation between scaled score and percentile, as shown on the following page. It should be noted that the percentile scale is adjusted and renormalized over time and thus can shift slightly from year to year.

Total Score	Percentile	Total Score	Percentile
528	100	499	44
527	100	498	41
526	100	497	37
525	100	496	34
524	100	495	31
523	99	494	28

522	99	493	25
521	99	492	23
520	98	491	20
519	97	490	18
518	96	489	16
517	95	488	14
516	93	487	12
515	92	486	10
514	90	485	8
513	88	484	7
512	85	483	6
511	83	482	5
510	80	481	4
509	77	480	3
508	74	479	2
507	71	478	2
506	68	477	1
505	65	476	1
504	61	475	<1
503	58	474	<1
502	54	473	<1
501	51	472	<1

500	47		
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Source: AAMC. 2019. *Summary of MCAT Total and Section Scores*. Accessed November 2019. <https://students-residents.aamc.org/advisors/article/percentile-ranks-for-the-mcat-exam/>

Further information on score reporting is included at the end of the next section (see *After Your Test*).

MCAT POLICIES AND PROCEDURES

We strongly encourage you to download the latest copy of *MCAT® Essentials*, available on the AAMC's website, to ensure that you have the latest information about registration and Test Day policies and procedures; this document is updated annually. A brief summary of some of the most important rules is provided here.

MCAT Registration

The only way to register for the MCAT is online. You can access AAMC's registration system at www.aamc.org/mcat.

You will be able to access the site approximately six months before Test Day. The AAMC designates three registration “Zones”—Gold, Silver, and Bronze. Registering during the Gold Zone (from the opening of registration until approximately one month before Test Day) provides the most flexibility and lowest test fees. The Silver Zone runs until approximately two to three weeks before Test Day and has less flexibility and higher fees; the Bronze Zone runs until approximately one to two weeks before Test Day and has the least flexibility and highest fees.

Fees and the Fee Assistance Program (FAP)

Payment for test registration must be made by MasterCard or VISA. As described earlier, the fees for registering for the MCAT—as well as rescheduling the exam or changing your testing center—increase as one approaches Test Day. In addition, it is not uncommon for test centers to fill up well in advance of the registration deadline. For these reasons, we recommend identifying your preferred Test Day as soon as possible and registering. There are ancillary benefits to having a set Test Day, as well: when you know the date you're working toward, you'll study harder and are less likely to keep pushing back the exam. The AAMC offers a Fee Assistance Program (FAP) for students with financial hardship to help reduce the cost of taking the MCAT, as well as for the American Medical College Application Service (AMCAS[®]) application. Further information on the FAP can be found at www.aamc.org/students/applying/fap.

Testing Security

On Test Day, you will be required to present a qualifying form of ID. Generally, a current driver's license or United States passport will be sufficient (consult the AAMC website for the full list of qualifying criteria). When registering, take care to spell your first and last names (middle names, suffixes, and prefixes are not required and will not be verified on Test Day) precisely the same as they appear on this ID; failure to provide this ID at the test center or differences in spelling between your registration and ID will be considered a “no-show,” and you will not receive a refund for the exam.

During Test Day registration, other identity data collected may include: a digital palm vein scan, a Test Day photo, a digitization of your valid ID, and signatures. Some testing centers may use a metal detection wand to ensure that no prohibited items are brought into the testing room. Prohibited items

include all electronic devices, including watches and timers, calculators, cell phones, and any and all forms of recording equipment; food, drinks (including water), and cigarettes or other smoking paraphernalia; hats and scarves (except for religious purposes); and books, notes, or other study materials. If you require a medical device, such as an insulin pump or pacemaker, you must apply for accommodated testing. During breaks, you are allowed to access food and drink, but not to electronic devices, including cell phones.

Testing centers are under video surveillance and the AAMC does not take potential violations of testing security lightly. The bottom line: *know the rules and don't break them.*

Accommodations

Students with disabilities or medical conditions can apply for accommodated testing. Documentation of the disability or condition is required, and requests may take two months—or more—to be approved. For this reason, it is recommended that you begin the process of applying for accommodated testing as early as possible. More information on applying for accommodated testing can be found at

www.aamc.org/students/applying/mcat/accommodations.

After Your Test

When your MCAT is all over, no matter how you feel you did, be good to yourself when you leave the test center. Celebrate! Take a nap. Watch a movie. Ride your bike. Plan a trip. Call up all of your neglected friends or stalk them on Facebook. Totally consume a cheesesteak and drink dirty martinis at night (assuming you're over 21). Whatever you do, make sure that it has absolutely nothing to do with thinking too hard—you deserve some rest and relaxation.

Perhaps most importantly, do not discuss specific details about the test with anyone. For one, it is important to let go of the stress of Test Day, and reliving your exam only inhibits you from being able to do so. But more significantly, the Examinee Agreement you sign at the beginning of your exam specifically prohibits you from discussing or disclosing exam content. The AAMC is known to seek out individuals who violate this agreement and retains the right to prosecute these individuals at their discretion. This means that you should not, under any circumstances, discuss the exam in person or over the phone with other individuals—including us at Kaplan—or post information or questions about exam content to Facebook, Student Doctor Network, or other online social media. You are permitted to comment on your “general exam experience,” including how you felt about the exam overall or an individual section, but this is a fine line. In summary: *if you're not certain whether you can discuss an aspect of the test or not, just don't do it!* Do not let a silly Facebook post stop you from becoming the doctor you deserve to be.

Scores are released approximately one month after Test Day. The release is staggered during the afternoon and evening, ending at 5 p.m. Eastern Standard Time. This means that not all examinees receive their scores at exactly the same time. Your score report will include a scaled score for each section between 118 and 132, as well as your total combined score between 472 and 528. These scores are given as confidence intervals. For each section, the confidence interval is approximately the given score ± 1 ; for the total score, it is approximately the given score ± 2 . You will also be given the corresponding percentile rank for each of these section scores and the total score.

AAMC CONTACT INFORMATION

For further questions, contact the MCAT team at the Association of American Medical Colleges:

MCAT Resource Center
Association of American Medical Colleges

www.aamc.org/mcat

(202) 828-0600

www.aamc.org/contactmcat

How This Book Was Created

The *Kaplan MCAT Review* project began shortly after the release of the *Preview Guide for the MCAT 2015 Exam*, 2nd edition. Through thorough analysis by our staff psychometricians, we were able to analyze the relative yield of the different topics on the MCAT, and we began constructing tables of contents for the books of the *Kaplan MCAT Review* series. A dedicated staff of 30 writers, 7 editors, and 32 proofreaders worked over 5,000 combined hours to produce these books. The format of the books was heavily influenced by weekly meetings with Kaplan's learning science team.

In the years since this book was created, a number of opportunities for expansion and improvement have occurred. The current edition represents the culmination of the wisdom accumulated during that time frame, and it also includes several new features designed to improve the reading and learning experience in these texts.

These books were submitted for publication in April 2020. For any updates after this date, please visit www.kaptest.com/retail-book-corrections-and-updates.

If you have any questions about the content presented here, email KaplanMCATfeedback@kaplan.com. For other questions not related to content, email booksupport@kaplan.com.

Each book has been vetted through at least ten rounds of review. To that end, the information presented in these books is true and accurate to the best of

our knowledge. Still, your feedback helps us improve our prep materials. Please notify us of any inaccuracies or errors in the books by sending an email to KaplanMCATfeedback@kaplan.com.

Using This Book

Kaplan MCAT Biology Review, and the other six books in the *Kaplan MCAT Review* series, bring the Kaplan classroom experience to you—right in your home, at your convenience. This book offers the same Kaplan content review, strategies, and practice that make Kaplan the #1 choice for MCAT prep.

This book is designed to help you review the biology topics covered on the MCAT. Please understand that content review—no matter how thorough—is not sufficient preparation for the MCAT! The MCAT tests not only your science knowledge but also your critical reading, reasoning, and problem-solving skills. Do not assume that simply memorizing the contents of this book will earn you high scores on Test Day; to maximize your scores, you must also improve your reading and test-taking skills through MCAT-style questions and practice tests.

LEARNING OBJECTIVES

At the beginning of each section, you'll find a short list of objectives describing the skills covered within that section. Learning objectives for these texts were developed in conjunction with Kaplan's learning science team, and have been designed specifically to focus your attention on tasks and concepts that are likely to show up on your MCAT. These learning objectives will function as a means to guide your study, and indicate what information and relationships you should be focused on within each section. Before starting each section, read these learning objectives carefully. They will not only allow

you to assess your existing familiarity with the content, but also provide a goal-oriented focus for your studying experience of the section.

MCAT CONCEPT CHECKS

At the end of each section, you'll find a few open-ended questions that you can use to assess your mastery of the material. These MCAT Concept Checks were introduced after numerous conversations with Kaplan's learning science team. Research has demonstrated repeatedly that introspection and self-analysis improve mastery, retention, and recall of material. Complete these MCAT Concept Checks to ensure that you've got the key points from each section before moving on!

PRACTICE QUESTIONS

At the end of each chapter, you'll find 15 MCAT-style practice questions. These are designed to help you assess your understanding of the chapter you just read. Most of these questions focus on the first of the *Scientific Inquiry and Reasoning Skills (Knowledge of Scientific Concepts and Principles)*, although there are occasional questions that fall into the second or fourth SIRS (*Scientific Reasoning and Problem-Solving* and *Data-Based and Statistical Reasoning*, respectively).

GUIDED EXAMPLES WITH EXPERT THINKING

Embedded in each chapter of this book is a Guided Example with Expert Thinking. These examples will be located adjacent to the content that they are related to, and contain an MCAT-like scientific article as a passage. Read

through the passage as you would on the real MCAT, referring to the Expert Thinking material on the right to clarify the key information you should be gathering from each passage. Read and attempt to answer the associated question once you have worked through the passage. There is a full explanation, including the correct answer, following the question given. These passages and questions are designed to help build your critical thinking, experimental reasoning, and data interpretation skills as preparation for the challenges you will face on the MCAT.

SIDEBARS

The following is a guide to the five types of sidebars you'll find in *Kaplan MCAT Biology Review*:

- **Bridge:** These sidebars create connections between science topics that appear in multiple chapters throughout the *Kaplan MCAT Review* series.
- **Key Concept:** These sidebars draw attention to the most important takeaways in a given topic, and they sometimes offer synopses or overviews of complex information. If you understand nothing else, make sure you grasp the Key Concepts for any given subject.
- **MCAT Expertise:** These sidebars point out how information may be tested on the MCAT or offer key strategy points and test-taking tips that you should apply on Test Day.
- **Mnemonic:** These sidebars present memory devices to help recall certain facts.
- **Real World:** These sidebars illustrate how a concept in the text relates to the practice of medicine or the world at large. While this is not information you need to know for Test Day, many of the topics in Real World sidebars are excellent examples of how a concept may appear in a passage or

discrete (stand-alone) question on the MCAT.

WHAT THIS BOOK COVERS

The information presented in the *Kaplan MCAT Review* series covers everything listed on the official MCAT content lists. Every topic in these lists is covered in the same level of detail as is common to the undergraduate and postbaccalaureate classes that are considered prerequisites for the MCAT. Note that your premedical classes may include topics not discussed in these books, or they may go into more depth than these books do. Additional exposure to science content is never a bad thing, but all of the content knowledge you are expected to have walking in on Test Day is covered in these books.

Chapter Profiles, on the first page of each chapter, represent a holistic look at the content within the chapter, and will include a pie chart as well as text information. The pie chart analysis is based directly on data released by the AAMC, and will give a rough estimate of the importance of the chapter in relation to the book as a whole. Further, the text portion of the Chapter Profiles includes which AAMC content categories are covered within the chapter. These are referenced directly from the AAMC MCAT exam content listing, available on the test maker's website.

You'll also see new High-Yield badges scattered throughout the sections of this book:

In This Chapter

1.1 Amino Acids Found in Proteins 	1.4 Primary and Secondary Protein Structure 
A Note on Terminology 4	Primary Structure 19
Stereochemistry of Amino Acids 5	Secondary Structure 20
Structures of the Amino Acids 5	1.5 Tertiary and Quaternary Protein Structure 
Hydrophobic and Hydrophilic Amino Acids 9	Tertiary Structure 22
Amino Acid Abbreviations 9	Folding and the Solvation Layer 23
1.2 Acid-Base Chemistry of Amino Acids	Quaternary Structure 24
Protonation and Deprotonation 12	Conjugated Proteins 25
Titration of Amino Acids 14	1.6 Denaturation 28
1.3 Peptide Bond Formation and Hydrolysis 	Concept Summary 30
Peptide Bond Formation 17	
Peptide Bond Hydrolysis 18	

1.1 Amino Acids Found in Proteins

High-Yield 

LEARNING OBJECTIVES

After Chapter 1.1, you will be able to:

These badges represent the top 100 topics most tested by the AAMC. In other words, according to the test maker and all our experience with their resources, a High-Yield badge means more questions on Test Day.

This book also contains a thorough glossary and index for easy navigation of the text.

In the end, this is your book, so write in the margins, draw diagrams, highlight the key points—do whatever is necessary to help you get that higher score. We look forward to working with you as you achieve your dreams and become the doctor you deserve to be!

STUDYING WITH THIS BOOK

In addition to providing you with the best practice questions and test strategies, Kaplan's team of learning scientists are dedicated to researching and testing the best methods for getting the most out of your study time. Here are their top four tips for improving retention:

Review multiple topics in one study session. This may seem counterintuitive—we're used to practicing one skill at a time in order to improve each skill. But research shows that weaving topics together leads to increased learning. Beyond that consideration, the MCAT often includes more than one topic in a single question. Studying in an integrated manner is the most effective way to prepare for this test.

Customize the content. Drawing attention to difficult or critical content can ensure you don't overlook it as you read and re-read sections. The best way to do this is to make it more visual—highlight, make tabs, use stickies, whatever works. We recommend highlighting only the most important or difficult sections of text. Selective highlighting of up to about 10 percent of text in a given chapter is great for emphasizing parts of the text, but over-highlighting can have the opposite effect.

Repeat topics over time. Many people try to memorize concepts by repeating them over and over again in succession. Our research shows that

retention is improved by spacing out the repeats over time and mixing up the order in which you study content. For example, try reading chapters in a different order the second (or third!) time around. Revisit practice questions that you answered incorrectly in a new sequence. Perhaps information you reviewed more recently will help you better understand those questions and solutions you struggled with in the past.

Take a moment to reflect. When you finish reading a section for the first time, stop and think about what you just read. Jot down a few thoughts in the margins or in your notes about why the content is important or what topics came to mind when you read it. Associating learning with a memory is a fantastic way to retain information! This also works when answering questions. After answering a question, take a moment to think through each step you took to arrive at a solution. What led you to the answer you chose? Understanding the steps you took will help you make good decisions when answering future questions.

ONLINE RESOURCES

In addition to the resources located within this text, you also have additional online resources awaiting you at www.kaptest.com/booksonline. Make sure to log on and take advantage of free practice and access to other resources!

Please note that access to the online resources is limited to the original owner of this book.

Studying for the MCAT

The first year of medical school is a frenzied experience for most students. To meet the requirements of a rigorous work schedule, students either learn to prioritize their time or else fall hopelessly behind. It's no surprise, then, that the MCAT, the test specifically designed to predict success in medical school, is a high-speed, time-intensive test. The MCAT demands excellent time-management skills, endurance, as well as grace under pressure both during the test as well as while preparing for it. Having a solid plan of attack and sticking with it are key to giving you the confidence and structure you need to succeed.

CREATING A STUDY PLAN

The best time to create a study plan is at the beginning of your MCAT preparation. If you don't already use a calendar, you will want to start. You can purchase a planner, print out a free calendar from the Internet, use a built-in calendar or app on one of your smart devices, or keep track using an interactive online calendar. Pick the option that is most PRACTICAL for you and that you are most likely to use consistently.

Once you have a calendar, you'll be able to start planning your study schedule with the following steps:

.. Fill in your obligations and choose a day off.

Write in all your school, extracurricular, and work obligations first: class sessions, work shifts, and meetings that you must attend. Then add in your personal obligations: appointments, lunch dates, family and social time, etc. Making an appointment in your calendar for hanging out with friends or going to the movies may seem strange at first, but planning social activities in advance will help you achieve a balance between personal and professional obligations even as life gets busy. Having a happy balance allows you to be more focused and productive when it comes time to study, so stay well-rounded and don't neglect anything that is important to you.

In addition to scheduling your personal and professional obligations, you should also plan your time off. Taking some time off is just as important as studying. Kaplan recommends taking at least one full day off per week, ideally from all your study obligations but at minimum from studying for the MCAT.

!. Add in study blocks around your obligations.

Once you have established your calendar's framework, add in study blocks around your obligations, keeping your study schedule as consistent as possible across days and across weeks. Studying at the same time of day as your official test is ideal for promoting recall, but if that's not possible, then fit in study blocks wherever you can.

To make your studying as efficient as possible, block out short, frequent periods of study time throughout the week. From a learning perspective, studying one hour per day for six days per week is much more valuable than studying for six hours all at once one day per week. Specifically, Kaplan recommends studying for no longer than three hours in one sitting. Within those three-hour blocks, also plan to take ten-minute breaks every

hour. Use these breaks to get up from your seat, do some quick stretches, get a snack and drink, and clear your mind. Although ten minutes of break for every 50 minutes of studying may sound like a lot, these breaks will allow you to deal with distractions and rest your brain so that, during the 50-minute study blocks, you can remain fully engaged and completely focused.

3. Add in your full-length practice tests.

Next, you'll want to add in full-length practice tests. You'll want to take one test very early in your prep and then spread your remaining full-length practice tests evenly between now and your test date. Staggering tests in this way allows you to form a baseline for comparison and to determine which areas to focus on right away, while also providing realistic feedback throughout your prep as to how you will perform on Test Day.

When planning your calendar, aim to finish your full-length practice tests and the majority of your studying by one week before Test Day, which will allow you to spend that final week completing a final, brief review of what you already know. In your online resources, you'll find sample study calendars for several different Test Day timelines to use as a starting point. The sample calendars may include more focus than you need in some areas, and less in others, and it may not fit your timeline to Test Day. You will need to customize your study calendar to your needs using the steps above.

The total amount of time you spend studying each week will depend on your schedule, your personal prep needs, and your time to Test Day, but it is recommended that you spend somewhere in the range of 300-350 hours preparing before taking the official MCAT. One way you could break this down is to study for three hours per day, six days per week, for four months, but this is just one approach. You might study six days per week for more than three

hours per day. You might study over a longer period of time if you don't have much time to study each week. No matter what your plan is, ensure you complete enough practice to feel completely comfortable with the MCAT and its content. A good sign you're ready for Test Day is when you begin to earn your goal score consistently in practice.

HOW TO STUDY

The MCAT covers a large amount of material, so studying for Test Day can initially seem daunting. To combat this, we have some tips on how to take control of your studying and make the most of your time.

Goal Setting

To take control of the amount of content and practice required to do well on the MCAT, break the content down into specific goals for each week instead of attempting to approach the test as a whole. A goal of "I want to increase my overall score by 5 points" is too big, abstract, and difficult to measure on the small scale. More reasonable goals are "I will read two chapters each day this week." Goals like this are much less overwhelming and help break studying into manageable pieces.

Active Reading

As you go through this book, much of the information will be familiar to you. After all, you have probably seen most of the content before. However, be very careful: Familiarity with a subject does not necessarily translate to knowledge or mastery of that subject. Do not assume that if you recognize a concept that you actually know it and can apply it quickly at an appropriate

level. Don't just passively read this book. Instead, read actively: Use the free margin space to jot down important ideas, draw diagrams, and make charts as you read. Highlighting can be an excellent tool, but use it sparingly: highlighting every sentence isn't active reading, it's coloring. Frequently stop and ask yourself questions while you read (e.g., *What is the main point? How does this fit into the overall scheme of things? Could I thoroughly explain this to someone else?*). *By making connections and focusing on the grander scheme, not only will you ensure you know the essential content, but you also prepare yourself for the level of critical thinking required by the MCAT.*

Focus on Areas of Greatest Opportunity

If you are limited by only having a minimal amount of time to prepare before Test Day, focus on your biggest areas of opportunity first. Areas of opportunity are topic areas that are highly tested and that you have not yet mastered. You likely won't have time to take detailed notes for every page of these books; instead, use your results from practice materials to determine which areas are your biggest opportunities and seek those out. After you've taken a full-length test, make sure you are using Smart Reports to best identify areas of opportunity. Skim over content matter for which you are already demonstrating proficiency, pausing to read more thoroughly when something looks unfamiliar or particularly difficult. Consider starting with the Review Problems at the end of each chapter. If you can get all of those questions correct within a reasonable amount of time, you may be able to quickly skim through that chapter, but if the questions prove to be more difficult, then you may need to spend time reading the chapter or certain subsections of the chapter more thoroughly.

Practice, Review, and Tracking

Leave time to review your practice questions and full-length tests. You may be tempted, after practicing, to push ahead and cover new material as quickly as possible, but failing to schedule ample time for review will actually throw away your greatest opportunity to improve your performance. The brain rarely remembers anything it sees or does only once. When you carefully review the questions you've solved (and the explanations for them), the process of retrieving that information reopens and reinforces the connections you've built in your brain. This builds long-term retention and repeatable skill sets—exactly what you need to beat the MCAT!

While reviewing, take notes about the specific reasons why you missed questions you got wrong or had to guess on, perhaps by using a spreadsheet like the one below. Keep adding to the same Why I Missed It Sheet (WIMIS) as you complete more practice questions, and periodically review your WIMIS to identify any patterns you see, such as consistently missing questions in certain content areas or falling for the same test maker traps. As you move through your MCAT prep, adjust your study plan based on your available study time and the results of your review. Your strengths and weaknesses are likely to change over the course of your prep. Keep addressing the areas that are most important to your score, shifting your focus as those areas change. For more help with reviewing and making the most of your full-length tests, including a WIMIS template, make sure to check out the videos and resources in your online syllabus.

Section	Q #	Topic or Type	Wrong answer chosen	Why I missed it
<i>Chem/Phys</i>	<i>42</i>	<i>Nuclear chem.</i>	<i>Opposite</i>	<i>Confused electron absorption and emission</i>

<i>Chem/Phys</i>	<i>47</i>	<i>K_{eq}</i>	<i>Miscalculation</i>	<i>Need to memorize equation</i>
<i>CARS</i>	<i>2</i>	<i>Detail</i>	<i>Opposite</i>	<i>Didn't read "not" in answer choice; slow down!</i>
<i>CARS</i>	<i>4</i>	<i>Inference</i>	<i>Out of Scope</i>	<i>Forgot to research answer</i>

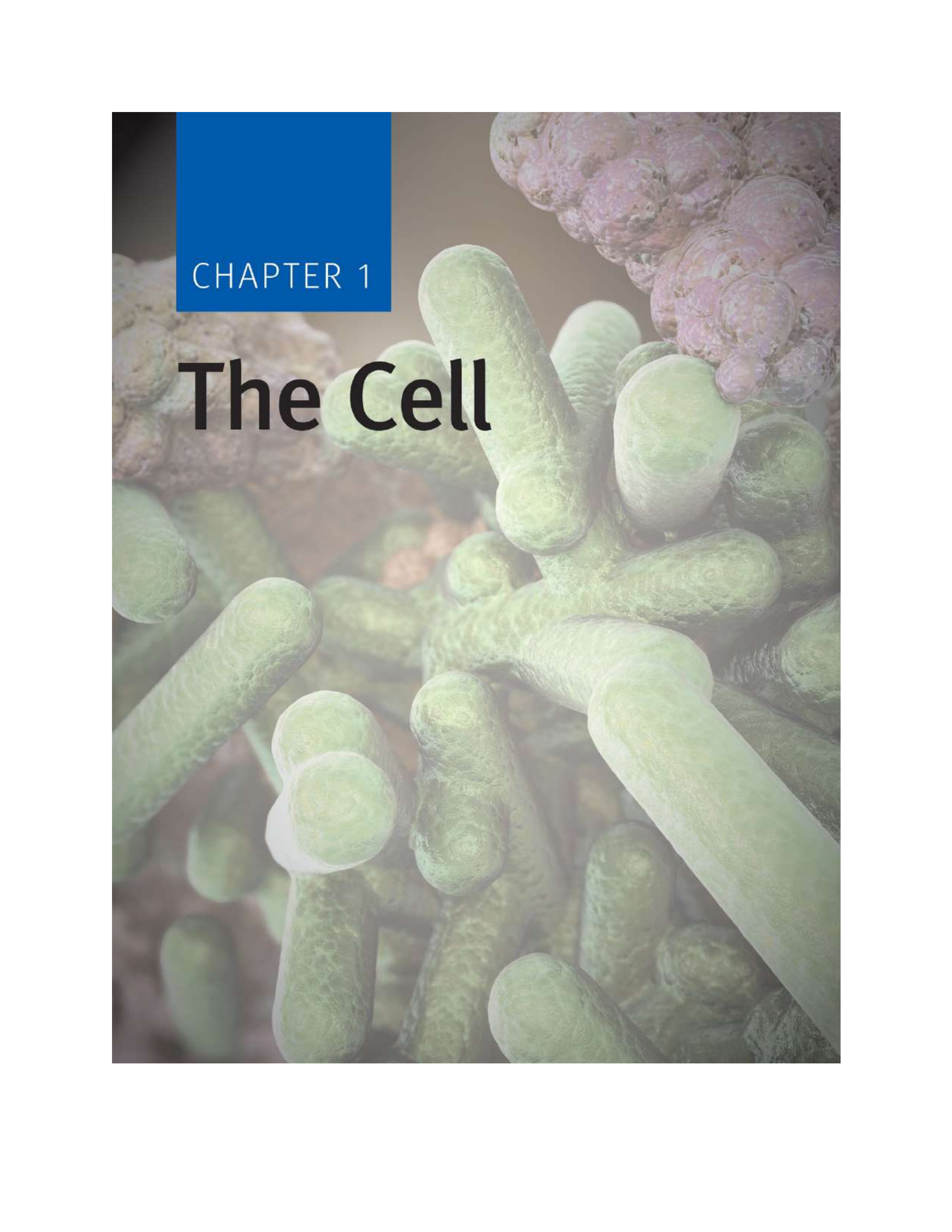
Where to Study

One often-overlooked aspect of studying is the environment where the learning actually occurs. Although studying at home is many students' first choice, several problems can arise in this environment, chief of which are distractions. Studying can be a mentally draining process, so as time passes, these distractions become ever more tempting as escape routes. Although you may have considerable willpower, there's no reason to make staying focused harder than it needs to be. Instead of studying at home, head to a library, quiet coffee shop, or another new location whenever possible. This will eliminate many of the usual distractions and also promote efficient studying; instead of studying off and on at home over the course of an entire day, you can stay at the library for three hours of effective studying and enjoy the rest of the day off from the MCAT.

No matter where you study, make your practice as much like Test Day as possible. Just as is required during the official test, don't have snacks or chew gum during your study blocks. Turn off your music, television, and phone. Practice on the computer with your online resources to simulate the computer-based test environment. When completing practice questions, do

your work on scratch paper or noteboard sheets rather than writing directly on any printed materials since you won't have that option on Test Day. Because memory is tied to all of your senses, the more test-like you can make your studying environment, the easier it will be on Test Day to recall the information you're putting in so much work to learn.



A microscopic image of green, rod-shaped bacteria, likely Bacillus subtilis, showing their characteristic shape and arrangement. The bacteria are in various orientations, some appearing as single rods and others in small clusters. The background is a soft, out-of-focus light green. In the top left corner, there is a solid blue rectangular box containing the text 'CHAPTER 1' in white, sans-serif, uppercase letters.

CHAPTER 1

The Cell

CHAPTER 1

The Cell

In This Chapter

1.1 Cell Theory

1.2 Eukaryotic Cells

Membrane-Bound Organelles

The Cytoskeleton

Tissue Formation

1.3 Classification and Structure of Prokaryotic Cells

Prokaryotic Domains

Classification of Bacteria by Shape

Aerobes and Anaerobes

Prokaryotic Cell Structure

1.4 Genetics and Growth of Prokaryotic Cells

Binary Fission

Genetic Recombination



Growth

1.5 Viruses and Subviral Particles



Viral Structure

Viral Genomes

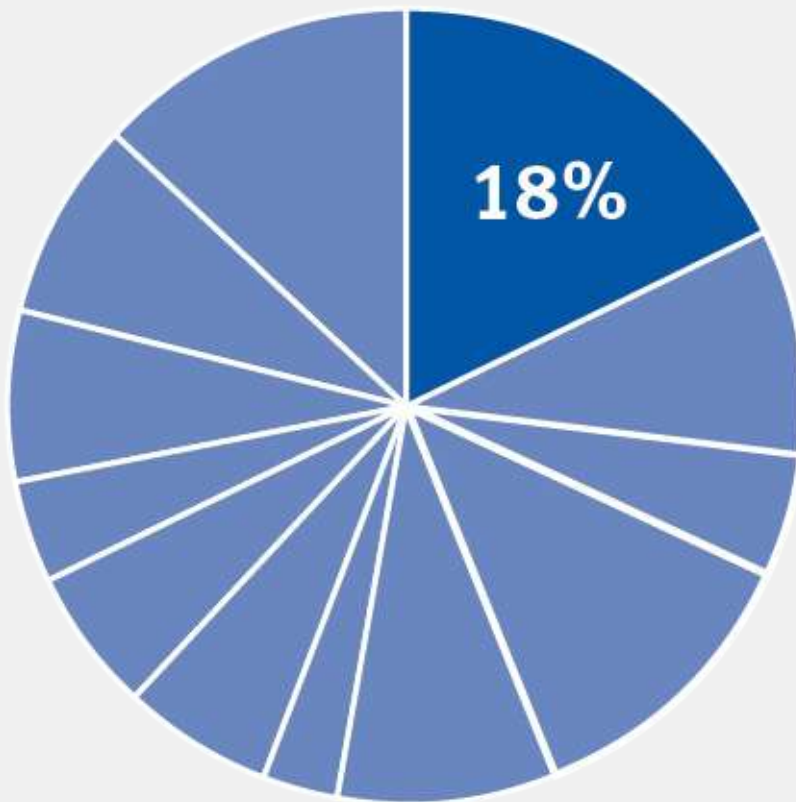
Viral Life Cycle

Prions and Viroids

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 18% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content categories:

1D: Principles of bioenergetics and fuel molecule metabolism

2A: Assemblies of molecules, cells, and groups of cells within single cellular and multicellular organisms

2B: The structure, growth, physiology, and genetics of prokaryotes and viruses

Introduction

The human body contains approximately 37 trillion cells, with bacterial cells outnumbering the eukaryotic cells by about 10 to 1. Our cells create tissues that form organs, and each cell serves a purpose, communicating and carrying out the reactions that make life possible.

The sheer number of cells that make up the human body is not nearly as impressive as the numerous functions these cells can perform—from the conduction of impulses through the nervous system that allows for memory and learning, to the simultaneous contraction of cardiac myocytes that pump blood through the entire human body. In order to understand the human organism as a whole and how the human body reacts to various pathogens, a thorough understanding of cell biology is required. It is not enough to simply memorize each part of the cell; the MCAT requires an understanding of how each cell structure carries out its functions and affects the entire organism.



MCAT EXPERTISE

This chapter represents 18% of all biology questions you would see on Test Day. That makes the cell, specifically eukaryotic cell structure and function, one of the single highest-yield subjects within any of the review books. Make sure to work sufficient study of these materials into your study plan.

1.1 Cell Theory

LEARNING OBJECTIVES

After Chapter 1.1, you will be able to:

- Recall the four fundamental tenets of cell theory

Prior to the 1600s, organisms were perceived as being complete and inseparable into smaller parts. This was due in part to the inability to see smaller structures through simple optical instruments like magnifying glasses. In 1665, Robert Hooke assembled a crude compound microscope and tested its properties on a piece of cork. He noticed a honeycomb-like structure and compared the spaces within the cork to the small rooms of a monastery, known as cells. Because cork consists of desiccated nonliving cells, Hooke was not able to see nuclei, organelles, or cell membranes. In 1674, Anton van Leeuwenhoek was the first to view a living cell under a microscope. Later researchers noted that cells could be separated, and that each cell was a distinct structure. Further research indicated that tissues were made of cells, and the function of a tissue was dependent upon the function of the cells that make up the tissue. Two centuries later, in 1850, Rudolph Virchow demonstrated that diseased cells could arise from normal cells in normal tissues.



Robert Hooke, who invented the first crude microscopes to look at cork, is also known for his characterization of springs. Hooke's law, $F = -kx$, describes the relationship between elastic force, the spring constant, and the displacement of a spring from equilibrium. While Hooke's law does not appear on the official MCAT content lists, the related topic of elastic potential energy, $U = \frac{1}{2} kx^2$, is testable content. This equation and other forms of energy are discussed in Chapter 2 of *MCAT Physics and Math Review*.

The original form of the **cell theory** consisted of three basic tenets:

- All living things are composed of cells.
- The cell is the basic functional unit of life.
- Cells arise only from preexisting cells.

Through advances in molecular biology, a fourth tenet has been added to the theory:

- Cells carry genetic information in the form of deoxyribonucleic acid (DNA). This genetic material is passed on from parent to daughter cell.

Cell theory has created an interesting dilemma with respect to viruses. Viruses are small structures that contain genetic material, but are unable to reproduce on their own. This violates the third and fourth tenets of the cell theory because virions can only replicate by invading other organisms and because they may use ribonucleic acid (RNA) as their genetic information.

Therefore, viruses, discussed later in this chapter, are not considered living organisms.



BRIDGE

Solutions to concept checks for a given chapter in *MCAT Biology Review* can be found near the end of the chapter in which the concept check is located, following the Concept Summary for that chapter.

MCAT CONCEPT CHECK 1.1:

Before you move on, assess your understanding of the material with this question.

1. What are the four fundamental tenets of the cell theory?

- _____
- _____
- _____
- _____

1.2 Eukaryotic Cells

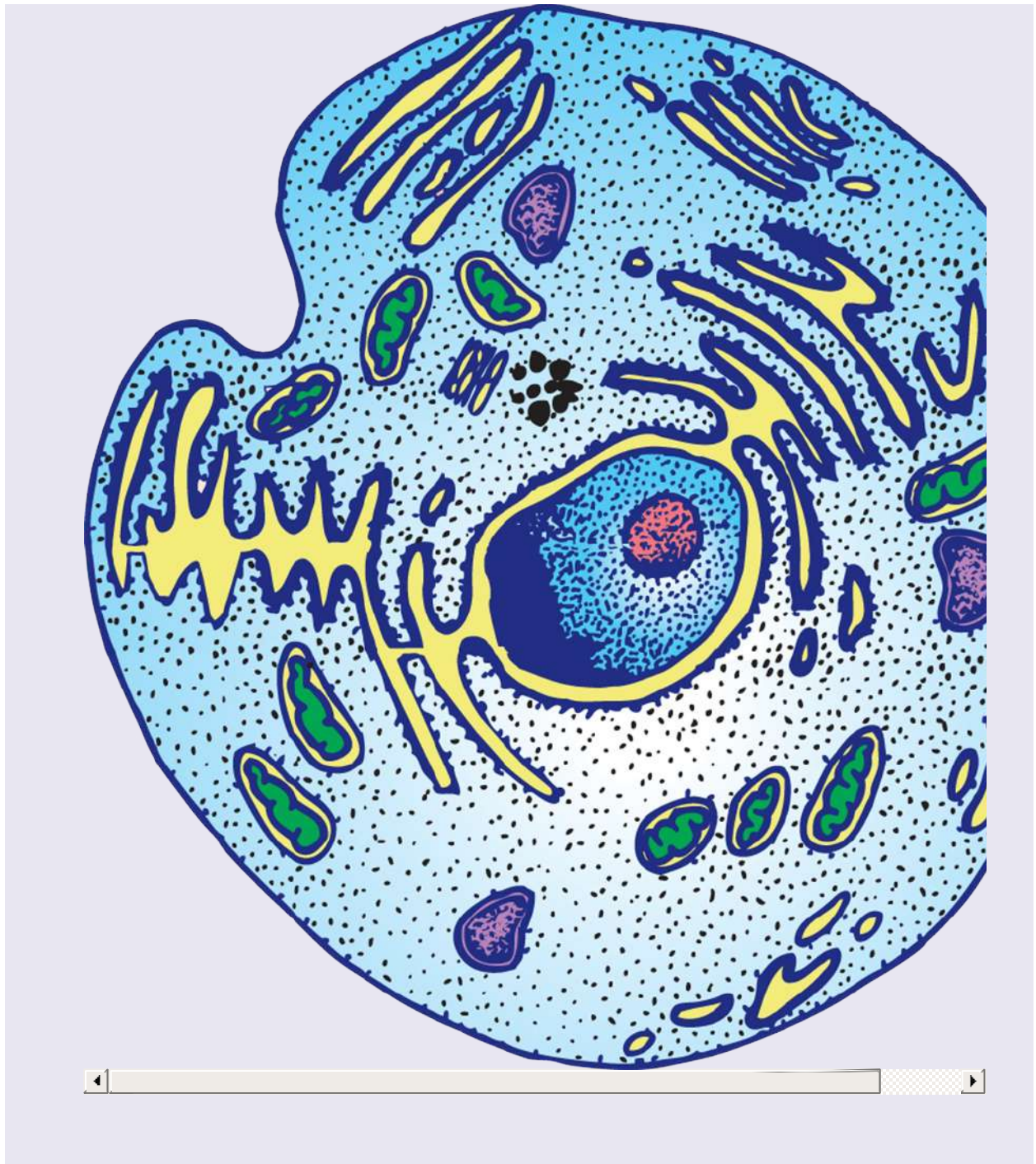
High-Yield



LEARNING OBJECTIVES

After Chapter 1.2, you will be able to:

- Explain the importance of hydrogen peroxide to cellular function
- Identify the predominant proteins found in microfilaments, microtubules, and intermediate filaments
- Distinguish between the properties of different cytoskeletal structures
- Classify cell types as epithelial or connective tissue
- Recall the names and functions of cellular organelles:



MCAT EXPERTISE

The “High-Yield” badge on this section indicates that the content is frequently tested on the MCAT.

The first major distinction we can make between living organisms is whether they are composed of prokaryotic or eukaryotic cells. Prokaryotic organisms are always single celled, while eukaryotic organisms can be unicellular or multicellular. Whereas **eukaryotic cells** contain a true nucleus enclosed in a membrane, **prokaryotic cells** do not contain a nucleus. The major organelles are identified in the eukaryotic cell in Figure 1.1.

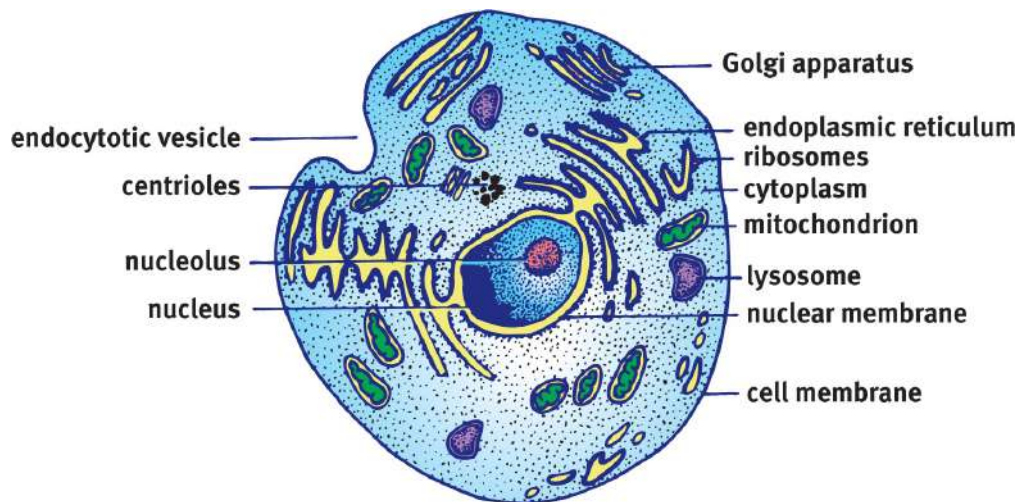


Figure 1.1. Eukaryotic Cell

Numerous membrane-bound organelles are found in the cytoplasm of a eukaryotic cell.

MEMBRANE-BOUND ORGANELLES

Each cell has a membrane enclosing a semifluid cytosol in which the **organelles** are suspended. In eukaryotic cells, most organelles are membrane

bound, allowing for compartmentalization of functions. Membranes of eukaryotic cells consist of a phospholipid bilayer. This membrane is unique in that its surfaces are hydrophilic, electrostatically interacting with the aqueous environments inside and outside of the cell, while its inner portion is hydrophobic, which helps to provide a highly selective barrier between the interior of the cell and the external environment. The cell membrane is such an important topic on the MCAT that an entire chapter—Chapter 8 of *MCAT Biochemistry Review*—is devoted solely to discussing the structure and physiology of biological membranes. The **cytosol** allows for the diffusion of molecules throughout the cell. Within the **nucleus**, genetic material is encoded in **deoxyribonucleic acid (DNA)**, which is organized into **chromosomes**. Eukaryotic cells reproduce by **mitosis**, allowing for the formation of two identical daughter cells.

The Nucleus

As the control center of the cell, the **nucleus** is the most heavily tested organelle on the MCAT. It contains all of the genetic material necessary for replication of the cell. The nucleus is surrounded by the **nuclear membrane** or **envelope**, a double membrane that maintains a nuclear environment separate and distinct from the cytoplasm. **Nuclear pores** in the nuclear membrane allow selective two-way exchange of material between the cytoplasm and the nucleus.



BRIDGE

The nuclear envelope creates two distinct environments within the cell because it separates the nucleus from the cytoplasm. This permits

compartmentalization of transcription (the formation of hnRNA from DNA, which is subsequently processed to form mRNA) and translation (the formation of a peptide from mRNA). These processes are discussed in Chapter 7 of *MCAT Biochemistry Review*.

The genetic material (DNA) contains coding regions called **genes**. Linear DNA is wound around organizing proteins known as **histones**, and is then further wound into linear strands called **chromosomes**. The location of DNA in the nucleus permits the compartmentalization of DNA transcription separate from RNA translation. Finally, there is a subsection of the nucleus known as the **nucleolus**, where the **ribosomal RNA (rRNA)** is synthesized. The nucleolus actually takes up approximately 25 percent of the volume of the entire nucleus and can often be identified as a darker spot in the nucleus.

Mitochondria

Mitochondria, shown in Figure 1.2, are often called *the power plants of the cell*, in reference to their important metabolic functions. The mitochondrion contains two layers: the outer and inner membranes. The **outer membrane** serves as a barrier between the cytosol and the inner environment of the mitochondrion. The inner membrane, which is arranged into numerous infoldings called **cristae**, contains the molecules and enzymes of the electron transport chain. The cristae are highly convoluted structures that increase the surface area available for electron transport chain enzymes. The space between the inner and outer membranes is called the **intermembrane space**; the space inside the inner membrane is called the mitochondrial **matrix**. As described in Chapter 10 of *MCAT Biochemistry Review*, the pumping of protons from the mitochondrial matrix to the intermembrane space establishes the

proton-motive force; ultimately, these protons flow through *ATP synthase* to generate ATP during oxidative phosphorylation.

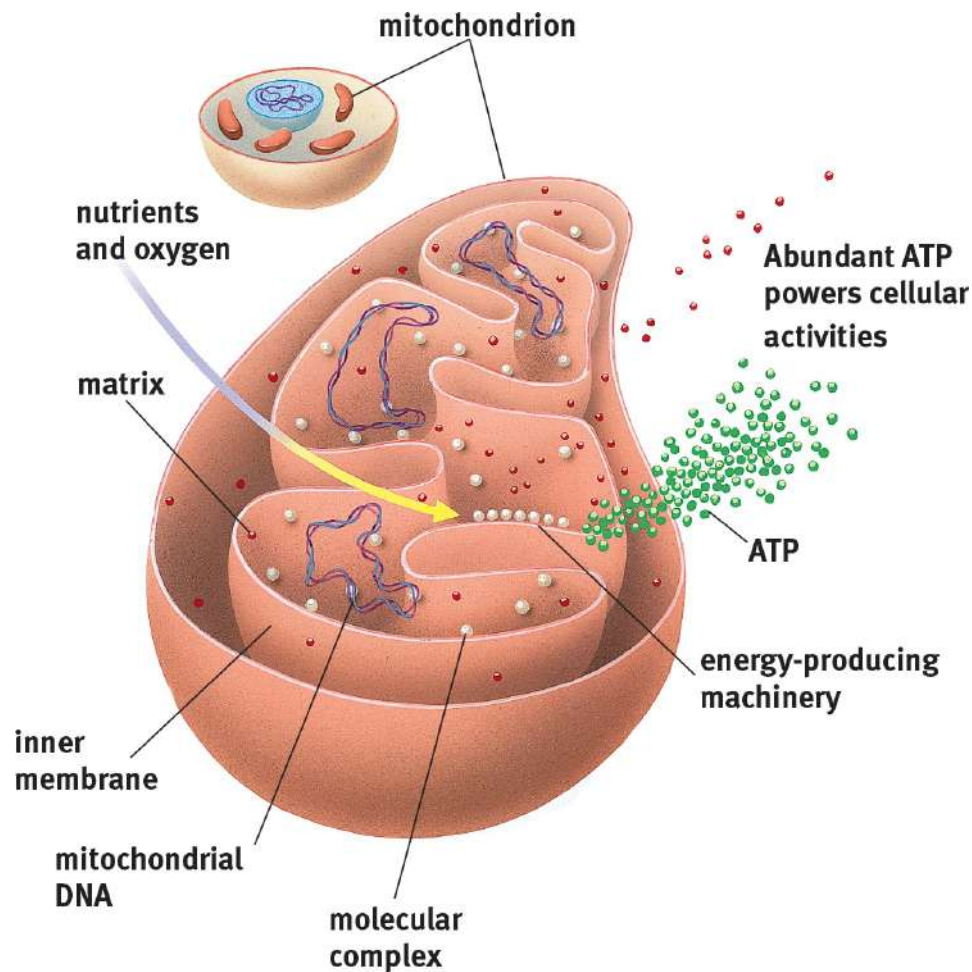


Figure 1.2. Mitochondrial Structure



REAL WORLD

The *serial endosymbiosis theory* attempts to explain the formation of some of the membrane-bound organelles; it posits that these organelles formed by the engulfing of one prokaryote by another and the establishment of a symbiotic relationship. In addition to

mitochondria, chloroplasts in plant cells and organelles of motility (such as flagella) are believed to have originated from this process.

Mitochondria are different from other parts of the cell in that they are semi-autonomous. They contain some of their own genes and replicate independently of the nucleus via binary fission. As such, they are paradigmatic examples of **cytoplasmic** or **extranuclear inheritance**—the transmission of genetic material independent of the nucleus. Mitochondria are thought to have originated when the engulfing of an aerobic prokaryote by an anaerobic prokaryote resulted in a symbiotic relationship.

In addition to keeping the cell alive by providing energy, the mitochondria are also capable of killing the cell by release of enzymes from the electron transport chain. This release kick-starts a process known as **apoptosis**, or programmed cell death.

Lysosomes

Lysosomes are membrane-bound structures containing hydrolytic enzymes that are capable of breaking down many different substrates, including substances ingested by endocytosis and cellular waste products. Lysosomes often function in conjunction with **endosomes**, which transport, package and sort cell material traveling to and from the membrane. Endosomes are capable of transporting materials to the *trans*-golgi, to the cell membrane, or to the lysosomal pathway for degradation. The lysosomal membrane sequesters these enzymes to prevent damage to the cell. However, release of these enzymes can occur in a process known as **autolysis**. Like mitochondria, when lysosomes release their hydrolytic enzymes, it results in apoptosis. In

this case, the released enzymes directly lead to the degradation of cellular components.

Endoplasmic Reticulum

The **endoplasmic reticulum (ER)** is a series of interconnected membranes that are actually contiguous with the nuclear envelope. The double membrane of the endoplasmic reticulum is folded into numerous invaginations, creating complex structures with a central lumen. There are two varieties of ER: smooth and rough. The **rough ER (RER)** is studded with **ribosomes**, which permit the translation of proteins destined for secretion directly into its lumen. On the other hand, the **smooth ER (SER)** lacks ribosomes and is utilized primarily for lipid synthesis (such as the phospholipids in the cell membrane) and the detoxification of certain drugs and poisons. The SER also transports proteins from the RER to the Golgi apparatus.

Golgi Apparatus

The **Golgi apparatus** consists of stacked membrane-bound sacs. Materials from the ER are transferred to the Golgi apparatus in vesicles. Once inside the Golgi apparatus, these cellular products may be modified by the addition of groups like carbohydrates, phosphates, and sulfates. The Golgi apparatus may also modify cellular products through the introduction of signal sequences, which direct the delivery of the product to a specific cellular location. After modification and sorting in the Golgi apparatus, cellular products are repackaged in vesicles, which are then directed to the correct cellular location. If the product is destined for secretion, the secretory vesicle merges with the cell membrane and its contents are released via **exocytosis**.

The relationships between lysosomes, the ER, and the Golgi apparatus are shown in Figure 1.3.

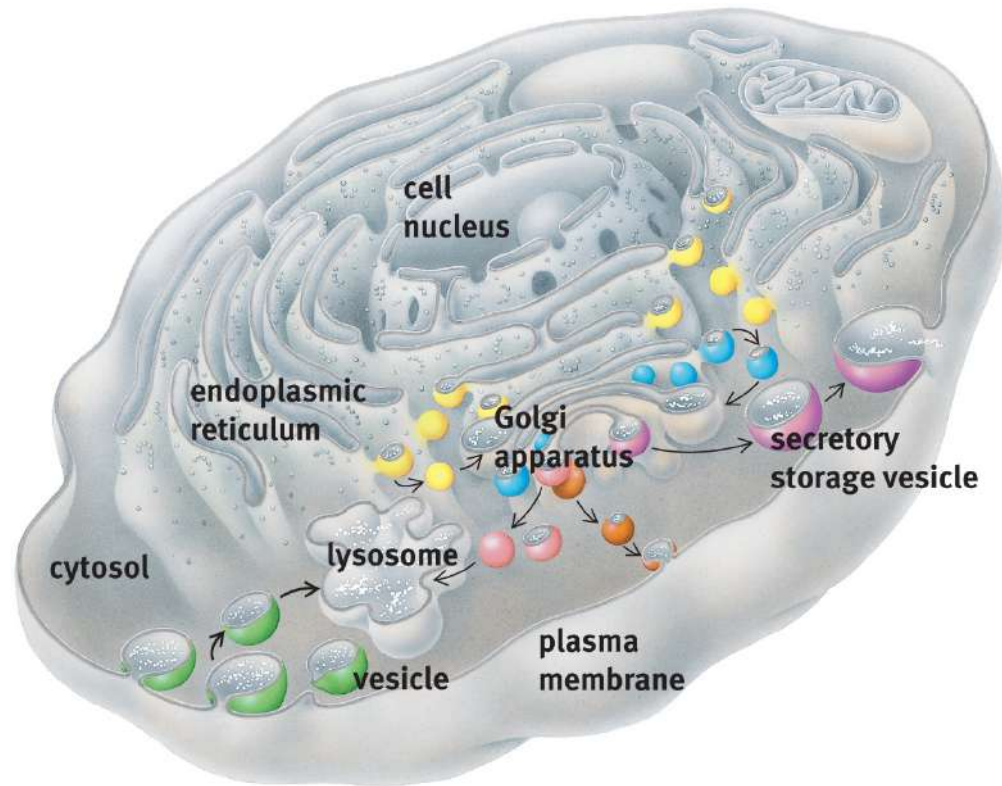


Figure 1.3. Lysosomes, the Endoplasmic Reticulum, and the Golgi Apparatus



KEY CONCEPT

Not all cells have the same relative distribution of organelles. Form will follow function. Cells that require a lot of energy for locomotion (such as sperm cells) have high concentrations of mitochondria. Cells involved in secretion (such as pancreatic islet cells and other endocrine tissues) have high concentrations of RER and Golgi

apparatuses. Other cells, such as red blood cells, which primarily serve a transport function, have no organelles at all.

Peroxisomes

Peroxisomes contain hydrogen peroxide. One of the primary functions of peroxisomes is the breakdown of very long chain fatty acids via β -oxidation. Peroxisomes participate in the synthesis of phospholipids and contain some of the enzymes involved in the pentose phosphate pathway, discussed in Chapter 9 of *MCAT Biochemistry Review*.

THE CYTOSKELETON

The **cytoskeleton**, shown in Figure 1.4, provides structure to the cell and helps it to maintain its shape. In addition, the cytoskeleton provides a conduit for the transport of materials around the cell. There are three components of the cytoskeleton: microfilaments, microtubules, and intermediate filaments.

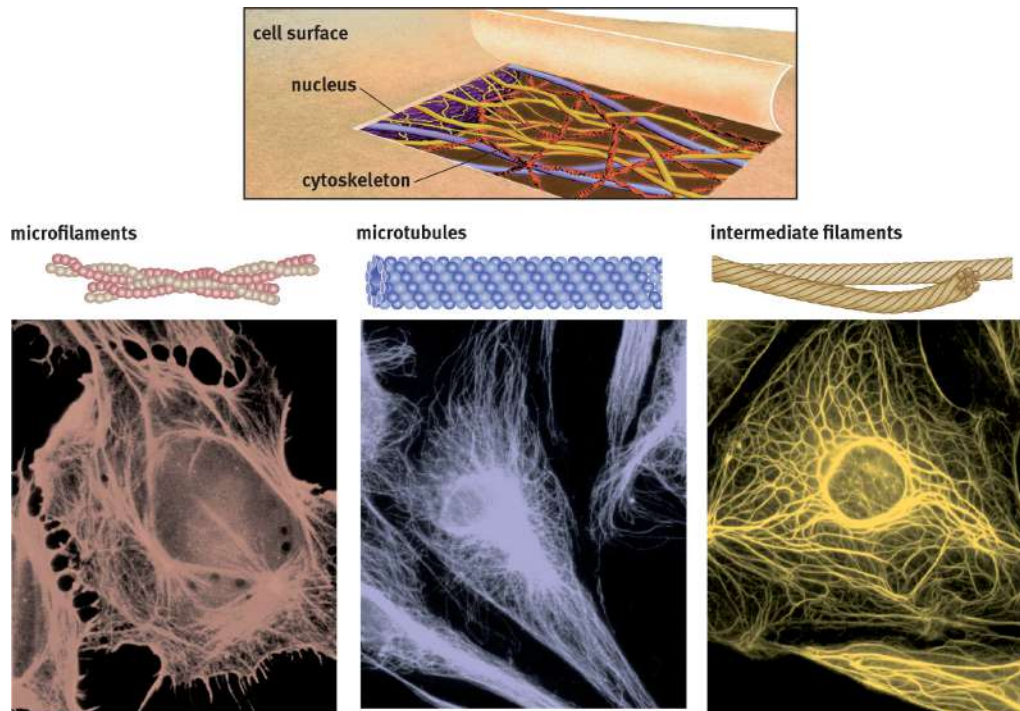


Figure 1.4. Cytoskeletal Elements

The rounded shape near the center in each of these photographs is the nucleus.

Microfilaments

Microfilaments are made up of solid polymerized rods of **actin**. The actin filaments are organized into bundles and networks and are resistant to both compression and fracture, providing protection for the cell. Actin filaments can also use ATP to generate force for movement by interacting with **myosin**, such as in muscle contraction.

Microfilaments also play a role in **cytokinesis**, or the division of materials between daughter cells. During mitosis, the **cleavage furrow** is formed from microfilaments, which organize as a ring at the site of division between the two new daughter cells. As the actin filaments within this ring contract, the

ring becomes smaller, eventually pinching off the connection between the two daughter cells.

Microtubules

Unlike microfilaments, **microtubules** are hollow polymers of **tubulin** proteins. Microtubules radiate throughout the cell, providing the primary pathways along which motor proteins like **kinesin** and **dynein** carry vesicles.



BRIDGE

Motor proteins like kinesin and dynein are classic examples of nonenzymatic protein function, along with binding proteins, cell adhesion molecules, immunoglobulins, and ion channels. Motor proteins often travel along cytoskeletal structures to accomplish their functions. Nonenzymatic protein functions are discussed in Chapter 3 of *MCAT Biochemistry Review*.

Cilia and flagella are motile structures composed of microtubules. **Cilia** are projections from a cell that are primarily involved in the movement of materials along the surface of the cell; for example, cilia line the respiratory tract and are involved in the movement of mucus. **Flagella** are structures involved in the movement of the cell itself, such as the movement of sperm cells through the reproductive tract. Cilia and flagella share the same structure, composed of nine pairs of microtubules forming an outer ring, with two microtubules in the center, as shown in Figure 1.5. This is known as a **9 + 2 structure** and is seen only in eukaryotic organelles of motility. Bacterial

flagella have a different structure with a different chemical composition, as discussed later in this chapter.

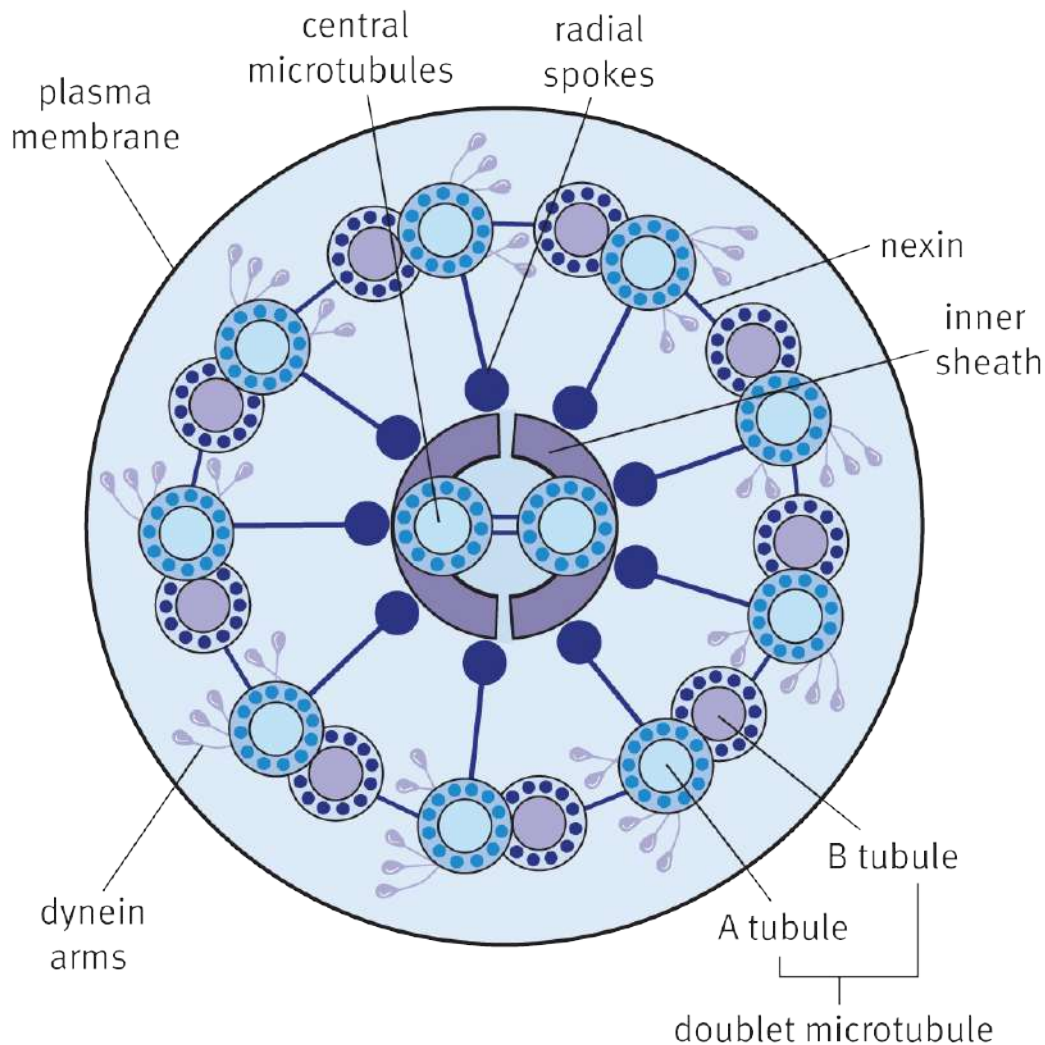


Figure 1.5. Cilium and Flagellum Structure

Microtubules are organized into a ring of 9 doublets with 2 central microtubules.

Centrioles are found in a region of the cell called the **centrosome**. They are the organizing centers for microtubules and are structured as nine triplets of microtubules with a hollow center. During mitosis, the centrioles migrate to opposite poles of the dividing cell and organize the mitotic spindle. The

microtubules emanating from the centrioles attach to the chromosomes via complexes called **kinetochores** and exert force on the sister chromatids, pulling them apart.

Intermediate Filaments

Intermediate filaments are a diverse group of filamentous proteins, which includes keratin, desmin, vimentin, and lamins. Many intermediate filaments are involved in cell–cell adhesion or maintenance of the overall integrity of the cytoskeleton. Intermediate filaments are able to withstand a tremendous amount of tension, increasing the structural rigidity of the cell. In addition, intermediate filaments help anchor other organelles, including the nucleus. The identity of the intermediate filament proteins within a cell is specific to the cell and tissue type.

TISSUE FORMATION

One of the unique characteristics of eukaryotic cells is the formation of tissues with division of labor, as different cells in a tissue may carry out different functions. For example, in the heart, some cells participate in the conduction pathways while others cause contraction; still others serve a supportive role like maintaining structural integrity of the organ. There are four tissue types: epithelial tissue, connective tissue, muscle, and nervous tissue. While muscle and nervous tissue are considered more extensively in subsequent chapters, we explore epithelial and connective tissues below.

Epithelial Tissue

Epithelial tissues cover the body and line its cavities, providing a means for protection against pathogen invasion and desiccation. In certain organs, epithelial cells are involved in absorption, secretion, and sensation. To remain as one cohesive unit, epithelial cells are tightly joined to each other and to an underlying layer of connective tissue known as the **basement membrane**. Epithelial cells are highly diverse and serve numerous functions depending on the identity of the organ in which they are found; in most organs, epithelial cells constitute the **parenchyma**, or the functional parts of the organ. For example, nephrons in the kidney are composed of epithelial cells, and hepatocytes in the liver, and acid-producing cells of the stomach are epithelial cells.

Epithelial cells are often polarized, meaning that one side faces a lumen (the hollow inside of an organ or tube) or the outside world, while the other side interacts with underlying blood vessels and structural cells. For example, in the small intestine, one side of the cell will be involved in absorption of nutrients from the lumen, while the other side will be involved in releasing those nutrients into circulation for use in the rest of the body.

We can classify different epithelia according to the number of layers they have and the shape of their cells. **Simple epithelia** have one layer of cells; **stratified epithelia** have multiple layers; and **pseudostratified epithelia** appear to have multiple layers due to differences in cell height but are, in reality, only one layer. Turning to shape, cells may be classified as cuboidal, columnar, or squamous. As their names imply, **cuboidal** cells are cube-shaped and **columnar** cells are long and thin. **Squamous** cells are flat and scale-like.

Connective Tissue

Connective tissue supports the body and provides a framework for the epithelial cells to carry out their functions. Whereas epithelial cells contribute to the parenchyma of an organ, connective tissues are the main contributors to the **stroma** or support structure. Bone, cartilage, tendons, ligaments, adipose tissue, and blood are all examples of connective tissues. Most cells in connective tissues produce and secrete materials such as collagen and elastin to form the **extracellular matrix**.

MCAT CONCEPT CHECK 1.2:

Before you move on, assess your understanding of the material with these questions.

1. Briefly describe the functions of each of the organelles listed below:

- Nucleus:

- Mitochondrion:

- Lysosome:

- Rough endoplasmic reticulum:

- Smooth endoplasmic reticulum:

- Golgi apparatus:

- Peroxisome:

-
2. A child is diagnosed with an enzyme deficiency that prevents the production of hydrogen peroxide. What would the likely outcome be of such a deficiency?
-

3. What are the predominant proteins in each cytoskeletal element?

- Microfilaments:

- Microtubules:

- Intermediate filaments:

4. How do the cytoskeletal structures of centrioles and flagella differ?
-

5. Classify each of the following cells as epithelial cells or connective tissue:

- Fibroblasts, which produce collagen in a number of organs:

-
- Endothelial cells, which line blood vessels:

-
- α -cells, which produce glucagon in the pancreas:

-
- Osteoblasts, which produce osteoid, the material that hardens into bone:

-
- Chondroblasts, which produce cartilage:
-

1.3 Classification and Structure of Prokaryotic Cells

LEARNING OBJECTIVES

After Chapter 1.3, you will be able to:

- Compare and contrast archaea, bacteria, and eukaryotes
- Identify the three common bacterial shapes
- Explain the differences between gram-positive and gram-negative bacteria
- Detail the structural differences between eukaryotic and prokaryotic flagella
- Differentiate between the metabolic processes of aerobic and anaerobic bacteria

Prokaryotes are the simplest of all organisms and include all bacteria. Prokaryotes do not contain any membrane-bound organelles, and their genetic material is organized into a single circular molecule of DNA concentrated in an area of the cell called the **nucleoid region**. Despite the simplicity of prokaryotes, they are incredibly diverse, and knowledge of this diversity is essential for the study of medicine because many prokaryotes can cause infection. In fact, choosing the appropriate antibiotic to fight an

infection requires knowledge about the basic structure of the bacteria causing the infection.

PROKARYOTIC DOMAINS

There are three overarching domains into which all life is classified: Archaea, Bacteria, and Eukarya. Two of these—Archaea and Bacteria—contain prokaryotes. Initially, Archaea and Bacteria were classified together into the kingdom of Monera. However, modern genetics and biochemical techniques have indicated that the differences in the evolutionary pathways between Archaea and Bacteria are at least as significant as between either of these domains and Eukarya.

Archaea

Archaea are single-celled organisms that are visually similar to bacteria, but contain genes and several metabolic pathways that are more similar to eukaryotes than to bacteria. Historically, Archaea were considered **extremophiles**, in that they were most commonly isolated from harsh environments with extremely high temperatures, high salinity, or no light. More recent research has demonstrated a greater variety of habitats for these organisms, including the human body. Archaea are notable for their ability to use alternative sources of energy. While some are photosynthetic, many are chemosynthetic and can generate energy from inorganic compounds, including sulfur- and nitrogen-based compounds such as ammonia.

Due to the similarities of this domain to eukaryotes, it is hypothesized that eukaryotes and the Archaea share a common origin. Both eukaryotes and Archaea start translation with methionine, contain similar *RNA polymerases*,

and associate their DNA with histones. However, Archaea contain a single circular chromosome, divide by binary fission or budding, and share a similar overall structure to bacteria. Interestingly, Archaea are resistant to many antibiotics.

Bacteria

All bacteria contain a **cell membrane** and **cytoplasm**, and some have **flagella** or **fimbriae** (similar to cilia), as shown in Figure 1.6. Because bacteria and eukaryotes often share analogous structures, it can be difficult to develop medicines that target only bacteria. However, in some cases, even seemingly similar structures have enough biochemical differences to allow the exclusive targeting of one kind of organism. For example, bacterial flagella and eukaryotic flagella are different enough that scientists are able to develop antibacterial vaccines that specifically target the bacterial flagellum. Also, many antibiotics target the bacterial ribosome, which is significantly smaller than the eukaryotic ribosome.

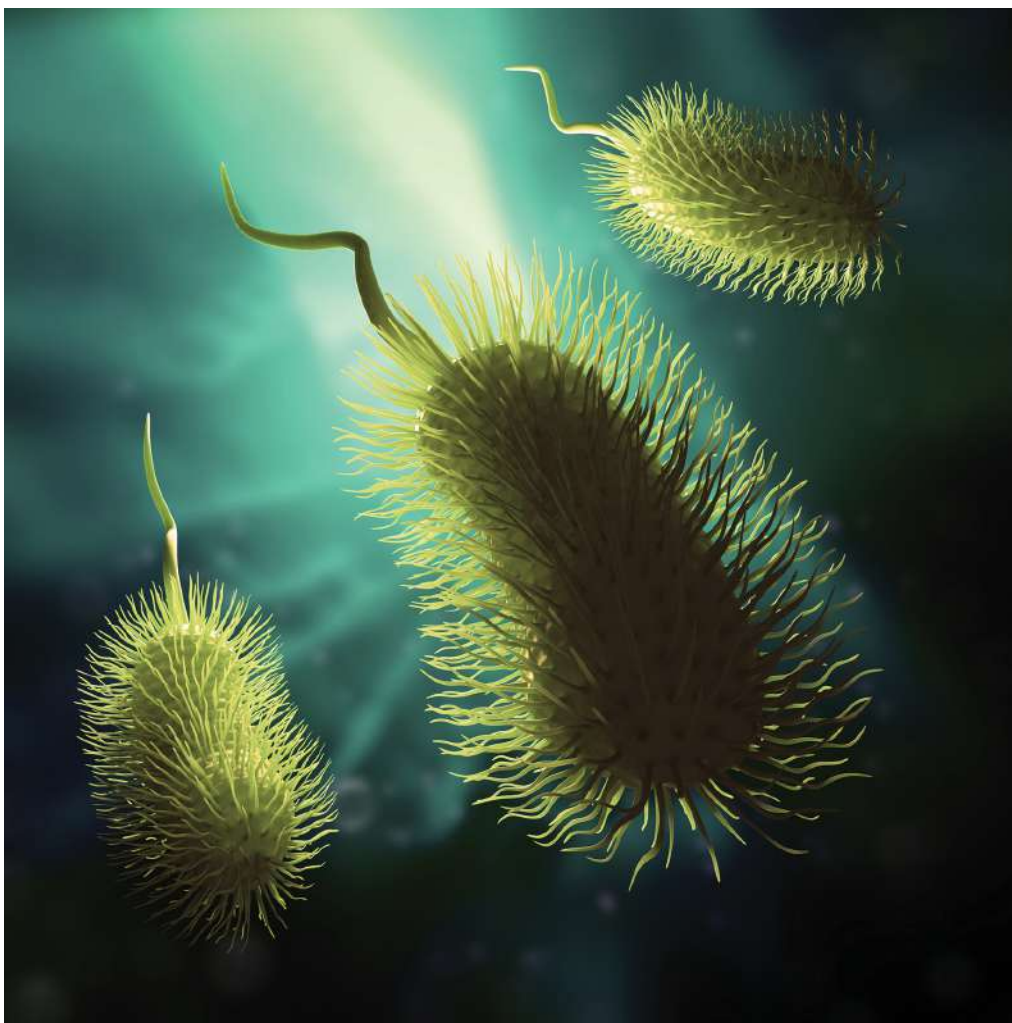


Figure 1.6. Prokaryotic Cell Specializations: Flagella and Fimbriae



REAL WORLD

Bacteria perform essential functions for human beings, including the production of vitamin K in the intestine. Vitamin K is required for production of the plasma proteins necessary for blood clotting. Newborn infants are not yet colonized by bacteria and cannot produce clotting factors, putting them at risk for hemorrhage. When babies are

born, they are given an injection of vitamin K to aid in the production of clotting factors until they have been colonized with bacteria.

There are approximately 5×10^{30} bacteria on Earth, outnumbering all of the plants and animals combined. As mentioned in the introduction to this chapter, bacteria outnumber human cells in the body by 10:1. The relationship between the human body and bacteria is complex. Some bacteria are **mutualistic symbiotes**, meaning that both humans and the bacteria benefit from the relationship. Examples include the bacteria in the human gut that produce vitamin K and biotin (vitamin B₇), and which also prevent the overgrowth of harmful bacteria. Other bacteria are **pathogens** or **parasites**, meaning that they provide no advantage or benefit to the host, but rather cause disease. Pathogenic bacteria may live intracellularly or extracellularly. For example, *Chlamydia trachomatis*, a common sexually transmitted infection, lives inside cells of the reproductive tract; *Clostridium tetani*, the cause of tetanus, lives outside of cells and produces toxins that enter the bloodstream.

CLASSIFICATION OF BACTERIA BY SHAPE

Classification of bacteria by shape provides scientists and pathologists (physicians who specialize in the identification and characterization of disease) a common language to talk about bacteria, as well as a way to identify different species of bacteria.

Most bacteria exist in one of three shapes, as shown in Figure 1.7. Spherical bacteria, known as **cocci**, include common pathogens such as *Streptococcus pyogenes*.

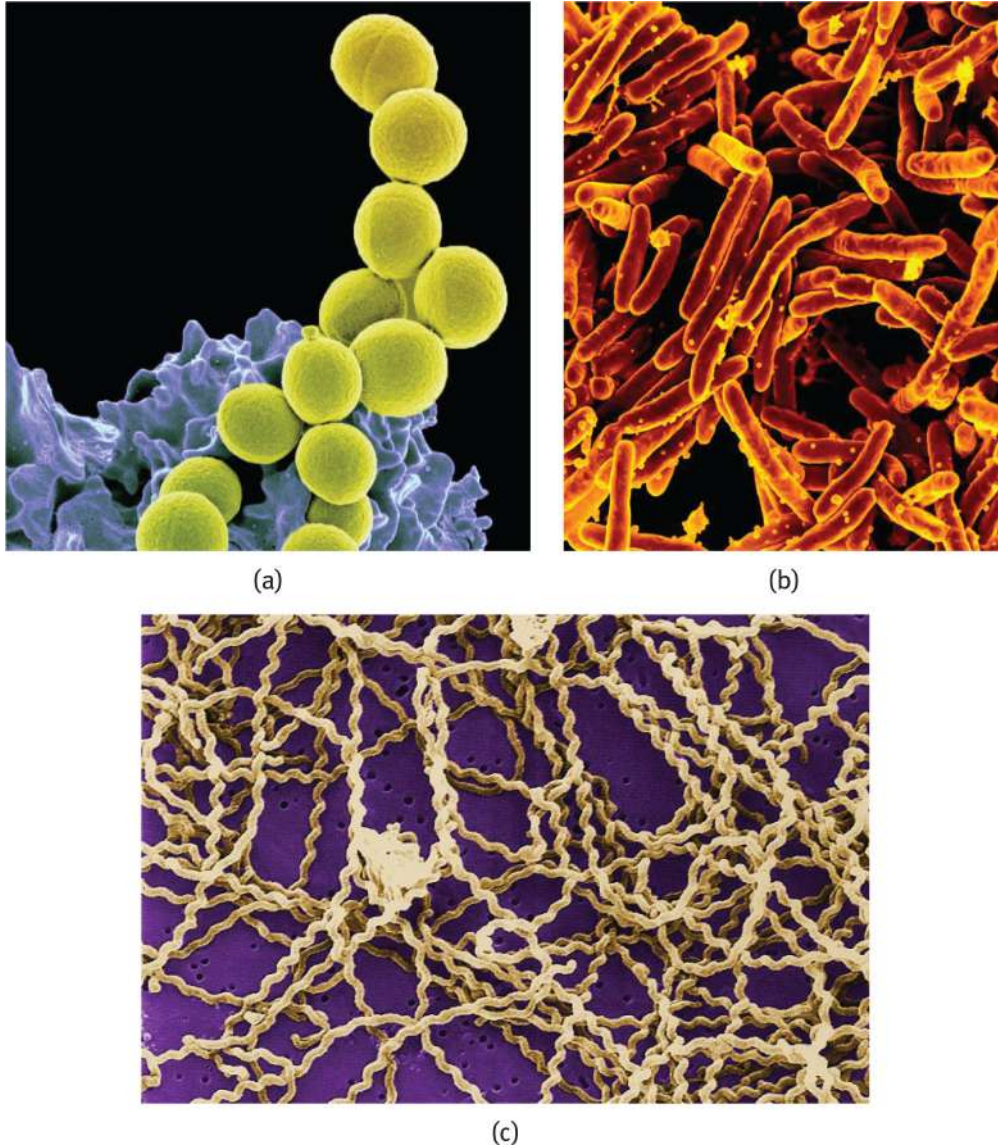


Figure 1.7. Prokaryotic Cell Shapes

(a) Cocci (*Staphylococcus aureus*), (b) Bacilli (*Mycobacterium tuberculosis*), (c) Spirilli (*Leptospira interrogans*).

Rod-shaped bacteria, like *Escherichia coli*, are known as **bacilli**. Finally, spiral-shaped bacteria, known as **spirilli**, include such species as *Treponema pallidum*, which causes syphilis.



REAL WORLD

Very few pathogenic bacteria are spiral shaped. The three most common are:

- *Treponema pallidum*, the cause of syphilis
- *Borrelia burgdorferi*, the cause of Lyme disease
- *Leptospira interrogans*, the cause of Weil's disease

AEROBES AND ANAEROBES

Some bacteria require oxygen for survival, while others do not. Bacteria that require oxygen for metabolism are termed **obligate aerobes**. Other bacteria that use fermentation, or some other form of cellular metabolism that does not require oxygen, are called **anaerobes**. There are different types of anaerobes. Anaerobes that cannot survive in an oxygen-containing environment are called **obligate anaerobes**; the presence of oxygen leads to the production of reactive oxygen-containing radicals in these species, which leads to cell death. Other bacteria can toggle between metabolic processes, using oxygen for aerobic metabolism if it is present, and switching to anaerobic metabolism if it is not. These bacteria are called **facultative anaerobes**. Finally, **aerotolerant anaerobes** are unable to use oxygen for metabolism, but are not harmed by its presence in the environment.

PROKARYOTIC CELL STRUCTURE

One of the main differences between prokaryotes and eukaryotes is that prokaryotes lack a nucleus and membrane-bound organelles, as shown in Figure 1.8. Prokaryotes are also single-celled organisms, meaning that each cell must be able to perform all of the functions necessary for life on its own. However, prokaryotes may live in colonies with other cells and may signal these cells to share information about the environment.

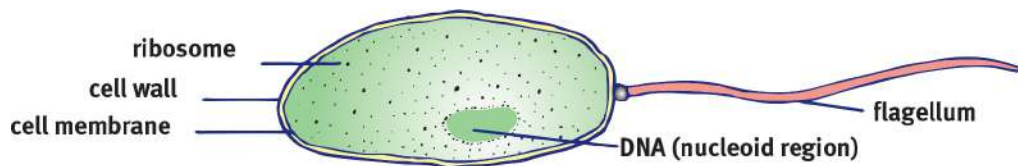


Figure 1.8. Prokaryotic Cell Structure

Cell Wall

Because prokaryotes do not form multicellular organisms, each bacterium is responsible for protecting itself from the environment. The **cell wall** forms the outer barrier of the cell. The next layer is the **cell membrane (plasma membrane)**, which is composed of phospholipids, similar to that of a eukaryote. Together, the cell wall and the cell membrane are known as the **envelope**.

The cell wall both provides structure and controls the movement of solutes into and out of the bacterium. This allows the cell to maintain a concentration gradient relative to the environment. In bacteria, there are two main types of cell wall: gram positive and gram negative. The type of cell wall is determined by the Gram staining process: a crystal violet stain, followed by a counterstain with a substance called *safranin*. If the envelope absorbs the crystal violet stain, it will appear deep purple, and the cell is said to be gram positive. If the

envelope does not absorb the crystal violet stain, but absorbs the safranin counterstain, then the cell will appear pink-red, and the cell is said to be gram negative.



REAL WORLD

The antibiotic penicillin targets the enzyme that catalyzes the cross-linking of peptidoglycan. If a gram-positive cell cannot cross-link its cell wall, it no longer serves as an effective barrier. The bacterium then becomes susceptible to osmotic damage and lyses. Most bacteria have developed resistance mechanisms to penicillin, although a few bacteria—including *Streptococcus pyogenes*, which causes strep throat and some skin infections, and *Treponema pallidum*, which causes syphilis—are still very sensitive to this antibiotic.

Gram-positive cell walls consist of a thick layer of **peptidoglycan**, a polymeric substance made from amino acids and sugars. In addition to its structural and barrier functions, the cell wall may also aid a bacterial pathogen by providing protection from a host organism's immune system. In addition to peptidoglycan, the gram-positive cell wall also contains **lipoteichoic acid**. It is not clear what role this acid serves for the bacterium, but the human immune system may be activated by exposure to these chemicals.

Gram-negative cell walls are very thin and also contain peptidoglycan, but in much smaller amounts. The peptidoglycan cell walls of these bacteria are adjacent to the cell membrane, and are separated from the membrane by the **periplasmic space**. In addition to the cell wall and cell membrane, gram-

negative bacteria also have **outer membranes** containing phospholipids and **lipopolysaccharides**. Interestingly, lipopolysaccharides are the part of gram-negative bacteria that triggers an immune response in human beings; the inflammatory response to lipopolysaccharides is much stronger than the response to lipoteichoic acid.



KEY CONCEPT

Bacteria contain a cell wall, the composition of which is different in gram-positive and gram-negative bacteria. Specific components of the cell wall can trigger an inflammatory response.

Flagella

Flagella are long, whip-like structures that can be used for propulsion; bacteria may have one, two, or many flagella, depending on the species. Flagella can be used to move toward food or away from toxins or immune cells. This ability of a cell to detect chemical stimuli and move toward or away from them is called **chemotaxis**. The flagella are composed of a filament, a basal body, and a hook, as shown in Figure 1.9. The **filament** is a hollow, helical structure composed of **flagellin**. The **basal body** is a complex structure that anchors the flagellum to the cytoplasmic membrane and is also the motor of the flagellum, which rotates at rates up to 300 Hz. The **hook** connects the filament and the basal body so that, as the basal body rotates, it exerts torque on the filament, which thereby spins and propels the bacterium forward. The overall structure of flagella is similar in both gram-positive and gram-negative bacteria, but there are slight differences due to the different

physical structure and chemical composition of the envelope in gram-positive and gram-negative bacteria. Archaea also contain flagella, but the structure of their flagella is quite different from that of bacteria and is unlikely to be asked about on Test Day.

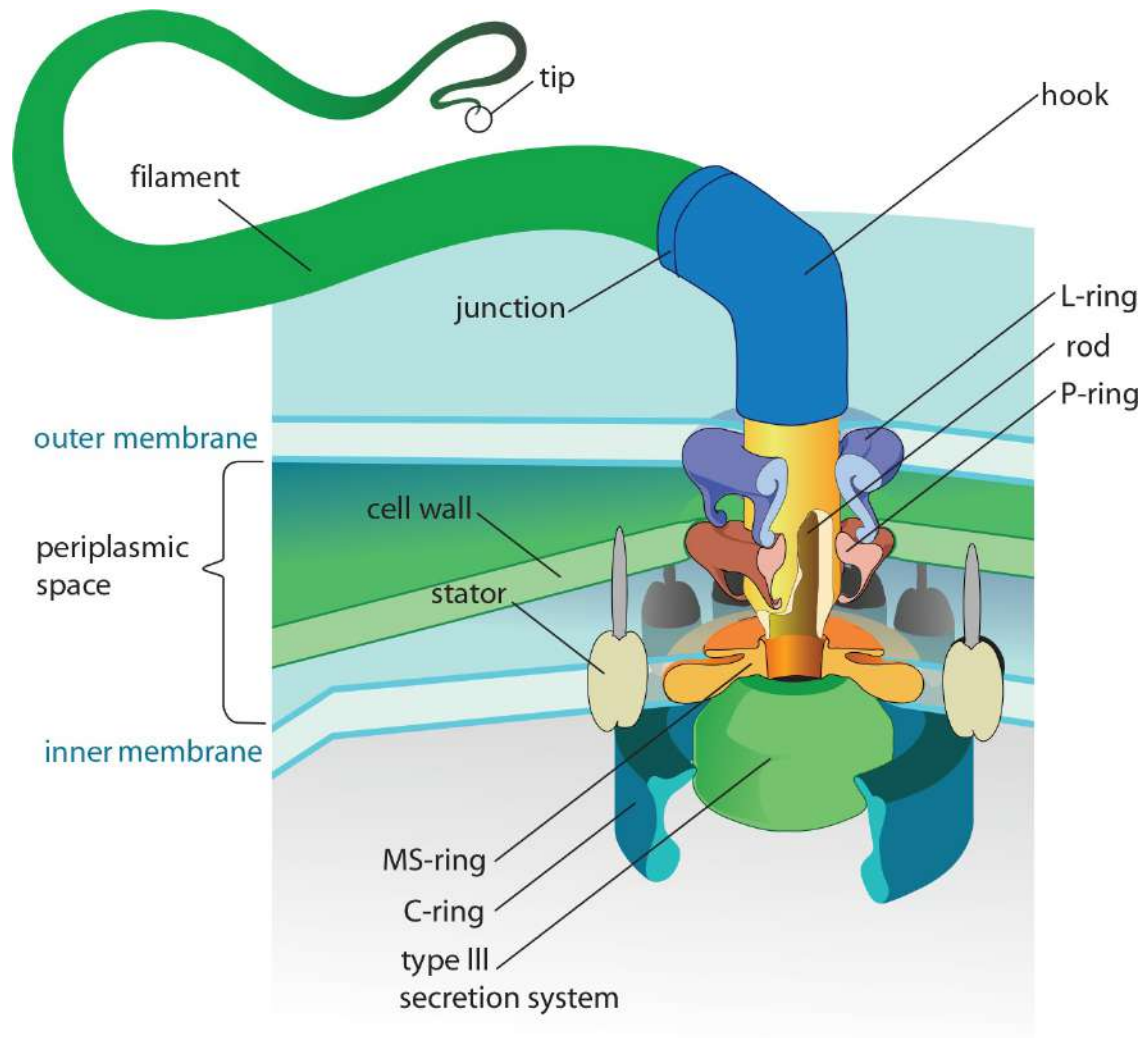


Figure 1.9. Prokaryotic Flagellum Structure

The hook connects the filament to the basal body (the complex structure of which is shown).

Other Organelles

As mentioned earlier, prokaryotes concentrate DNA in a region of the cell known as the nucleoid region, which does not contain a nuclear envelope. Prokaryotic DNA is carried on a single circular chromosome which can be found coiled around histone-like proteins in some bacteria. True histones are found in Archaea. In addition, DNA acquired from external sources may also be carried on smaller circular structures known as **plasmids**. Plasmids carry DNA that is not necessary for survival of the prokaryote—and therefore is not considered part of the genome of the bacterium—but may confer an advantage such as antibiotic resistance.



BRIDGE

The fact that prokaryotes and eukaryotes have different-sized ribosomes implies that they carry out protein synthesis in slightly different ways. These differences are highlighted in Chapter 7 of *MCAT Biochemistry Review*. This difference also allows us to target bacterial ribosomes with a number of antibiotics, including *tetracyclines*, *aminoglycosides*, and *macrolides*, while leaving the eukaryotic ribosome more or less unaffected.

Prokaryotes lack several key organelles, such as mitochondria. Instead, the cell membrane is used for the electron transport chain and generation of ATP. Prokaryotes do contain a primitive cytoskeleton, but it is not nearly as complex as the one found in eukaryotes. Prokaryotes also contain ribosomes, but this ribosome is a different size from that found in eukaryotes: prokaryotic ribosomes contain 30S and 50S subunits, whereas eukaryotic ribosomes contain 40S and 60S ribosomes.

MCAT CONCEPT CHECK 1.3:

Before you move on, assess your understanding of the material with these questions.

1. In what ways are Archaea similar to bacteria? In what ways are Archaea similar to eukaryotes?

- Similar to bacteria:

- Similar to eukaryotes:

2. What are the three common shapes of bacteria?

-

-

-

3. Compare and contrast the metabolisms of aerobic and anaerobic bacteria: (Note: Put “yes” or “no” in each box.)

	Oxygen Present	Oxygen Absent

Type of Bacteria	Can survive	Can carry out aerobic metabolism	Can survive	Can carry out anaerobic metabolism
Obligate aerobe				
Facultative anaerobe				
Obligate anaerobe				
Aerotolerant anaerobe				

4. What difference between the envelopes of gram-positive and gram-negative bacteria make gram-positive bacteria more susceptible to antibiotics such as penicillin?

5. How do the structures of eukaryotic and prokaryotic flagella differ?

- Eukaryotic:

- Prokaryotic:

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

To investigate the collective resistance property of bacteria, co-colonization experiments were conducted. Eight-weekold female CD1 mice were infected intratracheally with chloramphenicol sensitive (Cm^S) pneumococci or an equivalent quantity of $\text{Cm}^S:\text{Cm}^R$ (chloramphenicol resistant) pneumococci (1:1).

One hour post-infection, mice were treated with one intraperitoneal injection of chloramphenicol at a concentration of 75 mg kg^{-1} , followed by two additional doses spaced 5 hours apart. Control mice received vehicle injection only. Cm^R pneumococci are Cm resistant due to the expression of a resistance factor, chloramphenicol acetyltransferase (CAT).

Stable bacterial coexistence requires successful adjustment to several ecological constraints, such as limited resources and antibiotic concentration.

To better quantify bacterial behavior under these conditions, in vitro population analysis was conducted.

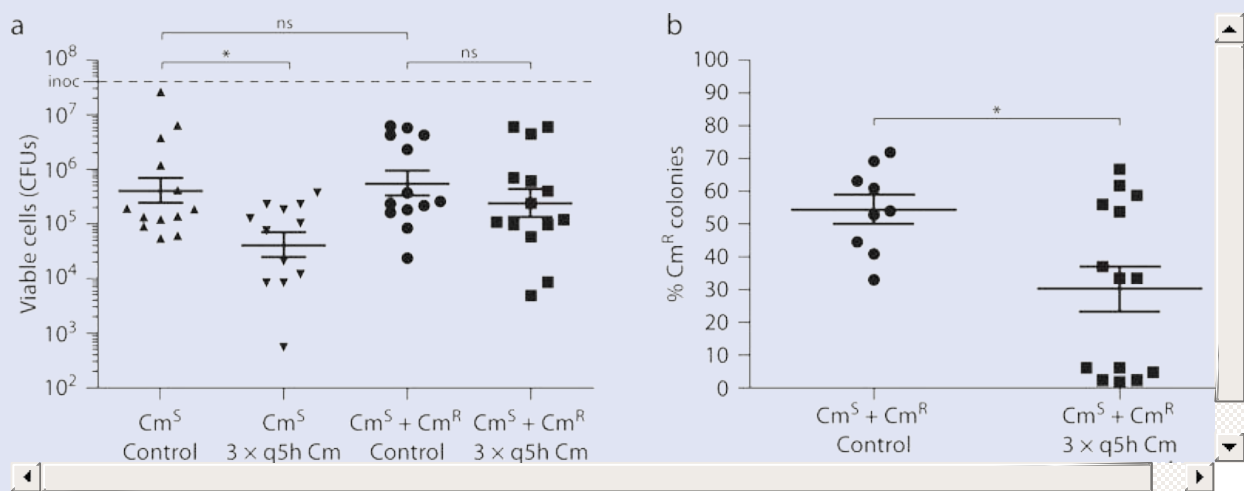


Figure 1 Mixed Culture Experiment. (a) Antibiotic stress on Cm^S alone and $Cm^S + Cm^R$ co-colonization condition. (b) Co-colonization colonies analysis of Cm^S to Cm^R ratio.

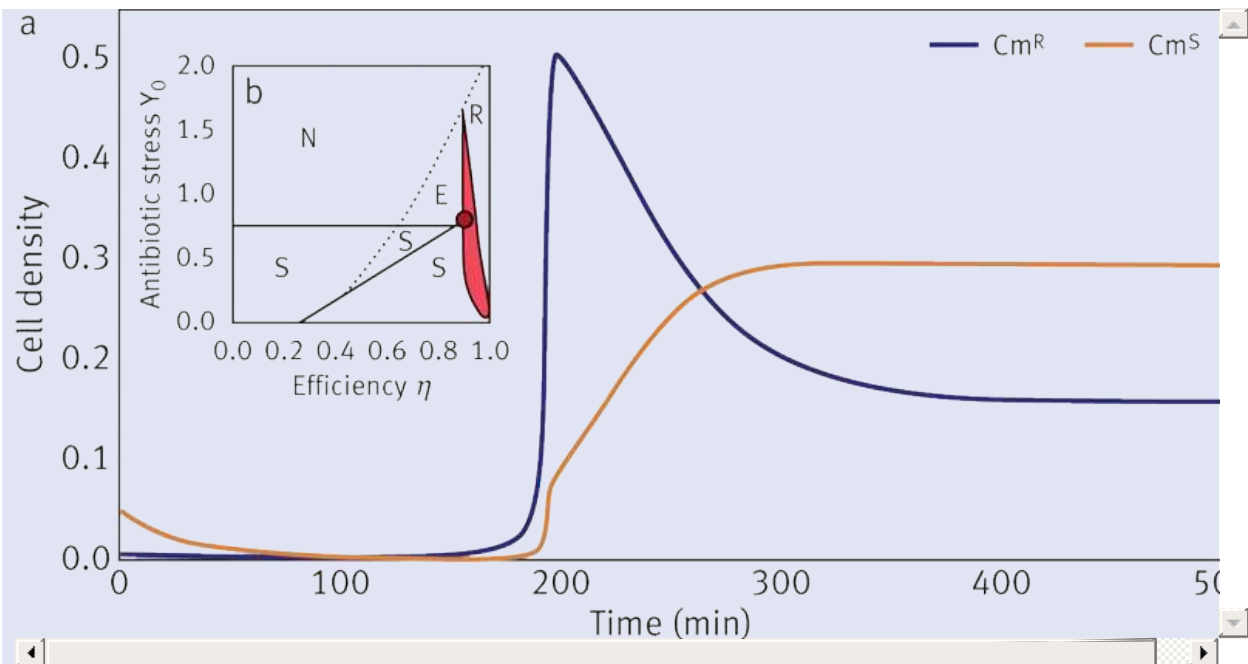


Figure 2 (a) Bacterial growth trajectories subject to antibiotic stress and resource competition. (b) Axes indicate co-colonization experiment parameters (efficiency refers to the growth rate efficiency of Cm^R cells). The dot indicates Figure 1 experiment antibiotic levels, with previous analysis establishing Cm^S only (area S), Cm^R only (area R), no bacterial growth (area N), and competition-induced extinction (area E, where Cm^S bacteria first outcompete Cm^R bacteria and are subsequently cleared by antibiotic).

adapted from Sorg, R. A., Lin, L., Van Doorn, G. S., Sorg, M., Olson, J., Nizet, V., & Veening, J. W. (2016). Collective resistance communities by intracellular antibiotic deactivation. *PLoS Biology*, 14(12), e2000631.

Would the following statement be a reasonable conclusion to draw from the given data?

‘For successful co-colonization with Cm^S , the growth efficiency of Cm^R must remain as a constant value in the presence or absence of antibiotic stress.’

This question asks us to analyze a conclusion's validity. First, we must find where the key term is located in the passage. In this case, we are asked about growth efficiency, which can be found in the figure description of Figure 2b. According to the description, growth efficiency is a reference to the growth rate of Cm^R , and is depicted as the x-axis of the graph. The conclusion given in the question stem states that the growth efficiency is unchanged by the presence of antibiotics. To determine if this conclusion is valid, we first need to decipher what the graph represents.

The figure description can be a rich source of crucial information. According to that same Figure 2b description, each area on the graph depicts a different outcome of Cm^S and Cm^R co-colonization. To ensure we're drawing valid conclusions, we should examine the graph one axis at a time. To analyze a graph, hold one variable constant first. Assume no antibiotics are present, meaning that the antibiotic stress is zero, looking at the x-axis along the zero line, as the growth efficiency of Cm^R decreases, the Cm^S strain starts to outcompete it. In fact, unless the growth efficiency of Cm^R is at its highest, the susceptible strain (Cm^S) will always outcompete Cm^R . Now let's hold the same variable at a different level. Let us assume the antibiotic stress is high, so we'll look to the right side of the x-axis. Under those conditions, there's no growth of the susceptible strain at all. That makes sense, as the susceptible strain wouldn't be expected to survive the antibiotics. We can also see that, for this x-axis value, there's an area, E, where at slightly lower Cm^R growth efficiency, the susceptible strain outcompetes Cm^R , removing all protection, and then succumbs to antibiotic. In fact, unless the growth efficiency of Cm^R is high, no bacteria will successfully grow. From these two observations, we can observe that the growth efficiency of the Cm^R strain has a large influence on how the co-colonized strain will grow.

The shaded area of Figure 2b is not explained in the figure description, but we

can deduce its meaning through context. All the other possibilities already have a designated area on the graph based on the figure label except for successful co-colonization. Also, the shaded area is between the area of Cm^S only and Cm^R only areas. From the figure description, we also learned that the large dot indicates the experiment conducted in Figure 1's conditions, and Figure 1's experiment demonstrates successful co-colonization. Thus, we can conclude that the shaded area is the region of successful co-colonization of the two strains.

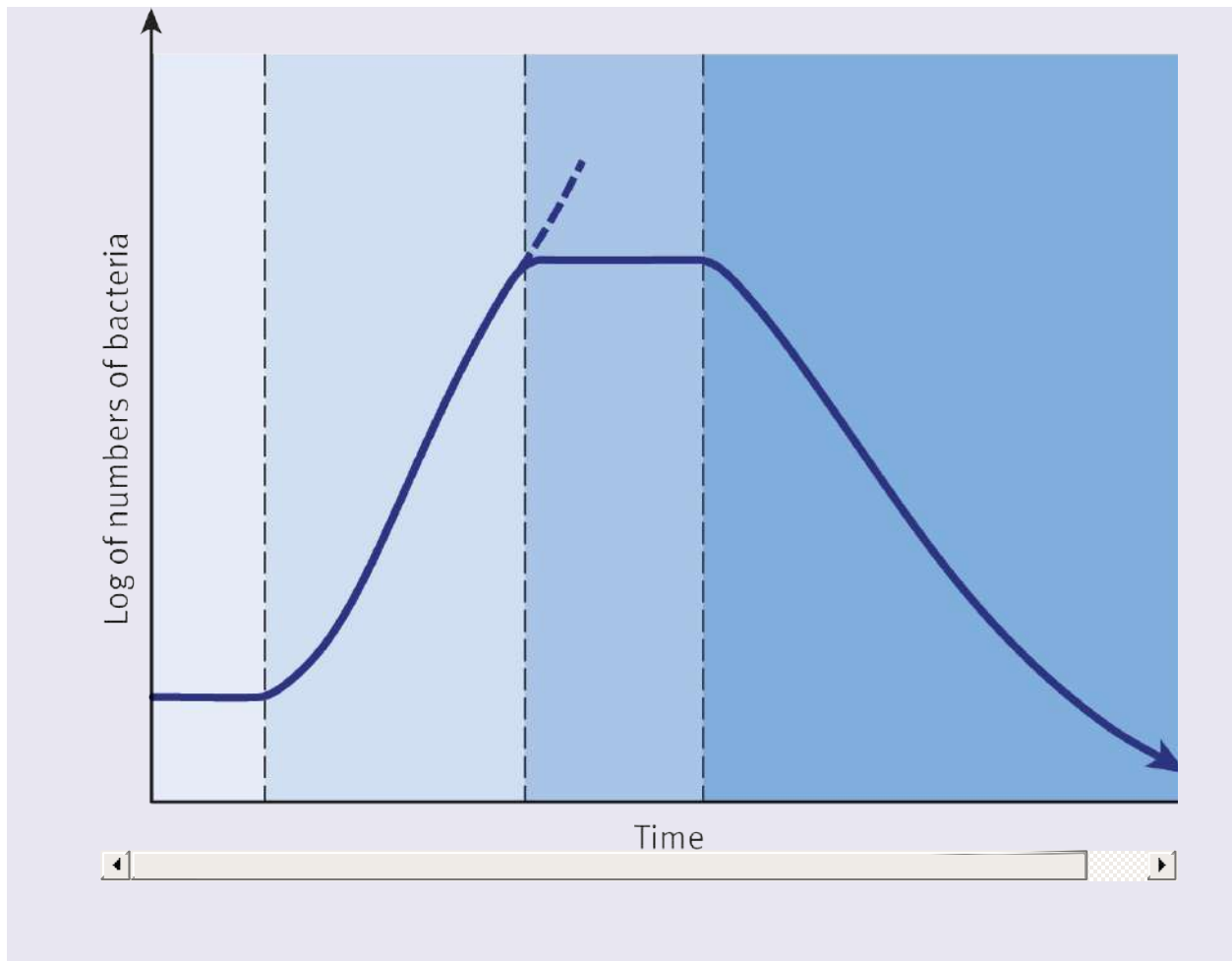
The conclusion presented in the question stem claims that growth efficiency of Cm^R does not change with or without the antibiotic presence. To assess this, we simply need to compare those two conditions. With no antibiotics, we can see that successful co-colonization only occurs when the growth efficiency of the Cm^R is at 1. If the growth efficiency is not that high, then the Cm^S strain will outcompete it. When we raise the antibiotic level to that of the red dot, we can see that if the growth efficiency remains at 1, the Cm^R bacteria will outcompete the Cm^S bacteria. In order for the two strains to coexist under antibiotic stress, the resistance strain's growth efficiency has to be lower than 1. Thus, we can reasonably argue that the conclusion is not valid.

1.4 Genetics and Growth of Prokaryotic Cells

LEARNING OBJECTIVES

After Chapter 1.4, you will be able to:

- Describe bacterial genetic recombination via transformation, conjugation, or transduction
- Recall the four phases of a bacterial growth curve and the major features of each phase:



As we have seen, prokaryotic cells differ from eukaryotic cells both structurally and biochemically. Prokaryotes reproduce via asexual reproduction in the form of binary fission. In addition, prokaryotes are capable of acquiring and using genetic material from outside the cell.

BINARY FISSION

Binary fission, shown in Figure 1.10, is a simple form of asexual reproduction seen in prokaryotes. The circular chromosome attaches to the cell wall and replicates while the cell continues to grow in size. Eventually, the plasma membrane and cell wall begin to grow inward along the midline of the cell to

produce two identical daughter cells. Because binary fission requires fewer events than mitosis, it can proceed more rapidly. In fact, some strains of *E. coli* can replicate every 20 minutes under ideal growth conditions.

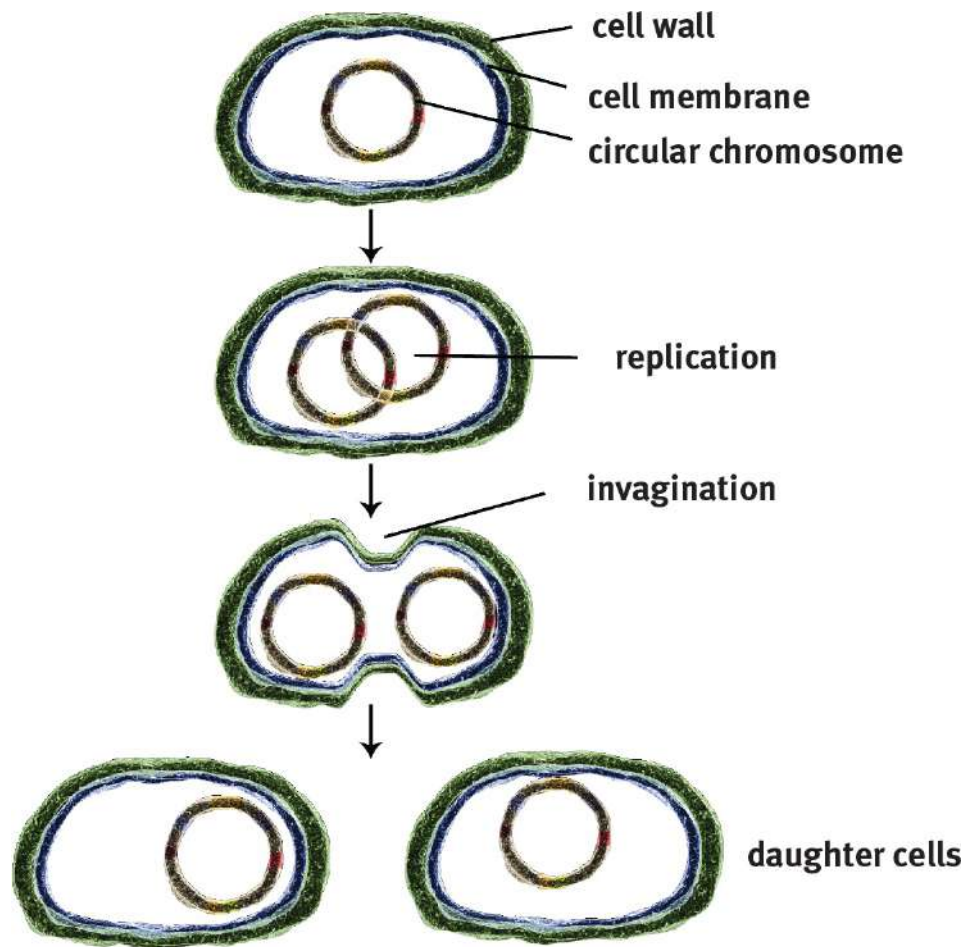


Figure 1.10. Stages of Binary Fission

GENETIC RECOMBINATION

This single circular chromosome of a prokaryotic cell contains the information that is necessary for the cell to survive and reproduce. However, many bacteria also contain extrachromosomal (extragenomic) material known as

plasmids. Plasmids often carry genes that impart some benefit to the bacterium, such as antibiotic resistance, some mechanisms of which are shown in Figure 1.11. Plasmids may also carry additional **virulence factors**, or traits that increase pathogenicity, such as toxin production, projections that allow attachment to certain kinds of cells, or features that allow evasion of the host's immune system. A subset of plasmids called **episomes** are capable of integrating into the genome of the bacterium.

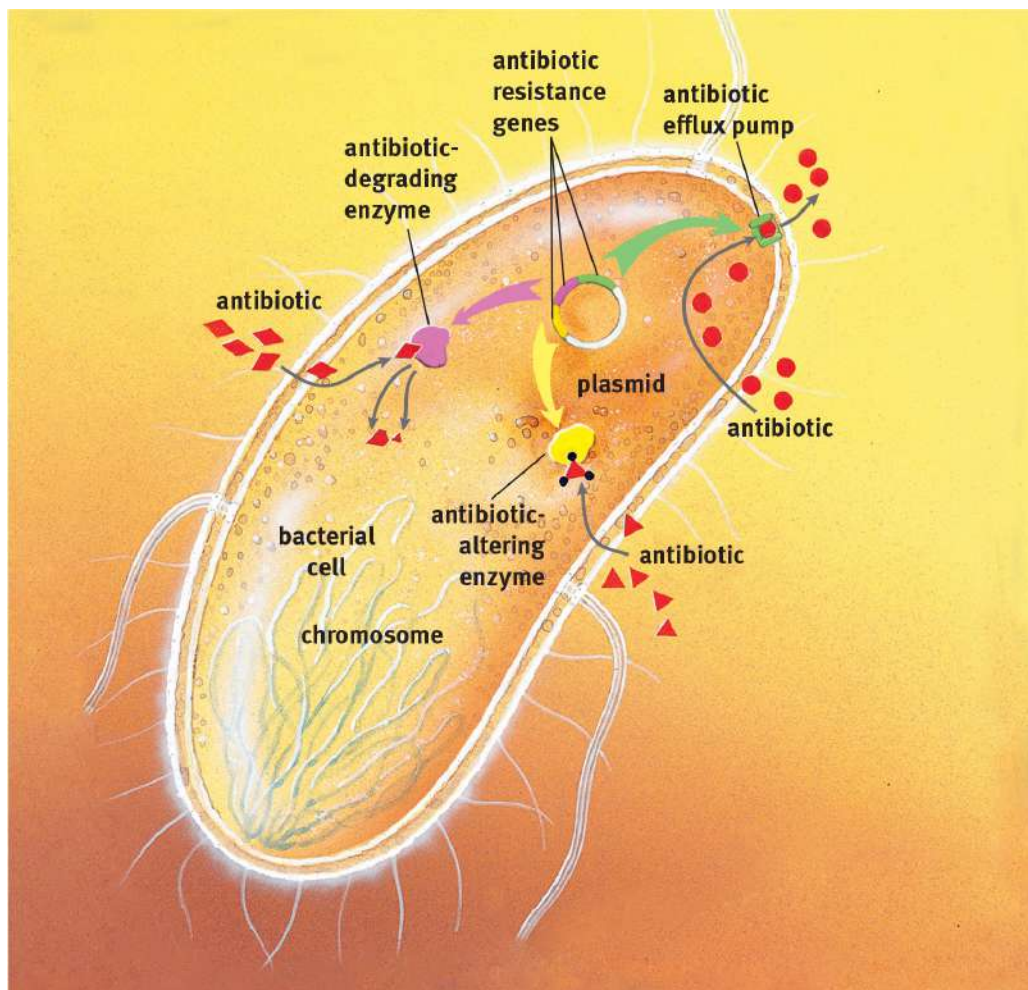


Figure 1.11. Mechanisms of Antibiotic Resistance

Bacterial genetic recombination helps increase bacterial diversity and thus permits evolution of a bacterial species over time. These recombination processes include transformation, conjugation, and transduction.

Transformation

Transformation results from the integration of foreign genetic material into the host genome. This foreign genetic material most frequently comes from other bacteria that, upon lysing, spill their contents into the vicinity of a bacterium capable of transformation. Many gram-negative rods are able to carry out this process.

Conjugation

Conjugation is the bacterial form of mating (sexual reproduction). It involves two cells forming a **conjugation bridge** between them that facilitates the transfer of genetic material. The transfer is unidirectional, from the **donor male (+)** to the **recipient female (-)**. The bridge is made from appendages called **sex pili** that are found on the donor male. To form the pilus, bacteria must contain plasmids known as **sex factors** that contain the necessary genes. The best-studied sex factor is the **F (fertility) factor** in *E. coli*. Bacteria possessing this plasmid are termed F^+ cells; those without are called F^- cells. During conjugation the F^+ cell replicates its F factor and donates the copy to the F^- cell, converting it to an F^+ cell. This enables the cell obtaining the new plasmid to then transfer copies to other cells. This method of genetic recombination allows for rapid acquisition of antibiotic resistance or virulence factors throughout a colony because other plasmids can also be passed through the conjugation bridge. The process of conjugation is illustrated in Figure 1.12.

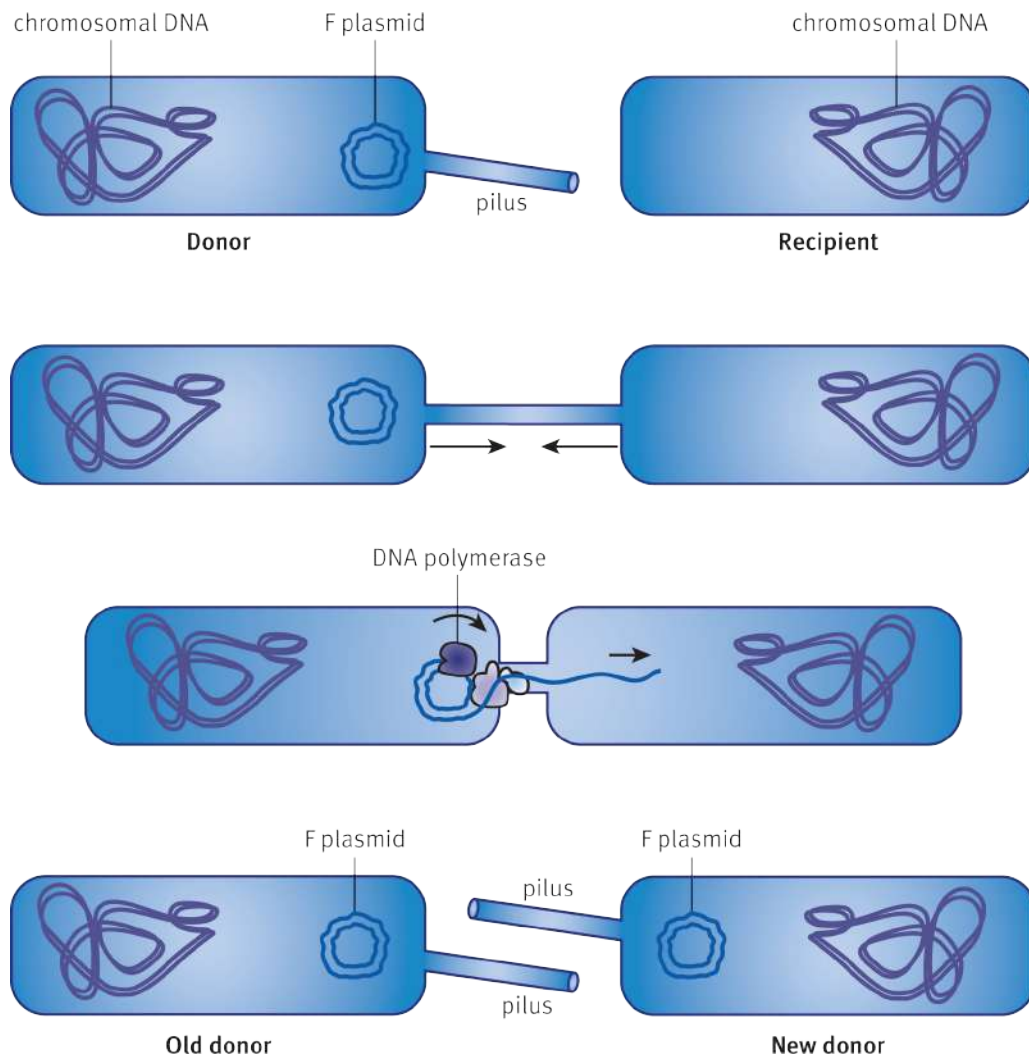


Figure 1.12. Bacterial Conjugation

The sex factor is a plasmid, but through processes such as transformation, it can become integrated into the host genome. In this case, when conjugation occurs, the entire genome replicates because it now contains the sex factor. The donor cell will then attempt to transfer an entire copy of its genome into the recipient; however, the bridge usually breaks before the full DNA sequence can be moved. Cells that have undergone this change are referred to by the abbreviation **Hfr** for **high frequency of recombination**.

Transduction

Transduction is the only genetic recombination process that requires a **vector**—a virus that carries genetic material from one bacterium to another. Viruses are obligate intracellular pathogens, which means that they cannot reproduce outside of a host cell. Because of this, **bacteriophages** (viruses that infect bacteria) can accidentally incorporate a segment of host DNA during assembly. When the bacteriophage infects another bacterium, it can release this trapped DNA into the new host cell. This transferred DNA can then integrate into the genome, giving the new host additional genes. The process of transduction is shown in Figure 1.13.

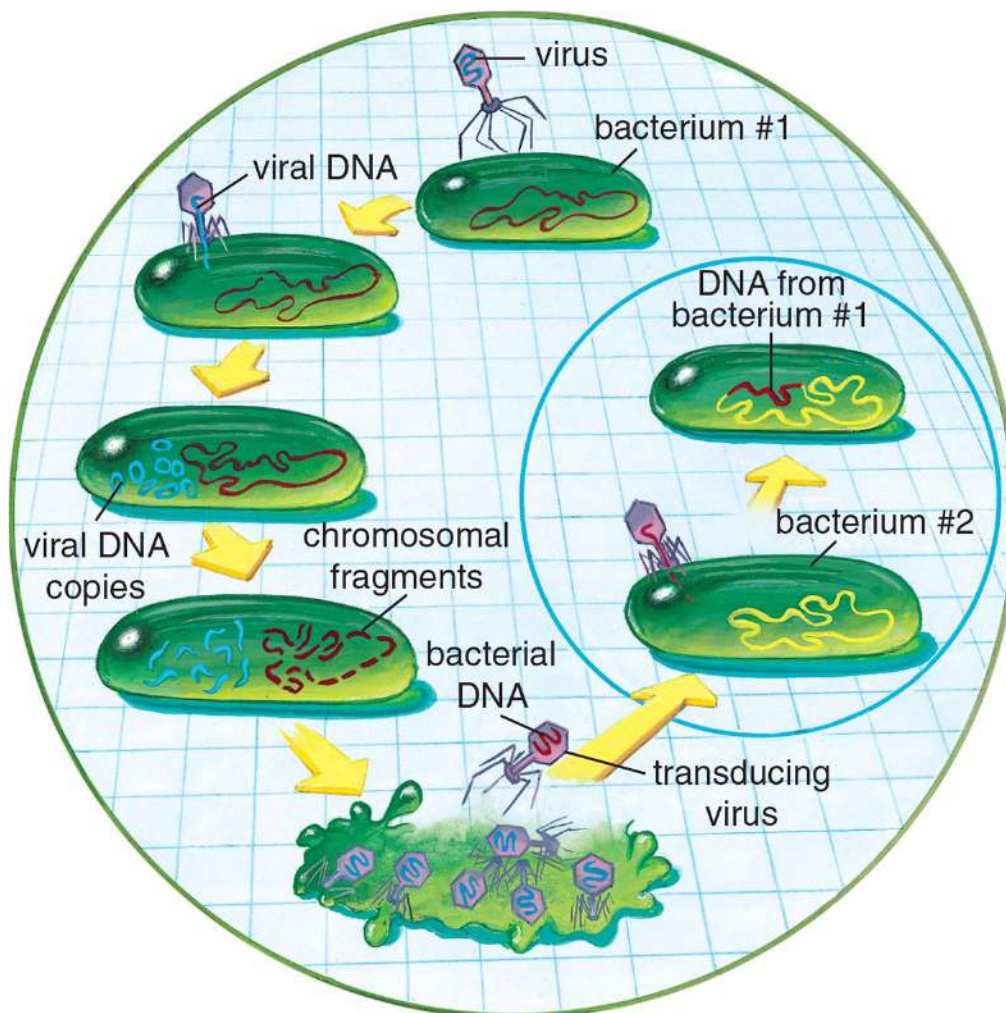


Figure 1.13. Bacterial Transduction

Transposons

Transposons are genetic elements capable of inserting and removing themselves from the genome. This phenomenon is not limited to prokaryotes; it has been seen in eukaryotes as well. If a transposon is inserted within a coding region of a gene, that gene may be disrupted.



REAL WORLD

One of the biggest challenges a doctor faces is patient compliance with treatment, especially antibiotics. Many patients fail to complete an entire course of antibiotics, often discontinuing the treatment because they feel better. Unfortunately, this breeds antibiotic resistance by killing off the bacteria that are nonresistant and leaving behind bacteria that are more resistant. These resistant bacteria then reproduce, resulting in recurrence of the infection. Over time, this practice has led to bacteria that are resistant to multiple antibiotics, making common infections more difficult to treat.

GROWTH

As discussed previously, bacteria reproduce via binary fission. This implies that all of the bacteria are exactly the same in a local colony (assuming no mutations or genetic recombination), and that no bacteria are dividing faster

than the others. Bacteria can be said to grow in a series of phases, as shown in Figure 1.14. In a new environment, bacteria first adapt to the new local conditions during the **lag phase**. As the bacteria adapt, the rate of division increases, causing an exponential increase in the number of bacteria in the colony during the **exponential phase**, also called the **log phase**. As the number of bacteria in the colony grows, resources are often reduced. The reduction of resources slows reproduction, and the **stationary phase** results. After the bacteria have exceeded the ability of the environment to support the number of bacteria, a **death phase** occurs, marking the depletion of resources.

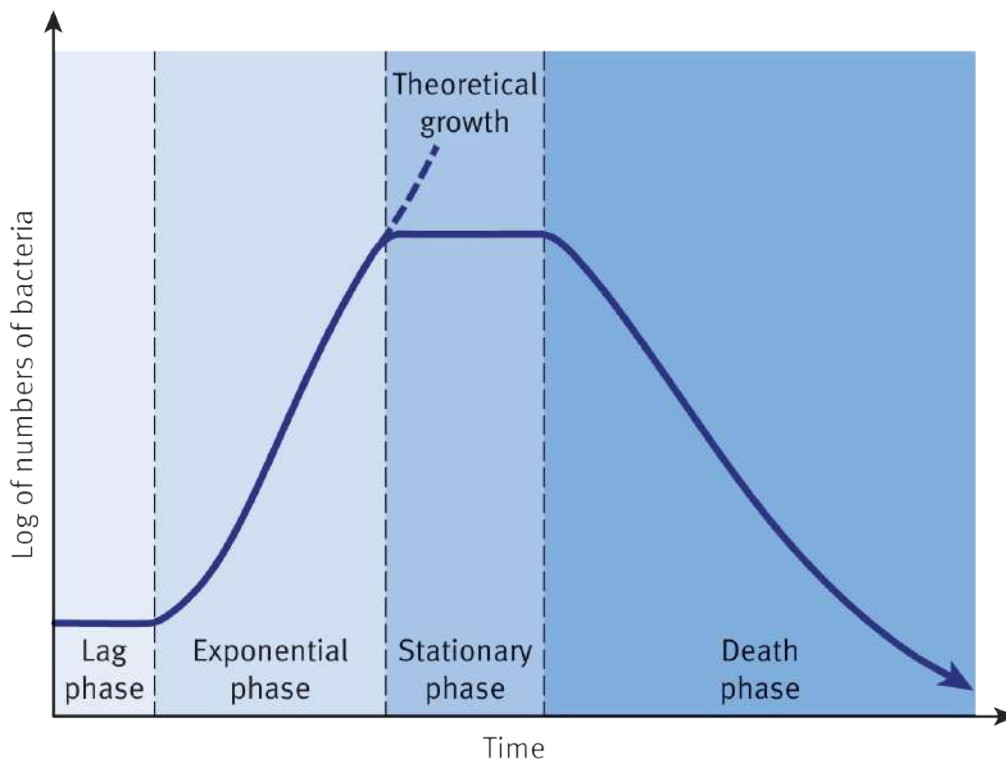


Figure 1.14. Bacterial Growth Curve



The bacterial growth curve is an example of a semilog plot. The fact that the y-axis is logarithmic means that a straight line (as seen during the exponential phase) actually represents an exponential increase in the number of bacteria, not a linear increase. Semilog and log-log plots are discussed in Chapter 12 of *MCAT Physics and Math Review*.

MCAT CONCEPT CHECK 1.4:

Before you move on, assess your understanding of the material with these questions.

1. Briefly describe the three mechanisms of bacterial genetic recombination:

- Transformation:

- Conjugation:

- Transduction:

2. What are the four phases of the bacterial growth curve?
What are the features of each phase?

Phase	Features

1.5 Viruses and Subviral Particles

High-Yield



LEARNING OBJECTIVES

After Chapter 1.5, you will be able to:

- Explain why viruses are defined as “obligate intracellular parasites”
- Recall key virus terminology, including retrovirus and positive/negative sense
- Describe how viruses are able to produce progeny via infection of a host cell
- Compare and contrast the lytic and lysogenic cycles
- Describe how prions and viroids are able to cause disease

At the beginning of this chapter, we discussed the cell theory and noted that viruses do not fit the definition of living things because they are acellular. Viruses may be as small as 20 nm or as large as 300 nm. For reference, prokaryotes are 1–10 μm , and eukaryotes are about ten times larger. Unlike eukaryotic cells, viruses lack organelles and a nucleus.

VIRAL STRUCTURE

Viruses are composed of genetic material, a protein coat, and sometimes an envelope containing lipids. The genetic information may be circular or linear, single- or double-stranded, and composed of either DNA or RNA. The protein coat is known as a **capsid**, which may be surrounded by an envelope composed of phospholipids and virus-specific proteins. The envelope is very sensitive to heat, detergents, and desiccation; thus, enveloped viruses are easier to kill. On the other hand, viruses that do not have an envelope are more resistant to sterilization and are likely to persist on surfaces for an extended period of time.

Because viruses cannot reproduce independently, they are considered obligate intracellular parasites. Viruses must express and replicate genetic information within a **host cell** because they lack ribosomes to carry out protein synthesis. After hijacking a cell's machinery, a virus will replicate and produce viral progeny, called **virions**, which can be released to infect additional cells.

Bacteriophages are viruses that specifically target bacteria. They do not actually enter bacteria; rather, they simply inject their genetic material, leaving the remaining structures outside the infected cell. In addition to a capsid, bacteriophages contain a tail sheath and tail fibers, as shown in Figure 1.15. The **tail sheath** can act like a syringe, injecting genetic material into a bacterium. The **tail fibers** help the bacteriophage recognize and connect to the correct host cell.

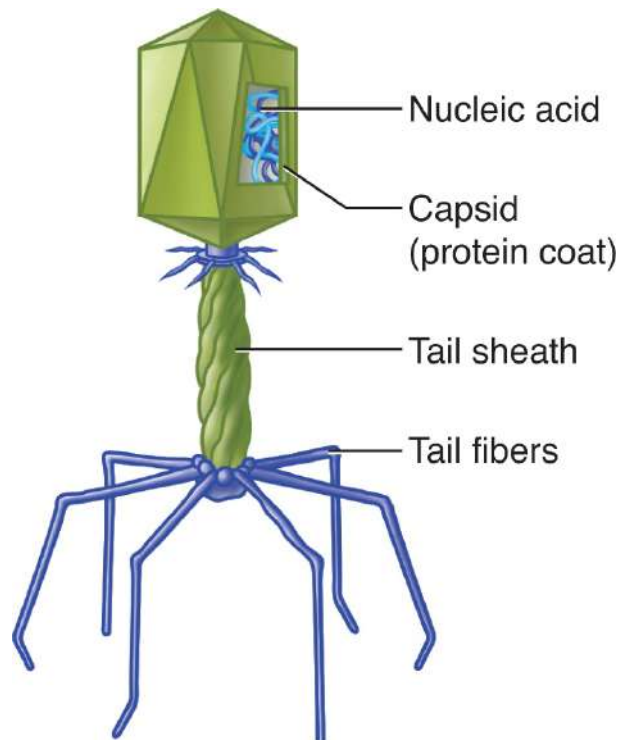


Figure 1.15. Structure of a Bacteriophage

VIRAL GENOMES

Viral genomes come in a variety of shapes and sizes. Some are made of only a few genes, while others have several hundred. In addition, viral genomes may be made of either single- or double-stranded DNA or RNA.

Single-stranded RNA viruses may be positive sense or negative sense.

Positive sense implies that the genome may be directly translated to functional proteins by the ribosomes of the host cell, just like mRNA.

Negative-sense RNA viruses are a bit more complicated: the negative-sense RNA strand acts as a template for synthesis of a complementary strand, which can then be used as a template for protein synthesis. Negative-sense

RNA viruses must carry an **RNA replicase** in the virion to ensure that the complementary strand is synthesized.

Retroviruses are enveloped, single-stranded RNA viruses in the family *Retroviridae*; usually, the virion contains two identical RNA molecules. These viruses carry an enzyme known as **reverse transcriptase**, which synthesizes DNA from single-stranded RNA. The DNA then integrates into the host cell genome, where it is replicated and transcribed as if it were the host cell's own DNA. This is a clever mechanism because the integration of the genetic material into the host cell genome allows the cell to be infected indefinitely, so the only way to remove the infection is to kill the infected cell. The human immunodeficiency virus (HIV) is a retrovirus that utilizes this life cycle, which is one of the characteristics that make HIV so difficult to treat, as shown in Figure 1.16.

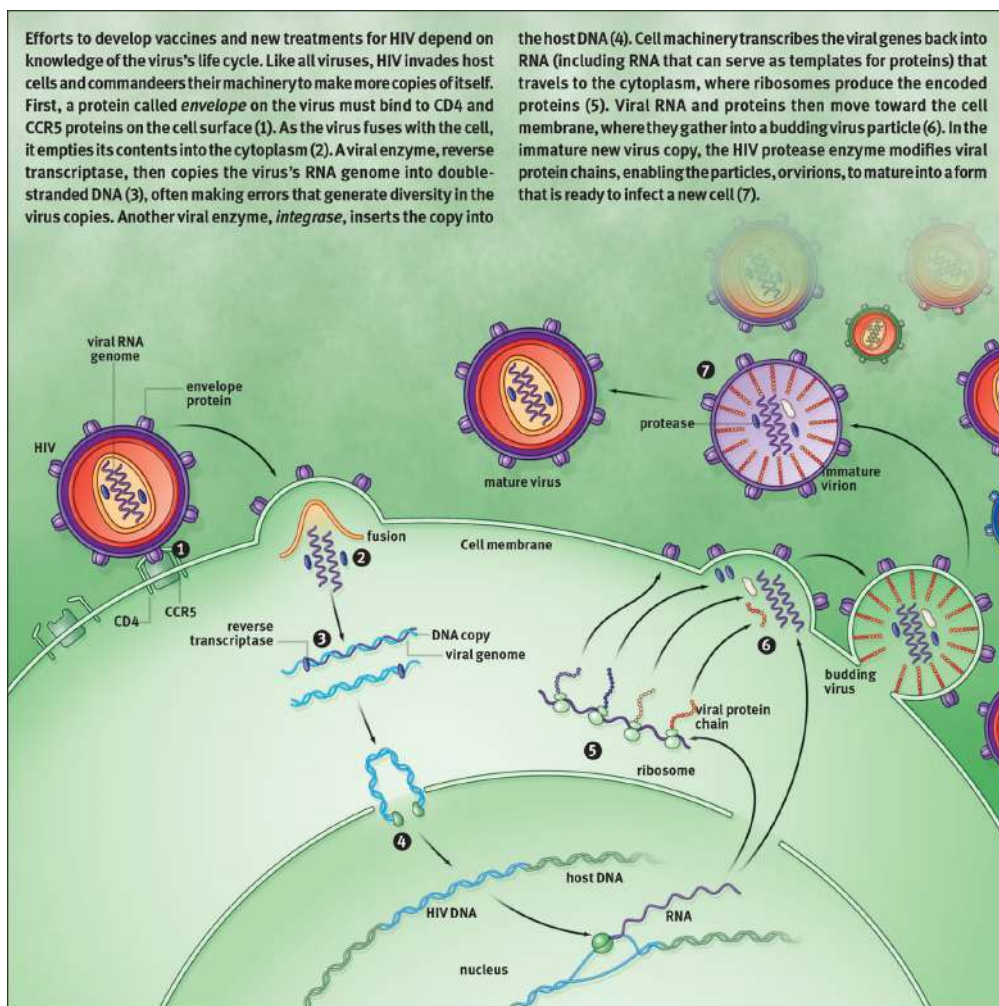


Figure 1.16. Life Cycle of the Human Immunodeficiency Virus (HIV)



REAL WORLD

Both retroviruses and transduction are under investigation as methods of gene therapy. It is theorized that retroviral and transduction methods can deliver functional versions of missing or altered genes, so that the correct proteins can be synthesized and certain diseases can be treated. Gene therapy is discussed in Chapter 6 of *MCAT Biochemistry Review*.

VIRAL LIFE CYCLE

As we have already discussed, viruses must infect a host cell and use the host cell's machinery in order to reproduce. Here, we will discuss the viral life cycle in detail.

Infection

Viruses can only infect a specific set of cells, because they must bind to specific receptors on the host cell. Without the proper receptors, a cell is essentially invisible to the virus. Once the virus binds the correct receptor, the virus and the cell are brought into close enough proximity to permit additional interactions. Enveloped viruses fuse with the plasma membrane of a cell, allowing entry of the virion into the host cell. Sometimes a host cell may mistake a virus bound to the membrane as nutrients or other useful molecules and will actually bring the virus into the cytoplasm via endocytosis. As mentioned earlier, bacteriophages use tail fibers to anchor themselves to the cell membrane and then inject their genome into the host bacterium through the tail sheath. Some tail fibers even have enzymatic activity, allowing for both penetration of the cell wall and the formation of pores in the cell membrane.



REAL WORLD

HIV must bind to a receptor called CCR5 on white blood cells in order to infect them. Some people lack this receptor and are thus immune to HIV. There was recently a case in which an HIV-positive man with

leukemia received a bone marrow transplant from a donor that lacked CCR5. Not only was his leukemia cured, but he experienced remission of his HIV infection because the white blood cells from his newly acquired bone marrow were not susceptible to HIV infection.

Depending on the virus, different portions of the virion will be inserted into host cells. Enveloped viruses such as HIV fuse with the membrane and enter the cell intact, whereas bacteriophages only insert their genetic material, leaving their capsids outside the host cell.

Translation and Progeny Assembly

After infection, translation of viral genetic material must occur in order for the virus to reproduce. This requires translocation of the genetic material to the correct location in the cell. Most DNA viruses must enter the nucleus in order to be transcribed into mRNA. The mRNA then goes to the cytoplasm, where it is translated into proteins. Genetic material from positive-sense RNA viruses stays in the cytoplasm, where it is directly translated into protein by host cell ribosomes. Negative-sense RNA viruses require synthesis of a complementary RNA strand via RNA replicase, which can then be translated to form proteins. DNA formed through reverse transcription in retroviruses also travels to the nucleus, where it can be integrated into the host genome.

Using the ribosomes, tRNA, amino acids, and enzymes of the host cell, viral RNA is translated into protein. Many of these proteins are structural capsid proteins and allow for the creation of new virions in the cytoplasm in the host cell. Once the viral genome has been replicated, it can be packaged within the capsid. Note that the viral genome must be returned to its original form

before packaging; for example, retroviruses must transcribe new copies of their single-stranded RNA from the DNA that entered the host genome. A single virus may create anywhere from hundreds to many thousands of new virions within a single host cell.

Progeny Release

Viral progeny may be released in multiple ways. First, the viral invasion may initiate cell death, which results in spilling of the viral progeny. Second, the host cell may lyse as a result of being filled with extremely large numbers of virions. Lysis is actually a disadvantage for the virus because the virus can no longer use the cell to carry out its life cycle. Finally, a virus can leave the cell by fusing with its plasma membrane as shown in Figure 1.17, a process known as **extrusion**. This process keeps the host cell alive, and thus allows for the continued use of the host cell by the virus. A virus in this state is said to be in a **productive cycle**.

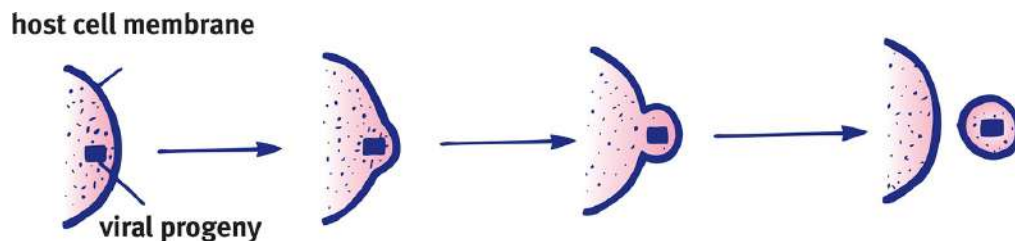


Figure 1.17. Viral Extrusion

Lytic and Lysogenic Cycles

Depending on growth conditions and the specific virus, bacteriophages may enter a lytic or lysogenic life cycle. These two phases are similar to the lysis and productive cycle methods of progeny release discussed above.

During a **lytic cycle**, the bacteriophage maximizes the use of the cell's machinery with little regard for the survival of the host cell. Once the host is swollen with new virions, the cell lyses, and other bacteria can be infected. Viruses in the lytic phase are termed **virulent**.

In the event that the virus does not lyse the bacterium, it may integrate into the host genome as a **provirus** or **prophage**, initiating the **lysogenic cycle**. In this case, the virus will be replicated as the bacterium reproduces because it is now a part of the host's genome. Although the virus may remain integrated into the host genome indefinitely, environmental factors (radiation, light, or chemicals) may cause the provirus to leave the genome and revert to a lytic cycle. As mentioned earlier, the provirus may extract bacterial genes as it leaves the genome, which allows transduction of genes from one bacterium to another. Although bacteriophages can kill their hosts, integration of the phage into the host genome may actually benefit the bacterium. Infection with one strain of phage generally makes the bacterium less susceptible to **superinfection** (simultaneous infection) with other phages. Because the provirus is relatively innocuous, this arrangement may confer an evolutionary advantage. The lytic and lysogenic cycles are contrasted in Figure 1.18.

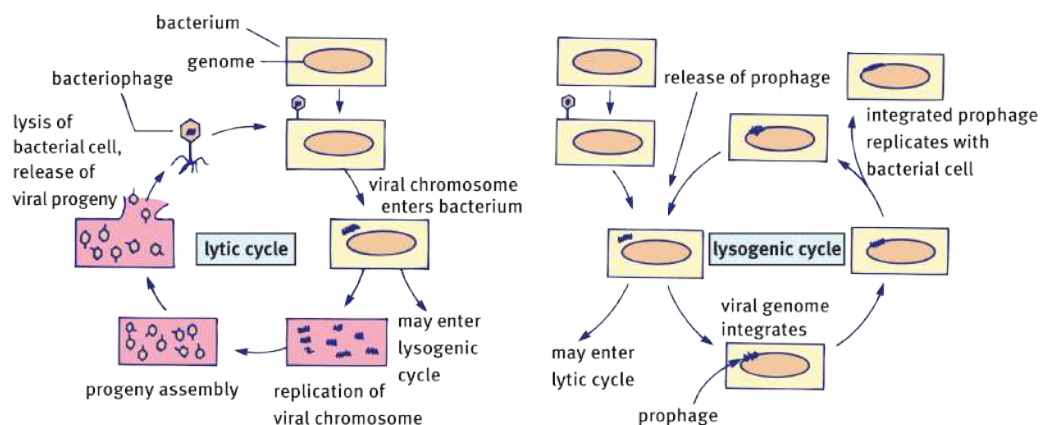


Figure 1.18. Lytic vs. Lysogenic Cycles of Bacteriophages

PRIONS AND VIROIDS

Prions and viroids are very small (subviral) particles that can cause disease under certain circumstances.

Prions

Prions are infectious proteins and are, therefore, nonliving things. They cause disease by triggering misfolding of other proteins, usually through the conversion of a protein from an α -helical structure to a β -pleated sheet. This drastically reduces the solubility of the protein, as well as the ability of the cell to degrade the protein. Eventually, protein aggregates form, interfering with cell function. Prions are known to cause *bovine spongiform encephalopathy* (mad cow disease), *Creutzfeldt–Jakob disease*, and *familial fatal insomnia* in humans.

Viroids

Viroids are small pathogens consisting of a very short circular single-stranded RNA that infect plants. Viroids can bind to a large number of RNA sequences and can silence genes in the plant genome. This prevents synthesis of necessary proteins, resulting in metabolic disruption and structural damage to the cell. Viroids are classically thought of as plant pathogens, but a few examples of human viroids do exist, including the hepatitis D virus (HDV). Alone, HDV is innocuous; however, when coinfecting with hepatitis B virus (HBV), HDV is able to exert its silencing effect on human hepatocytes.

MCAT CONCEPT CHECK 1.5:

Before you move on, assess your understanding of the material with these questions.

1. Why are viruses considered obligate intracellular parasites?

2. Coronavirus, which causes the common cold, is described as an enveloped, single-stranded positive-sense RNA virus. What does this indicate about the virus?

3. Briefly describe the pathway of retroviral nucleic acids from infection of a host cell to release of viral progeny:

4. What are the differences between the lytic and lysogenic cycles?

- Lytic cycle:

- Lysogenic cycle:

5. How do prions cause disease?



Conclusion

Our first chapter introduced the basis of all biology: cell theory. All living things are made of prokaryotic or eukaryotic cells. Prokaryotes are simpler and do not contain membrane-bound organelles. Eukaryotes contain membrane-bound organelles with highly specialized functions. While eukaryotic organisms (especially humans!) will be the primary focus on Test Day, an understanding of prokaryotic structure and physiology is necessary to understand infectious disease. Viruses are nonliving infectious particles that must invade cells in order to reproduce. Finally, we discussed the smallest infectious particles, prions and viroids.

Our discussions from here on out will focus primarily on eukaryotes, but you will spend significant time in your clinical career battling the pathogens we've presented in this chapter. Vaccines are available for a number of bacteria (*Bacillus anthracis* [anthrax], *Corynebacterium diphtheriae* [diphtheria], *Haemophilus influenzae* type B [many upper respiratory and ear infections], *Neisseria meningitidis* [some cases of bacterial meningitis], *Streptococcus pneumoniae* [many cases of bacterial pneumonia], *Clostridium tetani* [tetanus], *Salmonella typhi* [typhoid], and *Bordetella pertussis* [whooping cough]) and viruses (varicella zoster virus [chicken pox and shingles], hepatitis A and B viruses, human papillomavirus [HPV], influenza, measles, mumps, polio, rabies, rotavirus, rubella, and yellow fever). Those for which we do not have vaccines may be targeted with antibiotic and antiviral therapies; the appropriate pharmacotherapy usually depends on an understanding of bacterial or viral physiology and the known resistance patterns in the local

area. In other words, the principles presented in this chapter will show up in your everyday life as a physician!

We've discussed one method of cellular reproduction: the division of a bacterial cell into two cells by binary fission. Eukaryotic cells also must replicate, but use a different process: mitosis. In the next chapter, we will turn our attention to mitosis, as well as meiosis and human reproduction.

Concept Summary

Cell Theory

- The **cell theory** has four basic tenets:
 - All living things are composed of cells.
 - The cell is the basic functional unit of life.
 - Cells arise only from preexisting cells.
 - Cells carry genetic information in the form of DNA. This genetic material is passed on from parent to daughter cell.
- Viruses are not considered living things because they are acellular, cannot reproduce without the assistance of a host cell, and may use RNA as their genetic material.

Eukaryotic Cells

- **Eukaryotes** have membrane-bound organelles, a nucleus, and may form multicellular organisms.
- The cell membrane and membranes of organelles contain phospholipids, which organize to form hydrophilic interior and exterior surfaces with a hydrophobic core.
- The **cytosol** suspends the organelles and allows diffusion of molecules throughout the cell.
- The eukaryotic organelles each serve specific functions:

- The **nucleus** contains DNA organized into **chromosomes**. It is surrounded by the **nuclear membrane** or **envelope**, a double membrane that contains **nuclear pores** for two-way exchange of materials between the nucleus and cytosol. DNA is organized into coding regions called **genes**.
- The **nucleolus** is a subsection of the nucleus in which ribosomal RNA (rRNA) is synthesized.
- **Mitochondria** contain an outer and inner membrane. The **outer membrane** forms a barrier with the cytosol; the **inner membrane** is folded into **cristae** and contains enzymes for the electron transport chain. Between the membranes is the **intermembrane space**; inside the inner mitochondrial membrane is the mitochondrial **matrix**. Mitochondria can divide independently of the nucleus via binary fission and can trigger **apoptosis** by releasing mitochondrial enzymes into the cytoplasm.
- **Lysosomes** contain hydrolytic enzymes that can break down substances ingested by endocytosis and cellular waste products. When these enzymes are released, **autolysis** of the cell can occur.
- The **endoplasmic reticulum (ER)** is a series of interconnected membranes and is continuous with the nuclear envelope. The **rough ER (RER)** is studded with ribosomes, which permit translation of proteins destined for secretion. The **smooth ER (SER)** is used for lipid synthesis and detoxification.
- The **Golgi apparatus** consists of stacked membrane-bound sacs in which cellular products can be modified, packaged, and directed to specific cellular locations.
- **Peroxisomes** contain hydrogen peroxide and can break down very long chain fatty acids via β -oxidation. They also participate in phospholipid synthesis and the pentose phosphate pathway.

- The **cytoskeleton** provides stability and rigidity to the overall structure of the cell, while also providing transport pathways for molecules within the cell.
 - **Microfilaments** are composed of **actin**. They provide structural protection for the cell and can cause muscle contraction through interactions with **myosin**. They also help form the **cleavage furrow** during **cytokinesis** in mitosis.
 - **Microtubules** are composed of **tubulin**. They create pathways for motor proteins like **kinesin** and **dynein** to carry vesicles. They also contribute to the structure of **cilia** and **flagella**, where they are organized into nine pairs of microtubules in a ring with two microtubules at the center (**9 + 2 structure**). **Centrioles** are found in **centrosomes** and are involved in microtubule organization in the mitotic spindle.
 - **Intermediate filaments** are involved in cell–cell adhesion and maintenance of the integrity of the cytoskeleton; they help anchor organelles. Common examples include keratin and desmin.
- **Epithelial tissues** cover the body and line its cavities, protecting against pathogen invasion and desiccation. Some epithelial cells absorb or secrete substances, or participate in sensation.
 - In most organs, epithelial cells form the **parenchyma**, or the functional parts of the organ.
 - Epithelial cells may be polarized, with one side facing a lumen or the outside world, and the other side facing blood vessels and structural cells.
 - Epithelia can be classified by the number of layers: **simple epithelia** have one layer, **stratified epithelia** have many layers, and **pseudostratified epithelia** appear to have multiple layers due to

- differences in cell heights, but actually have only one layer.
- Epithelia can be classified by the shapes of the cells: **cuboidal cells** are cube-shaped, **columnar cells** are long and narrow, and **squamous cells** are flat and scale-like.
 - **Connective tissues** support the body and provide a framework for epithelial cells.
 - In most organs, connective tissues form the **stroma** or support structure by secreting materials to form an **extracellular matrix**.
 - Bone, cartilage, tendons, ligaments, adipose tissue, and blood are all connective tissues.

Classification and Structure of Prokaryotic Cells

- **Prokaryotes** do not contain membrane-bound organelles; they contain their genetic material in a single circular molecule of DNA located in the **nucleoid region**.
- There are three overarching domains of life; prokaryotes account for two of these:
 - **Archaea** are often extremophiles, living in harsh environments (high temperature, high salinity, no light) and often use chemical sources of energy (chemosynthesis) rather than light (photosynthesis). They have similarities to both eukaryotes (start translation with methionine, similar RNA polymerases, histones) and bacteria (single circular chromosome, divide by binary fission or budding).
 - **Bacteria** have many similar structures to eukaryotes, and have complex relationships with humans, including **mutualistic symbiosis** and

pathogenesis.

- **Eukarya** is the only non-prokaryotic domain.
- Bacteria can be classified by shape:
 - Spherical bacteria are called **cocci**.
 - Rod-shaped bacteria are called **bacilli**.
 - Spiral-shaped bacteria are called **spirilli**.
- Bacteria can be classified based on metabolic processes:
 - **Obligate aerobes** require oxygen for metabolism.
 - **Obligate anaerobes** cannot survive in oxygen-containing environments and can only carry out anaerobic metabolism.
 - **Facultative anaerobes** can survive in environments with or without oxygen and will toggle between metabolic processes based on the environment.
 - **Aerotolerant anaerobes** cannot use oxygen for metabolism, but can survive in an oxygen-containing environment.
- The cell wall and cell membrane of bacteria form the **envelope**. Together, they control the movement of solutes into and out of the cell.
 - Bacteria can be classified by the color their cell walls turn during Gram staining with a crystal violet stain, followed by a counterstain with safranin. Gram-positive bacteria turn purple, while gram-negative bacteria turn pink-red.
 - Gram-positive bacteria have a thick cell wall composed of **peptidoglycan** and **lipoteichoic acid**.
 - Gram-negative bacteria have a thin cell wall composed of peptidoglycan

and an outer membrane containing phospholipids and **lipopolysaccharides**.

- Bacteria may have one, two, or many flagella that generate propulsion to move the bacterium toward food or away from immune cells. Moving in response to chemical stimuli is called **chemotaxis**. Bacterial flagella contain a filament composed of flagellin, a basal body that anchors and rotates the flagellum, and a hook that connects the two.
- Prokaryotes carry out the electron transport chain using the cell membrane.
- Prokaryotic ribosomes are smaller than eukaryotic ribosomes (30S and 50S, rather than 40S and 60S).

Genetics and Growth of Prokaryotic Cells

- Prokaryotes multiply through **binary fission**, in which the chromosome replicates while the cell grows in size, until the cell wall begins to grow inward along the midline of the cell and divides it into two identical daughter cells.
- In addition to the single circular chromosome in prokaryotes, extrachromosomal material can be carried in **plasmids**. Plasmids may contain antibiotic resistance genes or **virulence factors**. Plasmids that can integrate into the genome are called **episomes**.
- Bacterial genetic recombination increases bacterial diversity.
 - **Transformation** occurs when genetic material from the surroundings is taken up by a cell, which can incorporate this material into its genome.
 - **Conjugation** is the transfer of genetic material from one bacterium to another across a **conjugation bridge**; a plasmid can be transferred from

F⁺ cells to **F⁻ cells**, or a portion of the genome can be transferred from an **Hfr cell** to a recipient.

- **Transduction** is the transfer of genetic material from one bacterium to another via a bacteriophage vector.
 - **Transposons** are genetic elements that can insert into or remove themselves from the genome.
- Bacterial growth follows a predictable pattern:
 - The bacteria adapt to new local conditions during the **lag phase**.
 - Growth then increases exponentially during the **exponential (log) phase**.
 - As resources are reduced, growth levels off during the **stationary phase**.
 - As resources are depleted, bacteria undergo a **death phase**.

Viruses and Subviral Particles

- Viruses contain genetic material, a protein coat (**capsid**), and sometimes a lipid-containing envelope.
- Viruses are obligate intracellular parasites, meaning that they cannot survive and replicate outside of a **host cell**. Individual virus particles are called **virions**.
- Bacteriophages are viruses that target bacteria. In addition to the other structures, they contain a **tail sheath**, which injects the genetic material into a bacterium, and **tail fibers**, which allow the bacteriophage to attach to the host cell.
- Viral genomes may be made of various nucleic acids:
 - They may be composed of DNA or RNA and may be single- or double-

stranded.

- Single-stranded RNA viruses may be **positive sense** (that can be translated by the host cell) or **negative sense** (which requires a complementary strand to be synthesized by **RNA replicase** before translation).
 - **Retroviruses** contain a single-stranded RNA genome, from which a complementary DNA strand is made using **reverse transcriptase**. The DNA strand can then be integrated into the genome.
-
- Viruses infect cells by attaching to specific receptors, and can then enter the cell by fusing with the plasma membrane, being brought in by endocytosis, or injecting their genome into the cell.
 - The virus reproduces by replicating and translating genetic material using the host cell's ribosomes, tRNA, amino acids, and enzymes.
 - Viral progeny are released through cell death, lysis, or **extrusion**.
 - Bacteriophages have two specific life cycles:
 - In the **lytic cycle**, the bacteriophage produces massive numbers of new virions until the cell lyses. Bacteria in the lytic phase are termed **virulent**.
 - In the **lysogenic cycle**, the virus integrates into the host genome as a pro-virus or prophage, which can then reproduce along with the cell. The provirus can remain in the genome indefinitely, or may leave the genome in response to a stimulus and enter the lytic cycle.
-
- **Prions** are infectious proteins that trigger misfolding of other proteins, usually converting an α -helical structure to a β -pleated sheet. This decreases the solubility of the protein and increases its resistance to degradation.
 - **Viroids** are plant pathogens that are small circles of complementary RNA

that can turn off genes, resulting in metabolic and structural changes and, potentially, cell death.

Answers to Concept Checks

1.1

- .. All living things are made of cells. The cell is the basic functional unit of life. All cells arise from other cells. Genetic information is carried in the form of deoxyribonucleic acid (DNA) and is passed from parent to daughter cell.

1.2

- .. The nucleus stores genetic information and is the site of transcription. The mitochondria are involved in ATP production and apoptosis. Lysosomes break down cellular waste products and molecules ingested through endocytosis, and can also be involved in apoptosis. The rough endoplasmic reticulum synthesizes proteins destined for secretion. The smooth endoplasmic reticulum is involved in lipid synthesis and detoxification. The Golgi apparatus packages, modifies, and distributes cellular products. Peroxisomes break down very long chain fatty acids, synthesize lipids, and contribute to the pentose phosphate pathway.
- !. Peroxisomes are dependent on hydrogen peroxide for their functions, so an enzyme deficiency that results in an inability to form hydrogen peroxide would likely result in an inability to digest very long chain fatty acids. These fatty acids would build up in peroxisomes until they displaced cellular contents, ultimately resulting in cell death.
- }. Microfilaments are composed of actin. Microtubules are composed of tubulin. Intermediate filaments differ by cell type, but may be composed of keratin, desmin, vimentin, and lamins.
- !. Centrioles consist of nine triplets of microtubules around a hollow center,

while flagella consist of nine doublets on the outside, with two microtubules on the inside.

- i. Endothelial cells and α -cells are epithelial cells. Fibroblasts, osteoblasts, and chondroblasts are connective tissue cells.

1.3

- .. Archaea are similar to bacteria in that both are single-celled organisms that lack a nucleus or membrane-bound organelles, contain a single circular chromosome, and divide by binary fission or budding. They are similar to eukaryotes in that they start translation with methionine, contain similar RNA polymerases, and contain DNA associated with histones.
- l. The three common shapes of bacteria are spherical (cocci), rod-shaped (bacilli), and spiral-shaped (spirilli).

- }.

	Oxygen Present		Oxygen Absent	
Type of Bacteria	Can survive	Can carry out aerobic metabolism	Can survive	Can carry out anaerobic metabolism
Obligate aerobe	Yes	Yes	No	No
Facultative anaerobe	Yes	Yes	Yes	Yes
Obligate anaerobe	No	No	Yes	Yes
Aerotolerant anaerobe	Yes	No	Yes	Yes

- l. The antibiotic penicillin targets the enzyme that catalyzes the cross-linking of peptidoglycan. Gram-positive bacteria have a thick layer of

peptidoglycan and lipoteichoic acid, and contain no outer membrane, whereas gram-negative bacteria have only a thin layer of peptidoglycan but also have an outer membrane containing lipopolysaccharides and phospholipids. Penicillin and antibiotics with similar function can more easily reach and weaken the peptidoglycan layer of gram-positive bacteria.

- i. Eukaryotic flagella contain microtubules composed of tubulin, organized in a 9 + 2 arrangement. Bacterial flagella are made of flagellin and consist of a filament, a basal body, and a hook.

1.4

- .. Transformation is the acquisition of exogenous genetic material that can be integrated into the bacterial genome. Conjugation is the transfer of genetic material from one bacterium to another across a conjugation bridge; a plasmid can be transferred from F⁺ cells to F⁻ cells, or a portion of the genome can be transferred from an Hfr cell to a recipient. Transduction is the transfer of genetic material from one bacterium to another by a bacteriophage.

!..	Phase	Features
	Lag phase	Bacteria get used to environment; little growth
	Exponential phase	Bacteria use available resources to multiply at an exponential rate
	Stationary phase	Bacterial multiplication slows as resources are used up
	Death phase	Bacteria die as resources become insufficient to support the colony

1.5

- .. Viruses do not contain organelles such as ribosomes; therefore, in order to reproduce and synthesize proteins, viruses must infect cells and hijack

their machinery.

2. This description indicates that the virus contains an outer layer of phospholipids with an inner capsid. Within the capsid, there is single-stranded RNA that can be immediately translated to protein by the ribosomes of the host cell.
3. The nucleic acid enters as single-stranded RNA, which undergoes reverse transcription (using reverse transcriptase) to form double-stranded DNA. This DNA enters the host genome and replicates with the host cell. The DNA is transcribed to mRNA, which can be used to make structural proteins. This mRNA doubles as the viral genome for new virions. Once new virions are assembled from the structural proteins and mRNA (single-stranded RNA) genome, the virions can be released to infect other cells.
4. In the lytic cycle, bacteriophages replicate in the host cell in extremely high numbers until the host cell lyses and releases the virions. In the lysogenic cycle, the bacteriophage genome enters the host genome and replicates with the host cell as a provirus. In response to an appropriate stimulus, the provirus may leave the host genome and can be used to synthesize new virions.
5. Prions cause disease by triggering a change in the conformation of a protein from an α -helix to a β -pleated sheet. This change reduces solubility of the protein and makes it highly resistant to degradation.



Shared Concepts

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biochemistry Chapter 8

Biological Membranes

Biochemistry Chapter 10

Carbohydrate Metabolism II

Biology Chapter 2

Reproduction

Biology Chapter 8

The Immune System

Biology Chapter 12

Genetics and Evolution

Discrete Practice Questions

1. Hyperbaric oxygen may be used as a treatment for certain types of bacterial infections. In this therapy, the patient is placed in a chamber in which the partial pressure of oxygen is significantly increased, increasing the partial pressure of oxygen in the patient's tissues. This treatment is most likely used for infections with:
 - (A) obligate aerobic bacteria.
 - (B) facultative anaerobic bacteria.
 - (C) aerotolerant anaerobic bacteria.
 - (D) obligate anaerobic bacteria.

2. Which of the following does NOT describe connective tissue cells?
 - (A) They account for most cells in muscles, bones, and tendons.
 - (B) They secrete substances to form the extracellular matrix.
 - (C) In organs, they tend to form the stroma.
 - (D) In organs, they provide support for epithelial cells.

3. Which of the following types of nucleic acid could form the genome of a virus?
 - I. Single-stranded RNA

II. Double-stranded DNA

III. Single-stranded DNA

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

4. Which of the following activities occurs in the Golgi apparatus?

- (A) Synthesis of proteins
- (B) Modification and distribution of proteins
- (C) Breakdown of lipids and carbohydrates
- (D) Production of ATP

5. Mitochondrial DNA is:

- I.** circular.
- II.** self-replicating.
- III.** single-stranded.

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

6. Which of the following is NOT a function of the smooth endoplasmic

reticulum?

- (A) Lipid synthesis
- (B) Poison detoxification
- (C) Protein synthesis
- (D) Transport of proteins

7. What is the main function of the nucleolus?

- (A) Ribosomal RNA synthesis
- (B) DNA replication
- (C) Cell division
- (D) Chromosome assembly

8. Which of the following organelles is surrounded by a single membrane?

- (A) Lysosomes
- (B) Mitochondria
- (C) Nuclei
- (D) Ribosomes

9. Which of the following is NOT a difference that would allow one to distinguish a prokaryotic and a eukaryotic cell?

- (A) Ribosomal subunit weight
- (B) Presence of a nucleus

- (C) Presence of a membrane on the outside surface of the cell
- (D) Presence of membrane-bound organelles

10. Which of the following does NOT contain tubulin?

- (A) Cilia
- (B) Flagella
- (C) Microfilaments
- (D) Centrioles

11. Herpes simplex virus (HSV) enters the human body and remains dormant in the nervous system until it produces an outbreak after exposure to heat, radiation, or other stimuli. Which of the following statements correctly describes HSV?

- (A) While it remains dormant in the nervous system, the virus is in its lytic cycle.
- (B) During an outbreak, the virus is in the lysogenic cycle.
- (C) Herpes simplex virus adds its genetic information to the genetic information of the cell.
- (D) The herpes simplex virus contains a tail sheath and tail fibers.

12. Resistance to antibiotics is a well-recognized medical problem. Which mechanism(s) can account for a bacterium's ability to increase its genetic variability and thus adapt itself to resist different antibiotics?

I. Binary fission

II. Conjugation

III. Transduction

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

13. A bacterial cell is noted to be resistant to penicillin. The bacterium is transferred to a colony that lacks the fertility factor, and the rest of the colony does not become resistant to penicillin. However, the penicillin-resistant cell has also started to exhibit other phenotypic characteristics, including secretion of a novel protein. Which of the following methods of bacterial recombination is NOT likely to account for this change?

- (A) Conjugation
- (B) Transformation
- (C) Transduction
- (D) Infection with a bacteriophage

14. In Alzheimer's disease, a protein called the amyloid precursor protein (APP) is cleaved to form a protein called β -amyloid. This protein has a β -pleated sheet structure and precipitates to form plaques in the brain. This mechanism of disease is most similar to which of the following pathogens?

- (A) Bacteria

- (B) Viruses
- (C) Prions
- (D) Viroids

15. After infection of a cell, a viral particle must transport itself to the nucleus in order to produce viral proteins. What is the likely genomic content of the virus?

- (A) Double-stranded DNA
- (B) Double-stranded RNA
- (C) Positive-sense RNA
- (D) Negative-sense RNA

Discrete Practice Answers

1. **D**

Obligate anaerobes cannot survive in the presence of oxygen and would likely be killed by such a therapy, treating the infection. The other types of bacteria listed can all survive in the presence of oxygen, so infections involving these bacteria would likely not be treated using this therapy.

2. **A**

While bones and tendons are composed predominantly of connective tissue cells, muscle tissue is considered a different tissue type. Other examples of connective tissue include cartilage, ligaments, adipose tissue, and blood. Connective tissue often secretes substances to form the extracellular matrix, such as collagen and elastin, eliminating **(B)**. **(C)** and **(D)** are essentially identical and can both be eliminated: in organs, connective tissue often forms the support structure for epithelial cells, called the stroma.

3. **D**

In a virus, the nucleic acid can be either DNA or RNA and—in both cases—can be either single- or double-stranded. Therefore, all of the types of nucleic acids listed here could be used for a viral genome, making **(D)** the correct answer.

4. **B**

The Golgi apparatus consists of a stack of membrane-enclosed sacs. It receives vesicles and their contents from the endoplasmic reticulum, modifies them (through glycosylation, phosphorylation, and other mechanisms), repackages them into vesicles, and distributes them to appropriate locations in the cell. Protein synthesis occurs in ribosomes and rough endoplasmic reticulum, eliminating **(A)**. Lipid and carbohydrate breakdown takes place in the peroxisomes and cytoplasm, eliminating **(C)**. ATP production occurs in the mitochondria, eliminating **(D)**.

5. **C**

Mitochondria are thought to have evolved from an anaerobic prokaryote engulfing an aerobic prokaryote and establishing a symbiotic relationship; therefore, mitochondrial DNA, or mDNA, is likely to be similar to bacterial DNA. Both mDNA and bacterial DNA are organized into a single circular chromosome of double-stranded DNA that can replicate during binary fission. Therefore, Statements I and II are correct, while Statement III is incorrect.

6. **C**

The smooth endoplasmic reticulum is involved in the transport of materials throughout the cell, in lipid synthesis, and in the detoxification of drugs and poisons. Proteins from the rough ER can cross into the smooth ER, where they are secreted into cytoplasmic vesicles and transported to the Golgi apparatus. However, protein synthesis is not a function of the smooth ER, but rather of the free ribosomes or the ribosomes associated with the rough ER. **(C)** is therefore the correct answer.

7. **A**

The nucleolus (not to be confused with the nucleus) is a dense structure within the nucleus where ribosomal RNA (rRNA) is synthesized. **(A)** is

therefore the correct answer.

8. **A**

Lysosomes are vesicular organelles that digest material using hydrolytic enzymes. They are surrounded by a single membrane. Both mitochondria and nuclei are surrounded by double membranes, eliminating **(B)** and **(C)**. Ribosomes must not be surrounded by membranes because they are found not only in eukaryotes but also in prokaryotes, which lack any membrane-bound organelles. This eliminates **(D)**.

9. **C**

Some of the main differences between prokaryotes and eukaryotes are that prokaryotes do not have a nucleus, while eukaryotes do, eliminating **(B)**; prokaryotes have ribosomal subunits of 30S and 50S, while eukaryotes have ribosomal subunits of 40S and 60S, eliminating **(A)**; and prokaryotes do not have membrane-bound organelles, whereas eukaryotes do, eliminating **(D)**. The presence of a membrane on the outer surface of the cell could not distinguish a prokaryotic cell from a eukaryotic one because both gram-negative bacteria and animal cells share this feature. Thus, **(C)** is the correct answer.

10. **C**

Tubulin is the primary protein in microtubules, which are responsible for the structure and movement of cilia and flagella, eliminating **(A)** and **(B)**. Centrioles organize microtubules into the mitotic spindle, eliminating **(D)**. Microfilaments are not composed of tubulin, but rather actin, making **(C)** the correct answer.

11. **C**

Viruses can exist in either the lytic or lysogenic cycle; they may even switch

between them. During the lytic cycle, the virus's DNA takes control of the host cell's genetic machinery, manufacturing numerous progeny. In the end, the host cell bursts (lyses) and releases new virions, each capable of infecting other cells. In the lysogenic cycle, viral DNA is added to the host cell's genome, where it can remain dormant for days or years. Either spontaneously or as a result of environmental circumstances, the provirus can reactivate and enter a lytic cycle. Thus, **(A)** and **(B)** are incorrect because the terms are reversed. **(D)** describes features of bacteriophages, which are viruses that infect bacteria—not the human nervous system. **(C)** accurately describes how HSV operates during the lysogenic cycle, making it the correct answer.

2. **C**

Bacterial cells reproduce by binary fission, an asexual process in which the progeny is identical to the parent. Therefore, binary fission (Statement I) does not increase genetic variability. Conjugation can be described as sexual mating in bacteria; it is the transfer of genetic material between two bacteria that are temporarily joined. Transduction occurs when fragments of the bacterial chromosome accidentally become packaged into viral progeny produced during a viral infection and are introduced into another bacterium by the viral vector. Therefore, both conjugation and transduction (Statements II and III) increase bacterial genetic variability.

3. **A**

A bacterial cell that does not rapidly cause a phenotypic change in the rest of the colony is likely not F^+ , meaning that this cell is not able to form a sex pilus for conjugation, making **(A)** correct. The expression of new phenotypic characteristics indicates that this bacterium may have acquired genetic material from the environment through transformation, **(B)**, or transduction (which occurs via bacteriophage infection), **(C)** and **(D)**.

4. **C**

Prions are infectious proteins that cause misfolding of other proteins. Prions generally cause a shift toward β -pleated sheet conformations, causing decreased solubility and increased resistance to degradation. This ultimately leads to disease. This mechanism is very similar to the one described here for Alzheimer's disease, making **(C)** the correct answer.

5. **A**

A virus that requires transport to the nucleus in order to produce viral proteins likely requires use of nuclear RNA polymerase in order to create mRNA that can be translated to protein. Therefore, only DNA viruses need to be transported to the nucleus to produce viral proteins, eliminating all answer choices but **(A)**.



A microscopic image of cells, likely from an onion skin, stained with a pinkish-purple dye. The cells are roughly rectangular and arranged in a grid-like pattern. A blue rectangular overlay is positioned in the top-left corner, containing the text 'CHAPTER 2'.

CHAPTER 2

Reproduction

CHAPTER 2

Reproduction

In This Chapter

2.1 The Cell Cycle and Mitosis

The Cell Cycle
Control of the Cell Cycle
Mitosis



2.2 Meiosis

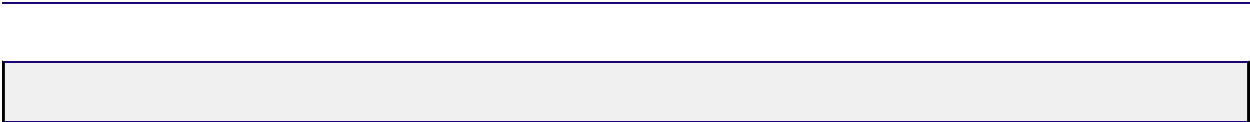
Meiosis I
Meiosis II

2.3 The Reproductive System

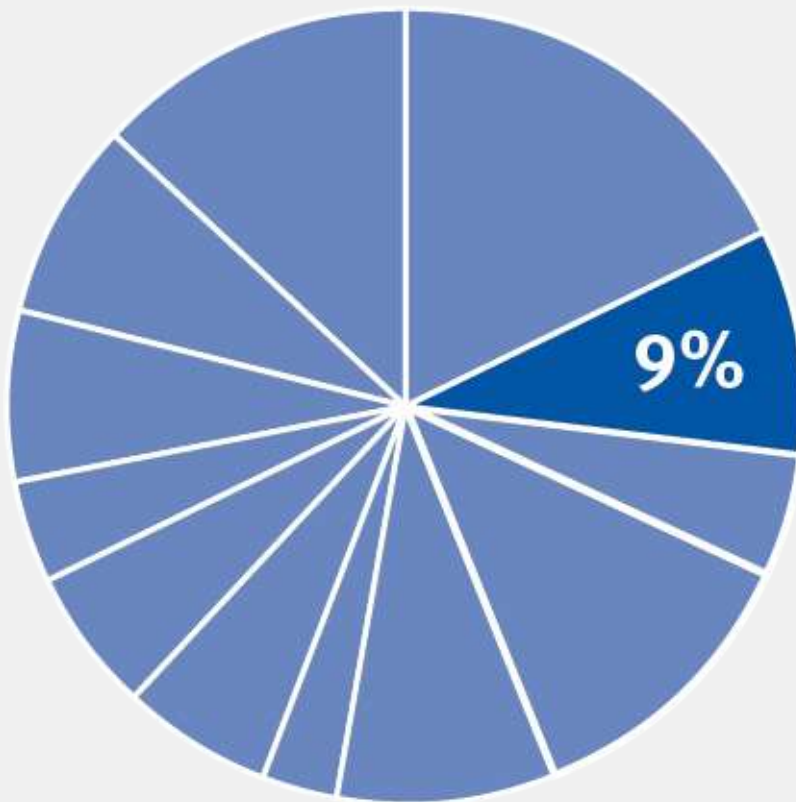
Male Reproductive Anatomy
Female Reproductive Anatomy
Sexual Development
The Menstrual Cycle



Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 9% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content categories:

1C: Transmission of heritable information from generation to generation and the processes that increase genetic diversity

2C: Processes of cell division, differentiation, and specialization

3B: Structure and integrative functions of the main organ systems

Introduction

All mammals share certain characteristics: milk-producing mammary glands, three bones in the middle ear and one in the lower jaw, fur or hair, heterodont dentition (different kinds of teeth), and both sebaceous (oil-producing) and sudoriferous (sweat) glands. What about placenta formation during embryonic development? This is a characteristic of humans, as we'll explore in Chapter 3 of *MCAT Biology Review*, but there are two groups of mammals that birth their young a bit differently: prototherians and metatherians.

Prototherians (monotremes), which include the duckbilled platypus and echidna (spiny anteater), encase their developing embryos within hard-shelled amniotic eggs and lay them to be hatched, like reptiles. This method of development is referred to as oviparity. Metatherians (marsupials) include koalas and kangaroos. A typical metatherian fetus (joey) undergoes some development in its mother's uterus and then climbs out of the birth canal and into her marsupium, or pouch. It might seem a bit strange that something as essential as reproduction can be so different between mammalian species, but there are, in fact, a wide variety of reproductive schemes in nature. Many organisms reproduce without a sexual partner. Others can reproduce sexually or asexually depending on environmental conditions. In Chapter 1 of *MCAT Biology Review*, we explored how bacteria and viruses reproduce. In this chapter, we'll explore eukaryotic and female reproductive systems.

2.1 The Cell Cycle and Mitosis

High-Yield

LEARNING OBJECTIVES

After Chapter 2.1, you will be able to:

- Describe the four phases of mitosis and the major events during each phase
- Identify the five stages of the cell cycle and the major events during each stage

In animals, autosomal cells are said to be **diploid** ($2n$), which means that they contain two copies of each chromosome. Germ cells, on the other hand, are **haploid** (n), containing only one copy of each chromosome. In humans, these numbers are 46 and 23, respectively; we inherit 23 chromosomes from each parent. Eukaryotic cells replicate through the **cell cycle**, a specific series of phases during which a cell grows, synthesizes DNA, and divides.

Derangements of the cell cycle can lead to unchecked cell division and may be responsible for the formation of cancer.

THE CELL CYCLE

The cell cycle, shown in Figure 2.1, is a perennial MCAT favorite. For actively dividing cells, the cell cycle consists of four stages: G_1 , S, G_2 , and M. The first three stages (G_1 , S, and G_2) are known collectively as **interphase**. Interphase is the longest part of the cell cycle; even actively dividing cells spend about 90 percent of their time in interphase. Cells that do not divide spend all of their time in an offshoot of G_1 called G_0 . During the **G_0 stage**, the cell is simply living and carrying out its functions, without any preparation for division.

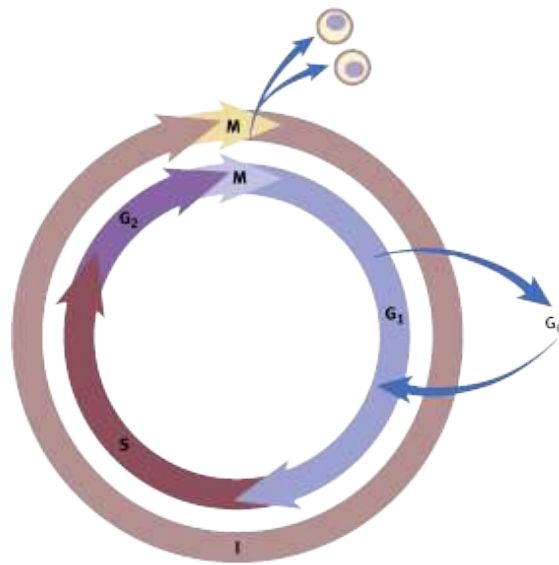


Figure 2.1. The Cell Cycle

During interphase, individual chromosomes are not visible with light microscopy because they are in a less condensed form known as **chromatin**. This is because the DNA must be available to *RNA polymerase* so that genes can be transcribed. During mitosis, however, it is preferable to condense the DNA into tightly coiled chromosomes to avoid losing any genetic material during cell division.

G_1 Stage: Presynthetic Gap

During the **G₁ stage**, cells create organelles for energy and protein production (mitochondria, ribosomes, and endoplasmic reticulum), while also increasing their size. In addition, passage into the S (synthesis) stage is governed by a **restriction point**. Certain criteria, such as containing the proper complement of DNA, must be met for the cell to pass the restriction point and enter the synthesis stage.



KEY CONCEPT

Each chromatid is composed of a complete double-stranded molecule of DNA. Sister chromatids are identical copies of each other. The term *chromosome* may be used to refer to either a single chromatid before S phase or the pair of chromatids attached at the centromere after S phase.

S Stage: Synthesis of DNA

During the **S stage**, the cell replicates its genetic material so that each daughter cell will have identical copies. After replication, each chromosome consists of two identical **chromatids** that are bound together at a specialized region known as the **centromere**, as shown in Figure 2.2. Note that the ploidy of the cell does not change even though the number of chromatids has doubled. In other words, humans in this stage still only have 46 chromosomes, even though 92 chromatids are present. Cells entering G₂ have twice as much DNA as cells in G₁.

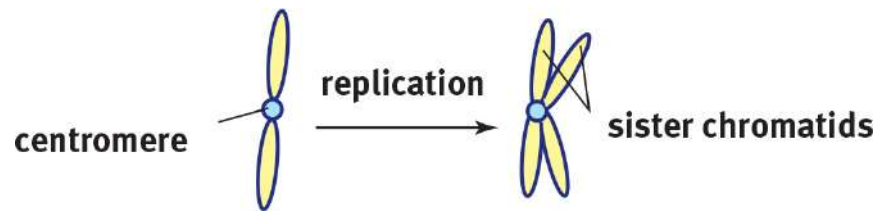


Figure 2.2. Chromosome Replication

A single chromatid replicates to form two sister chromatids.

G₂ Stage: Postsynthetic Gap

During the **G₂ stage**, the cell passes through another quality control checkpoint. DNA has already been duplicated, and the cell checks to ensure that there are enough organelles and cytoplasm for two daughter cells. Furthermore, the cell checks to make sure that DNA replication proceeded correctly to avoid passing on an error to daughter cells that may further pass on the error to their progeny.

M Stage: Mitosis

The **M stage** consists of mitosis itself along with cytokinesis. Mitosis is divided into four phases: prophase, metaphase, anaphase, and telophase. The features of each phase will be discussed in the next section. Cytokinesis is the splitting of the cytoplasm and organelles between the two daughter cells.



KEY CONCEPT

In autosomal cells, division results in two genetically identical daughter cells. In germ cells, the daughter cells are not equivalent.

CONTROL OF THE CELL CYCLE

The cell cycle is controlled by checkpoints, most notably between the G_1 and S phase and the G_2 and M phase. At the G_1 /S checkpoint, the cell determines if the condition of the DNA is good enough for synthesis. As mentioned previously, this checkpoint is also known as the restriction point. If there has been damage to the DNA, the cell cycle goes into arrest until the DNA has been repaired. The main protein in control of this is **p53**.

At the G_2 /M checkpoint, the cell is mainly concerned with ensuring that it has achieved adequate size and the organelles have been properly replicated to support two daughter cells. p53 also plays a role in the G_2 /M checkpoint.

The molecules responsible for the cell cycle are known as **cyclins** and **cyclin-dependent kinases (CDK)**. In order to be activated, CDKs require the presence of the right cyclins. During the cell cycle, concentrations of the various cyclins increase and decrease during specific stages. These cyclins bind to CDKs, creating an activated CDK–cyclin complex. This complex can then phosphorylate transcription factors. **Transcription factors** then promote transcription of genes required for the next stage of the cell cycle.

Cancer

Cell cycle control is essential to ensure that cells that are damaged or inadequately sized do not divide. When cell cycle control becomes deranged, and damaged cells are allowed to undergo mitosis, **cancer** may result. One of the most common mutations found in cancer is mutation of the gene that produces p53, called TP53. When this gene is mutated, the cell cycle is not

stopped to repair damaged DNA. This allows mutations to accumulate, eventually resulting in a cancerous cell that divides continuously and without regard to the quality or quantity of the new cells produced. Often, cancer cells undergo rapid cell division, creating **tumors**. Eventually, if the cell begins to produce the right factors (such as proteases that can digest basement membranes or factors that encourage blood vessel formation), the damaged cells are then able to reach other tissues. This may include both local invasion as well as distant spread of cancerous cells through the bloodstream or lymphatic systems. The latter result is known as **metastasis**.



BRIDGE

Cancer-causing genes can often be classified into oncogenes (genes that, when mutated, actively promote cell division) and tumor suppressor genes (genes that, when mutated, lose their ability to regulate or arrest the cell cycle). Different cancer types are often associated with specific mutations in either oncogenes or tumor suppressor genes, or both. The biochemistry of these genes is discussed in Chapter 6 of *MCAT Biochemistry Review*.

MITOSIS

Mitosis, shown in Figure 2.3, is the process by which two identical daughter cells are created from a single cell. Mitosis consists of four distinct phases—prophase, metaphase, anaphase, and telophase—and occurs in **somatic cells**, or cells that are not involved in sexual reproduction.

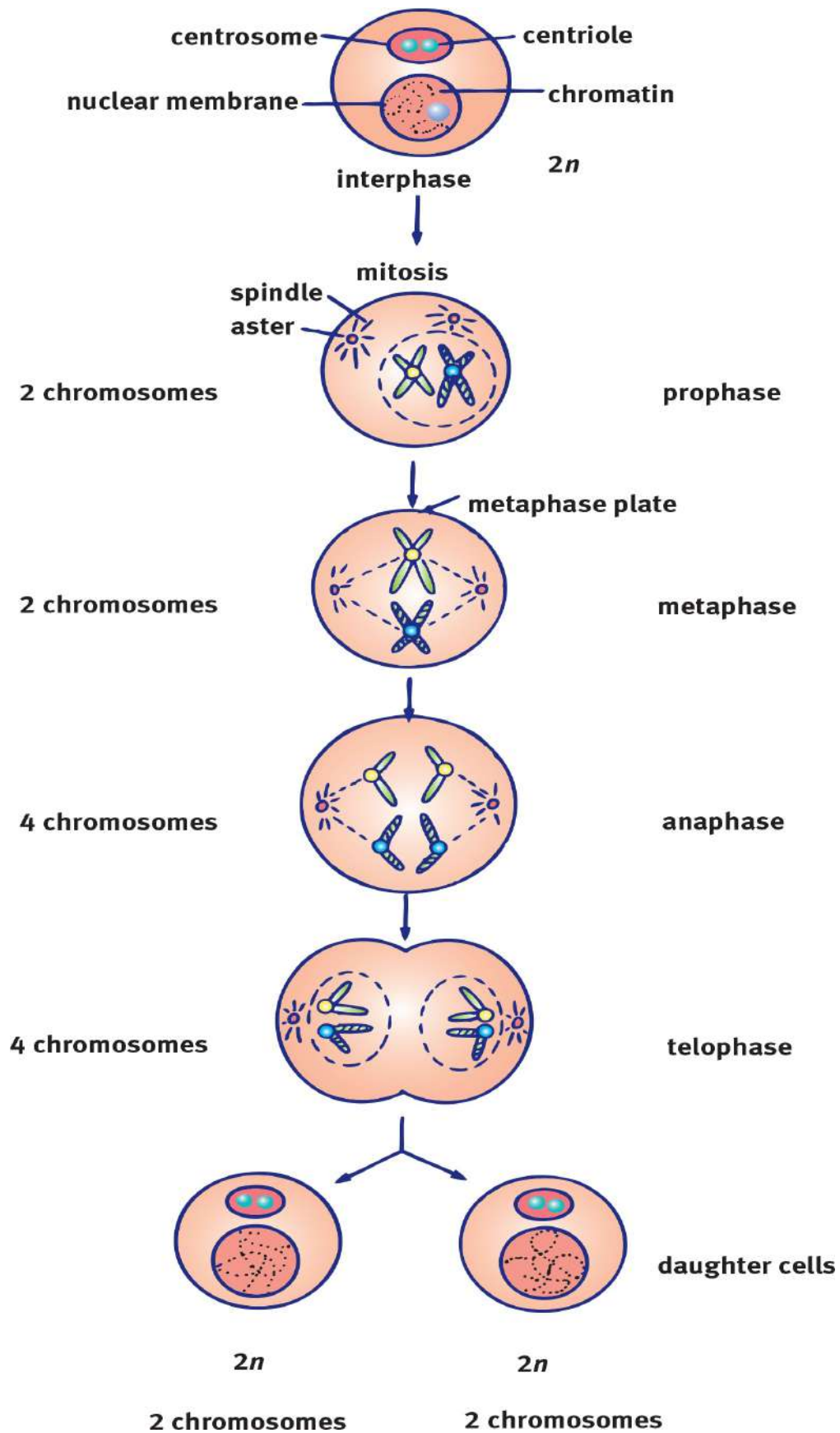


Figure 2.3. Mitosis

Mitosis results in two identical daughter cells.



KEY CONCEPT

The phases of mitosis:

- Prophase—chromosomes condense, spindle forms
- Metaphase—chromosomes align
- Anaphase—sister chromatids separate
- Telophase—new nuclear membranes form

Prophase

Prophase is the first phase in mitosis. The first step in prophase involves condensation of the chromatin into chromosomes. Also, the centriole pairs separate and move toward opposite poles of the cell. These paired cylindrical organelles, shown in Figure 2.4, are located outside the nucleus in a region known as the **centrosome** and are responsible for the correct division of DNA. Once the centrioles migrate to opposite poles of the cell, they begin to form **spindle fibers**, which are made of microtubules. This establishes the centrosome as one of the two **microtubule organizing centers** of the cell—the other being the basal body of a flagellum or cilium. Each of the fibers radiates outward from the centrioles. Some microtubules form **asters** that anchor the centrioles to the cell membrane. Others extend toward the middle of the cell. The nuclear membrane dissolves during prophase, allowing these

spindle fibers to contact the chromosomes. The nucleoli become less distinct and may disappear completely. **Kinetochores**, which appear at the centrosome, are protein structures located on the centromeres that serve as attachment points for specific fibers of the **spindle apparatus** (appropriately called **kinetochore fibers**).

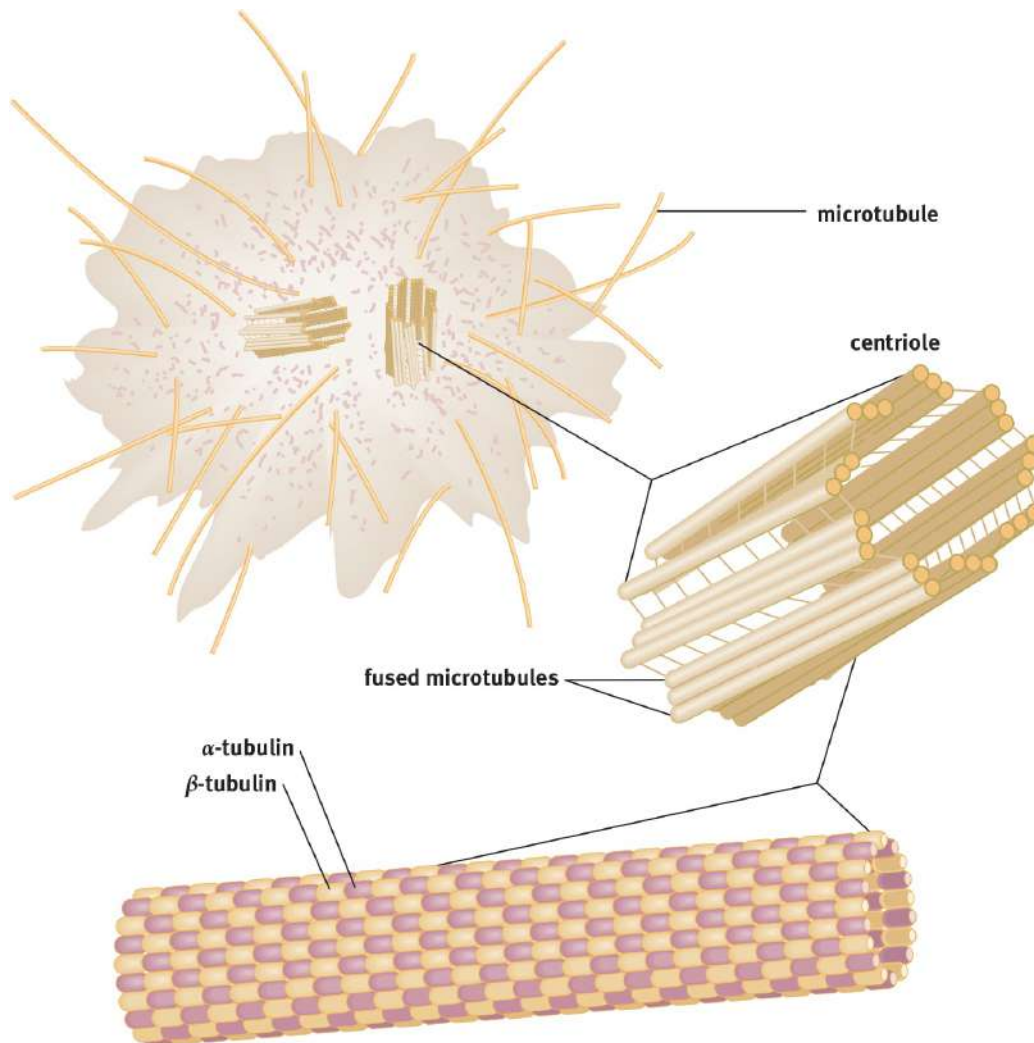


Figure 2.4. The Centrosome

Each centrosome contains two tubulin-based centrioles responsible for proper movement of the chromosomes during mitosis.

Metaphase

In **metaphase**, the centriole pairs are now at opposite ends of the cell. The kinetochore fibers interact with the fibers of the spindle apparatus to align the chromosomes at the **metaphase plate (equatorial plate)**, which is equidistant from the two poles of the cell.

Anaphase

During **anaphase**, the centromeres split so that each chromatid has its own distinct centromere, thus allowing the sister chromatids to separate. The sister chromatids are pulled toward the opposite poles of the cell by the shortening of the kinetochore fibers.

Telophase and Cytokinesis

Telophase is essentially the reverse of prophase. The spindle apparatus disappears. A nuclear membrane reforms around each set of chromosomes, and the nucleoli reappear. The chromosomes uncoil, resuming their interphase form. Each of the two new nuclei receives a complete copy of the genome identical to the original genome and to each other.

Cytokinesis, which occurs at the end of telophase, is the separation of the cytoplasm and organelles, giving each daughter cell enough material to survive on its own. Each cell undergoes a finite number of divisions before programmed death; for human somatic cells, this is usually between 20 and 50. After that, the cell can no longer divide continuously.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

The p53 tumor suppressor pathway has been heavily explored as a potential treatment for cancer patients. The p53 pathway induces growth arrest and apoptosis in response to cellular stress. Mutation of this pathway is thought to be nearly universal in human cancer.	Bac on mu pat cau
Researchers explored whether a retro-inverse p53C' peptide (termed RI-TATp53C') is a therapeutically effective means of activating the p53 tumor suppressor pathway in preclinical models of terminal metastatic cancer.	Hyp Doe TA1 act p53 anc tre
RI-TATp53C' contains a protein transduction domain (PTD), which is capable of traversing the plasma membrane, and a functional sequence of the p53 C-terminus.	RI- con pro tra dor anc fun ter
A non-functional RI-TATp53C' was developed by mutating the functional residues of the peptide.	The nor fun ver. tre TA1 this to k neg con

Researchers exposed TA3/St mammary carcinoma cells to Wild-Type RI-TATp53C' and the mutant peptide of RI-TATp53C'. The cells' DNA content was then analyzed to test if cell cycle arrest occurred (Figure 1).

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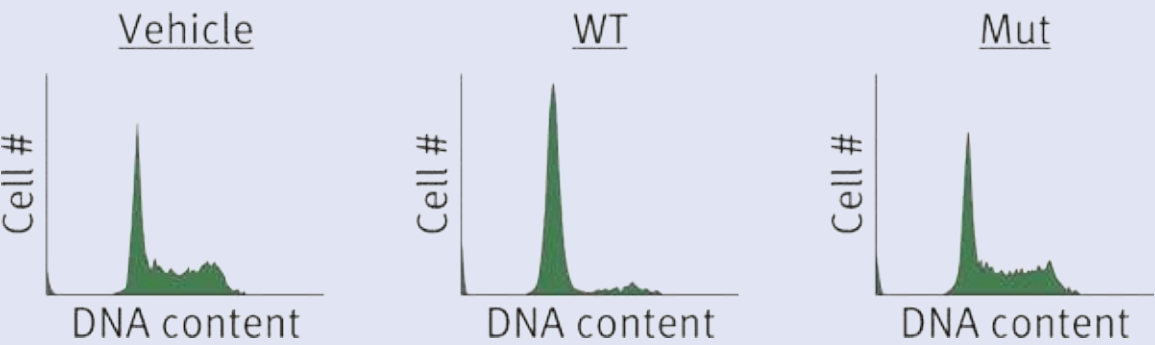


Figure 1

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Researchers then explored the dependency of RI-TATp53C' induction of growth arrest with endogenous p53 in cells (Figure 2). TA3/St and H1299 are both cancer cells lines.

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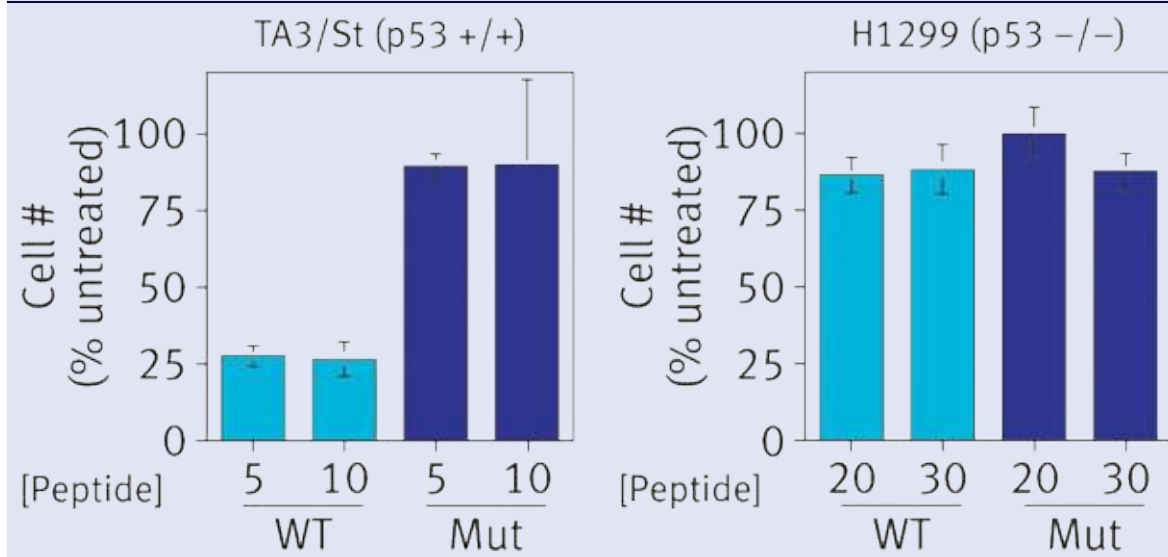


Figure 2

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adapted from: Snyder EL, Meade BR, Saenz CC, Dowdy SF (2004). Treatment of terminal peritoneal carcinomatosis by a trans-activating peptide. *PLOS Biology* 2(2): e36. <https://doi.org/10.1371/journal.pbio.0020036>.

What stage of the cell cycle does RI-TATp53C' arrest the cell in and does RI-TATp53C'-induced cell arrest require endogenous p53?

To answer this question we need to determine the relationship between RI-TATp53C' and the cell cycle, as well as the relationship between RI-TATp53C' and endogenous p53. With such an in-depth question it's worth considering the bigger picture to ensure that we understand the concepts and experiments in the passage. Start with the context: why are these experiments being done? As stated in paragraphs 1 and 2, p53 is a tumor-suppressing pathway that researchers are hoping to activate as a treatment to cancer. Both experiments, although distinct, will explore the activation of this pathway.

Keeping in mind the independent and dependent variables of the two experiments that we identified as we read, we can identify which experiment should be used to answer each question in the prompt. The first experiment has three conditions: Vehicle, WT, and Mut. The vehicle condition is something we should recognize as bio talk for “we did everything the same as the treatment condition, except the treatment,” so this is a negative control. The WT condition, according to the second to last sentence in paragraph 2, is functional RI-TATp53C'. This is the treatment group! If the hypothesis at the start of the 2nd paragraph is correct we should see interesting data here. The Mut condition is described in the third to last and second to last sentences of

paragraph 2 as a non-functional mutant RI-TATp53C'. In each of these conditions, the proportion of cells that have low/medium/high DNA content are measured, which, according to the last sentence of paragraph 2, determines if cell cycle arrest occurred. At this point, we know Exp 1 relates RI-TATp53C' (IV) and the cell cycle (DV) and should, therefore, be used to answer the first question of the prompt.

Let's analyze the data from the first experiment. Looking at Figure 1, notice that the Vehicle and Mutant conditions are identical, while the WT (treatment) condition differs greatly. Specifically, the RI-TATp53C' treatment (WT) results in a larger portion of cells having less DNA. Considering our background knowledge about the cell cycle and each phase's relative DNA content, we can deduce that G_1 has the lowest DNA content. This is because G_1 occurs after a division (decreases DNA) and before S phase (increases DNA). Thus, we can conclude RI-TATp53C' must arrest the cell in G_1 .

In order to answer the second question, the relationship between RI-TATp53C' functionality and endogenous p53 must be determined. This aligns pretty well with paragraph 3's description of experiment 2, and is a solid indication that we need to analyze Figure 2. Looking at the titles of the graphs there are two different cancerous cell cultures, TA3/St (p53 +/+) and H1299 (p53 -/-), which differ in their expression of p53. This is the first independent variable. Analyzing the graphs further, there are two additional independent variables, WT (light blue) vs. Mut (dark blue) and peptide concentration, [Peptide]. The dependent variable in Figure 2 is the Cell # or % untreated. In other words, it's the number of cancerous cells that are unarrested and thus still cancerous after peptide exposure.

With an understanding of the variables, let's take a look at the data in Figure 2. Between the two graphs, the most striking difference is WT RI-TATp53C' treatment in p53 +/+ cancerous cells results in a drastic drop in untreated (cancerous) cells (light blue, 1st graph), while in the p53 -/- condition WT RI-TATp53C' cancerous cell levels remain constant (light blue, 2nd graph). This shows us WT RI-TATp53C's ability to induce cell cycle arrest is dependent on endogenous p53. In addition, when both p53 +/+ and p53 -/- conditions are exposed to the mutant RI-TATp53C' (non-functional), there is no drop in cancerous cells. This second point is to be expected, but is still worth noting, as it shows us that nothing unexpected occurred during the experiment. Thus, given the data in Figure 2, we can conclude that in order for RI-TATp53C' to induce cell arrest, it requires endogenous p53 in the tumor.

Overall, Figure 1 shows that not only does WT RI-TATp53C' induce cell arrest, but based on the increased proportion of cells with lower DNA content, it induces cell arrest in G₁. From Figure 2, we were able to determine that RI-TATp53C' requires a functional p53 pathway in order to induce cell arrest due to the loss of RI-TATp53C' functionality in the p53 -/- condition.

MCAT CONCEPT CHECK 2.1:

Before you move on, assess your understanding of the material with these questions.

1. What are the five stages of the cell cycle? What happens in each stage?

Cell Cycle Stage	Features

2. What are the four phases of mitosis? What happens in each phase?

Mitotic Phase	Features

2.2 Meiosis

LEARNING OBJECTIVES

After Chapter 2.2, you will be able to:

- Predict the ploidy of daughter cells at the end of mitosis, meiosis I, and meiosis II
- Differentiate between homologous chromosomes and sister chromatids
- Compare and contrast mitosis and meiosis
- Explain the importance of crossing over events in relation to genetic diversity

Whereas mitosis occurs in somatic tissue and results in two identical daughter cells, meiosis occurs in **gametocytes (germ cells)** and results in up to four nonidentical sex cells (**gametes**). Meiosis shares some similarities with mitosis. In both processes, for instance, genetic material must be duplicated, chromatin is condensed to form chromosomes, and microtubules emanating from centrioles are involved in dividing genetic material. However, the MCAT tends to ask about the differences between these two processes.

In contrast to mitosis, which consists of one round each of replication and division, meiosis consists of one round of replication followed by two rounds

of division, as shown in Figure 2.5. **Meiosis I** results in homologous chromosomes being separated, generating haploid daughter cells; this is known as **reductional division**. **Meiosis II** is similar to mitosis, in that it results in the separation of sister chromatids without a change in ploidy, and is therefore known as **equational division**.

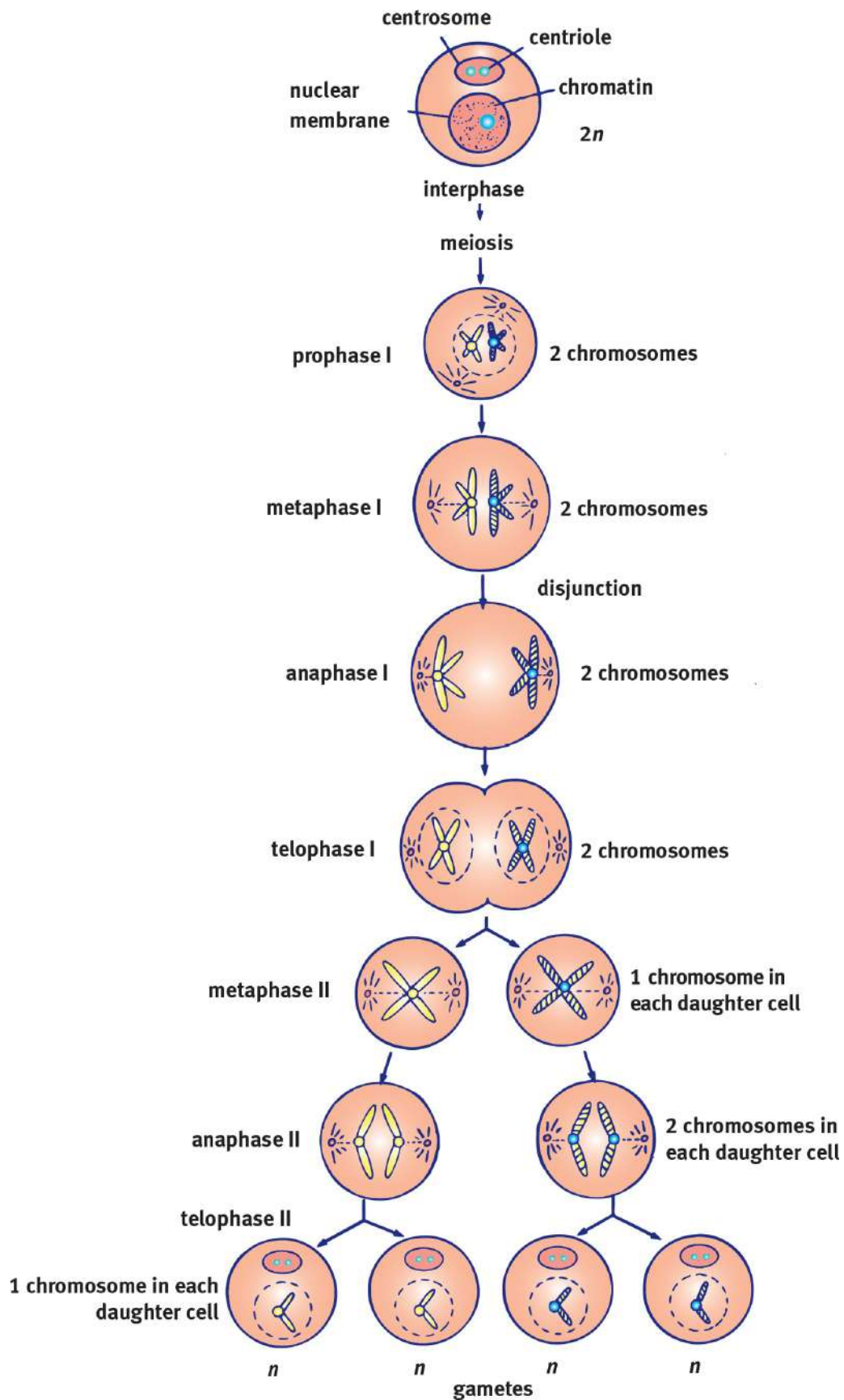


Figure 2.5. Meiosis

Meiosis results in up to four nonidentical daughter cells.

MEIOSIS I

The human genome is composed of 23 **homologous pairs** of chromosomes (**homologues**), each of which contains one chromosome inherited from each parent. This brings up an important note about terminology: whereas homologous pairs are considered separate chromosomes (such as maternal chromosome 15 and paternal chromosome 15), sister chromatids are identical strands of DNA connected at the centromere. Thus, after S phase, there are 92 chromatids organized into 46 chromosomes, which are organized into 23 homologous pairs.

Prophase I

During **prophase I**, the chromatin condenses into chromosomes, the spindle apparatus forms, and the nucleoli and nuclear membrane disappear. The first major difference between meiosis and mitosis occurs at this point: homologous chromosomes come together and intertwine in a process called **synapsis**. At this point, each chromosome consists of two sister chromatids, so each synaptic pair contains four chromatids and is referred to as a **tetrad**; the homologous chromosomes are held together by a group of proteins called the **synaptonemal complex**. Chromatids of homologous chromosomes may break at the point of contact, called the **chiasma** (plural: **chiasmata**) and exchange equivalent pieces of DNA, as shown in Figure 2.6. This process is called **crossing over**, and can be characterized by the number of crossover events that occur in one strand of DNA, including **single crossovers** and

double crossovers. Note that crossing over occurs between homologous chromosomes and not between sister chromatids of the same chromosome—the latter are identical, so crossing over would not produce any change. Those chromatids involved are left with an altered but structurally complete set of genes. Such genetic **recombination** can unlink linked genes, thereby increasing the variety of genetic combinations that can be produced via gametogenesis. Linkage refers to the tendency for genes to be inherited together; genes that are located farther from each other physically are less likely to be inherited together, and more likely to undergo crossing over relative to each other. Thus, as opposed to asexual reproduction, which produces identical offspring, sexual reproduction provides the advantage of great genetic diversity, which is believed to increase the ability of a species to evolve and adapt to a changing environment.

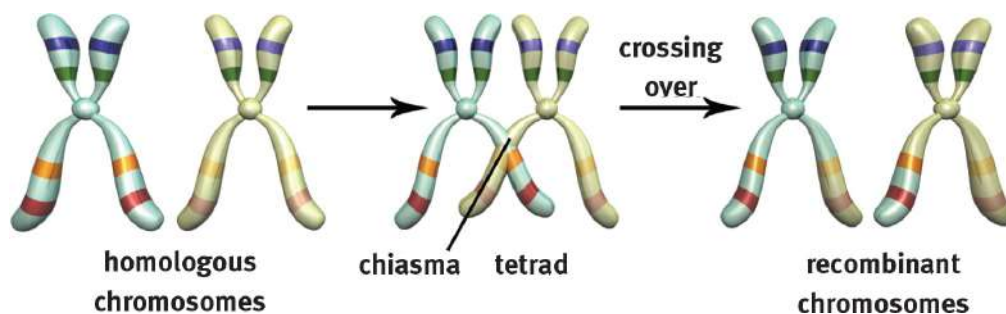


Figure 2.6. Synapsis

During prophase I, homologous chromosomes can exchange genetic material via crossing over.



REAL WORLD

The rate of gene unlinking is used to map differences between two genes on the same chromosome. The farther apart two genes are, the

more likely they are to become unlinked during crossing over. These statistics can then be used to determine the distance between genes on the chromosome, measured in units called centimorgans.

Because of crossing over, each daughter cell will have a unique pool of alleles (genes coding for alternative forms of a given trait) from a random mixture of maternal and paternal origin. In classical genetics, crossing over explains **Mendel's second law (of independent assortment)**, which states that the inheritance of one allele has no effect on the likelihood of inheriting certain alleles for other genes.

Metaphase I

During **metaphase I**, homologous pairs (tetrads) align at the metaphase plate, and each pair attaches to a separate spindle fiber by its kinetochore. Note the difference from mitosis: in mitosis, each chromosome is lined up on the metaphase plate by two spindle fibers (one from each pole); in meiosis, homologous chromosomes are lined up across from each other at the metaphase plate and are held by one spindle fiber.

Anaphase I

During **anaphase I**, homologous pairs separate and are pulled to opposite poles of the cell. This process is called **disjunction**, and it accounts for **Mendel's first law (of segregation)**. During disjunction, each chromosome of paternal origin separates (or disjoins) from its homologue of maternal origin, and either chromosome can end up in either daughter cell. Thus, the distribution of homologous chromosomes to the two intermediate daughter

cells is random with respect to parental origin. This separating of the two homologous chromosomes is referred to as **segregation**.



KEY CONCEPT

It is critical to understand how meiosis I is different from mitosis. The chromosome number is halved (reductional division) in meiosis I, and the daughter cells have the haploid number of chromosomes (23 in humans). Meiosis II is similar to mitosis in that sister chromatids are separated from one another; therefore, no change in ploidy is observed.

Telophase I

During **telophase I**, a nuclear membrane forms around each new nucleus. At this point, each chromosome still consists of two sister chromatids joined at the centromere. The cells are now haploid; once homologous chromosomes separate, only n chromosomes are found in each daughter cell (23 in humans). The cell divides into two daughter cells by cytokinesis. Between cell divisions, there may be a short rest period, or **interkinesis**, during which the chromosomes partially uncoil.



REAL WORLD

If, during anaphase I or II of meiosis, homologous chromosomes (anaphase I) or sister chromatids (anaphase II) fail to separate, one of

the resulting gametes will have two copies of a particular chromosome and the other gamete will have none. Subsequently, during fertilization, the resulting zygote may have too many or too few copies of that chromosome. Nondisjunction can affect both autosomal chromosomes (such as trisomy 21, resulting in Down syndrome) and the sex chromosomes (such as Klinefelter and Turner syndromes).

MEIOSIS II

Meiosis II is very similar to mitosis in that sister chromatids—rather than homologues—are separated from each other.

Prophase II

During **prophase II**, the nuclear envelope dissolves, nucleoli disappear, the centrioles migrate to opposite poles, and the spindle apparatus begins to form.

Metaphase II

During **metaphase II**, the chromosomes line up on the metaphase plate.

Anaphase II

During **anaphase II**, the centromeres divide, separating the chromosomes into sister chromatids. These chromatids are pulled to opposite poles by spindle fibers.

Telophase II

During **telophase II**, a nuclear membrane forms around each new nucleus. Cytokinesis follows, and two daughter cells are formed. Thus, by completion of meiosis II, up to four haploid daughter cells are produced per gametocyte. We use the phrase *up to* because oogenesis, discussed later in this chapter, may result in fewer than four cells if an egg remains unfertilized after ovulation.



KEY CONCEPT

Mitosis	Meiosis
$2n \rightarrow 2n$	$2n \rightarrow n$
Occurs in all dividing cells	Occurs in sex cells only
Homologous chromosomes do not pair	Homologous chromosomes align on opposite sides of the metaphase plate
No crossing over	Crossing over can occur

MCAT CONCEPT CHECK 2.2:

Before you move on, assess your understanding of the material with these questions.

1. What is the number and ploidy of the daughter cells produced from meiosis I? From meiosis II?

- Meiosis I:

- Meiosis II:

2. What is the difference between homologous chromosomes and sister chromatids?

- Homologous chromosomes:

- Sister chromatids:

3. For each phase of meiosis I listed below, what are the differences from the analogous phase of mitosis?

Meiotic Phase	Differences from Mitotic Phase
Prophase I	
Metaphase I	
Anaphase I	
Telophase I	

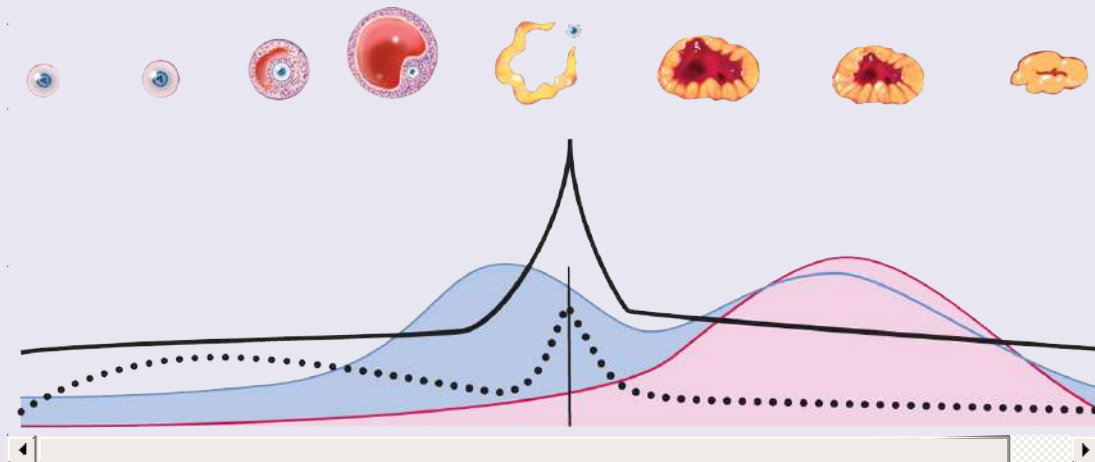
2.3 The Reproductive System

High-Yield

LEARNING OBJECTIVES

After Chapter 2.3, you will be able to:

- Recall the functions of the interstitial cells of Leydig and Sertoli cells
- Identify the phases of meiosis in which primary and secondary oocytes are arrested
- Describe the acrosome
- Differentiate between male and female sex organs and development
- Recall the phases of the menstrual cycle, including key features and relative hormone levels for each phase:



Biological **sex** is determined by the 23rd pair of chromosomes, with XX being female and XY being male. Ova can only carry the X chromosome, while sperm can carry either the X or Y chromosome. The **X chromosome** carries a sizeable amount of genetic information; mutations in these genes can cause **sex-linked** (X-linked) disorders. Males are termed **hemizygous** with respect to many of the genes on the X chromosome because they only have one copy. Therefore, a male with a disease-causing allele on the unpaired part of X chromosome will necessarily express that allele. Females, on the other hand, may be homozygous or heterozygous with respect to genes on the X chromosome. Most X-linked disorders are recessively inherited; therefore, females express these disorders far less frequently than males. Females carrying a diseased allele on an X-chromosome but not exhibiting the disease are said to be **carriers**.



MNEMONIC

Sex-linked is **X**-linked.

Comparatively, the **Y chromosome** contains very little genetic information. One notable gene on the Y chromosome is **SRY (sex-determining region Y)**, which codes for a transcription factor that initiates testis differentiation and, thus, the formation of male gonads. Therefore, in the absence of the Y chromosome, all zygotes will be female. In the presence of the Y chromosome a zygote will be male.



REAL WORLD

There are actually a handful of Y-linked diseases, most of which result in reduced fertility. A father will pass a Y-linked disease to all of his sons, assuming fertility has not been lost. These diseases are extremely rare and are not included on the official MCAT content lists.

MALE REPRODUCTIVE ANATOMY

The male reproductive system is shown in Figure 2.7.

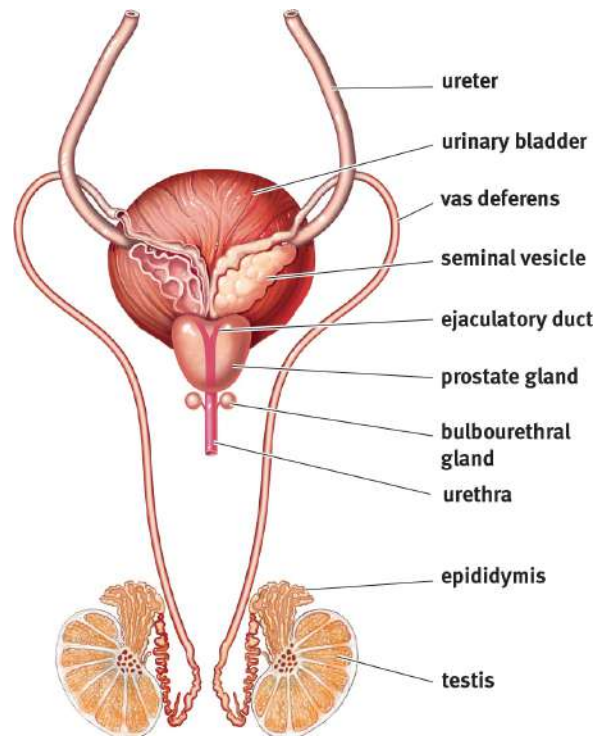


Figure 2.7. Male Reproductive System

In males, the primitive gonads develop into the **testes**. The testes have two functional components: the **seminiferous tubules** and the **interstitial cells of Leydig**. **Sperm** are produced in the highly coiled seminiferous tubules,

where they are nourished by **Sertoli cells**. The cells of Leydig secrete **testosterone** and other male sex hormones (**androgens**). The testes are located in the **scrotum**, an external pouch that hangs below the **penis**, a position that allows it to maintain a temperature 2°C to 4°C lower than the body. In fact, there is a layer of muscle around the vas deferens (ductus deferens) that can raise and lower the testis to maintain the proper temperature for sperm development.

As sperm are formed they are passed to the **epididymis**, where their flagella gain motility, and they are then stored until **ejaculation**. During **ejaculation**, sperm travel through the **vas deferens** and enter the **ejaculatory duct** at the posterior edge of the prostate gland. The two ejaculatory ducts then fuse to form the **urethra**, which carries sperm through the penis as they exit the body. In males, the reproductive and urinary systems share a common pathway; this is not the case in females.



MNEMONIC

Pathway of sperm through the male reproductive system: **SEVE(N) UP**

- **S**eminiferous tubules
- **E**pididymis
- **V**as deferens (also called the ductus deferens)
- **E**jaculatory duct
- **(N)**othing
- **U**rethra
- **P**enis

As sperm pass through the reproductive tract they are mixed with **seminal fluid**, which is produced through a combined effort by the seminal vesicles, prostate gland, and bulbourethral gland. The **seminal vesicles** contribute fructose to nourish sperm, and both the seminal vesicles and **prostate gland** give the fluid mildly alkaline properties so the sperm can survive in the relative acidity of the female reproductive tract. The **bulbourethral (Cowper's) glands** produce a clear viscous fluid that cleans out any remnants of urine and lubricates the urethra during sexual arousal. The combination of sperm and seminal fluid is known as **semen**.



REAL WORLD

The prostate enlarges with age and frequently causes problems in older males, including *benign prostatic hyperplasia*. Because the prostate surrounds the urethra, classic symptoms of this condition include urinary frequency, urgency, and nighttime awakenings to use the bathroom.

Spermatogenesis

As mentioned above, **spermatogenesis**, the formation of haploid sperm through meiosis, occurs in the seminiferous tubules. In males, the diploid stem cells are known as **spermatogonia**. After replicating their genetic material (S stage), they develop into diploid **primary spermatocytes**. The first meiotic division will result in haploid **secondary spermatocytes**, which then undergo meiosis II to generate haploid **spermatids**. Finally, the

spermatids undergo maturation to become mature **spermatozoa**. Spermatogenesis results in four functional sperm for each spermatogonium.

Mature sperm are very compact. They consist of a head (containing the genetic material), a midpiece (which generates ATP from fructose), and a flagellum (for motility), as shown in Figure 2.8. The **midpiece** is filled with mitochondria, which generate the energy for swimming through the female reproductive tract to reach the ovum in the fallopian tubes. Each sperm **head** is covered by a cap known as an **acrosome**. This structure is derived from the Golgi apparatus and is necessary to penetrate the **ovum**. Once a male reaches sexual maturity during puberty, approximately 3 million sperm are produced per day for the rest of his life.

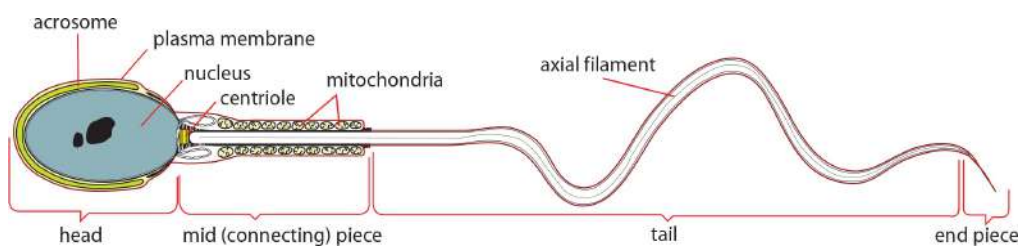


Figure 2.8. Structure of a Mature Sperm

FEMALE REPRODUCTIVE ANATOMY

Unlike the male reproductive system, all of the female reproductive organs are internal, as shown in Figure 2.9. The gonads, known as **ovaries**, produce estrogen and progesterone. The ovaries are located in the pelvic cavity; each consists of thousands of **follicles**, which are multilayered sacs that contain, nourish, and protect immature **ova** (eggs). Between puberty and menopause, one egg per month is **ovulated** into the **peritoneal sac**, which lines the abdominal cavity. It is then drawn into the **fallopian tube** or **oviduct**, which is

lined with cilia to propel the egg forward. The fallopian tubes are connected to the muscular **uterus**, which is the site of fetal development. The lower end of the uterus, known as the **cervix**, connects to the **vaginal canal**, where sperm are deposited during intercourse. The vagina is also the passageway through which childbirth occurs. The external female anatomy is known collectively as the **vulva**. As mentioned earlier, females have separate excretory and reproductive tracts.

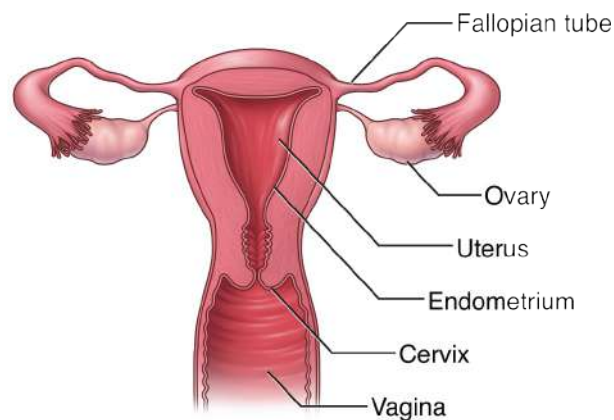


Figure 2.9. Female Reproductive System

Oogenesis

The production of female gametes is known as **oogenesis**. Although gametocytes undergo the same meiotic process in both females and males, there are some significant differences between the two sexes. First, there is no unending supply of stem cells analogous to spermatogonia in females; all of the oogonia a woman will ever have are formed during fetal development. By birth, all of the oogonia have already undergone DNA replication and are considered **primary oocytes**. These cells are $2n$, like primary spermatocytes, and are actually arrested in prophase I. Once a woman reaches **menarche** (her first menstrual cycle), one primary oocyte per month will complete

meiosis I, producing a **secondary oocyte** and a polar body. The division is characterized by unequal cytokinesis, which distributes ample cytoplasm to one daughter cell (the secondary oocyte) and nearly none to the other (the polar body). The polar body generally does not divide any further and will never produce functional gametes. The secondary oocyte, on the other hand, remains arrested in metaphase II and does not complete the remainder of meiosis II unless fertilization occurs.

Oocytes are surrounded by two layers: the zona pellucida and the corona radiata. The **zona pellucida** surrounds the oocyte itself and is an acellular mixture of glycoproteins that protect the oocyte and contain compounds necessary for sperm cell binding. The **corona radiata** lies outside the zona pellucida and is a layer of cells that adhered to the oocyte during ovulation. Meiosis II is triggered when a sperm cell penetrates these layers with the help of acrosomal enzymes. The secondary oocyte undergoes the second meiotic division to split into a mature ovum and another polar body, which will eventually be broken down.

A mature ovum is a very large cell consisting of large quantities of cytoplasm and organelles. The ovum contributes nearly everything to the zygote (half of the DNA and all of the cytoplasm, organelles [including mitochondria], and RNA for early cellular processes), whereas sperm only contribute half of the DNA. Upon completion of meiosis II, the haploid **pronuclei** of the sperm and the ovum join, creating a diploid **zygote**.

SEXUAL DEVELOPMENT

The ability to reproduce is under hormonal control. Prior to puberty, the **hypothalamus** restricts production of **gonadotropin-releasing hormone**

(**GnRH**). At the start of puberty, this restriction is lifted as the hypothalamus releases pulses of GnRH, which then triggers the **anterior pituitary gland** to synthesize and release **follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH)**. These hormones trigger the production of other sex hormones that develop and maintain the reproductive system.

Male Sexual Development

During the fetal period (from nine weeks after fertilization until birth), presence of the Y chromosome leads to production of androgens, resulting in male sexual differentiation. For the duration of infancy and childhood, androgen production is low. **Testosterone**, produced by the testes, increases dramatically during puberty, and sperm production begins. In order to achieve this, there is a delicate interplay of FSH and LH stimulation on two cell types in the testes. FSH stimulates the Sertoli cells and triggers sperm maturation, whereas LH causes the interstitial cells to produce testosterone. Testosterone not only develops and maintains the male reproductive system, but also results in the development of **secondary sexual characteristics** such as facial and axillary hair, deepening of the voice, and increased muscle and bone mass. Testosterone production remains high into adulthood and declines as men age. This hormone exerts negative feedback on the hypothalamus and anterior pituitary so that production is kept within an appropriate range.



REAL WORLD

If the receptors for testosterone are absent or defective, it cannot exert its effects. The result is a condition called *androgen insensitivity*

syndrome (AIS), in which a genetic male (XY) has female secondary sexual characteristics. In complete androgen insensitivity, a genetic male will appear female at birth. Oftentimes these individuals are not discovered until puberty, when they are examined for amenorrhea (failure to menstruate).

Female Sexual Development

The ovaries, which are derived from the same embryonic structures as the testes, are also under the control of FSH and LH secreted by the anterior pituitary. The ovaries produce estrogens and progesterone.

Estrogens are secreted in response to FSH and result in the development and maintenance of the female reproductive system and female secondary sexual characteristics (breast growth, widening of the hips, changes in fat distribution). In the embryo, estrogens stimulate development of the reproductive tract. In adults, estrogens lead to the thickening of the lining of the uterus (**endometrium**) each month in preparation for the implantation of a zygote.

Progesterone is secreted by the **corpus luteum**—the remains of the ovarian follicle following ovulation—in response to LH. Interestingly, progesterone is involved in the development and maintenance of the endometrium, but not in the initial thickening of the endometrium—this is the role of estrogen. This means that both estrogen and progesterone are required for the generation, development, and maintenance of an endometrium capable of supporting a zygote. By the end of the first trimester of a pregnancy, progesterone is

supplied by the placenta, while the corpus luteum atrophies and ceases to function.



MNEMONIC

Estrogen **e**stablishes and **p**rogestosterone **p**rotects the endometrium.

THE MENSTRUAL CYCLE

During the reproductive years (from menarche to menopause), estrogen and progesterone levels rise and fall in a cyclic pattern. In response, the endometrial lining will grow and be shed. This is known as the **menstrual cycle** and can be divided into four events, as shown in Figure 2.10: the follicular phase, ovulation, the luteal phase, and menstruation.



MCAT EXPERTISE

The MCAT likes to test your ability to identify graphs of the blood concentrations of FSH, LH, estrogen, and progesterone throughout the menstrual cycle. Be sure to know when each peaks by studying Figure 2.10.

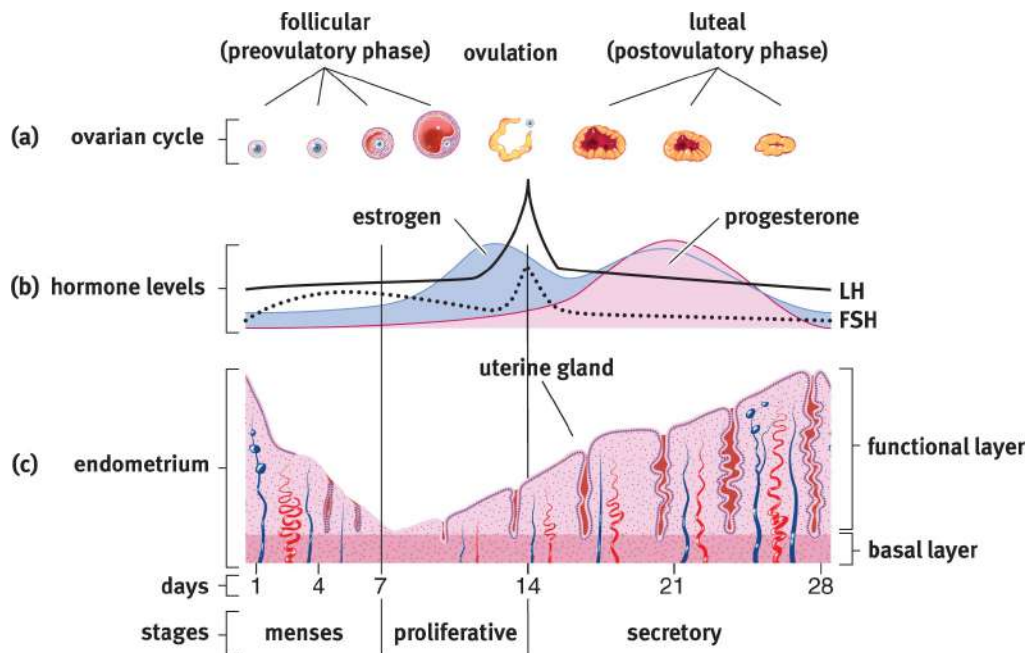


Figure 2.10. The Menstrual Cycle

(a) Follicle-stimulating hormone (FSH) facilitates the maturation of a single ovum; (b) The peak of luteinizing hormone (LH) around day 14 marks ovulation, the release of the oocyte from the follicle; (c) The endometrial lining of the uterus reaches its peak in the luteal phase and is shed at the beginning of the next cycle.

Follicular Phase

The **follicular phase** begins when the **menstrual flow**, which sheds the uterine lining of the previous cycle, begins. GnRH secretion from the hypothalamus increases in response to the decreased concentrations of estrogen and progesterone, which fall off toward the end of each cycle. The higher concentrations of GnRH cause increased secretions of both FSH and LH. These two hormones work in concert to develop several ovarian follicles. The follicles begin to produce estrogen, which has negative feedback effects and causes the GnRH, LH, and FSH concentrations to level off. Estrogen

stimulates regrowth of the endometrial lining, stimulating vascularization and glandularization of the **decidua**.



REAL WORLD

Oral contraceptive pills (OCPs) are simply estrogen/progesterone (or progesterone-only) preparations. These block conception by inhibiting LH and FSH release through negative feedback, thereby inhibiting ovulation. When a woman takes the placebo pills at the end of the month, withdrawal menstruation still occurs because estrogen and progesterone levels drop, but no egg passes with the menstrual flow.

Ovulation

Estrogen is interesting in that it can have both negative and positive feedback effects. Late in the follicular phase, the developing follicles secrete higher and higher concentrations of estrogen. Eventually, estrogen concentrations reach a threshold that paradoxically results in positive feedback, and GnRH, LH, and FSH levels spike. The surge in LH is important; it induces **ovulation**, the release of the ovum from the ovary into the abdominal (peritoneal) cavity.

Luteal Phase

After ovulation, LH causes the ruptured follicle to form the corpus luteum, which secretes progesterone. Remember that estrogen helps regenerate the uterine lining, but progesterone maintains it for implantation. Progesterone

levels begin to rise, while estrogen levels remain high. The high levels of progesterone cause negative feedback on GnRH, FSH, and LH, preventing the ovulation of multiple eggs.



KEY CONCEPT

Menstrual cycle:

- Follicles mature during the follicular phase (FSH, LH)
- LH surge at midcycle triggers ovulation
- Ruptured follicle becomes corpus luteum, which secretes estrogen and progesterone to build up uterine lining in preparation for implantation; LH and FSH are inhibited
- If fertilization does not occur, corpus luteum atrophies, progesterone and estrogen levels decrease, menses occurs, and LH and FSH levels begin to rise again

Menstruation

Assuming that implantation does not occur, the corpus luteum loses its stimulation from LH, progesterone levels decline, and the uterine lining is sloughed off. The loss of high levels of estrogen and progesterone removes the block on GnRH so that the next cycle can begin.

Pregnancy

On the other hand, if fertilization *has* occurred the resulting zygote will develop into a blastocyst that will implant in the uterine lining and secrete **human chorionic gonadotropin (hCG)**, an analog of LH—it looks very similar chemically and can stimulate LH receptors. This maintains the corpus luteum. hCG is critical during first trimester development because the estrogen and progesterone secreted by the corpus luteum keep the uterine lining in place. By the second trimester, hCG levels decline because the placenta has grown to a sufficient size to secrete enough progesterone and estrogen by itself. The high levels of estrogen and progesterone continue to serve as negative feedback on GnRH secretion.

Menopause

As a woman ages her ovaries become less sensitive to FSH and LH, resulting in ovarian atrophy. As estrogen and progesterone levels drop, the endometrium also atrophies, and menstruation stops. Also, because the negative feedback on FSH and LH is removed, the blood levels of these two hormones rise. This is called **menopause**. Profound physical and physiological changes usually accompany this process, including flushing, hot flashes, bloating, headaches, and irritability. Menopause usually occurs between the ages of 45 and 55.

MCAT CONCEPT CHECK 2.3:

Before you move on, assess your understanding of the material with these questions.

1. What are the functions of interstitial cells of Leydig and Sertoli cells?

- Interstitial cells of Leydig:

- Sertoli cells:

2. During which phase of meiosis is a primary oocyte arrested?
During which phase of meiosis is a secondary oocyte arrested?

- Primary oocyte:

- Secondary oocyte:

3. What is the acrosome? What organelle forms the acrosome?

4. Which hormones are key to sexual differentiation in a fetus with XY genotype? Describe the expected phenotype if receptors to these hormones are absent.

5. What are the four phases of the female menstrual cycle?
What are the features and relative hormone concentrations

of each phase? (Note: Draw in symbols to signify the levels of each hormone, such as ↑, =, and ↓)

Phase	Key Features	FSH	LH	Estrogen	Progesterone



Conclusion

In this chapter, we explored one of the key tenets of the cell theory—how cells produce more copies of themselves. We first examined mitosis, which results in genetically identical diploid daughter cells. We then moved on to meiosis, which results in genetically nonidentical haploid daughter cells, or gametes. We then looked at the male and female reproductive systems, which form these gametes, each of which contains half of the normal complement of genetic information. Finally, we explored basic reproductive endocrinology and saw how testosterone and estrogen are key in the development of the reproductive systems and the secondary sex characteristics that develop at puberty.

Formation of gametes is only half the story, of course. It serves us no good as a species to form sex cells if the cells cannot interact to form another human. Ultimately, gametes must accomplish their purpose: passing on the genes, the instructions for life, from one generation to another. Thus, we turn our attention in the next chapter to the next steps of fertilization, embryogenesis, and birth. The development of another human is a humbling and inspiring process. Indeed, it is through the union of one egg and one sperm that every human being on the planet today—and since the beginning of the human race—came into existence.

Concept Summary

The Cell Cycle and Mitosis

- **Diploid ($2n$)** cells have two copies of each chromosome; **haploid (n)** cells have one copy.
- The cell cycle contains five stages. The G_1 , S, and G_2 stages are collectively called **interphase**, during which the DNA is uncoiled in the form of **chromatin**.
 - In the **G_1** stage (**presynthetic gap**), cells create organelles for energy and protein production, and increase their size. The **restriction point**, during which the DNA is checked for quality, must be passed for the cell to move into the S stage.
 - In the **S** stage (**synthesis**), DNA is replicated. The strands of DNA, called **chromatids**, are held together at the centromere.
 - In the **G_2** stage (**postsynthetic gap**), there is further cell growth and replication of organelles in preparation for mitosis. Another quality checkpoint must be passed for the cell to enter into mitosis.
 - In the **M** stage (**mitosis**), mitosis and cytokinesis occur.
 - In the **G_0** stage, the cell performs its functions without preparing for division.
- **p53** plays a role in the two major checkpoints of the cell cycle (G_1 to S, and G_2 to M).

- **Cyclins** and **cyclin-dependent kinases (CDK)** rise and fall during the cell cycle. Cyclins bind to CDKs, phosphorylating and activating **transcription factors** for the next stage of the cell cycle.
- **Cancer** occurs when cell cycle control becomes deranged, allowing damaged cells to undergo mitosis without regard to quality or quantity of the new cells produced. Cancerous cells may begin to produce factors that allow them to delocalize and invade adjacent tissues or **metastasize** elsewhere.
- Mitosis produces two genetically identical diploid daughter cells from a single cell and occurs in **somatic cells**.
- Mitosis has four phases:
 - In **prophase**, the chromosomes condense, the nuclear membrane dissolves, nucleoli disappear, centrioles migrate to opposite sides of the cell, and the **spindle apparatus** begins to form. The **kinetochore** of each chromosome is contacted by a spindle fiber.
 - In **metaphase**, chromosomes line up along the **metaphase plate (equatorial plate)**.
 - In **anaphase**, sister chromatids are separated and pulled to opposite poles.
 - In **telophase**, the nuclear membrane reforms, spindle apparatus disappears, and cytosol and organelles are split between the two daughter cells through **cytokinesis**.

Meiosis

- **Meiosis** occurs in **gametocytes (germ cells)** and produces up to four nonidentical haploid sex cells (**gametes**).
- Meiosis has one round of replication and two rounds of division (the

reductional and **equational** divisions).

- In **meiosis I**, homologous pairs of chromosomes (homologues) are separated from each other. **Homologues** are chromosomes that are given the same number, but are of opposite parental origin.
 - In **prophase I**, the same events occur as in prophase of mitosis, except that homologues come together and intertwine in a process called **synapsis**. The four chromatids are referred to as a **tetrad**, and **crossing over** exchanges genetic material between one chromatid and material from a chromatid in the homologous chromosome. This accounts for **Mendel's second law (of independent assortment)**.
 - In **metaphase I**, homologous chromosomes line up on opposite sides of the metaphase plate.
 - In **anaphase I**, homologous chromosomes are pulled to opposite poles of the cell. This accounts for **Mendel's first law (of segregation)**.
 - In **telophase I**, the chromosomes may or may not fully decondense, and the cell may enter **interkinesis** after cytokinesis.
- In **meiosis II**, sister chromatids are separated from each other in a process that is functionally identical to mitosis. **Sister chromatids** are copies of the same DNA held together at the centromere.

The Reproductive System

- Biological sex is determined by the 23rd pair of chromosomes in humans, with XX being female and XY being male.
 - The **X chromosome** carries a sizeable amount of genetic information; mutations of X-linked genes can cause sex-linked disorders. Males are

hemizygous with respect to the unpaired genes on the X chromosome, so they will express sex-linked disorders, even if they only have one recessive disease-carrying allele. Women with one copy of the affected allele are called **carriers**.

- The **Y chromosome** carries little genetic information, but does contain the **SRY (sex-determining region Y)** gene, which causes the gonads to differentiate into testes.
- The male reproductive system contains both internal and external structures.
 - **Sperm** develop in the **seminiferous tubules** in the **testes**. They are nourished by **Sertoli cells**.
 - **Interstitial cells of Leydig**, in the testes, secrete **testosterone** and other male sex hormones (**androgens**).
 - The testes are located in the **scrotum**, which hangs outside of the abdominal cavity and has a temperature 2°C to 4°C lower than the rest of the body.
 - Once formed, sperm gain motility in the **epididymis** and are stored there until ejaculation.
 - During **ejaculation**, sperm travel through the **vas deferens** to the **ejaculatory duct**, and then to the **urethra** and out through the **penis**.
 - The **seminal vesicles** contribute fructose to nourish sperm and produce alkaline fluid.
 - The **prostate gland** also produces alkaline fluid.
 - The **bulbourethral glands** produce a clear viscous fluid that cleans out any remnants of urine and lubricates the urethra during sexual arousal.
 - **Semen** is composed of sperm and **seminal fluid** from the glands above.
- In **spermatogenesis**, four haploid sperm are produced from a

spermatogonium.

- After S stage, the germ cells are called **primary spermatocytes**.
 - After meiosis I, the germ cells are called **secondary spermatocytes**.
 - After meiosis II, the germ cells are called **spermatids**.
 - After maturation, the germ cells are called **spermatozoa**.
- Sperm contain a head, midpiece, and flagellum.
 - The **head** contains the genetic material and is covered with an **acrosome**—a modified Golgi apparatus that contains enzymes that help the sperm fuse with and penetrate the ovum.
 - The **midpiece** generates ATP from fructose and contains many mitochondria.
 - The **flagellum** promotes motility.
- The female reproductive system only contains internal structures.
 - **Ova** (eggs) are produced in **follicles** in the **ovaries**.
 - Once each month, an egg is **ovulated** into the **peritoneal sac** and is drawn into the **fallopian tube** or **oviduct**.
 - The fallopian tubes are connected to the **uterus**, the lower end of which is the **cervix**.
 - The **vaginal canal** lies below the cervix and is the site where sperm are deposited during intercourse.
 - The vaginal canal is also the site of natural childbirth.
 - The external female anatomy is known as the **vulva**.
- In **oogenesis**, one haploid ovum and a variable number of polar bodies are formed from an **oogonium**.

- At birth, all oogonia have already undergone replication and are considered **primary oocytes**. They are arrested in prophase I.
 - The ovulated egg each month is a **secondary oocyte**, which is arrested in metaphase II.
 - If the oocyte is fertilized, it will complete meiosis II to become a true ovum.
 - Cytokinesis is uneven in oogenesis. The cell receiving very little cytoplasm and organelles is called a **polar body**.
 - Oocytes are surrounded by the **zona pellucida**, an acellular mixture of glycoproteins that protect the oocyte and contain the compounds necessary for sperm binding; and the **corona radiata**, which is a layer of cells that adhered to the oocyte during ovulation.
- **Gonadotropin-releasing hormone (GnRH)** from the **hypothalamus** causes the release of **follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH)**, the functions of which depend on the sex of the individual.
 - In males, FSH stimulates the Sertoli cells and triggers spermatogenesis, while LH causes the interstitial cells to produce testosterone. **Testosterone** is responsible for the maintenance and development of the male reproductive system and male secondary sex characteristics (facial and axillary hair, deepening of the voice, and increased bone and muscle mass).
 - In females, FSH stimulates development of the ovarian follicles, while LH causes ovulation. These hormones also stimulate production of estrogens and progesterone.
 - The menstrual cycle is a periodic growth and shedding of the endometrial lining.

- In the **follicular phase**, GnRH secretion stimulates FSH and LH secretion, which promotes follicle development. Estrogen is released, stimulating vascularization and glandularization of the **decidua**.
 - **Ovulation** is stimulated by a sudden surge in LH. This surge is triggered when estrogen levels reach a threshold and switch from negative to positive feedback effects.
 - In the **luteal phase**, LH causes the ruptured follicle to become the corpus luteum, which secretes progesterone that maintains the uterine lining. High estrogen and progesterone levels cause negative feedback on GnRH, LH, and FSH.
 - **Menstruation** occurs if there is no fertilization. As the estrogen and progesterone levels drop, the endometrial lining is sloughed off, and the block on GnRH production is removed.
 - If fertilization does occur, the blastula produces **human chorionic gonadotropin (hCG)** which, as an LH analog, can maintain the corpus luteum. Near the end of the first trimester, hCG levels drop as the placenta takes over progesterone production.
- **Menopause** occurs when the ovaries stop producing estrogen and progesterone, usually between ages 45 and 55. Menstruation stops and FSH and LH levels rise. Physical and physiological changes accompanying menopause include flushing, hot flashes, bloating, headaches, and irritability.

Answers to Concept Checks

2.1

..

Cell Cycle Stage	Features
G₁	Cell grows and performs its normal functions. DNA is examined and repaired.
S	DNA is replicated.
G₂	Cell continues to grow and replicates organelles in preparation for mitosis. Cell continues to perform its normal functions.
M	Mitosis (cell division) occurs.
G₀	The cell performs its normal functions and is not preparing to divide.

!.

Mitotic Phase	Features
Prophase	Chromosomes condense, nuclear membrane dissolves, nucleoli disappear, centrioles migrate to opposite poles and begin forming the spindle apparatus
Metaphase	Chromosomes gather along the metaphase plate in the center of the cell under the guidance of the spindle apparatus
Anaphase	Sister chromatids separate, and a copy of each chromosome migrates to opposite poles
Telophase and Cytokinesis	Chromosomes decondense, nuclear membrane reforms, nucleoli reappear, spindle apparatus breaks down, cell divides into two identical daughter cells

2.2

- .. After meiosis I, there are two haploid daughter cells. After meiosis II, there are up to four haploid gametes.
- !. Homologous chromosomes are related chromosomes of opposite parental origin (such as maternal chromosome 15 and paternal chromosome 15, or—in males—the X and Y chromosomes). Sister chromatids are identical copies of the same DNA that are held together at the centromere. After S phase, a cell contains 92 chromatids, 46 chromosomes, and 23 homologous pairs.

}. Meiotic Phase	Differences from Mitotic Phase
Prophase I	Homologous chromosomes come together as tetrads during synapsis; crossing over
Metaphase I	Homologous chromosomes line up on opposite sides of the metaphase plate, rather than individual chromosomes lining up on the metaphase plate
Anaphase I	Homologous chromosomes separate from each other; centromeres do not break
Telophase I	Chromatin may or may not decondense; interkinesis occurs as the cell prepares for meiosis II

2.3

- .. The interstitial cells of Leydig secrete testosterone and other male sex hormones (androgens). Sertoli cells nourish sperm during their development.
- !. A primary oocyte is arrested in prophase I, while a secondary oocyte is arrested in metaphase II.
- }. The acrosome contains enzymes that are capable of penetrating the

corona radiata and zona pellucida of the ovum, permitting fertilization to occur. It is a modified Golgi apparatus.

- l. Androgens, such as testosterone, lead to male sexual differentiation. Absence of androgen receptors, a condition known as androgen insensitivity syndrome, leads to an XY genotype with phenotypically female characteristics.

- i.

Phase	Key Features	FSH	LH	Estrogen	Progesterone
Follicular	Egg develops, endometrial lining becomes vascularized and glandularized	↑	=	↓, then ↑	↓
Ovulation	Egg is released from follicle into peritoneal cavity	↑	↑↑	↑	↓
Luteal	Corpus luteum produces progesterone to maintain endometrium	↓	=	↑	↑
Menses	Shedding of endometrial lining	↓	↓	↓	↓



Shared Concepts

Behavioral Sciences Chapter 1

Biology and Behavior

Biochemistry Chapter 6

DNA and Biotechnology

Biology Chapter 1

The Cell

Biology Chapter 3

Embryogenesis and Development

Biology Chapter 5

The Endocrine System

Biology Chapter 12

Genetics and Evolution

Discrete Practice Questions

1. Which of the following is the correct sequence of the development of a mature sperm cell?
 - (A) Spermatid → spermatocyte → spermatogonium → 2° spermatocyte → spermatozoan
 - (B) Spermatogonium → 1° spermatocyte → 2° spermatocyte → spermatid → spermatozoan
 - (C) Spermatozoan → 1° spermatocyte → 2° spermatocyte → spermatogonium → spermatid
 - (D) Spermatogonium → 1° spermatocyte → 2° spermatocyte → spermatozoan → spermatid
2. Which of the following correctly pairs the stage of development of an egg cell with the relevant point in a woman's life cycle?
 - (A) From birth to menarche—prophase II
 - (B) At ovulation—metaphase I
 - (C) At ovulation—metaphase II
 - (D) At fertilization—prophase II
3. Some studies suggest that in patients with Alzheimer's disease, there is a defect in the way the spindle apparatus attaches to the

kinetochore fibers. At which stage of mitotic division would one first expect to be able to visualize this problem?

- (A) Prophase
- (B) Metaphase
- (C) Anaphase
- (D) Telophase

4. A researcher wishes to incorporate a radiolabeled deoxyadenine into the genome of one of the two daughter cells that would arise as a result of mitosis. What is the latest stage of cellular development during which the radiolabeled deoxyadenine could be added to achieve this result?

- (A) G_1
- (B) G_2
- (C) M
- (D) S

5. Certain ovarian tumors called granulosa cell tumors are known to produce excessive levels of estrogen. A physician who diagnoses a granulosa cell tumor should look for a secondary cancer in which of the following parts of the reproductive tract?

- (A) Fallopian tube
- (B) Cervix
- (C) Endometrium

(D) Vagina

6. Upon ovulation, the oocyte is released into the:

- (A) fallopian tube.
- (B) follicle.
- (C) abdominal cavity.
- (D) uterus.

7. Cancer cells are cells in which mitosis occurs continuously, without regard to quality or quantity of the cells produced. For this reason, most chemotherapies attack rapidly dividing cells. At which point(s) in the cell cycle could chemotherapy effectively prevent cancer cell division?

- I. S stage
- II. Prophase
- III. Metaphase

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

8. Which of the following INCORRECTLY pairs a structure of the male reproductive system with a feature of the structure?

- (A) Seminal vesicles—produce alkaline fructose-containing secretions
- (B) Epididymis—surrounded by muscle to raise and lower the testes
- (C) Vas deferens—tube connecting the epididymis to the ejaculatory duct
- (D) Cowper's glands—produce a fluid to clear traces of urine in the urethra

9. What is the last point in the meiotic cycle in which the cell has a diploid number of chromosomes?

- (A) During interphase
- (B) During telophase I
- (C) During interkinesis
- (D) During telophase II

10. Which of the following does NOT likely contribute to genetic variability?

- (A) Random fertilization of an egg by a sperm
- (B) Random segregation of homologous chromosomes
- (C) Crossing over between homologous chromosomes during meiosis
- (D) Replication of the DNA during S stage

11. Which of the following statements correctly identifies a key difference between mitosis and meiosis?

- (A) In metaphase of mitosis, replicated chromosomes line up in single file; in metaphase II of meiosis, replicated chromosomes line up on opposite sides of the metaphase plate.
- (B) During anaphase of mitosis, homologous chromosomes separate; during anaphase of meiosis I, sister chromatids separate.
- (C) At the end of telophase of mitosis, the daughter cells are identical to each other; at the end of meiosis I, the daughter cells are identical to the parent cell.
- (D) During metaphase of mitosis, centromeres are present directly on the metaphase plate; during metaphase of meiosis I, there are no centromeres on the metaphase plate.

12. Which of the following is true regarding prophase?

- (A) The chromosomes separate and move to opposite poles of the cell.
- (B) The spindle apparatus disappears.
- (C) The chromosomes uncoil.
- (D) The nucleoli disappear.

13. An individual who is phenotypically female is found to have only one copy of a disease-carrying recessive allele on the X chromosome, yet she demonstrates all of the classic symptoms of the disease. Geneticists determine she has a genotype that likely arose from nondisjunction in one of her parents. What is the likely genotype of this individual?

- (A) 46,XX (46 chromosomes, with XX for sex chromosomes)
- (B) 46,XY
- (C) 45,X
- (D) 47,XXY

14. During which phase of the menstrual cycle does progesterone concentration peak?

- (A) Follicular phase
- (B) Ovulation
- (C) Luteal phase
- (D) Menses

15. Which of the following would NOT be seen during pregnancy?

- (A) High levels of hCG in the first trimester
- (B) High levels of progesterone throughout the pregnancy
- (C) Low levels of FSH in the first trimester
- (D) High levels of GnRH throughout the pregnancy

Discrete Practice Answers

1. **B**

Diploid cells called spermatogonia differentiate into primary spermatocytes, which undergo the first meiotic division to yield two haploid secondary spermatocytes. These undergo a second meiotic division to become immature spermatids. The spermatids then undergo a series of changes leading to the production of mature sperm, or spermatozoa.

2. **C**

From the time of birth until shortly before ovulation, all egg cells are arrested at the prophase stage of meiosis I. These cells are referred to as primary oocytes. At ovulation, the egg cell has completed meiosis I and is now arrested in metaphase II as a haploid cell called a secondary oocyte. When a sperm penetrates the outer layers of the secondary oocyte, it completes meiosis II to become a mature ovum.

3. **A**

The spindle apparatus first interacts with the kinetochore fibers near the end of prophase. While the spindle apparatus aligns the chromosomes at the equatorial plate during metaphase, **(B)**, the initial connection of the microtubule to the kinetochore occurs in prophase.

4. **D**

To ensure that the labeled deoxyadenine will be incorporated into the DNA of one of the daughter cells, we have to insert the nucleotide before DNA replication has been completed. Because replication occurs during S stage, we could introduce the deoxyadenine during G₁ or S stage. Because G₁ precedes S, the latest point at which the deoxyadenine could be added is the S stage.

5. **C**

Estrogen is known to cause growth of the endometrial lining during the follicular phase of the menstrual cycle, and its levels stay high during the luteal phase to promote vascularization and glandularization of this tissue. Excessive levels of estrogen may provide a strong enough signal for cell growth to promote tumor formation or even cancer. The other tissues listed in this question require estrogen for development, but are not strongly dependent on estrogen for growth.

6. **C**

This subtle point about ovulation is missed by most students and remains hard to believe until the organs are examined in anatomy class in medical school. The ruptured ovarian follicle releases an oocyte into the abdominal cavity, close to the entrance of the fallopian tube. With the aid of beating cilia, the oocyte is drawn into the fallopian tube, through which it travels until it reaches the uterus. If it is fertilized in the fallopian tube, it will implant in the uterine wall. If fertilization does not occur, it will be expelled along with the uterine lining during menstruation.

7. **D**

The question is asking us to determine at which points in the cell cycle we can prevent or at least lower the number of cells undergoing mitosis. One

idea would be to prevent DNA synthesis during the S stage of the cell cycle. Without the DNA being replicated, two viable daughter cells could not be formed. Other ideas would be preventing the mitotic cycle from forming altogether in prophase by preventing spindle apparatus formation, preventing the nuclear membrane from dissolving, or interfering with other processes during this phase. Similarly, a treatment that would act on cells in the metaphase stage of the cell cycle would also interfere with the mitotic cycle. Therefore, any of the three solutions presented would be a viable option.

8. **B**

The epididymis is the site of sperm maturation at the posterior side of the testis. In the epididymis, sperm gain mobility and are stored until ejaculation. It is the vas deferens (ductus deferens) that is surrounded by muscle that raises and lowers the testis to maintain a constant temperature suitable for sperm production, not the epididymis.

9. **B**

The first meiotic division (reductional division) pulls homologous chromosomes to opposite poles of the cell during anaphase I. Near the end of telophase I, cytokinesis occurs, resulting in two haploid (n) daughter cells. Thus, during interkinesis and anaphase II, the daughter cells are already haploid, eliminating **(C)** and **(D)**. The cell is diploid during interphase, **(A)**, but remains diploid up until the end of telophase I.

10. **D**

The safest way to answer this question correctly is to go through each answer choice and eliminate the ones that contribute to genetic variability. The random fertilization of an egg by a sperm, the random segregation of homologous chromosomes during anaphase I, and crossing over between

homologous chromosomes during prophase I all contribute to genetic variability during sexual reproduction because they result in novel combinations of genetic material, eliminating **(A)**, **(B)**, and **(C)**. S stage, **(D)**, should *not* cause increased genetic variability; the DNA should be copied precisely, without error, meaning that both strands of DNA should be identical.

1. **D**

The key differences between mitosis and meiosis primarily appear during meiosis I. Of note, synapsis and crossing over occur during prophase I, and homologous chromosomes are separated during meiosis I (rather than sister chromatids, as in mitosis). While the location of the centromeres relative to the metaphase plate may seem trivial, it is representative of the fact that homologous chromosomes line up on opposite sides of the equatorial plate in meiosis, in contrast to the positioning of each chromosome directly upon the metaphase plate in mitosis.

2. **D**

In prophase, the chromatin condenses into chromosomes, the spindle apparatus forms, and the nucleoli and nuclear membrane disappear. **(A)** describes anaphase, whereas **(B)** and **(C)** describe telophase.

3. **C**

Nondisjunction refers to the incorrect segregation of homologous chromosomes during anaphase I, or of sister chromatids during anaphase II. In either case, one daughter cell ends up with two copies of related genetic material, while the other receives zero. Immediately, this should eliminate **(A)** and **(B)**, which show a normal complement of chromosomes (46). An individual who has only one recessive disease-carrying allele, and yet still expresses the disease, likely does not have a dominant allele for the

given trait. This is seen in males, who are hemizygous for many X-linked genes, and can also be seen in women with Turner syndrome (45,X), who have only one X chromosome. Thus, **(C)** is the answer.

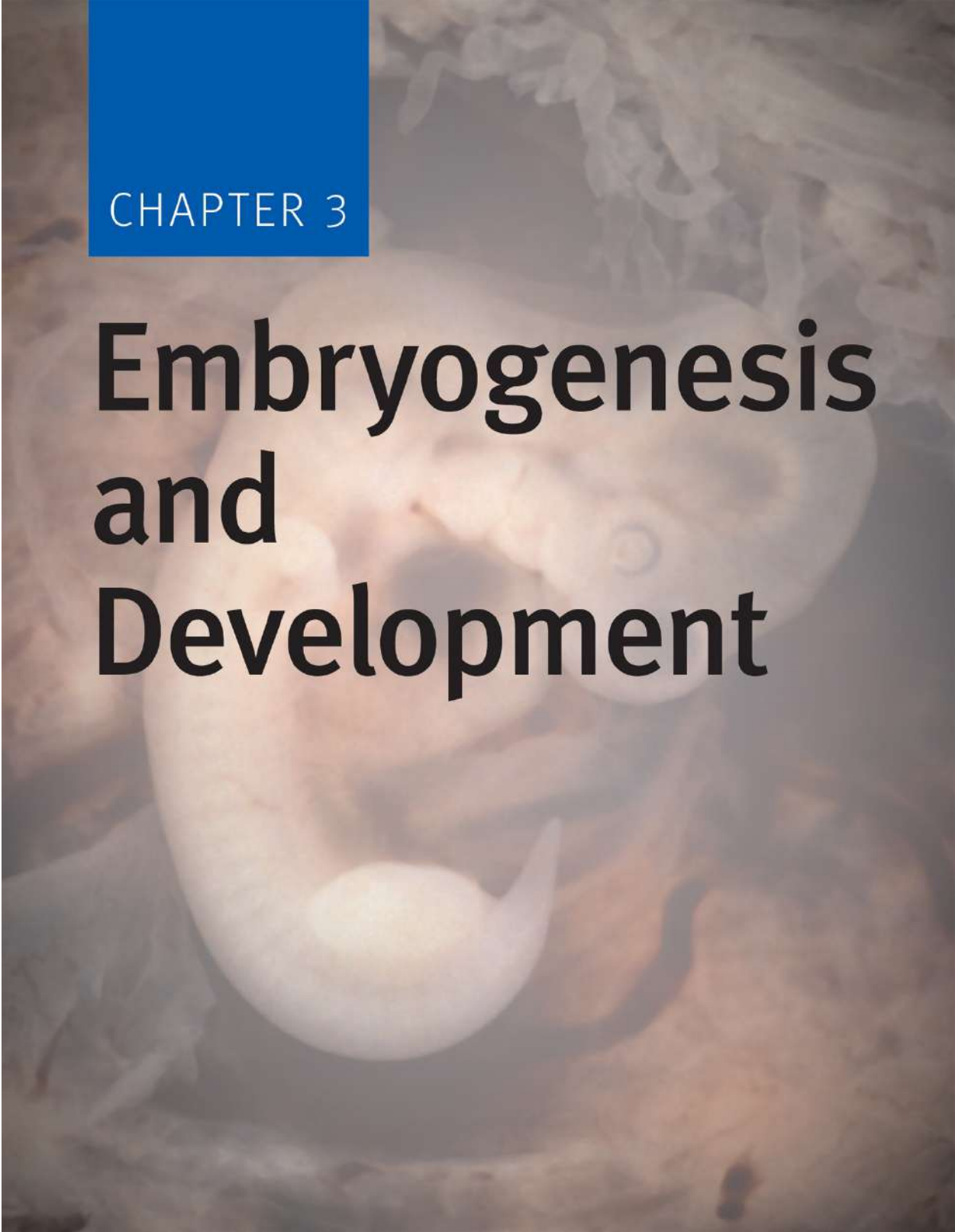
4. **C**

Progesterone peaks during the luteal phase, as it supports the endometrium for potential implantation of a blastula. Progesterone levels are relatively low during the follicular phase and ovulation, eliminating **(A)** and **(B)**. Withdrawal of progesterone actually causes menses, eliminating **(D)**.

5. **D**

During the first trimester of pregnancy, the corpus luteum is preserved by human chorionic gonadotropin (hCG); hence, progesterone secretion by the corpus luteum is maintained during the first trimester. This eliminates **(A)**. During the second trimester, hCG levels decline, but progesterone levels rise because the hormone is now secreted by the placenta itself, eliminating **(B)**. High levels of progesterone and estrogen inhibit GnRH secretion, thus preventing FSH and LH secretion and the onset of a new menstrual cycle. This eliminates **(C)** and validates **(D)**.





CHAPTER 3

Embryogenesis and Development

CHAPTER 3

Embryogenesis and Development

In This Chapter

3.1 Early Developmental Stages

Fertilization

Cleavage

Blastulation

Gastrulation

Neurulation

Problems in Early Development

3.2 Mechanisms of Development

Cell Specialization

Cell–Cell Communication

Cell Migration, Cell Death, and Regeneration

Senescence and Aging

3.3 Fetal Circulation

3.4 Gestation and Birth

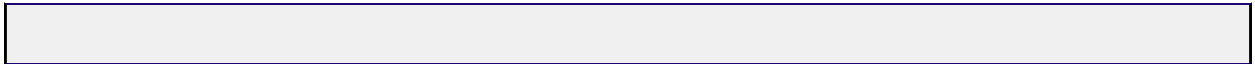
First Trimester

Second Trimester

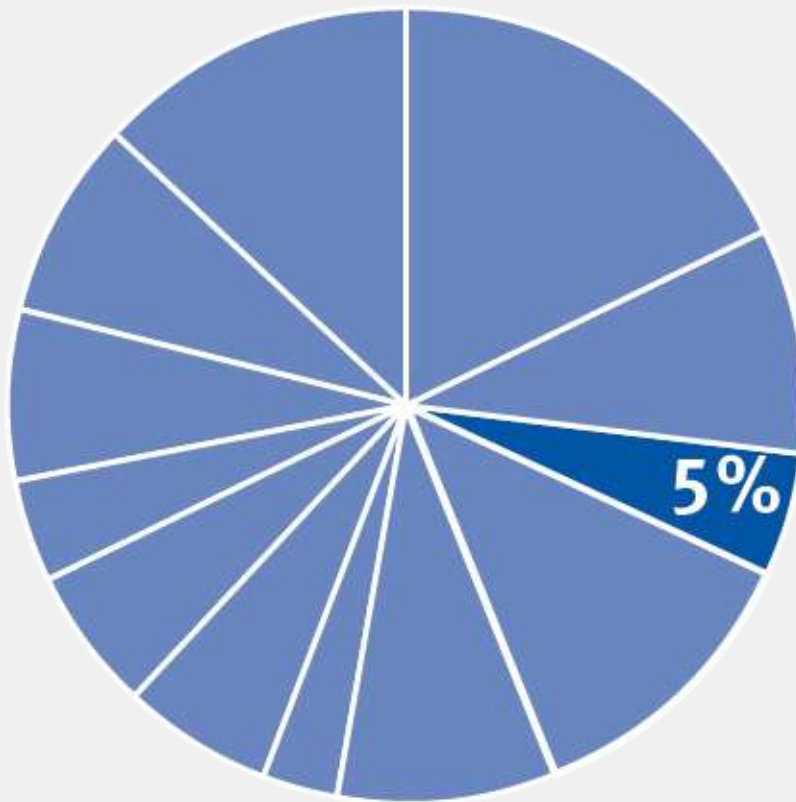
Third Trimester

Birth

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 5% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content categories:

1B: Transmission of genetic information from the gene to the protein

2C: Processes of cell division, differentiation, and specialization

3B: Structure and integrative functions of the main organ systems

7A: Individual influences on behavior

Introduction

Is it a boy or a girl? It's one of the most common questions asked of pregnant women. The suspense around learning the baby's sex never fails to excite family and friends. For centuries, members of older generations have offered advice on how to predict the outcome or even determine it. *If you dangle a needle over your belly by a thread, does it swing side to side or in circles?* In the modern world, needle dangling doesn't carry the same credibility as technological advances like ultrasonography. This radiographic technique is performed by placing a probe that emits high-frequency sound waves near the tissue to be examined. The probe transduces an image onto a computer screen to determine gestational age, screen for multiple pregnancies or anomalies, and identify the baby's sex.

So how early can we determine a baby's sex? All embryos are female by default—that is, for a male fetus to develop, it must undergo not only masculinization via the gene product of *SRY*, but also defeminization. These processes occur (or don't) around six to eight weeks post-fertilization. That said, prospective parents shouldn't expect any answers before 16 to 17 weeks without a blood test, because ultrasonography equipment does not have high enough resolution to give answers earlier in pregnancy.

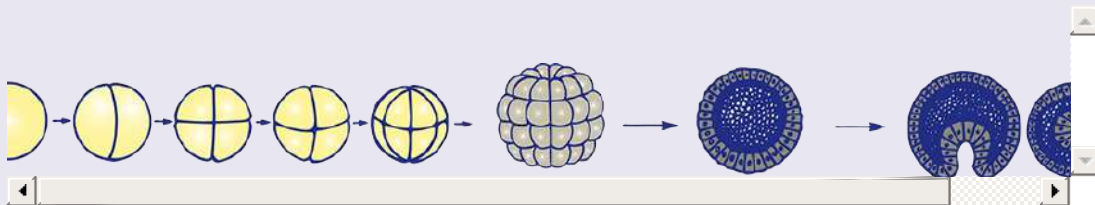
In this chapter, we'll continue the discussion from Chapter 2 by beginning with fertilization, the formation of a diploid zygote from the union of a sperm and an ovum. We'll then follow development from this point until the birth of an autonomously breathing baby. We'll examine how the cells of a developing human divide and differentiate. We'll also explore some specific system differences that exist between developing fetuses and adults as we present an overview of the stages of pregnancy and childbirth.

3.1 Early Developmental Stages

LEARNING OBJECTIVES

After Chapter 3.1, you will be able to:

- Distinguish between determinate and indeterminate cleavage of a zygote
- Describe the process of implantation and the stage of development at which it occurs
- Connect the ectoderm, mesoderm, neural crest, and endoderm to the organs they will form
- Describe how induction influences development
- Recall the stages of embryonic development up to the gastrula:



In this first section, we will explore development from the formation of a diploid zygote until neurulation, or the formation of the neural tube that will differentiate into the nervous system.

FERTILIZATION

As discussed in Chapter 2 of *MCAT Biology Review*, a secondary oocyte is ovulated from the follicle on approximately day 14 of the menstrual cycle. The secondary oocyte travels into the fallopian tube, where it can be fertilized up to 24 hours after ovulation. Fertilization, shown in Figure 3.1, usually occurs in the widest part of the fallopian tube, called the **ampulla**. When the sperm meets the secondary oocyte in the fallopian tube, it binds to the oocyte and releases acrosomal enzymes that enable the head of the sperm to penetrate the corona radiata and zona pellucida. The first sperm to come into direct contact with the secondary oocyte's cell membrane forms a tube-like structure known as the **acrosomal apparatus**, which extends to and penetrates the cell membrane. Its pronucleus may then freely enter the oocyte once meiosis II has come to completion.

After penetration of the sperm through the cell membrane, the **cortical reaction**, a release of calcium ions, occurs. These calcium ions depolarize the membrane of the ovum, which serves two purposes: depolarization prevents fertilization of the ovum by multiple sperm cells, and the increased calcium concentration increases the metabolic rate of the newly formed diploid **zygote**. The now depolarized and impenetrable membrane is called the **fertilization membrane**.

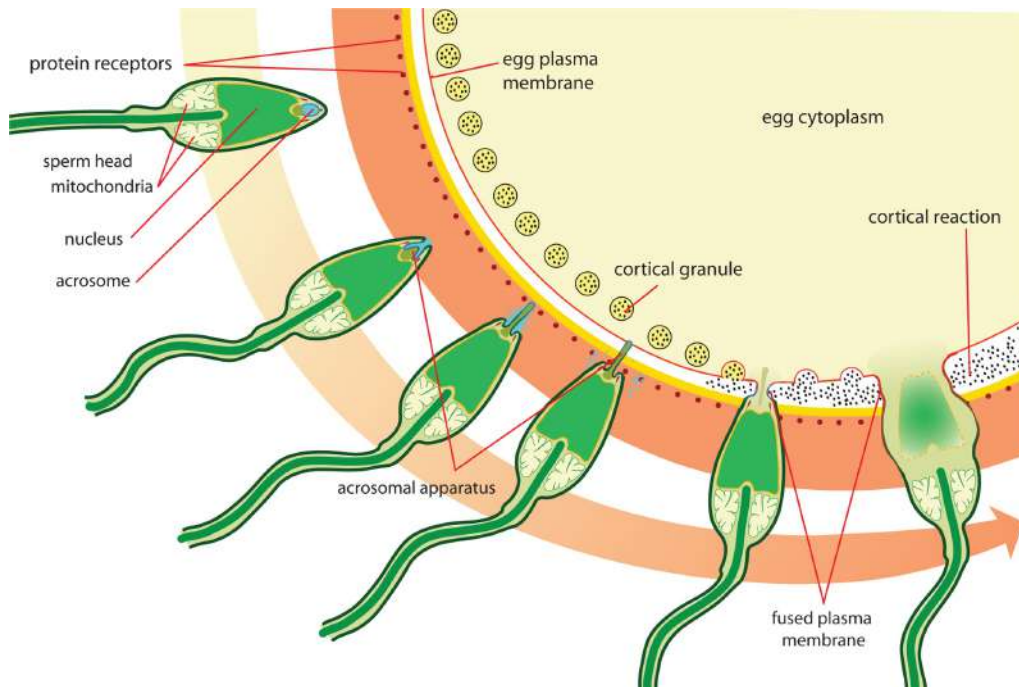


Figure 3.1. Fertilization

Twins

Twinning can occur by two different mechanisms. **Dizygotic (fraternal) twins** form from fertilization of two different eggs released during one ovulatory cycle by two different sperm. Each zygote will implant in the uterine wall, and each develops its own placenta, chorion, and amnion—these structures are discussed later in the chapter. If the zygotes implant close together the placentas may grow onto each other. Fraternal twins are no more genetically similar than any other pair of siblings.

Monozygotic (identical) twins form when a single zygote splits into two. Because the genetic material is identical, the genomes of the offspring will be too. If division is incomplete, **conjoined twins** may result, where the two offspring are physically attached. Monozygotic twins can be classified by the

number of structures they share. Monochorionic/monoamniotic twins share the same amnion and chorion. Monochorionic/diamniotic twins each have their own amnion, but share the same chorion. Dichorionic/diamniotic twins each have their own amnions and chorions. Which type of twinning occurs is a result of when the separation occurred. As more gestational structures are shared, there are more risks as the fetuses grow and develop.

CLEAVAGE

After fertilization in the fallopian tubes, the zygote must travel to the uterus for implantation. If it arrives too late, there will no longer be an endometrium capable of supporting the embryo. As it moves to the uterus for implantation, the zygote undergoes rapid mitotic cell divisions in a process called **cleavage**. The first cleavage officially creates an embryo, as it nullifies one of the zygote's defining characteristics: unicellularity. Although several rounds of mitosis occur, the total size of the embryo remains unchanged during the first few divisions, as shown in Figure 3.2. By dividing into progressively smaller cells, the cells increase two ratios: the nuclear-to-cytoplasmic (N:C) ratio and the surface area-to-volume ratio. Thus, the cells achieve increased area for gas and nutrient exchange relative to overall volume.

There are two types of cleavage: indeterminate and determinate.

Indeterminate cleavage results in cells that can still develop into complete organisms. In fact, monozygotic twins have identical genomes because they both originate from indeterminately cleaved cells of the same embryo.

Determinate cleavage results in cells with fates that are, as the term implies, already determined. In other words, these cells are committed to **differentiating** into a certain type of cell.



Figure 3.2. An 8-Cell Embryo

The embryo has undergone three cleavage events at this point.

BLASTULATION

Several divisions later, the embryo becomes a solid mass of cells known as a **morula**, as shown in Figure 3.3. This term comes from the Latin word for mulberry, which might help us grasp what an embryo at this stage looks like.

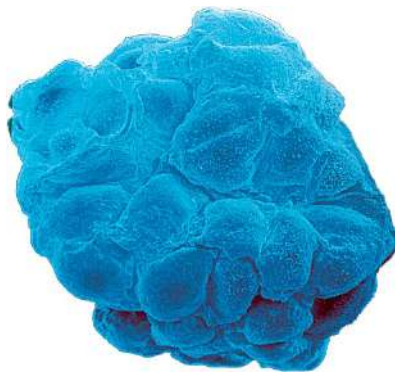


Figure 3.3. Morula

The morula is a solid ball of cells.

Once the morula is formed, it undergoes **blastulation**, which forms the **blastula**, a hollow ball of cells with a fluid-filled inner cavity known as a **blastocoel**. The mammalian blastula is known as a **blastocyst** and consists of two noteworthy cell groups, the trophoblast and inner cell mass (as shown in Figure 3.4). The **trophoblast cells** surround the blastocoel and give rise to the chorion and later the placenta, whereas the **inner cell mass** protrudes into the blastocoel and gives rise to the organism itself.



MNEMONIC

Remember that a **blastula** is an embryo with a **blasted**-out cavity.

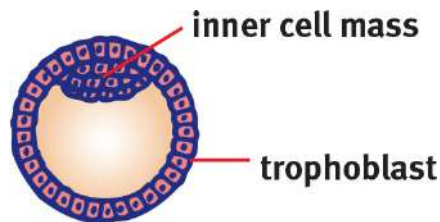


Figure 3.4. Blastula

The blastula contains a fluid-filled cavity called the blastocoel.

Implantation

The blastula moves through the fallopian tube to the uterus, where it burrows into the endometrium. The trophoblast cells are specialized to create an interface between the maternal blood supply and the developing embryo. These trophoblastic cells give rise to the **chorion**, an extraembryonic membrane that develops into the placenta. The trophoblasts form **chorionic**

villi, which are microscopic finger-like projections that penetrate the endometrium. As these chorionic villi develop into the placenta they support maternal–fetal gas exchange. The embryo is connected to the placenta by the **umbilical cord**, which consists of two arteries and one vein encased in a gelatinous substance. The vein carries freshly oxygenated blood rich with nutrients from the placenta to the embryo. The umbilical arteries carry deoxygenated blood and waste to the placenta for exchange.



REAL WORLD

Sometimes the blastula implants itself outside the uterus, a situation known as an ectopic pregnancy. Over 95% of ectopic pregnancies occur in the fallopian tube. Ectopic pregnancies are generally not viable because the narrow fallopian tube is not an environment in which an embryo can properly grow. If the embryo does not spontaneously abort, the tube may rupture and a considerable amount of hemorrhaging may result. In fact, a suspected ectopic pregnancy is often a surgical emergency.

Until the placenta is functional, the embryo is supported by the **yolk sac**, which is also the site of early blood cell development. There are two other extraembryonic membranes that require discussion: the allantois and the amnion. The **allantois** is involved in early fluid exchange between the embryo and the yolk sac. Ultimately, the umbilical cord is formed from remnants of the yolk sac and the allantois. The allantois is surrounded by the **amnion**, a thin, tough membrane filled with amniotic fluid. This fluid serves as a shock absorber during pregnancy, lessening the impact of maternal motion on the

developing embryo. In addition to forming the placenta, the chorion also forms an outer membrane around the amnion, adding an additional level of protection. The anatomy of these structures is shown in Figure 3.5.

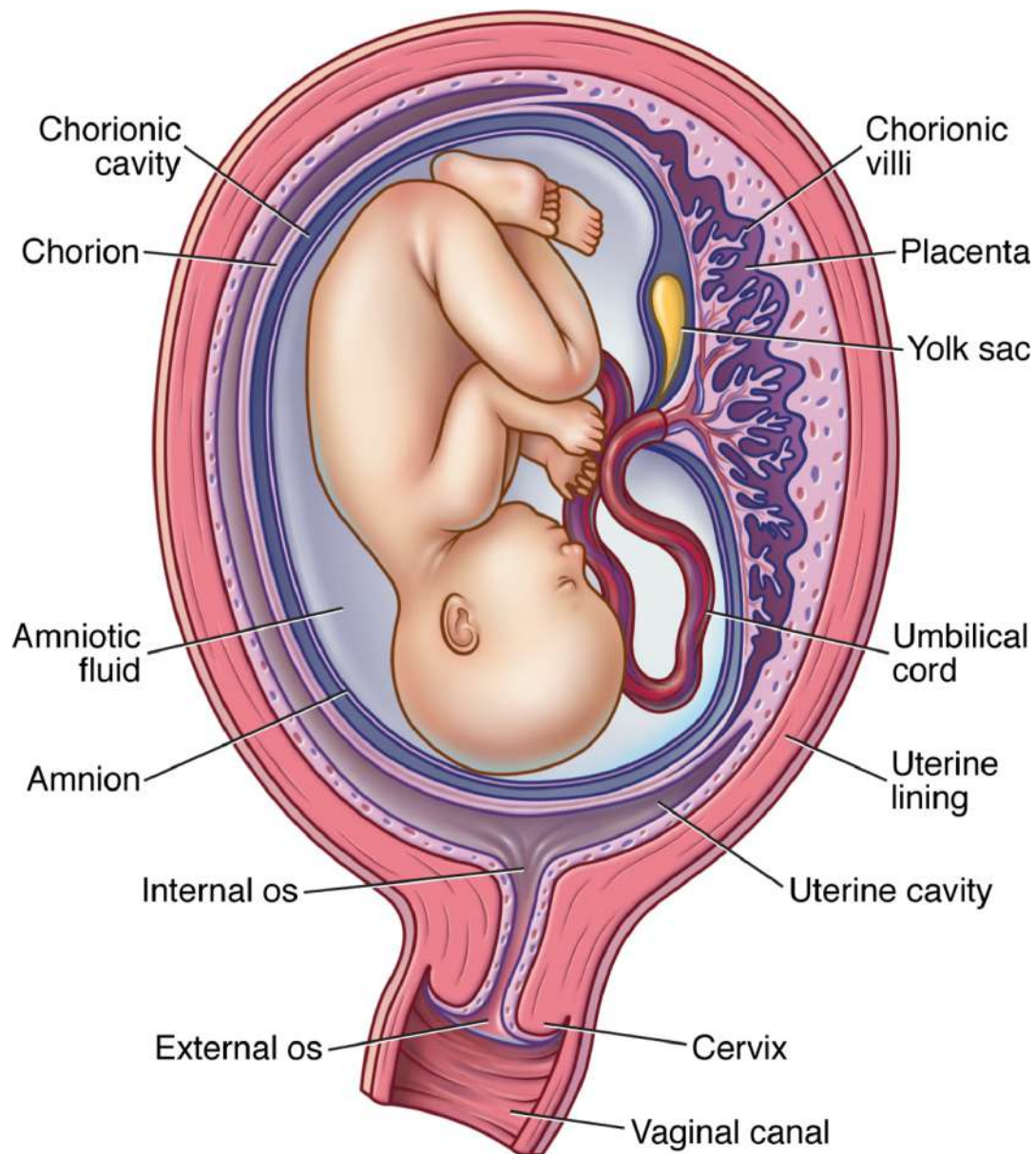


Figure 3.5. Anatomy of Pregnancy

GASTRULATION

Once the cell mass implants it can begin further developmental processes such as **gastrulation**, the generation of three distinct cell layers. The early developmental processes up to this point are shown in Figure 3.6. Much of our knowledge of development comes from the study of other organisms with varying degrees of similarity to human development. In sea urchins, gastrulation begins with a small invagination in the blastula. Cells continue moving toward the invagination, resulting in elimination of the blastocoel. To visualize this, imagine inflating a balloon and poking it with your finger. If you kept pushing, eventually the rubber from that side of the balloon would come into contact with the other side. If the two membranes could merge, as occurs in development, this would create a tube through the middle of the balloon. In living things, the result of this process is called a **gastrula**. The membrane invagination into the blastocoel is called the **archenteron**, which later develops into the gut. The opening of the archenteron is called the **blastopore**. In **deuterostomes**, such as humans, the blastopore develops into the anus. In **protostomes**, it develops into the mouth.



REAL WORLD

Amniocentesis is the process of aspirating amniotic fluid by inserting a thin needle into the amniotic sac. The amniotic fluid contains fetal cells that can be examined for chromosomal abnormalities as well as sex determination. Amniocentesis is recommended for pregnant women over 35 if earlier screening tests (blood tests and ultrasound) indicate a high chance of chromosomal abnormalities in the fetus. Women in this age group have a higher rate of meiotic nondisjunction, which can result in genetic aberrations such as Down syndrome.

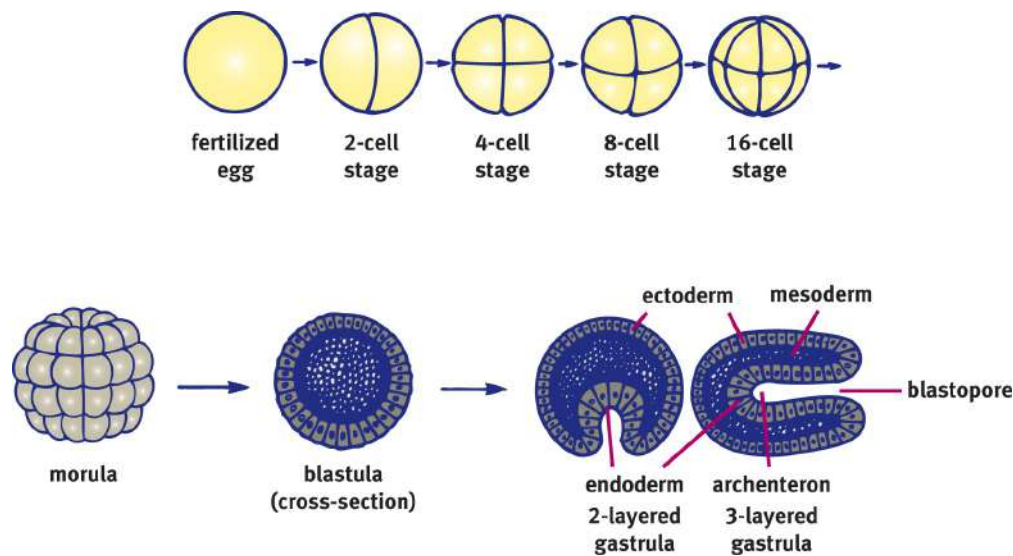


Figure 3.6. Early Stages of Embryonic Development



MNEMONIC

How can we remember the blastopore's fate in protostomes vs. deuterostomes? Think about how parents talk to toddlers—*deuterostome* starts with *deu*, which looks like *duo*, meaning *two*. Thus, **deu**terostomes develop the anus—the orifice associated with “**number two**”—from the blastopore. Protostomes must start at the other end (the mouth).

Primary Germ Layers

Eventually, some cells will also migrate into what remains of the blastocoel. This establishes three layers of cells called **primary germ layers**.



MNEMONIC

The primary germ layers:

- Ectoderm—“attracto”derm (things that attract us to others, such as cosmetic features and “smarts”)
- Mesoderm—“means”oderm (the means of getting around as an organism, such as bones and muscle; the means of getting around in the body, such as the circulatory system; the means of *getting around*, such as the gonads)
- Endoderm—linings of “endernal” (internal) organs (the digestive and respiratory tract, and accessory organs attached to these systems)

The outermost layer is called the **ectoderm** and gives rise to the integument, including the epidermis, hair, nails, and the epithelia of the nose, mouth, and lower anal canal. The lens of the eye, nervous system (including adrenal medulla), and inner ear are also derived from ectoderm.

The middle layer is called the **mesoderm** and develops into several different systems including musculoskeletal, circulatory, and most of the excretory systems. Mesoderm also gives rise to the gonads as well as the muscular and connective tissue layers of the digestive and respiratory systems and the adrenal cortex.

The innermost layer is called the **endoderm** and forms the epithelial linings of the digestive and respiratory tracts, including the lungs. The pancreas,

thyroid, bladder, and distal urinary tracts, as well as parts of the liver, are derived from endoderm.

Differentiation

So how is it that cells with the same genes are able to develop into such distinctly different cell types with highly specialized functions? Primarily, it is by **selective transcription** of the genome. In other words, only the genes needed for that particular cell type are transcribed. Thus, in pancreatic islet cells the genes to produce specific hormones (insulin, glucagon, or somatostatin) are turned on, while these same genes are turned off in other cell types. Selective transcription is often related to the concept of **induction**, which is the ability of one group of cells to influence the fate of nearby cells. This process is mediated by chemical substances called **inducers** which diffuse from the **organizing cells** to the **responsive cells**. These chemicals are responsible for processes such as the guidance of neuronal axons. Induction also ensures the proximity of different cell types that work together within an organ.



MCAT EXPERTISE

The MCAT likes to test on the dual embryonic origin of the adrenal glands. The adrenal cortex is derived from the mesoderm, but the adrenal medulla is derived from the ectoderm (because the adrenal medulla contains some nervous tissue).

NEURULATION

Once the three germ layers are formed, **neurulation**, or development of the nervous system, can begin. Remember that the nervous system is derived from the ectoderm. How, then, do cells originating on the surface of the embryo (ectoderm) end up inside the final organism? First, a rod of mesodermal cells known as the **notochord** forms along the long axis of the organism like a primitive spine (in fact, remnants of notochord persist in the intervertebral discs between vertebrae). The notochord induces a group of overlying ectodermal cells to slide inward to form **neural folds**, which surround a **neural groove**, as shown in Figure 3.7. The neural folds grow toward one another until they fuse into a **neural tube**, which gives rise to the central nervous system. At the tip of each neural fold are **neural crest cells**. These cells migrate outward to form the peripheral nervous system (including the sensory ganglia, autonomic ganglia, adrenal medulla, and Schwann cells) as well as specific cell types in other tissues (such as calcitonin-producing cells of the thyroid, melanocytes in the skin, and others). Finally, ectodermal cells will migrate over the neural tube and crest to cover the rudimentary nervous system.

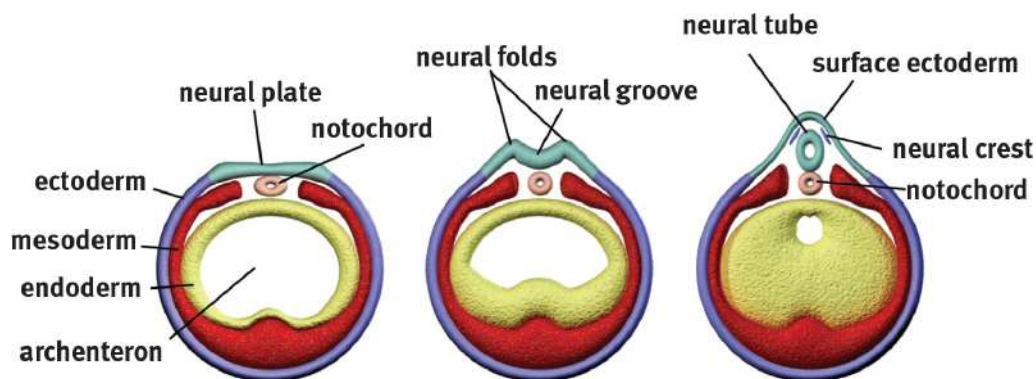


Figure 3.7. Formation of the Neural Tube

PROBLEMS IN EARLY DEVELOPMENT

Early development is a highly sensitive time. During this stage, as the germ layers are forming and as organogenesis (the production of organs) begins, teratogens may have far-reaching and highly detrimental effects. **Teratogens** are substances that interfere with development, causing defects or even death of the developing embryo. However, each teratogen will not have the same effect on every embryo or fetus. It is believed that the unique genetics of the embryo influences the effects of the teratogen. In addition to genetics, the route of exposure, length of exposure, rate of placental transmission of the teratogen, and the exact identity of the teratogen will also affect the outcome. Some common teratogens include alcohol, prescription drugs, viruses, bacteria, and environmental chemicals including polycyclic aromatic hydrocarbons.



REAL WORLD

Failure of the neural tube to close results in either spina bifida (in which some or all of the spinal cord may be exposed to the outside world) or anencephaly (in which the brain fails to develop and the skull is left open). The severity of spina bifida's effects range from no significant distress to death, whereas anencephaly is universally fatal. Women who wish to conceive are encouraged to take folate (folic acid) to prevent this complication; furthermore, it is recommended that all women of childbearing age supplement their diets with folate before getting pregnant because neurulation often occurs before pregnancy is detected.

In addition to teratogens, maternal health can also influence development. Certain conditions may cause changes in the overall physiology of the mother, resulting in overexposure or underexposure of the embryo or fetus to certain chemicals. For example, diabetic women with hyperglycemia (high blood glucose) can have poor birth outcomes. Overexposure to sugar *in utero* can lead to a fetus that is too large to be delivered and that suffers from hypoglycemia soon after birth (due to synthesizing very high levels of insulin to compensate). Maternal folic acid deficiency may prevent complete closure of the neural tube, resulting in spina bifida, in which parts of the nervous system are exposed to the outside world or covered with a thin membrane, or anencephaly, in which the brain fails to develop. However, like teratogens, maternal health issues can have variable effects on the developing fetus. Spina bifida may be so severe as to result in profound disability, or may be completely asymptomatic and only detected by a tuft of hair overlying the area. Overall, trends and associations can certainly be found between various environmental conditions and genes during development; however, outcomes are somewhat unpredictable and highly variable.

MCAT CONCEPT CHECK 3.1:

Before you move on, assess your understanding of the material with these questions.

1. What is the difference between determinate and indeterminate cleavage?
 - Determinate cleavage:

- Indeterminate cleavage:

2. From zygote to gastrula, what are the various stages of development?

3. During which stage of development does implantation occur?

4. What are the primary germ layers, and what organs are formed from each?

Germ Layer	Organs

5. What is induction and how does it influence development?

6. What tissues do neural crest cells develop into?

3.2 Mechanisms of Development

High-Yield



LEARNING OBJECTIVES

After Chapter 3.2, you will be able to:

- Explain the difference between determination and differentiation
- Connect totipotency, pluripotency, and multipotency to their respective levels of differentiation
- Recall the four types of cell–cell communication
- Distinguish between apoptosis and necrosis

As described earlier, cells undergo determinate cleavage to commit to a particular cell line, and inducers can be used for communication between one group of differentiating cells and another. In this section, we take a more specific look at the biochemical properties of these developmental mechanisms, as well as a few others.

CELL SPECIALIZATION

An adult human being is composed of approximately 37 trillion cells. These cells are organized into tissues that form organs within organ systems. In order to create an organism as complex as a human being, each cell must perform a specialized function. In addition, the cells in an organ must be organized such that the organ can function properly. For example, the pancreas must create both digestive enzymes (*trypsin*, *carboxypeptidases A* and *B*, *pancreatic lipase*, and others) and endocrine hormones (*insulin*, *glucagon*, and *somatostatin*). The cells that synthesize digestive enzymes must be located where cell products can enter ducts to ultimately empty into the duodenum. Likewise, the cells that synthesize endocrine hormones must be located near a blood vessel to put their products into systemic circulation. In order to accomplish this the cell must go through three stages: specification, determination, and differentiation.

Specification/Determination

The initial stage of cell specialization is **specification**, in which the cell is reversibly designated as a specific cell type. This is followed by **determination**, which was previously defined as the commitment of a cell to a particular function in the future. Prior to determination the cell can become any cell type, even if it has already gone through specification. After determination the cell is irreversibly committed to a specific lineage. There are multiple pathways by which determination may occur. During cleavage, where the existing mRNA and protein in the parent cell has been asymmetrically distributed between the daughter cells, the presence of specific mRNA and protein molecules may result in determination. Determination may also occur due to secretion of specific molecules from nearby cells. These molecules, also called **morphogens**, may cause neighboring cells to follow a particular developmental pathway.

Determination is a commitment to a particular cell type, but note that the cell has not yet actually produced what it needs to carry out the functions of that cell type—that is the goal of differentiation.

Differentiation

After a cell's fate has been determined, the cell must begin to undertake changes that cause the cell to develop into the determined cell type. This includes changing the structure, function, and biochemistry of the cell to match the cell type, a process called **differentiation**.



KEY CONCEPT

When a cell is determined, it is committed to a particular cell lineage. When the cell differentiates, it assumes the structure, function, and biochemistry of that cell type.

Cells that have not yet differentiated or that give rise to *other* cells that will differentiate are known as **stem cells**. Stem cells exist in embryonic tissues as well as adult tissues. The tissues a particular stem cell can differentiate into are determined by its **potency**. Cells with the greatest potency are called **totipotent** and include embryonic stem cells; totipotent cells can differentiate into any cell type, either in the fetus or in placental structures. After the 16-cell stage, the cells of the morula begin to differentiate into two groups: the inner cell mass and the trophoblast. After a few more cycles of cell division these totipotent cells start to differentiate into the three germ cell layers. At this stage, the cells are said to be **pluripotent**; these cells can

differentiate into any cell type except for those found in the placental structures. Finally, as the cells continue to become more specialized they are said to be **multipotent** stem cells, which can differentiate into multiple types of cells within a particular group. For example, hematopoietic stem cells are capable of differentiating into all of the cells found in blood, including the various types of white blood cells, red blood cells, and platelets—but not into skin cells, neurons, or muscle cells. While we use all of these different terms to describe potency, it is important to recognize that potency is a spectrum—not a series of strict definitions. Also, note that stem cells exist not only in embryos, but also in adults who have stem cells that give rise to skin, blood, and the epithelial lining of the digestive tract, among others.



KEY CONCEPT

Stem cells are able to differentiate into different cell types. The potency of the stem cell determines how many different cell types a stem cell can become. As cells become more differentiated, the potency of the cell decreases (from totipotent to pluripotent to multipotent).



BRIDGE

Stem cells are a cornerstone of biotechnology. Gene studies can be performed by introducing altered embryonic stem cells that contain transgenes into mice. Stem cells lacking a particular gene can be used

to create knockout mice. These processes are discussed in Chapter 6 of *MCAT Biochemistry Review*.

Over the last few decades, stem cell research has been a hotly contested issue. While harvesting of embryonic stem cells, highlighted in Figure 3.8, ultimately results in destruction of the embryo, it is thought that these cells could be used to regenerate human tissues, including the spinal cord (following injury) and the heart (following a heart attack). There are also immunologic concerns, as transplantation of stem cells of a different genetic makeup could evoke an immune response, resulting in rejection. In addition, once implanted, pluripotent cells may not necessarily differentiate into the desired tissue and may even become cancerous.

Embryonic stem (ES) cells are derived from the portion of a very early-stage embryo that would eventually give rise to an entire body. Because ES cells originate in this primordial stage, they retain pluripotency—the ability to form any cell type in the body.

CELL FATE

Less than a week after a human egg is fertilized, the developing embryo contains about 100 to 150 cells that have yet to differentiate. The embryo is a hollow ball, called a blastocyst, consisting only of trophoblasts, which in a pregnancy would later form the placenta, and an inner cell mass (ICM), which would become the fetus. Inside a womb, these cells would continue multiplying, beginning to specialize by the third week. The embryo, called a gastrula at this stage, would contain three distinctive germ layers, the descendants of which would ultimately form hundreds of different tissue types in the human body.

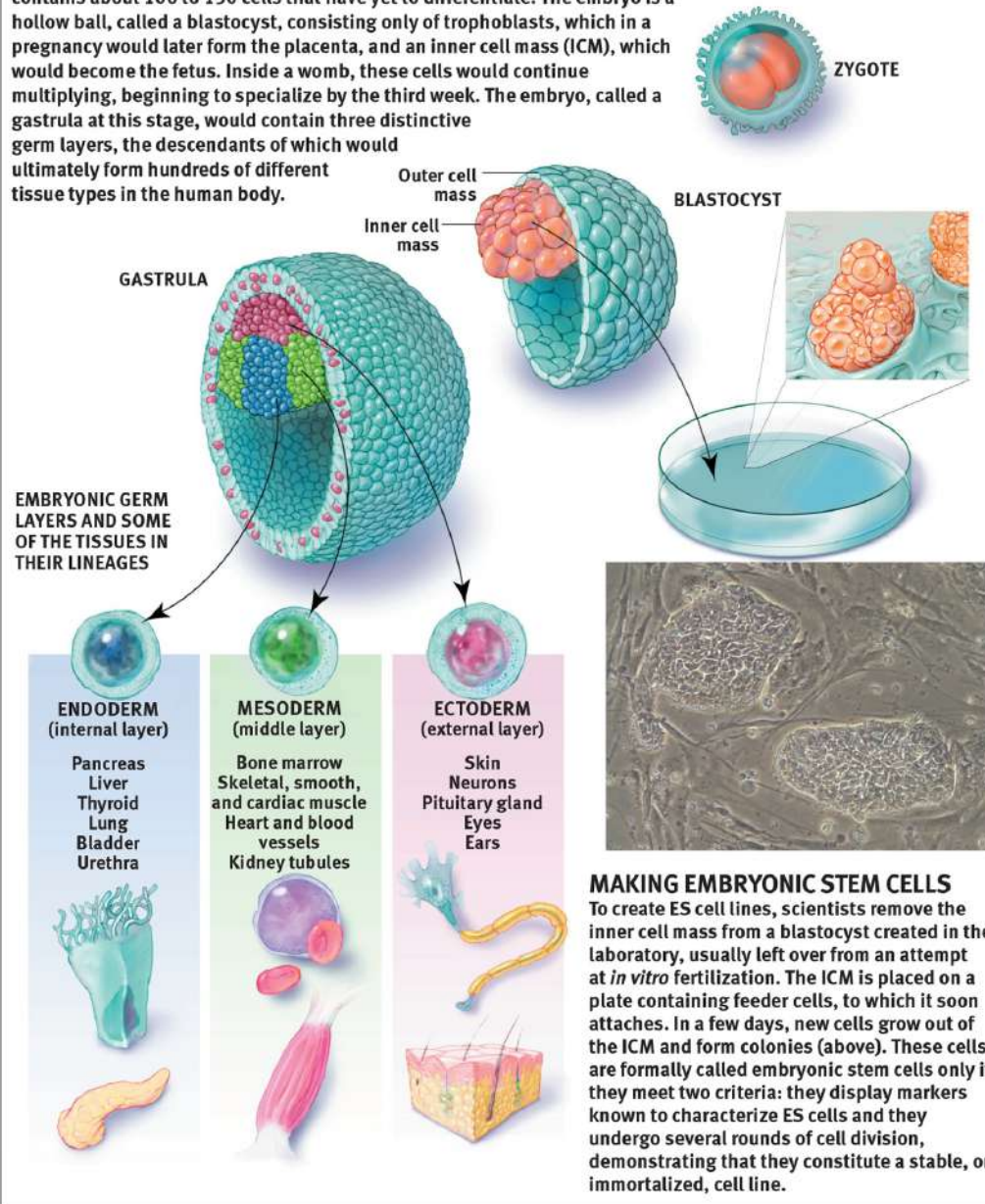


Figure 3.8. Embryonic Stem Cells

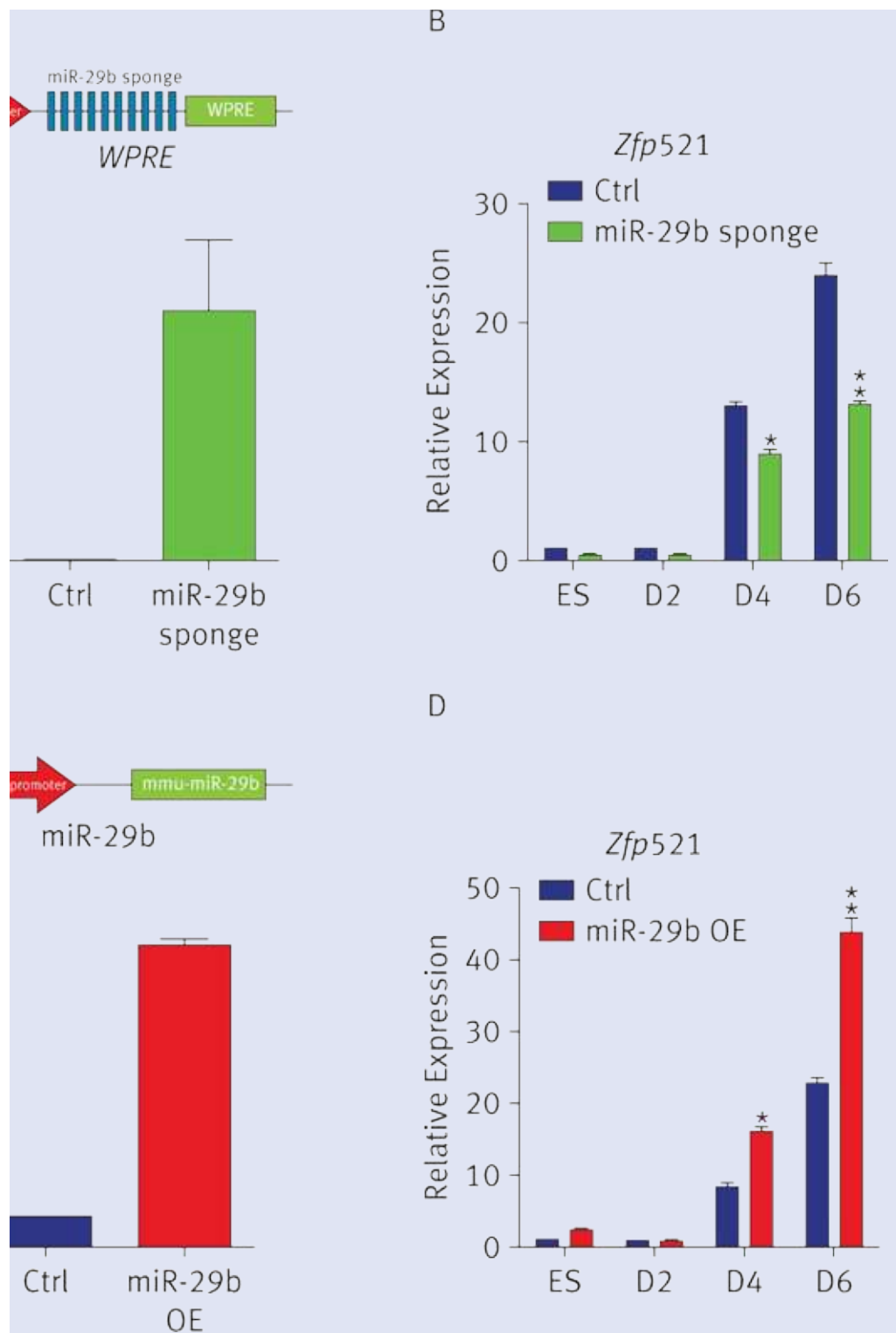
In order to address this controversy, many researchers have begun investigating adult stem cells. At best, these cells are multipotent, able to

differentiate into only a few different cell types. Researchers may take adult stem cells and use various transcription factors to increase potency in these cells. One of the potential advantages of this approach is that a stem cell can be taken from a patient (usually from blood, bone marrow, or adipose tissue), induced to become a different tissue type, and then implanted into that same patient. This offers reduced risk of rejection of foreign tissue. However, it is challenging to induce differentiation into the correct cell type, and most organs have a complex structure that depends on a number of different cell types, each of which requires different signals. Research is ongoing and holds promise, despite limited success.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

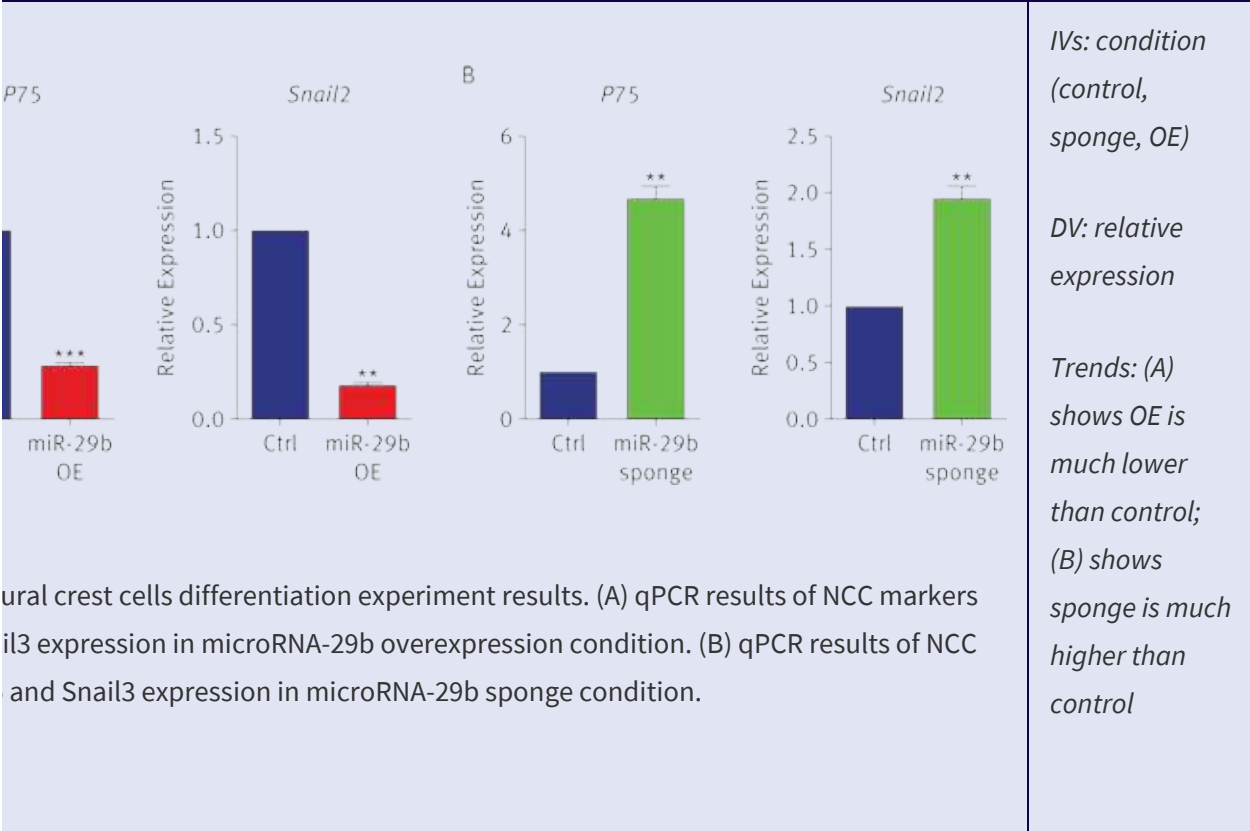
microRNA-29b in controlling the differentiation of neuroectoderm cell into neural cells (NTE) and neural crest cells (NCC) remains unclear.	<i>Study goal: clarify the role of microRNA-29b</i>
impact of microRNA-29b, researchers engineered microRNA sponges to contain dem binding sites for the target microRNA to competitively bind with microRNA-	<i>It sounds like these sponges will "pick up" the microRNA-29b that is present</i>
o the sponge, researchers also engineered microRNA-29b-overexpressed cells by microRNA-29b sequence into the embryonic stem cell genome.	<i>They made cells that overexpress microRNA-29b</i>
ven by a CAG promoter and inserted into the ROSA26 site.	<i>The terms "CAG" and "ROSA26" are</i>

	<i>unfamiliar, but both have the same promoter and insertion site</i>
	<i>IVs: condition (control, sponge, overexpression)</i> <i>DVs: relative expression</i> <i>Trends: (B) shows sponge is much lower than control, (D) shows OE is much higher than control</i>



ural tube epithelial cells differentiation experiment results. (A) The expression level
-29b sponge verified by qPCR. (B) qPCR results of NTE marker genes *Zfp521*
evels in microRNA-29b sponge condition. (C) The expression level of microRNA-29b

PCR. (D) qPCR results of NTE marker genes Zfp521 in microRNA-29b overexpression



, J., Wu, Y., Li, G., Ma, L., Feng, K., Guo, X., ... Kang, J. (2017). Mir-29b mediates the neural tube versus neural crest fate embryonic stem cell neural differentiation. *Stem Cell Reports*, 9(2), 571-586.

How does microRNA-29b influence the differentiation of neuroectoderm cells into neural epithelial cells and neural crest cells, respectively?

This question asks us to draw conclusions based on the data provided. To start we'll want to define the terms in the question stem and make sure we're clear on exactly what we're being asked for. We are told in the first paragraph that the neuroectoderm cells will differentiate into two types of cells, the neural epithelial cells and the neural crest cells. We are also told that microRNA-29b influences this process, but we do not know how it impacts differentiation.

Now we must parse out how the target microRNA are being manipulated to see how those changes alter the neuroectoderm cells' differentiation. The researchers created two experimental conditions, the microRNA sponge and the microRNA overexpression. Recall that microRNA works by binding to complementary sites on mRNA, which blocks translation mechanisms and prevents the synthesis of the gene product. As stated in the passage, the microRNA sponge has several sites that are complementary to the microRNA, which will act like a sponge by grabbing all the microRNA in the cell. As a result, the microRNA will bind to the sponge instead of just binding to local RNA, meaning that there is functionally less microRNA-29b binding to the RNA as compared to normal. In other words, it is the opposite of the overexpressed condition.

Now that we know what our action is (differentiation) and what the two conditions are (overexpression and inhibition), we can analyze the data. Figure 1 shows the neural epithelial cell results. In the microRNA sponge condition, the differentiation level is reduced compared to control, while in the microRNA overexpression condition the differentiation level is increased. This implies that microRNA-29b is critical for the differentiation of neuroectoderm cells to neural epithelial cells.

Looking at Figure 2, we see opposing effects. In the microRNA sponge condition, differentiation into neural crest cells is increased compared to control. In the microRNA overexpression condition, differentiation to neural crest cells is reduced. From these data, we can infer that microRNA suppresses neuroectoderm cell differentiation to neural crest cells.

In conclusion, the presence of microRNA-29b is important for neuroectoderm cell differentiation into neural epithelial cells, but the presence of microRNA-29b suppresses differentiation into neural crest cells.

CELL-CELL COMMUNICATION

The determination and differentiation of a cell depends on the location of the cell as well as the identity of the surrounding cells. The developing cell receives signals from organizing cells around it and may also secrete its own signaling molecules. As discussed previously, surrounding tissues induce a developing cell to become a particular cell type via inducers; the term *inducer* may also refer to the cell secreting the signal. The cell that is induced is called a **responder** (responsive cell); to be induced, a responder must be **competent**, or able to respond to the inducing signal.

Cell-cell communication can occur via autocrine, paracrine, juxtacrine, or endocrine signals. **Autocrine** signals act on the same cell that secreted the signal in the first place. **Paracrine** signals act on cells in the local area. **Juxtacrine** signals do not usually involve diffusion, but involve a cell directly stimulating receptors of an adjacent cell. Finally, **endocrine** signals involve secreted hormones that travel through the bloodstream to a distant target tissue.

Inducers

Inducers are often **growth factors**, which are peptides that promote differentiation and mitosis in certain tissues. Most growth factors only function on specific cell types or in certain areas, as determined by the competence of these cells. In this way, certain growth factors can code for particular tissues. For example, *PAX6* is expressed in the ectoderm of the

head, but in no other location. Therefore, as the optic vesicle approaches the overlying ectoderm producing this factor, development of the lens of the eye is induced. Interestingly, induction is not always a one-way pathway. To that end, differentiation of the lens then triggers the optic vesicle to form the optic cup, which ultimately becomes the retina. This is known as **reciprocal development**. Most tissues will be exposed to multiple inducers during the course of development.



REAL WORLD

In development of the eyes, lateral outpocketings from the brain (optic vesicles) grow out and touch the overlying ectoderm. The optic vesicle induces the ectoderm to form the lens placode. The lens placode in turn induces the optic vesicle to create the optic cup. The optic cup then induces the lens placode to develop into the cornea and lens. Experiments with frog embryos show that if this ectoderm is subsequently transplanted to the trunk (after the optic vesicles have grown out), a lens will develop in the trunk. If, however, the ectoderm is transplanted before the outgrowth of the optic vesicles, it will not.

One of the main methods of signaling occurs via the use of gradients. Morphogens, or molecules that cause determination of cells, diffuse throughout the organism. Locations closer to the origin of the morphogen will be exposed to higher concentrations, while areas further away will have less exposure. Multiple morphogens are secreted simultaneously, resulting in unique combinations of morphogen exposure throughout the organism, which can thereby induce the differentiation of specific cell types. Some

common morphogens include transforming growth factor beta (TGF- β), sonic hedgehog (Shh), and epidermal growth factor (EGF).

CELL MIGRATION, CELL DEATH, AND REGENERATION

Induction and differentiation lead to the creation of different types of cells; however, these cells are not always in the right location to carry out their function. Further, the sculpting of various anatomic structures requires not only differentiation, but also the death of some cells. Certain organs also have the ability to recreate injured or surgically removed portions of tissue.

Cell Migration

Cells must be able to disconnect from adjacent structures and migrate to their correct location. For example, the anterior pituitary gland originates from a segment of oral ectoderm and must migrate from the top of the mouth to its final location just below the hypothalamus. Neural crest cells also undergo extensive migration. These cells form at the edge of the neural folds during neurulation and then migrate throughout the body to form many different structures including the sensory ganglia, autonomic ganglia, adrenal medulla, and Schwann cells, as well as specific cell types in other tissues such as calcitonin-producing cells of the thyroid, melanocytes in the skin, and others.

Cell Death

Apoptosis, or programmed cell death, occurs at various times in development. For example, the fingers are originally webbed during development of the hand. The cells of the webbing later undergo apoptosis,

resulting in separation of each individual finger and toe. Apoptosis may occur via apoptotic signals or preprogramming.

During the process of apoptosis the cell undergoes changes in morphology and divides into many self-contained protrusions called **apoptotic blebs**, which can then be broken apart into **apoptotic bodies** and digested by other cells, as shown in Figure 3.9. This allows recycling of materials. Because the blebs are contained by a membrane, this also prevents the release of potentially harmful substances into the extracellular environment. This is different from **necrosis**, which is a process of cell death in which a cell dies as a result of injury. In necrosis, internal substances can be leaked, causing irritation of nearby tissues or even an immune response.

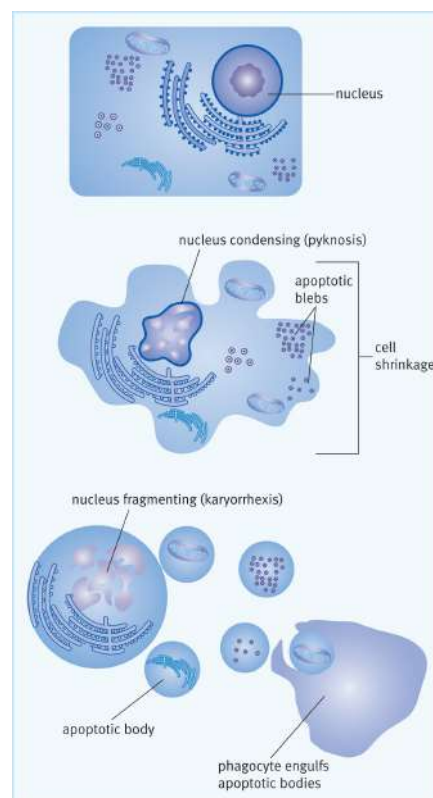


Figure 3.9. Apoptosis

An apoptotic cell disintegrates so that it can be absorbed and digested by other

cells.

Regeneration

Regenerative capacity, or the ability of an organism to regrow certain parts of the body, varies from species to species. Some species, such as salamanders and newts, have an enhanced capacity to regenerate because they retain extensive clusters of stem cells within their bodies. When regeneration is required, these stem cells can then migrate to the appropriate part of the body to initiate regrowth. These species are said to undergo **complete regeneration**, in that the lost or damaged tissues are replaced with identical tissues. In contrast, **incomplete regeneration** implies that the newly formed tissue is not identical in structure or function to the tissue that has been injured or lost.

Humans typically exhibit incomplete regeneration in response to injury. However, in humans regenerative capacity varies by the tissue type. Liver tissue has a high regenerative capacity, often able to undergo extensive regeneration following injury or loss. For example, living donors are often able to donate up to 50 percent of their liver tissue because their own livers will regenerate the missing portion. Unfortunately, the heart has little, if any, regenerative capacity, and scarring often results following an injury due to an event such as a heart attack. The kidneys have moderate regenerative capacity and are able to repair nephrons after injury to the tubules; however this regenerative capacity is easily overwhelmed, and kidney failure may result.

SENESCENCE AND AGING

As organisms age, changes occur in both molecular and cellular structure. This results in disruption of metabolism and, eventually, death of the organism. **Senescence**, or biological aging, can occur at the cellular and organismal level as these changes accumulate. At the cellular level, senescence results in the failure of cells to divide, normally after approximately 50 divisions *in vitro*. Research has demonstrated that this may be due to shortened **telomeres**, or the ends of chromosomes. Telomeres reduce the loss of genetic information from the ends of chromosomes and help prevent the DNA from unraveling—their high concentration of guanine and cytosine enables telomeres to “knot off” the end of the chromosome. Telomeres are difficult to replicate, however, so they shorten during each round of DNA synthesis. Eventually, the telomeres become too short, and the cell is no longer able to replicate. Some cells, including germ cells, fetal cells, and tumor cells, express an enzyme known as *telomerase*. This enzyme is a reverse transcriptase that is able to synthesize the ends of chromosomes, preventing senescence. Telomerase allows cells to divide indefinitely and may play a role in the survival of cancer cells.

At the organismal level, senescence represents changes in the body’s ability to respond to a changing environment. Aging is complex and often involves not only cellular senescence but also the accumulation of chemical and environmental insults over time.

MCAT CONCEPT CHECK 3.2:

Before you move on, assess your understanding of the material with these questions.

1. What is the difference between determination and differentiation?

- Determination:

- Differentiation:

2. What are the three types of potency? What lineages can a cell of each type differentiate into?

Type of Potency	Cell Lineages

3. What are the four types of cell–cell communication?

-

-

-

-

4. What is the difference between apoptosis and necrosis?

- Apoptosis:

- Necrosis:

3.3 Fetal Circulation

LEARNING OBJECTIVES

After Chapter 3.3, you will be able to:

- Recall the oxygenation status of blood in umbilical arteries and umbilical veins
- Identify the three fetal shunts, their locations, and the organs they bypass

Recall that the placenta, shown in Figure 3.10, is the organ where nutrient, gas, and waste exchange occurs. It is crucial that maternal and fetal blood do not mix because they may be different blood types. The simplest method to move nutrients and waste products is by diffusion, the preferred method for water, glucose, amino acids, and inorganic salts. Diffusion requires a gradient, which implies there is a higher partial pressure of oxygen in maternal blood than in fetal blood. To further enhance the transfer of oxygen from maternal to fetal circulation, fetal blood cells contain **fetal hemoglobin (HbF)**, which has a greater affinity for oxygen than maternal (adult) hemoglobin (primarily HbA). This also assists with the transfer (and retention) of oxygen into the fetal circulatory system. Waste material and carbon dioxide move in the opposite direction.

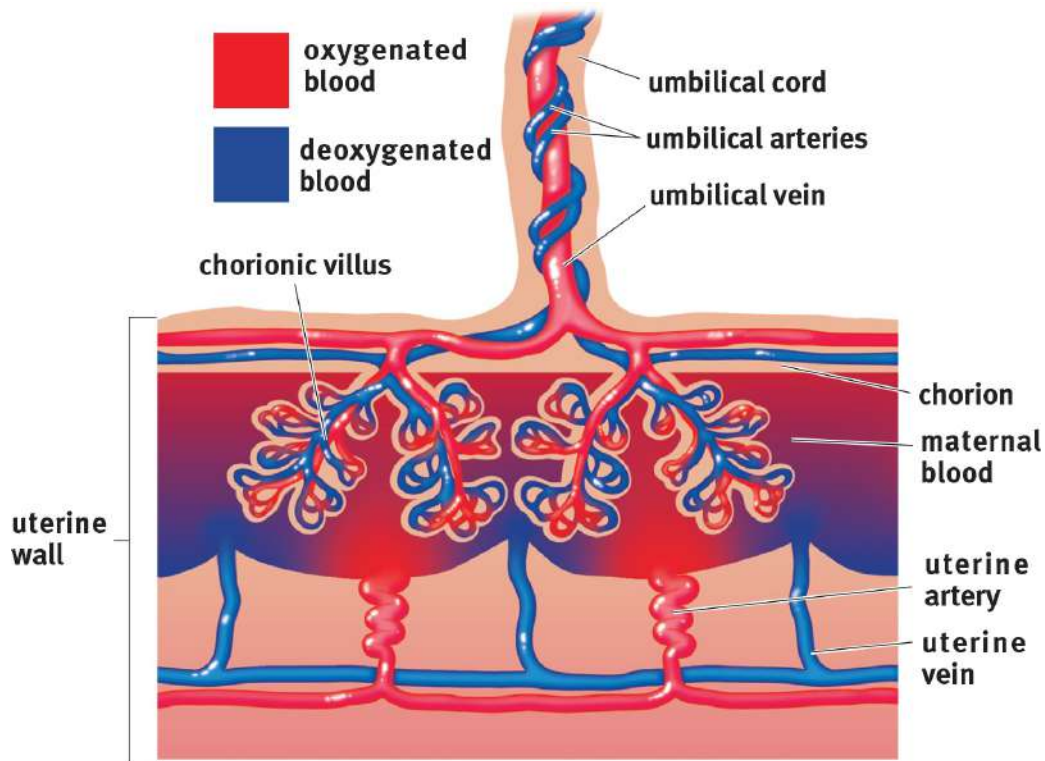


Figure 3.10. Placental Structure



KEY CONCEPT

Although the embryo obtains its nutrients and oxygen from the mother, there is no actual mixing of the blood. Instead, the placenta depends on the close proximity of the embryonic and maternal bloodstreams, facilitating diffusion between them.



KEY CONCEPT

Remember, gas exchange in the fetus occurs across the placenta. Fetal lungs do not function until birth.

The placental barrier also serves another function: immunity. The fetus is immunologically naïve because it has not yet been exposed to any pathogens; however, accidental exposure can happen *in utero*. Thus, the crossing of antibodies across the placental membrane serves a protective function. The placenta also qualifies as an endocrine organ because it produces progesterone, estrogen, and human chorionic gonadotropin (hCG), all which are essential for maintaining pregnancy.



REAL WORLD

Many pathogens are too large to cross the placental barrier by diffusion, but a set of pathogens called TORCHES infections can cross this barrier and cause significant birth defects. Therefore, screening for (and sometimes immunization against) these infections is recommended in pregnancy. TORCHES stands for **T***oxoplasma gondii*, **R**ubella, **C**ytomegalovirus, **H**Erpes or **H**IV, and **S**yphilis.

The umbilical vessels are commonly tested on the MCAT because they demonstrate the need to understand the proper biological definitions of artery and vein. Like all other arteries that carry blood away from the heart, the **umbilical arteries** carry blood away from the fetus toward the placenta. And, like all of the other veins that carry blood toward the heart, the **umbilical vein** carries blood toward the fetus from the placenta. Remember that

oxygenation occurs at the placenta, rather than in the fetal lungs. Therefore, the umbilical arteries carry deoxygenated blood and the umbilical vein carries oxygenated blood.



KEY CONCEPT

Unlike most other arteries, the umbilical arteries carry deoxygenated blood with waste products. Unlike most other veins, the umbilical vein carries oxygenated blood with nutrients.

There are several key differences between fetal and adult circulation that demonstrate important characteristics of the developing organism. The lungs and liver both do not serve significant functions prior to birth. Gas exchange does not occur at the lungs, but rather at the placenta. Detoxification and metabolism are primarily controlled by the mother's liver, and nutrient and waste exchange occurs at the placenta as well. Thus, the fetus does not depend on its own lungs and liver. Notably, these two organs are both underdeveloped and sensitive to the high blood pressures they will receive in postnatal life; thus, the fetus constructs three **shunts** to actively direct blood away from these organs while they develop, as shown in Figure 3.11.

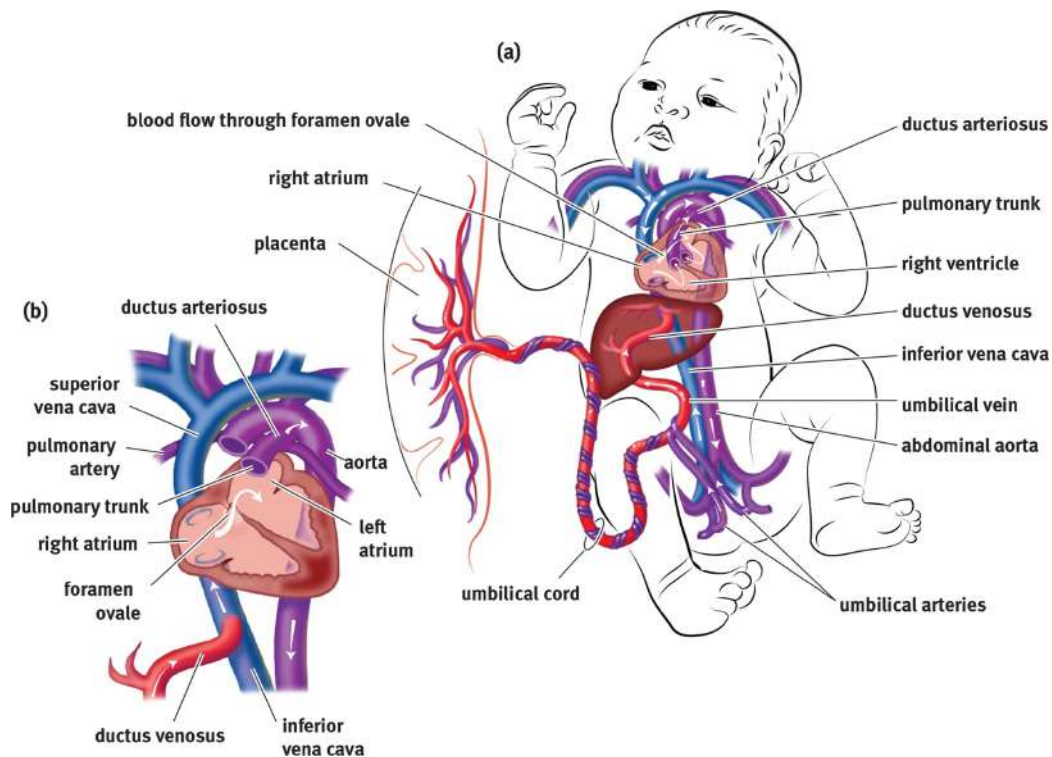


Figure 3.11. Fetal Circulation

(a) Systemic fetal circulation; (b) Enlarged view of fetal circulation highlighting the three fetal shunts.

Two different shunts are used to reroute blood from the lungs. The first, called the **foramen ovale**, is a one-way valve that connects the right atrium to the left atrium. This allows blood entering the right atrium from the inferior vena cava to flow into the left atrium instead of the right ventricle, and thereby be pumped through the aorta into systemic circulation directly. Unlike in adult circulation, the right side of the heart is at a higher pressure in the developing fetus than the left side, which pushes blood through the opening. After birth, this pressure differential reverses, shutting the foramen ovale. Second, the **ductus arteriosus** shunts leftover blood from the pulmonary artery to the aorta. Again, the pressure differential between the right and left sides of the heart pushes blood through this opening and into systemic circulation.

The liver is bypassed via the **ductus venosus**, which shunts blood returning from the placenta via the umbilical vein directly into the inferior vena cava. The liver still receives some blood supply from smaller hepatic arteries in the systemic circulation.

MCAT CONCEPT CHECK 3.3:

Before you move on, assess your understanding of the material with these questions.

1. What is the oxygenation status of the blood in the umbilical arteries? In the umbilical vein?

- Umbilical arteries:

- Umbilical vein:

2. What are the three fetal shunts? What vessels or heart chambers do they connect? What organ does each shunt bypass?

Shunt	Connected Vessels or Chambers	Organ Bypassed



3.4 Gestation and Birth

LEARNING OBJECTIVES

After Chapter 3.4, you will be able to:

- Identify the major developmental features of each trimester
- Describe the three stages of birth
- Recall key concepts and terms used for the birth process, including parturition, prostaglandins, oxytocin, and afterbirth

Human gestation lasts an estimated 280 days, which are divided into three trimesters. As a general rule, the larger the animal, the longer the gestational period and the fewer the offspring per pregnancy. For example, elephants usually have one calf and gestate for 22 months. In contrast, mice have 10 to 12 offspring per litter and gestate for only 20 days. Although you don't need to know every detail of gestation for the MCAT, there are key developmental events in each trimester with which you should be familiar.

FIRST TRIMESTER

The major organs begin to develop during the first few weeks. The heart begins to beat at approximately 22 days, and soon afterward the eyes,

gonads, limbs, and liver start to form. By five weeks the embryo is 10 mm in length, and by week six it has grown to 15 mm. The cartilaginous skeleton begins to harden into bone by the seventh week. By the end of eight weeks most of the organs have formed, the brain is fairly developed, and the embryo becomes known as a **fetus**. At the end of the third month the fetus is about 9 cm long.

SECOND TRIMESTER

During the second trimester the fetus undergoes a tremendous amount of growth. It begins to move within the amniotic fluid, its face takes on a human appearance, and its toes and fingers elongate. By the end of the sixth month the fetus measures 30 to 36 cm long.

THIRD TRIMESTER

The seventh and eighth months are characterized by continued rapid growth and further brain development. Antibodies are transported by highly selective active transport from the mother to the fetus for protection against foreign agents in preparation for life outside the womb; this transfer begins earlier in pregnancy, but is highest in the ninth month just before birth. The growth rate slows and the fetus becomes less active, as it has less room to move about.



REAL WORLD

Advances in medicine have allowed premature babies born as early as 24 weeks to survive—far short of the normal 40 weeks. While these

neonates may survive, there are often severe complications because fetal development is not complete at 24 weeks. These problems are most apparent in the respiratory, gastrointestinal, and nervous systems.

BIRTH

Vaginal childbirth, or **parturition**, is accomplished by rhythmic contractions of uterine smooth muscle, coordinated by **prostaglandins** and the peptide hormone **oxytocin**. Birth consists of three basic phases. First, the cervix thins out and the amniotic sac ruptures, which is commonly called *water breaking*. Next, strong uterine contractions result in the birth of the fetus. Finally, the placenta and umbilical cord are expelled; these are often referred to as the **afterbirth**.

MCAT CONCEPT CHECK 3.4:

Before you move on, assess your understanding of the material with these questions.

1. What are some of the key developmental features of each trimester?

- First trimester:

- Second trimester:

- Third trimester:

2. What occurs in each of the three phases of birth?

-

-

-



Conclusion

In this chapter, we have seen how a just-fertilized ovum (zygote) becomes an embryo. As organs develop and the body organizes into complex organ systems, that embryo will turn into a newborn baby. Development certainly does not stop there, however—humans nurture their young for years (sometimes decades!) as they undergo physical, cognitive, and sexual development. Embryonic development is extremely important because it lays the foundation for further development to proceed correctly. Most of the time, the process goes exactly as planned; however, this is not always the case. In medical school, you will study the wide spectrum of teratology—the study of birth defects.

Adult structures that arise from embryonic germ layers are of special importance to us because they are commonly tested on the MCAT. For the remainder of embryology—from the first cleavage event to the last uterine contraction—focus on the main terminology and highlights of each stage, in addition to the differences between fetal and adult physiology. Now that we have seen from where the organ systems derive, we will begin our survey of anatomy and physiology. For the next eight chapters (Chapters 4 to 11 of *MCAT Biology Review*), we will explore the cells, tissues, organs, and interactions of each of the major organ systems. Our discussion begins with the nervous system.

Concept Summary

Early Developmental Stages

- **Fertilization** is the joining of a sperm and an ovum.
 - It usually occurs in the **ampulla** of the fallopian tube.
 - The sperm uses acrosomal enzymes to penetrate the corona radiata and zona pellucida.
 - Once it contacts the oocyte's plasma membrane, the sperm establishes the **acrosomal apparatus** and injects its pronucleus.
 - When the first sperm penetrates it causes a release of calcium ions, which prevents additional sperm from fertilizing the egg and increases the metabolic rate of the resulting diploid **zygote**. This is called the **cortical reaction**.
- **Fraternal (dizygotic) twins** result from the fertilization of two eggs by two different sperm. **Identical (monozygotic) twins** result from the splitting of a zygote in two. Monozygotic twins can be classified by the placental structures they share (mono- vs. diamniotic, mono- vs. dichorionic).
- **Cleavage** refers to the early divisions of cells in the embryo. These mitotic divisions result in a larger number of smaller cells, as the overall volume does not change.
 - The zygote becomes an embryo after the first cleavage because it is no longer unicellular.

- **Indeterminate cleavage** results in cells that are capable of becoming any cell in the organism, while **determinate cleavage** results in cells that are committed to differentiating into a specific cell type.
- The **morula** is a solid mass of cells seen in early development.
- The **blastula (blastocyst)** has a fluid-filled center called a **blastocoel** and has two different structures: the **trophoblast** (which becomes placental structures) and the **inner cell mass** (which becomes the developing organism).
 - The blastula implants in the endometrial lining and forms the **placenta**.
 - The **chorion** contains **chorionic villi**, which penetrate the endometrium and create the interface between maternal and fetal blood.
 - Before the placenta is established, the embryo is supported by the **yolk sac**.
 - The **allantois** is involved in early fluid exchange between the embryo and the yolk sac.
 - The **amnion** lies just inside the chorion and produces amniotic fluid.
 - The developing organism is connected to the placenta via the **umbilical cord**.
- During **gastrulation**, the **archenteron** is formed with a **blastopore** at the end. As the archenteron grows through the blastocoel it contacts the opposite side, establishing three primary germ layers.
 - The **ectoderm** becomes epidermis, hair, nails, and the epithelia of the nose, mouth, and anal canal, as well as the nervous system (including adrenal medulla) and lens of the eye.
 - The **mesoderm** becomes much of the musculoskeletal, circulatory, and excretory systems. Mesoderm also gives rise to the gonads and the

- muscular and connective tissue layers of the digestive and respiratory systems, as well as the adrenal cortex.
- The **endoderm** becomes much of the epithelial linings of the respiratory and digestive tracts and parts of the pancreas, thyroid, bladder, and distal urinary tracts.
 - **Neurulation**, or development of the nervous system, begins after the formation of the three germ layers.
 - The **notochord** induces a group of overlying ectodermal cells to form **neural folds** surrounding a **neural groove**.
 - The neural folds fuse to form the **neural tube**, which becomes the central nervous system.
 - The tip of each neural fold contains **neural crest cells**, which become the peripheral nervous system (sensory ganglia, autonomic ganglia, adrenal medulla, and Schwann cells), as well as specific cell types in other tissues (calcitonin-producing cells of the thyroid, melanocytes in the skin, and others).
 - Teratogens are substances that interfere with development, causing defects or even death of the developing embryo. Teratogens include alcohol, certain prescription drugs, viruses, bacteria, and environmental chemicals.
 - Maternal conditions can affect development, including diabetes (increased fetal size and hypoglycemia after birth) and folic acid deficiency (neural tube defects).

Mechanisms of Development

- Cell specialization occurs as a result of determination and differentiation.
 - **Determination** is the commitment to a specific cell lineage, which may be accomplished by uneven segregation of cellular material during mitosis or with **morphogens**, which promote development down a specific cell line. To respond to a specific morphogen, a cell must have **competency**.
 - **Differentiation** refers to the changes a cell undergoes due to **selective transcription** to take on characteristics appropriate to its cell line.

- Stem cells are cells that are capable of developing into various cell types. They can be classified by potency.
 - **Totipotent cells** are able to differentiate into all cell types, including the three germ layers and placental structures.
 - **Pluripotent cells** are able to differentiate into all three of the germ layers and their derivatives.
 - **Multipotent cells** are able to differentiate only into a specific subset of cell types.

- Cells communicate through a number of different signaling methods. An **inducer** releases factors to promote the differentiation of a competent **responder**.
 - **Autocrine** signals act on the same cell that released the signal.
 - **Paracrine** signals act on local cells.
 - **Juxtacrine** signals act through direct stimulation of adjacent cells.
 - **Endocrine** signals act on distant tissues after traveling through the bloodstream.
 - These are often **growth factors**, which are peptides that promote

differentiation and mitosis in certain tissues.

- If two tissues both induce further differentiation in each other, this is **reciprocal induction**.
- Signaling often occurs via gradients.
- Cells may need to migrate to arrive at their correct location.
- **Apoptosis** is programmed cell death via the formation of **apoptotic blebs** that can subsequently be absorbed and digested by other cells. Apoptosis can be used for sculpting certain anatomical structures, such as removing the webbing between digits.
- **Regenerative capacity** is the ability of an organism to regrow certain parts of the body. The liver has high regenerative capacity, while the heart has low regenerative capacity.
- **Senescence** is the result of multiple molecular and metabolic processes, most notably, the shortening of telomeres during cell division.

Fetal Circulation

- Nutrient, gas, and waste exchange occurs at the placenta.
- Oxygen and carbon dioxide are passively exchanged due to concentration gradients.
- **Fetal hemoglobin (HbF)** has a higher affinity for oxygen than adult hemoglobin (primarily HbA); this affinity assists in the transfer (and retention) of oxygen into the fetal circulatory system.
- The placental barrier also serves as immune protection against many pathogens, and antibodies are transferred from mother to child.
- The placenta serves endocrine functions, secreting estrogen, progesterone, and human chorionic gonadotropin (hCG).
- The **umbilical arteries** carry deoxygenated blood from the fetus to the

placenta; the **umbilical vein** carries oxygenated blood from the placenta back to the fetus.

- The fetal circulatory system differs from its adult version by having three shunts:
 - The **foramen ovale** connects the right atrium to the left atrium, bypassing the lungs.
 - The **ductus arteriosus** connects the pulmonary artery to the aorta, bypassing the lungs.
 - The **ductus venosus** connects the umbilical vein to the inferior vena cava, bypassing the liver.

Gestation and Birth

- In the first trimester, organogenesis occurs (development of heart, eyes, gonads, limbs, liver, brain).
- In the second trimester, tremendous growth occurs, movement begins, the face becomes distinctly human, and the digits elongate.
- In the third trimester, rapid growth and brain development continue, and there is transfer of antibodies to the fetus.
- During birth the cervix thins out and the amniotic sac ruptures. Then, uterine contractions, coordinated by prostaglandins and oxytocin, result in birth of the fetus. Finally, the placenta and umbilical cord are expelled.

Answers to Concept Checks

3.1

- .. Determinate cleavage refers to cell division that results in cells having definitive lineages; that is, at least one daughter cell is programmed to differentiate into a particular cell type. Indeterminate cleavage refers to cell division that results in cells that can differentiate into any cell type (or a whole organism).
- 1. Zygote → 2-, 4-, 8-, and 16-cell embryo → morula → blastula (blastocyst) → gastrula
- 3. Implantation occurs during the blastula (blastocyst) stage.

4.

Germ Layer	Organs
Ectoderm	Integument (including the epidermis, hair, nails, and epithelia of the nose, mouth, and anal canal), lens of the eye, nervous system (including adrenal medulla), inner ear
Mesoderm	Musculoskeletal system, circulatory system, excretory system, gonads, muscular and connective tissue layers of the digestive and respiratory systems, adrenal cortex
Endoderm	Epithelial linings of digestive and respiratory tracts, and parts of the liver, pancreas, thyroid, bladder, and distal urinary and reproductive tracts

- i. Induction is the process by which nearby cells influence the differentiation of adjacent cells. This ensures proper spatial location and orientation of

cells that share a function or have complementary functions.

- i. Neural crest cells become the peripheral nervous system (including the sensory ganglia, autonomic ganglia, adrenal medulla, and Schwann cells) as well as specific cell types in other tissues (such as calcitonin-producing cells of the thyroid, melanocytes in the skin, and others).

3.2

- .. Determination is the commitment of a cell to a particular lineage. Differentiation refers to the actual changes that occur in order for the cell to assume the structure and function of the determined cell type.

!.

Type of Potency	Cell Lineages
Totipotency	Any cell type in the developing embryo (primary germ layers) or in extraembryonic tissues (amnion, chorion, placenta)
Pluripotency	Any cell type in the developing embryo (primary germ layers)
Multipotency	Any cell type within a particular lineage (for example, hematopoietic stem cells)

- i. Autocrine (the signal acts on the same cell that secreted it), paracrine (the signal acts on local cells), juxtacrine (a cell triggers adjacent cells through direct receptor stimulation), endocrine (the signal travels via the bloodstream to act on cells at distant sites)
- i. Apoptosis is programmed cell death and results in contained blebs of the dead cell that can be picked up and digested by other cells. Necrosis is cell death due to injury and results in spilling of cytoplasmic contents.

3.3

- .. The umbilical arteries carry deoxygenated blood. The umbilical vein carries oxygenated blood.

Shunt	Connected Vessels or Chambers	Organ Bypassed
Foramen ovale	Right atrium to left atrium	Lungs
Ductus arteriosus	Pulmonary artery to aorta	Lungs
Ductus venosus	Umbilical vein to inferior vena cava	Liver

3.4

- .. In the first trimester, organogenesis occurs (development of heart, eyes, gonads, limbs, liver, brain). In the second trimester, tremendous growth occurs, movement begins, the face becomes distinctly human, and the digits elongate. In the third trimester, rapid growth and brain development continue, and there is transfer of antibodies to the fetus.
- !. In the first phase of birth, the cervix thins out and the amniotic sac ruptures. In the second phase, uterine contractions, coordinated by prostaglandins and oxytocin, result in birth of the fetus. In the third phase, the placenta and umbilical cord are expelled.



Shared Concepts

Behavioral Sciences Chapter 1

Biology and Behavior

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biochemistry Chapter 6

DNA and Biotechnology

Biology Chapter 1

The Cell

Biology Chapter 2

Reproduction

Biology Chapter 5

The Endocrine System

Discrete Practice Questions

1. Which of the following developmental stages has the greatest nuclear-to-cytoplasmic ratio?
 - (A) Eight-cell embryo
 - (B) Morula
 - (C) Blastula
 - (D) Zygote
2. Which of the following associations of a primary germ layer and an adult organ is correct?
 - (A) Endoderm—cardiac muscle
 - (B) Endoderm—lens of the eye
 - (C) Ectoderm—fingernails
 - (D) Mesoderm—lining of digestive tract
3. From which of the following layers does the notochord form?
 - (A) Ectoderm
 - (B) Mesoderm
 - (C) Endoderm
 - (D) Archenteron

4. The influence of a specific group of cells on the differentiation of another group of cells is called:
- (A) competence.
 - (B) senescence.
 - (C) determination.
 - (D) induction.
5. Which of the following is likely to be found in maternal blood during pregnancy?
- (A) Immunoglobulins produced by the fetus
 - (B) Fetal hemoglobin released from fetal red blood cells
 - (C) Progesterone produced by placental cells
 - (D) Carbon dioxide exhaled from fetal lungs
6. A cell releases a substance that diffuses through the environment, resulting in differentiation of a nearby cell. This is an example of what type of cell–cell communication?
- (A) Autocrine
 - (B) Juxtacrine
 - (C) Paracrine
 - (D) Endocrine
7. A cancer cell is removed from a patient and cultured. The cells in this culture seem to be able to divide indefinitely with no cellular

senescence. Which protein is likely activated in these cells that accounts for this characteristic?

- (A) Epidermal growth factor
- (B) Sonic hedgehog
- (C) Transforming growth factor beta
- (D) Telomerase

8. A tumor is removed from a patient and when investigated by pathologists, appears to contain tissue resembling placenta, hair, thyroid tissue, and cardiac muscle. What was the likely potency of the cells from which this tumor originated?

- (A) Nonpotent
- (B) Multipotent
- (C) Pluripotent
- (D) Totipotent

9. Which of the following is FALSE with regard to adult stem cells?

- (A) They retain inherent pluripotency if harvested from selected organs.
- (B) They are less controversial than embryonic stem cells.
- (C) They require treatment with various transcription factors.
- (D) There is a reduced risk of rejection if the patient's own stem cells are used.

10. A child is born with an imperforate anus, in which the anal canal fails to form correctly and the rectum is not connected to the outside world. This pathology is most likely accounted for by a failure of:
- (A) cell differentiation.
 - (B) cell determination.
 - (C) apoptosis.
 - (D) neurulation.
11. Following a myocardial infarction, the heart often heals by the creation of a scar by fibroblasts. This is an example of:
- (A) complete regeneration.
 - (B) incomplete regeneration.
 - (C) competency.
 - (D) multipotency.
12. Neurofibromatosis type I, or von Recklinghausen's disease, is a disorder that causes formation of tumors in multiple nervous system structures as well as the skin. While all cells carry the same mutation on chromosome 17, selective transcription of the genome appears to cause the most significant tumorigenesis in which of the following primary germ tissue layers?
- (A) Ectoderm
 - (B) Mesoderm
 - (C) Endoderm
 - (D) Notochord

13. Which of the following shows the correct order of early developmental milestones during embryogenesis?

- (A) Blastula → gastrula → morula
- (B) Morula → gastrula → blastula
- (C) Morula → blastula → gastrula
- (D) Gastrula → blastula → morula

14. A pregnant woman is accidentally given a single dose of a teratogenic drug late in the third trimester. The baby is born three days later. Which of the following is the most likely outcome?

- (A) Complete failure of organ development and death of the fetus
- (B) Partial failure of organ development with survival of the fetus
- (C) Serious disfigurement of the fetus
- (D) Respiratory distress at birth, but no long-term effects

15. Which of the following statements regarding fetal circulation is FALSE?

- (A) In the umbilical cord, there are more arteries than veins.
- (B) The foramen ovale is the only shunt that connects two chambers of the heart.
- (C) Blood flow in the ductus arteriosus is from the aorta to the pulmonary artery.
- (D) The ductus venosus is the only shunt that bypasses the liver.

Discrete Practice Answers

1. **C**

The question is asking us to determine the developmental stage with the greatest nuclear-to-cytoplasmic material ratio. During the series of rapid mitotic divisions known as cleavage, the number of cells increases dramatically but the overall volume of the embryo does not change. In fact, the volume of the embryo does not significantly increase until after the blastula has already implanted. Therefore, a high ratio of nuclear to cytoplasmic material will be found at the stage with the greatest number of cells during early development. From the given choices, the stage with the greatest number of cells is the blastula. This is further supported by the fact that the cells are displaced from the center by the blastocoel.

2. **C**

To answer this question, it could be useful to review quickly the embryonic layers. The ectoderm gives rise to the integument (the epidermis, hair, nails, and the epithelia of the nose, mouth, and anal canal), the lens of the eye, and the nervous system (including the adrenal medulla). The endoderm gives rise to the epithelial linings of the digestive and respiratory tracts and parts of the liver, pancreas, thyroid, and bladder. Finally, the mesoderm gives rise to the musculoskeletal system, the circulatory system, the excretory system, the gonads, and the adrenal cortex. Therefore, the only correct association can be found in **(C)** because the fingernails are derived from ectoderm.

3. **B**

A rod of mesodermal cells called the notochord develops along the longitudinal axis just under the dorsal layer of ectoderm. Through inductive effects from the notochord, the overlying ectoderm starts bending inward and forms a groove on the dorsal surface of the embryo. The dorsal ectoderm will eventually pinch off and develop into the spinal cord and brain. While the neural tube forms from ectoderm, the notochord itself is mesodermal.

4. **D**

The influence of a specific group of cells on the differentiation of another group of cells is termed induction. For example, the eyes are formed through reciprocal induction between the brain and the ectoderm. Competence refers to the ability of a cell to respond to a given inducer, but not the influence of the group of organizing cells, eliminating **(A)**. Senescence is a term for biological aging, eliminating **(B)**. Determination may be the result of induction, but this term does not refer to the general concept of the effect of one group of cells on the differentiation of another group of cells, eliminating **(C)**.

5. **C**

During pregnancy, the placenta produces estrogen and progesterone to maintain the endometrium. These hormones are necessary for proper gestation of the fetus and should be measurable in maternal blood because they act on maternal organs. Prior to birth, the fetus is immunologically naïve and does not yet produce immunoglobulins, eliminating **(A)**. It is worth noting, though, that maternal immunoglobulins cross the placenta to enter fetal blood. Fetal hemoglobin is a large protein and, thus, cannot easily cross the placenta. Further, red blood cells are much too large to cross the barrier themselves, eliminating **(B)**. Carbon

dioxide from fetal metabolism can be found in maternal blood, but the fetal lungs are nonfunctional prior to birth as the fetus is suspended in amniotic fluid. Carbon dioxide is transferred across the placenta directly from the fetal bloodstream, eliminating **(D)**.

6. **C**

The question stem states that a cell releases a substance that diffuses through the environment and causes differentiation of a nearby cell. Because the cell is acting on a nearby cell and the molecule spreads by diffusion, this is an example of paracrine signaling. Autocrine signaling, **(A)**, occurs when a molecule secreted by a cell acts on the same cell. Juxtacrine signaling, **(B)**, occurs between adjacent cells, but the signal does not spread by diffusion. In endocrine signaling, **(D)**, a molecule is secreted that travels via the bloodstream to a distant target.

7. **D**

Cells that are able to divide indefinitely with no senescence are not exhibiting normal cell behavior. Normally, somatic cells divide a limited number of times until the telomeres become too short to be effective protectors of genomic material. When this occurs, the cells stop dividing. However, in this case, the cells have continued to divide indefinitely. It is likely that the enzyme telomerase has been activated, which allows for synthesis of telomeres to counteract shortening during replication.

8. **D**

This question stem describes a tumor composed of multiple types of embryonic tissues as well as tissues derived from all three germ cell layers. Because this tumor contains placental tissue, the tumor must have originated from a cell that was able to produce those types of tissues. Cells

capable of producing placental tissue as well as ectoderm, mesoderm, and endoderm are, by definition, totipotent.

9. **A**

Embryonic stem cells are controversial because they require destruction of an embryo to harvest, eliminating **(B)**. Adult stem cells are significantly less controversial, but require treatment with various transcription factors in order to increase the level of potency, eliminating **(C)**. Rejection is a concern when foreign cells are introduced into an individual; using one's own stem cells should remove this risk, eliminating **(D)**. Adult stem cells are not naturally pluripotent, unless pluripotency has been induced by strategic use of transcription factors. Therefore, **(A)** is the correct answer.

10. **C**

During development, programmed cell death occurs in multiple locations in order to ensure development of the correct adult structures. One of the places in which this occurs is between fingers and toes; another is the digestive tract, where a central lumen is formed. If apoptosis does not occur correctly in the digestive tract, an imperforate anus could result. Failure of determination or differentiation would likely result in the absence of anorectal structures altogether, eliminating **(A)** and **(B)**. Failure of neurulation would lead to the absence of a nervous system and would not be compatible with life, eliminating **(D)**.

11. **B**

After an injury, healing occurs by some sort of regenerative process. In humans, some tissues, such as the liver, are capable of regenerating tissue with much the same function and structure as the original tissue. However, the heart is not capable of this sort of regeneration, often forming a fibrous scar in an area of injury. This is an example of incomplete regeneration, in

which newly formed tissues are not identical in structure or function to the tissues that have been injured or lost.

2. **A**

Here, the mutation affects the skin and the nervous system, both of which are derived from ectoderm. The other germ tissue layers do not lead to skin or nervous system formation, eliminating **(B)** and **(C)**. The notochord is not actually a primary germ tissue layer, and thus cannot be an answer to the question, eliminating **(D)**.

3. **C**

After the first cell divisions occur, the embryo consists of a solid ball of cells known as a morula. Then a hollow center forms, creating the blastula. Finally, as the cells begin to differentiate into the three germ layers, the embryo is considered a gastrula. **(C)** is therefore the correct answer.

4. **D**

The question stem states that the pregnant woman was given the drug three days before the baby was born. It is important to remember that organogenesis occurs during the first trimester. The last structure to become fully functional is the lungs. Because the organs were already largely formed prior to the administration of the teratogenic drug, it is likely that there was no major effect on the development of most organs as a result of exposure to the teratogen, eliminating **(A)**, **(B)**, and **(C)**. However, because lung tissues are so sensitive and because they mature so late, it is likely that the infant may have some respiratory distress at birth.

5. **C**

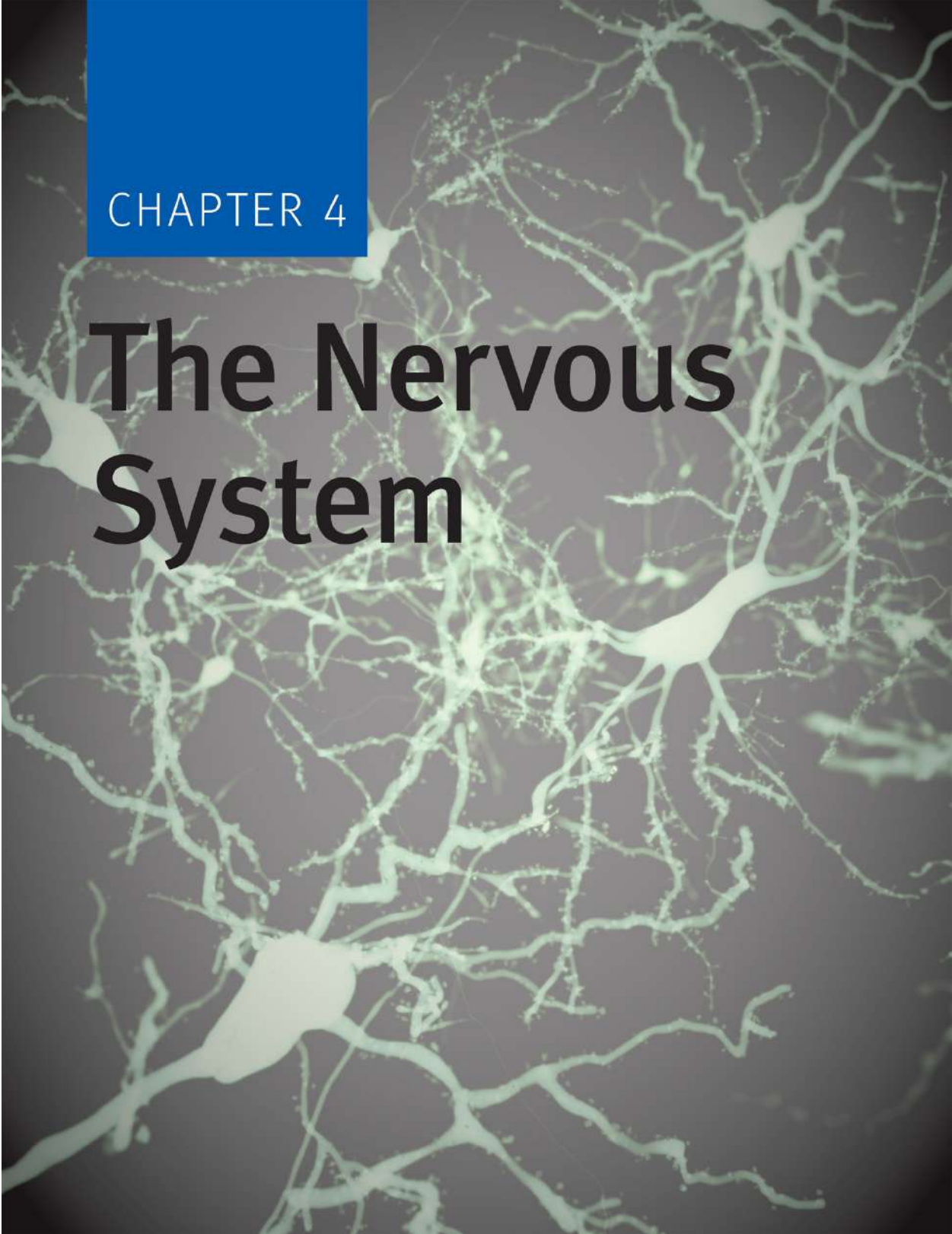
Blood flow in the ductus arteriosus is from the pulmonary artery to the aorta. The direction of flow is determined by the pressure differential

between the right side of the heart (and pulmonary circulation) and the left side of the heart (and systemic circulation). Unlike in adults, the right side of the heart is at a higher pressure during prenatal life than the left side, so blood will shunt from the pulmonary circulation to the systemic circulation through both the foramen ovale and ductus arteriosus.

GO ONLINE

Consult your online resources for additional practice.



A microscopic image of neurons, showing cell bodies and branching processes, serving as a background for the chapter title. The neurons are stained in a light greenish-yellow color against a dark background.

CHAPTER 4

The Nervous System

CHAPTER 4

The Nervous System

In This Chapter

4.1 Cells of the Nervous System

Neurons

Other Cells in the Nervous System



4.2 Transmission of Neural Impulses

The Action Potential

The Synapse



4.3 Organization of the Human Nervous System

Central and Peripheral Nervous Systems

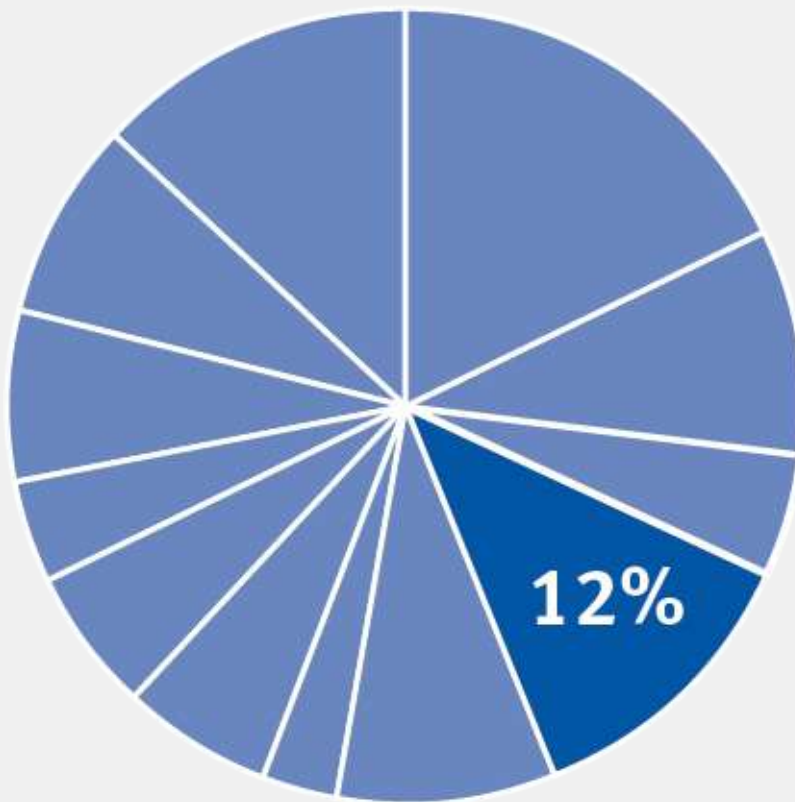
The Autonomic Nervous System

Reflexes

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 13% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content categories:

2A: Assemblies of molecules, cells, and groups of cells within single cellular and multicellular organisms

3A: Structure and functions of the nervous and endocrine systems and ways in which these systems coordinate the organ systems

4C: Electrochemistry and electrical circuits and their elements

6A: Sensing the environment

Introduction

For generations, the indigenous peoples of South America used blow darts laced with a paralytic plant extract to hunt their prey. In the 1800s, English physicians who interacted with these indigenous South Americans recognized the possible uses of this paralytic agent, now known as *tubocurarine*, as an anesthetic agent for surgeries. Physicians noticed that animals under the influence of *tubocurarine* would become temporarily immobilized, but would recover after a period of paralysis. According to these physicians, this anesthetic agent would revolutionize surgery. To test the effectiveness of the new drug, one of the physicians volunteered to demonstrate its effectiveness by being tested for pain perception while under the influence of *tubocurarine*. While the drug was an effective paralyzing agent, it did not have any effect on the sensory receptors of the body—he felt every test without being able to move or express his discomfort.

Organisms sense pain, temperature, and all aspects of their environment through the nervous system, which also coordinates this sensory information and responds to stimuli. Specifically, the nervous system is responsible for the control of muscular movement, neuromuscular reflexes, and glandular

secretions (such as salivation and lacrimation). In addition, the nervous system is responsible for higher-level thinking and mental function.

Despite all of its complex functions, the nervous system operates through basic electrical and chemical signals. Biomedical scientists have discovered much about the nervous system: its anatomical and functional divisions, the nature of the action potential, and its histological features under the microscope. However, there is so much more that we do not know. It is an inspirational challenge for future physicians to realize that the brain continues to be a vast frontier for human exploration and discovery.

4.1 Cells of the Nervous System High-Yield

LEARNING OBJECTIVES

After Chapter 4.1, you will be able to:

- Recall the different terms used for myelin-producing cells in the peripheral and the central nervous systems
- Identify the functions of the five main categories of glial cells
- Describe the purpose of each major structure of the neuron

Neurons are specialized cells capable of transmitting electrical impulses and then translating those electrical impulses into chemical signals. In this section, we will consider the structure of the neuron as well as how neurons communicate with other parts of the nervous system.

NEURONS

Each neuron has a shape that matches its function, as dictated by the other cells with which that neuron interacts. There are a variety of different types of neurons in the body, but they all share some specific features.

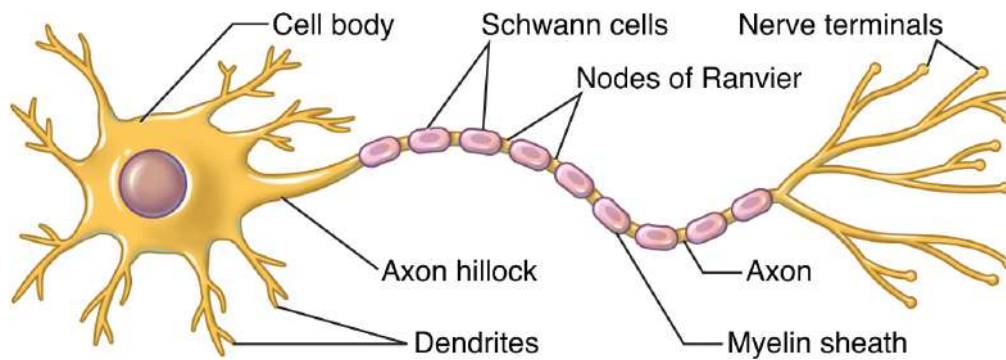


Figure 4.1. Structure of a Neuron

The anatomy of a neuron is shown in Figure 4.1. Like all other cells (besides mature red blood cells), neurons have nuclei. The nucleus is located in the **cell body**, also called the **soma**. The soma is also the location of the endoplasmic reticulum and ribosomes. The cell has many appendages emanating directly from the soma called **dendrites**, which receive incoming messages from other cells. The information received from the dendrites is transmitted through the cell body before it reaches the **axon hillock**, which integrates the incoming signals. The axon hillock plays an important role in **action potentials**, or the transmission of electrical impulses down the axon. Signals arriving from the dendrites can be either excitatory or inhibitory; the axon hillock sums up these signals, and if the result is excitatory enough (reaching threshold, as discussed later in this chapter), it will initiate an action potential. The **axon** is a long appendage that terminates in close proximity to a target structure (a muscle, a gland, or another neuron). Most mammalian nerve fibers are insulated by **myelin**, a fatty membrane, to prevent signal loss or crossing of signals. Just like insulation prevents wires next to each other from accidentally discharging each other, the **myelin sheath** maintains the electrical signal within one neuron. In addition, myelin increases the speed of conduction in the axon. Myelin is produced by **oligodendrocytes** in the central nervous system and **Schwann cells** in the peripheral nervous system.

At certain intervals along the axon, there are small breaks in the myelin sheath with exposed areas of axon membrane called **nodes of Ranvier**. As will be explored in the discussion of action potentials to follow, nodes of Ranvier are critical for rapid signal conduction. Finally, at the end of the axon is the **nerve terminal** or **synaptic bouton (knob)**. This structure is enlarged and flattened to maximize transmission of the signal to the next neuron and ensure proper release of **neurotransmitters**, the chemicals that transmit information between neurons.



MNEMONIC

Axons carry neural signals **a**way from the soma; dendrites carry signals toward the soma.



REAL WORLD

Sometimes the body mounts an immune response against its own myelin, leading to the destruction of this insulating substance (demyelination). Because myelin speeds the conduction of impulses along a neuron, the absence of myelin slows down information transfer. A common demyelinating disorder is multiple sclerosis (MS). In MS, the myelin of the brain and spinal cord is selectively targeted. Because so many different kinds of neurons are demyelinated, MS patients experience a wide variety of symptoms including weakness, lack of balance, vision problems, and incontinence.

Neurons are not physically connected to each other. Between the neurons, there is a small space into which the terminal portion of the axon releases neurotransmitters, which bind to the dendrites of the adjacent neuron (the postsynaptic neuron). This space is known as the **synaptic cleft**; together, the nerve terminal, synaptic cleft, and postsynaptic membrane are known as a **synapse**. Neurotransmitters released from the axon terminal traverse the synaptic cleft and bind to receptors on the postsynaptic neuron.

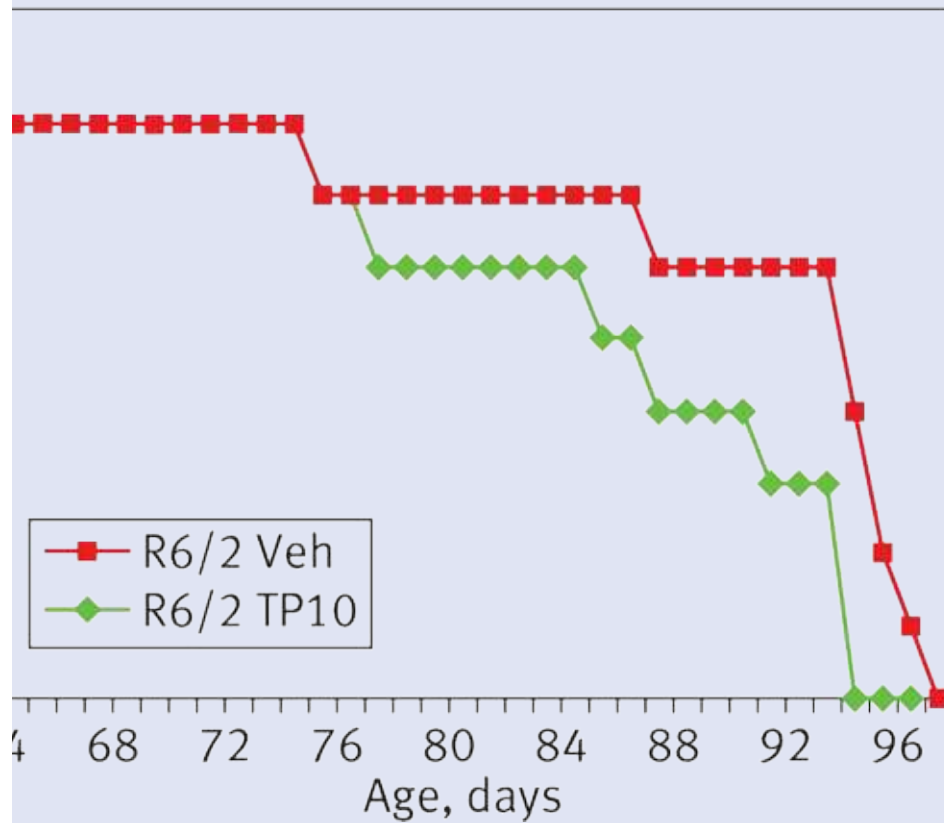
Multiple neurons may be bundled together to form a **nerve** in the peripheral nervous system. These nerves may be **sensory**, **motor**, or **mixed**, which refers to the type(s) of information they carry; mixed nerves carry both sensory and motor information. The cell bodies of neurons of the same type are clustered together into ganglia.

In the central nervous system, axons may be bundled together to form **tracts**. Unlike nerves, tracts only carry one type of information. The cell bodies of neurons in the same tract are grouped into **nuclei**.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING	
ase (HD) is a devastating neurodegenerative condition caused by expansion of on 1 of <i>IT15</i> , which encodes for the protein huntingtin.	<i>Background: Huntington's disease</i>
tin is widely expressed, HD is associated with the neurodegeneration of the piny neurons. This particular vulnerability is hypothesized to result from sregulation within the cAMP and CREB signaling cascades in these neurons.	<i>Hypothesis: dysregulation of transcription within cAMP and CREB leads to SMS neuron degeneration</i>
treatment would be to target phosphodiesterases (PDE) that inactivate cAMP	<i>Inactivate cAMP and</i>

es.	<i>CREB</i> <i>phosphodiesterase =</i> <i>they want to treat by</i> <i>increasing these</i> <i>cascades</i>
hesis, and the potential therapeutic approach, researchers investigated ration of TP-10, a highly specific phosphodiesterase inhibitor would alleviate icits	<i>IV: TP-10 (inhibitor),</i> <i>DV: neurological</i> <i>deficits reduced</i>
d HD model system, the R6/2 mouse.	<i>Model system for HD</i> <i>= these mice have</i> <i>Huntington's-like</i> <i>symptoms</i>
oss of body weight, and increased instances of clasping behavior in the mice n the TP-10 intervention group and the vehicle control group. Righting reflex is g the mice on their side and monitoring their ability to get back to the upright g is a behavior correlated to neurodegeneration.	<i>Behavior on these</i> <i>tests must correlate</i> <i>to symptom relief</i> <i>from HD.</i>
	<i>IV: age</i> <i>DV: percent retaining</i> <i>reflex</i> <i>Trend: treatment</i> <i>group (TP-10) loses</i> <i>reflex at a later age</i>

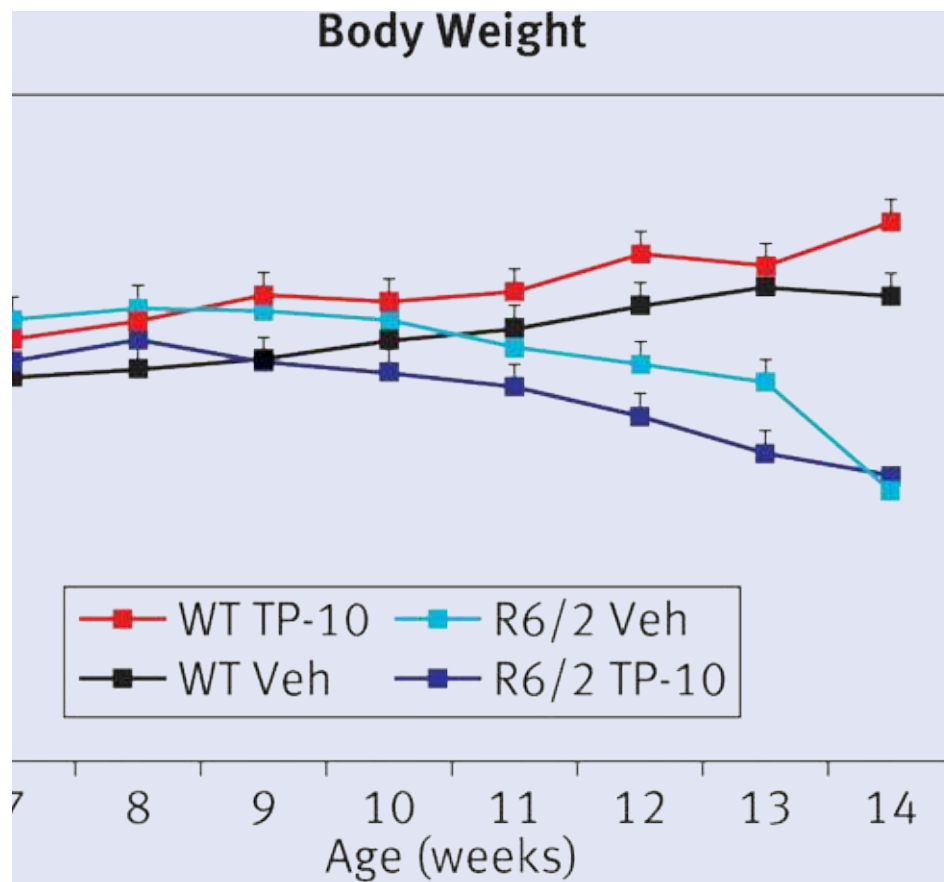
Loss of Righting Reflex



IV: age

DV: weight

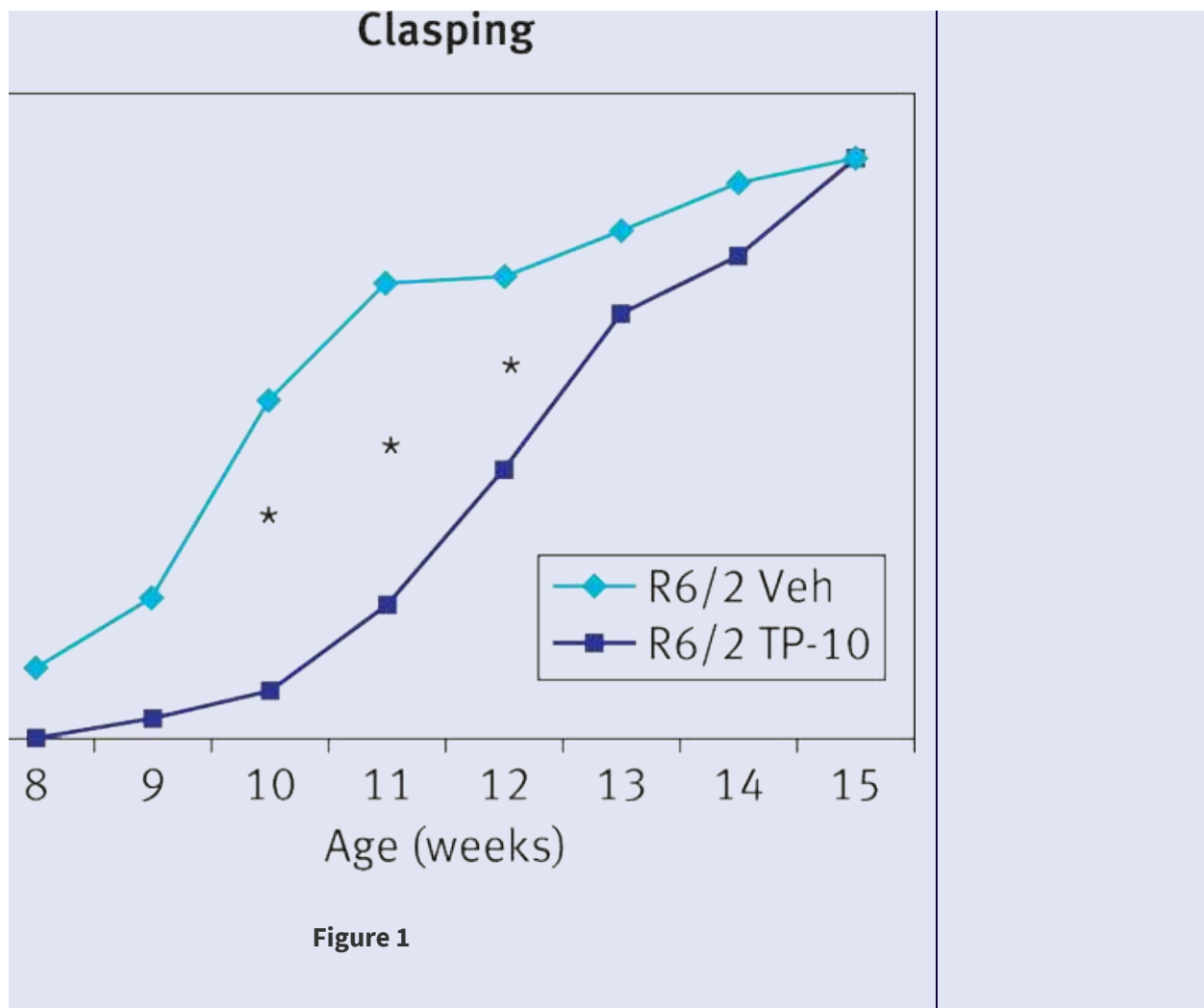
Trend: wild-type (WT) gain weight toward the end of the experiment; R6/2 lose weight



IV: age

DV: time

Trend: vehicle group
develops “clasping”
before treatment
(TP-10) group



, C., Laurenti, D., Anzilotti, S., Bernardi, G., Menniti, F. S., & Fusco, F. R. (2010). Inhibition of the striatal specific 10a ameliorates striatal and cortical pathology in r6/2 mouse model of Huntingtons disease. *PLoS One*, 5(10). ne.0013417.

Does TP-10 treatment alleviate neurological deficits associated with Huntington's disease?

This question asks about the results of the associated study, so we are going to have to use the information in the article and the results in the figures in order to answer. The article says that TP-10 is a possible treatment for Huntington's disease (HD), and the second paragraph describes the dependent variables

used to measure success in this experiment. There appears to be a figure associated with each dependent variable, so we will need to evaluate the results in each figure to reach a conclusion.

Based on the label and axes of the graph, Figure 1A shows the age at which mice lose their righting reflex. Specifically, we can see that R6/2 TP-10 (red) retains their righting reflex longer than R6/2 Vehicle (black). According to the passage, the loss of the righting reflex indicates neurodegeneration; therefore, TP-10 appears to be alleviating this particular symptom. However, Figure 1B shows that mice who had the HD genetic condition lost weight in both treatment and vehicle conditions, meaning TP-10 doesn't appear to have helped with weight maintenance. Finally, Figure 1C shows that clasping occurs more readily in R6/2 Vehicle than R6/2 TP-10. The term clasping alone doesn't imply positive or negative effects on the brain, but the article tells us that clasping is a sign of neurodegeneration. Thus, the data showing that R6/2 TP-10-treated mice have later onset of clasping demonstrates that neurological deficits are being alleviated.

Overall, Figures 1A and 1C indicate that TP-10 may have potential to treat neurological deficits associated with Huntington's disease. Figure 1B, however, indicates that this treatment may not be addressing all aspects of the disease.

OTHER CELLS IN THE NERVOUS SYSTEM

Neurons are not the only cells in the nervous system. Neurons must be supported and myelinated by other cells. These cells are often called **glial**

cells, or neuroglia. Glial cells play both structural and supportive roles, as shown in Figure 4.2.

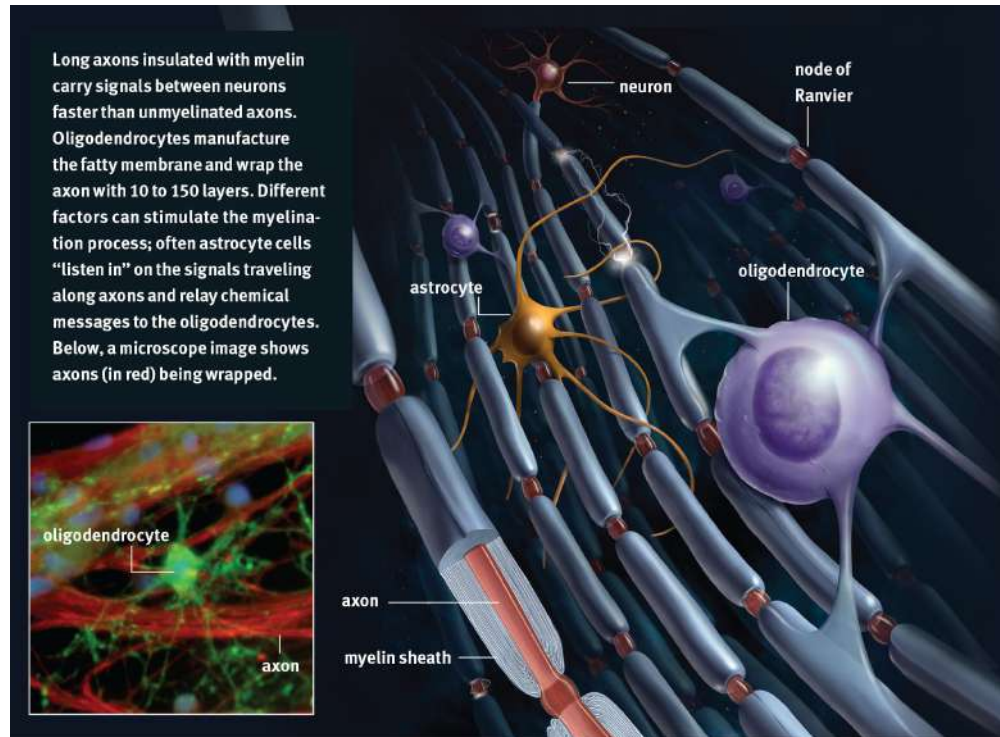


Figure 4.2. Glial Cells: Astrocytes and Oligodendrocytes

A detailed knowledge of these cell types is not necessary for the MCAT, so a familiarity with their basic functions will suffice:

- **Astrocytes** nourish neurons and form the blood–brain barrier, which controls the transmission of solutes from the bloodstream into nervous tissue.
- **Ependymal cells** line the ventricles of the brain and produce cerebrospinal fluid, which physically supports the brain and serves as a shock absorber.
- **Microglia** are phagocytic cells that ingest and break down waste products and pathogens in the central nervous system.

- **Oligodendrocytes** (CNS) and **Schwann cells** (PNS) produce myelin around axons.

MCAT CONCEPT CHECK 4.1:

Before you move on, assess your understanding of the material with these questions.

1. For each of the following neuron structures, provide a brief description of its purpose:

- Axon:

- Axon hillock:

- Dendrite:

- Myelin sheath:

- Soma:

- Synaptic bouton:

2. What is a collection of cell bodies called in the CNS? In the PNS?

- CNS:

-
- PNS:
-

3. Which two types of glial cells, if not properly functioning, will make an individual most susceptible to a CNS infection?

-
4. Guillain-Barré syndrome (GBS) is an autoimmune disease that causes demyelination in the peripheral nervous system. What type of glial cell is being targeted in GBS?
-

4.2 Transmission of Neural Impulses

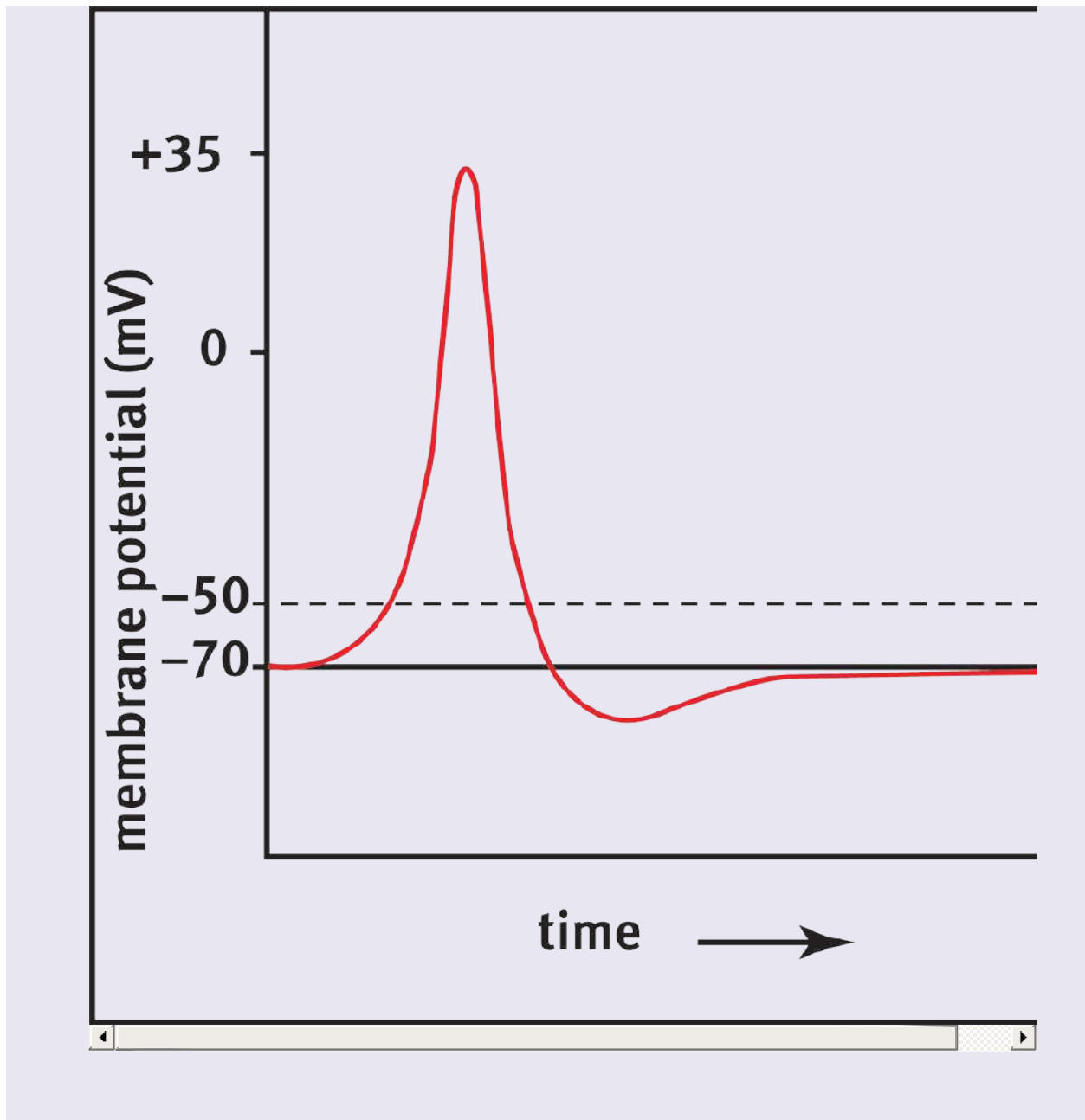
High-Yield



LEARNING OBJECTIVES

After Chapter 4.2, you will be able to:

- Explain the ion channels and regulatory steps involved in the process of initiating, propagating, and terminating an action potential
- Describe the resting membrane potential and how it is maintained
- Differentiate between temporal and spatial summation
- Identify the ion responsible for the fusion of neurotransmitter-containing vesicles at the nerve terminal membrane
- Recall the three main methods to block the action of a neurotransmitter
- Identify the ion channel changes that occur during the shifts in voltage associated with an action potential:



Now that we have discussed the basic anatomy of the neuron, we can turn to the physiology that underlies neuronal signaling.

THE ACTION POTENTIAL

Neurons use all-or-nothing messages called **action potentials** to relay electrical impulses down the axon to the synaptic bouton. As we will explore in the following section, action potentials ultimately cause the release of neurotransmitters into the synaptic cleft.

Resting Potential

A cell's **resting membrane potential** is the net electric potential difference that exists across the cell membrane, created by movement of charged molecules across that membrane. For neurons, this potential is about -70 mV, with the inside of the neuron being negative relative to the outside. The two most important ions involved in generating and maintaining the resting potential are potassium (K^+) and sodium (Na^+).

The potassium concentration inside the cell averages about 140 mM, as compared to 4 mM outside of the cell. This concentration difference makes it favorable for potassium to move to the outside of the cell. To facilitate the outward movement of potassium, the cell membrane has transmembrane **potassium leak channels**, which allow the slow leak of potassium out of the cell. As potassium continually leaks out of the cell, the cell loses a small amount of positive charge, leaving behind a small amount of negative charge and making the outside of the cell slightly positively charged.

However, as negative charge builds up inside the cell, some potassium will be drawn back into the cell due to the attraction between the positive potassium ions and the negative potential building inside the cell. As the potential difference continues to grow, potassium will also be more strongly drawn back into the cell. And at a certain potential, each potassium cation that is pushed out due to the concentration gradient will be matched by another

potassium cation pulled back in due to the electric potential. At this point, there is no more net movement of the ion, as the cell is in equilibrium with respect to potassium. The potential difference that represents this potassium equilibrium is called the **equilibrium potential of potassium**. Potassium's equilibrium potential is around -90 mV. The negative sign is assigned due to convention, and because a positive ion (potassium) is leaving the cell.



REAL WORLD

Even though we may think of these influxes and effluxes as big events, only a very small amount of potassium needs to exit the cell before the resulting electrostatic force equals the force of the concentration gradient. In fact, during an action potential the change to potassium's intracellular concentration is so small that it cannot even be accurately measured using current devices! The action potential is reliant only on local voltage changes at the membrane itself, so this overall lack of change in intracellular ion concentration does not impact transmission. This is the reason why you see membrane potentials reported in units of voltage, which are easily measurable, instead of concentration change, which is almost negligible. Because so little potassium needs to exit, the equilibrium potential with respect to potassium is established almost instantly.

Next, let's consider in isolation the other important ion, sodium. Sodium's concentration gradient is the reverse of potassium's, with a concentration of about 12 mM inside and 145 mM outside of the cell, meaning there is a driving force pushing sodium into the cell. This movement is facilitated by **sodium**

leak channels. The slow leak of sodium into the cell causes a build-up of electric potential. The **equilibrium potential of sodium** is around 60 mV, and is positive because sodium is moving into the cell.

In a living system, sodium and potassium are flowing across the cell's membrane at the same time. Potassium's concentration gradient causes potassium to leak out of the cell through potassium leak channels. At the same time, sodium is moving in the opposite direction, with the opposite effect. In a certain sense, sodium undoes the effect of potassium's movement. The resting potential is thus a tug-of-war: Potassium's movement pulls the cell potential toward -90 mV, while sodium's movement pulls the cell potential the opposite way, toward $+60$ mV. But neither ion ever "wins" the tug of war. Instead, a balance of these two effects is reached at around -70 mV for the average nerve cell, as can be seen in Figure 4.3. This balance, this net effect of sodium and potassium's equilibrium potentials, is the **resting membrane potential**. The resting potential is closer to potassium's equilibrium potential because the cell is slightly more permeable to potassium. Neither ion is ever able to establish its own equilibrium, so both ions continue leaking across the cell membrane.



BRIDGE

The resting membrane potential is dependent on the intra- and extracellular ion concentrations, relative permeability of the membrane to these different ions, and charges of these ions. The Goldman–Hodgkin–Katz voltage equation brings together these different factors into one equation that predicts the resting membrane

potential. This equation is discussed in Chapter 8 of *MCAT Biochemistry Review*.

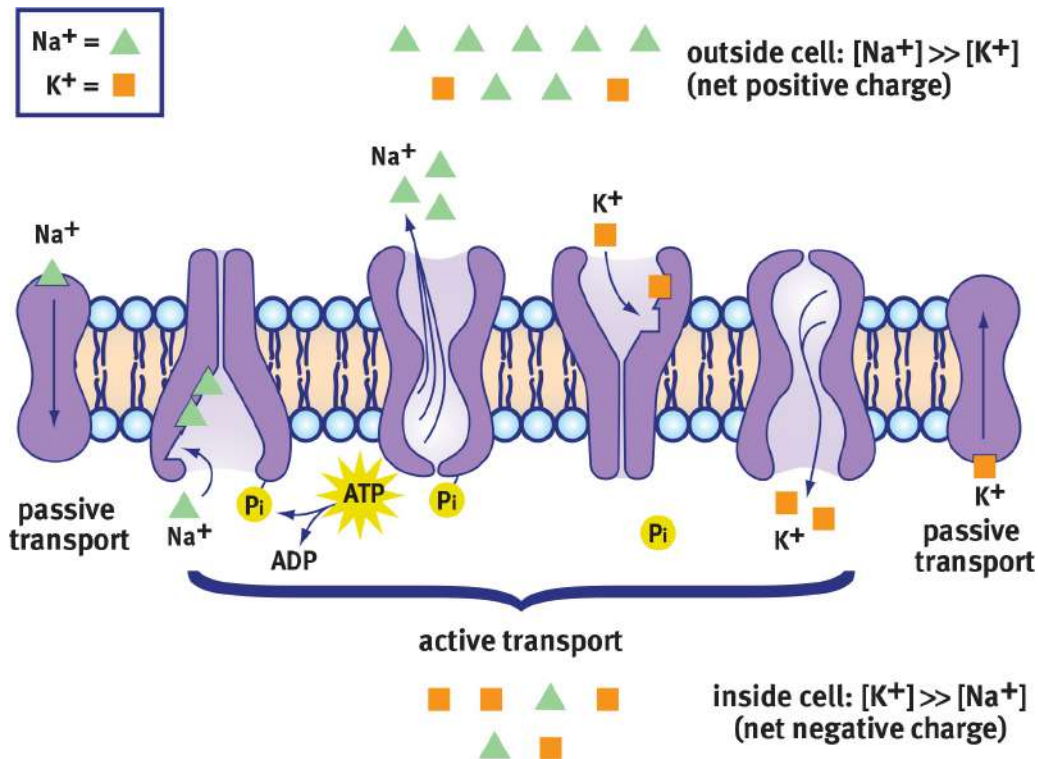


Figure 4.3. Maintenance of Resting Membrane Potential

The action of Na^+/K^+ ATPase, Na^+ leak channels, and K^+ leak channels creates and maintains a resting membrane potential of -70 mV .



MNEMONIC

To remember the direction of ion movement by Na/K ATPase, think *pumpKin*.

Given the continual ion leaking at the membrane, there must be a means of moving both sodium and potassium ions back against their gradients if a resting potential is to be maintained. **Na⁺/K⁺ ATPase** continually pumps sodium and potassium back to where they started: potassium into the cell and sodium out of the cell, to maintain their respective gradients. In fact, in your body more ATP is spent by the Na⁺/K⁺ ATPase to maintain these gradients than for any other single purpose.

The Axon Hillock

Neurons can receive both excitatory and inhibitory input. Excitatory input causes **depolarization** (raising the membrane potential, V_m , from its resting potential) and thus makes the neuron more likely to fire an action potential. Inhibitory input causes **hyperpolarization** (lowering the membrane potential from its resting potential) and thus makes the neuron less likely to fire an action potential. If the axon hillock receives enough excitatory input to be depolarized to the **threshold** value (usually in the range of -55 mV to -40 mV), an action potential will be triggered.

This implies that not every stimulus necessarily generates a response. A small excitatory signal may not be sufficient to bring the axon hillock to threshold. Further, a postsynaptic neuron may receive information from several different presynaptic neurons, some of which are excitatory and some of which are inhibitory. The additive effect of multiple signals is known as **summation**.

There are two types of summation: temporal and spatial. In **temporal summation**, multiple signals are integrated during a relatively short period of time. A number of small excitatory signals firing at nearly the same moment could bring a postsynaptic cell to threshold, enabling an action potential. In

spatial summation, the additive effects are based on the number and location of the incoming signals. A large number of inhibitory signals firing directly on the soma will cause more profound hyperpolarization of the axon hillock than the depolarization caused by a few excitatory signals firing on the dendrites of a neuron.

Ion Channels and Membrane Potential

A graph of membrane potential vs. time during an action potential is shown in Figure 4.4.

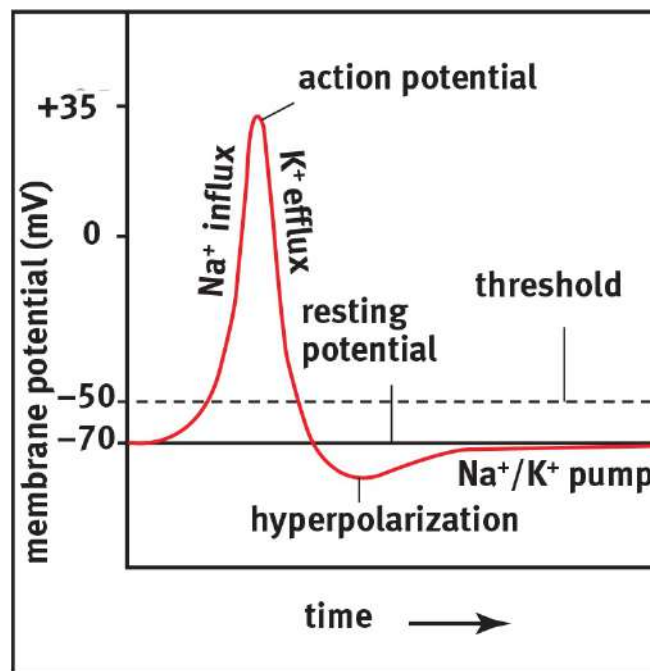


Figure 4.4. Action Potential Generation

Sufficient depolarization across the cell membrane to threshold leads to the generation of an action potential, followed by repolarization and hyperpolarization before returning to the resting membrane potential.

If the cell is brought to threshold, voltage-gated sodium channels open in the membrane. As the name implies, these ion channels open in response to the change in potential of the membrane (depolarization) and permit the passage of sodium ions. There is a strong **electrochemical gradient** that promotes the migration of sodium into the cell. From an electrical standpoint, the interior of the cell is more negative than the exterior of the cell, which favors the movement of positively charged sodium cations into the cell. From a chemical standpoint, there is a higher concentration of sodium outside the cell than inside, which also favors the movement of sodium into the cell. As sodium passes through these ion channels, the membrane potential becomes more positive; that is, the cell rapidly depolarizes. Sodium channels not only open in response to changes in membrane potential, but are also inactivated by them. When V_m approaches +35 mV, the sodium channels are **inactivated** and will have to be brought back near the resting potential to be **deinactivated**. Thus, these sodium channels can exist in three states: **closed** (before the cell reaches threshold, and after inactivation has been reversed), **open** (from threshold to approximately +35 mV), and **inactive** (from approximately +35 mV to the resting potential).



KEY CONCEPT

Na^+ wants to go into the cell because the cell is more negative inside (electrical gradient) and has a lower concentration of Na^+ inside (chemical gradient).

The positive potential inside the cell not only triggers the voltage-gated sodium channels to inactivate, but also triggers the voltage-gated potassium

channels to open. Once sodium has depolarized the cell, there is an electrochemical gradient favoring the efflux of potassium from the neuron. As positively charged potassium cations are driven out of the cell, there will be a restoration of the negative membrane potential called **repolarization**. The efflux of K^+ causes an overshoot of the resting membrane potential, hyperpolarizing the neuron. This hyperpolarization serves an important function: it makes the neuron refractory to further action potentials. There are two types of **refractory periods**. During the **absolute refractory period**, no amount of stimulation can cause another action potential to occur. During the **relative refractory period**, there must be *greater than normal* stimulation to cause an action potential because the membrane is starting from a potential that is more negative than its resting value.

The Na^+/K^+ ATPase acts to restore not only the resting potential, but also the sodium and potassium gradients that have been partially dissipated by the action potential.



KEY CONCEPT

Action potentials rely on both electrical and chemical gradients. The neuron starts at the resting potential, around -70 mV. At the resting potential, potassium is high inside the cell and low outside the cell, while sodium is high outside the cell and low inside the cell. Once the cell reaches threshold, sodium channels open and sodium floods the cell, making it more positive inside (depolarization). Then, sodium channels are inactivated and the potassium channels open. This allows potassium to flow out of the cell, bringing the potential into the negative range (repolarization), and actually overshooting the resting

potential (hyperpolarization). Na^+/K^+ ATPase then works to restore the resting potential.

Impulse Propagation

So far, we have discussed the movements of ions at one small segment of the axon. For a signal to be conveyed to another neuron, the action potential must travel down the axon and initiate neurotransmitter release. This movement is called **impulse propagation** and is shown in Figure 4.5. As sodium rushes into one segment of the axon, it will cause depolarization in the surrounding regions of the axon. This depolarization will bring subsequent segments of the axon to threshold, opening the sodium channels in those segments. Each of these segments then continues through the rest of the action potential in a wave-like fashion until the action potential reaches the nerve terminal. After the action potential has fired in one segment of axon, that segment becomes momentarily refractory, as described previously. The functional consequence of this is that information can only flow in one direction.

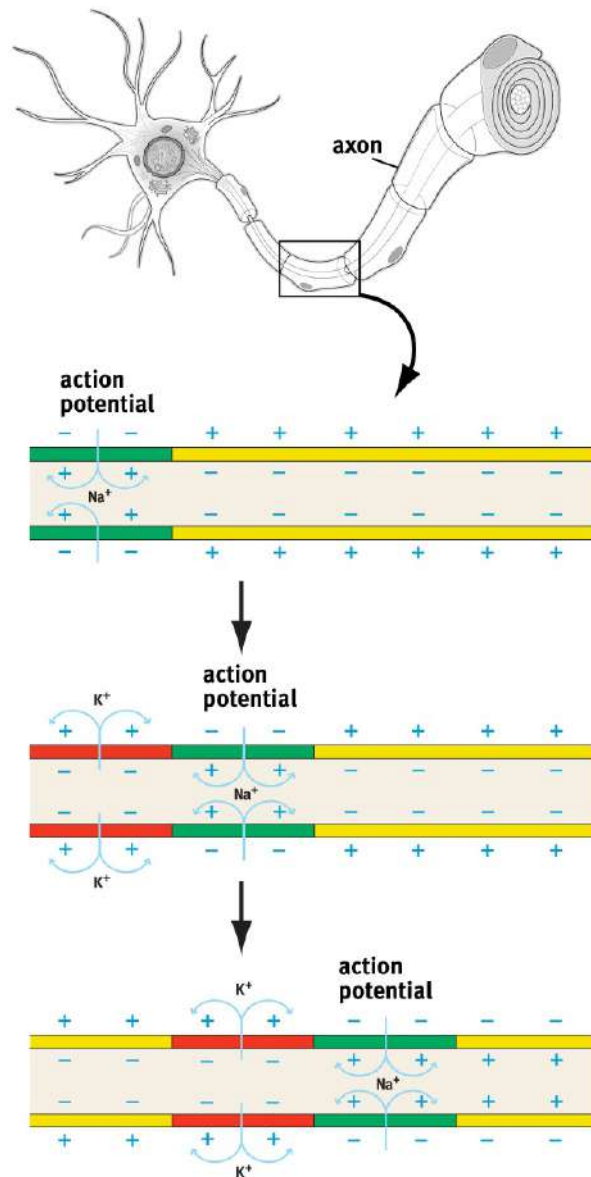


Figure 4.5. Action Potential Propagation

Action potentials are propagated down the axon when proximal sodium channels open and depolarize the membrane, inducing distal sodium channels to open as well; because of the refractory character of these channels, the action potential can move in only one direction.



REAL WORLD

A toxin called *tetrodotoxin* (TTX) is found in the pufferfish, a delicacy in Japan. TTX blocks voltage-gated Na^+ channels, blocking neuronal transmission. This can rapidly cause death because the phrenic nerves innervating the diaphragm can no longer depolarize, leading to paralysis of the muscle and cessation of breathing. For this reason, chefs who prepare pufferfish must be specially trained and licensed.



REAL WORLD

Local anesthetics work by blocking voltage-gated Na^+ channels. These drugs work particularly well on sensory neurons and therefore block the transmission of pain. They favor pain neurons because these neurons have small axonal diameters and little or no myelin, allowing easy access to the sodium channels. Anesthetic concentrations are kept sufficiently low to block pain neurons without significant effects on other sensory modalities or motor function.

The speed at which action potentials move depends on the length and cross-sectional area of the axon. Increased length of the axon results in higher resistance and slower conduction. Greater cross-sectional areas allow for faster propagation due to decreased resistance. The effect of cross-sectional area is more significant than the effect of length. In order to maximize the speed of transmission, mammals have myelin. Myelin is an extraordinarily good insulator, preventing the dissipation of the electric signal. The insulation is so effective that the membrane is only permeable to ion movement at the

nodes of Ranvier. Thus, the signal “hops” from node to node—what is called **saltatory conduction**.



MNEMONIC

Saltatory conduction can be recalled by thinking of the Spanish verb *saltar*, to jump.



REAL WORLD

Insulation by myelin is extremely effective. A human spinal cord is about the thickness of a finger. Without this insulation, the cord would have to be almost as wide as a telephone pole to prevent signal loss.

It is important to note that all action potentials within the same type of neuron have the same potential difference during depolarization. Increased intensity of a stimulus does not result in an increased potential difference of the action potential, but rather an increased frequency of firing.

THE SYNAPSE

As discussed previously, neurons are not actually in direct physical contact. There is a small space between neurons called the synaptic cleft into which neurotransmitters are secreted, as shown in Figure 4.6. To clarify the terminology, the neuron preceding the synaptic cleft is called the **presynaptic**

neuron; the neuron after the synaptic cleft is called the **postsynaptic neuron**. If a neuron signals to a gland or muscle, rather than another neuron, the postsynaptic cell is termed an **effector**. Most synapses are **chemical** in nature; they use small molecules referred to as **neurotransmitters** to send messages from one cell to the next.

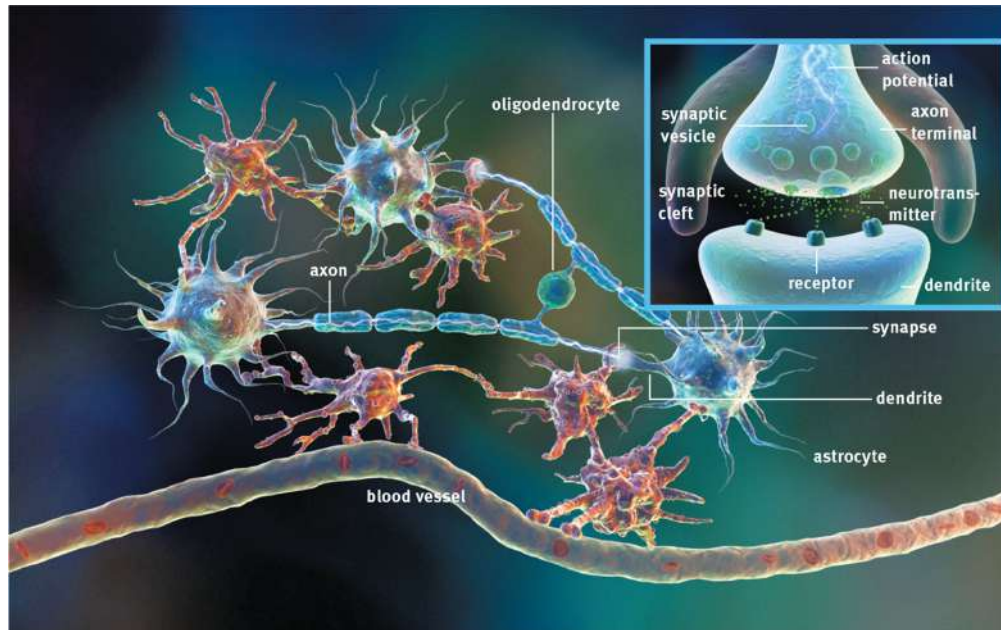


Figure 4.6. The Synapse

Synaptic vesicles are released from the presynaptic neuron and diffuse across the synaptic cleft to activate receptors on the postsynaptic neuron (or gland or muscle).

Neurotransmitters

Prior to release, neurotransmitter molecules are stored in membrane-bound vesicles in the nerve terminal. When the action potential reaches the nerve terminal, voltage-gated calcium channels open, allowing calcium to flow into the cell. This sudden increase in intracellular calcium triggers fusion of the

membrane-bound vesicles with the cell membrane at the synapse, causing exocytosis of the neurotransmitter.



KEY CONCEPT

It is critical to understand the difference between electrical and chemical transmission. Within a neuron, electricity is used to pass signals down the length of the axon. Between neurons, chemicals (neurotransmitters) are used to pass signals to the subsequent neuron (or gland or muscle).

Once released into the synapse, the neurotransmitter molecules diffuse across the cleft and bind to receptors on the postsynaptic membrane. This allows the message to be passed from one neuron to the next. As we stated earlier, neurons may be either excitatory or inhibitory; this distinction truly comes at the level of the neurotransmitter receptors. If the receptor is a ligand-gated ion channel, the postsynaptic cell will either be depolarized or hyperpolarized. If it is a G protein-coupled receptor, it will cause either changes in the levels of cyclic AMP (cAMP) or an influx of calcium. Note that the physiology of receptors is further discussed in Chapter 3 of *MCAT Biochemistry Review*.

Neurotransmission must be regulated—there are almost no circumstances under which constant signaling to the postsynaptic cell would be desirable. Therefore, the neurotransmitter must be removed from the synaptic cleft. There are three main mechanisms to accomplish this goal. First, neurotransmitters can be broken down by enzymatic reactions. The

breakdown of **acetylcholine (ACh)** by *acetylcholinesterase (AChE)*, shown in Figure 4.7, is a classic example.

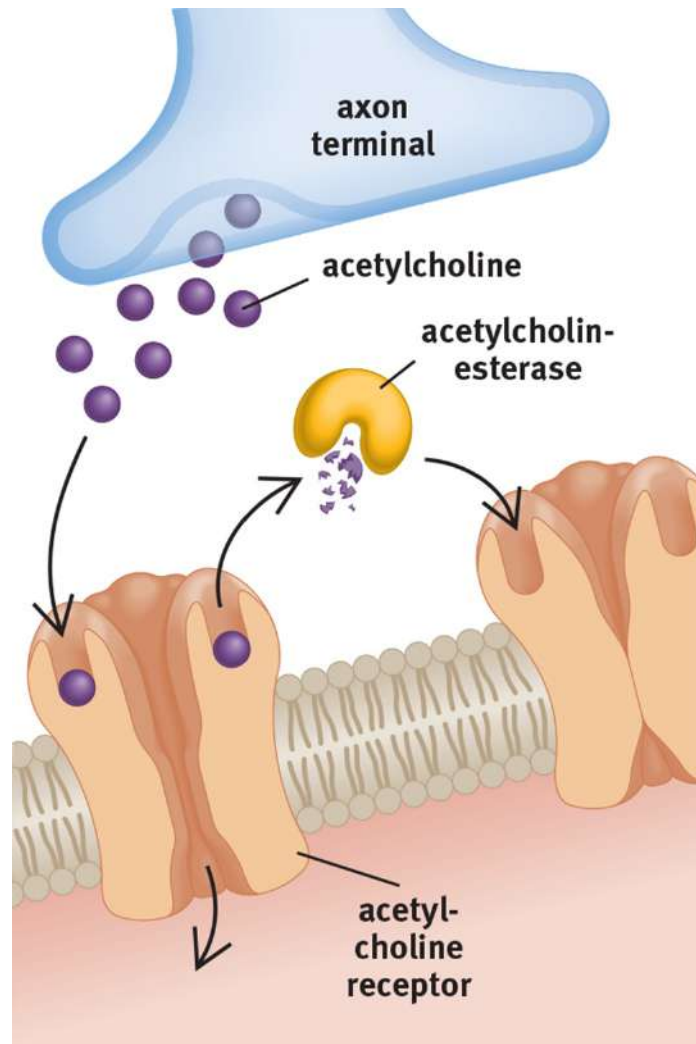


Figure 4.7. Breakdown of a Neurotransmitter by an Enzyme
Acetylcholine (ACh) can be broken down by acetylcholinesterase (AChE).

Second, neurotransmitters can be brought back into the presynaptic neuron using **reuptake carriers**. The reuptake of **serotonin (5-HT)**, shown in Figure 4.8, is a classic example of this mechanism. **Dopamine (DA)** and **norepinephrine (NE)** also use reuptake carriers.

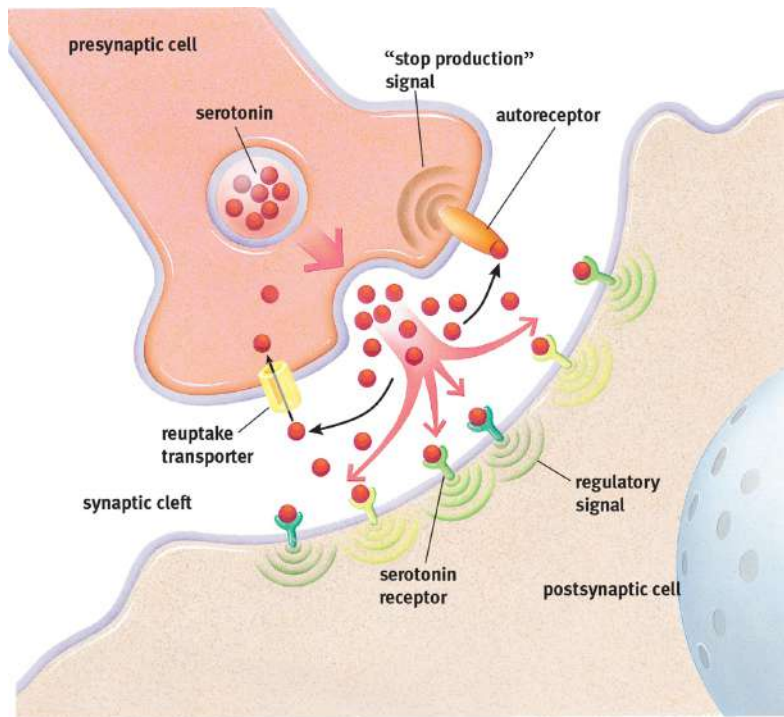


Figure 4.8. Reuptake of a Neurotransmitter

Serotonin (5-HT) can be taken back up by the presynaptic cell; an autoreceptor will signal the presynaptic cell to stop releasing serotonin and start the reuptake process.

Third, neurotransmitters may simply diffuse out of the synaptic cleft. **Nitric oxide (NO)**, a gaseous signaling molecule, fits into this category.



REAL WORLD

Many common drugs (either in clinical use or street drugs) modify processes that occur in the synapse. For instance, cocaine acts by blocking neuronal reuptake carriers, thus prolonging the action of neurotransmitters in the synapse. There are clinically useful drugs (some of which are used to treat Alzheimer's disease, glaucoma, and

myasthenia gravis) that inhibit acetylcholinesterase, thereby elevating synaptic levels of acetylcholine. Nerve gases, which have been used in warfare and terrorism, are extremely potent acetylcholinesterase inhibitors. Nerve gas causes rapid death by preventing the relaxation of skeletal muscle (most importantly, the diaphragm), leading to respiratory arrest.

MCAT CONCEPT CHECK 4.2:

Before you move on, assess your understanding of the material with these questions.

1. What neural structure initiates the action potential?

2. What entity maintains the resting membrane potential?
What is the approximate voltage of the resting membrane potential?

3. What is the difference between temporal and spatial summation?

- Temporal summation:

- Spatial summation:

4. During the action potential, which ion channel opens first? How is this ion channel regulated? What effect does the opening of this channel have on the polarization of the cell?

- Ion channel:

- Regulation:

- Effect on polarization:

5. During the action potential, which ion channel opens second? How is this ion channel regulated? What effect does the opening of this channel have on the polarization of the cell?

- Ion channel:

- Regulation:

- Effect on polarization:

6. What is the difference between the absolute and relative refractory period?

- Absolute refractory period:

- Relative refractory period:

7. What ion is primarily responsible for the fusion of neurotransmitter-containing vesicles with the nerve terminal membrane?

8. What are the three main methods by which a neurotransmitter's action can be stopped?

- ---

- ---

- ---

4.3 Organization of the Human Nervous System

High-Yield

LEARNING OBJECTIVES

After Chapter 4.3, you will be able to:

- Classify elements of the nervous system as components of either the central nervous system or the peripheral nervous system
- Differentiate between afferent and efferent neurons
- Describe the functions of the somatic and autonomic nervous systems
- Recall the physiological effects of activating the sympathetic nervous system and the parasympathetic nervous system
- Distinguish between the neural pathways for a monosynaptic and a polysynaptic reflex

The nervous system is a remarkable collection of cells that governs both involuntary and voluntary behavior, while also maintaining **homeostasis**. Functions of the nervous system include:

- Sensation and perception
- Motor function

- Cognition (thinking) and problem solving
- Executive function and planning
- Language comprehension and creation
- Memory
- Emotion and emotional expression
- Balance and coordination
- Regulation of endocrine organs
- Regulation of heart rate, breathing rate, vascular resistance, temperature, and exocrine glands

The human nervous system is a complex web of over 100 billion cells that communicate, coordinate, and regulate signals for the rest of the body. Action occurs when the body can react to external stimuli using the nervous system. In this section, we will look at the nervous system and its basic organization.

Note: Much of the information contained in this section is also discussed in Chapter 1 of MCAT Behavioral Sciences Review.

CENTRAL AND PERIPHERAL NERVOUS SYSTEMS

Generally speaking, there are three kinds of nerve cells in the nervous system: sensory neurons, motor neurons, and interneurons. **Sensory neurons** (also known as **afferent neurons**) transmit sensory information from sensory receptors to the spinal cord and brain. **Motor neurons** (also known as **efferent neurons**) transmit motor information from the brain and spinal cord to muscles and glands. **Interneurons** are found between other neurons and are the most numerous of the three types. Interneurons are located predominantly in the brain and spinal cord and are often linked to reflexive behavior.



MNEMONIC

Afferent neurons **a**scend in the spinal cord toward the brain; **e**fferent neurons **e**xit the spinal cord on their way to the rest of the body.

Different types of information require different types of processing. Processing of stimuli and response generation may happen at the level of the spinal cord, or may require input from the brainstem or cerebral cortex. Reflexes, discussed later in this section, only require processing at the level of the spinal cord. For example, when a reflex hammer hits the patellar tendon, the sensory information goes to the spinal cord, where a motor signal is sent to the quadriceps muscles, causing the leg to jerk forward at the knee. No input from the brain is required. However, some scenarios require input from the brain or brainstem. When this happens, **supraspinal** circuits are used.

Let's turn to the overall structure of the human nervous system, which is diagrammed in Figure 4.9.

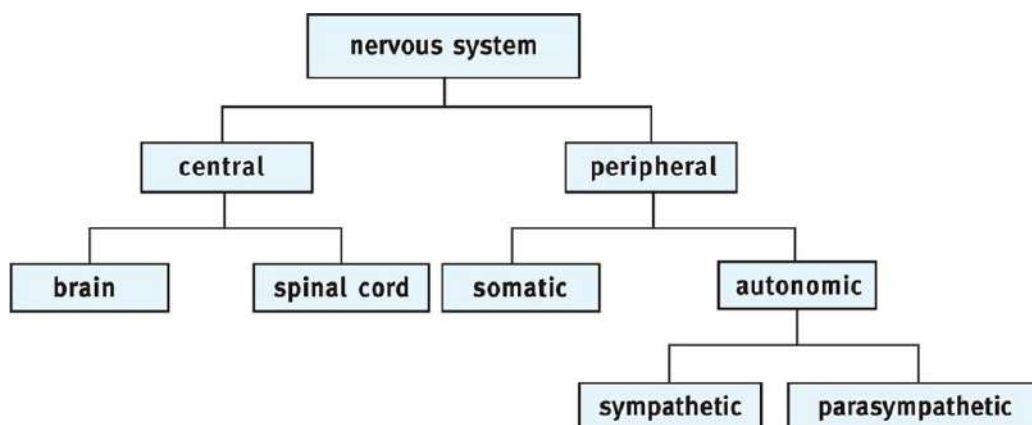


Figure 4.9. Major Divisions of the Nervous System

The nervous system can be broadly divided into two primary components: the central and peripheral nervous systems. The **central nervous system (CNS)** is composed of the brain and spinal cord. The brain consists of white matter and grey matter. The **white matter** consists of axons encased in myelin sheaths. The **grey matter** consists of unmyelinated cell bodies and dendrites. In the brain, the white matter lies deeper than the grey matter. At the base of the brain is the brainstem, which is largely responsible for basic life functions such as breathing. Note that the lobes of the brain and major brain structures are discussed in Chapter 1 of *MCAT Behavioral Sciences Review*.

The spinal cord extends downward from the brainstem and can be divided into four regions: **cervical**, **thoracic**, **lumbar**, and **sacral**. Almost all of the structures below the neck receive sensory and motor innervation from the spinal cord. The spinal cord is protected by the **vertebral column**, which transmits nerves at the space between adjacent vertebrae. Like the brain, the spinal cord also consists of white and grey matter. The white matter lies on the outside of the cord, and the grey matter is deep within it. The axons of motor and sensory neurons are in the spinal cord. The sensory neurons bring information in from the periphery and enter on the dorsal (back) side of the spinal cord. The cell bodies of these sensory neurons are found in the **dorsal root ganglia**. Motor neurons exit the spinal cord ventrally, or on the side closest to the front of the body. The structure of the spinal cord can be seen in Figure 4.10.

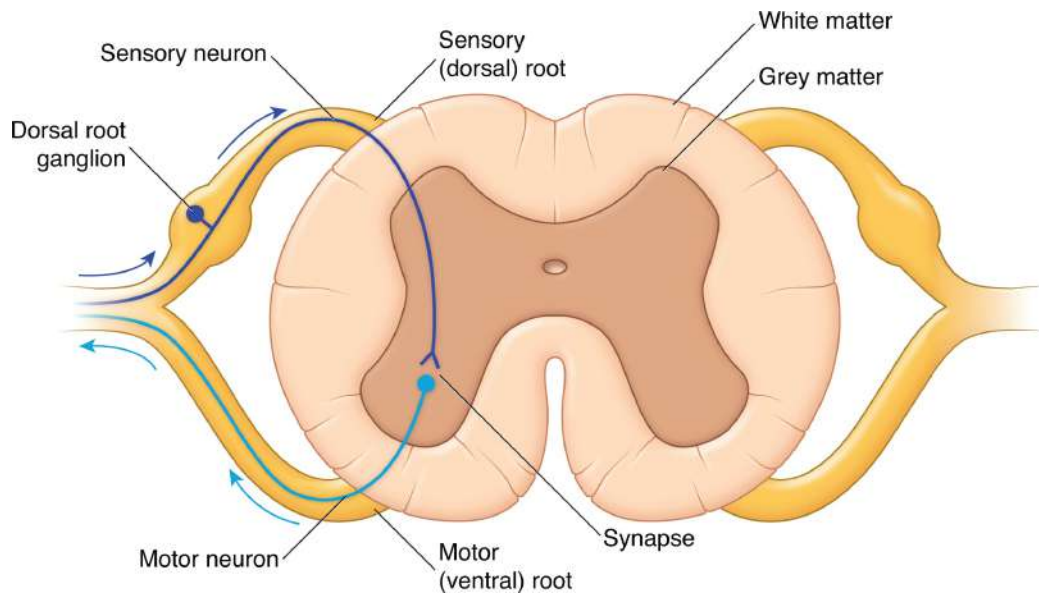


Figure 4.10. The Spinal Cord

Sensory neurons transmit information about pain, temperature, and vibration up to the brain and have cell bodies in the dorsal root ganglia toward the back of the spinal cord; the motor neurons run from the brain along the opposite side of the spinal cord and in the ventral root to control movements of skeletal muscle and glandular secretions.

The **peripheral nervous system (PNS)**, in contrast, is made up of nerve tissue and fibers outside the brain and spinal cord, including all 31 pairs of spinal nerves and 10 of the 12 pairs of cranial nerves (the olfactory and optic nerves are technically outgrowths of the central nervous system). The PNS thus connects the CNS to the rest of the body and can itself be subdivided into the somatic and autonomic nervous systems.

The **somatic nervous system** consists of sensory and motor neurons distributed throughout the skin, joints, and muscles. Sensory neurons transmit information through afferent fibers. Motor impulses, in contrast, travel along efferent fibers.

The **autonomic nervous system (ANS)** generally regulates heartbeat, respiration, digestion, and glandular secretions. In other words, the ANS manages the involuntary muscles associated with many internal organs and glands. The ANS also helps regulate body temperature by activating sweating or piloerection, depending on whether we are too hot or too cold, respectively. The main thing to understand about these functions is that they are automatic, or independent of conscious control. Note the similarity between the words autonomic and automatic. This association makes it easy to remember that the autonomic nervous system manages automatic functions such as heartbeat, respiration, digestion, and temperature control.

One primary difference between the somatic and autonomic nervous systems is that the peripheral component of the autonomic nervous system contains two neurons. By contrast, a motor neuron in the somatic nervous system goes directly from the spinal cord to the muscle without synapsing. In the autonomic nervous system, two neurons work in series to transmit messages from the spinal cord. The first neuron is known as the **preganglionic neuron**, whereas the second is the **postganglionic neuron**. The soma of the preganglionic neuron is in the CNS, and its axon travels to a ganglion in the PNS. Here it synapses on the cell body of the postganglionic neuron, which then stimulates the target tissue.



KEY CONCEPT

The first neuron in the autonomic nervous system is called the preganglionic neuron. The second neuron is the postganglionic neuron.

THE AUTONOMIC NERVOUS SYSTEM

The ANS has two subdivisions: the sympathetic nervous system and the parasympathetic nervous system. These two branches often act in opposition to one another, meaning that they are antagonistic. For example, the sympathetic nervous system acts to accelerate heart rate and inhibit digestion, while the parasympathetic nervous system decelerates heart rate and promotes digestion.

The main role of the **parasympathetic nervous system** is to conserve energy. It is associated with resting and sleeping states and acts to reduce heart rate and constrict the bronchi. The parasympathetic nervous system is also responsible for managing digestion by increasing peristalsis and exocrine secretions. Acetylcholine is the neurotransmitter responsible for parasympathetic responses in the body and is released by both preganglionic and postganglionic neurons. The vagus nerve (cranial nerve X) is responsible for much of the parasympathetic innervation of the thoracic and abdominal cavity. The functions of the parasympathetic nervous system are summarized in Figure 4.11.

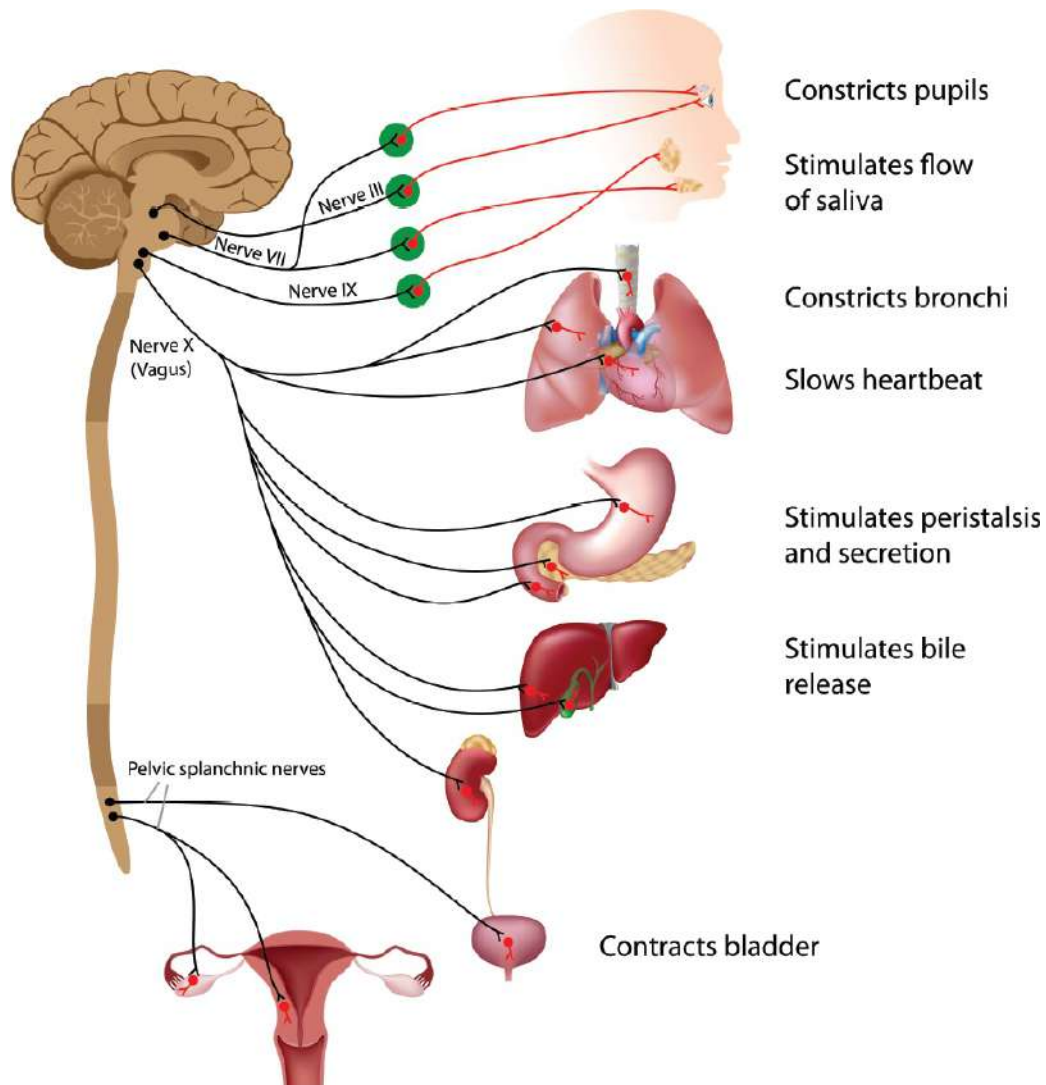


Figure 4.11. Functions of the Parasympathetic Nervous System

In contrast, the **sympathetic nervous system** is activated by stress. This can include everything from a mild stressor, such as keeping up with schoolwork, to emergencies that mean the difference between life and death. The sympathetic nervous system is closely associated with rage and fear reactions, also known as “fight-or-flight” reactions. When activated, the sympathetic nervous system:

- Increases heart rate

- Redistributes blood to muscles of locomotion
- Increases blood glucose concentration
- Relaxes the bronchi
- Decreases digestion and peristalsis
- Dilates the eyes to maximize light intake
- Releases epinephrine into the bloodstream



MNEMONIC

Sympathetic and parasympathetic nervous systems:

- Sympathetic: “**fight-or-flight**”
- Parasympathetic: “**rest-and-digest**”

The functions of the sympathetic nervous system are summarized in Figure 4.12. In the sympathetic nervous system, preganglionic neurons release acetylcholine, while most postganglionic neurons release norepinephrine.

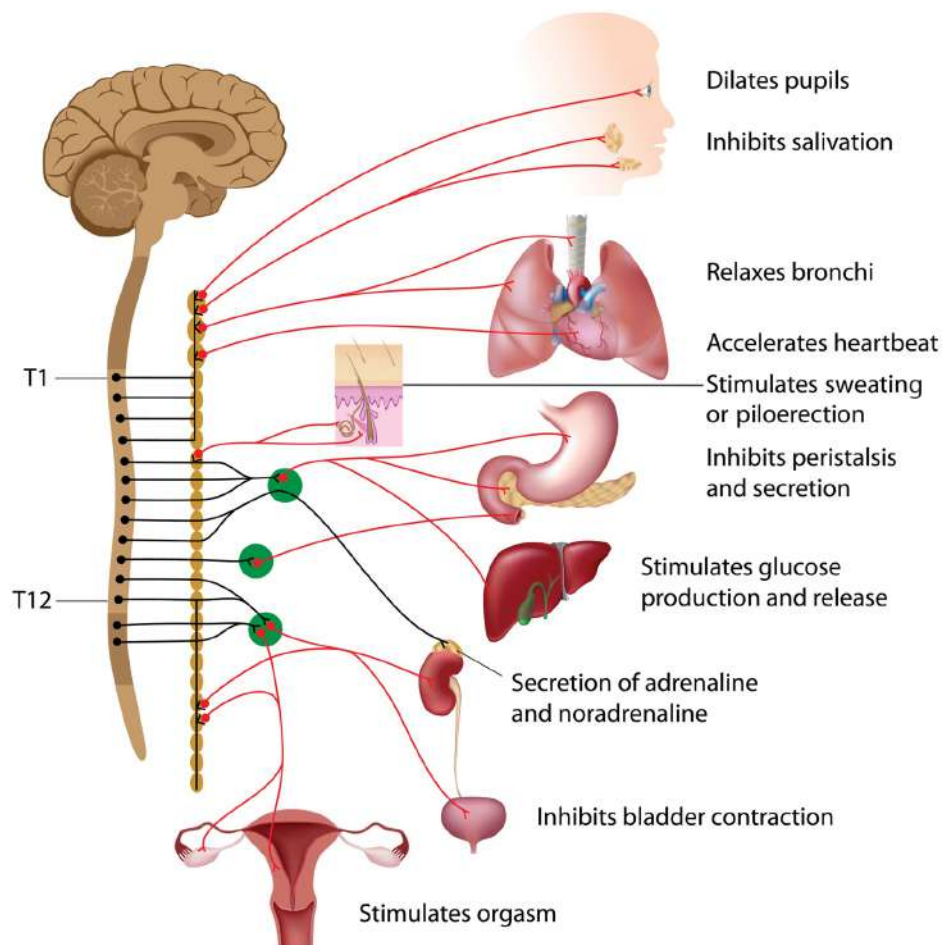


Figure 4.12. Functions of the Sympathetic Nervous System

REFLEXES

Neural circuits called **reflex arcs** control reflexive behavior. For example, consider what occurs when someone steps on a nail. Receptors in the foot detect pain, and the pain signal is transmitted by sensory neurons up to the spinal cord. At that point, the sensory neurons connect with interneurons, which can then relay pain impulses up to the brain. Rather than wait for the brain to send out a signal, interneurons in the spinal cord can also send

signals to the muscles of both legs directly, causing the individual to withdraw the foot with pain while supporting with the other foot. The original sensory information still makes its way up to the brain; however, by the time it arrives there, the muscles have already responded to the pain, thanks to the reflex arc. There are two types of reflex arcs: monosynaptic and polysynaptic.



KEY CONCEPT

Consider the purpose of reflexes. Although it may be amusing to make your friends' legs jump when you tap them, there is a more functional reason why this response occurs. The stretch on the patellar tendon makes the body think that the muscle may be getting overstretched. In response, the muscle contracts in order to prevent injury.

Monosynaptic

In a **monosynaptic reflex arc**, there is a single synapse between the sensory neuron that receives the stimulus and the motor neuron that responds to it. A classic example is the **knee-jerk reflex**, shown in Figure 4.13. When the patellar tendon is stretched, information travels up the sensory (afferent, presynaptic) neuron to the spinal cord, where it interfaces with the motor (efferent, postsynaptic) neuron that causes contraction of the quadriceps muscles. The net result is extension of the leg, which lessens the tension on the patellar tendon. Note that the reflex is simply a feedback loop and a response to potential injury. If the patellar tendon or quadriceps muscles are stretched too far, they may tear, damaging the knee joint. Thus, the reflex serves to protect the muscles.

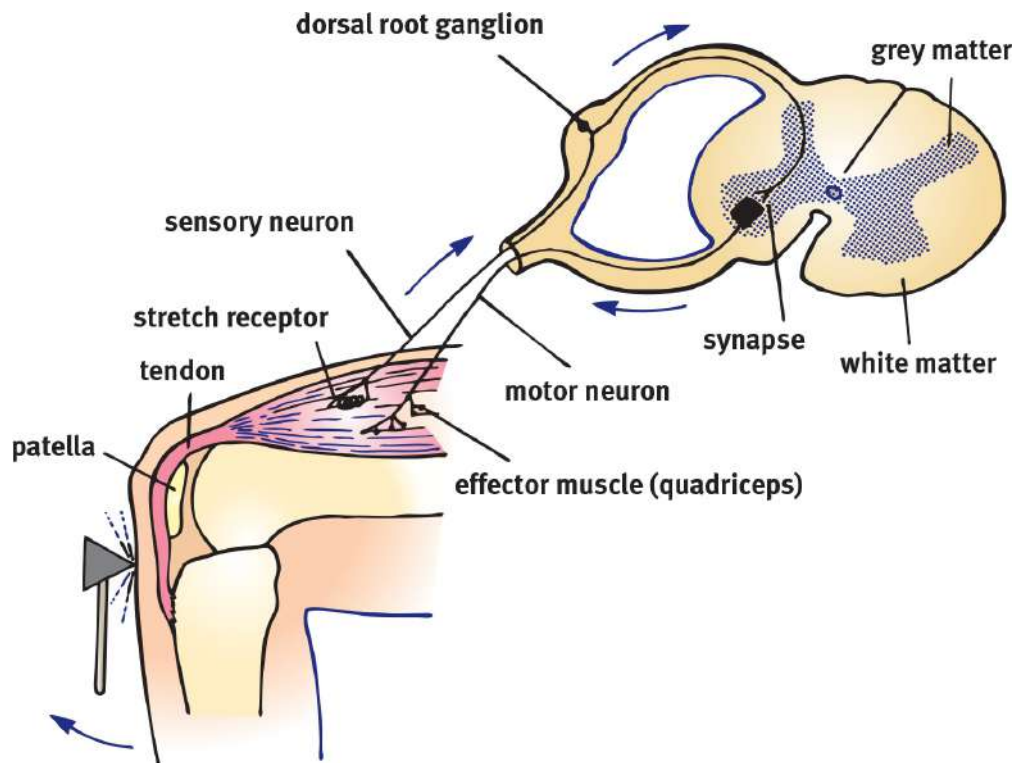


Figure 4.13. The Knee-Jerk Reflex

The knee-jerk or knee extension reflex may be elicited by swiftly stretching the patellar tendon with a reflex hammer.

Polysynaptic

In a **polysynaptic reflex arc**, there is at least one interneuron between the sensory and motor neurons. A real-life example is the reaction to stepping on a nail described earlier, which involves the **withdrawal reflex**. The extremity with which one steps on the nail will be stimulated to flex, using the hip muscles and hamstring muscles, pulling the foot away from the nail. This is a monosynaptic reflex, similar to the knee-jerk reflex described previously. However, if the person is to maintain balance, the other foot must be planted firmly on the ground. For this to occur, the motor neuron that controls the quadriceps muscles in the opposite limb must be stimulated, extending it.

Interneurons in the spinal cord provide the connections from the incoming sensory information to the motor neurons in the supporting limb.

MCAT CONCEPT CHECK 4.3:

Before you move on, assess your understanding of the material with these questions.

1. What parts of the nervous system are in the central nervous system (CNS)? Peripheral nervous system (PNS)?

- CNS:

- PNS:

2. What do afferent neurons do? Efferent neurons?

- Afferent:

- Efferent:

3. What functions are accomplished by the somatic nervous system? The autonomic nervous system?

- Somatic:

- Autonomic:
-

4. What are the effects of the sympathetic nervous system? The parasympathetic nervous system?

- Sympathetic:

- Parasympathetic:
-

5. What is the pathway of neural impulses in a monosynaptic reflex? In a polysynaptic reflex?

- Monosynaptic reflex:

- Polysynaptic reflex:
-



Conclusion

The nervous system is one of the most fascinating and complex systems of the human body; millions upon millions of cells allow for appropriate interactions in the everyday world. It is the seat of personality and, ultimately, the system that makes you *you*. In medical school, your courses on neuroscience will go into astounding detail about the nervous system, including the circuits that govern sensations such as pain and temperature, and circuits that allow your body to move and function.

In this chapter, we explored the nervous system at both the cellular and organizational level. Neurons are the primary cells of the nervous system, propagating impulses through both electrical and chemical means—action potentials and synaptic transmission, respectively. Neurons can be grouped together to form nerves, which are the primary organizational structures in one major branch of the nervous system, the peripheral nervous system. The central nervous system consists of the brain and spinal cord. The peripheral nervous system can be subdivided into the somatic and autonomic nervous systems, the latter of which can be further subdivided into the sympathetic and parasympathetic nervous systems.

The nervous system is heavily tested on the MCAT because it plays a role in the function of almost every other major organ system. Neurons cause muscles to move and digestive structures to carry food along through peristalsis, and they regulate breathing rate, heart rate, and glandular secretions. The nervous system is not the only system that has such a

profound effect throughout the body, however. The endocrine system, which we will explore in the next chapter, serves a similar role—but through chemical messengers carried in the blood called hormones.

Concept Summary

Cells of the Nervous System

- **Neurons** are highly specialized cells responsible for the conduction of impulses.
- Neurons communicate using both electrical and chemical forms of communication.
 - Electrical communication occurs via ion exchange and the generation of membrane potentials down the length of the axon.
 - Chemical communication occurs via neurotransmitter release from the presynaptic cell and the binding of these neurotransmitters to the postsynaptic cell.
- Neurons consist of many different parts.
 - **Dendrites** are appendages that receive signals from other cells.
 - The cell body or **soma** is the location of the nucleus as well as organelles such as the endoplasmic reticulum and ribosomes.
 - The **axon hillock** is where the cell body transitions to the axon, and where action potentials are initiated.
 - The **axon** is a long appendage down which an action potential travels.
 - The **nerve terminal** or **synaptic bouton** is the end of the axon from which neurotransmitters are released.
 - **Nodes of Ranvier** are exposed areas of myelinated axons that permit

saltatory conduction.

- The **synapse** consists of the nerve terminal of the presynaptic neuron, the membrane of the postsynaptic cell, and the space between the two, called the **synaptic cleft**.
- Many axons are coated in **myelin**, an insulating substance that prevents signal loss.
 - Myelin is created by **oligodendrocytes** in the central nervous system and **Schwann cells** in the peripheral nervous system.
 - Myelin prevents dissipation of the neural impulse and crossing of neural impulses from adjacent neurons.
- Individual axons are bundled into **nerves** or **tracts**.
 - A single nerve may carry multiple types of information, including sensory, motor, or both. Tracts contain only one type of information.
 - Cell bodies of neurons of the same type within a nerve cluster together in **ganglia** in the peripheral nervous system.
 - Cell bodies of the individual neurons within a tract cluster together in **nuclei** in the central nervous system.
- **Neuroglia** or **glial cells** are other cells within the nervous system in addition to neurons.
 - **Astrocytes** nourish neurons and form the blood–brain barrier, which controls the transmission of solutes from the bloodstream into nervous tissue.
 - **Ependymal cells** line the ventricles of the brain and produce cerebrospinal fluid, which physically supports the brain and serves as a

shock absorber.

- **Microglia** are phagocytic cells that ingest and break down waste products and pathogens in the central nervous system.
- Oligodendrocytes (CNS) and Schwann cells (PNS) produce myelin around axons.

Transmission of Neural Impulses

- All neurons exhibit a **resting membrane potential** of approximately -70 mV.
 - Resting potential is maintained using selective permeability of ions as well as the Na^+/K^+ ATPase.
 - The **Na^+/K^+ ATPase** pumps three sodium ions out of the cell for every two potassium ions pumped in.
- Incoming signals can be either excitatory or inhibitory.
 - Excitatory signals cause depolarization of the neuron.
 - Inhibitory signals cause hyperpolarization of the neuron.
 - **Temporal summation** refers to the integration of multiple signals near each other in time.
 - **Spatial summation** refers to the addition of multiple signals near each other in space.
- An **action potential** is used to propagate signals down the axon.
 - When enough excitatory stimulation occurs, the cell is **depolarized** to the **threshold voltage** and voltage-gated sodium channels open.

- Sodium flows into the neuron due to its strong **electrochemical gradient**. This continues depolarizing the neuron.
 - At the peak of the action potential (approximately +35 mV), sodium channels are inactivated and potassium channels open.
 - Potassium flows out of the neuron due to its strong electrochemical gradient, **repolarizing** the cell. Potassium channels stay open long enough to overshoot the action potential, resulting in a **hyperpolarized** neuron; then, the potassium channels close.
 - The Na^+/K^+ ATPase brings the neuron back to the resting potential and restores the sodium and potassium gradients.
 - While the axon is hyperpolarized, it is in its **refractory period**. During the **absolute refractory period**, the cell is unable to fire another action potential. During the **relative refractory period**, the cell requires a larger than normal stimulus to fire an action potential.
 - The impulse propagates down the length of the axon because the influx of sodium in one segment of the axon brings the subsequent segment of the axon to threshold. The fact that the preceding segment of the axon is in its refractory period means that the action potential can only travel in one direction.
- At the nerve terminal, neurotransmitters are released into the synapse.
 - When the action potential arrives at the nerve terminal, voltage-gated calcium channels open.
 - The influx of calcium causes fusion of vesicles filled with neurotransmitters with the presynaptic membrane, resulting in exocytosis of neurotransmitters into the synaptic cleft.
 - The neurotransmitters bind to receptors on the postsynaptic cell, which may be ligand-gated ion channels or G protein-coupled receptors.

- Neurotransmitters must be cleared from the postsynaptic receptors to stop the propagation of the signal. There are three ways this can happen:
 - The neurotransmitter can be enzymatically broken down.
 - The neurotransmitter can be absorbed back into the presynaptic cell by **reuptake channels**.
 - The neurotransmitter can diffuse out of the synaptic cleft.

Organization of the Human Nervous System

- There are three types of neurons in the nervous system: **motor (efferent)** neurons, **interneurons**, and **sensory (afferent)** neurons.
- The nervous system is made up of the **central nervous system (CNS: brain and spinal cord)** and **peripheral nervous system (PNS: cranial and spinal nerves)**.
 - In the CNS, **white matter** consists of myelinated axons, and **grey matter** consists of unmyelinated cell bodies and dendrites. In the brain, white matter is deeper than grey matter. In the spinal cord, grey matter is deeper than white matter.
 - The PNS is divided into the **somatic** (voluntary) and **autonomic** (automatic) nervous systems.
 - The autonomic nervous system is further divided into the **parasympathetic** (rest-and-digest) and **sympathetic** (fight-or-flight) branches.
- **Reflex arcs** use the ability of interneurons in the spinal cord to relay information to the source of a stimulus while simultaneously routing it to the brain.

- In a **monosynaptic reflex arc**, the sensory (afferent, presynaptic) neuron fires directly onto the motor (efferent, postsynaptic) neuron.
- In a **polysynaptic reflex arc**, the sensory neuron may fire onto a motor neuron as well as interneurons that fire onto other motor neurons.

Answers to Concept Checks

4.1

- .. The axon transmits an electrical signal (the action potential) from the soma to the synaptic knob. The axon hillock integrates excitatory and inhibitory signals from the dendrites and fires an action potential if the excitatory signals are strong enough to reach threshold. Dendrites receive incoming signals and carry them to the soma. The myelin sheath acts as insulation around the axon and speeds conduction. The soma is the cell body and contains the nucleus, endoplasmic reticulum, and ribosomes. The synaptic bouton lies at the end of the axon and releases neurotransmitters.
- !. A collection of cell bodies in the central nervous system is called a nucleus. In the peripheral nervous system, it is called a ganglion.
- }. Astrocytes nourish neurons and form the blood–brain barrier, which helps protect the brain from foreign pathogens gaining entrance. Microglia ingest and break down waste products and pathogens. Disruption of either of these mechanisms would increase susceptibility to a CNS infection.
- l. Oligodendrocytes produce myelin in the central nervous system while Schwann cells produce myelin in the peripheral nervous system. Since GBS causes demyelination in the PNS, it can be inferred that Schwann cells are targeted for immune destruction.

4.2

- .. The action potential is initiated at the axon hillock.
- !. The resting membrane potential is maintained by the $\text{Na}^+/\text{K}^+\text{ATPase}$ at approximately -70 mV .

- }. Temporal summation is the integration of multiple signals close to each other in time. Spatial summation is the integration of multiple signals close to each other in space.
- l. The sodium channel opens first at threshold (around -50 mV). It is regulated by inactivation, which occurs around $+35$ mV. Inactivation can only be reversed by repolarizing the cell. The opening of the sodium channel causes depolarization.
- i. The potassium channel opens second at approximately $+35$ mV. It is regulated by closing at low potentials (slightly below -70 mV). The opening of the potassium channel causes repolarization and, eventually, hyperpolarization.
- i. During the absolute refractory period, the cell is unable to fire an action potential regardless of the intensity of a stimulus. During the relative refractory period, the cell can fire an action potential only with a stimulus that is stronger than normal.
- l. Calcium is responsible for fusion of neurotransmitter vesicles with the nerve terminal membrane.
- }. A neurotransmitter's action can be stopped by enzymatic degradation, reuptake, or diffusion.

4.3

- .. The central nervous system includes the brain and spinal cord. The peripheral nervous system includes cranial and spinal nerves and sensory nerves.
- l. Afferent (sensory) neurons bring signals from a sensor to the central nervous system. Efferent (motor) neurons bring signals from the central nervous system to an effector.
- }. The somatic nervous system is responsible for voluntary actions—most notably, moving muscles. The autonomic nervous system is responsible for involuntary processes like heart rate, bronchial dilation, dilation of the

pupils, exocrine gland function, and peristalsis.

- l. The sympathetic nervous system promotes a “fight-or-flight” response, with increased heart rate and bronchial dilation, redistribution of blood to locomotor muscles, dilation of the pupils, and slowing of digestive and urinary function. The parasympathetic nervous system promotes “rest-and-digest” functions, slowing heart rate and constricting the bronchi, redistributing blood to the gut, promoting exocrine secretions, constricting the pupils, and promoting peristalsis and urinary function.
- i. In a monosynaptic reflex, a sensory (afferent, presynaptic) neuron fires directly onto a motor (efferent, postsynaptic) neuron. In a polysynaptic reflex, a sensory neuron may fire directly onto a motor neuron, but interneurons are used as well. These interneurons fire onto other motor neurons.



Shared Concepts

Behavioral Sciences Chapter 1

Biology and Behavior

Behavioral Sciences Chapter 2

Sensation and Perception

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biochemistry Chapter 8

Biological Membranes

Biology Chapter 11

The Musculoskeletal System

General Chemistry Chapter 12

Electrochemistry

Discrete Practice Questions

1. Resting membrane potential depends on:
 - I. differential distribution of ions across the axon membrane.
 - II. the opening of voltage-gated calcium channels.
 - III. active transport of ions across the membrane.
 - (A) I only
 - (B) I and II only
 - (C) I and III only
 - (D) II and III only

2. All of the following are associated with the myelin sheath EXCEPT:
 - (A) faster conduction of nerve impulses.
 - (B) nodes of Ranvier forming gaps along the axon.
 - (C) increased magnitude of the potential difference during an action potential.
 - (D) saltatory conduction of action potentials.

3. Which of the following is true with regard to the action potential?
 - (A) All hyperpolarized stimuli will be carried to the axon terminal

without a decrease in size.

- (B) The size of the action potential is proportional to the size of the stimulus that produced it.
- (C) Increasing the intensity of the depolarization increases the size of the impulse.
- (D) Once an action potential is triggered, an impulse of a given magnitude and speed is produced.

4. Which of the following correctly describes a difference between nerves and tracts?

- (A) Nerves are seen in the central nervous system; tracts are seen in the peripheral nervous system.
- (B) Nerves have cell bodies in nuclei; tracts have cell bodies in ganglia.
- (C) Nerves may carry more than one type of information; tracts can only carry one type of information.
- (D) Nerves contain only one neuron; tracts contain many neurons.

5. Which of the following accurately describes sensory neurons?

- (A) Sensory neurons are afferent and enter the spinal cord on the dorsal side.
- (B) Sensory neurons are efferent and enter the spinal cord on the dorsal side.
- (C) Sensory neurons are afferent and enter the spinal cord on the ventral side.
- (D) Sensory neurons are efferent and enter the spinal cord on the

ventral side.

6. When a sensory neuron receives a stimulus that brings it to threshold, it will do all of the following EXCEPT:
 - (A) become depolarized.
 - (B) transduce the stimulus to an action potential.
 - (C) inhibit the spread of the action potential to other sensory neurons.
 - (D) cause the release of neurotransmitters onto cells in the central nervous system.

7. When the potential across the axon membrane is more negative than the normal resting potential, the neuron is said to be in a state of:
 - (A) depolarization.
 - (B) hyperpolarization.
 - (C) repolarization.
 - (D) polarization.

8. Which of the following statements concerning the somatic division of the peripheral nervous system is INCORRECT?
 - (A) Its pathways innervate skeletal muscle.
 - (B) Its pathways are usually voluntary.
 - (C) Some of its pathways are referred to as reflex arcs.
 - (D) Its pathways always involve more than two neurons.

9. Which of the following is a function of the parasympathetic nervous system?
- (A) Increasing blood sugar during periods of stress
 - (B) Dilating the pupils to enhance vision
 - (C) Increasing oxygen delivery to muscles
 - (D) Decreasing heart rate and blood pressure
10. Which of the following neurotransmitters is used in the ganglia of both the sympathetic and parasympathetic nervous systems?
- (A) Acetylcholine
 - (B) Dopamine
 - (C) Norepinephrine
 - (D) Serotonin
11. In which neural structure are ribosomes primarily located?
- (A) Dendrites
 - (B) Soma
 - (C) Axon hillock
 - (D) Axon
12. An autoimmune disease attacks the voltage-gated calcium channels in the synaptic terminal of an excitatory neuron. What is a likely

symptom of this condition?

- (A) Spastic paralysis (inability to relax the muscles)
- (B) Flaccid paralysis (inability to contract the muscles)
- (C) Inability to reuptake neurotransmitters once released
- (D) Retrograde flow of action potentials

13. A neuron only fires an action potential if multiple presynaptic cells release neurotransmitters onto the dendrites of the neuron. This is an example of:

- (A) saltatory conduction.
- (B) summation.
- (C) a feedback loop.
- (D) inhibitory transmission.

14. A disease results in the death of Schwann cells. Which portion of the nervous system is NOT likely to be affected?

- (A) Central nervous system
- (B) Somatic nervous system
- (C) Autonomic nervous system
- (D) Parasympathetic nervous system

15. A surgeon accidentally clips a dorsal root ganglion during a spinal surgery. What is a likely consequence of this error?

- I. Loss of motor function at that level
- II. Loss of sensation at that level
- III. Loss of cognitive function

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

Discrete Practice Answers

1. **C**

The polarization of the neuron at rest is the result of an uneven distribution of ions between the inside and outside of the cell. This difference is achieved through the active pumping of ions into and out of the neuron (using the Na^+/K^+ ATPase). Voltage-gated calcium channels are important in the nerve terminal, where the influx of calcium triggers the fusion of vesicles containing neurotransmitters with the membrane, but not in maintaining resting membrane potential.

2. **C**

Myelin is a white lipid-containing material surrounding the axons of many neurons in the central and peripheral nervous systems. It is arranged on the axon discontinuously; the gaps between the segments of myelin are called nodes of Ranvier, eliminating **(B)**. Myelin increases the conduction velocity by insulating segments of the axon so that the membrane is permeable to ions only at the nodes of Ranvier, eliminating **(A)**. The action potential jumps from node to node, a process known as saltatory conduction, eliminating **(D)**. Action potentials are often described as being “all-or-nothing”; the magnitude of the potential difference of an action potential is fixed, regardless of the intensity of the stimulus. Thus, myelin does not affect the magnitude of the potential difference in an action potential, making **(C)** the correct answer.

3. **D**

As in the previous question, the action potential is often described as an all-or-nothing response. This means that, whenever the threshold membrane potential is reached, an action potential with a consistent size and duration is produced. Neuronal information is coded by the frequency and number of action potentials, not the size of the action potential, eliminating **(B)** and **(C)** and making **(D)** the correct answer. Hyperpolarizing (inhibitory) signals are not transmitted to the nerve terminal, eliminating **(A)**.

4. **C**

Nerves are collections of neurons in the peripheral nervous system and may contain multiple types of information (sensory or motor); they contain cell bodies in ganglia. Tracts are collections of neurons in the central nervous system and contain only one type of information; they contain cell bodies in nuclei.

5. **A**

Sensory neurons are considered afferent (carrying signals from the periphery to the central nervous system) and enter the spinal cord on the dorsal side. Motor neurons are considered efferent (carrying signals from the central nervous system to the periphery) and exit the spinal cord on the ventral side.

6. **C**

When a sensory neuron receives a signal that is strong enough to bring it to threshold, one can assume that the receptor becomes depolarized, allowing it to transduce the stimulus to an action potential. The action potential will then be carried by sensory neurons to the central nervous system, where the cell will release neurotransmitters. Therefore, among the given choices, the only incorrect statement is found in **(C)**. If a receptor

is stimulated, it will promote the spread of the action potential to postsynaptic sensory neurons in the spinal cord, which can send the signal toward the brain.

7. **B**

When the potential across the axon membrane is more negative than the normal resting potential, the neuron is referred to as hyperpolarized. Hyperpolarization occurs right after an action potential and is caused by excess potassium exiting the neuron.

8. **D**

The somatic division of the peripheral nervous system innervates skeletal muscles and is responsible for voluntary movement. Some of the pathways in this part of the nervous system are reflex arcs, which are reflexive responses to certain stimuli that involve only a sensory and a motor neuron. These neurons synapse in the spinal cord and do not require signaling from the brain. The pathways of the somatic division can involve two, three, or more neurons, depending on the type of signal. The correct answer therefore is **(D)**.

9. **D**

The parasympathetic nervous system governs the “rest-and-digest” response. The parasympathetic nervous system slows the heart rate, decreases blood pressure, promotes blood flow to the GI tract, and constricts the pupils, among other functions. The sympathetic nervous system governs the fight-or-flight response, including increased heart rate and blood pressure, decreased bloodflow to the digestive tract, and increased bloodflow to the muscles. **(D)** is the only answer choice that represents a function of the parasympathetic nervous system.

0. **A**

Acetylcholine is the neurotransmitter released by the preganglionic neuron in both the sympathetic and parasympathetic nervous systems. The postganglionic neuron in the sympathetic nervous system usually releases norepinephrine, while the postganglionic neuron in the parasympathetic nervous system releases acetylcholine.

1. **B**

Neurons contain very specialized structures, including dendrites, axons, and the axon hillock. However, neurons are still cells and must carry out cellular functions including protein synthesis. The cell body or soma contains the nucleus, endoplasmic reticulum, and ribosomes.

2. **B**

First, consider the function of voltage-gated calcium channels. When the nerve terminal depolarizes, voltage-gated calcium channels open, allowing for influx of calcium. This influx of calcium triggers fusion of the synaptic vesicles containing neurotransmitters with the membrane of the neuron at the nerve terminal. This allows for exocytosis of the neurotransmitters into the synapse. If a disease blocked the influx of calcium, there would be no release of neurotransmitters. A lack of neurotransmitters means that the neuron cannot send excitatory signals. Thus, any symptoms resulting from this disease would be due to an inability of neurons to transmit excitatory signals to the muscle. If neurons cannot communicate, flaccid paralysis may be one of the results.

3. **B**

Some neurons require multiple instances of excitatory transmission to be brought to threshold. These excitatory signals may be close to each other

in time (temporal) or in space (spatial); either way, this pattern of excitation is termed summation.

4. **A**

Schwann cells are responsible for myelination of cells in the peripheral nervous system. Thus, the central nervous system is unlikely to be affected. The peripheral nervous system includes the somatic nervous system and the autonomic nervous system. The autonomic nervous system is composed of both the parasympathetic and sympathetic nervous systems. Thus, **(A)** is the right answer.

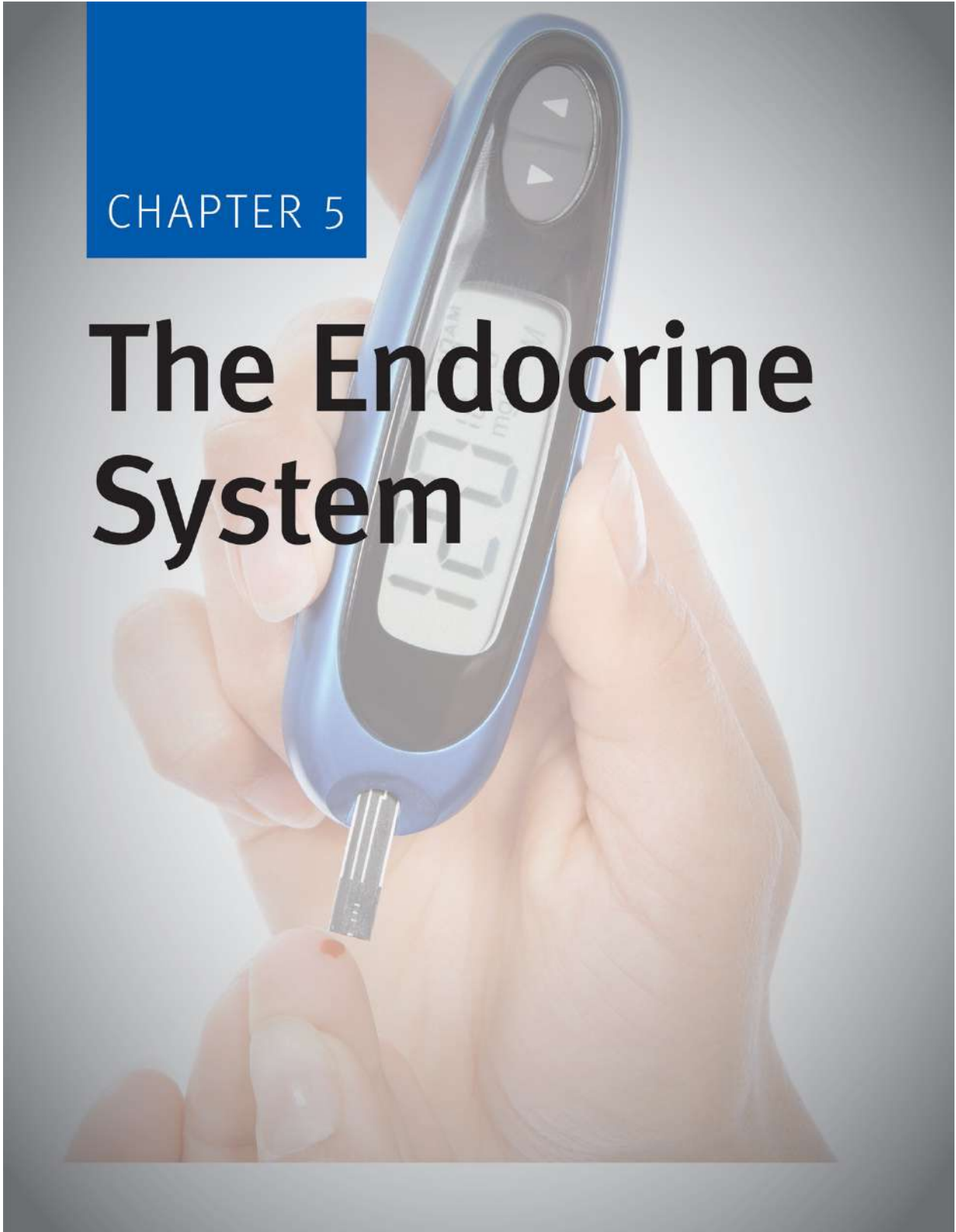
5. **B**

The dorsal root ganglion contains cell bodies of sensory neurons only. If a dorsal root ganglion is disrupted at a certain level, there will be a loss of sensation at that level.



CHAPTER 5

The Endocrine System



CHAPTER 5

The Endocrine System

In This Chapter

5.1 Mechanisms of Hormone Action



Classification of Hormones by Chemical Structure
Classification of Hormones by Target Tissue

5.2 Endocrine Organs and Hormones

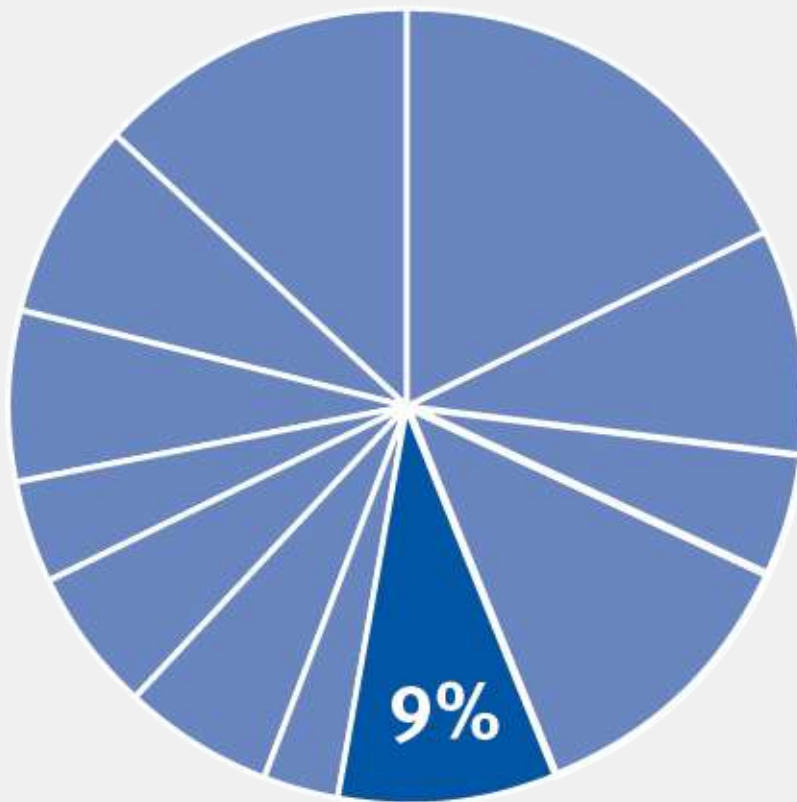


Hypothalamus
Anterior Pituitary
Posterior Pituitary
Thyroid
Parathyroid Glands
Adrenal Cortex
Adrenal Medulla
Pancreas
Gonads
Pineal Gland
Other Organs

Concept Summary

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Chapter Profile



The content in this chapter should be relevant to about 9% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content categories:

3A: Structure and functions of the nervous and endocrine systems and ways in which these systems coordinate the organ systems

Introduction

The human body consists of many trillions of cells that must work together to sustain life. In order to work effectively, fuel resources must be conserved or used appropriately, such as when responding to stressful situations. In addition, organs must be able to communicate with the brain to cause changes in behavior and physiology to maintain homeostasis. As the messengers in the endocrine system, hormones play an essential role in this communication. For example, the pancreas produces insulin and glucagon. Insulin induces the transport of glucose into organs and the storage of excess glucose when blood glucose concentrations are high. Conversely, glucagon triggers the release of sugar stores which raises blood glucose concentration. Working together, these hormones ensure that there are adequate levels of glucose for organ function, but that glucose levels are not so high as to cause damage to organ systems.

Diabetes mellitus is one of the most common diseases in the United States and a major cause of morbidity and mortality. Type 1 diabetes mellitus is an autoimmune disease in which insulin-producing cells in the islets of Langerhans are destroyed. Type 2 diabetes mellitus is caused by end-organ insensitivity to insulin. In both cases, blood glucose concentrations rise to dangerous levels (sometimes up to ten times the normal concentration), which can cause significant damage to organs, including the retina of the eye, the glomeruli of the kidneys, the coronary vessels of the heart and cerebral vessels of the brain, and peripheral nerves. Left untreated (or, to be frank,

even if treated in many cases), diabetes can lead to blindness, kidney failure, heart attacks, strokes, and limb amputations. Regardless of the field you enter, you will spend a significant amount of time working with diabetic patients and will have to think about the effects of this disease on other conditions and their treatment.

In this chapter, we will explore the different types of hormones and how they work. We'll survey the various endocrine organs and discuss the hormones each one produces. This is an extremely high-yield chapter: the MCAT frequently tests not only the make-up of the endocrine system (hormones and their functions), but also the processes of the endocrine system (feedback loops and their regulation). Return to this chapter frequently; a thorough knowledge of this system will definitely pay off on Test Day.

5.1 Mechanisms of Hormone Action

High-Yield



LEARNING OBJECTIVES

After Chapter 5.1, you will be able to:

- Compare and contrast the traits and actions of peptide vs. steroid hormones
- Recall the process for synthesizing amino acid-derivative hormones
- Distinguish between direct and tropic hormones

The endocrine system consists of organs, known as **glands**, that secrete hormones. **Hormones** are signaling molecules that are secreted directly into the bloodstream to distant target tissues. At target tissues, hormones bind to receptors, inducing a change in gene expression or cellular functioning. Not all hormones share the same structure and function. In order to understand how each hormone functions, it is first important to understand basic hormone structure.

CLASSIFICATION OF HORMONES BY CHEMICAL STRUCTURE

Hormones can be subdivided into categories based on different criteria. First, hormones can be classified by their chemical identities. Hormones can be **peptides, steroids, or amino acid derivatives.**

Peptide Hormones

Peptide hormones are made up of amino acids, ranging in size from quite small (such as anti-diuretic hormone, ADH) to relatively large (such as insulin). Peptide hormones are all derived from larger precursor polypeptides that are cleaved during posttranslational modification. These smaller units are transported to the Golgi apparatus for further modifications that activate the hormones and direct them to the correct locations in the cell. Such hormones are released by exocytosis after being packaged into vesicles.

Because peptide hormones are charged and cannot pass through the plasma membrane, they must bind to an extracellular receptor. The peptide hormone is considered the **first messenger**; it binds to the receptor and triggers the transmission of a second signal, known as the **second messenger**. There are many different receptor subtypes, and the type of receptor determines what happens once the hormone has stimulated the receptor.

The connection between the hormone at the surface and the effect brought about by second messengers within the cell is known as a **signaling cascade**. At each step, there is the possibility of **amplification**. For example, one hormone molecule may bind to multiple receptors before it is degraded. Also, each receptor may activate multiple enzymes, each of which will trigger the production of large quantities of second messengers. Thus, each step can result in an increase in signal intensity. Some common second messengers are **cyclic adenosine monophosphate (cAMP)**, **inositol triphosphate (IP₃)**,

and calcium. The activation of a G protein–coupled receptor is shown in Figure 5.1. In this system, the binding of a peptide hormone triggers the receptor to either activate or inhibit an enzyme called ***adenylate cyclase***, raising or lowering the levels of cAMP accordingly. cAMP can bind to intracellular targets, such as *protein kinase A*, which phosphorylates transcription factors like cAMP response element-binding protein (CREB) to exert the hormone’s ultimate effect. Keep in mind that protein kinase A can modify other enzymes as well as transcription factors, and therefore it can have a rapid or slow effect on the cell.

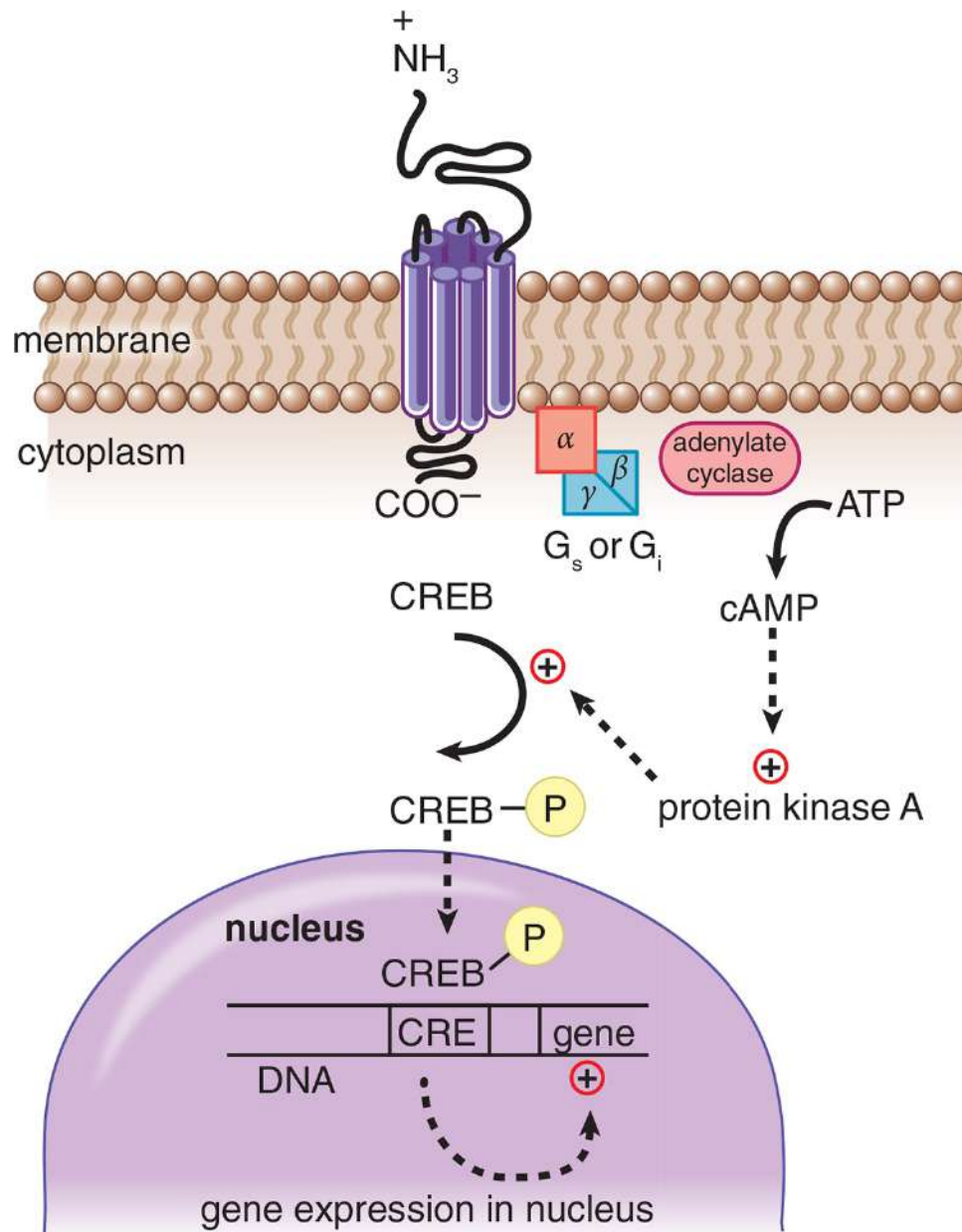


Figure 5.1. Mechanism of Action of a Peptide Hormone

Peptide hormones bind to membrane-bound receptors to initiate a signal cascade, using second messengers like cAMP.

The effects of peptide hormones are usually rapid but short-lived because these hormones act through second messenger cascades, which are transient. It is quicker to turn them on and off, compared with steroid

hormones, but their effects do not last without relatively constant stimulation.

Because peptides are generally water-soluble, peptide hormones can travel freely in the bloodstream and usually do not require carriers. This is in stark contrast to steroid hormones, which are lipid-soluble.

Steroid Hormones

Steroid hormones are derived from cholesterol and are produced primarily by the gonads and adrenal cortex. Because steroid hormones are derived from nonpolar molecules, they can easily cross the cell membrane. Hence, their receptors are usually intracellular (in the cytosol) or intranuclear (in the nucleus). Upon binding to the receptor, steroid hormone–receptor complexes undergo conformational changes. The receptor can then bind directly to DNA, resulting in either increased or decreased transcription of particular genes, depending on the identity of the hormone, as shown in Figure 5.2. One common form of conformational change is **dimerization**, or pairing of two receptor–hormone complexes. The effects of steroid hormones are slower but longer-lived than peptide hormones because steroid hormones participate in gene regulation, causing alterations in the amount of mRNA and protein present in a cell by direct action on DNA.

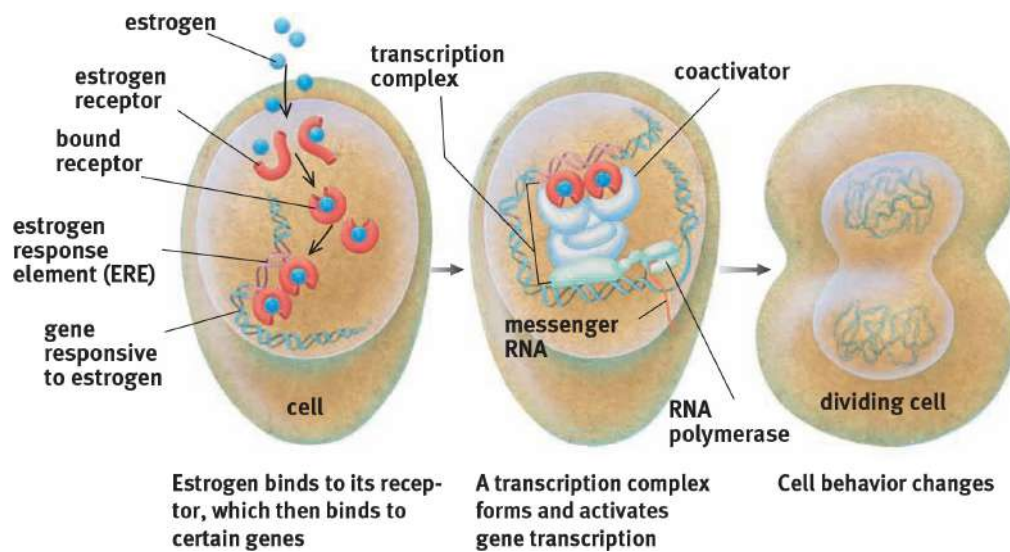


Figure 5.2. Mechanism of Action of a Steroid Hormone

Estrogen, like all steroid hormones, influences cell behavior by modifying transcription.



KEY CONCEPT

Peptide hormones have surface receptors and act via second messenger systems. Steroid hormones bind to intracellular receptors and function by binding to DNA to alter gene transcription.



MNEMONIC

Insulin is a peptide hormone, and it has to be released at every meal in order to be active. Thus, it has fast onset but is short-acting (like most peptide hormones). Estrogen and testosterone are steroid hormones

that promote sexual maturation. This is a slower, but longer-lasting change (as is true for most steroid hormones).

Steroid hormones are not water-soluble, so they must be carried by proteins in the bloodstream to travel around the body. Some of these proteins are very specific and carry only one hormone (such as *sex hormone-binding globulin*), while other proteins are nonspecific (such as **albumin**). Note that hormones are generally inactive while attached to a carrier protein and must dissociate from the carrier to function. Therefore, levels of carrier proteins can change the levels of *active* hormone. For example, some conditions increase the quantity of a protein that carries thyroid hormones, *thyroxine-binding globulin* (TBG). This causes the body to perceive a lower level of thyroid hormone because the increased quantity of TBG binds a larger proportion of the hormone, meaning there is less free hormone available.



REAL WORLD

During pregnancy, high levels of estrogen and progesterone cause increased production of TBG, thyroxine-binding globulin. In order to compensate, pregnant women secrete much higher levels of the thyroid hormones. Thus, in order to diagnose thyroid disease in a pregnant woman, different reference values must be used.

Amino Acid-Derivative Hormones

Finally, **amino acid–derivative hormones** are less common than peptide and steroid hormones, but include some of the most important hormones discussed in this chapter, including epinephrine, norepinephrine, triiodothyronine, and thyroxine. These hormones are derived from one or two amino acids, usually with a few additional modifications. For example, thyroid hormones are made from tyrosine modified by the addition of several iodine atoms.

The chemistry of this family of hormones is considerably less predictable and is one of the few instances where overt memorization may be the best strategy. For instance, the **catecholamines** (epinephrine and norepinephrine) bind to G protein–coupled receptors, while thyroid hormones bind intracellularly.



MCAT EXPERTISE

The mechanism of action of the amino acid–derivative hormones should be memorized because it is so unpredictable. Epinephrine and norepinephrine have extremely fast onset but are short-lived, like peptide hormones—think of an *adrenaline rush*. Thyroxine and triiodothyronine, on the other hand, have slower onset but a longer duration, like steroid hormones—they regulate metabolic rate over a long period of time.

CLASSIFICATION OF HORMONES BY TARGET TISSUE

Some hormones, known as **direct hormones**, are secreted and then act directly on a target tissue. For example, insulin released by the pancreas causes increased uptake of glucose by muscles. Other hormones, known as **tropic hormones**, require an intermediary to act. For example, as discussed in Chapter 2 of *MCAT Biology Review*, gonadotropin-releasing hormone (GnRH) from the hypothalamus stimulates the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH). LH then acts on the gonads to stimulate testosterone production in the male and estrogen production in the female. GnRH and LH do not cause direct changes in the physiology of muscle, bone, and hair follicles; rather, they stimulate the production of another hormone by another endocrine gland that acts on these target tissues. Tropic hormones usually originate in the brain and anterior pituitary gland, allowing for the coordination of multiple processes within the body.



MNEMONIC

Most peptide and amino acid–derivative hormones have names that end in **–in** or **–ine** (insul**in**, vasopress**in**, thyrox**ine**, triiodothyron**ine**, and so on). Most steroid hormones have names that end in **–one**, **–ol**, or **–oid** (testoster**one**, aldoster**one** and other mineralocortic**oids**, cortisol**ol** and other glucocortic**oids**, and so on). This is not exhaustive, but may help you identify the chemistry of a hormone on Test Day.

MCAT CONCEPT CHECK 5.1:

Before you move on, assess your understanding of the material with these questions.

1. Compare and contrast peptide and steroid hormones based on the following criteria:

Criterion	Peptide Hormones	Steroid Hormones
Chemical precursor		
Location of receptor		
Mechanism of action		
Method of travel in the bloodstream		
Speed of onset		
Duration of action		

2. How are amino acid–derivative hormones synthesized?

3. What is the difference between a direct and a tropic hormone?

- Direct hormone:

- Tropic hormone:

5.2 Endocrine Organs and Hormones

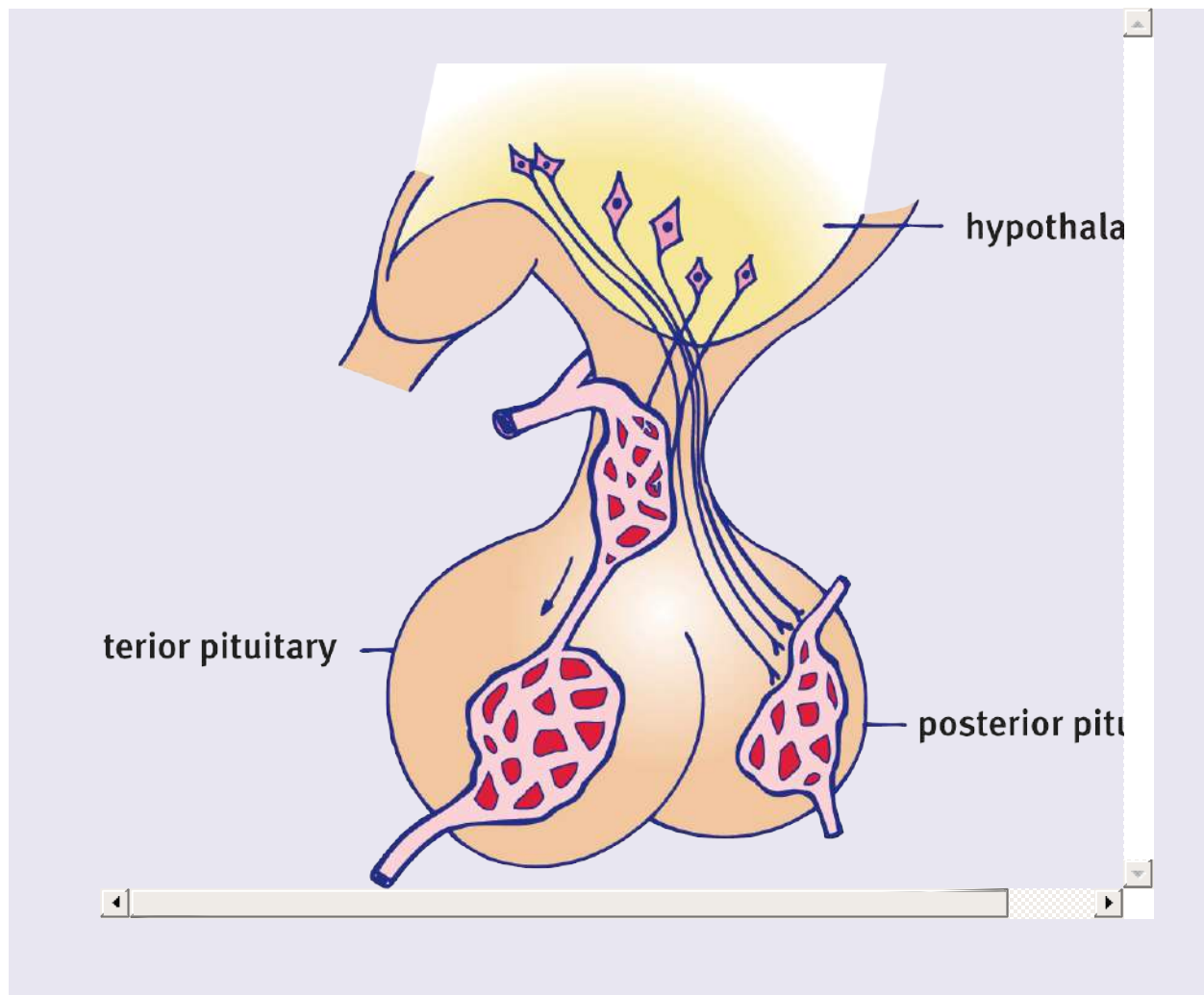
High-Yield



LEARNING OBJECTIVES

After Chapter 5.2, you will be able to:

- Recall the hormones involved in calcium homeostasis and their impact on blood calcium
- Identify the tissue that synthesizes catecholamines and the major catecholamines it produces
- List the pancreatic hormones, their regulators, and their impact on blood glucose concentration
- Recall details about the hormones involved in water homeostasis, including their production, action, and ultimate impact on blood volume and osmolarity
- Recall the releasing hormones produced by the hypothalamus, their pituitary targets, and the end result of each signaling pathway on the final target organ:



The hypothalamus, the pituitary, the thyroid, the parathyroid glands, the adrenal glands, the pancreas, the gonads (testes and ovaries), and the pineal gland are all endocrine glands, as shown in Figure 5.3. Each of these organs is capable of synthesizing and secreting one or more hormones. Furthermore, there are collections of cells within organs, such as the kidneys, gastrointestinal glands, heart, and thymus, that serve important endocrine roles. The organs in this second group are traditionally not called endocrine organs because hormone production is not their main function.

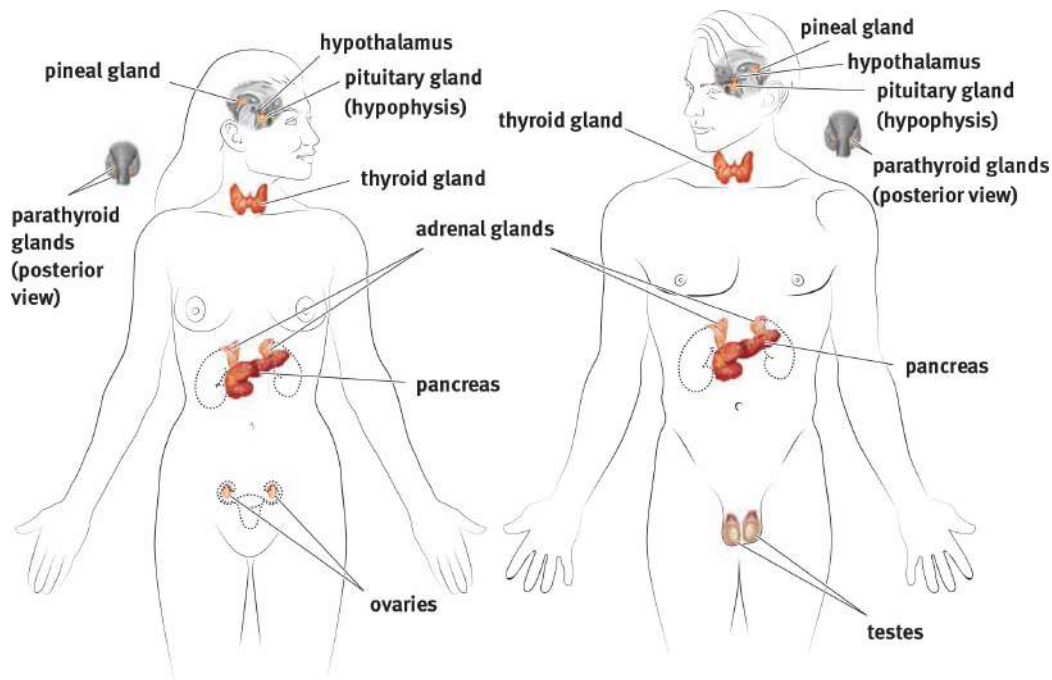


Figure 5.3. Organs of the Endocrine System

Endocrine organs produce hormones that are secreted into the bloodstream to act on distant target tissues.

Now that we have discussed the mechanisms through which hormones act and their classification, we can discuss the individual endocrine organs and the hormones secreted by each.

HYPOTHALAMUS

Let's begin with the **hypothalamus**, the bridge between the nervous and endocrine systems. By regulating the pituitary gland through tropic hormones, the hypothalamus is capable of having organism-wide effects. The hypothalamus is located in the forebrain, directly above the pituitary gland and below the thalamus (hence the name *hypothalamus*). Because the hypothalamus and the pituitary are close to each other, the hypothalamus

controls the pituitary through paracrine release of hormones into a portal system that directly connects the two organs. The hypothalamus receives input from a wide variety of sources. For example, a part of the hypothalamus called the *suprachiasmatic nucleus* receives some of the light input from the retinae and helps to control sleep–wake cycles. Other parts of the hypothalamus respond to increases in blood osmolarity. Still other parts of the hypothalamus regulate appetite and satiety.



BRIDGE

The hypothalamus contains a number of nuclei in its three sections, called the lateral, ventromedial, and anterior hypothalamus. These nuclei play roles in emotional experience, aggressive behavior, sexual behavior, metabolism, temperature regulation, and water balance. The parts of the hypothalamus are discussed in Chapter 1 of *MCAT Behavioral Sciences Review*.

The release of hormones by the hypothalamus is regulated by negative feedback. **Negative feedback** occurs when a hormone (or product) later in the pathway inhibits hormones (or enzymes) earlier in the pathway. This type of feedback maintains homeostasis and conserves energy by restricting production of substances that are already present in sufficient quantities. The hypothalamus and pituitary gland are inextricably linked. The pituitary gland has an anterior and posterior component, each with a unique interaction with the hypothalamus. We will discuss each in turn.

Interactions with the Anterior Pituitary

The hypothalamus secretes compounds into the **hypophyseal portal system**, which is a blood vessel system that directly connects the hypothalamus with the anterior pituitary, as shown in Figure 5.4. Thus, hormones released from the hypothalamus travel directly to the anterior pituitary and cannot be found in appreciable concentrations in the systemic circulation. Note that **hypophysis** is an alternative term for the pituitary. Once hormones have been released from the hypothalamus into this portal bloodstream, they travel down the pituitary stalk and bind to receptors in the anterior pituitary, stimulating the release of other hormones.

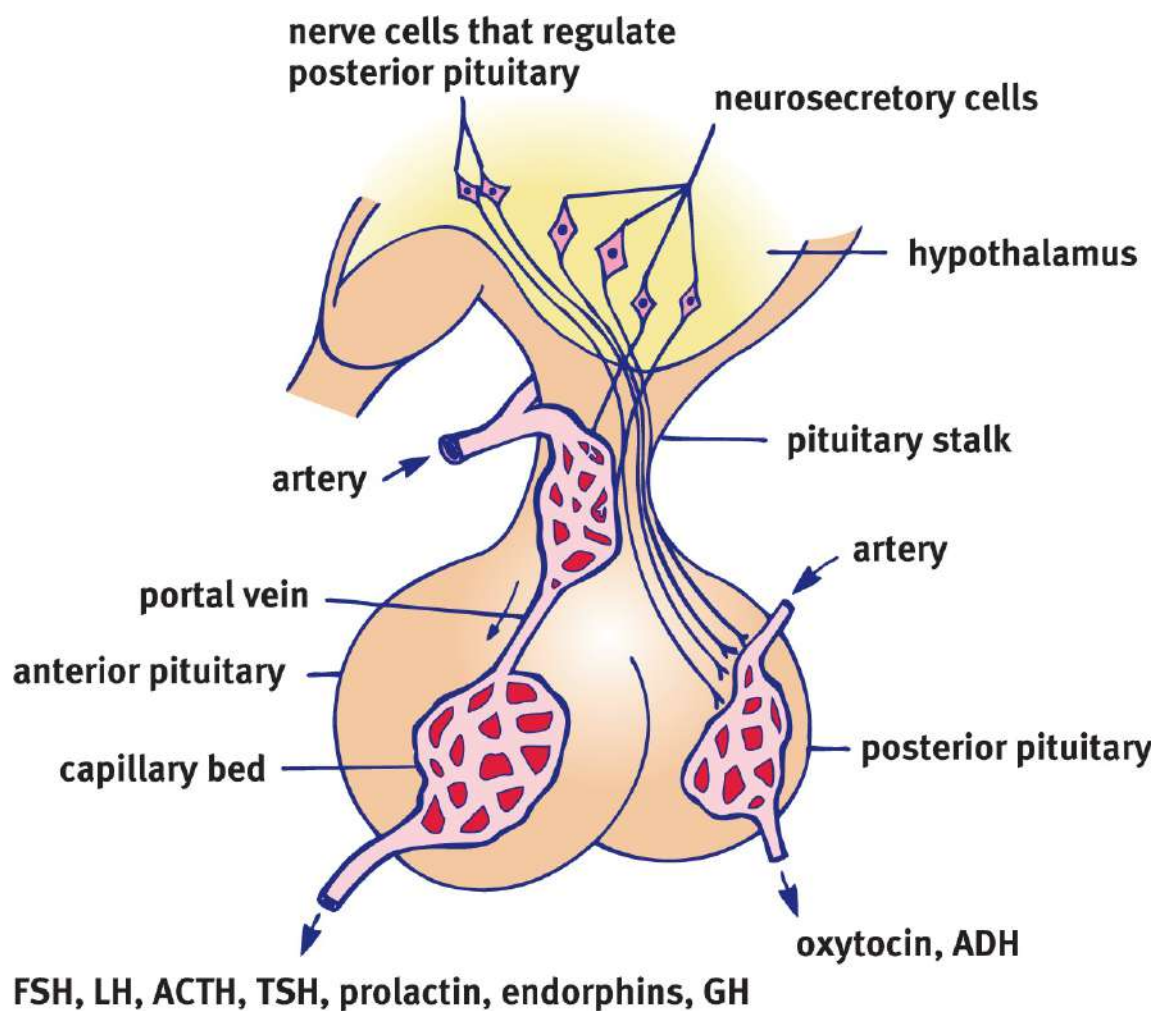


Figure 5.4. The Hypophyseal Portal System

A system of blood vessels connects the hypothalamus to the pituitary.

The hypothalamus secretes several different tropic hormones. The following indicates each hormone released by the hypothalamus and the hormone(s) released by the anterior pituitary in response:

- Gonadotropin-releasing hormone (GnRH) → follicle-stimulating hormone (FSH) and luteinizing hormone (LH)
- Growth hormone-releasing hormone (GHRH) → growth hormone (GH)
- Thyroid-releasing hormone (TRH) → thyroid-stimulating hormone (TSH)
- Corticotropin-releasing factor (CRF) → adrenocorticotrophic hormone (ACTH)



KEY CONCEPT

Although it seems as if the anterior pituitary has all the power in the endocrine system, it is controlled by the hypothalamus, which is located directly above it.

There is one exception to this pattern—prolactin-inhibiting factor (PIF), which is actually dopamine, is released by the hypothalamus and causes a *decrease* in prolactin secretion.



KEY CONCEPT

Whereas most of the hormones in the anterior pituitary require a factor from the hypothalamus to be released, prolactin is the

exception. As long as the hypothalamus releases PIF (which is actually dopamine), no prolactin will be released. It is the *absence* of PIF that allows prolactin to be released.

Each of the tropic hormones then causes the release of another hormone from an endocrine gland that has negative feedback effects. For example, release of CRF from the hypothalamus will stimulate the anterior pituitary to secrete ACTH. ACTH will then cause the adrenal cortex to increase the level of **cortisol** being secreted into the blood. However, cortisol is detrimental when levels become too high. To prevent excess cortisol secretion, cortisol inhibits the hypothalamus and anterior pituitary from releasing CRF and ACTH, respectively, as shown in Figure 5.5. This makes sense because CRF and ACTH have already accomplished their desired effect: getting more cortisol into the blood. What does this mean in terms of receptors in the hypothalamus and pituitary? Cortisol receptors must be present in these organs; otherwise, they wouldn't be able to recognize that cortisol levels had increased. Three-organ systems like these are commonly referred to as **axes**; for example, the hypothalamic–pituitary–adrenal (HPA) axis, the hypothalamic–pituitary–ovarian (HPO) axis, and so on.

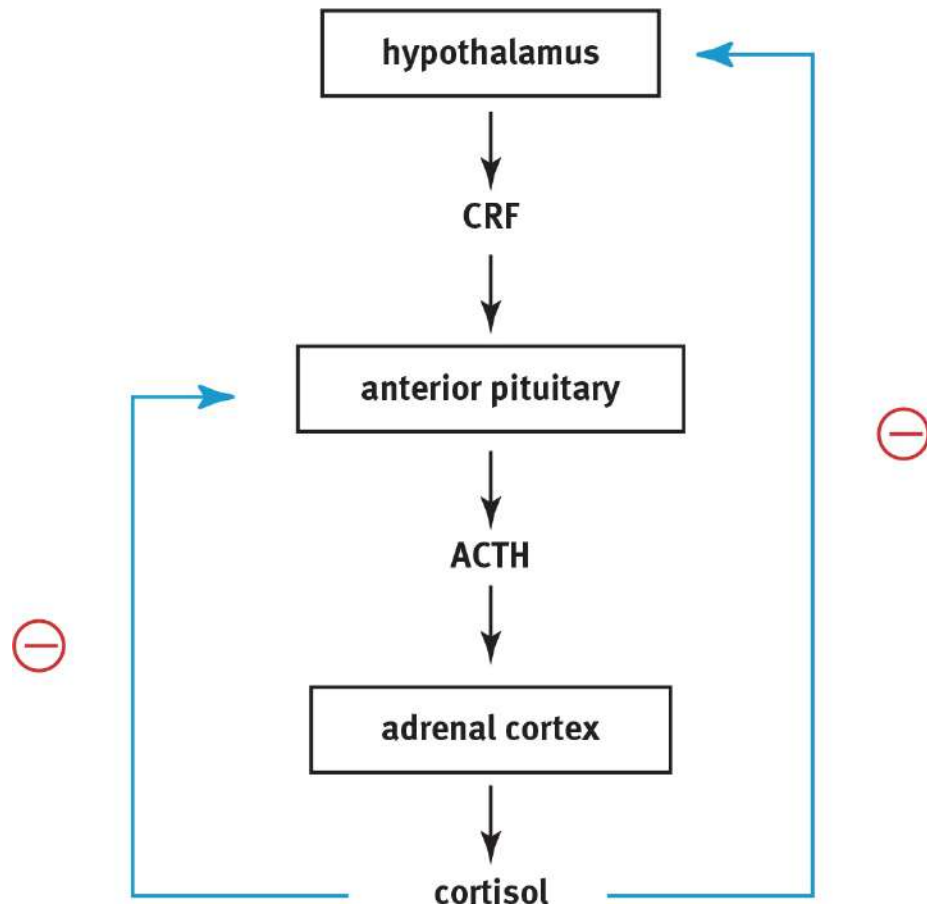


Figure 5.5. Hypothalamic Negative Feedback Mechanism

Interactions with the Posterior Pituitary

The posterior pituitary does not receive tropic hormones through the hypophyseal portal system. Rather, neurons in the hypothalamus send their axons down the pituitary stalk directly into the posterior pituitary, which can then release oxytocin and antidiuretic hormone. **Oxytocin** stimulates uterine contractions during labor, as well as milk letdown during lactation. There is evidence that oxytocin is also involved in bonding behavior. **Antidiuretic hormone (ADH, also called vasopressin)** increases reabsorption of water in the collecting ducts of the kidneys. ADH is secreted in response to increased plasma osmolarity, or increased concentration of solutes within the blood.



REAL WORLD

A tumor of the pituitary gland may result in compression of the portal system that connects the pituitary to the hypothalamus. This will block the ability of PIF to reach the pituitary and exert its effect. Thus, more prolactin will be released. This may result in lactation (galactorrhea), even in males. Thus, milk production in a male or nonpregnant female should lead a physician to suspect the presence of a pituitary tumor.

ANTERIOR PITUITARY

As described earlier, the pituitary gland is divided into anterior and posterior sections. Because this distinction has already been covered, our discussion of the pituitary gland here will focus on the hormones released by each section.



MNEMONIC

Products of the anterior pituitary:

FLAT PEG

- **F**ollicle-stimulating hormone (FSH)
- **L**uteinizing hormone (LH)
- **A**drenocorticotrophic hormone (ACTH)
- **T**hyroid-stimulating hormone (TSH)
- **P**rolactin

- **Endorphins**
- **Growth hormone (GH)**

Additionally, the four hormones in **FLAT** are all tropic hormones, while the three hormones in **PEG** are all direct hormones.

The **anterior pituitary** synthesizes and secretes seven different products. Four of these are tropic hormones, while the other three are direct hormones.

Tropic Hormones

We are going to mention the tropic hormones only briefly here. These hormones work by causing the release of another hormone at the organ level. Thus, we will discuss the tropic hormones in tandem with the endocrine organ on which they act. The release of both follicle-stimulating hormone (FSH) and luteinizing hormone (LH) is stimulated by gonadotropin-releasing hormone (GnRH) from the hypothalamus. These two hormones act on the gonads (testes and ovaries). The release of adrenocorticotrophic hormone (ACTH) is stimulated by corticotropin-releasing factor (CRF) from the hypothalamus; ACTH acts on the adrenal cortex. The release of thyroid-stimulating hormone (TSH) is stimulated by thyroid-releasing hormone (TRH) from the hypothalamus; TSH acts on the thyroid.

Direct Hormones

Prolactin is more important in females than in males; it stimulates milk production in the **mammary glands**. Milk production in the male is always pathologic. During pregnancy, estrogen and progesterone levels are high. In

addition, prolactin, a hormone that increases milk production, is also secreted by the anterior pituitary. Prolactin is an unusual hormone in that the release of dopamine from the hypothalamus *decreases* its secretion. The high levels of estrogen and progesterone allow for the development of milk ducts in preparation for lactation, but it is not until shortly after the expulsion of the placenta, when estrogen, progesterone, and dopamine levels drop, that the block on milk production is removed and lactation actually begins.

Milk ejection occurs when the newborn infant latches on to the breast. Nipple stimulation causes activation of the hypothalamus, resulting in two different reactions. First, oxytocin is released from the posterior pituitary, resulting in contraction of the smooth muscle of the breast and ejection of milk through the nipple. Second, the hypothalamus stops releasing dopamine onto the anterior pituitary, which allows prolactin release, causing production of milk and regulation of the milk supply.

Endorphins decrease the perception of pain. For example, after completing a marathon, many people will say they are on an endorphin “high” or “rush.” Endorphins mask the pain from having run 26.2 miles and can even induce a sense of euphoria. Many pharmaceutical agents, such as morphine, mimic the effect of these naturally occurring painkillers.

Growth hormone (GH) is named for exactly what it does: it promotes the growth of bone and muscle. This sort of growth is energetically expensive and requires large quantities of glucose. Growth hormone prevents glucose uptake in certain tissues (those that are not growing) and stimulates the breakdown of fatty acids. This increases the availability of glucose overall, allowing muscle and bone to use it. GH release is stimulated by growth hormone–releasing hormone (GHRH) from the hypothalamus.

Bone growth originates in special regions of the bone known as epiphyseal plates, which seal shut during puberty. An excess of GH released in childhood (before this closure) can cause **gigantism**, and a deficit results in **dwarfism**. In adults, the situation is slightly different. Because the long bones are sealed, GH still has an effect, but it is primarily in the smaller bones. The resulting medical condition is known as **acromegaly**. The bones most commonly affected are those in the hands, feet, and head. Patients with acromegaly tend to seek medical help because they have had to buy larger shoes, cannot wear their rings, and can no longer fit into their hats.

POSTERIOR PITUITARY

The **posterior pituitary** contains the nerve terminals of neurons with cell bodies in the hypothalamus. As mentioned earlier, the posterior pituitary receives and stores two hormones produced by the hypothalamus: ADH and oxytocin.

ADH is secreted in response to low blood volume (as sensed by baroreceptors) or increased blood osmolarity (as sensed by osmoreceptors), shown in Figure 5.6. Its action is at the level of the collecting duct, where it increases the permeability of the duct to water. This increases the reabsorption of water from the filtrate in the nephron. This results in greater retention of water, which results in increased blood volume and higher blood pressure.

Oxytocin is secreted during childbirth and allows for coordinated contraction of uterine smooth muscle. Its secretion may also be stimulated by suckling, as it promotes milk ejection through contraction of smooth muscle in the breast. Finally, oxytocin may be involved in bonding behavior. Oxytocin is unusual in that it has a **positive feedback** loop: the release of oxytocin promotes uterine

contraction, which promotes more oxytocin release, which promotes stronger uterine contractions, and so on. Positive feedback loops can usually be identified by a “spiraling forward” scheme and usually have a definitive endpoint—in this case, delivery.



KEY CONCEPT

The two hormones released from the posterior pituitary are actually synthesized in the hypothalamus and simply released from the posterior pituitary gland. The posterior pituitary does not synthesize any hormones itself.

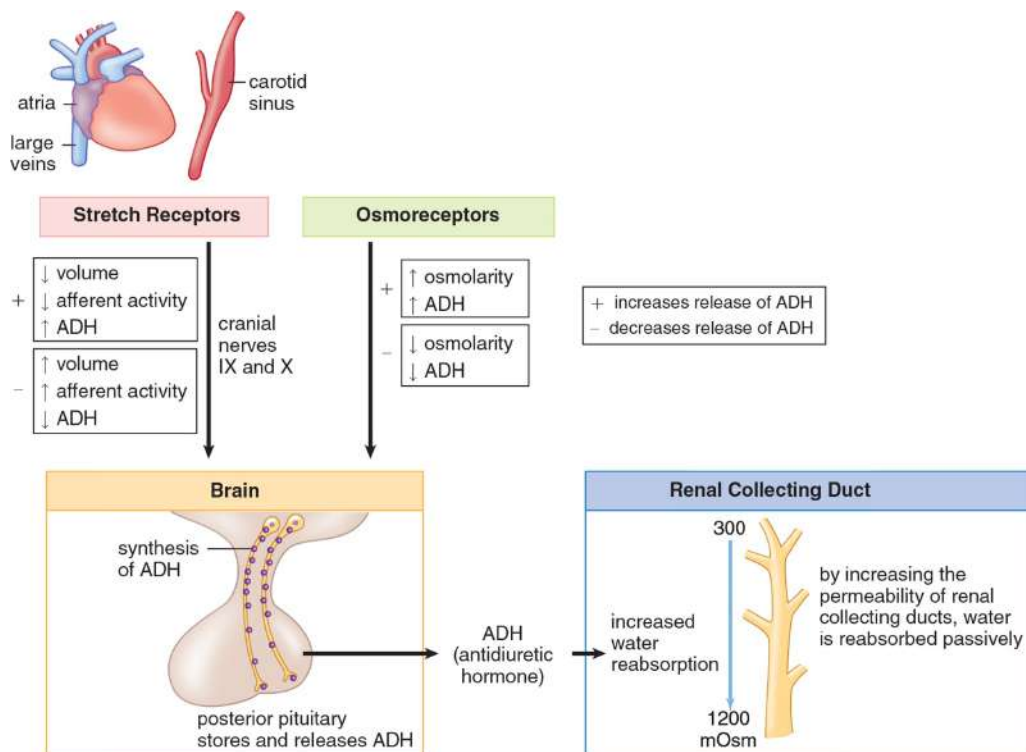


Figure 5.6. Antidiuretic Hormone (ADH)

THYROID

The **thyroid** is controlled by **thyroid-stimulating hormone** from the anterior pituitary. The thyroid is on the front surface of the trachea; it can be palpated (felt) as an organ near the base of the neck that moves up and down with swallowing. The thyroid has two major functions: setting basal metabolic rate and promoting calcium homeostasis. It mediates the first effect by releasing triiodothyronine (T_3) and thyroxine (T_4), while it carries out the second effect through the release of calcitonin.

Triiodothyronine and Thyroxine

Triiodothyronine (T_3) and **thyroxine** (T_4) are both produced by the iodination of the amino acid tyrosine in the **follicular cells** of the thyroid. The numbers 3 and 4 refer to the number of iodine atoms attached to the tyrosine. Thyroid hormones are capable of resetting the basal metabolic rate of the body by making energy production more or less efficient, as well as altering the utilization of glucose and fatty acids. Increased amounts of T_3 and T_4 will lead to increased cellular respiration. This leads to increased protein and fatty acid turnover by speeding up both synthesis and degradation of these compounds. High plasma levels of thyroid hormones will lead to decreased TSH and TRH synthesis; negative feedback prevents excessive secretion of T_3 and T_4 , as shown in Figure 5.7.

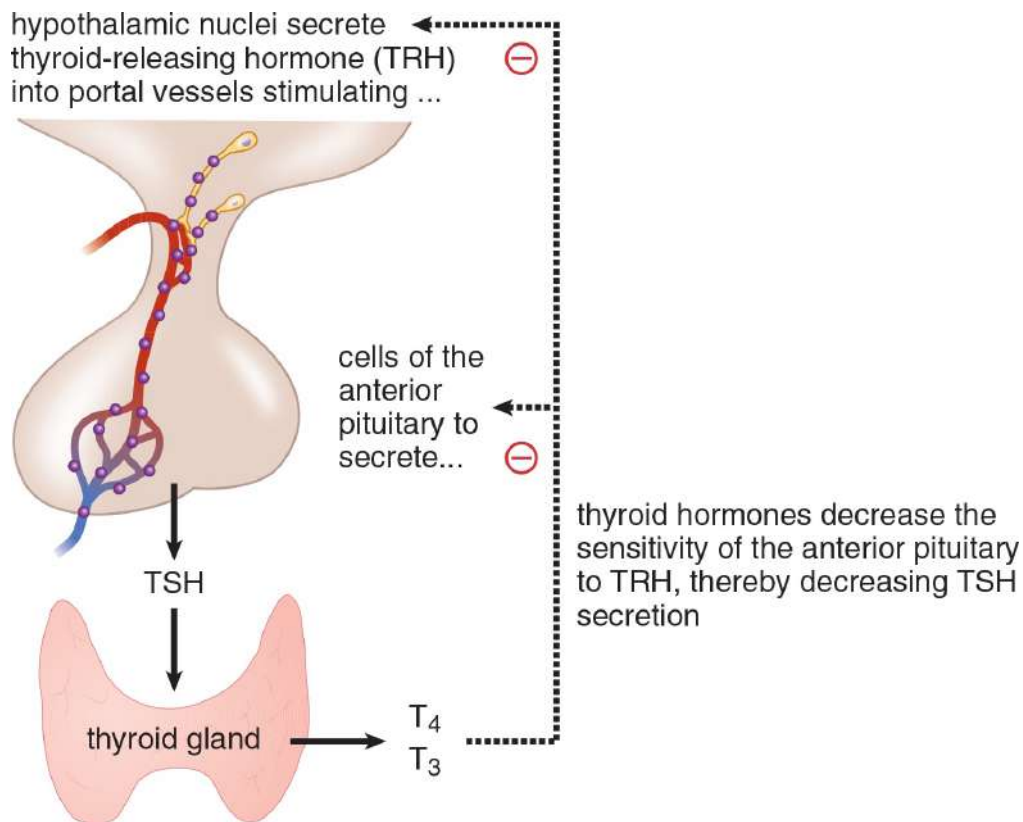


Figure 5.7. Thyroid Hormone Negative Feedback Mechanism

A deficiency of iodine or inflammation of the thyroid may result in **hypothyroidism**, in which thyroid hormones are secreted in insufficient amounts or not at all. The condition is characterized by lethargy, decreased body temperature, slowed respiratory and heart rate, cold intolerance, and weight gain. Thyroid hormones are required for appropriate neurological and physical development in children. Most children are tested at birth for appropriate levels because a deficiency will result in intellectual disability and developmental delay (**cretinism**).

An excess of thyroid hormone, which may result from a tumor or thyroid overstimulation, is called **hyperthyroidism**. We can predict the clinical course of this syndrome by considering the opposite of each of the effects seen in

hypothyroidism: heightened activity level, increased body temperature, increased respiratory and heart rate, heat intolerance, and weight loss.

Calcitonin

If we were to examine thyroid tissue under a light microscope, we would see two distinct cell populations within the gland. Follicular cells produce thyroid hormones and **C-cells** (also called **parafollicular cells**) produce **calcitonin**. Calcitonin decreases plasma calcium levels in three ways: by increasing calcium excretion from the kidneys, by decreasing calcium absorption from the gut, and by increasing storage of calcium in the bone. High levels of calcium in the blood stimulate secretion of calcitonin from the C-cells.



MNEMONIC

Calcitonin **tones** down calcium levels in the blood.

PARATHYROID GLANDS

The parathyroids are four small pea-sized structures that sit on the posterior surface of the thyroid. The hormone produced by the parathyroid glands is aptly named **parathyroid hormone (PTH)**. PTH serves as an antagonistic hormone to calcitonin, raising blood calcium levels; specifically, it decreases excretion of calcium by the kidneys, increases absorption of calcium in the gut (via vitamin D), and increases bone resorption, thereby freeing up calcium, as shown in Figure 5.8. Like the hormones we have already seen, PTH is also subject to feedback inhibition. As levels of plasma calcium rise, PTH secretion

is decreased. Parathyroid hormone also promotes phosphorus homeostasis by increasing the resorption of phosphate from bone and reducing reabsorption of phosphate in the kidney (thus promoting its excretion in the urine).



KEY CONCEPT

Calcium is an exceptionally important ion. The critically important functions of calcium include:

- Bone structure and strength
- Release of neurotransmitters from neurons
- Regulation of muscle contraction
- Clotting of blood (calcium is a cofactor)

In addition, calcium also plays a role in cell movement and exocytosis of cellular materials.

PTH also activates **vitamin D**, which is required for the absorption of calcium and phosphate in the gut. The overall effect of parathyroid hormone, therefore, is a significant increase in blood calcium levels with little effect on phosphate (the absorption of phosphate in the gut and its excretion in the kidney somewhat cancel each other).



KEY CONCEPT

Just like glucagon and insulin, PTH and calcitonin are antagonistic to each other. We should think of these hormones as a pair with the primary function of regulating calcium levels in the blood. PTH increases serum calcium levels, whereas calcitonin decreases calcium levels.

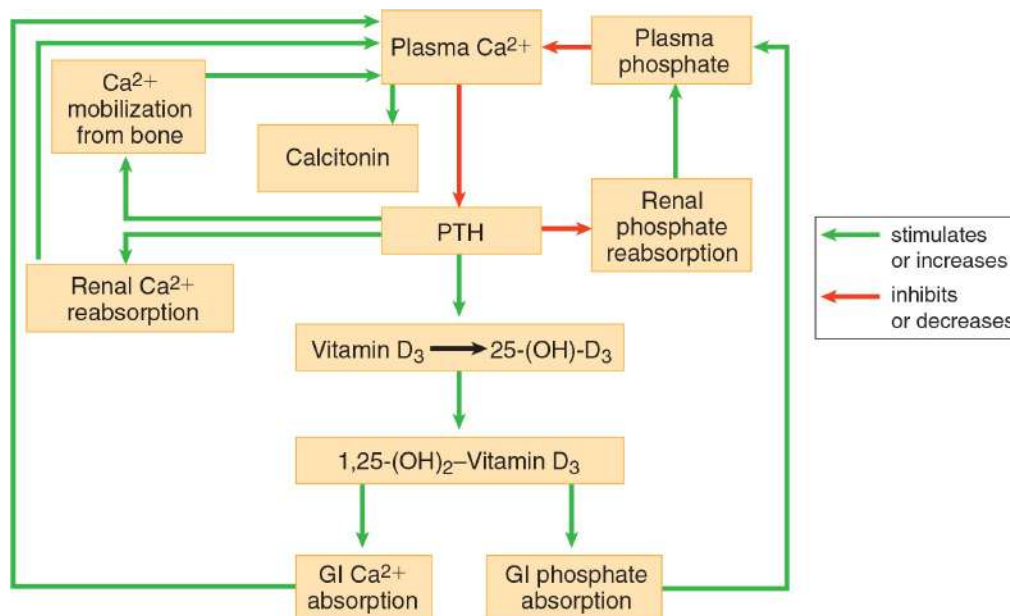


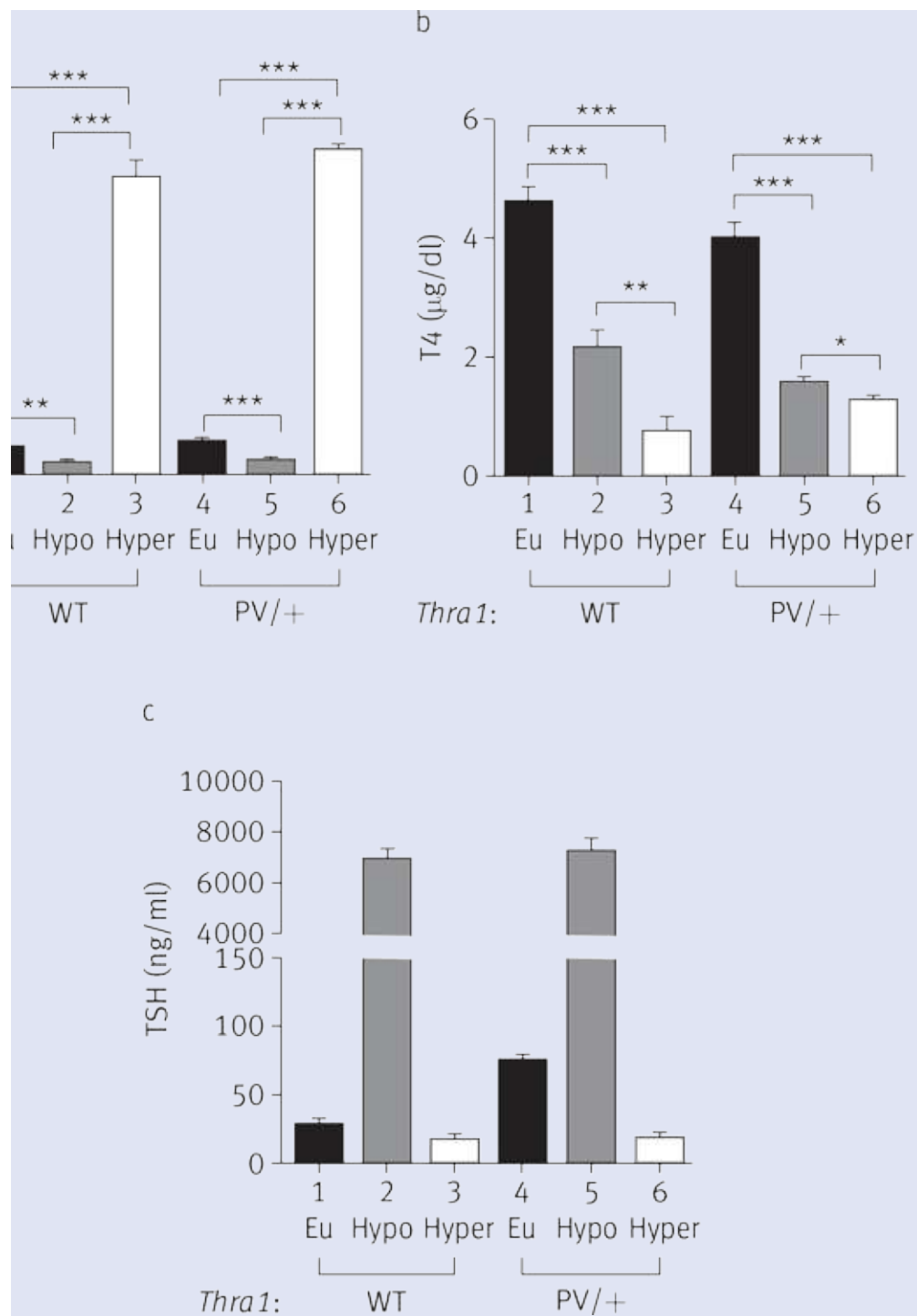
Figure 5.8. Calcium and Phosphorus Homeostasis

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

mutations of the THRA (thyroid hormone receptor alpha) gene exhibit classical hypothyroidism. Researchers created a mouse expressing a mutated TRα1 (denoted V/+ mouse) that faithfully reproduces the classical hypothyroidism seen in

*Topic:
hypothyroidism;
some outside
knowledge of
the endocrine
system might be
necessary.*

ominant negative mutant and cannot bind T3. Researchers then rendered mice hypothyroidic by treating them with propylthiouracil (PTU).	<i>Specific mutation of THRA cannot bind T3 and mice with mutation have hypothyroidism</i>
cation used to treat hyperthyroidism by inhibiting the enzyme thyroperoxidase. Mice were then treated with T3 to induce symptoms of hyperthyroidism. Results are shown in Figure 1.	<i>PTU: medicine to treat hyperthyroidism, used to induce hypothyroidism in these mice</i>
Findings are consistent with earlier reports that the feedback loop in the pituitary-thyroid axis is not affected by expressing TRα1PV mutant in Thra1PV/+ mice.	<i>Findings are consistent with idea that the feedback loop is not affected</i>
	<i>IVs: Eu, Hypo, Hyper</i> <i>DVs: TT3, TT4, TSH</i> <i>Trends:</i> <i>Euthyroid means the thyroid is kept normal, so this is the control. The changes in T3, T4, and TSH are pretty consistent</i>



for each condition (Eu, Hypo, Hyper) between WT and PV/+ mice.

roid experiment results. Eu = euthyroid condition.

k, S., Han, C. R., Park, J. W., Zhao, L., Zhu, X., Willingham, M., ... Cheng, S. Y. (2017). Defective erythropoiesis caused by thyroid hormone receptor α gene. *PLoS Genetics*, 13(9), e1006991. doi:10.1371/journal.pgen.1006991.

Why are TSH levels elevated in mice given propylthiouracil (PTU)?

A question like this is asking us to explain the underlying biological cause for an observed phenomenon. The MCAT often tests our ability to reason with a concept or mechanism that we are already familiar with. For this question, the stated change to the system is “propylthiouracil” and the effect observed is “elevated TSH levels”. It is now our job to determine how the two are related.

First, we'll want to identify what information we'll need, both from the passage and from our science background. MCAT questions will often intentionally include terms that we aren't familiar with. The key to answering these questions, even if we're not familiar with the phenomenon described, is finding the relevant information from the passage and using that information in conjunction with our scientific understanding to solve the problem. We already identified the change to the system in this question as the application of PTU. According to the passage, PTU is a medication that inhibits thyroperoxidase and it is used to treat hyperthyroidism. In the experiment, they used PTU to “render[ed] Thra1PV/+ mice hypothyroidic”. If we look to Figure 1A, we can see that in the T3 and T4 graphs, the hypothyroid (gray) bar is significantly smaller than the euthyroid (black) bar. If we weren't sure what euthyroid referred to, we could infer that, given that the other two categories in each graph are hypothyroid and hyperthyroid, this must be wild-type or normal function. Based on these data and the context from the passage, it is reasonable to infer that PTU is a drug that functions by lowering the production of thyroid hormone (T3 and T4).

Now we must consider how having low thyroid hormones would be related to elevated TSH levels. TSH is a tropic hormone released from the anterior

pituitary, and it is regulated by a negative feedback mechanism involving thyroid hormone. Specifically, we should know that the hypothalamus releases thyroid-releasing hormone (TRH), which causes the anterior pituitary to release thyroid-stimulating hormone (TSH). TSH acts upon the thyroid to stimulate the release of thyroid hormone triiodothyronine (T3) and thyroxine (T4). T3 and T4 then act as negative feedback on the hypothalamus and the pituitary gland, lowering TRH and TSH levels. We can reason that, if T3 and T4 levels were low, TRH and TSH would be continuously produced, as they would not be inhibited by the presence of T3 and T4.

The mice in the experiment have normally functioning endocrine glands prior to treatment. We can always assume this to be the case unless the passage explicitly states otherwise: subjects should be assumed to be normal, healthy individuals. When PTU is administered, the thyroid hormone levels in the mice decreased. As a result, there is now less thyroid hormone in circulation. As we predicted, there is now little to no negative feedback on the hypothalamus and the anterior pituitary gland. TRH and TSH levels in the mice would be elevated as a result of this lack of feedback. Normally, that would work to create more T3 and T4, but in this case, the mice are under the influence of PTU, and despite TSH stimulation, the thyroid will not be able to produce more T3 and T4. The cycle will continue, and more TRH and TSH will be secreted.

In short, the reason why TSH level is elevated in mice administered with PTU is that there is no T3 or T4 to activate the negative feedback mechanism on TSH and TRH, which “tricks” the brain into secreting more TSH (and TRH).

ADRENAL CORTEX

The **adrenal glands** are located on top of the kidneys. Adrenal actually translates to *near or next to the kidney*. Each adrenal gland consists of a cortex and a medulla. This distinction is more than anatomical. Each part of the gland is responsible for the secretion of different hormones. The **adrenal cortex** secretes **corticosteroids**. These are steroid hormones that can be divided into three functional classes: **glucocorticoids**, **mineralocorticoids**, and **cortical sex hormones**.

Glucocorticoids

Glucocorticoids are steroid hormones that regulate glucose levels. In addition, these hormones also affect protein metabolism. The two glucocorticoids most likely to be tested on the MCAT are **cortisol** and **cortisone**. These hormones raise blood glucose by increasing gluconeogenesis and decreasing protein synthesis. Cortisol and cortisone can also decrease inflammation and immunologic responses. Cortisol is known as a stress hormone because it is released in times of physical or emotional stress. This increases blood sugar and provides a ready source of fuel in case the body must react quickly to a dangerous stimulus.



REAL WORLD

Many people who suffer from joint pain are treated with glucocorticoid injections into the joint space to decrease inflammation. In addition, glucocorticoids are used to treat systemic inflammation caused by allergic reactions or autoimmune disease.

Glucocorticoid release is under the control of adrenocorticotrophic hormone (ACTH) as described earlier. Corticotropin-releasing factor (CRF) from the hypothalamus promotes release of adrenocorticotrophic hormone (ACTH) from the anterior pituitary, which promotes release of glucocorticoids from the adrenal cortex.

Mineralocorticoids

Mineralocorticoids are used in salt and water homeostasis; their most profound effects are on the kidneys. The most noteworthy mineralocorticoid is **aldosterone**, which increases sodium reabsorption in the distal convoluted tubule and collecting duct of the nephron. Water follows the sodium cations into the bloodstream, increasing blood volume and pressure. Since water and sodium ions flow together, plasma osmolarity remains unchanged; this is in contrast to ADH, which only increases water reabsorption (decreasing plasma osmolarity). Aldosterone also decreases the reabsorption of potassium and hydrogen ions in these same segments of the nephron, promoting their excretion in the urine.



REAL WORLD

In addition to stimulating the secretion of aldosterone, which increases blood volume and hence blood pressure, angiotensin II also increases blood pressure directly through a powerful vasoconstrictive effect and increases heart rate. Angiotensin-converting enzyme (ACE) inhibitors block the conversion of angiotensin I to angiotensin II, inhibiting vasoconstriction. Therefore, ACE inhibitors are frequently prescribed for the treatment of high blood pressure and congestive

heart failure. In the latter case, vasodilation helps reduce vascular resistance, decreasing the pressure against which the failing heart must pump. ACE inhibitors' generic names end with *-pril* (*lisinopril*, *enalapril*, *ramipril*).

Unlike the glucocorticoids, aldosterone is primarily under the control of the **renin-angiotensin-aldosterone** system, as shown in Figure 5.9. Decreased blood pressure causes the **juxtaglomerular cells** of the kidney to secrete **renin**, which cleaves an inactive plasma protein, **angiotensinogen**, to its active form, **angiotensin I**. Angiotensin I is then converted to **angiotensin II** by **angiotensin-converting enzyme (ACE)** in the lungs. Angiotensin II stimulates the adrenal cortex to secrete aldosterone. Once blood pressure is restored, there is a decreased drive to stimulate renin release, thus serving as the negative feedback mechanism for this system.

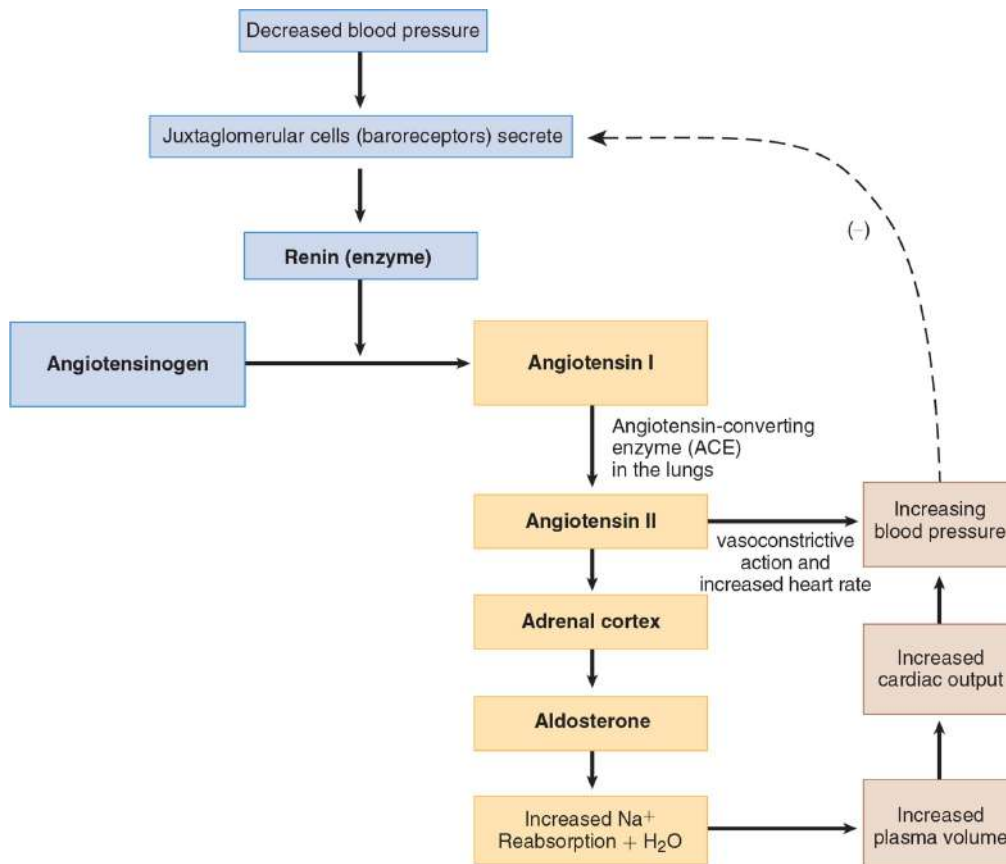


Figure 5.9. The Renin-Angiotensin-Aldosterone System

Cortical Sex Hormones

The adrenal glands also make **cortical sex hormones (androgens and estrogens)**. Because the testes already secrete large quantities of androgens, adrenal testosterone plays a relatively small role in male physiology. But because the ovaries secrete far smaller amounts of androgens, females are much more sensitive to disorders of cortical sex hormone production. For example, certain enzyme deficiencies in the synthetic pathways of other adrenal cortex hormones result in excess androgen production in the adrenal cortex. Such a deficiency would result in no obvious phenotypic effects in a male fetus; however, a genotypic female may be born with ambiguous or

masculinized genitalia due to the presence of excess cortical sex hormones. Males can be affected by similar disorders if they lead to excessive production of estrogens.



MNEMONIC

Functions of the corticosteroids:

The 3 **S**'s

- **S**alt (mineralocorticoids)
- **S**ugar (glucocorticoids)
- **S**ex (cortical sex hormones)

ADRENAL MEDULLA

Nestled inside the adrenal cortex is the adrenal medulla. A derivative of the nervous system, this organ is responsible for the production of the sympathetic hormones **epinephrine** and **norepinephrine**. The specialized nerve cells in the medulla are capable of secreting these compounds directly into the bloodstream. Both epinephrine and norepinephrine are amino acid-derivative hormones that belong to a larger class of molecules known as **catecholamines**.

Much like the sympathetic component of the autonomic nervous system, the hormones released from the adrenal medulla have diverse system-wide effects, all centered on the fight-or-flight response. Epinephrine can increase the breakdown of glycogen to glucose (glycogenolysis) in both liver and

muscle, as well as increase the basal metabolic rate. Both epinephrine and norepinephrine will increase heart rate, dilate the bronchi, and shunt blood flow to the systems that would be used in a sympathetic response. That is, there is vasodilation of blood vessels leading to increased bloodflow to the skeletal muscle, heart, lungs, and brain. Concurrently, vasoconstriction decreases bloodflow to the gut, kidneys, and skin. Note that the stress response involves both cortisol and epinephrine. Classically, cortisol is understood to mediate long-term (slow) stress responses, while catecholamines are understood to control short-term (fast) stress responses. In fact, cortisol actually increases the synthesis of catecholamines as well, resulting in an increase in catecholamine release.

PANCREAS

The pancreas has both exocrine and endocrine functions. Exocrine tissues secrete substances directly into ducts; the pancreas produces a number of digestive enzymes, as discussed in Chapter 9 of *MCAT Biology Review*. From an endocrine standpoint, small clusters of hormone-producing cells are grouped together into **islets of Langerhans** throughout the pancreas, as shown in Figure 5.10. Islets contain three distinct types of cells: **alpha (α)**, **beta (β)**, and **delta (δ) cells**. Each cell type secretes a different hormone: α -cells secrete **glucagon**, β -cells secrete **insulin**, and δ -cells secrete **somatostatin**.

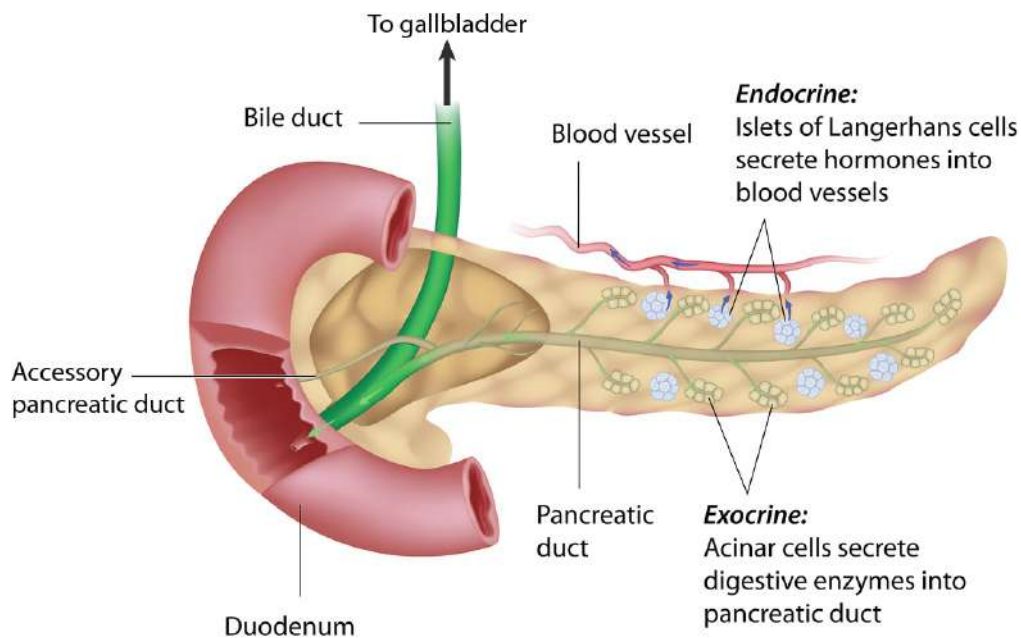


Figure 5.10. Anatomy of the Pancreas

Islets of Langerhans are scattered throughout the pancreas and carry out the endocrine function of the organ.

Glucagon

Glucagon is secreted during times of fasting. When glucose levels are low, glucagon increases glucose production by triggering glycogenolysis, gluconeogenesis, and the degradation of protein and fat. In addition to low blood glucose concentrations, certain gastrointestinal hormones (such as cholecystikinin and gastrin) increase glucagon release from α -cells. When blood glucose concentrations are high, glucagon release is inhibited.



MNEMONIC

Glucagon levels are high when **glucose** is **gone**.

Insulin

Insulin is antagonistic to glucagon and is therefore secreted when blood glucose levels are high, as shown in Figure 5.11. Insulin induces muscle and liver cells to take up glucose and store it as glycogen for later use. In addition, because it is active when glucose levels are high, insulin stimulates anabolic processes such as fat and protein synthesis.

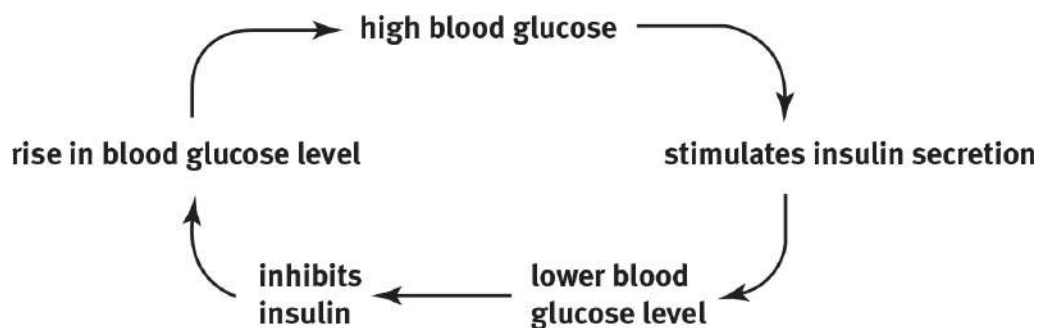


Figure 5.11. Insulin Has a Direct Relationship with Blood Glucose Concentration

When blood glucose levels are high, insulin levels rise, causing cells to take up glucose from the blood; when blood glucose levels are low, insulin levels remain low as well.

In excess, insulin will cause **hypoglycemia**, which is characterized by low blood glucose concentration. Underproduction, insufficient secretion, or insensitivity to insulin all can result in **diabetes mellitus**, which is clinically characterized by **hyperglycemia** (excess glucose in the blood). In the kidneys, excessive glucose in the filtrate will overwhelm the nephron's ability to reabsorb glucose, resulting in its presence in the urine. Because it is an osmotically active particle and does not readily cross the cell membrane, the presence of glucose in the filtrate leads to excess excretion of water and an

increase—that is sometimes quite dramatic—in the urine volume. Therefore, diabetics often report **polyuria** (increased frequency of urination) and **polydipsia** (increased thirst). There are two types of diabetes mellitus. **Type I** (insulin-dependent) diabetes mellitus is caused by autoimmune destruction of the β -cells of the pancreas, resulting in low or absent insulin production. Type I diabetics require regular injections of insulin to prevent hyperglycemia and to permit uptake of glucose into cells. **Type II** (non-insulin-dependent) diabetes mellitus is the result of receptor-level resistance to the effects of insulin. Type II diabetes mellitus is partially inherited and partially due to environmental factors, such as high-carbohydrate diets and obesity. Certain pharmaceutical agents can be taken orally to help the body more effectively use the insulin it produces. These individuals require insulin only when their bodies can no longer control glucose levels, even when aided by these medications.



KEY CONCEPT

Insulin decreases plasma glucose. Glucagon increases plasma glucose. Growth hormone, glucocorticoids, and epinephrine are also capable of increasing plasma glucose. These hormones that raise blood glucose levels are commonly called counterregulatory hormones.

Somatostatin

Somatostatin is an inhibitor of both insulin and glucagon secretion. High blood glucose and amino acid concentrations stimulate its secretion.

Somatostatin is also produced by the hypothalamus, where it decreases growth hormone secretion in addition to its effects on insulin and glucagon.

GONADS

Because reproductive endocrinology is discussed in detail in Chapter 2 of *MCAT Biology Review*, we offer only a brief overview in this chapter. The **testes** secrete testosterone in response to stimulation by gonadotropins (LH and FSH). Testosterone causes sexual differentiation of the male during gestation and also promotes the development and maintenance of secondary sex characteristics in males, such as axillary and pubic hair, deepening of the voice, and muscle growth.

The **ovaries** secrete estrogen and progesterone in response to gonadotropins. Estrogen is involved in development of the female reproductive system during gestation and also promotes the development and maintenance of secondary sex characteristics in females, such as axillary and pubic hair, breast growth, and body fat redistribution. These two steroid hormones also govern the menstrual cycle as well as pregnancy.

PINEAL GLAND

The **pineal gland** is located deep within the brain, where it secretes the hormone **melatonin**. The precise mechanism of this hormone is unclear, although it has been demonstrated to be involved in **circadian rhythms**. Blood levels of melatonin are at least partially responsible for the sensation of sleepiness. The pineal gland receives projections directly from the retina, but

is not involved in vision; it is hypothesized that the pineal gland responds to decreases in light intensity by releasing melatonin.



REAL WORLD

Melatonin has enjoyed somewhat of a wonder drug status over the past decade as it is touted as a remedy for everything from jet lag to aging. Certainly, one effect of melatonin is to cause profound drowsiness, hence its use in “resetting” one’s daily rhythm. Melatonin is available over the counter in health food stores, but the long-term effects of melatonin therapy are currently unknown.

OTHER ORGANS

In addition to the organs listed above, specific cells and tissues in other organs exhibit endocrine functions. In the gastrointestinal tract, endocrine tissue can be found in both the stomach and intestine. Many gastrointestinal peptides have been identified; important ones include secretin, gastrin, and cholecystokinin. The specific functions of these hormones are discussed in Chapter 9 of *MCAT Biology Review*, but as we might expect for the digestive system, the stimulus for release of most of these peptides is the presence of specific nutrients.

The kidneys play a role in water balance. As mentioned earlier, ADH increases water permeability in the collecting duct, and the renin–angiotensin–aldosterone system increases sodium and water reabsorption in the distal convoluted tubule and collecting duct. The kidneys also produce

erythropoietin, which stimulates bone marrow to increase production of erythrocytes (red blood cells). It is secreted in response to low oxygen levels in the blood.

The heart releases **atrial natriuretic peptide (ANP)** to help regulate salt and water balance. When cells in the atria are stretched from excess blood volume, they release ANP. This hormone promotes excretion of sodium and therefore increases urine volume. This effect is functionally antagonistic to aldosterone because it lowers blood volume and pressure, and has no effect on blood osmolarity.

The thymus, located directly behind the sternum, releases **thymosin**, which is important for proper T-cell development and differentiation. The thymus atrophies by adulthood, and thymosin levels drop accordingly. The function of T-cells is discussed in Chapter 8 of *MCAT Biology Review*.

A full list of hormones and their actions can be found in Table 5.1 below.

Source	Hormone	Type	Action
Anterior pituitary	Follicle-stimulating hormone (FSH)	Peptide	Stimulates follicle maturation in females; spermatogenesis in males
	Luteinizing hormone (LH)	Peptide	Stimulates ovulation in females; testosterone synthesis in males
	Adrenocorticotrophic hormone (ACTH)	Peptide	Stimulates the adrenal cortex to synthesize and secrete glucocorticoids
	Thyroid-stimulating hormone (TSH)	Peptide	Stimulates the thyroid to produce thyroid hormones
	Prolactin	Peptide	Stimulates milk production and secretion

	Endorphins	Peptide	Decrease sensation of pain; can promote euphoria
	Growth hormone (GH)	Peptide	Stimulates bone and muscle growth; raises blood glucose levels
Hypothalamus (released by posterior pituitary)	Antidiuretic hormone (ADH; vasopressin)	Peptide	Stimulates water reabsorption in kidneys by increasing permeability of collecting duct
	Oxytocin	Peptide	Stimulates uterine contractions during labor and milk secretion during lactation; may promote bonding behavior
Thyroid (follicular cells)	Triiodothyronine (T ₃) and thyroxine (T ₄)	Amino acid-derivative	Stimulate metabolic activity
Thyroid (parafollicular or C cells)	Calcitonin	Peptide	Decreases blood calcium concentrations
Parathyroids	Parathyroid hormone (PTH)	Peptide	Increases blood calcium concentrations
Adrenal cortex	Glucocorticoids (cortisol and cortisone)	Steroid	Increase blood glucose concentrations; decrease protein synthesis; anti-inflammatory
	Mineralocorticoids (aldosterone)	Steroid	Increase water reabsorption in the kidneys by increasing sodium reabsorption; promote potassium and hydrogen ion excretion
Adrenal medulla	Epinephrine and norepinephrine	Amino acid-derivative	Increase blood glucose concentrations and heart rate; dilate bronchi; alter blood flow patterns
Pancreas (α -)	Glucagon	Peptide	Stimulates glycogen breakdown

cells)			(glycogenolysis); increases blood glucose concentrations
Pancreas (β -cells)	Insulin	Peptide	Lowers blood glucose concentrations and promotes anabolic processes
Pancreas (δ -cells)	Somatostatin	Peptide	Suppresses secretion of glucagon and insulin
Testis (and adrenal cortex)	Testosterone	Steroid	Induces the development and maintenance of male reproductive system and male secondary sex characteristics
Ovary (and placenta)	Estrogen	Steroid	Induces the development and maintenance of female reproductive system and female secondary sex characteristics
	Progesterone	Steroid	Promotes maintenance of the endometrium
Pineal gland	Melatonin	Peptide	Involved in circadian rhythms
Kidney	Erythropoietin	Peptide	Stimulates bone marrow to produce erythrocytes
Heart (atria)	Atrial natriuretic peptide (ANP)	Peptide	Promotes salt and water excretion
Thymus	Thymosin	Peptide	Stimulates T-cell development

Table 5.1. Major Hormones in Humans

MCAT CONCEPT CHECK 5.2:

Before you move on, assess your understanding of the material with these questions.

1. For each of the hypothalamic releasing hormones listed below, what hormone(s) does each affect in the anterior pituitary? On which organ does each pituitary hormone act? What hormone(s) are released by the target organs?

Hypothalamic Releasing Hormone	Hormone(s) from Anterior Pituitary	Target Organ	Hormone(s) Released by Target Organ
Gonadotropin-releasing hormone (GnRH)			
Corticotropin-releasing factor (CRF)			
Thyroid-releasing hormone (TRH)			
Dopamine			N/A
Growth hormone-releasing hormone (GHRH)			N/A

2. Which two hormones are primarily involved in calcium homeostasis? Where does each come from, and what effect does each have on blood calcium concentrations?

• _____

-

3. Which endocrine tissue synthesizes catecholamines? What are the two main catecholamines it produces?

4. Which two pancreatic hormones are the major drivers of glucose homeostasis? Where does each come from, and what effect does each have on blood glucose concentrations?

-

-

5. Which three hormones are primarily involved in water homeostasis? Where does each come from, and what effect does each have on blood volume and osmolarity?

-

-

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Conclusion

The endocrine system is unique because its organs are not anatomically related. Hormones are produced in a wide variety of locations and can have far-reaching effects throughout the entire organism. The endocrine system allows for integration and execution of the homeostatic parameters that are necessary to ensure proper functioning of the body. For example, we learned that calcium levels are maintained within a narrow concentration range in the plasma by the antagonistic actions of calcitonin and parathyroid hormone (and vitamin D). Each hormone manipulates the steady state of the organism. As you continue your study of the human body, you will find that the endocrine system has effects on every system of the body by regulating fuel metabolism, blood flow, growth, and development.

In the last section of this chapter, there was a small note on erythropoietin, the hormone that stimulates the production of red blood cells in the bone marrow. Don't interpret this brief mention as an indication that it is relatively unimportant. In fact, every cell of the body (except red blood cells themselves) needs a constant supply of oxygen to accomplish its function. Our bodies take in this oxygen through the respiratory system and then distribute the oxygen to tissues via the circulatory system. In the next two chapters, we will explore each of these systems separately. Recognize, however, that this division is artificial; indeed, like the body as a whole, the respiratory and circulatory systems are inseparable as they serve a common function: providing oxygen for every working cell in the body.

Concept Summary

Mechanisms of Hormone Action

- Endocrine signaling involves the secretion of **hormones** directly into the bloodstream. The hormones travel to distant target tissues, where they bind to receptors and induce a change in gene expression or cell function.
- **Peptide hormones** are composed of amino acids and are derived from larger precursor proteins that are cleaved during posttranslational modification.
 - Peptide hormones are polar and cannot pass through the plasma membrane.
 - These hormones bind to extracellular receptors, where they trigger the transmission of a **second messenger**.
 - Each step of the **signaling cascade** can induce **amplification** of the signal.
 - Peptide hormones exert effects that usually have rapid onset but are short-lived.
 - These hormones are water-soluble, so they travel freely in the bloodstream and do not require a special carrier.
- **Steroid hormones** are derived from cholesterol.
 - Steroid hormones are minimally polar and can pass through the plasma membrane.

- These hormones bind to and promote a conformational change in cytosolic or intranuclear receptors; the hormone–receptor complex binds to DNA, altering the transcription of a particular gene.
 - Steroid hormones exert effects that usually have slow onset but are long-lived.
 - Because these hormones are lipid-soluble, they cannot dissolve in the bloodstream and must be carried by specific proteins.
- **Amino acid–derivative hormones** are modified amino acids.
 - Their chemistry shares some features with peptide hormones and some features with steroid hormones; different amino acid–derivative hormones share different features with these other hormone classes.
 - Common examples are epinephrine, norepinephrine, triiodothyronine, and thyroxine.
- Hormones can be classified by their target tissues.
 - **Direct hormones** have major effects on non-endocrine tissues.
 - **Tropic hormones** have major effects on other endocrine tissues.

Endocrine Organs and Hormones

- The hypothalamus is the bridge between the nervous and endocrine systems.
 - The release of hormones from the hypothalamus is mediated by a number of factors, including projections from other parts of the brain, chemo- and baroreceptors in the blood vessels, and negative feedback

from other hormones.

- In **negative feedback**, the final hormone (or product) of a pathway inhibits hormones (or enzymes) earlier in the pathway, maintaining **homeostasis**.
 - The hypothalamus stimulates the anterior pituitary gland through paracrine release of hormones into the hypophyseal portal system, which directly connects the two organs.
 - **Gonadotropin-releasing hormone (GnRH)** promotes the release of follicle-stimulating hormone (FSH) and luteinizing hormone (LH).
 - **Growth hormone-releasing hormone (GHRH)** promotes the release of growth hormone.
 - **Thyroid-releasing hormone (TRH)** promotes the release of thyroid-stimulating hormone (TSH).
 - **Corticotropin-releasing factor (CRF)** promotes the release of adrenocorticotrophic hormone (ACTH).
 - **Prolactin-inhibiting factor (PIF or dopamine)** inhibits the release of prolactin.
 - Interactions with the posterior pituitary occur via the axons of nerves projected by the hypothalamus. Antidiuretic hormone (ADH or vasopressin) and oxytocin are synthesized in the hypothalamus and then travel down these axons to the posterior pituitary, where they are released into the bloodstream.
- The **anterior pituitary** releases hormones in response to stimulation from the hypothalamus. Four of these (FSH, LH, ACTH, and TSH) are tropic hormones, while three (prolactin, endorphins, and growth hormone) are direct hormones.
- **Follicle-stimulating hormone (FSH)** promotes the development of ovarian follicles in females and spermatogenesis in males.

- **Luteinizing hormone (LH)** promotes ovulation in females and testosterone production in males.
 - **Adrenocorticotrophic hormone (ACTH)** promotes the synthesis and release of glucocorticoids from the adrenal cortex.
 - **Thyroid-stimulating hormone (TSH)** promotes the synthesis and release of triiodothyronine and thyroxine from the thyroid.
 - **Prolactin** promotes milk production.
 - **Endorphins** decrease perception of pain and can produce euphoria.
 - **Growth hormone (GH)** promotes growth of bone and muscle and shunts glucose to these tissues. It raises blood glucose concentrations.
- The **posterior pituitary** releases two hormones produced in the hypothalamus.
 - **Antidiuretic hormone (ADH or vasopressin)** is secreted in response to low blood volume or increased blood osmolarity and increases reabsorption of water in the collecting duct of the nephron, increasing blood volume and decreasing blood osmolarity.
 - **Oxytocin** is secreted during childbirth and promotes uterine contractions. It also promotes milk ejection and may be involved in bonding behavior. It is unusual in that it has a **positive feedback** loop, not negative.
 - The **thyroid** is located at the base of the neck in front of the trachea; it produces three key hormones.
 - **Triiodothyronine (T₃)** and **thyroxine (T₄)** are produced by **follicular cells** and contain iodine. They increase basal metabolic rate and alter the utilization of glucose and fatty acids. Thyroid hormones are required

for proper neurological and physical development in children.

- **Calcitonin** is produced by **parafollicular (C) cells**. It decreases plasma calcium concentration by promoting calcium excretion in the kidneys, decreasing calcium absorption in the gut, and promoting calcium storage in bone.
- The **parathyroid glands** release **parathyroid hormone (PTH)**, which increases blood calcium concentration.
 - PTH decreases excretion of calcium by the kidneys and increases bone resorption directly to increase blood calcium concentration.
 - PTH activates vitamin D, which is necessary for calcium and phosphate absorption from the gut.
 - PTH promotes resorption of phosphate from bone and reduces reabsorption of phosphate in the kidney, but vitamin D promotes absorption of phosphate from the gut; these two effects on phosphate concentration somewhat cancel each other out.
- The **adrenal cortex** produces three classes of steroid hormones.
 - **Glucocorticoids** such as **cortisol** and **cortisone** increase blood glucose concentration, reduce protein synthesis, inhibit the immune system, and participate in the stress response. Glucocorticoid release is stimulated by ACTH.
 - **Mineralocorticoids** such as **aldosterone** promote sodium reabsorption in the distal convoluted tubule and collecting duct, thus increasing water reabsorption. Aldosterone also increases potassium and hydrogen ion excretion. Aldosterone activity is regulated by the **renin-angiotensin-aldosterone system**, not ACTH.
 - **Cortical sex hormones** include **androgens** (like **testosterone**) and

estrogens in both males and females.

- The **adrenal medulla** is derived from the nervous system and secretes catecholamines into the bloodstream.
 - **Catecholamines** include **epinephrine** and **norepinephrine**, which are involved in the fight-or-flight (sympathetic) response.
 - These hormones promote glycogenolysis, increase the basal metabolic rate, increase heart rate, dilate the bronchi, and alter blood flow.
- The endocrine **pancreas** produces hormones that regulate glucose homeostasis.
 - **Glucagon** is produced by **α -cells** and raises blood glucose levels by stimulating protein and fat degradation, glycogenolysis, and gluconeogenesis.
 - **Insulin** is produced by **β -cells** and lowers blood glucose levels by stimulating glucose uptake by cells and promoting anabolic processes, like glycogen, fat, and protein synthesis.
 - **Somatostatin** is produced by **δ -cells** and inhibits insulin and glucagon secretion.
- The gonads produce hormones that are involved in the development and maintenance of the reproductive systems and secondary sex characteristics.
 - The **testes** secrete **testosterone**.
 - The **ovaries** secrete **estrogen** and **progesterone**.
- The **pineal gland** releases **melatonin**, which helps to regulate **circadian**

rhythms.

- Other organs may release hormones, even if they are not primarily considered part of the endocrine system.
 - Cells in the stomach and intestine produce hormones like **secretin**, **gastrin**, and **cholecystokinin**.
 - The kidneys secrete **erythropoietin**, which stimulates bone marrow to produce erythrocytes (red blood cells) in response to low oxygen levels in the blood.
 - The atria of the heart secrete **atrial natriuretic peptide (ANP)**, which promotes excretion of salt and water in the kidneys in response to stretching of the atria (high blood volume).
 - The thymus secretes **thymosin**, which is important for proper T-cell development and differentiation.

Answers to Concept Checks

5.1

Criterion	Peptide Hormones	Steroid Hormones
Chemical precursor	Amino acids (polypeptides)	Cholesterol
Location of receptor	Extracellular (cell membrane)	Intracellular or intranuclear
Mechanism of action	Stimulates a receptor (usually a G protein–coupled receptor), affecting levels of second messengers (commonly cAMP). Initiates a signal cascade	Binds to a receptor, induces conformational change, and regulates transcription at the level of the DNA
Method of travel in the bloodstream	Dissolves and travels freely	Binds to a carrier protein
Speed of onset	Quick	Slow
Duration of action	Short-lived	Long-lived

1. Amino acid–derivative hormones are made by modifying amino acids, such as the addition of iodine to tyrosine (in thyroid hormone production).
2. Direct hormones are secreted into the bloodstream and travel to a target tissue, where they have direct effects. Tropic hormones cause secretion of another hormone that then travels to the target tissue to cause an effect.

5.2

Hypothalamic Releasing Hormone	Hormone(s) from Anterior Pituitary	Target Organ	Hormone(s) Released by Target Organ
Gonadotropin-releasing hormone (GnRH)	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)	Gonads (testes or ovaries)	Testosterone (testes) or estrogen and progesterone (ovaries)
Corticotropin-releasing factor (CRF)	Adrenocorticotrophic hormone (ACTH)	Adrenal cortex	Glucocorticoids (cortisol and cortisone)
Thyroid-releasing hormone (TRH)	Thyroid-stimulating hormone (TSH)	Thyroid	Triiodothyronine (T ₃), thyroxine (T ₄)
Dopamine	Prolactin*	Breast tissue	N/A
Growth hormone-releasing hormone (GHRH)	Growth hormone	Bone, muscle	N/A
*Note that a <i>decrease</i> in dopamine from the hypothalamus promotes prolactin secretion.			

- l. Calcitonin from the parafollicular (C-) cells of the thyroid decreases blood calcium concentration. Parathyroid hormone from the parathyroid glands increases blood calcium concentration.
- }. The adrenal medulla synthesizes catecholamines, including epinephrine and norepinephrine.
- l. Glucagon from the α -cells of the pancreas increases blood glucose concentration. Insulin from the β -cells of the pancreas decreases blood glucose concentration.
- i. Antidiuretic hormone (ADH or vasopressin) from the hypothalamus (released by the posterior pituitary) increases blood volume and decreases

blood osmolarity. Aldosterone from the adrenal cortex increases blood volume with no effect on blood osmolarity. Atrial natriuretic peptide (ANP) from the heart decreases blood volume with no effect on blood osmolarity.



Shared Concepts

Behavioral Sciences Chapter 5

Motivation, Emotion, and Stress

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biochemistry Chapter 8

Biological Membranes

Biochemistry Chapter 12

Bioenergetics and Regulation of Metabolism

Biology Chapter 4

The Nervous System

Biology Chapter 10

Homeostasis

Discrete Practice Questions

1. Which of the following associations between a hormone and its category is INCORRECT?
 - (A) Aldosterone—mineralocorticoid
 - (B) Testosterone—cortical sex hormone
 - (C) ADH—mineralocorticoid
 - (D) Cortisone—glucocorticoid
2. Which of the following hormones directly stimulates a target tissue that is NOT an endocrine organ?
 - (A) ACTH
 - (B) TSH
 - (C) LH
 - (D) GH
3. Increased synthetic activity of the parathyroid glands would lead to:
 - (A) an increase in renal calcium reabsorption.
 - (B) a decrease in the rate of bone resorption.
 - (C) a decrease in basal metabolic rate.
 - (D) a decrease in blood glucose concentration.

4. Which of the following statements concerning growth hormone is NOT true?

- (A) Overproduction of growth hormone in adults results in acromegaly.
- (B) It promotes growth of bone and muscle.
- (C) It is produced by the hypothalamus but secreted by the pituitary.
- (D) A childhood deficiency in growth hormone results in dwarfism.

5. Iodine deficiency may result in:

- (A) galactorrhea.
- (B) cretinism.
- (C) gigantism.
- (D) hyperthyroidism.

6. A patient has a very high TSH level. Which of the following would NOT cause a high TSH level?

- (A) Autoimmune destruction of thyroid cells that produce T_3 and T_4 .
- (B) A tumor in the hypothalamus that secretes high levels of TRH.
- (C) High levels of T_4 from thyroid replacement medications.
- (D) Cancerous growth of parafollicular cells in the thyroid, destroying other cell types in the organ.

7. Which of the following hormones is NOT derived from cholesterol?

- (A) Aldosterone
- (B) Estrogen
- (C) Oxytocin
- (D) Progesterone

8. Which of the following is true regarding pancreatic somatostatin?

- (A) Its secretion is increased by low blood glucose.
- (B) It is always inhibitory.
- (C) It is regulated by cortisol levels.
- (D) It stimulates insulin and glucagon secretion.

9. Destruction of all β -cells in the pancreas would cause:

- (A) glucagon secretion to stop and an increase in blood glucose concentration.
- (B) glucagon secretion to stop and a decrease in blood glucose concentration.
- (C) insulin secretion to stop and an increase in blood glucose concentration.
- (D) insulin secretion to stop and a decrease in blood glucose concentration.

10. Which of the following is FALSE regarding aldosterone regulation?

- (A) Renin converts the plasma protein angiotensinogen to angiotensin I.

- (B) Angiotensin II stimulates the adrenal cortex to secrete aldosterone.
- (C) Angiotensin I is converted to angiotensin II by angiotensin-converting enzyme.
- (D) A decrease in blood oxygen concentrations stimulates renin production.

11. A scientist discovers a new hormone that is relatively large in size and triggers the conversion of ATP to cAMP. Which of the following best describes the type of hormone that was discovered?

- (A) Amino acid-derivative hormone
- (B) Peptide hormone
- (C) Steroid hormone
- (D) Tropic hormone

12. A patient presents with muscle weakness, slow movement, and calcium deposits in his tissues. A blood test reveals very low calcium levels in the blood. Administration of which of the following would be an appropriate treatment for the blood test findings?

- (A) Calcitonin
- (B) Parathyroid hormone
- (C) Aldosterone
- (D) Thymosin

13. Oxytocin and antidiuretic hormone are:

- (A) peptide hormones produced and released by the pituitary.
- (B) steroid hormones produced and released by the pituitary.
- (C) peptide hormones produced by the hypothalamus and released by the pituitary.
- (D) steroid hormones produced by the hypothalamus and released by the pituitary.

14. Excessive levels of dopamine in the brain are associated with psychosis. Accordingly, many antipsychotic medications block dopamine receptors. Which of the following effects may be seen in an individual taking antipsychotics?

- (A) Increased secretion of growth hormone
- (B) Decreased secretion of growth hormone
- (C) Increased secretion of prolactin
- (D) Decreased secretion of prolactin

15. A genotypically female infant is born with ambiguous genitalia. Soon after birth, she suffers from hyponatremia, or low blood concentrations of sodium. Which endocrine organ is most likely to be affected?

- (A) Hypothalamus
- (B) Pituitary
- (C) Kidneys
- (D) Adrenal cortex

Discrete Practice Answers

1. **C**

Unlike the other hormones listed here, ADH is not secreted by the adrenal cortex and is therefore not a *-corticoid*. Rather, ADH is a peptide hormone produced by the hypothalamus and released by the posterior pituitary that promotes water reabsorption. The other associations are all correct.

2. **D**

A hormone that directly stimulates a non-endocrine target tissue is referred to as a direct hormone. Glancing at the answer choices, we notice that all of the hormones are secreted by the anterior pituitary gland. The direct hormones secreted by the anterior pituitary are prolactin, endorphins, and growth hormone (GH). All of the other answer choices are tropic hormones. The tropic hormones of the anterior pituitary include follicle-stimulating hormone (FSH), luteinizing hormone (LH), adrenocorticotrophic hormone (ACTH), and thyroid-stimulating hormone (TSH).

3. **A**

The parathyroid glands secrete parathyroid hormone (PTH), a hormone that functions to increase blood calcium levels. An increase in synthetic activity of the parathyroid glands would lead to an increase in PTH and, therefore, an increase in blood calcium levels through three mechanisms: increased calcium reabsorption in the kidneys (decreased excretion), increased bone resorption, and increased absorption of calcium from the gut (via activation of vitamin D).

4. **C**

Growth hormone is a direct hormone secreted by the anterior pituitary. Among its many functions, GH promotes growth of bone and muscle, eliminating **(B)**. An overproduction of growth hormone in children results in gigantism, whereas in adults it results in acromegaly (enlargement of the small bones in the extremities and of certain facial bones, such as the jaw), eliminating **(A)**. On the other hand, a childhood deficiency of growth hormone results in dwarfism, eliminating **(D)**. GH is synthesized and secreted in the anterior pituitary; **(C)** describes antidiuretic hormone and oxytocin, not GH.

5. **B**

Inflammation of the thyroid or iodine deficiency can cause hypothyroidism, in which the thyroid hormones are undersecreted or not secreted at all. Hypothyroidism in newborn infants causes cretinism, which is characterized by poor neurological and physical development (including mental retardation, short stature, and coarse facial features). While iodine deficiency can result in a swelling of the thyroid gland (called a goiter), which can also be seen in causes of hyperthyroidism, iodine deficiency does not cause hyperthyroidism, eliminating **(D)**. Galactorrhea, **(A)**, is associated with prolactin; gigantism, **(C)**, is associated with growth hormone.

6. **C**

The hypothalamic–pituitary–thyroid axis includes the secretion of thyroid-releasing hormone (TRH) from the hypothalamus triggering the secretion of thyroid-stimulating hormone (TSH) from the anterior pituitary, which stimulates the secretion of triiodothyronine (T_3) and thyroxine (T_4) from the thyroid. Overproduction of TRH would promote overproduction of TSH,

eliminating **(B)**. Destruction of the follicular cells that produce T_3 and T_4 would remove negative feedback, allowing TSH levels to rise, eliminating **(A)** and **(D)**. High levels of T_4 would cause too much negative feedback and lower TSH levels, making **(C)** the correct answer.

7. **C**

Steroid hormones are derived from cholesterol. Many of their names end with *-one*, *-ol*, or *-oid*, and they include the hormones of the adrenal cortex and gonads. Aldosterone is a mineralocorticoid from the adrenal cortex, while estrogen and progesterone are sex hormones from the ovaries, eliminating **(A)**, **(B)**, and **(D)**. Oxytocin is a peptide hormone and is thus not derived from cholesterol. Peptide and amino acid-derivative hormones often end with *-in* or *-ine*.

8. **B**

Pancreatic somatostatin secretion is increased by high blood glucose or amino acid levels, leading to decreased insulin and glucagon secretion, eliminating **(A)** and **(D)**. Somatostatin is thus always an inhibitory hormone, confirming **(B)**. The stimuli for somatostatin release include high blood glucose or amino acids, as mentioned above, as well as high levels of certain gastrointestinal hormones, as discussed in Chapter 9 of *MCAT Biology Review*—but not cortisol, eliminating **(C)**.

9. **C**

β -cells are responsible for insulin production. The function of insulin is to lower blood glucose levels by promoting the influx of glucose into cells and by stimulating anabolic processes, such as glycogenesis or fat and protein synthesis. Thus, destruction of the β -cells would result in a cessation of

insulin production, which would lead to hyperglycemia, or high blood glucose concentration.

0. **D**

The stimulus for renin production is low blood pressure, which causes the juxtaglomerular cells of the kidney to produce renin, an enzyme that converts the plasma protein angiotensinogen to angiotensin I. Angiotensin I is then converted to angiotensin II by an enzyme in the lungs; angiotensin II then stimulates the adrenal cortex to secrete aldosterone. Aldosterone helps to restore blood volume by increasing sodium reabsorption in the kidney, leading to an increase in water reabsorption. This removes the initial stimulus for renin production. Thus, **(A)**, **(B)**, and **(C)** correctly describe the renin–angiotensin–aldosterone system, while **(D)** describes the stimulus for erythropoietin secretion.

1. **B**

The question stem indicates that the newly discovered hormone functions as a first messenger, stimulating the conversion of ATP to cAMP; cAMP functions as a second messenger, triggering a signaling cascade in the cell. Hormones that act via second messengers and are relatively large in size (short peptides or complex polypeptides) are peptide hormones. This hormone could be a tropic hormone, but it is also entirely possible for it to be a direct hormone; thus, **(D)** can be eliminated.

2. **B**

Regardless of the cause, the low levels of calcium in the blood require treatment. While other therapies are more frequently used to treat hypocalcemia (low blood calcium levels), such as calcium gluconate or calcium chloride, administration of parathyroid hormone would also raise blood calcium concentration. Calcitonin would be a poor choice in this

case, as this hormone lowers blood calcium concentrations, eliminating **(A)**. Aldosterone and thymosin play no role in calcium homeostasis, eliminating **(C)** and **(D)**.

3. **C**

Both oxytocin and vasopressin (another name for antidiuretic hormone) end with the suffix *-in*; this should hint that they are peptide or amino acid-derivative hormones. These two hormones are both synthesized by the hypothalamus, but released by the posterior pituitary. Remember that the posterior pituitary does not actually synthesize any hormones itself; rather, it contains the axons that originate in cells in the hypothalamus and is the site of release for these hormones.

4. **C**

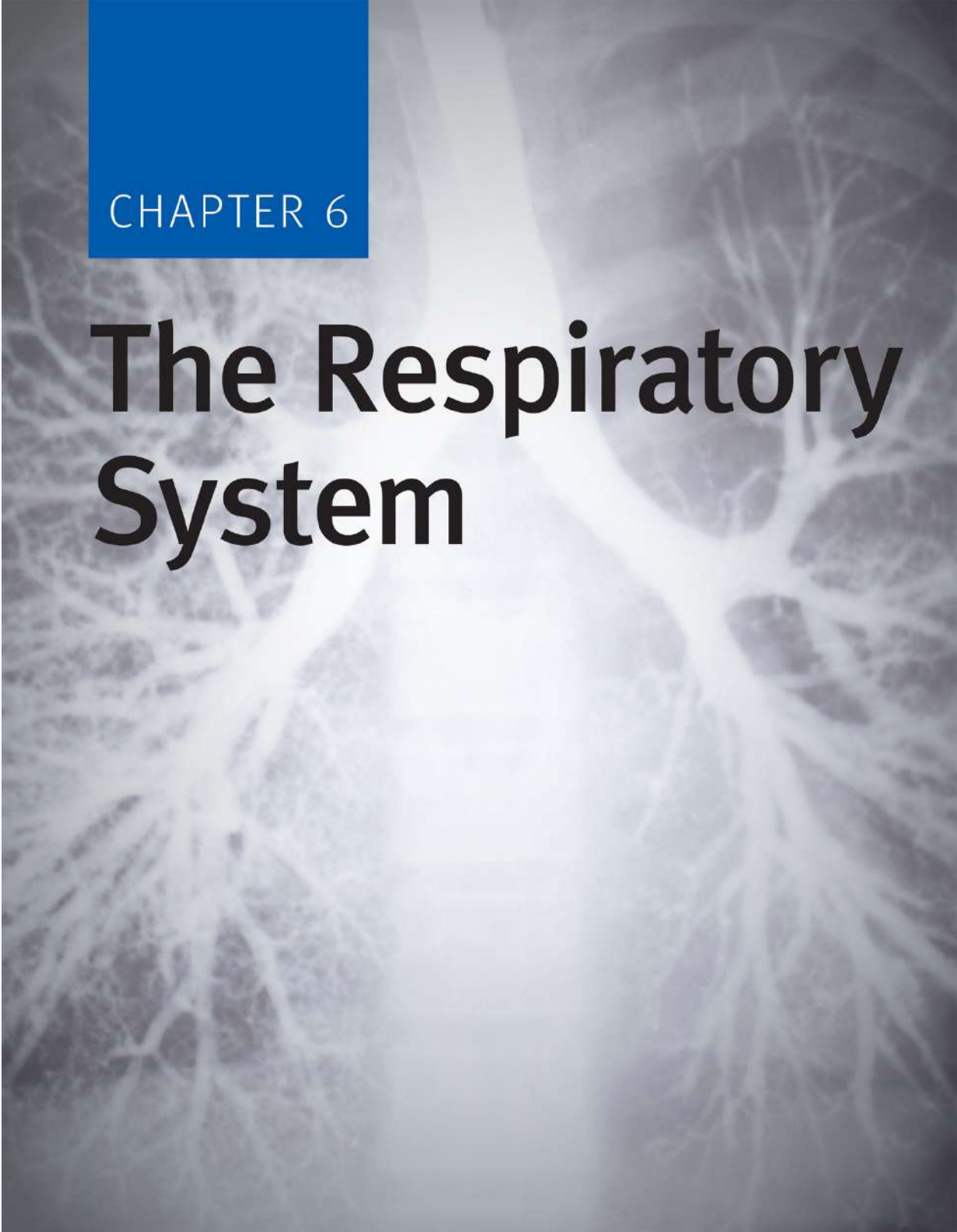
Dopamine is used in a number of neurological systems; most relevant to the endocrine system is the fact that dopamine secretion prevents prolactin release. Thus, an individual taking medications that block dopamine receptors would lose this inhibition on prolactin release and have elevated prolactin levels.

5. **D**

The question stem states that an infant who is genotypically female is born with ambiguous genitalia, meaning that the genitalia do not appear to be specifically female or specifically male. In a genotypic female, this indicates that she was exposed to androgens during the fetal period. In addition, she is also losing sodium, causing hyponatremia. This indicates that she has two defects. First, she has excess androgens. Second, she is lacking in aldosterone, a hormone required for proper reuptake of sodium in the kidneys. Both of these hormones are synthesized in the adrenal cortex, making **(D)** the correct answer. Note that neither of these hormones is

regulated by the hypothalamic–pituitary–adrenal axis, eliminating **(A)** and **(B)**.





CHAPTER 6

The Respiratory System

CHAPTER 6

The Respiratory System

In This Chapter

6.1 Anatomy and Mechanism of Breathing

Anatomy

Breathing

Lung Capacities and Volumes

Regulation of Breathing

6.2 Functions of the Respiratory System

Gas Exchange

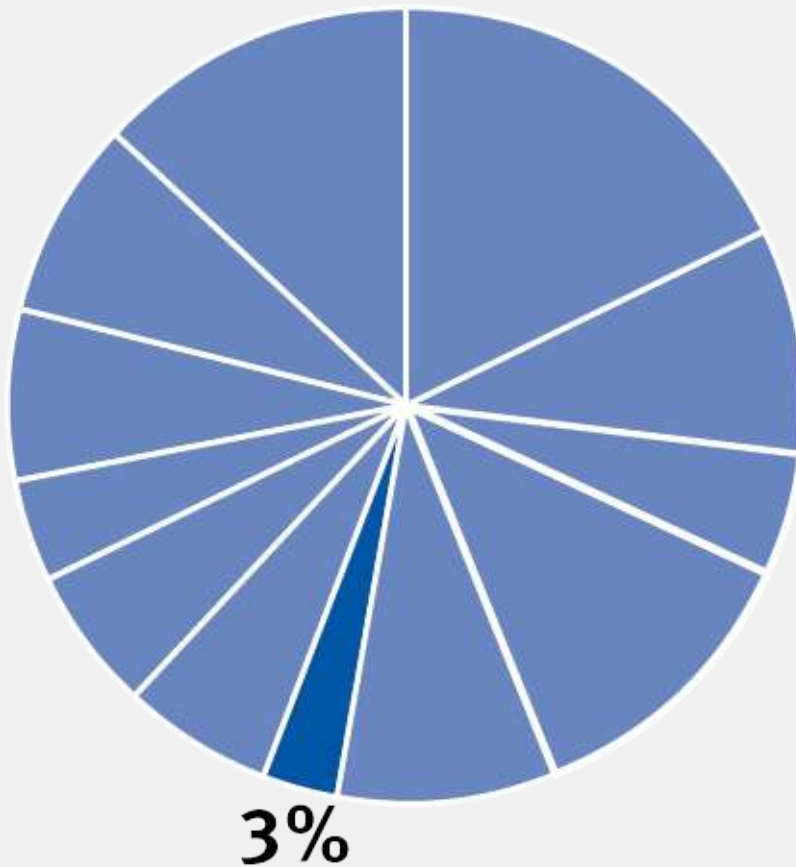
Thermoregulation

Immune Function

Control of pH

Concept Summary

Chapter Profile



The content in this chapter should be relevant to about 3% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems



MCAT EXPERTISE

You may be surprised at the low percentage of questions associated with this chapter, given that this chapter touches on several classic MCAT topics including pressure, gases, and equilibrium. And you may see questions on your MCAT that relate to the work done on air in the lungs, for example. However, such a question would depend much more on your general knowledge of work, energy, and gases (regardless of context) than your knowledge of the respiratory system specifically.

Introduction

Coughing. Fever. Shortness of breath. Hypoxia. All are symptoms of a number of pulmonary diseases, from a flare-up of *chronic obstructive pulmonary disease* (COPD), to *Streptococcus pneumoniae* (pneumococcal) pneumonia, to a type of hypersensitivity pneumonitis known as *extrinsic allergic alveolitis* (EAA). This last example is a bit more esoteric and can be brought on by hypersensitivity to anything from dried grass, to rat urine, to mold that grows in hot tubs—what is sometimes called *hot tub lung*. Not all cases of hot tub lung are severe, but certainly none are enjoyable. They are often misdiagnosed as asthma or bronchitis and may be treated with steroids, which quell the immune system and reduce inflammation. Because hot tub lung can potentially go away by itself, antibiotic therapy is not always recommended. As a physician, you may end up simply having to tell your

patients that the best way to avoid hot tub lung is to make sure that the tub is cleaned properly and routinely before use.

The lesson here isn't to avoid hot tubs. It's that the lungs are essential, sensitive organs with delicate membranes that must be protected. Many types of stressors (pathogens, particles, or chemicals) can irritate them and cause respiratory distress. In this chapter, we'll look at the structure of the lungs and the microanatomy of respiration. We'll also talk about the mechanics of breathing, as well as the overall function of the lungs.

6.1 Anatomy and Mechanism of Breathing

LEARNING OBJECTIVES

After Chapter 6.1, you will be able to:

- Identify the muscles involved in inhalation and exhalation
- Explain the purpose and function of surfactant
- Recall the mathematical relationships between vital capacity, inspiratory reserve volume, expiratory reserve volume, and tidal volume
- Predict how the brain will alter respiratory rate in response to changing blood levels of O₂ and CO₂
- Order the structures in the pathway that air uses to enter the body, from the nares to the alveoli

The lungs are located in the **thoracic cavity**, the structure of which is specially designed to perform breathing.

ANATOMY

The anatomy of the respiratory system is summarized in Figure 6.1. Gas exchange occurs in the lungs. Air enters the respiratory tract through the external **nares** of the nose and then passes through the nasal cavity, where it is filtered by mucous membranes and nasal hairs (**vibrissae**).



KEY CONCEPT

The nose and mouth serve several important roles in breathing by removing dirt and particulate matter from the air and warming and humidifying it before it reaches the lungs.

Next, air passes into the pharynx and the larynx. The **pharynx** resides behind the nasal cavity and at the back of the mouth; it is a common pathway for both air destined for the lungs and food destined for the esophagus. In contrast, the **larynx** lies below the pharynx and is only a pathway for air. To keep food out of the respiratory tract, the opening of the larynx (**glottis**) is covered by the **epiglottis** during swallowing. The larynx contains two **vocal cords** that are maneuvered using skeletal muscle and cartilage. From the larynx, air passes into the cartilaginous **trachea** and then into one of the two mainstem **bronchi**. The bronchi and trachea contain ciliated epithelial cells to catch material that has made it past the mucous membranes in the nose and mouth.

In the **lungs**, the bronchi continue to divide into smaller structures known as **bronchioles**, which divide further until they end in the tiny balloon-like structures in which gas exchange occurs (**alveoli**). Each alveolus is coated with **surfactant**, a detergent that lowers surface tension and prevents the

alveolus from collapsing on itself. A network of capillaries surrounds each alveolus to carry oxygen and carbon dioxide. The branching and minute size of the alveoli allow for an exceptionally large surface area for gas exchange—approximately 100 m² in total.

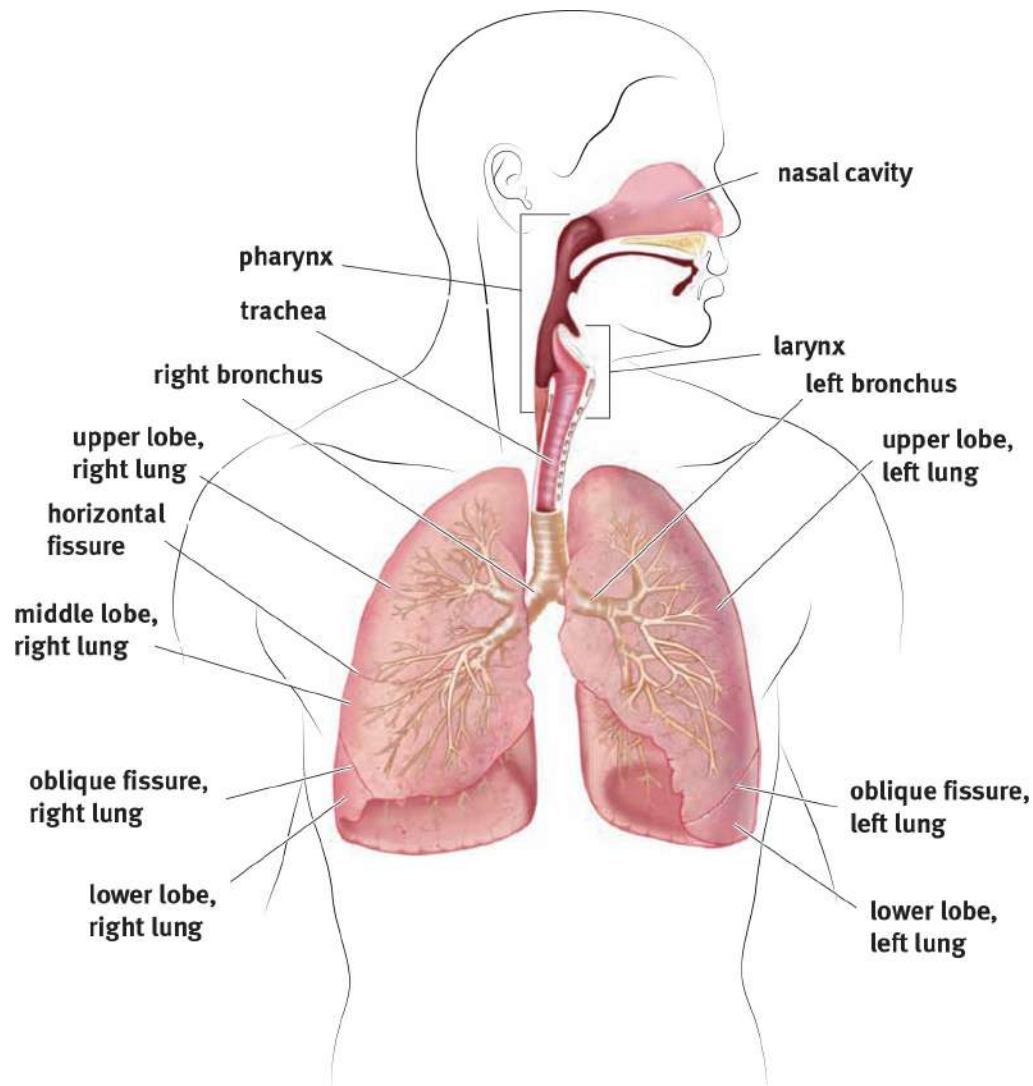


Figure 6.1. Anatomy of the Respiratory System



REAL WORLD

The left lung has a small indentation, making it slightly smaller than the right lung. It also contains only two lobes, while the right lung contains three. Why might this be? It is due to the position of the heart in the thoracic cavity.

The lungs themselves are contained in the thoracic cavity, which also contains the heart. The chest wall forms the outside of the thoracic cavity. Membranes known as **pleurae** surround each lung, as shown in Figure 6.2. The pleura forms a closed sac against which the lung expands. The surface adjacent to the lung is the **visceral pleura**, and the outer part is the **parietal pleura**.

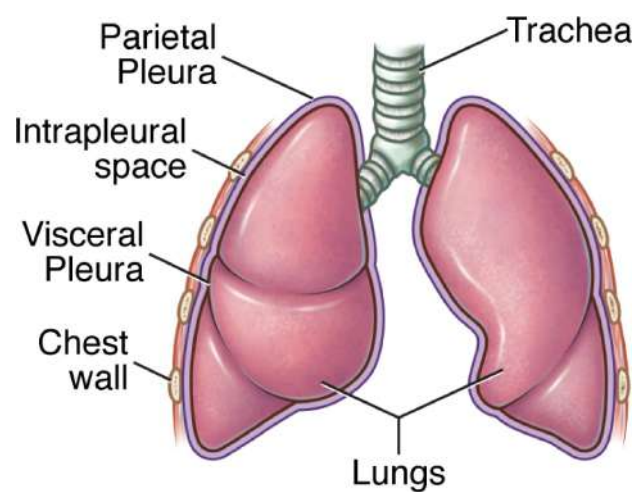


Figure 6.2. Lung Membranes

The lungs do not fill passively, and require skeletal muscle to generate the negative pressure for expansion. The most important of these muscles is the **diaphragm**, a thin, muscular structure that divides the thoracic (chest) cavity from the abdominal cavity. The diaphragm is under somatic control, even though breathing itself is under autonomic control. In addition, muscles of

the chest wall, back, and neck may also participate in breathing, especially when breathing is labored due to a pathologic condition.

BREATHING

Before we discuss breathing itself, it is worth taking a closer look at the relationship between the pleurae and the lungs. Imagine that you have a large, partially deflated balloon. Now, imagine taking your fist and pushing it against the balloon so that the balloon comes up and surrounds your hand. This is analogous to a lung and its pleura. Our fist is the lung, and the balloon represents both pleural layers. The side directly touching our fist is the visceral pleura, and the outer layer is the parietal pleura, which is associated with the chest wall in real life. The space within the sac is referred to as the **intrapleural space**, which contains a thin layer of fluid. This pleural fluid helps lubricate the two pleural surfaces. The pressure differentials that can be created across the pleura ultimately drive breathing, as we explore in the next section.



REAL WORLD

The intrapleural space is an example of a potential space—a space that is normally empty or collapsed. In some pathologic states, that potential space can be expanded by fluid or air that accumulates between the two pleural layers. For example, in a pleural effusion, fluid accumulates in the intrapleural space. In a pneumothorax, air collects here. Each of these states disturbs the normal mechanics of the breathing apparatus and can cause atelectasis, or lung collapse.

Let's turn to the mechanics of ventilation, which are grounded in physics. As discussed in Chapters 2 and 3 of *MCAT Physics and Math Review*, we can use pressure to do useful work in a system. Here, we use pressure differentials between the intrapleural space and the lungs to drive air into the lungs.

Inhalation

Inhalation is an active process. We use our diaphragm as well as the **external intercostal muscles** (one of the layers of muscles between the ribs) to expand the thoracic cavity, as shown in Figure 6.3. As the diaphragm flattens and the chest wall expands outward, the **intrathoracic volume** (the volume of the chest cavity) increases. Specifically, because the intrapleural space most closely abuts the chest wall, its volume increases first. Can we predict what will happen to intrapleural pressure? From our understanding of Boyle's law, an increase in intrapleural volume leads to a *decrease* in intrapleural pressure.



BRIDGE

Boyle's law states that, at a constant temperature, the pressure and volume of gases are inversely related. This is the principle underlying negative-pressure breathing: as the chest wall expands, the pressure in the lungs drops, and air is drawn into the lungs.

Now we have low pressure in the intrapleural space. What about inside the lungs? The gas in the lungs is initially at atmospheric pressure, which is now higher than the pressure in the intrapleural space. The lungs will therefore

expand into the intrapleural space, and the pressure in the lungs will drop. Air will then be sucked in from a higher-pressure environment—the outside world. This mechanism is referred to as **negative-pressure breathing** because the driving force is the lower (relatively negative) pressure in the intrapleural space compared with the lungs.

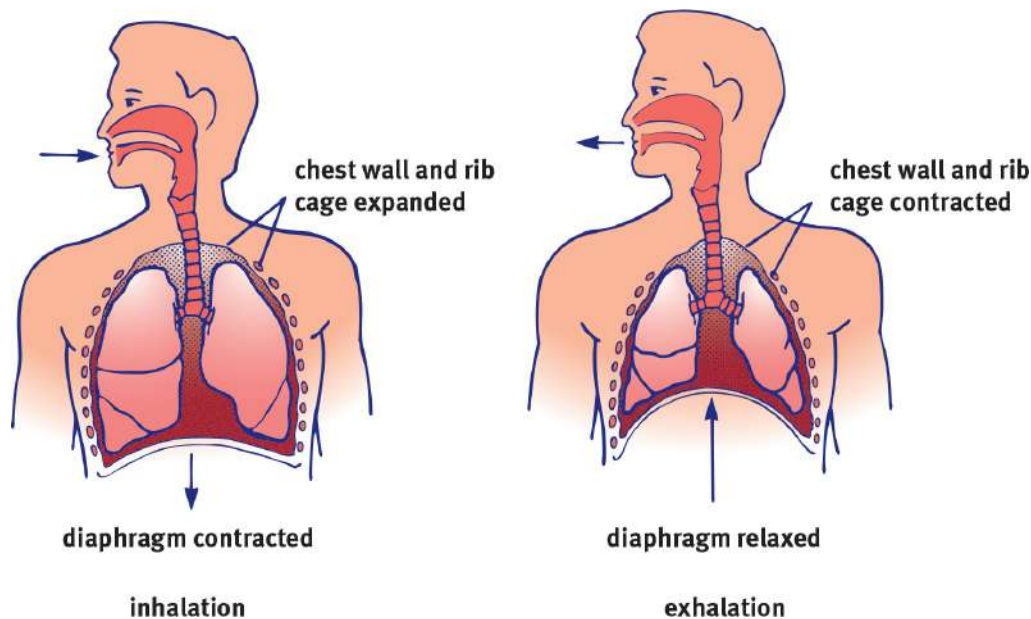


Figure 6.3. Stages of Ventilation

The diaphragm contracts during inhalation and relaxes during exhalation.

Exhalation

Unlike inhalation, exhalation does not have to be an active process. Simple relaxation of the external intercostal muscles will reverse the processes we discussed in the last paragraph. As the diaphragm and external intercostals relax, the chest cavity decreases in volume. What will happen to pressure in the intrapleural space? It will go up, again explained by Boyle's law. Now pressure in the intrapleural space is higher than in the lungs, which is still at atmospheric pressure. Thus, air will be pushed out, resulting in exhalation.

During active tasks, we can speed this process up by using the **internal intercostal muscles** and abdominal muscles, which oppose the external intercostals and pull the rib cage down. This actively decreases the volume of the thoracic cavity. Finally, recall that surfactant prevents the complete collapse of the alveoli during exhalation by reducing surface tension at the alveolar surface.



KEY CONCEPT

Inhalation and exhalation require different amounts of energy expenditure. Muscle contraction is required to create the negative pressure in the thoracic cavity that forces air into the lungs during inspiration. Expiration during calm states is entirely due to elastic recoil of the lungs and the musculature. During more active states, the muscles can be used to force air out and speed up the process of ventilation.

Remember the balloon analogy from before. The lungs have a resilient, elastic quality and are attached via the pleurae to the chest wall. The chest wall expands on inhalation, pulling the lungs with it and creating the pressure differential required for inhalation. As the chest wall relaxes, the lungs recoil due to the intrinsic elastic quality of the lungs and surface tension in the alveoli, accentuating the relaxation process. When the lungs recoil, their volume becomes smaller, and the pressure increases. Now the pressure inside the lungs is higher than the outside pressure, and exhalation occurs. Note that the indirect connection of the lungs to the chest wall also prevents them from collapsing completely on recoil, like surfactant.



REAL WORLD

Emphysema is a disease characterized by the destruction of the alveolar walls. This results in reduced elastic recoil of the lungs, making the process of exhalation extremely difficult. Most cases of emphysema are caused by cigarette smoking.

LUNG CAPACITIES AND VOLUMES

In pulmonology (the medical field associated with the lungs and breathing), we frequently must assess lung capacities and volumes—but we don't have the luxury of removing an individual's lungs to do so. One instrument used to measure these quantities is a **spirometer**. While a spirometer cannot measure the amount of air remaining in the lung after complete exhalation (residual volume), it provides a number of measures that are useful in clinical medicine.

Commonly tested lung volumes include:

- **Total lung capacity (TLC):** The maximum volume of air in the lungs when one inhales completely; usually around 6 to 7 liters
- **Residual volume (RV):** The volume of air remaining in the lungs when one exhales completely
- **Vital capacity (VC):** The difference between the minimum and maximum volume of air in the lungs ($TLC - RV$)
- **Tidal volume (TV):** The volume of air inhaled or exhaled in a normal breath
- **Expiratory reserve volume (ERV):** The volume of additional air that can be forcibly exhaled after a normal exhalation

- **Inspiratory reserve volume (IRV):** The volume of additional air that can be forcibly inhaled after a normal inhalation

These different lung volumes and capacities can be seen in Figure 6.4.

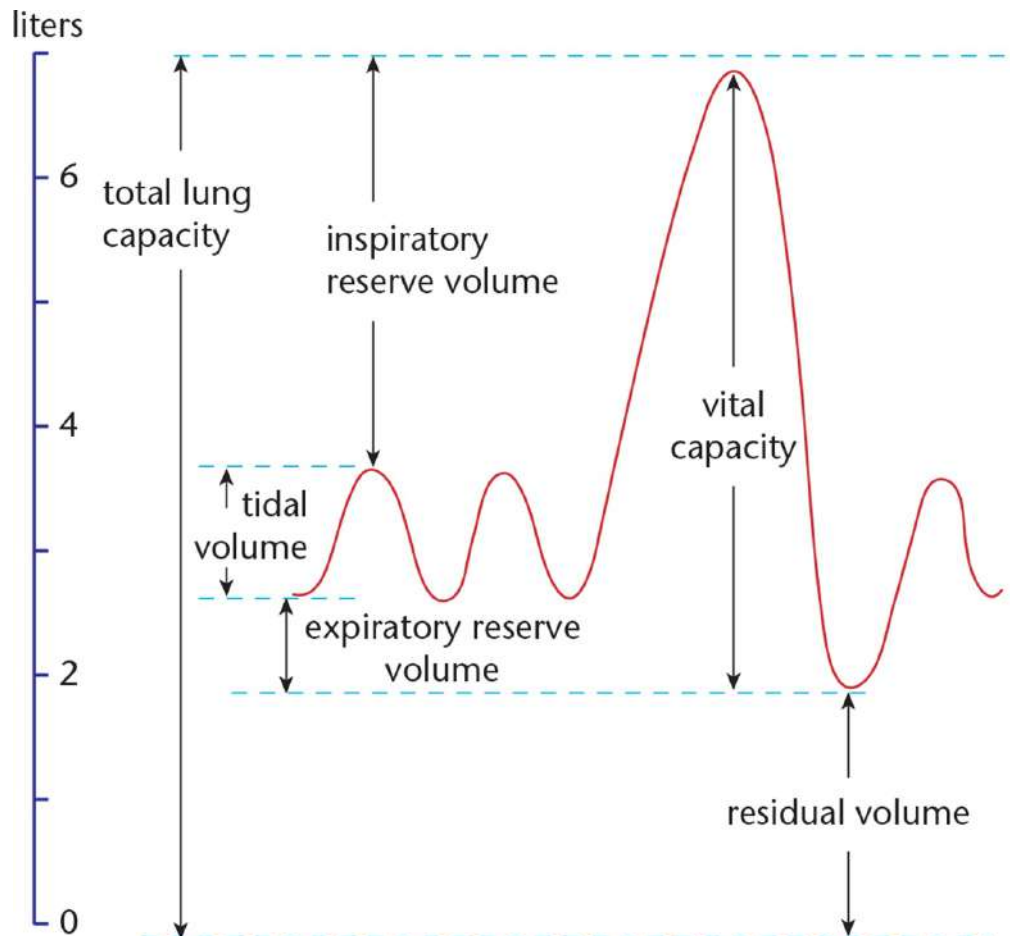


Figure 6.4. Lung Volumes

REGULATION OF BREATHING

Breathing requires input from our nervous control center. Ventilation is primarily regulated by a collection of neurons in the medulla oblongata called the **ventilation center** that fire rhythmically to cause regular contraction of

respiratory muscles. These neurons contain **chemoreceptors** that are primarily sensitive to carbon dioxide concentration. As the partial pressure of carbon dioxide in the blood rises (**hypercarbia** or **hypercapnia**), the **respiratory rate** will increase so that more carbon dioxide is exhaled, causing carbon dioxide levels in the blood to fall. These cells also respond to changes in oxygen concentration, although this tends to have significance only during periods of significant **hypoxemia** (low oxygen concentration in the blood).

We can, to a limited extent, control our breathing consciously. We can choose to breathe more rapidly or slowly; however, extended periods of hypoventilation would lead to increased carbon dioxide levels and an override by the medulla oblongata (which would jump-start breathing). The opposite process (hyperventilation) would blow off too much carbon dioxide and ultimately inhibit ventilation.

MCAT CONCEPT CHECK 6.1:

Before you move on, assess your understanding of the material with these questions.

1. List the structures in the respiratory pathway, from where air enters the nares to the alveoli.

2. Which muscle(s) are involved in inhalation? Exhalation?

- Inhalation:

- Exhalation:

3. What is the purpose of surfactant?

4. What is the mathematical relationship between vital capacity (VC), inspiratory reserve volume (IRV), expiratory reserve volume (ERV), and tidal volume (TV)?

5. If blood levels of CO_2 become too low, how does the brain alter the respiratory rate to maintain homeostasis?

6.2 Functions of the Respiratory System

LEARNING OBJECTIVES

After Chapter 6.2, you will be able to:

- Describe the mechanisms used in the respiratory system to prevent infection
- Recall the chemical equation for the bicarbonate buffer system
- Predict how blood pH will change in response to changing concentrations of O₂ or CO₂

No organ system functions alone. The lungs function in gas exchange, but this is only part of the respiratory story. The lungs are lined with a tremendous number of capillaries that can also be used in thermoregulation. The lungs also represent a pathway into the body and serve an immune function to prevent invaders from gaining access to the bloodstream. Finally, the lungs also allow for control of blood pH by controlling carbon dioxide concentrations. Therefore, the lungs are integrated with many other body systems, including the cardiovascular, immune, renal, and nervous systems.

GAS EXCHANGE

Gas exchange is, of course, the primary function of the lungs. Each alveolus is surrounded by a network of capillaries. The capillaries bring deoxygenated blood from the **pulmonary arteries**, which originate from the right ventricle of the heart. The walls of the alveoli are only one cell thick, which facilitates the diffusion of carbon dioxide from the blood into the lungs, and oxygen into the blood. The oxygenated blood returns to the left atrium of the heart via the **pulmonary veins**.



REAL WORLD

Diffusion of gases occurs across a very thin membrane between the alveolus and the capillary. However, certain diseases may cause fibrosis, or scarring, of this membrane, resulting in less effective diffusion. Other diseases may cause a limitation of ventilation (gas flow) or perfusion (blood flow) to the lung. All of these mechanisms can cause hypoxia—low blood oxygen levels—although they accomplish that same end result through different means.

The driving force for gas exchange is the pressure differential of the gases. When it initially arrives at the alveoli, blood has a relatively low partial pressure of oxygen and a relatively high partial pressure of carbon dioxide, facilitating transfer of each down its respective concentration gradient, as shown in Figure 6.5. Because the gradient between the blood and air in the lungs is already present as the blood enters the lungs, no energy is required for gas transfer.

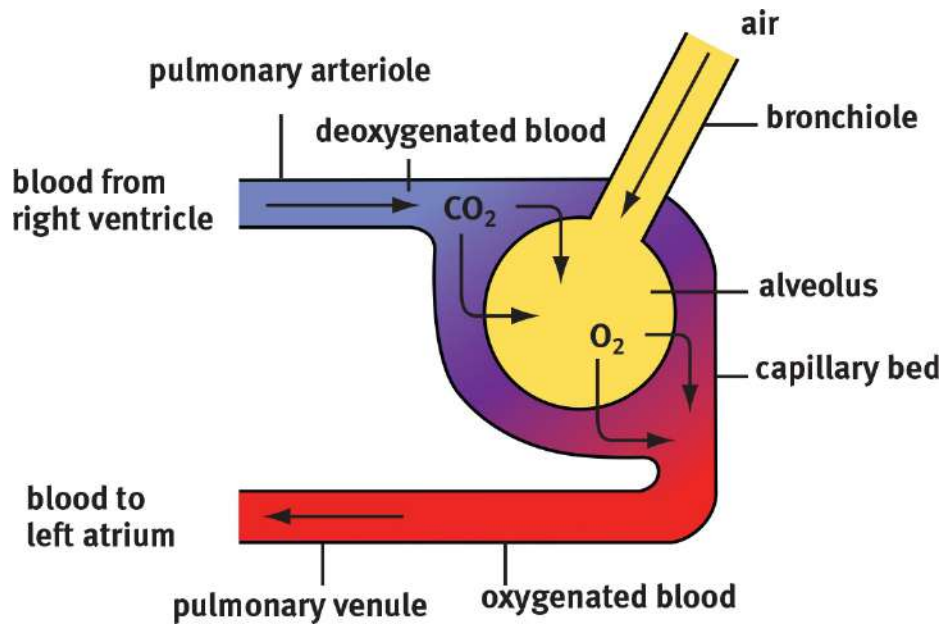


Figure 6.5. Gas Exchange in the Alveolus



KEY CONCEPT

O_2 in the alveoli flows down its partial pressure gradient from the alveoli into the pulmonary capillaries, where it can bind to hemoglobin for transport. Meanwhile, CO_2 flows down its partial pressure gradient from the capillaries into the alveoli for expiration.

How would our respiratory systems adjust if we moved to higher altitudes where less oxygen is available? First, we would breathe more rapidly to try to avoid hypoxia; second, the binding dynamics of hemoglobin to oxygen would be altered to facilitate the unloading of oxygen at the tissues. As we will discuss in Chapter 7 of *MCAT Biology Review*, the natural response of hemoglobin to the decreased carbon dioxide concentration in the

environment would actually be to *decrease* the unloading of oxygen to tissues, so other mechanisms must counteract and override this phenomenon. In the short term, the body can make more red blood cells to ensure the adequate delivery of oxygen. In the long term, the body could develop more blood vessels (vascularization), which would facilitate the distribution of oxygen to tissues.

THERMOREGULATION

In order to maximize gas exchange, there is a tremendous surface area over which the alveoli and capillaries interact. Because the entire respiratory tract is highly vascular, it can also be used for **thermoregulation**, or the regulation of body temperature. Heat—the transfer of thermal energy—is regulated via the body surfaces by **vasodilation** and **vasoconstriction**. As capillaries expand, more blood can pass through these vessels, and a larger amount of thermal energy can be dissipated. As capillaries contract, less blood can pass through them, conserving thermal energy. Nasal and tracheal capillaries are most frequently used for these purposes within the respiratory system. While these capillary beds provide a mechanism for thermoregulation, humans predominantly regulate temperature using capillaries and sweat glands in the skin, or rapid muscle contraction (shivering). The respiratory system can also transfer heat to the environment through evaporation of water in mucous secretions. Other animals, such as dogs, take advantage of this cooling mechanism by **panting**.

IMMUNE FUNCTION

As mentioned above, the lungs provide a large interface for the body to interact with the outside world. While this is important for gas exchange and thermoregulation, it also comes with potential risks—pathogens such as bacteria, viruses, and fungi can cause infections in the lung, or can gain access to the body through the rich vascularity of the alveolar membranes. By necessity, the lungs must be able to fight off potential invaders. The first line of defense is the nasal cavity, which has small hairs (vibrissae) that help to trap particulate matter and potentially infectious particles. The nasal cavity also contains an enzyme called **lysozyme**. Also found in tears and saliva, lysozyme is able to attack the peptidoglycan walls of gram-positive bacteria. The internal airways are lined with mucus, which traps particulate matter and larger invaders. Underlying cilia then propel the mucus up the respiratory tract to the oral cavity, where it can be expelled or swallowed; this mechanism is called the **mucociliary escalator**.



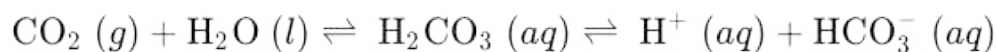
REAL WORLD

Pneumonia is an infection of the lung most often caused by bacteria or viruses. *Atypical pneumonia*, commonly called walking pneumonia (because the infection does not require hospitalization and does not leave the patient bedridden), is often caused by a very small bacterium called *Mycoplasma pneumoniae*. This bacterium causes a prolonged cough because it damages epithelial cells lining the lung and paralyzes the cilia lining the respiratory tract. The lack of cilia makes it much more difficult to clear mucus from the lungs. The cough lasts until the respiratory epithelial cells have recovered and the cilia are once again functional.

The lungs, especially the alveoli, also contain numerous immune cells, including macrophages. **Macrophages** can engulf and digest pathogens and signal to the rest of the immune system that there is an invader. Mucosal surfaces also contain IgA antibodies that help to protect against pathogens that contact the mucous membranes. Finally, **mast cells** also populate the lungs. These cells have preformed antibodies on their surfaces. When the right substance attaches to the antibody, the mast cell releases inflammatory chemicals into the surrounding area to promote an immune response. Unfortunately, these antibodies are often reactive to substances such as pollen and molds, so mast cells also provide the inflammatory chemicals that mediate allergic reactions.

CONTROL OF pH

The respiratory system plays a role in pH balance through the **bicarbonate buffer system** in the blood:



MCAT EXPERTISE

The division within the sciences is largely artificial; the MCAT often contains questions that integrate multiple science disciplines. A question located in the *Biological and Biochemical Foundations of Living Systems* section may require knowledge of general chemistry. In fact, 10% of this section is chemistry: 5% general chemistry and 5% organic chemistry.

Questions regarding the bicarbonate buffer system are MCAT favorites, and you are very likely to see it in some form on Test Day. This equation represents an opportunity for the MCAT to test understanding of basic chemistry concepts, such as Le Châtelier's principle, as well as how disturbances in pH may affect respiration.

The body attempts to maintain a pH between 7.35 and 7.45. When the pH is lower, and hydrogen ion concentration is higher (**acidemia**), acid-sensing chemoreceptors just outside the blood–brain barrier send signals to the brain to increase the respiratory rate. Further, an increasing hydrogen ion concentration will cause a shift in the bicarbonate buffer system, generating additional carbon dioxide. As described earlier, the respiratory centers in the brain are sensitive to this increasing partial pressure of carbon dioxide and will also promote an increase in respiratory rate.



REAL WORLD

Metabolic acidosis—a condition of excess acid by any mechanism besides hypoventilation—is a common occurrence in medicine. Anaerobic respiration can generate lactic acid; individuals with type 1 diabetes mellitus can produce ketoacids when they are hypoinsulinemic; certain poisons, like methanol and formaldehyde, can produce organic acids. In each of these cases, one of the primary methods of compensation is increasing respiration rate.

As the respiratory rate increases, more carbon dioxide is blown off. This will also push the buffer equation to the left, but notice the difference: the shift to

the left in the previous paragraph was caused by an increase in hydrogen ion concentration, which elevated the concentration of carbon dioxide. Here, the removal of carbon dioxide causes a shift to the left that allows the hydrogen ion concentration to drop back to normal.

If the blood is too basic (**alkalemia**), then the body will seek to increase acidity. How can the lungs contribute to this? If the respiratory rate is slowed, then more carbon dioxide will be retained, shifting the buffer equation to the right and producing more hydrogen ions and bicarbonate ions. This results in a lower pH.



BRIDGE

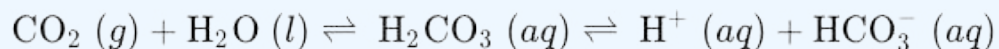
If H^+ is an acid and HCO_3^- is a base, then why doesn't increasing both of them yield a constant pH? The reason is that H^+ is a strong acid, while HCO_3^- is a weak base. Just like a titration, discussed in Chapter 10 of *MCAT General Chemistry Review*, this combination will shift the pH of the solution toward the acidic range.

Overall, the lungs play a role in the immediate adjustment of carbon dioxide levels and, by extension, hydrogen ion levels. However, the lungs do not work alone to maintain proper pH. The kidneys also play a role by modulating secretion and reabsorption of acid and base within the nephron. This is a much slower response, however, and represents long-term compensation. For more information on kidney function and homeostasis, see Chapter 10 of *MCAT Biology Review*.



KEY CONCEPT

This equation is essential to Test Day success:



It is likely to be tested in both the *Biological and Biochemical Foundations of Living Systems* and the *Chemical and Physical Foundations of Biological Systems* sections.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

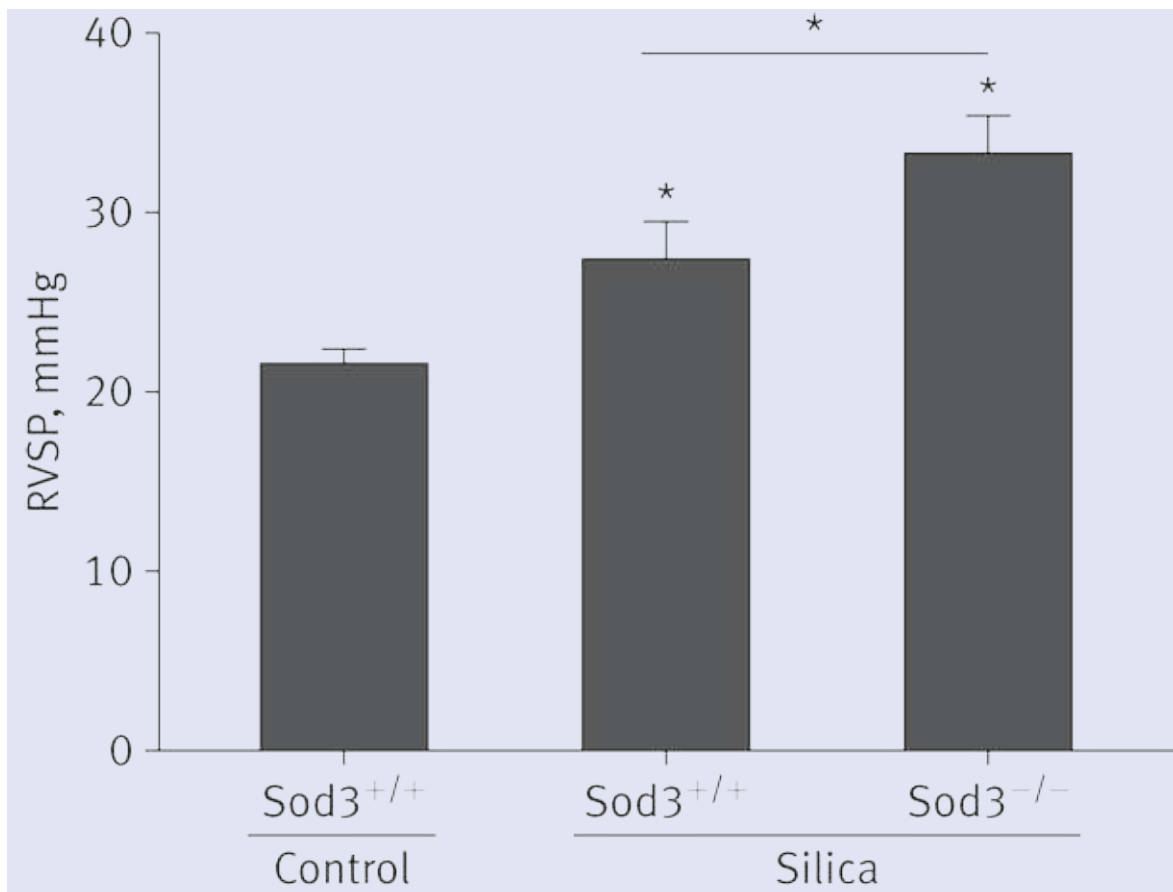
Chronic inhalation of crystalline silica has been shown to result in silicosis and pulmonary fibrosis and may also play a role in the development of pulmonary hypertension. The mechanism by which these pathologies develop is not clearly understood, but oxidant/antioxidant imbalances have been suggested.

Background information regarding lung health through oxidative

Extracellular superoxide dismutase (Sod3) is one of the most abundant antioxidant enzymes in the pulmonary vasculature.

Research shows that proinflammatory enzymes promote the development of pulmonary hypertension. Let's first

To study the role of Sod3 in the development of silica-mediated pathologies, scientists injected a crystalline silica suspension into the trachea of WT and Sod3^{-/-} mice.	Exp set con silic ind pat bet anc (kn mic
28 days later, a pressure catheter was inserted into the right ventricle through the jugular vein to measure the right ventricular systolic pressure (RVSP) (Figure 1). Elevated RVSP is associated with pulmonary hypertension.	Exp me RVSP RVSP pul hyp
Following the RVSP measurement, mice were sacrificed. mRNA was extracted from lung tissue and assayed for Sod3 expression (Figure 2).	Exp me Sod3 exp
Lung sections were stained with Mason's trichrome to visualize collagen, a marker of pulmonary fibrosis (Figure 3).	Exp qu col con the inci col lun fibr
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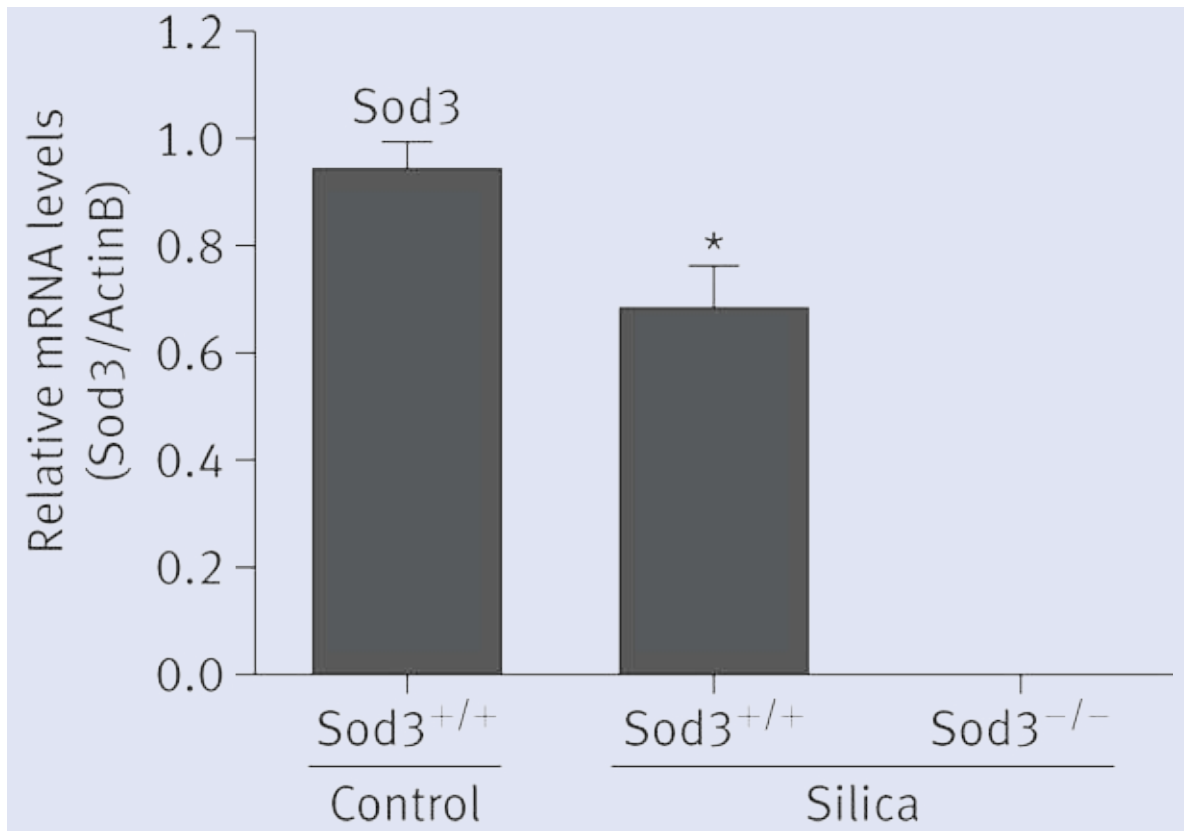
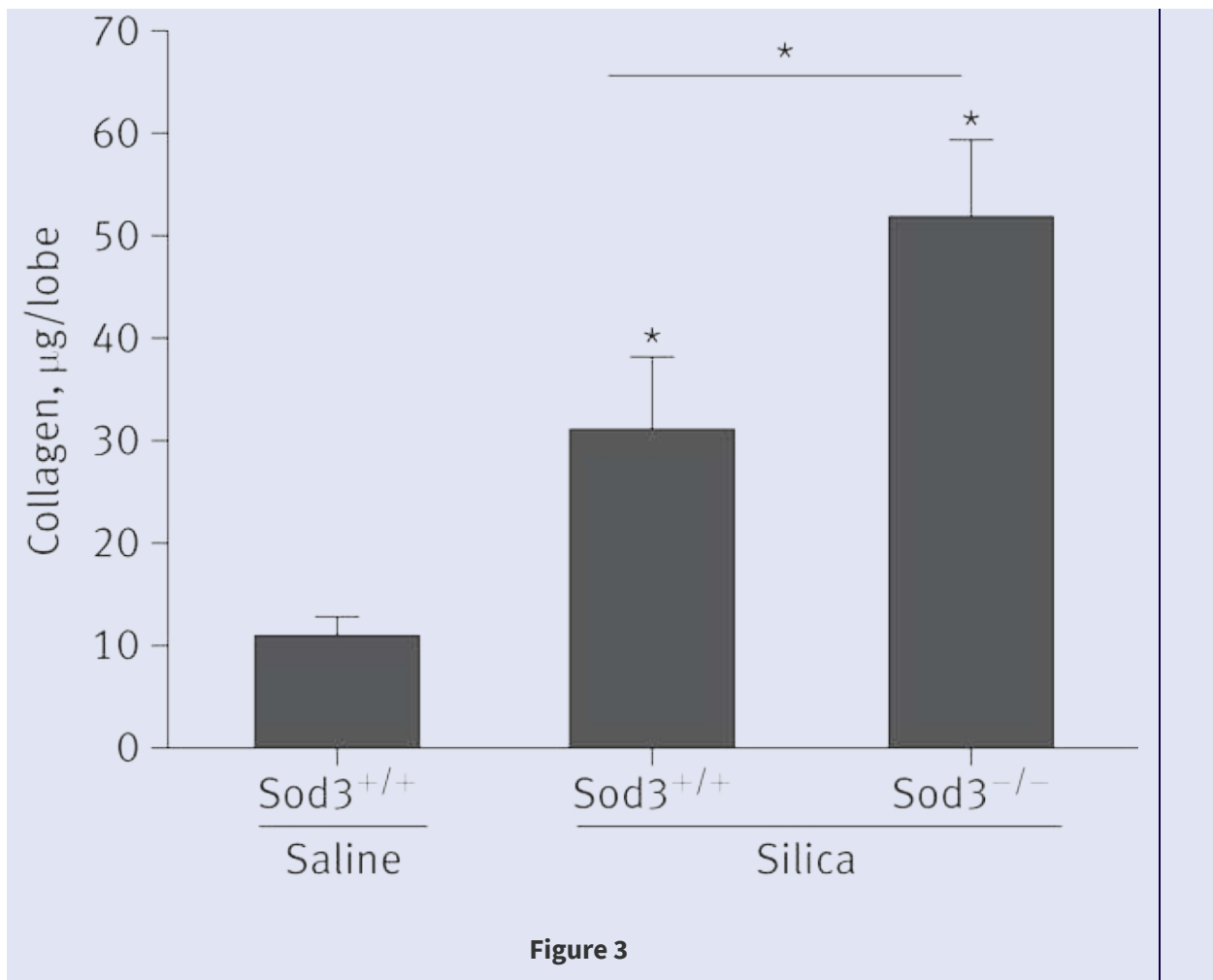


Figure 2

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adapted from Zelko, I. N., Zhu, J., & Roman, J. (2018). Role of SOD3 in silica-related lung fibrosis and pulmonary vascular remodeling. *Respiratory Research*, 19(1), 221. doi:10.1186/s12931-018-0933-6.

Based on the findings of this study, does Sod3 protect against or exacerbate crystalline silica-induced pathologies?

This question is really testing our understanding of how the presence or absence of Sod3 affects the development of silica-induced lung pathologies. Finding an answer will require a strong understanding of the experimental design and careful analysis of the data. Let's start with the experimental design. Paragraph 2 tells us that we have three experimental groups: normal

mice not treated with silica, normal mice treated with silica, and Sod3 knockout (KO) mice treated with silica. The normal mice lacking treatment serve as a negative control, whereas the normal mice treated with silica serve as the positive control. Paragraph 1 tells us that crystalline silica has already been shown to cause pulmonary fibrosis and silicosis, so we should expect to see indicators of these pathologies in normal mice treated with silica, but not in the untreated mice. If Sod3 has a protective role, deleting it should lead to an increase in lung damage. If Sod3 exacerbates silica-induced lung damage, deleting it should lead to a decrease in lung damage.

Time to analyze the data! Figure 1 shows that in normal mice, exposure to silica leads to an increase in RVSP, indicating a potential role for silica in the development of pulmonary hypertension. Furthermore, deletion of Sod3 leads to a greater elevation in RVSP, suggesting that Sod3 may have a protective role. Figure 2 shows the expression of the Sod3 transcript in lung tissue. The Sod3 KO mice show no expression, which is consistent with a gene knockout. We also note that the silica exposure leads to a decrease in Sod3 expression. Figure 3 shows an elevation in collagen content in the lungs of silica-treated mice compared to untreated mice. This elevation is significantly higher in Sod3 KO mice. Taken together, these results suggest that Sod3 protects against both silica-induced pulmonary fibrosis and pulmonary hypertension.

Because knockout of Sod3 leads to an increase in pulmonary fibrosis and pulmonary hypertension after exposure to crystalline silica, Sod3 most likely serves a protective role in the lungs against silica-mediated damage.

MCAT CONCEPT CHECK 6.2:

Before you move on, assess your understanding of the material with these questions.

1. What are some of the mechanisms used in the respiratory system to prevent infection?

2. What is the chemical equation for the bicarbonate buffer system?

3. Respiratory failure refers to inadequate ventilation to provide oxygen to the tissues. How would the pH change in respiratory failure?



Conclusion

As we learn about the human body, it may be easy to reduce the complex and varied functions of the lungs to breathing and providing a supply of oxygen. The lungs do indeed perform gas exchange, which relies on differences in partial pressures of gases between the alveoli and the blood. Oxygen is taken up by the blood, while carbon dioxide is released for exhalation. Inhalation and exhalation also require pressure differentials created by anatomical structures such as the chest wall, diaphragm, pleurae, and lungs.

However, the lungs are so much more than just *bags of air*; gas exchange is not the only function of the respiratory system. The respiratory system also serves essential roles in thermoregulation, immunity, and pH regulation. As we go through the individual systems within the human body, take special note of how each system is integrated with the other systems. One of the more clear connections is the binding of oxygen to hemoglobin in the lungs and in the rest of the circulatory system—a concept we will expand upon in the next chapter, along with the effects of altitude, pH, and chemicals on this binding.

Concept Summary

Anatomy and Mechanism of Breathing

- Air is drawn in through the **nares**, and through the nasal cavity and **pharynx**, where it is warmed and humidified. It is filtered by nasal hairs (**vibrissae**) and mucous membranes. It then enters the **larynx**, followed by the **trachea**. The trachea divides into two mainstem **bronchi**, which divide into **bronchioles**, which divide into continually smaller passages until they reach the alveoli.
- **Alveoli** are small sacs that interface with the pulmonary capillaries, allowing gases to diffuse across a one-cell-thick membrane.
- **Surfactant** in the alveoli reduces surface tension at the liquid–gas interface, preventing collapse.
- The pleurae cover the lungs and line the chest wall.
 - The **visceral pleura** lies adjacent to the lung itself.
 - The **parietal pleura** lines the chest wall.
 - The **intrapleural space** lies between these two layers and contains a thin layer of fluid that lubricates the two pleural surfaces.
- The **diaphragm** is a thin skeletal muscle that helps to create the pressure differential required for breathing.
- Inhalation is an active process.
 - The diaphragm and **external intercostal muscles** expand the thoracic

cavity, increasing the volume of the intrapleural space. This decreases the intrapleural pressure.

- This pressure differential ultimately expands the lungs, dropping the pressure within and drawing in air from the environment. This mechanism is termed **negative-pressure breathing**.
- Exhalation may be passive or active.
 - In passive exhalation, relaxation of the muscles of inspiration and elastic recoil of the lungs allow the chest cavity to decrease in volume, reversing the pressure differentials seen in inhalation.
 - In active exhalation, the internal intercostal muscles and abdominal muscles can be used to forcibly decrease the volume of the thoracic cavity, pushing out air.
- A **spirometer** can be used to measure lung capacities and volumes.
 - **Total lung capacity (TLC)** is the maximum volume of air in the lungs when one inhales completely.
 - **Residual volume (RV)** is the volume of air remaining in the lungs when one exhales completely.
 - **Vital capacity (VC)** is the difference between the minimum and maximum volume of air in the lungs.
 - **Tidal volume (TV)** is the volume of air inhaled or exhaled in a normal breath.
 - **Expiratory reserve volume (ERV)** is the volume of additional air that can be forcibly exhaled after a normal exhalation.
 - **Inspiratory reserve volume (IRV)** is the volume of additional air that can be forcibly inhaled after a normal inhalation.

- Ventilation is regulated by the **ventilation center**, a collection of neurons in the medulla oblongata.
 - **Chemoreceptors** respond to carbon dioxide concentrations, increasing the respiratory rate when there is a high concentration of carbon dioxide in the blood (**hypercarbia** or **hypercapnia**).
 - The ventilation center can also respond to low oxygen concentrations in the blood (**hypoxemia**) by increasing ventilation rate.
 - Ventilation can also be controlled consciously through the cerebrum, although the medulla oblongata will override the cerebrum during extended periods of hypo- or hyperventilation.

Functions of the Respiratory System

- The lungs perform gas exchange with the blood through simple diffusion across concentration gradients.
 - Deoxygenated blood with a high carbon dioxide concentration is brought to the lungs via the **pulmonary arteries**.
 - Oxygenated blood with a low carbon dioxide concentration leaves the lungs via the **pulmonary veins**.
- The large surface area of interaction between the alveoli and capillaries allows the respiratory system to assist in thermoregulation through **vasodilation** and **vasoconstriction** of capillary beds.
- The respiratory system must be protected from potential pathogens.
 - Multiple mechanisms, including vibrissae, mucous membranes, and the **mucociliary escalator**, help filter the incoming air and trap particulate

matter.

- **Lysozyme** in the nasal cavity and saliva attacks peptidoglycan cell walls of gram-positive bacteria.
 - **Macrophages** can engulf and digest pathogens and signal to the rest of the immune system that there is an invader.
 - Mucosal surfaces are covered with IgA antibodies.
 - **Mast cells** have antibodies on their surface that, when triggered, can promote the release of inflammatory chemicals. Mast cells are often involved in allergic reactions as well.
- The respiratory system is involved in pH control through the bicarbonate buffer system.
 - When blood pH decreases, respiration rate increases to compensate by blowing off carbon dioxide. This causes a left shift in the buffer equation, reducing hydrogen ion concentration.
 - When blood pH increases, respiration rate decreases to compensate by trapping carbon dioxide. This causes a right shift in the buffer equation, increasing hydrogen ion concentration.

Answers to Concept Checks

6.1

- .. Nares → nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli
- 1. Inhalation uses the diaphragm and external intercostal muscles; in labored breathing, muscles of the neck and back may also be involved. Passive exhalation uses the recoil of these same muscles; active exhalation also uses the internal intercostal muscles and abdominal muscles.
- 2. Surfactant reduces surface tension at the air–liquid interface in the alveoli. This prevents their collapse.
- 3. Vital capacity is the sum of the inspiratory reserve volume, expiratory reserve volume, and tidal volume: $VC = IRV + ERV + TV$
- 4. When CO_2 levels become too low, the brain can decrease the respiratory rate in order to raise CO_2 levels.

6.2

- .. Immune mechanisms in the respiratory system include vibrissae in the nares, lysozyme in the mucous membranes, the mucociliary escalator, macrophages in the lungs, mucosal IgA antibodies, and mast cells.
- 1. $CO_2 (g) + H_2O (l) \rightleftharpoons H_2CO_3 (aq) \rightleftharpoons H^+ (aq) + HCO_3^- (aq)$
- 2. In respiratory failure, ventilation slows, and less carbon dioxide is blown off. As this occurs, the buffer equation shifts to the right, and more hydrogen ions are generated. This results in a lower pH of the blood.



Shared Concepts

Biology Chapter 7

The Cardiovascular System

Biology Chapter 10

Homeostasis

General Chemistry Chapter 6

Equilibrium

General Chemistry Chapter 8

The Gas Phase

General Chemistry Chapter 10

Acids and Bases

Physics and Math Chapter 3

Thermodynamics

Discrete Practice Questions

1. All of the following facilitate gas exchange in the lungs EXCEPT:
 - (A) thin alveolar walls.
 - (B) multiple subdivisions of the respiratory tree.
 - (C) differences in the partial pressures of O_2 and CO_2 .
 - (D) active transporters in alveolar cells.

2. Which of the following associations correctly pairs a stage of respiration with the muscle actions occurring during that stage?
 - (A) Inhalation—diaphragm relaxes
 - (B) Inhalation—internal intercostal muscles contract
 - (C) Exhalation—diaphragm contracts
 - (D) Exhalation—external intercostal muscles relax

3. Total lung capacity is equal to the vital capacity plus the:
 - (A) tidal volume.
 - (B) expiratory reserve volume.
 - (C) residual volume.
 - (D) inspiratory reserve volume.

4. The intrapleural pressure is necessarily lower than the atmospheric pressure during:
- (A) inhalation, because the expansion of the chest cavity causes compression of the intrapleural space, decreasing its pressure.
 - (B) inhalation, because the expansion of the chest cavity causes expansion of the intrapleural space, decreasing its pressure.
 - (C) exhalation, because the compression of the chest cavity causes compression of the intrapleural space, decreasing its pressure.
 - (D) exhalation, because the compression of the chest cavity causes expansion of the intrapleural space, decreasing its pressure.
5. A patient presents to the emergency room with a stab wound to the left side of the chest. On a chest X-ray, blood is noted to be collecting in the chest cavity, causing collapse of both lobes of the left lung. The blood is most likely located between:
- (A) the parietal pleura and the chest wall.
 - (B) the parietal pleura and the visceral pleura.
 - (C) the visceral pleura and the lung.
 - (D) the alveolar walls and the lung surface.
6. Each of the following statements regarding the anatomy of the respiratory system is true EXCEPT:
- (A) the epiglottis covers the glottis during swallowing to ensure that food does not enter the trachea.
 - (B) the trachea and bronchi are lined by ciliated epithelial cells.

- (C) the pharynx contains two vocal cords, which are controlled by skeletal muscle and cartilage.
- (D) the nares are lined with vibrissae, which help filter out particulate matter from inhaled air.

7. Which of the following is a correct sequence of passageways through which air travels during inhalation?

- (A) Pharynx → trachea → bronchioles → bronchi → alveoli
- (B) Pharynx → trachea → larynx → bronchi → alveoli
- (C) Larynx → pharynx → trachea → bronchi → alveoli
- (D) Pharynx → larynx → trachea → bronchi → alveoli

8. Idiopathic pulmonary fibrosis (IPF) is a disease in which scar tissue forms in the alveolar walls, making the lung tissue significantly more stiff. Which of the following findings would likely be detected through spirometry in a patient with IPF?

- I.** Decreased total lung capacity
- II.** Decreased inspiratory reserve volume
- III.** Increased residual volume

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

9. Studies have indicated that premature babies are often deficient in lysozyme. What is a possible consequence of this deficiency?
- (A) Respiratory distress and alveolar collapse shortly after birth
 - (B) Increased susceptibility to certain infections
 - (C) Inability to humidify air as it passes through the nasal cavity
 - (D) Slowing of the respiratory rate in response to acidemia
10. Some forms of pneumonia cause an excess of fluids such as mucus or pus to build up within an entire lobe of the lung. How will this affect the diffusion of gases within the affected area?
- (A) Carbon dioxide can diffuse out, but oxygen will not be able to enter the blood.
 - (B) Oxygen can diffuse into the blood, but carbon dioxide cannot diffuse out.
 - (C) No change in diffusion will occur.
 - (D) No diffusion will occur in the affected area.
11. Some people with anxiety disorders respond to stress by hyperventilating. It is recommended that these people breathe into a paper bag and then rebreathe this air. Why is this treatment appropriate?
- (A) Hyperventilation causes an increase in blood carbon dioxide, and breathing the air in the bag helps to readjust blood levels of carbon dioxide.
 - (B) Hyperventilation causes a decrease in blood carbon dioxide, and

breathing the air in the bag helps to readjust blood levels of carbon dioxide.

- (C) Hyperventilation causes an increase in blood oxygen, and breathing the air in the bag helps to readjust blood levels of oxygen.
- (D) Hyperventilation causes a decrease in blood oxygen, and breathing the air in the bag helps to readjust blood levels of oxygen.

12. A patient presents to the emergency room with an asthma attack. The patient has been hyperventilating for the past hour and has a blood pH of 7.52. The patient is given treatment and does not appear to respond, but a subsequent blood pH reading is 7.41. Why might this normal blood pH NOT be a reassuring sign?

- (A) The patient's kidneys may have compensated for the alkalemia.
- (B) The normal blood pH reading is likely inaccurate.
- (C) The patient may be descending into respiratory failure.
- (D) The patient's blood should ideally become acidemic for some time to compensate for the alkalemia.

13. Premature infants with respiratory distress are often placed on ventilators. Often, the ventilators are set to provide positive end-expiratory pressure. Why might this setting be useful for a premature infant?

- (A) Premature infants lack surfactant.
- (B) Premature infants lack lysozyme.

- (C) Premature infants cannot thermoregulate.
- (D) Premature infants are unable to control pH.

14. In emphysema, the alveolar walls are destroyed, decreasing the recoil of the lung tissue. Which of the following changes may be seen in a patient with emphysema?

- (A) Increased residual volume
- (B) Decreased total lung capacity
- (C) Increased blood concentration of oxygen
- (D) Decreased blood concentration of carbon dioxide

15. Allergic reactions occur due to an overactive immune response to a substance. Which cells within the respiratory tract play the largest role in the generation of allergic reactions?

- (A) Alveolar epithelial cells
- (B) Macrophages
- (C) Mast cells
- (D) Ciliated epithelial cells

Discrete Practice Answers

1. **D**

Gas exchange in the lungs relies on passive diffusion of oxygen and carbon dioxide. This is accomplished easily because there is always a difference in the partial pressures of these two gases and because the subdivision of the respiratory tree creates a large surface area of interaction between the alveoli and the circulatory system. In addition, the thin alveolar walls allow for fast diffusion and gas exchange. Therefore, **(A)**, **(B)**, and **(C)** can be eliminated. **(D)** is the correct answer because active transport is not used in the gas exchange process in the lungs.

2. **D**

The muscles involved in ventilation are the diaphragm, which separates the thoracic cavity from the abdominal cavity, and the intercostal muscles. During inhalation, the diaphragm contracts and flattens, while the external intercostal muscles contract, pulling the rib cage up and out. These actions cause an overall increase in the volume of the thoracic cavity. During exhalation, both the diaphragm and the external intercostals relax, causing a decrease in the volume of the thoracic cavity because of the recoil of these tissues. In forced exhalation, the internal intercostals and abdominal muscles may contract to force out air. Thus, the only correct association from the given answers is **(D)**.

3. **C**

Total lung capacity is equal to the vital capacity (the maximum volume of air that can be forcibly inhaled and exhaled from the lungs) plus the residual volume (the air that always remains in the lungs, preventing the alveoli from collapsing).

4. **B**

During inhalation, the chest cavity expands, causing expansion of the intrapleural space. According to Boyle's law, an increase in volume at a constant temperature is accompanied by a decrease in pressure. When the intrapleural pressure (and, by extension, the alveolar pressure) is less than atmospheric pressure, air enters the lungs. During exhalation, these pressure gradients reverse; thus, during exhalation, intrapleural pressure is higher than atmospheric pressure, not lower.

5. **B**

The intrapleural space, bounded by the parietal and visceral pleurae, is a potential space. As such, it is normally collapsed and contains a small amount of fluid. However, introduction of fluid or air into the intrapleural space can fill the space, causing collapse of the lung. The other options listed are too firmly apposed to permit blood to collect in these spaces.

6. **C**

The pharynx, which lies behind the nasal cavity and oral cavity, is a common pathway for food entering the digestive system and air entering the respiratory system. It is the larynx that contains the vocal cords, not the pharynx.

7. **D**

Air enters the respiratory tract through the external nares (nostrils) and travels through the nasal cavities. It then passes through the pharynx and

into the larynx. Ingested food also passes through the pharynx on its way to the esophagus; to ensure that food does not accidentally enter the larynx, the epiglottis covers the larynx during swallowing. After the larynx, air goes to the trachea, which eventually divides into two bronchi, one for each lung. The bronchi branch into smaller bronchioles, which terminate in clusters of alveoli. From the given sequences, only **(D)** correctly describes the sequence of the passages through which air travels.

8. **B**

In a patient with IPF, the increased stiffness of the lungs would likely decrease the volume of air the individual could inhale, which would decrease both the total lung capacity and inspiratory reserve volume. However, spirometry cannot measure the total lung capacity accurately because it cannot determine the residual volume—the volume of air left in the lungs when an individual has maximally exhaled. Because the residual volume makes up a portion of the total lung capacity (total lung capacity = vital capacity + residual volume), a spirometer cannot be used to determine the total lung capacity. Therefore, while Statement I is a true statement about individuals with IPF, it cannot appear in the answer choice. Finally, increased stiffness of the lungs would be expected to decrease the residual volume, not increase it; further, residual volume, as described above, cannot be measured with a spirometer.

9. **B**

Lysozyme is an enzyme present in the nasal cavity, saliva, and tears that degrades peptidoglycan, preventing infection by gram-positive bacteria. Thus, premature infants who lack lysozyme are more likely to suffer from infections with these organisms.

10. **D**

If an area of the lung becomes filled with mucus and inflammatory cells, the area will not be able to participate in gas exchange. Because no air will enter or leave the area, the concentration gradient will no longer exist, and neither oxygen nor carbon dioxide will be able to diffuse across the alveolar wall.

1. **B**

When people hyperventilate, their respiratory rate increases. When the respiratory rate increases, more carbon dioxide is blown off. This causes a shift to the left in the bicarbonate buffer equation, and the blood becomes more alkaline. Breathing into the bag allows some of this carbon dioxide to be returned to the bloodstream in order to maintain the proper pH.

2. **C**

When a patient with an asthma attack does not respond to treatment and has been hyperventilating for over an hour, he or she may become fatigued and may not be able to maintain hyperventilation. In this case, the patient begins to decrease his or her breathing rate and is not receiving adequate oxygen. By extension, carbon dioxide is trapped in the blood, and the pH begins to drop. Despite the fact that this pH is normal at the moment, this patient is crashing and may start demonstrating acidemia in the near future. While the kidneys should compensate for alkalemia, this is a slow process and would not normalize the blood pH within an hour; further, adequate compensation by the kidneys would actually be a reassuring sign, eliminating **(A)**. There is no evidence to suggest that the measurement was inaccurate, eliminating **(B)**. Finally, after treatment, the patient should return to a normal blood pH with adequate ventilation and would not be expected to overcompensate by becoming acidemic, eliminating **(D)**.

3. **A**

This question requires a few different levels of thinking. The question stem states that premature infants often require ventilation using positive end-expiratory pressures. While you are not expected to know ventilator settings for the MCAT, you should be able to decode what this phrase means: at the end of expiration, the ventilator will provide a higher pressure than normal, which forces extra air into the alveoli. This pressure must be used to prevent alveolar collapse, which should remind you that surfactant serves the same purpose by reducing surface tension. Thus, it makes sense that if premature babies lack surfactant, providing extra air pressure at the end of expiration would be beneficial.

4. **A**

The intrinsic elastic properties of the lung are important during exhalation as the passive recoil of lung tissue helps decrease lung volume. With decreased recoil, the patient will have difficulty exhaling completely, increasing the residual volume. The total lung capacity would be expected to increase in this case because there would be less recoil opposing inhalation, eliminating **(B)**. With decreased alveolar surface area, one would expect decreased gas exchange, which would decrease blood concentrations of oxygen while increasing blood concentrations of carbon dioxide, eliminating **(C)** and **(D)**.

5. **C**

Allergic reactions occur when a substance binds to an antibody and promotes an overactive immune response with inflammatory chemicals. The antibody is already attached to a mast cell. Thus, when the substance binds to the antibody, the mast cell can release the inflammatory mediators that cause allergic reactions.



CHAPTER 7

The Cardiovascular System

CHAPTER 7

The Cardiovascular System

In This Chapter

7.1 **Anatomy of the Cardiovascular System**

The Heart
The Vasculature
Circulation



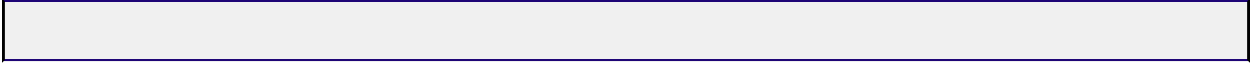
7.2 **Blood**

Composition
Blood Antigens

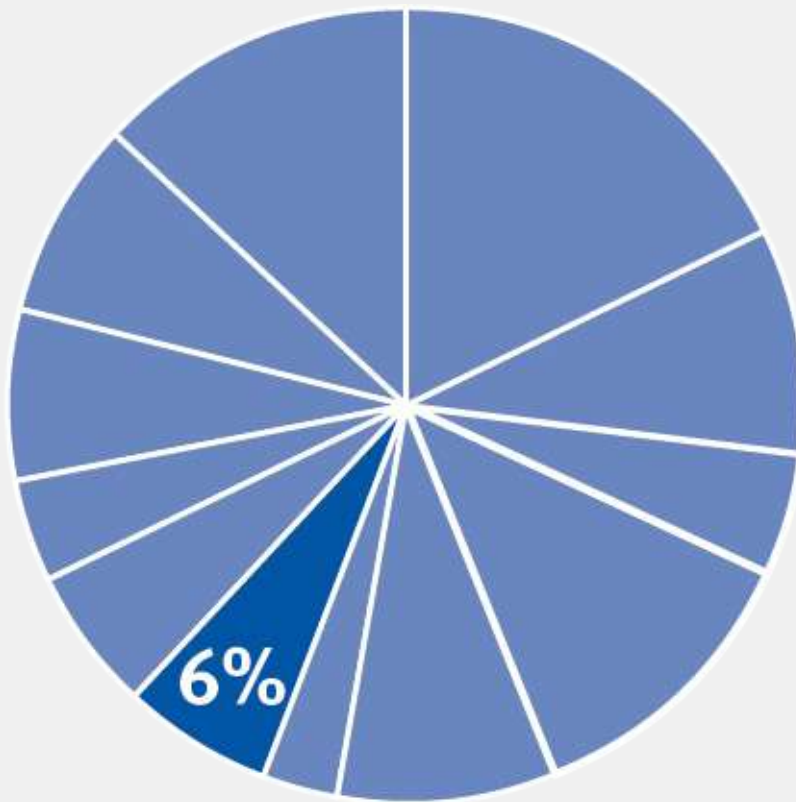
7.3 **Physiology of the Cardiovascular System**

Blood Pressure
Gas and Solute Exchange
Coagulation

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 6% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems

Introduction

As late as the 19th century, physicians adhered to a doctrine of health known as *humoralism*. This theory, developed by Greek and Roman physicians and philosophers and adopted by Islamic physicians, remained dominant in medical thought and practice until it was displaced by modern medical research in the 1800s. The theory of humoralism holds that the human body is composed of four fluids or substances called humors: black bile, yellow bile, phlegm, and blood. In the healthy state, these four humors are in balance, but excess or deficiency of any one of them would cause illness, disease, and even maladaptive personality characteristics. Over the course of a lifetime the levels of each of the four humors would rise and fall in accordance with diet and activity, resulting in maladies reflective of the imbalance. Treatments were intended to restore this balance.

Perhaps one of the most well-known treatments associated with humoralism is the practice of bloodletting. Because many diseases were associated with an excess of blood, physicians would withdraw significant amounts of blood from their patients to restore balance to the four humors. Methods for bloodletting were many, and some were dramatic—including drawing blood from major veins in the arm or neck and puncturing arteries. Devices known as scarificators were developed to cut into the superficial vessels. Most famously, leeches were used (especially in the early 19th century) to draw out excess blood. In fact, in the early decades of the 1800s, hundreds of millions of leeches were used by European physicians; in the 1830s, France alone imported about 40 million leeches per year for medical treatments.

While the humoral theory has been completely discredited by modern science, some practices associated with humoralism are still being used,

albeit based on very different medical understanding and for different purposes. For example, new research has shown that medicinal leeches can be used effectively in microsurgery to help prevent blood coagulation, and in reconstructive surgery to stimulate circulation to the reattached tissue.

The cardiovascular system is one of the most commonly tested organ systems on the MCAT. It serves a variety of functions, including the movement of respiratory gases, nutrients, and wastes. We will review the structures and functional anatomy of the cardiovascular system and then discuss blood and its functional components. We'll also trace the pathways created by the electrically excitable cells that initiate and spread contractions through the heart. A quick recap of genetics and inheritance will help explain the consequences of ABO and Rh antigens. In addition, the binding of oxygen and carbon dioxide to hemoglobin will be discussed in detail.

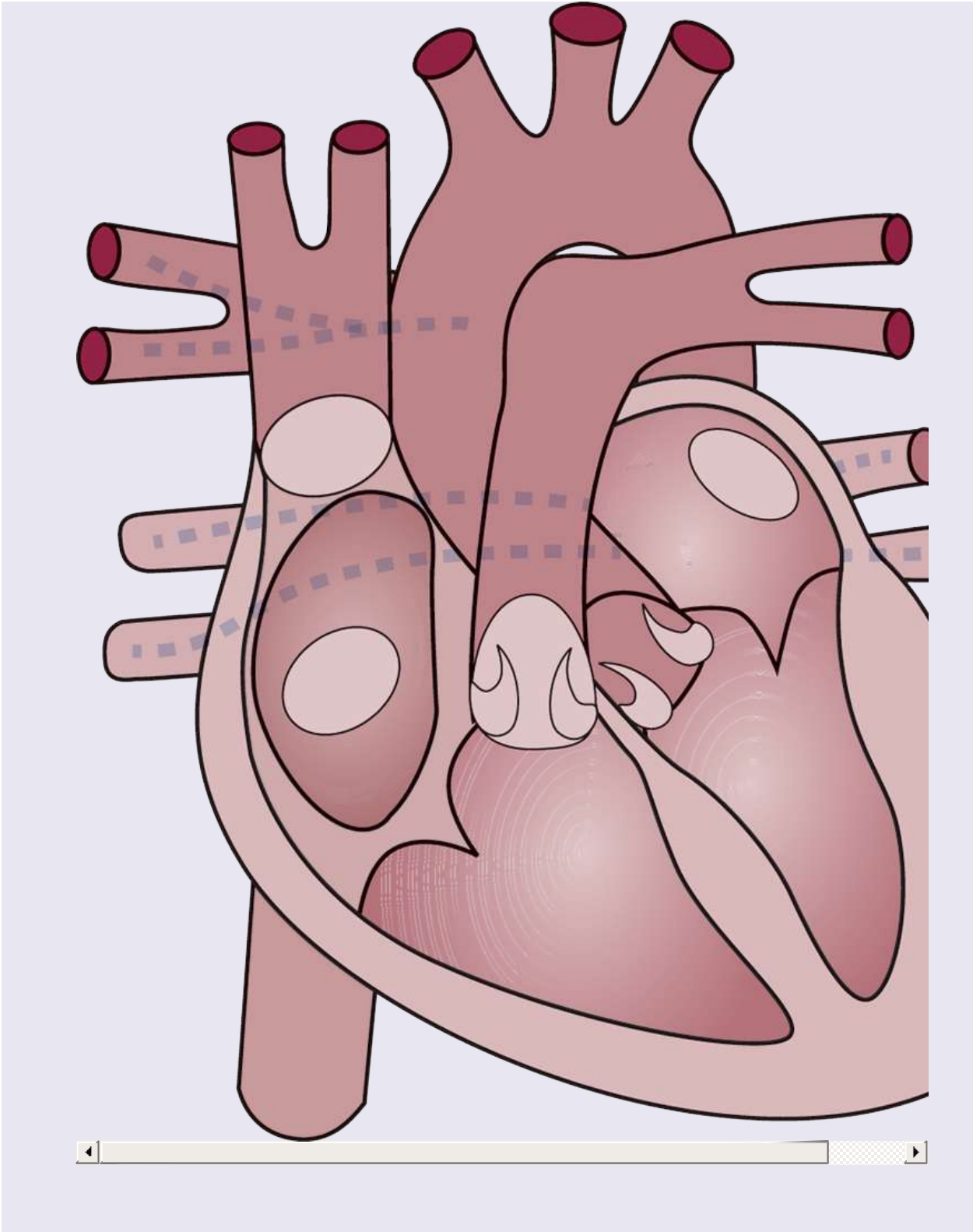
7.1 Anatomy of the Cardiovascular System

High-Yield

LEARNING OBJECTIVES

After Chapter 7.1, you will be able to:

- Recall the names of the chambers and valves of the heart
- Describe the chain of events in the conduction system of the heart that leads to heartbeat generation
- Identify autonomic inputs into the heart and their effects
- Distinguish between arteries, capillaries, and veins
- Explain why the right side of the heart is less muscular than the left side
- Trace the flow of blood through the heart:



The **cardiovascular system** consists of a muscular four-chambered **heart**, **blood vessels**, and **blood**, as shown in Figure 7.1. The heart acts as a pump, distributing blood through the vasculature. The vasculature consists of arteries, capillaries, and veins. After blood travels through veins, it is returned to the right side of the heart where it is pumped to the lungs to be reoxygenated. Then, the oxygenated blood returns to the left side of the heart where it is once again pumped to the rest of the body.

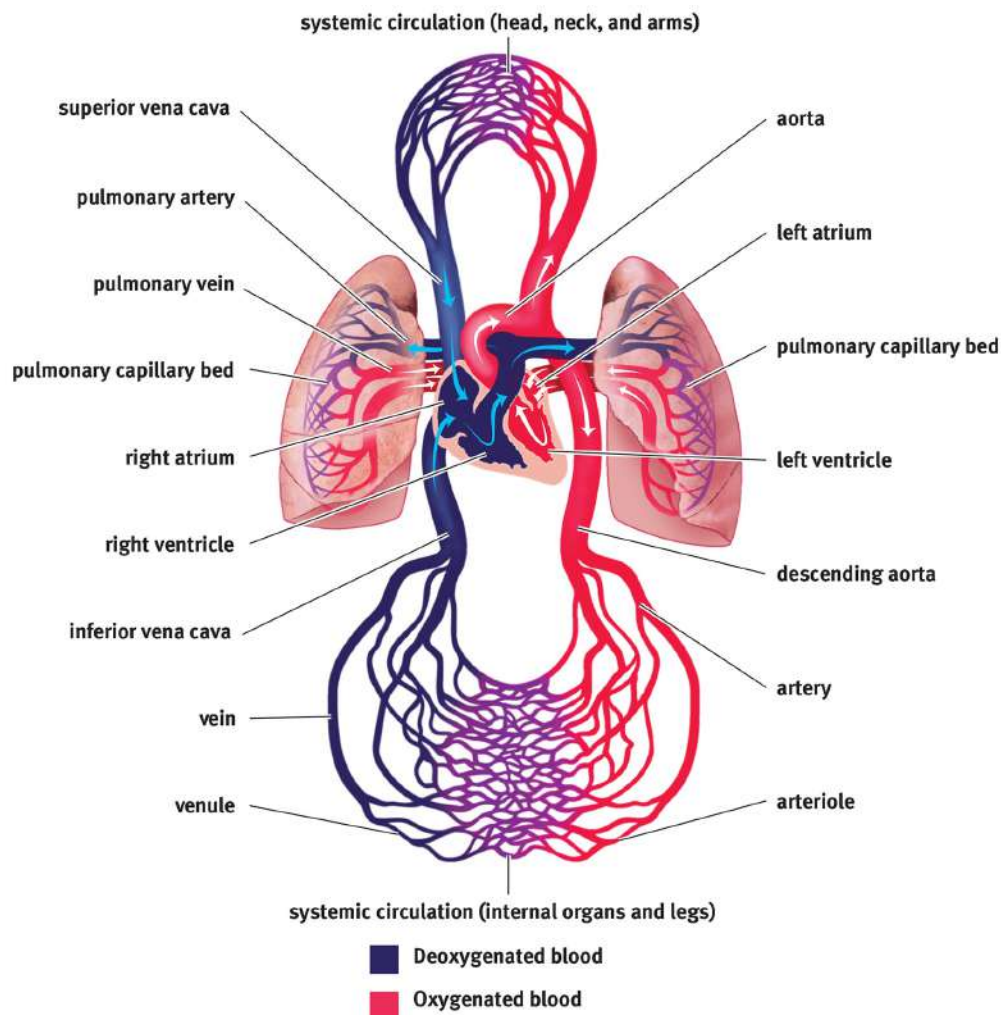


Figure 7.1. Anatomy of the Cardiovascular System

THE HEART

The **heart** is a four-chambered structure composed predominantly of cardiac muscle. We often speak of the heart as a pump, supporting a single circulatory system. However, the heart is actually composed of two pumps supporting two different circulations in series. The right side of the heart accepts deoxygenated blood returning from the body and moves it to the lungs by way of the pulmonary arteries; this constitutes the first pump (**pulmonary circulation**). The second pump is the left side of the heart, which receives oxygenated blood from the lungs by way of the pulmonary veins and forces it out to the body through the aorta (**systemic circulation**).

Each side of the heart consists of an atrium and a ventricle. The **atria** are thin-walled structures where blood is received from either the **venae cavae** (deoxygenated blood entering the right side of the heart) or the **pulmonary veins** (oxygenated blood entering the left side of the heart). The atria contract to push blood into the **ventricles**. After the ventricles fill, they contract to send blood to the lungs (right ventricle) and the systemic circulation (left ventricle). Note that the ventricles are far more muscular than the atria, allowing for more powerful contractions that are necessary to push blood through the rest of the body.

The atria are separated from the ventricles by the **atrioventricular valves**. Likewise, the ventricles are separated from the vasculature by the **semilunar valves**. These valves allow the heart muscle to create the pressure within the ventricles necessary to propel the blood forward within the circulation, while also preventing backflow of blood. The valve between the right atrium and the right ventricle is known as the **tricuspid valve** (three leaflets), while the valve between the left atrium and the left ventricle is known as the **mitral** or **bicuspid valve** (two leaflets). The valve that separates the right ventricle from

the pulmonary circulation is known as the **pulmonary valve**, while the valve that separates the left ventricle from the aorta is known as the **aortic valve**. Both semilunar valves have three leaflets.



MNEMONIC

Atrioventricular valves: **LAB RAT**

Left **A**trium = **B**icuspid

Right **A**trium = **T**ricuspid

One of the central themes in biology is something that we've touched on already in previous chapters: structure and function are related. The right and left sides of the heart are two different pumps, with the right side of the heart pumping blood to the lungs, and the left side of the heart pumping blood into the systemic circulation. Blood leaving the left side of the heart must travel a considerable distance, so blood pressure must be maintained as far away as the feet. Thus, the left side of the heart is more muscular than the right side of the heart. In fact, if the right side of the heart were as muscular as the left and pumped blood as forcefully, this would damage the lungs.

Electrical Conduction in the Heart

The coordinated, rhythmic contraction of cardiac muscle originates in an electrical impulse generated by and traveling through a pathway formed by four electrically excitable structures, as shown in Figure 7.2. This commonly tested pathway consists of, in order of excitation: the sinoatrial (SA) node, the atrioventricular (AV) node, the bundle of His (AV bundle) and its branches, and

the Purkinje fibers. Impulse initiation occurs at the **SA node**, which generates 60–100 signals per minute without requiring any neurological input. This small collection of cells is located in the wall of the right atrium. As the depolarization wave spreads from the SA node, it causes the two atria to contract simultaneously. While most ventricular filling is passive (that is, blood moves from the atria to the ventricles based solely on ventricular relaxation), atrial **systole** (contraction) results in an increase in atrial pressure that forces a little more blood into the ventricles. This additional volume of blood is called the **atrial kick** and accounts for about 5–30 percent of cardiac output. Next, the signal reaches the **AV node**, which sits at the junction of the atria and ventricles. The signal is delayed here to allow the ventricles to fill completely before they contract. The signal then travels down the **bundle of His** and its branches, embedded in the **interventricular septum** (wall), and to the **Purkinje fibers**, which distribute the electrical signal through the ventricular muscle. The muscle cells are connected by **intercalated discs**, which contain many gap junctions directly connecting the cytoplasm of adjacent cells. This allows for coordinated ventricular contraction.

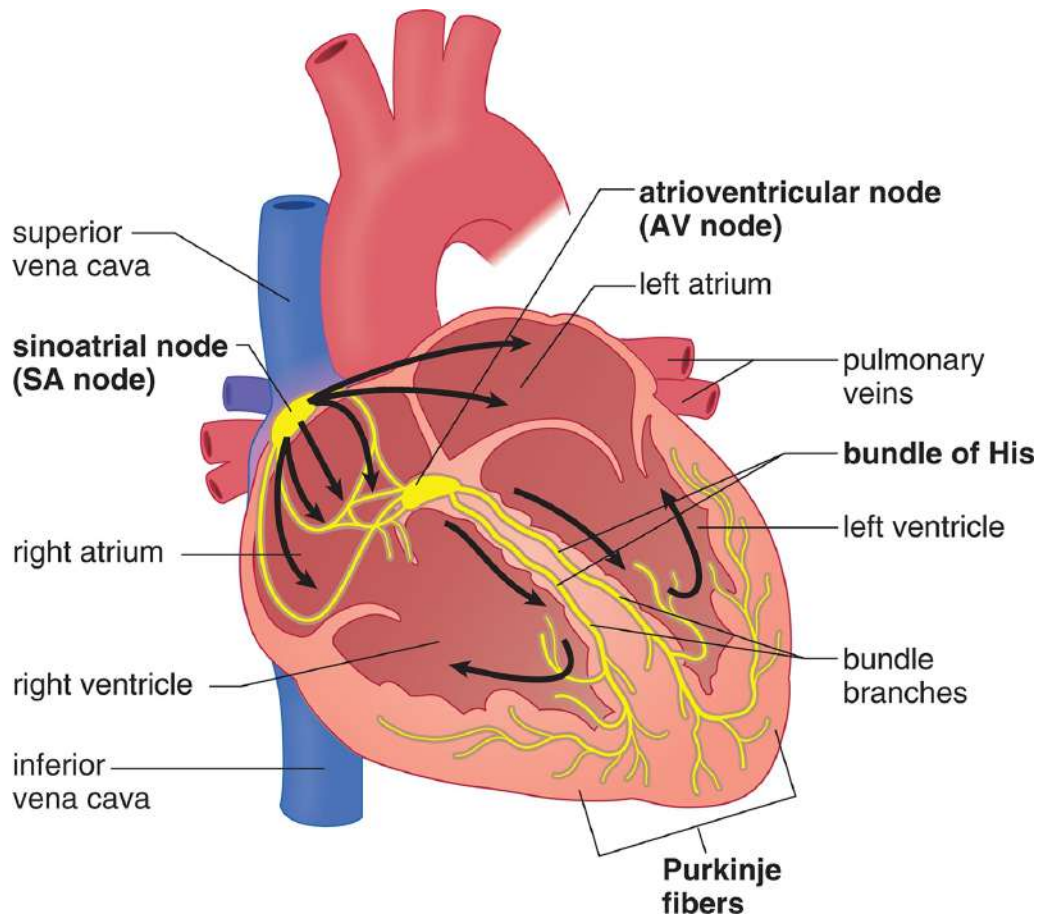


Figure 7.2. Electrical Conduction System of the Heart

Electrical impulses travel from the SA node to the AV node, through the bundle of His, and finally to the Purkinje fibers.



KEY CONCEPT

Cardiac muscle has myogenic activity, meaning that it can contract without any neurological input. The SA node generates about 60–100 beats per minute, even if all innervation to the heart is cut. The neurological input to the heart is important in speeding up and slowing the rate of contraction, but not generating it in the first place.

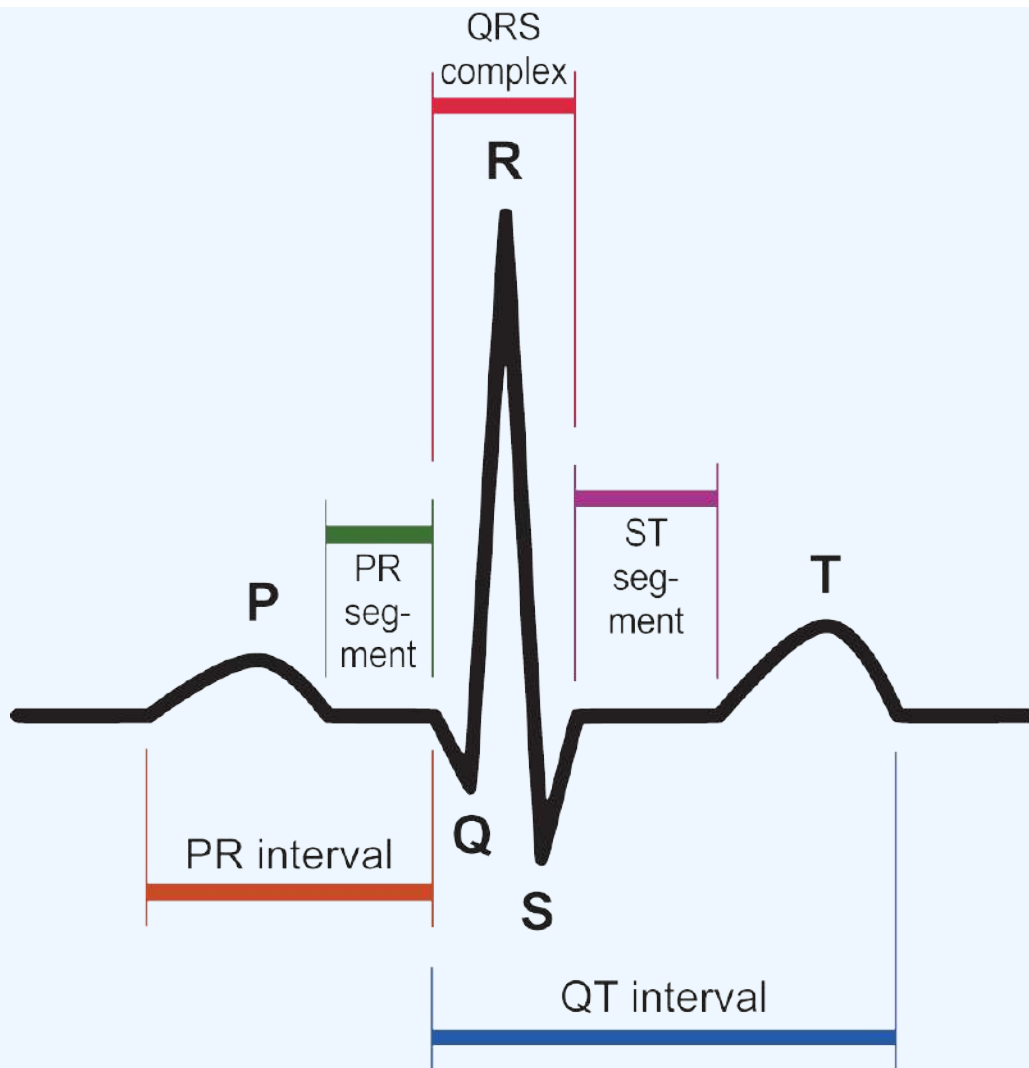
The SA node has an intrinsic rhythm of 60–100 signals per minute, so the normal human heart rate is 60–100 beats per minute. Highly conditioned athletes may have heart rates significantly lower than 60, in the range of 40–50 beats per minute. Stress, exercise, excitement, surprise, or danger can cause the heart rate to rise significantly above 100.



REAL WORLD

The heart's electrical impulses can be detected on the body's surface by placing electrodes on the skin on opposite sides of the heart. A recording of these currents is called an electrocardiogram (ECG or EKG; the *K*, by the way, reflects the German spelling).

Electrocardiograms are incredibly powerful tools for assessing the status of a patient's heart. A normal EKG is shown below.



Depolarization precedes cardiac muscle contraction, so the electrical spikes of the EKG occur just before a cardiac contractile event. The P-wave occurs immediately before the atria contract, and the QRS complex occurs just before the ventricles contract. The T-wave represents ventricular repolarization.

The circulatory system is under autonomic control. The autonomic system consists of the sympathetic (“fight-or-flight”) and parasympathetic (“rest-and-”

digest”) branches, controls the heart and affects the vasculature. Sympathetic signals speed up the heart rate and increase the contractility of cardiac muscle, while parasympathetic signals, provided by the **vagus nerve**, slow down the heart rate.

Contraction

The heart is a muscle that must contract in order to move blood. Each heartbeat is composed of two phases, known as systole and diastole. During **systole**, ventricular contraction and closure of the AV valves occurs and blood is pumped out of the ventricles. During **diastole**, the ventricles are relaxed, the semilunar valves are closed, and blood from the atria fills the ventricles. Contraction of the ventricles generates a higher pressure during systole, whereas their relaxation during diastole causes the pressure to decrease. The elasticity of the walls of the large arteries, which stretch to receive the volume of blood from the heart, allows the vessels to maintain sufficient pressure while the ventricular muscles are relaxed. In fact, if it weren’t for the elasticity of the large arteries, diastolic blood pressure would plummet to zero. The normal events of one heartbeat, including pressures in the left atrium, left ventricle, and aorta; left ventricular volume; normal and pathologic heart sounds; and an EKG are shown in Figure 7.3.

A measure to be aware of is **cardiac output**, or the total blood volume pumped by a ventricle in a minute. Does it matter which ventricle one chooses? As mentioned previously, the two pumps are connected in series, so the volumes of blood passing through each side must be the same, much like the electrical current between two resistors in series must be the same. Cardiac output (CO) is the product of **heart rate** (HR, beats per minute) and **stroke volume** (SV, volume of blood pumped per beat):

$$CO = HR \times SV$$

Equation 7.1

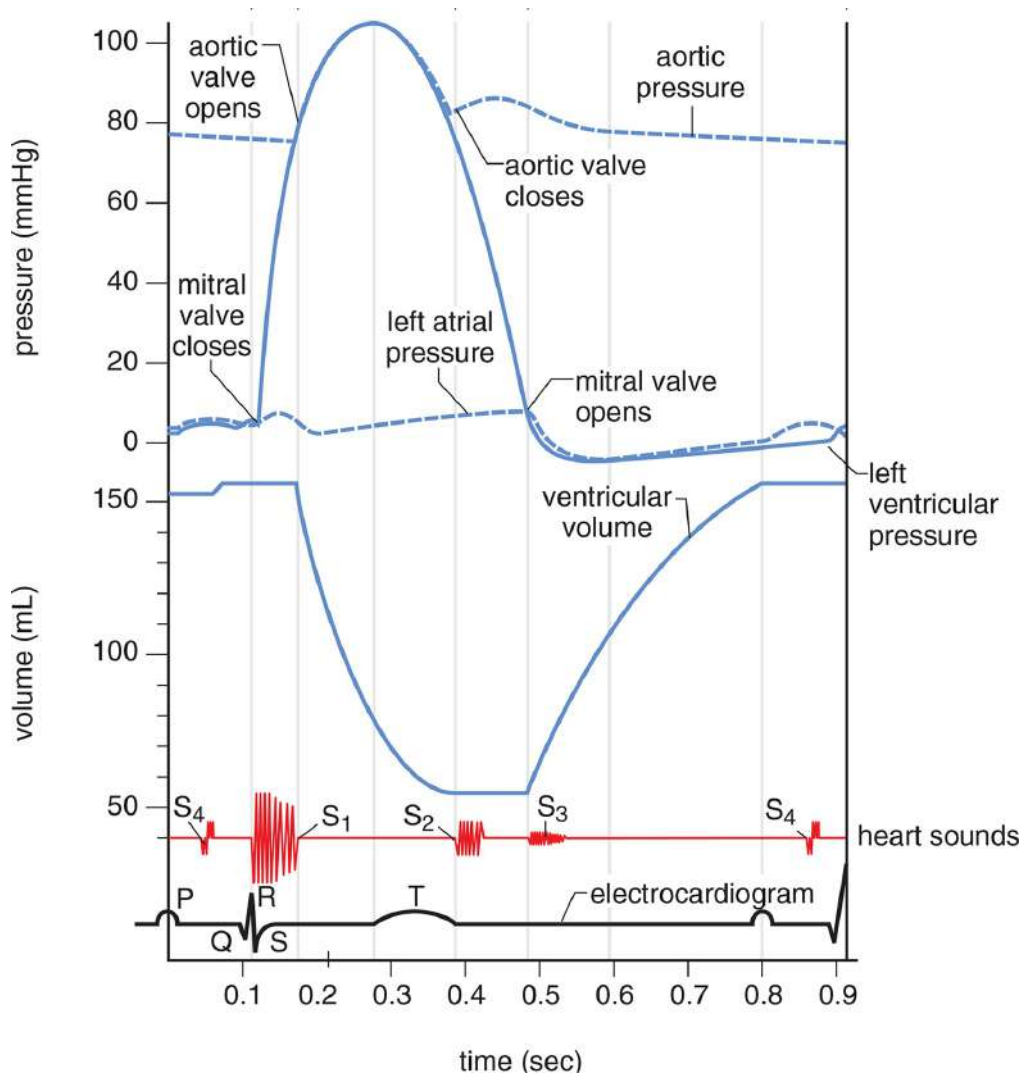


Figure 7.3. The Cardiac Cycle

The MCAT will not expect a thorough understanding of every detail of this diagram; it is more important to see how the changes in valves (open or closed), pressures, and volumes are related.

For humans, cardiac output is about 5 liters per minute. During periods of exercise or rest, the autonomic nervous system will increase (sympathetic) or decrease (parasympathetic) cardiac output, respectively.



REAL WORLD

The rhythmic impulses we hear when we listen to someone's heart with a stethoscope (*lub dub*) are referred to as the heart sounds. The first sound, S_1 , is produced when the two AV valves close at the start of systole to prevent backflow into the atria. The second sound, S_2 , is produced when the two semilunar valves close at the end of systole to prevent backflow into the ventricles. Extra heart sounds (S_3 and S_4) can result from stiffness of the heart muscle or high blood pressure. Heart murmurs, which may be so loud as to be audible without a stethoscope, may arise when the valves malfunction and become either narrow and stiff or wide and floppy, resulting in abnormal flow patterns across the valve.



REAL WORLD

There is a limit to how fast the heart can beat and still pump blood effectively. Because the heart fills with blood when it is relaxing (diastole), the faster it beats, the less time there is for blood to enter the heart during relaxation. Thus, a faster heartbeat means diminishing returns in terms of the amount of blood supplied to the body. A dangerous condition called ventricular tachycardia (often

abbreviated *v-tach*) features a very rapid rate of ventricular contraction. The heart in v-tach cannot properly fill with blood and, paradoxically, stops pumping blood despite its fast rate. Systemic pressures drop precipitously. Death will result unless the heart is forced out of this abnormal rhythm.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

To further investigate the effects of atorvastatin on post-myocardial infarction recovery, left ventricular myocardial infarction was artificially induced in rats.	Thi full just res fair con the
Rats were randomly assigned to atorvastatin or vehicle treatment. After four weeks, the left ventricles of all rats were examined. Results are summarized in Figures 1A–E below.	Veh to g bas wit act ing itse tim sali ser con
	IVs: ato trec

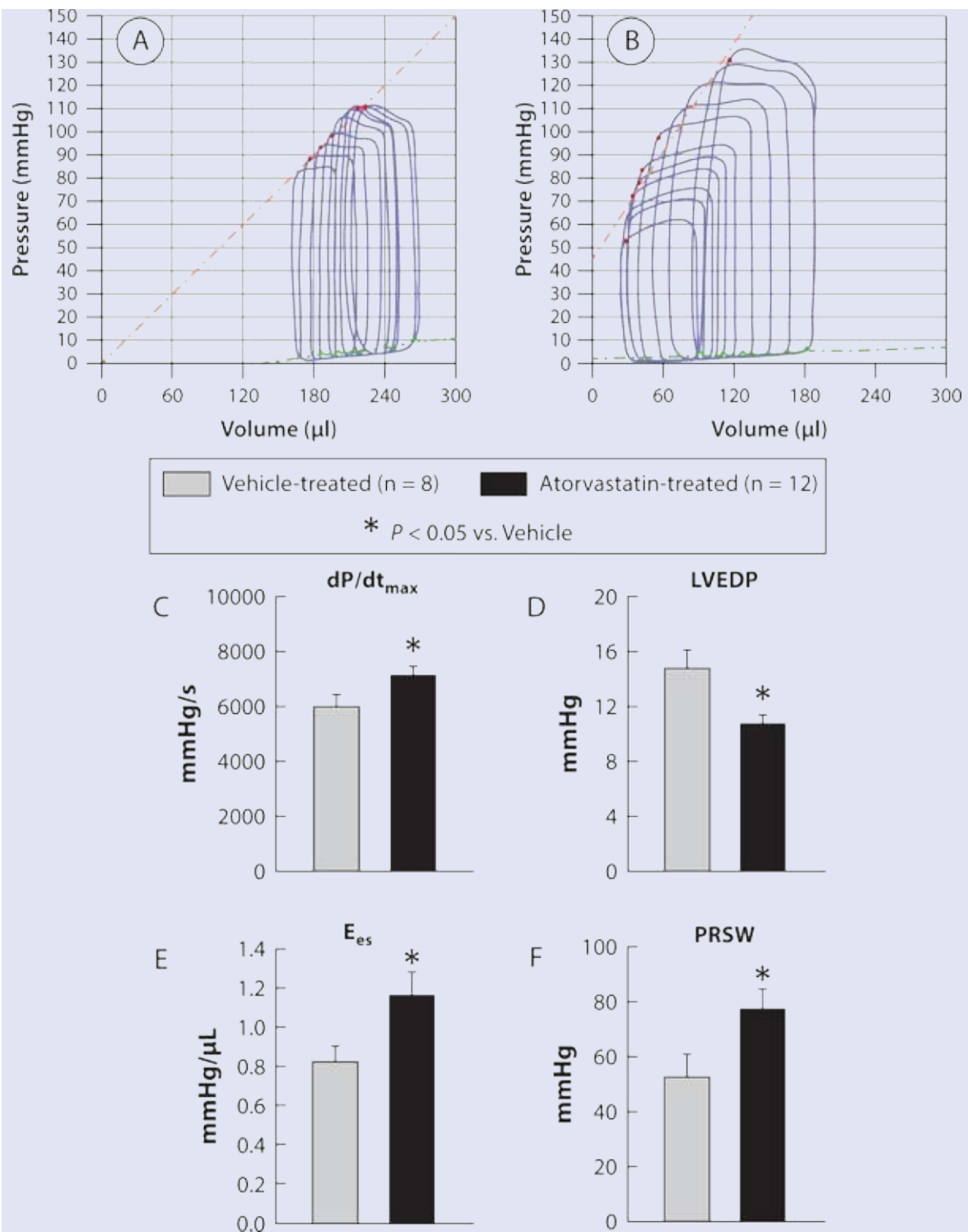


Figure 1

Representative pressure-volume cardiac cycle loops from a vehicle-treated rat (A) and an atorvastatin-treated rat (B) recorded during preload manipulation by a brief period of inferior vena cava occlusion. Panels C–F illustrate the quantitative analysis of hemodynamic variables of

the rats including dP/dt (C), LV end-diastolic pressure (D), end-systolic elastance (E), and preload recruitable stroke work (F).

adapted from Tang X-L, Sanganalmath SK, Sato H, Bi Q, Hunt G, Vincent RJ, et al. (2011) Atorvastatin therapy during the p
eriod attenuates left ventricular dysfunction and remodeling after myocardial infarction. *PLoS One* 6(9): e25320.
<https://doi.org/10.1371/journal.pone.0025320>.

Which treatment condition is correlated with a greater stroke volume in rats following myocardial infarction?

This question asks us to interpret the given figure in order to determine which condition is associated with a particular outcome. The MCAT will require us to bring in testable content knowledge, but it will also expect us to know how to use logical reasoning to analyze an experiment.

The first step is to think about what the question is asking for us to analyze from within the given passage and figure. We know from our outside content knowledge that stroke volume is defined as the volume of blood pumped per heartbeat. Simply speaking, all we need to do is to identify what information on the figure is related to the volume of blood pumped.

If we're not sure how to do this, a good place to start is to look at the units of each graph in the figure. In a novel figure, units can serve as a great clue as to what the figure is about. Since we're looking for stroke volume, we should be looking for units of volume like liters or cm^3 . Parts C through F have units of pressure per seconds, pressure, or pressure per volume, which are not what we are looking for. We'll need to refer to images A and B, both of which have a unit of volume on their x-axes.

According to the figure label, graph A shows the vehicle-treated rat, and graph B shows the atorvastatin-treated rat. The label also tells us that these graphs show pressure-volume cardiac cycle loops of two rats, one in each condition. Now we must relate these graphs to stroke volume. Remember that stroke volume is the volume of blood pumped per heartbeat. In other words, it is the amount of blood pumped after one cardiac cycle. On a pressure-volume graph, a cardiac cycle is seen here as a rectangular loop, with the height and width of the loop representing the pressure change and volume change, respectively. Looking at the two graphs in the figure, we must compare the changes in the volume of the loops of the two conditions. Even without explicitly measuring the dimensions, we can see that the atorvastatin graph has much wider loops, and thus a greater stroke volume.

Based on parts A and B of Figure 1, we can reasonably conclude that treatment with atorvastatin is correlated with greater stroke volume in rats post-infarction as compared to the control (vehicle) treatment.

THE VASCULATURE

In order to deliver blood to the entire body, the circulatory system utilizes vessels of different sizes. The three major types of vessels are arteries, veins, and capillaries. Blood travels away from the heart in **arteries**, the largest of which is the aorta (in the systemic circulation). Major arteries, such as the common carotids, subclavians, and renal arteries, branch off of the aorta to distribute the bloodflow toward different peripheral tissues. Arteries then undergo further divisions and name changes as they divert blood to specific tissues and organs until, upon reaching their target, they branch into

arterioles, which ultimately lead to **capillaries** that perfuse the tissues. There is also a set of coronary arteries at the base of the aorta that sends blood to perfuse the heart musculature. On the venous side of a capillary network, the capillaries join together into **venules**, which join to form **veins**. Venous blood empties into the superior and inferior venae cavae for entry into the right side of the heart. All blood vessels are lined with **endothelial cells**. This special type of cell helps to maintain the vessel by releasing chemicals that aid in vasodilation and vasoconstriction. In addition, endothelial cells can allow white blood cells to pass through the vessel wall and into the tissues during an inflammatory response. Finally, endothelial cells release certain chemicals when damaged that are involved in the formation of blood clots to repair the vessel and stop bleeding.



REAL WORLD

A heart attack, or myocardial infarction, is caused by a lack of bloodflow through the coronary arteries, which results in decreased oxygen delivery to the cardiac muscle itself. Anaerobic respiration cannot produce enough ATP to keep up with demand, so the muscle tissue begins to die. A person suffering a heart attack is often given a β -blocker, which blocks the sympathetic stimulation of the heart, resulting in lower heart rate and lower contractility. With a β -blocker, the heart does not work as hard, so its oxygen demand is diminished, which helps to prevent further damage to cardiac tissue.



MNEMONIC

Arteries carry blood **away** from the heart.

Cross-sections of the different blood vessels are shown in Figure 7.4. Don't worry about the names of the layers; simply be able to recognize that the same types of cells comprise the different vessels and that arteries have much more smooth muscle than veins.

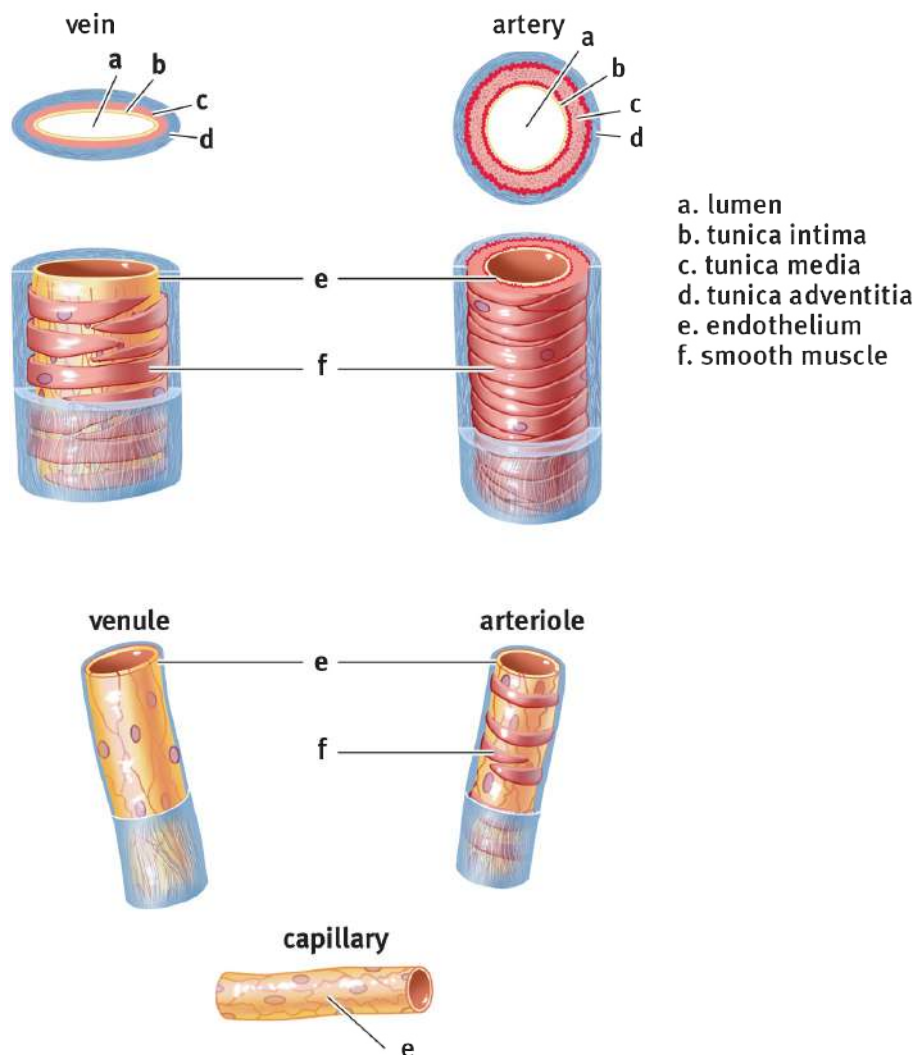


Figure 7.4. Structure of Blood Vessels

Arteries

Arteries move blood away from the heart to the lungs and other parts of the body. Most arteries contain oxygenated blood; in fact, only the pulmonary arteries and umbilical arteries contain deoxygenated blood. Smaller, muscular arteries are known as arterioles.

Arteries are highly muscular and elastic, creating tremendous resistance to the flow of blood. This is one of the reasons why the left side of the heart must generate much higher pressures: to overcome the resistance caused by systemic arteries. After arteries are filled with blood, the elastic recoil from their walls maintains a high pressure and forces blood forward.

Capillaries

Capillaries are vessels with a single endothelial cell layer and are so small that red blood cells must pass through the capillaries in a single-file line. The thin wall of the capillary allows easy diffusion of gases (O_2 and CO_2), nutrients (most notably, glucose), and wastes (ammonia and urea, among others). Capillaries are therefore the interface for communication of the circulatory system with the tissues. Remember, too, that blood also carries hormones, so capillaries allow endocrine signals to arrive at their target tissues.

Capillaries can be quite delicate. When capillaries are damaged, blood can leave the capillaries and enter the interstitial space. If this occurs in a closed space, it results in a bruise.

Veins

Veins are thin-walled, inelastic vessels that transport blood to the heart. Except for the pulmonary and umbilical veins, all veins carry deoxygenated blood. **Venules** are smaller venous structures that connect capillaries to the larger veins of the body.

The smaller amount of smooth muscle in the walls of veins gives them less recoil than arteries. Furthermore, veins are able to stretch to accommodate larger quantities of blood. Indeed, three-fourths of our total blood volume may be in venous circulation at any one time. Note that, even though the volume of arterial blood is normally much less than the volume of venous blood, the total volume passing through either side of the heart per unit time (cardiac output) is the same.

Given that the heart is located in the chest, bloodflow in most veins is upward from the lower body back to the heart, against gravity. In the inferior vena cava, this translates into a large amount of blood in a vertical column. The pressure at the bottom of this venous column in the large veins of the legs can be quite high. In fact, it can exceed systolic pressure (120 mmHg), going as high as 200 mmHg or more. Thus, veins must have structures to push the blood forward and prevent backflow. Larger veins contain valves; as blood flows forward in the veins, the valves open. When blood tries to move backward, the valves will slam shut. Failure of the venous valves can result in the formation of varicose veins, which are distended where blood has pooled. Pregnant women are especially susceptible to the formation of varicose veins due to an increase in the total blood volume during pregnancy and compression of the inferior vena cava by the fetus.



REAL WORLD

Blood clots may form in the deep veins of the legs as a result of injury, inactivity (blood stasis), or a hypercoagulable state (a tendency for the blood to clot excessively). The clots may dislodge and travel through the right atrium and right ventricle, through the pulmonary artery, and into the lungs. Such clots, called pulmonary emboli (or, more specifically, thromboemboli), block segments of the pulmonary arteries and produce rapid, labored breathing and chest pain. Death may occur if the thromboemboli are large.

In addition to high pressure in the lower extremities, the small amount of smooth muscle also creates a challenge for propelling blood forward. Thus, the veins must rely on an external force to generate the pressure to push blood toward the heart. Most veins are surrounded by skeletal muscles, which squeeze the veins as the muscles contract, forcing the blood up against gravity in much the same way that squeezing the bottom of a tube of toothpaste causes the contents to be expelled through the top of the tube. This is why sitting motionless for long periods of time, such as in the cramped middle seat on a long transoceanic flight or after surgery, can increase the risk of blood clot formation in the veins of the legs and pelvis. Blood pools in the lower extremities, and sluggish blood coagulates more easily. A clot in the deep veins of the leg is called a deep vein thrombosis (DVT). This clot may become dislodged and travel through the right side of the heart to the lungs, where it can cause a life-threatening condition called a pulmonary embolus.



REAL WORLD

Many patients may be immobile following a surgical procedure or during a long hospital stay. This introduces the threat of DVT and pulmonary emboli, which are very undesirable complications in someone who is already sick. Thus, in hospitals, special wraps are placed on the legs that contract rhythmically in order to prevent pooling of blood and the formation of blood clots. In addition, many patients receive medications like *heparin* or *warfarin* to prevent the formation of clots.

CIRCULATION

Circulation is, by definition, circular. Let's return to the anatomy of the heart and vasculature, shown in Figure 7.1, to trace the flow of blood through the body. A close-up of bloodflow through the heart is also shown in Figure 7.5. Here, we begin with the return of blood to the right atrium. Blood returns to the heart from the body via the venae cavae, which are divided into the **superior vena cava (SVC)** and the **inferior vena cava (IVC)**. The superior vena cava returns blood from the portions of the body above the heart, while the inferior vena cava returns blood from portions of the body below the heart. Deoxygenated blood enters the right atrium, travels through the tricuspid valve, and enters the right ventricle. On contraction, the blood from the right ventricle passes through the pulmonary valve and enters the pulmonary arteries, where it travels to the lungs and breaks up into continuously smaller vessels. Once the blood reaches the capillaries that line the alveoli, it participates in gas exchange, with carbon dioxide leaving the blood and oxygen entering the blood. The blood then travels into pulmonary venules and into the pulmonary veins, which carry the blood to the left side of the heart. Oxygenated blood enters the left atrium, travels through the mitral

valve, and enters the left ventricle. On contraction, the blood from the left ventricle passes through the aortic valve and enters the aorta. From the aorta, blood enters arteries, then arterioles, and then capillaries. After gas and nutrient exchange occurs at the capillaries, the blood enters the venules, which lead to the larger veins. The veins then empty into either the SVC or IVC for return to the right side of the heart.

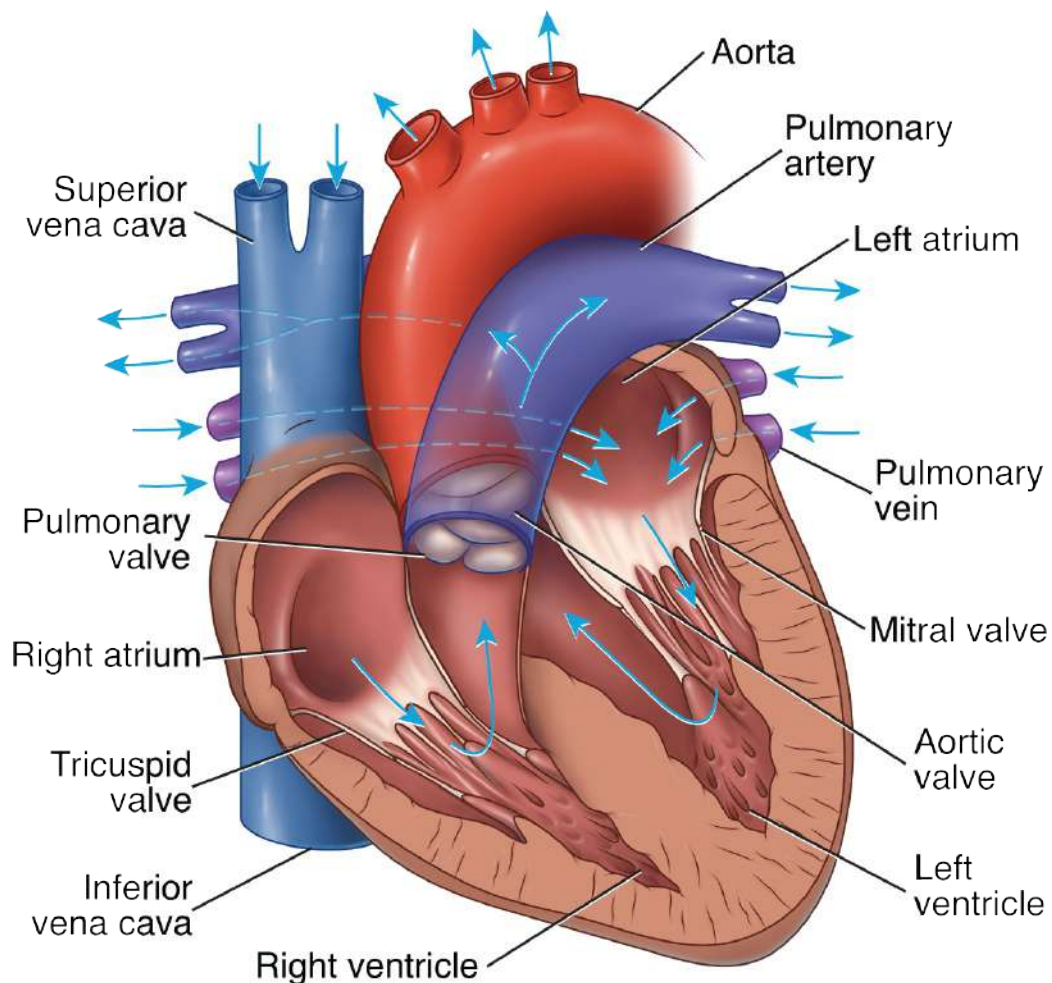


Figure 7.5. Bloodflow through the Heart

Written in shorthand, the pathway appears like this:

right atrium $\xrightarrow[\text{valve}]{\text{tricuspid}}$ right ventricle $\xrightarrow[\text{valve}]{\text{pulmonary}}$ pulmonary artery $\rightarrow$ lungs $\rightarrow$
 pulmonary veins $\rightarrow$ left atrium $\xrightarrow[\text{valve}]{\text{mitral}}$ left ventricle $\xrightarrow[\text{valve}]{\text{aortic}}$ aorta $\rightarrow$ arteries
 $\rightarrow$ arterioles $\rightarrow$ capillaries $\rightarrow$ venules $\rightarrow$ veins $\rightarrow$ venae cavae $\rightarrow$ right atrium



REAL WORLD

While *bicuspid valve* is an acceptable alternative name for *mitral valve*, it is rarely used in medicine. Most practitioners will refer to valves by one-letter abbreviations (*M* for mitral). The name *mitral* refers to a *miter*, which is the name for the hat the pope and bishops wear in the Catholic church. The two large leaflets of the mitral valve somewhat resemble this headdress.

In most cases, blood will pass through only one capillary bed before returning to the heart. However, there are three **portal systems** in the body, in which blood will pass through two capillary beds in series before returning to the heart. In the **hepatic portal system**, blood leaving capillary beds in the walls of the gut passes through the hepatic portal vein before reaching the capillary beds in the liver. In the **hypophyseal portal system**, blood leaving capillary beds in the hypothalamus travels to a capillary bed in the anterior pituitary to allow for paracrine secretion of releasing hormones. In the **renal portal system**, blood leaving the glomerulus travels through an efferent arteriole before surrounding the nephron in a capillary network called the vasa recta.

MCAT CONCEPT CHECK 7.1:

Before you move on, assess your understanding of the material with these questions.

1. Starting from entering the heart from the venae cavae, what are the four chambers through which blood passes in the heart? Which valve prevents backflow into each chamber?

Heart Chamber	Valve That Prevents Backflow

2. Starting with the site of impulse initiation, what are the structures in the conduction system of the heart?

3. Compare and contrast arteries, capillaries, and veins:

Vessel	Carries Blood in Which Direction?	Relative Wall Thickness	Smooth Muscle Present?	Contains Valves?
Artery				
Capillary				

Vein				
-------------	--	--	--	--

4. Why does the right side of the heart contain less cardiac muscle than the left side?

5. If all autonomic input to the heart were cut, what would happen?

7.2 Blood

LEARNING OBJECTIVES

After Chapter 7.2, you will be able to:

- Recall the components of plasma
- Predict compatible blood types given a blood type
- Identify the purpose of hematocrit measurements, as well as the relevant unit of measurement
- Recognize the different types of leukocytes and their functions
- Describe how platelets are produced
- Distinguish between cell types within blood that contain or do not contain a nucleus

Now that we have examined the pump and the pipes through which blood travels, let's take a look at this fluid.

COMPOSITION

In the pathology lab, we frequently study the composition of the blood using a centrifuge. By spinning the blood at a rapid rate, we can separate this complex fluid into its components based on density. By volume, blood is

about 55% liquid and 45% cells, as shown in Figure 7.6. **Plasma** is the liquid portion of blood, an aqueous mixture of nutrients, salts, respiratory gases, hormones, and blood proteins. Plasma can be further refined via the removal of clotting factors into serum. The cellular portion of blood consists of three major categories: **erythrocytes**, **leukocytes**, and **platelets**. All blood cells are formed from hematopoietic stem cells, which originate in the bone marrow.



REAL WORLD

Serum (plural sera) is used in a variety of medical testing procedures such as antibody testing and blood typing. Serum is considered preferable to plasma for many applications due to the lack of clotting factors and fibrinogens.

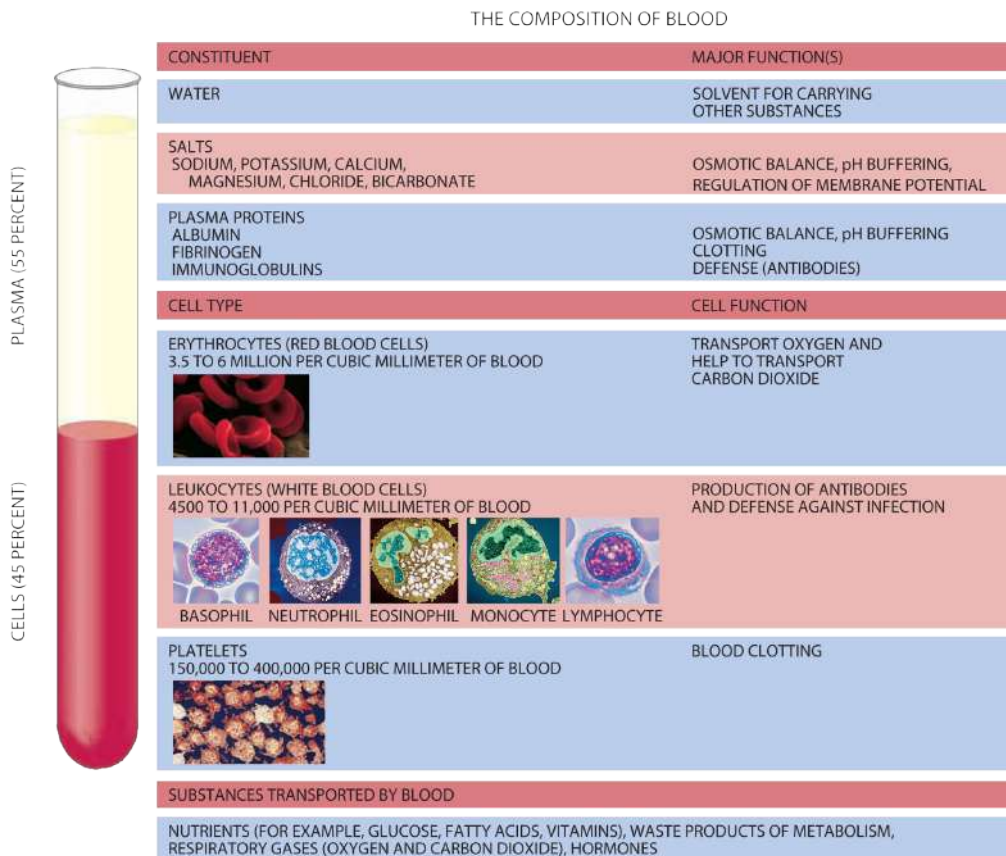


Figure 7.6. The Composition of Blood

Erythrocytes—Red Blood Cells

In the body, oxygen and nutrients are delivered to the peripheral tissues, and carbon dioxide and other wastes (such as hydrogen ions and ammonia) are picked up from the peripheral tissues and delivered to the organs that process this waste: the lungs, liver, and kidneys. The **erythrocyte** or **red blood cell** is a specialized cell designed for oxygen transport. Oxygen does not simply dissolve in the cytoplasm of the red blood cell—remember, molecular oxygen is nonpolar and therefore has low solubility in aqueous environments. Rather, each erythrocyte contains about 250 million molecules of **hemoglobin**, each

of which can bind four molecules of oxygen. Therefore, each red blood cell can carry approximately 1 billion molecules of oxygen.

Red blood cells are unique in a number of ways, and their modifications reflect the special role they play in the human body. Red blood cells are biconcave, or indented on both sides, which serves a dual purpose. First, this shape assists them in traveling through tiny capillaries. Second, it increases the cell's surface area, which increases gas exchange. Red blood cells are also unique in that, when they mature, the nuclei, mitochondria, and other membrane-bound organelles are lost. The loss of organelles makes space for the molecules of hemoglobin. In addition, the loss of mitochondria in particular means that the red blood cell does not consume the oxygen it is carrying before it is delivered to peripheral tissues. In other words, red blood cells do not carry out oxidative phosphorylation to generate ATP; rather, they rely entirely on glycolysis for ATP, with lactic acid (arising from fermentation) as the main by-product. Because red blood cells lack nuclei, they are unable to divide. Erythrocytes can live for 120 days in the bloodstream before cells in the liver and spleen phagocytize senescent (old) red blood cells to recycle them for their parts.

In medicine, a complete blood count measures the quantity of each cell type in the blood. For red blood cells, two commonly given measures are the hemoglobin and hematocrit. Hemoglobin, of course, measures the quantity of hemoglobin in the blood, giving a result in grams per deciliter. Hematocrit is a measure of how much of the blood sample consists of red blood cells, given as a percentage. A normal hemoglobin is considered to be between 13.5 and $17.5 \frac{\text{g}}{\text{dL}}$ for males and between 12.0 and $16.0 \frac{\text{g}}{\text{dL}}$ for females. A normal hematocrit is considered to be between 41 and 53% for males and between 36

and 46% for females. For example, a patient may have a hemoglobin of $13.8 \frac{\text{g}}{\text{dL}}$ and a hematocrit of 41.2%.

Leukocytes—White Blood Cells

Leukocytes or **white blood cells** usually comprise less than 1 percent of total blood volume. This translates into about 4,500–11,000 leukocytes per microliter of blood, which is a small number relative to the erythrocyte concentration. This number can massively increase under certain conditions when we need more white blood cells, most notably during infection. White blood cells are a crucial part of the immune system, acting as our defenders against pathogens, foreign cells, cancer, and other materials not recognized as *self*. Let's briefly discuss five basic types of leukocytes, which are all categorized into two classes: granulocytes and agranulocytes.

The granular leukocytes or **granulocytes** (**neutrophils**, **eosinophils**, and **basophils**) are so named because they contain cytoplasmic granules that are visible by microscopy. These granules contain a variety of compounds that are toxic to invading microbes; these compounds can be released through exocytosis. Granular leukocytes are involved in inflammatory reactions, allergies, pus formation, and destruction of bacteria and parasites. The specific functions of the three granulocytes are discussed in Chapter 8 of *MCAT Biology Review*.

The agranulocytes, which do not contain granules that are released by exocytosis, consist of **lymphocytes** and **monocytes**. Lymphocytes are important in the **specific immune response**, the body's targeted fight against particular pathogens, such as viruses and bacteria. Some lymphocytes act as primary responders against an infection, while others function to maintain a

long-term memory bank of pathogen recognition. These cells, in a very real sense, help our body learn from experience and are prepared to mount a fast response upon repeated exposure to familiar pathogens. Many vaccines work by training these cells. Through exposure to a weakened pathogen, or an antigenic protein (a protein that can be recognized by the immune system) of the pathogen, memory cells can be created. For example, most children in the United States receive the varicella (chickenpox) vaccine, which includes a live but weakened strain of the varicella–zoster virus that causes chickenpox. When the vaccine is administered, the virus is recognized as foreign and an immune response is activated. During this process, certain immune cells form a memory of the virus; in other words, our body learns to remember the virus and prepares itself to ward off the virus if it appears again later in life.

Lymphocyte maturation takes place in one of three locations. Lymphocytes that mature in the bone marrow are referred to as B-cells, and those that mature in the thymus are called T-cells. B-cells are responsible for antibody generation, whereas T-cells kill virally infected cells and activate other immune cells. The details of these two components of the specific immune response are discussed in Chapter 8 of *MCAT Biology Review*.

The other agranulocytes are monocytes, which phagocytize foreign matter such as bacteria. Most organs of the body contain a collection of these phagocytic cells; once they leave the bloodstream and enter an organ, monocytes are renamed **macrophages**. Each organ's macrophage population may have a specific name, as well. In the central nervous system, for example, they are called **microglia**; in the skin, they are called **Langerhans cells**; in bone, they are called **osteoclasts**.

Thrombocytes—Platelets

Thrombocytes or **platelets** are cell fragments or shards released from cells in bone marrow known as **megakaryocytes**. Their function is to assist in blood clotting and they are present in high concentrations (150,000–400,000 per microliter of blood). The enzymatic reactions involved in the formation of a clot (the clotting cascade) will be discussed shortly.

As mentioned above, all of the cellular elements of blood originate in the bone marrow. The production of blood cells and platelets is called **hematopoiesis**, and is triggered by a number of hormones, growth factors, and cytokines. The most notable of these are **erythropoietin**, which is secreted by the kidney and stimulates mainly red blood cell development, and **thrombopoietin**, which is secreted by the liver and kidney and stimulates mainly platelet development. The hematopoietic pathways are shown in Figure 7.7.

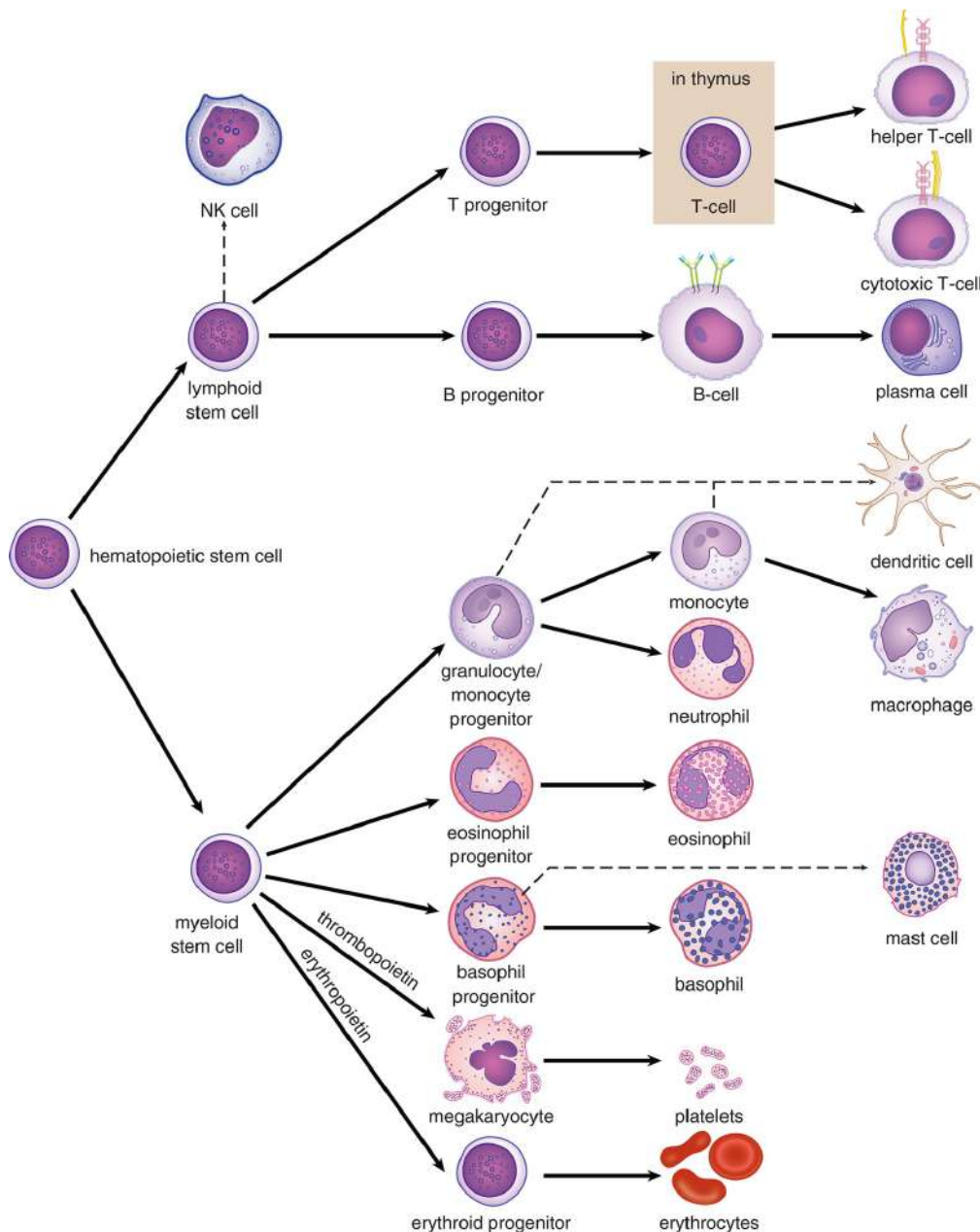


Figure 7.7. Hematopoiesis

BLOOD ANTIGENS

Red blood cells express surface proteins called **antigens**. In general, an antigen is any specific target (usually a protein) to which the immune system

can react. The two major antigen families relevant for blood groups are the **ABO antigens** and the **Rh factor**.

ABO Antigens

The ABO system is comprised of three alleles for blood type. In this particular class of erythrocyte cell-surface proteins, the A and B alleles are codominant, which means that a person may express one, both, or none of the ABO antigens. If the A allele (I^A or simply A) is present on one chromosome and the B allele (I^B or B) is present on the other chromosome, both will be expressed, and the person's blood type will be AB. The O allele (i or O) is recessive to both the A and B alleles. People with type O blood do not express either variant (A or B antigen) of this protein and have a homozygous recessive genotype. The naming system of blood types is based on the presence or absence of these protein variants. The four blood types are: A, B, AB, and O. Because the A and B alleles are dominant, the genotypes for A may be $I^A I^A$ or $I^A i$, while the genotypes for B may be $I^B I^B$ or $I^B i$.



MCAT EXPERTISE

There are two different ways of writing the ABO alleles: I^A , I^B , and i; or A, B, and O. Be comfortable using either system because the MCAT could show one or the other. Because the former system is more common in medical practice, we'll be using it throughout this book.

The ABO classification has important implications for medical practice; it is critical to match blood types for transfusions. It is no exaggeration to say that

blood-type matching is a life and death matter, given the severe hemolysis that can result if the donor blood antigen is recognized as foreign by the recipient's immune system. For example, a person with type A blood will recognize the type A protein as *self* but the type B protein as *foreign* and will make antibodies to types B and AB. Because type O blood cells express neither antigen variant, they will not initiate any immune response, regardless of the recipient's actual blood type; people with type O blood are therefore considered **universal donors** because their blood will not cause ABO-related hemolysis in any recipient. However, a recipient who is type O will produce both anti-A *and* anti-B antibodies and can only receive blood from other type O individuals. On the other hand, people with type AB blood are considered **universal recipients** because they can receive blood from all blood types: no blood antigen is foreign to AB individuals, so no adverse reactions will occur upon transfusion. A more thorough description of each blood type is given in Table 7.1. Note that whole blood is almost never given in a transfusion; rather, packed red blood cells (with no plasma) are generally given. Thus, we care only about the donor's red blood cell antigens (and not his or her plasma antibodies) when determining whether hemolysis will occur.



MCAT EXPERTISE

You are almost guaranteed to see at least one question on blood groups on Test Day. It is critical that you learn how the system works (I^A and I^B are codominant, i is recessive).

Another important point needs to be made here about antibodies. Antibodies are created in response to an antigen, and they specifically target that

antigen. You would not expect to have antibodies to the Ebola virus if you had never been exposed to it. This is true for the Rh factor as well—an Rh-negative individual would not have anti-Rh antibodies prior to exposure to Rh-positive blood. Why, then, does an individual lacking the A allele automatically have an anti-A antibody? The reason may lie in the gut: research has demonstrated that *E. coli* that inhabit the colon may have proteins that match the A and B alleles. This would serve as a source of exposure and would allow one to develop anti-A (or anti-B) antibodies prior to exposure to another person's blood. This is why ABO compatibility is so important during blood transfusions—giving the wrong ABO blood type would lead to rapid hemolysis.

Blood Type	Genotype(s)	Antigens Produced	Antibodies Produced	Can Donate to...	Can Receive from...
A	$I^A I^A, I^A i$	A	anti-B	A, AB	A, O
B	$I^B I^B, I^B i$	B	anti-A	B, AB	B, O
AB	$I^A I^B$	A and B	none	AB only	A, B, AB, O (universal recipient)
O	ii	none	anti-A and anti-B	A, B, AB, O (universal donor)	O only

Table 7.1 ABO Blood Types



REAL WORLD

The most common blood type in the United States is O⁺. The least common is AB⁻.



KEY CONCEPT

Antigens are the stimuli for B-cells to make antibodies. After exposure of a B-cell to its specific antigen, the cell becomes an antibody-producing factory.

Rh Factor

The **Rh factor** (so named because it was first described in rhesus monkeys) is also a surface protein expressed on red blood cells. Although at one time it was thought to be a single antigen, it has since been found to exist as several variants. When left unmodified, Rh-positive (Rh^+) or Rh-negative (Rh^-) refers to the presence or absence of a specific allele called D. The presence or absence of D can also be indicated with a plus or minus superscript on the ABO blood type (such as O^+ or AB^-). Rh-positivity follows autosomal dominant inheritance; one positive allele is enough for the protein to be expressed.



REAL WORLD

In blood bank pathology, the Rh factor is often referred to as D. Note that while the A, B, and D proteins are the most important for blood typing, dozens of other antigens can (and should) be matched as well, including C, E, Kell, Lewis, Duffy, and others. The more antigens that are correctly matched, the lower the probability of hemolysis.

The Rh factor status is particularly important in maternal–fetal medicine. During childbirth, no matter how good the obstetrician is, women are exposed to a small amount of fetal blood. If a woman is Rh[−] and her fetus is Rh⁺, she will become sensitized to the Rh factor, and her immune system will begin making antibodies against it. This is not a problem for the first child; by the time the mother starts producing antibodies, the child has already been born. However, any subsequent pregnancy in which the fetus is Rh⁺ will present a problem because maternal anti-Rh antibodies can cross the placenta and attack the fetal blood cells, resulting in hemolysis of the fetal cells. This condition is known as **erythroblastosis fetalis** and can be fatal to the fetus. Today, we can use medicine to prevent this condition. There is less concern with ABO mismatching between mother and fetus because these maternal antibodies against AB antigens are of a class called IgM, which does not readily cross the placenta (unlike anti-Rh IgG antibodies, which can).



REAL WORLD

When an Rh[−] woman is pregnant with her first Rh⁺ fetus, the risk of erythroblastosis fetalis in subsequent Rh-mismatched pregnancies can usually be avoided by giving the Rh[−] mother Rh-immunoglobulin (RhoGAM) during pregnancy and immediately after delivery. Administration of immunoglobulin (which is a type of passive immunization) will absorb the fetus's Rh⁺ cells, preventing the production of anti-Rh antibodies by the mother.

MCAT CONCEPT CHECK 7.2:

Before you move on, assess your understanding of the material with these questions.

1. What are the components of plasma?

2. An individual with B⁺ blood is in an automobile accident and requires a blood transfusion. What blood types could he receive? The same individual is so thankful that, after recovery, he decides to donate blood. To which blood types could he donate?

- Could receive from:

- Could donate to:

3. What does a hematocrit measure? What are the units for hematocrit?

4. Which types of leukocytes are involved in the specific immune response?

5. Where do platelets come from?

6. Which cell type(s) in blood contain nuclei? Which do not?

- Contain nuclei:

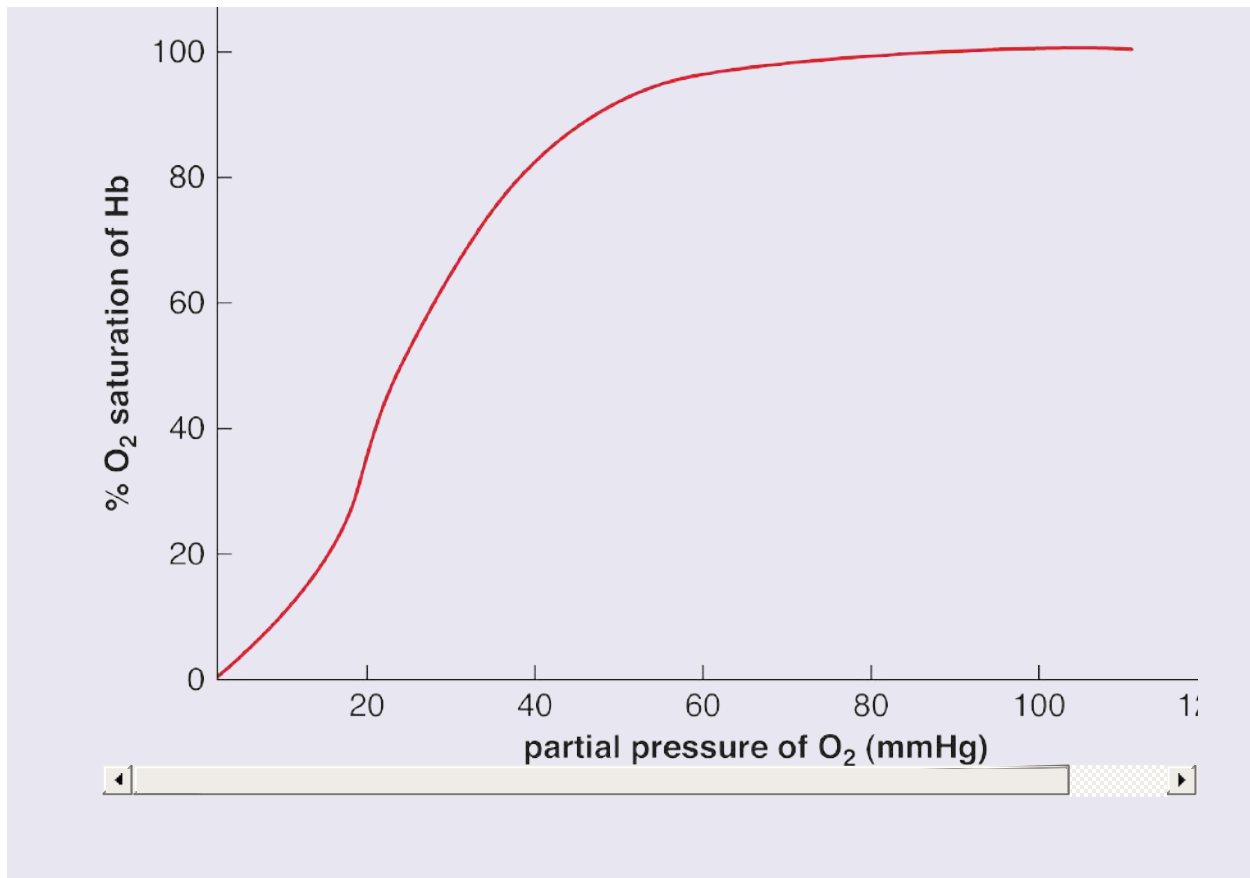
- Do not contain nuclei:

7.3 Physiology of the Cardiovascular System

LEARNING OBJECTIVES

After Chapter 7.3, you will be able to:

- Predict the impact of changing bloodflow through arteries, capillaries, and veins on blood pressure and heart function
- Recall the bicarbonate buffer chemical equation, including its catalyzing enzyme
- Explain how the oxyhemoglobin dissociation curve can be shifted to the left or right
- Recall the series of events and compounds in a coagulation cascade
- Identify the regions of the body associated with different parts of the oxyhemoglobin dissociation curve:



The cardiovascular system transports many compounds, including gases, nutrients, and waste products, to and from the body's tissues via red blood cells and plasma. Furthermore, it serves an important role in immunity through specialized cells, such as leukocytes, which help the body fight localized or systemic pathogens. Capillaries within the body can dilate and constrict to maintain proper body temperature. In addition, the circulatory system mediates the formation of blood clots to repair damaged vessels. These functions reflect the important jobs of the cardiovascular system, which include maintenance of blood pressure, gas and solute exchange, coagulation, and thermoregulation.

BLOOD PRESSURE

Before we can even discuss gas and solute exchange, it is important to recognize that, for the circulatory system to serve its predominant functions, blood pressure must be kept sufficiently high to propel blood forward. Blood pressure, therefore, provides healthcare professionals with information regarding the health of the circulatory system. In addition, high blood pressure, or *hypertension*, is a pathological state that may result in damage to the blood vessels and organs. Blood pressure is a measure of the force per unit area exerted on the wall of the blood vessels and is measured with a **sphygmomanometer**. Sphygmomanometers measure the gauge pressure in the systemic circulation, which is the pressure above and beyond atmospheric pressure (760 mmHg at sea level), as discussed in Chapter 4 of *MCAT Physics and Math Review*. Blood pressure is expressed as a ratio of the systolic (ventricular contraction) to diastolic (ventricular relaxation) pressures. Pressure gradually drops from the arterial to venous circulation, with the largest drop occurring across the arterioles, as shown in Figure 7.8. Normal blood pressure is considered to be between 90/60 and 120/80.

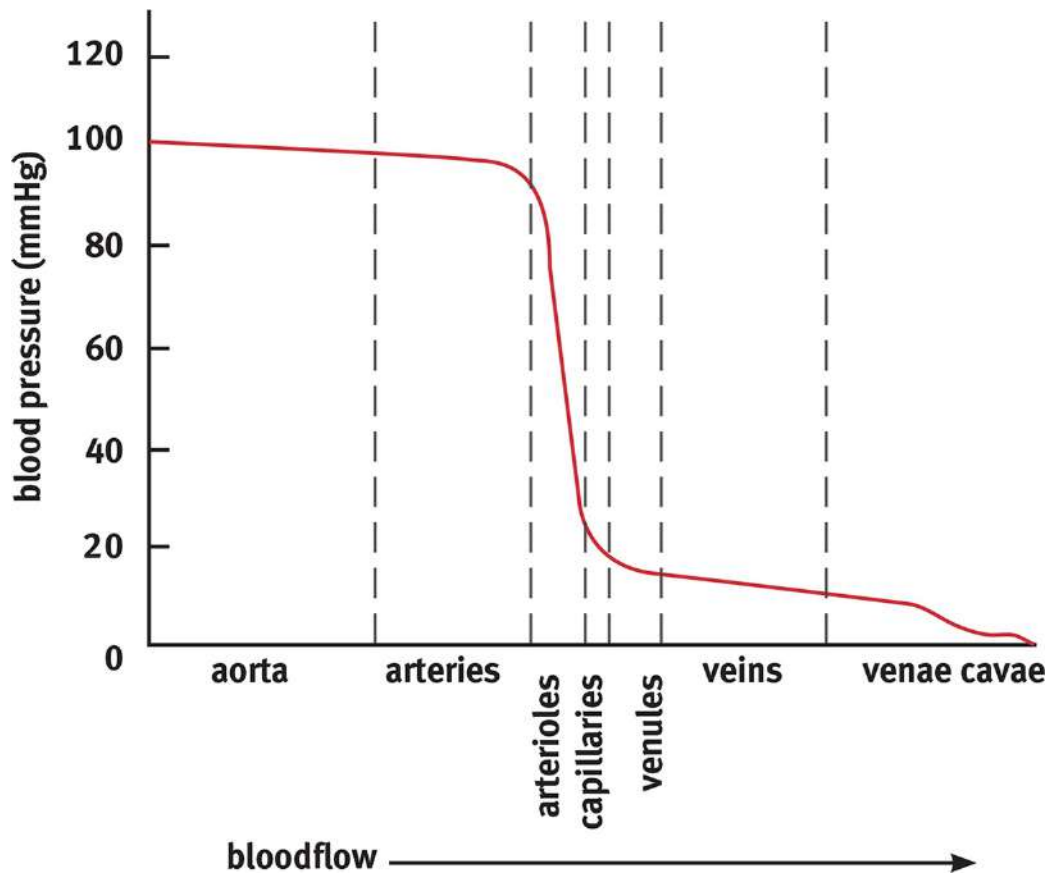


Figure 7.8. Mean Arterial Pressure at Different Locations in the Cardiovascular System



KEY CONCEPT

The largest drop in blood pressure occurs across the arterioles. This is critical because the capillaries are thin-walled and unable to withstand the pressure of the arterial side of the vasculature.

An analogy can be drawn between circulation and an electric circuit. Much like an electromotive force (voltage) drives a current through a given electrical

resistance, the pressure gradient across the circulatory system drives cardiac output through a given vascular resistance. This analogy is an important one to remember because the equations of electric circuits can be applied to the cardiovascular system. For example, Ohm's law ($V = IR$) can be translated into the following equation for circulation:

$$\Delta P = CO \times TPR$$

Equation 7.2

where ΔP is the pressure differential across the circulation, CO is the cardiac output, and TPR is the total peripheral (vascular) resistance.

It is also important to note that arterioles and capillaries act much like resistors in a circuit. When electricity travels through a wire, the wire itself provides an intrinsic level of resistance that limits the flow of electricity through it. Resistance is based on three factors: resistivity, length, and cross-sectional area. Resistivity has no obvious correlate in physiology, but the other two factors certainly do. The longer a blood vessel is, the more resistance it offers. The larger the cross-sectional area of a blood vessel, the less resistance it offers. In addition, arteries are highly muscular and are able to expand and contract as needed to change vascular resistance and maintain blood pressure. Arterioles can also contract to limit the amount of blood entering a given capillary bed (much like increasing resistance will decrease current flow to a given branch in a circuit). Finally, with the exception of the three portal systems, all systemic capillary beds are in parallel with each other. Therefore, opening capillary beds will decrease vascular resistance (like

adding another resistor in parallel) and, assuming the body can compensate, increase cardiac output.



REAL WORLD

Vital signs include temperature, heart rate, respiratory rate, and blood pressure. They provide a quick snapshot of a person's condition at a given moment. High blood pressure may indicate clinical hypertension, anxiety, catecholamine-producing tumors, or stimulant use, while low blood pressure may indicate hemorrhage, a heart attack, sepsis (severe bloodstream infection), or neurological damage. These are very broad differential diagnoses; however, vital signs combined with a quick clinical history provide a way for clinicians to identify the patients that require immediate intervention.

Blood pressure is regulated using baroreceptors in the walls of the vasculature. Baroreceptors are specialized neurons that detect changes in the mechanical forces on the walls of the vessel. When the blood pressure is too low, they can stimulate the sympathetic nervous system, which causes vasoconstriction, thereby increasing the blood pressure. In addition, chemoreceptors can sense when the osmolarity of the blood is too high, which could indicate dehydration. This promotes the release of antidiuretic hormone (ADH or vasopressin), a peptide hormone made in the hypothalamus but stored in the posterior pituitary, which increases the reabsorption of water, thereby increasing blood volume and pressure (while also diluting the blood). Low perfusion to the juxtaglomerular cells of the kidney stimulates aldosterone release through the renin–angiotensin–

aldosterone system; aldosterone increases the reabsorption of sodium and, by extension, water, thereby increasing the blood volume and pressure.

So, what if blood pressure is too high? Neurologically, sympathetic impulses could decrease, permitting relaxation of the vasculature with a concurrent drop in blood pressure. Within the heart, specialized atrial cells are able to secrete a hormone called **atrial natriuretic peptide (ANP)**. This hormone aids in the loss of salt within the nephron, acting as a natural diuretic with loss of fluid. Interestingly, ANP is a fairly weak diuretic. Some fluid is lost, but it is often not enough to counter the effects of a high-salt diet on blood pressure. Indeed, the human body has many different ways to raise blood pressure, but very few ways to lower it.

GAS AND SOLUTE EXCHANGE

Blood pressure ensures sufficient forward flow of blood through the system. However, what happens when the blood reaches the capillaries? Here, oxygen and nutrients diffuse out of the blood into tissues, while waste products like carbon dioxide, hydrogen ions, urea, and ammonia diffuse into the blood. In addition, hormones are secreted into the capillaries, travel with the circulation, and diffuse into their target tissue. Ions and fluid must also be returned to the blood to ensure that no area becomes too swollen with fluid. Regardless of the substance being exchanged, there is one fundamental concept to be considered in this process: concentration gradients. In each case, one side of the capillary wall has a higher concentration of a given substance than the other. This allows for movement of gases and solutes by diffusion.

Oxygen

Oxygen is carried primarily by **hemoglobin** in the blood. Hemoglobin is a protein composed of four cooperative subunits, each of which has a prosthetic heme group that binds to an oxygen molecule. The binding of oxygen occurs at the heme group's central iron atom, which can undergo changes in its oxidation state. The binding or releasing of oxygen to or from the iron atom in the heme group is an oxidation–reduction reaction. It is also important to note that some oxygen does diffuse into the blood and dissolve into the plasma, but this amount is negligible compared to the quantity of oxygen bound to hemoglobin. The level of oxygen in the blood is often measured as the partial pressure of O_2 within the blood, or P_aO_2 . A normal P_aO_2 is approximately 70–100 mmHg. However, taking this measurement is inconvenient because it involves taking a sample of blood from an artery. By contrast, **oxygen saturation**—that is, the percentage of hemoglobin molecules carrying oxygen—is easily measured using a finger probe. Most healthy people have an oxygen saturation above 97 percent.

In the lungs, oxygen diffuses into the alveolar capillaries. As the first oxygen binds to a heme group, it induces a conformational shift in the shape of hemoglobin from taut to relaxed. This shift increases hemoglobin's affinity for oxygen, making it easier for subsequent molecules of oxygen to bind to the remaining three unoccupied heme groups. As other heme groups acquire an oxygen molecule, the affinity continues to increase, thus creating a positive feedback-like (spiraling forward) mechanism. Once all of the hemoglobin subunits are bound to oxygen, the removal of one molecule of oxygen will induce a conformational shift, decreasing the overall affinity for oxygen, and making it easier for the other molecules of oxygen to leave the heme groups. This is again a positive feedback process; as oxygen molecules leave

hemoglobin, it becomes progressively easier for more oxygen to be removed. This phenomenon is a form of allosteric regulation referred to as **cooperative binding** and results in the classic sigmoidal (S-shaped) oxyhemoglobin dissociation curve shown in Figure 7.9.

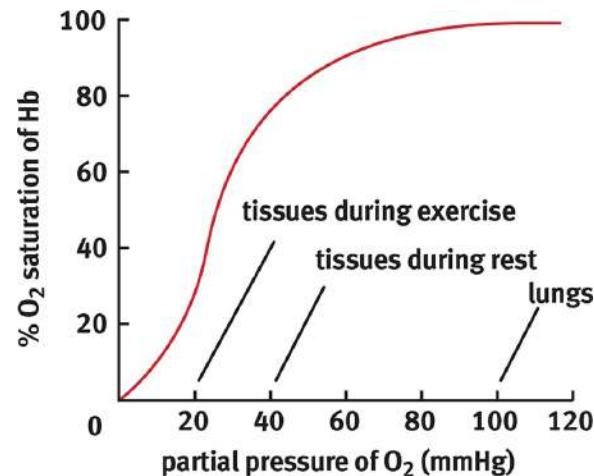


Figure 7.9. Oxyhemoglobin Dissociation Curve

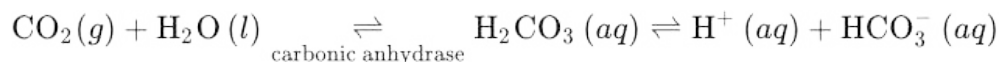
Before looking at shifts in this curve, let's make sure we understand what everything means. According to the curve, the blood is 100 percent saturated in the lungs, at a partial pressure of 100 mmHg O₂. The tissues are at a lower partial pressure of oxygen, around 40 mmHg during rest; at this lower partial pressure, the hemoglobin is approximately 80 percent saturated. Therefore, about $100 - 80 = 20\%$ of the oxygen has been released from the hemoglobin. Where did this oxygen go? Into the tissues, of course.

During exercise, the partial pressure of oxygen in the tissues is even lower—around 20 mmHg. At this lower partial pressure, the hemoglobin is approximately 30 percent saturated. Therefore, about $100 - 30 = 70\%$ of the oxygen has been released to the tissues. In reality, unloading of oxygen is also

facilitated by shifts in the hemoglobin curve that occur during exercise, as described later.

Carbon Dioxide

Delivering oxygen to tissues is only part of the job of transporting respiratory gases; removing carbon dioxide gas (CO₂), the primary waste product of cellular respiration, is also important. Carbon dioxide gas, like oxygen gas, is nonpolar and therefore has low solubility in the aqueous plasma; only a small percentage of the total CO₂ being transported in the blood to the lungs will be dissolved in the plasma. Carbon dioxide can be carried by hemoglobin, but hemoglobin has a much lower affinity for carbon dioxide than for oxygen. The vast majority of CO₂ exists in the blood as the bicarbonate ion (HCO₃⁻). When CO₂ enters a red blood cell, it encounters the enzyme *carbonic anhydrase*, which catalyzes the combination reaction between carbon dioxide and water to form carbonic acid (H₂CO₃). Carbonic acid, a weak acid, will dissociate into a proton and the bicarbonate anion. The hydrogen ion (proton) and bicarbonate ion both have high solubilities in water, making them a more effective method of transporting metabolic waste products to the lungs for excretion. Upon reaching the alveolar capillaries in the lungs, the same reactions that led to the formation of the proton and bicarbonate anion can be reversed, allowing us to breathe out carbon dioxide:



This chemical reaction is important, not only because it provides an effective means of ridding the body's tissues of carbon dioxide gas, but also because the concentration of free protons in the blood affects pH; the pH, in turn, can have allosteric effects on the oxyhemoglobin dissociation curve. Increased

carbon dioxide production will cause a right shift in the bicarbonate buffer equation, resulting in increased $[H^+]$ (decreased pH). These protons can bind to hemoglobin, reducing hemoglobin's affinity for oxygen. This decreased affinity can be seen in the oxyhemoglobin curve as a shift to the right; this is known as the **Bohr effect**. Note that the triggers for this right shift (increased P_aCO_2 , increased $[H^+]$, decreased pH) are often associated with oxygen demand; higher rates of cellular metabolism result in increased carbon dioxide production and accumulation of lactic acid, both of which decrease pH. This decreased affinity allows more oxygen to be unloaded at the tissues, as shown in Figure 7.10. Looking at the red and green lines, we see that hemoglobin is nearly 100 percent saturated in the lungs (at a partial pressure of 100 mmHg O_2) for both lines. However, the green line is significantly lower than the red one when we reach a partial pressure of 20 mmHg O_2 , around that of exercising muscle. Therefore, the right shift represents greater unloading of oxygen into the tissues.

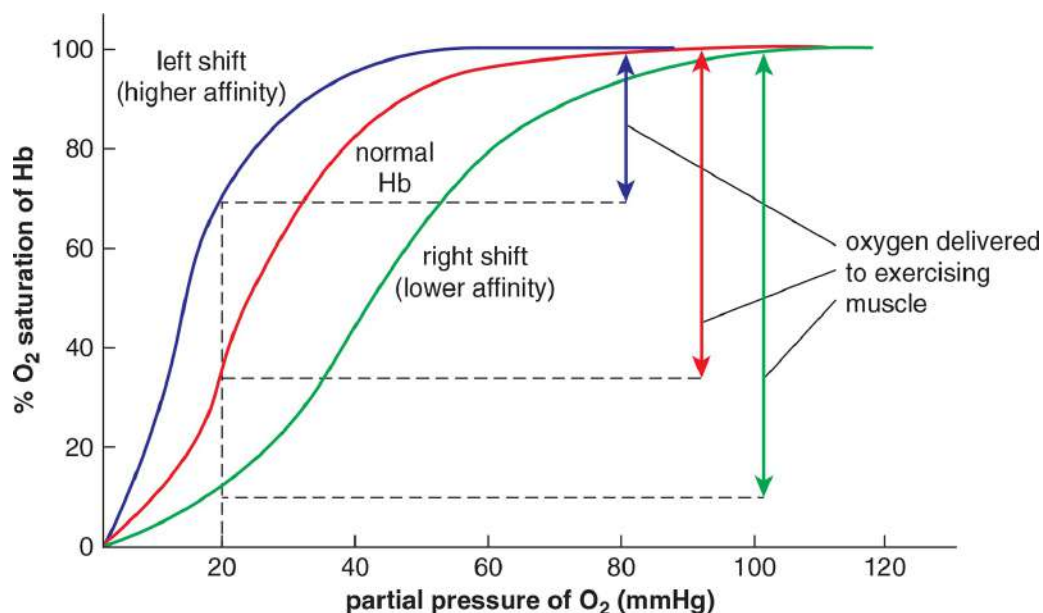


Figure 7.10. Shifts in the Oxyhemoglobin Dissociation Curve



MNEMONIC

Causes of a **right** shift of the oxyhemoglobin curve: **Exercise** is the **right** thing to do.

The following occur during **exercise**:

- Increased $P_a\text{CO}_2$
- Increased $[\text{H}^+]$ (decreased pH)
- Increased temperature

Other causes of a right shift in the oxyhemoglobin curve include increased temperature and increased 2,3-bisphosphoglycerate (2,3-BPG), a side product of glycolysis in red blood cells.

A left shift, like the blue line in Figure 7.10, may occur due to decreased $P_a\text{CO}_2$, decreased $[\text{H}^+]$, increased pH, decreased temperature, and decreased 2,3-BPG. In addition, **fetal hemoglobin (HbF)** has a higher affinity for oxygen than **adult hemoglobin (HbA)**. This should make sense because fetal red blood cells must literally pull oxygen off of maternal hemoglobin and onto fetal hemoglobin.



KEY CONCEPT

Fetal hemoglobin has a higher affinity for oxygen than adult hemoglobin, and therefore has a left-shifted curve compared to adult hemoglobin.

The bicarbonate buffer system is also important because it links the respiratory and renal systems. Disturbances in either of these systems can lead to changes in the pH of the blood. For example, if an individual hyperventilates, excess CO_2 will be blown off, shifting the bicarbonate buffer system to the left and decreasing the concentration of protons. This leads to an increase in pH, or what is known as respiratory alkalosis. The kidney can compensate for this change by increasing excretion of bicarbonate, which brings the pH back to normal. In contrast, in *renal tubular acidosis type I*, the kidney is unable to excrete acid effectively. This leads to a buildup of protons in the blood (metabolic acidosis), which causes the buffer system to shift to the left. The excess CO_2 formed in the process can be exhaled, and the person may increase respiratory rate to compensate, bringing the pH back to normal.

Nutrients, Waste, and Hormones

In addition to respiratory gases, blood also carries nutrients, waste products, and hormones to the appropriate location for use or disposal. As discussed earlier, concentration gradients guide much of the movement of these substances to and from the tissues.

Carbohydrates and amino acids are absorbed into the capillaries of the small intestine and enter the systemic circulation via the hepatic portal system. Fats are absorbed into lacteals in the small intestine, bypassing the hepatic portal circulation to enter systemic circulation via the thoracic duct. When released from intestinal cells, fats are packaged into lipoproteins, which are water-soluble. The absorption of nutrients is covered more extensively in Chapter 9 of *MCAT Biology Review*.

Wastes, such as carbon dioxide, ammonia, and urea, enter the bloodstream by traveling down their respective concentration gradients from the tissues to the capillaries. The blood eventually travels to the kidneys, where these waste products are filtered or secreted for elimination from the body.

Hormones enter the circulation in or near the organ where the hormone is produced. This usually occurs by exocytosis, allowing for secretion of hormones into the bloodstream. Certain hormones are carried by proteins in the blood and are released under specific conditions. Once hormones reach their target tissues, they can activate cell-surface receptors (peptide hormones) or diffuse into the cell to activate intracellular or intranuclear receptors (steroid hormones).

Fluid Balance

In the bloodstream, two pressure gradients are essential for maintaining a proper balance of fluid volume and solute concentrations between the blood and the interstitium (the cells surrounding the blood vessels). These are the opposing but related hydrostatic and osmotic (oncotic) pressures.

Hydrostatic pressure is the force per unit area that the blood exerts against the vessel walls. This is generated by the contraction of the heart and the elasticity of the arteries, and can be measured upstream in the large arteries as blood pressure. Hydrostatic pressure pushes fluid out of the bloodstream and into the interstitium through the capillary walls, which are somewhat leaky by design. **Osmotic pressure**, on the other hand, is the “sucking” pressure generated by solutes as they attempt to draw water into the bloodstream. Because most of this osmotic pressure is attributable to plasma proteins, it is usually called **oncotic pressure**.

At the arteriole end of a capillary bed, hydrostatic pressure (pushing fluid out) is much larger than oncotic pressure (drawing fluid in), and there is a net efflux of water from the circulation, as shown in Figure 7.11. As fluid moves out of the vessels, the hydrostatic pressure drops significantly, but the osmotic pressure stays about the same. Therefore, at the venule end of the capillary bed, hydrostatic pressure (pushing fluid out) has dropped below oncotic pressure (drawing fluid in), and there is a net influx of water back into the circulation.

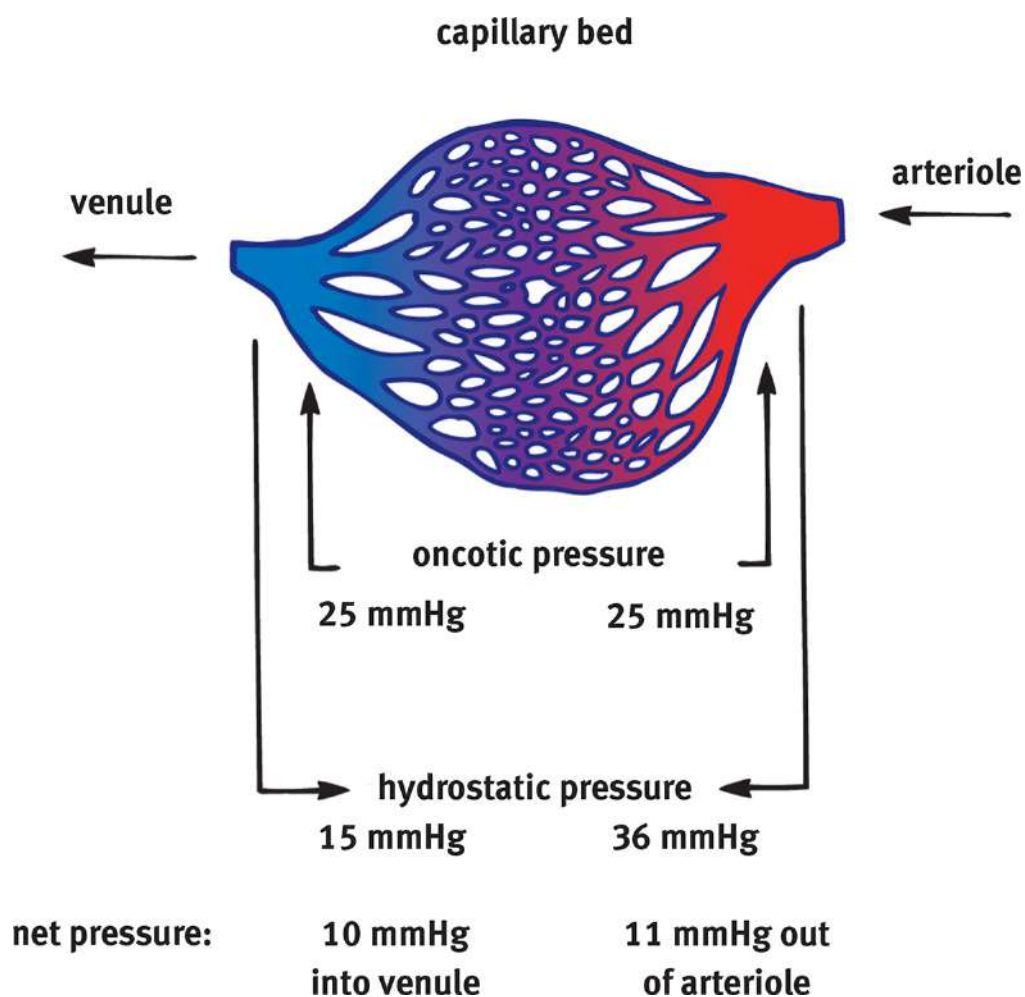


Figure 7.11. Starling Forces at a Capillary Bed



KEY CONCEPT

Hydrostatic pressure *pushes* fluid *out* of vessels and is dependent on blood pressure generated by the heart and the elastic arteries.

Osmotic pressure *pulls* fluid back *into* the vessels and is dependent on the number of particles dissolved in the plasma; most are proteins, so we often refer to this as oncotic pressure.

The balance of these opposing pressures, also called **Starling forces**, is essential for maintaining the proper fluid volumes and solute concentrations inside and outside the vasculature. Imbalance of these pressures can result in too much or too little fluid in the tissues. For example, accumulation of excess fluid in the interstitium results in a condition called **edema**. We should note that some interstitial fluid is also taken up by the lymphatic system. Most lymphatic fluid (**lymph**) is returned to the central circulatory system by way of a channel called the **thoracic duct**. Blockage of lymph nodes by infection or surgery can also result in edema. Although you do not need to learn or memorize the Starling equation, which quantifies the net filtration rate between two fluid compartments, you should understand that the movement of solutes and fluid at the capillary level is governed by pressure differentials, just like the movement of carbon dioxide and oxygen in the lungs.

COAGULATION

We have now covered most of the functions of red blood cells and plasma. We've briefly touched on white blood cells, which we'll explore more extensively in Chapter 8 of *MCAT Biology Review*. This leaves us with platelets,

which protect the vascular system in the event of damage by forming a clot. **Clots** are composed of both coagulation factors (proteins) and platelets, and they prevent (or at least minimize) blood loss. When the endothelium of a blood vessel is damaged, it exposes the underlying connective tissue, which contains collagen and a protein called **tissue factor**. When platelets come into contact with exposed collagen, they sense this as evidence of injury. In response, they release their contents and begin to aggregate, or clump together. Simultaneously, **coagulation factors**, most of which are secreted by the liver, sense tissue factor and initiate a complex activation cascade. While the details of the coagulation cascade are beyond the scope of the MCAT, it is important to know that the endpoint of the cascade is the activation of **prothrombin** to form **thrombin** by **thromboplastin**. Thrombin can then convert **fibrinogen** into **fibrin**. Fibrin ultimately forms small fibers that aggregate and cross-link into a woven structure, like a net, that captures red blood cells and other platelets, forming a stable clot over the area of damage, as shown in Figure 7.12. A clot that forms on a surface vessel that has been cut is called a scab.



REAL WORLD

Certain genetic diseases, such as hemophilia, cause malfunctions in the cascade of clotting reactions and increase the risk of life-threatening blood loss from even relatively minor injuries. Hemophilia A is the most common form and, as a sex-linked trait, is far more common in males than females.

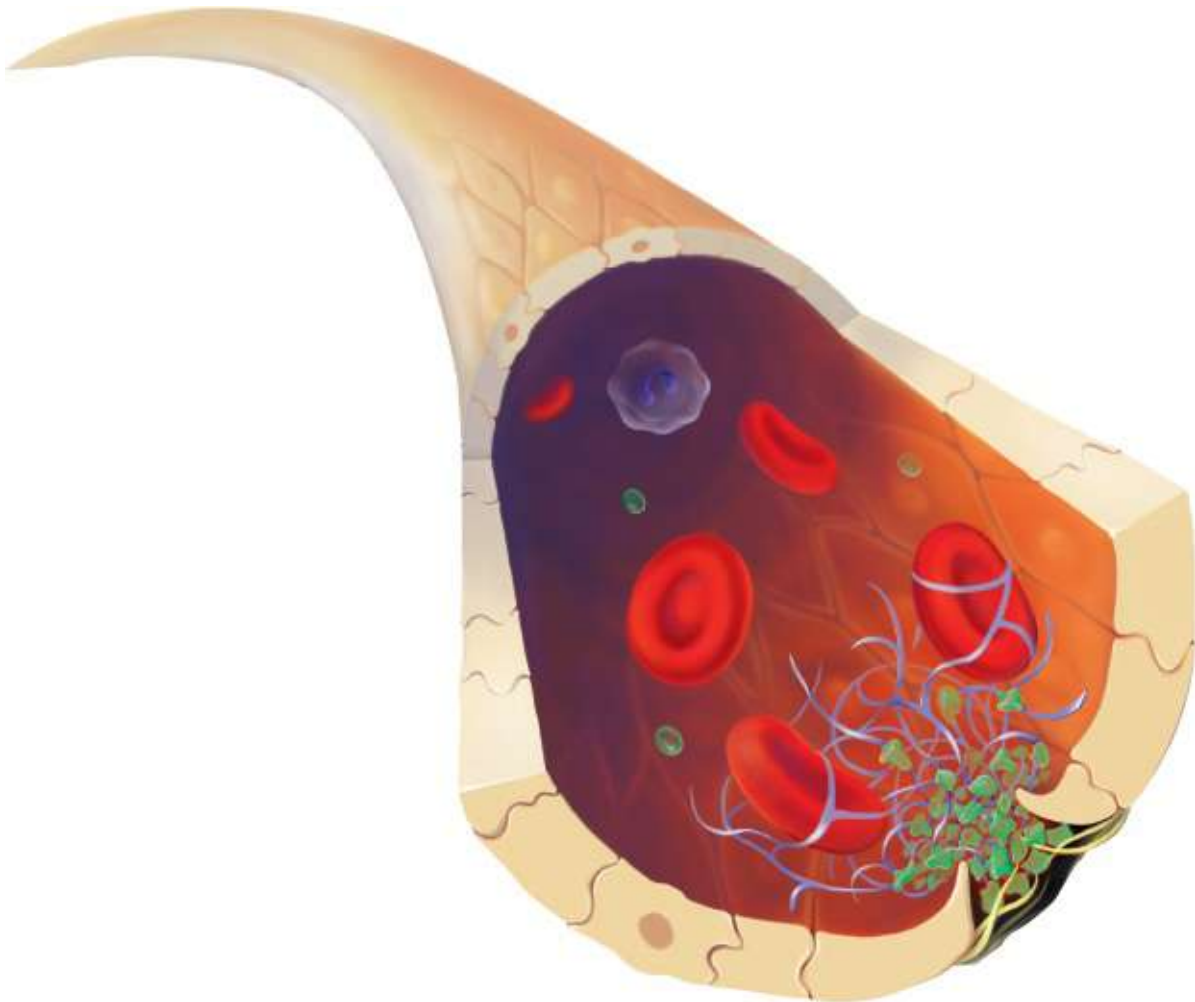


Figure 7.12. Thrombus (Clot) Formation

Thrombus formation, or blood clotting, occurs when blood vessels are injured. The process begins when platelets attach to the matrix that becomes exposed when the endothelial cells lining blood vessels are disrupted. This attachment then activates quiescent $\alpha_{IIb}\beta_3$ integrin molecules, causing them to adhere to circulating proteins—including fibrinogen, which forms bridges to additional platelets. Together the cells and proteins ultimately form a network of cells and fibers dense enough to plug the injury and prevent blood loss until the wound can be repaired.

Ultimately, the clot will have to be broken down. This task is accomplished predominantly by **plasmin**, which is generated from **plasminogen**.

MCAT CONCEPT CHECK 7.3:

Before you move on, assess your understanding of the material with these questions.

1. In bacterial sepsis (overwhelming bloodstream infection), a number of capillary beds throughout the body open simultaneously. What effect would this have on the blood pressure? Besides the risk of infection, why might sepsis be dangerous for the heart?

2. What is the chemical equation for the bicarbonate buffer system? What enzyme catalyzes this reaction?

3. Where should you look on the oxyhemoglobin dissociation curve to determine the amount of oxygen that has been delivered to tissues?

4. What direction does the oxyhemoglobin dissociation curve shift as a result of exercise? What physiological changes cause this shift and why?

5. Exposure of which subendothelial compounds start the coagulation cascade? What protein helps stabilize the clot?

- Starts the cascade:

- Stabilizes the clot:

Conclusion

The cardiovascular system is one of the most commonly tested MCAT topics. You should be familiar with its basic structure: a system with two pumps in series. The right ventricle pumps blood into the pulmonary circulation, while the left ventricle pumps blood into the systemic circulation. We discussed the myogenic activity of cardiac muscle and the pathway that electricity follows in the heart through the SA node, AV node, bundle of His, and Purkinje fibers. The movement of blood through the vascular system is a result of the heart's pumping to generate pressure. Blood pressure is a measure of the blood's force per unit area on the vessel walls and is recorded as a gauge pressure (pressure above and beyond atmospheric pressure). We discussed the differences in structure between arteries, capillaries, and veins, and how these anatomical differences are reflective of their different functions. We then reviewed the composition of blood along with the three major blood cell types. We examined the ABO and Rh antigen systems, which frequently appear on the MCAT due to their widespread clinical relevance. The blood's ability to carry oxygen and carbon dioxide was also described; recall that carbon dioxide is primarily carried as bicarbonate ions in the blood. The conversion of carbon dioxide to and from this ion is accomplished by the enzyme carbonic anhydrase.

In this chapter, we focused on the functions of red blood cells, plasma, and platelets. We briefly examined the immune system, which is primarily driven by the actions of white blood cells and their products. Immunology is considered one of the most challenging courses in medical school, as you'll

learn about dozens of cytokines, clusters of differentiation (CD), and specialized cell types. In the next chapter, we'll focus on the basics of immunology, discussing the major components of the innate (nonspecific) and adaptive (specific) immune responses.

Concept Summary

Anatomy of the Cardiovascular System

- The **cardiovascular system** consists of a muscular four-chambered heart, blood vessels, and blood.
- The **heart** is composed of **cardiac muscle** and supports two different circulations: the **pulmonary circulation** and the **systemic circulation**.
 - Each side of the heart consists of an **atrium** and a **ventricle**.
 - The atria are separated from the ventricles by the **atrioventricular valves** (**tricuspid** on the right, bicuspid [mitral] on the left).
 - The ventricles are separated from the vasculature by the **semilunar valves** (**pulmonary** on the right, **aortic** on the left).
 -
 - The left side of the heart contains more muscle than the right side because the systemic circulation has a much higher resistance and pressure.
 - Electrical conduction of the heart starts at the **sinoatrial (SA) node** and then goes to the **atrioventricular (AV) node**. From the AV node, electrical impulses travel to the **bundle of His** before traveling through the **Purkinje fibers**.
 - **Systole** refers to the period during ventricular contraction when the AV

valves are closed. During **diastole**, the heart is relaxed and the semilunar valves are closed.

- The **cardiac output** is the product of **heart rate** and **stroke volume**.
- The sympathetic nervous system increases the heart rate and contractility. The parasympathetic nervous system decreases heart rate.
- The vasculature consists of arteries, veins, and capillaries.
 - **Arteries** are thick, highly muscular structures with an elastic quality. This allows for recoil and helps to propel blood forward within the system. Small muscular arteries are **arterioles**, which control flow into capillary beds.
 - **Capillaries** have walls that are one cell thick, making them so narrow that red blood cells must travel through them single-file. Capillaries are the sites of gas and solute exchange.
 - **Veins** are inelastic, thin-walled structures that transport blood to the heart. They are able to stretch in order to accommodate large volumes of blood but do not have recoil capability. Veins are compressed by surrounding skeletal muscles and have **valves** to maintain one-way flow. Small veins are called **venules**.
- A portal system is one in which blood passes through two capillary beds in series.
 - In the **hepatic portal system**, blood travels from the gut capillary beds to the liver capillary bed via the hepatic portal vein.
 - In the **hypophyseal portal system**, blood travels from the capillary bed in the hypothalamus to the capillary bed in the anterior pituitary.
 - In the **renal portal system**, blood travels from the glomerulus to the vasa recta through an efferent arteriole.

Blood

- Blood is composed of cells and plasma, an aqueous mixture of nutrients, salts, respiratory gases, hormones, and blood proteins.
- **Erythrocytes (red blood cells)** lack mitochondria, a nucleus, and organelles in order to make room for **hemoglobin**, a protein that carries oxygen. Common measurements include hemoglobin concentration and **hematocrit**, the percentage of blood composed of erythrocytes.
- **Leukocytes (white blood cells)** are formed in the bone marrow. They are a crucial part of the immune system.
 - Granular leukocytes such as neutrophils, eosinophils, and basophils play a role in nonspecific immunity.
 - Agranulocytes, including lymphocytes and monocytes, also play a role in immunity, with lymphocytes playing a large role in specific immunity.
- **Thrombocytes (platelets)** are cell fragments from **megakaryocytes** that are required for coagulation.
- Blood antigens include the surface antigens A, B, and O, as well as Rh factor (D).
 - The I^A (A) and I^B (B) alleles are codominant, while the i (O) allele is recessive. An individual has antibodies for any AB alleles he or she does not have.
 - Positive Rh factor is dominant. An Rh-negative individual will only create anti-Rh antibodies after exposure to Rh-positive blood.

Physiology of the Cardiovascular System

- **Blood pressure** refers to the force per unit area that is exerted on the walls of blood vessels by blood. It is divided into systolic and diastolic components.
 - It must be high enough to overcome the resistance created by arterioles and capillaries, but low enough to avoid damaging the vasculature and surrounding structures.
 - It can be measured with a **sphygmomanometer**.
 - Blood pressure is maintained by baroreceptor and chemoreceptor reflexes. Low blood pressure promotes **aldosterone** and **antidiuretic hormone (ADH or vasopressin)** release. High blood osmolarity also promotes ADH release. High blood pressure promotes **atrial natriuretic peptide (ANP)** release.

- Gas and solute exchange occurs at the level of the capillaries and relies on the existence of concentration gradients to facilitate diffusion across the capillary walls. Capillaries are also leaky, which aids in the transport of gases and solutes.
 - **Starling forces** consist of **hydrostatic pressure** and **osmotic (oncotic) pressure**. Hydrostatic pressure is the pressure of the fluid within the blood vessel, while osmotic pressure is the “sucking” pressure drawing water toward solutes. Oncotic pressure is osmotic pressure due to proteins. Hydrostatic pressure forces fluid out at the arteriolar end of a capillary bed; oncotic pressure draws it back in at the venule end.
 - Oxygen is carried by hemoglobin, which exhibits **cooperative binding**. In the lungs, there is a high partial pressure of oxygen, resulting in loading of oxygen onto hemoglobin. In the tissues, there is a low partial pressure of oxygen, resulting in unloading. With cooperative binding, each successive oxygen bound to hemoglobin increases the affinity of

the other subunits, while each successive oxygen released decreases the affinity of the other subunits.

- Carbon dioxide is largely carried in the blood in the form of carbonic acid, or bicarbonate and hydrogen ions. Carbon dioxide is nonpolar and not particularly soluble, while bicarbonate, hydrogen ions, and carbonic acid are polar and highly soluble.
 - A high $P_a\text{CO}_2$, high $[\text{H}^+]$, low pH, high temperature, and high concentration of 2,3-BPG can cause a right shift in the **oxyhemoglobin dissociation curve**, reflecting a decreased affinity for oxygen.
 - In addition to the opposites of the causes of a right shift, a left shift can also be seen in the dissociation curve for fetal hemoglobin compared to adult hemoglobin.
 - Nutrients, wastes, and hormones are carried in the bloodstream to tissues for use or disposal.
- **Coagulation** results from an activation cascade.
 - When the endothelial lining of a blood vessel is damaged, the collagen and **tissue factor** underlying the endothelial cells are exposed. This results in a cascade of events known as the **coagulation cascade**, ultimately resulting in the formation of a clot over the damaged area.
 - Platelets bind to the collagen and are stabilized by **fibrin**, which is activated by **thrombin**.
 - Clots can be broken down by **plasmin**.

Answers to Concept Checks

7.1

..

Heart Chamber	Valve That Prevents Backflow
Right atrium	Tricuspid valve
Right ventricle	Pulmonary valve
Left atrium	Mitral (bicuspid) valve
Left ventricle	Aortic valve

1. Sinoatrial (SA) node → atrioventricular (AV) node → bundle of His (AV bundle) and its branches → Purkinje fibers

3.

Vessel	Carries Blood Which Direction?	Relative Wall Thickness	Smooth Muscle Present?	Contains Valves?
Artery	Away from heart	Thick	Yes, a lot	No
Capillary	From arterioles to venules	Very thin (one cell layer)	No	No
Vein	Toward heart	Thin	Yes, a little	Yes

4. The right side of the heart pumps blood into a lower-resistance circuit and must do so at lower pressures; therefore, it requires less muscle. The left side of the heart pumps blood into a higher-resistance circuit at higher pressures; therefore, it requires more muscle.

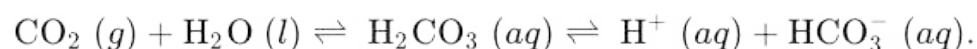
- i. If all autonomic innervation to the heart were lost, the heart would continue beating at the intrinsic rate of the pacemaker (SA node). The individual would be unable to change his or her heart rate via the sympathetic or parasympathetic nervous system, but the heart would not stop beating.

7.2

- .. Plasma is an aqueous mixture of nutrients, salts, respiratory gases, hormones, and blood proteins (clotting proteins, immunoglobulins, and so on).
- !. A B⁺ person could receive blood from a B⁺, B⁻, O⁺, or O⁻ person. A B⁺ person could donate blood to a B⁺ or AB⁺ person.
- }. Hematocrit measures the percentage of a blood sample occupied by red blood cells. It is measured in percentage points.
- l. Lymphocytes are involved in specific immune defense.
- i. Platelets are cellular fragments or shards that are given off by megakaryocytes in the bone marrow.
- }. Only leukocytes (including neutrophils, eosinophils, basophils, monocytes/macrophages, and lymphocytes) contain nuclei. Erythrocytes and platelets do not.

7.3

- .. Opening up more capillary beds (which are in parallel) will decrease the overall resistance of the circuit. The cardiac output will therefore increase in an attempt to maintain constant blood pressure. This is a risk to the heart because the increased demand on the heart can eventually tire it, leading to a heart attack or a precipitous drop in blood pressure.
- !. The bicarbonate buffer system equation is



The combining of carbon dioxide and water is catalyzed by carbonic anhydrase.

- h. The amount of oxygen delivery can be seen as a drop in the y-value (percent hemoglobin saturation) on an oxyhemoglobin dissociation curve. For example, if the blood is 100% saturated while in the lungs (at 100 mmHg O_2) and only 80% saturated while in tissues (at 40 mmHg O_2), then 20% of the oxygen has been released to tissues.
- i. The oxyhemoglobin curve shifts to the right during exercise in response to increased arterial CO_2 , increased $[H^+]$, decreased pH, and increased temperature. This right shift represents hemoglobin's decreased affinity for oxygen, which allows more oxygen to be unloaded at the tissues.
- j. The coagulation cascade can be started by the exposure of collagen and tissue factor to platelets and coagulation factors. The clot is stabilized by fibrin.

Equations to Remember

(7.1) **Cardiac output:** $CO = HR \times SV$

(7.2) **Ohm's law applied to circulation:** $\Delta P = CO \times TPR$



Shared Concepts

Biochemistry Chapter 9

Carbohydrate Metabolism I

Biology Chapter 6

The Respiratory System

Biology Chapter 8

The Immune System

General Chemistry Chapter 12

Electrochemistry

Physics and Math Chapter 4

Fluids

Physics and Math Chapter 6

Circuits

Discrete Practice Questions

1. Which of the following is a FALSE statement regarding erythrocytes?

- (A) Erythrocytes contain hemoglobin.
- (B) Erythrocytes are anaerobic.
- (C) The nuclei of erythrocytes are located in the middle of the biconcave disc.
- (D) Erythrocytes are phagocytized in the spleen and liver after a certain period of time.

2. Which of the following is the correct sequence of a cardiac impulse?

- (A) SA node → AV node → Purkinje fibers → bundle of His → ventricles
- (B) AV node → bundle of His → Purkinje fibers → ventricles → atria
- (C) SA node → atria → AV node → bundle of His → Purkinje fibers → ventricles
- (D) SA node → AV node → atria → bundle of His → Purkinje fibers → ventricles

3. Hemoglobin's affinity for O₂:

- (A) increases in exercising muscle tissue.
- (B) decreases as blood P_aCO₂ decreases.

- (C) decreases as blood pH decreases.
- (D) is higher in maternal blood than in fetal blood.

4. Which of the following correctly traces the circulatory pathway?

- (A) Superior vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left ventricle → aorta
- (B) Superior vena cava → left atrium → left ventricle → pulmonary artery → lungs → pulmonary veins → right atrium → right ventricle → aorta
- (C) Aorta → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left ventricle → superior vena cava
- (D) Superior vena cava → right atrium → right ventricle → pulmonary veins → lungs → pulmonary artery → left atrium → left ventricle → aorta

5. At the venous end of a capillary bed, the osmotic pressure:

- (A) is greater than the hydrostatic pressure.
- (B) results in a net outflow of fluid.
- (C) is significantly higher than the osmotic pressure at the arterial end.
- (D) causes proteins to enter the interstitium.

6. A patient's chart reveals that he has a cardiac output of 7500 mL per

minute and a stroke volume of 50 mL. What is his pulse, in beats per minute?

- (A) 50
- (B) 100
- (C) 150
- (D) 400

7. An unconscious patient is rushed into the emergency room and needs an immediate blood transfusion. Because there is no time to check her medical history or determine her blood type, which type of blood should she receive?

- (A) AB⁺
- (B) AB⁻
- (C) O⁺
- (D) O⁻

8. Which of the following is true regarding arteries and veins?

- (A) Arteries are thin-walled, muscular, and elastic, whereas veins are thick-walled and inelastic.
- (B) Arteries always conduct oxygenated blood, whereas veins always carry deoxygenated blood.
- (C) The blood pressure in the aorta is always higher than the pressure in the superior vena cava.
- (D) Arteries facilitate blood transport by using skeletal muscle contractions, whereas veins make use of the pumping of the

heart to push blood.

9. At any given time, there is more blood in the venous system than the arterial system. Which of the following features of veins allows for this?
- (A) Relative lack of smooth muscle in the wall
 - (B) Presence of valves
 - (C) Proximity of veins to lymphatic vessels
 - (D) Thin endothelial lining
10. Which of the following is involved in the body's primary blood-buffering mechanism?
- (A) Fluid intake
 - (B) Absorption of nutrients in the gastrointestinal system
 - (C) Carbon dioxide produced from metabolism
 - (D) Hormones released by the kidneys
11. Due to kidney disease, a person is losing albumin into the urine. What effect is this likely to have within the capillaries?
- (A) Increased oncotic pressure
 - (B) Increased hydrostatic pressure
 - (C) Decreased oncotic pressure
 - (D) Decreased hydrostatic pressure

12. The tricuspid valve prevents backflow of blood from the:
- (A) left ventricle into the left atrium.
 - (B) aorta into the left ventricle.
 - (C) pulmonary artery into the right ventricle.
 - (D) right ventricle into the right atrium.
13. The world record for the longest-held breath is 22 minutes and 0 seconds. If a sample were taken from this individual during the last minute of breath-holding, which of the following might be observed?
- (A) Increased hemoglobin affinity for oxygen
 - (B) Decreased $P_a\text{CO}_2$
 - (C) Increased hematocrit
 - (D) Decreased pH
14. A person has a heart attack that primarily affects the wall between the two ventricles. Which portion of the electrical conduction system is most likely affected?
- (A) AV node
 - (B) SA node
 - (C) Bundle of His
 - (D) Left ventricular muscle
15. Which vascular structure creates the most resistance to blood flow?

- (A) Aorta
- (B) Arterioles
- (C) Capillaries
- (D) Veins

Discrete Practice Answers

1. **C**

Erythrocytes, or red blood cells, are produced in the red bone marrow and circulate in the blood for about 120 days, after which they are phagocytized in the spleen and the liver, eliminating **(D)**. Red blood cells have a disc-like shape and lose their membranous organelles (like mitochondria and nuclei) during maturation. This makes **(C)** the correct answer. Erythrocytes are filled with hemoglobin; their lack of mitochondria makes their metabolism solely anaerobic, eliminating **(A)** and **(B)**.

2. **C**

An ordinary cardiac contraction originates in, and is regulated by, the sinoatrial (SA) node. The impulse travels through both atria, stimulating them to contract simultaneously. The impulse then arrives at the atrioventricular (AV) node, which momentarily slows conduction, allowing for completion of atrial contraction and ventricular filling. The impulse is then carried by the bundle of His and its branches through the Purkinje fibers in the walls of both ventricles, generating a strong contraction.

3. **C**

According to the Bohr effect, decreasing the pH in the blood decreases hemoglobin's affinity for O_2 . This makes **(C)** the correct answer. The affinity is generally lowered in exercising muscle to facilitate unloading of oxygen to tissues, eliminating **(A)**. A decrease in the P_aCO_2 would cause a decrease in $[H^+]$ or increased pH—which increases hemoglobin's affinity for O_2 ,

eliminating **(B)**. Finally, **(D)** is incorrect because hemoglobin's affinity for O₂ is higher in fetal blood than in adult blood.

4. **A**

Blood drains from the superior and inferior venae cavae into the right atrium. It passes through the tricuspid valve and into the right ventricle, and then through the pulmonary valve into the pulmonary artery, which leads to the lungs. Oxygenated blood returns to the left atrium via the pulmonary veins. It flows through the mitral valve into the left ventricle. From the left ventricle, it is pumped through the aortic valve into the aorta for distribution throughout the body.

5. **A**

The exchange of fluid is greatly influenced by the differences in the hydrostatic and osmotic pressures of blood and tissues. The osmotic (oncotic) pressure remains relatively constant; however, the hydrostatic pressure at the arterial end is greater than the hydrostatic pressure at the venous end. As a result, fluid moves out of the capillaries at the arterial end and back in at the venous end. Fluid is reabsorbed at the venous end because the osmotic pressure exceeds the hydrostatic pressure. Proteins should not cross the capillary wall under normal circumstances.

6. **C**

The first step in solving this problem is to define cardiac output: cardiac output = heart rate × stroke volume. We can therefore divide the cardiac output by the stroke volume to determine heart rate:

$$\text{HR} = \frac{\text{CO}}{\text{SV}} = \frac{7500 \frac{\text{mL}}{\text{min}}}{50 \frac{\text{mL}}{\text{beat}}} = 150 \frac{\text{beats}}{\text{min}}$$

Note that this heart rate is actually pathologically fast; a normal heart rate is considered to be between 60 and 100 beats per minute.

7. **D**

Without knowing a patient's blood type, the only type of transfusion that we can safely give is O⁻. People with O⁻ blood are considered universal donors because their blood cells contain no surface antigens. Therefore, O⁻ blood can be given to anyone without potentially life-threatening consequences from ABO or Rh incompatibility.

8. **C**

The only answer choice that correctly describes arteries and veins is **(C)**; the pressure in the aorta usually ranges between 120 and 80 mmHg, depending on whether the heart is in systole or diastole, whereas the pressure in the superior vena cava is near zero. **(A)** is incorrect because arteries are thick-walled and veins are thin-walled. **(B)** is also incorrect; this relationship is reversed in pulmonary and umbilical circulation. **(D)** is reversed as well; arteries make use of the pumping of the heart and the “snapping back” of their elastic walls to transport blood, whereas venous blood is propelled along by skeletal muscle contractions.

9. **A**

The relative lack of smooth muscle in venous walls allows stretching to store most of the blood in the body. Valves in the veins allow for one-way flow of blood toward the heart, not stretching. Both arteries and veins are close to lymphatic vessels, which has no bearing on their relative difference in volume. Both arteries and veins have a single-cell endothelial lining.

10. **C**

Carbon dioxide is a byproduct of metabolism in cells that later combines

with water to form bicarbonate in a reaction catalyzed by carbonic anhydrase. This system is blood plasma's most important buffer system. Food and fluid absorption are not significant sources of buffering, eliminating **(A)** and **(B)**. While the kidneys can be involved in acid–base balance, they carry out this function through their filtration, secretion, and reabsorption mechanisms, not through hormone release, eliminating **(D)**.

1. **C**

In circulation, plasma proteins play an important role in generating osmotic (oncotic) pressure. This allows water that is displaced at the arterial end of a capillary bed by hydrostatic pressure to be reabsorbed at the venule end. Loss of these plasma proteins would cause a decrease in the plasma osmotic (oncotic) pressure.

2. **D**

The atrioventricular valves are located between the atria and the ventricles on both sides of the heart. Their role is to prevent backflow of blood into the atria. The valve on the right side of the heart has three cusps and is called the tricuspid valve. It prevents backflow of blood from the right ventricle into the right atrium.

3. **D**

Holding one's breath for a prolonged period would result in a drop in oxygenation and an increase in $P_a\text{CO}_2$. The increased carbon dioxide would associate with water to form carbonic acid, which would dissociate into a proton and a bicarbonate anion. Further, the low oxygen saturation would eventually lead to anaerobic metabolism in some tissues, causing an increase in lactic acid. These would all lead to a decreased pH.

4. **C**

The cardiac conduction system starts at the SA node, which is located near the top of the right atrium, and continues down to the AV node, which is located between the two AV valves. The bundle of His is located within the wall between the ventricles, and is likely to be affected if the wall between the ventricles has been damaged by a heart attack. This may affect the left ventricle, but the left ventricular muscle itself is not part of the cardiac conduction system.

5. **B**

The greatest amount of resistance is provided by the arterioles. Arterioles are highly muscular and have the ability to contract and dilate in order to regulate blood pressure.



The background of the slide is a microscopic view of a blood sample. It features numerous red blood cells, which are small, biconcave, and reddish-orange. Interspersed among these cells are several white Y-shaped structures, representing antibodies, which are translucent and have a textured surface. The overall color palette is a mix of warm browns, oranges, and a prominent blue square in the top-left corner.

CHAPTER 8

The Immune System

CHAPTER 8

The Immune System

In This Chapter

8.1 Structure of the Immune System

Innate and Adaptive Immunity
Anatomy

8.2 The Innate Immune System

Noncellular Nonspecific Defenses
Cells of the Innate Immune System

8.3 The Adaptive Immune System

Cells of the Adaptive Immune System
Activation of the Adaptive Immune System
Recognition of Self and Nonself
Immunization



8.4 The Lymphatic System

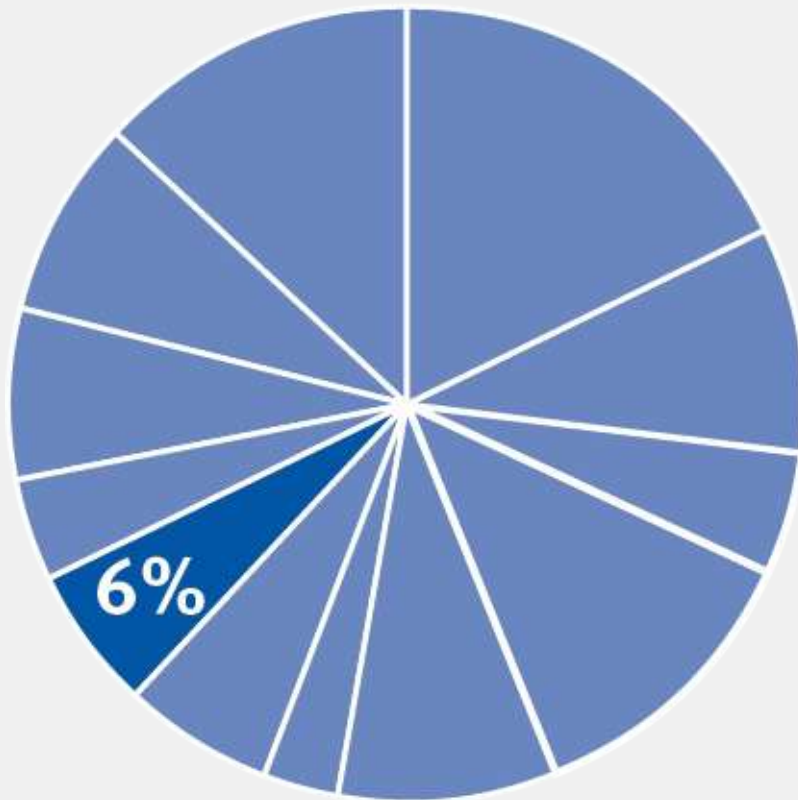
Structure

Function

Concept Summary

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Chapter Profile



The content in this chapter should be relevant to about 6% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems

Introduction

Over the past few decades, the public's imagination has been captured by alarming reports of “flesh-eating” bacteria and diseases. While these bacteria do not actually eat flesh, that doesn't make them any less dangerous. “Flesh-eating” bacteria cause a condition called *necrotizing fasciitis*, a serious disease that requires aggressive medical and surgical treatment including intravenous antibiotics, surgical debridement (removal) of the necrotic tissue, and sometimes even amputation. Many different types of bacteria, including group A *Streptococcus*, *Clostridium perfringens*, and methicillin-resistant *Staphylococcus aureus* (MRSA), can cause necrotizing fasciitis. The massive destruction of skin, muscle, and connective tissue by the release of bacterial toxins called superantigens is life-threatening. These superantigens cause the immune system to become nonspecifically overactivated. Necrotizing fasciitis is dangerous not only because of the bacteria and subsequent inflammatory response, but also because the destruction of skin—a nonspecific immune defense—leaves the body susceptible to superinfection (infection with another pathogen).

In this chapter, we will consider this type of nonspecific defense, as well as specific immune defenses. The human body relies on the interaction between the innate (nonspecific) and adaptive (specific) immune systems in order to protect itself from disease. However, the immune system is not always perfect in its responses. Sometimes the immune system may become nonspecifically overactivated, as is the case with necrotizing fasciitis, or it may become activated against the same human it is supposed to protect. We will discuss the individual parts of the immune system and how these parts work together to protect from disease. In addition, we will discuss the concept of autoimmunity, or disease that results from immune attack against oneself.

We'll briefly touch on vaccines, which take advantage of our understanding of the immune system to protect us from life-threatening infections. The immune system is largely integrated with the lymphatic system, which will be discussed in this chapter as well.

8.1 Structure of the Immune System

LEARNING OBJECTIVES

After Chapter 8.1, you will be able to:

- Distinguish between innate and adaptive immunity
- Draw comparisons between B- and T-cells, including their development, maturation, functions, specificity, and control mechanisms
- Identify immune cells as granulocytes or agranulocytes

Each day, the human body is exposed to numerous bacteria, viruses, fungi, and even parasites. Yet our bodies are able to protect us from infection most of the time. Even when we do get sick, the immune system is usually able to contain and eliminate the infection.

INNATE AND ADAPTIVE IMMUNITY

In order to fight infection, the human body has two different divisions of the immune system: innate and adaptive immunity, compared in Figure 8.1.

Innate immunity is composed of defenses that are always active against infection, but lack the ability to target specific invaders; for this reason, it is

also called **nonspecific immunity**. **Adaptive** or **specific immunity** refers to the defenses that target a specific pathogen. This system is slower to act, but can maintain immunological memory of an infection to mount a faster attack in subsequent infections.

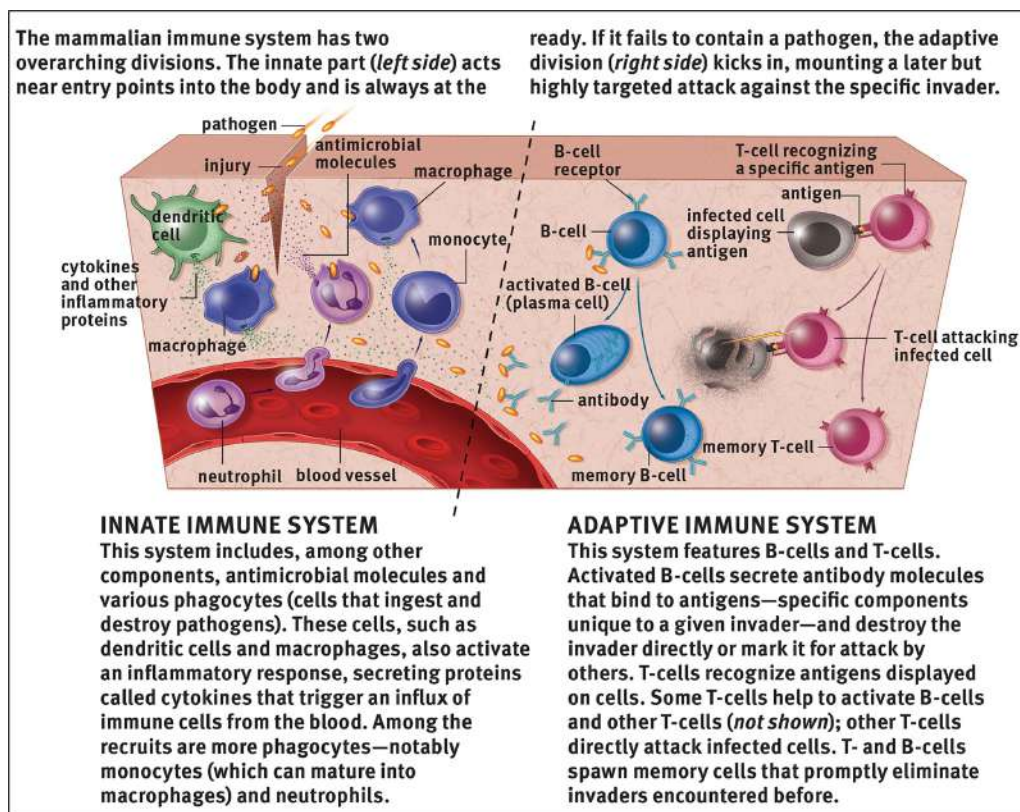


Figure 8.1. Divisions of the Immune System



MCAT EXPERTISE

The immune system is a topic where cell biology, biochemistry, anatomy, and biology interact. Thus, it is a topic that allows the MCAT to ask questions that integrate these four topics. As we discuss these complex ideas, focus on the big picture first, creating a mental image

to help you put these concepts together. Then go back and plug in the details. This approach will foster your ability to associate structure with function and to think critically about the topic on Test Day.

ANATOMY

The immune system is not housed in a single organ, as shown in Figure 8.2. The structure and components that serve as nonspecific defenses often serve functions in other organ systems. The **bone marrow** produces all of the **leukocytes (white blood cells)** that participate in the immune system through the process of hematopoiesis, discussed in Chapter 7 of *MCAT Biology Review*. The **spleen** is a location of blood storage and activation of **B-cells**, which turn into **plasma cells** to produce antibodies as part of adaptive immunity. Note that when B-cells leave the bone marrow, they are considered mature but naïve (because they have not yet been exposed to an antigen). Because these antibodies dissolve and act in the blood (rather than within cells), this division of adaptive immunity is called **humoral immunity**. **T-cells**, another class of adaptive immune cells, mature in the **thymus**, a small gland just in front of the pericardium, the sac that protects the heart. T-cells are the agents of **cell-mediated immunity** because they coordinate the immune system and directly kill virally infected cells. Finally, **lymph nodes**, a major component of the lymphatic system, provide a place for immune cells to communicate and mount an attack; B-cells can be activated here as well. Other immune tissue is found in close proximity to the digestive system, which is a site of potential invasion by pathogens. These tissues are commonly called **gut-associated lymphoid tissue (GALT)** and include the **tonsils** and **adenoids** in the head, **Peyer's patches** in the small intestine, and lymphoid aggregates in the **appendix**.

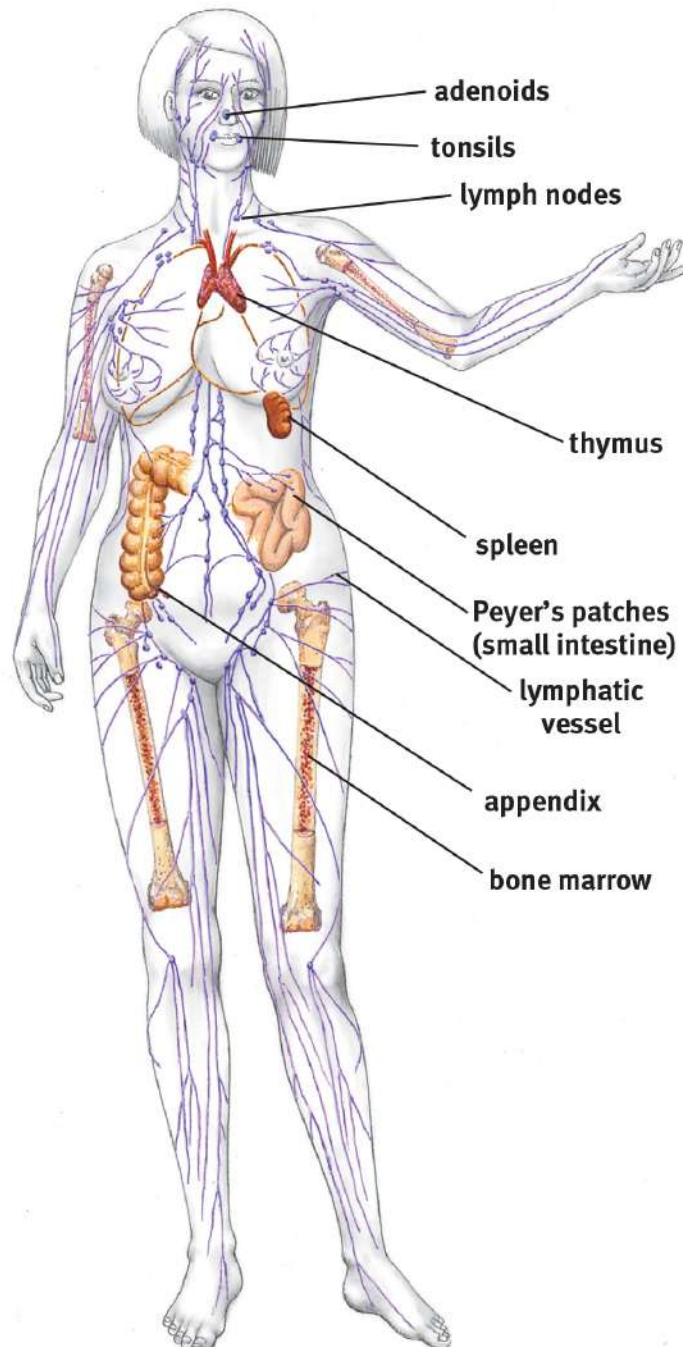


Figure 8.2. Anatomy of the Immune and Lymphatic Systems



REAL WORLD

Lay people not involved in medicine often say that their “glands” are swollen when they feel a sensation of fullness around the throat. From a biological standpoint, lymph nodes are not glands—they do not secrete products into the bloodstream (endocrine) or ducts (exocrine). What they are actually experiencing is known medically as *lymphadenopathy*, or the swelling of the lymph nodes that occurs with activation of the immune system.



KEY CONCEPT

Organs of the immune system:

- Lymph nodes filter lymph and are a site where immune responses can be mounted.
- Bone marrow is the site of immune cell production.
- The thymus is the site of T-cell maturation.
- The spleen acts as a storage area for white blood cells and platelets, a recycling center for red blood cells, and a filter of blood and lymph for the immune system.

Leukocytes are produced in the bone marrow through hematopoiesis, shown in Figure 8.3. (Note: This is the same image shown in Figure 7.7, but is copied here for convenience.) Leukocytes are divided into two groups of cells: **granulocytes** and **agranulocytes**. These names refer to the presence or absence of **granules** in the cytoplasm. These granules contain toxic enzymes and chemicals, which can be released by exocytosis, and are particularly

effective against bacterial, fungal, and parasitic pathogens. Both granulocytes and agranulocytes come from a common precursor: **hematopoietic stem cells**. Remember from Chapter 7 of *MCAT Biology Review* that hematopoietic stem cells are also the cell type that gives rise to red blood cells and platelets. Granulocytes include cells such as **neutrophils**, **eosinophils**, and **basophils**. The names of these cells actually refer to the way that the cells appear after staining with certain chemicals. Agranulocytes include the **lymphocytes**, which are responsible for antibody production, immune system modulation, and targeted killing of infected cells. **Monocytes**, which are phagocytic cells in the bloodstream, are also considered agranulocytes. They become **macrophages** in tissues; many tissues have resident populations of macrophages with specific names (such as **microglia** in the central nervous system, **Langerhans cells** in the skin, and **osteoclasts** in bone).

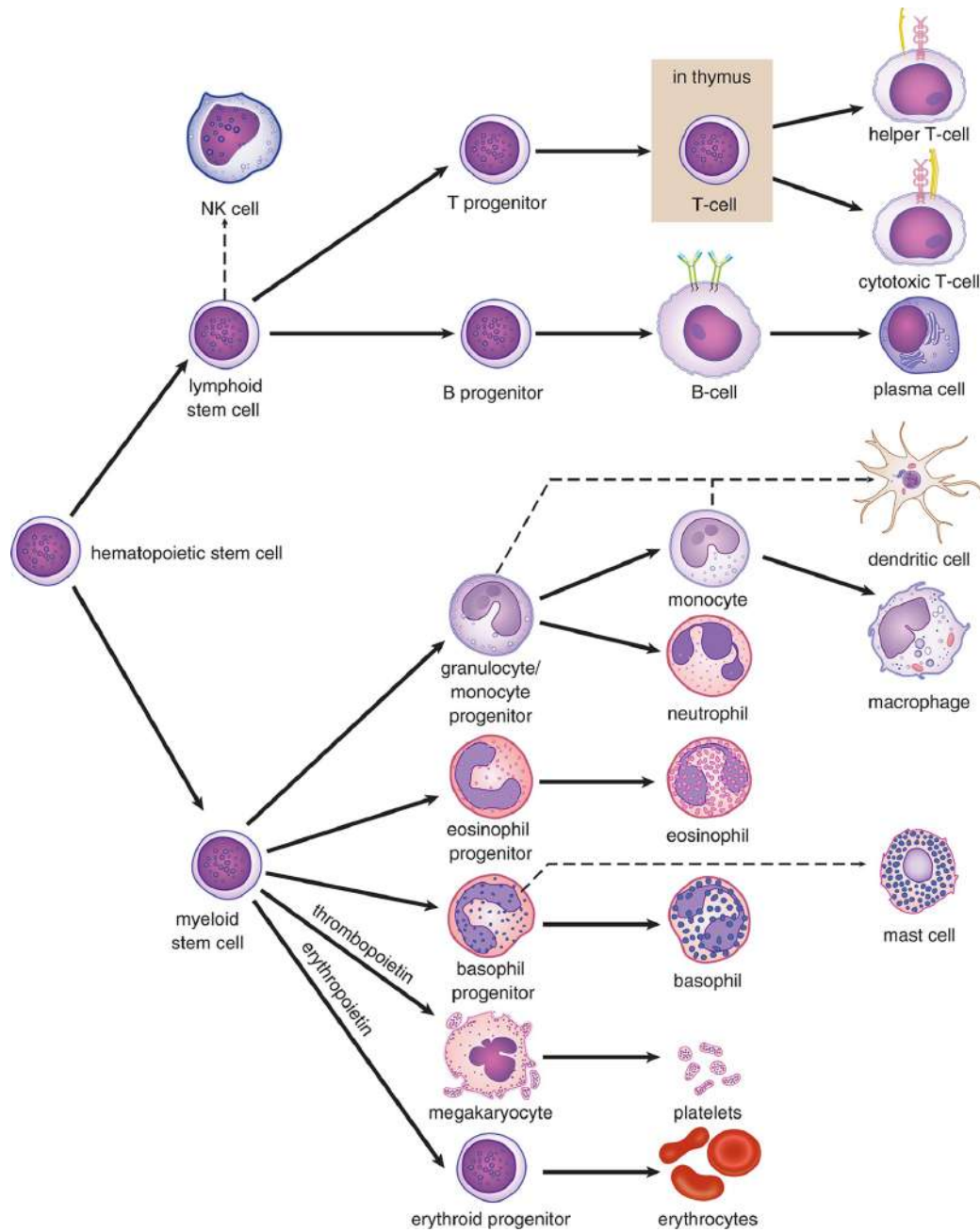


Figure 8.3. Hematopoiesis



REAL WORLD

One of the tests often ordered by physicians is known as a complete blood count (CBC). This test counts the number of red blood cells, platelets, and white blood cells. If the CBC is ordered with a *differential*, it means that each type of white blood cell is counted. Any cell populations that are abnormal may indicate a particular type of infection given the right clinical conditions. For example, if 90 percent of the white blood cells are neutrophils, it is possible that a bacterial infection is present. If 20 percent of the white blood cells are eosinophils, it is possible that an invasive parasitic infection is present.

Innate immunity refers to the responses cells can carry out without learning; for this reason, it is also known as the **nonspecific immune response**. Conversely, adaptive immunity is developed as immune cells learn to recognize and respond to particular antigens, and is often aptly referred to as the **specific immune response**. We can also divide the specific immune system into **humoral immunity** (driven by B-cells and antibodies) and **cell-mediated immunity** (driven by T-cells).

MCAT CONCEPT CHECK 8.1:

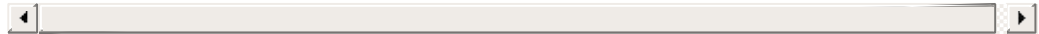
Before you move on, assess your understanding of the material with these questions.

1. What are the differences between innate and adaptive immunity?
 - Innate immunity:

- Adaptive immunity:

2. Compare and contrast B- and T-cells:

Cell Type	Site of Development	Site of Maturation	Major Functions	Specific or Nonspecific?	Humoral or Cell-Mediated?
B-cell					
T-cell					



3. Which cells are considered granulocytes? Which are considered agranulocytes?

- Granulocytes:

- Agranulocytes:

8.2 The Innate Immune System

LEARNING OBJECTIVES

After Chapter 8.2, you will be able to:

- Describe the immunologic functions of each of the noncellular nonspecific immune defenses, including defensins and mucus
- Identify the immune cell type used for antigen presentation
- Differentiate between MHC-I and MHC-II
- Recall the stimuli required for activating natural killer cells, neutrophils, eosinophils, basophils, and mast cells

The innate immune system consists of cells and structures that offer nonspecific protection.

NONCELLULAR NONSPECIFIC DEFENSES

Our first line of defense is the **skin (integument)**. In Chapter 10 of *MCAT Biology Review*, we will discuss the specific homeostatic functions of the skin; for now, we'll focus on how skin protects the body. The skin provides a physical barrier between the outside world and our internal organs, preventing most bacteria, viruses, fungi, and parasites from entering the

body. Additionally, antibacterial enzymes called **defensins** can be found on the skin. Sweat also has antimicrobial properties. The skin is an important first line of defense: a cut or abrasion on the skin provides an entry point for pathogens into the body. Deeper wounds allow pathogens to penetrate deeper into the body.

As discussed in Chapter 6 of *MCAT Biology Review*, the respiratory system also has mechanisms to prevent pathogens from entering the body. The respiratory passages are mucous membranes, lined with cilia to trap particulate matter and push it up toward the oropharynx, where it can be swallowed or expelled. While mucus helps to trap particulates like smoke and dirt, it also helps to prevent bacteria and viruses from gaining access to the lung tissue below. Several other mucous membranes, including those around the eye and in the oral cavity, produce a nonspecific bacterial enzyme called **lysozyme**, which is secreted in tears and saliva, respectively.

The Gastrointestinal Tract

The gastrointestinal tract also plays a role in nonspecific immunity. First, the stomach secretes acid, resulting in the elimination of most pathogens. In addition, the gut is colonized by bacteria. Most of these bacteria lack the necessary characteristics to cause infection. Because there is already such a large bacterial population in the gut, many potential invaders are not able to compete and are thus kept at bay. Many antibiotics reduce the population of gut flora, providing an opportunity for the growth of pathogens resistant to that antibiotic.



REAL WORLD

The GI tract of a newborn baby is particularly susceptible to infection because the newborn's immune system is underdeveloped and the GI tract is not yet colonized. Breast milk contains a family of antibodies that are particularly effective on mucosal surfaces and help to defend newborn babies against gastrointestinal infections.

Complement

The **complement** system consists of a number of proteins in the blood that act as a nonspecific defense against bacteria. Complement can be activated through a **classical pathway** (which requires the binding of an antibody to a pathogen) or an **alternative pathway** (which does not require antibodies). The complement proteins punch holes in the cell walls of bacteria, making them osmotically unstable. Despite the association with antibodies, complement is considered a nonspecific defense because it cannot be modified to target a specific organism over others.

Interferons

To protect against viruses, cells that have been infected with viruses produce **interferons**, proteins that prevent viral replication and dispersion. Interferons cause nearby cells to decrease production of both viral and cellular proteins. They also decrease the permeability of these cells, making it harder for a virus to infect them. In addition, interferons upregulate MHC class I and class II molecules, resulting in increased antigen presentation and better detection of the infected cells by the immune system, as described in the next section. Interferons are responsible for many “flu-like” symptoms that occur during viral infection, including malaise, tiredness, muscle soreness, and fever.

CELLS OF THE INNATE IMMUNE SYSTEM

So, what happens when bacteria, viruses, fungi, or parasites breach noncellular defenses? The cells of the innate immune system are always poised and ready to attack.

Macrophages

Macrophages, a type of agranulocyte, reside within the tissues. These cells derive from blood-borne **monocytes** and can become a **resident population** within a tissue (becoming a permanent, rather than transient, cell group in the tissue). Many of these resident macrophages are highlighted throughout *MCAT Biology Review*, including microglia in the central nervous system, Langerhans cells in the skin, and osteoclasts in bone. When a bacterial invader enters a tissue, the macrophages become activated. The activated macrophage does three things. First, it phagocytizes the invader through endocytosis. Then, it digests the invader using enzymes. Finally, it presents little pieces of the invader (mostly peptides) to other cells using a protein called **major histocompatibility complex (MHC)**. MHC binds to a pathogenic peptide (also called an **antigen**) and carries it to the cell surface, where it can be recognized by cells of the adaptive immune system. In addition, macrophages release **cytokines**, chemical substances that stimulate inflammation and recruit additional immune cells to the area.

MHC molecules come in two main classes: class I and class II. All nucleated cells in the body display **MHC class I** molecules. Any protein produced within a cell can be loaded onto MHC-I and presented on the surface of the cell, as shown in Figure 8.4. This allows the immune system to monitor the health of these cells and to detect if the cells have been infected with a virus or another intracellular pathogen; only those cells that are infected would be expected to

present an unfamiliar (nonself) protein on their surfaces. Therefore, the MHC-I pathway is often called the **endogenous pathway** because it binds antigens that come from inside the cell. Cells that have been invaded by intracellular pathogens can then be killed by a certain group of T-cells (cytotoxic T-lymphocytes) to prevent infection of other cells.

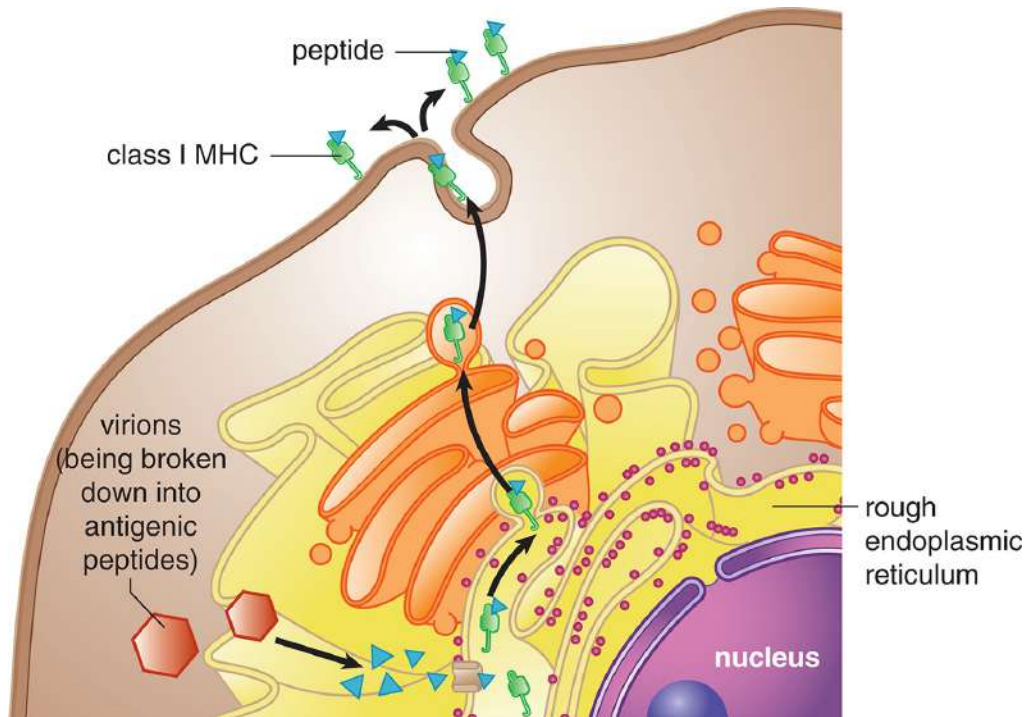


Figure 8.4. Endogenous Pathway for Antigen Presentation (MHC Class I)

MHC-I exists in all nucleated cells.



REAL WORLD

MHC molecules are highly variable among humans, but can usually be divided into particular subtypes. People with certain MHC subtypes have different susceptibility to disease, and individuals with certain subtypes are considerably more susceptible to autoimmune diseases.

It is also important to match the MHC types of the donor and recipient as closely as possible during transplants to avoid rejection.

MHC class II molecules are mainly displayed by professional **antigen-presenting cells** like macrophages, as shown in Figure 8.5. Remember that these phagocytic cells pick up pathogens from the environment, process them, and then present them on MHC-II. An **antigen** is a substance (usually a pathogenic protein) that can be targeted by an antibody. While antibody production is the domain of the adaptive immune system, it is important to understand that cells of the innate immune system also present antigens. Because these antigens originated outside the cell, the MHC-II pathway is often called the **exogenous pathway**. The presentation of an antigen by an immune cell may result in the activation of both the innate and adaptive immune systems. Professional antigen-presenting cells include macrophages, **dendritic cells** in the skin, some B-cells, and certain activated epithelial cells.

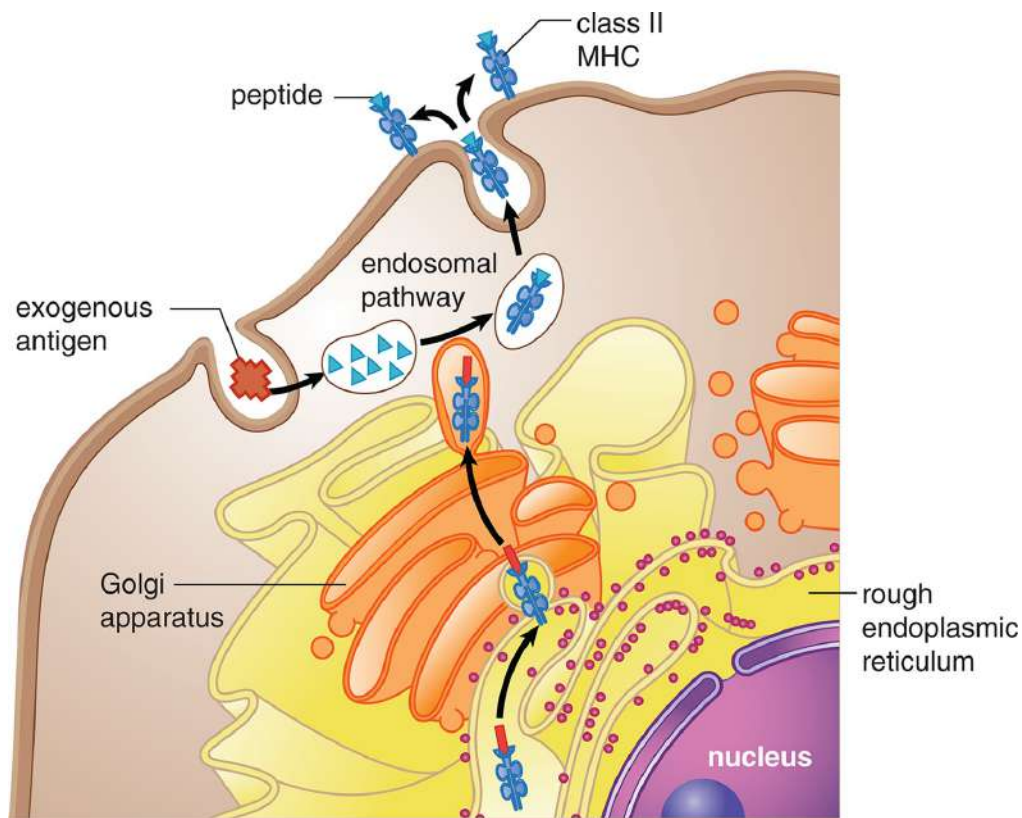


Figure 8.5. Exogenous Pathway for Antigen Presentation (MHC Class II)

MHC-II exists only in professional antigen-presenting cells, like macrophages, dendritic cells, some B-cells, and some activated epithelial cells.



KEY CONCEPT

Major histocompatibility complex molecules are joined with antigens. The MHC–antigen complex then goes to the cell surface to display the antigen. This allows the immune system to monitor the health of cells. MHC-I is in all nucleated cells and presents endogenous antigens. MHC-II is in antigen-presenting cells and presents exogenous antigens.

Macrophages and dendritic cells also have special receptors known as **pattern recognition receptors (PRR)**, the best-described of which are **toll-like receptors (TLR)**. PRRs are able to recognize the category of the invader (bacterium, virus, fungus, or parasite). This allows for the production of appropriate cytokines to recruit the right type of immune cells; each immune cell has different weapons that can target particular groups of pathogens.

Natural Killer Cells

In the arms race between the human immune system and pathogens, some pathogens have found ways to avoid certain defenses. For example, some viruses cause downregulation of MHC molecules, making it harder for T-cells to recognize the presence of an infection. **Natural killer (NK) cells**, a type of nonspecific lymphocyte, are able to detect the downregulation of MHC and induce apoptosis in these virally infected cells. Cancer cells may also downregulate MHC production, so NK cells also offer protection from the growth of cancer as well.



KEY CONCEPT

The innate immune cells are nonspecific and form the first line of defense against pathogens.

INNATE

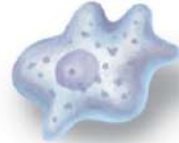
MACROPHAGE

This immune defender engulfs and consumes pathogens.



MAST CELL

This cell releases histamine and other chemicals that promote inflammation.



GRANULOCYTES

Three cell types with tiny granules in their interiors—neutrophils, eosinophils, and basophils—participate in the inflammatory response.



DENDRITIC CELL

It presents antigens—fragments of protein or other molecules from pathogens or cancer cells—to adaptive immune cells, inducing them to attack bearers of the displayed antigens.



NATURAL KILLER CELL

This cell destroys the body's own cells that have become infected with pathogens; it also goes after cancer cells.



Granulocytes

In addition to macrophages, the granulocytes, which include neutrophils, eosinophils, and basophils (and closely related mast cells), are also involved in nonspecific defense. **Neutrophils** are the most populous leukocyte in blood and are very short-lived (a bit more than five days). These cells are phagocytic, like macrophages, and target bacteria. Neutrophils can follow bacteria using chemotaxis—the movement of an organism according to chemical stimuli; in this case, the neutrophil senses products given off by bacteria, moving up the concentration gradient to the source. Neutrophils can also detect bacteria once they have been **opsonized** (marked with an antibody from a B-cell). Other cells, like natural killer cells, macrophages, monocytes, and eosinophils, also contain receptors for antibodies and can attack opsonized bacteria. Dead neutrophil collections are responsible for the formation of **pus** during an infection.

Eosinophils contain bright red-orange granules and are primarily involved in allergic reactions and invasive parasitic infections. Upon activation, eosinophils release large amounts of **histamine**, an inflammatory mediator. This results in vasodilation and increased leakiness of the blood vessels, allowing additional immune cells (especially macrophages and neutrophils) to move out of the bloodstream and into the tissue. **Inflammation** is particularly useful against extracellular pathogens, including bacteria, fungi, and parasites.



KEY CONCEPT

Histamine causes inflammation by inducing vasodilation and the movement of fluid and cells from the bloodstream into tissues.

Finally, **basophils** contain large purple granules and are involved in allergic responses. They are the least populous leukocyte in the bloodstream under normal conditions. **Mast cells** are closely related to basophils, but have smaller granules and exist in the tissues, mucosa, and epithelium. Both basophils and mast cells release large amounts of histamine in response to allergens, leading to inflammatory responses.

MCAT CONCEPT CHECK 8.2:

Before you move on, assess your understanding of the material with these questions.

1. For each of the noncellular nonspecific immune defenses listed below, provide a brief description of its immunologic function:

- Skin:

- Defensins:

- Lysozyme:

- Mucus:

- Stomach acid:

- Normal gastrointestinal flora:

- Complement:

2. Which cells are professional antigen-presenting cells?

3. What are the differences between MHC-I and MHC-II?

- MHC-I:

- MHC-II:

4. What stimulus activates each of the following types of cells?

- Natural killer cells:

- Neutrophils:

- Eosinophils:

- Basophils and mast cells:

8.3 The Adaptive Immune System

High-Yield

LEARNING OBJECTIVES

After Chapter 8.3, you will be able to:

- Recall the major classes of lymphocytes and their functions
- Describe the three main effects that circulating antibodies can have on a pathogen
- Describe the effects of positive and negative selection on T-cell maturation
- Explain why the secondary response to a pathogen is more efficient than the primary response
- Differentiate between passive and active immunity
- Explain how antibodies become specific for a given antigen

The adaptive immune system can identify specific invaders and mount an attack against that pathogen. The response is variable and depends on the identity of the pathogen. The adaptive immune system can be divided into two divisions: humoral immunity and cell-mediated (cytotoxic) immunity. Each involves the identification of the specific pathogen and organization of an appropriate immune response.

CELLS OF THE ADAPTIVE IMMUNE SYSTEM

The adaptive immune system consists mainly of two types of lymphocytes, B-cells and T-cells. B-cells govern the humoral response, while T-cells mount the cell-mediated response. All cells of the immune system are created in the bone marrow, but B- and T-cells mature in different locations. B-cells mature in the bone marrow (although the B in their name originally stood for the bursa of Fabricius, an organ found in birds), and T-cells mature in the thymus. When we are exposed to a pathogen, it may take a few days for the physical symptoms to be relieved. This occurs because the adaptive immune response takes time to form specific defenses against the pathogen.



KEY CONCEPT

B-cells mature in the **b**one marrow. **T**-cells mature in the **t**hymus.

Humoral Immunity

Humoral immunity, which involves the production of **antibodies**, may take as long as a week to become fully effective after initial infection. These antibodies are specific to the antigens of the invading microbe. Antibodies are produced by B-cells, which are lymphocytes that originate and mature in the bone marrow and are activated in the spleen and lymph nodes.



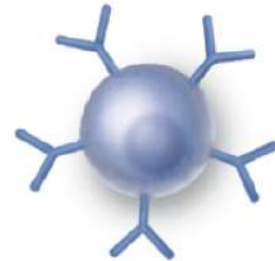
KEY CONCEPT

The adaptive immune cells target invaders with greater specificity, but respond slower since they are replicated in response to an antigen.

ADAPTIVE

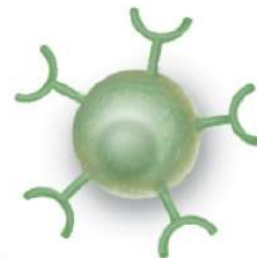
B-CELL

Antigens stimulate this cell to divide and produce antibodies that neutralize invaders or tag them for killing.



T-CELL

A killer T-cell destroys an infected cell in which it detects the presence of antigens. Other T-cells—such as helper and regulatory types—coordinate the immune response.



Antibodies (also called **immunoglobulins [Ig]**) can carry out many different jobs in the body. Just as antigens can be displayed on the surface of cells or can float freely in blood, chyle (lymphatic fluid), or air, so too can antibodies be present on the surface of a cell or secreted into body fluids. When an antibody binds to an antigen, the response will depend on the location. For antibodies secreted into body fluids, there are three main possibilities: first, once bound to a specific antigen, antibodies may attract other leukocytes to phagocytize those antigens immediately. This is called opsonization, as

described earlier. Second, antibodies may cause pathogens to clump together or **agglutinate**, forming large insoluble complexes that can be phagocytized. Third, antibodies can block the ability of a pathogen to invade tissues, essentially neutralizing it. For cell-surface antibodies, the binding of antigen to a B-cell causes activation of that cell, resulting in its proliferation and formation of plasma and memory cells, as described later in this chapter. In contrast, when antigen binds to antibodies on the surface of a mast cell, it causes **degranulation** (exocytosis of granule contents), releasing histamine and causing an inflammatory allergic reaction.

Antibodies are Y-shaped molecules that are made up of two identical **heavy chains** and two identical **light chains**, as shown in Figure 8.6. Disulfide linkages and noncovalent interactions hold the heavy and light chains together. Each antibody has an **antigen-binding region** at the end of which is called the **variable region (domain)**, at the tips of the Y. Within this region, there are specific polypeptide sequences that will bind one, and only one, specific antigenic sequence. Part of the reason it takes so long to initiate the antibody response is that each B-cell undergoes **hypermutation** of its antigen-binding region, trying to find the best match for the antigen. Only those B-cells that can bind the antigen with high affinity survive, providing a mechanism for generating specificity called **clonal selection**. The remaining part of the antibody molecule is known as the **constant region (domain)**. It is this region that cells such as natural killer cells, macrophages, monocytes, and eosinophils have receptors for, and that can initiate the complement cascade. Each B-cell makes only one type of antibody, but we have many B-cells, so our immune system can recognize many antigens. Further, antibodies come in five different isotypes (IgM, IgD, IgG, IgE, and IgA). While the specific purposes of each antibody isotype is outside the scope of the MCAT, you should know that the different types can be used at different times during the adaptive immune response, for different types of pathogens, or in

different locations in the body. Cells can change which isotype of antibody they produce when stimulated by specific cytokines in a process called **isotype switching**.

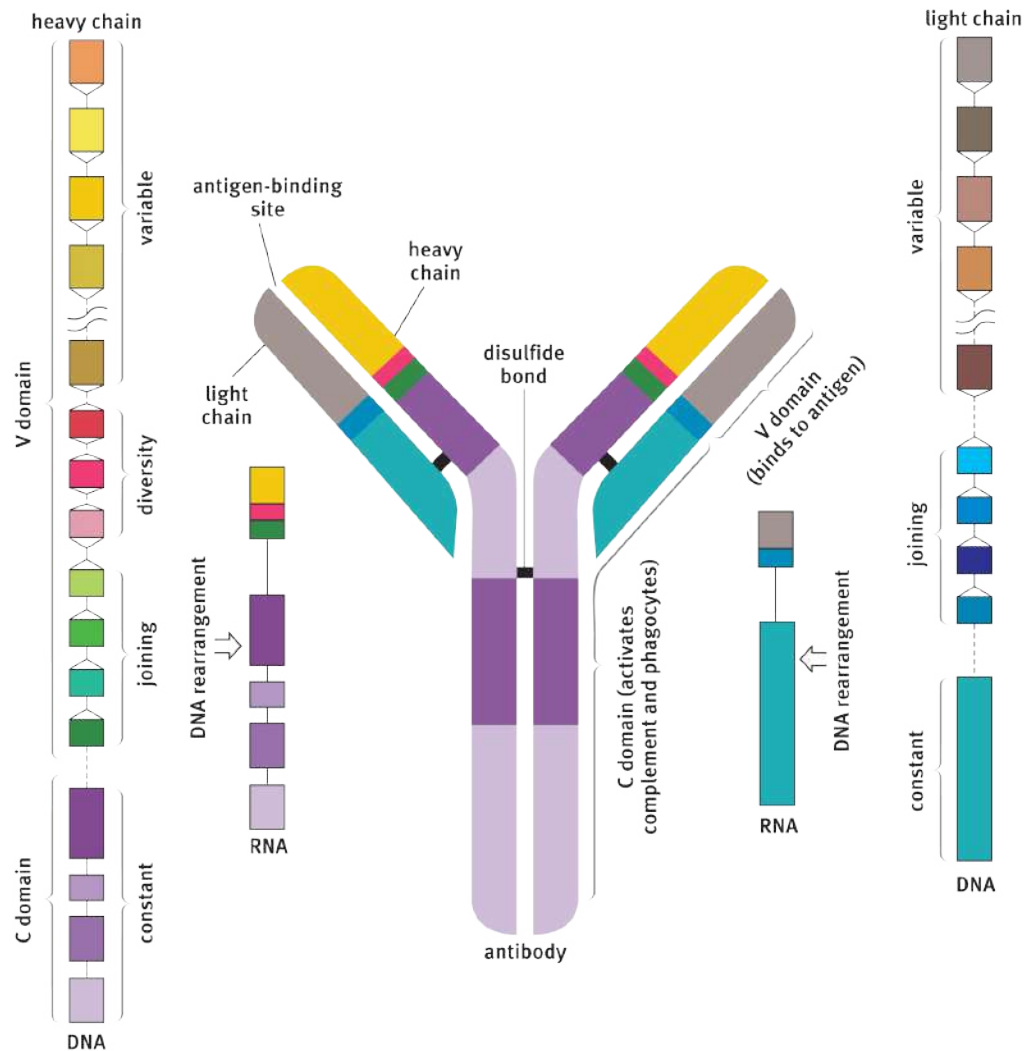


Figure 8.6. Structure of an Antibody Molecule

Not all B-cells that are generated actively or constantly produce antibodies. Antibody production is an energetically expensive process, and there is no reason to expend energy to produce antibodies that are not needed. Instead, **naïve** B-cells (those that have not yet been exposed to an antigen) wait in the

lymph nodes for their particular antigen to come along. Upon exposure to the correct antigen, a B-cell will proliferate and produce two types of daughter cells. **Plasma cells** produce large amounts of antibodies, whereas **memory B-cells** stay in the lymph node, awaiting reexposure to the same antigen. This initial activation takes approximately seven to ten days and is known as the **primary response**. The plasma cells will eventually die, but the memory cells may last the lifetime of the organism. If the same microbe is ever encountered again, the memory cells jump into action and produce the antibodies specific to that pathogen. This immune response, called the **secondary response**, will be more rapid and robust. The development of these lasting memory cells is the basis of the efficacy of **vaccination**.

Cytotoxic Immunity

Whereas humoral immunity is based on the activity of B-cells, cell-mediated immunity involves the T-cells. T-cells mature in the thymus, where they undergo both positive and negative selection. **Positive selection** refers to allowing only the maturation of cells that can respond to the presentation of antigen on MHC (cells that cannot respond to MHC undergo apoptosis because they will not be able to respond in the periphery). **Negative selection** refers to causing apoptosis in cells that are self-reactive (activated by proteins produced by the organism itself). The maturation of T-cells is facilitated by **thymosin**, a peptide hormone secreted by thymic cells. Once the T-cell has left the thymus, it is mature but naïve. Upon exposure to antigen, T-cells will also undergo clonal selection so that only those with the highest affinity for a given antigen proliferate.

There are three major types of T-cells: helper T-cells, suppressor T-cells, and killer (cytotoxic) T-cells. **Helper T-cells** (T_H), also called **CD4⁺ T-cells**,

coordinate the immune response by secreting chemicals known as **lymphokines**. These molecules are capable of recruiting other immune cells (such as plasma cells, cytotoxic T-cells, and macrophages) and increasing their activity. The loss of these cells, as occurs in **human immunodeficiency virus (HIV)** infection, prevents the immune system from mounting an adequate response to infection; in advanced HIV infection, also called **acquired immunodeficiency syndrome (AIDS)**, even weak pathogens can cause devastating consequences as opportunistic infections. CD4⁺ T-cells respond to antigens presented on MHC-II molecules. Because MHC-II presents exogenous antigens, CD4⁺ T-cells are most effective against bacterial, fungal, and parasitic infections.



REAL WORLD

“CD” in immunology stands for *cluster of differentiation* and includes cell-surface markers that can be detected by the lab technique called flow cytometry; these markers give an indication of the types of leukocytes under investigation, how many are present, and their state of maturity.

Cytotoxic T-cells (T_c or **CTL**, for cytotoxic T-lymphocytes), also called **CD8⁺ T-cells**, are capable of directly killing virally infected cells by injecting toxic chemicals that promote apoptosis into the infected cell. CD8⁺ T-cells respond to antigens presented on MHC-I molecules. Because MHC-I presents endogenous antigens, CD8⁺ T-cells are most effective against viral (and intracellular bacterial or fungal) infections.



KEY CONCEPT

CD4⁺ T-cells are better at fighting extracellular infections, while CD8⁺ T-cells are better at targeting intracellular infections.

Suppressor or **regulatory T-cells** (T_{reg}) also express CD4, but can be differentiated from helper T-cells because they also express a protein called *Foxp3*. These cells help to tone down the immune response once infection has been adequately contained. These cells also turn off self-reactive lymphocytes to prevent autoimmune diseases: this is termed **self-tolerance**.



MNEMONIC

CD × MHC = 8

CD4⁺ cells respond to MHC-II ($4 \times 2 = 8$)

CD8⁺ cells respond to MHC-I ($8 \times 1 = 8$)

Finally, **memory T-cells** can be generated. Similar to memory B-cells, these cells lie in wait until the next exposure to the same antigen. When activated, they carry out a more robust and rapid response.



REAL WORLD

Many suppressor T-cells were formerly self-reactive T-cells that have been turned off. When a suppressor T-cell inactivates another lymphocyte, it can either target it for destruction or promote its conversion into another suppressor T-cell.

A summary of the different types of lymphocytes in adaptive (specific) immunity is shown in Figure 8.7.

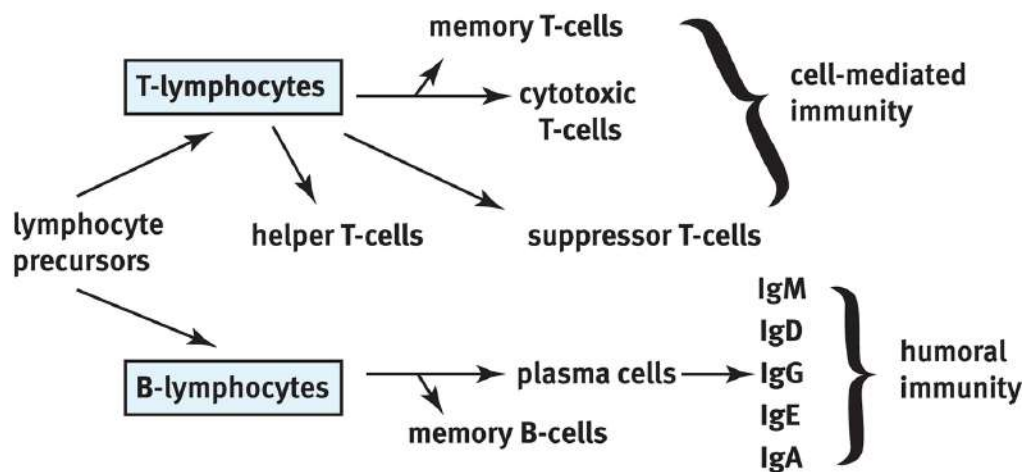


Figure 8.7. Lymphocytes of Specific Immunity

This diagram shows the differentiation of lymphocyte precursors and the cell types involved in specific immunity.

ACTIVATION OF THE ADAPTIVE IMMUNE SYSTEM

It is important to note that the innate and adaptive immune systems are not really disparate entities that function separately. The proper functioning of the entire immune system depends on the interactions between these two systems. There are five types of infectious pathogens: bacteria, viruses, fungi, parasites (including protozoa, worms, and insects), and prions (for which

there are no immune defenses). While the immune system's response depends on the specific identity of the pathogen, we present two classic examples: a bacterial (extracellular pathogen) infection and a viral (intracellular pathogen) infection. Keep in mind that this categorization is imperfect; for example, some bacteria, like *Mycobacterium tuberculosis* and *Listeria monocytogenes*, are actually intracellular pathogens.

Bacterial (Extracellular Pathogen) Infections

Macrophages are like the sentinels of the human body, always on the lookout for potential invaders. Let's say a person suffers a laceration and bacteria are introduced into the body via this laceration. First, macrophages (and other antigen-presenting cells) engulf the bacteria and subsequently release inflammatory mediators. These cells also digest the bacteria and present antigens from the pathogen on their surfaces in conjunction with MHC-II. The cytokines attract inflammatory cells, including neutrophils and additional macrophages. Mast cells are activated by the inflammation and degranulate, resulting in histamine release and increased leakiness of the capillaries. This augments the ability of the immune cells to leave the bloodstream to travel to the affected tissue. The dendritic cell then leaves the affected tissue and travels to the nearest lymph node, where it presents the antigen to B-cells. B-cells that produce the correct antibody proliferate through clonal selection to create plasma cells and memory cells. Antibodies then travel through the bloodstream to the affected tissue, where they tag the bacteria for destruction.

At the same time, dendritic cells are also presenting the antigen to T-cells, activating a T-cell response. In particular, $CD4^+$ T-cells are activated. These cells come in two types, called T_h1 and T_h2 . T_h1 cells release interferon

gamma (IFN- γ), which activates macrophages and increases their ability to kill bacteria. T_H2 cells help activate B-cells and are more common in parasitic infections.

After the pathogen has been eliminated, plasma cells die, but memory B- and T-cells remain. These memory cells allow for a much faster secondary response upon exposure to the pathogen at a later time.

Viral (Intracellular Pathogen) Infections

In a viral-infection, the virally infected cell will begin to produce interferons, which reduce the permeability of nearby cells (decreasing the ability of the virus to infect these cells), reduce the rate of transcription and translation in these cells (decreasing the ability of the virus to multiply), and cause systemic symptoms (malaise, muscle aching, fever, and so on). These infected cells also present intracellular proteins on their surface in conjunction with MHC-I; in a virally infected cell, at least some of these intracellular proteins will be viral proteins.

CD8⁺ T-cells will recognize the MHC-I and antigen complex as foreign and will inject toxins into the cell to promote apoptosis. In this way, the infection can be shut down before it is able to spread to nearby cells. In the event that the virus downregulates the production and presentation of MHC-I molecules, natural killer cells will recognize the absence of MHC-I and will accordingly cause apoptosis of these cells.

Again, once the pathogen has been cleared, memory T-cells will be generated that can allow a much faster response to be mounted upon a second exposure.

RECOGNITION OF SELF AND NONSELF

Self-antigens are the proteins and carbohydrates present on the surface of every cell of the body. Under normal circumstances, these self-antigens signal to immune cells that the cell is not foreign and should not be attacked.

However, when the immune system fails to make the distinction between *self* and *foreign*, it may attack cells expressing particular self-antigens, a condition known as **autoimmunity**. Note that autoimmunity is only one potential problem with immune functioning: another problem arises when the immune system misidentifies a foreign antigen as dangerous when, in fact, it is not. Pet dander, pollen, and peanuts are not inherently threatening to human life, yet some people's immune systems are hypersensitive to these antigens and become overactivated when these antigens are encountered. This is called an allergic reaction. Allergies and autoimmunity are part of a family of immune reactions classified as **hypersensitivity reactions**.

The human body strives to prevent autoimmune reactions very early in T-cell and B-cell maturation processes. T-cells are educated in the thymus. Part of this education involves the elimination of T-cells that respond to self-antigens, called negative selection. Immature B-cells that respond to self-antigens are eliminated before they leave the bone marrow. However, this process is not perfect, and occasionally a cell that responds to self-antigens is allowed to survive. Most autoimmune diseases can be treated with a number of therapies; one common example is administration of **glucocorticoids** (modified versions of cortisol), which have potent immunosuppressive qualities.



REAL WORLD

Autoimmune diseases can result in destruction of tissues, which may cause various deficiencies. Type I diabetes mellitus results from autoimmune destruction of the β -cells of the pancreas. This results in an inability to produce insulin, characterized by high blood sugars and excessive utilization of fats and proteins for energy. Other examples of autoimmune diseases include multiple sclerosis, myasthenia gravis, psoriasis, systemic lupus erythematosus, rheumatoid arthritis, Graves' disease, and Guillain–Barré syndrome.

IMMUNIZATION

Often, diseases can have significant, long-term consequences. Infection with the poliovirus, for example, can leave a person disabled for the remainder of his or her life. Polio used to be a widespread illness; however, today we hardly hear about it outside of the Indian subcontinent because of a highly effective vaccination program that led to the elimination of polio from the Western hemisphere.

Immunization can be achieved in an active or passive fashion. In **active immunity**, the immune system is stimulated to produce antibodies against a specific pathogen. The means by which we are exposed to this pathogen may either be natural or artificial. Through natural exposure, antibodies are generated by B-cells once an individual becomes infected. Artificial exposure (through vaccines) also results in the production of antibodies; however, the individual never experiences true infection. Instead, he or she receives an injection or intranasal spray containing an antigen that will activate B-cells to produce antibodies to fight the specific infection. The antigen may be a

weakened or killed form of the microbe, or it may be a part of the microbe's protein structure.

Immunization may also be achieved passively. **Passive immunity** results from the transfer of antibodies to an individual. The immunity is transient because only the antibodies, and not the plasma cells that produce them, are given to the individual. Natural examples are the transfer of antibodies across the placenta during pregnancy to protect the fetus and the transfer of antibodies from a mother to her nursing infant through breast milk. In some cases of exposure, such as to the rabies virus or tetanus, intravenous immunoglobulin may be given to prevent the pathogen from spreading.



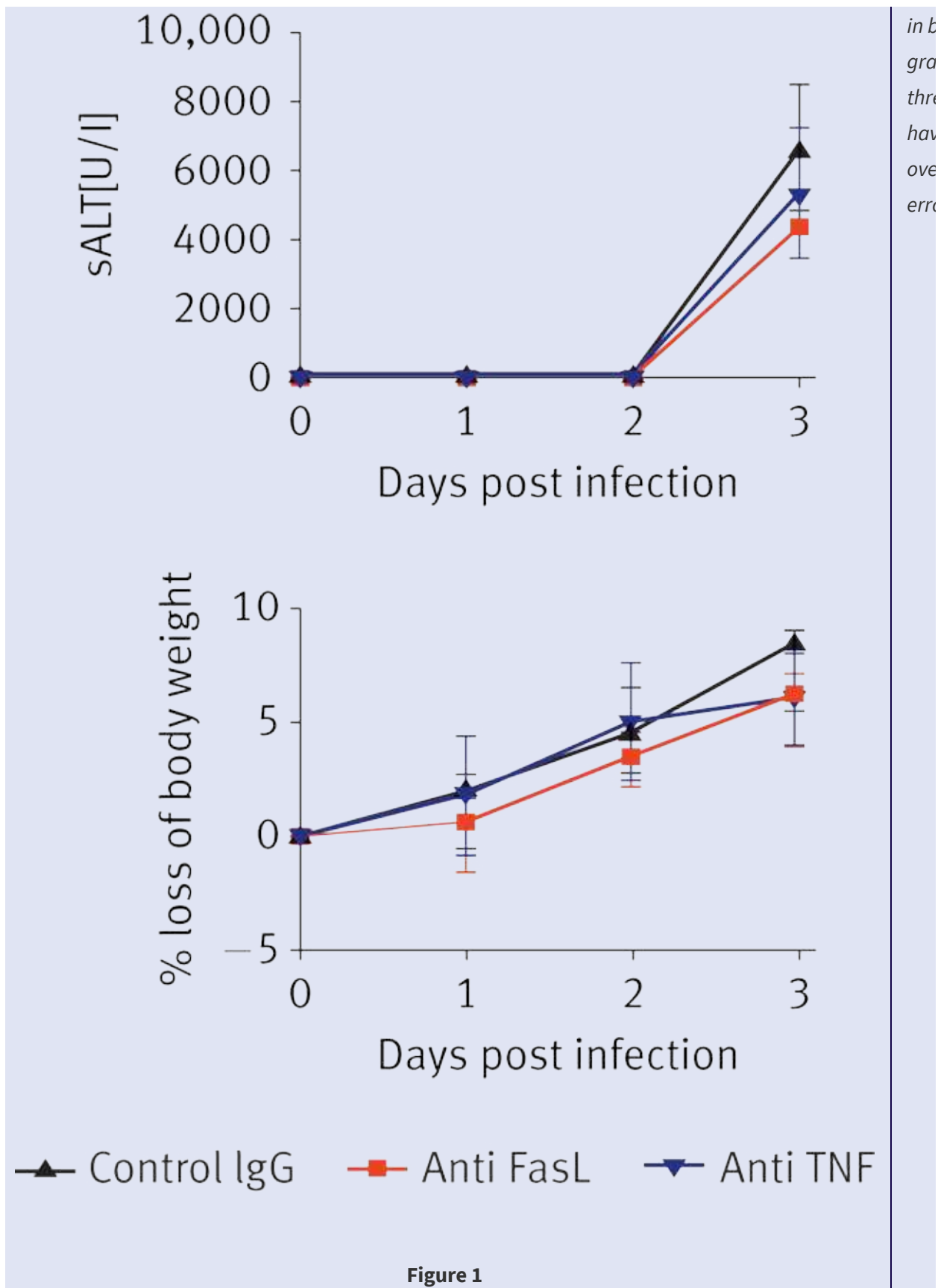
REAL WORLD

In 1998, a paper published in *The Lancet* claimed to have found a link between vaccines and autism. This paper was withdrawn from *The Lancet* after it was demonstrated to be fraudulent and scientifically inaccurate. In fact, no well-designed scientific study has yet shown this link to exist. However, the sensationalist reporting of this connection in the lay media has led many parents to avoid immunizing their children. Since 1998, outbreaks of measles and mumps in the United States and other industrialized nations have raised concerns about the resurgence of illnesses that were previously thought to have been eliminated from these countries. Vaccines do carry risks, including rare cases of encephalitis (brain inflammation) and Guillain–Barré syndrome (an autoimmune disease in which the myelin of peripheral nerves is attacked), but so too do the pathogens these vaccines protect against.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

[New Paragraph]The development of immune-mediated liver failure during viral hepatitis has recently been linked to the action of CD8 ⁺ T-cells.	Top T-c hep
Several effector mechanisms have been identified in preclinical models, including the cytokines interferon (IFN) and tumor necrosis factor (TNF), as well as apoptosis-inducing molecules such as Fas Ligand (FasL) and perforin-1 (Prf1).	The con of t imr sys ma imr me live
To better study these factors, a mouse model of acute hepatitis was developed by immunizing mice with OVA epitope (a short peptide sequence derived from the protein ovalbumin) and then infected with recombinant adenovirus expressing OVA peptide 30 days later.	Nev to s imr me live dar dur hep
Acute liver damage was observed, marked by an increase in serum alanine aminotransferase (sALT), a marker of liver damage, and a decrease in body weight.	Dar live ele sAL dec bod
[New Paragraph]To study the contribution of FasL and TNF to liver pathology, mice were injected with Anti-FasL, anti-TNF, or IgG control antibodies before adenovirus challenge.	Exp Fas

	<i>TNI con ant adc will fac exe the bio fun</i>
Researchers measured sALT levels and body weight in the days post challenge (Figure 1).	
To better elucidate the role of perforin-1, OVA-specific T-cells (OT-1 cells) with either WT perforin or perforin-1 KO (OT-1 x Prf1-/- cells), were injected into naive mice prior to adenovirus challenge.	<i>Exp role per adc wit wit per me sAL boc</i>
Serum ALT and weight measurements were taken following infection (Figure 2).	
[New Paragraph]	<i>Fig ant stu IV: c tree DVs leve loss wei Tre</i>



[New Paragraph]

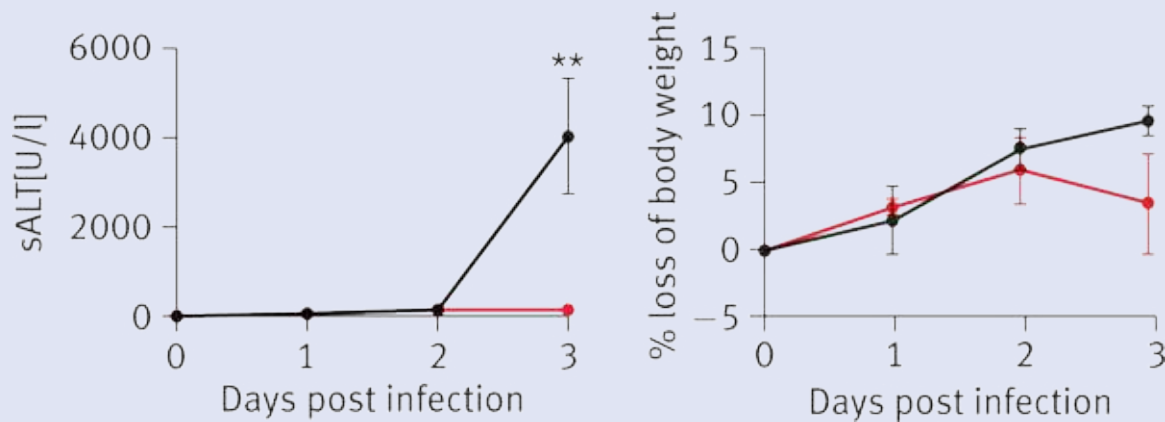


Figure 2

adapted from Welz, M., Eickhoff, S., Abdullah, Z., Trebicka, J., Gartlan, K. H., Spicer, J. A., & Kastenmüller, W. (2018). Perforin inhibition protects from lethal endothelial damage during fulminant viral hepatitis. *Nature Communications*, 9(1), 4805. doi:10.1038/s41467-018-07213-x.

Based on the results, which of the three parameters tested, FAS Ligand, Tumor Necrosis Factor, or Perforin is most likely involved in the destruction of liver tissue during acute infection?

This question is really testing our understanding of the experimental design and comprehension of the results. Let's start with the experimental design. Paragraph 1 tells us that there are four main factors that are believed to be involved in immune-mediated liver damage: FasL, TNF, perforin-1, and INF. To

test the involvement of these factors, the researchers designed a new experimental model. They first injected mice with OVA peptide to induce a primary immune response and the development of memory cells. Thirty days later, the researchers infected the mice with adenovirus expressing OVA. This will result in activation of memory cells and a secondary immune response, which the researchers showed was able to induce immune-controlled liver damage.

In the first experiment, the mice were injected with antibodies against FasL and TNF prior to adenovirus infection. These antibodies will bind to FasL and TNF, rendering them unable to function, similar to knocking out the gene (but significantly cheaper than developing transgenic mice!). Figure 1 shows the results. We notice that there is no significant difference in sALT levels nor body weight loss between mice treated with the control antibody and mice treated with either FasL or TNF antibody. We see the same increase in sALT and the same decrease in weight loss, indicating that removing TNF and FasL function did not prevent liver damage. This suggests that these two factors are not required for the immune-mediated liver damage.

In the second experiment, we test the involvement of perforin-1. The experimental setup is slightly different. Instead of injecting OVA peptide, OVA specific T-cells were injected into mice. Two different types of cells were used, normal and perforin-1 knockout. The mice were then infected with adenovirus as before. This enables us to examine the contribution of perforin-1 in liver damage. Figure 2 shows the results of the experiment. We can note that there is a significant difference in sALT levels between control and perforin-1 knockout (KO) cells. There is no elevation in sALT in the knockout cells, whereas there is an increase in control cells. This suggests that perforin-1 contributes to liver damage. Furthermore, mice given perforin-1 KO cells lose less weight compared to mice given control cells. Because a knockout of the perforin-1

gene led to a decrease in immune-mediated liver damage in mice, perforin-1 plays an important role in immune-mediated liver damage.

Because the results show no significant difference between the control and TNF and FasL antibody treated mice, TNF and FasL do not play a role in immune-mediated liver damage. However, a statistically significant difference between control and perforin-1 knockout cells was observed, indicating a role for perforin-1 in immune-mediated liver damage.

MCAT CONCEPT CHECK 8.3:

Before you move on, assess your understanding of the material with these questions.

1. For each of the lymphocytes listed below, what are its main functions?

- Plasma cell:

- Memory B-cell:

- Helper T-cell:

- Cytotoxic T-cell:

- Suppressor (regulatory) T-cell:

- Memory T-cell:

2. What are the three main effects circulating antibodies can have on a pathogen?

-

-

-

3. How do antibodies become specific for a given antigen?

4. A T-cell appropriately passes through positive selection, but then inappropriately passes through negative selection. What will this T-cell be reactive toward?

5. Which cells account for the fact that the secondary response to a pathogen is much more rapid and robust than the

primary response?

6. What is the difference between active and passive immunity?

- Active immunity:

- Passive immunity:

8.4 The Lymphatic System

LEARNING OBJECTIVES

After Chapter 8.4, you will be able to:

- Predict how a blockage of flow from a lymph node would impact the lymphatic system and the body as a whole
- Describe the factors and structures linking lymphatic and cardiovascular circulation

The immune system and the lymphatic system are intimately related. B-cells proliferate and develop within the lymphatic system, especially the lymph nodes. This system also serves other functions for the body.

STRUCTURE

The lymphatic system, along with the cardiovascular system, is a type of circulatory system. It is made up of one-way vessels that become larger as they move toward the center of the body. These vessels carry lymphatic fluid (**lymph**) and most join to form a large **thoracic duct** in the posterior chest, which then delivers the fluid into the left subclavian vein (near the heart).

Lymph nodes are small, bean-shaped structures along the lymphatic vessels. Lymph nodes contain a lymphatic channel, as well as an artery and a vein. The lymph nodes provide a space for the cells of the immune system to be exposed to possible pathogens.



REAL WORLD

Certain cancers, especially breast cancer, are prone to spread via lymphatic channels. Mastectomy, a surgery that removes the breast, is one of the treatments for breast cancer. In order to ensure that all of the cancer has been removed, local lymph nodes are often removed at the same time.

FUNCTION

The lymphatic system serves many different purposes for the body by providing a secondary system for circulation.

Equalization of Fluid Distribution

At the capillaries, fluid leaves the bloodstream and goes into the tissues. The quantity of fluid that leaves the tissues at the arterial end of the capillary bed depends on both hydrostatic and oncotic pressures (Starling forces).

Remember that the oncotic pressure of the blood draws water back into the vessel at the venule end, once hydrostatic pressure has decreased. Because the net pressure drawing fluid in at the venule end is slightly less than the net pressure pushing fluid out at the arterial end, a small amount of fluid remains

in the tissues. Lymphatic vessels drain these tissues and subsequently return the fluid to the bloodstream.

The lymphatics offer some protection against pathology. For example, if the blood has a low concentration of albumin (a key plasma protein), the oncotic pressure of the blood is decreased, and less water is driven back into the bloodstream at the venule end. Thus, this fluid will collect in the tissues. Provided that the lymphatic channels are not blocked, much of this fluid may eventually return to the bloodstream via the lymphatics. Only when the lymphatics are overwhelmed does **edema** occur—swelling due to fluid collecting in tissue.

Transportation of Biomolecules

The lymphatic system also transports fats from the digestive system into the bloodstream. **Lacteals**, small lymphatic vessels, are located at the center of each villus in the small intestine. Fats, packaged into chylomicrons by intestinal mucosal cells, enter the lacteal for transport. Lymphatic fluid carrying many chylomicrons takes on a milky white appearance and is called **chyle**.

Immunity

As stated previously in this chapter, lymph nodes are a place for antigen-presenting cells and lymphocytes to interact. B-cells proliferate and mature in the lymph nodes in collections called **germinal centers**.

MCAT CONCEPT CHECK 8.4:

Before you move on, assess your understanding of the material with these questions.

1. *Filariasis* is the name for an infection with a member of a certain group of parasites, most notably *Wuchereria bancrofti*. This parasite resides in lymph nodes and causes blockage of flow. If an individual had a *W. bancrofti* infection in the lymph nodes of his or her thigh, what would likely happen?

2. What structure is primarily responsible for returning materials from lymphatic circulation to the cardiovascular system?

Conclusion

The ability to fend off microbial invasion is critical to our survival. The immune system is throughout the body and involves multiple different organs and cell types. Nonspecific mechanisms, such as intact skin, mucous membranes, defensins, lysozyme, complement, interferons, natural killer cells, neutrophils, eosinophils, basophils, and monocytes/macrophages, constitute a complex first line of defense; these mechanisms comprise the innate immune system, which is capable of an immediate response but cannot target a specific pathogen or maintain immunologic memory. The adaptive immune system, comprised of B- and T-cells (lymphocytes), allows our immune system to target specific pathogens and learn from past exposure. Thus, once we are infected with a certain strain of virus, activation of specific immunity confers long-term protection against that particular virus. We take advantage of this secondary response through immunization, and we can see the problems specificity can have when a self-antigen is labeled as foreign, leading to autoimmune disease.

If the immune system is focused on destroying pathogens, including bacteria, then it's an interesting transition we make in the next chapter. We're going from a "sterilization" system to one in which bacterial colonization is the norm. From oral flora to the normal gut bacteria, our ability to digest and absorb nutrients is intimately linked to symbiotic bacteria throughout the digestive tract. In the next chapter, we will explore the anatomy and physiology of the digestive system, which provides us with the raw materials to generate energy, build proteins, and carry out activities of daily living.

Concept Summary

Structure of the Immune System

- The immune system can be divided into innate and adaptive immunity.
 - **Innate immunity** is composed of defenses that are always active, but that cannot target a specific invader and cannot maintain immunologic memory; also called **nonspecific immunity**.
 - **Adaptive immunity** is composed of defenses that take time to activate, but that target a specific invader and can maintain immunologic memory; also called **specific immunity**.
- The immune system is dispersed in the body.
 - Immune cells come from the **bone marrow**.
 - The **spleen** and **lymph nodes** are sites where immune responses can be mounted, and in which B-cells are activated.
 - The **thymus** is the site of T-cell maturation.
 - **Gut-associated lymphoid tissue (GALT)** includes the **tonsils** and **adenoids**.
- **Leukocytes**, or white blood cells, are involved in immune defenses.

The Innate Immune System

- Many of the nonspecific defenses are noncellular.
 - The **skin** acts as a physical barrier and secretes antimicrobial compounds, like **defensins**.
 - **Mucus** on mucous membranes traps pathogens; in the respiratory system, the mucus is propelled upward by cilia and can be swallowed or expelled.
 - Tears and saliva contain **lysozyme**, an antibacterial compound.
 - The stomach produces acid, killing most pathogens. Colonization of the gut helps prevent overgrowth by pathogenic bacteria through competition.
 - The **complement** system can punch holes in the cell walls of bacteria, making them osmotically unstable.
 - **Interferons** are given off by virally infected cells and help prevent viral replication and dispersion to nearby cells.

- Many of the nonspecific defenses are also cellular.
 - **Macrophages** ingest pathogens and present them on **major histocompatibility complex (MHC)** molecules. They also secrete **cytokines**.
 - **MHC class I (MHC-I)** is present in all nucleated cells and displays **endogenous antigen** (proteins from within the cell) to cytotoxic T-cells (CD8⁺ cells).
 - **MHC class II (MHC-II)** is present in professional antigen-presenting cells (macrophages, dendritic cells, some B-cells, and certain activated epithelial cells) and displays **exogenous antigen** (proteins from outside the cell) to helper T-cells (CD4⁺ cells).
 - **Dendritic cells** are antigen-presenting cells in the skin.
 - **Natural killer cells** attack cells not presenting MHC molecules,

including virally infected cells and cancer cells.

- **Granulocytes** include neutrophils, eosinophils, and basophils.
- **Neutrophils** ingest bacteria, particularly opsonized bacteria (those marked with antibodies). They can follow bacteria using **chemotaxis**.
- **Eosinophils** are used in allergic reactions and invasive parasitic infections. They release **histamine**, causing an inflammatory response.
- **Basophils** are used in allergic reactions. **Mast cells** are related cells found in the skin.

The Adaptive Immune System

- **Humoral immunity** is centered on antibody production by plasma cells, which are activated **B-cells**.
 - **Antibodies** target a particular **antigen**. They contain two heavy chains and two light chains. They have a **constant region** and a **variable region**; the tip of the variable region is the **antigen-binding region**.
 - When activated, the antigen-binding region undergoes **hypermutation** to improve the specificity of the antibody produced. Cells may be given signals to switch **isotypes** of antibody (IgM, IgD, IgG, IgE, IgA).
 - Circulating antibodies can **opsonize** pathogens (mark them for destruction), cause **agglutination** (clumping) into insoluble complexes that are ingested by phagocytes, or neutralize pathogens.
 - Cell-surface antibodies can activate immune cells or mediate allergic reactions.
 - **Memory B-cells** lie in wait for a second exposure to a pathogen and can then mount a more rapid and vigorous immune response (**secondary response**).

- **Cell-mediated (cytotoxic) immunity** is centered on the functions of **T-cells**.
 - T-cells undergo maturation in the thymus through **positive selection** (only selecting for T-cells that can react to antigen presented on MHC) and **negative selection** (causing apoptosis in self-reactive T-cells). The peptide hormone **thymosin** promotes T-cell development.
 - **Helper T-cells** (T_h or $CD4^+$) respond to antigen on MHC-II and coordinate the rest of the immune system, secreting **lymphokines** to activate various arms of immune defense. **T_h1 cells** secrete **interferon gamma**, which activates macrophages. **T_h2 cells** activate B-cells, primarily in parasitic infections.
 - **Cytotoxic T-cells** (T_c , **CTL**, or $CD8^+$) respond to antigen on MHC-I and kill virally infected cells.
 - **Suppressor (regulatory) T-cells** (T_{reg}) tone down the immune response after an infection and promote self-tolerance.
 - **Memory T-cells** serve a similar function to memory B-cells.
- In **autoimmune** conditions, a self-antigen is identified as foreign, and the immune system attacks the body's own cells.
- In **allergic** reactions, nonthreatening exposures incite an inflammatory response.
- Immunization is a method of inducing **active immunity** (activation of B-cells that produce antibodies to an antigen) prior to exposure to a particular pathogen.
- **Passive immunity** is the transfer of antibodies to an individual.

The Lymphatic System

- The **lymphatic system** is a circulatory system that consists of one-way vessels with intermittent lymph nodes.
- The lymphatic system connects to the cardiovascular system via the **thoracic duct** in the posterior chest.
- The lymphatic system equalizes fluid distribution, transports fats and fat-soluble compounds in **chylomicrons**, and provides sites for mounting immune responses.

Answers to Concept Checks

8.1

- .. Innate immunity consists of defenses that are always active against pathogens, but that are not capable of targeting specific invaders. It takes longer to mount a response with adaptive immunity, but the response targets a specific pathogen and maintains immunologic memory of the infection to mount a faster response during subsequent infections.

1.

Cell Type	Site of Development	Site of Maturation	Major Functions	Specific or Nonspecific?	Humoral or Cell-Mediated?
B-cell	Bone marrow	Bone marrow (but are activated in spleen or lymph nodes)	Produce antibodies	Specific	Humoral
T-cell	Bone marrow	Thymus	Coordinate immune system and directly kill infected cells	Specific	Cell-mediated

- }. Granulocytes include neutrophils, eosinophils, and basophils. Agranulocytes include B- and T-cells (lymphocytes) and monocytes (macrophages).

8.2

- .. Skin provides a physical barrier and secretes antimicrobial enzymes.

Defensins are examples of antibacterial enzymes on the skin. Lysozyme is antimicrobial and is present in tears and saliva. Mucus is present on mucous membranes and traps incoming pathogens; in the respiratory system, cilia propel the mucus upward so it can be swallowed or expelled. Stomach acid is an antimicrobial substance in the digestive system. The normal gastrointestinal flora provides competition, making it hard for pathogenic bacteria to grow in the gut. Complement is a set of proteins in the blood that can create holes in bacteria.

1. Professional antigen-presenting cells include macrophages, dendritic cells in the skin, some B-cells, and certain activated epithelial cells.
2. MHC-I is found in all nucleated cells and presents pieces of proteins (peptides) created within the cell (endogenous antigens); this can allow for detection of cells infected with intracellular pathogens (especially viruses). MHC-II is only found in antigen-presenting cells and presents proteins that result from the digestion of extracellular pathogens that have been brought in by endocytosis (exogenous antigens).
3. Natural killer cells are activated by cells that do not present MHC (such as virally infected cells and cancer cells). Neutrophils are activated by bacteria, especially those that have been opsonized (tagged with an antibody on their surface). Eosinophils are activated by invasive parasites and allergens. Basophils and mast cells are activated by allergens.

8.3

1. Plasma cells form from B-cells exposed to antigen and produce antibodies. Memory B-cells also form from B-cells exposed to antigen and lie in wait for a second exposure to a given antigen to mount a rapid, robust response. Helper T-cells coordinate the immune system through lymphokines and respond to antigen bound to MHC-II. Cytotoxic T-cells directly kill virally infected cells and respond to antigen bound to MHC-I. Suppressor (regulatory) T-cells quell the immune response after a pathogen has been

cleared and promote self-tolerance. Memory T-cells, like memory B-cells, lie in wait until a second exposure to a pathogen to mount a rapid, robust response.

- 1. Circulating antibodies can mark a pathogen for destruction by phagocytic cells (opsonization), cause agglutination of the pathogen into insoluble complexes that can be taken up by phagocytic cells, or neutralize the pathogen by preventing it from invading tissues.
- 2. B-cells originally mature in the bone marrow and have some specificity at that point; however, antibodies that can respond to a given antigen undergo hypermutation, or rapid mutation of their antigen-binding sites. Only those B-cells that have the highest affinity for the antigen survive and proliferate, increasing the specificity for the antigen over time.
- 3. Positive selection occurs when T-cells in the thymus that are able to respond to antigen presented on MHC are allowed to survive (those that do not respond undergo apoptosis). Negative selection occurs when T-cells that respond to self-antigens undergo apoptosis before leaving the thymus. A T-cell that appropriately passes through positive selection, but then inappropriately passes through negative selection will be reactive to self-antigens.
- 4. Memory cells allow the immune system to carry out a much more rapid and robust secondary response.
- 5. Active immunity refers to the stimulation of the immune system to produce antibodies against a pathogen. Passive immunity refers to the transfer of antibodies to prevent infection, without stimulation of the plasma cells that produce these antibodies.

8.4

- .. Fluid would be unable to return from the lower leg, and edema would result. This infection leads to *elephantiasis*, severe swelling of the limb with thickening of the skin.

!. The thoracic duct carries lymphatic fluid into the left subclavian vein.



Shared Concepts

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biology Chapter 1

The Cell

Biology Chapter 6

The Respiratory System

Biology Chapter 7

The Cardiovascular System

Biology Chapter 9

The Digestive System

Biology Chapter 10

Homeostasis

Discrete Practice Questions

1. In DiGeorge syndrome, the thymus can be completely absent. The absence of the thymus would leave an individual unable to mount specific defenses against which of the following types of pathogens?
 - (A) Viruses
 - (B) Bacteria
 - (C) Parasites
 - (D) Fungi

2. Which of the following are NOT involved in cell-mediated immunity?
 - (A) Memory cells
 - (B) Plasma cells
 - (C) Cytotoxic cells
 - (D) Suppressor cells

3. The lymphatic system:
 - (A) transports hormones throughout the body.
 - (B) transports chylomicrons to the circulatory system.
 - (C) causes extravasation of fluid into tissues.
 - (D) is the site of mast cell activation.

4. Which of the following are involved in antibody production?
- (A) Plasma cells
 - (B) Memory cells
 - (C) Helper T-cells
 - (D) Cytotoxic cells
5. Which of the following is NOT an example of a nonspecific defense mechanism?
- (A) Skin provides a physical barrier against invasion.
 - (B) Macrophages engulf and destroy foreign particles.
 - (C) An inflammatory response is initiated in response to physical damage.
 - (D) Cytotoxic T-cells destroy cells displaying foreign antigens.
6. Which of the following is an example of adaptive immunity?
- (A) PRRs recognize that a pathogen is an invasive parasite and eosinophils are recruited to the area.
 - (B) Complement is activated, causing osmotic instability in a bacterium.
 - (C) Memory B-cells generated through vaccination are activated when their antigen is encountered.
 - (D) Dendritic cells sample bacteria within a laceration and travel to the lymph nodes to present the antigen.

7. Which of the following is true regarding passive and active immunity?

- (A) Active immunity requires weeks to build, whereas passive immunity is acquired immediately.
- (B) Active immunity is short-lived, whereas passive immunity is long-lived.
- (C) Active immunity may be acquired during pregnancy through the placenta.
- (D) Passive immunity may be acquired through vaccination.

8. Where are most self-reactive T-cells eliminated?

- (A) Spleen
- (B) Lymph nodes
- (C) Bone marrow
- (D) Thymus

9. What is the response of the immune system to downregulation of MHC molecules on somatic cells?

- (A) B-cells are activated and antibodies are released.
- (B) T-cells are activated, resulting in a cytotoxic response.
- (C) Natural killer cells induce apoptosis of affected cells.
- (D) Macrophages engulf the pathogen and display its antigens.

10. Which of the following correctly indicates the response of CD8⁺ T-cells when activated?

- (A) Secretion of cytotoxic chemicals
- (B) Causing isotype switching
- (C) Presentation of antigens
- (D) Activation of B-cells

11. Lymphoma is cancer of the cells of lymphoid lineage. These cells often reside within lymph nodes. What type of cell is NOT likely to cause a lymphoma?

- (A) CD8⁺ T-cells
- (B) B-cells
- (C) Macrophages
- (D) T_h1 cells

12. Upon encountering an antigen, only T-cells with a specific T-cell receptor are activated. This is an example of:

- (A) innate immunity.
- (B) a cytotoxic T-cell response.
- (C) humoral immunity.
- (D) clonal selection.

13. Which cell type is a phagocyte that attacks bacterial pathogens in the bloodstream?

- (A) Neutrophils
- (B) Eosinophils
- (C) Basophils
- (D) Dendritic cells

14. What type of immunity is likely to be affected by removal of the spleen?

- (A) Cytotoxic immunity
- (B) Humoral immunity
- (C) Innate immunity
- (D) Passive immunity

15. Which of the following cell types is NOT a granulocyte?

- (A) Natural killer cell
- (B) Mast cell
- (C) Eosinophil
- (D) Basophil

Discrete Practice Answers

1. **A**

T-lymphocytes, which mature in the thymus, are the only specific defense against intracellular pathogens. While some bacteria, fungi, and parasites can live intracellularly, viruses—by definition—must replicate within cells. The absence of T-cells would leave an individual unable to fight viral infections with specific defenses.

2. **B**

The lymphocytes involved in cell-mediated immunity are the T-lymphocytes, or T-cells. There are four types of T-cells, each playing a different role in cell-mediated immunity: cytotoxic T-cells, helper T-cells, memory T-cells, and suppressor T-cells. Thus, from the answer choices, the only cells not involved in cell-mediated immunity are the plasma cells, which are differentiated immunoglobulin-secreting B-lymphocytes involved in humoral immunity. **(B)** is therefore the correct answer.

3. **B**

The main function of the lymphatic system is to collect excess interstitial fluid and return it to the circulatory system, maintaining the balance of body fluids. However, this is not one of the answer choices. In addition, the lymphatic system accepts chylomicrons from the small intestine and delivers them to the cardiovascular circulation. Transport of hormones is a function of the cardiovascular system, eliminating **(A)**. The lymphatic system absorbs fluid that has been pushed into tissues, but does not cause

the extravasation of the fluid, eliminating **(C)**. Mast cells reside in (and are activated in) the skin and mucous membranes, eliminating **(D)**.

4. **A**

Antibodies are produced by plasma cells derived from B-lymphocytes. The other cells are all types of T-lymphocytes, although memory B-cells can also exist. Still, memory B-cells do not produce antibodies; plasma cells do.

5. **D**

The body employs a number of nonspecific defense mechanisms against foreign invasion. The skin and mucous membranes provide a physical barrier against bacterial invasion. In addition, sweat contains enzymes that attack bacterial cell walls. Certain passages, such as the respiratory tract, are lined with ciliated mucus-coated epithelia, which filter and trap foreign particles. Macrophages engulf and destroy foreign particles. The inflammatory response is initiated in response to physical damage. The only choice that is not a nonspecific defense mechanism is **(D)**, the correct answer. Cytotoxic T-cells are involved in (specific) cell-mediated immunity.

6. **C**

Adaptive immunity involves the activation of B-cells and T-cells specific to the encountered antigen. Any choice that conforms to this paradigm will be correct. **(C)** indicates that B-cells are activated. Pattern recognition receptors, or PRRs, in **(A)** recognize patterns common to certain pathogens, but do not identify the specific pathogen. Complement is an example of a blood-borne nonspecific defense against bacteria, eliminating **(B)**. Dendritic cells traveling to the lymph nodes **(D)** are a part of the interaction between the innate and adaptive immune systems, but the dendritic cells themselves are nonspecific.

7. **A**

Active immunity refers to the production of antibodies during an immune response. Active immunity may be conferred on an individual by vaccination, such as when an individual is injected with a weakened, inactive, or modified form of a particular antigen that stimulates the immune system to produce antibodies. Active immunity may require weeks to build. Passive immunity, on the other hand, involves the transfer of antibodies through, for example, breast milk. Another example of passive immunity would be during pregnancy, when some maternal antibodies cross the placenta and enter fetal circulation, conferring passive immunity to the fetus. Although passive immunity is acquired immediately, it is very short-lived, lasting only as long as the antibodies circulate in blood.

8. **D**

T-cells mature in the thymus, where they are “educated.” This education involves the elimination of T-cells with improper binding to MHC–antigen complexes (positive selection) and self-reactive T-cells (negative selection). Thus, self-reactive T-cells are eliminated in the thymus.

9. **C**

Healthy cells exhibit MHC class I molecules. Natural killer cells monitor the expression of MHC molecules on the surface of cells. Viral infection and cancer often cause a reduction in the expression of MHC class I molecules on the cell surface. Natural killer cells detect this lack of MHC and induce apoptosis in the affected cells.

10. **A**

CD8⁺ T-cells are largely responsible for the cytotoxic immune response. By releasing toxic chemicals into virally infected cells, CD8⁺ T-cells are able to kill these cells in an effort to contain viral infections. Isotype switching

refers to changes in the isotype of antibody produced, which is not caused by CD8⁺ cells, eliminating **(B)**. Antigens are presented by macrophages, dendritic cells, certain epithelial cells, and some B-cells, eliminating **(C)**. B-cells are not activated by cytotoxic T-lymphocytes, eliminating **(D)**.

1. **C**

Lymphocytes arise from the lymphoid lineage, which includes B-cells and T-cells. Thus, all types of B- and T-cells are capable of causing lymphoma. Macrophages, however, are not lymphocytes and are not likely to cause lymphoma.

2. **D**

When the adaptive immune system encounters an antigen, only the cells with antibodies or T-cell receptors specific to that antigen are activated. This is known as clonal selection. While a T-cell response may be a cytotoxic response, it could also be the activation of helper T-cells; plus, this does not explain the specificity of the response, eliminating **(B)**.

3. **A**

The only phagocytes that attack bacteria on this list are neutrophils and dendritic cells. Dendritic cells are able to sample and present any type of material, and reside in the skin. Neutrophils, on the other hand, are present in the bloodstream and can attack bacteria present there or in tissues. Eosinophils and basophils are involved in the development of allergies; eosinophils also defend against parasites.

4. **B**

The spleen is a location where B-cells mature and proliferate. Therefore, removal of the spleen is likely to result in a reduction of humoral immunity. In fact, many people receive vaccinations prior to removal of the spleen in

order to bolster their immunity.

5. **A**

Granulocytes are cells with granules in their cytoplasm. These granules are released in immune responses to attack a pathogen. Granulocytes include neutrophils, basophils, eosinophils, and mast cells. Natural killer cells are agranulocytes, which include lymphocytes (B-, T-, and NK cells) and monocytes/macrophages.



An endoscopic view of the interior of a human colon, showing the mucosal lining and the haustra. The image is used as a background for the chapter title. A solid blue rectangle is positioned in the top-left corner, containing the chapter number.

CHAPTER 9

The Digestive System

CHAPTER 9

The Digestive System

In This Chapter

9.1 Anatomy of the Digestive System

9.2 Ingestion and Digestion

Oral Cavity

Pharynx

Esophagus

Stomach

Duodenum

9.3 Accessory Organs of Digestion

Pancreas

Liver

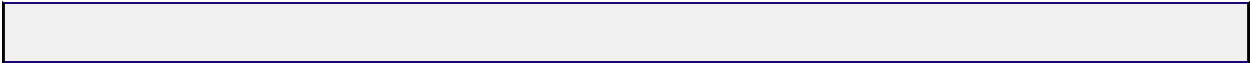
Gallbladder

9.4 Absorption and Defecation

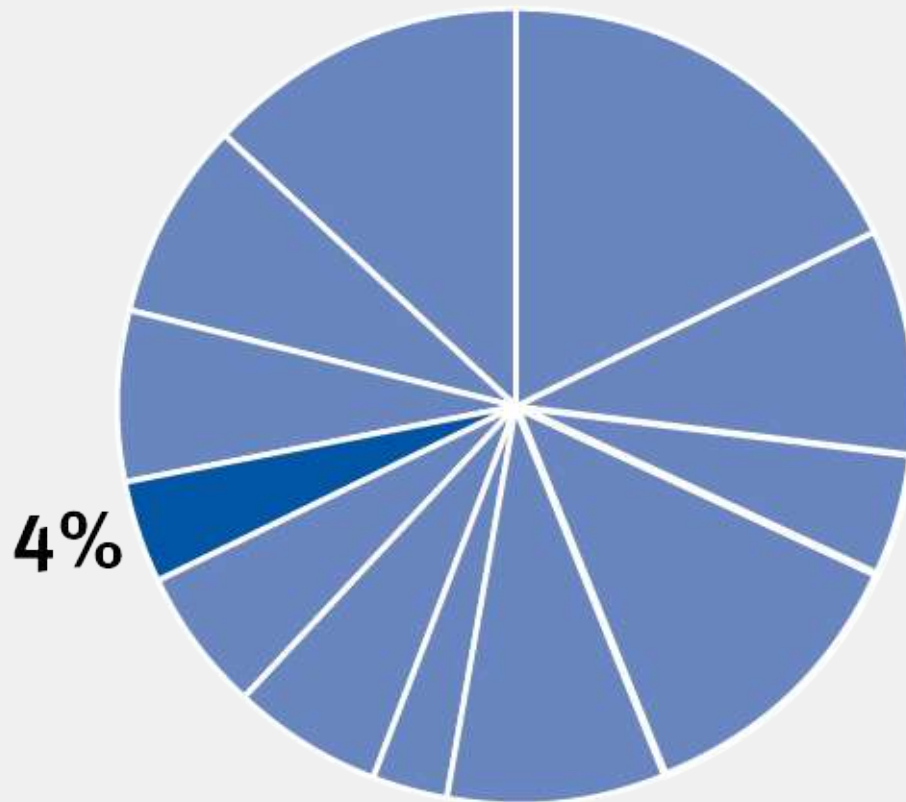
Jejunum and Ileum

Large Intestine

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 4% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems

Introduction

As we continue our survey of organ systems, we now come to the digestive system. As with our previous reviews of other organ systems, we will start with a basic anatomical overview of the organs of digestion (including the accessory organs) and then move on to discuss how these organs function to provide nutrition to the individual. The food we eat is complex, incorporating meats, grains, vegetables and fruits, dairy products, and nuts. The job of the digestive system is to take these complex foods—composed of polysaccharides, fats, and proteins—and turn large macromolecules into smaller, simpler monosaccharides, fatty acids, and amino acids. In order to cleave all of these bonds, the body requires a complex system of mechanical and chemical agents. These compounds can then be absorbed from the gut, transported to the tissues by the circulatory system, and used by cells. In this chapter, we will consider the organs that make up the digestive system as well as the processes by which the foods we eat become the fuel we need for energy, growth, development, and maintenance of other essential activities.

9.1 Anatomy of the Digestive System

LEARNING OBJECTIVES

After Chapter 9.1, you will be able to:

- Describe mechanical and chemical digestion and differentiate between them
- Identify the interactions between the sympathetic and parasympathetic nervous systems and the digestive system
- Trace the path of food through the body, naming major organs and valves in the digestive tract

There are two types of digestion that occur. First, **intracellular digestion**, as a part of **metabolism**, involves the oxidation of glucose and fatty acids for energy. However, our diets do not consist of pure glucose and fatty acids; rather, these substances must be extracted from our food. The process by which these nutrients are obtained from food occurs within the lumen of the **alimentary canal** and is known as **extracellular digestion**. This is technically “outside” the body, because the lumen of the gastrointestinal tract communicates directly with the outside world. The alimentary canal runs from the **mouth** to the **anus** and is sectioned off by **sphincters**, or circular smooth muscles around the canal that can contract to allow compartmentalization of function.

The human digestive tract has specialized sections with different functional roles. The most basic functional distinction is between digestion and absorption. **Digestion** involves the breakdown of food into its constituent organic molecules: starches and other carbohydrates into monosaccharides, lipids (fats) into free fatty acids and glycerol, and proteins into amino acids. Digestion can be subdivided into mechanical and chemical processes.

Mechanical digestion is the physical breakdown of large food particles into smaller food particles, but does not involve breaking chemical bonds.

Chemical digestion is the enzymatic cleavage of chemical bonds, such as the peptide bonds of proteins or the glycosidic bonds of starches. **Absorption** involves the transport of products of digestion from the digestive tract into the circulatory system for distribution to the body's tissues and cells.

The digestive tract, shown in Figure 9.1, begins with the **oral cavity** (mouth) followed by the **pharynx**, a shared pathway for both food entering the digestive system and air entering the respiratory system. From the pharynx, food enters the **esophagus**, which transports it to the **stomach**. From the stomach, food travels to the **small intestine**, and then to the **large intestine**. Finally, waste products of digestion enter the **rectum**, where feces are stored until an appropriate time of release. In addition to the digestive tract itself, the **salivary glands**, **pancreas**, **liver**, and **gallbladder** help to provide the enzymes and lubrication necessary to aid in the digestion of food.

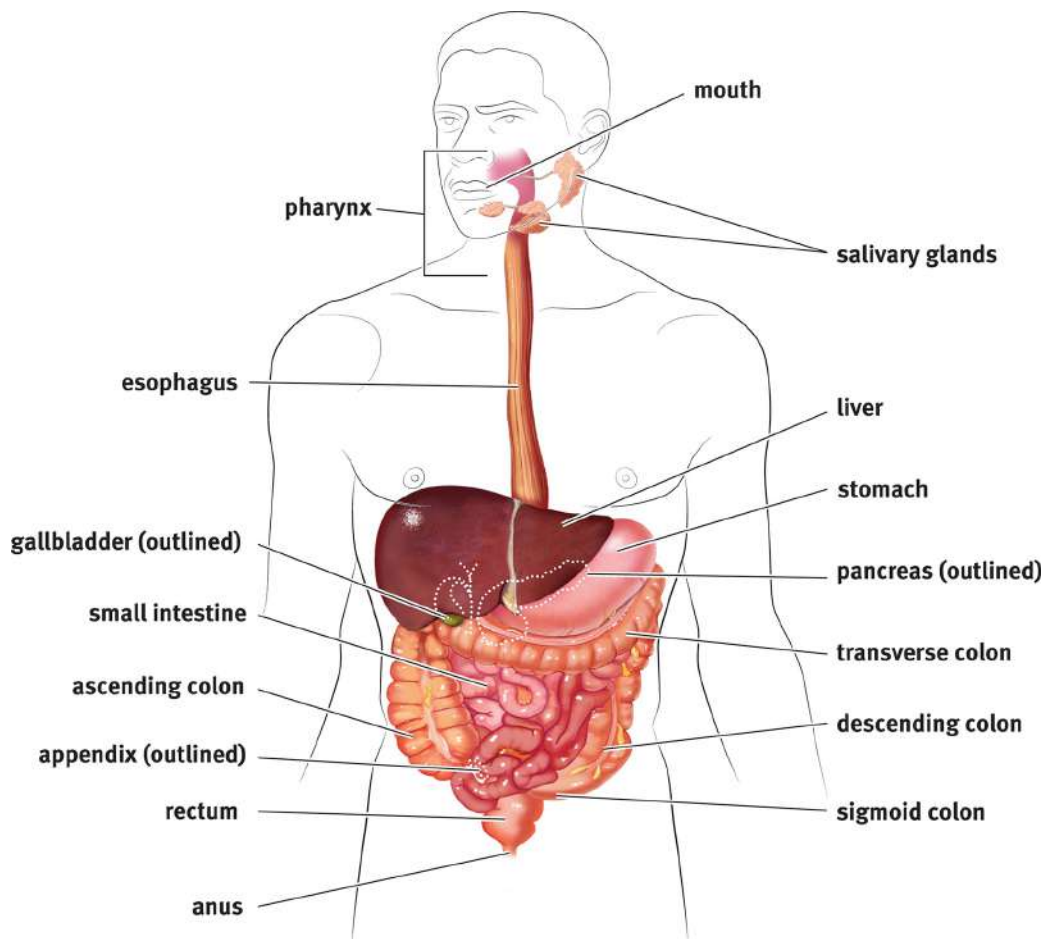


Figure 9.1. Anatomy of the Digestive System

The **enteric nervous system** is a collection of one hundred million neurons that govern the function of the gastrointestinal system. These neurons are present in the walls of the digestive tract and trigger **peristalsis**, or rhythmic contractions of the gut tube, in order to move materials through the system. This system can function independently of the brain and spinal cord, although it is heavily regulated by the autonomic nervous system. The parasympathetic division is involved in stimulation of digestive activities, increasing secretions from exocrine glands and promoting peristalsis. The sympathetic division is involved in the inhibition of these activities. The fact that so often we feel sleepy and lethargic after eating a big meal (often called a *food coma*

colloquially) is due, in part, to parasympathetic activity. On the other hand, during periods of high sympathetic activity, bloodflow is decreased to the digestive tract, and gut motility slows significantly.



KEY CONCEPT

All of the glands of the body (except sweat glands) are innervated by the parasympathetic nervous system.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

Bile acids are crucial for the intestinal absorption of dietary fatty acids, of cholesterol, and of fat-soluble vitamins.

This background information

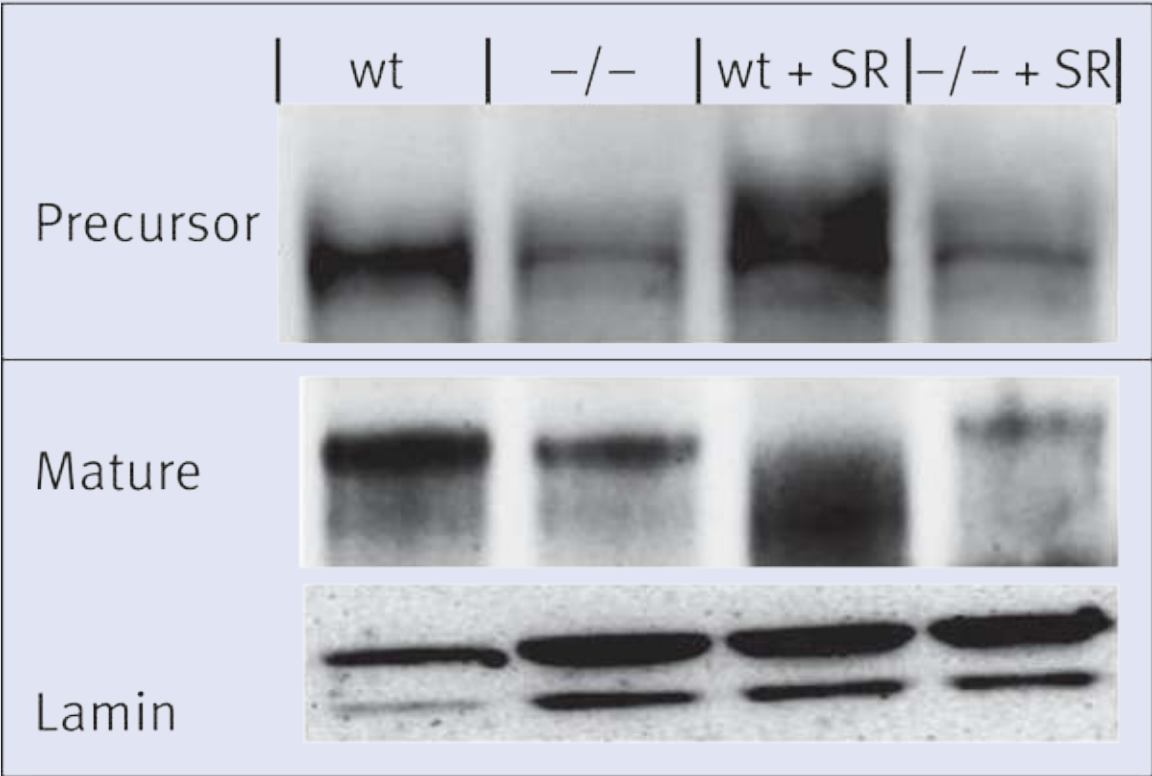
To determine whether the interruption of the enterohepatic circulation of bile acids alters triglyceride and glucose metabolism, researchers tested ileal sodium-dependent bile acid transporter (Slc10a2) KO mice.

Purpose of the study was to determine whether removal of bile acids from the system affects metabolism. A notable finding was that "KO mice" had the same triglyceride and glucose levels as the control mice. "KO mice" had the same triglyceride and glucose levels as the control mice.

Over 95% of intestinal bile acid is absorbed and returned to the liver; thus, knocking out Slc10a2 should dramatically reduce the total pool of bile acid in the body (approx. 80% reduction). In addition, some mice were fed a sucrose-rich (SR) diet to simulate the metabolic stress of an unhealthy diet.

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Srebp1 Protein



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Figure 1 Expression of sterol regulatory element-binding protein 1 (Srebp1) Protein Immunoblot. The transcription factor Srebp1c is crucial for optimal activation of most genes in fatty acid synthesis and gluconeogenesis.

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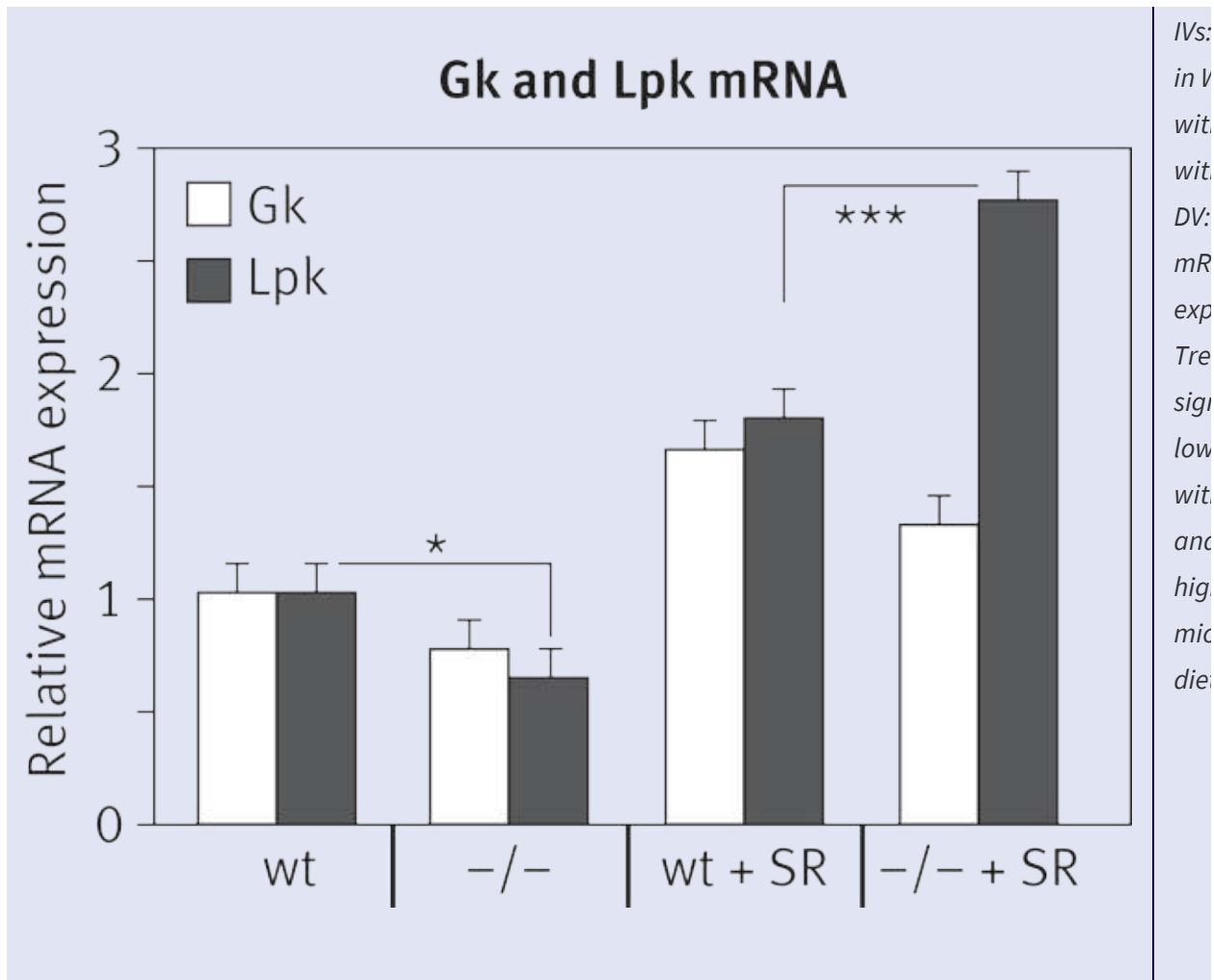


Figure 2 Hepatic mRNA expression of Glucokinase (Gk) and Liver Pyruvate Kinase (Lpk)

adapted from Lundåsen T, Andersson E-M, Snaith M, Lindmark H, Lundberg J, Östlund-Lindqvist A-M, et al. (2012) Inhibitory acid transporter slc10a2 improves triglyceride metabolism and normalizes elevated plasma glucose levels in mice. *PLoS* <https://doi.org/10.1371/journal.pone.0037787>.

Would reducing the total pool of bile acid be a good strategy to combat the hypertriglyceridemia and diabetes induced by an unhealthy diet?

This question asks whether an action, reducing the total pool of bile acid, would have a specified impact on two conditions, hypertriglyceridemia and diabetes. To answer this question, we must assess the data we are given in the

article regarding how changing bile acid quantity affects traits and outcomes associated with those conditions.

We should start by making sure we're comfortable with the terms used in the question, relying on either the article or our critical thinking skills to fill our knowledge gaps. Hypertriglyceridemia is a condition when blood triglyceride levels are too high, and diabetes is a condition when blood glucose levels are too high. However, that information wasn't given to us in the passage, and it's not entirely in scope for Test Day—so how could we figure out what we need to know to answer this question? Even if we didn't know this, though, we know the mice were fed food to simulate an unhealthy diet that would supply them with too much sugar, and we should recognize at least diabetes as a metabolic disorder linked with sugar metabolism. We should also recognize that triglycerides are fats, so this is probably a condition involving fat metabolism. Thus, we can assume that any action that lowers blood triglycerides (relating to hypertriglyceridemia) and blood glucose (relating to diabetes) could constitute a potentially beneficial therapy for these two conditions. So, when we look at the data for the experimental mice as compared to controls, we should be looking for changes related to these two metabolic pathways to determine if there are possible treatment implications for reducing bile salt concentrations in order to treat these conditions.

According to the passage, the knockout mice do not have the bile acid transporter, meaning that the pool of available bile acid in their system should be significantly lower. There's also a second variable, diet, which should be simulating mice that are experiencing metabolic stressors in the sucrose-rich condition as compared to normal metabolic function. Figure 1 shows the concentration of a protein involved in fatty acid synthesis in the KO mice versus wild-type mice, with both normal and sucrose-rich diets. In the wild-type mice, we can see that the addition of a metabolically stressful, high-sugar diet (lane

3) leads to a big increase in production of that protein as compared to regular diet (lane 1). However, in the knockout mice, there's no perceivable change from the regular diet (lane 2) to the sucrose-rich diet (lane 4). This is true both for the actual protein and the precursor, which means that the KO mouse is just not transcribing as much of that metabolic protein. The lamin lane down at the bottom seems like it's just a control for protein expression, and it looks pretty much the same across the board, so we can ignore it in our data analysis. Based on this figure, which is all about a metabolic protein in the fatty acid synthesis chain, it looks like the knockout mice just aren't experiencing the same amount of metabolic stress; they're not trying to turn that extra sucrose into fatty acids. Based on this piece of data, it looks like reducing bile acid might be one way to relieve the metabolic stress associated with hypertriglyceridemia, or at least, it's worth more investigation!

Moving on, Figure 2 shows the same conditions, but is measuring mRNA production of 2 different proteins: glucokinase and pyruvate kinase. Even if we aren't exactly sure of what those enzymes do, they have glucose and pyruvate in their names, so it sounds like they are probably involved in the sugar metabolism pathway. Looking at the figure, we can see in the first 2 pairs of columns that wild-type mice are expressing more mRNA for both of these proteins than KO mice. After metabolic stress is added, it looks like pyruvate kinase production specifically spikes within the KO mice. That seems to indicate that these mice are responding to the metabolic stress to an even greater extent (in terms of Lpk) than the wild-type mice. Without directly measuring the level of blood glucose, we cannot determine whether that increased metabolic response represents a positive or negative trend. So, unlike the data in Figure 1, it looks like bile salt reduction may not be a good treatment option for diabetic individuals, and we may even need to reevaluate whether it would be truly helpful for those with triglyceridemia, as the

energetic processing of fats can't be considered as fully separate from the processing of sugars.

Overall, our takeaway is that we have mixed results. Figure 1 made the reduction of bile salts look like a promising treatment for triglyceridemia, but Figure 2 indicates that reducing bile salts may not be effective in treating diabetes. We can conclude that more data is needed to assess this therapy's effect on sugar and lipid metabolism.

MCAT CONCEPT CHECK 9.1:

Before you move on, assess your understanding of the material with these questions.

1. What is the difference between mechanical and chemical digestion?

- Mechanical digestion:

- Chemical digestion:

2. Trace the path of food through the body, starting with ingestion and ending with excretion of feces:

3. What effect does the parasympathetic nervous system have on the digestive system? What effect does the sympathetic nervous system have?

- Parasympathetic nervous system:

- Sympathetic nervous system:

9.2 Ingestion and Digestion

LEARNING OBJECTIVES

After Chapter 9.2, you will be able to:

- Identify the secretory cells of the digestive tract, their secretions, and the function of their secretions
- Explain how bile and pancreatic lipase work together to digest fats
- Recall the key digestive enzymes and hormones produced in saliva, the stomach, and the intestines, and summarize the function of each

To supply the body with nutrients, we must **ingest** (eat) food. Several hormones are involved with feeding behavior, including antidiuretic hormone (ADH or vasopressin), aldosterone, glucagon, ghrelin, leptin, and cholecystokinin. ADH and aldosterone trigger the sensation of thirst, encouraging the behavior of fluid consumption. Glucagon, secreted by the pancreas, and ghrelin, secreted by the stomach and pancreas, stimulate feelings of hunger. Leptin and cholecystokinin do the opposite, stimulating feelings of satiety. Digestion begins in the oral cavity and continues in the stomach and the first part of the small intestine, known as the duodenum.

ORAL CAVITY

The **oral cavity** plays a role in both mechanical and chemical digestion of food. Mechanical digestion in the mouth involves the breaking up of large food particles into smaller particles using the teeth, tongue, and lips. This process is called **mastication** (chewing). Chewing helps to increase the surface area-to-volume ratio of the food, creating more surface area for enzymatic digestion as it passes through the gut tube. It also moderates the size of food particles entering the lumen of the alimentary canal; food particles that are too large create an obstruction risk in the tract.



KEY CONCEPT

The chemical digestion of carbohydrates occurs in both the mouth and small intestine, with the two amylases targeting complementary starches. Salivary amylase (active in the mouth) has a higher specificity for rapidly soluble starches, while pancreatic amylase (active in the small intestine) has a higher specificity for less soluble, more nonpolar starches.

Chemical digestion is the breakdown of chemical bonds in the macromolecules that make up food. This relies on enzymes from **saliva** produced by the three pairs of **salivary glands**. Saliva also aids mechanical digestion by moistening and lubricating food. The salivary glands, like all glands of the digestive tract, are innervated by the parasympathetic nervous system. The presence of food in the oral cavity triggers a neural circuit that ultimately leads to increased parasympathetic stimulation of these glands. Salivation can also be triggered by signals that food is near, such as smell or

sight. Saliva contains *salivary amylase*, also known as *ptyalin*, and *lipase*. **Salivary amylase** is capable of hydrolyzing starch into smaller sugars (maltose and dextrins), while **lipase** catalyzes the hydrolysis of lipids. The amount of chemical digestion that occurs in the mouth is minimal, though, because the food does not stay in the mouth for long. Our muscular tongue forms the food into a **bolus**, which is forced back to the pharynx and swallowed.



REAL WORLD

There are three pairs of major salivary glands: the parotid, submandibular, and sublingual glands. Other microscopic salivary glands are scattered throughout the upper digestive system. While the parasympathetic nervous system is responsible for promoting salivation, the sympathetic nervous system has some input into the glands as well. The sympathetic nervous system increases the viscosity of saliva, which is why dry mouth and even a tacky sensation in the mouth occurs during a fight-or-flight response.

PHARYNX

The **pharynx** is the cavity that leads from the mouth and posterior nasal cavity to the esophagus. The pharynx connects not only to the esophagus, but also to the larynx, which is a part of the respiratory tract. The pharynx can be divided into three parts: the **nasopharynx** (behind the nasal cavity), the **oropharynx** (at the back of the mouth), and the **laryngopharynx** (above the vocal cords). Food is prevented from entering the larynx during swallowing by

the **epiglottis**, a cartilaginous structure that folds down to cover the laryngeal inlet. Failure of this mechanism can lead to aspiration of food and choking.

ESOPHAGUS

The **esophagus** is a muscular tube that connects the pharynx to the stomach. The top third of the esophagus is composed of skeletal muscle, the bottom third is composed of smooth muscle, and the middle third is a mix of both. What does this mean in terms of nervous control? While the top of the esophagus is under somatic (voluntary) motor control, the bottom—and most of the rest of the gastrointestinal tract, for that matter—is under autonomic (involuntary) nervous control. The rhythmic contraction of smooth muscle that propels food toward the stomach is called **peristalsis**. Under normal circumstances, peristalsis proceeds down the digestive tract. However, certain factors such as exposure to chemicals, infectious agents, physical stimulation in the posterior pharynx, and even cognitive stimulation, can lead to reversal of peristalsis in the process of **emesis (vomiting)**.

Swallowing is initiated in the muscles of the oropharynx, which constitute the **upper esophageal sphincter**. Peristalsis squeezes, pushes, and propels the bolus toward the stomach. As the bolus approaches the stomach, a muscular ring known as the **lower esophageal sphincter (cardiac sphincter)** relaxes and opens to allow the passage of food.



REAL WORLD

Weakness in the lower esophageal sphincter is a key feature of gastroesophageal reflux disease (GERD), which causes classic

heartburn symptoms after eating. As food and acid reflux into the lower esophagus, they irritate the less-protected mucosa, stimulating pain receptors. These receptors localize pain poorly, so a general burning sensation is felt in the lower chest—leading to the common term *heartburn*.

STOMACH

There are three main energy sources: carbohydrates, fats, and proteins. As mentioned earlier, the chemical digestion of carbohydrates and fats is initiated in the mouth. No mechanical or chemical digestion takes place in the esophagus, except for the continued enzymatic activity initiated in the mouth by salivary enzymes. Thus, digestion that occurs prior to the entrance of the bolus into the stomach is minimal compared to the digestion that occurs in the stomach and small intestine.

The **stomach** is a highly muscular organ with a capacity of approximately two liters. In humans, the stomach is located in the upper left quadrant of the abdominal cavity, underneath the diaphragm. This organ uses hydrochloric acid and enzymes to digest food, creating a fairly harsh environment. Therefore, its mucosa is quite thick to prevent autodigestion. The stomach can be divided into four main anatomical divisions, as shown in Figure 9.2: the **fundus** and **body**, which contain mostly gastric glands, and the **antrum** and **pylorus**, which contain mostly pyloric glands. The internal curvature of the stomach is called the **lesser curvature**; the external curvature is called the **greater curvature**. The lining of the stomach is thrown into folds called **rugae**.

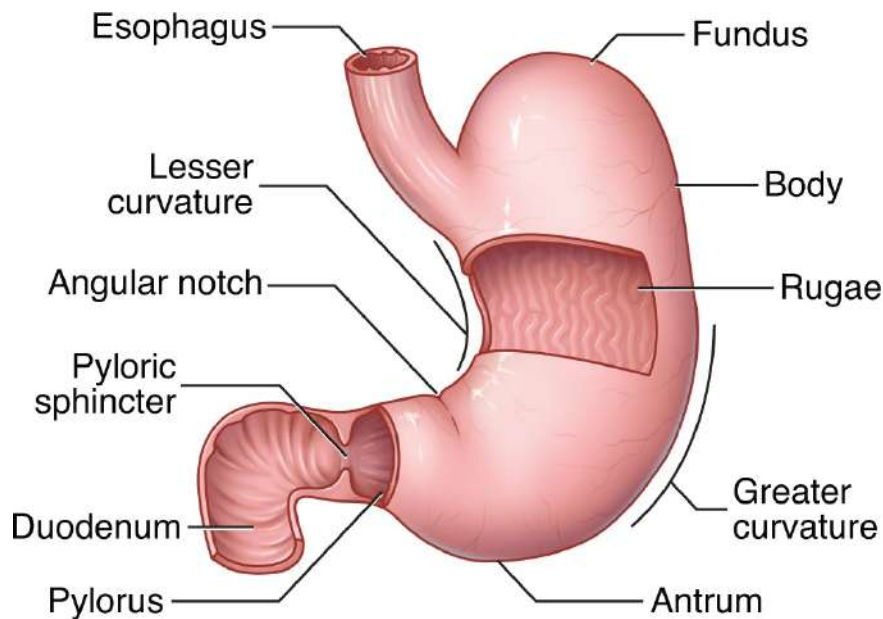


Figure 9.2. Anatomy of the Stomach

The mucosa of the stomach contains gastric glands and pyloric glands. The **gastric glands** respond to signals from the **vagus nerve** of the parasympathetic nervous system, which is activated by the brain in response to the sight, taste, and smell of food. Gastric glands have three different cell types: mucous cells, chief cells, and parietal cells. **Mucous cells** produce the bicarbonate-rich mucus that protects the muscular wall from the harshly acidic ($\text{pH} = 2$) and proteolytic environment of the stomach.



REAL WORLD

Nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen, naproxen, and aspirin, are very common drugs. However, these drugs are not without side effects. One such side effect is the disruption of mucus production within the stomach. This leaves the mucosa of the stomach unprotected, and irritation results. This irritation can be in

the form of *gastritis*, inflammation of the mucosa of the stomach, or *gastric ulcers*, deep wounds in the lining of the stomach.

Gastric juice is a combination of secretions from the other two cell types in the gastric glands: chief cells and parietal cells. The **chief cells** secrete **pepsinogen**. This is the inactive, **zymogen** form of *pepsin*, a proteolytic enzyme. Hydrogen ions in the stomach, secreted by **parietal cells** as hydrochloric acid, cleave pepsinogen to pepsin. **Pepsin** digests proteins by cleaving peptide bonds near aromatic amino acids, resulting in short peptide fragments. Because pepsin is activated by the acidic environment, it follows that pepsin is most active at a low pH. This is a unique characteristic among human enzymes, as most human enzymes are most active at physiological pH. Stomach acid also kills most harmful bacteria (with the exception of *Helicobacter pylori*, infection with which is usually asymptomatic but can cause inflammation, ulcers, and even certain gastric cancers). The acidic environment also helps to denature proteins and can break down some intramolecular bonds that hold food together. In addition to HCl, parietal cells secrete **intrinsic factor**, a glycoprotein involved in the proper absorption of vitamin B₁₂.



KEY CONCEPT

The stomach secretes six products:

1. HCl (kills microbes, denatures proteins, and converts pepsinogen into pepsin)
2. Pepsinogen (cleaved in the stomach to pepsin; an enzyme that

- partially digests proteins)
- 3. Mucus (protects mucosa)
- 4. Bicarbonate (protects mucosa)
- 5. Water (dissolves and dilutes ingested material)
- 6. Intrinsic factor (required for normal absorption of vitamin B₁₂)

The **pyloric glands** contain **G-cells** that secrete gastrin, a peptide hormone. **Gastrin** induces the parietal cells in the stomach to secrete more HCl and signals the stomach to contract, mixing its contents. The digestion of solid food in the stomach results in an acidic, semifluid mixture known as **chyme**. The combined mechanical and chemical digestive activities of the stomach result in a significant increase in the surface area of the now unrecognizable food particles, so when the chyme reaches the small intestine, the absorption of nutrients from it can be maximized. There are a few substances that are absorbed directly from the stomach (such as alcohol and aspirin), but the stomach is mainly an organ of digestion.



REAL WORLD

Zollinger–Ellison syndrome is a rare disease resulting from a gastrin-secreting tumor (gastrinoma). Typically, this tumor is found in the pancreas. The excess gastrin leads to excessive HCl production by parietal cells. Not surprisingly, one of the most common signs of Zollinger–Ellison syndrome is the presence of intractable ulcer disease.

DUODENUM

The **small intestine** consists of three segments: the duodenum, the jejunum, and the ileum. The small intestine is quite long, up to seven meters. The duodenum is responsible for the majority of chemical digestion and has some minor involvement in absorption. However, most of the absorption in the small intestine takes place in the jejunum and ileum.

Food leaves the stomach through the **pyloric sphincter** and enters the duodenum. The presence of chyme in the duodenum causes the release of brush-border enzymes like *disaccharidases* (*maltase*, *isomaltase*, *lactase*, and *sucrase*) and *peptidases* (including *dipeptidase*). **Brush-border enzymes** are present on the luminal surface of cells lining the duodenum and break down dimers and trimers of biomolecules into absorbable monomers. The duodenum also secretes *enteropeptidase*, which is involved in the activation of other digestive enzymes from the accessory organs of digestion. Finally, it secretes hormones like secretin and cholecystokinin (CCK) into the bloodstream.

The **disaccharidases** digest disaccharides. **Maltase** digests maltose, **isomaltase** digests isomaltose, **lactase** digests lactose, and **sucrase** digests sucrose. Lack of a particular disaccharidase causes an inability to break down the corresponding disaccharide. Then bacteria in the intestines are able to hydrolyze that disaccharide, producing methane gas as a byproduct. In addition, undigested disaccharides can have an osmotic effect, pulling water into the stool and causing diarrhea. This is why people who are lactose intolerant have symptoms of bloating, flatulence, and possibly diarrhea after ingesting dairy products.

Peptidases break down proteins (or peptides, as the name implies).

Aminopeptidase is a peptidase secreted by glands in the duodenum that removes the N-terminal amino acid from a peptide. **Dipeptidases** cleave the peptide bonds of dipeptides to release free amino acids. Unlike carbohydrates, which must be broken down into monosaccharides for absorption, proteins can be broken down into di- and even tripeptides and can be absorbed across the small intestine wall.

Enteropeptidase (formerly called **enterokinase**) is an enzyme critical for the activation of **trypsinogen**, a pancreatic protease, to **trypsin**. Trypsin then initiates an activation cascade, as described later in this chapter.

Enteropeptidase can also activate **procarboxypeptidases A** and **B** to their active forms.



REAL WORLD

Celiac disease results from an immune reaction against gluten, a protein found in grains, especially wheat. In this condition, the immune system develops antibodies against certain components of gluten. These antibodies then cross-react with elements of the small intestine, causing damage to the mucosa. This results in diarrhea and discomfort. Sometimes, this condition also results in malabsorptive syndromes, including the inability to absorb fat and fat-soluble vitamins. Contrary to popular belief, celiac disease and gluten sensitivity are immune conditions, but not true allergies.

Secretin is a peptide hormone that causes pancreatic enzymes to be released into the duodenum. It also regulates the pH of the digestive tract by reducing HCl secretion from parietal cells and increasing bicarbonate secretion from the pancreas. Secretin is also an **enterogastrone**, a hormone that slows motility through the digestive tract. Slowing of motility allows increased time for digestive enzymes to act on chyme—especially fats.

Finally, **cholecystikinin (CCK)** is secreted in response to the entry of chyme (specifically, amino acids and fat in the chyme) into the duodenum. This peptide hormone stimulates the release of both bile and pancreatic juices and also acts in the brain, where it promotes satiety. **Bile** is a complex fluid composed of bile salts, pigments, and cholesterol. **Bile salts** are derived from cholesterol. They are not enzymes and therefore do not directly perform chemical digestion (the enzymatic cleavage of chemical bonds). However, bile salts serve an important role in the mechanical digestion of fats and ultimately facilitate the chemical digestion of lipids. Bile salts have hydrophobic and hydrophilic regions, allowing them to serve as a bridge between aqueous and lipid environments. In fact, bile salts are much like the common soaps and detergents we use to wash our hands, clothes, and dishes. In the small intestine, bile salts **emulsify** fats and cholesterol into **micelles**. Without bile, fats would spontaneously separate out of the aqueous mixture in the duodenum and would not be accessible to *pancreatic lipase*, which is water-soluble. In addition, these micelles increase the surface area of the fats, increasing the rate at which lipase can act. Ultimately, proper fat digestion depends on both bile and lipase. Bile gets the fats into the solution and increases their surface area by placing them in micelles (mechanical digestion). Then, lipase can come in to hydrolyze the ester bonds holding the lipids together (chemical digestion).

CCK also promotes the secretion of pancreatic juices into the duodenum, as

shown in Figure 9.3. **Pancreatic juices** are a complex mixture of several enzymes in a bicarbonate-rich alkaline solution. This bicarbonate helps to neutralize acidic chyme, as well as provide an ideal working environment for the digestive enzymes, which are most active around pH 8.5. Pancreatic juices contain enzymes that can digest all three types of nutrients: carbohydrates, fats, and proteins. The identities and functions of these enzymes will be discussed in the next section of this chapter.

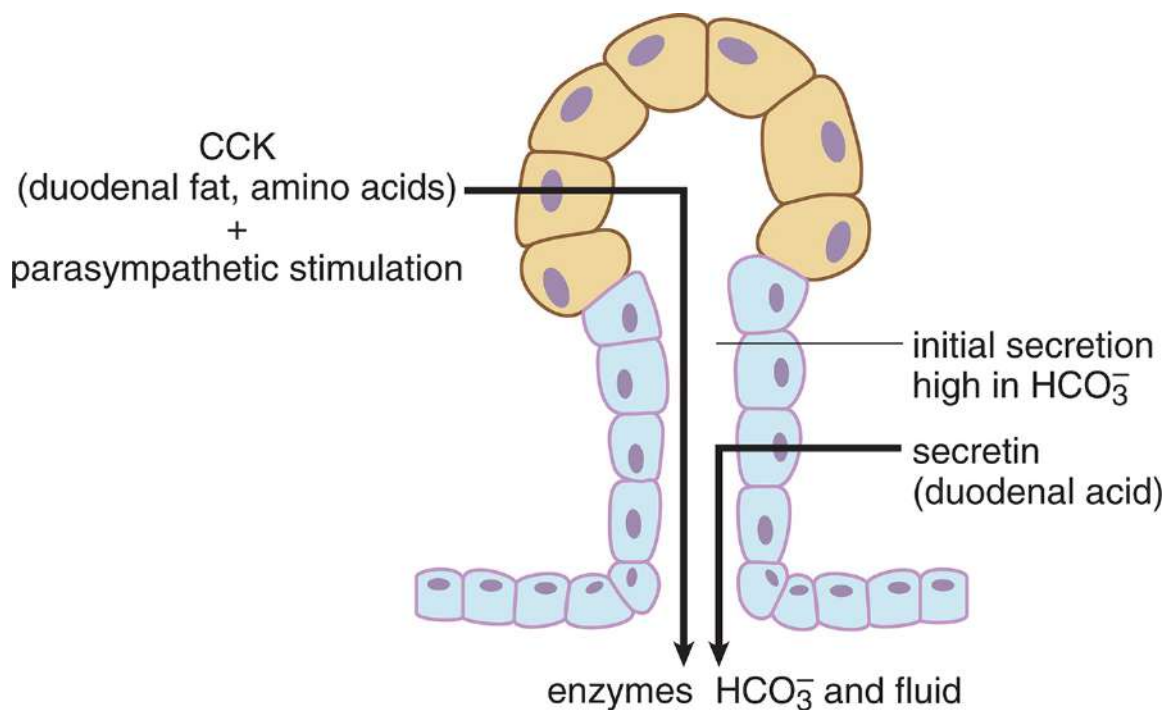


Figure 9.3. Hormonal Control of the Exocrine Pancreas



BRIDGE

The acidic environment of the stomach and basic environment of the duodenum point out the important correlation between pH and enzyme activity. Pepsin is active in the stomach and works best at pH

2. Pancreatic enzymes are active in the duodenum and work best at pH 8.5. pH, temperature, and salinity can all affect enzyme function, as highlighted in Chapter 2 of *MCAT Biochemistry Review*.

MCAT CONCEPT CHECK 9.2:

Before you move on, assess your understanding of the material with these questions.

1. What two main enzymes are found in saliva? What do these enzymes do?

- _____
- _____

2. For each of the cell types below, list the major secretions of the cell and the functions of these secretions.

Cell	Secretions	Functions
Mucous cell		
Chief cell		
Parietal cell		
G-cell		

3. For each of the following substances, determine whether it is a digestive enzyme or a hormone and briefly summarize its functions.

Substance	Enzyme or Hormone?	Functions
Sucrase		
Secretin		
Dipeptidase		
Cholecystokinin		
Enteropeptidase		

4. How do bile and pancreatic lipase work together to digest fats?

9.3 Accessory Organs of Digestion

LEARNING OBJECTIVES

After Chapter 9.3, you will be able to:

- Recall the pancreatic enzymes and the molecules they help to digest
- Describe the significance and the function of bile, including its production, components, and release
- List the major functions of the liver
- Associate the accessory organs of digestion with their germ layer of origin

Digestion is a complex process that requires the release of enzymes not only from the cells directly lining the alimentary canal, but also from the pancreas, liver, and gallbladder. Collectively, these organs—which all originate as outgrowths of endoderm from the gut tube during development—are called **accessory organs of digestion**.

PANCREAS

The pancreas serves two quite different roles in the body, reflecting its exocrine and endocrine functions. As discussed in Chapter 5 of *MCAT Biology*

Review, the endocrine functions of the pancreas include the release of insulin, glucagon, and somatostatin—peptide hormones necessary for the maintenance of proper blood sugar levels. The hormonal function of the pancreas is limited to cells residing in islets of Langerhans scattered throughout the organ. The bulk of the pancreas, however, is made of exocrine cells called **acinar cells** that produce pancreatic juices. As mentioned earlier, pancreatic juices are bicarbonate-rich alkaline secretions containing many digestive enzymes that work on all three classes of biomolecules. **Pancreatic amylase** breaks down large polysaccharides into small disaccharides and is therefore responsible for carbohydrate digestion. The pancreatic peptidases (**trypsinogen**, **chymotrypsinogen**, and **carboxypeptidases A and B**) are released in their zymogen form, but once activated are responsible for protein digestion. Enteropeptidase, produced by the duodenum, is the master switch. It converts trypsinogen to trypsin, which can then activate the other zymogens, and also activates procarboxypeptidases A and B. Finally, the pancreas secretes **pancreatic lipase**, which is capable of breaking down fats into free fatty acids and glycerol.

Pancreatic juices are transferred to the duodenum via a duct system that runs along the middle of the pancreas, as shown in Figure 9.4. Like all exocrine cells, acinar cells secrete their products into ducts. These ducts then empty into the duodenum through the **major** and **minor duodenal papillae**.

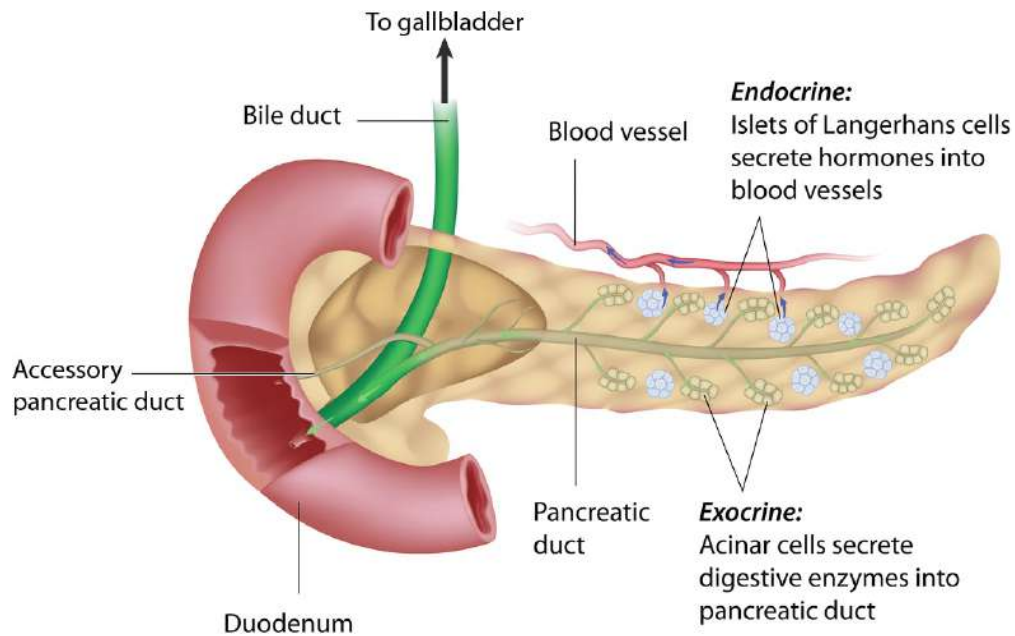


Figure 9.4. Anatomy of the Pancreas



REAL WORLD

Pancreatitis, or inflammation of the pancreas, is usually caused by gallstones or excessive consumption of alcohol. Regardless of the cause, pancreatitis results from premature activation of pancreatic enzymes and autodigestion of the pancreatic tissue. This is a very painful condition that may result in a long hospital stay and long-term consequences such as diabetes and the reduced digestion of proteins and fats.

LIVER

The **liver** is located in the upper right quadrant of the abdomen and contains two unique structures for communicating with the digestive system. First, **bile ducts** connect the liver with both the gallbladder and small intestine. Bile is produced in the liver and travels down these bile ducts where it may be stored in the gallbladder or secreted into the duodenum. The liver also receives all blood draining from the abdominal portion of the digestive tract through the **hepatic portal vein**. This nutrient-rich blood can be processed by the liver before draining into the inferior vena cava on its way to the right side of the heart. For example, the liver takes up excess sugar to create **glycogen**, the storage form of glucose, and stores fats as **triacylglycerols**. The liver can also reverse these processes, producing glucose for the rest of the body through **glycogenolysis** and **gluconeogenesis** and mobilizing fats in **lipoproteins**. The liver detoxifies both endogenous compounds (those made in the body) and exogenous compounds (those brought in from the environment). For example, the liver converts ammonia, a toxic waste product of amino acid metabolism, into urea, which can be excreted by the kidneys. The liver also detoxifies and metabolizes alcohol and medications. Some drugs actually require activation by the enzymes of the liver. In addition, some drugs cannot be taken orally because modification of these drugs by the liver renders them inactive.



KEY CONCEPT

The functions of the liver include processing and synthesis of nutrients (glycogenesis and glycogenolysis, storage and mobilization of fats, gluconeogenesis), production of urea, detoxification of chemicals, production of bile, and synthesis of albumin and clotting factors.

Bile production is one of the most significant jobs of the liver *vis-à-vis* the digestive system. As mentioned earlier, bile is composed of bile salts, pigments, and cholesterol. Bile salts are amphipathic molecules that can emulsify fat in the digestive system. The major pigment in bile is **bilirubin**, which is a byproduct of the breakdown of hemoglobin. Bilirubin travels to the liver, where it is **conjugated** (attached to a protein) and secreted into the bile for excretion. If the liver is unable to process or excrete bilirubin (due to liver damage, excessive red blood cell destruction, or blockage of the bile ducts), **jaundice** or yellowing of the skin may occur.



KEY CONCEPT

The major components of bile are bile salts, which emulsify fats; pigments (especially bilirubin, from the breakdown of hemoglobin); and cholesterol.

In addition to bile production, processing of nutrients, and detoxification and drug metabolism, the liver also synthesizes certain proteins necessary for proper body function. These proteins include **albumin**, a protein that maintains plasma oncotic pressure and also serves as a carrier for many drugs and hormones, and **clotting factors** used during blood coagulation.



REAL WORLD

Cirrhosis of the liver can result from many different processes, including chronic alcohol consumption, hepatitis C infection,

autoimmune hepatitis, and fatty liver disease. However, the outcome is the same. Cirrhosis is scarring of the liver, and this scar tissue builds up, creating increased resistance within the portal vein, resulting in portal hypertension. This causes a backup of fluid within the portal system, resulting in swollen veins in the digestive system, especially the esophagus, which may rupture and cause life-threatening bleeding. This often manifests as *hematemesis*, or vomiting of blood. Cirrhosis also causes bleeding disorders because production of clotting factors is disrupted. The inability to properly dispose of ammonia results in increased ammonia in the blood, which affects mentation. Finally, cirrhosis may also cause hepatocellular carcinoma, or cancer of the hepatocytes.

GALLBLADDER

The **gallbladder** is located just beneath the liver and both stores and concentrates bile. Upon release of CCK, the gallbladder contracts and pushes bile out into the **biliary tree**. The bile duct system merges with the pancreatic duct, as shown in Figure 9.4 earlier, before emptying into the duodenum.

The gallbladder is a common site of cholesterol or bilirubin stone formation. This painful condition causes inflammation of the gallbladder. The stones may also travel into the bile ducts and get stuck in the biliary tree. In some cases, stones can get caught just before entering the duodenum, resulting in blockage of not only the biliary tree, but the pancreatic duct as well, causing pancreatitis.



Stone formation is nothing more than precipitation of a solid once its concentration has reached the K_{sp} of the compound. This is particularly common in the gallbladder because bile is concentrated there. Solubility and K_{sp} are discussed in Chapter 9 of *MCAT General Chemistry Review*.

The functions of the various digestive enzymes (and bile) are summarized in Table 9.1.

Nutrient	Enzyme	Site of Production	Site of Function	Function
Carbohydrates	Salivary amylase (ptyalin)	Salivary glands	Mouth	Hydrolyzes starch to maltose and dextrins
	Pancreatic amylase	Pancreas (acinar cells)	Duodenum	Hydrolyzes starch to maltose and dextrins
	Maltase	Intestinal glands	Duodenum	Hydrolyzes maltose to two glucose molecules
	Isomaltase	Intestinal glands	Duodenum	Hydrolyzes isomaltose to two glucose molecules
	Sucrase	Intestinal glands	Duodenum	Hydrolyzes sucrose to glucose and fructose
	Lactase	Intestinal glands	Duodenum	Hydrolyzes lactose to glucose and galactose

Proteins	Pepsin(ogen)	Gastric glands (chief cells)	Stomach	Hydrolyzes specific peptide bonds; activated by HCl
	Trypsin(ogen)	Pancreas (acinar cells)	Duodenum	Hydrolyzes specific peptide bonds; converts chymotrypsinogen to chymotrypsin; activated by enteropeptidase
	Chymotrypsin (ogen)	Pancreas (acinar cells)	Duodenum	Hydrolyzes specific peptide bonds; activated by trypsin
	(Pro)carboxy peptidases A and B	Pancreas (acinar cells)	Duodenum	Hydrolyzes terminal peptide bond at carboxy end; activated by enteropeptidase
	Aminopeptidase	Intestinal glands	Duodenum	Hydrolyzes terminal peptide bond at amino end
	Dipeptidases	Intestinal glands	Duodenum	Hydrolyzes pairs of amino acids
	Enteropeptidase	Intestinal glands	Duodenum	Converts trypsinogen to trypsin and procarboxypeptidases A and B to carboxypeptidases A and B
Lipids	Bile*	Liver (stored in gallbladder)	Duodenum	Emulsifies fat
	Lipase	Pancreas (acinar cells)	Duodenum	Hydrolyzes lipids

*Note: Bile is not an enzyme, but is involved in mechanical digestion of fats.

Table 9.1. Digestive Enzymes

A summary of the digestion of each major class of biomolecules is provided in Figure 9.5.

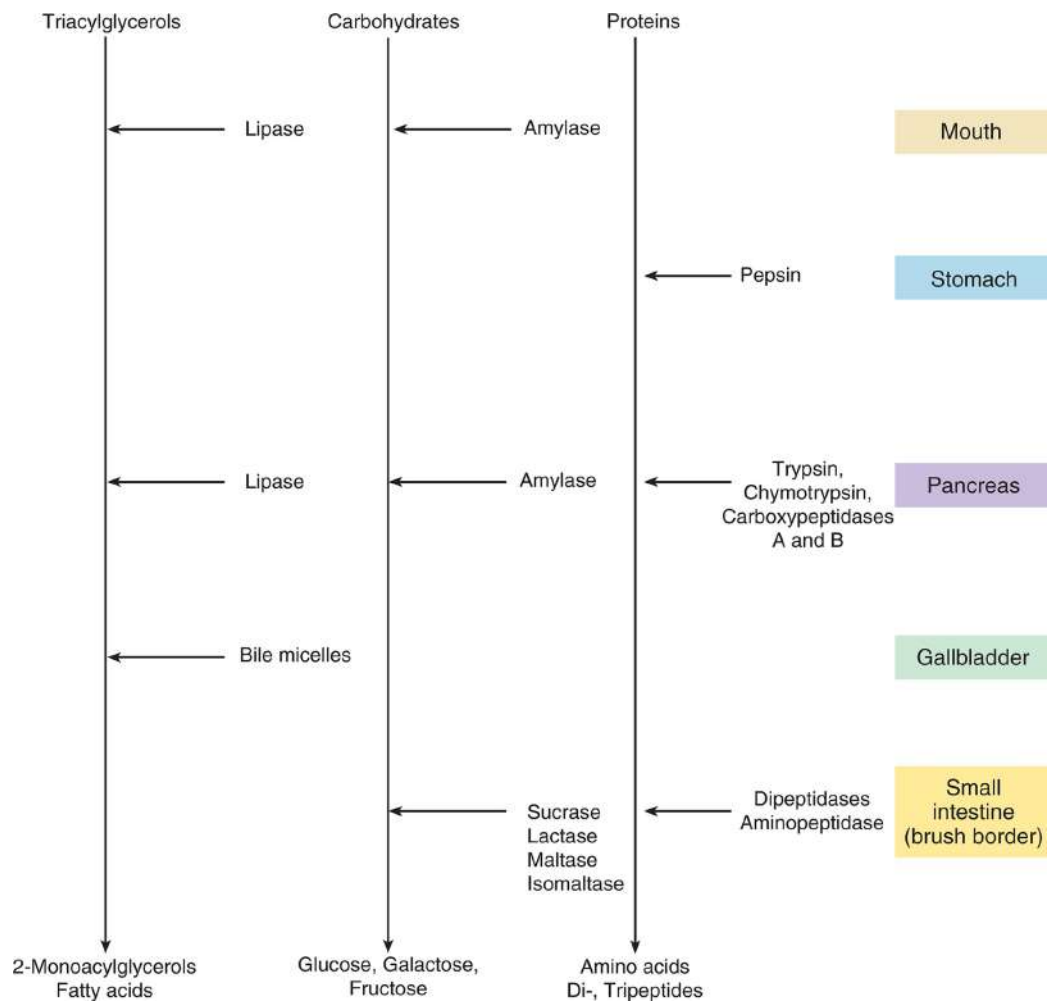


Figure 9.5. Summary of Digestive Processes

MCAT CONCEPT CHECK 9.3:

Before you move on, assess your understanding of the material with these questions.

1. List at least one pancreatic enzyme that digests each of the three major classes of biomolecules:

- Carbohydrates:

- Proteins:

- Fats:

2. What are the main components of bile?

3. Where is bile synthesized? Where is bile stored? Where does bile carry out its digestive function?

- Synthesized:

- Stored:

- Carries out function:

4. List at least four functions of the liver:

-

-

-

-

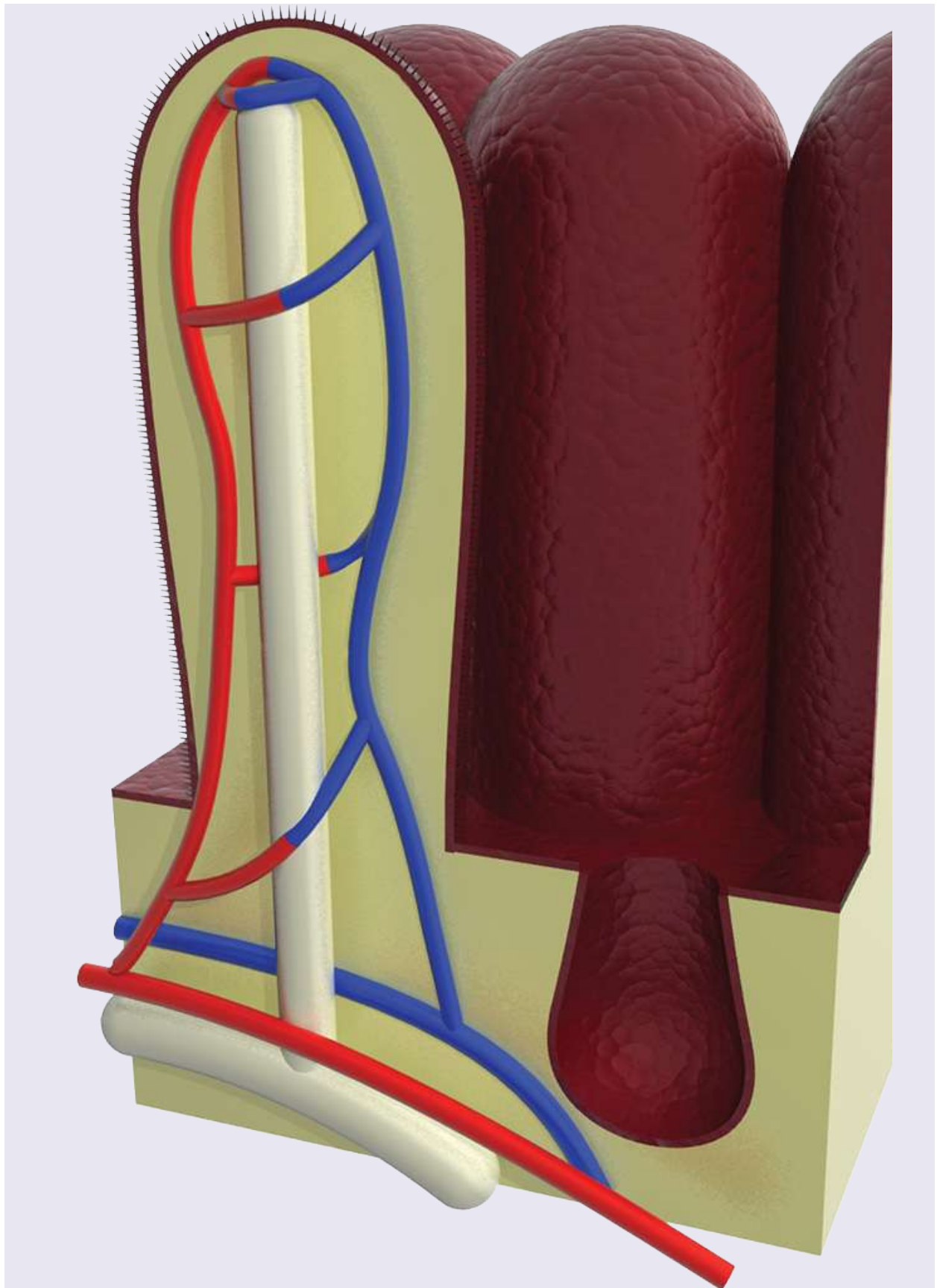
5. The accessory organs of digestion originate from which primary germ layer?

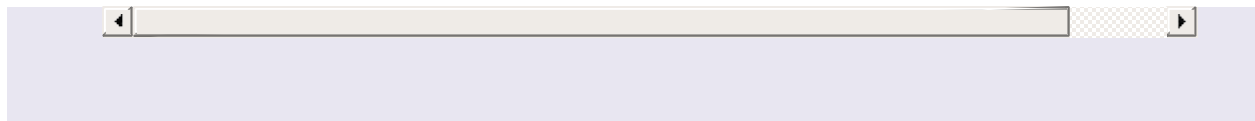
9.4 Absorption and Defecation

LEARNING OBJECTIVES

After Chapter 9.4, you will be able to:

- Recall the four fat-soluble vitamins
- Order the three sections of the small intestine and the three sections of the large intestine
- Predict the portions of the gut impacted by a disease when given a digestive symptom, such as watery stool
- Identify the biomolecules typically absorbed into each vessel of the villus:





Absorption of nutrients primarily occurs in the small intestine, especially in the jejunum and ileum. The large intestine largely absorbs water.

JEJUNUM AND ILEUM

The small intestine consists of three segments: the duodenum, the jejunum, and the ileum. As discussed previously, the **duodenum** is primarily involved in digestion. The **jejunum** and **ileum** are involved in the absorption of nutrients. The small intestine is lined with **villi**, which are small, finger-like projections from the epithelial lining, as shown in Figure 9.6. Each villus has many **microvilli**, drastically increasing the surface area available for absorption. In addition, at the middle of each villus there is a capillary bed for the absorption of water-soluble nutrients and a **lacteal**, a lymphatic channel that takes up fats for transport into the lymphatic system.

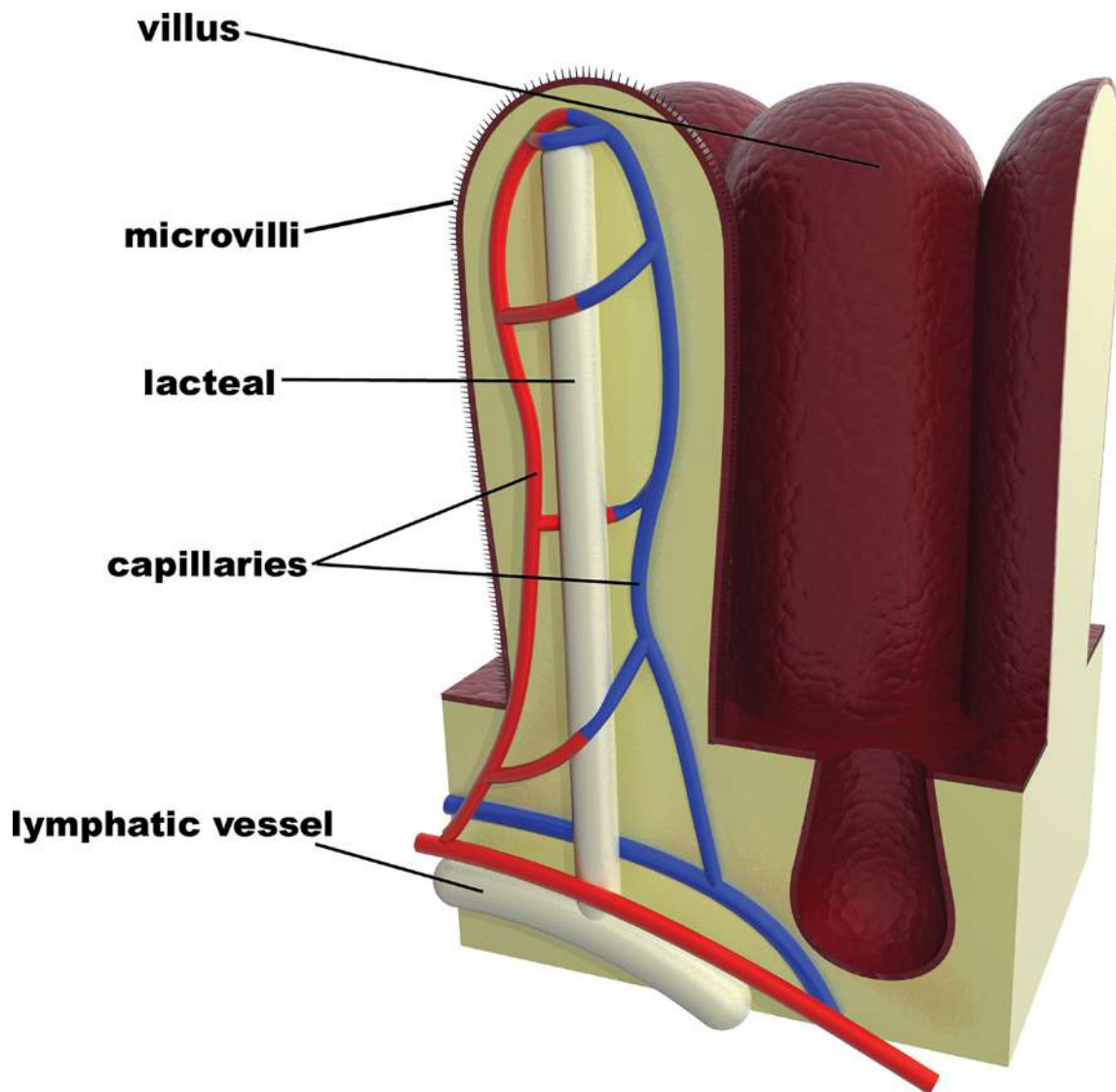


Figure 9.6. Structure of a Villus



MNEMONIC

Segments of the small intestine: **Dow Jones Industrial**

- **D**uodenum
- **J**ejunum

- Ileum

Simple sugars, such as glucose, fructose, and galactose, and amino acids are absorbed by secondary active transport and facilitated diffusion into the epithelial cells lining the small intestine, as shown in Figure 9.7. Then, these substances move across the epithelial cell membrane into the intestinal capillaries. Blood is constantly passing by the epithelial cells, carrying the carbohydrate and amino acid molecules away. This creates a concentration gradient such that the blood always has a lower concentration of monosaccharides and amino acids than inside the epithelial cells. Thus, simple carbohydrates and amino acids diffuse from the epithelial cells into the capillaries. The absorbed molecules then go to the liver via the hepatic portal circulation.

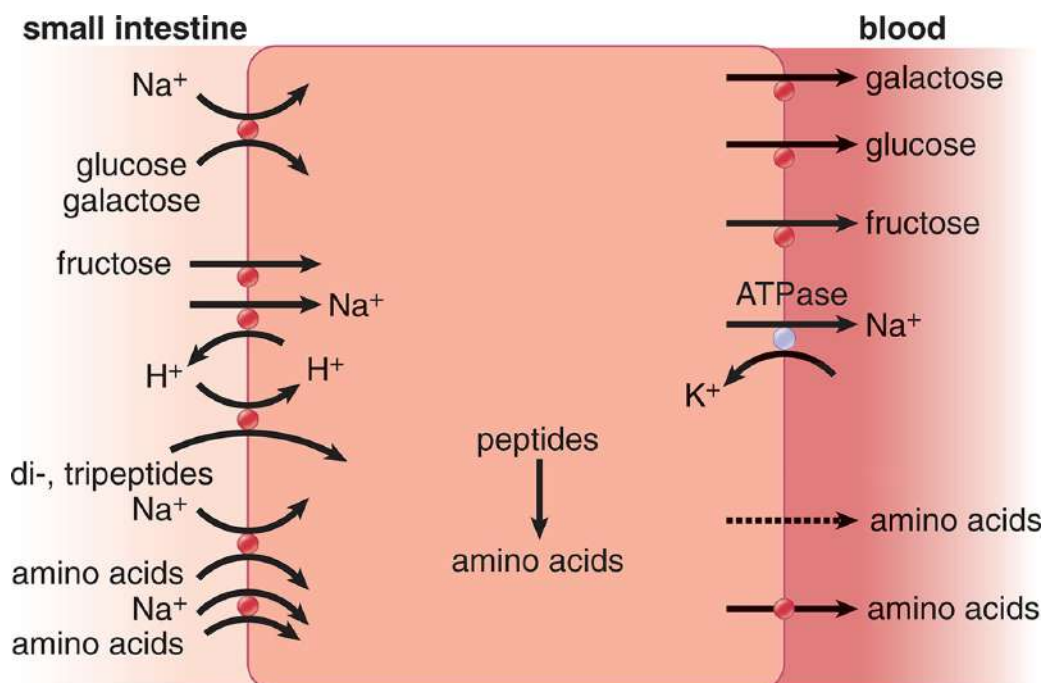


Figure 9.7. Absorption of Carbohydrates and Amino Acids in the Small Intestine

What about fats? Short-chain fatty acids will follow the same process as carbohydrates and amino acids by diffusing directly into the intestinal capillaries. These fatty acids do not require transporters because they are nonpolar, so they can easily traverse the cellular membrane. Larger fats, glycerol, and cholesterol move separately into the intestinal cells but then reform into triglycerides, as shown in Figure 9.8. The triglycerides and esterified cholesterol molecules are packaged into **chylomicrons**. Rather than entering the bloodstream, chylomicrons enter the lymphatic circulation through **lacteals**, small vessels that form the beginning of the lymphatic system. These lacteals converge and enter the venous circulation at the **thoracic duct** in the base of the neck, which empties into the left subclavian vein.

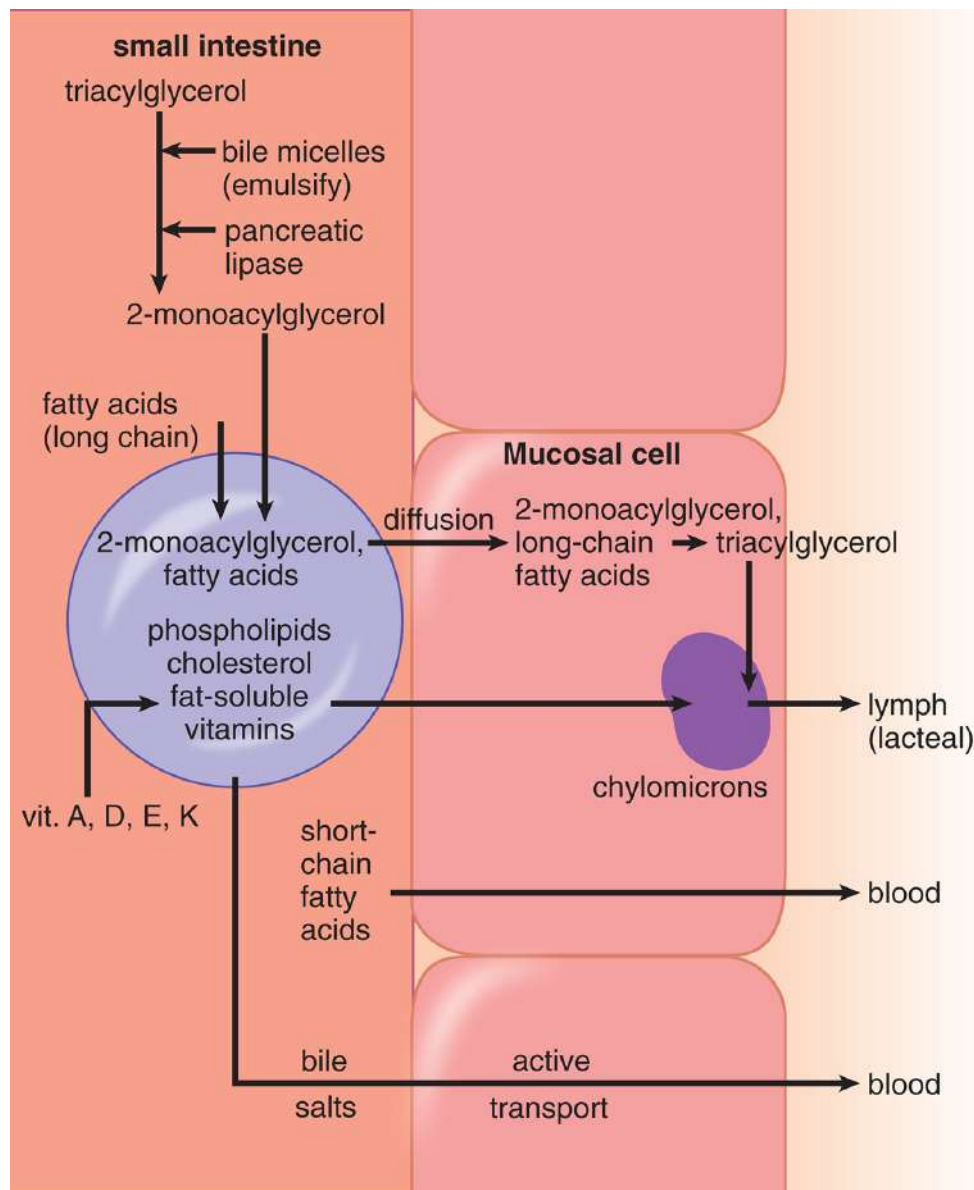


Figure 9.8. Absorption of Lipids

Vitamins are also absorbed in the small intestine. Vitamins can be categorized as either fat-soluble or water-soluble. Because there are only four **fat-soluble vitamins** (A, D, E, and K), these can be easily memorized. All other vitamins (B complex and C) are water-soluble. Fat-soluble vitamins dissolve directly into chylomicrons to enter the lymphatic circulation. Failure to digest and absorb fat properly, which can be due to pathologies in the liver,

gallbladder, pancreas, or small intestine, may lead to deficiencies of fat-soluble vitamins. The **water-soluble vitamins** are taken up, along with water, amino acids, and carbohydrates, across the endothelial cells of the small intestine, passing directly into the plasma.

In addition to fats, carbohydrates, amino acids, and vitamins, the small intestine also absorbs water. Much of the water in chyme is actually the result of secretions. The average person may consume up to two liters of fluid per day, but secretions into the upper gastrointestinal tract may total up to seven liters of fluid per day. In order to maintain proper fluid levels within the body, much of this fluid must be reabsorbed by osmosis. As solutes are absorbed into the bloodstream, water is drawn with them, eventually reaching the capillaries. Water passes not only **transcellularly** (across the cell membrane), but also **paracellularly** (squeezing between cells) to reach the blood.

LARGE INTESTINE

The final part of the gastrointestinal tract is the **large intestine**. It is primarily involved in water absorption. The large intestine has a larger diameter but shorter length than the small intestine. It is divided into three major sections: the cecum, the colon, and the rectum. The **cecum** is simply an outpocketing that accepts fluid exiting the small intestine through the **ileocecal valve** and is the site of attachment of the **appendix**. The appendix is a small finger-like projection that was once thought to be **vestigial**. Recent evidence, however, suggests that it may have a role in warding off certain bacterial infections and repopulating the large intestine with normal flora after episodes of diarrhea. Inflammation of the appendix (appendicitis) is a surgical emergency; in fact, it is the most common reason for an unscheduled surgery in the United States.

The **colon** itself is divided into the ascending, transverse, descending, and sigmoid colons. Its main function is to absorb water and salts (such as sodium chloride) from the undigested material left over from the small intestine. The small intestine actually absorbs much more water than the colon, so the colon primarily concentrates the remaining material to form **feces**. Too little or too much water absorption can cause diarrhea or constipation, respectively.



BRIDGE

Note the similarity between the muscles that control voiding of urine and feces. In both cases, there is an internal sphincter under autonomic control (internal urethral sphincter and internal anal sphincter) and an external sphincter under somatic control (external urethral sphincter and external anal sphincter). Urination is discussed in Chapter 10 of *MCAT Biology Review*.

Finally, the rectum serves as a storage site for feces, which consists of indigestible material, water, bacteria (*E. coli* and others), and certain digestive secretions that are not reabsorbed (enzymes and some bile). The **anus** is the opening through which wastes are eliminated and consists of two sphincters: the **internal** and **external anal sphincters**. The external sphincter is under voluntary control (somatic), but the internal sphincter is under involuntary control (autonomic).

The large intestine—and even the small intestine—is home to many different species of bacteria. In fact, 30 percent of the dry matter in stool consists of

bacteria. Most of these bacteria are anaerobes, but the cecum is also home to many aerobic bacteria. The presence of bacteria in the colon represents a symbiotic relationship: the bacteria are provided with a steady source of food, and the byproducts produced by the bacteria are beneficial to humans. For example, bacteria in the gut produce vitamin K, which is essential for the production of clotting factors, and biotin (vitamin B₇), which is a coenzyme for many metabolic enzymes.

MCAT CONCEPT CHECK 9.4:

Before you move on, assess your understanding of the material with these questions.

1. What are the two circulatory vessels in a villus? What biomolecules are absorbed into each?

-

-

2. What are the four fat-soluble vitamins?

3. What are the three sections of the small intestine, in order?
What are the three sections of the large intestine, in order?

,

- Small intestine: _____

,

- Large intestine: _____

4. *Vibrio cholera* causes a severe infection in the intestines, leading to massive volumes of watery diarrhea—up to 20 liters per day. Given these symptoms, does cholera likely impact the small intestine or the large intestine?



Conclusion

In this chapter, we have reviewed a lot of information about the digestive system that we can use to our advantage on Test Day. We began with an overview of the anatomy, keeping in mind that the system is designed to carry out extracellular digestion. Considering that all our foodstuffs are made up of fats, proteins, and carbohydrates, these compounds have to be broken down to their simplest molecular forms before they can be absorbed and distributed to the tissues and cells of the body. As we moved through the gastrointestinal tract, we discussed whether each organ was a site of absorption, digestion, or both. We spent a good bit of time discussing each of the enzymes involved in digestion and their specific purposes. While digestion occurs primarily in the oral cavity, stomach, and duodenum, absorption occurs primarily in the jejunum and ileum, where the method of transport into the circulatory system is slightly different depending on the compound. Finally, we discussed the three segments of the large intestine and their roles in water and salt absorption, as well as the temporary storage of waste products. Although the amount of information about the digestive system may seem overwhelming, the underlying concepts are relatively straightforward, and a systematic approach (like charts, tables, or flashcards) will help you manage this content.

In the end, the digestive system's main purpose is to break down energy-containing compounds and get them into the circulation so they can be used by the rest of the body. Equally important are the systems the body has for getting rid of compounds from the blood. Buildup of waste products like

ammonia, urea, potassium, and hydrogen ions can lead to serious pathology. For instance, hyperammonemia (buildup of ammonia in the blood) can lead to severe, permanent neurological damage. Hyperkalemia (buildup of potassium in the blood) can quickly cause a fatal heart attack. Temperature regulation is similarly important; both hyperthermia and hypothermia can lead to organ dysfunction and, ultimately, death. In the next chapter, we turn our attention to these regulatory systems: the renal system and the skin.

Concept Summary

Anatomy of the Digestive System

- **Intracellular digestion** involves the oxidation of glucose and fatty acids to make energy. **Extracellular digestion** occurs in the lumen of the **alimentary canal**.
 - **Mechanical digestion** is the physical breakdown of large food particles into smaller food particles.
 - **Chemical digestion** is the enzymatic cleavage of chemical bonds, such as the peptide bonds of proteins or the glycosidic bonds of starches.
- The pathway of the digestive tract is: oral cavity → pharynx → esophagus → stomach → small intestine → large intestine → rectum
- The **accessory organs of digestion** are the salivary glands, pancreas, liver, and gallbladder.
- The **enteric nervous system** is in the wall of the alimentary canal and controls peristalsis. Its activity is upregulated by the parasympathetic nervous system and downregulated by the sympathetic nervous system.

Ingestion and Digestion

- Multiple hormones regulate feeding behavior, including antidiuretic hormone (ADH or vasopressin) and aldosterone, which promote thirst; glucagon and ghrelin, which promote hunger; and leptin and

cholecystokinin, which promote satiety.

- In the **oral cavity**, **mastication** starts the mechanical digestion of food, while **salivary amylase** and **lipase** start the chemical digestion of food. Food is formed into a **bolus** and swallowed.
- The **pharynx** connects the mouth and posterior nasal cavity to the esophagus.
- The **esophagus** propels food to the stomach using peristalsis. Food enters the stomach through the **lower esophageal (cardiac) sphincter**.
- The stomach has four parts: **fundus**, **body**, **antrum**, and **pylorus**. The stomach has a **lesser** and **greater curvature** and is thrown into folds called **rugae**. Numerous secretory cells line the stomach.
 - **Mucous cells** produce bicarbonate-rich mucus to protect the stomach.
 - **Chief cells** secrete **pepsinogen**, a protease activated by the acidic environment of the stomach.
 - **Parietal cells** secrete hydrochloric acid and **intrinsic factor**, which is needed for vitamin B₁₂ absorption.
 - **G-cells** secrete **gastrin**, a peptide hormone that increases HCl secretion and gastric motility.
- After mechanical and chemical digestion in the stomach, the food particles are now called **chyme**. Food passes into the duodenum through the **pyloric sphincter**.
- The **duodenum** is the first part of the small intestine and is primarily involved in chemical digestion.
 - **Disaccharidases** are brush-border enzymes that break down maltose, isomaltose, lactose, and sucrose into monosaccharides.
 - Brush-border **peptidases** include **aminopeptidase** and **dipeptidases**.

- **Enteropeptidase** activates trypsinogen and procarboxypeptidases, initiating an activation cascade.
- **Secretin** stimulates the release of pancreatic juices into the digestive tract and slows motility.
- **Cholecystokinin** stimulates bile release from the gallbladder, release of pancreatic juices, and satiety.

Accessory Organs of Digestion

- **Acinar cells** in the pancreas produce pancreatic juices that contain bicarbonate, **pancreatic amylase**, pancreatic peptidases (**trypsinogen**, **chymotrypsinogen**, **carboxypeptidases A and B**), and **pancreatic lipase**.
- The **liver** synthesizes **bile**, which can be stored in the gallbladder or secreted into the duodenum directly.
 - Bile emulsifies fats, making them soluble and increasing their surface area.
 - The main components of bile are **bile salts**, pigments (especially **bilirubin** from the breakdown of hemoglobin), and cholesterol.
- The liver also processes nutrients (through glycogenesis and glycogenolysis, storage and mobilization of fats, and gluconeogenesis), produces urea, detoxifies chemicals, activates or inactivates medications, produces bile, and synthesizes albumin and clotting factors.
- The **gallbladder** stores and concentrates bile.

Absorption and Defecation

- The **jejunum** and **ileum** of the small intestine are primarily involved in absorption.
 - The small intestine is lined with **villi**, which are covered with **microvilli**, increasing the surface area available for absorption.
 - Villi contain a capillary bed and a **lacteal**, a vessel of the lymphatic system.
 - Water-soluble compounds, such as monosaccharides, amino acids, water-soluble vitamins, small fatty acids, and water, enter the capillary bed.
 - Fat-soluble compounds, such as fats, cholesterol, and fat-soluble vitamins, enter the lacteal.
- The **large intestine** absorbs water and salts, forming semisolid feces.
 - The **cecum** is an outpocketing that accepts fluid from the small intestine through the **ileocecal valve** and is the site of attachment of the **appendix**.
 - The **colon** is divided into ascending, transverse, descending, and sigmoid portions.
 - The **rectum** stores feces, which are then excreted through the **anus**.
 - Gut bacteria produce vitamin K and biotin (vitamin B₇).

Answers to Concept Checks

9.1

- .. Mechanical digestion, such as chewing, physically breaks down food into smaller pieces. Chemical digestion involves hydrolysis of bonds and breakdown of food into smaller biomolecules.
- 1. Oral cavity (mouth) → pharynx → esophagus → stomach → small intestine → large intestine → rectum → anus
- 2. The parasympathetic nervous system increases secretions from all of the glands of the digestive system and promotes peristalsis. The sympathetic nervous system slows peristalsis.

9.2

- .. Saliva contains salivary amylase (ptyalin), which digests starch into smaller sugars (maltose and dextrin); and lipase, which digests fats.

1.	Cell	Secretions	Functions
	Mucous cell	Mucus	Protects lining of stomach, increases pH (bicarbonate)
	Chief cell	Pepsinogen	Digests proteins, activated by H^+
	Parietal cell	HCl, intrinsic factor	HCl: decreases pH, kills microbes, denatures proteins, carries out some chemical digestion; intrinsic factor: absorption of vitamin B ₁₂
	G-cell	Gastrin	Increases HCl production, increases gastric motility

j.

Substance	Enzyme or Hormone?	Functions
Sucrase	Enzyme	Brush-border enzyme; breaks down sucrose into monosaccharides
Secretin	Hormone	Increases pancreatic secretions, especially bicarbonate; reduces HCl secretion; decreases motility
Dipeptidase	Enzyme	Brush-border enzyme; breaks down dipeptides into free amino acids
Cholecystokinin	Hormone	Recruits secretions from gallbladder and pancreas; promotes satiety
Enteropeptidase	Enzyme	Activates trypsinogen, which initiates an activation cascade

- k. Bile accomplishes mechanical digestion of fats, emulsifying them and increasing their surface area. Pancreatic lipase accomplishes chemical digestion of fats, breaking their ester bonds.

9.3

- .. Carbohydrates: pancreatic amylase; proteins: trypsin, chymotrypsin, carboxypeptidases A and B; fats: pancreatic lipase
- 1. Bile is composed of bile salts (amphipathic molecules derived from cholesterol that emulsify fats), pigments (especially bilirubin from the breakdown of hemoglobin), and cholesterol.
- j. Bile is synthesized in the liver, stored in the gallbladder, and serves its function in the duodenum.
- k. The liver processes nutrients (through glycogenesis and glycogenolysis, storage and mobilization of fats, and gluconeogenesis), produces urea,

detoxifies chemicals, activates or inactivates medications, produces bile, and synthesizes albumin and clotting factors.

- i. As outgrowths of the gut tube, the accessory organs of digestion arise from embryonic endoderm.

9.4

- .. The two circulatory vessels are capillaries and lacteals. The capillary absorbs water-soluble nutrients like monosaccharides, amino acids, short-chain fatty acids, water-soluble vitamins, and water itself. The lacteal absorbs fat-soluble nutrients, like fats, cholesterol, and fat-soluble vitamins.
- !. The fat-soluble vitamins are A, D, E, and K.
- }. The small intestine consists of the duodenum, jejunum, and ileum. The large intestine consists of the cecum, colon, and rectum.
- l. While the large intestine's main function is to absorb water, the small intestine actually absorbs a much larger volume of water. Thus, massive volumes of watery diarrhea are more likely to arise from infections in the small intestine than in the large intestine.



Shared Concepts

Biochemistry Chapter 2

Enzymes

Biochemistry Chapter 9

Carbohydrate Metabolism I

Biochemistry Chapter 11

Lipid and Amino Acid Metabolism

Biology Chapter 5

The Endocrine System

Biology Chapter 7

The Cardiovascular System

Biology Chapter 8

The Immune System

Discrete Practice Questions

1. Which of the following associations correctly matches a gastric cell with a compound it secretes?
 - (A) G-cells—HCl
 - (B) Chief cells—pepsinogen
 - (C) Parietal cells—alkaline mucus
 - (D) Mucous cells—intrinsic factor
2. Which of the following is NOT part of the small intestine?
 - (A) Ileum
 - (B) Cecum
 - (C) Jejunum
 - (D) Duodenum
3. In an experiment, enteropeptidase secretion was blocked. As a direct result, levels of all of the following active enzymes would likely be affected EXCEPT:
 - (A) trypsin.
 - (B) aminopeptidase.
 - (C) chymotrypsin.

(D) carboxypeptidase A.

4. Which of the following INCORRECTLY pairs a digestive enzyme with its function?

- (A) Trypsin—hydrolyzes specific peptide bonds
- (B) Lactase—hydrolyzes lactose to glucose and galactose
- (C) Pancreatic amylase—hydrolyzes starch to maltose
- (D) Lipase—emulsifies fats

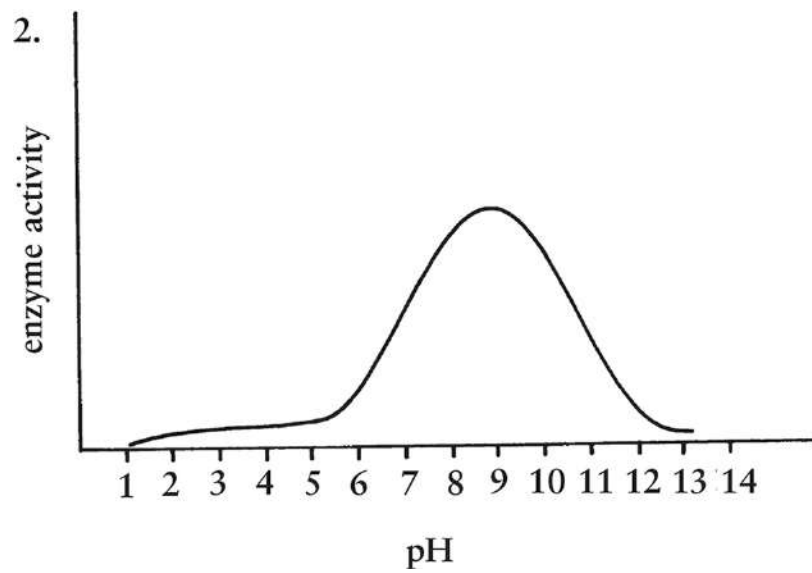
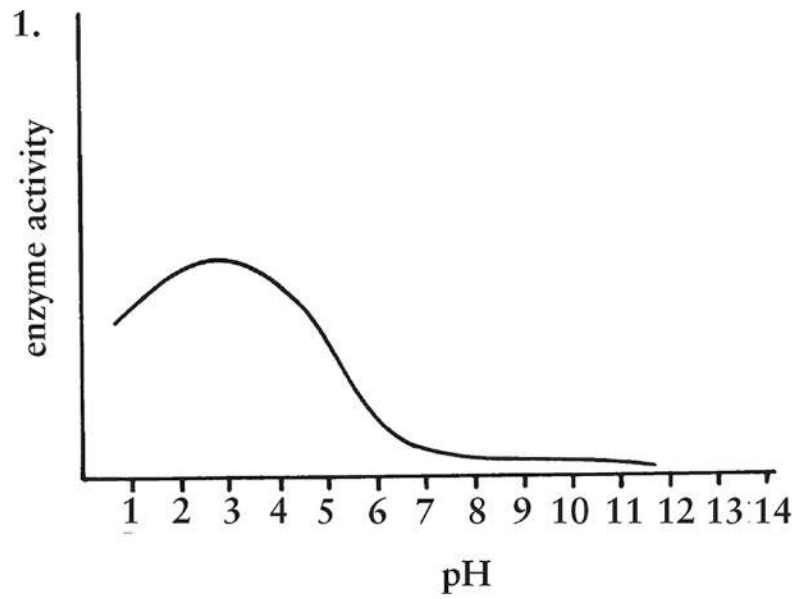
5. Which of the following correctly lists two organs in which proteins are digested?

- (A) Mouth and stomach
- (B) Stomach and large intestine
- (C) Stomach and small intestine
- (D) Small intestine and large intestine

6. Which of the following choices INCORRECTLY pairs a digestive enzyme with its site of secretion?

- (A) Sucrase—salivary glands
- (B) Carboxypeptidase—pancreas
- (C) Trypsin—pancreas
- (D) Lactase—duodenum

7. A two-week-old male infant is brought to the emergency room. His mother reports that he has been unable to keep any milk down; shortly after he nurses, he has sudden projectile vomiting. During exam, an olive-shaped mass can be felt in his upper abdomen. It is determined that there is a constriction in the digestive system that prevents food from reaching the small intestine from the stomach. Which structure is most likely the site of the problem?
- (A) Cardiac sphincter
 - (B) Pyloric sphincter
 - (C) Ileocecal valve
 - (D) Internal anal sphincter
8. Many medications have anticholinergic side effects, which block the activity of parasympathetic neurons throughout the body. Older individuals may be on many such medications simultaneously, exacerbating the side effects. Which of the following would NOT be expected in an individual taking medications with anticholinergic activity?
- (A) Dry mouth
 - (B) Diarrhea
 - (C) Slow gastric emptying
 - (D) Decreased gastric acid production
9. The two graphs below show the relative activities of two enzymes in solutions of varying pH. Which of the following choices correctly identifies the two enzymes?



- (A) 1—chymotrypsin; 2—pepsin
- (B) 1—pepsin; 2—carboxypeptidase B
- (C) 1—lactase; 2—aminopeptidase
- (D) 1—enteropeptidase; 2—amylase

10. Which of the following would NOT likely lead to elevated levels of bilirubin in the blood?

- (A) Cholangiocarcinoma, a cancer of the bile ducts that can ultimately lead to full occlusion of the duct lumen
- (B) Autoimmune hemolytic anemia, a disease in which the red blood cells are attacked by antibodies and are lysed
- (C) Ménétrier's disease, in which rugae thicken and overlying glands lose secretory ability
- (D) Acetaminophen (Tylenol) overdose, in which the accumulation of toxic metabolites can cause rapid liver failure

11. Which of the following correctly pairs the molecule with its primary site of absorption?

- (A) Chylomicrons—lacteals
- (B) Amino acids—large intestine
- (C) Vitamins A and E—stomach
- (D) Cholesterol—ascending colon

12. Starch is hydrolyzed into maltose by enzymes from the:

- I. salivary glands.
- II. brush border.
- III. pancreas.

- (A) I only
- (B) I and III only

- (C) II and III only
- (D) I, II, and III

13. Which of the following biomolecules does NOT drain into the liver before arriving at the right side of the heart?

- (A) Cholecalciferol (vitamin D)
- (B) Threonine (an amino acid)
- (C) Fructose (a monosaccharide)
- (D) Pantothenic acid (vitamin B₅)

14. Which of the following hormones increases feeding behavior?

- (A) Leptin
- (B) Cholecystokinin
- (C) Ghrelin
- (D) Gastrin

15. Which of the following is likely to be seen in a patient with liver failure?

- (A) High concentrations of urea in the blood
- (B) High concentrations of albumin in the blood
- (C) Low concentrations of ammonia in the blood
- (D) Low concentrations of clotting factors in the blood

Discrete Practice Answers

1. **B**

Chief cells secrete pepsinogen, a protease secreted as a zymogen that is activated by the acidic environment of the stomach. G-cells secrete gastrin, parietal cells secrete hydrochloric acid and intrinsic factor, and mucous cells secrete alkaline mucus, eliminating the other answer choices.

2. **B**

The small intestine is divided into three sections: the duodenum, the jejunum, and the ileum. The cecum is part of the large intestine, making **(B)** the correct answer.

3. **B**

Aminopeptidase is a brush-border peptidase secreted by the cells lining the duodenum; it does not require enteropeptidase for activation. Both trypsinogen and procarboxypeptidases A and B are activated by enteropeptidase, eliminating **(A)** and **(D)**. Once activated, trypsin can activate chymotrypsinogen; if trypsinogen cannot be activated, then chymotrypsinogen will not be activated either, eliminating **(C)**.

4. **D**

Lipase is involved in the digestion of fats, but its function is not to emulsify fats—this is the job of bile. Rather, lipase chemically digests fats in the duodenum, allowing them to be brought into duodenal cells and packaged into chylomicrons. The other associations given here are all correct.

5. **C**

Protein digestion begins in the stomach, where pepsin (secreted as pepsinogen) hydrolyzes specific peptide bonds. Protein digestion continues in the small intestine as trypsin (secreted as trypsinogen), chymotrypsin (secreted as chymotrypsinogen), carboxypeptidases A and B (secreted as procarboxypeptidases A and B), aminopeptidase, and dipeptidases hydrolyze specific parts of the peptide. No protein digestion occurs in the mouth or large intestine.

6. **A**

Sucrase is a brush-border enzyme found on duodenal cells and is not secreted by the salivary glands. This enzyme hydrolyzes sucrose (a disaccharide) to form glucose and fructose (monosaccharides). The other associations are all correct.

7. **B**

The question is basically asking us to identify the structure that lies between the stomach and the small intestine. This is the pyloric sphincter; the presentation given in the question is a classic example of what is called pyloric stenosis, in which the pyloric sphincter is thickened and cannot relax to permit chyme through. The cardiac sphincter, **(A)**, lies between the esophagus and the stomach. The ileocecal valve, **(C)**, lies between the ileum of the small intestine and the cecum of the large intestine. The internal anal sphincter, **(D)**, lies at the end of the rectum.

8. **B**

The parasympathetic nervous system has many roles in the digestive system. It promotes motility of the gut tube and secretion from glands. Therefore, blocking the parasympathetic nervous system would likely result in dry mouth (from reduced secretion of saliva), slow gastric

emptying (from decreased peristalsis), and decreased gastric acid production (from reduced HCl secretion from the parietal cells in the gastric glands), eliminating **(A)**, **(C)**, and **(D)**. **(B)** is the correct answer because we would expect constipation in such an individual, not diarrhea: slowed motility through the colon would lead to increased water reabsorption, making the feces too firm and causing constipation.

9. **B**

The first graph shows maximal activity at a very acidic pH, implying that this is an enzyme acting in the stomach. The second graph shows maximal activity around pH 8.5, implying that this is an enzyme acting in the duodenum. The only choice that matches the first graph with a stomach enzyme (pepsin) and the second with a duodenal enzyme (carboxypeptidase B) is **(B)**.

10. **C**

Elevated bilirubin implies a blockage to bile flow, increased production of bilirubin (from massive hemoglobin release), or an inability of the liver to produce bile. If the bile duct were occluded, as in **(A)**, then bile would not be able to flow into the digestive tract and would build up, increasing bilirubin levels in the blood. If many red blood cells were lysed, as in **(B)**, then bilirubin levels would rise in accordance with the increased hemoglobin release. If liver failure occurred, as in **(D)**, then the liver would be unable to produce bile, and bilirubin would again build up. **(C)** refers to a pathology in the stomach—the key word given here is rugae, which are the folds in the stomach wall. Lack of gastric function would have no effect on bilirubin levels, making this the correct choice.

11. **A**

Chylomicrons contain triacylglycerols, cholesteryl esters, and fat-soluble

vitamins and are secreted by intestinal cells into lacteals. Amino acids, fat-soluble vitamins (like vitamins A and E), and cholesterol are all absorbed in the small intestine.

2. **B**

Starch is hydrolyzed to maltose by two enzymes: salivary amylase (secreted by the salivary glands) in the mouth and pancreatic amylase (secreted by the pancreas) in the duodenum. Brush-border disaccharidases can further break down maltose, but do not break down starch.

3. **A**

While the capillaries from the intestine come together to form the portal vein, which drains to the liver, the lacteals come together to form the thoracic duct, which drains directly into the left subclavian vein. Therefore, fat-soluble compounds do not pass through the liver before reaching the right side of the heart. Only **(A)**, vitamin D, is fat-soluble.

4. **C**

Ghrelin promotes a sensation of hunger, increasing feeding behavior. Both leptin and cholecystokinin promote satiety, decreasing feeding behavior—eliminating **(A)** and **(B)**. Gastrin increases acid production and gastric motility, but does not have any significant relationship with feeding behavior, eliminating **(D)**.

5. **D**

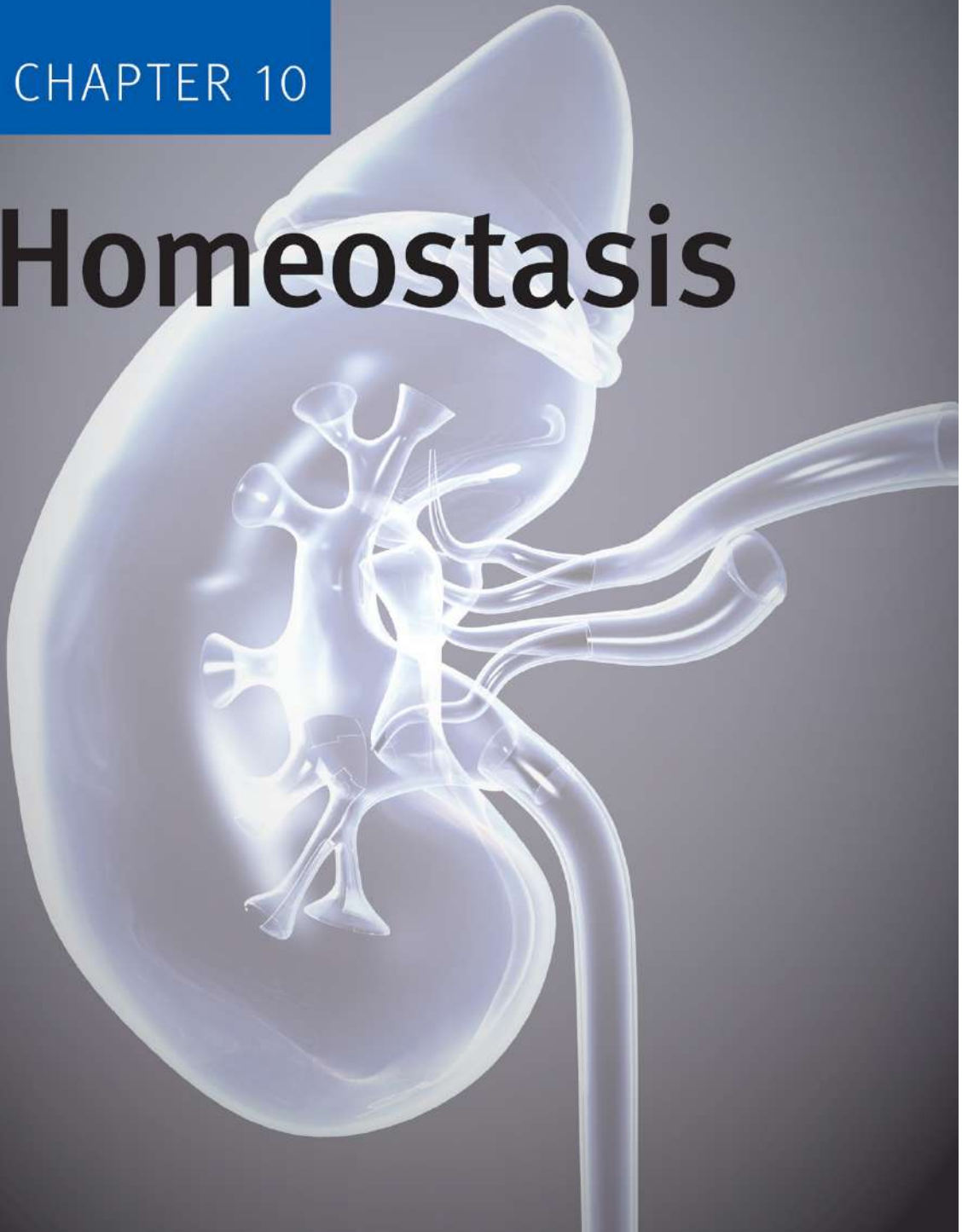
The liver serves many functions, including carrying out metabolic processes (glycogenesis and glycogenolysis, fat storage, and gluconeogenesis), detoxification or activation of medications, and synthesis of bile. Germane to this question are the roles of converting ammonia into urea through the urea cycle and synthesis of proteins,

including albumin and clotting factors. A patient with liver failure would thus not be able to convert ammonia into urea and would have high concentrations of ammonia and low concentrations of urea in the blood, eliminating **(A)** and **(C)**. With decreased synthetic activity, both albumin and clotting factor concentrations would be low, eliminating **(B)** and making **(D)** the correct answer.



CHAPTER 10

Homeostasis



CHAPTER 10

Homeostasis

In This Chapter

10.1 The Excretory System

Anatomy of the Excretory System

Osmoregulation

Nephron Function

Functions of the Excretory System

10.2 Skin

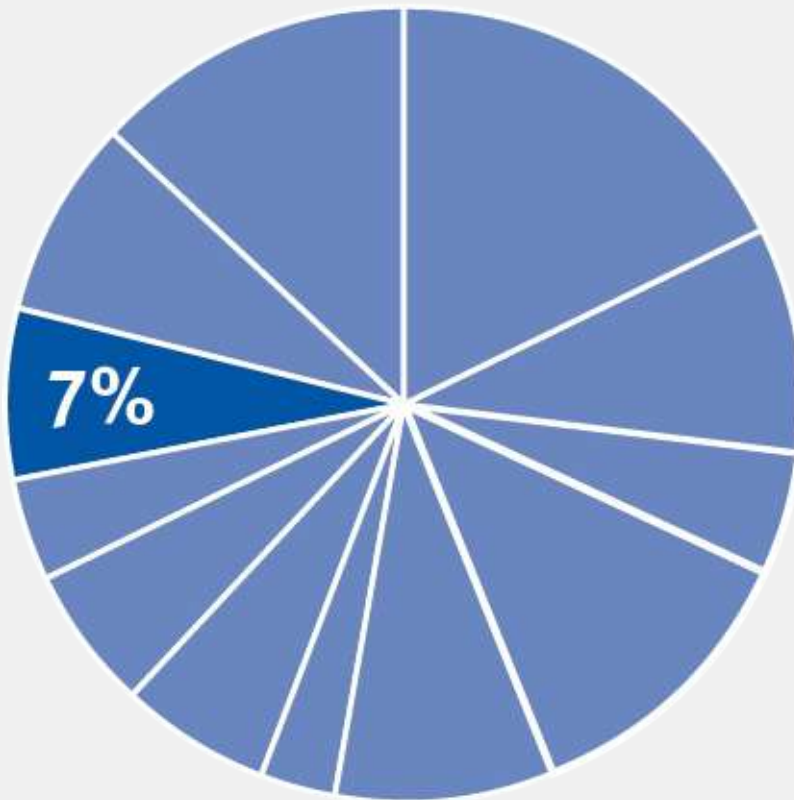
Structure

Thermoregulation

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 7% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems

Introduction

Have a headache? Pop an ibuprofen. Backache? Works for that, too.

Ibuprofen, which has been around for over 50 years, is an inexpensive, over-the-counter, nonsteroidal anti-inflammatory drug (NSAID). Ibuprofen is known as an analgesic, or pain reliever. When used in moderation, ibuprofen carries relatively little risk. However, taking multiple doses for many years can profoundly affect the kidneys. Years of analgesic use (usually as self-therapy) can lead to kidney failure, known as analgesic nephropathy. If untreated, kidney failure is universally fatal.

If kidney failure is detected, however, dialysis (or a kidney transplant) could save an individual's life. How does dialysis save a patient with kidney failure? Dialyzing fluid has many of the same solutes as blood, in strategic concentrations, and it is kept separate from blood by a semipermeable membrane. As blood is filtered through the dialysis machine, fluid and solutes diffuse down their concentration gradients, limited only by size (as determined by the membrane). The dialysis machine therefore performs filtration to purify the blood and excrete wastes, a crucial function that the kidneys would normally perform.

In this chapter, we'll learn more about filtration, as well as reabsorption and secretion. These processes are collectively responsible for osmoregulation. Osmoregulation is just one mechanism the body uses to maintain homeostasis in its fluids and tissues. We'll also discuss the skin, which plays a significant role in temperature homeostasis (thermoregulation).

10.1 The Excretory System

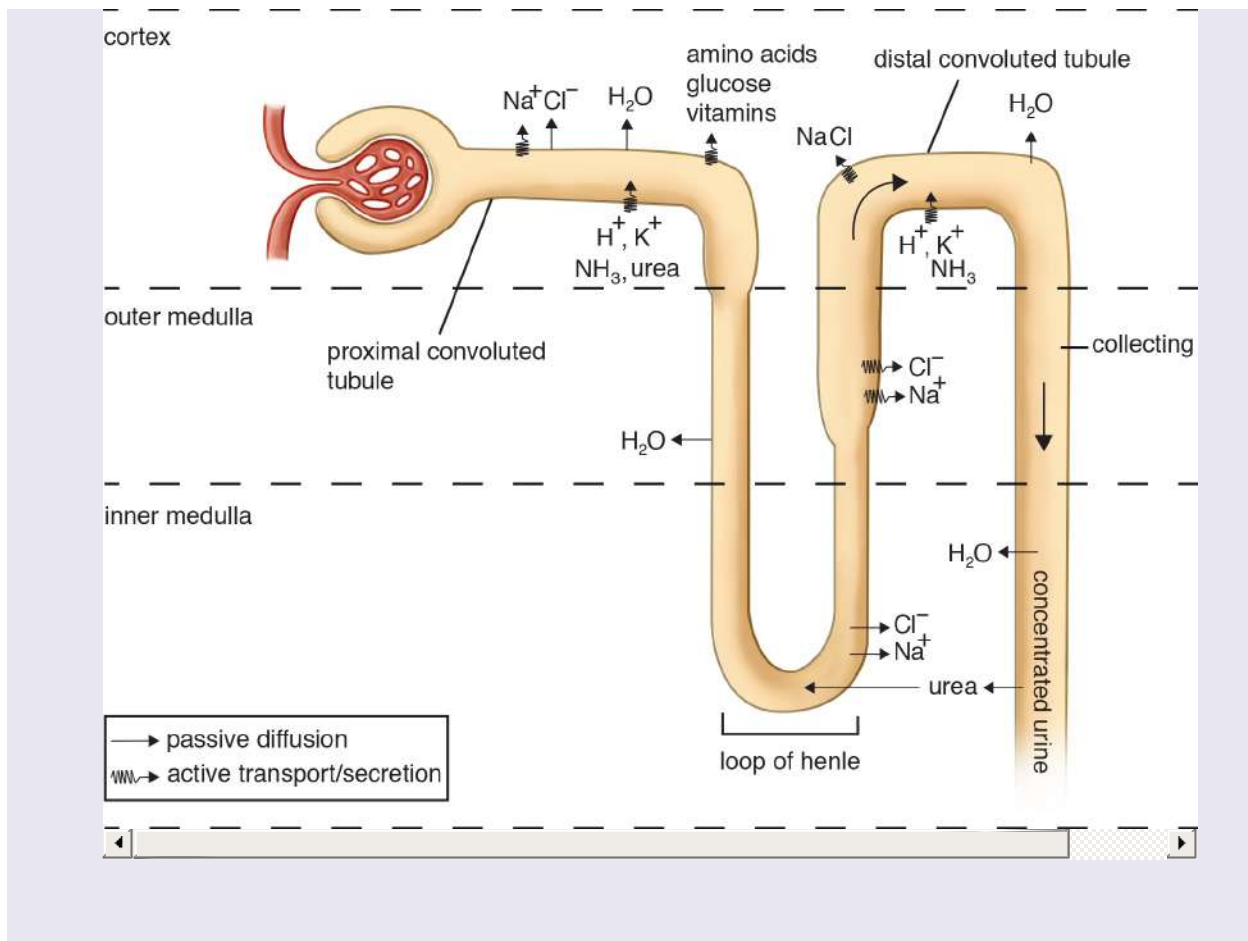
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LEARNING OBJECTIVES

After Chapter 10.1, you will be able to:

- List the structures of the secretory pathway
- Order the vessels in the renal vascular pathway
- Identify the nervous system components used to control the detrusor muscle
- Describe the processes by which components of the kidney are able to exchange solutes between the filtrate and the blood
- Identify the function(s) of each segment of the nephron:



The **excretory system** serves many functions, including the regulation of blood pressure, blood osmolarity, acid–base balance, and removal of nitrogenous wastes. The kidneys play an essential role in these functions.

ANATOMY OF THE EXCRETORY SYSTEM

The excretory system consists of the kidneys, ureters, bladder, and urethra, as shown in Figure 10.1. The kidneys are two bean-shaped structures located behind the digestive organs at the level of the bottom rib. The functional unit of the kidney is the nephron; each kidney has approximately 1 million nephrons. All of the nephrons eventually empty into the renal pelvis, which narrows to form the the ureter. Urine travels through the ureter to the

bladder. From the bladder, urine is transported through the urethra to exit the body.

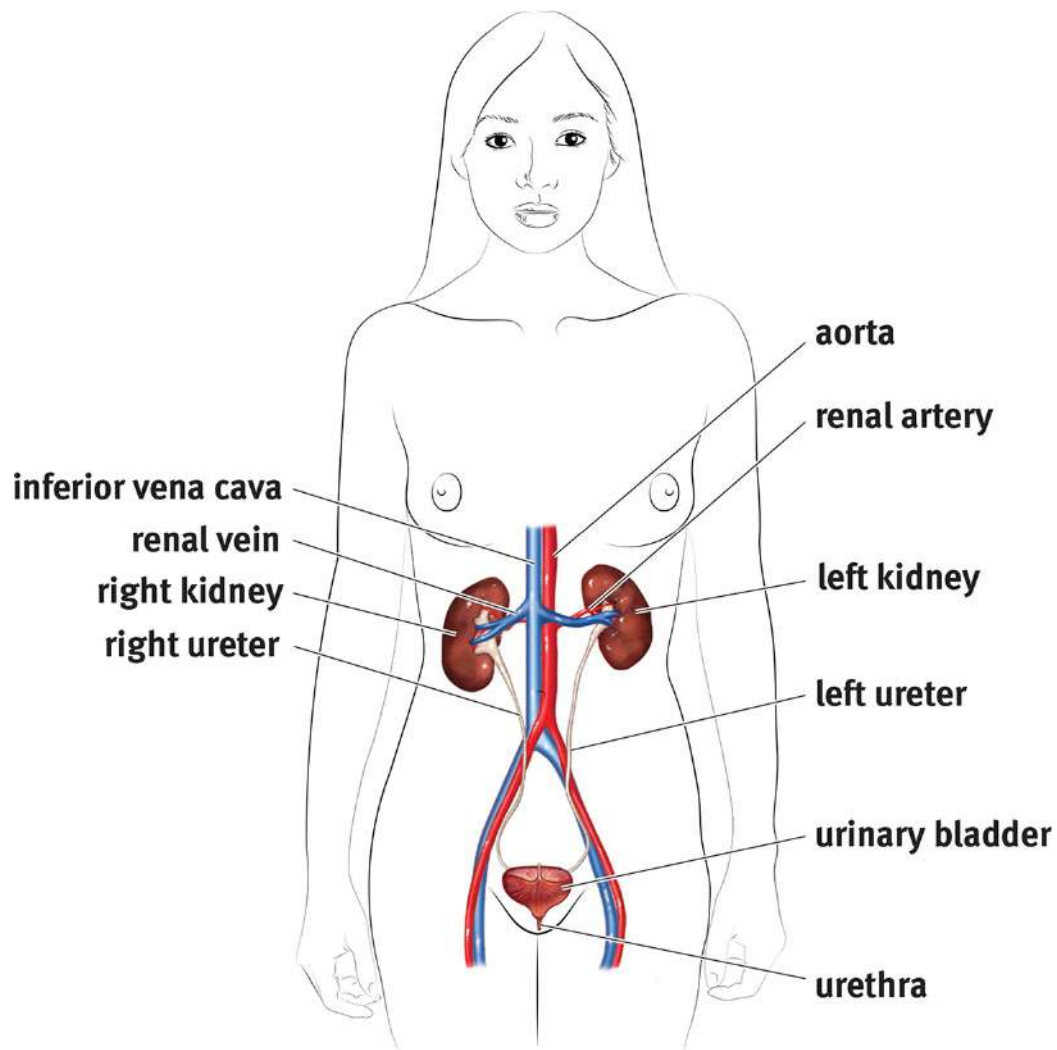


Figure 10.1. The Excretory System

Once it leaves the kidneys, urine moves through the ureters to be stored in the urinary bladder until it is excreted through the urethra.

Kidney Structure

Each kidney is subdivided into a cortex and a medulla, as shown in Figure 10.2. The **cortex** is the kidney's outermost layer, while the **medulla** of the kidney sits within the cortex. Each kidney also has a renal **hilum**, which is a deep slit in the center of its medial surface. The widest part of the ureter, the **renal pelvis**, spans almost the entire width of the renal hilum. The **renal artery**, **renal vein**, and **ureter** enter and exit through the renal hilum.

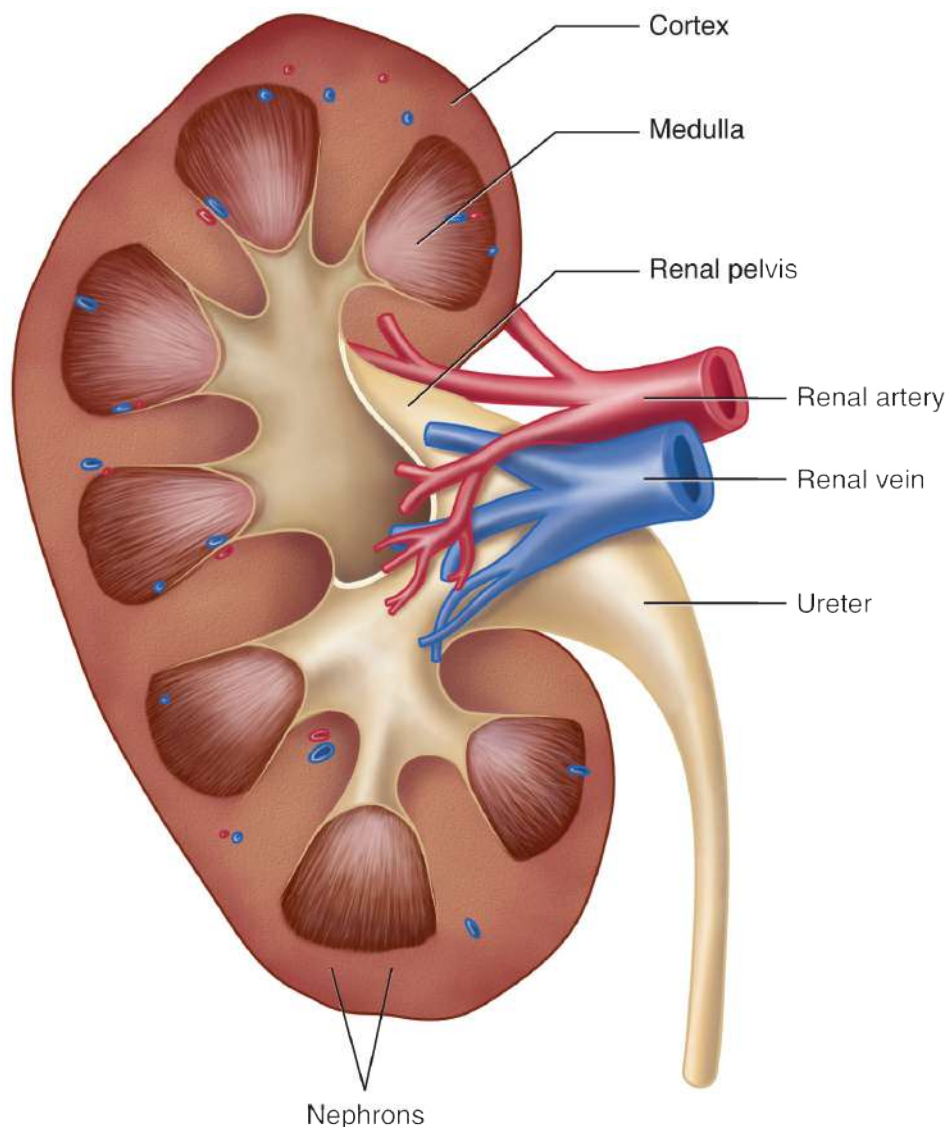


Figure 10.2. Gross Anatomy of the Kidney

The kidney has one of the few portal systems in the body. A **portal system** consists of two capillary beds in series through which blood must travel before returning to the heart. The renal artery branches out, passes through the medulla, and enters the cortex as **afferent arterioles**. The highly convoluted capillary tufts derived from these afferent arterioles are known as **glomeruli**. After blood passes through a glomerulus, the **efferent arterioles** then form a second capillary bed. These capillaries surround the loop of Henle and are known as **vasa recta**. The renal vascular system is shown in Figure 10.3.



BRIDGE

The portal system in the kidney is one of three major portal systems in the body. The other two are the hypophyseal portal system, which connects the hypothalamus and the anterior pituitary (discussed in Chapter 5 of *MCAT Biology Review*), and the hepatic portal system, which connects the gut tube and the liver (discussed in Chapter 9 of *MCAT Biology Review*). In each case, blood passes through two capillary beds before returning to the heart.

Also visible in Figure 10.3 is the structure of the nephron. Around the glomerulus is a cup-like structure known as **Bowman's capsule**. Bowman's capsule leads to a long tubule with many distinct areas; in order, these are the proximal convoluted tubule, descending and ascending limbs of the Loop of Henle, distal convoluted tubule, and collecting duct. The kidney's ability to excrete waste is intricately tied to the specific placement of these structures and their physiology.

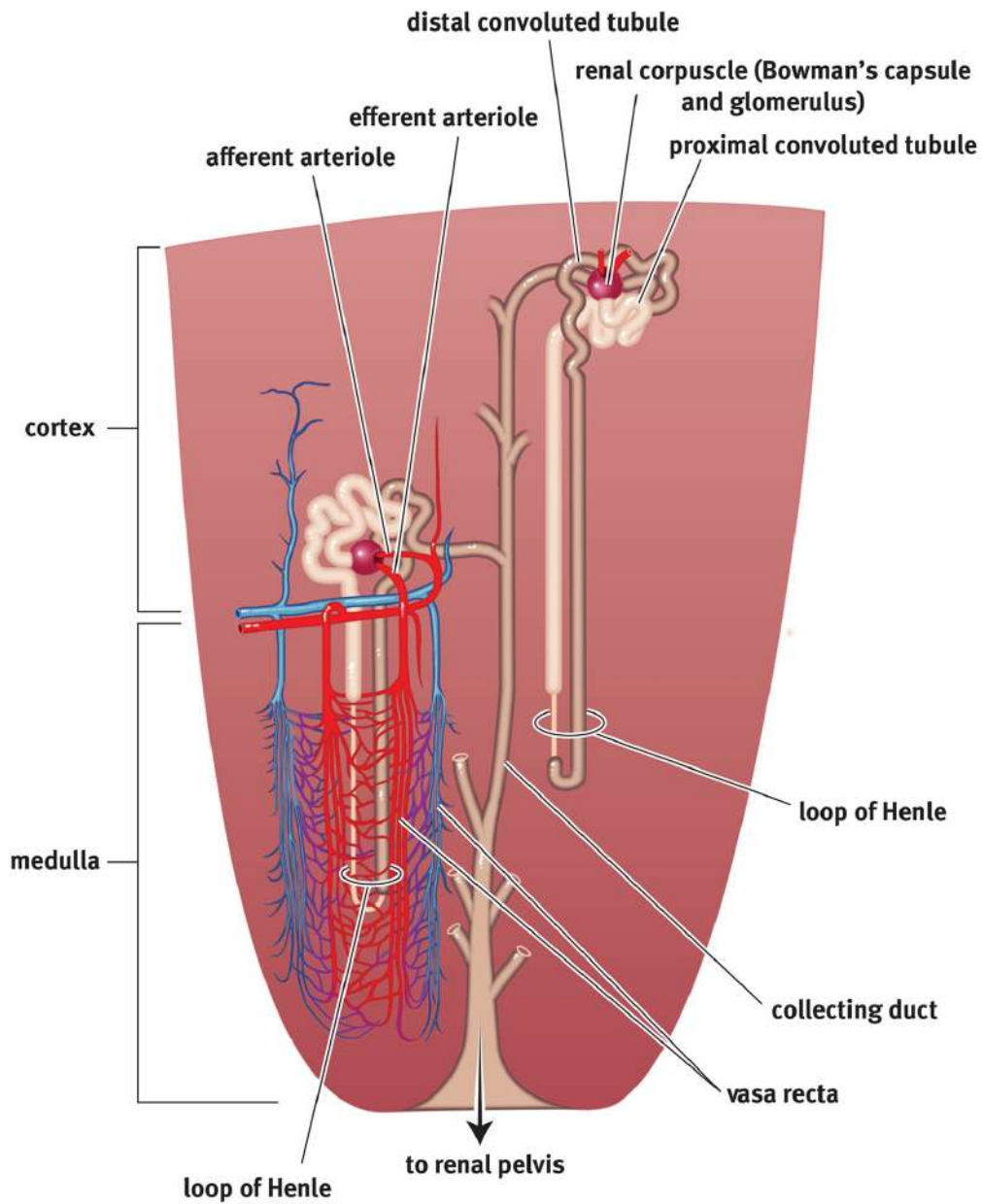


Figure 10.3. Microanatomy of the Renal Vascular System and the Nephron



Our discussion of neurons in Chapter 4 of *MCAT Biology Review* used some of the same terms as those that describe the organization of blood vessels in the kidney. Afferent neurons carry sensory information toward the central nervous system much as afferent arterioles carry blood toward the glomeruli. Efferent neurons relay signals away from the central nervous system just as efferent arterioles carry blood away from the glomerulus.

Bladder Structure

The bladder has a muscular lining known as the **detrusor muscle**. Parasympathetic activity causes the detrusor muscle to contract. However, in order to leave the body, urine must pass through two sphincters—the internal and external urethral sphincters. The **internal urethral sphincter**, consisting of smooth muscle, is contracted in its normal state. Because the internal sphincter is made of smooth muscle, it is under involuntary control. The **external urethral sphincter** consists of skeletal muscle and is under voluntary control. When the bladder is full, stretch receptors convey to the nervous system that the bladder requires emptying. This causes parasympathetic neurons to fire, and the detrusor muscle contracts. This contraction also causes the internal sphincter to relax. This reflex is known as the **micturition reflex**. The next step is up to the individual. The person can choose to relax the external sphincter to urinate, or can maintain the tone of the external sphincter to prevent urination. This can cause a few moments of discomfort, but the reflex usually dissipates in a few minutes. However, if the bladder is not emptied, then the process will begin anew shortly thereafter. Urination itself is facilitated by the contraction of the abdominal musculature,

which increases pressure within the abdominal cavity, resulting in compression of the bladder and increased urine flow rate.

OSMOREGULATION

The kidney filters the blood to form urine. The composition and quantity of urine is determined by the present state of the body. For example, if blood volume is low and blood osmolarity is high, then it is most beneficial to the body to maximally retain water. This results in low-volume, highly concentrated urine. Likewise, a patient receiving large amounts of intravenous fluids is likely to produce a larger volume of less concentrated urine. Thus, the primary job of the kidneys is to regulate blood volume and osmolarity. In order to do this, kidney function may be divided into three different processes: filtration, secretion, and reabsorption.

Filtration

The nephron's first function is **filtration**. In the kidneys, approximately 20 percent of the blood that passes through the glomerulus is filtered as fluid into Bowman's space. The collected fluid is known as the **filtrate**. The movement of fluid into Bowman's space is governed by **Starling forces**, which account for the pressure differentials in both hydrostatic and oncotic pressures between the blood and Bowman's space, as shown in Figure 10.4. The hydrostatic pressure in the glomerulus is significantly higher than that in Bowman's space, which causes fluid to move into the nephron. On the other hand, the osmolarity of blood is higher than that of Bowman's space, resulting in pressure opposing the movement of fluid into the nephron. However, the hydrostatic pressure is much larger than the oncotic pressure, so the net flow is still from blood into the nephron.

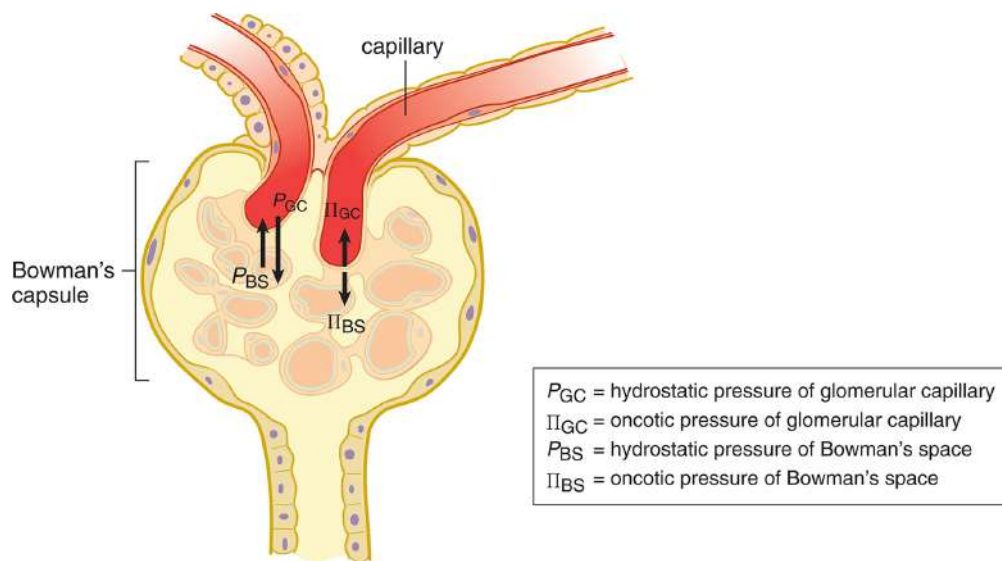


Figure 10.4. Starling Forces

The relative hydrostatic and oncotic pressure gradients determine the direction and rate of filtration.



BRIDGE

Like the electromotive force discussed in Chapter 6 of *MCAT Physics and Math Review*, Starling forces are really a misnomer. Rather than forces, it is a pressure differential that causes the net movement of fluid from the glomerulus to Bowman's space. Pressure is discussed in Chapter 4 of *MCAT Physics and Math Review*.

Under most circumstances, fluid will flow from the glomerulus into Bowman's space. However, various pathologies can cause derangements of this flow. Consider what might happen if the ureter was obstructed by a kidney stone. An obstruction would result in a buildup of urine behind the stone. Eventually, enough fluid will build up and cause distention of the renal pelvis and the

nephrons. What will happen to filtration in this case? The hydrostatic pressure in Bowman's space would increase to the point that filtration could no longer occur because there would be excessive pressure opposing movement of fluid into the nephron.

The filtrate is similar in composition to blood but does not contain cells or proteins due to the filter's ability to select based on size. In other words, molecules or cells that are larger than glomerular pores will remain in the blood. As described earlier, the blood remaining in the glomerulus then travels into the efferent arterioles, which empty into the vasa recta. The filtrate is isotonic to blood so that neither the capsule nor the capillaries swell. Our kidneys filter about 180 liters per day, which is approximately 36 times our blood volume. This means that the entire volume of a person's blood is filtered about every 40 minutes.



KEY CONCEPT

Imagine that the glomerulus is like a sieve or colander. Small molecules dissolved in the blood will pass through the tiny pores (such as glucose, which is later reabsorbed), whereas large molecules such as proteins and blood cells will not. If blood cells or proteins are found in the urine, this indicates a health problem at the level of the glomerulus.

Secretion

In addition to filtering blood, the nephrons are able to **secrete** salts, acids, bases, and urea directly into the tubule by either active or passive transport. The quantity and identity of the substances secreted into the nephron are directly related to the needs of the body at that time. For example, a diet heavy in meat results in the intake of large amounts of protein, which contains a significant amount of nitrogen. Ammonia (NH_3) is a byproduct of the metabolism of nitrogen-containing compounds and, as a base, can disturb the pH of blood and cells. The liver converts the ammonia to **urea**, a neutral compound, which travels to the kidney and is secreted into the nephron for excretion in the urine. The kidneys are capable of eliminating ions or other substances when present in relative excess in the blood, such as potassium cations, hydrogen ions, or metabolites of medications. Secretion is also a mechanism for excreting wastes that are simply too large to pass through glomerular pores.

Reabsorption

Some compounds that are filtered or secreted may be taken back up for use via **reabsorption**. Certain substances are almost always reabsorbed, such as glucose, amino acids, and vitamins. In addition, hormones such as antidiuretic hormone (ADH or vasopressin) and aldosterone can alter the quantity of water reabsorbed within the kidney in order to maintain blood pressure.



KEY CONCEPT

- Filtration: movement of solutes from blood to filtrate at Bowman's capsule

- Secretion: movement of solutes from blood to filtrate anywhere besides Bowman's capsule
- Reabsorption: movement of solutes from filtrate to blood

NEPHRON FUNCTION

The kidney uses mechanisms such as filtration, secretion, and reabsorption to produce urine and to regulate the blood volume and osmolarity. However, the function of the nephron isn't quite that simple. In fact, renal physiology is often considered one of the most difficult topics covered in medical school.

In order to simplify this topic, it is important to understand that the kidney has two main goals: keep what the body needs and lose what it doesn't, and concentrate the urine to conserve water. The kidney allows the human body to reabsorb certain materials for reuse, while also selectively eliminating waste. For example, glucose and amino acids are not usually present in the urine because the kidney is able to reabsorb these substances for later use. On the contrary, waste products like hydrogen and potassium ions, ammonia, and urea remain in the filtrate and are excreted. Finally, water is reabsorbed in large quantities in order to maintain blood pressure and adequate hydration.



KEY CONCEPT

Anything that makes it into the filtrate and that is not reabsorbed will be lost from the body in the urine.

In order to understand this complex organ, we will study the nephron piece-by-piece, discussing exactly what is occurring in each segment. Follow along with the nephron diagram shown in Figure 10.5. As a theme, note that segments that are horizontal in the diagram (Bowman's capsule, the proximal convoluted tubule, and the distal convoluted tubule) are primarily focused on the *identity* of the particles in the urine (*keep what the body needs and lose what it doesn't*). In contrast, the segments that are vertical in the diagram (the loop of Henle and collecting duct) are primarily focused on the volume and concentration of the urine (*concentrate the urine to conserve water*).

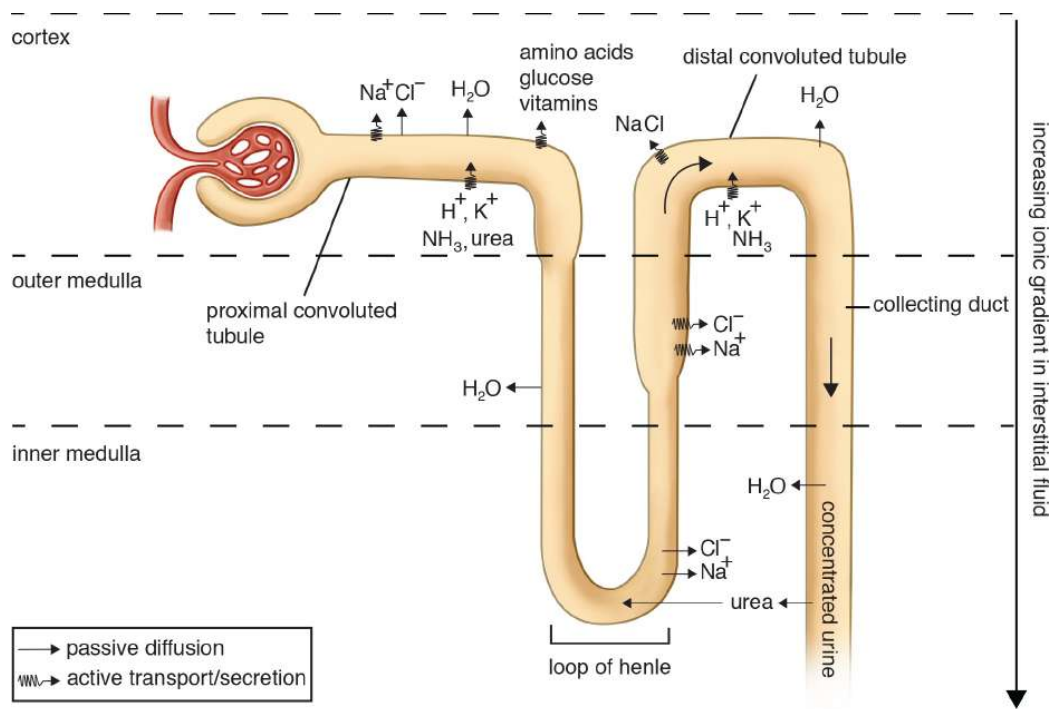


Figure 10.5. Reabsorption and Secretion in the Nephron

Proximal Convoluted Tubule

The filtrate first enters the **proximal convoluted tubule (PCT)**. In this region, amino acids, glucose, water-soluble vitamins, and the majority of salts are

reabsorbed along with water. Almost 70 percent of filtered sodium will be reabsorbed here, but the filtrate remains isotonic to the interstitium, as other solutes and a large volume of water are also reabsorbed. Solutes that enter the **interstitium**—the connective tissue surrounding the nephron—are picked up by the vasa recta to be returned to the bloodstream for reuse within the body. The PCT is also the site of secretion for a number of waste products, including hydrogen ions, potassium ions, ammonia, and urea.



MNEMONIC

Major waste products excreted in the urine: **Dump** the **HUNK**

- H^+
- Urea
- NH_3
- K^+

Loop of Henle

Filtrate from the proximal convoluted tubule then enters the **descending limb of the loop of Henle**, which dives deep into the medulla before turning around to become the **ascending limb of the loop of Henle**. The descending limb is permeable only to water, and the medulla has an ever-increasing osmolarity as the descending limb travels deeper into it. Think for a moment how this would affect the flow of water. As the descending limb traverses deeper into the medulla, the increasing interstitial concentration favors the

outflow of water from the descending limb, which is reabsorbed into the vasa recta.

The kidney is capable of altering the osmolarity of the interstitium. This creates a gradient that, coupled with selective permeability of the nephron, allows maximal reabsorption and conservation of water. In the normal physiological state, the osmolarity in the cortex is approximately the same as that of the blood and remains at that level. Deeper in the medulla, the osmolarity in the interstitium can range from isotonic with blood (when trying to excrete water) to four times as concentrated (when trying to conserve water). If the concentration is the same in the tubule and in the interstitium, there is no driving force (gradient), and the water will be lost in urine. If the interstitium is more concentrated, then water will move out of the tubule, into the interstitium, and eventually back into the blood.

Together, the vasa recta and nephron create a **countercurrent multiplier system**. This means that the flow of filtrate through the loop of Henle is in the opposite direction from the flow of blood through the vasa recta. If the two flowed in the same direction, they would quickly reach equilibrium and the kidney would be unable to reabsorb as much water. By making the two flow in opposite directions, the filtrate is constantly being exposed to hypertonic blood, which allows maximal reabsorption of water.

As the descending limb transitions to become the ascending limb of the loop of Henle, a change in permeability occurs. The ascending limb is only permeable to salts and is impermeable to water. So while the descending limb maximizes water reabsorption by taking advantage of increasing medullary osmolarity, the ascending limb maximizes salt reabsorption by taking advantage of *decreasing* medullary osmolarity.

At the transition from the inner to outer medulla, the loop of Henle becomes thicker in what is termed the **diluting segment**. This is not because the lumen within the tube has enlarged, but because the cells lining the tube are larger. These cells contain large amounts of mitochondria, which allow the reabsorption of sodium and chloride by active transport. Indeed, because so much salt is reabsorbed while water is stuck in the nephron, the filtrate actually becomes hypotonic compared to the interstitium. While we tend to focus on the concentrating abilities of the nephron, this segment is noteworthy because it is the only portion of the nephron that can produce urine that is more dilute than the blood. This is important during periods of *overhydration* and provides a mechanism for eliminating excess water.

At the beginning of the loop of Henle, the filtrate is isotonic to the interstitium. Thus, from the beginning of the loop of Henle to the end, there is a slight degree of dilution. Far more important, however, is the fact that the volume of the filtrate has been significantly reduced, demonstrating a net reabsorption of a large volume of water.

Distal Convoluted Tubule

Next, the filtrate enters the **distal convoluted tubule (DCT)**. The DCT responds to aldosterone, which promotes sodium reabsorption. Because sodium ions are osmotically active particles, water will follow the sodium, concentrating the urine and decreasing its volume. The DCT is also a site of waste product secretion, like the PCT.



REAL WORLD

In certain conditions, such as congestive heart failure, the body accumulates excess fluid in the lungs or peripheral tissues (edema). The judicious use of a diuretic drug can help the body get rid of excess fluid. Diuretics typically inhibit the reabsorption of sodium in one or more regions of the nephron, thereby increasing sodium excretion. As an osmotically active particle, sodium will pull water with it, thereby relieving the body of some of its excess fluid.

Collecting Duct

The final concentration of the urine will depend largely on the permeability of the collecting duct, which is responsive to both aldosterone and antidiuretic hormone (ADH or vasopressin). As permeability of the collecting duct increases, so too does water reabsorption, resulting in further concentration of the urine. The reabsorbed water enters the interstitium and makes its way to the vasa recta, where it reenters the bloodstream to once again become part of the plasma. The collecting duct almost always reabsorbs water, but the amount is variable. When the body is very well hydrated, the collecting duct will be fairly impermeable to salt and water. When in conservation mode, ADH and aldosterone will each act to increase reabsorption of water in the collecting duct, allowing for greater water retention and more concentrated urine output.

Ultimately, anything that is not reabsorbed from the tubule by the end of the collecting duct will be excreted; the collecting duct is the point of no return. After that, there are no further opportunities for reabsorption. As the filtrate leaves the tubule, it collects in the renal pelvis. The fluid, which carries mostly urea, uric acid, and excess ions (sodium, potassium, magnesium, and

calcium), flows through the ureter to the bladder where it is stored until voiding.

FUNCTIONS OF THE EXCRETORY SYSTEM

The kidneys use osmolarity gradients and selective permeability to filter, secrete, and reabsorb materials in the process of making urine. However, these processes have larger implications for the human body as a whole. The selective elimination of water and solutes allows the kidneys, in conjunction with the endocrine, cardiovascular, and respiratory systems, to control blood pressure, blood osmolarity, and acid–base balance.

Blood Pressure

In Chapter 5 of *MCAT Biology Review*, we discussed two hormones that are very important for the maintenance of proper blood pressure: aldosterone and antidiuretic hormone (ADH or vasopressin).

Aldosterone is a steroid hormone that is secreted by the adrenal cortex in response to decreased blood pressure. Decreased blood pressure stimulates the release of **renin** from **juxtaglomerular cells** in the kidney. Renin then cleaves **angiotensinogen**, a liver protein, to form **angiotensin I**. This peptide is then metabolized by **angiotensin-converting enzyme** in the lungs to form **angiotensin II**, which promotes the release of aldosterone from the adrenal cortex.



REAL WORLD

Aldosterone increases blood pressure by increasing the reabsorption of sodium. In medicine, we exploit this characteristic by using drugs that actually block angiotensin-converting enzyme or the angiotensin II receptor. In Chapter 5 of *MCAT Biology Review*, we mentioned the ACE inhibitors, which mostly end with *-pril*. The angiotensin II receptor blockers mostly end with *-sartan* (*losartan*, *valsartan*, *irbesartan*). Blocking this receptor limits aldosterone release, which limits salt and water reabsorption, and therefore results in lowered blood pressure.

Aldosterone works by altering the ability of the distal convoluted tubule and collecting duct to reabsorb sodium. Remember that water does not move on its own, but rather travels down its osmolarity gradient. Thus, if we reabsorb more sodium, water will flow with it. This reabsorption of isotonic fluid has the net effect of increasing blood volume and therefore blood pressure. Aldosterone will also increase potassium and hydrogen ion excretion.

Antidiuretic hormone (ADH), also known as **vasopressin** is a peptide hormone synthesized by the hypothalamus and released by the posterior pituitary in response to high blood osmolarity. It directly alters the permeability of the collecting duct, allowing more water to be reabsorbed by making the cell junctions of the duct leaky. Increased concentration in the interstitium (hypertonic to the filtrate) will then cause the reabsorption of water from the tubule. Alcohol and caffeine both inhibit ADH release and lead to the frequent excretion of dilute urine.



KEY CONCEPT

ADH only governs water reabsorption and thus results in a lower blood osmolarity. Aldosterone causes both salt and water reabsorption and does not change blood osmolarity.

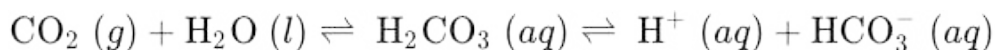
In addition to the kidneys, the cardiovascular system also regulates blood pressure, specifically by vasoconstricting or vasodilating in order to maintain blood pressure. Constriction of the afferent arteriole will lead to a lower pressure of blood reaching the glomeruli, which are adjacent to the juxtaglomerular cells. Therefore, this vasoconstriction will secondarily lead to renin release, which will also help raise blood pressure.

Osmoregulation

The osmolarity of the blood must be tightly controlled to ensure correct oncotic pressures within the vasculature. A note on terminology: **osmotic pressure** is the “sucking” pressure that draws water into the vasculature caused by all dissolved particles. **Oncotic pressure**, on the other hand, is the osmotic pressure that is attributable to dissolved proteins specifically. Blood osmolarity is usually maintained at approximately 290 milliosmoles (mOsm) per liter. As described earlier, the kidneys control osmolarity by modulating the reabsorption of water and by filtering and secreting dissolved particles. When blood osmolarity is low, excess water will be excreted, while solutes will be reabsorbed in higher concentrations. In contrast, when blood osmolarity is high, water reabsorption increases and solute excretion increases.

Acid–Base Balance

The **bicarbonate buffer system** is the major regulator of blood pH. Remind yourself of the buffer equation:



In Chapter 6 of *MCAT Biology Review*, we talked about how the respiratory system can contribute to acid–base balance by increasing or decreasing the respiratory rate. If the blood pH is too low, then increasing the respiratory rate blows off more CO_2 and favors the conversion of H^+ and HCO_3^- to water and carbon dioxide, increasing the pH. If the blood pH is too high, then decreasing the respiratory rate causes the opposite effects. The respiratory system can react to derangements of pH quickly. What can the excretory system do to contribute? The kidneys are able to selectively increase or decrease the secretion of hydrogen ions and bicarbonate. When blood pH is too low, the kidneys excrete more hydrogen ions and increase reabsorption of bicarbonate, resulting in a higher pH. Likewise, when blood pH is too high, the kidneys can excrete more bicarbonate and increase the reabsorption of hydrogen ions. This is slower than the respiratory response, but it is a highly effective way for the body to maintain acid–base balance.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

Carbonic anhydrases are a group of zinc metalloenzymes that catalyze the reversible hydration of carbon dioxide into carbonic acid, which in solution rapidly dissociates into bicarbonate and

CA=enzyme that catalyzes CO_2 into bicarbonate ion

The α isoform, carbonic anhydrase II (CAII) is soluble and is expressed in most tissues and blood cells, osteoclasts, and the kidney.

CAII isoform is expressed in

	<i>most tissues</i>
ally and functionally interacts with aquaporin-1 (AQP1) to increase permeability of water.	<i>CAII interacts with AQP1→more water reabsorption</i>
expressed in the lumen facing and the interstitium facing membranes of the proximal convoluted tubule as well as the thin descending limb (TDL) of the loop of Henle.	<i>AQP1 is found in the proximal convoluted tubule and descending limb of loop of Henle</i>
the hypothesis that CAII augments water flux through AQP1 led us to posit that CAII may be important to water reabsorption and consequently to urinary concentrating ability.	<i>Hypothesis: CAII helps with water reabsorption and concentrating urine</i>
to test this hypothesis, we examined water homeostasis in CAII-deficient (CAII def) mice.	<i>Knocked out CAII in mice, looked at water homeostasis</i>
CAII-def mice were found to be polydipsic (high water intake) and polyuric (high urine volume).	<i>CAII-def mice drank lots of water, had high volume of urine</i>
the osmolarity of the urine is seen in Figure 1.	<i>Fig 1: urine</i>

	<i>osmolarity</i>
asured the following properties in the interstitium of the renal cortex, inner strip of medulla (ISOM), which lies between the cortex and the medulla, and the inner medulla	<i>Looked at three different kidney areas: cortex, between cortex and medulla, and medulla</i>
(Figure 2) and	<i>Fig 2: osmolarity in three kidney regions</i>
n ion concentration (Figure 3). *Represents $p < 0.05$	<i>Fig 3: sodium ion concentration in three kidney regions</i>
	<i>IV: wild-type or CAII-def mice DV: urine osmolarity Trend: CAII-def mice have significantly lower urine osmolarity</i>

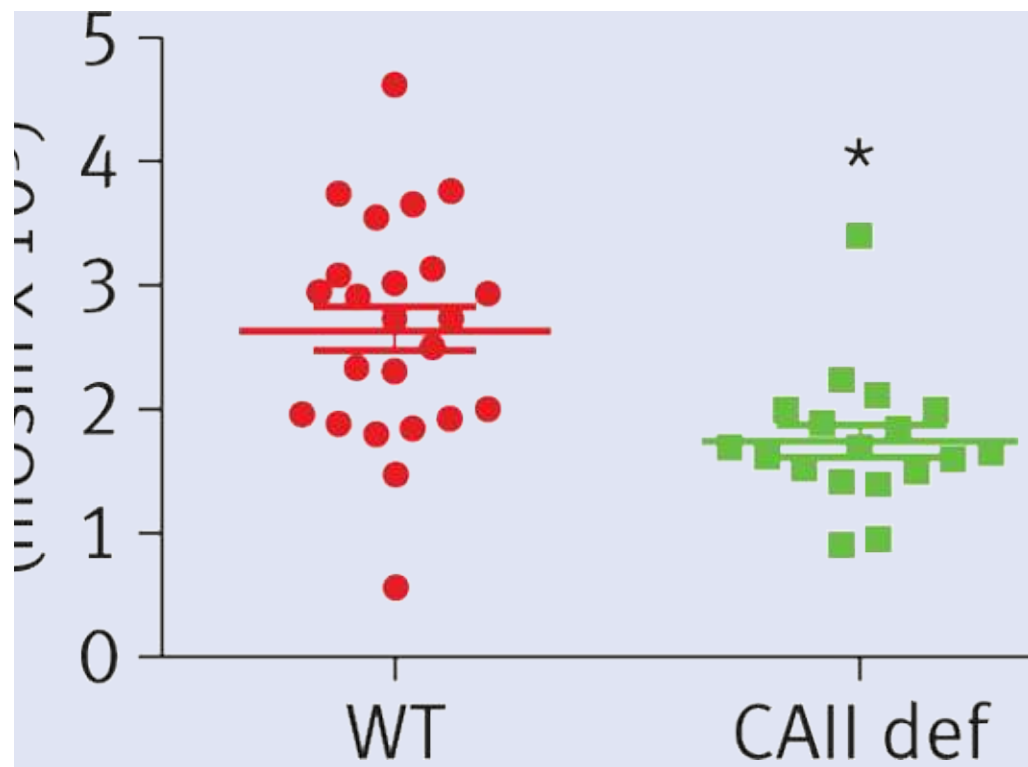
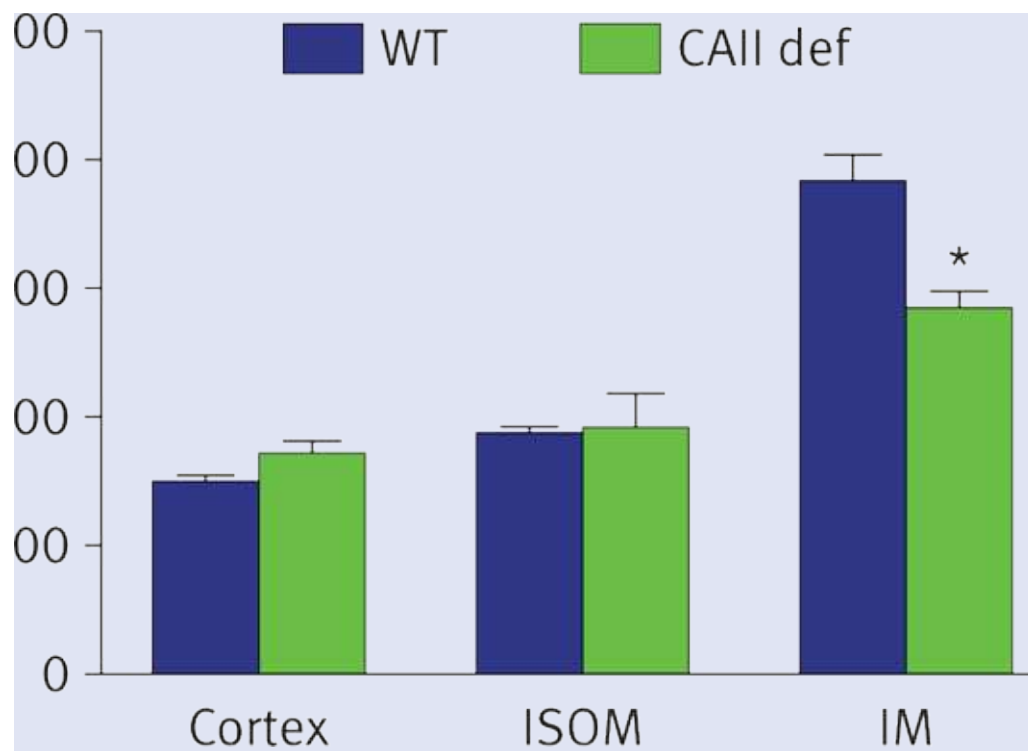


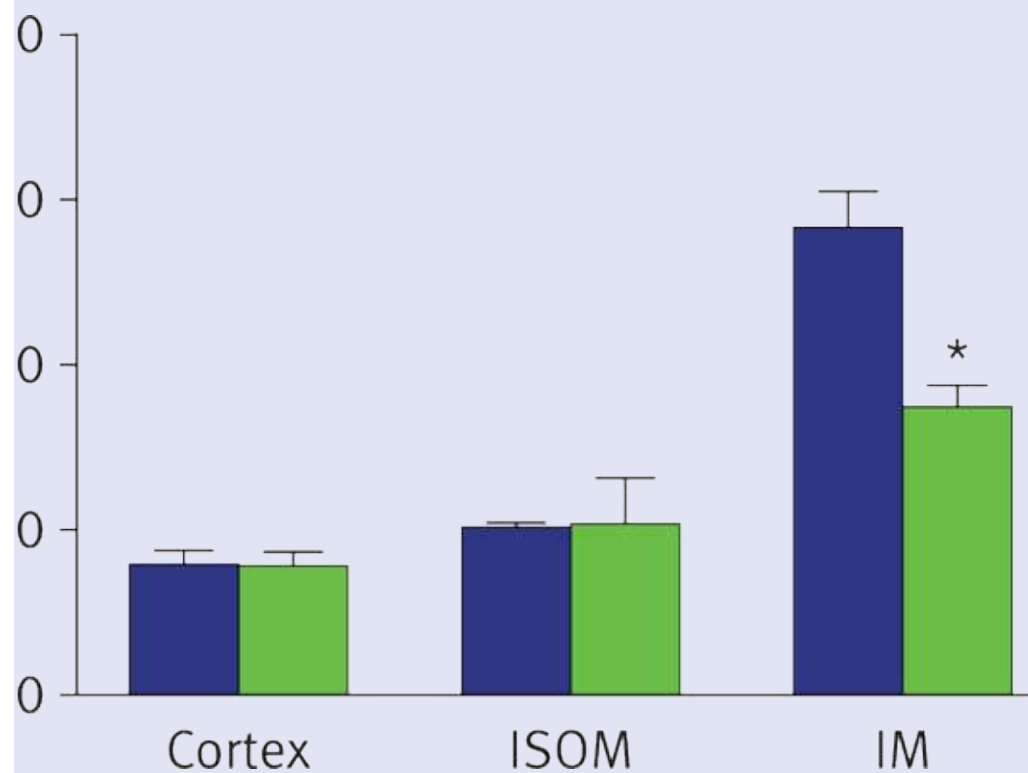
Figure 1

IV: region of kidney, type of mice (WT or CAII-def)
DV: interstitium osmolarity
Trend: inner medulla of kidney interstitium has significantly lower osmolarity in CAII-def mice



compared to
WT

Figure 2



IV: region of
kidney, type of
mice (WT or
CAII-def)
DV:
interstitium
sodium
concentration
Trend: inner
medulla of
kidney
interstitium
has
significantly
lower sodium
concentration
in CAII-def
mice

Figure 3

compared to
WT

Krishnan, D., Pan, W., Beggs, M. R., Trepiccione, F., Chambrey, R., Eladari, D., Cordat, E., Dimke, H., ... Alexander, R. T. Deficiency of carbonic anhydrase ii results in a urinary concentrating defect. *Frontiers in Physiology* 8, 1108. doi:10.3389/fphys.2017.01108.

According to the data, lack of CAII has the greatest impact on which portion of the nephron?

Since the question is asking us to use the data to pinpoint a specific area of the nephron being affected, it's important to understand the experimental setup and the results, and to be prepared to use our outside content knowledge. The researcher's goal is to better understand the role CAII plays in water resorption through its interaction with AQP1. We should know from our content background that aquaporin (AQP) is a water-specific membrane channel found in many parts of the nephron, and AQP availability on the membrane allows for movement of water from the filtrate back into the body. We're also given some information about CAII—it's the enzyme responsible for creating bicarbonate ions, and we're told it interacts with AQP1 to increase water permeability.

In creating mice with deficient CAII, the researchers are looking for a change in the osmolarity of the urine and the kidney. We're told in paragraph 2 that the CAII-deficient mice have a high urine volume and high water intake, supporting the hypothesis that the mice are not reabsorbing enough water, but instead are excreting it as urine. Figure 1 also supports this conclusion by showing that urine is more dilute in CAII-deficient mice than in wild-type (controls).

However, this data doesn't answer the question of where specifically in the nephron lack of CAII is having the greatest impact. Figures 2 and 3 both have information about specific parts of the kidney, which is where our answer is

going to come from. Recall that cortex refers to the outer region of the organ, while medulla refers to the inner part. We're given an additional area called the SOM, which is the boundary between the cortex and medulla. We can see from those two figures that the osmolarity of the interstitium in the inner medulla (IM) is significantly lower than in normal mice, indicating that the part of the nephron existing in the inner medulla is the problem area. Paragraph 1 states that AQP1 is found in the proximal tubule and the thin descending limb of the loop of Henle. The proximal tubule would be in the outer part of the kidney in the cortex, and the thin descending limb would go deep into the kidney, moving into the inner medulla, making the descending limb the part of the nephron that is being most affected. While this satisfies the requirements of the question, this brings up another issue: why does less water being reabsorbed into the interstitium result in less osmolarity? Shouldn't the expected result be a higher osmolarity? This ultimately comes down to how the nephron works as a whole to concentrate your urine. Lack of water being reabsorbed by the thin descending limb will result in a decrease of ions being passively reabsorbed in the ascending limb, giving us that drop in osmolarity.

Using the results of the data, we can conclude that the CAII-deficient mice are being most impacted by the reduced efficacy of AQP1 in the thin descending limb of the loop of Henle.

MCAT CONCEPT CHECK 10.1:

Before you move on, assess your understanding of the material with these questions.

1. List the structures in the excretory pathway, from where filtrate enters the nephron to the excretion of urine from the body.

2. List the vessels in the renal vascular pathway, starting from the renal artery and ending at the renal vein.

3. What arm of the nervous system is responsible for contraction of the detrusor muscle?

4. What are the three processes by which solutes are exchanged between the filtrate and the blood? What happens in each process?

-

-

-

5. For each segment of the nephron listed below, what are its major functions?

- Bowman's capsule:

- Proximal convoluted tubule:

- Descending limb of the loop of Henle:

- Ascending limb of the loop of Henle:

- Distal convoluted tubule:

- Collecting duct:

10.2 Skin

LEARNING OBJECTIVES

After Chapter 10.2, you will be able to:

- Recall the predominant cell type in the epidermis
- Order the layers of the epidermis and dermis
- Describe the mechanisms the body uses to regulate temperature using the integument

By both weight and size, the **skin (integument)** is the largest organ in our bodies. It makes up about 16 percent of total body weight, on average. Skin is one of the major components of nonspecific immune defense, protecting us from exposure to the elements and invasion by pathogens.

STRUCTURE

The skin has several layers. Starting from the deepest layer and working outward, these layers are the **hypodermis (subcutaneous layer)**, **dermis**, and **epidermis**, as shown in Figure 10.6. Skin is derived from the ectoderm.

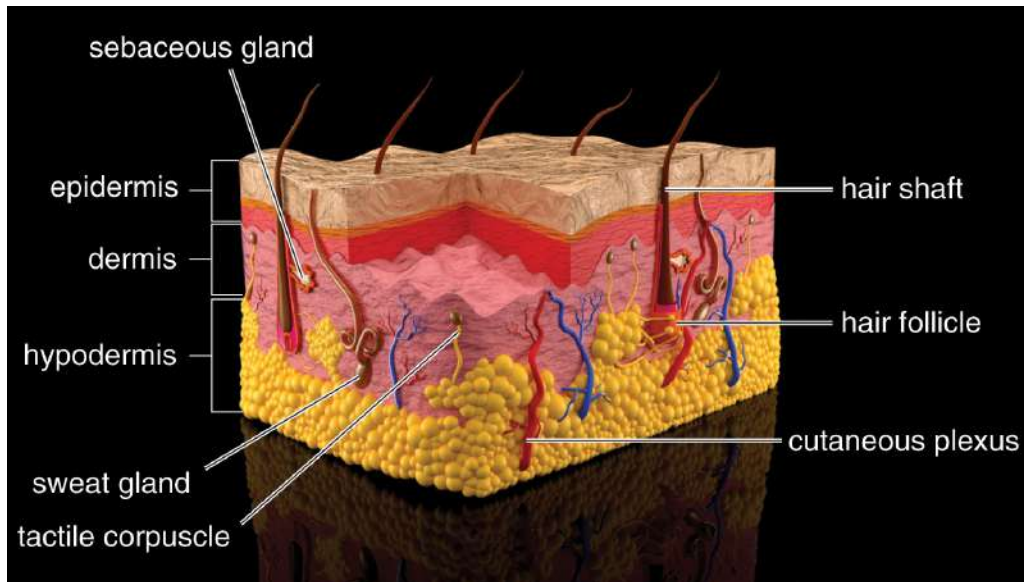


Figure 10.6. Anatomy of the Skin and Subcutaneous Tissue

The Epidermis

The **epidermis** is also subdivided into layers called **strata**. From the deepest layer outward, these are the stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum, as shown in Figure 10.7. The **stratum basale** contains stem cells and is responsible for proliferation of **keratinocytes**, the predominant cells of the skin, that produce **keratin**. In the **stratum spinosum**, these cells become connected to each other; this layer is also the site of Langerhans cells, described below. In the **stratum granulosum**, the keratinocytes die and lose their nuclei. The **stratum lucidum** is only present in thick, hairless skin, such as the skin on the sole of the foot or the palms, and is nearly transparent. Finally, the **stratum corneum** contains up to several dozen layers of flattened keratinocytes, forming a barrier that prevents invasion by pathogens and that helps to prevent loss of fluids and salt. Hair projects above the skin, and there are openings for sweat and sebaceous glands.

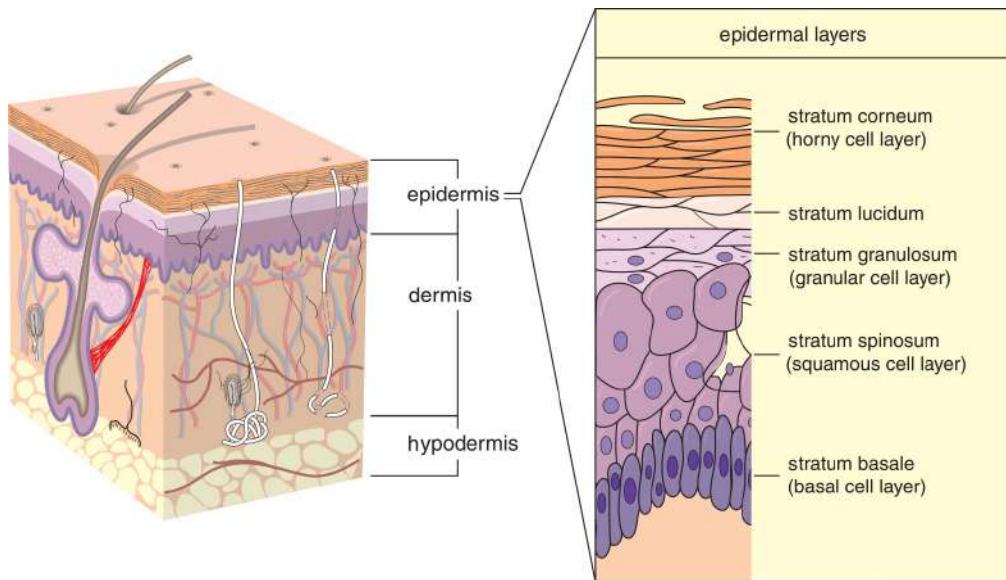


Figure 10.7. Layers of the Epidermis



MNEMONIC

Layers of the skin (superficial to deep): **C**ome, **L**et's **G**et **S**un **B**urned

- Stratum **c**orneum
- Stratum **l**ucidum
- Stratum **g**ranulosum
- Stratum **s**pinosum
- Stratum **b**asale

Other Cells of the Epidermis

In the epidermis, the main cells are keratinocytes. Keratin, mentioned above, is resistant to damage and provides protection against injury, water, and

pathogens. **Calluses** form from excessive keratin deposition in areas of repeated strain due to friction; they provide protection to avoid damage in the future. **Fingernails** and **hair** are also formed from keratin and are produced by specialized cells in the skin.



REAL WORLD

People who have albinism suffer from a genetic metabolic disorder characterized by the inability to synthesize melanin. Typically, albinism is inherited in an autosomal recessive fashion. As would be expected, patients with albinism are extremely sensitive to the harmful effects of the sun. Depigmentation can also occur in vitiligo, an autoimmune disorder in which melanocytes are killed by the individual's immune system.

Melanocytes are a cell type derived from neural crest cells and found in the stratum basale. These cells produce **melanin**, a pigment that serves to protect the skin from DNA damage caused by ultraviolet radiation. Once produced, the pigment is transferred to the keratinocytes. All humans actually have comparable numbers of melanocytes; skin color is caused by varying levels of activity of the melanocytes. More active melanocytes result in darker skin tones. Upon exposure to ultraviolet radiation, melanocytes increase activity, resulting in a darker skin color.

Langerhans cells are actually special macrophages that reside within the stratum spinosum. These cells are capable of presenting antigens to T-cells in order to activate the immune system.

The Dermis

The dermis also consists of multiple layers. The upper layer (right below the epidermis) is the **papillary layer**, which consists of loose connective tissue. Below the papillary layer is the denser **reticular layer**. Sweat glands, blood vessels, and hair follicles originate in the dermis.

Most sensory receptors are also located in the dermis. **Merkel cells (discs)**, for example, are sensory receptors present at the epidermal–dermal junction. These cells are connected to sensory neurons and are responsible for deep pressure and texture sensation within the skin. Other sensory organs in the skin include free nerve endings, which respond to pain; **Meissner's corpuscles**, which respond to light touch; **Ruffini endings**, which respond to stretch; and **Pacinian corpuscles**, which respond to deep pressure and vibration.

The Hypodermis

Finally, the **hypodermis** is a layer of connective tissue that connects the skin to the rest of the body. This layer contains fat and fibrous tissue.

THERMOREGULATION

We have already mentioned that the skin protects us from the elements and microbes. It also has other functions, including ultraviolet protection (via melanin) and transduction of sensory information from the outside world. In this section, we look at another function of the skin: thermoregulation.



Evaporation is an endothermic process and, thus, substances absorb energy from the surroundings to undergo this phase change. Further, the presence of dissolved solutes in sweat increases the boiling point of sweat slightly in comparison to pure water; this allows the absorption of even more heat energy. Boiling point elevation is a colligative property that is discussed in Chapter 9 of *MCAT General Chemistry Review*.

Thermoregulation is achieved by sweating, piloerection, vasodilation, and vasoconstriction. **Sweating** is an excellent cooling mechanism that is controlled by the autonomic nervous system. When body temperature rises above the set point determined by the hypothalamus, thermoregulation processes must occur in order to rid the body of heat. Postganglionic sympathetic neurons that utilize acetylcholine innervate sweat glands and promote the secretion of water with certain ions onto the skin. Heat is then absorbed from the body as the water molecules undergo a phase change to evaporate. The production of sweat itself is not the main mechanism of cooling; it is the evaporation of water from the skin which absorbs body heat. At the same time, arteriolar vasodilation occurs to maximize heat loss. This brings a large quantity of blood to the skin, which accelerates the evaporation of sweat by maximizing the heat energy available for the liquid–gas phase change.



The neurons that innervate sweat glands are actually very unusual. Unlike all other postganglionic sympathetic neurons, these neurons are cholinergic—not noradrenergic (that is, they release acetylcholine, not norepinephrine). All preganglionic neurons in the autonomic nervous system and postganglionic neurons in the parasympathetic nervous system are cholinergic as well. These neurons are discussed in Chapter 4 of *MCAT Biology Review*.

In cold conditions, **arrector pili** muscles contract, causing the hairs of the skin to stand up on end (**piloerection**). This helps to trap a layer of heated air near the skin. The arterioles that feed the capillaries of the skin constrict, limiting the quantity of blood reaching the skin. Skeletal muscle may also begin to contract rapidly, causing shivering. **Shivering** requires a sizeable amount of ATP; however, a significant portion of the energy from ATP is converted into thermal energy. In addition to these mechanisms, humans possess a layer of fat just below the skin. This fat helps to insulate the body. In addition to this fat, which is called **white fat**, **brown fat** may also be present, especially in infants. Brown fat has a much less efficient electron transport chain, which means that more heat energy is released as fuel is burned.

The skin also helps to maintain the osmolarity of the body. This is because the skin is relatively impermeable to water. This prevents not only the entrance of water through the skin, but also the loss of water from the tissues. This becomes very important in cases such as burns or large losses of skin as dehydration of the tissues becomes a real threat to survival.

MCAT CONCEPT CHECK 10.2:

Before you move on, assess your understanding of the material with these questions.

1. What is the predominant cell type in the epidermis?

2. What are the layers of the epidermis, from superficial to deep?

- ---
- ---
- ---
- ---
- ---

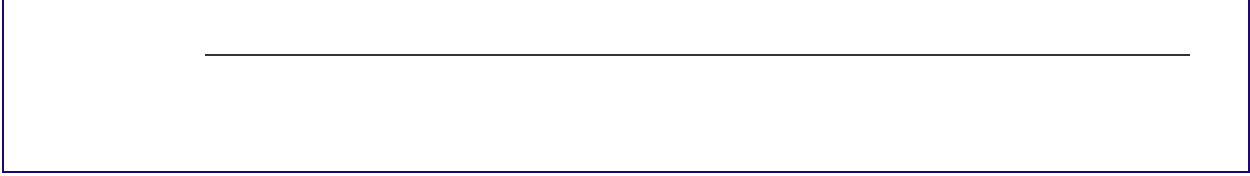
3. What are the layers of the dermis, from superficial to deep?

- ---
- ---

4. What are some mechanisms the body uses to cool itself?
What are some mechanisms the body uses to retain heat?

- Cooling:

- Retaining heat:





Conclusion

Two main organ systems were discussed in this chapter: the excretory system and the skin. Both of these systems play an essential role in homeostasis. However, this chapter also demonstrated a very MCAT-worthy concept: no system works alone. Multiple systems participate in homeostasis. While the kidneys are the major players in salt, water, and acid–base balance, their function depends on the endocrine system, circulatory system, and respiratory system. While the skin is an important immune organ in and of itself, its function as a thermoregulatory organ is dependent on the nervous system—and it also sends sensory signals *to* the nervous system. As you move on to the last organ system in the next chapter—the musculoskeletal system—notice how each system interacts with other systems in order to produce a fully functioning organism. The MCAT is far more focused on how you conceptualize the big picture, and not on how many details you memorize. As you study, focus on understanding these systems and how each system influences the rest of the body.

Concept Summary

The Excretory System

- The **excretory system** serves many functions, including the regulation of blood pressure, blood osmolarity, acid–base balance, and removal of nitrogenous wastes.
- The **kidney** produces urine, which flows into the **ureter** at the **renal pelvis**. Urine is then collected in the **bladder** until it is excreted through the **urethra**.
- The kidney contains a cortex and a medulla. Each kidney has a **hilum**, which contains a renal artery, renal vein, and ureter.
- The kidney contains a **portal system** with two capillary beds in series.
 - Blood from the renal artery flows into **afferent arterioles**, which form **glomeruli** in Bowman's capsule (the first capillary bed).
 - Blood then flows through the **efferent arteriole** to the **vasa recta** (the second capillary bed), which surround the nephron, before leaving the kidney through the renal vein.
- The bladder has a muscular lining known as the **detrusor muscle**, which is under parasympathetic control. It also has two muscular sphincters.
 - The **internal urethral sphincter** consists of smooth muscle and is under involuntary (parasympathetic) control.
 - The **external urethral sphincter** consists of skeletal muscle and is

under voluntary control.

- The kidney participates in solute movement through three processes:
 - **Filtration** is the movement of solutes from blood to filtrate at Bowman's capsule. The direction and rate of filtration is determined by **Starling forces**, which account for the hydrostatic and oncotic pressure differentials between the glomerulus and Bowman's space.
 - **Secretion** is the movement of solutes from blood to filtrate anywhere other than Bowman's capsule.
 - **Reabsorption** is the movement of solutes from filtrate to blood.
- Each segment of the nephron has a specific function.
 - The **proximal convoluted tubule (PCT)** is the site of bulk reabsorption of glucose, amino acids, soluble vitamins, salt, and water. It is also the site of secretion for hydrogen ions, potassium ions, ammonia, and urea.
 - The **descending limb of the loop of Henle** is permeable to water but not salt; therefore, as the filtrate moves into the more osmotically concentrated renal medulla, water is reabsorbed from the filtrate. The vasa recta and nephron flow in opposite directions, creating a **countercurrent multiplier system** that allows maximal reabsorption of water.
 - The **ascending limb of the loop of Henle** is permeable to salt but not water; therefore, salt is reabsorbed both passively and actively. The **diluting segment** is in the outer medulla; because salt is actively reabsorbed in this site, the filtrate actually becomes hypotonic compared to the blood.
 - The **distal convoluted tubule (DCT)** is responsive to aldosterone and is a site of salt reabsorption and waste product excretion, like the PCT.

- The **collecting duct** is responsive to both aldosterone and antidiuretic hormone and has variable permeability, which allows reabsorption of the right amount of water depending on the body's needs.
- The kidney is under hormonal control. When blood pressure (and volume) are low, two different hormonal systems are activated.
 - **Aldosterone** is a steroid hormone regulated by the renin–angiotensin–aldosterone system that increases sodium reabsorption in the distal convoluted tubule and collecting duct, thereby increasing water reabsorption. This results in an increased blood volume (and pressure), but no change in blood osmolarity.
 - **Antidiuretic hormone (ADH or vasopressin)** is a peptide hormone synthesized by the hypothalamus and released by the posterior pituitary. Its release is stimulated not only by low blood volume but also by high blood osmolarity. It increases the permeability of the collecting duct to water, increasing water reabsorption. This results in an increased blood volume (and pressure) and a decreased blood osmolarity.
- The kidney can regulate pH by selective reabsorption or secretion of bicarbonate or hydrogen ions.

Skin

- The skin acts as a barrier, protecting us from the elements and invasion by pathogens.
- The skin is composed of three major layers: the **hypodermis** (**subcutaneous** layer), **dermis**, and **epidermis**.

- The epidermis is composed of five layers: the **stratum basale**, **stratum spinosum**, **stratum granulosum**, **stratum lucidum**, and **stratum corneum**. The stratum basale contains stem cells that proliferate to form keratinocytes. Keratinocyte nuclei are lost in the stratum granulosum, and many thin layers form in the stratum corneum.
 - **Melanocytes** produce **melanin**, which protects the skin from DNA damage caused by ultraviolet radiation; melanin is passed to keratinocytes.
 - **Langerhans cells** are special macrophages that serve as antigen-presenting cells in the skin.
 - The dermis is composed of two layers: the **papillary layer** and the **reticular layer**.
 - Many sensory cells are located in the dermis, including **Merkel cells** (deep pressure and texture), free nerve endings (pain), **Meissner's corpuscles** (light touch), **Ruffini endings** (stretch), and **Pacinian corpuscles** (deep pressure and vibration).
 - The hypodermis contains fat and connective tissue and connects the skin to the rest of the body.
- The skin is important for thermoregulation, or the maintenance of a constant internal temperature.
 - Cooling mechanisms include **sweating**, which draws heat from the body through evaporation of water from sweat, and vasodilation. Sweat glands are innervated by postganglionic cholinergic sympathetic neurons.
 - Warming mechanisms include: piloerection, in which arrector pili muscles contract, causing hairs to stand on end (trapping a layer of warmed air around the skin); vasoconstriction; shivering; and insulation provided by fat.

- The skin also prevents dehydration and salt loss from the body.

Answers to Concept Checks

10.1

- .. Bowman's space → proximal convoluted tubule → descending limb of the loop of Henle → ascending limb of the loop of Henle → distal convoluted tubule → collecting duct → renal pelvis → ureter → bladder → urethra
- !. Renal artery → afferent arteriole → glomerulus → efferent arteriole → vasa recta → renal vein
- }. The parasympathetic nervous system causes contraction of the detrusor muscle.
- l. Filtration is the movement of solutes from blood into filtrate at Bowman's capsule. Secretion is the movement of solutes from blood into filtrate anywhere besides Bowman's capsule. Reabsorption is the movement of solutes from filtrate into blood.
- i. Bowman's capsule is the site of filtration, through which water, ions, amino acids, vitamins, and glucose pass (essentially everything besides cells and proteins). The proximal convoluted tubule controls solute identity, reabsorbing vitamins, amino acids, and glucose, while secreting potassium and hydrogen ions, ammonia, and urea. The descending limb of the loop of Henle is important for water reabsorption, and uses the medullary concentration gradient. The ascending limb of the loop of Henle is important for salt reabsorption and dilution of the urine in the diluting segment. The distal convoluted tubule, like the PCT, is important for solute identity by reabsorbing salts while secreting potassium and hydrogen ions, ammonia, and urea. The collecting duct is important for urine concentration; its variable permeability allows water to be reabsorbed

based on the needs of the body.

10.2

- .. Keratinocytes are the primary cells of the epidermis.
- !. Stratum corneum, stratum lucidum, stratum granulosum, stratum spinosum, stratum basale
- }. The papillary layer and the reticular layer
- l. The body can cool itself through sweating and vasodilation. The body can warm itself through vasoconstriction, piloerection, and shivering.



Shared Concepts

Behavioral Sciences Chapter 2

Sensation and Perception

Biochemistry Chapter 8

Biological Membranes

Biology Chapter 6

The Respiratory System

Biology Chapter 7

The Cardiovascular System

Biology Chapter 8

The Immune System

General Chemistry Chapter 9

Solutions

Discrete Practice Questions

1. Which of the following would most likely be filtered through the glomerulus into Bowman's space?
 - (A) Erythrocytes
 - (B) Monosaccharides
 - (C) Platelets
 - (D) Proteins
2. In which of the following segments of the nephron is sodium NOT actively transported out of the nephron?
 - (A) Proximal convoluted tubule
 - (B) Thin portion of the ascending limb of the loop of Henle
 - (C) Distal convoluted tubule
 - (D) Thick portion of the ascending limb of the loop of Henle
3. Which region of the kidney has the lowest solute concentration under normal physiological circumstances?
 - (A) Cortex
 - (B) Outer medulla
 - (C) Inner medulla

(D) Renal pelvis

4. Which of the following sequences correctly shows the passage of blood through the vessels of the kidney?

- (A) Renal artery → afferent arterioles → glomerulus → efferent arterioles → vasa recta → renal vein
- (B) Afferent arterioles → renal artery → glomerulus → vasa recta → renal vein → efferent arterioles
- (C) Glomerulus → renal artery → afferent arterioles → efferent arterioles → renal vein → vasa recta
- (D) Renal vein → efferent arterioles → glomerulus → afferent arterioles → vasa recta → renal artery

5. Which of the following statements is FALSE?

- (A) ADH increases water reabsorption in the kidney.
- (B) Aldosterone indirectly increases water reabsorption in the kidney.
- (C) ADH acts directly on the proximal convoluted tubule.
- (D) Aldosterone stimulates reabsorption of sodium from the collecting duct.

6. In the nephron, amino acids enter the vasa recta via the process of:

- (A) filtration.
- (B) secretion.

- (C) excretion.
- (D) reabsorption.

7. On a very cold day, a man waits for over an hour at the bus stop.
Which of the following structures helps his body set and maintain a normal temperature?

- (A) Hypothalamus
- (B) Kidneys
- (C) Posterior pituitary
- (D) Brainstem

8. Glucose reabsorption in the nephron occurs in the:

- (A) loop of Henle.
- (B) distal convoluted tubule.
- (C) proximal convoluted tubule.
- (D) collecting duct.

9. Under normal physiological circumstances, the primary function of the nephron is to create urine that is:

- (A) hypertonic to the blood.
- (B) hypotonic to the blood.
- (C) isotonic to the filtrate.
- (D) hypotonic to the vasa recta.

10. Diabetic nephropathy is commonly detected by finding protein in the urine of a patient. In such a disease, where is the likely defect in the nephron?

- (A) Glomerulus
- (B) Proximal convoluted tubule
- (C) Loop of Henle
- (D) Collecting duct

11. A laceration cuts down into a layer of loose connective tissue in the skin. Which layer of the skin is this?

- (A) Stratum corneum
- (B) Stratum lucidum
- (C) Papillary layer
- (D) Reticular layer

12. When the pH of the blood is high, which substance is likely to be excreted in larger quantities in the urine?

- (A) Urea
- (B) Ammonia
- (C) Hydrogen ions
- (D) Bicarbonate ions

13. In which layer of the skin can the stem cells of keratinocytes be found?

- (A) Stratum lucidum
- (B) Stratum granulosum
- (C) Stratum basale
- (D) Stratum corneum

14. A drug is used that prevents the conversion of angiotensin I to angiotensin II. What is a likely effect of this drug?

- (A) Increased sodium reabsorption
- (B) Increased potassium reabsorption
- (C) Increased blood pressure
- (D) Increased blood pH

15. Sarin is a potent organophosphate that can be used in chemical warfare. As an inhibitor of acetylcholinesterase, sarin causes excessive buildup of acetylcholine in all synapses where it is the neurotransmitter. Which of the following symptoms would most likely be seen in an individual with sarin poisoning?

- (A) Increased urination and increased sweating
- (B) Increased urination and decreased sweating
- (C) Decreased urination and increased sweating
- (D) Decreased urination and decreased sweating

Discrete Practice Answers

1. **B**

The glomerulus functions like a sieve; small molecules dissolved in the fluid will pass through the glomerulus, including glucose, which is later reabsorbed. Large molecules, such as proteins, and cells, such as erythrocytes and platelets, will not be able to pass through the glomerular filter.

2. **B**

Sodium is actively transported out of the nephron in the proximal and distal convoluted tubules, where the concentration of sodium outside of the nephron is higher than inside; thus, energy is required to transport the sodium molecules against their concentration gradient, eliminating **(A)** and **(C)**. In the inner medulla, however, sodium and other ions (such as chloride) diffuse passively down their concentration gradients at the thin ascending limb of the loop of Henle, making **(B)** the correct answer. The thick ascending limb of the loop of Henle is thick because its cells contain many mitochondria—which produce the ATP needed for active transport of sodium and chloride out of the filtrate, eliminating **(D)**.

3. **A**

The region of the kidney that has the lowest solute concentration is the cortex, where the proximal convoluted tubule and a part of the distal convoluted tubule are found. The solute concentration increases as one descends into the medulla, and concentrated urine can be found in the

renal pelvis.

4. **A**

Blood enters the kidney through the renal artery, which divides into many afferent arterioles that run through the medulla and into the cortex. Each afferent arteriole branches into a convoluted network of capillaries called a glomerulus. Rather than converging directly into a vein, the capillaries converge into an efferent arteriole, which divides into a fine capillary network known as the vasa recta. The vasa recta capillaries envelop the nephron tubule, where they reabsorb various ions, and then converge into the renal vein. The arrangement of tandem capillary beds is known as a portal system.

5. **C**

All of the answer choices describe ADH or aldosterone. These two hormones ultimately act to increase water reabsorption in the kidney; their respective mechanisms of action, however, are different. ADH increases water reabsorption by increasing the permeability of the collecting duct to water, whereas aldosterone stimulates reabsorption of sodium from the distal convoluted tubule and collecting duct. Using this knowledge, we can now attack the answer choices. **(C)** is the correct answer because ADH does not act on the proximal convoluted tubule, but rather on the collecting duct.

6. **D**

Essential substances, such as glucose, salts, amino acids, and water, are reabsorbed from the filtrate and returned to the blood in the vasa recta. In general, reabsorption refers to the movement of solutes from the filtrate back into the blood.

7. **A**

The hypothalamus functions as a thermostat that regulates body temperature. When it's cold outside, nervous stimulation to the blood vessels in the skin is increased, causing the vessels to constrict. This constriction diminishes blood flow to the skin surface and prevents heat loss. Sweat glands are turned off to prevent heat loss through evaporation. Skeletal muscles are stimulated to shiver (rapidly contract), which increases the metabolic rate and produces heat. The hypothalamus is also involved in other processes, including the release of endocrine hormones, regulation of appetite, and circadian rhythms.

8. **C**

The filtrate enters Bowman's capsule and then flows into the proximal convoluted tubule, where virtually all glucose, amino acids, and other important organic molecules are reabsorbed via active transport.

9. **A**

The kidneys function to eliminate wastes such as urea, while reabsorbing various important substances such as glucose and amino acids for reuse by the body. Generation of a solute concentration gradient from the cortex to the medulla allows a considerable amount of water to be reabsorbed. Excretion of concentrated urine serves to limit water losses from the body and helps to preserve blood volume. Thus, the primary function of the nephron is to create urine that is hypertonic to the blood, making **(A)** the correct answer and eliminating **(B)** and **(D)**. Water should be reabsorbed from the filtrate, so urine should be hypertonic to the filtrate, eliminating **(C)**.

10. **A**

The glomerulus is the most likely location of pathology if large proteins are

detected in the urine. This is because large proteins should not be able to pass through the filter of the glomerulus in the first place. Once large proteins are in the filtrate, no other nephron structure can reabsorb them. Thus, the only likely source of protein in the urine is glomerular pathology.

1. **C**

The layer of the skin that is predominantly loose connective tissue is the papillary layer of the dermis. The stratum corneum and stratum lucidum, **(A)** and **(B)**, contain dead keratinocytes, while the reticular layer, **(D)**, consists of dense connective tissue.

2. **D**

When the pH of the blood is high, this indicates that the blood is alkalemic. In order to correct the pH of the blood, the kidney will increase the excretion of a base, namely bicarbonate. Excretion of urea would have little effect on the pH, eliminating **(A)**. While ammonia is a base, it is quite toxic and is generally converted into urea before excretion, eliminating **(B)**. Excretion of hydrogen ions would exacerbate the alkalemia, eliminating **(C)**.

3. **C**

The stratum basale contains the stem cells that proliferate to form keratinocytes, which then ascend through the other layers of skin until they are shed from the stratum corneum.

4. **B**

Normally, angiotensin II causes secretion of aldosterone from the adrenal cortex. Aldosterone serves to increase reabsorption of sodium, while promoting excretion of potassium and hydrogen ions. Thus, blocking the release of aldosterone should result in decreased reabsorption of sodium,

while decreasing excretion of potassium and hydrogen ions. This eliminates **(A)** and **(D)** and makes **(B)** the correct answer. In the absence of aldosterone, less sodium reabsorption will occur, leading to less water reabsorption, eliminating **(C)**.

5. **A**

An excess of acetylcholine will lead to activation of all parasympathetic neurons, preganglionic sympathetic neurons, and the postganglionic sympathetic neurons that innervate sweat glands. Because the parasympathetic nervous system causes contraction of the bladder, one would expect increased urination. The increased activation of sweat glands would lead to increased sweating as well.



CHAPTER 11

The Musculoskeletal System

CHAPTER 11

The Musculoskeletal System

In This Chapter

11.1 The Muscular System



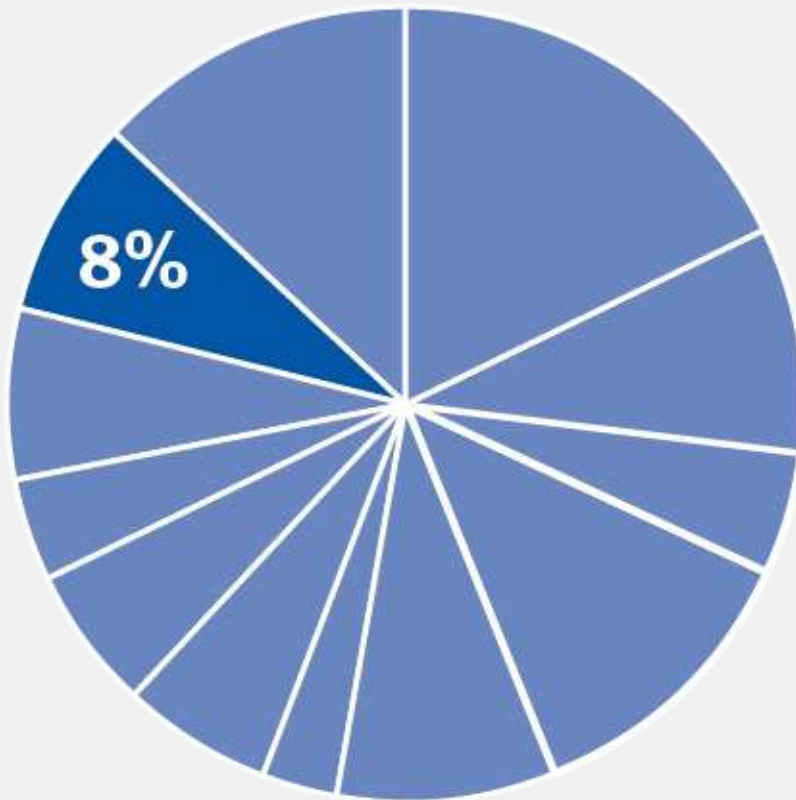
Types of Muscle
Microscopic Structure of Skeletal Muscle
Muscle Contraction
Stimulation, Summation, and Muscle Fatigue

11.2 The Skeletal System

Skeletal Structure
Bone Composition
Cartilage
Joints and Movement

Concept Summary

Chapter Profile



The content in this chapter should be relevant to about 8% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

3B: Structure and integrative functions of the main organ systems

Introduction

Populations affected by large disasters or traumatic events like wars or earthquakes are often fertile ground for unique medical discoveries. During World War II, Nazi Germany bombed London for 57 consecutive days during the beginning of what came to be known as the *Blitzkrieg*, or an eight-month "lightning war." Victims of the Blitz, as it is known in London, included those afflicted with a specific set of symptoms: pain and swelling with accompanying effects of depleted blood volume (shock, weakness, low blood pressure, and decreased urine output). Less obvious was acute kidney failure, which could lead quickly to death if left untreated.

What caused the Blitz victims to suffer from these symptoms? Extreme physical trauma to muscles—namely, compression—destroys skeletal muscle tissue. This condition is called *rhabdomyolysis* (*rhabdo*– refers to striation, *myo*– to muscle, and *-lysis* to breakdown). The products of skeletal muscle destruction, some of which are toxic, circulate in the blood until they are filtered out. *Creatine kinase* is one of these products; in fact, rhabdomyolysis is diagnosed with a creatine kinase level five times the normal upper limit. Myoglobin is another. Much like hemoglobin, myoglobin uses heme to carry oxygen; it is not, however, housed within a red blood cell. Thus, an erythrocyte-free urine sample that tests positive for heme points compellingly toward rhabdomyolysis. Myoglobin oxygen reserves are just one of the specialized features of muscles, as we will see in this chapter.

Skeletal muscles are only able to exert an effect on the body by moving bony structures around joints. Further, skeletal muscle isn't the only form of muscle in the body; smooth muscle plays roles in the cardiovascular, respiratory, reproductive, and digestive systems, and cardiac muscle comprises the

contractile tissue of the heart. Bones are more than simply a support structure, however; they also provide protection to internal organs, serve as a storage reserve of calcium and other minerals, and are the site of hematopoiesis. In this chapter, we'll explore the biology of all of these tissues, completing our tour of systems anatomy and physiology.

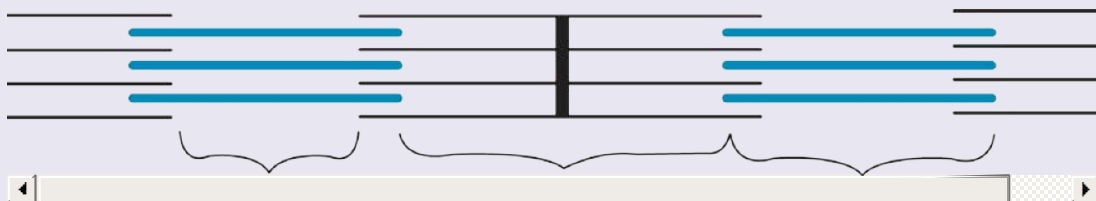
11.1 The Muscular System

High-Yield

LEARNING OBJECTIVES

After Chapter 11.1, you will be able to:

- Categorize muscles as skeletal, smooth, or cardiac based on their innervation and structure
- Order the series of events in muscular contraction
- Explain the relationship between ATP binding and muscular contraction
- Recall the meaning of the terms summation, simple twitch, and tetanus
- Identify the zones and bands within a sarcomere and predict their length changes during contraction:



The muscular system is composed of not only skeletal muscle, but also smooth muscle and cardiac muscle. Skeletal muscle is essential for supporting the body and facilitating movement. The contraction of skeletal

muscle also compresses venous structures and helps propel blood through the low-pressure venous system toward the heart, as well as lymph through the lymphatic system. Rapid muscle contraction also leads to shivering, which is important in thermoregulation. Smooth muscle is responsible for involuntary movement, such as the rhythmic contractions of smooth muscle in the digestive system called peristalsis. Smooth muscle also aids in the regulation of blood pressure by constricting and relaxing the vasculature. Cardiac muscle is a special type of muscle that is able to maintain rhythmic contraction of the heart without nervous system input. In this section, we will discuss each type of muscle as well as the physiology of muscles.

TYPES OF MUSCLE

Muscle can be divided into the three different subtypes: skeletal muscle, smooth muscle, and cardiac muscle. Each muscle type performs specific functions, although they share some similarities. All muscle is capable of contraction, which relies on calcium ions. All muscle is innervated, although—as we will see—the part of the nervous system that innervates the muscle and the ability of the muscle to contract without nervous input varies from type to type.

Skeletal Muscle

Skeletal muscle is responsible for voluntary movement and is therefore innervated by the somatic nervous system. Due to the arrangement of actin and myosin into repeating units called **sarcomeres**, it appears striped or **striated** when viewed microscopically. Skeletal muscle is multinucleated because it is formed as individual muscle cells fuse into long rods during development.

There are multiple different types of fibers within skeletal muscle. **Red fibers**, also known as **slow-twitch fibers**, have high myoglobin content and primarily derive their energy aerobically. **Myoglobin** is an oxygen carrier that uses iron in a heme group to bind oxygen, imparting a red color. Red fibers also contain many mitochondria to carry out oxidative phosphorylation. **White fibers**, also known as **fast-twitch fibers**, contain much less myoglobin. Because there is less myoglobin, and therefore less iron, the color is lighter. These two types of fibers can be mixed in muscles. Muscles that contract slowly, but that can sustain activity (such as the muscles that support posture), contain a predominance of red fibers. Muscles that contract rapidly, but fatigue quickly, contain mostly white fibers.



REAL WORLD

Poultry provides a great example of the difference between red and white fibers. Most muscles of support, such as the thigh, are considered dark meat and contain a high concentration of red fibers. The pectoral muscles (breast meat), used by some poultry for only short bursts of flight, are considered white meat and have a high concentration of white fibers.

Smooth Muscle

Smooth muscle is responsible for involuntary action. Thus, smooth muscle is controlled by the autonomic nervous system. It is found in the respiratory tree, digestive tract, bladder, uterus, blood vessel walls, and many other locations. Smooth muscle cells have a single nucleus located in the center of

the cell. Just like skeletal muscle, smooth muscle cells contain actin and myosin, but the fibers are not as well-organized, so striations cannot be seen. Compared to skeletal muscle, smooth muscle is capable of more sustained contractions; a constant state of low-level contraction, as may be seen in the blood vessels, is called **tonus**. Smooth muscle can actually contract without nervous system input in what is known as **myogenic activity**. In this case, the muscle cells contract directly in response to stretch or other stimuli.



MCAT EXPERTISE

The MCAT loves to test the fact that both smooth and cardiac muscle exhibit myogenic activity. These muscle cells will respond to nervous input, but do not require external signals to undergo contraction.

Cardiac Muscle

Cardiac muscle has characteristics of both smooth and skeletal muscle types. Cardiac muscle is primarily uninucleated, but cells may contain two nuclei. Like smooth muscle, cardiac muscle contraction is involuntary and innervated by the autonomic nervous system. Unlike smooth muscle, cardiac muscle appears striated like skeletal.

One of the unique characteristics of cardiac muscle is how each cardiac myocyte communicates. Cardiac muscle cells are connected by **intercalated discs**, which contain many **gap junctions**. These gap junctions are connections between the cytoplasm of adjacent cells, allowing for the flow of

ions directly between cells. This allows for rapid and coordinated depolarization of muscle cells and efficient contraction of cardiac muscle.

Cardiac muscle cells are able to define and maintain their own rhythm, termed myogenic activity. Starting at the **sinoatrial (SA) node**, depolarization spreads using conduction pathways to the **atrioventricular (AV) node**. From there, the depolarization spreads to the **bundle of His** and its branches, and then to the **Purkinje fibers**. The gap junctions allow for progressive depolarizations to spread via ion flow across the gap junctions between cells. The nervous and endocrine systems also play a role in the regulation of cardiac muscle contraction. The vagus nerve provides parasympathetic outflow to the heart and slows the heart rate. Norepinephrine from sympathetic neurons or epinephrine from the adrenal medulla binds to adrenergic receptors in the heart, causing an increased heart rate and greater contractility. One of the ways epinephrine does this is by increasing intracellular calcium levels within cardiac myocytes. Ultimately, cardiac contraction—like that of all types of muscle—relies on calcium.

The main characteristics of each muscle type are summarized in Table 11.1.

Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Striated	Striated	Nonstriated
Voluntary	Involuntary	Involuntary
Somatic innervation	Autonomic innervation	Autonomic innervation
Many nuclei per cell	1–2 nuclei per cell	1 nucleus per cell
Ca ²⁺ required for contraction	Ca ²⁺ required for contraction	Ca ²⁺ required for contraction

Table 11.1. Types of Muscle

MICROSCOPIC STRUCTURE OF SKELETAL MUSCLE

Each type of muscle has a specific microscopic structure. However, the MCAT tends to focus on the contractile elements and microscopic structure of skeletal muscle. In order to accurately answer those questions on Test Day, let's take a moment to discuss skeletal muscle in detail.

The Sarcomere

The **sarcomere** is the basic contractile unit of skeletal muscle. Sarcomeres are made of **thick** and **thin filaments**. The thick filaments are organized bundles of **myosin**, whereas the thin filaments are made of **actin** along with two other proteins: **troponin** and **tropomyosin**. These proteins help to regulate the interaction between the actin and myosin filaments. Another protein, **titin**, acts as a spring and anchors the actin and myosin filaments together, preventing excessive stretching of the muscle.



MNEMONIC

To remember which filament is associated with **t**roponin and **t**ropomyosin, remember that **a**ctin has a **T** in it.



MNEMONIC

To remember which is the thin and thick filament, remember **acthin**: **actin** filaments are **thin** (while myosin filaments are thick).

Each sarcomere is divided into different lines, zones, and bands, as shown in Figure 11.1. **Z-lines** define the boundaries of each sarcomere. The **M-line** runs down the center of the sarcomere, through the middle of the myosin filaments. The **I-band** is the region containing exclusively thin filaments, whereas the **H-zone** contains only thick filaments. The **A-band** contains the thick filaments in their entirety, including any overlap with thin filaments. During contraction, the H-zone, I-band, the distance between Z-lines, and the distance between M-lines all become smaller, whereas the A-band's size remains constant.

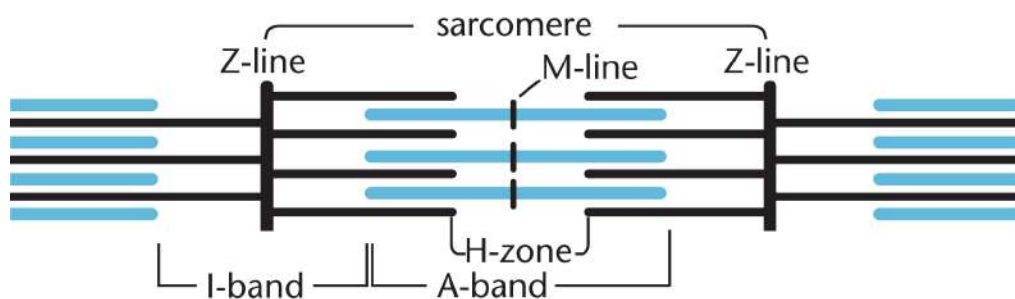


Figure 11.1. The Sarcomere

Sarcomeres are the functional units of striated muscle.



MNEMONIC

Parts of the sarcomere:

- **Z—Z** is the end of the alphabet, and the end of the sarcomere
- **M—M**iddle of the **m**yosin filaments

- **I—I** is a **thin** letter (**thin** filaments only)
- **H—H** is a **thick** letter (**thick** filaments only)
- **A—A** All of the thick filament, whether or not it is overlapping

Gross Structure of Myocytes

Sarcomeres are attached end-to-end to form **myofibrils**. Myofibrils are surrounded by a covering known as the **sarcoplasmic reticulum (SR)**, a modified endoplasmic reticulum that contains a high concentration of Ca^{2+} ions. The **sarcoplasm** is a modified cytoplasm located just outside the sarcoplasmic reticulum. The cell membrane of a myocyte is known as the **sarcolemma**. The sarcolemma is capable of propagating an action potential and can distribute the action potential to all sarcomeres in a muscle using a system of **transverse tubules (T-tubules)** that are oriented perpendicularly to the myofibrils, as shown in Figure 11.2. Each **myocyte**, or muscle cell, contains many myofibrils arranged in parallel and can also be called a **muscle fiber**. The nuclei, of which there are many, are usually found at the periphery of the cell. Finally, many myocytes in parallel form a muscle.

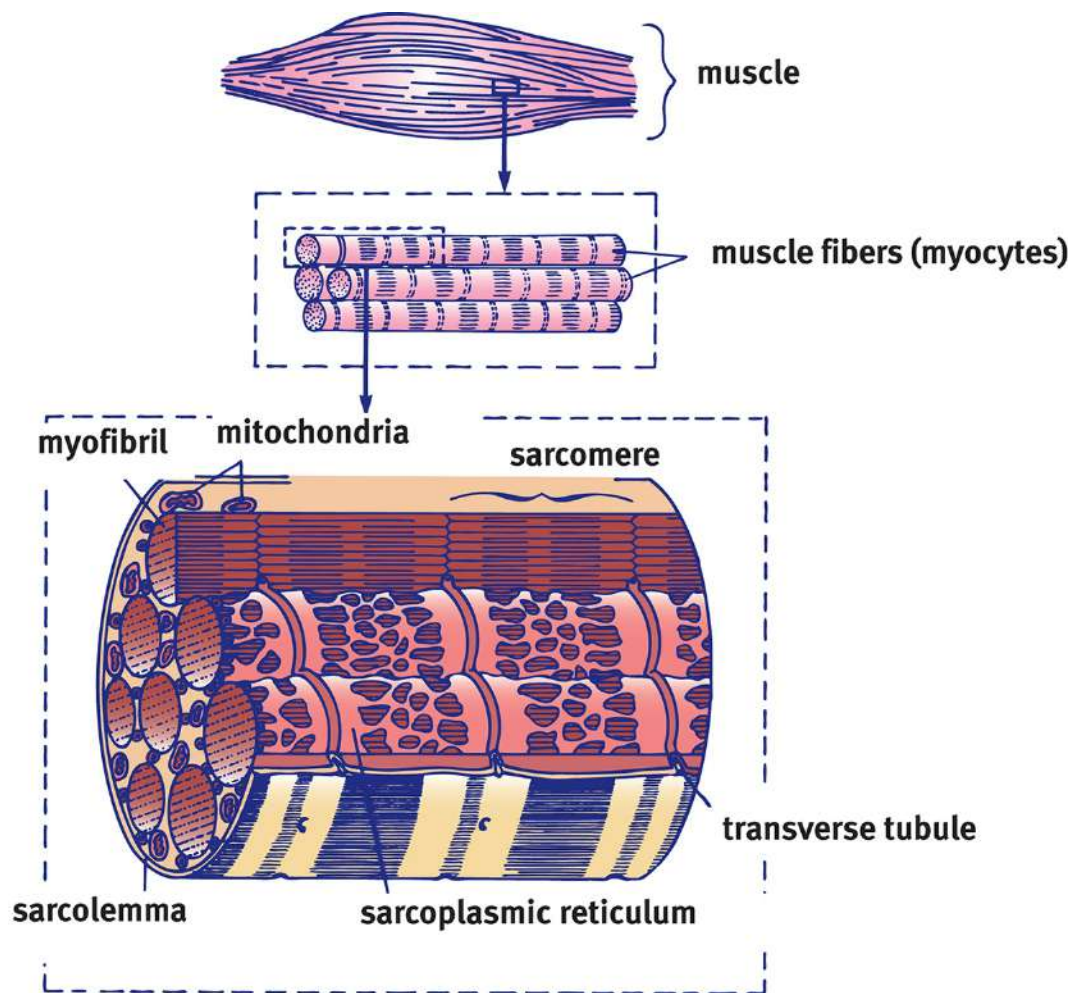


Figure 11.2. Architecture of Skeletal Muscle

A muscle is composed of parallel myocytes (muscle fibers), which are composed of parallel myofibrils.



KEY CONCEPT

The sarcoplasmic reticulum is just a fancy name for the specialized endoplasmic reticulum in muscle cells.



KEY CONCEPT

A myofibril is an arrangement of many sarcomeres in series. A muscle fiber (or myocyte, or muscle cell) contains many myofibrils within it, arranged in parallel. A muscle is made up of parallel muscle fibers. These names are very similar; pay careful attention to terminology when reading about muscle structure!

MUSCLE CONTRACTION

Contraction of muscle requires a series of coordinated steps that are repeated to induce further shortening. This process depends on both ATP and calcium.

Initiation

Contraction starts at the **neuromuscular junction**, where the nervous system communicates with muscles via **motor (efferent) neurons**. This signal travels down the neuron until it reaches the **nerve terminal (synaptic bouton)**, where acetylcholine is released into the synapse. In the case of the neuromuscular junction, the nerve terminal can also be called the **motor end plate**. Acetylcholine binds to receptors on the sarcolemma, causing depolarization. Each nerve terminal controls a group of myocytes; together, the nerve terminal and its myocytes constitute a **motor unit**.

Depolarization triggers an action potential, which spreads down the sarcolemma to the T-tubules. The action potential travels down the T-tubules into the muscle tissues to the sarcoplasmic reticulum. When the action

potential reaches the sarcoplasmic reticulum, Ca^{2+} is ultimately released. The calcium ions bind to a regulatory subunit in troponin, triggering a change in the conformation of tropomyosin, to which troponin is bound. This change exposes the **myosin-binding sites** on the actin thin filament, as shown in Figure 11.3.

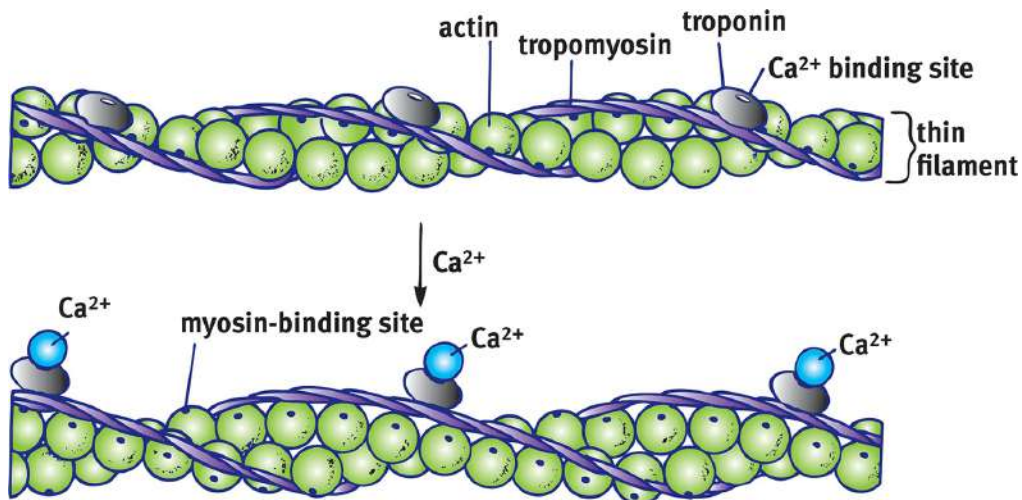


Figure 11.3. Regulation of Contraction with Calcium

Calcium binds to troponin, leading to a conformational change in tropomyosin, which exposes the myosin-binding sites of actin.

Shortening of the Sarcomere

The free globular heads of the myosin molecules move toward and bind with the exposed sites on actin. The newly formed actin–myosin cross bridges then allow myosin to pull on actin, which draws the thin filaments toward the M-line, resulting in shortening of the sarcomere. The actin–myosin cross-bridge cycle is illustrated in Figure 11.4.

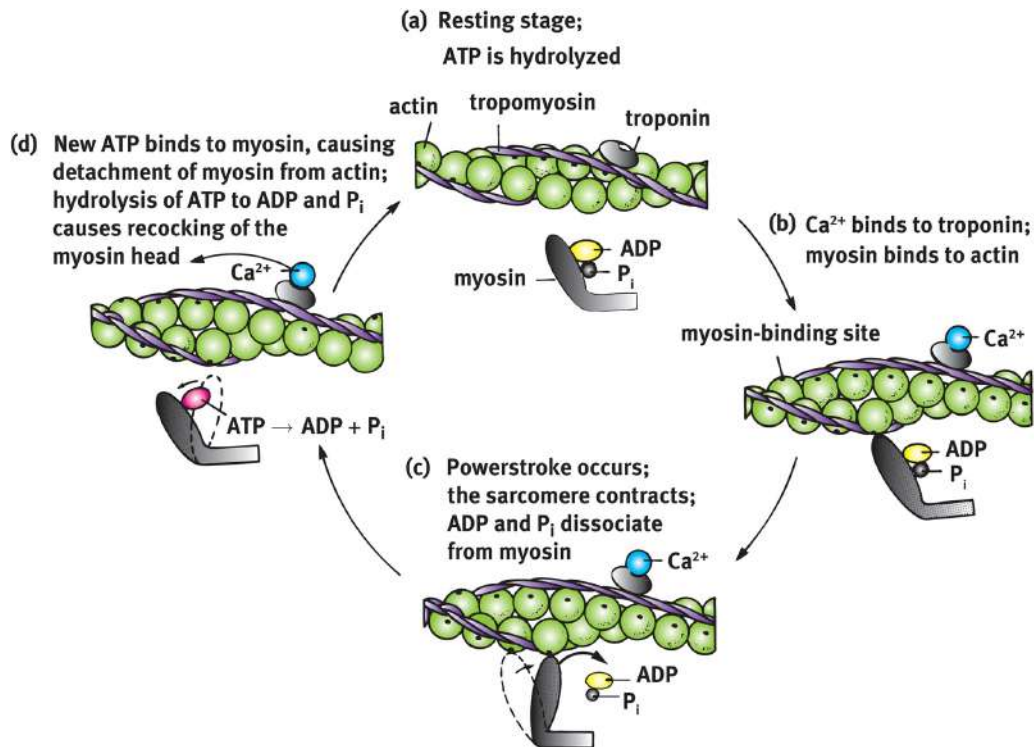


Figure 11.4. The Actin–Myosin Cross-Bridge Cycle

Calcium and ATP are essential for muscle contraction and relaxation.



KEY CONCEPT

It is the dissociation of ADP and P_i from myosin that is responsible for the powerstroke, not the hydrolysis of ATP. The binding of ATP is required for releasing the myosin head from the actin filament.

Starting at the top of the diagram, myosin carrying hydrolyzed ATP (ADP and an inorganic phosphate, P_i) is able to bind with the myosin-binding site. The release of the inorganic phosphate and ADP in rapid succession provides the energy for the powerstroke and results in sliding of the actin filament over the

myosin filament. Then, ATP binds to the myosin head, releasing it from actin. This ATP is hydrolyzed to ADP and P_i , which **recocks** the myosin head so that it is in position to initiate another cross-bridge cycle. The repetitive binding and releasing of myosin heads on actin filaments allows the thin filament to slide along the thick filament, causing sequential shortening of the sarcomere. This is known as the **sliding filament model**, as shown in Figure 11.5.

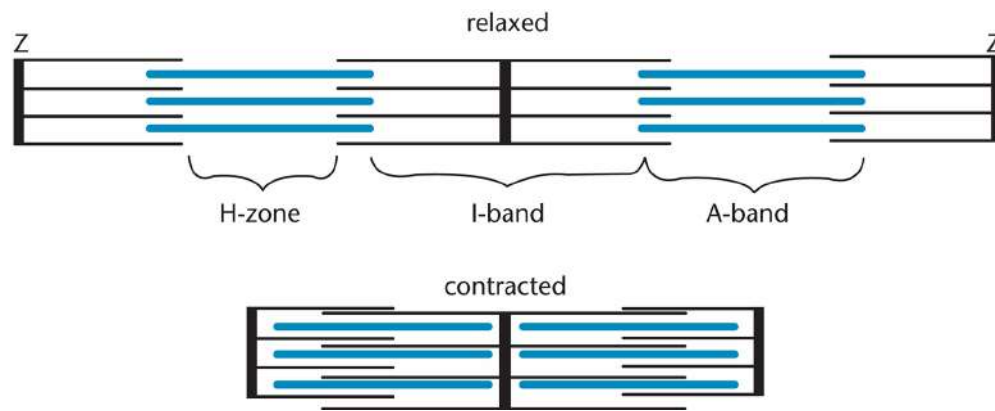


Figure 11.5. Sarcomere Contraction

When a sarcomere contracts, both the H-zone and I-band shorten while the A-band is unchanged.

Relaxation

Acetylcholine is degraded in the synapse by the enzyme known as **acetylcholinesterase**. This results in termination of the signal at the neuromuscular junction and allows the sarcolemma to repolarize. As the signal decays, calcium release ceases, and the SR takes up calcium from the sarcoplasm. The SR tightly controls intracellular calcium concentrations so that muscles are contracted only when necessary. ATP binds to the myosin heads, freeing them from actin. Once the myosin and actin disconnect, the

sarcomere can return to its original width. Without calcium, the myosin-binding sites are covered by tropomyosin and contraction is prevented.



REAL WORLD

After death, ATP production ceases. Myosin heads cannot detach from actin, making it impossible for muscles to relax and lengthen. This condition is known as rigor mortis.

STIMULATION, SUMMATION, AND MUSCLE FATIGUE

Muscle cells, like neurons, exhibit an all-or-nothing response; either they respond completely to a stimulus or not at all. For muscle cells to respond, stimuli must reach a threshold value. The strength of a response from one muscle cell cannot be changed because the only options are all or nothing. Therefore, nerves control overall force by the number of motor units they recruit to respond. Maximal response occurs when all fibers within a muscle are stimulated to contract simultaneously.



BRIDGE

Why do muscle fibers contract in an all-or-nothing fashion? Because they are innervated by neurons that deliver signals using action potentials, which are all-or-nothing phenomena as well. Action potentials are discussed in Chapter 4 of *MCAT Biology Review*.

Simple Twitch

A simple twitch is the response of a single muscle fiber to a brief stimulus at or above threshold, as shown in Figure 11.6a. It consists of a latent period, contraction period, and relaxation period. The **latent period** is the time between reaching threshold and the onset of contraction. It is during this time that the action potential spreads along the muscle and allows for calcium to be released from the sarcoplasmic reticulum. The muscle then contracts, and, assuming calcium is cleared from the sarcoplasm, it then relaxes.



REAL WORLD

The disease *tetanus* is caused by a bacterium called *Clostridium tetani* that releases the toxin *tetanospasmin*. Tetanospasmin blocks the release of GABA from the neurons that *inhibit* motor neurons, making the motor neurons overexcitable. This leads to constant contraction of muscles, which can be so strong as to fracture bones. Tetanus can usually be prevented after exposure by administration of a tetanus immunoglobulin.

Summation and Tetanus

If a muscle fiber is exposed to frequent and prolonged stimulation, it will have insufficient time to relax. The contractions will combine, become stronger and more prolonged. This is known as **frequency summation**, as shown in Figure 11.6b. If the contractions become so frequent that the muscle is unable to relax at all, this is known as **tetanus**. Prolonged tetanus will result in muscle

fatigue, explained below. Note that tetanus (the disease) includes tetanus (the physiological phenomenon) as one of its primary clinical features, but that tetanic physiology also occurs under normal circumstances with multiple simple twitches in succession.

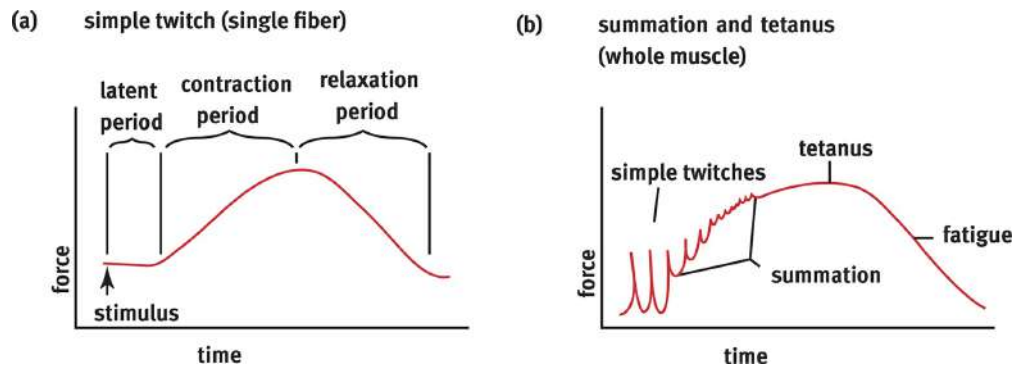


Figure 11.6. Force of Muscle Contraction Due to a Simple Twitch and Tetanus

(a) A simple twitch contraction; (b) Summation of frequent simple twitches can lead to tetanus.

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING

Study 1

Previous research has shown muscle tension generation is influenced by temperature of the muscle fibers.

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To further establish the connection between how temperature interacts with muscle tension generation, researchers measured the rate of tension rise at various temperatures. More specifically, researchers established change in the rate of tension rise is different at high

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temperatures (25°C to 35°C) and low temperatures (20°C to 10°C).

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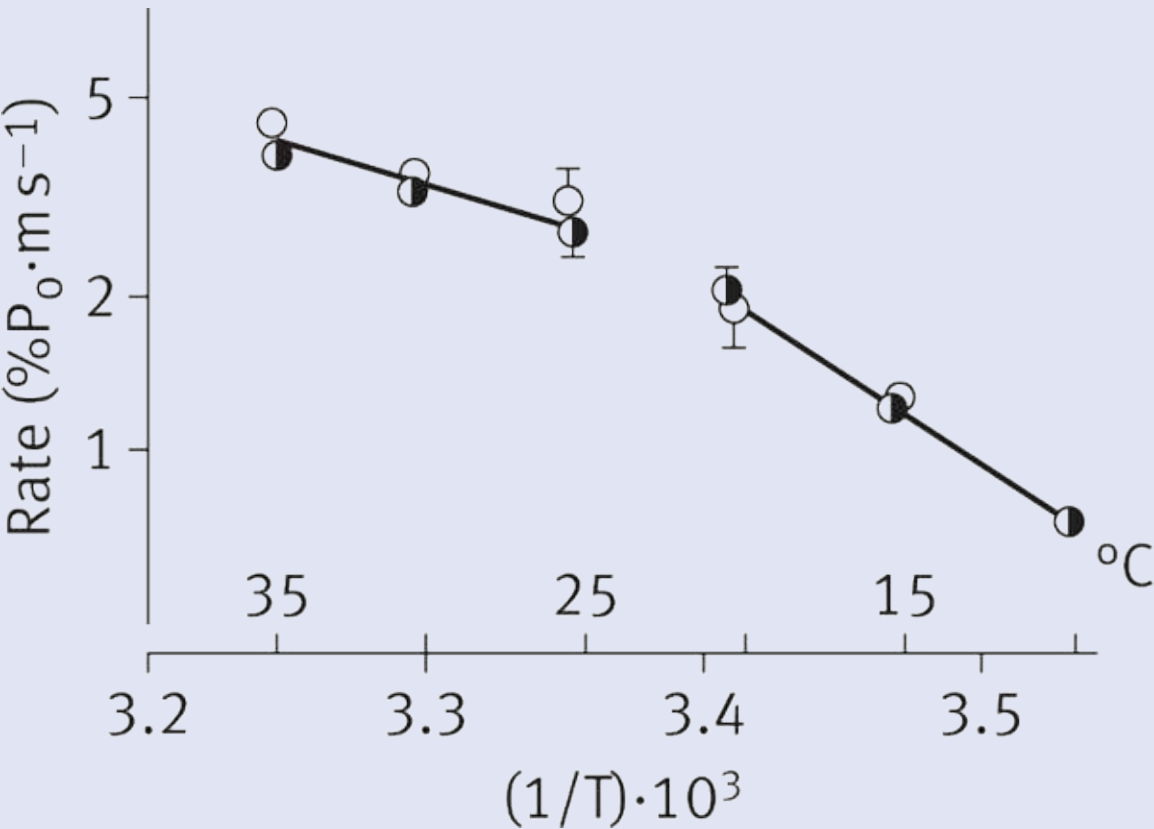


Figure 1 Tension generation rate versus temperature experiment results.

Study 2

Another study is conducted that investigates the relationship between tetanic tension responses and temperature of the same type of muscle fibers. In this experiment, researchers directly observe the tension generation of the muscle plotted over time with electromyography sensor.

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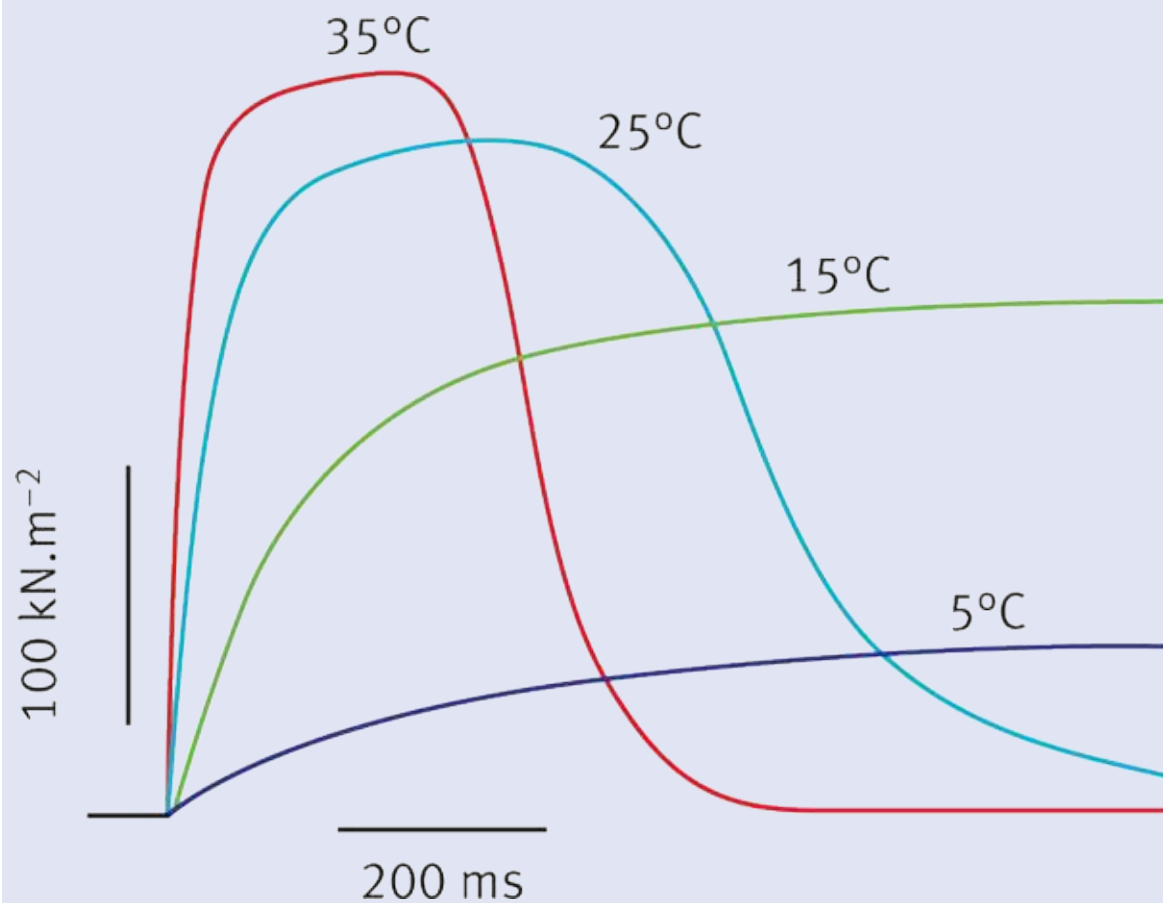


Figure 2 Tetanic tension responses obtained from one intact fiber bundle at four different temperatures, using suitable stimulation frequencies and duration.

adapted from Ranatunga K. W. (2018). Temperature effects on force and actin-myosin interaction in muscle: a look back on

Does Study 2's data conflict with Study 1's data?

This question asks us whether two sets of data conflict with one another. To analyze this relationship, we need to draw the proper conclusions from each data set, and see how these two data sets are related. Conflict is when one data set reveals a relationship that is either not validated by similar data sets, or directly contradicted by similar data sets.

To start, we'll want to make sure we can understand each data set separately. The first study seeks to investigate how temperature affects the rate of tension generation by the muscle tissues. The results of this experiment are displayed as a line graph. One of the first things to look at when analyzing any figure are axes labels and their corresponding units. On the y-axis, we see the rate of tension rise. So, a higher value on that axis means the muscle can generate tension faster, or in other words, have a higher rate. For the x-axis, things are a bit more confusing. The axis displays temperature, but in two ways. Technically, it is the reciprocal of temperature, but fortunately for us the graph is also labeled with the corresponding real temperature location. Because of the reciprocal, higher temperatures are found closer to the left, and temperature decreases going toward the right. Besides this complication, the graph shows a pretty straightforward relationship. At higher temperatures (close to physiological temperature), the rate of tension rise is high, with the drop in rate from 35°C to 25°C being relatively minor. At lower temperatures, the rate of tension rise is much lower, and the change in rate as temperature decreases is more dramatic.

Study 2 examined a similar relationship between temperature and muscle tension. Here the graph shows a more visual representation of how the tension is generated in those muscles. From the shape of the curve, it appears that the measurement is for one stimulation, as the tension rises once and quickly tapers off. From the graph, we can make some quick observations. At higher temperatures, the max tension generated is higher, as well as the time required to reach peak tension. When temperature is lowered, muscle tension generation drops significantly.

To evaluate whether Study 2's data conflicts with Study 1's data, we need to compare and assess how the two data sets are related. Both of these graphs have temperature as the independent variable, so we must look at what each data set demonstrates with that change in temperature. Study 1 shows a rise in rate of tension. While Study 2 does not report rate of tension rise explicitly, it does show tension over time, which is essentially rate! At 35°C and 25°C, the slope of tension increase is sharp, meaning that the rate of tension increase is high. At lower temperature, the slope is more gentle, and tension generation takes significantly longer. This means that Study 2 corroborates Study 1's data in that higher temperature is correlated with a higher rate, while lower temperature is correlated with lower rate.

What about the *change* in rate? How is that visualized? Study 1 shows that change in the rate of tension rise at high temperature is more mild than the change in rate at lower temperatures. From Study 2's graph, we can see that from 35°C to 25°C, the change in the slope (and therefore rate) of the tension is rather small. However, the change in slope between 15°C and 5°C is much more dramatic. Since we know that the slope represents the rate of tension rise, the change in slope must represent the change in rate.

Thus, Study 2 does not conflict with Study 1. In fact, the two studies

corroborate each other!

Oxygen Debt and Muscle Fatigue

Muscles require ATP to function. Slow-twitch (red) muscle fibers have high levels of mitochondria and thus use oxidative phosphorylation to make ATP. However, this means that high concentrations of oxygen are required to generate the large amounts of ATP muscle cells need. There are two supplemental energy reserves in muscle. **Creatine phosphate** is created by transferring a phosphate group from ATP to creatine during times of rest. This reaction can then be reversed during muscle use to quickly generate ATP from ADP:



Muscle also contains myoglobin, which binds oxygen with high affinity. As exercising muscles run out of oxygen, they use myoglobin reserves to keep aerobic metabolism going. Fast-twitch (white) muscle fibers have fewer mitochondria and must rely on glycolysis and fermentation to make ATP under most circumstances. When a person exercises, heart rate and respiratory rate increase in order to move more oxygen to actively respiring muscles. The oxyhemoglobin dissociation curve undergoes a right shift in the presence of increased carbon dioxide concentration, increased hydrogen ion concentration (decreased pH), and increased temperature. However, even with these adaptations, muscle use can quickly overwhelm the ability of the body to deliver oxygen. Then, even red muscle fibers must switch to

anaerobic metabolism and produce lactic acid, at which point the muscle begins to fatigue. The difference between the amount of oxygen needed by the muscles and the actual amount present is called the **oxygen debt**. After the cessation of strenuous exercise, the body must metabolize all of the lactic acid it has produced. Most lactic acid is converted back into pyruvate, which can enter the citric acid cycle. This process requires oxygen, and the amount of oxygen required to recover from strenuous exercise is equal to the oxygen debt.

MCAT CONCEPT CHECK 11.1:

Before you move on, assess your understanding of the material with these questions.

1. What type(s) of muscle (skeletal, smooth, or cardiac) does each of the following describe? (Note: Circle the correct response(s) next to each item.)

• Striated:	Skeletal	Smooth	Cardiac
• Always uninucleated:	Skeletal	Smooth	Cardiac
• Always polynucleated:	Skeletal	Smooth	Cardiac
• Voluntary:	Skeletal	Smooth	Cardiac
• Innervated by the autonomic nervous system:	Skeletal	Smooth	Cardiac
• Exhibits myogenic activity:	Skeletal	Smooth	Cardiac

2. Which zone or band in the sarcomere does NOT change its length during muscle contraction? Why?

3. What are the events that initiate muscle contraction, in order? Start with neurotransmitter release and trace the pathway to the point where myosin binds with actin.

4. What role does the binding of ATP to the myosin head play in the cross-bridge cycle? What about the dissociation of ADP and inorganic phosphate from the myosin head?

- Binding of ATP:

- Dissociation of ADP and inorganic phosphate:

5. What is tetanus (the physiological phenomenon, not the disease)?

11.2 The Skeletal System

LEARNING OBJECTIVES

After Chapter 11.2, you will be able to:

- Distinguish between compact and spongy bone
- Identify the three structural parts of a bone and their relative contributions to growth
- Recall the major chemical component of bone
- Describe the function(s) of osteoblasts, osteoclasts, and chondrocytes
- Name the substance that lubricates movable joints and where it is produced

There are two types of skeletons: exoskeletons and endoskeletons.

Exoskeletons encase whole organisms and are usually found in arthropods, such as crustaceans and insects. Vertebrates, including humans, have **endoskeletons**. Endoskeletons are internal, but are not able to protect the soft tissue structures as well as exoskeletons. However, exoskeletons must be shed and regrown to accommodate growth. Endoskeletons are much better able to accommodate the growth of a larger organism.

SKELETAL STRUCTURE

The components of our skeletal system are divided into axial and appendicular skeletons. The **axial skeleton** consists of the skull, vertebral column, rib cage, and hyoid bone (a small bone in the anterior neck used for swallowing); it provides the basic central framework for the body. The **appendicular skeleton** consists of the bones of the limbs (humerus, radius and ulna, carpals, metacarpals, and phalanges in the upper limb; and femur, tibia and fibula, tarsals, metatarsals, and phalanges in the lower limb); the pectoral girdle (scapula and clavicle); and pelvis. Both skeleton types are covered by other structures (muscle, connective tissue, and vasculature). The structure of the skeleton is shown in Figure 11.7, with many of the bones labeled (individual bones are labeled in red).

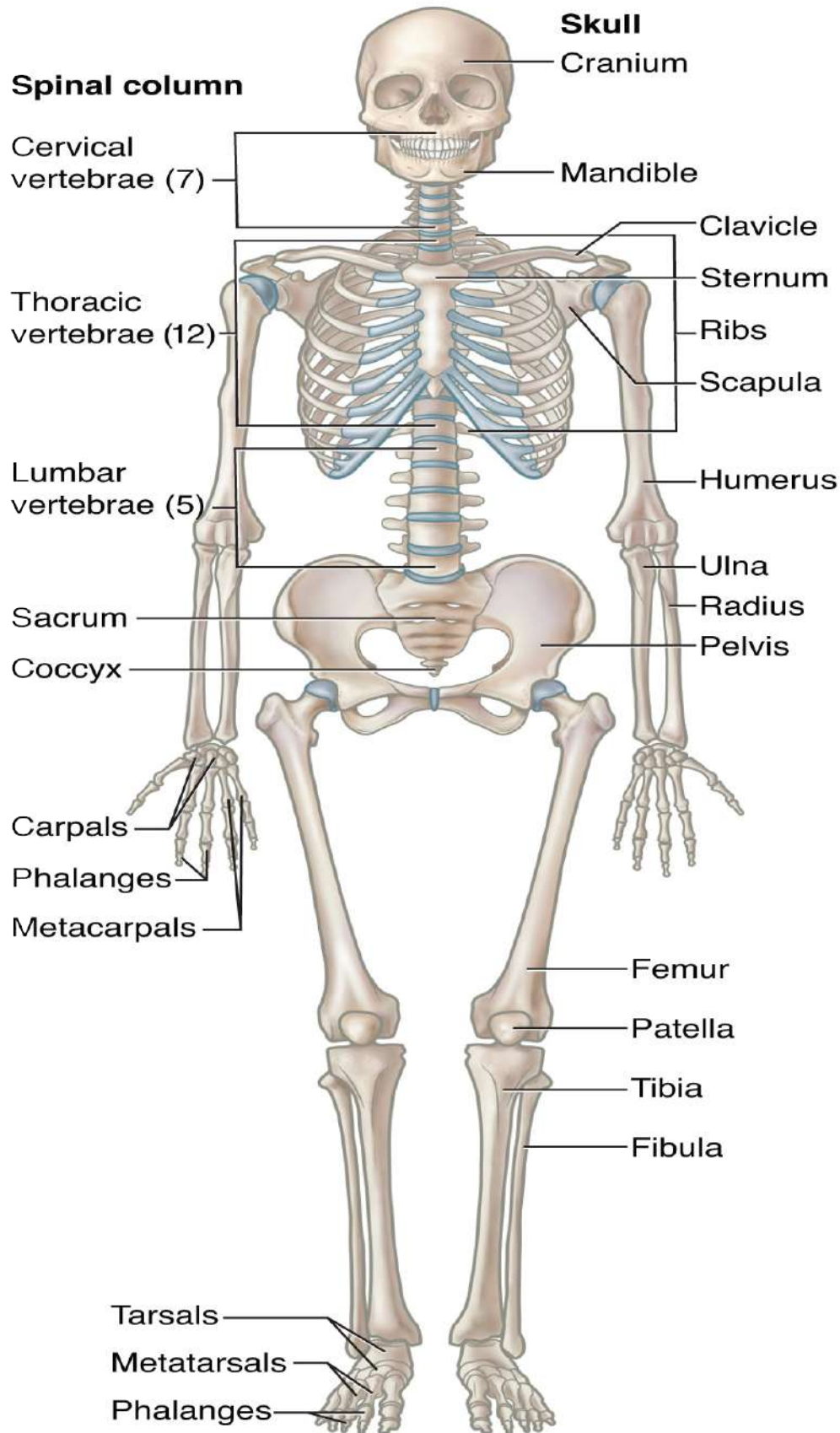


Figure 11.7. Anatomy of the Human Skeleton

The skeleton is created from two major components: bone and cartilage.



REAL WORLD

An adult human has 206 bones. Over 100 of these are in the hands and feet.

BONE COMPOSITION

Bone is a connective tissue derived from embryonic mesoderm. Bone is much harder than cartilage, but is relatively lightweight.

Macroscopic Bone Structure

The structure of bone can be seen in Figure 11.8.

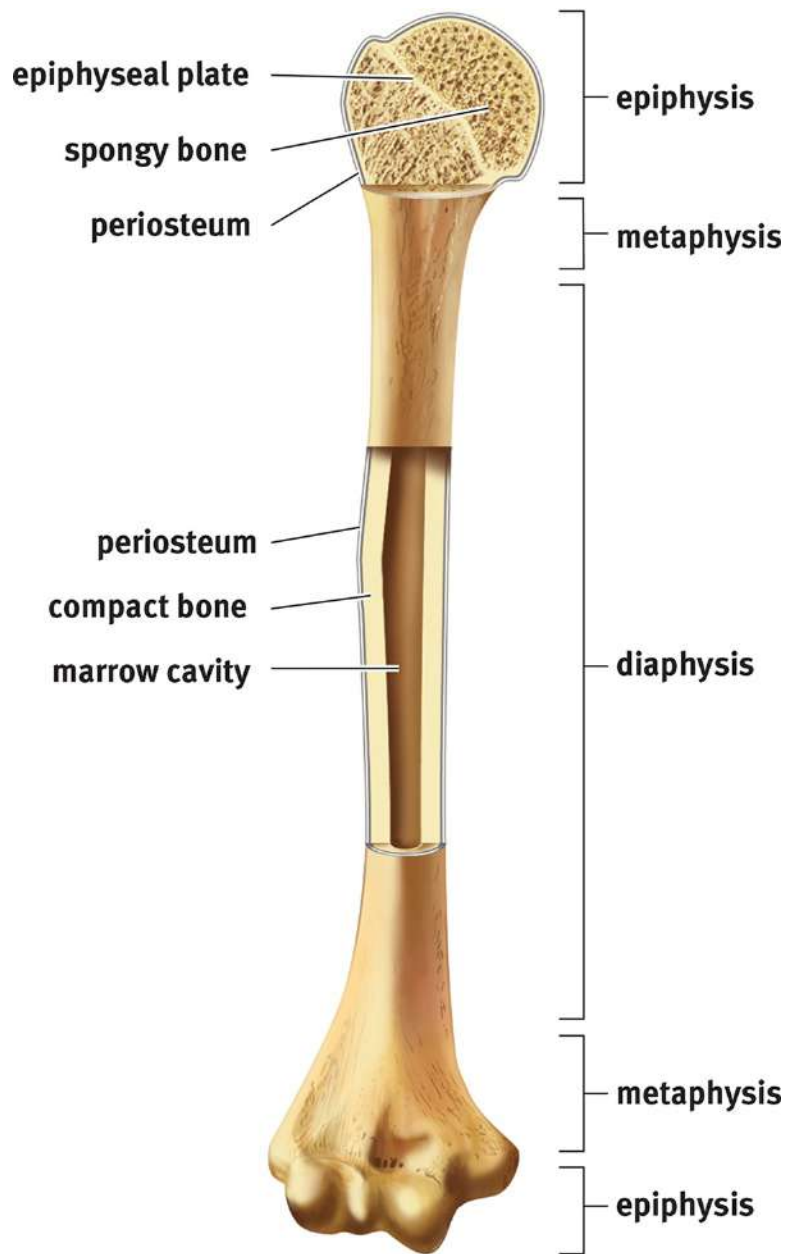


Figure 11.8. Anatomy of a Long Bone (Humerus)

Bone's characteristic strength comes specifically from **compact bone**. It lives up to its name, as it is both dense and strong. The other type of bone structure is **spongy** or **cancellous** bone. The lattice structure of spongy bone is visible under microscopy and consists of bony spicules (points) known as **trabeculae**. The cavities between trabeculae are filled with **bone marrow**,

which may be either red or yellow. **Red marrow** is filled with hematopoietic stem cells, which are responsible for the generation of all the cells in our blood; **yellow marrow** is composed primarily of fat and is relatively inactive.

Bones in the appendicular skeleton are typically **long bones**, which are characterized by cylindrical shafts called **diaphyses** that swell at each end to form **metaphyses**, and that terminate in **epiphyses**. The outermost portions of bone are composed of compact bone, whereas the internal core is made of spongy bone. Long bone diaphyses and metaphyses are full of bone marrow. The epiphyses, on the other hand, use their spongy cores for more effective dispersion of force and pressure at the joints. At the internal edge of the epiphysis is an **epiphyseal (growth) plate**, which is a cartilaginous structure and the site of longitudinal growth. Prior to adulthood, the epiphyseal plate is filled with mitotic cells that contribute to growth; during puberty, these epiphyseal plates close and vertical growth is halted. Finally, a fibrous sheath called the **periosteum** surrounds the long bone to protect it as well as serve as a site for muscle attachment. Some periosteal cells are capable of differentiating into bone-forming cells; a healthy periosteum is necessary for bone growth and repair.

Structures in the musculoskeletal system are held together with dense connective tissue. **Tendons** attach muscle to bone and **ligaments** hold bones together at joints.



BRIDGE

The root *lig-* comes from Latin, where it means to “to tie; bind.” Think of *DNA ligase*, discussed in Chapter 6 of *MCAT Biochemistry Review*.

Think of ligands in complex ions, discussed in Chapter 9 of *MCAT General Chemistry Review*. In this case, ligaments tie bones to each other to stabilize joints.

Microscopic Bone Structure

The strength of compact bone comes from the **bone matrix**, which has both organic and inorganic components. The organic components include collagen, glycoproteins, and other peptides. The inorganic components include calcium, phosphate, and hydroxide ions, which harden together to form **hydroxyapatite** crystals ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$). Minerals such as sodium, magnesium, and potassium are also stored in bone.

Strong bones require uniform distribution of organic and inorganic materials. The bony matrix is ordered into structural units known as **osteons** or **Haversian systems**, as shown in Figure 11.9. Each of these osteons contains concentric circles of bony matrix called **lamellae** surrounding a central microscopic channel. Longitudinal channels (those with an axis parallel to the bone) are known as **Haversian canals**, while transverse channels (those with an axis perpendicular to the bone) are known as **Volkman's canals**. These canals contain the blood vessels, nerve fibers, and lymph vessels that maintain the health of the bone. Between the lamellar rings are small spaces called **lacunae**, which house mature bone cells known as **osteocytes**. The lacunae are interconnected by tiny channels called **canaliculi** that allow for the exchange of nutrients and wastes between osteocytes and the Haversian and Volkman's canals.

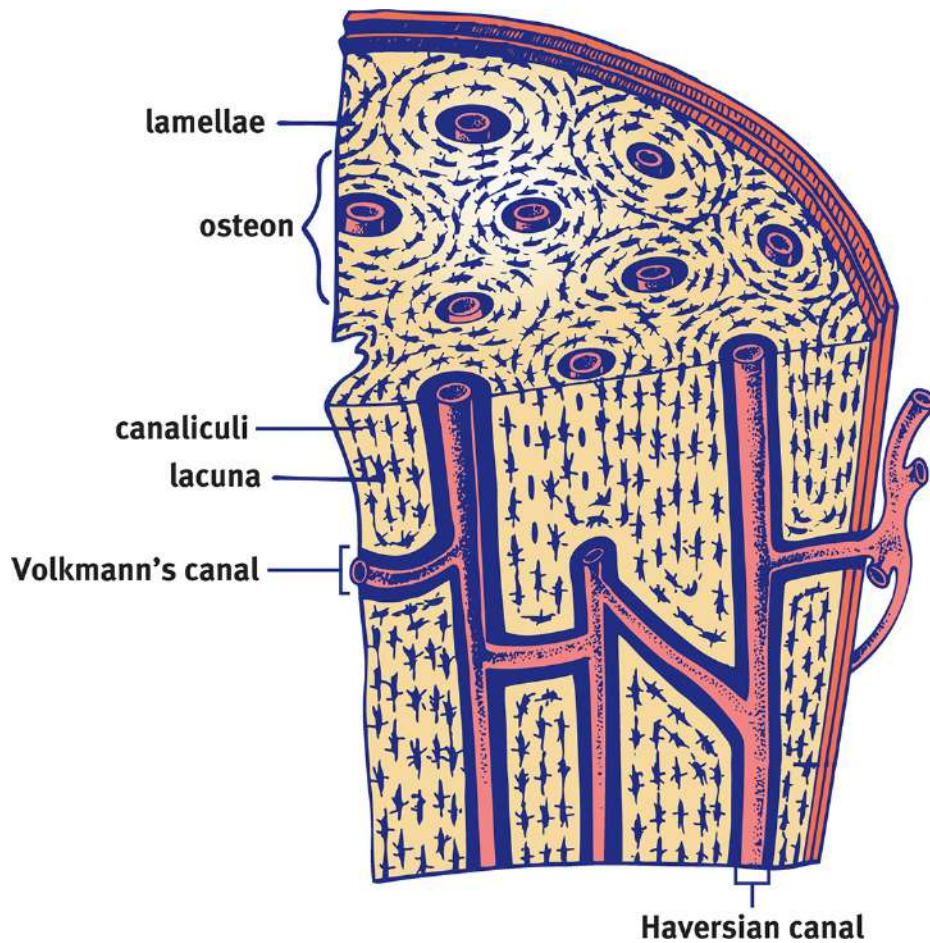


Figure 11.9. Bone Matrix

Cross-sectional and longitudinal views highlighting Haversian systems.



KEY CONCEPT

Bone appears to be rigid and static, but it is actually quite dynamic. It is both vascular and innervated, which is why it hurts so much to break a bone. In addition, bone remains in a vigorous equilibrium between construction and destruction, known as bone remodeling.



REAL WORLD

Osteoporosis is the most common bone disease in the United States. It is thought to be the result of increased osteoclast resorption and some concomitant slowing of bone formation, both of which lead to loss of bone mass. Estrogen is believed to help prevent osteoporosis by stimulating osteoblast activity.

Bone Remodeling

Two cell types are largely responsible for building and maintaining strong bones: osteoblasts and osteoclasts. **Osteoblasts** build bone, whereas **osteoclasts**, polynucleated resident macrophages of bone, resorb it. These processes together contribute to the constant turnover of bone, as shown in Figure 11.10. During bone formation, essential ingredients such as calcium and phosphate are obtained from the blood. During bone resorption, these ions are released back into the bloodstream. Bone remodeling occurs in response to stress, and bone actually remodels in such a way as to accommodate the repetitive stresses faced by the body. Endocrine hormones may also affect bone metabolism. **Parathyroid hormone**, a peptide hormone released by the parathyroid glands in response to low blood calcium, promotes resorption of bone, increasing the concentration of calcium and phosphate in the blood. **Vitamin D**, which is activated by parathyroid hormone, also promotes the resorption of bone. This may seem counterintuitive at first— isn't vitamin D used to promote bone growth? Indeed, the resorption of bone in response to vitamin D actually encourages the growth of new, stronger bone, thus overcompensating for the effect of resorbing bone in the first place. Finally, **calcitonin**, a peptide hormone

released by the parafollicular cells of the thyroid in response to high blood calcium, promotes bone formation, lowering blood calcium levels.

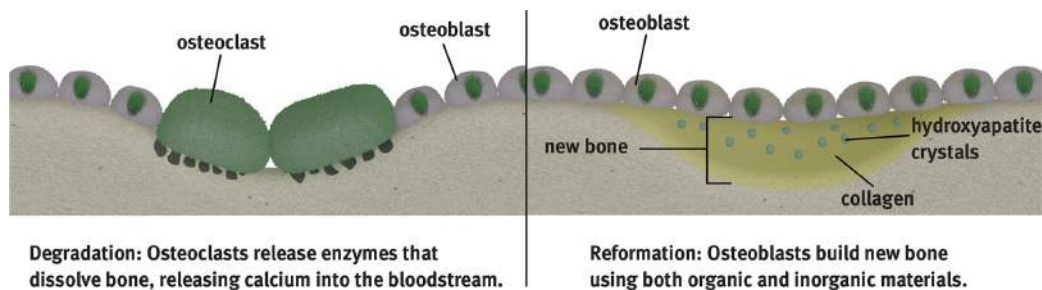


Figure 11.10. Bone Remodeling



MNEMONIC

Osteo**b**lasts **b**uild bone. Osteo**c**lasts **c**hew bone.

CARTILAGE

Cartilage is softer and more flexible than bone. Cartilage consists of a firm but elastic matrix called **chondrin** that is secreted by cells called **chondrocytes**. Fetal skeletons are mostly made up of cartilage. This is advantageous because fetuses must grow and develop in a confined environment and then must traverse the birth canal. Adults have cartilage only in body parts that need a little extra flexibility or cushioning (external ear, nose, walls of the larynx and trachea, intervertebral discs, and joints). Cartilage also differs from bone in that it is avascular (without blood and lymphatic vessels) and is not innervated.

Most of the bones of the body are created by the hardening of cartilage into bone. This process is known as **endochondral ossification** and is responsible for the formation of most of the long bones of the body. Bones may also be formed through **intramembranous ossification**, in which undifferentiated embryonic connective tissue (**mesenchymal tissue**) is transformed into, and replaced by, bone. This occurs in bones of the skull.

JOINTS AND MOVEMENT

Like bone and cartilage, joints are also made of connective tissue and come in two major varieties: immovable and movable. **Immovable joints** consist of bones that are fused together to form **sutures** or similar fibrous joints. These joints are found primarily in the head, where they anchor bones of the skull together.

Movable joints, structures of which are shown in Figure 11.11, include hinge joints (like the elbow or knee), ball-and-socket joints (like the shoulder or hip), and others. They permit bones to shift relative to one another. Movable joints are strengthened by **ligaments**, which are pieces of fibrous tissue that connect bones to one another, and consist of a **synovial capsule**, which encloses the actual **joint cavity (articular cavity)**. A layer of soft tissue called the **synovium** secretes **synovial fluid**, which lubricates the movement of structures in the joint space. The **articular cartilage** contributes to the joint by coating the articular surfaces of the bones so that impact is restricted to the lubricated joint cartilage, rather than to the bones.

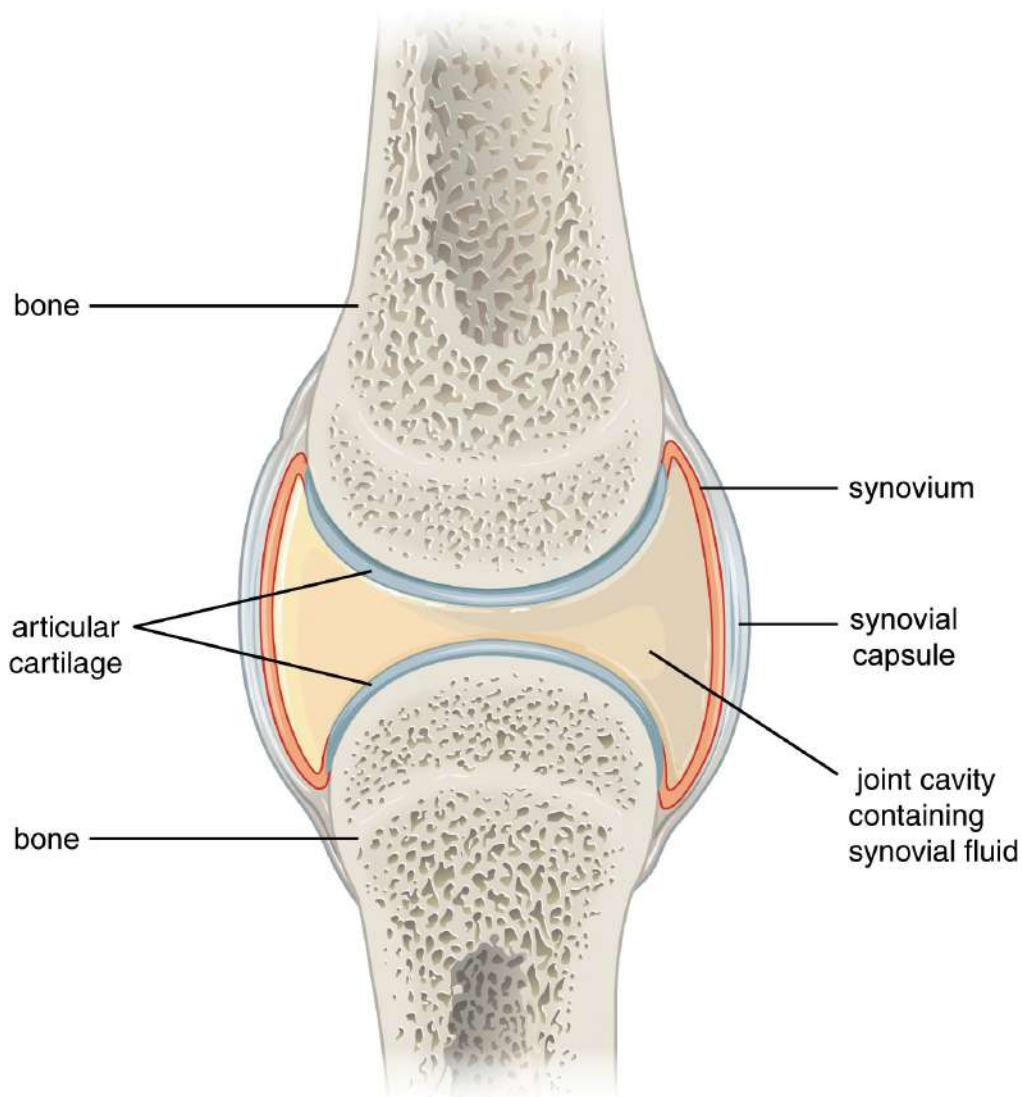


Figure 11.11. Structures in a Movable Joint



REAL WORLD

Degradation of articular cartilage (cartilage in joints) can lead to medical issues like osteoarthritis. Osteoarthritis (or “arthritis” in the lay population) is painful because a lack of cartilage in joints leads to bones rubbing directly on one another.

When a muscle is attached to two bones, its contraction will cause one of the bones to move. The end of the muscle with a larger attachment to bone (usually the proximal connection) is called the **origin**. The end with the smaller attachment to bone (usually the distal connection) is called the **insertion**. Often, our muscles work in **antagonistic pairs**; one relaxes while the other contracts. Such is the case in the arm, where the biceps brachii and triceps brachii work antagonistically, as shown in Figure 11.12. When the biceps contracts and the triceps relaxes, the elbow is flexed; when the triceps contracts and the biceps relaxes, the elbow is extended. Muscles can also be **synergistic**—working together to accomplish the same function.

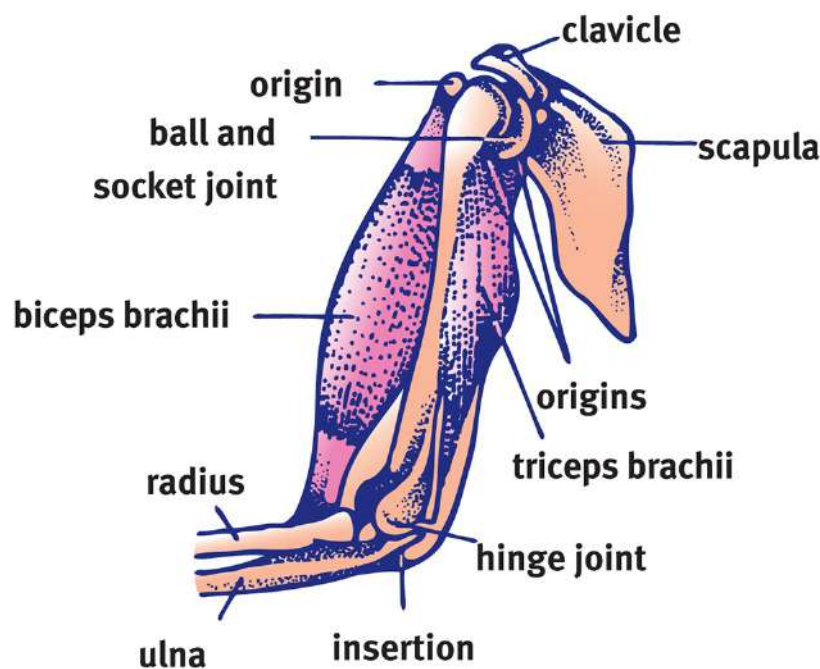


Figure 11.12. Antagonistic Muscle Pairs

The biceps brachii and triceps brachii are an example of a muscle pair that works antagonistically; the contraction of one causes the other to elongate.

Muscles may also be classified by the types of movements they coordinate. A **flexor** muscle decreases the angle across a joint (like the biceps brachii); an **extensor** increases or straightens this angle (like the triceps brachii). An **abductor** moves a part of the body away from the midline (like the deltoid); an **adductor** moves a part of the body toward the midline (like the pectoralis major). **Medial** and **lateral rotation** describe motions that occur in limbs; a medial rotator rotates the axis of the limb toward the midline (like the subscapularis), whereas a lateral rotator rotates the axis of the limb away from the midline (like the infraspinatus).

MCAT CONCEPT CHECK 11.2:

Before you move on, assess your understanding of the material with these questions.

1. What is the difference between compact and spongy bone?

- Compact bone:

- Spongy bone:

2. What are the three structural parts of a bone? Which part contributes most to linear growth?

3. What chemical forms most of the inorganic component of bone?

4. What are the functions of osteoblasts, osteoclasts, and chondrocytes?

- Osteoblast:

- Osteoclast:

- Chondrocyte:

5. What liquid provides the lubrication for movable joints?
What tissue produces it?



Conclusion

One concept that has been emphasized throughout the past eight chapters on anatomy and physiology is the notion that organ systems work together in order to achieve a desired effect. The musculoskeletal system is no different. Usually, we think of the musculoskeletal system as being responsible for movement, but to limit the musculoskeletal system to that function would be shortsighted. The bones are reservoirs of calcium and other minerals that can be released through hormonal signaling. They protect the internal organs and provide support for the body. Muscle tissue not only moves these bones, but pumps blood through the body and is key to the function of a number of other systems, including respiration, digestion, blood pressure and vascular tone, and reproduction and childbirth. As you review anatomy and physiology and master the fundamentals of each organ system, be sure to pay special attention to how each organ system interacts with the others. While the MCAT expects you to understand each individual organ system, it will also challenge you by asking you to think critically about how one system impacts another. By spending some time in your studies looking at these interactions, you will be one step ahead on Test Day.

In our final chapter of *MCAT Biology Review*, we will switch gears and look at the transfer of information from generation to generation. This picks up on a discussion from the first three chapters of the book in which we explored the organization of cells and their genetic material, reproduction, and embryogenesis and development. In the next chapter, we'll describe classical

(Mendelian) inheritance and conclude with a note on how the gene pool can change over time with the topic of evolution.

Concept Summary

The Muscular System

- There are three main types of muscle: skeletal muscle, smooth muscle, and cardiac muscle.
 - **Skeletal muscle** is involved in support and movement, propulsion of blood in the venous system, and thermoregulation. It appears **striated**, is under voluntary (somatic) control, is polynucleated, and can be divided into **red (slow-twitch) fibers** that carry out oxidative phosphorylation and **white (fast-twitch) fibers** that rely on anaerobic metabolism.
 - **Smooth muscle** is in the respiratory, reproductive, cardiovascular, and digestive systems. It appears nonstriated, is under involuntary (autonomic) control, and is uninucleated. It can display **myogenic activity**, or contraction without neural input.
 - **Cardiac muscle** comprises the contractile tissue of the heart. It appears striated, is under involuntary (autonomic) control, and is uninucleated (sometimes binucleated). It can also display myogenic activity. Cells are connected with **intercalated discs** that contain **gap junctions**.
- The **sarcomere** is the basic contractile unit of striated muscle.
 - Sarcomeres are made of thick (**myosin**) and thin (**actin**) filaments.
 - **Troponin** and **tropomyosin** are found on the thin filament and regulate

actin–myosin interactions.

- The sarcomere can be divided into different lines, zones, and bands.
 - The boundaries of each sarcomere are defined by **Z-lines**.
 - The **M-line** is located in the middle of the sarcomere.
 - The **I-band** contains only thin filaments.
 - The **H-zone** consists of only thick filaments.
 - The **A-band** contains the thick filaments in their entirety. It is the only part of the sarcomere that maintains a constant size during contraction.
- Sarcomeres attach end-to-end to become **myofibrils**, and each **myocyte** (muscle cell or **muscle fiber**) contains many myofibrils.
 - Myofibrils are surrounded by the **sarcoplasmic reticulum**, a calcium-containing modified endoplasmic reticulum, and the cell membrane of a myocyte is known as the **sarcolemma**.
 - A system of **T-tubules** is connected to the sarcolemma and oriented perpendicularly to the myofibrils, allowing the action potential to reach all parts of the muscle.
- Muscle contraction begins at the **neuromuscular junction**, where the motor neuron releases acetylcholine that binds to receptors on the sarcolemma, causing depolarization.
 - This depolarization spreads down the sarcolemma to the T-tubules, triggering the release of calcium ions.
 - Calcium binds to troponin, causing a shift in tropomyosin and exposure of the myosin-binding sites on the actin thin filament.
 - Shortening of the sarcomere occurs as myosin heads bind to the

exposed sites on actin, forming cross bridges and pulling the actin filament along the thick filament, which results in contraction. This is known as the **sliding filament model**.

- The muscle relaxes when acetylcholine is degraded by acetylcholinesterase, terminating the signal and allowing calcium to be brought back into the SR. ATP binds to the myosin head, allowing it to release from actin.
- Muscle cells exhibit an all-or-nothing response called a **simple twitch**.
 - Addition of multiple simple twitches before the muscle has an opportunity to fully relax is called **frequency summation**.
 - Simple twitches that occur so frequently as to not let the muscle relax at all can lead to **tetanus**, a more prolonged and stronger contraction.
- Muscle cells have additional energy reserves to reduce **oxygen debt** (the difference between the amount of oxygen needed and the amount present) and forestall fatigue.
 - **Creatine phosphate** can transfer a phosphate group to ADP, forming ATP.
 - **Myoglobin** is a heme-containing protein that is a muscular oxygen reserve.

The Skeletal System

- Internal skeletons (like those in humans) are called **endoskeletons**; external skeletons (like those in arthropods) are called **exoskeletons**.

- The human skeletal system can be divided into axial and appendicular skeletons.
 - The **axial skeleton** consists of structures in the midline such as the skull, vertebral column, rib cage, and hyoid bone.
 - The **appendicular skeleton** consists of the bones of the limbs, the pectoral girdle, and the pelvis.

- Bone is derived from embryonic mesoderm and includes both compact and spongy (cancellous) types.
 - **Compact bone** provides strength and is dense.
 - **Spongy** or **cancellous bone** has a lattice-like structure consisting of bony spicules known as **trabeculae**. The cavities are filled with bone marrow.
 - Long bones contain shafts called **diaphyses** that flare to form **metaphyses** and terminate in **epiphyses**. The epiphysis contains an **epiphyseal (growth) plate** that causes linear growth of the bone.
 - Bone is surrounded by a layer of connective tissue called **periosteum**.
 - Bones are attached to **muscles** by tendons and to each other by **ligaments**.

- **Bone matrix** has both organic components, like collagen, glycoproteins, and other peptides; and inorganic components, like **hydroxyapatite**.
 - Bone is organized into concentric rings called **lamellae** around a central **Haversian** or **Volkman's canal**. This structural unit is called an **osteon** or **Haversian system**.
 - Between lamellar rings are **lacunae**, where osteocytes reside, which are connected with **canaliculi** to allow for nutrient and waste transfer.

- Bone remodeling is carried out by osteoblasts and osteoclasts.
Osteoblasts build bone, while **osteoclasts** resorb bone.
 - **Parathyroid hormone** increases resorption of bone, increasing calcium and phosphate concentrations in the blood.
 - **Vitamin D** also increases resorption of bone, leading to increased turnover and, subsequently, the production of stronger bone.
 - **Calcitonin** increases bone formation, decreasing calcium concentrations in the blood.
- Cartilage is a firm, elastic material secreted by **chondrocytes**. Its matrix is called **chondrin**.
 - Cartilage is usually found in areas that require more flexibility or cushioning.
 - Cartilage is avascular and is not innervated.
- In fetal life, bone forms from cartilage through **endochondral ossification**. Some bones, especially those of the skull, form directly from undifferentiated tissue (**mesenchyme**) in **intramembranous ossification**.
- Joints may be classified as immovable or movable.
 - **Immovable joints** are fused together to form sutures or similar fibrous joints.
 - **Movable joints** are usually strengthened by ligaments and contain a **synovial capsule**.
 - **Synovial fluid**, secreted by the **synovium**, aids in motion by lubricating the joint.
 - Each bone in the joint is coated with **articular cartilage** to aid in

movement and provide cushioning.

- Muscles that serve opposite functions come in **antagonistic pairs**; when one muscle contracts, the other lengthens.

Answers to Concept Checks

11.1

- .. Skeletal and cardiac muscle are striated. Smooth muscle is always uninucleated. Skeletal muscle is always polynucleated. Skeletal muscle is voluntary. Smooth and cardiac muscle are innervated by the autonomic nervous system. Smooth and cardiac muscle exhibit myogenic activity.
- 1. The A-band does not change length during muscle contraction because it is the entire length of the myosin filament. The filaments do not change length, but rather slide over each other; thus, the A-band should remain a constant length during contraction.
- 2. Release of acetylcholine from motor neuron → activation of acetylcholine receptors in sarcolemma → depolarization of sarcolemma → spreading of signal using T-tubules → release of calcium from sarcoplasmic reticulum (SR) → binding of calcium to troponin → conformational shift in tropomyosin → exposure of myosin-binding sites → myosin binds to actin
- 3. ATP binding allows the myosin filament to disconnect from actin. Dissociation of ADP and inorganic phosphate from myosin causes the powerstroke.
- 4. Tetanus is the summation of multiple simple twitches that occur too quickly for the muscle to relax. This leads to a stronger and more prolonged contraction of the muscle.

11.2

- .. Compact bone is dense and is used for its strength; it forms most of the outer layers of a bone. Spongy (cancellous) bone has many spaces between

bony spicules called trabeculae and is the site of marrow production. It is found in the interior core of the bone and also helps distribute forces or pressures on the bone.

- 2. The three parts of a bone are the diaphysis, metaphysis, and epiphysis. Growth plates are found in epiphyses and contribute to linear growth.
- 3. Most inorganic bone is composed of hydroxyapatite crystals.
- 4. Osteoblasts build bone. Osteoclasts “chew” bone (break it down). Chondrocytes form cartilage.
- 5. Synovial fluid, produced by the synovium, lubricates movable joints.



Shared Concepts

Biology Chapter 4

The Nervous System

Biology Chapter 5

The Endocrine System

Biology Chapter 6

The Respiratory System

Biology Chapter 7

The Cardiovascular System

General Chemistry Chapter 9

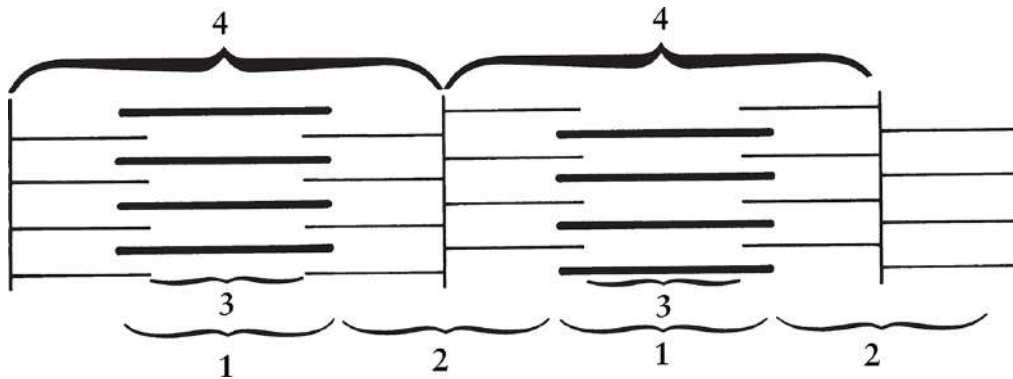
Solutions

Physics and Math Chapter 1

Kinematics and Dynamics

Discrete Practice Questions

Questions 1, 2, and 3 are based on the following diagram:



1. During muscle contraction, which of the following regions decrease(s) in length?

(A) 1 only
(B) 1 and 2 only
(C) 3 and 4 only
(D) 2, 3, and 4 only

2. Region 1 refers to:

(A) the thick filaments only.

- (B) the thin filaments only.
- (C) the A-band.
- (D) the I-band.

3. Which region represents one sarcomere?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

4. With which of the following molecules does Ca^{2+} bind after its release from the sarcoplasmic reticulum to regulate muscle contraction?

- (A) Myosin
- (B) Actin
- (C) Troponin
- (D) Tropomyosin

5. Which of the following cells is correctly coupled with its definition?

- (A) Osteoblasts—bone cells involved in the secretion of bone matrix
- (B) Osteoclasts—immature bone cells
- (C) Osteocytes—polynucleated cells actively involved in bone resorption
- (D) Chondrocytes—undifferentiated bone marrow cells

6. An X-ray of the right femur in a child shows that it is shorter than the opposite femur, and below the average length for a child of this age. Which region of the bone is most likely to have caused this abnormality?
- (A) Diaphysis
 - (B) Metaphysis
 - (C) Epiphysis
 - (D) Periosteum
7. Which of the following INCORRECTLY pairs a type of muscle fiber with a characteristic of that fiber?
- (A) Red fibers—rich in mitochondria
 - (B) Red fibers—high levels of myoglobin
 - (C) White fibers—fast-twitching
 - (D) White fibers—predominantly use aerobic respiration
8. When the knee moves back and forth during walking, what prevents the surfaces of the leg bones from rubbing against each other?
- (A) Articular cartilage
 - (B) Epiphyses
 - (C) Synovial fluid
 - (D) Smooth muscle
9. Which type(s) of muscle is/are always multinucleated?

- I. Cardiac muscle
- II. Skeletal muscle
- III. Smooth muscle

- (A) I only
- (B) II only
- (C) III only
- (D) I and II only

10. Which type(s) of muscle has/have myogenic activity?

- I. Cardiac muscle
- II. Skeletal muscle
- III. Smooth muscle

- (A) I only
- (B) II only
- (C) III only
- (D) I and III only

11. Red bone marrow is involved in erythrocyte formation. In contrast, yellow bone marrow:

- (A) is involved in leukocyte formation.
- (B) is responsible for drainage of lymph.
- (C) causes the formation of spicules.
- (D) contains predominantly adipose tissue.

12. Which of the following statements regarding the periosteum is INCORRECT?
- (A) The periosteum serves as a site of attachment of bone to muscle.
 - (B) Cells of the periosteum may differentiate into osteoblasts.
 - (C) The periosteum is a fibrous sheath that surrounds long bones.
 - (D) The periosteum secretes fluid into the joint cavity.
13. Which of the following bones is NOT a part of the appendicular skeleton?
- (A) The triquetrum, one of the carpal bones
 - (B) The calcaneus, which forms the heel
 - (C) The ischium, one of the fused pelvic bones
 - (D) The sternum, or breastbone
14. To facilitate the process of birth, an infant's head is somewhat flexible. This flexibility is due in part to the two fontanelles, which are soft spots of connective tissue in the infant's skull. With time, the fontanelles will close through a process known as:
- (A) endochondral ossification.
 - (B) intramembranous ossification.
 - (C) bone resorption.
 - (D) longitudinal growth.

15. A young woman presents to the emergency room with a broken hip. She denies any recent history of trauma to the joint. Blood tests reveal a calcium concentration of $11.5 \frac{\text{mg}}{\text{dL}}$ (normal: 8.4–10.2). Which tissue is likely responsible for these findings?

- (A) Thyroid
- (B) Cartilage
- (C) Parathyroid
- (D) Smooth muscle

Discrete Practice Answers

1. **D**

We are given a diagram of a sarcomere and asked to determine which regions shorten during muscle contraction. All bands and zones of the sarcomere shorten during contraction except the A-band, which is the full length of the thick filaments. In this diagram, that is region 1. Thus, the remaining regions all shorten, making **(D)** the correct answer. Region 2 represents the I-band, region 3 represents the H-zone, and region 4 is the length of the sarcomere between Z-lines.

2. **C**

Region 1 contains both thick and thin filaments overlapping each other. This region refers to the A-band and is measured from one end of the thick filaments to the other. This is also the only portion of the sarcomere that does not change length during muscle contraction.

3. **D**

The sarcomere is the contractile unit in striated muscle cells. One sarcomere is represented by the area between the two vertical lines, referred to as the Z-lines. In addition, the Z-lines anchor the thin filaments. In the diagram, a sarcomere is therefore defined by region 4.

4. **C**

Calcium is released from the sarcoplasmic reticulum into the sarcoplasm. It binds the troponin molecules on the thin filaments, causing the strands of

tropomyosin to shift, thereby exposing the myosin-binding sites on the thin filaments.

5. **A**

Let's quickly define each one of the four cells discussed in the answer choices. Osteoblasts are bone cells involved in the secretion of bone matrix, as **(A)** states. Osteoclasts are large, polynucleated cells involved in bone resorption. Osteocytes are mature bone cells that eventually become surrounded by their matrix; their primary role is bone maintenance. Finally, chondrocytes are cells that secrete chondrin, an elastic matrix that makes up cartilage.

6. **C**

This question is essentially asking where longitudinal growth occurs in bones. The most likely site of abnormalities in this child's femur is the epiphyseal plate, a disc of cartilaginous cells at the internal border of the epiphysis, because the epiphyseal plate is the site of longitudinal growth. Damage to the epiphysis (with or without metaphysis involvement) can imply damage to the epiphyseal plate.

7. **D**

Red fibers are slow-twitching fibers that have high levels of myoglobin and many mitochondria. They derive their energy from aerobic respiration and are capable of sustained vigorous activity. This eliminates **(A)** and **(B)**. White fibers, on the other hand, are fast-twitching fibers and contain lower levels of myoglobin and fewer mitochondria. Because of their composition, they derive more of their energy anaerobically and fatigue more easily. This eliminates **(C)** and makes **(D)** the correct answer.

8. **A**

The articular surfaces of the bones are covered with a layer of smooth articular cartilage. The epiphysis is a portion of the bone itself, eliminating **(B)**. Synovial fluid lubricates the movement in the joint space, but does not stop the bones from contacting one another; this is the job of articular cartilage, eliminating **(C)**. There is no appreciable function for smooth muscle in the joint space, eliminating **(D)**.

9. **B**

The only type of muscle that is always multinucleated is skeletal muscle, making **(B)** the correct answer. Cardiac muscle may contain one or two centrally located nuclei, so Statement I is incorrect. Smooth muscle, on the other hand, always has only one centrally located nucleus.

10. **D**

Myogenic activity refers to the ability of a muscle to contract without nervous stimulation, such as in response to other stimuli like stretching. Smooth and cardiac muscle both possess myogenic activity.

1. **D**

Yellow marrow is largely inactive and is infiltrated by adipose tissue, making **(D)** the correct answer.

2. **D**

The periosteum, a fibrous sheath that surrounds long bones, is the site of attachment to muscle tissue. Some periosteum cells are capable of differentiating into bone-forming cells called osteoblasts. This eliminates choices **(A)**, **(B)**, and **(C)**. It is the synovium that secretes fluid into the joint cavity (joint space), not the periosteum, making **(D)** the correct answer.

3. **D**

The axial skeleton includes the skull, vertebral column, rib cage, and hyoid bone. The sternum is a point of attachment of the rib cage and is thus a part of the axial, not appendicular, skeleton. The limb bones, pectoral girdle, and pelvis are all part of the appendicular skeleton.

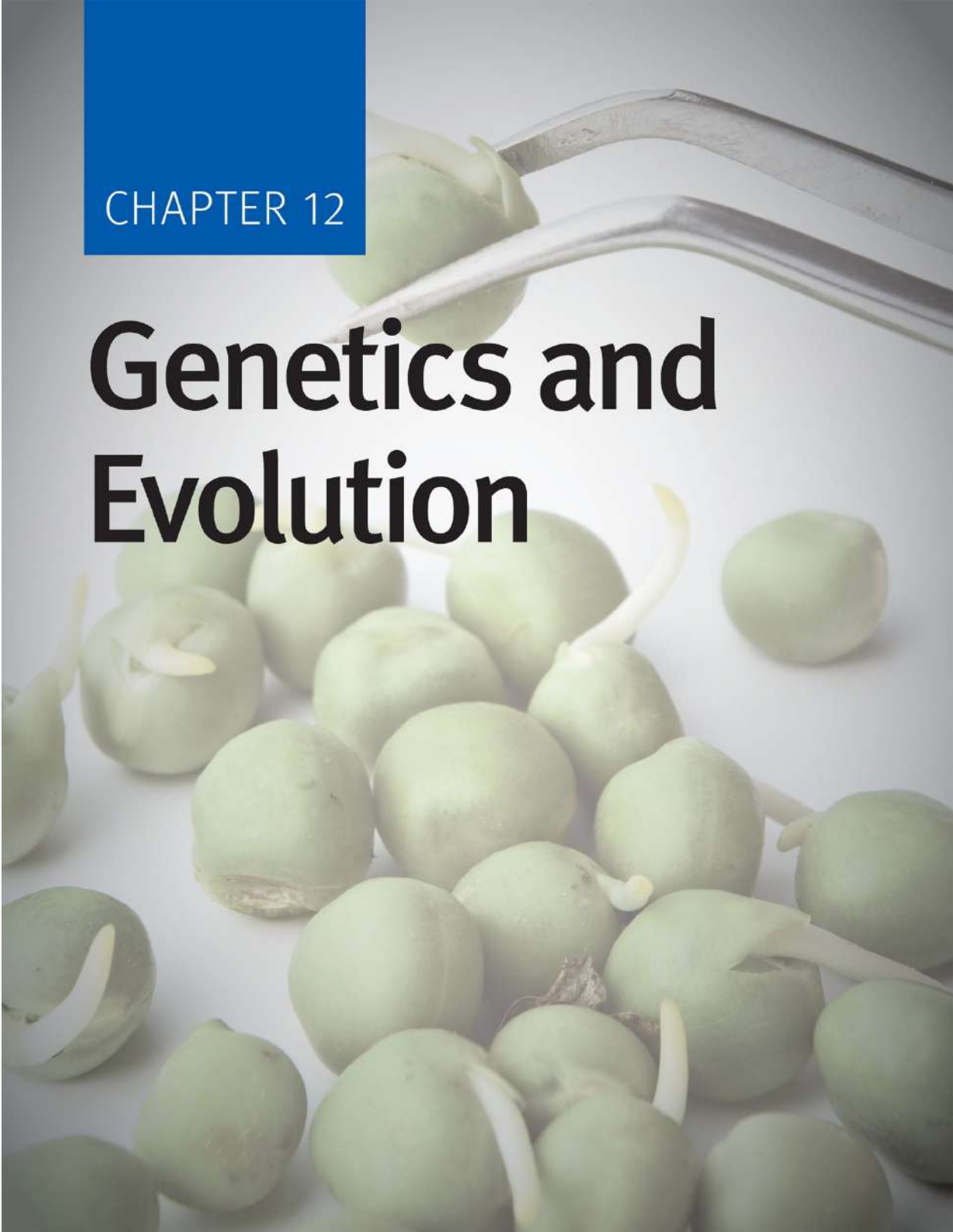
4. **B**

Bones form in one of two ways: endochondral ossification and intramembranous ossification. Endochondral ossification is the replacement of cartilage with bone and occurs mostly in long bones, eliminating **(A)**. Intramembranous ossification is the formation of bone from undifferentiated connective tissue cells (mesenchyme) and occurs mostly in the skull, making **(B)** the correct answer. Bone resorption is the breakdown of bone, not its formation, eliminating **(C)**. Longitudinal growth occurs in long bones and is responsible for increasing height over time, but does not play a role in fontanelle ossification, eliminating **(D)**.

5. **C**

An unprovoked fracture of the hip is not a normal finding in a young woman. Given that she has a high calcium level, it is likely that she has an increased level of bone resorption that is causing her bones to be more fragile. Parathyroid hormone causes calcium release from bones. If this woman had an overactive parathyroid gland—or even cancer in this gland—then it is likely that calcium could still be resorbed from her bones even though her blood calcium levels are already high.



The background of the slide features a photograph of pea plants. In the upper portion, a pair of metal tweezers is shown holding a small, light-green pea. Below this, a cluster of similar green peas is scattered on a white surface. Some of these peas have small, pale, curved structures (likely developing roots or shoots) emerging from them. The overall scene is brightly lit, with soft shadows.

CHAPTER 12

Genetics and Evolution

CHAPTER 12

Genetics and Evolution

In This Chapter

12.1 Fundamental Concepts of Genetics



Patterns of Dominance
Penetrance and Expressivity
Mendelian Concepts
DNA as Genetic Material

12.2 Changes in the Gene Pool

Mutations
Leakage
Genetic Drift

12.3 Analytical Approaches in Genetics

Punnett Squares
Gene Mapping
Hardy–Weinberg Principle

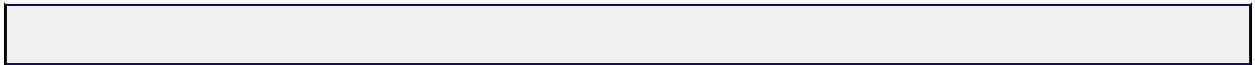
12.4 Evolution

Natural Selection

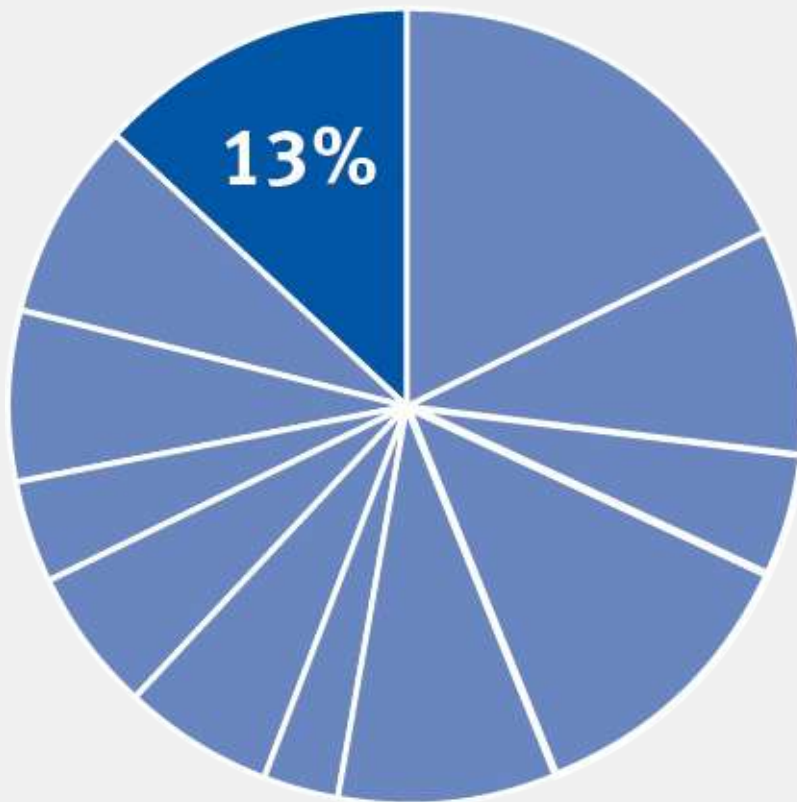
Speciation

Measuring Evolutionary Time

Concept Summary



Chapter Profile



The content in this chapter should be relevant to about 13% of all questions about biology on the MCAT.

This chapter covers material from the following AAMC content category:

1C: Transmission of heritable information from generation to generation and the processes that increase genetic diversity

Introduction

For generations, European royal families practiced what is known as royal intermarriage. For purposes of establishing or continuing political alliances, maintaining bloodline purity, or smoothing out diplomatic relations, marriages between royal families were arranged, resulting in such an interweaving of bloodlines that eventually most European royalty was—and still is—genetically related.

Such marriage unions led to rather severe restrictions on the gene pool, or all the alleles represented in the royal family lines. Offspring of parents who were also related to each other through blood lineage (consanguinity) came to have greater similarities in their genotypes, and certain alleles became so frequent that their phenotypic expression became almost a hallmark of royal descent. The House of Habsburg, which ruled a number of European kingdoms from the 11th century until the late 18th century, was perhaps the most infamous for its inbreeding practices. Members of this royal family bore the unmistakable mark of their restricted genes through a jaw malformation that even came to be known as the *Habsburg lip*. Medically termed *prognathism* (Greek: “forward jaw”), the condition is a misalignment of the mandible and maxilla. The Habsburg family portraits present individuals with prominent, forward-thrusting lower jaws and chins characteristic of mandibular prognathism. The genetic condition has more than just aesthetic implications; it can lead to serious disfigurement and disability. Charles II of Spain suffered from the worst case of the Habsburg lip on record—his lower teeth protruded so much farther than his upper teeth that he was not able to chew his food.

In this chapter, we will explore the concepts of classical genetics, which were originally described in the mid-1800s. Consider this chapter in tandem with the discussions of molecular genetics in Chapters 6 and 7 of *MCAT*

Biochemistry Review. Then, we will explore the changes in the gene pool that occur over time with a discussion of evolution. We'll also quantify the genetics of populations that are *not* undergoing evolution with the Hardy–Weinberg principle.

12.1 Fundamental Concepts of Genetics

High-Yield

LEARNING OBJECTIVES

After Chapter 12.1, you will be able to:

- Recall what traits make alleles dominant or recessive
- Differentiate between homozygous, heterozygous, and hemizygous genotypes
- Compare and contrast complete dominance, codominance, and incomplete dominance
- Explain the difference between penetrance and expressivity
- Connect Mendel's laws with the phase of meiosis to which they are most closely correlated

The physical and biochemical characteristics of every living organism are determined by **genes**, which are DNA sequences that code for heritable traits that can be passed from one generation to the next. Taken together, all genes (as well as a large supply of noncoding DNA) are organized into **chromosomes** to ensure that genetic material is passed easily to daughter cells during mitosis and meiosis. Each gene may have alternative forms called **alleles**. We've already explored the ABO blood antigens as an example of three alleles

for the same gene (I^A , I^B , and i). The genetic combination possessed by an individual is known as a **genotype**, and the manifestation of a given genotype as an observable trait is known as a **phenotype**.



BRIDGE

Blood type A individuals carry the A antigen on their erythrocytes and have circulating anti-B antibodies. Blood type B individuals carry the B antigen on their erythrocytes and have circulating anti-A antibodies. Those with type AB have both antigens and neither antibody; those with type O have neither antigen and both antibodies. That makes type O individuals universal donors and type AB individuals universal recipients. Blood typing is discussed in Chapter 7 of *MCAT Biology Review*.

Each human being possesses two copies of each chromosome, called **homologues**, except for male sex chromosomes (in which there is one X chromosome and one Y chromosome). Each gene has a particular **locus**, or location on a specific chromosome. The normal locus of a particular gene is consistent among human beings—a gene can be described by its location. Because each chromosome is part of a homologous pair, a person will inherit two alleles for all genes (again, except for male sex chromosomes). Alleles can be categorized based on their expression. If only one copy of an allele is needed to express a given phenotype, the allele is said to be **dominant** and is usually represented with a capital letter. If two copies are needed, the allele is said to be **recessive** and is usually represented with a lowercase letter. If both alleles are the same for a given gene, the individual is said to have a

homozygous genotype. If the alleles are different, the individual has a **heterozygous** genotype. A **hemizygous** genotype describes a situation in which only one allele is present for a given gene, as is the case for parts of the X chromosome in males.

PATTERNS OF DOMINANCE

When only one dominant and one recessive allele exist for a given gene, there is said to be **complete dominance**. In this case, the presence of one dominant allele will mask the recessive allele, if present. When more than one dominant allele exists for a given gene, there is **codominance**. For example, a person with one allele for the A blood antigen and one allele for the B blood antigen will express both antigens simultaneously. Finally, **incomplete dominance** occurs when a heterozygote expresses a phenotype that is intermediate between the two homozygous genotypes. A classic example of incomplete dominance is the mating of certain flowers, in which a red flower crossed with a white flower results in pink flowers, as shown in Figure 12.1.

R = allele for red flowers

r = allele for white flowers

parental: **RR** × **rr** (red × white)

	R	R
r	Rr	Rr
r	Rr	Rr

F₁ genotypic ratio: 100% Rr

F₁ phenotypic ratio: 100% pink

F₁: Rr × **Rr** (pink × pink)

	R	r
R	RR	Rr
r	Rr	rr

F₂ genotypic ratio: 1 RR:2 Rr:1 rr

F₂ phenotypic ratio: 1 red:2 pink:1 white

Figure 12.1. Incomplete Dominance

Snapdragons display incomplete dominance, in which neither allele is dominant and the heterozygous phenotype is a mixture of the two homozygous phenotypes.

PENETRANCE AND EXPRESSIVITY

Penetrance and expressivity both reveal the complex interplay between genes and the environment. **Penetrance** is a population measure defined as the proportion of individuals in the population carrying the allele who actually express the phenotype. In other words, it is the probability that, given a

particular genotype, a person will express the phenotype. Alleles can be classified by their degree of penetrance; Huntington's disease, caused by an expansion of a repetitive sequence in the *huntingtin* gene, is a classic example. Individuals with more than 40 sequence repeats have **full penetrance**—100 percent of individuals with this allele show symptoms of Huntington's disease. Individuals with fewer sequence repeats show **high penetrance**, wherein most (but not all) of those with the allele show symptoms of the disease. With fewer sequence repeats, the gene comes to have **reduced penetrance**, **low penetrance**, or even **nonpenetrance**.

A related, but distinct, concept is **expressivity**, which is defined as varying phenotypes despite identical genotypes. If expressivity is **constant**, then all individuals with a given genotype express the same phenotype. However, if expressivity is **variable**, then individuals with the same genotype may have different phenotypes. Whereas penetrance is a population parameter (what percentage of individuals with a given genotype express the phenotype?), expressivity reflects the gray area in expression and is more commonly considered at the individual level. For example, the disease *neurofibromatosis type II* is an autosomal dominant disease that results from a mutation of the gene *NF2* (*merlin*). Interestingly, a range of phenotypes is associated with carrying the defective allele. Many patients have debilitating tumors of the vestibulocochlear nerve, which is needed for hearing and balance. Some have cataracts, while others have tumors in the skin called neuromas; still others have spinal lesions. A small proportion of the population is nonpenetrant. The disease shows variable expressivity because presentations range from no clinical effect to severe disability.



KEY CONCEPT

- **Penetrance**—the proportion of the population with a given genotype who actually express the phenotype
- **Expressivity**—the different manifestations of the same genotype across the population

MENDELIAN CONCEPTS

Gregor Mendel, an Augustinian friar, developed several of the tenets of genetics in the 1860s based on his work with pea plants. While the study of genetics has come a long way, from pedigree analysis to DNA probes to whole-genome sequencing, many of Mendel's original ideas still hold.

Mendel's First Law: Law of Segregation

There are four basic tenets of the modern interpretation of **Mendel's first law (of segregation)**, some which have already been discussed:

- Genes exist in alternative forms (alleles).
- An organism has two alleles for each gene—one inherited from each parent.
- The two alleles segregate during meiosis, resulting in gametes that carry only one allele for any inherited trait.
- If two alleles of an organism are different, only one will be fully expressed and the other will be silent. The expressed allele is said to be dominant, while the silent allele is recessive. (Keep in mind that codominance and incomplete dominance are exceptions to this rule.)

The key cellular correlate to draw here is the separation of homologous chromosomes during anaphase I of meiosis. By separating—segregating—these chromosomes into different cells, each gamete carries only one allele for any given trait.

Mendel's Second Law: Law of Independent Assortment

Mendel's second law (of independent assortment) states that the inheritance of one gene does not affect the inheritance of another gene. Remember from Chapter 2 of *MCAT Biology Review* that spermatogonia and oogonia undergo genome replication before meiosis I. The daughter DNA strand is held to the parent strand at the **centromere**. Together, these DNA strands are known as **sister chromatids**. During prophase I of meiosis, homologous chromosomes pair up to form **tetrads**, which derive their name from the four chromatids involved (two chromatids in each of two homologous chromosomes). Small segments of genetic material are swapped between chromatids in homologous chromosomes, resulting in novel combinations of alleles that were not present in the original chromosomes (**recombination**). This allows the inheritance of one gene to be independent of the inheritance of all others.

Mendel's second law has been complicated by the discovery of linked genes. We will discuss nonindependent assortment and linkage later in this chapter in the section on analytical approaches in genetics.

Both segregation of homologous chromosomes and independent assortment of alleles increase the genetic diversity of gametes and, subsequently, the genetic diversity of offspring. This has been demonstrated to improve the ability of a species to evolve and adapt to environmental stresses.



KEY CONCEPT

Segregation and independent assortment allow for greater genetic diversity in the offspring.



BRIDGE

The transforming principle described in Griffith's experiment is the same as bacterial transformation discussed in Chapter 1 of *MCAT Biology Review*. Remember that transformation is one of three main ways bacteria increase genetic variability; the other two are conjugation and transduction.

DNA AS GENETIC MATERIAL

While Mendel noticed there were certain patterns of inheritance, what he did not know was that DNA was the genetic material transferred to offspring and that genes were made of DNA. In fact, the scientific community rejected Mendel's initial papers on inheritance. It wasn't until the early 1900s that his work was rediscovered. In the early to mid-1900s, it was largely believed that protein was the heritable material. In the mid-1900s, there were three experiments conducted that largely pointed to DNA's role in genetic inheritance.

In the 1920s, Frederick Griffith was a scientist working for the British government, studying *Streptococcus pneumoniae*, a bacteria that causes pneumonia. Two strains of *S. pneumoniae* were identified: a virulent (disease-causing) strain and a nonvirulent strain. In successive trials, Griffith exposed mice to these strains of bacteria under different conditions and observed whether the mice lived or died, as shown in Figure 12.2. The virulent *S. pneumonia* has a smooth capsule that helps the bacterium evade the immune system and cause disease. He injected this strain into mice, which resulted in death of the mice. Naturally, if the virulent bacteria were killed prior to injection, no disease resulted. Likewise, exposure of the mice to the nonvirulent strain, which has a rough capsule, did not cause disease. However, when both dead virulent bacteria and live nonvirulent bacteria were injected into the mouse, the mouse died and live bacteria with smooth capsules could be found in the mice. He theorized that the live, nonvirulent bacteria must have acquired the ability to form smooth capsules from the dead virulent bacteria. This was known as the *transforming principle*.

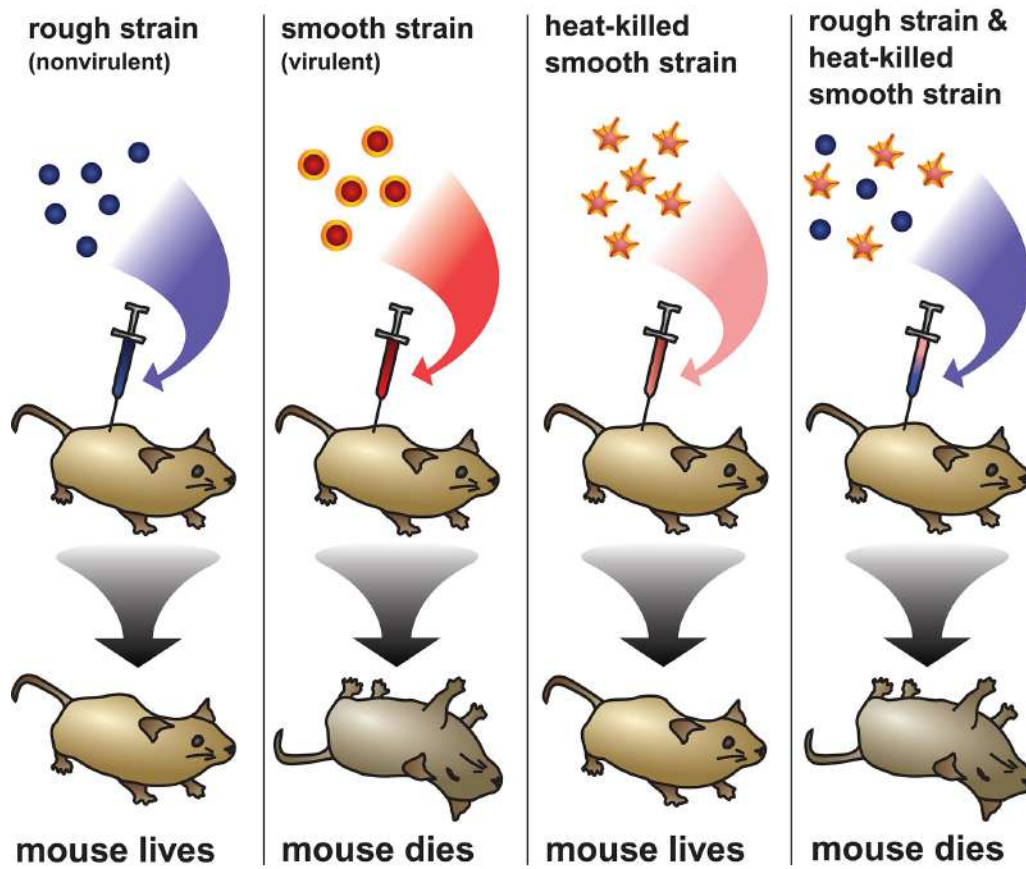


Figure 12.2. The Griffith Experiment

Researchers at the Rockefeller Institute confirmed the transformation principle. Three American scientists—Oswald Avery, Colin MacLeod, and Maclyn McCarty—were attempting to identify the exact material underlying the transformation principle. These scientists purified a very large quantity of heat-killed virulent *S. pneumoniae* bacteria and separated the subcellular components of the bacteria into different extracts. It was noted that the addition of one particular extract to nonvirulent *S. pneumoniae* transformed the bacteria and enabled them to kill the mouse when injected. When this substance was treated with enzymes known to degrade DNA, the bacteria were not transformed and the mice lived. However, when the substance was treated with enzymes known to degrade proteins, the bacteria were still

transformed and the mice died. Thus, the group concluded that the transforming substance must be DNA.

In 1952 (one year before the description of the Watson–Crick model), Alfred Hershey and Martha Chase worked to confirm the idea that DNA could independently carry genetic information. These scientists created bacteriophages with radiolabeled DNA and protein. One group of bacteriophages contained radiolabeled sulfur, which is found in protein but not in DNA. Another group contained radiolabeled phosphorus, which is found in DNA but not in protein. Each of these bacteriophages was permitted to infect a group of nonlabeled bacteria. Recall from Chapter 1 of *MCAT Biology Review* that when bacteriophages infect a bacterium, they inject their genetic material into the cell and leave their capsid outside. After the phages and bacteria were incubated, the sample was centrifuged to separate the material that remained outside the cell from the bacterial cells themselves. It was determined that while no radiolabeled protein entered the cells, radiolabeled DNA had. It was known that viruses must enter a cell to cause disease and replicate, so this experiment once again helped confirm that DNA was the heritable genetic material.



BRIDGE

Radiolabeling sulfur was an appropriate choice to tag proteins in the Hershey–Chase experiment. Recall from Chapter 1 of *MCAT Biochemistry Review* that two amino acids—cysteine and methionine—contain sulfur in their R group, while no nucleotides contain any sulfur.

MCAT CONCEPT CHECK 12.1:

Before you move on, assess your understanding of the material with these questions.

1. What does it mean for an allele to be dominant? Recessive?

- Dominant:

- Recessive:

2. What does it mean for a genotype to be homozygous? Heterozygous? Hemizygous?

- Homozygous:

- Heterozygous:

- Hemizygous:

3. What is the difference between complete dominance, codominance, and incomplete dominance?

- Complete dominance:

- Codominance:

- Incomplete dominance:

4. What is the difference between penetrance and expressivity?

- Penetrance:

- Expressivity:

5. With which phase of meiosis does each of Mendel's laws most closely correlate?

- Mendel's first law:

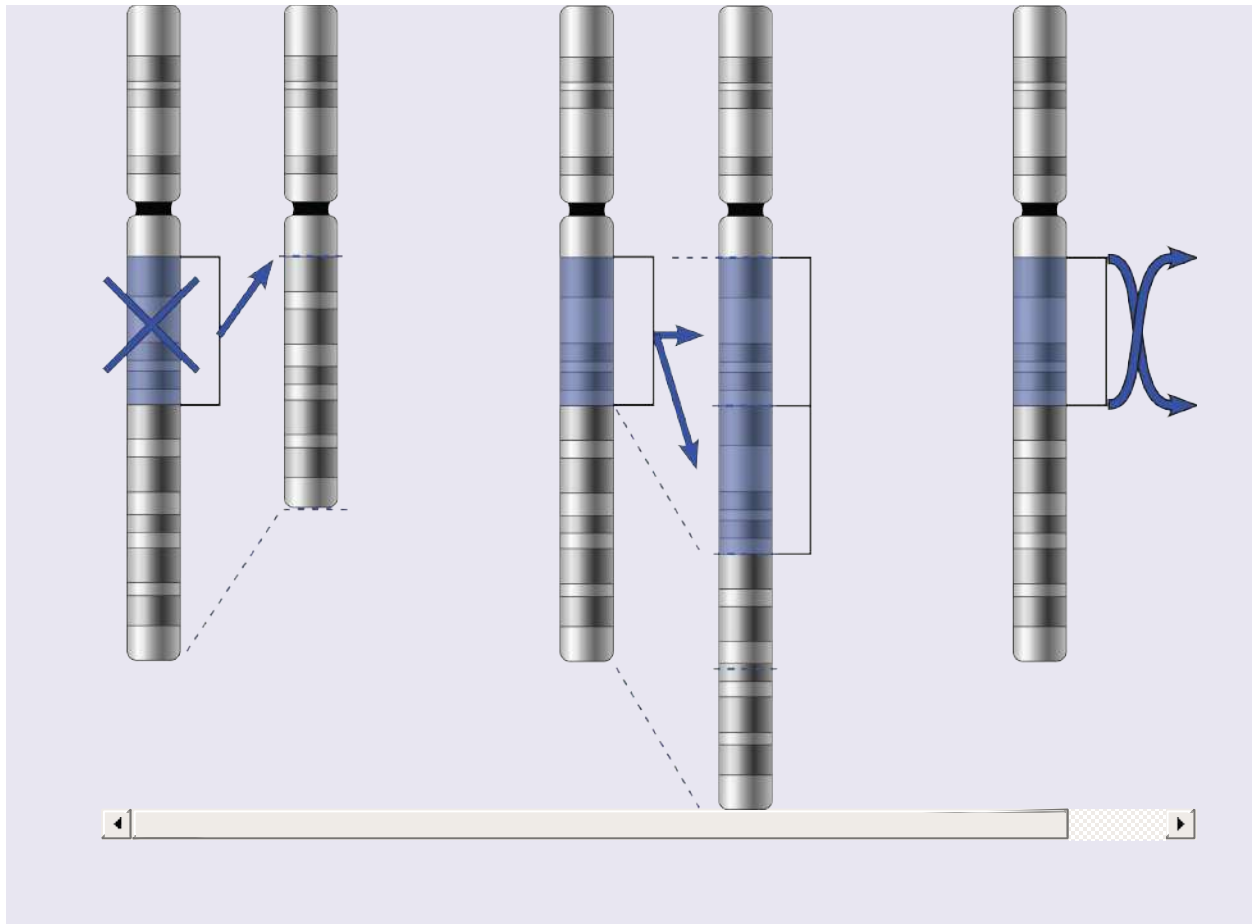
- Mendel's second law:

12.2 Changes in the Gene Pool

LEARNING OBJECTIVES

After Chapter 12.2, you will be able to:

- Identify the three main types of point mutations and the genetic changes they are associated with
- Recall the two main types of frameshift mutation
- Explain why genetic leakage would increase within the last century
- Describe the relationship between genetic drift, small population, and the founder effect
- Identify the main types of chromosomal mutation and the impact they have at the chromosomal level:



All of the alleles that exist within a species are known as the **gene pool**. When mutations or genetic leakage occur, new genes are introduced into the gene pool. Genetic variability is essential for the survival of a species because it allows it to evolve to adapt to changing environmental stresses. Certain traits may be more desirable than others and confer a selective advantage that allows for an individual to produce more viable, fertile offspring. In this section, we will consider genetic diversity and mutations, leakage, and genetic drift, which cause changes to the alleles or their frequency in the gene pool.

MUTATIONS

A **mutation** is a change in DNA sequence, and it results in a mutant allele. Mutant alleles can be contrasted with their **wild-type** counterparts, which are alleles that are considered “normal” or “natural” and are ubiquitous in the study population. New mutations may be introduced in a variety of ways. Ionizing radiation, such as ultraviolet rays from the sun, and chemical exposures can damage DNA; substances that can cause mutations are called **mutagens**. *DNA polymerase* is subject to making mistakes during DNA replication, albeit at a very low rate; proofreading mechanisms help prevent mutations from occurring through this mechanism. Elements known as **transposons** can insert and remove themselves from the genome. If a transposon inserts in the middle of a coding sequence, the mutation will disrupt the gene.

Flawed proteins can arise in other ways without an underlying change in DNA sequence. Incorrect pairing of nucleotides during transcription or translation, or a tRNA molecule charged with the incorrect amino acid for its anticodon, can result in derangements of the normal amino acid sequence.

The major types of nucleotide-level mutations are discussed in great detail in Chapter 7 of *MCAT Biochemistry Review*, so we offer just a brief overview here of each type.

Nucleotide-Level Mutations

Many mutations occur at the level of a single nucleotide (or a very small number of nucleotides). These mutations are shown in Figure 12.3 and are summarized below.

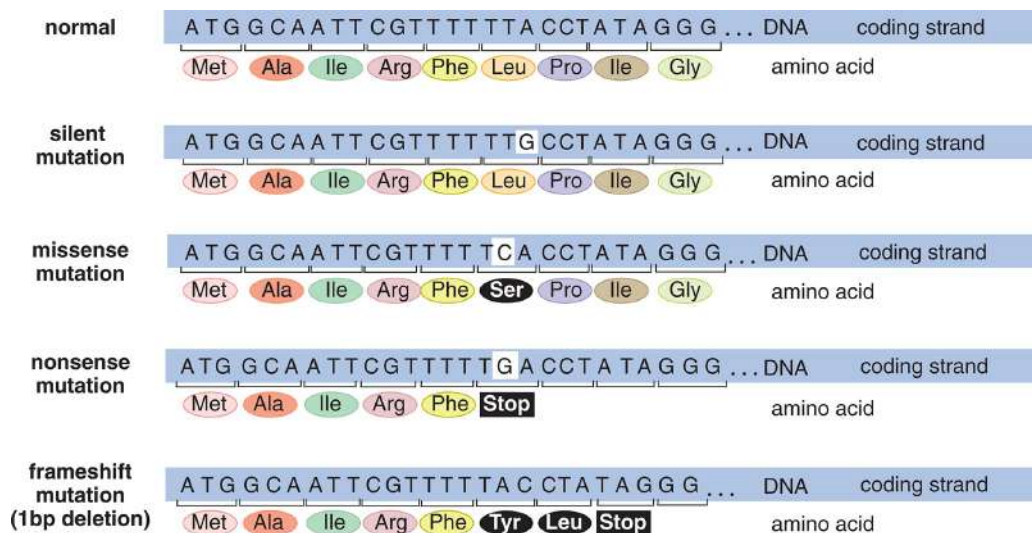


Figure 12.3. Common Nucleotide-Level Mutations

Point mutations occur when one nucleotide in DNA (A, C, T, or G) is swapped for another. These can be subcategorized as silent, missense, or nonsense mutations:

- **Silent mutations** occur when the change in nucleotide has no effect on the final protein synthesized from the gene. This most commonly occurs when the changed nucleotide is transcribed to be the third nucleotide in a codon because there is **degeneracy (wobble)** in the genetic code.
- **Missense mutations** occur when the change in nucleotide results in substituting one amino acid for another in the final protein.
- **Nonsense mutations** occur when the change in nucleotide results in substituting a **stop codon** for an amino acid in the final protein.

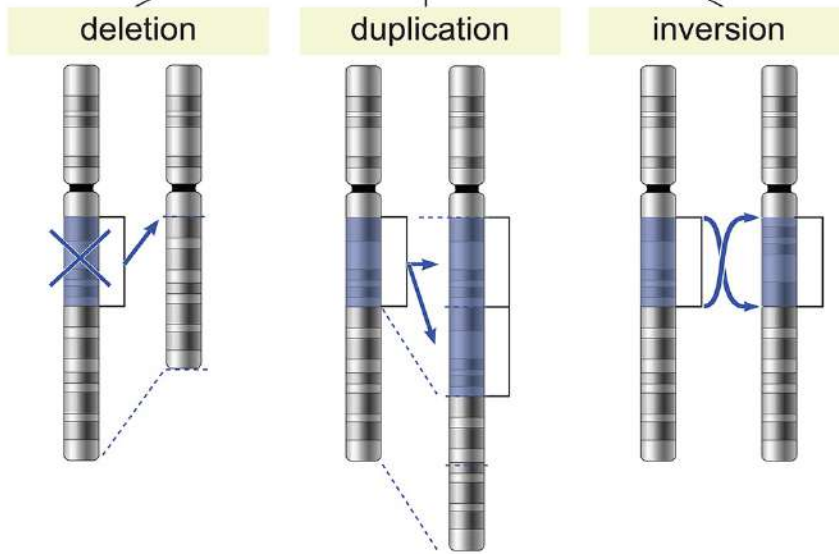
Frameshift mutations occur when nucleotides are inserted into or deleted from the genome. Because mRNA transcribed from DNA is always read in three-letter sequences called **codons**, insertion or deletion of nucleotides can shift the **reading frame**, usually resulting in either changes in the amino acid

sequence or premature truncation of the protein (due to the generation of a nonsense mutation). These can be subcategorized as **insertion** or **deletion mutations**.

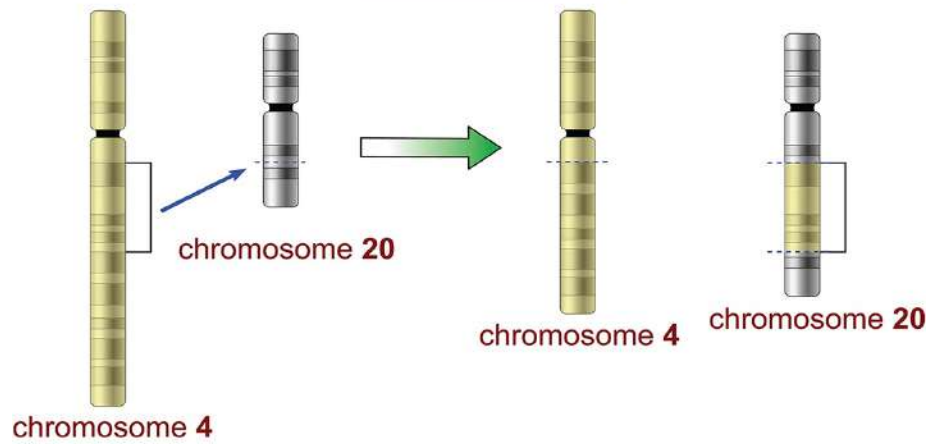
Chromosomal Mutations

Chromosomal mutations are larger-scale mutations in which large segments of DNA are affected, as demonstrated in Figure 12.4 and summarized below.

single chromosome mutations



insertion



translocation

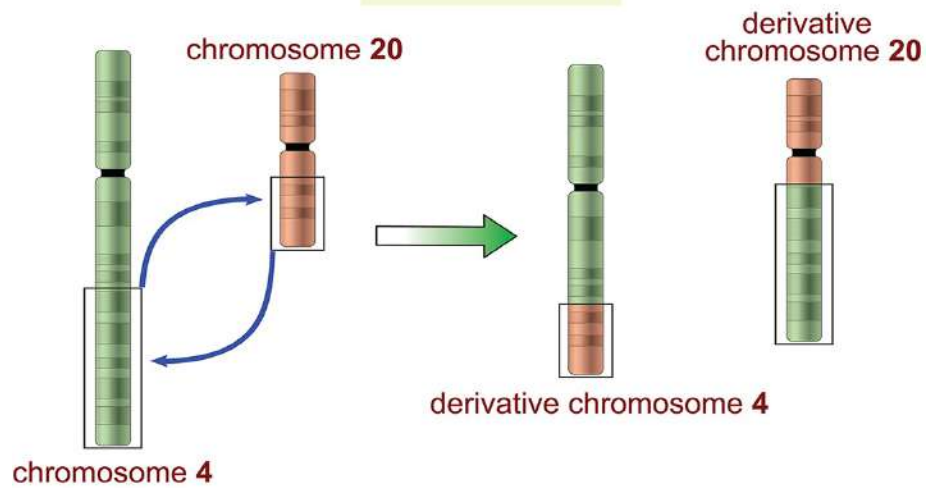


Figure 12.4. Common Chromosomal Mutations

- **Deletion mutations** occur when a large segment of DNA is lost from a chromosome. Small deletion mutations are considered frameshift mutations, as described previously.
- **Duplication mutations** occur when a segment of DNA is copied multiple times in the genome.
- **Inversion mutations** occur when a segment of DNA is reversed within the chromosome.
- **Insertion mutations** occur when a segment of DNA is moved from one chromosome to another. Small insertion mutations (including those where the inserted DNA is not from another chromosome) are considered frameshift mutations, as described previously.
- **Translocation mutations** occur when a segment of DNA from one chromosome is swapped with a segment of DNA from another chromosome.

Consequences of Mutations

Mutations can have many different consequences. Some mutations can be **advantageous**, conferring a positive selective advantage that may allow the organism to produce fitter offspring. For example, sickle cell disease is a single nucleotide mutation that causes sickled hemoglobin. While the disease itself is detrimental to life, heterozygotes for sickle cell disease usually have minor symptoms, if any, and have natural resistance to malaria because their red blood cells have a slightly shorter lifespan—just short enough that the parasitic *Plasmodium* species that causes malaria cannot reproduce in them.

Thus, heterozygotes for sickle cell disease actually have a selective advantage because they are less likely to die from malaria.

On the other hand, some mutations can be detrimental or **deleterious**. For example, *xeroderma pigmentosum* (XP) is an inherited defect in the nucleotide excision repair mechanism. In patients with XP, DNA that has been damaged by ultraviolet radiation cannot be repaired appropriately. Ultraviolet radiation can introduce cancer-causing mutations; since they lack a repair mechanism, patients with XP are frequently diagnosed with malignancies, especially of the skin.

One important class of deleterious mutations is known as **inborn errors of metabolism**. These are defects in genes required for metabolism. Children born with these defects often require very early intervention in order to prevent permanent damage from the buildup of metabolites in various pathways. For example, in *phenylketonuria* (PKU), the enzyme *phenylalanine hydrolase*, which completes the metabolism of the amino acid phenylalanine, is defective. In the absence of this enzyme, toxic metabolites of phenylalanine accumulate, causing seizures, impairment of cerebral function, and learning disabilities, as well as a musty odor to bodily secretions. However, if the disease is discovered shortly after birth, then dietary phenylalanine can be eliminated and treatments can be administered to aid in metabolizing any remaining phenylalanine.

LEAKAGE

Genetic **leakage** is a flow of genes between species. In some cases, individuals from different (but closely related) species can mate to produce **hybrid** offspring. Many hybrid offspring, such as the mule (hybrid of a male

horse and a female donkey), are not able to reproduce because they have odd numbers of chromosomes—horses have 64 chromosomes and donkeys have 62, so mules, with 63 chromosomes, cannot undergo normal homologous pairing in meiosis and cannot form gametes. In some cases, however, a hybrid can reproduce with members of one species or the other, such as the *beefalo* (a cross between cattle and American bison). The hybrid carries genes from both parent species, so this can result in a net flow of genes from one species to the other.

GENETIC DRIFT

Genetic drift refers to changes in the composition of the gene pool due to chance. Genetic drift tends to be more pronounced in small populations. The **founder effect** is a more extreme case of genetic drift in which a small population of a species finds itself in reproductive isolation from other populations as a result of natural barriers, catastrophic events, or other **bottlenecks** that drastically and suddenly reduce the size of the population available for breeding. Because the breeding group is small, **inbreeding**, or mating between two genetically related individuals, may occur in later generations. Inbreeding encourages homozygosity, which increases the prevalence of both homozygous dominant and recessive genotypes. Ultimately, genetic drift, the founder effect, and inbreeding cause a reduction in genetic diversity, which is often the reason why a small population may have increased prevalence of certain traits and diseases. For example, *branched-chain ketoacid dehydrogenase deficiency* (also called *maple syrup urine disease*) is especially common in Mennonite communities; this implies a common origin of the mutation, which may have been in a very small original population.

This loss of genetic variation may cause reduced fitness of the population, a condition known as **inbreeding depression**. On the opposite end of the spectrum, **outbreeding** or **outcrossing** is the introduction of unrelated individuals into a breeding group. Theoretically, this could result in increased variation within a gene pool and increased fitness of the population.

MCAT CONCEPT CHECK 12.2:

Before you move on, assess your understanding of the material with these questions.

1. What are the three main types of point mutations? What change occurs in each?

-

-

-

2. What are the two main types of frameshift mutations?

-

-

3. What are the three main types of chromosomal mutations that do NOT share their name with a type of frameshift mutation? What change occurs in each?

-

-

-

4. Why would genetic leakage in animals be rare prior to the last century?

5. Why is genetic drift more common in small populations? What relationship does this have to the founder effect?

12.3 Analytical Approaches in Genetics

LEARNING OBJECTIVES

After Chapter 12.3, you will be able to:

- Predict the phenotype ratio in the offspring of a cross, such as AaBB x AAbb
- Order genes on a chromosome given their recombination frequencies
- Recall the five criteria of the Hardy–Weinberg principle and the underlying population characteristic each one implies
- Solve calculation problems requiring use of the Hardy–Weinberg equations: $p + q = 1$, and $p^2 + 2pq + q^2 = 1$

Genetics is a field in which a number of **biometric techniques**, or quantitative approaches to biological data, have been developed. These range from the Punnett square to mapping of chromosomes with recombinant frequencies to Hardy–Weinberg equilibrium.

PUNNETT SQUARES

Punnett squares are diagrams that predict the relative genotypic and phenotypic frequencies that will result from the crossing of two individuals. The alleles of the two parents are arranged on the top and side of the square, with the genotypes of the progeny represented at the intersections of these alleles. The genotypes of the progeny will be the product of the two parental alleles.



MCAT EXPERTISE

Pedigree (or family tree) analysis was once a mainstay of MCAT passages and questions. While this topic no longer appears on the exam, it will appear in your medical school genetics studies. The symbology of pedigree analysis is complex and intricate, but a great deal of information can be gleaned from a well-drawn pedigree.

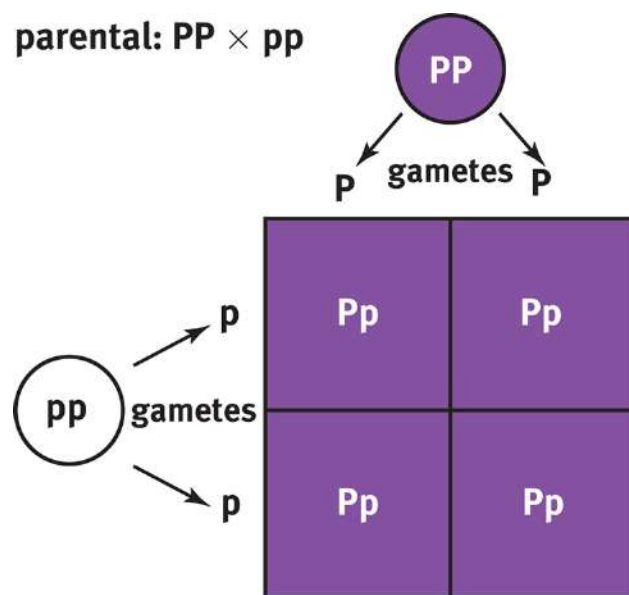
Monohybrid Cross

In genetics problems, including those on the MCAT, dominant alleles are assigned capital letters and recessive alleles are assigned lowercase letters. If both copies of the allele are the same, that individual is said to be homozygous; if they are different, the individual is heterozygous.

A cross in which only one trait is being studied is said to be **monohybrid**. The **parent** or **P generation** refers to the individuals being crossed; the offspring are the **filial** or **F generation**. Multiple generations can be denoted F generations by using numeric subscripts. If you think of your grandparents as

the P generation, then your parents are in the F₁ generation, and you are in the F₂ generation.

Mendel worked with pea plants that had either purple or white flowers. Before crossing the different plants, each group contained homozygotes; subsequent experimentation revealed that the allele for purple color was dominant (P) and the allele for white color was recessive (p). Thus, crossing a homozygous purple flower with a white flower would be crossing PP with pp, resulting in an F₁ generation that contained 100 percent Pp or heterozygotes, as shown in Figure 12.5. All of the flowers in this generation would be purple because P is a dominant allele.

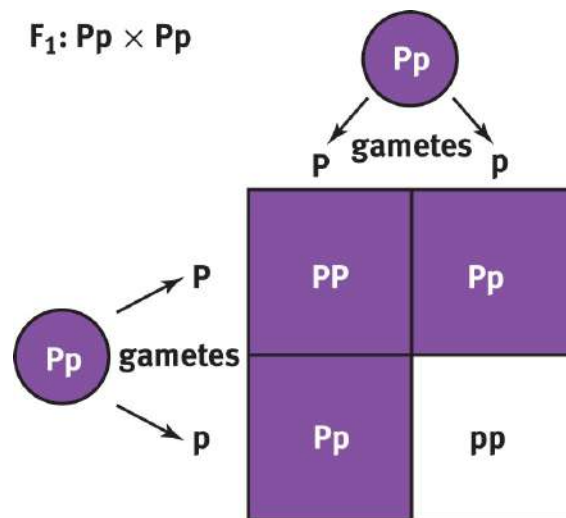


F₁ genotypic ratio: 100% Pp (heterozygous)

F₁ phenotypic ratio: 100% purple

Figure 12.5. Punnett Square of Homozygous Parents

If two members of the F_1 generation were crossed, the resulting offspring in the F_2 generation would be more genotypically and phenotypically diverse than their parents. Crossing two plants with the genotype Pp would result in 25 percent PP , 50 percent Pp , and 25 percent pp offspring, as shown in Figure 12.6. Phenotypically, this would be a 3:1 distribution because both the homozygous dominant and heterozygous dominant offspring would be purple-flowering plants. Thus, crossing two heterozygotes in a case of complete dominance will result in a 1:2:1 distribution of genotypes (homozygous dominant:heterozygous dominant:homozygous recessive) and a 3:1 distribution of phenotypes (dominant:recessive). These ratios are, of course, theoretical probabilities and will not always hold true—especially in a small population of offspring. Usually, the more offspring parents have, the closer their phenotypic ratios will be to the expected ratios.



F_2 genotypic ratio: 1:2:1; 1 PP : 2 Pp :1 pp

F_2 phenotypic ratio: 3:1; 3 purple:1 white

Figure 12.6. Punnett Square of Heterozygous Parents



KEY CONCEPT

Crossing two heterozygotes for a trait with complete dominance results in a 1:2:1 ratio of genotypes and a 3:1 ratio of phenotypes. Know these ratios cold for Test Day!



MCAT EXPERTISE

The ability to create and read a Punnett square quickly on Test Day is one of the most useful skills for questions involving Mendelian inheritance. Often, an entire passage in the *Biological and Biochemical Foundations of Living Systems* section will be devoted to classical and molecular genetics and will require the use of at least one Punnett square.

Test Cross

A **test cross** is used to determine an unknown genotype, as shown in Figure 12.7. In a test cross, the organism with an unknown genotype is crossed with an organism known to be homozygous recessive. If all of the offspring (100 percent) are of the dominant phenotype, then the unknown genotype is likely to be homozygous dominant. If there is a 1:1 distribution of dominant to recessive phenotypes, then the unknown genotype is likely to be heterozygous. Because a test cross is used to determine the genotype of the

parent based on the phenotypes of its offspring, test crosses are sometimes called **back crosses**.

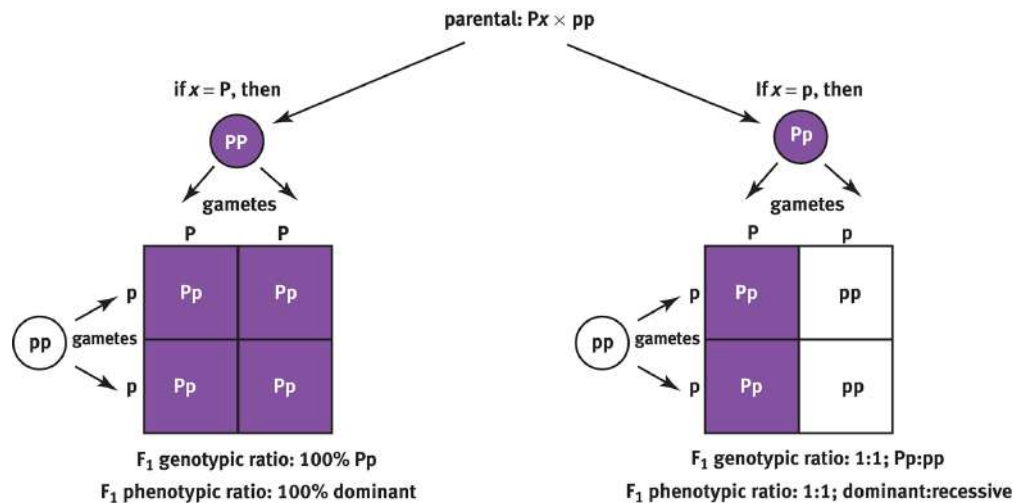


Figure 12.7. Test Cross

An organism with an unknown genotype is crossed with a homozygous recessive organism to identify the unknown genotype using the phenotypes of the resulting offspring.

Dihybrid Cross

We can extend a Punnett square to account for the inheritance of two different genes using a **dihybrid cross**. Remember, according to Mendel's second law (of independent assortment) the inheritance of one gene is independent of the inheritance of the other. This will hold true for **unlinked genes**, although it will be more complicated for linked genes, as described later in this chapter.

If we expand the previous crosses to consider not only flower color, but also plant height, then we can create a 4×4 Punnett square as shown in Figure 12.8. Remember that purple is dominant (P) and white is recessive (p);

similarly, tall is dominant (T) and short or dwarf is recessive (t). If we cross two plants that are heterozygous for both traits, then the offspring have a phenotypic ratio of 9:3:3:1 (9 tall and purple:3 tall and white:3 dwarf and purple:1 dwarf and white). Note that the 3:1 phenotypic ratio still holds for each trait (12 tall:4 dwarf and 12 purple:4 white), reflecting Mendel's second law.

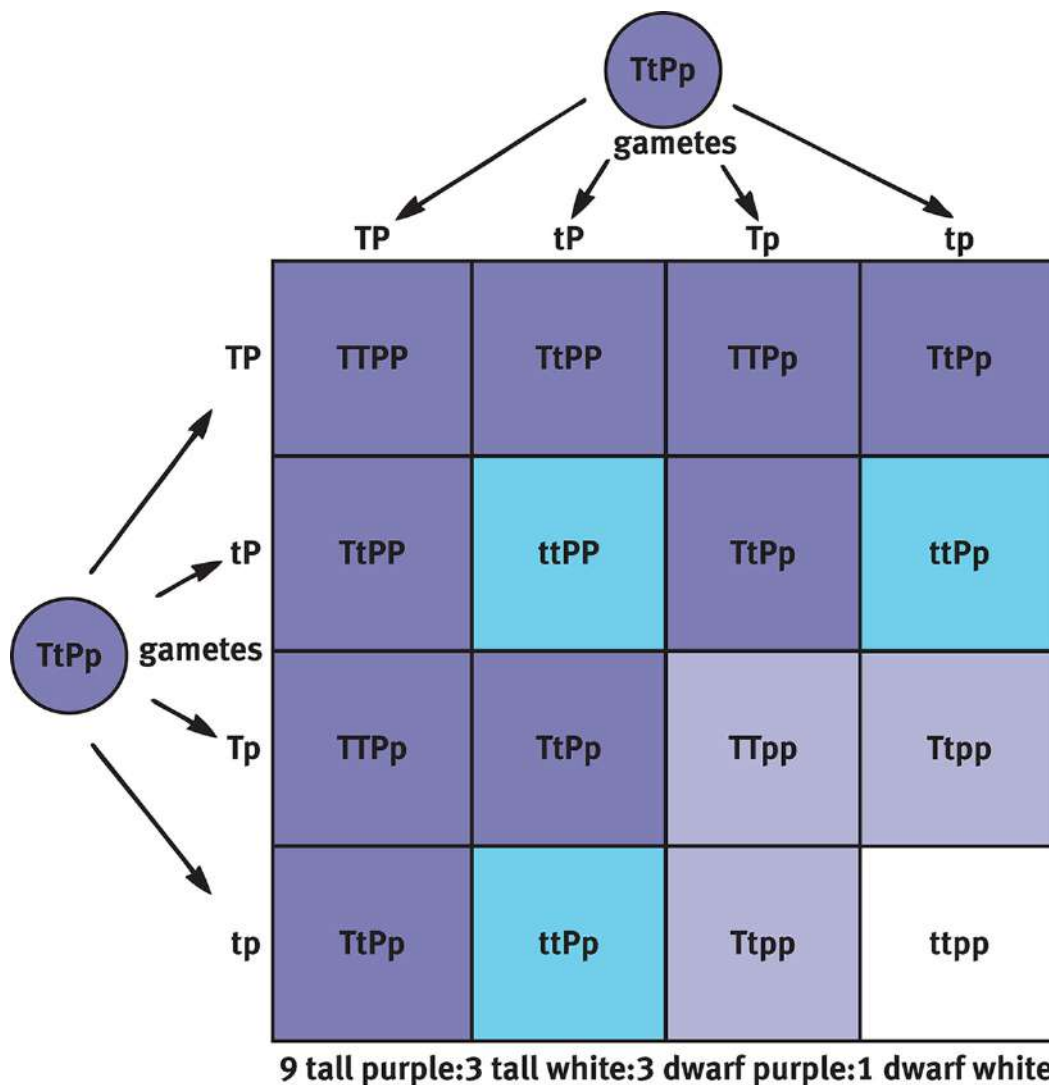


Figure 12.8. Dihybrid Cross



MCAT EXPERTISE

Just as with the 3:1 phenotypic ratio of a monohybrid cross, it is worth memorizing the 9:3:3:1 distribution for dihybrid crosses between two heterozygotes with complete dominance.

Sex-Linked Crosses

When considering **sex-linked (X-linked) traits**, a slightly different system is used to symbolize the various alleles because females have two X chromosomes and thus may be homozygous or heterozygous for a condition carried on the X chromosome. Males, on the other hand, have only one X chromosome (and one Y chromosome) and are hemizygous for many genes carried on the X chromosome. This is why sex-linked traits are much more common in males; having only one recessive allele is sufficient for expression of the recessive phenotype.



MNEMONIC

On the MCAT, **sex**-linked is **X**-linked. Y-linked diseases exist, but are exceedingly rare. Further, unless told otherwise, assume that sex-linked traits are recessive.

When writing genotypes for sex-linked traits, we use X and Y to symbolize normal X and Y chromosomes. An X chromosome carrying a defective allele is

commonly given a subscript, such as X_h , to indicate the presence of the disease-carrying allele. Hemophilia is a particularly common example of a sex-linked trait; Punnett squares for a heterozygous (**carrier**) female and both a normal male and affected (hemophiliac) male are shown in Figure 12.9.

cross between a carrier female (X_hX) and a normal male (XY)

	X_h	X	
X	X_hX	XX	offspring 25% X_hX = 25% carrier female 25% XX = 25% normal female 25% X_hY = 25% hemophiliac male 25% XY = 25% normal male
Y	X_hY	XY	

cross between a carrier female (X_hX) and a hemophiliac male (X_hY)

	X_h	X	
X_h	X_hX_h	X_hX	offspring 25% X_hX_h = 25% hemophiliac female 25% X_hX = 25% carrier female 25% X_hY = 25% hemophiliac male 25% XY = 25% normal male
Y	X_hY	XY	

Figure 12.9. Sex-Linked Cross

Unless stated otherwise, assume that all sex-linked traits on the MCAT are X-linked recessive.



KEY CONCEPT

Because an egg necessarily carries an X chromosome, the sperm determines the sex of a child. It follows that men with a sex-linked trait will have daughters who are all either carriers of the trait or who express the trait, and that a man can never pass down a sex-linked trait to his son.

GENE MAPPING

Genes are organized in a linear fashion on chromosomes. As discussed earlier, crossing over during prophase I of meiosis causes alleles to be swapped between homologous chromosomes, supporting Mendel's second law (of independent assortment). However, genes that are located very close together on a chromosome are less likely to be separated from each other during crossing over. In other words, the further apart two genes are, the more likely it is that there will be a point of crossing over, called a **chiasma**, between them. The likelihood that two alleles are separated from each other during crossing over, called the **recombination frequency** (θ), is roughly proportional to the distance between the genes on the chromosome. We can also describe the strength of linkage between genes based on the recombination frequency: tightly linked genes have recombination frequencies close to 0 percent; weakly linked genes have recombination frequencies approaching 50 percent, as expected from independent assortment.

By analyzing recombination frequencies, a **genetic map** representing the relative distance between genes on a chromosome can be constructed. By convention, one **map unit** or **centimorgan** corresponds to a 1 percent chance of recombination occurring between two genes. Thus, if two genes were 25

map units apart, we would expect 25 percent of the total gametes examined to show recombination somewhere between these two genes. Recombination frequencies can be added in a crude approximation to determine the order of genes in the chromosome, as shown in Figure 12.10.

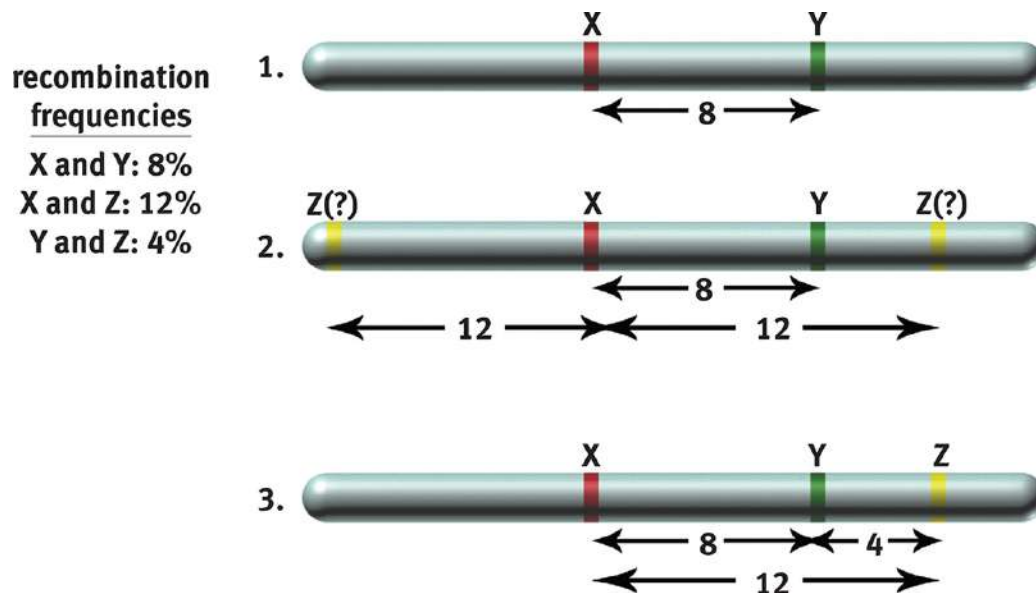


Figure 12.10. Genetic Maps from Recombination Frequencies

If the recombination frequencies are known, one can deduce the order of genes on the chromosome because map units are roughly additive.

HARDY-WEINBERG PRINCIPLE

How often an allele appears in a population is known as its **allele frequency**. For example, if we took a one-cell sample from 50 of Mendel's plants, we could collect 100 copies of alleles for flower color (two from each cell). If 75 of these alleles were the dominant allele, we could say that the allele frequency of P is $75 \div 100 = 0.75$. Note that this does not indicate which flowers contain the allele or if those flowers are homozygous or heterozygous; it only tells us the representation of the allele across all chromosomes in the population.

Evolution results from changes in these gene frequencies in reproducing populations over time. However, when the gene frequencies of a population are not changing, the gene pool is stable and evolution is ostensibly *not* occurring. The following five criteria are mandatory for this to be possible:

- The population is very large (no genetic drift).
- There are no mutations that affect the gene pool.
- Mating between individuals in the population is random (no sexual selection).
- There is no migration of individuals into or out of the population.
- The genes in the population are all equally successful at being reproduced.

Provided that all of these conditions are met, the population is said to be in **Hardy-Weinberg equilibrium**, and a pair of equations can be used to predict the allelic and phenotypic frequencies.

Let us define a particular gene as having only two possible alleles, T and t. We will define p to be the frequency of the dominant allele T and q to be the frequency of the recessive allele t. Because there are only these two choices at the same gene locus, $p + q = 1$. That is, the combined allele frequencies of T and t must equal 100 percent. If we square both sides of the equation, we get:

$$(p + q)^2 = 1^2$$
$$p^2 + 2pq + q^2 = 1$$

where p^2 is the frequency of the TT (homozygous dominant) genotype, $2pq$ is the frequency of the Tt (heterozygous dominant) genotype, and q^2 is the frequency of the tt (homozygous recessive) genotype. Note that the sum $p^2 +$

$2pq$ would represent the frequency of the dominant *phenotype* (both homozygous and heterozygous dominant genotypes).



KEY CONCEPT

All you need to know to solve any MCAT Hardy–Weinberg problem is the value of p (or p^2) or q (or q^2). From there, you can calculate everything else using $p + q = 1$ and $p^2 + 2pq + q^2 = 1$.

For Test Day, you should know the two key Hardy–Weinberg equations demonstrated above:

$$p + q = 1$$
$$p^2 + 2pq + q^2 = 1$$

Equation 12.1

Each equation provides us with different information. The first tells us about the frequency of *alleles* in the population, whereas the second provides information about the frequency of *genotypes* and *phenotypes* in the population.



KEY CONCEPT

The Hardy–Weinberg equations allow you to find two pieces of information: first, the relative frequency of alleles in a population, and

second, the frequency of a given genotype or phenotype in the population. Remember that there will be twice as many alleles as individuals in a population because each individual has two autosomal copies of each gene.

These equations can also be used to demonstrate that evolution is *not* occurring in a population. Assuming that the conditions listed earlier are met, the allele frequencies will remain constant from generation to generation. For example, imagine that we have a population of Mendel's pea plants in which the frequency of the tall allele, T, is 0.80. This value is represented by p . This means that q (the short allele, t) is 0.20 by subtraction. Setting up our F_1 cross for two heterozygotes, we can see the results of such a mating:

	$p = 0.80$ (T = 80%)	$q = 0.20$ (t = 20%)
$p = 0.80$ (T = 80%)	$p^2 = 0.64$ (TT = 64%)	$pq = 0.16$ (Tt = 16%)
$q = 0.20$ (t = 20%)	$pq = 0.16$ (Tt = 16%)	$q^2 = 0.04$ (tt = 4%)

We see that the filial generation contains 64 percent homozygous tall, 32 percent heterozygous tall, and 4 percent homozygous short plants. These are the genotypic frequencies. We can determine the allele frequencies in this generation as follows:

$$\begin{array}{rclcl}
 64\% \text{ TT} & = & 64\% \text{ T} & \text{and} & 0\% \text{ t} \\
 32\% \text{ Tt} & = & 16\% \text{ T} & \text{and} & 16\% \text{ t} \\
 4\% \text{ tt} & = & 0\% \text{ T} & \text{and} & 4\% \text{ t} \\
 \hline
 \text{allele frequencies} & = & 80\% \text{ T} & \text{and} & 20\% \text{ t}
 \end{array}$$

Notice that the allele frequencies are unchanged compared to the parent generation. T is still 0.80 and t is still 0.20. Populations in Hardy–Weinberg equilibrium will exhibit this property.

MCAT CONCEPT CHECK 12.3:

Before you move on, assess your understanding of the material with these questions.

- For each of the crosses below, what is the phenotypic ratio seen in the offspring?

Cross	Phenotypic Ratio
Bb × Bb	
Aa × aa	
DdEe × ddEE	
X_qX × XY	
X_rX × X_rY	

2. If genes Q and R have a recombination frequency of 2%, genes R and S have a recombination frequency of 6%, genes S and T have a recombination frequency of 23%, and genes Q and T have a recombination frequency of 19%, then what is the order of these four genes in the chromosome?
-

3. All five criteria of the Hardy–Weinberg principle are required to imply what characteristic of the study population?
-

4. Assume that a population is in Hardy–Weinberg equilibrium. If 9% of the population is homozygous dominant, then solve for the following:

- The frequency of the dominant allele:
-

- The frequency of the recessive allele:
-

- The portion of the population that is heterozygous:
-

- The portion of the population with a homozygous recessive genotype:
-

- The portion of the population with a dominant
-

phenotype:

12.4 Evolution

LEARNING OBJECTIVES

After Chapter 12.4, you will be able to:

- Describe the key tenets of the major theories of evolution, including punctuated equilibrium, natural selection, inclusive fitness, and the modern synthesis model
- Identify the three patterns of selection and the changes they create in the population phenotype
- Recall the three patterns of evolution between species and the outcome of each
- Recall the biological definition of a species

Evolutionary thought has a relatively short history; the first theories suggesting that new species may arise from older ones were proposed in the 19th century. Significant alterations to these initial theories have been made since then.

NATURAL SELECTION

Natural selection, sometimes called *survival of the fittest*, is the theory that certain characteristics or traits possessed by individuals within a species may help those individuals have greater reproductive success, thus passing on those traits to offspring. This theory was originally proposed by Charles Darwin in his 1859 publication *On the Origin of Species*. His theory was built on several basic tenets:

- Organisms produce offspring, few of which survive to reproductive maturity.
- Chance variations within individuals in a population may be heritable. If these variations give an organism even a slight survival advantage, the variation is termed **favorable**.
- Individuals with a greater preponderance of these favorable variations are more likely to survive to reproductive age and produce offspring; the overall result will be an increase in these traits in future generations. This level of reproductive success is termed **fitness**, and an organism's fitness is directly related to the relative genetic contribution of this individual to the next generation.

Darwin's theory was ultimately proven to be correct in many ways—although not completely. In the 20th century, modern genetics led to the development of the currently accepted theory.



KEY CONCEPT

Evolution is not equivalent to natural selection. The MCAT likes to test your ability to understand that natural selection is simply a

mechanism for evolution. Natural selection is, however, equivalent to *survival of the fittest*.

Modern Theories

The **modern synthesis model**, sometimes called **neo-Darwinism**, adds knowledge of genetic inheritance and changes in the gene pool to Darwin's original theory. Once scientists showed that inheritance occurs through the passing of genes from parent to child and that genes ultimately change due to mutation or recombination, Darwin's theory was updated to its current form: when mutation or recombination results in a change that is favorable to the organism's reproductive success, that change is more likely to pass on to the next generation. The opposite is also true. This process is termed **differential reproduction**. Over time, those traits passed on by the more successful organisms will become ubiquitous in the gene pool. Because the gene pool changes over time, it is important to note that populations evolve, not individuals.

Also germane to the modernization of Darwin's theory is a shift in scope to focus on inclusive fitness over the fitness of an individual organism. **Inclusive fitness** is a measure of an organism's success in the population, based on the number of offspring, success in supporting offspring, and the ability of the offspring to then support others. Early descriptions of evolutionary success, like those of Darwin, were based solely on the number of viable offspring of an organism. However, contemporary theories take into account the benefits of certain behaviors on the population at large. For example, the existence of altruism could be supported by the observation that close relatives of an individual will share many of the same genes; thus, promoting the

reproduction and survival of related or similar individuals can also lead to genetic success. Other species show examples of inclusive fitness by protecting the offspring of the group at large. By endangering themselves to protect the young, these organisms ensure the passing of genes to future generations. Inclusive fitness therefore promotes the idea that altruistic behavior can improve the fitness and success of a species as a whole.

One final theory to consider was proposed as a result of research into the fossil record. Upon examination, it was discovered that little evolution would occur within a lineage of related lifeforms for long periods of time, followed by an explosion in evolutionary change. Niles Eldredge and Stephen Jay Gould proposed the theory of **punctuated equilibrium** to explain this in 1972. In contrast to Darwin's theory, punctuated equilibrium suggests that changes in some species occur in rapid bursts rather than evenly over time.

Modes of Natural Selection

Natural selection may occur as stabilizing selection, directional selection, or disruptive selection, as shown in Figure 12.11.

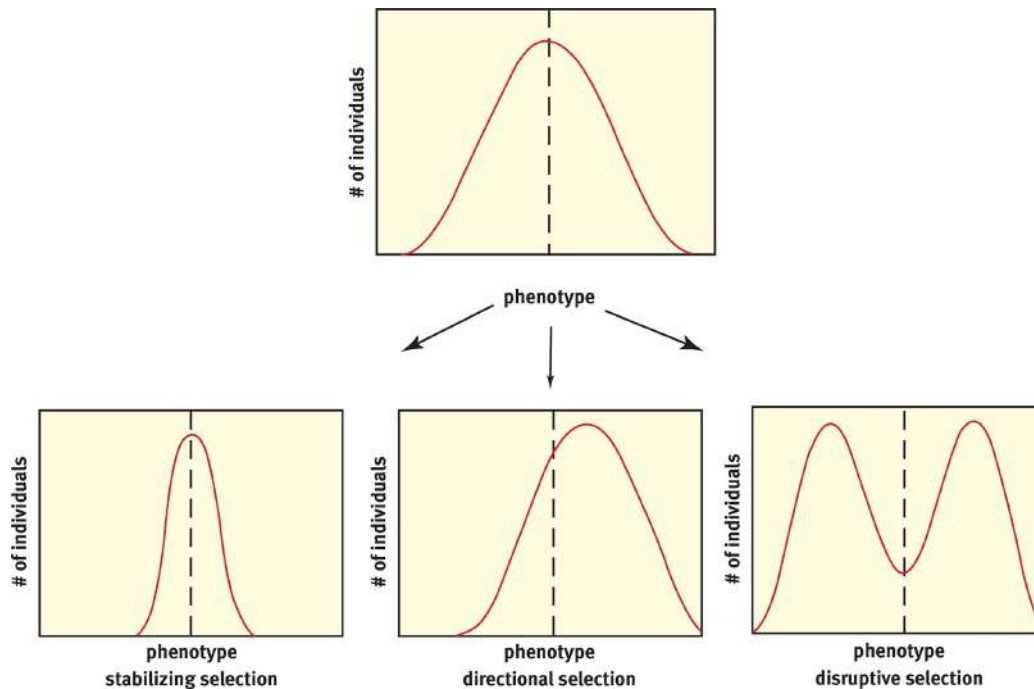


Figure 12.11. Modes of Natural Selection

Stabilizing selection keeps phenotypes within a specific range by selecting against extremes. For instance, human birth weight is maintained within a narrow band by stabilizing selection. Fetuses that weigh too little may not be healthy enough to survive, and fetuses that weigh too much can experience trauma during delivery through the relatively narrow birth canal. In addition, the larger the fetus, the more resources it requires from the mother. For all of these reasons, it is advantageous to keep birth weights within a narrow range.

Adaptive pressure can lead to the emergence and dominance of an initially extreme phenotype through **directional selection**. For example, if we have a heterogeneous plate of bacteria, very few may have resistance to antibiotics. If the plate is then treated with *ampicillin* (an antibiotic), only those colonies that exhibit resistance to this antibiotic will survive. A new standard phenotype emerges as a result of differential survivorship. Natural selection is

the history of differential survivorship over time. The emergence of mosquitoes resistant to *dichlorodiphenyltrichloroethane* (DDT), a type of pesticide, is attributed to directional selection.

In **disruptive selection**, two extreme phenotypes are selected over the norm. When Darwin studied finches on the Galapagos Islands, he noted that although there were many species, all the species arguably had a common ancestor, given their similar appearances. However, when he compared beak sizes they were all either large or small, as shown in Figure 12.12. No animals exhibited the intermediate phenotype of medium-size beaks. Darwin hypothesized that the sizes of seeds on the island (the finches' food) led to this effect. Seeds were either quite large or fairly small, requiring a large or small beak, respectively. Thus, if the original ancestor had a medium-size beak, over time the animals with slightly larger or smaller beaks would be selected for. Disruptive selection is facilitated by the existence of **polymorphisms**—naturally occurring differences in form between members of the same population, such as light and dark coloration in the same species of butterfly. **Adaptive radiation** is a related concept that describes the rapid rise of a number of different species from a common ancestor. The benefit of adaptive radiation is that it allows for various species to occupy different niches. A **niche** is a specific environment, including habitat, available resources, and predators, for which a species is specifically adapted. Adaptive radiation is favored by environmental changes or isolation of small groups of the ancestral species.

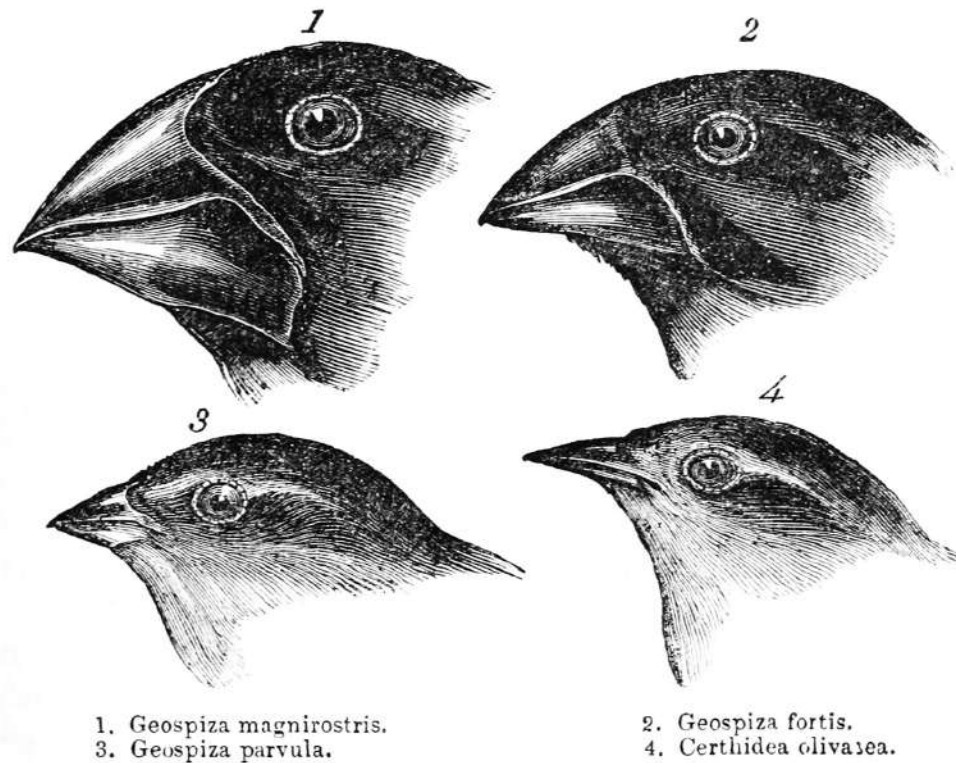


Figure 12.12. Darwin's Finches

Image of Darwin's finches as drawn by John Gould

SPECIATION

A **species** is defined as the largest group of organisms capable of breeding to form fertile offspring; the formation of a new species through evolution is called **speciation**. If we took two populations from the same species and separated them geographically for a long period of time, different evolutionary pressures would lead to different adaptive changes. If enough time passed, the changes would be sufficient to lead to **isolation**, which means the progeny of these populations could no longer freely interbreed. We would now consider the two groups separate species.

Reproductive isolation may occur either prezygotically or postzygotically. **Prezygotic mechanisms** prevent formation of the zygote completely; **postzygotic mechanisms** allow for gamete fusion but yield either nonviable or sterile offspring. Examples of prezygotic mechanisms include temporal isolation (breeding at different times), ecological isolation (living in different niches within the same territory), behavioral isolation (a lack of attraction between members of the two species due to differences in pheromones, courtship displays, and so on), reproductive isolation (incompatibility of reproductive anatomy), or gametic isolation (intercourse can occur, but fertilization cannot). Postzygotic mechanisms include hybrid inviability (formation of a zygote that cannot develop to term), hybrid sterility (forming hybrid offspring that cannot reproduce), and hybrid breakdown (forming first-generation hybrid offspring that are viable and fertile, but second-generation hybrid offspring that are inviable or infertile). As described earlier in this chapter, mules are an example of postzygotic hybrid sterility. Although a horse and donkey can produce a viable mule, the mule will be sterile and thus unable to establish a self-perpetuating mule lineage.

Patterns of Evolution

When we look at similarities between two species, we must be careful to determine whether those similarities are due to sharing a common ancestor or sharing a common environment with the same evolutionary pressures. When analyzing species this way, three patterns of evolution emerge: divergent evolution, parallel evolution, and convergent evolution, as shown in Figure 12.13.

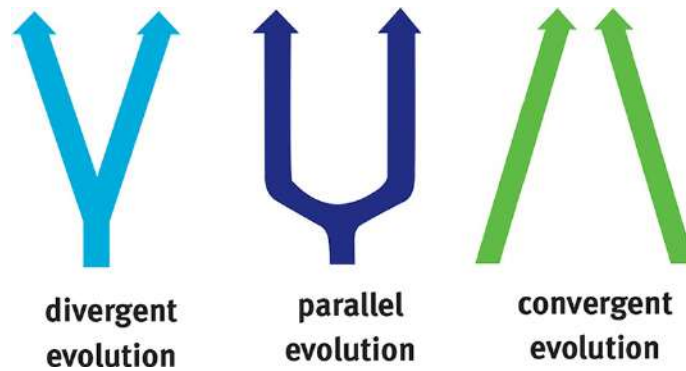


Figure 12.13. Patterns of Evolution

Divergent evolution refers to the independent development of dissimilar characteristics in two or more lineages sharing a common ancestor. For example, seals and cats are both mammals in the order Carnivora, yet they differ markedly in general appearance. These two species live in very different environments and adapted to different selection pressures while evolving.

Parallel evolution refers to the process whereby related species evolve in similar ways for a long period of time in response to analogous environmental selection pressures.

Convergent evolution refers to the independent development of similar characteristics in two or more lineages not sharing a recent common ancestor. For example, fish and dolphins have come to resemble one another physically, though they belong to different classes of vertebrates. They evolved certain similar features in adapting to the conditions of aquatic life.

MEASURING EVOLUTIONARY TIME

Evolution is a slow process, featuring changes in the environment and subsequent changes in genotypes and phenotypes of a population over time. The rate of evolution is measured by the rate of change of a genotype over a period of time and is related to the severity of the evolutionary pressures on the species. In other words, if a species is already perfectly suited to its habitat and there are no changes to the conditions in which it lives, the rate of evolution will be exceedingly slow—although there will still be some small base rate of genetic mutation. On the other hand, if an organism lives in a rapidly changing environment, the rate of evolution will be greater, as selection for and against certain traits will be actively occurring within that population.

By comparing DNA sequences between different species, scientists can quantify the degree of similarity between two organisms. For example, chimpanzees share over 95 percent of their genome with humans, whereas mice share only about 70 percent. As species become more taxonomically distant, the proportion of the shared genome will decrease. Molecular evolutionists correlate the degree of genomic similarity with the amount of time since two species split off from the same common ancestor; the more similar the genomes, the more recently the two species separated from each other. This is sometimes called the **molecular clock model**.

MCAT CONCEPT CHECK 12.4:

Before you move on, assess your understanding of the material with these questions.

1. What are the key tenets of each of the following theories of evolution?

- Natural selection:

- Modern synthesis model:

- Inclusive fitness:

- Punctuated equilibrium:

2. What are the three patterns of selection? What changes would each create to the population phenotype?

Pattern of Selection	Change to Population Phenotype

3. What are the three patterns of evolution between species? What is the outcome of each one?

Pattern of Evolution	Outcome

4. What is the biological definition of a species?

BIOLOGY GUIDED EXAMPLE WITH EXPERT THINKING	
While the genus <i>Saccharomyces</i> is best known for the model and industrial yeast <i>S. cerevisiae</i> , it also includes eight closely related additional species.	<i>Sac</i> <i>has</i> <i>spe</i>
The repeated isolation of <i>S. cerevisiae</i> from wine, beer, and other fermented beverages, and the difficulty in finding its truly natural habitats, has led to the common view that this species was a product of domestication. However, the unusual degree of shared physiological characteristics among <i>Saccharomyces</i> species along with the recent isolation of <i>S. cerevisiae</i> from a natural environment suggests <i>S. cerevisiae</i> may be a result of natural evolution.	<i>Issu</i> <i>for</i> <i>cer</i> <i>to</i> <i>dor</i> <i>or r</i> <i>evo</i>
In the present work we used a comparative genomics approach and publicly available complete genome sequences of five <i>Saccharomyces</i> species to search for proteins exhibiting molecular patterns of evolution. We used the dN/dS ratio [rate of nonsynonymous substitutions per nonsynonymous site (dN)/rate of synonymous substitutions per synonymous site (dS)] of the complete <i>Saccharomyces</i> ORFeomes to measure protein divergence and corrected for phylogenetic distance. From this data, we created the phylogenetic tree shown in Figure 1 and	<i>The</i> <i>tec</i> <i>jarg</i> <i>to r</i> <i>the</i> <i>tak</i>

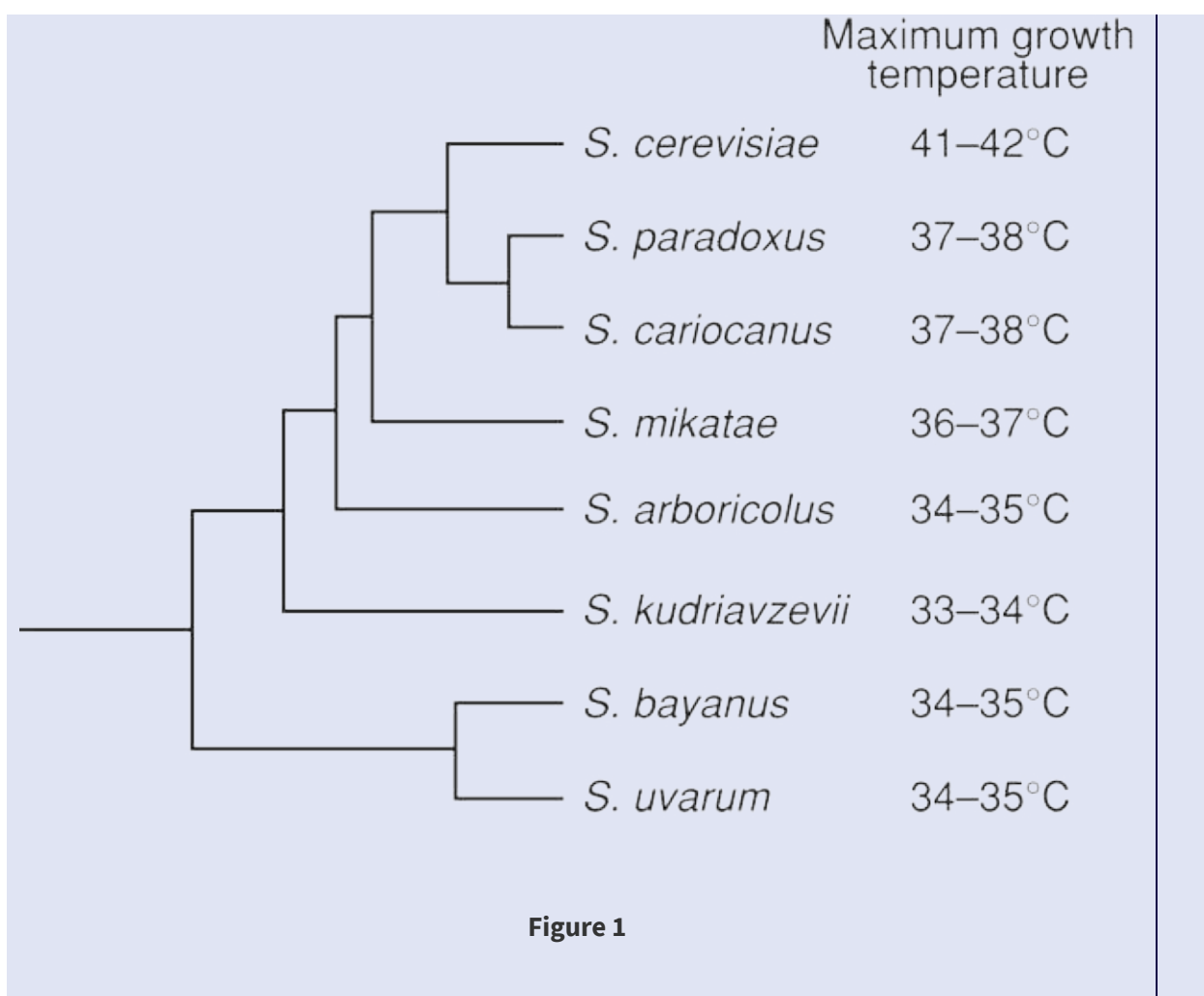
searched for genes associated with divergence.

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Some of the identified genes had been previously associated with adaptation to growth at suboptimal temperatures. This prompted us to examine catabolic fluxes at different temperatures in *Saccharomyces* species with different growth temperature preferences. Figure 1 depicts schematically the phylogenetic relationships of the species in the genus *Saccharomyces* and assigns them to the thermotolerant or cryotolerant groups.

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adapted from Gonçalves, P., Valério, E., Correia, C., de Almeida, J. M., & Sampaio, J. P. (2011). Evidence for divergent evolution of temperature preference in sympatric *Saccharomyces* species. *PloS One*, 6(6), e20739.

Based on the information provided, could temperature have provided the selective pressure for the divergent evolution of *Saccharomyces* species?

The question is asking us if there is support in the passage and figure for the hypothesis that temperature can explain the divergence of the various *Saccharomyces* species in the phylogenetic tree. Notice the phrasing of the question—it's not asking if we have proof that the evolution was due to temperature selection. This analysis is retrospective, so we would have no way

to verify whether or not the relationship between growth and temperature leading up to speciation was causal.

In paragraph 2, the researchers used fairly complex techniques to analyze the genomes of *Saccharomyces* species and arranged them in the phylogenetic tree. We don't need to understand all of the details behind this method. The takeaway is that the phylogeny was generated prior to the analysis of optimal growth temperatures. The researchers then searched for genes associated with divergence and noticed that some of those genes corresponded to adaptations for growth at nonideal temperatures. Finally, an analysis of the optimal growth temperature for the listed *Saccharomyces* species gave us a workable pattern—species that are close together on the tree also have very similar optimal growth temperatures. We know from content background that changes in environment can definitely apply selection pressures strong enough to lead to speciation, or the formation of new species. Recall also that divergent evolution is the development of species with different characteristics despite a shared common ancestor.

Since temperature is an environmental factor, we can say that based on the given information, there is plausible evidence to support that temperature could have provided the selection pressure for divergent evolution of *Saccharomyces* species.



Conclusion

Genetics and the mechanisms of evolution are becoming increasingly important in medicine, as we unintentionally breed strains of highly resistant bacteria. Antibiotic stewardship, or the use of the appropriate antibiotics only as necessary, is very important as the medical community seeks to preserve the effectiveness of antibiotics. In order to understand and apply the concepts of antibiotic stewardship, one must understand that creating environmental pressures leads to directional selection in microorganisms which can increase the frequency of the resistant phenotype. In this chapter, we covered genetics and mutations, as well as evolution. We also gave you some tools to analyze the biometric (statistical) side of genetics through the use of Punnett squares, recombinant frequencies, and the Hardy–Weinberg equations.

It seems fitting to complete this book with a discussion of evolution. You've spent hundreds of pages (and hours!) preparing for the MCAT, learning the basics of cell biology, embryogenesis and development, anatomy and physiology, genetics, and evolution. Our understanding of these topics relies on generations and generations of scientists who came before us, who passed down their knowledge through books, letters, articles, lectures, and—more recently—television, film, and popular media. But science is a field that is constantly evolving itself. At the beginning of medical school, students are often told that no more than 25 percent of what they learn during the first year will remain true by the time they enter practice. We're not sure if this statistic actually holds, but it does speak to the importance of staying on top of the latest research—not only as a medical student, but also as a

practitioner. Every day, new discoveries about the human body and the practice of medicine are being made—soon, you'll be one of those making these very discoveries and bringing them to your practice, improving your patients' lives. And at the end of it all, as a provider, an attending physician, or a researcher, you too will pass on your knowledge to future generations of physicians who will also help medical science to evolve and improve. The human body is astoundingly complex. Take a moment to genuinely think about that—*the human body is astoundingly complex*. There's so much more to learn. Medical school and your future awaits!

Concept Summary

Fundamental Concepts of Genetics

- **Chromosomes** contain **genes** in a linear sequence.
- **Alleles** are alternative forms of a gene.
 - A **dominant** allele requires only one copy to be expressed.
 - A **recessive** allele requires two copies to be expressed.
- A **genotype** is the combination of alleles one has at a given genetic **locus**.
 - Having two of the same allele is termed **homozygous**.
 - Having two different alleles is termed **heterozygous**.
 - Having only one allele is termed **hemizygous** (such as in male sex chromosomes).
 - A **phenotype** is the observable manifestation of a genotype.
- There are different patterns of dominance.
 - **Complete dominance** when the effect of one allele completely masks the effect of another.
 - **Codominance** has more than one dominant allele.
 - **Incomplete dominance** has no dominant alleles; heterozygotes have intermediate phenotypes.

- **Penetrance** is the proportion of a population with a given genotype who express the phenotype.
- **Expressivity** refers to the varying phenotypic manifestations of a given genotype.
- The modern interpretations of Mendel's laws help explain the inheritance of genes from parent to offspring.
 - **Mendel's first law (of segregation)** states that an organism has two alleles for each gene, which segregate during meiosis, resulting in gametes carrying only one allele for a trait.
 - **Mendel's second law (of independent assortment)** states that the inheritance of one allele does not influence the probability of inheriting an allele for a different trait.
- Support for DNA as genetic material came through a number of experiments.
 - The Griffith experiment demonstrated the transforming principle, converting nonvirulent live bacteria into virulent bacteria by exposure to heat-killed virulent bacteria.
 - The Avery–MacLeod–McCarty experiment demonstrated that DNA is the genetic material because degradation of DNA led to a cessation of bacterial transformation.
 - The Hershey–Chase experiment confirmed that DNA is the genetic material because only radiolabeled DNA could be found in bacteriophage-infected bacteria.

Changes in the Gene Pool

- All of the alleles in a given population constitute the **gene pool**.
- **Mutations** are changes in DNA sequence.
- Nucleotide mutations include **point mutations** (the substituting of one nucleotide for another) and **frameshift mutations** (moving the three-letter transcriptional reading frame).
 - A **silent mutation** has no effect on the protein.
 - A **missense mutation** results in the substitution of one amino acid for another.
 - A **nonsense mutation** results in the substitution of a stop codon for an amino acid.
 - **Insertions** and **deletions** result in a shift in the **reading frame**, leading to changes for all downstream amino acids.
- Chromosomal mutations include larger-scale mutations affecting whole segments of DNA.
 - **Deletion mutations** occur when a large segment of DNA is lost.
 - **Duplication mutations** occur when a segment of DNA is copied multiple times.
 - **Inversion mutations** occur when a segment of DNA is reversed.
 - **Insertion mutations** occur when a segment of DNA is moved from one chromosome to another.
 - **Translocation mutations** occur when a segment of DNA is swapped with a segment of DNA from another chromosome.
- Genetic **leakage** is a flow of genes between species through hybrid offspring.
- **Genetic drift** occurs when the composition of the gene pool changes as a result of chance.

- The **founder effect** results from **bottlenecks** that suddenly isolate a small population, leading to **inbreeding** and increased prevalence of certain homozygous genotypes.

Analytical Approaches in Genetics

- **Punnett squares** visually represent the crossing of gametes from parents to show relative genotypic and phenotypic frequencies.
 - The **parent generation** is represented by P; **filial** (offspring) **generations** are represented by F_1 , F_2 , and so on in sequence.
 - A **monohybrid cross** accounts for one gene; a **dihybrid cross** accounts for two genes.
 - In **sex-linked crosses**, sex chromosomes are usually used to indicate sex as well as genotype.
- The **recombination frequency** (θ) is the likelihood of two alleles being separated during crossing over in meiosis. **Genetic maps** can be made using recombination frequency as the scale in **centimorgans**.
- The **Hardy-Weinberg principle** states that if a population meets certain criteria (aimed at a lack of evolution), then the **allele frequencies** will remain constant (**Hardy-Weinberg equilibrium**).

Evolution

- **Natural selection** states that chance variations exist between individuals and that advantageous variations—those that increase an individual's **fitness** for survival or adaptation to the environment—afford the most opportunities for reproductive success.

- The **modern synthesis model (neo-Darwinism)** accounts for mutation and recombination as mechanisms of variation and considers **differential reproduction** to be the mechanism for reproductive success.
- **Inclusive fitness** considers an organism's success to be based on the number of offspring, success in supporting offspring, and the ability of the offspring to then support others; survival of offspring or relatives ensures appearance of genes in subsequent generations.
- **Punctuated equilibrium** considers evolution to be a very slow process with intermittent rapid bursts of evolutionary activity.
- Different types of selection lead to changes in phenotypes.
 - **Stabilizing selection** keeps phenotypes in a narrow range, excluding extremes.
 - **Directional selection** moves the average phenotype toward one extreme.
 - **Disruptive selection** moves the population toward two different phenotypes at the extremes and can lead to **speciation**.
 - **Adaptive radiation** is the rapid emergence of multiple species from a common ancestor, each of which occupies its own ecological **niche**.
- A **species** is the largest group of organisms capable of breeding to form fertile offspring. Species are **reproductively isolated** from each other by **pre- or postzygotic mechanisms**.
- Two species can evolve with different relationship patterns.
 - **Divergent evolution** occurs when two species sharing a common ancestor become more different.
 - **Parallel evolution** occurs when two species sharing a common ancestor evolve in similar ways due to analogous selection pressures.
 - **Convergent evolution** occurs when two species not sharing a recent

ancestor evolve to become more similar due to analogous selection pressures.

- According to the **molecular clock model**, the degree of difference in the genome between two species is related to the amount of time since the two species broke off from a common ancestor.

Answers to Concept Checks

12.1

- .. A dominant allele is one that requires only one copy for expression. A recessive allele requires two copies for expression.
- 1. A homozygous genotype is one in which the two alleles are the same. A heterozygous genotype is one in which the two alleles are different. A hemizygous genotype is one in which only one allele is present for a given gene (such as parts of the X chromosome in males).
- 2. Complete dominance occurs one allele (the dominant one) completely masks the expression of the other (the recessive one). Codominance occurs when a gene has more than one dominant allele, and two different dominant alleles can be expressed simultaneously. Incomplete dominance occurs when a gene has no dominant alleles, and heterozygotes have phenotypes that are intermediate between homozygotes.
- 3. Penetrance describes the proportion of the population that expresses a phenotype, given a particular genotype. Expressivity describes the differences in expression (severity, location, and so on) of a phenotype in individuals.
- 4. Mendel's first law (of segregation) most aligns with anaphase I of meiosis. Mendel's second law (of independent assortment) most aligns with prophase I of meiosis.

12.2

- .. Silent point mutations occur when one nucleotide is changed for another, but there is no change in the protein coded for by this DNA sequence (due

to redundancy in the genetic code). Missense mutations occur when one nucleotide is changed for another, and one amino acid is substituted for another in the final protein. Nonsense mutations occur when one nucleotide is changed for another, and a stop codon substitutes for an amino acid in the final protein.

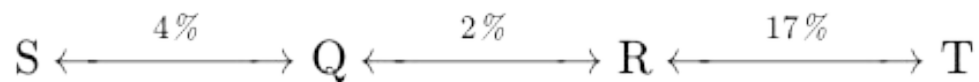
- l. The two types of frameshift mutations are insertion and deletion mutations.
- l. Duplication mutations occur when a segment of DNA is copied multiple times in the genome. Inversion mutations occur when a segment of DNA is reversed in the genome. Translocation mutations occur when a segment of DNA from one chromosome is swapped with a segment of DNA from another chromosome.
- l. Genetic leakage requires the formation of a hybrid organism that can then mate with members of one or the other parent species. While hybrids existed historically (especially mules), fertile hybrids were certainly rare before a more modern understanding of genetics (and before a commercial, financial, or academic impetus existed to create these organisms).
- i. Genetic drift occurs due to chance, so its effects will be more pronounced with a smaller sample size (in smaller populations). The founder effect occurs when a small group is reproductively isolated from the larger population, allowing certain alleles to take on a higher prevalence in the group than in the rest of the population.

12.3

Cross	Phenotypic Ratio
Bb × Bb	3 dominant:1 recessive
Aa × aa	1 dominant:1 recessive

DdEe × ddEE	1 dominant (for D)/dominant (for E): 1 recessive (for D)/dominant (for E)
X_qX × XY	Female: all unaffected; male: 1 unaffected:1 affected
X_rX × X_rY	Both male and female: 1 unaffected:1 affected

1. The genes must be in the order SQRT:



3. The criteria for the Hardy–Weinberg principle all imply that the study population is *not* undergoing evolution; thus, the allele frequencies will remain stable over time.
4. The frequency of the dominant allele (p) is 0.3. The frequency of the recessive allele (q) is 0.7. The fraction of the population with a heterozygous genotype ($2pq$) is $2 \times 0.3 \times 0.7 = 0.42$ (42%). The fraction of the population with a homozygous recessive genotype (q^2) is $(0.7)^2 = 0.49$ (49%). The fraction of the population with a dominant phenotype ($p^2 + 2pq$) is $0.09 + 0.42 = 0.51 = 51\%$.

12.4

- .. Natural selection states that certain traits that arise from chance are more favorable for reproductive success in a given environment, and that those traits will be passed on to future generations. The modern synthesis model takes natural selection and explains that selection is for specific alleles, which are passed on to future generations through formation of gametes; the alleles for these favorable traits arise from mutations. Inclusive fitness explains that the reproductive success of an organism is not only due to the number of offspring it creates, but also the ability to care for young (that can then care for others); it explains changes not only at the individual

level, but changes based on the survival of the species (and that individual's alleles within the species, including in other related individuals). Punctuated equilibrium states that for some species, little evolution occurs for a long period, which is interrupted by rapid bursts of evolutionary change.

2.

Pattern of Selection	Change to Population Phenotype
Stabilizing	Loss of extremes, maintenance of phenotype in a small window
Directional	Movement toward one extreme or the other
Disruptive	Movement toward both extremes with loss of the norm; speciation may occur

3.

Pattern of Evolution	Outcome
Divergent	Two species with a common ancestor become less similar because of different evolutionary pressures
Parallel	Two species with a common ancestor remain similar because of similar evolutionary pressures
Convergent	Two species with no recent common ancestor become more similar because of similar evolutionary pressures

4. A species is defined as the largest group of organisms capable of breeding to form fertile offspring.

Equations to Remember

(12.1) **Hardy–Weinberg equations:**

$$\begin{aligned}p + q &= 1 \\p^2 + 2pq + q^2 &= 1\end{aligned}$$



Shared Concepts

Behavioral Sciences Chapter 10

Social Thinking

Biochemistry Chapter 6

DNA and Biotechnology

Biochemistry Chapter 7

RNA and the Genetic Code

Biology Chapter 1

The Cell

Physics and Math Chapter 11

Reasoning About the Design and Execution of Research

Physics and Math Chapter 12

Data-Based and Statistical Reasoning

Discrete Practice Questions

1. What is the gene order of linked genes M, N, O, and P, given the following recombination frequencies?

MN: 6%

NO: 18%

MO: 12%

NP: 1%

MP: 5%

OP: 17%

- (A) MOPN
(B) NPMO
(C) ONPM
(D) PNMO
2. Suppose that in a mammalian species, the allele for black hair (B) is dominant to the allele for brown hair (b), and the allele for curly hair (C) is dominant to the allele for straight hair (c). When an organism of unknown genotype is crossed against one with straight, brown hair, the phenotypic ratio is as follows:
- I. 25% curly black hair
 - II. 25% straight black hair
 - III. 25% curly brown hair
 - IV. 25% straight brown hair

What is the genotype of the unknown parent?

- (A) BbCC
- (B) bbCc
- (C) Bbcc
- (D) BbCc

3. If a male with hemophilia (X_hY) is crossed with a female carrier of both color blindness and hemophilia (X_cX_h), what is the probability that a female child will be phenotypically normal?

- (A) 0%
- (B) 25%
- (C) 50%
- (D) 100%

4. If a test cross on a species of plant reveals the appearance of a recessive phenotype in the offspring, what must be true of the phenotypically dominant parent?

- (A) It must be genotypically heterozygous.
- (B) It must be genotypically homozygous.
- (C) It could be either genotypically heterozygous or homozygous.
- (D) It must have the same genotype as the test cross control parent.

5. Which of the following definitions is FALSE?

- (A) Penetrance—the percentage of individuals in the population carrying the allele who actually express the phenotype associated with it
- (B) Expressivity—the percentage of individuals in the population carrying the allele who do not express the phenotype associated with it
- (C) Incomplete dominance—occurs when the phenotype of the heterozygote is an intermediate of the phenotypes of the homozygotes
- (D) Codominance—occurs when multiple alleles exist for a given gene and more than one of them is dominant

6. In a species of plant, a homozygous red flower (RR) is crossed with a homozygous yellow flower (rr). If the F_1 generation is self-crossed and the F_2 generation has a phenotypic ratio of red:orange:yellow of 1:2:1, which characteristic accounts for these results?

- (A) Codominance
- (B) Incomplete dominance
- (C) Penetrance
- (D) Expressivity

7. Which of the following statements is INCORRECT regarding inheritance of traits according to the modern synthesis model?

- (A) A mutation due to excessive amounts of ultraviolet light occurs in an unfertilized egg; this will affect the child who is born from that

egg.

- (B) The muscular strength gained by a weight lifter during his lifetime is inherited by his children.
- (C) A green-feathered bird that survived all of the predators in the forest will pass on the green feather genes to its offspring.
- (D) A flower with a tasty nectar eaten by a butterfly is more likely to pass on its genes through the pollen spread by the butterfly than a flower with less desirable nectar.

8. Which of the following statements is FALSE based on Darwin's theory of natural selection?

- (A) Natural selection is the driving force for evolution.
- (B) Favorable genetic variations become more and more common in individuals throughout their lives.
- (C) Natural selection can drive organisms living in groups to ultimately become distinct species.
- (D) Fitness is measured by reproductive success.

9. Which of the following is NOT a necessary condition for Hardy-Weinberg equilibrium?

- (A) Large population size
- (B) No mutations
- (C) Monogamous mating partners
- (D) No migration into or out of the population

10. As the climate became colder during the Ice Age, a particular species of mammal evolved a thicker layer of fur. What kind of selection occurred in this population?
- (A) Stabilizing selection
 - (B) Directional selection
 - (C) Disruptive selection
 - (D) Speciation selection
11. At what point are two populations descended from the same ancestral stock considered to be separate species?
- (A) When they can no longer produce viable, fertile offspring.
 - (B) When they look significantly different from each other.
 - (C) When they can interbreed successfully and produce offspring.
 - (D) When their habitats are separated by a significantly large distance so that they cannot meet.
12. In a nonevolving population, there are two alleles, R and r, which code for the same trait. The frequency of R is 30 percent. What are the frequencies of all the possible genotypes?
- (A) 49% RR, 42% Rr, 9% rr
 - (B) 30% RR, 21% Rr, 49% rr
 - (C) 0.09% RR, 0.42% Rr, 0.49% rr
 - (D) 9% RR, 42% Rr, 49% rr

13. In a particular Hardy–Weinberg population, there are only two eye colors: brown and blue. Of the population, 36% have blue eyes, the recessive trait. What percentage of the population is heterozygous?
- (A) 24%
 - (B) 48%
 - (C) 60%
 - (D) 64%
14. Which of the following was NOT a belief of Darwin's?
- (A) Evolution of species occurs gradually and evenly over time.
 - (B) There is a struggle for survival among organisms.
 - (C) Genetic recombination and mutation are the driving forces of evolution.
 - (D) Those individuals with fitter variations will survive and reproduce.
15. A child is born with a number of unusual phenotypic features and genetic testing is performed. The child is determined to have partial trisomy 21, with three copies of some segments of DNA from chromosome 21, and partial monosomy 4, with only one copy of some segments of DNA from chromosome 4. Which of the following mutations could have occurred in one of the parental gametes during development to explain both findings?
- (A) Deletion
 - (B) Insertion

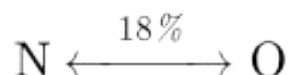
(C) Translocation

(D) Inversion

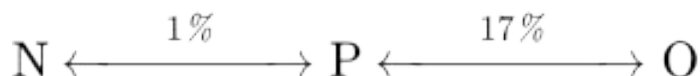
Discrete Practice Answers

1. B

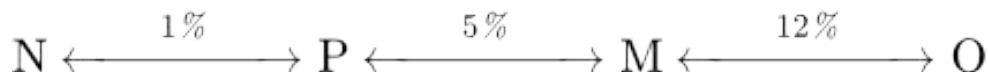
This is a gene-mapping problem. Because there is a correlation between the frequency of recombination and the distance between genes on a chromosome, if we are given the frequencies, we can determine gene order. Remember that one map unit equals 1 percent recombination frequency. The easiest way to begin is to determine the two genes that are farthest apart; in this case, N and O recombine with a frequency of 18%, so they are 18 map units apart on the chromosome:



N and P recombine with 1% frequency, and P and O recombine with 17% frequency, so P must be between N and O:



Finally, M and P recombine with 5% frequency, and M and O recombine with 12% frequency, so M must be between P and O:



2. D

In this dihybrid problem, a doubly recessive individual is crossed with an individual of unknown genotype; this is known as a test cross. The straight-

and brown-haired organism has the genotype $bbcc$ and can thus only produce gametes carrying bc . Looking at the F_1 offspring, there is a 1:1:1:1 phenotypic ratio. The fact that both the dominant and recessive traits are present in the offspring means that the unknown parental genotype must contain both dominant and recessive alleles for each trait. The unknown parental genotype must therefore be $BbCc$. If you want to double-check the answer, you can work out the Punnett square for the cross $BbCc \times bbcc$:

	BC	Bc	bC	bc
bc	BbCc	Bbcc	bbCc	bbcc
bc	BbCc	Bbcc	bbCc	bbcc
bc	BbCc	Bbcc	bbCc	bbcc
bc	BbCc	Bbcc	bbCc	bbcc
	↓	↓	↓	↓
	25% black curly	25% black straight	25% brown curly	25% brown straight

3. C

The female in this example is a carrier of two sex-linked traits; based on her genotype, the affected alleles are found on different X chromosomes. Drawing out a Punnett square, we see that 25% of the offspring will be female hemophiliacs (X_hX_h) and 25% will be female carriers of both alleles

(X_cX_h). This question is asking what percentage of females will have a normal phenotype, which would be half of the females (those who are carriers for both traits).

	X_c	X_h	
X_h	X_cX_h	X_hX_h	<u>offspring</u> 25% female hemophiliac 25% female carrier of both traits (phenotypically normal)
Y	X_cY	X_hY	

4. **A**

The control parent in a test cross is always recessive. Therefore, if the test parent is phenotypically dominant, yet can provide a recessive allele (as evidenced by the presence of recessive children), then the parent must have both a dominant and recessive allele. Therefore, this test parent must be heterozygous.

5. **B**

The definition given here for expressivity is a much better match for defining penetrance (or, really, one minus the penetrance). Expressivity refers to the variable manifestations of a given genotype as different phenotypes; the degree to which various phenotypes are expressed. All of the other definitions given are accurate.

6. **B**

Some progeny in the second generation are apparently blends of the parental phenotypes. The orange color is the result of the combined effects

of the red and yellow alleles. An allele is incompletely dominant if the phenotype of the heterozygotes is an intermediate of the phenotypes of the homozygotes.

7. **B**

To find the correct answer, we have to read each choice and eliminate the ones that fit with the modern-day theories of inheritance, which state that the genes that make an organism most fit for its environment will be passed to offspring. This can be seen in **(C)** and **(D)**, both of which demonstrate that an organism with improved fitness will pass those genes to offspring. **(A)** mentions a gamete being exposed to mutagens; a zygote created from this gamete would contain any mutations that were present in the egg and would be affected by them. Thus, **(B)** must be the correct answer: indeed, acquired characteristics not encoded in the genome should not be passed to offspring according to the modern synthesis model.

8. **B**

Darwin's theory of natural selection argues that chance variations between organisms can help certain organisms survive to reproductive age and produce many offspring, transmitting their variations to the next generation. Thus, natural selection would drive the process of evolution forward, enabling the persistence of characteristics that impart an advantage in the environment, eliminating **(A)**. In Darwin's theory, fitness is measured in terms of reproductive success, as **(D)** states. Through natural selection, organisms may be separated into groups depending on environmental pressures, and these groups can eventually separate to the point of becoming distinct species, eliminating **(C)**. **(B)** is the correct answer because the theory of natural selection applies to a population of organisms, not to a particular individual. As such, favorable genetic

variations become more and more common from generation to generation, not during the lifetime of an individual.

9. **C**

Hardy–Weinberg equilibrium exists under certain ideal conditions that, when satisfied, allow one to calculate the gene frequencies within a population. The Hardy–Weinberg equation can be applied only under these five conditions: (1) the population is very large; (2) there are no mutations that affect the gene pool; (3) mating between individuals in the population is random; (4) there is no migration of individuals into or out of the population; (5) the genes in the population are all equally successful at being reproduced. Thus, from the given choices, only **(C)** is false: monogamy is not a necessary condition for Hardy–Weinberg equilibrium.

0. **B**

The situation described in the question stem is an example of directional selection. In directional selection, the phenotypic norm of a particular species shifts toward an extreme to adapt to a selective pressure, such as an increasingly colder environment. Only those individuals with a thicker layer of fur were able to survive during the Ice Age, thus shifting the phenotypic norm.

1. **A**

A species is defined as the largest group of organisms that can interbreed to produce viable, fertile offspring. Therefore, two populations are considered separate species when they can no longer do so.

2. **D**

Let's use the information provided by the question stem to set up our equations. We are told that the frequency of R equals 30%, and as such, $p =$

0.30. The frequency of the recessive gene is $r = 100\% - 30\% = 70\%$; thus, $q = 0.70$. The frequency of the genotypes, according to the Hardy–Weinberg equilibrium, are given by $p^2 = RR$, $2pq = Rr$, and $q^2 = rr$. Therefore, the frequency of the genotypes are $(0.3)^2 = 0.09 = 9\%$ RR , $2 \times 0.3 \times 0.7 = 0.42 = 42\%$ Rr , and $(0.7)^2 = 0.49 = 49\%$ rr .

3. **B**

Using the information given in the question stem, we can determine that the percentage of the population with blue eyes (genotype bb) = $36\% = 0.36 = q^2$; therefore, $q = 0.6$. Because this is a Hardy–Weinberg population, we can assume that $p + q = 1$, so $p = 1 - 0.6 = 0.4$. The frequency of heterozygous brown eyes is therefore $2pq = 2 \times 0.4 \times 0.6 = 0.48 = 48\%$.

4. **C**

Darwin's main argument was that natural selection is the driving force for evolution. Darwin did not consider the role of genetic mutation and recombination, which were unknown at the time. These aspects were added to Darwin's theory as part of the modern synthesis model, which was not developed until significantly after Darwin's time.

5. **C**

This scenario—a deletion of some DNA and a duplication of other DNA—would be consistent with a translocation between chromosomes 4 and 21 during development of an egg or sperm. If part of chromosome 21 was swapped with part of chromosome 4, then a gamete resulting from meiosis in this cell would result in a daughter cell with two copies of some of the DNA from 21 and no copies of some of the DNA from 4. Therefore, after fertilization, there would be partial trisomy 21 and partial monosomy 4. While a deletion or insertion could explain one of the findings, it cannot explain both, eliminating **(A)** and **(B)**. An inversion should not lead to

partial trisomy or partial monosomy because the DNA is simply reversed, eliminating **(D)**.

Glossary

Abductor—A muscle that moves a limb away from the center of the body.

Absorption—The process by which substances are taken up into or across tissues.

Acetylcholine—A neurotransmitter found throughout the nervous system (somatic motor neurons, preganglionic parasympathetic and sympathetic nerves, and postganglionic parasympathetic neurons); metabolized by acetylcholinesterase.

Acrosome—The large vesicle at the head of a sperm cell containing enzymes that degrade the ovum cell membrane to allow fertilization.

Actin—A protein found in the cytoskeleton and muscle cells; it is the principal constituent of the thin filaments and microfilaments.

Action potential—An abrupt change in the membrane potential of a nerve or muscle caused by changes in membrane ionic permeability; results in conduction of an impulse in nerves or contraction in muscles.

Active immunity—An immune response (antibody production or cellular immunity) acquired in response to exposure to an antigen.

Active site—Substrate-binding region of an enzyme.

Adaptation—The development of characteristics that enable an organism to survive and reproduce in its habitat.

Adaptive immunity—Highly specific form of immunity that retains chemical memory of each invader encountered and is able to tailor the immune response to the specific pathogen.

Adaptive radiation—The evolutionary process by which one species gives rise to several species, each specialized for different niches.

Adductor—A muscle that moves a limb toward the center of the body.

Adenine—A purine base present in DNA and RNA; it forms hydrogen bonds with thymine and uracil.

Adenosine triphosphate (ATP)—A nucleotide molecule consisting of adenine, ribose, and three phosphate moieties; the outer two phosphates are bound by high-energy bonds.

Adipose—Refers to fatty tissue, fat-storing tissue, or fat within cells.

Aerobic—Refers to a biological process that occurs in the presence of molecular oxygen (O₂) or to organisms that cannot live without molecular oxygen.

Afferent (sensory) neuron—A neuron that picks up impulses from sensory receptors and transmits them toward the central nervous system.

Agranulocyte—Type of leukocyte that does not contain cytoplasmic granules, including lymphocytes and monocytes.

Albumin—Protein synthesized in the liver that maintains the oncotic pressure of the blood and serves as a carrier for many drugs and hormones.

Allantois—One of four embryonic membranes; it contains the growing embryo's waste products.

Allele—Alternative forms of the same gene coding for a particular trait; alleles segregate during meiosis.

Allergy—A type of autoimmunity in which a person's immune system becomes overactivated by common substances in the environment.

Alveolus—Basic functional unit of the lung; a tiny sac specialized for passive gas exchange between the lungs and the blood.

Amino acids—The building blocks of proteins, each containing an amino group, a carboxylic acid group, and a side chain (or R group) attached to the α -carbon.

Amnion—The innermost fluid-filled embryonic membrane; it forms a protective sac surrounding the embryos of birds, reptiles, and mammals.

Amplification—Characteristic of a signaling cascade, in which the binding of a single peptide hormone to a membrane-bound receptor results in a signal that increases in strength through the signaling cascade.

Anaerobic—Refers to a biological process that can occur without oxygen or to

organisms that can live without molecular oxygen.

Anaphase—The stage of mitosis or meiosis characterized by the migration of chromatids or homologous chromosomes to opposite poles of the dividing cell.

Androgen—Any male sex hormone, such as testosterone.

Antibiotic—Substance that kills or inhibits the growth of bacteria or fungi (usually by disrupting cell wall assembly or by binding to ribosomes, thus inhibiting protein synthesis).

Antibody—Immune or protective protein whose synthesis is induced by the presence of foreign substances (antigens) in the body; each antibody binds to a specific antigen in an immune response; also called immunoglobulin.

Antigen—A substance that binds to an antibody; may be foreign or a self-antigen.

Antigen-binding region—Portion of an antibody that is specific for a particular antigen; the area of the antibody to which the antigen binds.

Aortic valve—One of the semilunar valves, separating the left ventricle from the aorta.

Apoptosis—Process by which a cell undergoes programmed cell death in a highly organized manner in response to either external or internal signals.

Appendicular skeleton—Peripheral portion of the skeleton consisting of arms, legs, and pelvic and pectoral girdles.

Archenteron—The central cavity in the gastrula stage of embryological development; it is lined by endoderm and ultimately gives rise to the adult digestive tract.

Arterioles—Small arterial structures that link the arteries to the capillaries.

Artery—Thick-walled, muscular blood vessel that generally carries blood away from the heart.

Articular cartilage—Cartilaginous coating at the ends of bones that provides a smooth surface for articulation of bones within a joint.

Asexual reproduction—Any reproductive process that does not involve the fusion of gametes (such as budding).

Asters—Star-shaped structures that form around the centrosome during mitosis.

Atrium—One of two paired structures on either side of the heart, into which blood returning from either the body (right atrium) or the lungs (left atrium) flows.

Autocrine—Form of cell–cell communication in which a cell releases a substance that then binds to the membrane of the releasing cell to either inhibit or activate a cellular activity.

Autoimmunity—Inappropriate immune response that targets self-antigens.

Autonomic nervous system—Subdivision of the peripheral nervous system

responsible for involuntary activities, which is further subdivided into the parasympathetic and sympathetic nervous systems.

Autosome—Any chromosome other than a sex chromosome.

Axial skeleton—Midline structures of the skeleton including the skull, vertebral column, and rib cage; provides the central framework of the body.

Axon—The long fiber of a neuron; it conducts impulses away from the cell body toward the synapse.

Axon hillock—Transition point between the cell body (soma) and the axon of a neuron; the site of action potential initiation.

Bacillus—Rod-shaped bacterium.

Bacteriophage—A virus that invades bacteria and sometimes uses bacterial RNA and ribosomes to self-replicate.

Basophil—Type of granulocytic leukocyte that largely participates in allergic reactions and local inflammation.

Bile—A solution of salts, pigments, and cholesterol produced by the liver and stored in the gallbladder; it emulsifies large fat droplets when secreted into the small intestine via the bile duct.

Bilirubin—Product of the breakdown of hemoglobin that is modified to a more soluble form in the liver.

Binary fission—A type of asexual reproduction characteristic of prokaryotes

in which there is equal nuclear and cytoplasmic division.

Blastocoel—The fluid-filled central cavity of the blastula.

Blastocyst—A mammalian blastula, consisting of the trophoblastic cells and an inner cell mass.

Blastopore—Opening of the archenteron to the external environment in the gastrula stage of embryonic development.

Blastula—The early embryonic stage during which the embryo is a hollow, fluid-filled sphere of undifferentiated cells.

Blastulation—Process by which a solid mass of early embryonic cells, known as the morula, becomes the blastula, a hollow fluid-filled sphere of undifferentiated cells.

Bohr effect—Changes in the affinity of hemoglobin for oxygen caused by changes in the environment; when pH is low (increased concentration of hydrogen ions), the oxyhemoglobin dissociation curve shifts right, indicating a decreased affinity of hemoglobin for oxygen and more efficient off-loading of oxygen from hemoglobin.

Bolus—An initial dose of medication; in the digestive system, chewed food leaving the mouth, traveling through the esophagus, and entering the stomach.

Bone marrow—Central portion of bones, especially long bones, that contains fat and developing blood cells, including erythrocytes, leukocytes, and megakaryocytes.

Bone matrix—Organic and inorganic minerals that provide strength to compact bone; organic components include collagen, glycoproteins, and other peptides; inorganic components include calcium, phosphate, and hydroxide ions (in hydroxyapatite).

Bowman's capsule—The cup-like structure of the nephron; it collects the glomerular filtrate and channels it into the proximal convoluted tubule.

Bronchi—Tube-like passages for air that connect the trachea to the bronchioles.

Bronchioles—Passageways for air that start at the bronchi, dividing into continuously smaller passageways that eventually lead to the alveoli, where gas exchange occurs.

Brush-border enzymes—Group of enzymes present on the luminal surface of cells lining the duodenum that break down larger biomolecules into monomers that are able to be absorbed.

Bundle of His—Part of the conduction system of the heart; it carries impulses from the AV node to the ventricles.

Callus—Area of excessive deposition of keratin in response to repeated strain due to friction.

Canaliculi—Small canals connecting lacunae within the bone matrix with Haversian canals, allowing for the flow of nutrients and wastes.

Capillary—Small, thin-walled blood vessel where gas, nutrient, and waste

exchange occurs between blood and tissues.

Capsid—Protein coat surrounding a virus.

Cardiac output—Total blood volume pumped by the left ventricle in one minute, found by multiplying the heart rate by the stroke volume.

Cartilage—A firm, elastic, translucent connective tissue produced by cells called chondrocytes.

Catabolism—The chemical breakdown of complex substances (macromolecules) to yield simpler substances and energy.

Cecum—The first part of the large intestine; accepts material flowing through the ileocecal valve and is the point of attachment of the appendix.

Cell body—Portion of a neuron where the nucleus, endoplasmic reticulum, and ribosomes are located; also known as the soma.

Cell theory—A foundational belief in modern biology that all living things are composed of cells, that the cell is the basic functional unit of life, that all cells arise from preexisting cells, and that DNA is the genetic material.

Cell-mediated immunity—Type of immunity that uses cytotoxic chemicals released from cells to cause death of cells that have been infected by viruses.

Central nervous system (CNS)—The brain and spinal cord.

Centriole—A small organelle in the cytoplasm of animal cells; it organizes the spindle apparatus during mitosis or meiosis.

Centromere—The area of a chromosome where sister chromatids are joined; it is also the point of attachment to the spindle fiber during mitosis and meiosis.

Centrosomes—Paired cylindrical organelles, located in the cytoplasm, that contain the centrioles.

Cerebellum—The section of the mammalian hindbrain that controls muscle coordination and equilibrium.

Cerebral cortex—The outer layer of the forebrain, consisting of grey matter; it is the site of higher cognitive functions in humans.

Cervix—Lower end of the uterus that marks the transition between the vagina and the uterus.

Chemical digestion—Enzymatic cleavage of chemical bonds within foodstuffs, resulting in smaller molecules.

Chemotaxis—Movement of cells toward or away from a chemical within the environment.

Chiasmata—Sites where crossing over occurs between homologous chromosomes during meiosis.

Chief cells—Cells within the stomach that secrete pepsinogen, a zymogen that is converted to its active form, pepsin, by the acidic environment of the stomach.

Chondrin—Elastic cartilage matrix substance secreted by chondrocytes.

Chondrocyte—A differentiated cartilage cell that synthesizes the cartilaginous matrix.

Chromatid—Each of the two chromosomal strands formed by DNA replication in the S phase of the cell cycle; held together by the centromere.

Chromosome—A filamentous body found within the nucleus of a eukaryotic cell or nucleoid region of a prokaryotic cell, composed of DNA.

Chylomicron—Soluble lipid molecule that consists of triglycerides and esterified cholesterol molecules; absorbed into lacteals from the digestive tract.

Chyme—Aqueous mixture of food and secretions that leaves the stomach to enter the duodenum.

Cilia—Projection from a cell involved in movement of materials on the outside of the cell.

Circadian rhythm—A behavioral pattern based on a 24-hour cycle, related to cycling of hormones such as cortisol and melatonin.

Cleavage—A series of mitotic divisions of the zygote immediately following fertilization, resulting in progressively smaller cells with increased nucleus-to-cytoplasm ratios.

Clonal selection—Phenomenon in which only B- or T-cells specific to a particular pathogen are activated.

Coccus—Spherically shaped bacterium.

Codominance—A genetic effect in which the phenotype of a heterozygote is a distinct reflection of both alleles at a particular locus.

Competent—Describes a cell capable of responding to induction signals.

Conjugation—The temporary joining of two organisms via a tube called a pilus, through which genetic material is exchanged; a form of sexual reproduction used by bacteria.

Connective tissue—Animal tissue composed of cells lying in an extracellular proteinaceous network that supports, connects, and surrounds the organs and structures of the body.

Constant region—Portion of an antibody molecule that is not variable and participates in the binding of other immune modulators.

Convergent evolution—The process by which unrelated organisms living in a similar environment develop analogous structures.

Corona radiata—Layer of cells surrounding an oocyte that aids in the development of the ovum.

Corpus luteum—The remnant of the ovarian follicle, which after ovulation continues to secrete progesterone. Its degeneration leads to menstruation; it also maintains the uterine lining during pregnancy.

Cortex—The external layer found in many organs of the body, including the

brain, adrenal glands, and kidney.

Cortical reaction—Release of calcium ions by an ovum after fertilization, resulting in the creation of a fertilization membrane, a structure that prevents fertilization of an ovum by multiple sperm cells.

Corticosteroids—Steroid hormones produced in the adrenal cortex, including glucocorticoids (cortisol), mineralocorticoids (aldosterone), and cortical sex hormones.

Crossing over—The exchange of genetic material between homologous chromosomes during meiosis.

Cyclic adenosine monophosphate(cAMP)—An intracellular second messenger in the signaling cascade initiated by a peptide hormone; synthesized from ATP by adenylate cyclase.

Cytokine—Chemical substance that stimulates inflammation and recruits additional immune cells to a specific area.

Cytokinesis—The division and distribution of parent cell cytoplasm to the two daughter cells during mitotic and meiotic cell division.

Cytoplasm—The fluid and solutes within a cell membrane, external to the nucleus and cellular organelles.

Cytotoxic T-cell—T-cell that seeks out infected cells and induces apoptosis in these cells to prevent spread of the pathogen.

Deletion—A type of genetic mutation in which some variable amount of DNA

is removed.

Dendrite—The portion of a neuron that receives stimuli and conveys them toward the cell body.

Dermis—The layer of skin cells under the epidermis. Contains sweat glands, hair follicles, fat, and blood vessels.

Determinate cleavage—Rapid mitotic divisions occurring in an embryo that result in cells with predetermined fates; these cells are only capable of differentiating into certain kinds of tissues within an organism.

Determination—Designation of a cell within an embryo as having a particular future function.

Diaphragm—Thin, muscular structure that divides the thorax from the abdomen and provides the driving force for inhalation.

Diaphysis—Cylindrical shaft of a long bone.

Diastole—The period of relaxation of cardiac muscle during which the atrioventricular valves open and the ventricles fill with blood.

Differentiation—The process by which unspecialized cells become specialized. Involves selective transcription of the genome.

Digestion—The breakdown of macromolecular nutrient material via mechanical and chemical means to simple molecular building blocks; this facilitates absorption.

Diploid—Having two chromosomes of each type per cell; symbolized by $2n$.

Direct hormone—Substance secreted into the bloodstream that causes a change in the physiological activity of cells without requiring an intermediary.

Directional selection—Selective pressures favor the development of an extreme phenotype that provides a selective advantage; this phenotype emerges as the primary phenotype over time.

Disruptive selection—Type of selection in which selective pressures favor extreme phenotypes over the norm.

Divergent evolution—A process of change whereby organisms with a common ancestor evolve dissimilar structures (such as dolphin flippers and human arms).

Dominant—Refers to an allele that requires only one copy for expression.

Ductus arteriosus—Fetal structure that shunts blood from the pulmonary artery to the aorta to bypass the developing lungs.

Ductus venosus—Shunt from the umbilical vein to the inferior vena cava, allowing oxygenated blood returning from the placenta to bypass the liver and enter the systemic circulation.

Duodenum—First segment of the small intestine; the contents of the stomach and the pancreatic and bile ducts empty into it; site of digestion and some absorption.

Ectoderm—Outermost embryonic germ layer; it gives rise to the skin, nervous

system, inner ear, lens of the eye, and other structures.

Effector—An organ, muscle, or gland used by an organism to respond to a stimulus.

Efferent (motor) neuron—A neuron that transmits nervous impulses from the central nervous system to an effector.

Embryo—An organism at the early developmental stage; in humans, the term refers to the first eight weeks after fertilization.

Endocrine—A form of cell–cell communication that involves the secretion of hormones into the bloodstream by ductless glands; these hormones then travel to distant locations within the organism to cause a change in cellular activity.

Endoderm—Innermost embryonic germ layer; it later gives rise to the linings of the alimentary canal and of the digestive and respiratory organs.

Endometrium—Uterine lining that is regenerated each month in preparation for implantation of an embryo; absence of an embryo results in sloughing off of the endometrium in a process known as menstruation.

Endoplasmic reticulum—Membrane-bound channels in the cytoplasm that transport proteins and lipids to various parts of the cell.

Endothelium—Lining of blood vessels consisting of endothelial cells.

Enteric nervous system—Collection of neurons within the gastrointestinal tract that governs peristalsis.

Eosinophil—Type of granulocytic leukocyte that largely participates in the immune response against parasites; also involved in the pathogenesis of allergies.

Epidermis—The outermost layer of the skin.

Epididymis—The coiled tube in which sperm gain motility and are stored after production in the testes.

Epiglottis—The small flap of cartilage that covers the glottis during swallowing, preventing food from entering the larynx.

Epinephrine—A hormone synthesized by the adrenal medulla; it stimulates the fight-or-flight response; also a neurotransmitter in the sympathetic nervous system.

Epiphyseal plate—Cartilaginous structure in the epiphysis where growth occurs.

Epiphysis—Dilated end of a long bone.

Episomes—A specialized subset of plasmids capable of integrating into the genome of bacteria under specific circumstances.

Epithelium—The cellular layer that covers internal and external surfaces of body structures and cavities.

Erythrocyte—Red blood cell; a biconcave, disc-shaped cell that contains hemoglobin and has no nucleus.

Esophagus—Portion of the alimentary canal connecting the pharynx and the stomach.

Eukaryote—A unicellular or multicellular organism composed of cells that contain a membrane-bound nucleus and other membrane-bound organelles.

Evolution—The changes in the gene pool from one generation to the next caused by mutation, nonrandom mating, natural selection, and genetic drift.

Exocrine glands—Glands that release their secretions into ducts (such as the parts of the liver and sweat glands).

Expressivity—Varying expression of disease symptoms despite identical genotypes.

Extensor—A muscle used in the straightening of a limb.

Facultative anaerobes—Prokaryotes that can exist with or without oxygen.

Fertilization—Fusion of two gametes.

Fertilization membrane—Structure created by the cortical reaction after fertilization of an ovum by a sperm cell; prevents fertilization of an ovum by multiple sperm cells.

Fetus—A developing organism that has passed the early developmental stages. In humans, the term refers to an embryo from the ninth week after fertilization until birth.

Fibrin—The insoluble protein that forms the bulk of a blood clot.

Filial generation—Offspring in a genetic cross; may be supplemented with a subscript to show how many generations out from the parents.

Filtration—In the nephron, the process by which blood plasma is forced (under high pressure) out of the glomerulus into Bowman's capsule.

Fitness—Reproductive success of an individual, measured in increased number and survival of offspring.

Flagellum—A microscopic, whip-like filament that functions in the locomotion of sperm cells and some unicellular organisms; composed of microtubules.

Flexor—A muscle used in the bending of a limb.

Follicle—The set of cells surrounding a developing or mature ovum. Secretes nutrients and estrogen and atrophies into the corpus luteum after ovulation.

Foramen ovale—Shunt within the fetal heart between the right and left atria that allows the circulation to largely bypass the developing lungs.

Gallbladder—Organ below the liver that stores bile; contracts in response to stimulation by cholecystokinin, resulting in release of bile into the biliary system and eventually into the duodenum.

Gamete—Sperm or ovum; a cell that has half the number of chromosomes of a somatic cell (haploid) and can fuse with another gamete to form a zygote.

Ganglion—A mass of neuron cell bodies outside the central nervous system.

Gastrula—The embryonic stage characterized by the presence of ectoderm, mesoderm, and endoderm.

Gene—The basic unit of heredity; a region on a chromosome that codes for a specific product.

Gene pool—All of the alleles for every gene in every individual in a given population.

Genetic drift—Variations in the gene pool caused by chance.

Genetic map—A diagrammatic representation of a chromosome indicating distance between two genes on a chromosome as determined by recombination frequencies.

Genome—An organism's complete set of chromosomes.

Genotype—The genetic composition of an entire organism or in reference to a particular trait.

Glomerulus—The network of capillaries encapsulated by Bowman's capsule. Acts as a filter for blood entering the nephron.

Glottis—The opening to the trachea.

Golgi apparatus—Organelle that plays a role in the packaging and secretion of proteins and other molecules produced intracellularly.

Gonad—Ovary or testis; the reproductive organ in which gametes are produced.

Gram staining—A process of staining bacterial cells such that cells containing large amounts of peptidoglycan within the cell wall are stained purple, while cells with less peptidoglycan within their cell walls appear pink-red after counterstaining.

Granulocyte—Type of leukocyte with cytoplasmic granules that are visible under a microscope, such as neutrophils, basophils, or eosinophils.

Grey matter—Any region in the central nervous system that consists largely of neuron cell bodies, dendrites, and synapses.

Growth factors—Substances that cause induction during embryonic development and ensure the development of the correct structure in the right location.

Haploid—Having only one of each type of chromosome per cell; symbolized by n .

Hardy-Weinberg principle—States that gene ratios and allelic frequencies remain constant through the generations in a non-evolving population.

Haversian canal—Central channel within the osteon (Haversian system) containing blood vessels, nerve fibers, and lymph vessels.

Heavy chain—One of two types of chains, made of peptides, that create an antibody; each antibody consists of two heavy chains and two light chains.

Helper T-cells—Type of T-cell that secretes lymphokines; the specific combination of lymphokines secreted will determine the nature of the immune response; activation of T_h1 cells will result in a cytotoxic response, while a T_h2 response will rely on B-cells.

Hematocrit—Measurement of how much of a blood sample consists of red blood cells, expressed as a percentage.

Hemoglobin—Iron-containing protein found in red blood cells that binds O_2 and transports it throughout the body.

Heterozygous—Having two different alleles for a particular trait.

Hilum—Area of an organ where large vessels or other structures enter or exit; the renal hilum is where the renal artery enters the kidney, the renal vein leaves the kidney, and the ureter exits the kidney to transport urine to the bladder.

Histamine—An inflammatory mediator that causes vasodilation and results in increased movement of fluid and cells out of the blood vessels and into the tissues.

Homeostasis—Maintenance of a stable internal physiological environment in an organism.

Homologous chromosomes—Chromosomes in a diploid cell that carry corresponding genes for the same traits at corresponding loci.

Homozygous—Having two identical alleles for a given trait.

Hormones—Chemical messengers secreted by cells of one part of the body and carried by the bloodstream to cells elsewhere in the body, where they regulate biochemical activity.

Humoral immunity—Form of adaptive immunity that takes place within body fluids, driven by B-cells and antibodies.

Hybrid—The resultant offspring of a cross (mating) either between two different gene types or between two different species.

Hypodermis—Subcutaneous layer beneath the dermis in the skin.

Hypothalamus—The region of the vertebrate forebrain that controls the autonomic nervous system and is the control center for hunger, thirst, body temperature, and other visceral functions; also secretes factors that stimulate or inhibit pituitary secretions.

Ileum—The terminal portion of the small intestine.

Impulse propagation—Movement of an action potential down an axon, resulting in neurotransmitter release at the synaptic bouton and transmission of the impulse to the target neuron or organ.

Inborn error of metabolism—Genetic mutation that causes a change in an enzyme required for metabolism; early intervention is necessary to prevent the development of life-threatening conditions; some inborn errors of metabolism are ultimately incompatible with life.

Incomplete dominance—A genetic effect in which the phenotype of a heterozygote is a mixture of the two parental phenotypes.

Independent assortment—Unlinked genes within a primary germ cell separate randomly during gametogenesis.

Indeterminate cleavage—Rapid mitotic divisions resulting in cells that are individually capable of becoming complete organisms.

Inducer—A chemical substance passed from an organizing cell to a responsive cell, resulting in differentiation of the responsive cell.

Induction—The initiation of cell differentiation in a developing embryo due to the influence of other cells.

Innate immunity—Form of immunity that is nonspecific and does not require learning.

Integument—The outer layer of the body (skin); provides function for thermoregulation and innate immunity.

Intermediate filament—Collection of fibers that help to maintain the overall integrity of the cytoskeleton.

Interneuron—A neuron that has its cell body and nerve terminals confined to one specific area; often involved in spinal reflexes.

Interphase—The stage between successive nuclear divisions; it is divided into the G₁, S, and G₂ stages; cell growth and DNA replication occur during interphase.

Intracellular digestion—Oxidation of fatty acids and glucose for energy within cells.

Intrapleural space—Fluid-filled potential space between the parietal and visceral pleura that lubricates the two pleural surfaces and allows for a pressure differential between the intrapleural space and the lungs.

Inversion—A chromosomal mutation in which a section of a chromosome breaks off, flips over, and then reattaches in its original spot.

Isolation—Mechanism that prevents genetic exchange between individuals of different species or populations.

Jejunum—The middle portion of the small intestine.

Joint cavity—Space between two bones in a joint; enclosed and maintained by fibrous tissues.

Juxtacrine—A form of cell–cell communication in which a cell releases a substance that binds to receptors on cells directly adjacent to the releasing cell.

Keratin—Protein present in the outermost layer of the skin that is largely responsible for preventing the loss of fluids and salts as well as the entry of foreign substances into the body; also present as an intermediate filament within cells.

Keratinocytes—Cells within the epidermis that produce keratin.

Kidney—Vertebrate organ that regulates water and salt concentration in the

blood and is responsible for urine formation.

Kinetochores—A protein structure, located at the centromere, that provides a place for spindle fibers to attach to the chromosome.

Lacteal—Small lymphatic vessel that runs in the center of the villi in the small intestine; site of lipid absorption into the lymphatic system.

Lacunae—Small spaces within the bone matrix where osteocytes reside.

Lamellae—Concentric circles of bony matrix within the Haversian systems of bone.

Langerhans cells—Specialized macrophages that reside within the skin.

Large intestine—Tube-like structure, shorter but wider than the small intestine, largely responsible for resorption of water and the formation of semisolid feces; consists of the cecum, ascending colon, transverse colon, descending colon, sigmoid colon, and rectum.

Larynx—Pathway for air between the pharynx and the trachea. The epiglottis closes to prevent food from entering the larynx.

Latent period—The short interval between the application of a stimulus to a muscle and the contraction of the muscle.

Leakage—Flow of genes between closely related species.

Leukocytes—White blood cells; can be subdivided into granulocytes and agranulocytes.

Ligament—Connective tissue that joins two bones.

Light chain—One of two types of chains, made of peptides, that create antibodies; each antibody consists of two heavy chains and two light chains.

Linkage—Tendency for certain alleles to be inherited together due to proximity on the same chromosome.

Lipase—Enzyme that specifically cleaves the bonds in lipids.

Locus—In genetics, an area or region of a chromosome.

Loop of Henle—The U-shaped section of a mammalian nephron.

Lower esophageal sphincter—Ring-shaped muscular structure that separates the esophagus from the stomach; also known as the cardiac sphincter.

Lumen—The space within a tube or a sac.

Lymph—Clear fluid derived from blood plasma and transported through lymph vessels to the lymphatic ducts, which empty into the circulatory system.

Lymph node—Small, bean-shaped structure that provides a location for antigen presentation and mounting of an attack by the adaptive immune system.

Lymphocyte—A type of white blood cell involved in an organism's specific

immune response.

Lysogenic cycle—Bacteriophage infection involving the integration of viral DNA into the bacterial genome without disrupting or destroying the host. The virus may subsequently reemerge and enter a lytic cycle.

Lysosome—A membrane-bound organelle that stores hydrolytic enzymes.

Lytic cycle—Bacteriophage infection involving the destruction (lysis) of the host bacterium.

Macrophage—A phagocytic white blood cell.

Map unit—A unit used to denote a 1 percent recombination frequency between two genes when creating a genetic map; corresponds to one centimorgan.

Mast cell—A granulocyte that releases histamine and causes inflammation.

Mastication—Breaking up of large food particles using the teeth, tongue, and lips; chewing.

Mechanical digestion—Physical breakdown of large food particles into smaller food particles.

Medulla—The internal section of an organ (such as the adrenal glands and the kidney); may generally refer to the medulla oblongata of the mammalian hindbrain.

Medulla oblongata—The part of the brainstem closest to the spinal cord. It

controls vital functions, such as breathing and heartbeat.

Megakaryocyte—Precursor cell that gives off platelets.

Meiosis—A process of cell division in which two successive nuclear divisions produce up to four haploid gametes from one diploid germ cell.

Melanin—Skin pigment produced by melanocytes that protects the skin from UV radiation and provides color to skin.

Melanocytes—Melanin-producing cells of the skin.

Memory cell—Lymphocyte of B- or T-cell lineage that remains after an infection is gone in order to recognize the previous invader and rapidly induce a humoral immune response.

Menstruation—The shedding of the uterine lining that occurs every four weeks in a nonpregnant, human female of reproductive age.

Mesoderm—The middle embryonic germ layer; it gives rise to the muscular, skeletal, urogenital, and circulatory systems.

Metabolism—The sum of all biochemical reactions that occur in an organism.

Metaphase—The stage of mitosis or meiosis during which single chromosomes or tetrads line up on the central axis of the dividing cell and become attached to spindle fibers.

Microfilaments—Small polymerized rods of actin that participate in muscle contraction, movement of material within the cellular membrane, and

amoeboid movement.

Microglia—Phagocytic white blood cells that reside in the central nervous system.

Microtubule—A small, hollow tube composed of two types of protein subunits; serves numerous functions in the cell, such as comprising the internal structures of cilia and flagella and allowing vesicle movement in the cell.

Missense mutation—Type of mutation that results in the substitution of one amino acid for another.

Mitochondria—Membrane-bound cellular organelles in which the reactions of aerobic respiration and ATP synthesis occur.

Mitosis—Cellular division that results in the formation of two daughter cells that are genetically identical to each other and to the parent cell.

Mitral valve—The atrioventricular valve separating the left atrium from the left ventricle.

Mixed nerve—Nerve carrying both afferent (sensory) and efferent (motor) fibers.

Monocyte—A white blood cell that transforms into a macrophage or dendritic cell once it enters tissues.

Monohybrid cross—A cross between two members of a species that seeks to study only one trait.

Monosaccharide—A sugar consisting of one monomer (glucose, fructose, or galactose).

Morphogen—Molecule that causes nearby cells to proceed in a specific developmental pathway during embryonic development.

Morula—The solid ball of cells that results from the early stages of cleavage in an embryo.

Mucosa—The type of epithelial tissue that lines moist body cavities; a mucous membrane.

Mucous cells—Type of epithelial cell that secretes mucus.

Multipotent—Stem cell that is able to differentiate into various cells within a particular lineage.

Mutagen—An agent, either chemical or physical, that can cause mutations.

Mutation—A change in DNA sequence.

Myelin—The white, lipid-containing material surrounding the axons of many neurons in the central and peripheral nervous systems.

Myogenic activity—Ability of a muscle cell to contract without input from the nervous system; found in smooth and cardiac muscle types.

Myoglobin—Heme-containing protein that binds molecular oxygen in muscle cells.

Myosin—A protein found in muscle cells that functions in muscle contraction; myosin fibers are also called thick filaments.

Natural selection—An ongoing evolutionary process resulting in changes in gene frequencies, leading to the differential development of phenotypes in a population.

Negative sense—Describes the genome of an RNA virus that contains an RNA sequence that is complementary to the actual transcript for viral protein synthesis.

Nephron—The functional unit of the vertebrate kidney.

Nerve—A bundle of neurons.

Nerve impulse—The self-propagating change in electrical potential across the axon membrane.

Nerve terminal—End of the axon from which neurotransmitter molecules are released; also called a synaptic bouton.

Neural crest cells—Cells that originate at the tip of the neural fold and then migrate outward to form the peripheral nervous system, melanocytes, C-cells of the thyroid, and others.

Neural fold—Group of ectodermal cells that slide together to create a fold, which later becomes the neural tube.

Neural tube—Embryonic hollow tube that subsequently gives rise to the

central nervous system.

Neuroglia—Support cells for neurons; responsible for functions such as holding neurons in place, supplying neurons with oxygen and nutrients, insulating neurons from other neurons, destroying pathogens, and removing dead neurons.

Neuron—A cell that conducts electrical impulses; the functional unit of the nervous system.

Neurotransmitter—A chemical agent released into the synaptic cleft by the synaptic bouton of a neuron; binds to receptor sites on postsynaptic neurons or effector membranes to alter activity.

Neutrophil—Type of granulocytic leukocyte that largely participates in the nonspecific immune response against bacteria.

Niche—The specific way of life occupied by a given organism within the environment, including its interactions with other organisms and with the physical environment.

Nodes of Ranvier—Points on a myelinated axon that are not covered by myelin.

Nondisjunction—Failure of homologous chromosomes to separate during meiosis.

Nonsense mutation—A change in nucleotide sequence of DNA that results in a premature stop codon in the mRNA sequence.

Norepinephrine—A hormone synthesized by the adrenal medulla; it stimulates the fight-or-flight response; also, a neurotransmitter in the sympathetic nervous system.

Notochord—A supportive rod running just ventral to the neural tube in vertebrate embryos that induces neurulation.

Nuclear membrane—Double membrane enveloping the nucleus, interrupted periodically by pores; found in eukaryotic cells only; also known as the nuclear envelope.

Nuclear pore—Small hole in the nuclear membrane that allows for two-way exchange of material between the cytoplasm and nucleus.

Nucleoid region—Location in prokaryotic cells where the chromosome is found.

Nucleolus—Dense body visible in a nondividing nucleus; site of ribosomal RNA synthesis.

Nucleus—The eukaryotic membrane-bound organelle that contains the cell's chromosomes; in neuroscience, a collection of cell bodies in the central nervous system.

Oligodendrocyte—Myelin-producing cells in the central nervous system.

Oocyte—An undifferentiated cell that undergoes meiosis to produce an egg cell (ovum).

Oogenesis—Gametogenesis in the ovary leading to the formation of mature

ova.

Osmotic pressure—A “sucking” pressure generated by the presence of solutes drawing in water.

Osteoblast—Bone cell responsible for the generation of new bone due to bone remodeling or storage of minerals within the bone matrix.

Osteoclast—Bone cell responsible for the resorption of bone due to bone remodeling or mobilization of minerals from the bone matrix.

Osteocytes—Mature bone cells housed within the bone matrix.

Ovary—The female egg-producing gonad.

Oviduct—The tube leading from the ovary to the uterus; generally, the site of fertilization; also called the fallopian tube.

Ovulation—The release of the mature ovum from the ovarian follicle.

Ovum—The female gamete; egg cell.

Oxygen debt—The amount of oxygen needed to reconvert lactic acid to pyruvate following strenuous exercise; the difference between the amount of oxygen needed by the tissue and the amount of oxygen available.

Pancreas—A gland that secretes digestive enzymes into the duodenum via a duct and synthesizes and secretes the hormones insulin, glucagon, and somatostatin; located between the stomach and the duodenum.

Papillary layer—Upper layer of the dermis, right below the epidermis, that consists of loose connective tissue.

Paracrine—A form of cell–cell communication in which a cell releases a substance into the extracellular fluid and the substance binds to receptors on nearby cells to cause a change in cellular activities.

Parasympathetic nervous system—The subdivision of the autonomic nervous system involved in rest and homeostasis; it is generally antagonistic to the sympathetic nervous system.

Parathyroid glands—Two pairs of glands located on the thyroid that secrete hormones that regulate calcium and phosphorous metabolism.

Parietal cells—Cells within the stomach that are responsible for the secretion of acid into the lumen of the stomach.

Passive immunity—Immunity conferred by the transfer or injection of previously formed antibodies.

Pathogen—An infectious disease-causing agent; includes bacteria, viruses, fungi, parasites, and prions.

Pattern recognition receptor—Type of receptor on macrophages and dendritic cells that is able to recognize the nature of the invader (bacteria, virus, or fungi) and release the appropriate cytokines to attract the right immune cells to the area.

Penetrance—Percent of individuals with a particular genotype that actually express the associated phenotype.

Peptidase—Enzyme that cleaves peptide bonds.

Periosteum—Fibrous sheath surrounding long bones.

Peripheral nervous system—Includes all neurons outside the central nervous system, including sensory and motor neurons; it is subdivided into the somatic and autonomic nervous systems.

Peristalsis—Rhythmic waves of muscular contraction that move a substance through a tube (most commonly, food through the digestive tract).

Peroxisome—Organelle that contains hydrogen peroxide and participates in the breakdown of very long chain fatty acids.

Pharynx—Pathway for food from the mouth to the esophagus, and for air from the nose and mouth to the larynx.

Phenotype—The physical manifestation of an organism's genotype.

Pineal gland—Structure within the brain that secretes melatonin, a hormone that aids in the regulation of sleep–wake cycles.

Pituitary—The bilobed endocrine gland that lies just below the hypothalamus; because many of its hormones regulate other endocrine glands, it is known as the “master gland.”

Placenta—The structure formed by the wall of the uterus and the chorion of the embryo; contains a network of capillaries through which exchange between maternal and fetal circulation occurs.

Plasma—The fluid component of blood containing dissolved solutes, minus the cells.

Plasma cells—Derived from B-lymphocytes; have the ability to produce and secrete antibodies.

Plasmid—Small circular ring of extrachromosomal DNA found in bacteria.

Platelets—Small, enucleated, disc-shaped shards of blood cells that play an important role in clotting.

Pleura—Connective tissue that surrounds each lung and aids in providing attachment of the lungs to the chest wall; the parietal pleura lies along the chest wall, while the visceral pleura is adherent to the lungs.

Pluripotent—Stem cell that has undergone gastrulation and is able to differentiate into any cell type within the same primary germ layer.

Polar body—A small, nonfunctional haploid cell created during oogenesis.

Population—A group of organisms of the same species living together in a given location.

Portal system—A circuit of blood in which there are two capillary beds in tandem connected by an artery or vein; examples include the hypophyseal, hepatic, and renal portal systems.

Positive sense—Describes the genome of an RNA virus containing RNA that serves directly as the transcript for viral protein production.

Potency—Term used to describe the ability or inability of a stem cell to differentiate into different cell types.

Primary response—Humoral immune response against an invader during the first encounter; takes seven to ten days to become effective.

Prion—Infectious protein that causes disease by causing changes in the three-dimensional structure of other proteins from α -helices to β -pleated sheets.

Prokaryote—Cell lacking a nuclear membrane and membrane-bound organelles, such as a bacterium.

Prophase—The stage of mitosis or meiosis during which the DNA strands condense to form visible chromosomes; during prophase I of meiosis, homologous chromosomes align.

Prostate—A gland in the mammalian male that secretes alkaline seminal fluid.

Pulmonary valve—One of the semilunar valves, separating the right ventricle from the pulmonary arteries.

Purkinje fibers—The terminal fibers of the heart's conduction system; located in the walls of ventricles.

Pyloric sphincter—The valve that regulates the flow of chyme from the stomach into the small intestine.

Recessive—An allele that requires two copies to be expressed.

Recombination—New gene combinations achieved by sexual reproduction or crossing over in eukaryotes and by transformation, transduction, or conjugation in prokaryotes.

Recombination frequency—Measurement of how often genes recombine in different combinations; genes that are closer together have lower recombination frequencies.

Rectum—Terminal portion of the large intestine, where feces is stored until defecation.

Reflex—An involuntary nervous pathway consisting of sensory neurons, interneurons, motor neurons, and effectors; it occurs in response to a specific stimulus.

Refractory period—The period of time following an action potential during which the neuron is incapable of depolarization.

Releasing hormones—Proteins synthesized and secreted by the hypothalamus that stimulate the pituitary to synthesize and release its hormones. Also known as tropic hormones.

Renal pelvis—The widest part of the ureter, located within the kidney; location into which all collecting ducts eventually empty.

Renin-angiotensin-aldosterone system—Hormonal pathway that, among other functions, raises blood pressure.

Repolarization—Restoration of the resting membrane potential in neurons

from being depolarized by both active and passive processes.

Respiration—In biochemistry, the series of oxygen-requiring biochemical reactions that lead to ATP synthesis; in physiology, the inhalation and exhalation of gases and their exchange in the lungs.

Responder—Embryonic cell that is undergoing induction.

Resting potential—The electrical potential of a cell at rest, approximately -70 mV in most excitable cells.

Restriction point—A point in the cell cycle that prevents the cell from entering the next portion of the cell cycle unless certain criteria are met.

Reticular layer—Lower layer of the dermis, consisting of dense connective tissue.

Retrovirus—An RNA virus that contains the enzyme reverse transcriptase, which transcribes RNA into DNA.

Rh factor—An antigen on a red blood cell, the presence or absence of which is indicated by + or –, respectively, in blood type notation; may also be called the D allele.

Ribosome—Organelle composed of RNA and protein; it translates mRNA during protein synthesis.

Rough endoplasmic reticulum—Portion of the endoplasmic reticulum that appears rough microscopically due to the presence of ribosomes attached to

the outer surface; site of protein synthesis for proteins destined to be membrane-bound or secreted.

Saltatory conduction—Process by which an electrical signal jumps across the nodes of Ranvier to travel down the axon.

Sarcolemma—Muscle cell membrane capable of propagating action potentials.

Sarcomere—The functional contractile unit of striated muscle.

Sarcoplasmic reticulum—The endoplasmic reticulum of a muscle cell; it envelops myofibrils.

Schwann cell—Myelin-producing cell in the peripheral nervous system.

Second messenger—Substance that is mobilized within a cell after the binding of a hormone to its receptor.

Secondary response—Humoral immune response against a previously encountered invader; results in activation of memory cells and an immediate response.

Semen—Fluid released during ejaculation consisting of sperm cells suspended in seminal fluid.

Seminal vesicle—A gland found in mammalian males that produces seminal fluid.

Sex factor—Plasmid containing genetic material for the formation of a sex

pili, required for conjugation.

Sex pilus—Appendage extending from the donor male (+) bacterial cell to the recipient female (–), allowing for the formation of the cytoplasmic bridge and transfer of genetic material.

Sex-linked gene—A gene located only on a sex chromosome (almost always the X chromosome); such genes exhibit different inheritance patterns in males and females.

Sexual reproduction—Any reproductive process that involves the fusion of gametes, resulting in the passage of combined genetic information to offspring.

Silent mutation—Change of one nucleotide for another that does not result in a change in the protein due to the degenerative nature of the genetic code (multiple codons code for the same amino acid).

Small intestine—Long tube-like structure; longer, but narrower than the large intestine, largely responsible for chemical digestion of foodstuffs and absorption of nutrients; consists of the duodenum, jejunum, and ileum.

Smooth endoplasmic reticulum—Portion of the endoplasmic reticulum that lacks ribosomes on its surface; location of lipid synthesis and detoxification of drugs and poisons.

Somatic cells—All cells in the body except germ cells and gametes.

Somatic nervous system—Subdivision of the peripheral nervous system that governs all voluntary actions.

Species—The largest group of organisms capable of mating to produce viable, fertile offspring.

Specific immune response—An organism's targeted fight against a specific pathogen using both antibodies and cytotoxic immunity.

Sperm—The mature male gamete or sex cell.

Spermatids—Immature haploid sperm cells.

Spermatogenesis—Gametogenesis in the testes leading to sperm formation.

Spermatogonia—Diploid stem cells in males that eventually give rise to sperm cells.

Spermatozoa—Mature haploid sperm cells.

Sphincter—A ring-shaped muscle that closes and opens a tube (such as the pyloric sphincter).

Sphygmomanometer—Device used to measure blood pressure, consisting of an inflatable cuff and a gauge that measures pressure.

Spindle—A structure within dividing cells composed of microtubules; it is involved in the separation of chromosomes during mitosis and meiosis.

Spirilli—Spiral-shaped bacteria.

Spleen—Highly vascular organ in the left upper quadrant of the abdomen;

serves as a location for disposal of aged red blood cells and the presentation of antigens to B-cells.

Stabilizing selection—Selective pressure resulting in the elimination of extremes.

Starling forces—A sum of the forces generated by hydrostatic and osmotic pressures; results in a greater attraction of fluid to one side of a membrane.

Stroke volume—Amount of blood ejected from a ventricle with each heartbeat.

Summation—Process that occurs when the postsynaptic neuron or target organ requires stimulation from multiple presynaptic neurons in order to respond to the stimulus; may be spatial or temporal.

Suppressor T-cells—Also known as regulatory T-cells (T_{reg}), these T-cells limit the immune response to prevent detrimental immune reactions, such as autoimmunity.

Surfactant—A detergent that lowers surface tension and prevents collapse of the alveoli.

Symbiote—Organism living closely with a host and engaging in a mutually beneficial relationship.

Sympathetic nervous system—The subdivision of the autonomic nervous system that produces the “fight-or-flight” response.

Synapse—The junction between two neurons into which neurotransmitters are released.

Synapsis—The pairing of homologous chromosomes during prophase I of meiosis.

Synovial capsule—Fluid-filled space between bones in a joint; enclosed by fibrous tissue; synovial fluid lubricates the joint.

Systole—The period of the cardiac cycle during which the ventricles contract and pump blood into the aorta and pulmonary arteries.

T-cells—Type of leukocyte that matures in the thymus and participates in adaptive immunity.

Telophase—The final stage of mitosis or meiosis during which the chromosomes uncoil, nuclear membranes reform, and cytokinesis occurs.

Tendon—A fibrous connective tissue that connects a bone to a muscle.

Test cross—A cross between an organism showing a dominant trait and an organism showing a recessive trait to determine whether the former organism is homozygous or heterozygous for that trait.

Testis—The sperm-producing organ; also secretes testosterone.

Tetanus—Sustained muscle contraction that results from continuous stimulation.

Tetrad—A pair of homologous chromosomes synapsing during prophase I of

meiosis. Each chromosome consists of two sister chromatids; thus, each tetrad consists of four chromatids.

Thermoregulation—Process by which an organism regulates its internal temperature by using the respiratory, integumentary, and circulatory systems.

Thoracic duct—The main lymphatic vessel that empties lymph into the bloodstream.

Threshold—The lowest magnitude of stimulus strength that will induce a response.

Thrombin—An enzyme that participates in blood clotting; it converts fibrinogen into fibrin.

Thymus—A ductless gland in the upper chest region of vertebrates; it functions in the development of the immune system.

Thyroid—A vertebrate endocrine gland located in the neck that synthesizes triiodothyronine, thyroxine, and calcitonin.

Tissue—A mass of similar cells and support structures organized into a functional unit.

Tonus—A continuous state of low-level muscle contraction.

Totipotent—Type of stem-cell potency describing cells that are able to differentiate into all cell types within an organism.

Trachea—The tube that connects the pharynx to the bronchi.

Transduction—The transposition of genetic material from one organism to another by a virus.

Transformation—Uptake and incorporation of DNA from the environment by a recipient bacterial cell.

Transposon—Genetic element capable of inserting and removing itself from the genome.

Tricuspid valve—One of the atrioventricular valves, separating the right atrium from the right ventricle.

Trophoblast—Embryonic cells that line the blastocoel and give rise to the chorion and the placenta.

Tropic hormone—Hormone that is secreted and travels to a target cell or organ, where it triggers release of another hormone, which causes changes in the physiological activity of target cells.

Tubulin—Protein constituent of microtubules.

Universal donor—A person (O⁻ blood) whose blood is able to be given to all types without inducing an immune response.

Universal recipient—A person (AB⁺ blood) who is able to receive all blood types without undergoing an immune response.

Urea—A nitrogenous waste product produced in the liver from ammonia.

Ureter—The tube that carries urine from the kidneys to the bladder.

Urethra—The tube that carries urine from the bladder to the exterior.

Urine—Liquid waste resulting from the filtration, reabsorption, and secretion of filtrate in the nephron.

Uterus—Organ in the mammalian female reproductive system that is the site of embryonic development.

Vaccine—A solution of fractionated, dead, or attenuated live pathogenic material that is introduced into an individual for the purpose of stimulating a primary immune response or “boosting” a previously produced anamnestic state.

Vagina—Passageway through which childbirth occurs; location into which sperm is deposited during sexual intercourse.

Vagus nerve—The tenth cranial nerve; it innervates the palate, pharynx, larynx, heart, lungs, and abdominal viscera; responsible for maintaining homeostatic activity through the parasympathetic response.

Vas deferens—The tube carrying sperm from the testis to the urethra in mammalian males.

Vasa recta—Second capillary bed within the kidney that removes substances from the interstitium of the kidney to be returned to the systemic circulation.

Vein—Thin-walled blood vessel that carries blood toward the heart.

Venae cavae—Two large veins (superior and inferior) that return deoxygenated blood from the periphery to the right atrium of the heart.

Ventilation center—Groups of neurons in the medulla oblongata that regulate respiration.

Ventricles—The chambers of the heart that pump blood into pulmonary and systemic circulation.

Venule—Small venous structure that links the capillaries to the veins.

Vestigial—Referring to an organ or limb that has no apparent function now, but was functional at some time in the organism's evolutionary past.

Villus—A small projection from the wall of the small intestine that increases the surface area for digestion and absorption.

Viroid—A small plant pathogen consisting of a very short, circular, single strand of RNA.

Virus—A tiny, organism-like particle composed of protein-encased nucleic acid; viruses are obligate parasites.

Vitamin—An organic nutrient that an organism cannot produce itself and that is required by the organism in small amounts to aid in proper metabolic functioning; vitamins often function as cofactors for enzymes.

White matter—The portion of the central nervous system consisting primarily of myelinated axons.

Zona pellucida—One of two layers of cells surrounding an oocyte.

Zygote—The diploid ($2n$) cell that results from the fusion of two haploid (n) gametes.

Zymogen—An inactive enzyme precursor that is converted into an active enzyme.

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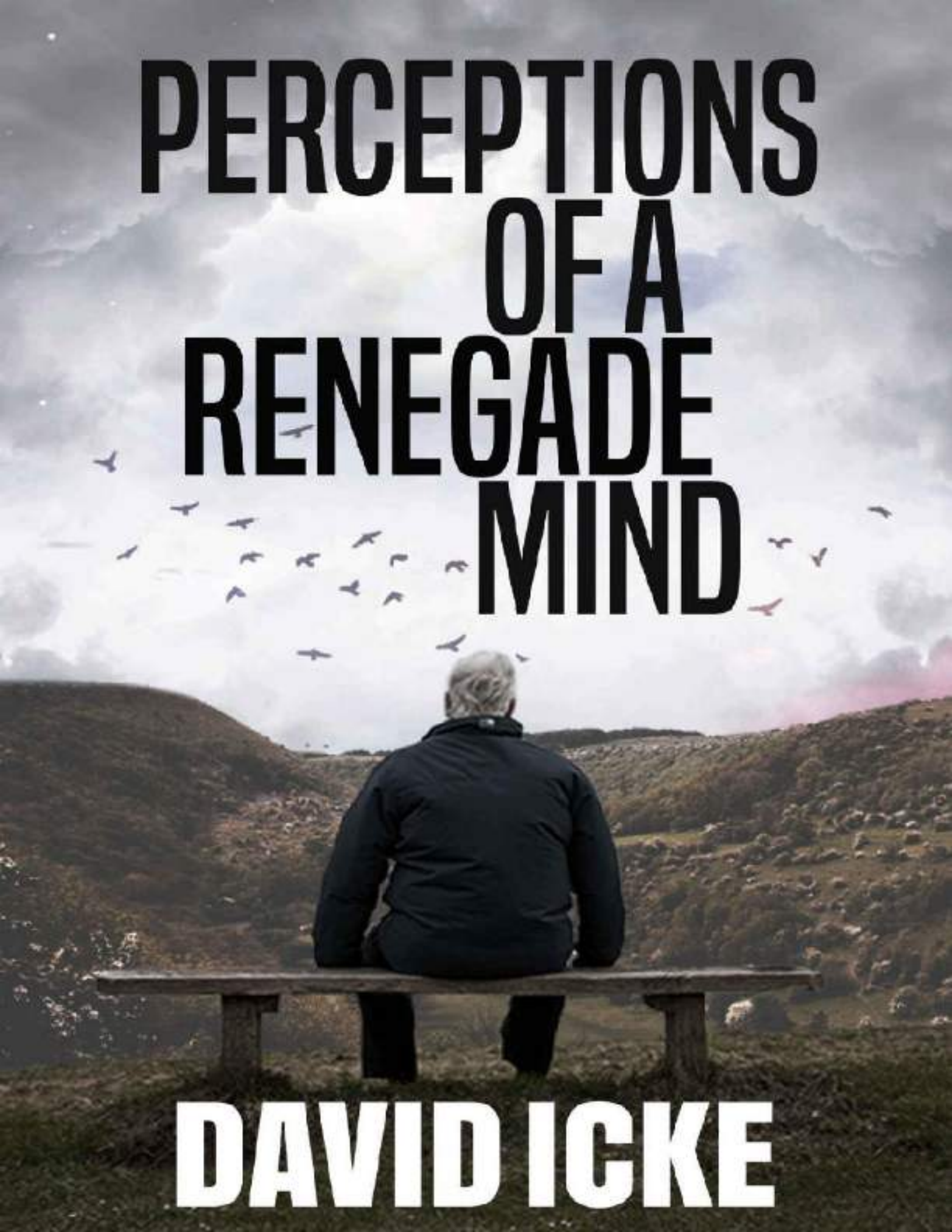
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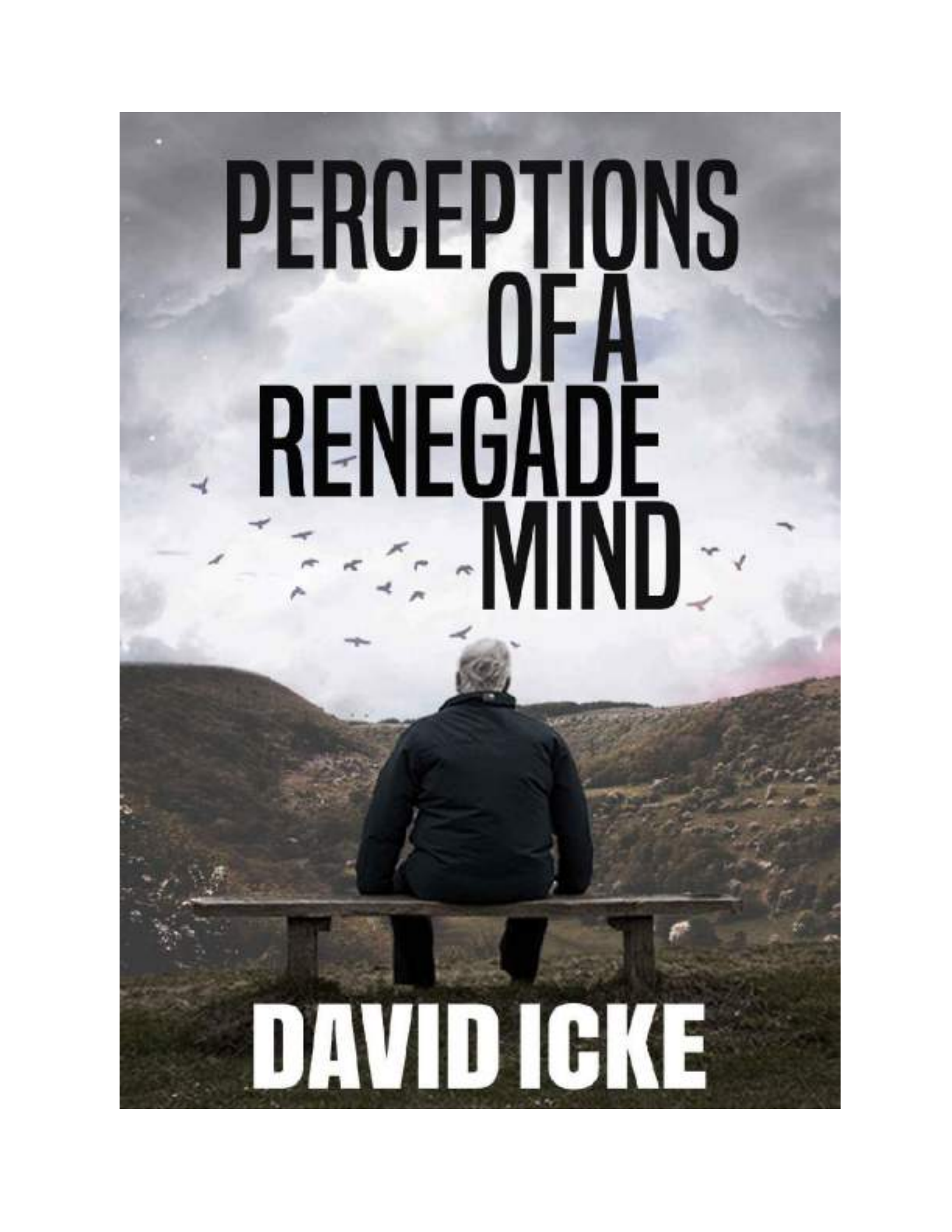


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A person with grey hair, wearing a dark jacket, is seen from behind, sitting on a wooden bench. They are looking out over a landscape of rolling hills covered in low-lying vegetation. The sky is filled with large, white, fluffy clouds, and a large flock of birds is flying in the air. The overall mood is contemplative and serene.

PERCEPTIONS OF A RENEGADE MIND

DAVID ICKE



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Renegade:

Adjective

‘Having rejected tradition: Unconventional.’

Merriam-Webster Dictionary

Acquiescence to tyranny is the death of the spirit

You may be 38 years old, as I happen to be. And one day, some great opportunity stands before you and calls you to stand up for some great principle, some great issue, some great cause. And you refuse to do it because you are afraid ... You refuse to do it because you want to live longer ... You're afraid that you will lose your job, or you are afraid that you will be criticised or that you will lose your popularity, or you're afraid that somebody will stab you, or shoot at you or bomb your house; so you refuse to take the stand.

Well, you may go on and live until you are 90, but you're just as dead at 38 as you would be at 90. And the cessation of breathing in your life is but the belated announcement of an earlier death of the spirit.

Martin Luther King

**How the few control the many and always have – the many do
whatever they're told**

'Forward, the Light Brigade!'
Was there a man dismayed?
Not though the soldier knew
Someone had blundered.
Theirs not to make reply,
Theirs not to reason why,
Theirs but to do and die.
Into the valley of Death
Rode the six hundred.

Cannon to right of them,
Cannon to left of them,
Cannon in front of them
Volleyed and thundered;
Stormed at with shot and shell,
Boldly they rode and well,
Into the jaws of Death,
Into the mouth of hell
Rode the six hundred

Alfred Lord Tennyson (1809-1892)

The mist is lifting slowly
I can see the way ahead
And I've left behind the empty streets
That once inspired my life
And the strength of the emotion
Is like thunder in the air
'Cos the promise that we made each other
Haunts me to the end

The secret of your beauty
And the mystery of your soul
I've been searching for in everyone I meet
And the times I've been mistaken
It's impossible to say
And the grass is growing
Underneath our feet

The words that I remember
From my childhood still are true
That there's none so blind
As those who will not see
And to those who lack the courage
And say it's dangerous to try
Well they just don't know
That love eternal will not be denied

I know you're out there somewhere
Somewhere, somewhere
I know you're out there somewhere

Somewhere you can hear my voice
I know I'll find you somehow
Somehow, somehow
I know I'll find you somehow
And somehow I'll return again to you

The Moody Blues

Are you a gutless wonder - or a Renegade Mind?

Monuments put from pen to paper,
Turns me into a gutless wonder,
And if you tolerate this,
Then your children will be next.
Gravity keeps my head down,
Or is it maybe shame ...

Manic Street Preachers

Rise like lions after slumber
In unvanquishable number.
Shake your chains to earth like dew
Which in sleep have fallen on you.
Ye are many – they are few.

Percy Shelley

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CHAPTER ONE

I'm thinking' – Oh, but *are* you?

Think for yourself and let others enjoy the privilege of doing so too
Voltaire

French-born philosopher, mathematician and scientist René Descartes became famous for his statement in Latin in the 17th century which translates into English as: 'I think, therefore I am.'

On the face of it that is true. Thought reflects perception and perception leads to both behaviour and self-identity. In that sense 'we' are what we think. But who or what is doing the thinking and is thinking the only route to perception? Clearly, as we shall see, 'we' are not always the source of 'our' perception, indeed with regard to humanity as a whole this is rarely the case; and thinking is far from the only means of perception. Thought is the village idiot compared with other expressions of consciousness that we all have the potential to access and tap into. This has to be true when we *are* those other expressions of consciousness which are infinite in nature. We have forgotten this, or, more to the point, been manipulated to forget.

These are not just the esoteric musings of the navel. The whole foundation of human control and oppression is control of perception. Once perception is hijacked then so is behaviour which is dictated by perception. Collective perception becomes collective behaviour and collective behaviour is what we call human society. Perception is all and those behind human control know that which is

why perception is the target 24/7 of the psychopathic manipulators that I call the Global Cult. They know that if they dictate perception they will dictate behaviour and collectively dictate the nature of human society. They are further aware that perception is formed from information received and if they control the circulation of information they will to a vast extent direct human behaviour. Censorship of information and opinion has become globally Nazi-like in recent years and never more blatantly than since the illusory 'virus pandemic' was triggered out of China in 2019 and across the world in 2020. Why have billions submitted to house arrest and accepted fascistic societies in a way they would have never believed possible? Those controlling the information spewing from government, mainstream media and Silicon Valley (all controlled by the same Global Cult networks) told them they were in danger from a 'deadly virus' and only by submitting to house arrest and conceding their most basic of freedoms could they and their families be protected. This monumental and provable lie became the *perception* of the billions and therefore the *behaviour* of the billions. In those few words you have the whole structure and modus operandi of human control. Fear is a perception – False Emotion Appearing Real – and fear is the currency of control. In short ... get them by the balls (or give them the impression that you have) and their hearts and minds will follow. Nothing grips the dangly bits and freezes the rear-end more comprehensively than fear.

World number 1

There are two 'worlds' in what appears to be one 'world' and the prime difference between them is knowledge. First we have the mass of human society in which the population is maintained in coldly-calculated ignorance through control of information and the 'education' (indoctrination) system. That's all you really need to control to enslave billions in a perceptual delusion in which what are perceived to be *their* thoughts and opinions are ever-repeated mantras that the system has been downloading all their lives through 'education', media, science, medicine, politics and academia

in which the personnel and advocates are themselves overwhelmingly the perceptual products of the same repetition. Teachers and academics in general are processed by the same programming machine as everyone else, but unlike the great majority they never leave the 'education' program. It gripped them as students and continues to grip them as programmers of subsequent generations of students. The programmed become the programmers – the programmed programmers. The same can largely be said for scientists, doctors and politicians and not least because as the American writer Upton Sinclair said: 'It is difficult to get a man to understand something when his salary depends upon his not understanding it.' If your career and income depend on thinking the way the system demands then you will – bar a few free-minded exceptions – concede your mind to the Perceptual Mainframe that I call the Postage Stamp Consensus. This is a tiny band of perceived knowledge and possibility 'taught' (downloaded) in the schools and universities, pounded out by the mainstream media and on which all government policy is founded. Try thinking, and especially speaking and acting, outside of the 'box' of consensus and see what that does for your career in the Mainstream Everything which bullies, harasses, intimidates and ridicules the population into compliance. Here we have the simple structure which enslaves most of humanity in a perceptual prison cell for an entire lifetime and I'll go deeper into this process shortly. Most of what humanity is taught as fact is nothing more than programmed belief. American science fiction author Frank Herbert was right when he said: 'Belief can be manipulated. Only knowledge is dangerous.' In the 'Covid' age belief is promoted and knowledge is censored. It was always so, but never to the extreme of today.

World number 2

A 'number 2' is slang for 'doing a poo' and how appropriate that is when this other 'world' is doing just that on humanity every minute of every day. World number 2 is a global network of secret societies and semi-secret groups dictating the direction of society via

governments, corporations and authorities of every kind. I have spent more than 30 years uncovering and exposing this network that I call the Global Cult and knowing its agenda is what has made my books so accurate in predicting current and past events. Secret societies are secret for a reason. They want to keep their hoarded knowledge to themselves and their chosen initiates and to hide it from the population which they seek through ignorance to control and subdue. The whole foundation of the division between World 1 and World 2 is *knowledge*. What number 1 knows number 2 must not. Knowledge they have worked so hard to keep secret includes (a) the agenda to enslave humanity in a centrally-controlled global dictatorship, and (b) the nature of reality and life itself. The latter (b) must be suppressed to allow the former (a) to prevail as I shall be explaining. The way the Cult manipulates and interacts with the population can be likened to a spider's web. The 'spider' sits at the centre in the shadows and imposes its will through the web with each strand represented in World number 2 by a secret society, satanic or semi-secret group, and in World number 1 – the world of the seen – by governments, agencies of government, law enforcement, corporations, the banking system, media conglomerates and Silicon Valley (Fig 1 overleaf). The spider and the web connect and coordinate all these organisations to pursue the same global outcome while the population sees them as individual entities working randomly and independently. At the level of the web governments *are* the banking system *are* the corporations *are* the media *are* Silicon Valley *are* the World Health Organization working from their inner cores as one unit. Apparently unconnected countries, corporations, institutions, organisations and people are on the *same team* pursuing the same global outcome. Strands in the web immediately around the spider are the most secretive and exclusive secret societies and their membership is emphatically restricted to the Cult inner-circle emerging through the generations from particular bloodlines for reasons I will come to. At the core of the core you would get them in a single room. That's how many people are dictating the direction of human society and its transformation

through the 'Covid' hoax and other means. As the web expands out from the spider we meet the secret societies that many people will be aware of – the Freemasons, Knights Templar, Knights of Malta, Opus Dei, the inner sanctum of the Jesuit Order, and such like. Note how many are connected to the Church of Rome and there is a reason for that. The Roman Church was established as a revamp, a rebranding, of the relocated 'Church' of Babylon and the Cult imposing global tyranny today can be tracked back to Babylon and Sumer in what is now Iraq.



Figure 1: The global web through which the few control the many. (Image Neil Hague.)

Inner levels of the web operate in the unseen away from the public eye and then we have what I call the cusp organisations located at the point where the hidden meets the seen. They include a series of satellite organisations answering to a secret society founded in London in the late 19th century called the Round Table and among them are the Royal Institute of International Affairs (UK, founded in 1920); Council on Foreign Relations (US, 1921); Bilderberg Group (worldwide, 1954); Trilateral Commission (US/worldwide, 1972); and the Club of Rome (worldwide, 1968) which was created to exploit environmental concerns to justify the centralisation of global power to 'save the planet'. The Club of Rome instigated with others the human-caused climate change hoax which has led to all the 'green

new deals' demanding that very centralisation of control. Cusp organisations, which include endless 'think tanks' all over the world, are designed to coordinate a single global policy between political and business leaders, intelligence personnel, media organisations and anyone who can influence the direction of policy in their own sphere of operation. Major players and regular attenders will know what is happening – or some of it – while others come and go and are kept overwhelmingly in the dark about the big picture. I refer to these cusp groupings as semi-secret in that they can be publicly identified, but what goes on at the inner-core is kept very much 'in house' even from most of their members and participants through a fiercely-imposed system of compartmentalisation. Only let them know what they need to know to serve your interests and no more. The structure of secret societies serves as a perfect example of this principle. Most Freemasons never get higher than the bottom three levels of 'degree' (degree of knowledge) when there are 33 official degrees of the Scottish Rite. Initiates only qualify for the next higher 'compartment' or degree if those at that level choose to allow them. Knowledge can be carefully assigned only to those considered 'safe'. I went to my local Freemason's lodge a few years ago when they were having an 'open day' to show how cuddly they were and when I chatted to some of them I was astonished at how little the rank and file knew even about the most ubiquitous symbols they use. The mushroom technique – keep them in the dark and feed them bullshit – applies to most people in the web as well as the population as a whole. Sub-divisions of the web mirror in theme and structure transnational corporations which have a headquarters somewhere in the world dictating to all their subsidiaries in different countries. Subsidiaries operate in their methodology and branding to the same centrally-dictated plan and policy in pursuit of particular ends. The Cult web functions in the same way. Each country has its own web as a subsidiary of the global one. They consist of networks of secret societies, semi-secret groups and bloodline families and their job is to impose the will of the spider and the global web in their particular country. Subsidiary networks control and manipulate the national political system, finance, corporations, media, medicine, etc. to

ensure that they follow the globally-dictated Cult agenda. These networks were the means through which the 'Covid' hoax could be played out with almost every country responding in the same way.

The 'Yessir' pyramid

Compartmentalisation is the key to understanding how a tiny few can dictate the lives of billions when combined with a top-down sequence of imposition and acquiescence. The inner core of the Cult sits at the peak of the pyramidal hierarchy of human society (Fig 2 overleaf). It imposes its will – its agenda for the world – on the level immediately below which acquiesces to that imposition. This level then imposes the Cult will on the level below them which acquiesces and imposes on the next level. Very quickly we meet levels in the hierarchy that have no idea there even is a Cult, but the sequence of imposition and acquiescence continues down the pyramid in just the same way. 'I don't know why we are doing this but the order came from "on-high" and so we better just do it.' Alfred Lord Tennyson said of the cannon fodder levels in his poem *The Charge of the Light Brigade*: 'Theirs not to reason why; theirs but to do and die.' The next line says that 'into the valley of death rode the six hundred' and they died because they obeyed without question what their perceived 'superiors' told them to do. In the same way the population capitulated to 'Covid'. The whole hierarchical pyramid functions like this to allow the very few to direct the enormous many.

Eventually imposition-acquiescence-imposition-acquiescence comes down to the mass of the population at the foot of the pyramid. If they acquiesce to those levels of the hierarchy imposing on them (governments/law enforcement/doctors/media) a circuit is completed between the population and the handful of super-psychopaths in the Cult inner core at the top of the pyramid.

Without a circuit-breaking refusal to obey, the sequence of imposition and acquiescence allows a staggeringly few people to impose their will upon the entirety of humankind. We are looking at the very sequence that has subjugated billions since the start of 2020. Our freedom has not been taken from us. Humanity has given it

away. Fascists do not impose fascism because there are not enough of them. Fascism is imposed by the population acquiescing to fascism. Put another way allowing their perceptions to be programmed to the extent that leads to the population giving their freedom away by giving their perceptions – their mind – away. If this circuit is not broken by humanity ceasing to cooperate with their own enslavement then nothing can change. For that to happen people have to critically think and see through the lies and window dressing and then summon the backbone to act upon what they see. The Cult spends its days working to stop either happening and its methodology is systematic and highly detailed, but it can be overcome and that is what this book is all about.

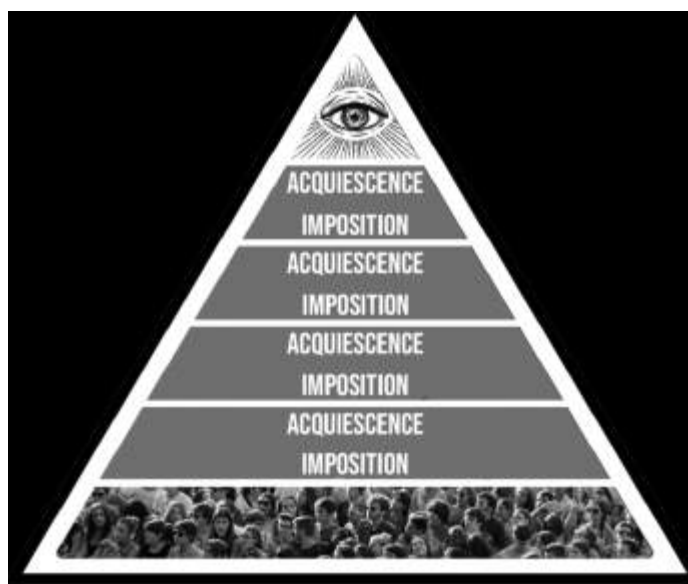


Figure 2: The simple sequence of imposition and compliance that allows a handful of people at the peak of the pyramid to dictate the lives of billions.

The Life Program

Okay, back to world number 1 or the world of the 'masses'. Observe the process of what we call 'life' and it is a perceptual download from cradle to grave. The Cult has created a global structure in which perception can be programmed and the program continually topped-up with what appears to be constant confirmation that the program is indeed true reality. The important word here is 'appears'.

This is the structure, the fly-trap, the Postage Stamp Consensus or Perceptual Mainframe, which represents that incredibly narrow band of perceived possibility delivered by the 'education' system, mainstream media, science and medicine. From the earliest age the download begins with parents who have themselves succumbed to the very programming their children are about to go through. Most parents don't do this out of malevolence and mostly it is quite the opposite. They do what they believe is best for their children and that is what the program has told them is best. Within three or four years comes the major transition from parental programming to full-blown state (Cult) programming in school, college and university where perceptually-programmed teachers and academics pass on their programming to the next generations. Teachers who resist are soon marginalised and their careers ended while children who resist are called a problem child for whom Ritalin may need to be prescribed. A few years after entering the 'world' children are under the control of authority figures representing the state telling them when they have to be there, when they can leave and when they can speak, eat, even go to the toilet. This is calculated preparation for a lifetime of obeying authority in all its forms. Reflex-action fear of authority is instilled by authority from the start. Children soon learn the carrot and stick consequences of obeying or defying authority which is underpinned daily for the rest of their life. Fortunately I daydreamed through this crap and never obeyed authority simply because it told me to. This approach to my alleged 'betters' continues to this day. There can be consequences of pursuing open-minded freedom in a world of closed-minded conformity. I spent a lot of time in school corridors after being ejected from the classroom for not taking some of it seriously and now I spend a lot of time being ejected from Facebook, YouTube and Twitter. But I can tell you that being true to yourself and not compromising your self-respect is far more exhilarating than bowing to authority for authority's sake. You don't have to be a sheep to the shepherd (authority) and the sheep dog (fear of not obeying authority).

The perceptual download continues throughout the formative years in school, college and university while script-reading 'teachers', 'academics' 'scientists', 'doctors' and 'journalists' insist that ongoing generations must be as programmed as they are. Accept the program or you will not pass your 'exams' which confirm your 'degree' of programming. It is tragic to think that many parents pressure their offspring to work hard at school to download the program and qualify for the next stage at college and university. The late, great, American comedian George Carlin said: 'Here's a bumper sticker I'd like to see: We are proud parents of a child who has resisted his teachers' attempts to break his spirit and bend him to the will of his corporate masters.' Well, the best of luck finding many of those, George. Then comes the moment to leave the formal programming years in academia and enter the 'adult' world of work. There you meet others in your chosen or prescribed arena who went through the same Postage Stamp Consensus program before you did. There is therefore overwhelming agreement between almost everyone on the basic foundations of Postage Stamp reality and the rejection, even contempt, of the few who have a mind of their own and are prepared to use it. This has two major effects. Firstly, the consensus confirms to the programmed that their download is really how things are. I mean, everyone knows that, right? Secondly, the arrogance and ignorance of Postage Stamp adherents ensure that anyone questioning the program will have unpleasant consequences for seeking their own truth and not picking their perceptions from the shelf marked: 'Things you must believe without question and if you don't you're a dangerous lunatic conspiracy theorist and a harebrained nutter'.

Every government, agency and corporation is founded on the same Postage Stamp prison cell and you can see why so many people believe the same thing while calling it their own 'opinion'. Fusion of governments and corporations in pursuit of the same agenda was the definition of fascism described by Italian dictator Benito Mussolini. The pressure to conform to perceptual norms downloaded for a lifetime is incessant and infiltrates society right

down to family groups that become censors and condemners of their own 'black sheep' for not, ironically, being sheep. We have seen an explosion of that in the 'Covid' era. Cult-owned global media unleashes its propaganda all day every day in support of the Postage Stamp and targets with abuse and ridicule anyone in the public eye who won't bend their mind to the will of the tyranny. Any response to this is denied (certainly in my case). They don't want to give a platform to expose official lies. Cult-owned-and-created Internet giants like Facebook, Google, YouTube and Twitter delete you for having an unapproved opinion. Facebook boasts that its AI censors delete 97-percent of 'hate speech' before anyone even reports it. Much of that 'hate speech' will simply be an opinion that Facebook and its masters don't want people to see. Such perceptual oppression is widely known as fascism. Even Facebook executive Benny Thomas, a 'CEO Global Planning Lead', said in comments secretly recorded by investigative journalism operation Project Veritas that Facebook is 'too powerful' and should be broken up:

I mean, no king in history has been the ruler of two billion people, but Mark Zuckerberg is ... And he's 36. That's too much for a 36-year-old ... You should not have power over two billion people. I just think that's wrong.

Thomas said Facebook-owned platforms like Instagram, Oculus, and WhatsApp needed to be separate companies. 'It's too much power when they're all one together'. That's the way the Cult likes it, however. We have an executive of a Cult organisation in Benny Thomas that doesn't know there is a Cult such is the compartmentalisation. Thomas said that Facebook and Google 'are no longer companies, they're countries'. Actually they are more powerful than countries on the basis that if you control information you control perception and control human society.

I love my oppressor

Another expression of this psychological trickery is for those who realise they are being pressured into compliance to eventually

convince themselves to believe the official narratives to protect their self-respect from accepting the truth that they have succumbed to meek and subservient compliance. Such people become some of the most vehement defenders of the system. You can see them everywhere screaming abuse at those who prefer to think for themselves and by doing so reminding the compliers of their own capitulation to conformity. 'You are talking dangerous nonsense you Covidiot!!' Are you trying to convince me or yourself? It is a potent form of Stockholm syndrome which is defined as: 'A psychological condition that occurs when a victim of abuse identifies and attaches, or bonds, positively with their abuser.' An example is hostages bonding and even 'falling in love' with their kidnappers. The syndrome has been observed in domestic violence, abused children, concentration camp inmates, prisoners of war and many and various Satanic cults. These are some traits of Stockholm syndrome listed at goodtherapy.org:

- Positive regard towards perpetrators of abuse or captor [see 'Covid'].
- Failure to cooperate with police and other government authorities when it comes to holding perpetrators of abuse or kidnapping accountable [or in the case of 'Covid' cooperating with the police to enforce and defend their captors' demands].
- Little or no effort to escape [see 'Covid'].
- Belief in the goodness of the perpetrators or kidnappers [see 'Covid'].
- Appeasement of captors. This is a manipulative strategy for maintaining one's safety. As victims get rewarded – perhaps with less abuse or even with life itself – their appeasing behaviours are reinforced [see 'Covid'].
- Learned helplessness. This can be akin to 'if you can't beat 'em, join 'em'. As the victims fail to escape the abuse or captivity, they may start giving up and soon realize it's just easier for everyone if they acquiesce all their power to their captors [see 'Covid'].

- Feelings of pity toward the abusers, believing they are actually victims themselves. Because of this, victims may go on a crusade or mission to 'save' [protect] their abuser [see the venom unleashed on those challenging the official 'Covid' narrative].
- Unwillingness to learn to detach from their perpetrators and heal. In essence, victims may tend to be less loyal to themselves than to their abuser [*definitely* see 'Covid'].

Ponder on those traits and compare them with the behaviour of great swathes of the global population who have defended governments and authorities which have spent every minute destroying their lives and livelihoods and those of their children and grandchildren since early 2020 with fascistic lockdowns, house arrest and employment deletion to 'protect' them from a 'deadly virus' that their abusers' perceptually created to bring about this very outcome. We are looking at mass Stockholm syndrome. All those that agree to concede their freedom will believe those perceptions are originating in their own independent 'mind' when in fact by conceding their reality to Stockholm syndrome they have by definition conceded any independence of mind. Listen to the 'opinions' of the acquiescing masses in this 'Covid' era and what gushes forth is the repetition of the official version of everything delivered unprocessed, unfiltered and unquestioned. The whole programming dynamic works this way. I must be free because I'm told that I am and so I think that I am.

You can see what I mean with the chapter theme of 'I'm thinking – Oh, but *are* you?' The great majority are not thinking, let alone for themselves. They are repeating what authority has told them to believe which allows them to be controlled. Weaving through this mentality is the fear that the 'conspiracy theorists' are right and this again explains the often hysterical abuse that ensues when you dare to contest the official narrative of anything. Denial is the mechanism of hiding from yourself what you don't want to be true. Telling people what they want to hear is easy, but it's an infinitely greater challenge to tell them what they would rather not be happening.

One is akin to pushing against an open door while the other is met with vehement resistance no matter what the scale of evidence. I don't want it to be true so I'll convince myself that it's not. Examples are everywhere from the denial that a partner is cheating despite all the signs to the reflex-action rejection of any idea that world events in which country after country act in exactly the same way are centrally coordinated. To accept the latter is to accept that a force of unspeakable evil is working to destroy your life and the lives of your children with nothing too horrific to achieve that end. Who the heck wants that to be true? But if we don't face reality the end is duly achieved and the consequences are far worse and ongoing than breaking through the walls of denial today with the courage to make a stand against tyranny.

Connect the dots – but how?

A crucial aspect of perceptual programming is to portray a world in which everything is random and almost nothing is connected to anything else. Randomness cannot be coordinated by its very nature and once you perceive events as random the idea they could be connected is waved away as the rantings of the tinfoil-hat brigade. You can't plan and coordinate random you idiot! No, you can't, but you can hide the coldly-calculated and long-planned behind the *illusion* of randomness. A foundation manifestation of the Renegade Mind is to scan reality for patterns that connect the apparently random and turn pixels and dots into pictures. This is the way I work and have done so for more than 30 years. You look for similarities in people, modus operandi and desired outcomes and slowly, then ever quicker, the picture forms. For instance: There would seem to be no connection between the 'Covid pandemic' hoax and the human-caused global-warming hoax and yet they are masks (appropriately) on the same face seeking the same outcome. Those pushing the global warming myth through the Club of Rome and other Cult agencies are driving the lies about 'Covid' – Bill Gates is an obvious one, but they are endless. Why would the same people be involved in both when they are clearly not connected? Oh, but they

are. Common themes with personnel are matched by common goals. The 'solutions' to both 'problems' are centralisation of global power to impose the will of the few on the many to 'save' humanity from 'Covid' and save the planet from an 'existential threat' (we need 'zero Covid' and 'zero carbon emissions'). These, in turn, connect with the 'dot' of globalisation which was coined to describe the centralisation of global power in every area of life through incessant political and corporate expansion, trading blocks and superstates like the European Union. If you are the few and you want to control the many you have to centralise power and decision-making. The more you centralise power the more power the few at the centre will have over the many; and the more that power is centralised the more power those at the centre have to centralise even quicker. The momentum of centralisation gets faster and faster which is exactly the process we have witnessed. In this way the hoaxed 'pandemic' and the fakery of human-caused global warming serve the interests of globalisation and the seizure of global power in the hands of the Cult inner-circle which is behind 'Covid', 'climate change' and globalisation. At this point random 'dots' become a clear and obvious picture or pattern.

Klaus Schwab, the classic Bond villain who founded the Cult's Gates-funded World Economic Forum, published a book in 2020, *The Great Reset*, in which he used the 'problem' of 'Covid' to justify a total transformation of human society to 'save' humanity from 'climate change'. Schwab said: 'The pandemic represents a rare but narrow window of opportunity to reflect, reimagine, and reset our world.' What he didn't mention is that the Cult he serves is behind both hoaxes as I show in my book *The Answer*. He and the Cult don't have to reimagine the world. They know precisely what they want and that's why they destroyed human society with 'Covid' to 'build back better' in their grand design. Their job is not to imagine, but to get humanity to imagine and agree with their plans while believing it's all random. It must be pure coincidence that 'The Great Reset' has long been the Cult's code name for the global imposition of fascism and replaced previous code-names of the 'New World

Order' used by Cult frontmen like Father George Bush and the 'New Order of the Ages' which emerged from Freemasonry and much older secret societies. New Order of the Ages appears on the reverse of the Great Seal of the United States as 'Novus ordo seclorum' underneath the Cult symbol used since way back of the pyramid and all seeing-eye (Fig 3). The pyramid is the hierarchy of human control headed by the illuminated eye that symbolises the force behind the Cult which I will expose in later chapters. The term 'Annuit Coeptis' translates as 'He favours our undertaking'. We are told the 'He' is the Christian god, but 'He' is not as I will be explaining.



Figure 3: The all-seeing eye of the Cult 'god' on the Freemason-designed Great Seal of the United States and also on the dollar bill.

Having you on

Two major Cult techniques of perceptual manipulation that relate to all this are what I have called since the 1990s Problem-Reaction-Solution (PRS) and the Totalitarian Tiptoe (TT). They can be uncovered by the inquiring mind with a simple question: Who benefits? The answer usually identifies the perpetrators of a given action or happening through the concept of 'he who most benefits from a crime is the one most likely to have committed it'. The Latin 'Cue bono?' – Who benefits? – is widely attributed to the Roman orator and statesman Marcus Tullius Cicero. No wonder it goes back so far when the concept has been relevant to human behaviour since

history was recorded. Problem-Reaction-Solution is the technique used to manipulate us every day by covertly creating a problem (or the illusion of one) and offering the solution to the problem (or the illusion of one). In the first phase you create the problem and blame someone or something else for why it has happened. This may relate to a financial collapse, terrorist attack, war, global warming or pandemic, anything in fact that will allow you to impose the 'solution' to change society in the way you desire at that time. The 'problem' doesn't have to be real. PRS is manipulation of perception and all you need is the population to believe the problem is real. Human-caused global warming and the 'Covid pandemic' only have to be *perceived* to be real for the population to accept the 'solutions' of authority. I refer to this technique as NO-Problem-Reaction-Solution. Billions did not meekly accept house arrest from early 2020 because there was a real deadly 'Covid pandemic' but because they perceived – believed – that to be the case. The antidote to Problem-Reaction-Solution is to ask who benefits from the proposed solution. Invariably it will be anyone who wants to justify more control through deletion of freedom and centralisation of power and decision-making.

The two world wars were Problem-Reaction-Solutions that transformed and realigned global society. Both were manipulated into being by the Cult as I have detailed in books since the mid-1990s. They dramatically centralised global power, especially World War Two, which led to the United Nations and other global bodies thanks to the overt and covert manipulations of the Rockefeller family and other Cult bloodlines like the Rothschilds. The UN is a stalking horse for full-blown world government that I will come to shortly. The land on which the UN building stands in New York was donated by the Rockefellers and the same Cult family was behind Big Pharma scalpel and drug 'medicine' and the creation of the World Health Organization as part of the UN. They have been stalwarts of the eugenics movement and funded Hitler's race-purity expert' Ernst Rudin. The human-caused global warming hoax has been orchestrated by the Club of Rome through the UN which is

manufacturing both the 'problem' through its Intergovernmental Panel on Climate Change and imposing the 'solution' through its Agenda 21 and Agenda 2030 which demand the total centralisation of global power to 'save the world' from a climate hoax the United Nations is itself perpetrating. What a small world the Cult can be seen to be particularly among the inner circles. The bedfellow of Problem-Reaction-Solution is the Totalitarian Tiptoe which became the Totalitarian Sprint in 2020. The technique is fashioned to hide the carefully-coordinated behind the cover of apparently random events. You start the sequence at 'A' and you know you are heading for 'Z'. You don't want people to know that and each step on the journey is presented as a random happening while all the steps strung together lead in the same direction. The speed may have quickened dramatically in recent times, but you can still see the incremental approach of the Tiptoe in the case of 'Covid' as each new imposition takes us deeper into fascism. Tell people they have to do this or that to get back to 'normal', then this and this and this. With each new demand adding to the ones that went before the population's freedom is deleted until it disappears. The spider wraps its web around the flies more comprehensively with each new diktat. I'll highlight this in more detail when I get to the 'Covid' hoax and how it has been pulled off. Another prime example of the Totalitarian Tiptoe is how the Cult-created European Union went from a 'free-trade zone' to a centralised bureaucratic dictatorship through the Tiptoe of incremental centralisation of power until nations became mere administrative units for Cult-owned dark suits in Brussels.

The antidote to ignorance is knowledge which the Cult seeks vehemently to deny us, but despite the systematic censorship to that end the Renegade Mind can overcome this by vociferously seeking out the facts no matter the impediments put in the way. There is also a method of thinking and perceiving – *knowing* – that doesn't even need names, dates, place-type facts to identify the patterns that reveal the story. I'll get to that in the final chapter. All you need to know about the manipulation of human society and to what end is still out there – *at the time of writing* – in the form of books, videos

and websites for those that really want to breach the walls of programmed perception. To access this knowledge requires the abandonment of the mainstream media as a source of information in the awareness that this is owned and controlled by the Cult and therefore promotes mass perceptions that suit the Cult. Mainstream media lies all day, every day. That is its function and very reason for being. Where it does tell the truth, here and there, is only because the truth and the Cult agenda very occasionally coincide. If you look for fact and insight to the BBC, CNN and virtually all the rest of them you are asking to be conned and perceptually programmed.

Know the outcome and you'll see the journey

Events seem random when you have no idea where the world is being taken. Once you do the random becomes the carefully planned. Know the outcome and you'll see the journey is a phrase I have been using for a long time to give context to daily happenings that appear unconnected. Does a problem, or illusion of a problem, trigger a proposed 'solution' that further drives society in the direction of the outcome? Invariably the answer will be yes and the random – *abracadabra* – becomes the clearly coordinated. So what is this outcome that unlocks the door to a massively expanded understanding of daily events? I will summarise its major aspects – the fine detail is in my other books – and those new to this information will see that the world they thought they were living in is a very different place. The foundation of the Cult agenda is the incessant centralisation of power and all such centralisation is ultimately in pursuit of Cult control on a global level. I have described for a long time the planned world structure of top-down dictatorship as the Hunger Games Society. The term obviously comes from the movie series which portrayed a world in which a few living in military-protected hi-tech luxury were the overlords of a population condemned to abject poverty in isolated 'sectors' that were not allowed to interact. 'Covid' lockdowns and travel bans anyone? The 'Hunger Games' pyramid of structural control has the inner circle of the Cult at the top with pretty much the entire

population at the bottom under their control through dependency for survival on the Cult. The whole structure is planned to be protected and enforced by a military-police state ([Fig 4](#)).

Here you have the reason for the global lockdowns of the fake pandemic to coldly destroy independent incomes and livelihoods and make everyone dependent on the 'state' (the Cult that controls the 'states'). I have warned in my books for many years about the plan to introduce a 'guaranteed income' – a barely survivable pittance – designed to impose dependency when employment was destroyed by AI technology and now even more comprehensively at great speed by the 'Covid' scam. Once the pandemic was played and lockdown consequences began to delete independent income the authorities began to talk right on cue about the need for a guaranteed income and a 'Great Reset'. Guaranteed income will be presented as benevolent governments seeking to help a desperate people – desperate as a direct result of actions of the same governments. The truth is that such payments are a trap. You will only get them if you do exactly what the authorities demand including mass vaccination (genetic manipulation). We have seen this theme already in Australia where those dependent on government benefits have them reduced if parents don't agree to have their children vaccinated according to an insane health-destroying government-dictated schedule. Calculated economic collapse applies to governments as well as people. The Cult wants rid of countries through the creation of a world state with countries broken up into regions ruled by a world government and super states like the European Union. Countries must be bankrupted, too, to this end and it's being achieved by the trillions in 'rescue packages' and furlough payments, trillions in lost taxation, and money-no-object spending on 'Covid' including constant all-medium advertising (programming) which has made the media dependent on government for much of its income. The day of reckoning is coming – as planned – for government spending and given that it has been made possible by printing money and not by production/taxation there is inflation on the way that has the

potential to wipe out monetary value. In that case there will be no need for the Cult to steal your money. It just won't be worth anything (see the German Weimar Republic before the Nazis took over). Many have been okay with lockdowns while getting a percentage of their income from so-called furlough payments without having to work. Those payments are dependent, however, on people having at least a theoretical job with a business considered non-essential and ordered to close. As these business go under because they are closed by lockdown after lockdown the furlough stops and it will for everyone eventually. Then what? The 'then what?' is precisely the idea.



Figure 4: The Hunger Games Society structure I have long warned was planned and now the 'Covid' hoax has made it possible. This is the real reason for lockdowns.

Hired hands

Between the Hunger Games Cult elite and the dependent population is planned to be a vicious military-police state (a fusion of the two into one force). This has been in the making for a long time with police looking ever more like the military and carrying weapons to match. The pandemic scam has seen this process accelerate so fast as

lockdown house arrest is brutally enforced by carefully recruited fascist minds and gormless system-servers. The police and military are planned to merge into a centrally-directed world army in a global structure headed by a world government which wouldn't be elected even by the election fixes now in place. The world army is not planned even to be human and instead wars would be fought, primarily against the population, using robot technology controlled by artificial intelligence. I have been warning about this for decades and now militaries around the world are being transformed by this very AI technology. The global regime that I describe is a particular form of fascism known as a technocracy in which decisions are not made by clueless and co-opted politicians but by unelected technocrats – scientists, engineers, technologists and bureaucrats. Cult-owned-and-controlled Silicon Valley giants are examples of technocracy and they already have far more power to direct world events than governments. They are with their censorship *selecting* governments. I know that some are calling the 'Great Reset' a Marxist communist takeover, but fascism and Marxism are different labels for the same tyranny. Tell those who lived in fascist Germany and Stalinist Russia that there was a difference in the way their freedom was deleted and their lives controlled. I could call it a fascist technocracy or a Marxist technocracy and they would be equally accurate. The Hunger Games society with its world government structure would oversee a world army, world central bank and single world cashless currency imposing its will on a microchipped population ([Fig 5](#)). Scan its different elements and see how the illusory pandemic is forcing society in this very direction at great speed. Leaders of 23 countries and the World Health Organization (WHO) backed the idea in March, 2021, of a global treaty for 'international cooperation' in 'health emergencies' and nations should 'come together as a global community for peaceful cooperation that extends beyond this crisis'. Cut the Orwellian bullshit and this means another step towards global government. The plan includes a cashless digital money system that I first warned about in 1993. Right at the start of 'Covid' the deeply corrupt Tedros

Adhanom Ghebreyesus, the crooked and merely gofer 'head' of the World Health Organization, said it was possible to catch the 'virus' by touching cash and it was better to use cashless means. The claim was ridiculous nonsense and like the whole 'Covid' mind-trick it was nothing to do with 'health' and everything to do with pushing every aspect of the Cult agenda. As a result of the Tedros lie the use of cash has plummeted. The Cult script involves a single world digital currency that would eventually be technologically embedded in the body. China is a massive global centre for the Cult and if you watch what is happening there you will know what is planned for everywhere. The Chinese government is developing a digital currency which would allow fines to be deducted immediately via AI for anyone caught on camera breaking its fantastic list of laws and the money is going to be programmable with an expiry date to ensure that no one can accrue wealth except the Cult and its operatives.



Figure 5: The structure of global control the Cult has been working towards for so long and this has been enormously advanced by the 'Covid' illusion.

Serfdom is so smart

The Cult plan is far wider, extreme, and more comprehensive than even most conspiracy researchers appreciate and I will come to the true depths of deceit and control in the chapters 'Who controls the

Cult?’ and ‘Escaping Wetiko’. Even the world that we know is crazy enough. We are being deluged with ever more sophisticated and controlling technology under the heading of ‘smart’. We have smart televisions, smart meters, smart cards, smart cars, smart driving, smart roads, smart pills, smart patches, smart watches, smart skin, smart borders, smart pavements, smart streets, smart cities, smart communities, smart environments, smart growth, smart planet ... smart *everything* around us. Smart technologies and methods of operation are designed to interlock to create a global Smart Grid connecting the entirety of human society including human minds to create a centrally-dictated ‘hive’ mind. ‘Smart cities’ is code for densely-occupied megacities of total surveillance and control through AI. Ever more destructive frequency communication systems like 5G have been rolled out without any official testing for health and psychological effects (colossal). 5G/6G/7G systems are needed to run the Smart Grid and each one becomes more destructive of body and mind. Deleting independent income is crucial to forcing people into these AI-policed prisons by ending private property ownership (except for the Cult elite). The Cult’s Great Reset now openly foresees a global society in which no one will own any possessions and everything will be rented while the Cult would own literally everything under the guise of government and corporations. The aim has been to use the lockdowns to destroy sources of income on a mass scale and when the people are destitute and in unrepayable amounts of debt (problem) Cult assets come forward with the pledge to write-off debt in return for handing over all property and possessions (solution). Everything – literally everything including people – would be connected to the Internet via AI. I was warning years ago about the coming Internet of Things (IoT) in which all devices and technology from your car to your fridge would be plugged into the Internet and controlled by AI. Now we are already there with much more to come. The next stage is the Internet of Everything (IoE) which is planned to include the connection of AI to the human brain and body to replace the human mind with a centrally-controlled AI mind. Instead of perceptions

being manipulated through control of information and censorship those perceptions would come direct from the Cult through AI. What do you think? You think whatever AI decides that you think. In human terms there would be no individual 'think' any longer. Too incredible? The ravings of a lunatic? Not at all. Cult-owned crazies in Silicon Valley have been telling us the plan for years without explaining the real motivation and calculated implications. These include Google executive and 'futurist' Ray Kurzweil who highlights the year 2030 for when this would be underway. He said:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and 'think in the cloud' ... We're going to put gateways to the cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations.

As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

The sales-pitch of Kurzweil and Cult-owned Silicon Valley is that this would make us 'super-human' when the real aim is to make us post-human and no longer 'human' in the sense that we have come to know. The entire global population would be connected to AI and become the centrally-controlled 'hive-mind' of externally-delivered perceptions. The Smart Grid being installed to impose the Cult's will on the world is being constructed to allow particular locations – even one location – to control the whole global system. From these prime control centres, which absolutely include China and Israel, anything connected to the Internet would be switched on or off and manipulated at will. Energy systems could be cut, communication via the Internet taken down, computer-controlled driverless autonomous vehicles driven off the road, medical devices switched off, the potential is limitless given how much AI and Internet connections now run human society. We have seen nothing yet if we allow this to continue. Autonomous vehicle makers are working with law enforcement to produce cars designed to automatically pull over if they detect a police or emergency vehicle flashing from up to 100 feet away. At a police stop the car would be unlocked and the

window rolled down automatically. Vehicles would only take you where the computer (the state) allowed. The end of petrol vehicles and speed limiters on all new cars in the UK and EU from 2022 are steps leading to electric computerised transport over which ultimately you have no control. The picture is far bigger even than the Cult global network or web and that will become clear when I get to the nature of the 'spider'. There is a connection between all these happenings and the instigation of DNA-manipulating 'vaccines' (which aren't 'vaccines') justified by the 'Covid' hoax. That connection is the unfolding plan to transform the human body from a biological to a synthetic biological state and this is why synthetic biology is such a fast-emerging discipline of mainstream science. 'Covid vaccines' are infusing self-replicating synthetic genetic material into the cells to cumulatively take us on the Totalitarian Tiptoe from Human 1.0 to the synthetic biological Human 2.0 which will be physically and perceptually attached to the Smart Grid to one hundred percent control every thought, perception and deed. Humanity needs to wake up and *fast*.

This is the barest explanation of where the 'outcome' is planned to go but it's enough to see the journey happening all around us. Those new to this information will already see 'Covid' in a whole new context. I will add much more detail as we go along, but for the minutiae evidence see my mega-works, *The Answer*, *The Trigger* and *Everything You Need to Know But Have Never Been Told*.

Now – how does a Renegade Mind see the 'world'?

CHAPTER TWO

Renegade Perception

It is one thing to be clever and another to be wise

George R.R. Martin

A simple definition of the difference between a programmed mind and a Renegade Mind would be that one sees only dots while the other connects them to see the picture. Reading reality with accuracy requires the observer to (a) know the planned outcome and (b) realise that everything, but *everything*, is connected.

The entirety of infinite reality is connected – that's its very nature – and with human society an expression of infinite reality the same must apply. Simple cause and effect is a connection. The effect is triggered by the cause and the effect then becomes the cause of another effect. Nothing happens in isolation because it *can't*. Life in whatever reality is simple choice and consequence. We make choices and these lead to consequences. If we don't like the consequences we can make different choices and get different consequences which lead to other choices and consequences. The choice and the consequence are not only connected they are indivisible. You can't have one without the other as an old song goes. A few cannot control the world unless those being controlled allow that to happen – cause and effect, choice and consequence. Control – who has it and who doesn't – is a two-way process, a symbiotic relationship, involving the controller and controlled. 'They took my freedom away!!' Well, yes, but you also gave it to them. Humanity is

subjected to mass control because humanity has acquiesced to that control. This is all cause and effect and literally a case of give and take. In the same way world events of every kind are connected and the Cult works incessantly to sell the illusion of the random and coincidental to maintain the essential (to them) perception of dots that hide the picture. Renegade Minds know this and constantly scan the world for patterns of connection. This is absolutely pivotal in understanding the happenings in the world and without that perspective clarity is impossible. First you know the planned outcome and then you identify the steps on the journey – the day-by-day apparently random which, when connected in relation to the outcome, no longer appear as individual events, but as the proverbial *chain* of events leading in the same direction. I'll give you some examples:

Political puppet show

We are told to believe that politics is 'adversarial' in that different parties with different beliefs engage in an endless tussle for power. There may have been some truth in that up to a point – and only a point – but today divisions between 'different' parties are rhetorical not ideological. Even the rhetorical is fusing into one-speak as the parties eject any remaining free thinkers while others succumb to the ever-gathering intimidation of anyone with the 'wrong' opinion. The Cult is not a new phenomenon and can be traced back thousands of years as my books have documented. Its intergenerational initiatives have been manipulating events with increasing effect the more that global power has been centralised. In ancient times the Cult secured control through the system of monarchy in which 'special' bloodlines (of which more later) demanded the right to rule as kings and queens simply by birthright and by vanquishing others who claimed the same birthright. There came a time, however, when people had matured enough to see the unfairness of such tyranny and demanded a say in who governed them. Note the word – *governed* them. Not served them – *governed* them, hence government defined as 'the political direction and control exercised over the

actions of the members, citizens, or inhabitants of communities, societies, and states; direction of the affairs of a state, community, etc.' Governments exercise control over rather than serve just like the monarchies before them. Bizarrely there are still countries like the United Kingdom which are ruled by a monarch *and* a government that officially answers to the monarch. The UK head of state and that of Commonwealth countries such as Canada, Australia and New Zealand is 'selected' by who in a *single family* had unprotected sex with whom and in what order. Pinch me it can't be true. Ouch! Shit, it is. The demise of monarchies in most countries offered a potential vacuum in which some form of free and fair society could arise and the Cult had that base covered. Monarchies had served its interests but they couldn't continue in the face of such widespread opposition and, anyway, replacing a 'royal' dictatorship that people could see with a dictatorship 'of the people' hiding behind the concept of 'democracy' presented far greater manipulative possibilities and ways of hiding coordinated tyranny behind the illusion of 'freedom'.

Democracy is quite wrongly defined as government selected by the population. This is not the case at all. It is government selected by *some* of the population (and then only in theory). This 'some' doesn't even have to be the majority as we have seen so often in first-past-the-post elections in which the so-called majority party wins fewer votes than the 'losing' parties combined. Democracy can give total power to a party in government from a minority of the votes cast. It's a sleight of hand to sell tyranny as freedom. Seventy-four million Trump-supporting Americans didn't vote for the 'Democratic' Party of Joe Biden in the distinctly dodgy election in 2020 and yet far from acknowledging the wishes and feelings of that great percentage of American society the Cult-owned Biden government set out from day one to destroy them and their right to a voice and opinion. Empty shell Biden and his Cult handlers said they were doing this to 'protect democracy'. Such is the level of lunacy and sickness to which politics has descended. Connect the dots and relate them to the desired outcome – a world government run by self-appointed technocrats and no longer even elected

politicians. While operating through its political agents in government the Cult is at the same time encouraging public disdain for politicians by putting idiots and incompetents in theoretical power on the road to deleting them. The idea is to instil a public reaction that says of the technocrats: 'Well, they couldn't do any worse than the pathetic politicians.' It's all about controlling perception and Renegade Minds can see through that while programmed minds cannot when they are ignorant of both the planned outcome and the manipulation techniques employed to secure that end. This knowledge can be learned, however, and fast if people choose to get informed.

Politics may at first sight appear very difficult to control from a central point. I mean look at the 'different' parties and how would you be able to oversee them all and their constituent parts? In truth, it's very straightforward because of their structure. We are back to the pyramid of imposition and acquiescence. Organisations are structured in the same way as the system as a whole. Political parties are not open forums of free expression. They are hierarchies. I was a national spokesman for the British Green Party which claimed to be a different kind of politics in which influence and power was devolved; but I can tell you from direct experience – and it's far worse now – that Green parties are run as hierarchies like all the others however much they may try to hide that fact or kid themselves that it's not true. A very few at the top of all political parties are directing policy and personnel. They decide if you are elevated in the party or serve as a government minister and to do that you have to be a yes man or woman. Look at all the maverick political thinkers who never ascended the greasy pole. If you want to progress within the party or reach 'high-office' you need to fall into line and conform. Exceptions to this are rare indeed. Should you want to run for parliament or Congress you have to persuade the local or state level of the party to select you and for that you need to play the game as dictated by the hierarchy. If you secure election and wish to progress within the greater structure you need to go on conforming to what is acceptable to those running the hierarchy

from the peak of the pyramid. Political parties are perceptual gulags and the very fact that there are party 'Whips' appointed to 'whip' politicians into voting the way the hierarchy demands exposes the ridiculous idea that politicians are elected to serve the people they are supposed to represent. Cult operatives and manipulation has long seized control of major parties that have any chance of forming a government and at least most of those that haven't. A new party forms and the Cult goes to work to infiltrate and direct. This has reached such a level today that you see video compilations of 'leaders' of all parties whether Democrats, Republicans, Conservative, Labour and Green parroting the same Cult mantra of 'Build Back Better' and the 'Great Reset' which are straight off the Cult song-sheet to describe the transformation of global society in response to the Cult-instigated hoaxes of the 'Covid pandemic' and human-caused 'climate change'. To see Caroline Lucas, the Green Party MP that I knew when I was in the party in the 1980s, speaking in support of plans proposed by Cult operative Klaus Schwab representing the billionaire global elite is a real head-shaker.

Many parties – one master

The party system is another mind-trick and was instigated to change the nature of the dictatorship by swapping 'royalty' for dark suits that people believed – though now ever less so – represented their interests. Understanding this trick is to realise that a single force (the Cult) controls all parties either directly in terms of the major ones or through manipulation of perception and ideology with others. You don't need to manipulate Green parties to demand your transformation of society in the name of 'climate change' when they are obsessed with the lie that this is essential to 'save the planet'. You just give them a platform and away they go serving your interests while believing they are being environmentally virtuous. America's political structure is a perfect blueprint for how the two or multi-party system is really a one-party state. The Republican Party is controlled from one step back in the shadows by a group made up of billionaires and their gofers known as neoconservatives or Neocons.

I have exposed them in fine detail in my books and they were the driving force behind the policies of the imbecilic presidency of Boy George Bush which included 9/11 (see *The Trigger* for a comprehensive demolition of the official story), the subsequent 'war on terror' (war of terror) and the invasions of Afghanistan and Iraq. The latter was a No-Problem-Reaction-Solution based on claims by Cult operatives, including Bush and British Prime Minister Tony Blair, about Saddam Hussein's 'weapons of mass destruction' which did not exist as war criminals Bush and Blair well knew.



Figure 6: Different front people, different parties – same control system.

The Democratic Party has its own 'Neocon' group controlling from the background which I call the 'Democons' and here's the penny-drop – the Neocons and Democons answer to the same masters one step further back into the shadows (Fig 6). At that level of the Cult the Republican and Democrat parties are controlled by the same people and no matter which is in power the Cult is in power. This is how it works in almost every country and certainly in Britain with Conservative, Labour, Liberal Democrat and Green parties now all on the same page whatever the rhetoric may be in their feeble attempts to appear different. Neocons operated at the time of Bush through a think tank called The Project for the New American Century which in September, 2000, published a document entitled *Rebuilding America's Defenses: Strategies, Forces, and Resources*

For a New Century demanding that America fight ‘multiple, simultaneous major theatre wars’ as a ‘core mission’ to force regime-change in countries including Iraq, Libya and Syria. Neocons arranged for Bush (‘Republican’) and Blair (‘Labour Party’) to front-up the invasion of Iraq and when they departed the Democons orchestrated the targeting of Libya and Syria through Barack Obama (‘Democrat’) and British Prime Minister David Cameron (‘Conservative Party’). We have ‘different’ parties and ‘different’ people, but the same unfolding script. The more the Cult has seized the reigns of parties and personnel the more their policies have transparently pursued the same agenda to the point where the fascist ‘Covid’ impositions of the Conservative junta of Jackboot Johnson in Britain were opposed by the Labour Party because they were not fascist enough. The Labour Party is likened to the US Democrats while the Conservative Party is akin to a British version of the Republicans and on both sides of the Atlantic they all speak the same language and support the direction demanded by the Cult although some more enthusiastically than others. It’s a similar story in country after country because it’s all centrally controlled. Oh, but what about Trump? I’ll come to him shortly. Political ‘choice’ in the ‘party’ system goes like this: You vote for Party A and they get into government. You don’t like what they do so next time you vote for Party B and they get into government. You don’t like what they do when it’s pretty much the same as Party A and why wouldn’t that be with both controlled by the same force? Given that only two, sometimes three, parties have any chance of forming a government to get rid of Party B that you don’t like you have to vote again for Party A which ... you don’t like. This, ladies and gentlemen, is what they call ‘democracy’ which we are told – wrongly – is a term interchangeable with ‘freedom’.

The cult of cults

At this point I need to introduce a major expression of the Global Cult known as Sabbatian-Frankism. Sabbatian is also spelt as Sabbatean. I will summarise here. I have published major exposés

and detailed background in other works. Sabbatian-Frankism combines the names of two frauds posing as 'Jewish' men, Sabbatai Zevi (1626-1676), a rabbi, black magician and occultist who proclaimed he was the Jewish messiah; and Jacob Frank (1726-1791), the Polish 'Jew', black magician and occultist who said he was the reincarnation of 'messiah' Zevi and biblical patriarch Jacob. They worked across two centuries to establish the Sabbatian-Frankist cult that plays a major, indeed central, role in the manipulation of human society by the Global Cult which has its origins much further back in history than Sabbatai Zevi. I should emphasise two points here in response to the shrill voices that will scream 'anti-Semitism': (1) Sabbatian-Frankists are NOT Jewish and only pose as such to hide their cult behind a Jewish façade; and (2) my information about this cult has come from Jewish sources who have long realised that their society and community has been infiltrated and taken over by interloper Sabbatian-Frankists. Infiltration has been the foundation technique of Sabbatian-Frankism from its official origin in the 17th century. Zevi's Sabbatian sect attracted a massive following described as the biggest messianic movement in Jewish history, spreading as far as Africa and Asia, and he promised a return for the Jews to the 'Promised Land' of Israel. Sabbatianism was not Judaism but an inversion of everything that mainstream Judaism stood for. So much so that this sinister cult would have a feast day when Judaism had a fast day and whatever was forbidden in Judaism the Sabbatians were encouraged and even commanded to do. This included incest and what would be today called Satanism. Members were forbidden to marry outside the sect and there was a system of keeping their children ignorant of what they were part of until they were old enough to be trusted not to unknowingly reveal anything to outsiders. The same system is employed to this day by the Global Cult in general which Sabbatian-Frankism has enormously influenced and now largely controls.

Zevi and his Sabbatians suffered a setback with the intervention by the Sultan of the Islamic Ottoman Empire in the Middle East and what is now the Republic of Turkey where Zevi was located. The

Sultan gave him the choice of proving his 'divinity', converting to Islam or facing torture and death. Funnily enough Zevi chose to convert or at least appear to. Some of his supporters were disillusioned and drifted away, but many did not with 300 families also converting – only in theory – to Islam. They continued behind this Islamic smokescreen to follow the goals, rules and rituals of Sabbatianism and became known as 'crypto-Jews' or the 'Dönme' which means 'to turn'. This is rather ironic because they didn't 'turn' and instead hid behind a fake Islamic persona. The process of appearing to be one thing while being very much another would become the calling card of Sabbatianism especially after Zevi's death and the arrival of the Satanist Jacob Frank in the 18th century when the cult became Sabbatian-Frankism and plumbed still new depths of depravity and infiltration which included – still includes – human sacrifice and sex with children. Wherever Sabbatians go paedophilia and Satanism follow and is it really a surprise that Hollywood is so infested with child abuse and Satanism when it was established by Sabbatian-Frankists and is still controlled by them? Hollywood has been one of the prime vehicles for global perceptual programming and manipulation. How many believe the version of 'history' portrayed in movies when it is a travesty and inversion (again) of the truth? Rabbi Marvin Antelman describes Frankism in his book, *To Eliminate the Opiate*, as 'a movement of complete evil' while Jewish professor Gershom Scholem said of Frank in *The Messianic Idea in Judaism*: 'In all his actions [he was] a truly corrupt and degenerate individual ... one of the most frightening phenomena in the whole of Jewish history.' Frank was excommunicated by traditional rabbis, as was Zevi, but Frank was undeterred and enjoyed vital support from the House of Rothschild, the infamous banking dynasty whose inner-core are Sabbatian-Frankists and not Jews. Infiltration of the Roman Church and Vatican was instigated by Frank with many Dönme 'turning' again to convert to Roman Catholicism with a view to hijacking the reins of power. This was the ever-repeating modus operandi and continues to be so. Pose as an advocate of the religion, culture or country that you want to control and then

manipulate your people into the positions of authority and influence largely as advisers, administrators and Svengalis for those that appear to be in power. They did this with Judaism, Christianity (Christian Zionism is part of this), Islam and other religions and nations until Sabbatian-Frankism spanned the world as it does today.

Sabbatian Saudis and the terror network

One expression of the Sabbatian-Frankist Dönme within Islam is the ruling family of Saudi Arabia, the House of Saud, through which came the vile distortion of Islam known as Wahhabism. This is the violent creed followed by terrorist groups like Al-Qaeda and ISIS or Islamic State. Wahhabism is the hand-chopping, head-chopping 'religion' of Saudi Arabia which is used to keep the people in a constant state of fear so the interloper House of Saud can continue to rule. Al-Qaeda and Islamic State were lavishly funded by the House of Saud while being created and directed by the Sabbatian-Frankist network in the United States that operates through the Pentagon, CIA and the government in general of whichever 'party'. The front man for the establishment of Wahhabism in the middle of the 18th century was a Sabbatian-Frankist 'crypto-Jew' posing as Islamic called Muhammad ibn Abd al-Wahhab. His daughter would marry the son of Muhammad bin Saud who established the first Saudi state before his death in 1765 with support from the British Empire. Bin Saud's successors would establish modern Saudi Arabia in league with the British and Americans in 1932 which allowed them to seize control of Islam's major shrines in Mecca and Medina. They have dictated the direction of Sunni Islam ever since while Iran is the major centre of the Shiite version and here we have the source of at least the public conflict between them. The Sabbatian network has used its Wahhabi extremists to carry out Problem-Reaction-Solution terrorist attacks in the name of 'Al-Qaeda' and 'Islamic State' to justify a devastating 'war on terror', ever-increasing surveillance of the population and to terrify people into compliance. Another insight of the Renegade Mind is the streetwise understanding that

just because a country, location or people are attacked doesn't mean that those apparently representing that country, location or people are not behind the attackers. Often they are *orchestrating* the attacks because of the societal changes that can be then justified in the name of 'saving the population from terrorists'.

I show in great detail in *The Trigger* how Sabbatian-Frankists were the real perpetrators of 9/11 and not '19 Arab hijackers' who were blamed for what happened. Observe what was justified in the name of 9/11 alone in terms of Middle East invasions, mass surveillance and control that fulfilled the demands of the Project for the New American Century document published by the Sabbatian Neocons. What appear to be enemies are on the deep inside players on the same Sabbatian team. Israel and Arab 'royal' dictatorships are all ruled by Sabbatians and the recent peace agreements between Israel and Saudi Arabia, the United Arab Emirates (UAE) and others are only making formal what has always been the case behind the scenes. Palestinians who have been subjected to grotesque tyranny since Israel was bombed and terrorised into existence in 1948 have never stood a chance. Sabbatian-Frankists have controlled Israel (so the constant theme of violence and war which Sabbatians love) and they have controlled the Arab countries that Palestinians have looked to for real support that never comes. 'Royal families' of the Arab world in Saudi Arabia, Bahrain, UAE, etc., are all Sabbatians with allegiance to the aims of the cult and not what is best for their Arabic populations. They have stolen the oil and financial resources from their people by false claims to be 'royal dynasties' with a genetic right to rule and by employing vicious militaries to impose their will.

Satanic 'illumination'

The Satanist Jacob Frank formed an alliance in 1773 with two other Sabbatians, Mayer Amschel Rothschild (1744-1812), founder of the Rothschild banking dynasty, and Jesuit-educated fraudulent Jew, Adam Weishaupt, and this led to the formation of the Bavarian Illuminati, firstly under another name, in 1776. The Illuminati would

be the manipulating force behind the French Revolution (1789-1799) and was also involved in the American Revolution (1775-1783) before and after the Illuminati's official creation. Weishaupt would later become (in public) a Protestant Christian in archetypal Sabbatian style. I read that his name can be decoded as Adam-Weishaupt or 'the first man to lead those who know'. He wasn't a leader in the sense that he was a subordinate, but he did lead those below him in a crusade of transforming human society that still continues today. The theme was confirmed as early as 1785 when a horseman courier called Lanz was reported to be struck by lightning and extensive Illuminati documents were found in his saddlebags. They made the link to Weishaupt and detailed the plan for world takeover. Current events with 'Covid' fascism have been in the making for a very long time. Jacob Frank was jailed for 13 years by the Catholic Inquisition after his arrest in 1760 and on his release he headed for Frankfurt, Germany, home city and headquarters of the House of Rothschild where the alliance was struck with Mayer Amschel Rothschild and Weishaupt. Rothschild arranged for Frank to be given the title of Baron and he became a wealthy nobleman with a big following of Jews in Germany, the Austro-Hungarian Empire and other European countries. Most of them would have believed he was on their side.

The name 'Illuminati' came from the Zohar which is a body of works in the Jewish mystical 'bible' called the Kabbalah. 'Zohar' is the foundation of Sabbatian-Frankist belief and in Hebrew 'Zohar' means 'splendour', 'radiance', 'illuminated', and so we have 'Illuminati'. They claim to be the 'Illuminated Ones' from their knowledge systematically hidden from the human population and passed on through generations of carefully-chosen initiates in the global secret society network or Cult. Hidden knowledge includes an awareness of the Cult agenda for the world and the nature of our collective reality that I will explore later. Cult 'illumination' is symbolised by the torch held by the Statue of Liberty which was gifted to New York by French Freemasons in Paris who knew exactly what it represents. 'Liberty' symbolises the goddess worshipped in

Babylon as Queen Semiramis or Ishtar. The significance of this will become clear. Notice again the ubiquitous theme of inversion with the Statue of 'Liberty' really symbolising mass control (Fig 7). A mirror-image statute stands on an island in the River Seine in Paris from where New York Liberty originated (Fig 8). A large replica of the Liberty flame stands on top of the Pont de l'Alma tunnel in Paris where Princess Diana died in a Cult ritual described in *The Biggest Secret*. Lucifer 'the light bringer' is related to all this (and much more as we'll see) and 'Lucifer' is a central figure in Sabbatian-Frankism and its associated Satanism. Sabbatians reject the Jewish Torah, or Pentateuch, the 'five books of Moses' in the Old Testament known as Genesis, Exodus, Leviticus, Numbers, and Deuteronomy which are claimed by Judaism and Christianity to have been dictated by 'God' to Moses on Mount Sinai. Sabbatians say these do not apply to them and they seek to replace them with the Zohar to absorb Judaism and its followers into their inversion which is an expression of a much greater global inversion. They want to delete all religions and force humanity to worship a one-world religion – Sabbatian Satanism that also includes worship of the Earth goddess. Satanic themes are being more and more introduced into mainstream society and while Christianity is currently the foremost target for destruction the others are planned to follow.



Figure 7: The Cult goddess of Babylon disguised as the Statue of Liberty holding the flame of Lucifer the 'light bringer'.



Figure 8: Liberty's mirror image in Paris where the New York version originated.

Marx brothers

Rabbi Marvin Antelman connects the Illuminati to the Jacobins in *To Eliminate the Opiate* and Jacobins were the force behind the French Revolution. He links both to the Bund der Gerechten, or League of the Just, which was the network that inflicted communism/Marxism on the world. Antelman wrote:

The original inner circle of the Bund der Gerechten consisted of born Catholics, Protestants and Jews [Sabbatian-Frankist infiltrators], and those representatives of respective subdivisions formulated schemes for the ultimate destruction of their faiths. The heretical Catholics laid plans which they felt would take a century or more for the ultimate destruction of the church; the apostate Jews for the ultimate destruction of the Jewish religion.

Sabbatian-created communism connects into this anti-religion agenda in that communism does not allow for the free practice of religion. The Sabbatian 'Bund' became the International Communist Party and Communist League and in 1848 'Marxism' was born with the Communist Manifesto of Sabbatian assets Karl Marx and Friedrich Engels. It is absolutely no coincidence that Marxism, just a different name for fascist and other centrally-controlled tyrannies, is being imposed worldwide as a result of the 'Covid' hoax and nor that Marxist/fascist China was the place where the hoax originated. The reason for this will become very clear in the chapter 'Covid: The calculated catastrophe'. The so-called 'Woke' mentality has hijacked

traditional beliefs of the political left and replaced them with far-right make-believe 'social justice' better known as Marxism. Woke will, however, be swallowed by its own perceived 'revolution' which is really the work of billionaires and billionaire corporations feigning being 'Woke'. Marxism is being touted by Wokers as a replacement for 'capitalism' when we don't have 'capitalism'. We have cartelism in which the market is stitched up by the very Cult billionaires and corporations bankrolling Woke. Billionaires love Marxism which keeps the people in servitude while they control from the top. Terminally naïve Wokers think they are 'changing the world' when it's the Cult that is doing the changing and when they have played their vital part and become surplus to requirements they, too, will be targeted. The Illuminati-Jacobins were behind the period known as 'The Terror' in the French Revolution in 1793 and 1794 when Jacobin Maximillian de Robespierre and his Orwellian 'Committee of Public Safety' killed 17,000 'enemies of the Revolution' who had once been 'friends of the Revolution'. Karl Marx (1818-1883), whose Sabbatian creed of Marxism has cost the lives of at least 100 million people, is a hero once again to Wokers who have been systematically kept ignorant of real history by their 'education' programming. As a result they now promote a Sabbatian 'Marxist' abomination destined at some point to consume them. Rabbi Antelman, who spent decades researching the Sabbatian plot, said of the League of the Just and Karl Marx:

Contrary to popular opinion Karl Marx did not originate the Communist Manifesto. He was paid for his services by the League of the Just, which was known in its country of origin, Germany, as the Bund der Geächteten.

Antelman said the text attributed to Marx was the work of other people and Marx 'was only repeating what others already said'. Marx was 'a hired hack – lackey of the wealthy Illuminists'. Marx famously said that religion was the 'opium of the people' (part of the Sabbatian plan to demonise religion) and Antelman called his books, *To Eliminate the Opiate*. Marx was born Jewish, but his family converted to Christianity (Sabbatian modus operandi) and he

attacked Jews, not least in his book, *A World Without Jews*. In doing so he supported the Sabbatian plan to destroy traditional Jewishness and Judaism which we are clearly seeing today with the vindictive targeting of orthodox Jews by the Sabbatian government of Israel over 'Covid' laws. I don't follow any religion and it has done much damage to the world over centuries and acted as a perceptual straightjacket. Renegade Minds, however, are always asking *why* something is being done. It doesn't matter if they agree or disagree with what is happening – *why* is it happening is the question. The 'why?' can be answered with regard to religion in that religions create interacting communities of believers when the Cult wants to dismantle all discourse, unity and interaction (see 'Covid' lockdowns) and the ultimate goal is to delete all religions for a one-world religion of Cult Satanism worshipping their 'god' of which more later. We see the same 'why?' with gun control in America. I don't have guns and don't want them, but why is the Cult seeking to disarm the population at the same time that law enforcement agencies are armed to their molars and why has every tyrant in history sought to disarm people before launching the final takeover? They include Hitler, Stalin, Pol Pot and Mao who followed confiscation with violent seizing of power. You know it's a Cult agenda by the people who immediately race to the microphones to exploit dead people in multiple shootings. Ultra-Zionist Cult lackey Senator Chuck Schumer was straight on the case after ten people were killed in Boulder, Colorado in March, 2121. Simple rule ... if Schumer wants it the Cult wants it and the same with his ultra-Zionist mate the wild-eyed Senator Adam Schiff. At the same time they were calling for the disarmament of Americans, many of whom live a long way from a police response, Schumer, Schiff and the rest of these pampered clowns were sitting on Capitol Hill behind a razor-wired security fence protected by thousands of armed troops in addition to their own armed bodyguards. Mom and pop in an isolated home? They're just potential mass shooters.

Zion Mainframe

Sabbatian-Frankists and most importantly the Rothschilds were behind the creation of 'Zionism', a political movement that demanded a Jewish homeland in Israel as promised by Sabbatai Zevi. The very symbol of Israel comes from the German meaning of the name Rothschild. Dynasty founder Mayer Amschel Rothschild changed the family name from Bauer to Rothschild, or 'Red-Shield' in German, in deference to the six-pointed 'Star of David' hexagram displayed on the family's home in Frankfurt. The symbol later appeared on the flag of Israel after the Rothschilds were centrally involved in its creation. Hexagrams are not a uniquely Jewish symbol and are widely used in occult ('hidden') networks often as a symbol for Saturn (see my other books for why). Neither are Zionism and Jewishness interchangeable. Zionism is a political movement and philosophy and not a 'race' or a people. Many Jews oppose Zionism and many non-Jews, including US President Joe Biden, call themselves Zionists as does Israel-centric Donald Trump. America's support for the Israel government is pretty much a gimme with ultra-Zionist billionaires and corporations providing fantastic and dominant funding for both political parties. Former Congresswoman Cynthia McKinney has told how she was approached immediately she ran for office to 'sign the pledge' to Israel and confirm that she would always vote in that country's best interests. All American politicians are approached in this way. Anyone who refuses will get no support or funding from the enormous and all-powerful Zionist lobby that includes organisations like mega-lobby group AIPAC, the American Israel Public Affairs Committee. Trump's biggest funder was ultra-Zionist casino and media billionaire Sheldon Adelson while major funders of the Democratic Party include ultra-Zionist George Soros and ultra-Zionist financial and media mogul, Haim Saban. Some may reel back at the suggestion that Soros is an Israel-firster (Sabbatian-controlled Israel-firster), but Renegade Minds watch the actions not the words and everywhere Soros donates his billions the Sabbatian agenda benefits. In the spirit of Sabbatian inversion Soros pledged \$1 billion for a new university network to promote 'liberal values and tackle intolerance'. He made the announcement during his annual speech

at the Cult-owned World Economic Forum in Davos, Switzerland, in January, 2020, after his 'harsh criticism' of 'authoritarian rulers' around the world. You can only laugh at such brazen mendacity. How *he* doesn't laugh is the mystery. Translated from the Orwellian 'liberal values and tackle intolerance' means teaching non-white people to hate white people and for white people to loathe themselves for being born white. The reason for that will become clear.

The 'Anti-Semitism' fraud

Zionists support the Jewish homeland in the land of Palestine which has been the Sabbatian-Rothschild goal for so long, but not for the benefit of Jews. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. This is nothing more than a Sabbatian protection racket to stop legitimate investigation and exposure of their agendas and activities. The official definition of 'anti-Semitism' has more recently been expanded to include criticism of Zionism – a *political movement* – and this was done to further stop exposure of Sabbatian infiltrators who created Zionism as we know it today in the 19th century. Renegade Minds will talk about these subjects when they know the shit that will come their way. People must decide if they want to know the truth or just cower in the corner in fear of what others will say. Sabbatians have been trying to label me as 'anti-Semitic' since the 1990s as I have uncovered more and more about their background and agendas. Useless, gutless, fraudulent 'journalists' then just repeat the smears without question and on the day I was writing this section a pair of unquestioning repeaters called Ben Quinn and Archie Bland (how appropriate) outright called me an 'anti-Semite' in the establishment propaganda sheet, the London *Guardian*, with no supporting evidence. The

Sabbatian Anti-Semitism Industry said so and who are they to question that? They wouldn't dare. Ironically 'Semitic' refers to a group of languages in the Middle East that are almost entirely Arabic. 'Anti-Semitism' becomes 'anti-Arab' which if the consequences of this misunderstanding were not so grave would be hilarious. Don't bother telling Quinn and Bland. I don't want to confuse them, bless 'em. One reason I am dubbed 'anti-Semitic' is that I wrote in the 1990s that Jewish operatives (Sabbatians) were heavily involved in the Russian Revolution when Sabbatians overthrew the Romanov dynasty. This apparently made me 'anti-Semitic'. Oh, really? Here is a section from *The Trigger*:

British journalist Robert Wilton confirmed these themes in his 1920 book *The Last Days of the Romanovs* when he studied official documents from the Russian government to identify the members of the Bolshevik ruling elite between 1917 and 1919. The Central Committee included 41 Jews among 62 members; the Council of the People's Commissars had 17 Jews out of 22 members; and 458 of the 556 most important Bolshevik positions between 1918 and 1919 were occupied by Jewish people. Only 17 were Russian. Then there were the 23 Jews among the 36 members of the vicious Cheka Soviet secret police established in 1917 who would soon appear all across the country.

Professor Robert Service of Oxford University, an expert on 20th century Russian history, found evidence that ['Jewish'] Leon Trotsky had sought to make sure that Jews were enrolled in the Red Army and were disproportionately represented in the Soviet civil bureaucracy that included the Cheka which performed mass arrests, imprisonment and executions of 'enemies of the people'. A US State Department Decimal File (861.00/5339) dated November 13th, 1918, names [Rothschild banking agent in America] Jacob Schiff and a list of ultra-Zionists as funders of the Russian Revolution leading to claims of a 'Jewish plot', but the key point missed by all is they were not 'Jews' – they were Sabbatian-Frankists.

Britain's Winston Churchill made the same error by mistake or otherwise. He wrote in a 1920 edition of the *Illustrated Sunday Herald* that those behind the Russian revolution were part of a 'worldwide conspiracy for the overthrow of civilisation and for the reconstitution of society on the basis of arrested development, of envious malevolence, and impossible equality' (see 'Woke' today because that has been created by the same network). Churchill said there was no need to exaggerate the part played in the creation of Bolshevism and in the actual bringing about of the Russian

Revolution 'by these international and for the most part atheistical Jews' ['atheistical Jews' = Sabbatians]. Churchill said it is certainly a very great one and probably outweighs all others: 'With the notable exception of Lenin, the majority of the leading figures are Jews.' He went on to describe, knowingly or not, the Sabbatian modus operandi of placing puppet leaders nominally in power while they control from the background:

Moreover, the principal inspiration and driving power comes from the Jewish leaders. Thus Tchitcherin, a pure Russian, is eclipsed by his nominal subordinate, Litvinoff, and the influence of Russians like Bukharin or Lunacharski cannot be compared with the power of Trotsky, or of Zinovieff, the Dictator of the Red Citadel (Petrograd), or of Krassin or Radek – all Jews. In the Soviet institutions the predominance of Jews is even more astonishing. And the prominent, if not indeed the principal, part in the system of terrorism applied by the Extraordinary Commissions for Combatting Counter-Revolution has been taken by Jews, and in some notable cases by Jewesses.

What I said about seriously disproportionate involvement in the Russian Revolution by Jewish 'revolutionaries' (Sabbatians) is provable fact, but truth is no defence against the Sabbatian Anti-Semitism Industry, its repeater parrots like Quinn and Bland, and the now breathtaking network of so-called 'Woke' 'anti-hate' groups with interlocking leaderships and funding which have the role of discrediting and silencing anyone who gets too close to exposing the Sabbatians. We have seen 'truth is no defence' confirmed in legal judgements with the Saskatchewan Human Rights Commission in Canada decreeing this: 'Truthful statements can be presented in a manner that would meet the definition of hate speech, and not all truthful statements must be free from restriction.' Most 'anti-hate' activists, who are themselves consumed by hatred, are too stupid and ignorant of the world to know how they are being used. They are far too far up their own virtue-signalling arses and it's far too dark for them to see anything.

The 'revolution' game

The background and methods of the 'Russian' Revolution are straight from the Sabbatian playbook seen in the French Revolution

and endless others around the world that appear to start as a revolution of the people against tyrannical rule and end up with a regime change to more tyrannical rule overtly or covertly. Wars, terror attacks and regime overthrows follow the Sabbatian cult through history with its agents creating them as Problem-Reaction-Solutions to remove opposition on the road to world domination. Sabbatian dots connect the Rothschilds with the Illuminati, Jacobins of the French Revolution, the 'Bund' or League of the Just, the International Communist Party, Communist League and the Communist Manifesto of Karl Marx and Friedrich Engels that would lead to the Rothschild-funded Russian Revolution. The sequence comes under the heading of 'creative destruction' when you advance to your global goal by continually destroying the status quo to install a new status quo which you then also destroy. The two world wars come to mind. With each new status quo you move closer to your planned outcome. Wars and mass murder are to Sabbatians a collective blood sacrifice ritual. They are obsessed with death for many reasons and one is that death is an inversion of life. Satanists and Sabbatians are obsessed with death and often target churches and churchyards for their rituals. Inversion-obsessed Sabbatians explain the use of inverted symbolism including the *inverted* pentagram and *inverted* cross. The inversion of the cross has been related to targeting Christianity, but the cross was a religious symbol long before Christianity and its inversion is a statement about the Sabbatian mentality and goals more than any single religion.

Sabbatians operating in Germany were behind the rise of the occult-obsessed Nazis and the subsequent Jewish exodus from Germany and Europe to Palestine and the United States after World War Two. The Rothschild dynasty was at the forefront of this both as political manipulators and by funding the operation. Why would Sabbatians help to orchestrate the horrors inflicted on Jews by the Nazis and by Stalin after they organised the Russian Revolution? Sabbatians hate Jews and their religion, that's why. They pose as Jews and secure positions of control within Jewish society and play the 'anti-Semitism' card to protect themselves from exposure

through a global network of organisations answering to the Sabbatian-created-and-controlled globe-spanning intelligence network that involves a stunning web of military-intelligence operatives and operations for a tiny country of just nine million. Among them are Jewish assets who are not Sabbatians but have been convinced by them that what they are doing is for the good of Israel and the Jewish community to protect them from what they have been programmed since childhood to believe is a Jew-hating hostile world. The Jewish community is just a highly convenient cover to hide the true nature of Sabbatians. Anyone getting close to exposing their game is accused by Sabbatian place-people and gofers of 'anti-Semitism' and claiming that all Jews are part of a plot to take over the world. I am not saying that. I am saying that Sabbatians – the *real* Jew-haters – have infiltrated the Jewish community to use them both as a cover and an 'anti-Semitic' defence against exposure. Thus we have the Anti-Semitism Industry targeted researchers in this way and most Jewish people think this is justified and genuine. They don't know that their 'Jewish' leaders and institutions of state, intelligence and military are not controlled by Jews at all, but cultists and stooges of Sabbatian-Frankism. I once added my name to a pro-Jewish freedom petition online and the next time I looked my name was gone and text had been added to the petition blurb to attack me as an 'anti-Semite' such is the scale of perceptual programming.

Moving on America

I tell the story in *The Trigger* and a chapter called 'Atlantic Crossing' how particularly after Israel was established the Sabbatians moved in on the United States and eventually grasped control of government administration, the political system via both Democrats and Republicans, the intelligence community like the CIA and National Security Agency (NSA), the Pentagon and mass media. Through this seriously compartmentalised network Sabbatians and their operatives in Mossad, Israeli Defense Forces (IDF) and US agencies pulled off 9/11 and blamed it on 19 'Al-Qaeda hijackers' dominated by men from, or connected to, Sabbatian-ruled Saudi

Arabia. The '19' were not even on the planes let alone flew those big passenger jets into buildings while being largely incompetent at piloting one-engine light aircraft. 'Hijacker' Hani Hanjour who is said to have flown American Airlines Flight 77 into the Pentagon with a turn and manoeuvre most professional pilots said they would have struggled to do was banned from renting a small plane by instructors at the Freeway Airport in Bowie, Maryland, just *six weeks* earlier on the grounds that he was an incompetent pilot. The Jewish population of the world is just 0.2 percent with even that almost entirely concentrated in Israel (75 percent Jewish) and the United States (around two percent). This two percent and globally 0.2 percent refers to *Jewish* people and not Sabbatian interlopers who are a fraction of that fraction. What a sobering thought when you think of the fantastic influence on world affairs of tiny Israel and that the Project for the New America Century (PNAC) which laid out the blueprint in September, 2000, for America's war on terror and regime change wars in Iraq, Libya and Syria was founded and dominated by Sabbatians known as 'Neocons'. The document conceded that this plan would not be supported politically or publicly without a major attack on American soil and a Problem-Reaction-Solution excuse to send troops to war across the Middle East. Sabbatian Neocons said:

... [The] process of transformation ... [war and regime change] ... is likely to be a long one, absent some catastrophic and catalysing event – like a new Pearl Harbor.

Four months later many of those who produced that document came to power with their inane puppet George Bush from the long-time Sabbatian Bush family. They included Sabbatian Dick Cheney who was officially vice-president, but really de-facto president for the entirety of the 'Bush' government. Nine months after the 'Bush' inauguration came what Bush called at the time 'the Pearl Harbor of the 21st century' and with typical Sabbatian timing and symbolism 2001 was the 60th anniversary of the attack in 1941 by the Japanese Air Force on Pearl Harbor, Hawaii, which allowed President Franklin Delano Roosevelt to take the United States into a Sabbatian-

instigated Second World War that he said in his election campaign that he never would. The evidence is overwhelming that Roosevelt and his military and intelligence networks knew the attack was coming and did nothing to stop it, but they did make sure that America's most essential naval ships were not in Hawaii at the time. Three thousand Americans died in the Pearl Harbor attacks as they did on September 11th. By the 9/11 year of 2001 Sabbatians had widely infiltrated the US government, military and intelligence operations and used their compartmentalised assets to pull off the 'Al-Qaeda' attacks. If you read *The Trigger* it will blow your mind to see the utterly staggering concentration of 'Jewish' operatives (Sabbatian infiltrators) in essential positions of political, security, legal, law enforcement, financial and business power before, during, and after the attacks to make them happen, carry them out, and then cover their tracks – and I do mean *staggering* when you think of that 0.2 percent of the world population and two percent of Americans which are Jewish while Sabbatian infiltrators are a fraction of that. A central foundation of the 9/11 conspiracy was the hijacking of government, military, Air Force and intelligence computer systems in real time through 'back-door' access made possible by Israeli (Sabbatian) 'cyber security' software. Sabbatian-controlled Israel is on the way to rivalling Silicon Valley for domination of cyberspace and is becoming the dominant force in cyber-security which gives them access to entire computer systems and their passcodes across the world. Then add to this that Zionists head (officially) Silicon Valley giants like Google (Larry Page and Sergey Brin), Google-owned YouTube (Susan Wojcicki), Facebook (Mark Zuckerberg and Sheryl Sandberg), and Apple (Chairman Arthur D. Levinson), and that ultra-Zionist hedge fund billionaire Paul Singer has a \$1 billion stake in Twitter which is only nominally headed by 'CEO' pothead Jack Dorsey. As cable news host Tucker Carlson said of Dorsey: 'There used to be debate in the medical community whether dropping a ton of acid had permanent effects and I think that debate has now ended.' Carlson made the comment after Dorsey told a hearing on Capitol Hill (if you cut through his bullshit) that he

believed in free speech so long as he got to decide what you can hear and see. These 'big names' of Silicon Valley are only front men and women for the Global Cult, not least the Sabbatians, who are the true controllers of these corporations. Does anyone still wonder why these same people and companies have been ferociously censoring and banning people (like me) for exposing any aspect of the Cult agenda and especially the truth about the 'Covid' hoax which Sabbatians have orchestrated?

The Jeffrey Epstein paedophile ring was a Sabbatian operation. He was officially 'Jewish' but he was a Sabbatian and women abused by the ring have told me about the high number of 'Jewish' people involved. The Epstein horror has Sabbatian written all over it and matches perfectly their modus operandi and obsession with sex and ritual. Epstein was running a Sabbatian blackmail ring in which famous people with political and other influence were provided with young girls for sex while everything was being filmed and recorded on hidden cameras and microphones at his New York house, Caribbean island and other properties. Epstein survivors have described this surveillance system to me and some have gone public. Once the famous politician or other figure knew he or she was on video they tended to do whatever they were told. Here we go again ...when you've got them by the balls their hearts and minds will follow. Sabbatians use this blackmail technique on a wide scale across the world to entrap politicians and others they need to act as demanded. Epstein's private plane, the infamous 'Lolita Express', had many well-known passengers including Bill Clinton while Bill Gates has flown on an Epstein plane and met with him four years after Epstein had been jailed for paedophilia. They subsequently met many times at Epstein's home in New York according to a witness who was there. Epstein's infamous side-kick was Ghislaine Maxwell, daughter of Mossad agent and ultra-Zionist mega-crooked British businessman, Bob Maxwell, who at one time owned the *Daily Mirror* newspaper. Maxwell was murdered at sea on his boat in 1991 by Sabbatian-controlled Mossad when he became a liability with his

business empire collapsing as a former Mossad operative has confirmed (see *The Trigger*).

Money, money, money, funny money ...

Before I come to the Sabbatian connection with the last three US presidents I will lay out the crucial importance to Sabbatians of controlling banking and finance. Sabbatian Mayer Amschel Rothschild set out to dominate this arena in his family's quest for total global control. What is freedom? It is, in effect, choice. The more choices you have the freer you are and the fewer your choices the more you are enslaved. In the global structure created over centuries by Sabbatians the biggest decider and restrictor of choice is ... money. Across the world if you ask people what they would like to do with their lives and why they are not doing that they will reply 'I don't have the money'. This is the idea. A global elite of multi-billionaires are described as 'greedy' and that is true on one level; but control of money – who has it and who doesn't – is not primarily about greed. It's about control. Sabbatians have seized ever more control of finance and sucked the wealth of the world out of the hands of the population. We talk now, after all, about the 'One-percent' and even then the wealthiest are a lot fewer even than that. This has been made possible by a money scam so outrageous and so vast it could rightly be called the scam of scams founded on creating 'money' out of nothing and 'loaning' that with interest to the population. Money out of nothing is called 'credit'. Sabbatians have asserted control over governments and banking ever more completely through the centuries and secured financial laws that allow banks to lend hugely more than they have on deposit in a confidence trick known as fractional reserve lending. Imagine if you could lend money that doesn't exist and charge the recipient interest for doing so. You would end up in jail. Bankers by contrast end up in mansions, private jets, Malibu and Monaco.

Banks are only required to keep a fraction of their deposits and wealth in their vaults and they are allowed to lend 'money' they don't have called 'credit'. Go into a bank for a loan and if you succeed

the banker will not move any real wealth into your account. They will type into your account the amount of the agreed 'loan' – say £100,000. This is not wealth that really exists; it is non-existent, fresh-air, created-out-of-nothing 'credit' which has never, does not, and will never exist except in theory. Credit is backed by nothing except wind and only has buying power because people think that it has buying power and accept it in return for property, goods and services. I have described this situation as like those cartoon characters you see chasing each other and when they run over the edge of a cliff they keep running forward on fresh air until one of them looks down, realises what's happened, and they all crash into the ravine. The whole foundation of the Sabbatian financial system is to stop people looking down except for periodic moments when they want to crash the system (as in 2008 and 2020 ongoing) and reap the rewards from all the property, businesses and wealth their borrowers had signed over as 'collateral' in return for a 'loan' of fresh air. Most people think that money is somehow created by governments when it comes into existence from the start as a debt through banks 'lending' illusory money called credit. Yes, the very currency of exchange is a *debt* from day one issued as an interest-bearing loan. Why don't governments create money interest-free and lend it to their people interest-free? Governments are controlled by Sabbatians and the financial system is controlled by Sabbatians for whom interest-free money would be a nightmare come true. Sabbatians underpin their financial domination through their global network of central banks, including the privately-owned US Federal Reserve and Britain's Bank of England, and this is orchestrated by a privately-owned central bank coordination body called the Bank for International Settlements in Basle, Switzerland, created by the usual suspects including the Rockefellers and Rothschilds. Central bank chiefs don't answer to governments or the people. They answer to the Bank for International Settlements or, in other words, the Global Cult which is dominated today by Sabbatians.

Built-in disaster

There are so many constituent scams within the overall banking scam. When you take out a loan of thin-air credit only the amount of that loan is theoretically brought into circulation to add to the amount in circulation; but you are paying back the principle plus interest. The additional interest is not created and this means that with every 'loan' there is a shortfall in the money in circulation between what is borrowed and what has to be paid back. There is never even close to enough money in circulation to repay all outstanding public and private debt including interest. Coldly weaved in the very fabric of the system is the certainty that some will lose their homes, businesses and possessions to the banking 'lender'. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts it becomes painfully obvious that there is not enough money to service all debt and interest. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts and it becomes painfully obvious – as in 2008 and currently – that there is not enough money to service all debt and interest. Sabbatian banksters have been leading the human population through a calculated series of booms (more debt incurred) and busts (when the debt can't be repaid and the banks get the debtor's tangible wealth in exchange for non-existent 'credit'). With each 'bust' Sabbatian bankers have absorbed more of the world's tangible wealth and we end up with the One-percent. Governments are in bankruptcy levels of debt to the same system and are therefore owned by a system they do not control. The Federal Reserve, 'America's central bank', is privately-owned and American presidents only nominally appoint its chairman or woman to maintain the illusion that it's an arm of government. It's not. The 'Fed' is a cartel of private banks which handed billions to its associates and friends after the crash of 2008 and has been Sabbatian-controlled since it was manipulated into being in 1913 through the covert trickery of Rothschild banking agents Jacob Schiff and Paul

Warburg, and the Sabbatian Rockefeller family. Somehow from a Jewish population of two-percent and globally 0.2 percent (Sabbatian interlopers remember are far smaller) ultra-Zionists headed the Federal Reserve for 31 years between 1987 and 2018 in the form of Alan Greenspan, Bernard Bernanke and Janet Yellen (now Biden's Treasury Secretary) with Yellen's deputy chairman a Israeli-American dual citizen and ultra-Zionist Stanley Fischer, a former governor of the Bank of Israel. Ultra-Zionist Fed chiefs spanned the presidencies of Ronald Reagan ('Republican'), Father George Bush ('Republican'), Bill Clinton ('Democrat'), Boy George Bush ('Republican') and Barack Obama ('Democrat'). We should really add the pre-Greenspan chairman, Paul Adolph Volcker, 'appointed' by Jimmy Carter ('Democrat') who ran the Fed between 1979 and 1987 during the Carter and Reagan administrations before Greenspan took over. Volcker was a long-time associate and business partner of the Rothschilds. No matter what the 'party' officially in power the United States economy was directed by the same force. Here are members of the Obama, Trump and Biden administrations and see if you can make out a common theme.

Barack Obama ('Democrat')

Ultra-Zionists Robert Rubin, Larry Summers, and Timothy Geithner ran the US Treasury in the Clinton administration and two of them reappeared with Obama. Ultra-Zionist Fed chairman Alan Greenspan had manipulated the crash of 2008 through deregulation and jumped ship just before the disaster to make way for ultra-Zionist Bernard Bernanke to hand out trillions to Sabbatian 'too big to fail' banks and businesses, including the ubiquitous ultra-Zionist Goldman Sachs which has an ongoing staff revolving door operation between itself and major financial positions in government worldwide. Obama inherited the fallout of the crash when he took office in January, 2009, and fortunately he had the support of his ultra-Zionist White House Chief of Staff Rahm Emmanuel, son of a terrorist who helped to bomb Israel into being in 1948, and his ultra-Zionist senior adviser David Axelrod, chief strategist in Obama's two

successful presidential campaigns. Emmanuel, later mayor of Chicago and former senior fundraiser and strategist for Bill Clinton, is an example of the Sabbatian policy after Israel was established of migrating insider families to America so their children would be born American citizens. 'Obama' chose this financial team throughout his administration to respond to the Sabbatian-instigated crisis:

Timothy Geithner (ultra-Zionist) Treasury Secretary; Jacob J. Lew, Treasury Secretary; Larry Summers (ultra-Zionist), director of the White House National Economic Council; Paul Adolph Volcker (Rothschild business partner), chairman of the Economic Recovery Advisory Board; Peter Orszag (ultra-Zionist), director of the Office of Management and Budget overseeing all government spending; Penny Pritzker (ultra-Zionist), Commerce Secretary; Jared Bernstein (ultra-Zionist), chief economist and economic policy adviser to Vice President Joe Biden; Mary Schapiro (ultra-Zionist), chair of the Securities and Exchange Commission (SEC); Gary Gensler (ultra-Zionist), chairman of the Commodity Futures Trading Commission (CFTC); Sheila Bair (ultra-Zionist), chair of the Federal Deposit Insurance Corporation (FDIC); Karen Mills (ultra-Zionist), head of the Small Business Administration (SBA); Kenneth Feinberg (ultra-Zionist), Special Master for Executive [bail-out] Compensation. Feinberg would be appointed to oversee compensation (with strings) to 9/11 victims and families in a campaign to stop them having their day in court to question the official story. At the same time ultra-Zionist Bernard Bernanke was chairman of the Federal Reserve and these are only some of the ultra-Zionists with allegiance to Sabbatian-controlled Israel in the Obama government. Obama's biggest corporate donor was ultra-Zionist Goldman Sachs which had employed many in his administration.

Donald Trump ('Republican')

Trump claimed to be an outsider (he wasn't) who had come to 'drain the swamp'. He embarked on this goal by immediately appointing ultra-Zionist Steve Mnuchin, a Goldman Sachs employee for 17

years, as his Treasury Secretary. Others included Gary Cohn (ultra-Zionist), chief operating officer of Goldman Sachs, his first Director of the National Economic Council and chief economic adviser, who was later replaced by Larry Kudlow (ultra-Zionist). Trump's senior adviser throughout his four years in the White House was his sinister son-in-law Jared Kushner, a life-long friend of Israel Prime Minister Benjamin Netanyahu. Kushner is the son of a convicted crook who was pardoned by Trump in his last days in office. Other ultra-Zionists in the Trump administration included: Stephen Miller, Senior Policy Adviser; Avrahm Berkowitz, Deputy Adviser to Trump and his Senior Adviser Jared Kushner; Ivanka Trump, Adviser to the President, who converted to Judaism when she married Jared Kushner; David Friedman, Trump lawyer and Ambassador to Israel; Jason Greenblatt, Trump Organization executive vice president and chief legal officer, who was made Special Representative for International Negotiations and the Israeli-Palestinian Conflict; Rod Rosenstein, Deputy Attorney General; Elliot Abrams, Special Representative for Venezuela, then Iran; John Eisenberg, National Security Council Legal Adviser and Deputy Council to the President for National Security Affairs; Anne Neuberger, Deputy National Manager, National Security Agency; Ezra Cohen-Watnick, Acting Under Secretary of Defense for Intelligence; Elan Carr, Special Envoy to monitor and combat anti-Semitism; Len Khodorkovsky, Deputy Special Envoy to monitor and combat anti-Semitism; Reed Cordish, Assistant to the President, Intragovernmental and Technology Initiatives. Trump Vice President Mike Pence and Secretary of State Mike Pompeo, both Christian Zionists, were also vehement supporters of Israel and its goals and ambitions.

Donald 'free-speech believer' Trump pardoned a number of financial and violent criminals while ignoring calls to pardon Julian Assange and Edward Snowden whose crimes are revealing highly relevant information about government manipulation and corruption and the widespread illegal surveillance of the American people by US 'security' agencies. It's so good to know that Trump is on the side of freedom and justice and not mega-criminals with

allegiance to Sabbatian-controlled Israel. These included a pardon for Israeli spy Jonathan Pollard who was jailed for life in 1987 under the Espionage Act. Aviem Sella, the Mossad agent who recruited Pollard, was also pardoned by Trump while Assange sat in jail and Snowden remained in exile in Russia. Sella had 'fled' (was helped to escape) to Israel in 1987 and was never extradited despite being charged under the Espionage Act. A Trump White House statement said that Sella's clemency had been 'supported by Benjamin Netanyahu, Ron Dermer, Israel's US Ambassador, David Friedman, US Ambassador to Israel and Miriam Adelson, wife of leading Trump donor Sheldon Adelson who died shortly before. Other friends of Jared Kushner were pardoned along with Sholom Weiss who was believed to be serving the longest-ever white-collar prison sentence of more than 800 years in 2000. The sentence was commuted of Ponzi-schemer Eliyahu Weinstein who defrauded Jews and others out of \$200 million. I did mention that Assange and Snowden were ignored, right? Trump gave Sabbatians almost everything they asked for in military and political support, moving the US Embassy from Tel Aviv to Jerusalem with its critical symbolic and literal implications for Palestinian statehood, and the 'deal of the Century' designed by Jared Kushner and David Friedman which gave the Sabbatian Israeli government the green light to substantially expand its already widespread program of building illegal Jewish-only settlements in the occupied land of the West Bank. This made a two-state 'solution' impossible by seizing all the land of a potential Palestinian homeland and that had been the plan since 1948 and then 1967 when the Arab-controlled Gaza Strip, West Bank, Sinai Peninsula and Syrian Golan Heights were occupied by Israel. All the talks about talks and road maps and delays have been buying time until the West Bank was physically occupied by Israeli real estate. Trump would have to be a monumentally ill-informed idiot not to see that this was the plan he was helping to complete. The Trump administration was in so many ways the Kushner administration which means the Netanyahu administration which means the Sabbatian administration. I understand why many opposing Cult fascism in all its forms gravitated to Trump, but he

was a crucial part of the Sabbatian plan and I will deal with this in the next chapter.

Joe Biden ('Democrat')

A barely cognitive Joe Biden took over the presidency in January, 2021, along with his fellow empty shell, Vice-President Kamala Harris, as the latest Sabbatian gofers to enter the White House. Names on the door may have changed and the 'party' – the force behind them remained the same as Zionists were appointed to a stream of pivotal areas relating to Sabbatian plans and policy. They included: Janet Yellen, Treasury Secretary, former head of the Federal Reserve, and still another ultra-Zionist running the US Treasury after Mnuchin (Trump), Lew and Geithner (Obama), and Summers and Rubin (Clinton); Anthony Blinken, Secretary of State; Wendy Sherman, Deputy Secretary of State (so that's 'Biden's' Sabbatian foreign policy sorted); Jeff Zients, White House coronavirus coordinator; Rochelle Walensky, head of the Centers for Disease Control; Rachel Levine, transgender deputy health secretary (that's 'Covid' hoax policy under control); Merrick Garland, Attorney General; Alejandro Mayorkas, Secretary of Homeland Security; Cass Sunstein, Homeland Security with responsibility for new immigration laws; Avril Haines, Director of National Intelligence; Anne Neuberger, National Security Agency cybersecurity director (note, cybersecurity); David Cohen, CIA Deputy Director; Ronald Klain, Biden's Chief of Staff (see Rahm Emanuel); Eric Lander, a 'leading geneticist', Office of Science and Technology Policy director (see Smart Grid, synthetic biology agenda); Jessica Rosenworcel, acting head of the Federal Communications Commission (FCC) which controls Smart Grid technology policy and electromagnetic communication systems including 5G. How can it be that so many pivotal positions are held by two-percent of the American population and 0.2 percent of the world population administration after administration no matter who is the president and what is the party? It's a coincidence? Of course it's not and this is why Sabbatians have built their colossal global web of interlocking 'anti-

hate' hate groups to condemn anyone who asks these glaring questions as an 'anti-Semite'. The way that Jewish people horrifically abused in Sabbatian-backed Nazi Germany are exploited to this end is stomach-turning and disgusting beyond words.

Political fusion

Sabbatian manipulation has reversed the roles of Republicans and Democrats and the same has happened in Britain with the Conservative and Labour Parties. Republicans and Conservatives were always labelled the 'right' and Democrats and Labour the 'left', but look at the policy positions now and the Democrat-Labour 'left' has moved further to the 'right' than Republicans and Conservatives under the banner of 'Woke', the Cult-created far-right tyranny. Where once the Democrat-Labour 'left' defended free speech and human rights they now seek to delete them and as I said earlier despite the 'Covid' fascism of the Jackboot Johnson Conservative government in the UK the Labour Party of leader Keir Starmer demanded even more extreme measures. The Labour Party has been very publicly absorbed by Sabbatians after a political and media onslaught against the previous leader, the weak and inept Jeremy Corbyn, over made-up allegations of 'anti-Semitism' both by him and his party. The plan was clear with this 'anti-Semite' propaganda and what was required in response was a swift and decisive 'fuck off' from Corbyn and a statement to expose the Anti-Semitism Industry (Sabbatian) attempt to silence Labour criticism of the Israeli government (Sabbatians) and purge the party of all dissent against the extremes of ultra-Zionism (Sabbatians). Instead Corbyn and his party fell to their knees and appeased the abusers which, by definition, is impossible. Appeasing one demand leads only to a new demand to be appeased until takeover is complete. Like I say – 'fuck off' would have been a much more effective policy and I have used it myself with great effect over the years when Sabbatians are on my case which is most of the time. I consider that fact a great compliment, by the way. The outcome of the Labour Party capitulation is that we now have a Sabbatian-controlled

Conservative Party 'opposed' by a Sabbatian-controlled Labour Party in a one-party Sabbatian state that hurtles towards the extremes of tyranny (the Sabbatian cult agenda). In America the situation is the same. Labour's Keir Starmer spends his days on his knees with his tongue out pointing to Tel Aviv, or I guess now Jerusalem, while Boris Johnson has an 'anti-Semitism czar' in the form of former Labour MP John Mann who keeps Starmer company on his prayer mat.

Sabbatian influence can be seen in Jewish members of the Labour Party who have been ejected for criticism of Israel including those from families that suffered in Nazi Germany. Sabbatians despise real Jewish people and target them even more harshly because it is so much more difficult to dub them 'anti-Semitic' although in their desperation they do try.

CHAPTER THREE

The Pushbacker sting

*Until you realize how easy it is for your mind to be manipulated, you
remain the puppet of someone else's game*

Evita Ochel

I will use the presidencies of Trump and Biden to show how the manipulation of the one-party state plays out behind the illusion of political choice across the world. No two presidencies could – on the face of it – be more different and apparently at odds in terms of direction and policy.

A Renegade Mind sees beyond the obvious and focuses on outcomes and consequences and not image, words and waffle. The Cult embarked on a campaign to divide America between those who blindly support its agenda (the mentality known as 'Woke') and those who are pushing back on where the Cult and its Sabbatians want to go. This presents infinite possibilities for dividing and ruling the population by setting them at war with each other and allows a perceptual ring fence of demonisation to encircle the Pushbackers in a modern version of the Little Big Horn in 1876 when American cavalry led by Lieutenant Colonel George Custer were drawn into a trap, surrounded and killed by Native American tribes defending their land of thousands of years from being seized by the government. In this modern version the roles are reversed and it's those defending themselves from the Sabbatian government who are surrounded and the government that's seeking to destroy them. This trap was set years ago and to explain how we must return to 2016

and the emergence of Donald Trump as a candidate to be President of the United States. He set out to overcome the best part of 20 other candidates in the Republican Party before and during the primaries and was not considered by many in those early stages to have a prayer of living in the White House. The Republican Party was said to have great reservations about Trump and yet somehow he won the nomination. When you know how American politics works – politics in general – there is no way that Trump could have become the party's candidate unless the Sabbatian-controlled 'Neocons' that run the Republican Party wanted that to happen. We saw the proof in emails and documents made public by WikiLeaks that the Democratic Party hierarchy, or Democons, systematically undermined the campaign of Bernie Sanders to make sure that Sabbatian gofer Hillary Clinton won the nomination to be their presidential candidate. If the Democons could do that then the Neocons in the Republican Party could have derailed Trump in the same way. But they didn't and at that stage I began to conclude that Trump could well be the one chosen to be president. If that was the case the 'why' was pretty clear to see – the goal of dividing America between Cult agenda-supporting Wokers and Pushbackers who gravitated to Trump because he was telling them what they wanted to hear. His constituency of support had been increasingly ignored and voiceless for decades and profoundly through the eight years of Sabbatian puppet Barack Obama. Now here was someone speaking their language of pulling back from the incessant globalisation of political and economic power, the exporting of American jobs to China and elsewhere by 'American' (Sabbatian) corporations, the deletion of free speech, and the mass immigration policies that had further devastated job opportunities for the urban working class of all races and the once American heartlands of the Midwest.

Beware the forked tongue

Those people collectively sighed with relief that at last a political leader was apparently on their side, but another trait of the Renegade Mind is that you look even harder at people telling you

what you want to hear than those who are telling you otherwise. Obviously as I said earlier people wish what they want to hear to be true and genuine and they are much more likely to believe that than someone saying what they don't want to hear and don't want to be true. Sales people are taught to be skilled in eliciting by calculated questioning what their customers want to hear and repeating that back to them as their own opinion to get their targets to like and trust them. Assets of the Cult are also sales people in the sense of selling perception. To read Cult manipulation you have to play the long and expanded game and not fall for the Vaudeville show of party politics. Both American parties are vehicles for the Cult and they exploit them in different ways depending on what the agenda requires at that moment. Trump and the Republicans were used to be the focus of dividing America and isolating Pushbackers to open the way for a Biden presidency to become the most extreme in American history by advancing the full-blown Woke (Cult) agenda with the aim of destroying and silencing Pushbackers now labelled Nazi Trump supporters and white supremacists.

Sabbatians wanted Trump in office for the reasons described by ultra-Zionist Saul Alinsky (1909-1972) who was promoting the Woke philosophy through 'community organising' long before anyone had heard of it. In those days it still went by its traditional name of Marxism. The reason for the manipulated Trump phenomenon was laid out in Alinsky's 1971 book, *Rules for Radicals*, which was his blueprint for overthrowing democratic and other regimes and replacing them with Sabbatian Marxism. Not surprisingly his to-do list was evident in the Sabbatian French and Russian 'Revolutions' and that in China which will become very relevant in the next chapter about the 'Covid' hoax. Among Alinsky's followers have been the deeply corrupt Barack Obama, House Speaker Nancy Pelosi and Hillary Clinton who described him as a 'hero'. All three are Sabbatian stooges with Pelosi personifying the arrogant corrupt idiocy that so widely fronts up for the Cult inner core. Predictably as a Sabbatian advocate of the 'light-bringer' Alinsky features Lucifer on the dedication page of his book as the original radical who gained

his own kingdom ('Earth' as we shall see). One of Alinsky's golden radical rules was to pick an individual and focus all attention, hatred and blame on them and not to target faceless bureaucracies and corporations. *Rules for Radicals* is really a Sabbatian handbook with its contents repeatedly employed all over the world for centuries and why wouldn't Sabbatians bring to power their designer-villain to be used as the individual on which all attention, hatred and blame was bestowed? This is what they did and the only question for me is how much Trump knew that and how much he was manipulated. A bit of both, I suspect. This was Alinsky's Trump technique from a man who died in 1972. The technique has spanned history:

Pick the target, freeze it, personalize it, polarize it. Don't try to attack abstract corporations or bureaucracies. Identify a responsible individual. Ignore attempts to shift or spread the blame.

From the moment Trump came to illusory power everything was about him. It wasn't about Republican policy or opinion, but all about Trump. Everything he did was presented in negative, derogatory and abusive terms by the Sabbatian-dominated media led by Cult operations such as CNN, MSNBC, *The New York Times* and the Jeff Bezos-owned *Washington Post* – 'Pick the target, freeze it, personalize it, polarize it.' Trump was turned into a demon to be vilified by those who hated him and a demi-god loved by those who worshipped him. This, in turn, had his supporters, too, presented as equally demonic in preparation for the punchline later down the line when Biden was about to take office. It was here's a Trump, there's a Trump, everywhere a Trump, Trump. Virtually every news story or happening was filtered through the lens of 'The Donald'. You loved him or hated him and which one you chose was said to define you as Satan's spawn or a paragon of virtue. Even supporting some Trump policies or statements and not others was enough for an assault on your character. No shades of grey were or are allowed. Everything is black and white (literally and figuratively). A Californian I knew had her head utterly scrambled by her hatred for Trump while telling people they should love each other. She was so totally consumed by

Trump Derangement Syndrome as it became to be known that this glaring contradiction would never have occurred to her. By definition anyone who criticised Trump or praised his opponents was a hero and this lady described Joe Biden as 'a kind, honest gentleman' when he's a provable liar, mega-crook and vicious piece of work to boot. Sabbatians had indeed divided America using Trump as the fall-guy and all along the clock was ticking on the consequences for his supporters.

In hock to his masters

Trump gave Sabbatians via Israel almost everything they wanted in his four years. Ask and you shall receive was the dynamic between himself and Benjamin Netanyahu orchestrated by Trump's ultra-Zionist son-in-law Jared Kushner, his ultra-Zionist Ambassador to Israel, David Friedman, and ultra-Zionist 'Israel adviser', Jason Greenblatt. The last two were central to the running and protecting from collapse of his business empire, the Trump Organisation, and colossal business failures made him forever beholding to Sabbatian networks that bailed him out. By the start of the 1990s Trump owed \$4 billion to banks that he couldn't pay and almost \$1 billion of that was down to him personally and not his companies. This mega-disaster was the result of building two new casinos in Atlantic City and buying the enormous Taj Mahal operation which led to crippling debt payments. He had borrowed fantastic sums from 72 banks with major Sabbatian connections and although the scale of debt should have had him living in a tent alongside the highway they never foreclosed. A plan was devised to lift Trump from the mire by BT Securities Corporation and Rothschild Inc. and the case was handled by Wilber Ross who had worked for the Rothschilds for 27 years. Ross would be named US Commerce Secretary after Trump's election. Another crucial figure in saving Trump was ultra-Zionist 'investor' Carl Icahn who bought the Taj Mahal casino. Icahn was made special economic adviser on financial regulation in the Trump administration. He didn't stay long but still managed to find time to make a tidy sum of a reported \$31.3 million when he sold his

holdings affected by the price of steel three days before Trump imposed a 235 percent tariff on steel imports. What amazing bits of luck these people have. Trump and Sabbatian operatives have long had a close association and his mentor and legal adviser from the early 1970s until 1986 was the dark and genetically corrupt ultra-Zionist Roy Cohn who was chief counsel to Senator Joseph McCarthy's 'communist' witch-hunt in the 1950s. *Esquire* magazine published an article about Cohn with the headline 'Don't mess with Roy Cohn'. He was described as the most feared lawyer in New York and 'a ruthless master of dirty tricks ... [with] ... more than one Mafia Don on speed dial'. Cohn's influence, contacts, support and protection made Trump a front man for Sabbatians in New York with their connections to one of Cohn's many criminal employers, the 'Russian' Sabbatian Mafia. Israel-centric media mogul Rupert Murdoch was introduced to Trump by Cohn and they started a long friendship. Cohn died in 1986 weeks after being disbarred for unethical conduct by the Appellate Division of the New York State Supreme Court. The wheels of justice do indeed run slow given the length of Cohn's crooked career.

QAnon-sense

We are asked to believe that Donald Trump with his fundamental connections to Sabbatian networks and operatives has been leading the fight to stop the Sabbatian agenda for the fascistic control of America and the world. Sure he has. A man entrapped during his years in the White House by Sabbatian operatives and whose biggest financial donor was casino billionaire Sheldon Adelson who was Sabbatian to his DNA?? Oh, do come on. Trump has been used to divide America and isolate Pushbackers on the Cult agenda under the heading of 'Trump supporters', 'insurrectionists' and 'white supremacists'. The US Intelligence/Mossad Psyop or psychological operation known as QAnon emerged during the Trump years as a central pillar in the Sabbatian campaign to lead Pushbackers into the trap set by those that wished to destroy them. I knew from the start that QAnon was a scam because I had seen the same scenario many

times before over 30 years under different names and I had written about one in particular in the books. 'Not again' was my reaction when QAnon came to the fore. The same script is pulled out every few years and a new name added to the letterhead. The story always takes the same form: 'Insiders' or 'the good guys' in the government-intelligence-military 'Deep State' apparatus were going to instigate mass arrests of the 'bad guys' which would include the Rockefellers, Rothschilds, Barack Obama, Hillary Clinton, George Soros, etc., etc. Dates are given for when the 'good guys' are going to move in, but the dates pass without incident and new dates are given which pass without incident. The central message to Pushbackers in each case is that they don't have to do anything because there is 'a plan' and it is all going to be sorted by the 'good guys' on the inside. 'Trust the plan' was a QAnon mantra when the only plan was to misdirect Pushbackers into putting their trust in a Psyop they believed to be real. Beware, beware, those who tell you what you want to hear and always check it out. Right up to Biden's inauguration QAnon was still claiming that 'the Storm' was coming and Trump would stay on as president when Biden and his cronies were arrested and jailed. It was never going to happen and of course it didn't, but what did happen as a result provided that punchline to the Sabbatian Trump/QAnon Psyop.

On January 6th, 2021, a very big crowd of Trump supporters gathered in the National Mall in Washington DC down from the Capitol Building to protest at what they believed to be widespread corruption and vote fraud that stopped Trump being re-elected for a second term as president in November, 2020. I say as someone that does not support Trump or Biden that the evidence is clear that major vote-fixing went on to favour Biden, a man with cognitive problems so advanced he can often hardly string a sentence together without reading the words written for him on the Teleprompter. Glaring ballot discrepancies included serious questions about electronic voting machines that make vote rigging a comparative cinch and hundreds of thousands of paper votes that suddenly appeared during already advanced vote counts and virtually all of

them for Biden. Early Trump leads in crucial swing states suddenly began to close and disappear. The pandemic hoax was used as the excuse to issue almost limitless numbers of mail-in ballots with no checks to establish that the recipients were still alive or lived at that address. They were sent to streams of people who had not even asked for them. Private organisations were employed to gather these ballots and who knows what they did with them before they turned up at the counts. The American election system has been manipulated over decades to become a sick joke with more holes than a Swiss cheese for the express purpose of dictating the results. Then there was the criminal manipulation of information by Sabbatian tech giants like Facebook, Twitter and Google-owned YouTube which deleted pro-Trump, anti-Biden accounts and posts while everything in support of Biden was left alone. Sabbatians wanted Biden to win because after the dividing of America it was time for full-on Woke and every aspect of the Cult agenda to be unleashed.

Hunter gatherer

Extreme Silicon Valley bias included blocking information by the *New York Post* exposing a Biden scandal that should have ended his bid for president in the final weeks of the campaign. Hunter Biden, his monumentally corrupt son, is reported to have sent a laptop to be repaired at a local store and failed to return for it. Time passed until the laptop became the property of the store for non-payment of the bill. When the owner saw what was on the hard drive he gave a copy to the FBI who did nothing even though it confirmed widespread corruption in which the Joe Biden family were using his political position, especially when he was vice president to Obama, to make multiple millions in countries around the world and most notably Ukraine and China. Hunter Biden's one-time business partner Tony Bobulinski went public when the story broke in the *New York Post* to confirm the corruption he saw and that Joe Biden not only knew what was going on he also profited from the spoils. Millions were handed over by a Chinese company with close

connections – like all major businesses in China – to the Chinese communist party of President Xi Jinping. Joe Biden even boasted at a meeting of the Cult's World Economic Forum that as vice president he had ordered the government of Ukraine to fire a prosecutor. What he didn't mention was that the same man just happened to be investigating an energy company which was part of Hunter Biden's corrupt portfolio. The company was paying him big bucks for no other reason than the influence his father had. Overnight Biden's presidential campaign should have been over given that he had lied publicly about not knowing what his son was doing. Instead almost the entire Sabbatian-owned mainstream media and Sabbatian-owned Silicon Valley suppressed circulation of the story. This alone went a mighty way to rigging the election of 2020. Cult assets like Mark Zuckerberg at Facebook also spent hundreds of millions to be used in support of Biden and vote 'administration'.

The Cult had used Trump as the focus to divide America and was now desperate to bring in moronic, pliable, corrupt Biden to complete the double-whammy. No way were they going to let little things like the will of the people thwart their plan. Silicon Valley widely censored claims that the election was rigged because it *was* rigged. For the same reason anyone claiming it was rigged was denounced as a 'white supremacist' including the pathetically few Republican politicians willing to say so. Right across the media where the claim was mentioned it was described as a 'false claim' even though these excuses for 'journalists' would have done no research into the subject whatsoever. Trump won seven million more votes than any sitting president had ever achieved while somehow a cognitively-challenged soon to be 78-year-old who was hidden away from the public for most of the campaign managed to win more votes than any presidential candidate in history. It makes no sense. You only had to see election rallies for both candidates to witness the enthusiasm for Trump and the apathy for Biden. Tens of thousands would attend Trump events while Biden was speaking in empty car parks with often only television crews attending and framing their shots to hide the fact that no one was there. It was pathetic to see

footage come to light of Biden standing at a podium making speeches only to TV crews and party fixers while reading the words written for him on massive Teleprompter screens. So, yes, those protestors on January 6th had a point about election rigging, but some were about to walk into a trap laid for them in Washington by the Cult Deep State and its QAnon Psyop. This was the Capitol Hill riot ludicrously dubbed an 'insurrection'.

The spider and the fly

Renegade Minds know there are not two 'sides' in politics, only one side, the Cult, working through all 'sides'. It's a stage show, a puppet show, to direct the perceptions of the population into focusing on diversions like parties and candidates while missing the puppeteers with their hands holding all the strings. The Capitol Hill 'insurrection' brings us back to the Little Big Horn. Having created two distinct opposing groupings – Woke and Pushbackers – the trap was about to be sprung. Pushbackers were to be encircled and isolated by associating them all in the public mind with Trump and then labelling Trump as some sort of Confederate leader. I knew immediately that the Capitol riot was a set-up because of two things. One was how easy the rioters got into the building with virtually no credible resistance and secondly I could see – as with the 'Covid' hoax in the West at the start of 2020 – how the Cult could exploit the situation to move its agenda forward with great speed. My experience of Cult techniques and activities over more than 30 years has showed me that while they do exploit situations they haven't themselves created this never happens with events of fundamental agenda significance. Every time major events giving cultists the excuse to rapidly advance their plan you find they are manipulated into being for the specific reason of providing that excuse – Problem-Reaction-Solution. Only a tiny minority of the huge crowd of Washington protestors sought to gain entry to the Capitol by smashing windows and breaching doors. That didn't matter. The whole crowd and all Pushbackers, even if they did not support Trump, were going to be lumped together as dangerous

insurrectionists and conspiracy theorists. The latter term came into widespread use through a CIA memo in the 1960s aimed at discrediting those questioning the nonsensical official story of the Kennedy assassination and it subsequently became widely employed by the media. It's still being used by inept 'journalists' with no idea of its origin to discredit anyone questioning anything that authority claims to be true. When you are perpetrating a conspiracy you need to discredit the very word itself even though the dictionary definition of conspiracy is merely 'the activity of secretly planning with other people to do something bad or illegal' and 'a general agreement to keep silent about a subject for the purpose of keeping it secret'. On that basis there are conspiracies almost wherever you look. For obvious reasons the Cult and its lapdog media have to claim there are no conspiracies even though the word appears in state laws as with conspiracy to defraud, to murder, and to corrupt public morals.

Agent provocateurs are widely used by the Cult Deep State to manipulate genuine people into acting in ways that suit the desired outcome. By genuine in this case I mean protestors genuinely supporting Trump and claims that the election was stolen. In among them, however, were agents of the state wearing the garb of Trump supporters and QAnon to pump-prime the Capital riot which some genuine Trump supporters naively fell for. I described the situation as 'Come into my parlour said the spider to the fly'. Leaflets appeared through the Woke paramilitary arm Antifa, the anti-fascist fascists, calling on supporters to turn up in Washington looking like Trump supporters even though they hated him. Some of those arrested for breaching the Capitol Building were sourced to Antifa and its stable mate Black Lives Matter. Both organisations are funded by Cult billionaires and corporations. One man charged for the riot was according to his lawyer a former FBI agent who had held top secret security clearance for 40 years. Attorney Thomas Plofchan said of his client, 66-year-old Thomas Edward Caldwell:

He has held a Top Secret Security Clearance since 1979 and has undergone multiple Special Background Investigations in support of his clearances. After retiring from the Navy, he

worked as a section chief for the Federal Bureau of Investigation from 2009-2010 as a GS-12 [mid-level employee].

He also formed and operated a consulting firm performing work, often classified, for U.S government customers including the US. Drug Enforcement Agency, Department of Housing and Urban Development, the US Coast Guard, and the US Army Personnel Command.

A judge later released Caldwell pending trial in the absence of evidence about a conspiracy or that he tried to force his way into the building. *The New York Post* reported a 'law enforcement source' as saying that 'at least two known Antifa members were spotted' on camera among Trump supporters during the riot while one of the rioters arrested was John Earle Sullivan, a seriously extreme Black Lives Matter Trump-hater from Utah who was previously arrested and charged in July, 2020, over a BLM-Antifa riot in which drivers were threatened and one was shot. Sullivan is the founder of Utah-based Insurgence USA which is an affiliate of the Cult-created-and-funded Black Lives Matter movement. Footage appeared and was then deleted by Twitter of Trump supporters calling out Antifa infiltrators and a group was filmed changing into pro-Trump clothing before the riot. Security at the building was *pathetic* – as planned. Colonel Leroy Fletcher Prouty, a man with long experience in covert operations working with the US security apparatus, once described the tell-tale sign to identify who is involved in an assassination. He said:

No one has to direct an assassination – it happens. The active role is played secretly by permitting it to happen. This is the greatest single clue. Who has the power to call off or reduce the usual security precautions?

This principle applies to many other situations and certainly to the Capitol riot of January 6th, 2021.

The sting

With such a big and potentially angry crowd known to be gathering near the Capitol the security apparatus would have had a major police detail to defend the building with National Guard troops on

standby given the strength of feeling among people arriving from all over America encouraged by the QAnon Psyop and statements by Donald Trump. Instead Capitol Police 'security' was flimsy, weak, and easily breached. The same number of officers was deployed as on a regular day and that is a blatant red flag. They were not staffed or equipped for a possible riot that had been an obvious possibility in the circumstances. No protective and effective fencing worth the name was put in place and there were no contingency plans. The whole thing was basically a case of standing aside and waving people in. Once inside police mostly backed off apart from one Capitol police officer who ridiculously shot dead unarmed Air Force veteran protestor Ashli Babbitt without a warning as she climbed through a broken window. The 'investigation' refused to name or charge the officer after what must surely be considered a murder in the circumstances. They just lifted a carpet and swept. The story was endlessly repeated about five people dying in the 'armed insurrection' when there was no report of rioters using weapons. Apart from Babbitt the other four died from a heart attack, strokes and apparently a drug overdose. Capitol police officer Brian Sicknick was reported to have died after being bludgeoned with a fire extinguisher when he was alive after the riot was over and died later of what the Washington Medical Examiner's Office said was a stroke. Sicknick had no external injuries. The lies were delivered like rapid fire. There was a narrative to build with incessant repetition of the lie until the lie became the accepted 'everybody knows that' truth. The 'Big Lie' technique of Nazi Propaganda Minister Joseph Goebbels is constantly used by the Cult which was behind the Nazis and is today behind the 'Covid' and 'climate change' hoaxes. Goebbels said:

If you tell a lie big enough and keep repeating it, people will eventually come to believe it. The lie can be maintained only for such time as the State can shield the people from the political, economic and/or military consequences of the lie. It thus becomes vitally important for the State to use all of its powers to repress dissent, for the truth is the mortal enemy of the lie, and thus by extension, the truth is the greatest enemy of the State.

Most protestors had a free run of the Capitol Building. This allowed pictures to be taken of rioters in iconic parts of the building including the Senate chamber which could be used as propaganda images against all Pushbackers. One Congresswoman described the scene as 'the worst kind of non-security anybody could ever imagine'. Well, the first part was true, but someone obviously did imagine it and made sure it happened. Some photographs most widely circulated featured people wearing QAnon symbols and now the Psyop would be used to dub all QAnon followers with the ubiquitous fit-all label of 'white supremacist' and 'insurrectionists'. When a Muslim extremist called Noah Green drove his car at two police officers at the Capitol Building killing one in April, 2021, there was no such political and media hysteria. They were just disappointed he wasn't white.

The witch-hunt

Government prosecutor Michael Sherwin, an aggressive, dark-eyed, professional Rottweiler led the 'investigation' and to call it over the top would be to understate reality a thousand fold. Hundreds were tracked down and arrested for the crime of having the wrong political views and people were jailed who had done nothing more than walk in the building, committed no violence or damage to property, took a few pictures and left. They were labelled a 'threat to the Republic' while Biden sat in the White House signing executive orders written for him that were dismantling 'the Republic'. Even when judges ruled that a mother and son should not be in jail the government kept them there. Some of those arrested have been badly beaten by prison guards in Washington and lawyers for one man said he suffered a fractured skull and was made blind in one eye. Meanwhile a woman is shot dead for no reason by a Capitol Police officer and we are not allowed to know who he is never mind what has happened to him although that will be *nothing*. The Cult's QAnon/Trump sting to identify and isolate Pushbackers and then target them on the road to crushing and deleting them was a resounding success. You would have thought the Russians had

invaded the building at gunpoint and lined up senators for a firing squad to see the political and media reaction. Congresswoman Alexandria Ocasio-Cortez is a child in a woman's body, a terrible-tuos, me, me, me, Woker narcissist of such proportions that words have no meaning. She said she thought she was going to die when 'insurrectionists' banged on her office door. It turned out she wasn't even in the Capitol Building when the riot was happening and the 'banging' was a Capitol Police officer. She referred to herself as a 'survivor' which is an insult to all those true survivors of violent and sexual abuse while she lives her pampered and privileged life talking drivel for a living. Her Woke colleague and fellow mega-narcissist Rashida Tlaib broke down describing the devastating effect on her, too, of *not being* in the building when the rioters were there. Ocasio-Cortez and Tlaib are members of a fully-Woke group of Congresswomen known as 'The Squad' along with Ilhan Omar and Ayanna Pressley. The Squad from what I can see can be identified by its vehement anti-white racism, anti-white men agenda, and, as always in these cases, the absence of brain cells on active duty.

The usual suspects were on the riot case immediately in the form of Democrat ultra-Zionist senators and operatives Chuck Schumer and Adam Schiff demanding that Trump be impeached for 'his part in the insurrection'. The same pair of prats had led the failed impeachment of Trump over the invented 'Russia collusion' nonsense which claimed Russia had helped Trump win the 2016 election. I didn't realise that Tel Aviv had been relocated just outside Moscow. I must find an up-to-date map. The Russia hoax was a Sabbatian operation to keep Trump occupied and impotent and to stop any rapport with Russia which the Cult wants to retain as a perceptual enemy to be pulled out at will. Puppet Biden began attacking Russia when he came to office as the Cult seeks more upheaval, division and war across the world. A two-year stage show 'Russia collusion inquiry' headed by the not-very-bright former 9/11 FBI chief Robert Mueller, with support from 19 lawyers, 40 FBI agents plus intelligence analysts, forensic accountants and other

staff, devoured tens of millions of dollars and found no evidence of Russia collusion which a ten-year-old could have told them on day one. Now the same moronic Schumer and Schiff wanted a second impeachment of Trump over the Capitol 'insurrection' (riot) which the arrested development of Schumer called another 'Pearl Harbor' while others compared it with 9/11 in which 3,000 died and, in the case of CNN, with the Rwandan genocide in the 1990s in which an estimated 500,000 to 600,000 were murdered, between 250, 000 and 500,000 women were raped, and populations of whole towns were hacked to death with machetes. To make those comparisons purely for Cult political reasons is beyond insulting to those that suffered and lost their lives and confirms yet again the callous inhumanity that we are dealing with. Schumer is a monumental idiot and so is Schiff, but they serve the Cult agenda and do whatever they're told so they get looked after. Talking of idiots – another inane man who spanned the Russia and Capitol impeachment attempts was Senator Eric Swalwell who had the nerve to accuse Trump of collusion with the Russians while sleeping with a Chinese spy called Christine Fang or 'Fang Fang' which is straight out of a Bond film no doubt starring Klaus Schwab as the bloke living on a secret island and controlling laser weapons positioned in space and pointing at world capitals. Fang Fang plays the part of Bond's infiltrator girlfriend which I'm sure she would enjoy rather more than sharing a bed with the brainless Swalwell, lying back and thinking of China. The FBI eventually warned Swalwell about Fang Fang which gave her time to escape back to the Chinese dictatorship. How very thoughtful of them. The second Trump impeachment also failed and hardly surprising when an impeachment is supposed to remove a sitting president and by the time it happened Trump was no longer president. These people are running your country America, well, officially anyway. Terrifying isn't it?

Outcomes tell the story - always

The outcome of all this – and it's the *outcome* on which Renegade Minds focus, not the words – was that a vicious, hysterical and

obviously pre-planned assault was launched on Pushbackers to censor, silence and discredit them and even targeted their right to earn a living. They have since been condemned as 'domestic terrorists' that need to be treated like Al-Qaeda and Islamic State. 'Domestic terrorists' is a label the Cult has been trying to make stick since the period of the Oklahoma bombing in 1995 which was blamed on 'far-right domestic terrorists'. If you read *The Trigger* you will see that the bombing was clearly a Problem-Reaction-Solution carried out by the Deep State during a Bill Clinton administration so corrupt that no dictionary definition of the term would even nearly suffice. Nearly 30, 000 troops were deployed from all over America to the empty streets of Washington for Biden's inauguration. Ten thousand of them stayed on with the pretext of protecting the capital from insurrectionists when it was more psychological programming to normalise the use of the military in domestic law enforcement in support of the Cult plan for a police-military state. Biden's fascist administration began a purge of 'wrong-thinkers' in the military which means anyone that is not on board with Woke. The Capitol Building was surrounded by a fence with razor wire and the Land of the Free was further symbolically and literally dismantled. The circle was completed with the installation of Biden and the exploitation of the QAnon Psyop.

America had never been so divided since the civil war of the 19th century, Pushbackers were isolated and dubbed terrorists and now, as was always going to happen, the Cult immediately set about deleting what little was left of freedom and transforming American society through a swish of the hand of the most controlled 'president' in American history leading (officially at least) the most extreme regime since the country was declared an independent state on July 4th, 1776. Biden issued undebated, dictatorial executive orders almost by the hour in his opening days in office across the whole spectrum of the Cult wish-list including diluting controls on the border with Mexico allowing thousands of migrants to illegally enter the United States to transform the demographics of America and import an election-changing number of perceived Democrat

voters. Then there were Biden deportation amnesties for the already illegally resident (estimated to be as high as 20 or even 30 million). A bill before Congress awarded American citizenship to anyone who could prove they had worked in agriculture for just 180 days in the previous two years as 'Big Ag' secured its slave labour long-term. There were the plans to add new states to the union such as Puerto Rico and making Washington DC a state. They are all parts of a plan to ensure that the Cult-owned Woke Democrats would be permanently in power.

Border – what border?

I have exposed in detail in other books how mass immigration into the United States and Europe is the work of Cult networks fuelled by the tens of billions spent to this and other ends by George Soros and his global Open Society (open borders) Foundations. The impact can be seen in America alone where the population has increased by *100 million* in little more than 30 years mostly through immigration. I wrote in *The Answer* that the plan was to have so many people crossing the southern border that the numbers become unstoppable and we are now there under Cult-owned Biden. El Salvador in Central America puts the scale of what is happening into context. A third of the population now lives in the United States, much of it illegally, and many more are on the way. The methodology is to crush Central and South American countries economically and spread violence through machete-wielding psychopathic gangs like MS-13 based in El Salvador and now operating in many American cities. Biden-imposed lax security at the southern border means that it is all but open. He said before his 'election' that he wanted to see a surge towards the border if he became president and that was the green light for people to do just that after election day to create the human disaster that followed for both America and the migrants. When that surge came the imbecilic Alexandria Ocasio-Cortez said it wasn't a 'surge' because they are 'children, not insurgents' and the term 'surge' (used by Biden) was a claim of 'white supremacists'.

This disingenuous lady may one day enter the realm of the most basic intelligence, but it won't be any time soon.

Sabbatians and the Cult are in the process of destroying America by importing violent people and gangs in among the genuine to terrorise American cities and by overwhelming services that cannot cope with the sheer volume of new arrivals. Something similar is happening in Europe as Western society in general is targeted for demographic and cultural transformation and upheaval. The plan demands violence and crime to create an environment of intimidation, fear and division and Soros has been funding the election of district attorneys across America who then stop prosecuting many crimes, reduce sentences for violent crimes and free as many violent criminals as they can. Sabbatians are creating the chaos from which order – their order – can respond in a classic Problem-Reaction-Solution. A Freemasonic moto says 'Ordo Ab Chao' (Order out of Chaos) and this is why the Cult is constantly creating chaos to impose a new 'order'. Here you have the reason the Cult is constantly creating chaos. The 'Covid' hoax can be seen with those entering the United States by plane being forced to take a 'Covid' test while migrants flooding through southern border processing facilities do not. Nothing is put in the way of mass migration and if that means ignoring the government's own 'Covid' rules then so be it. They know it's all bullshit anyway. Any pushback on this is denounced as 'racist' by Wokers and Sabbatian fronts like the ultra-Zionist Anti-Defamation League headed by the appalling Jonathan Greenblatt which at the same time argues that Israel should not give citizenship and voting rights to more Palestinian Arabs or the 'Jewish population' (in truth the Sabbatian network) will lose control of the country.

Society-changing numbers

Biden's masters have declared that countries like El Salvador are so dangerous that their people must be allowed into the United States for humanitarian reasons when there are fewer murders in large parts of many Central American countries than in US cities like

Baltimore. That is not to say Central America cannot be a dangerous place and Cult-controlled American governments have been making it so since way back, along with the dismantling of economies, in a long-term plan to drive people north into the United States. Parts of Central America are very dangerous, but in other areas the story is being greatly exaggerated to justify relaxing immigration criteria. Migrants are being offered free healthcare and education in the United States as another incentive to head for the border and there is no requirement to be financially independent before you can enter to prevent the resources of America being drained. You can't blame migrants for seeking what they believe will be a better life, but they are being played by the Cult for dark and nefarious ends. The numbers since Biden took office are huge. In February, 2021, more than 100,000 people were known to have tried to enter the US illegally through the southern border (it was 34,000 in the same month in 2020) and in March it was 170,000 – a 418 percent increase on March, 2020. These numbers are only known people, not the ones who get in unseen. The true figure for migrants illegally crossing the border in a single month was estimated by one congressman at 250,000 and that number will only rise under Biden's current policy. Gangs of murdering drug-running thugs that control the Mexican side of the border demand money – thousands of dollars – to let migrants cross the Rio Grande into America. At the same time gun battles are breaking out on the border several times a week between rival Mexican drug gangs (which now operate globally) who are equipped with sophisticated military-grade weapons, grenades and armoured vehicles. While the Capitol Building was being 'protected' from a non-existent 'threat' by thousands of troops, and others were still deployed at the time in the Cult Neocon war in Afghanistan, the southern border of America was left to its fate. This is not incompetence, it is cold calculation.

By March, 2021, there were 17,000 unaccompanied children held at border facilities and many of them are ensnared by people traffickers for paedophile rings and raped on their journey north to America. This is not conjecture – this is fact. Many of those designated

children are in reality teenage boys or older. Meanwhile Wokers posture their self-purity for encouraging poor and tragic people to come to America and face this nightmare both on the journey and at the border with the disgusting figure of House Speaker Nancy Pelosi giving disingenuous speeches about caring for migrants. The woman's evil. Wokers condemned Trump for having children in cages at the border (so did Obama, *Shhhh*), but now they are sleeping on the floor without access to a shower with one border facility 729 percent over capacity. The Biden insanity even proposed flying migrants from the southern border to the northern border with Canada for 'processing'. The whole shambles is being overseen by ultra-Zionist Secretary of Homeland Security, the moronic liar Alejandro Mayorkas, who banned news cameras at border facilities to stop Americans seeing what was happening. Mayorkas said there was not a ban on news crews; it was just that they were not allowed to film. Alongside him at Homeland Security is another ultra-Zionist Cass Sunstein appointed by Biden to oversee new immigration laws. Sunstein despises conspiracy researchers to the point where he suggests they should be banned or *taxed* for having such views. The man is not bonkers or anything. He's perfectly well-adjusted, but adjusted to what is the question. Criticise what is happening and you are a 'white supremacist' when earlier non-white immigrants also oppose the numbers which effect their lives and opportunities. Black people in poor areas are particularly damaged by uncontrolled immigration and the increased competition for work opportunities with those who will work for less. They are also losing voting power as Hispanics become more dominant in former black areas. It's a downward spiral for them while the billionaires behind the policy drone on about how much they care about black people and 'racism'. None of this is about compassion for migrants or black people – that's just wind and air. Migrants are instead being mercilessly exploited to transform America while the countries they leave are losing their future and the same is true in Europe. Mass immigration may now be the work of Woke Democrats, but it can be traced back to the 1986 Immigration Reform and Control Act (it

wasn't) signed into law by Republican hero President Ronald Reagan which gave amnesty to millions living in the United States illegally and other incentives for people to head for the southern border. Here we have the one-party state at work again.

Save me syndrome

Almost every aspect of what I have been exposing as the Cult agenda was on display in even the first days of 'Biden' with silencing of Pushbackers at the forefront of everything. A Renegade Mind will view the Trump years and QAnon in a very different light to their supporters and advocates as the dots are connected. The QAnon/Trump Psyop has given the Cult all it was looking for. We may not know how much, or little, that Trump realised he was being used, but that's a side issue. This pincer movement produced the desired outcome of dividing America and having Pushbackers isolated. To turn this around we have to look at new routes to empowerment which do not include handing our power to other people and groups through what I will call the 'Save Me Syndrome' – 'I want someone else to do it so that I don't have to'. We have seen this at work throughout human history and the QAnon/Trump Psyop is only the latest incarnation alongside all the others. Religion is an obvious expression of this when people look to a 'god' or priest to save them or tell them how to be saved and then there are 'save me' politicians like Trump. Politics is a diversion and not a 'saviour'. It is a means to block positive change, not make it possible.

Save Me Syndrome always comes with the same repeating theme of handing your power to whom or what you believe will save you while your real 'saviour' stares back from the mirror every morning. Renegade Minds are constantly vigilant in this regard and always asking the question 'What can I do?' rather than 'What can someone else do for me?' Gandhi was right when he said: 'You must be the change you want to see in the world.' We are indeed the people we have been waiting for. We are presented with a constant raft of reasons to concede that power to others and forget where the real power is. Humanity has the numbers and the Cult does not. It has to

use diversion and division to target the unstoppable power that comes from unity. Religions, governments, politicians, corporations, media, QAnon, are all different manifestations of this power-diversion and dilution. Refusing to give your power to governments and instead handing it to Trump and QAnon is not to take a new direction, but merely to recycle the old one with new names on the posters. I will explore this phenomenon as we proceed and how to break the cycles and recycles that got us here through the mists of repeating perception and so repeating history.

For now we shall turn to the most potent example in the entire human story of the consequences that follow when you give your power away. I am talking, of course, of the 'Covid' hoax.

CHAPTER FOUR

'Covid': Calculated catastrophe

Facts are threatening to those invested in fraud
DaShanne Stokes

We can easily unravel the real reason for the 'Covid pandemic' hoax by employing the Renegade Mind methodology that I have outlined this far. We'll start by comparing the long-planned Cult outcome with the 'Covid pandemic' outcome. Know the outcome and you'll see the journey.

I have highlighted the plan for the Hunger Games Society which has been in my books for so many years with the very few controlling the very many through ongoing dependency. To create this dependency it is essential to destroy independent livelihoods, businesses and employment to make the population reliant on the state (the Cult) for even the basics of life through a guaranteed pittance income. While independence of income remained these Cult ambitions would be thwarted. With this knowledge it was easy to see where the 'pandemic' hoax was going once talk of 'lockdowns' began and the closing of all but perceived 'essential' businesses to 'save' us from an alleged 'deadly virus'. Cult corporations like Amazon and Walmart were naturally considered 'essential' while mom and pop shops and stores had their doors closed by fascist decree. As a result with every new lockdown and new regulation more small and medium, even large businesses not owned by the Cult, went to the wall while Cult giants and their frontmen and women grew financially fatter by the second. Mom and pop were

denied an income and the right to earn a living and the wealth of people like Jeff Bezos (Amazon), Mark Zuckerberg (Facebook) and Sergei Brin and Larry Page (Google/Alphabet) have reached record levels. The Cult was increasing its own power through further dramatic concentrations of wealth while the competition was being destroyed and brought into a state of dependency. Lockdowns have been instigated to secure that very end and were never anything to do with health. My brother Paul spent 45 years building up a bus repair business, but lockdowns meant buses were running at a fraction of normal levels for months on end. Similar stories can be told in their hundreds of millions worldwide. Efforts of a lifetime coldly destroyed by Cult multi-billionaires and their lackeys in government and law enforcement who continued to earn their living from the taxation of the people while denying the right of the same people to earn theirs. How different it would have been if those making and enforcing these decisions had to face the same financial hardships of those they affected, but they never do.

Gates of Hell

Behind it all in the full knowledge of what he is doing and why is the psychopathic figure of Cult operative Bill Gates. His puppet Tedros at the World Health Organization declared 'Covid' a pandemic in March, 2020. The WHO had changed the definition of a 'pandemic' in 2009 just a month before declaring the 'swine flu pandemic' which would not have been so under the previous definition. The same applies to 'Covid'. The definition had included... 'an infection by an infectious agent, occurring simultaneously in different countries, with a significant mortality rate relative to the proportion of the population infected'. The new definition removed the need for 'significant mortality'. The 'pandemic' has been fraudulent even down to the definition, but Gates demanded economy-destroying lockdowns, school closures, social distancing, mandatory masks, a 'vaccination' for every man, woman and child on the planet and severe consequences and restrictions for those that refused. Who gave him this power? The

Cult did which he serves like a little boy in short trousers doing what his daddy tells him. He and his psychopathic missus even smiled when they said that much worse was to come (what they knew was planned to come). Gates responded in the matter-of-fact way of all psychopaths to a question about the effect on the world economy of what he was doing:

Well, it won't go to zero but it will shrink. Global GDP is probably going to take the biggest hit ever [Gates was smiling as he said this] ... in my lifetime this will be the greatest economic hit. But you don't have a choice. People act as if you have a choice. People don't feel like going to the stadium when they might get infected ... People are deeply affected by seeing these stats, by knowing they could be part of the transmission chain, old people, their parents and grandparents, could be affected by this, and so you don't get to say ignore what is going on here.

There will be the ability to open up, particularly in rich countries, if things are done well over the next few months, but for the world at large normalcy only returns when we have largely vaccinated the entire population.

The man has no compassion or empathy. How could he when he's a psychopath like all Cult players? My own view is that even beyond that he is very seriously mentally ill. Look in his eyes and you can see this along with his crazy flailing arms. You don't do what he has done to the world population since the start of 2020 unless you are mentally ill and at the most extreme end of psychopathic. You especially don't do it when to you know, as we shall see, that cases and deaths from 'Covid' are fakery and a product of monumental figure massaging. 'These stats' that Gates referred to are based on a 'test' that's not testing for the 'virus' as he has known all along. He made his fortune with big Cult support as an infamously ruthless software salesman and now buys global control of 'health' (death) policy without the population he affects having any say. It's a breathtaking outrage. Gates talked about people being deeply affected by fear of 'Covid' when that was because of *him* and his global network lying to them minute-by-minute supported by a lying media that he seriously influences and funds to the tune of hundreds of millions. He's handed big sums to media operations including the BBC, NBC, Al Jazeera, Univision, *PBS NewsHour*,

ProPublica, National Journal, The Guardian, The Financial Times, The Atlantic, Texas Tribune, USA Today publisher Gannett, Washington Monthly, Le Monde, Center for Investigative Reporting, Pulitzer Center on Crisis Reporting, National Press Foundation, International Center for Journalists, Solutions Journalism Network, the Poynter Institute for Media Studies, and many more. Gates is everywhere in the 'Covid' hoax and the man must go to prison – or a mental facility – for the rest of his life and his money distributed to those he has taken such enormous psychopathic pleasure in crushing.

The Muscle

The Hunger Games global structure demands a police-military state – a fusion of the two into one force – which viciously imposes the will of the Cult on the population and protects the Cult from public rebellion. In that regard, too, the 'Covid' hoax just keeps on giving. Often unlawful, ridiculous and contradictory 'Covid' rules and regulations have been policed across the world by moronic automatons and psychopaths made faceless by face-nappy masks and acting like the Nazi SS and fascist blackshirts and brownshirts of Hitler and Mussolini. The smallest departure from the rules decreed by the psychos in government and their clueless gofers were jumped upon by the face-nappy fascists. Brutality against public protestors soon became commonplace even on girls, women and old people as the brave men with the batons – the Face-Nappies as I call them – broke up peaceful protests and handed out fines like confetti to people who couldn't earn a living let alone pay hundreds of pounds for what was once an accepted human right. Robot Face-Nappies of Nottingham police in the English East Midlands fined one group £11,000 for attending a child's birthday party. For decades I charted the transformation of law enforcement as genuine, decent officers were replaced with psychopaths and the brain dead who would happily and brutally do whatever their masters told them. Now they were let loose on the public and I would emphasise the point that none of this just happened. The step-by-step change in the dynamic between police and public was orchestrated from the shadows by

those who knew where this was all going and the same with the perceptual reframing of those in all levels of authority and official administration through 'training courses' by organisations such as Common Purpose which was created in the late 1980s and given a massive boost in Blair era Britain until it became a global phenomenon. Supposed public 'servants' began to view the population as the enemy and the same was true of the police. This was the start of the explosion of behaviour manipulation organisations and networks preparing for the all-war on the human psyche unleashed with the dawn of 2020. I will go into more detail about this later in the book because it is a core part of what is happening.

Police desecrated beauty spots to deter people gathering and arrested women for walking in the countryside alone 'too far' from their homes. We had arrogant, clueless sergeants in the Isle of Wight police where I live posting on Facebook what they insisted the population must do or else. A schoolmaster sergeant called Radford looked young enough for me to ask if his mother knew he was out, but he was posting what he *expected* people to do while a Sergeant Wilkinson boasted about fining lads for meeting in a McDonald's car park where they went to get a lockdown takeaway. Wilkinson added that he had even cancelled their order. What a pair of prats these people are and yet they have increasingly become the norm among Jackboot Johnson's Yellowshirts once known as the British police. This was the theme all over the world with police savagery common during lockdown protests in the United States, the Netherlands, and the fascist state of Victoria in Australia under its tyrannical and again moronic premier Daniel Andrews. Amazing how tyrannical and moronic tend to work as a team and the same combination could be seen across America as arrogant, narcissistic Woke governors and mayors such as Gavin Newsom (California), Andrew Cuomo (New York), Gretchen Whitmer (Michigan), Lori Lightfoot (Chicago) and Eric Garcetti (Los Angeles) did their Nazi and Stalin impressions with the full support of the compliant brutality of their enforcers in uniform as they arrested small business owners defying

fascist shutdown orders and took them to jail in ankle shackles and handcuffs. This happened to bistro owner Marlena Pavlos-Hackney in Gretchen Whitmer's fascist state of Michigan when police arrived to enforce an order by a state-owned judge for 'putting the community at risk' at a time when other states like Texas were dropping restrictions and migrants were pouring across the southern border without any 'Covid' questions at all. I'm sure there are many officers appalled by what they are ordered to do, but not nearly enough of them. If they were truly appalled they would not do it. As the months passed every opportunity was taken to have the military involved to make their presence on the streets ever more familiar and 'normal' for the longer-term goal of police-military fusion.

Another crucial element to the Hunger Games enforcement network has been encouraging the public to report neighbours and others for 'breaking the lockdown rules'. The group faced with £11,000 in fines at the child's birthday party would have been dobbed-in by a neighbour with a brain the size of a pea. The technique was most famously employed by the Stasi secret police in communist East Germany who had public informants placed throughout the population. A police chief in the UK says his force doesn't need to carry out 'Covid' patrols when they are flooded with so many calls from the public reporting other people for visiting the beach. Dorset police chief James Vaughan said people were so enthusiastic about snitching on their fellow humans they were now operating as an auxiliary arm of the police: 'We are still getting around 400 reports a week from the public, so we will respond to reports ... We won't need to be doing hotspot patrols because people are very quick to pick the phone up and tell us.' Vaughan didn't say that this is a pillar of all tyrannies of whatever complexion and the means to hugely extend the reach of enforcement while spreading distrust among the people and making them wary of doing anything that might get them reported. Those narcissistic Isle of Wight sergeants Radford and Wilkinson never fail to add a link to their Facebook posts where the public can inform on their fellow slaves.

Neither would be self-aware enough to realise they were imitating the Stasi which they might well never have heard of. Government psychologists that I will expose later laid out a policy to turn communities against each other in the same way.

A coincidence? Yep, and I can knit fog

I knew from the start of the alleged pandemic that this was a Cult operation. It presented limitless potential to rapidly advance the Cult agenda and exploit manipulated fear to demand that every man, woman and child on the planet was 'vaccinated' in a process never used on humans before which infuses self-replicating *synthetic* material into human cells. Remember the plan to transform the human body from a biological to a synthetic biological state. I'll deal with the 'vaccine' (that's not actually a vaccine) when I focus on the genetic agenda. Enough to say here that mass global 'vaccination' justified by this 'new virus' set alarms ringing after 30 years of tracking these people and their methods. The 'Covid' hoax officially beginning in China was also a big red flag for reasons I will be explaining. The agenda potential was so enormous that I could dismiss any idea that the 'virus' appeared naturally. Major happenings with major agenda implications never occur without Cult involvement in making them happen. My questions were twofold in early 2020 as the media began its campaign to induce global fear and hysteria: Was this alleged infectious agent released on purpose by the Cult or did it even exist at all? I then did what I always do in these situations. I sat, observed and waited to see where the evidence and information would take me. By March and early April synchronicity was strongly – and ever more so since then – pointing me in the direction of *there is no 'virus'*. I went public on that with derision even from swathes of the alternative media that voiced a scenario that the Chinese government released the 'virus' in league with Deep State elements in the United States from a top-level bio-lab in Wuhan where the 'virus' is said to have first appeared. I looked at that possibility, but I didn't buy it for several reasons. Deaths from the 'virus' did not in any way match what they

would have been with a 'deadly bioweapon' and it is much more effective if you sell the *illusion* of an infectious agent rather than having a real one unless you can control through injection who has it and who doesn't. Otherwise you lose control of events. A made-up 'virus' gives you a blank sheet of paper on which you can make it do whatever you like and have any symptoms or mutant 'variants' you choose to add while a real infectious agent would limit you to what it actually does. A phantom disease allows you to have endless ludicrous 'studies' on the 'Covid' dollar to widen the perceived impact by inventing ever more 'at risk' groups including one study which said those who walk slowly may be almost four times more likely to die from the 'virus'. People are in psychiatric wards for less.

A real 'deadly bioweapon' can take out people in the hierarchy that are not part of the Cult, but essential to its operation. Obviously they don't want that. Releasing a real disease means you immediately lose control of it. Releasing an illusory one means you don't. Again it's vital that people are extra careful when dealing with what they want to hear. A bioweapon unleashed from a Chinese laboratory in collusion with the American Deep State may fit a conspiracy narrative, but is it true? Would it not be far more effective to use the excuse of a 'virus' to justify the real bioweapon – the 'vaccine'? That way your disease agent does not have to be transmitted and arrives directly through a syringe. I saw a French virologist Luc Montagnier quoted in the alternative media as saying he had discovered that the alleged 'new' severe acute respiratory syndrome coronavirus , or SARS-CoV-2, was made artificially and included elements of the human immunodeficiency 'virus' (HIV) and a parasite that causes malaria. SARS-CoV-2 is alleged to trigger an alleged illness called Covid-19. I remembered Montagnier's name from my research years before into claims that an HIV 'retrovirus' causes AIDs – claims that were demolished by Berkeley virologist Peter Duesberg who showed that no one had ever proved that HIV causes acquired immunodeficiency syndrome or AIDS. Claims that become accepted as fact, publicly and medically, with no proof whatsoever are an ever-recurring story that profoundly applies to

‘Covid’. Nevertheless, despite the lack of proof, Montagnier’s team at the Pasteur Institute in Paris had a long dispute with American researcher Robert Gallo over which of them discovered and isolated the HIV ‘virus’ and with *no evidence* found it to cause AIDS. You will see later that there is also no evidence that any ‘virus’ causes any disease or that there is even such a thing as a ‘virus’ in the way it is said to exist. The claim to have ‘isolated’ the HIV ‘virus’ will be presented in its real context as we come to the shocking story – and it is a story – of SARS-CoV-2 and so will Montagnier’s assertion that he identified the full SARS-CoV-2 genome.

Hoax in the making

We can pick up the ‘Covid’ story in 2010 and the publication by the Rockefeller Foundation of a document called ‘Scenarios for the Future of Technology and International Development’. The inner circle of the Rockefeller family has been serving the Cult since John D. Rockefeller (1839-1937) made his fortune with Standard Oil. It is less well known that the same Rockefeller – the Bill Gates of his day – was responsible for establishing what is now referred to as ‘Big Pharma’, the global network of pharmaceutical companies that make outrageous profits dispensing scalpel and drug ‘medicine’ and are obsessed with pumping vaccines in ever-increasing number into as many human arms and backsides as possible. John D. Rockefeller was the driving force behind the creation of the ‘education’ system in the United States and elsewhere specifically designed to program the perceptions of generations thereafter. The Rockefeller family donated exceptionally valuable land in New York for the United Nations building and were central in establishing the World Health Organization in 1948 as an agency of the UN which was created from the start as a Trojan horse and stalking horse for world government. Now enter Bill Gates. His family and the Rockefellers have long been extremely close and I have seen genealogy which claims that if you go back far enough the two families fuse into the same bloodline. Gates has said that the Bill and Melinda Gates Foundation was inspired by the Rockefeller Foundation and why not

when both are serving the same Cult? Major tax-exempt foundations are overwhelmingly criminal enterprises in which Cult assets fund the Cult agenda in the guise of 'philanthropy' while avoiding tax in the process. Cult operatives can become mega-rich in their role of front men and women for the psychopaths at the inner core and they, too, have to be psychopaths to knowingly serve such evil. Part of the deal is that a big percentage of the wealth gleaned from representing the Cult has to be spent advancing the ambitions of the Cult and hence you have the Rockefeller Foundation, Bill and Melinda Gates Foundation (and *so* many more) and people like George Soros with his global Open Society Foundations spending their billions in pursuit of global Cult control. Gates is a global public face of the Cult with his interventions in world affairs including Big Tech influence; a central role in the 'Covid' and 'vaccine' scam; promotion of the climate change shakedown; manipulation of education; geoengineering of the skies; and his food-control agenda as the biggest owner of farmland in America, his GMO promotion and through other means. As one writer said: 'Gates monopolizes or wields disproportionate influence over the tech industry, global health and vaccines, agriculture and food policy (including biopiracy and fake food), weather modification and other climate technologies, surveillance, education and media.' The almost limitless wealth secured through Microsoft and other not-allowed-to-fail ventures (including vaccines) has been ploughed into a long, long list of Cult projects designed to enslave the entire human race. Gates and the Rockefellers have been working as one unit with the Rockefeller-established World Health Organization leading global 'Covid' policy controlled by Gates through his mouth-piece Tedros. Gates became the WHO's biggest funder when Trump announced that the American government would cease its donations, but Biden immediately said he would restore the money when he took office in January, 2021. The Gates Foundation (the Cult) owns through limitless funding the world health system and the major players across the globe in the 'Covid' hoax.

Okay, with that background we return to that Rockefeller Foundation document of 2010 headed 'Scenarios for the Future of Technology and International Development' and its 'imaginary' epidemic of a virulent and deadly influenza strain which infected 20 percent of the global population and killed eight million in seven months. The Rockefeller scenario was that the epidemic destroyed economies, closed shops, offices and other businesses and led to governments imposing fierce rules and restrictions that included mandatory wearing of face masks and body-temperature checks to enter communal spaces like railway stations and supermarkets. The document predicted that even after the height of the Rockefeller-envisaged epidemic the authoritarian rule would continue to deal with further pandemics, transnational terrorism, environmental crises and rising poverty. Now you may think that the Rockefellers are our modern-day seers or alternatively, and rather more likely, that they well knew what was planned a few years further on. Fascism had to be imposed, you see, to 'protect citizens from risk and exposure'. The Rockefeller scenario document said:

During the pandemic, national leaders around the world flexed their authority and imposed airtight rules and restrictions, from the mandatory wearing of face masks to body-temperature checks at the entries to communal spaces like train stations and supermarkets. Even after the pandemic faded, this more authoritarian control and oversight of citizens and their activities stuck and even intensified. In order to protect themselves from the spread of increasingly global problems – from pandemics and transnational terrorism to environmental crises and rising poverty – leaders around the world took a firmer grip on power.

At first, the notion of a more controlled world gained wide acceptance and approval. Citizens willingly gave up some of their sovereignty – and their privacy – to more paternalistic states in exchange for greater safety and stability. Citizens were more tolerant, and even eager, for top-down direction and oversight, and national leaders had more latitude to impose order in the ways they saw fit.

In developed countries, this heightened oversight took many forms: biometric IDs for all citizens, for example, and tighter regulation of key industries whose stability was deemed vital to national interests. In many developed countries, enforced cooperation with a suite of new regulations and agreements slowly but steadily restored both order and, importantly, economic growth.

There we have the prophetic Rockefellers in 2010 and three years later came their paper for the Global Health Summit in Beijing, China, when government representatives, the private sector, international organisations and groups met to discuss the next 100 years of 'global health'. The Rockefeller Foundation-funded paper was called 'Dreaming the Future of Health for the Next 100 Years and more prophecy ensued as it described a dystopian future: 'The abundance of data, digitally tracking and linking people may mean the 'death of privacy' and may replace physical interaction with transient, virtual connection, generating isolation and raising questions of how values are shaped in virtual networks.' Next in the 'Covid' hoax preparation sequence came a 'table top' simulation in 2018 for another 'imaginary' pandemic of a disease called Clade X which was said to kill 900 million people. The exercise was organised by the Gates-funded Johns Hopkins University's Center for Health Security in the United States and this is the very same university that has been compiling the disgustingly and systematically erroneous global figures for 'Covid' cases and deaths. Similar Johns Hopkins health crisis scenarios have included the Dark Winter exercise in 2001 and Atlantic Storm in 2005.

Nostradamus 201

For sheer predictive genius look no further prophecy-watchers than the Bill Gates-funded Event 201 held only six weeks before the 'coronavirus pandemic' is supposed to have broken out in China and Event 201 was based on a scenario of a global 'coronavirus pandemic'. Melinda Gates, the great man's missus, told the BBC that he had 'prepared for years' for a coronavirus pandemic which told us what we already knew. Nostradamugates had predicted in a TED talk in 2015 that a pandemic was coming that would kill a lot of people and demolish the world economy. My god, the man is a machine – possibly even literally. Now here he was only weeks before the real thing funding just such a simulated scenario and involving his friends and associates at Johns Hopkins, the World Economic Forum Cult-front of Klaus Schwab, the United Nations,

Johnson & Johnson, major banks, and officials from China and the Centers for Disease Control in the United States. What synchronicity – Johns Hopkins would go on to compile the fraudulent ‘Covid’ figures, the World Economic Forum and Schwab would push the ‘Great Reset’ in response to ‘Covid’, the Centers for Disease Control would be at the forefront of ‘Covid’ policy in the United States, Johnson & Johnson would produce a ‘Covid vaccine’, and everything would officially start just weeks later in China. Spooky, eh? They were even accurate in creating a simulation of a ‘virus’ pandemic because the ‘real thing’ would also be a simulation. Event 201 was not an exercise preparing for something that might happen; it was a rehearsal for what those in control knew was *going* to happen and very shortly. Hours of this simulation were posted on the Internet and the various themes and responses mirrored what would soon be imposed to transform human society. News stories were inserted and what they said would be commonplace a few weeks later with still more prophecy perfection. Much discussion focused on the need to deal with misinformation and the ‘anti-vax movement’ which is exactly what happened when the ‘virus’ arrived – was said to have arrived – in the West.

Cult-owned social media banned criticism and exposure of the official ‘virus’ narrative and when I said there *was* no ‘virus’ in early April, 2020, I was banned by one platform after another including YouTube, Facebook and later Twitter. The mainstream broadcast media in Britain was in effect banned from interviewing me by the Tony-Blair-created government broadcasting censor Ofcom headed by career government bureaucrat Melanie Dawes who was appointed just as the ‘virus’ hoax was about to play out in January, 2020. At the same time the Ickonic media platform was using Vimeo, another ultra-Zionist-owned operation, while our own player was being created and they deleted in an instant hundreds of videos, documentaries, series and shows to confirm their unbelievable vindictiveness. We had copies, of course, and they had to be restored one by one when our player was ready. These people have no class. Sabbatian Facebook promised free advertisements for the Gates-

controlled World Health Organization narrative while deleting ‘false claims and conspiracy theories’ to stop ‘misinformation’ about the alleged coronavirus. All these responses could be seen just a short while earlier in the scenarios of Event 201. Extreme censorship was absolutely crucial for the Cult because the official story was so ridiculous and unsupportable by the evidence that it could never survive open debate and the free-flow of information and opinion. If you can’t win a debate then don’t have one is the Cult’s approach throughout history. Facebook’s little boy front man – front boy – Mark Zuckerberg equated ‘credible and accurate information’ with official sources and exposing their lies with ‘misinformation’.

Silencing those that can see

The censorship dynamic of Event 201 is now the norm with an army of narrative-supporting ‘fact-checker’ organisations whose entire reason for being is to tell the public that official narratives are true and those exposing them are lying. One of the most appalling of these ‘fact-checkers’ is called NewsGuard founded by ultra-Zionist Americans Gordon Crovitz and Steven Brill. Crovitz is a former publisher of *The Wall Street Journal*, former Executive Vice President of Dow Jones, a member of the Council on Foreign Relations (CFR), and on the board of the American Association of Rhodes Scholars. The CFR and Rhodes Scholarships, named after Rothschild agent Cecil Rhodes who plundered the gold and diamonds of South Africa for his masters and the Cult, have featured widely in my books. NewsGuard don’t seem to like me for some reason – I really can’t think why – and they have done all they can to have me censored and discredited which is, to quote an old British politician, like being savaged by a dead sheep. They are, however, like all in the censorship network, very well connected and funded by organisations themselves funded by, or connected to, Bill Gates. As you would expect with anything associated with Gates NewsGuard has an offshoot called HealthGuard which ‘fights online health care hoaxes’. How very kind. Somehow the NewsGuard European Managing Director Anna-Sophie Harling, a remarkably young-

looking woman with no broadcasting experience and little hands-on work in journalism, has somehow secured a position on the 'Content Board' of UK government broadcast censor Ofcom. An executive of an organisation seeking to discredit dissidents of the official narratives is making decisions for the government broadcast 'regulator' about content?? Another appalling 'fact-checker' is Full Fact funded by George Soros and global censors Google and Facebook.

It's amazing how many activists in the 'fact-checking', 'anti-hate', arena turn up in government-related positions – people like UK Labour Party activist Imran Ahmed who heads the Center for Countering Digital Hate founded by people like Morgan McSweeney, now chief of staff to the Labour Party's hapless and useless 'leader' Keir Starmer. Digital Hate – which is what it really is – uses the American spelling of Center to betray its connection to a transatlantic network of similar organisations which in 2020 shapeshifted from attacking people for 'hate' to attacking them for questioning the 'Covid' hoax and the dangers of the 'Covid vaccine'. It's just a coincidence, you understand. This is one of Imran Ahmed's hysterical statements: 'I would go beyond calling anti-vaxxers conspiracy theorists to say they are an extremist group that pose a national security risk.' No one could ever accuse this prat of understatement and he's including in that those parents who are now against vaccines after their children were damaged for life or killed by them. He's such a nice man. Ahmed does the rounds of the Woke media getting soft-ball questions from spineless 'journalists' who never ask what right he has to campaign to destroy the freedom of speech of others while he demands it for himself. There also seems to be an overrepresentation in Ofcom of people connected to the narrative-worshipping BBC. This incredible global network of narrative-support was super-vital when the 'Covid' hoax was played in the light of the mega-whopper lies that have to be defended from the spotlight cast by the most basic intelligence.

Setting the scene

The Cult plays the long game and proceeds step-by-step ensuring that everything is in place before major cards are played and they don't come any bigger than the 'Covid' hoax. The psychopaths can't handle events where the outcome isn't certain and as little as possible – preferably nothing – is left to chance. Politicians, government and medical officials who would follow direction were brought to illusory power in advance by the Cult web whether on the national stage or others like state governors and mayors of America. For decades the dynamic between officialdom, law enforcement and the public was changed from one of service to one of control and dictatorship. Behaviour manipulation networks established within government were waiting to impose the coming 'Covid' rules and regulations specifically designed to subdue and rewire the psyche of the people in the guise of protecting health. These included in the UK the Behavioural Insights Team part-owned by the British government Cabinet Office; the Scientific Pandemic Insights Group on Behaviours (SPI-B); and a whole web of intelligence and military groups seeking to direct the conversation on social media and control the narrative. Among them are the cyberwarfare (on the people) 77th Brigade of the British military which is also coordinated through the Cabinet Office as civilian and military leadership continues to combine in what they call the Fusion Doctrine. The 77th Brigade is a British equivalent of the infamous Israeli (Sabbatian) military cyberwarfare and Internet manipulation operation Unit 8200 which I expose at length in *The Trigger*. Also carefully in place were the medical and science advisers to government – many on the payroll past or present of Bill Gates – and a whole alternative structure of unelected government stood by to take control when elected parliaments were effectively closed down once the 'Covid' card was slammed on the table. The structure I have described here and so much more was installed in every major country through the Cult networks. The top-down control hierarchy looks like this: The Cult – Cult-owned Gates – the World Health Organization and Tedros – Gates-funded or controlled chief medical officers and science 'advisers' (dictators) in each country –

political 'leaders' – law enforcement – The People. Through this simple global communication and enforcement structure the policy of the Cult could be imposed on virtually the entire human population so long as they acquiesced to the fascism. With everything in place it was time for the button to be pressed in late 2019/early 2020.

These were the prime goals the Cult had to secure for its will to prevail:

- 1) Locking down economies, closing all but designated 'essential' businesses (Cult-owned corporations were 'essential'), and putting the population under house arrest was an imperative to destroy independent income and employment and ensure dependency on the Cult-controlled state in the Hunger Games Society. Lockdowns had to be established as the global blueprint from the start to respond to the 'virus' and followed by pretty much the entire world.
- 2) The global population had to be terrified into believing in a deadly 'virus' that didn't actually exist so they would unquestioningly obey authority in the belief that authority must know how best to protect them and their families. Software salesman Gates would suddenly morph into the world's health expert and be promoted as such by the Cult-owned media.
- 3) A method of testing that wasn't testing for the 'virus', but was only claimed to be, had to be in place to provide the illusion of 'cases' and subsequent 'deaths' that had a very different cause to the 'Covid-19' that would be scribbled on the death certificate.
- 4) Because there was no 'virus' and the great majority testing positive with a test not testing for the 'virus' would have no symptoms of anything the lie had to be sold that people without symptoms (without the 'virus') could still pass it on to others. This was crucial to justify for the first time quarantining – house arresting – healthy people. Without this the economy-destroying lockdown of *everybody* could not have been credibly sold.
- 5) The 'saviour' had to be seen as a vaccine which beyond evil drug companies were working like angels of mercy to develop as quickly as possible, with all corners cut, to save the day. The public must absolutely not know that the 'vaccine' had nothing to do with a 'virus' or that the contents were ready and waiting with a very different motive long before the 'Covid' card was even lifted from the pack.

I said in March, 2020, that the 'vaccine' would have been created way ahead of the 'Covid' hoax which justified its use and the following December an article in the New York *Intelligencer* magazine said the Moderna 'vaccine' had been 'designed' by

January, 2020. This was 'before China had even acknowledged that the disease could be transmitted from human to human, more than a week before the first confirmed coronavirus case in the United States'. The article said that by the time the first American death was announced a month later 'the vaccine had already been manufactured and shipped to the National Institutes of Health for the beginning of its Phase I clinical trial'. The 'vaccine' was actually 'designed' long before that although even with this timescale you would expect the article to ask how on earth it could have been done that quickly. Instead it asked why the 'vaccine' had not been rolled out then and not months later. Journalism in the mainstream is truly dead. I am going to detail in the next chapter why the 'virus' has never existed and how a hoax on that scale was possible, but first the foundation on which the Big Lie of 'Covid' was built.

The test that doesn't test

Fraudulent 'testing' is the bottom line of the whole 'Covid' hoax and was the means by which a 'virus' that did not exist *appeared* to exist. They could only achieve this magic trick by using a test not testing for the 'virus'. To use a test that *was* testing for the 'virus' would mean that every test would come back negative given there was no 'virus'. They chose to exploit something called the RT-PCR test invented by American biochemist Kary Mullis in the 1980s who said publicly that his PCR test ... *cannot detect infectious disease*. Yes, the 'test' used worldwide to detect infectious 'Covid' to produce all the illusory 'cases' and 'deaths' compiled by Johns Hopkins and others *cannot detect infectious disease*. This fact came from the mouth of the man who invented PCR and was awarded the Nobel Prize in Chemistry in 1993 for doing so. Sadly, and incredibly conveniently for the Cult, Mullis died in August, 2019, at the age of 74 just before his test would be fraudulently used to unleash fascism on the world. He was said to have died from pneumonia which was an irony in itself. A few months later he would have had 'Covid-19' on his death certificate. I say the timing of his death was convenient because had he lived Mullis, a brilliant, honest and decent man, would have been

vociferously speaking out against the use of his test to detect 'Covid' when it was never designed, or able, to do that. I know that to be true given that Mullis made the same point when his test was used to 'detect' – not detect – HIV. He had been seriously critical of the Gallo/Montagnier claim to have isolated the HIV 'virus' and shown it to cause AIDS for which Mullis said there was no evidence. AIDS is actually not a disease but a series of diseases from which people die all the time. When they die from those *same diseases* after a positive 'test' for HIV then AIDS goes on their death certificate. I think I've heard that before somewhere. Countries instigated a policy with 'Covid' that anyone who tested positive with a test not testing for the 'virus' and died of any other cause within 28 days and even longer 'Covid-19' had to go on the death certificate. Cases have come from the test that can't test for infectious disease and the deaths are those who have died of *anything* after testing positive with a test not testing for the 'virus'. I'll have much more later about the death certificate scandal.

Mullis was deeply dismissive of the now US 'Covid' star Anthony Fauci who he said was a liar who didn't know anything about anything – 'and I would say that to his face – nothing.' He said of Fauci: 'The man thinks he can take a blood sample, put it in an electron microscope and if it's got a virus in there you'll know it – he doesn't understand electron microscopy and he doesn't understand medicine and shouldn't be in a position like he's in.' That position, terrifyingly, has made him the decider of 'Covid' fascism policy on behalf of the Cult in his role as director since 1984 of the National Institute of Allergy and Infectious Diseases (NIAID) while his record of being wrong is laughable; but being wrong, so long as it's the *right kind* of wrong, is why the Cult loves him. He'll say anything the Cult tells him to say. Fauci was made Chief Medical Adviser to the President immediately Biden took office. Biden was installed in the White House by Cult manipulation and one of his first decisions was to elevate Fauci to a position of even more control. This is a coincidence? Yes, and I identify as a flamenco dancer called Lola. How does such an incompetent criminal like Fauci remain in that

pivotal position in American health since *the 1980s*? When you serve the Cult it looks after you until you are surplus to requirements. Kary Mullis said prophetically of Fauci and his like: 'Those guys have an agenda and it's not an agenda we would like them to have ... they make their own rules, they change them when they want to, and Tony Fauci does not mind going on television in front of the people who pay his salary and lie directly into the camera.' Fauci has done that almost daily since the 'Covid' hoax began. Lying is in Fauci's DNA. To make the situation crystal clear about the PCR test this is a direct quote from its inventor Kary Mullis:

It [the PCR test] doesn't tell you that you're sick and doesn't tell you that the thing you ended up with was really going to hurt you ...'

Ask yourself why governments and medical systems the world over have been using this very test to decide who is 'infected' with the SARS-CoV-2 'virus' and the alleged disease it allegedly causes, 'Covid-19'. The answer to that question will tell you what has been going on. By the way, here's a little show-stopper – the 'new' SARS-CoV-2 'virus' was 'identified' as such right from the start using ... *the PCR test not testing for the 'virus'*. If you are new to this and find that shocking then stick around. I have hardly started yet. Even worse, other 'tests', like the 'Lateral Flow Device' (LFD), are considered so useless that they have to be *confirmed* by the PCR test! Leaked emails written by Ben Dyson, adviser to UK 'Health' Secretary Matt Hancock, said they were 'dangerously unreliable'. Dyson, executive director of strategy at the Department of Health, wrote: 'As of today, someone who gets a positive LFD result in (say) London has at best a 25 per cent chance of it being a true positive, but if it is a self-reported test potentially as low as 10 per cent (on an optimistic assumption about specificity) or as low as 2 per cent (on a more pessimistic assumption).' These are the 'tests' that schoolchildren and the public are being urged to have twice a week or more and have to isolate if they get a positive. Each fake positive goes in the statistics as a 'case' no matter how ludicrously inaccurate and the

'cases' drive lockdown, masks and the pressure to 'vaccinate'. The government said in response to the email leak that the 'tests' were accurate which confirmed yet again what shocking bloody liars they are. The real false positive rate is *100 percent* as we'll see. In another 'you couldn't make it up' the UK government agreed to pay £2.8 billion to California's Innova Medical Group to supply the irrelevant lateral flow tests. The company's primary test-making centre is in China. Innova Medical Group, established in March, 2020, is owned by Pasaca Capital Inc, chaired by Chinese-American millionaire Charles Huang who was born in Wuhan.

How it works – and how it doesn't

The RT-PCR test, known by its full title of Polymerase chain reaction, is used across the world to make millions, even billions, of copies of a DNA/RNA genetic information sample. The process is called 'amplification' and means that a tiny sample of genetic material is amplified to bring out the detailed content. I stress that it is not testing for an infectious disease. It is simply amplifying a sample of genetic material. In the words of Kary Mullis: 'PCR is ... just a process that's used to make a whole lot of something out of something.' To emphasise the point companies that make the PCR tests circulated around the world to 'test' for 'Covid' warn on the box that it can't be used to detect 'Covid' or infectious disease and is for research purposes only. It's okay, rest for a minute and you'll be fine. This is the test that produces the 'cases' and 'deaths' that have been used to destroy human society. All those global and national medical and scientific 'experts' demanding this destruction to 'save us' *KNOW* that the test is not testing for the 'virus' and the cases and deaths they claim to be real are an almost unimaginable fraud. Every one of them and so many others including politicians and psychopaths like Gates and Tedros must be brought before Nuremburg-type trials and jailed for the rest of their lives. The more the genetic sample is amplified by PCR the more elements of that material become sensitive to the test and by that I don't mean sensitive for a 'virus' but for elements of the genetic material which

is *naturally* in the body or relates to remnants of old conditions of various kinds lying dormant and causing no disease. Once the amplification of the PCR reaches a certain level *everyone* will test positive. So much of the material has been made sensitive to the test that everyone will have some part of it in their body. Even lying criminals like Fauci have said that once PCR amplifications pass 35 cycles everything will be a false positive that cannot be trusted for the reasons I have described. I say, like many proper doctors and scientists, that 100 percent of the 'positives' are false, but let's just go with Fauci for a moment.

He says that any amplification over 35 cycles will produce false positives and yet the US Centers for Disease Control (CDC) and Food and Drug Administration (FDA) have recommended up to 40 *cycles* and the National Health Service (NHS) in Britain admitted in an internal document for staff that it was using 45 *cycles* of amplification. A long list of other countries has been doing the same and at least one 'testing' laboratory has been using 50 *cycles*. Have you ever heard a doctor, medical 'expert' or the media ask what level of amplification has been used to claim a 'positive'. The 'test' comes back 'positive' and so you have the 'virus', end of story. Now we can see how the government in Tanzania could send off samples from a goat and a pawpaw fruit under human names and both came back positive for 'Covid-19'. Tanzania president John Magufuli mocked the 'Covid' hysteria, the PCR test and masks and refused to import the DNA-manipulating 'vaccine'. The Cult hated him and an article sponsored by the Bill Gates Foundation appeared in the London *Guardian* in February, 2021, headed 'It's time for Africa to rein in Tanzania's anti-vaxxer president'. Well, 'reined in' he shortly was. Magufuli appeared in good health, but then, in March, 2021, he was dead at 61 from 'heart failure'. He was replaced by Samia Hassan Suhulu who is connected to Klaus Schwab's World Economic Forum and she immediately reversed Magufuli's 'Covid' policy. A sample of cola tested positive for 'Covid' with the PCR test in Germany while American actress and singer-songwriter Erykah Badu tested positive in one nostril and negative in the other. Footballer Ronaldo called

the PCR test 'bullshit' after testing positive three times and being forced to quarantine and miss matches when there was nothing wrong with him. The mantra from Tedros at the World Health Organization and national governments (same thing) has been test, test, test. They know that the more tests they can generate the more fake 'cases' they have which go on to become 'deaths' in ways I am coming to. The UK government has its Operation Moonshot planned to test multiple millions every day in workplaces and schools with free tests for everyone to use twice a week at home in line with the Cult plan from the start to make testing part of life. A government advertisement for an 'Interim Head of Asymptomatic Testing Communication' said the job included responsibility for delivering a 'communications strategy' (propaganda) 'to support the expansion of asymptomatic testing that *'normalises testing as part of everyday life'*'. More tests means more fake 'cases', 'deaths' and fascism. I have heard of, and from, many people who booked a test, couldn't turn up, and yet got a positive result through the post for a test they'd never even had. The whole thing is crazy, but for the Cult there's method in the madness. Controlling and manipulating the level of amplification of the test means the authorities can control whenever they want the number of apparent 'cases' and 'deaths'. If they want to justify more fascist lockdown and destruction of livelihoods they keep the amplification high. If they want to give the illusion that lockdowns and the 'vaccine' are working then they lower the amplification and 'cases' and 'deaths' will appear to fall. In January, 2021, the Cult-owned World Health Organization suddenly warned laboratories about over-amplification of the test and to lower the threshold. Suddenly headlines began appearing such as: 'Why ARE "Covid" cases plummeting?' This was just when the vaccine rollout was underway and I had predicted months before they would make cases appear to fall through amplification tampering when the 'vaccine' came. These people are so predictable.

Cow vaccines?

The question must be asked of what is on the test swabs being poked far up the nose of the population to the base of the brain? A nasal swab punctured one woman's brain and caused it to leak fluid. Most of these procedures are being done by people with little training or medical knowledge. Dr Lorraine Day, former orthopaedic trauma surgeon and Chief of Orthopaedic Surgery at San Francisco General Hospital, says the tests are really a '*vaccine*'. Cows have long been vaccinated this way. She points out that masks have to cover the nose and the mouth where it is claimed the '*virus*' exists in saliva. Why then don't they take saliva from the mouth as they do with a DNA test instead of pushing a long swab up the nose towards the brain? The ethmoid bone separates the nasal cavity from the brain and within that bone is the cribriform plate. Dr Day says that when the swab is pushed up against this plate and twisted the procedure is '*depositing things back there*'. She claims that among these '*things*' are nanoparticles that can enter the brain. Researchers have noted that a team at the Gates-funded Johns Hopkins have designed tiny, star-shaped micro-devices that can latch onto intestinal mucosa and release drugs into the body. Mucosa is the thin skin that covers the inside surface of parts of the body such as *the nose* and mouth and produces mucus to protect them. The Johns Hopkins micro-devices are called '*theragrippers*' and were '*inspired*' by a parasitic worm that digs its sharp teeth into a host's intestines. Nasal swabs are also coated in the sterilisation agent ethylene oxide. The US National Cancer Institute posts this explanation on its website:

At room temperature, ethylene oxide is a flammable colorless gas with a sweet odor. It is used primarily to produce other chemicals, including antifreeze. In smaller amounts, ethylene oxide is used as a pesticide and a sterilizing agent. The ability of ethylene oxide to damage DNA makes it an effective sterilizing agent but also accounts for its cancer-causing activity.

The Institute mentions lymphoma and leukaemia as cancers most frequently reported to be associated with occupational exposure to ethylene oxide along with stomach and breast cancers. How does anyone think this is going to work out with the constant testing

regime being inflicted on adults and children at home and at school that will accumulate in the body anything that's on the swab?

Doctors know best

It is vital for people to realise that 'hero' doctors 'know' only what the Big Pharma-dominated medical authorities tell them to 'know' and if they refuse to 'know' what they are told to 'know' they are out the door. They are mostly not physicians or healers, but repeaters of the official narrative – or else. I have seen alleged professional doctors on British television make shocking statements that we are supposed to take seriously. One called 'Dr' Amir Khan, who is actually telling patients how to respond to illness, said that men could take the birth pill to 'help slow down the effects of Covid-19'. In March, 2021, another ridiculous 'Covid study' by an American doctor proposed injecting men with the female sex hormone progesterone as a 'Covid' treatment. British doctor Nighat Arif told the BBC that face coverings were now going to be part of ongoing normal. Yes, the vaccine protects you, she said (evidence?) ... but the way to deal with viruses in the community was always going to come down to hand washing, face covering and keeping a physical distance. That's not what we were told before the 'vaccine' was circulating. Arif said she couldn't imagine ever again going on the underground or in a lift without a mask. I was just thanking my good luck that she was not my doctor when she said – in March, 2021 – that if 'we are *behaving* and we are doing all the right things' she thought we could 'have our nearest and dearest around us at home ... around *Christmas* and *New Year*! Her patronising delivery was the usual school teacher talking to six-year-olds as she repeated every government talking point and probably believed them all. If we have learned anything from the 'Covid' experience surely it must be that humanity's perception of doctors needs a fundamental rethink. NHS 'doctor' Sara Kayat told her television audience that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Not even Big Pharma claimed that. We have to stop taking 'experts' at their word without question when so many of them are

clueless and only repeating the party line on which their careers depend. That is not to say there are not brilliant doctors – there are and I have spoken to many of them since all this began – but you won't see them in the mainstream media or quoted by the psychopaths and yes-people in government.

Remember the name – Christian Drosten

German virologist Christian Drosten, Director of Charité Institute of Virology in Berlin, became a national star after the pandemic hoax began. He was feted on television and advised the German government on 'Covid' policy. Most importantly to the wider world Drosten led a group that produced the 'Covid' testing protocol for the PCR test. What a remarkable feat given the PCR cannot test for infectious disease and even more so when you think that Drosten said that his method of testing for SARS-CoV-2 was developed 'without having virus material available'. *He developed a test for a 'virus' that he didn't have and had never seen.* Let that sink in as you survey the global devastation that came from what he did. The whole catastrophe of Drosten's 'test' was based on the alleged genetic sequence published by Chinese scientists on the Internet. We will see in the next chapter that this alleged 'genetic sequence' has never been produced by China or anyone and cannot be when there *is no* SARS-CoV-2. Drosten, however, doesn't seem to let little details like that get in the way. He was the lead author with Victor Corman from the same Charité Hospital of the paper 'Detection of 2019 novel coronavirus (2019-nCoV) by real-time PCR' published in a magazine called *Eurosurveillance*. This became known as the Corman-Drosten paper. In November, 2020, with human society devastated by the effects of the Corman-Drosten test baloney, the protocol was publicly challenged by 22 international scientists and independent researchers from Europe, the United States, and Japan. Among them were senior molecular geneticists, biochemists, immunologists, and microbiologists. They produced a document headed 'External peer review of the RTPCR test to detect SARS-Cov-2 Reveals 10 Major Flaws At The Molecular and Methodological Level: Consequences

For False-Positive Results'. The flaws in the Corman-Drosten test included the following:

- The test is non-specific because of erroneous design
- Results are enormously variable
- The test is unable to discriminate between the whole 'virus' and viral fragments
- It doesn't have positive or negative controls
- The test lacks a standard operating procedure
- It is unsupported by proper peer view

The scientists said the PCR 'Covid' testing protocol was not founded on science and they demanded the Corman-Drosten paper be retracted by *Eurosurveillance*. They said all present and previous Covid deaths, cases, and 'infection rates' should be subject to a massive retroactive inquiry. Lockdowns and travel restrictions should be reviewed and relaxed and those diagnosed through PCR to have 'Covid-19' should not be forced to isolate. Dr Kevin Corbett, a health researcher and nurse educator with a long academic career producing a stream of peer-reviewed publications at many UK universities, made the same point about the PCR test debacle. He said of the scientists' conclusions: 'Every scientific rationale for the development of that test has been totally destroyed by this paper. It's like Hiroshima/Nagasaki to the Covid test.' He said that China hadn't given them an isolated 'virus' when Drosten developed the test. Instead they had developed the test from *a sequence in a gene bank*.' Put another way ... *they made it up!* The scientists were supported in this contention by a Portuguese appeals court which ruled in November, 2020, that PCR tests are unreliable and it is unlawful to quarantine people based solely on a PCR test. The point about China not providing an isolated virus must be true when the 'virus' has never been isolated to this day and the consequences of that will become clear. Drosten and company produced this useless 'protocol' right on cue in January, 2020, just as the 'virus' was said to

be moving westward and it somehow managed to successfully pass a peer-review in 24 hours. In other words there was no peer-review for a test that would be used to decide who had 'Covid' and who didn't across the world. The Cult-created, Gates-controlled World Health Organization immediately recommended all its nearly 200 member countries to use the Drosten PCR protocol to detect 'cases' and 'deaths'. The sting was underway and it continues to this day.

So who is this Christian Drosten that produced the means through which death, destruction and economic catastrophe would be justified? His education background, including his doctoral thesis, would appear to be somewhat shrouded in mystery and his track record is dire as with another essential player in the 'Covid' hoax, the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College in London of whom more shortly. Drosten predicted in 2003 that the alleged original SARS 'virus' (SARS-1) was an epidemic that could have serious effects on economies and an effective vaccine would take at least two years to produce. Drosten's answer to every alleged 'outbreak' is a vaccine which you won't be shocked to know. What followed were just 774 official deaths worldwide and none in Germany where there were only nine cases. That is even if you believe there ever was a SARS 'virus' when the evidence is zilch and I will expand on this in the next chapter. Drosten claims to be co-discoverer of 'SARS-1' and developed a test for it in 2003. He was screaming warnings about 'swine flu' in 2009 and how it was a widespread infection far more severe than any dangers from a vaccine could be and people should get vaccinated. It would be helpful for Drosten's vocal chords if he simply recorded the words 'the virus is deadly and you need to get vaccinated' and copies could be handed out whenever the latest made-up threat comes along. Drosten's swine flu epidemic never happened, but Big Pharma didn't mind with governments spending hundreds of millions on vaccines that hardly anyone bothered to use and many who did wished they hadn't. A study in 2010 revealed that the risk of dying from swine flu, or H1N1, was no higher than that of the annual seasonal flu which is what at least most of 'it' really was as in

the case of 'Covid-19'. A media investigation into Drosten asked how with such a record of inaccuracy he could be *the* government adviser on these issues. The answer to that question is the same with Drosten, Ferguson and Fauci – they keep on giving the authorities the 'conclusions' and 'advice' they want to hear. Drosten certainly produced the goods for them in January, 2020, with his PCR protocol garbage and provided the foundation of what German internal medicine specialist Dr Claus Köhnlein, co-author of *Virus Mania*, called the 'test pandemic'. The 22 scientists in the *Eurosurveillance* challenge called out conflicts of interest within the Drosten 'protocol' group and with good reason. Olfert Landt, a regular co-author of Drosten 'studies', owns the biotech company TIB Molbiol Syntheselabor GmbH in Berlin which manufactures and sells the tests that Drosten and his mates come up with. They have done this with SARS, Enterotoxigenic E. coli (ETEC), MERS, Zika 'virus', yellow fever, and now 'Covid'. Landt told the *Berliner Zeitung* newspaper:

The testing, design and development came from the Charité [Drosten and Corman]. We simply implemented it immediately in the form of a kit. And if we don't have the virus, which originally only existed in Wuhan, we can make a synthetic gene to simulate the genome of the virus. That's what we did very quickly.

This is more confirmation that the Drosten test was designed without access to the 'virus' and only a synthetic simulation which is what SARS-CoV-2 really is – a computer-generated synthetic fiction. It's quite an enterprise they have going here. A Drosten team decides what the test for something should be and Landt's biotech company flogs it to governments and medical systems across the world. His company must have made an absolute fortune since the 'Covid' hoax began. Dr Reiner Fuellmich, a prominent German consumer protection trial lawyer in Germany and California, is on Drosten's case and that of Tedros at the World Health Organization for crimes against humanity with a class-action lawsuit being prepared in the United States and other legal action in Germany.

Why China?

Scamming the world with a 'virus' that doesn't exist would seem impossible on the face of it, but not if you have control of the relatively few people that make policy decisions and the great majority of the global media. Remember it's not about changing 'real' reality it's about controlling *perception* of reality. You don't have to make something happen you only have to make people *believe* that it's happening. Renegade Minds understand this and are therefore much harder to swindle. 'Covid-19' is not a 'real' 'virus'. It's a mind virus, like a computer virus, which has infected the minds, not the bodies, of billions. It all started, publically at least, in China and that alone is of central significance. The Cult was behind the revolution led by its asset Mao Zedong, or Chairman Mao, which established the People's Republic of China on October 1st, 1949. It should have been called The Cult's Republic of China, but the name had to reflect the recurring illusion that vicious dictatorships are run by and for the people (see all the 'Democratic Republics' controlled by tyrants). In the same way we have the 'Biden' Democratic Republic of America officially ruled by a puppet tyrant (at least temporarily) on behalf of Cult tyrants. The creation of Mao's merciless communist/fascist dictatorship was part of a frenzy of activity by the Cult at the conclusion of World War Two which, like the First World War, it had instigated through its assets in Germany, Britain, France, the United States and elsewhere. Israel was formed in 1948; the Soviet Union expanded its 'Iron Curtain' control, influence and military power with the Warsaw Pact communist alliance in 1955; the United Nations was formed in 1945 as a Cult precursor to world government; and a long list of world bodies would be established including the World Health Organization (1948), World Trade Organization (1948 under another name until 1995), International Monetary Fund (1945) and World Bank (1944). Human society was redrawn and hugely centralised in the global Problem-Reaction-Solution that was World War Two. All these changes were significant. Israel would become the headquarters of the Sabbatians

and the revolution in China would prepare the ground and control system for the events of 2019/2020.

Renegade Minds know there are no borders except for public consumption. The Cult is a seamless, borderless global entity and to understand the game we need to put aside labels like borders, nations, countries, communism, fascism and democracy. These delude the population into believing that countries are ruled within their borders by a government of whatever shade when these are mere agencies of a global power. America's illusion of democracy and China's communism/fascism are subsidiaries – vehicles – for the same agenda. We may hear about conflict and competition between America and China and on the lower levels that will be true; but at the Cult level they are branches of the same company in the way of the McDonald's example I gave earlier. I have tracked in the books over the years support by US governments of both parties for Chinese Communist Party infiltration of American society through allowing the sale of land, even military facilities, and the acquisition of American business and university influence. All this is underpinned by the infamous stealing of intellectual property and technological know-how. Cult-owned Silicon Valley corporations waive their fraudulent 'morality' to do business with human-rights-free China; Cult-controlled Disney has become China's PR department; and China in effect owns 'American' sports such as basketball which depends for much of its income on Chinese audiences. As a result any sports player, coach or official speaking out against China's horrific human rights record is immediately condemned or fired by the China-worshipping National Basketball Association. One of the first acts of China-controlled Biden was to issue an executive order telling federal agencies to stop making references to the 'virus' by the 'geographic location of its origin'. Long-time Congressman Jerry Nadler warned that criticising China, America's biggest rival, leads to hate crimes against Asian people in the United States. So shut up you bigot. China is fast closing in on Israel as a country that must not be criticised which is apt, really, given that Sabbatians control them both. The two countries have

developed close economic, military, technological and strategic ties which include involvement in China's 'Silk Road' transport and economic initiative to connect China with Europe. Israel was the first country in the Middle East to recognise the establishment of Mao's tyranny in 1950 months after it was established.

Project Wuhan – the 'Covid' Psyop

I emphasise again that the Cult plays the long game and what is happening to the world today is the result of centuries of calculated manipulation following a script to take control step-by-step of every aspect of human society. I will discuss later the common force behind all this that has spanned those centuries and thousands of years if the truth be told. Instigating the Mao revolution in China in 1949 with a 2020 'pandemic' in mind is not only how they work – the 71 years between them is really quite short by the Cult's standards of manipulation preparation. The reason for the Cult's Chinese revolution was to create a fiercely-controlled environment within which an extreme structure for human control could be incubated to eventually be unleashed across the world. We have seen this happen since the 'pandemic' emerged from China with the Chinese control-structure founded on AI technology and tyrannical enforcement sweep across the West. Until the moment when the Cult went for broke in the West and put its fascism on public display Western governments had to pay some lip-service to freedom and democracy to not alert too many people to the tyranny-in-the-making. Freedoms were more subtly eroded and power centralised with covert government structures put in place waiting for the arrival of 2020 when that smokescreen of 'freedom' could be dispensed with. The West was not able to move towards tyranny before 2020 anything like as fast as China which was created as a tyranny and had no limits on how fast it could construct the Cult's blueprint for global control. When the time came to impose that structure on the world it was the same Cult-owned Chinese communist/fascist government that provided the excuse – the 'Covid pandemic'. It was absolutely crucial to the Cult plan for the Chinese response to the 'pandemic' –

draconian lockdowns of the entire population – to become the blueprint that Western countries would follow to destroy the livelihoods and freedom of their people. This is why the Cult-owned, Gates-owned, WHO Director-General Tedros said early on:

The Chinese government is to be congratulated for the extraordinary measures it has taken to contain the outbreak. China is actually setting a new standard for outbreak response and it is not an exaggeration.

Forbes magazine said of China: ‘... those measures protected untold millions from getting the disease’. The Rockefeller Foundation ‘epidemic scenario’ document in 2010 said ‘prophetically’:

However, a few countries did fare better – China in particular. The Chinese government’s quick imposition and enforcement of mandatory quarantine for all citizens, as well as its instant and near-hermetic sealing off of all borders, saved millions of lives, stopping the spread of the virus far earlier than in other countries and enabling a swifter post-pandemic recovery.

Once again – *spooky*.

The first official story was the ‘bat theory’ or rather the bat diversion. The source of the ‘virus outbreak’ we were told was a ‘wet market’ in Wuhan where bats and other animals are bought and eaten in horrifically unhygienic conditions. Then another story emerged through the alternative media that the ‘virus’ had been released on purpose or by accident from a BSL-4 (biosafety level 4) laboratory in Wuhan not far from the wet market. The lab was reported to create and work with lethal concoctions and bioweapons. Biosafety level 4 is the highest in the World Health Organization system of safety and containment. Renegade Minds are aware of what I call designer manipulation. The ideal for the Cult is for people to buy its prime narrative which in the opening salvos of the ‘pandemic’ was the wet market story. It knows, however, that there is now a considerable worldwide alternative media of researchers sceptical of anything governments say and they are often given a version of events in a form they can perceive as credible while misdirecting them from the real truth. In this case let them

think that the conspiracy involved is a 'bioweapon virus' released from the Wuhan lab to keep them from the real conspiracy – *there is no 'virus'*. The WHO's current position on the source of the outbreak at the time of writing appears to be: 'We haven't got a clue, mate.' This is a good position to maintain mystery and bewilderment. The inner circle will know where the 'virus' came from – *nowhere*. The bottom line was to ensure the public believed there *was* a 'virus' and it didn't much matter if they thought it was natural or had been released from a lab. The belief that there was a 'deadly virus' was all that was needed to trigger global panic and fear. The population was terrified into handing their power to authority and doing what they were told. They had to or they were 'all gonna die'.

In March, 2020, information began to come my way from real doctors and scientists and my own additional research which had my intuition screaming: 'Yes, that's it! *There is no virus.*' The 'bioweapon' was not the 'virus'; it was the '*vaccine*' already being talked about that would be the bioweapon. My conclusion was further enhanced by happenings in Wuhan. The 'virus' was said to be sweeping the city and news footage circulated of people collapsing in the street (which they've never done in the West with the same 'virus'). The Chinese government was building 'new hospitals' in a matter of ten days to 'cope with demand' such was the virulent nature of the 'virus'. Yet in what seemed like no time the 'new hospitals' closed – even if they even opened – and China declared itself 'virus-free'. It was back to business as usual. This was more propaganda to promote the Chinese draconian lockdowns in the West as the way to 'beat the virus'. Trouble was that we subsequently had lockdown after lockdown, but never business as usual. As the people of the West and most of the rest of the world were caught in an ever-worsening spiral of lockdown, social distancing, masks, isolated old people, families forced apart, and livelihood destruction, it was party-time in Wuhan. Pictures emerged of thousands of people enjoying pool parties and concerts. It made no sense until you realised there never was a 'virus' and the

whole thing was a Cult set-up to transform human society out of one of its major global strongholds – China.

How is it possible to deceive virtually the entire world population into believing there is a deadly virus when there is not even a 'virus' let alone a deadly one? It's nothing like as difficult as you would think and that's clearly true because it happened.

Postscript: See end of book Postscript for more on the 'Wuhan lab virus release' story which the authorities and media were pushing heavily in the summer of 2021 to divert attention from the truth that the 'Covid virus' is pure invention.

CHAPTER FIVE

There is no 'virus'

You can fool some of the people all of the time, and all of the people some of the time, but you cannot fool all of the people all of the time
Abraham Lincoln

The greatest form of mind control is repetition. The more you repeat the same mantra of alleged 'facts' the more will accept them to be true. It becomes an 'everyone knows that, mate'. If you can also censor any other version or alternative to your alleged 'facts' you are pretty much home and cooking.

By the start of 2020 the Cult owned the global mainstream media almost in its entirety to spew out its 'Covid' propaganda and ignore or discredit any other information and view. Cult-owned social media platforms in Cult-owned Silicon Valley were poised and ready to unleash a campaign of ferocious censorship to obliterate all but the official narrative. To complete the circle many demands for censorship by Silicon Valley were led by the mainstream media as 'journalists' became full-out enforcers for the Cult both as propagandists and censors. Part of this has been the influx of young people straight out of university who have become 'journalists' in significant positions. They have no experience and a headful of programmed perceptions from their years at school and university at a time when today's young are the most perceptually-targeted generations in known human history given the insidious impact of technology. They enter the media perceptually prepared and ready to repeat the narratives of the system that programmed them to

repeat its narratives. The BBC has a truly pathetic 'specialist disinformation reporter' called Marianna Spring who fits this bill perfectly. She is clueless about the world, how it works and what is really going on. Her role is to discredit anyone doing the job that a proper journalist would do and system-serving hacks like Spring wouldn't dare to do or even see the need to do. They are too busy licking the arse of authority which can never be wrong and, in the case of the BBC propaganda programme, *Panorama*, contacting payments systems such as PayPal to have a donations page taken down for a film company making documentaries questioning vaccines. Even the BBC soap opera *EastEnders* included a disgracefully biased scene in which an inarticulate white working class woman was made to look foolish for questioning the 'vaccine' while a well-spoken black man and Asian woman promoted the government narrative. It ticked every BBC box and the fact that the black and minority community was resisting the 'vaccine' had nothing to do with the way the scene was written. The BBC has become a disgusting tyrannical propaganda and censorship operation that should be defunded and disbanded and a free media take its place with a brief to stop censorship instead of demanding it. A BBC 'interview' with Gates goes something like: 'Mr Gates, sir, if I can call you sir, would you like to tell our audience why you are such a great man, a wonderful humanitarian philanthropist, and why you should absolutely be allowed as a software salesman to decide health policy for approaching eight billion people? Thank you, sir, please sir.' Propaganda programming has been incessant and merciless and when all you hear is the same story from the media, repeated by those around you who have only heard the same story, is it any wonder that people on a grand scale believe absolute mendacious garbage to be true? You are about to see, too, why this level of information control is necessary when the official 'Covid' narrative is so nonsensical and unsupportable by the evidence.

Structure of Deceit

The pyramid structure through which the 'Covid' hoax has been manifested is very simple and has to be to work. As few people as possible have to be involved with full knowledge of what they are doing – and why – or the real story would get out. At the top of the pyramid are the inner core of the Cult which controls Bill Gates who, in turn, controls the World Health Organization through his pivotal funding and his puppet Director-General mouthpiece, Tedros. Before he was appointed Tedros was chair of the Gates-founded Global Fund to 'fight against AIDS, tuberculosis and malaria', a board member of the Gates-funded 'vaccine alliance' GAVI, and on the board of another Gates-funded organisation. Gates owns him and picked him for a specific reason – Tedros is a crook and worse. 'Dr' Tedros (he's not a medical doctor, the first WHO chief not to be) was a member of the tyrannical Marxist government of Ethiopia for decades with all its human rights abuses. He has faced allegations of corruption and misappropriation of funds and was exposed three times for covering up cholera epidemics while Ethiopia's health minister. Tedros appointed the mass-murdering genocidal Zimbabwe dictator Robert Mugabe as a WHO goodwill ambassador for public health which, as with Tedros, is like appointing a psychopath to run a peace and love campaign. The move was so ridiculous that he had to drop Mugabe in the face of widespread condemnation. American economist David Steinman, a Nobel peace prize nominee, lodged a complaint with the International Criminal Court in The Hague over alleged genocide by Tedros when he was Ethiopia's foreign minister. Steinman says Tedros was a 'crucial decision maker' who directed the actions of Ethiopia's security forces from 2013 to 2015 and one of three officials in charge when those security services embarked on the 'killing' and 'torturing' of Ethiopians. You can see where Tedros is coming from and it's sobering to think that he has been the vehicle for Gates and the Cult to direct the global response to 'Covid'. Think about that. A psychopathic Cult dictates to psychopath Gates who dictates to psychopath Tedros who dictates how countries of the world must respond to a 'Covid virus' never scientifically shown to exist. At the same time psychopathic Cult-owned Silicon Valley information

giants like Google, YouTube, Facebook and Twitter announced very early on that they would give the Cult/Gates/Tedros/WHO version of the narrative free advertising and censor those who challenged their intelligence-insulting, mendacious story.

The next layer in the global 'medical' structure below the Cult, Gates and Tedros are the chief medical officers and science 'advisers' in each of the WHO member countries which means virtually all of them. Medical officers and arbiters of science (they're not) then take the WHO policy and recommended responses and impose them on their country's population while the political 'leaders' say they are deciding policy (they're clearly not) by 'following the science' on the advice of the 'experts' – the same medical officers and science 'advisers' (dictators). In this way with the rarest of exceptions the entire world followed the same policy of lockdown, people distancing, masks and 'vaccines' dictated by the psychopathic Cult, psychopathic Gates and psychopathic Tedros who we are supposed to believe give a damn about the health of the world population they are seeking to enslave. That, amazingly, is all there is to it in terms of crucial decision-making. Medical staff in each country then follow like sheep the dictates of the shepherds at the top of the national medical hierarchies – chief medical officers and science 'advisers' who themselves follow like sheep the shepherds of the World Health Organization and the Cult. Shepherds at the national level often have major funding and other connections to Gates and his Bill and Melinda Gates Foundation which carefully hands out money like confetti at a wedding to control the entire global medical system from the WHO down.

Follow the money

Christopher Whitty, Chief Medical Adviser to the UK Government at the centre of 'virus' policy, a senior adviser to the government's Scientific Advisory Group for Emergencies (SAGE), and Executive Board member of the World Health Organization, was gifted a grant of \$40 million by the Bill and Melinda Gates Foundation for malaria research in Africa. The BBC described the unelected Whitty as 'the

official who will probably have the greatest impact on our everyday lives of any individual policymaker in modern times' and so it turned out. What Gates and Tedros have said Whitty has done like his equivalents around the world. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of Big Pharma giant GlaxoSmithKline with its fundamental financial and business connections to Bill Gates. In September, 2020, it was revealed that Vallance owned a deferred bonus of shares in GlaxoSmithKline worth £600,000 while the company was 'developing' a 'Covid vaccine'. Move along now – nothing to see here – what could possibly be wrong with that? Imperial College in London, a major player in 'Covid' policy in Britain and elsewhere with its 'Covid-19' Response Team, is funded by Gates and has big connections to China while the now infamous Professor Neil Ferguson, the useless 'computer modeller' at Imperial College is also funded by Gates. Ferguson delivered the dramatically inaccurate excuse for the first lockdowns (much more in the next chapter). The Institute for Health Metrics and Evaluation (IHME) in the United States, another source of outrageously false 'Covid' computer models to justify lockdowns, is bankrolled by Gates who is a vehement promotor of lockdowns. America's version of Whitty and Vallance, the again now infamous Anthony Fauci, has connections to 'Covid vaccine' maker Moderna as does Bill Gates through funding from the Bill and Melinda Gates Foundation. Fauci is director of the National Institute of Allergy and Infectious Diseases (NIAID), a major recipient of Gates money, and they are very close. Deborah Birx who was appointed White House Coronavirus Response Coordinator in February, 2020, is yet another with ties to Gates. Everywhere you look at the different elements around the world behind the coordination and decision making of the 'Covid' hoax there is Bill Gates and his money. They include the World Health Organization; Centers for Disease Control (CDC) in the United States; National Institutes of Health (NIH) of Anthony Fauci; Imperial College and Neil Ferguson; the London School of Hygiene where Chris Whitty worked; Regulatory agencies like the UK Medicines & Healthcare products Regulatory Agency (MHRA)

which gave emergency approval for 'Covid vaccines'; Wellcome Trust; GAVI, the Vaccine Alliance; the Coalition for Epidemic Preparedness Innovations (CEPI); Johns Hopkins University which has compiled the false 'Covid' figures; and the World Economic Forum. A [Nationalfile.com](https://www.nationalfile.com) article said:

Gates has a lot of pull in the medical world, he has a multi-million dollar relationship with Dr. Fauci, and Fauci originally took the Gates line supporting vaccines and casting doubt on [the drug hydroxychloroquine]. Coronavirus response team member Dr. Deborah Birx, appointed by former president Obama to serve as United States Global AIDS Coordinator, also sits on the board of a group that has received billions from Gates' foundation, and Birx reportedly used a disputed Bill Gates-funded model for the White House's Coronavirus effort. Gates is a big proponent for a population lockdown scenario for the Coronavirus outbreak.

Another funder of Moderna is the Defense Advanced Research Projects Agency (DARPA), the technology-development arm of the Pentagon and one of the most sinister organisations on earth. DARPA had a major role with the CIA covert technology-funding operation In-Q-Tel in the development of Google and social media which is now at the centre of global censorship. Fauci and Gates are extremely close and openly admit to talking regularly about 'Covid' policy, but then why wouldn't Gates have a seat at every national 'Covid' table after his Foundation committed \$1.75 billion to the 'fight against Covid-19'. When passed through our Orwellian Translation Unit this means that he has bought and paid for the Cult-driven 'Covid' response worldwide. Research the major 'Covid' response personnel in your own country and you will find the same Gates funding and other connections again and again. Medical and science chiefs following World Health Organization 'policy' sit atop a medical hierarchy in their country of administrators, doctors and nursing staff. These 'subordinates' are told they must work and behave in accordance with the policy delivered from the 'top' of the national 'health' pyramid which is largely the policy delivered by the WHO which is the policy delivered by Gates and the Cult. The whole 'Covid' narrative has been imposed on medical staff by a climate of fear although great numbers don't even need that to comply. They do so through breathtaking levels of ignorance and

include doctors who go through life simply repeating what Big Pharma and their hierarchical masters tell them to say and believe. No wonder Big Pharma 'medicine' is one of the biggest killers on Planet Earth.

The same top-down system of intimidation operates with regard to the Cult Big Pharma cartel which also dictates policy through national and global medical systems in this way. The Cult and Big Pharma agendas are the same because the former controls and owns the latter. 'Health' administrators, doctors, and nursing staff are told to support and parrot the dictated policy or they will face consequences which can include being fired. How sad it's been to see medical staff meekly repeating and imposing Cult policy without question and most of those who can see through the deceit are only willing to speak anonymously off the record. They know what will happen if their identity is known. This has left the courageous few to expose the lies about the 'virus', face masks, overwhelmed hospitals that aren't, and the dangers of the 'vaccine' that isn't a vaccine. When these medical professionals and scientists, some renowned in their field, have taken to the Internet to expose the truth their articles, comments and videos have been deleted by Cult-owned Facebook, Twitter and YouTube. What a real head-shaker to see YouTube videos with leading world scientists and highly qualified medical specialists with an added link underneath to the notorious Cult propaganda website *Wikipedia* to find the 'facts' about the same subject.

HIV – the 'Covid' trial-run

I'll give you an example of the consequences for health and truth that come from censorship and unquestioning belief in official narratives. The story was told by PCR inventor Kary Mullis in his book *Dancing Naked in the Mind Field*. He said that in 1984 he accepted as just another scientific fact that Luc Montagnier of France's Pasteur Institute and Robert Gallo of America's National Institutes of Health had independently discovered that a 'retrovirus' dubbed HIV (human immunodeficiency virus) caused AIDS. They

were, after all, Mullis writes, specialists in retroviruses. This is how the medical and science pyramids work. Something is announced or *assumed* and then becomes an everybody-knows-that purely through repetition of the assumption as if it is fact. Complete crap becomes accepted truth with no supporting evidence and only repetition of the crap. This is how a 'virus' that doesn't exist became the 'virus' that changed the world. The HIV-AIDS fairy story became a multi-billion pound industry and the media poured out propaganda terrifying the world about the deadly HIV 'virus' that caused the lethal AIDS. By then Mullis was working at a lab in Santa Monica, California, to detect retroviruses with his PCR test in blood donations received by the Red Cross. In doing so he asked a virologist where he could find a reference for HIV being the cause of AIDS. 'You don't need a reference,' the virologist said ... '*Everybody knows it.*' Mullis said he wanted to quote a reference in the report he was doing and he said he felt a little funny about not knowing the source of such an important discovery when everyone else seemed to. The virologist suggested he cite a report by the Centers for Disease Control and Prevention (CDC) on morbidity and mortality. Mullis read the report, but it only said that an organism had been identified and did not say how. The report did not identify the original scientific work. Physicians, however, *assumed* (key recurring theme) that if the CDC was convinced that HIV caused AIDS then proof must exist. Mullis continues:

I did computer searches. Neither Montagnier, Gallo, nor anyone else had published papers describing experiments which led to the conclusion that HIV probably caused AIDS. I read the papers in *Science* for which they had become well known as AIDS doctors, but all they had said there was that they had found evidence of a past infection by something which was probably HIV in some AIDS patients.

They found antibodies. Antibodies to viruses had always been considered evidence of past disease, not present disease. Antibodies signaled that the virus had been defeated. The patient had saved himself. There was no indication in these papers that this virus caused a disease. They didn't show that everybody with the antibodies had the disease. In fact they found some healthy people with antibodies.

Mullis asked why their work had been published if Montagnier and Gallo hadn't really found this evidence, and why had they been fighting so hard to get credit for the discovery? He says he was hesitant to write 'HIV is the probable cause of AIDS' until he found published evidence to support that. 'Tens of thousands of scientists and researchers were spending billions of dollars a year doing research based on this idea,' Mullis writes. 'The reason had to be there somewhere; otherwise these people would not have allowed their research to settle into one narrow channel of investigation.' He said he lectured about PCR at numerous meetings where people were always talking about HIV and he asked them how they knew that HIV was the cause of AIDS:

Everyone said something. Everyone had the answer at home, in the office, in some drawer. They all knew, and they would send me the papers as soon as they got back. But I never got any papers. Nobody ever sent me the news about how AIDS was caused by HIV.

Eventually Mullis was able to ask Montagnier himself about the reference proof when he lectured in San Diego at the grand opening of the University of California AIDS Research Center. Mullis says this was the last time he would ask his question without showing anger. Montagnier said he should reference the CDC report. 'I read it', Mullis said, and it didn't answer the question. 'If Montagnier didn't know the answer who the hell did?' Then one night Mullis was driving when an interview came on National Public Radio with Peter Duesberg, a prominent virologist at Berkeley and a California Scientist of the Year. Mullis says he finally understood why he could not find references that connected HIV to AIDS – *there weren't any!* No one had ever proved that HIV causes AIDS even though it had spawned a multi-billion pound global industry and the media was repeating this as fact every day in their articles and broadcasts terrifying the shit out of people about AIDS and giving the impression that a positive test for HIV (see 'Covid') was a death sentence. Duesberg was a threat to the AIDS gravy train and the agenda that underpinned it. He was therefore abused and castigated after he told the Proceedings of the National Academy of Sciences

there was no good evidence implicating the new 'virus'. Editors rejected his manuscripts and his research funds were deleted. Mullis points out that the CDC has defined AIDS as one of more than 30 diseases *if accompanied* by a positive result on a test that detects antibodies to HIV; but those same diseases are not defined as AIDS cases when antibodies are not detected:

If an HIV-positive woman develops uterine cancer, for example, she is considered to have AIDS. If she is not HIV positive, she simply has uterine cancer. An HIV-positive man with tuberculosis has AIDS; if he tests negative he simply has tuberculosis. If he lives in Kenya or Colombia, where the test for HIV antibodies is too expensive, he is simply presumed to have the antibodies and therefore AIDS, and therefore he can be treated in the World Health Organization's clinic. It's the only medical help available in some places. And it's free, because the countries that support WHO are worried about AIDS.

Mullis accuses the CDC of continually adding new diseases (see ever more 'Covid symptoms') to the grand AIDS definition and of virtually doctoring the books to make it appear as if the disease continued to spread. He cites how in 1993 the CDC enormously broadened its AIDS definition and county health authorities were delighted because they received \$2,500 per year from the Federal government for every reported AIDS case. Ladies and gentlemen, I have just described, via Kary Mullis, the 'Covid pandemic' of 2020 and beyond. Every element is the same and it's been pulled off in the same way by the same networks.

The 'Covid virus' exists? Okay – prove it. Er ... still waiting

What Kary Mullis described with regard to 'HIV' has been repeated with 'Covid'. A claim is made that a new, or 'novel', infection has been found and the entire medical system of the world repeats that as fact exactly as they did with HIV and AIDS. No one in the mainstream asks rather relevant questions such as 'How do you know?' and 'Where is your proof?' The SARS-Cov-2 'virus' and the 'Covid-19 disease' became an overnight 'everybody-knows-that'. The origin could be debated and mulled over, but what you could not suggest was that 'SARS-Cov-2' didn't exist. That would be

ridiculous. 'Everybody knows' the 'virus' exists. Well, I didn't for one along with American proper doctors like Andrew Kaufman and Tom Cowan and long-time American proper journalist Jon Rappaport. We dared to pursue the obvious and simple question: 'Where's the evidence?' The overwhelming majority in medicine, journalism and the general public did not think to ask that. After all, *everyone knew* there was a new 'virus'. Everyone was saying so and I heard it on the BBC. Some would eventually argue that the 'deadly virus' was nothing like as deadly as claimed, but few would venture into the realms of its very existence. Had they done so they would have found that the evidence for that claim had gone AWOL as with HIV causes AIDS. In fact, not even that. For something to go AWOL it has to exist in the first place and scientific proof for a 'SARS-Cov-2' can be filed under nothing, nowhere and zilch.

Dr Andrew Kaufman is a board-certified forensic psychiatrist in New York State, a Doctor of Medicine and former Assistant Professor and Medical Director of Psychiatry at SUNY Upstate Medical University, and Medical Instructor of Hematology and Oncology at the Medical School of South Carolina. He also studied biology at the Massachusetts Institute of Technology (MIT) and trained in Psychiatry at Duke University. Kaufman is retired from allopathic medicine, but remains a consultant and educator on natural healing. I saw a video of his very early on in the 'Covid' hoax in which he questioned claims about the 'virus' in the absence of any supporting evidence and with plenty pointing the other way. I did everything I could to circulate his work which I felt was asking the pivotal questions that needed an answer. I can recommend an excellent pull-together interview he did with the website The Last Vagabond entitled *Dr Andrew Kaufman: Virus Isolation, Terrain Theory and Covid-19* and his website is andrewkaufmanmd.com. Kaufman is not only a forensic psychiatrist; he is forensic in all that he does. He always reads original scientific papers, experiments and studies instead of second-third-fourth-hand reports about the 'virus' in the media which are repeating the repeated repetition of the narrative. When he did so with the original Chinese 'virus' papers Kaufman

realised that there was no evidence of a 'SARS-Cov-2'. They had never – from the start – shown it to exist and every repeat of this claim worldwide was based on the accepted existence of proof that was nowhere to be found – see Kary Mullis and HIV. Here we go again.

Let's postulate

Kaufman discovered that the Chinese authorities immediately concluded that the cause of an illness that broke out among about 200 initial patients in Wuhan was a 'new virus' when there were no grounds to make that conclusion. The alleged 'virus' was not isolated from other genetic material in their samples and then shown through a system known as Koch's postulates to be the causative agent of the illness. The world was told that the SARS-Cov-2 'virus' caused a disease they called 'Covid-19' which had 'flu-like' symptoms and could lead to respiratory problems and pneumonia. If it wasn't so tragic it would almost be funny. *'Flu-like' symptoms? Pneumonia? Respiratory disease?* What in *CHINA* and particularly in *Wuhan*, one of the most polluted cities in the world with a resulting epidemic of respiratory disease?? Three hundred thousand people get pneumonia in China every year and there are nearly a billion cases worldwide of 'flu-like symptoms'. These have a whole range of causes – including pollution in Wuhan – but no other possibility was credibly considered in late 2019 when the world was told there was a new and deadly 'virus'. The global prevalence of pneumonia and 'flu-like systems' gave the Cult networks unlimited potential to re-diagnose these other causes as the mythical 'Covid-19' and that is what they did from the very start. Kaufman revealed how Chinese medical and science authorities (all subordinates to the Cult-owned communist government) took genetic material from the lungs of only a few of the first patients. The material contained their own cells, bacteria, fungi and other microorganisms living in their bodies. The only way you could prove the existence of the 'virus' and its responsibility for the alleged 'Covid-19' was to isolate the virus from all the other material – a process also known as 'purification' – and

then follow the postulates sequence developed in the late 19th century by German physician and bacteriologist Robert Koch which became the 'gold standard' for connecting an alleged causation agent to a disease:

1. The microorganism (bacteria, fungus, virus, etc.) must be present in every case of the disease and all patients must have the same symptoms. It must also *not be present in healthy individuals*.
2. The microorganism must be isolated from the host with the disease. If the microorganism is a bacteria or fungus it must be grown in a pure culture. If it is a virus, it must be purified (i.e. containing no other material except the virus particles) from a clinical sample.
3. The specific disease, with all of its characteristics, must be reproduced when the infectious agent (the purified virus or a pure culture of bacteria or fungi) is inoculated into a healthy, susceptible host.
4. The microorganism must be recoverable from the experimentally infected host as in step 2.

Not one of these criteria has been met in the case of 'SARS-Cov-2' and 'Covid-19'. Not ONE. EVER. Robert Koch refers to bacteria and not viruses. What are called 'viral particles' are so minute (hence masks are useless by any definition) that they could only be seen after the invention of the electron microscope in the 1930s and can still only be observed through that means. American bacteriologist and virologist Thomas Milton Rivers, the so-called 'Father of Modern Virology' who was very significantly director of the Rockefeller Institute for Medical Research in the 1930s, developed a less stringent version of Koch's postulates to identify 'virus' causation known as 'Rivers criteria'. 'Covid' did not pass that process either. Some even doubt whether any 'virus' can be isolated from other particles containing genetic material in the Koch method. Freedom of Information requests in many countries asking for scientific proof that the 'Covid virus' has been purified and isolated and shown to exist have all come back with a 'we don't have that' and when this happened with a request to the UK Department of Health they added this comment:

However, outside of the scope of the [Freedom of Information Act] and on a discretionary basis, the following information has been advised to us, which may be of interest. Most infectious diseases are caused by viruses, bacteria or fungi. Some bacteria or fungi have the capacity to grow on their own in isolation, for example in colonies on a petri dish. Viruses are different in that they are what we call 'obligate pathogens' – that is, they cannot survive or reproduce without infecting a host ...

... For some diseases, it is possible to establish causation between a microorganism and a disease by isolating the pathogen from a patient, growing it in pure culture and reintroducing it to a healthy organism. These are known as 'Koch's postulates' and were developed in 1882. However, as our understanding of disease and different disease-causing agents has advanced, these are no longer the method for determining causation [Andrew Kaufman asks why in that case are there two published articles falsely claiming to satisfy Koch's postulates].

It has long been known that viral diseases cannot be identified in this way as viruses cannot be grown in 'pure culture'. When a patient is tested for a viral illness, this is normally done by looking for the presence of antigens, or viral genetic code in a host with molecular biology techniques [Kaufman asks how you could know the origin of these chemicals without having a pure culture for comparison].

For the record 'antigens' are defined so:

Invading microorganisms have antigens on their surface that the human body can recognise as being foreign – meaning not belonging to it. When the body recognises a foreign antigen, lymphocytes (white blood cells) produce antibodies, which are complementary in shape to the antigen.

Notwithstanding that this is open to question in relation to 'SARS-Cov-2' the presence of 'antibodies' can have many causes and they are found in people that are perfectly well. Kary Mullis said: 'Antibodies ... had always been considered evidence of past disease, not present disease.'

'Covid' really is a *computer* 'virus'

Where the UK Department of Health statement says 'viruses' are now 'diagnosed' through a 'viral genetic code in a host with molecular biology techniques', they mean ... *the PCR test* which its inventor said cannot test for infectious disease. They have no credible method of connecting a 'virus' to a disease and we will see that there is no scientific proof that any 'virus' causes any disease or there is any such thing as a 'virus' in the way that it is described. Tenacious Canadian researcher Christine Massey and her team made

some 40 Freedom of Information requests to national public health agencies in different countries asking for proof that SARS-CoV-2 has been isolated and not one of them could supply that information. Massey said of her request in Canada: 'Freedom of Information reveals Public Health Agency of Canada has no record of 'SARS-COV-2' isolation performed by anyone, anywhere, ever.' If you accept the comment from the UK Department of Health it's because they can't isolate a 'virus'. Even so many 'science' papers claimed to have isolated the 'Covid virus' until they were questioned and had to admit they hadn't. A reply from the Robert Koch Institute in Germany was typical: 'I am not aware of a paper which purified isolated SARS-CoV-2.' So what the hell was Christian Drosten and his gang using to design the 'Covid' testing protocol that has produced all the illusory Covid' cases and 'Covid' deaths when the head of the Chinese version of the CDC admitted there was a problem right from the start in that the 'virus' had never been isolated/purified? Breathe deeply: What they are calling 'Covid' is actually created by a *computer program* i.e. *they made it up* – er, that's it. They took lung fluid, with many sources of genetic material, from one single person alleged to be infected with Covid-19 by a PCR test which they *claimed*, without clear evidence, contained a 'virus'. They used several computer programs to create a model of a theoretical virus genome sequence from more than fifty-six million small sequences of RNA, each of an unknown source, assembling them like a puzzle with no known solution. The computer filled in the gaps with sequences from bits in the gene bank to make it look like a bat SARS-like coronavirus! A wave of the magic wand and poof, an *in silico* (computer-generated) genome, a scientific fantasy, was created. UK health researcher Dr Kevin Corbett made the same point with this analogy:

... It's like giving you a few bones and saying that's your fish. It could be any fish. Not even a skeleton. Here's a few fragments of bones. That's your fish ... It's all from gene bank and the bits of the virus sequence that weren't there they made up.

They synthetically created them to fill in the blanks. That's what genetics is; it's a code. So it's ABBCCDDDD and you're missing some what you think is EEE so you put it in. It's all

synthetic. You just manufacture the bits that are missing. This is the end result of the geneticization of virology. This is basically a computer virus.

Further confirmation came in an email exchange between British citizen journalist Frances Leader and the government's Medicines & Healthcare Products Regulatory Agency (the Gates-funded MHRA) which gave emergency permission for untested 'Covid vaccines' to be used. The agency admitted that the 'vaccine' is not based on an isolated 'virus', but comes from a *computer-generated model*. Frances Leader was naturally banned from Cult-owned fascist Twitter for making this exchange public. The process of creating computer-generated alleged 'viruses' is called 'in silico' or 'in silicon' – computer chips – and the term 'in silico' is believed to originate with biological experiments using only a computer in 1989. 'Vaccines' involved with 'Covid' are also produced 'in silico' or by computer not a natural process. If the original 'virus' is nothing more than a made-up computer model how can there be 'new variants' of something that never existed in the first place? They are not new 'variants'; they are new *computer models* only minutely different to the original program and designed to further terrify the population into having the 'vaccine' and submitting to fascism. You want a 'new variant'? Click, click, enter – there you go. Tell the medical profession that you have discovered a 'South African variant', 'UK variants' or a 'Brazilian variant' and in the usual HIV-causes-AIDS manner they will unquestioningly repeat it with no evidence whatsoever to support these claims. They will go on television and warn about the dangers of 'new variants' while doing nothing more than repeating what they have been told to be true and knowing that any deviation from that would be career suicide. Big-time insiders will know it's a hoax, but much of the medical community is clueless about the way they are being played and themselves play the public without even being aware they are doing so. What an interesting 'coincidence' that AstraZeneca and Oxford University were conducting 'Covid vaccine trials' in the three countries – the UK, South Africa and Brazil – where the first three 'variants' were claimed to have 'broken out'.

Here's your 'virus' – it's a unicorn

Dr Andrew Kaufman presented a brilliant analysis describing how the 'virus' was imagined into fake existence when he dissected an article published by *Nature* and written by 19 authors detailing *alleged* 'sequencing of a complete viral genome' of the 'new SARS-CoV-2 virus'. This computer-modelled *in silico* genome was used as a template for all subsequent genome sequencing experiments that resulted in the so-called variants which he said now number more than 6,000. The fake genome was constructed from more than 56 million individual short strands of RNA. Those little pieces were assembled into longer pieces by finding areas of overlapping sequences. The computer programs created over two million possible combinations from which the authors simply chose the longest one. They then compared this to a 'bat virus' and the computer 'alignment' rearranged the sequence and filled in the gaps! They called this computer-generated abomination the 'complete genome'. Dr Tom Cowan, a fellow medical author and collaborator with Kaufman, said such computer-generation constitutes scientific fraud and he makes this superb analogy:

Here is an equivalency: A group of researchers claim to have found a unicorn because they found a piece of a hoof, a hair from a tail, and a snippet of a horn. They then add that information into a computer and program it to re-create the unicorn, and they then claim this computer re-creation is the real unicorn. Of course, they had never actually seen a unicorn so could not possibly have examined its genetic makeup to compare their samples with the actual unicorn's hair, hooves and horn.

The researchers claim they decided which is the real genome of SARS-CoV-2 by 'consensus', sort of like a vote. Again, different computer programs will come up with different versions of the imaginary 'unicorn', so they come together as a group and decide which is the real imaginary unicorn.

This is how the 'virus' that has transformed the world was brought into fraudulent 'existence'. Extraordinary, yes, but as the Nazis said the bigger the lie the more will believe it. Cowan, however, wasn't finished and he went on to identify what he called the real blockbuster in the paper. He quotes this section from a paper written

by virologists and published by the CDC and then explains what it means:

Therefore, we examined the capacity of SARS-CoV-2 to infect and replicate in several common primate and human cell lines, including human adenocarcinoma cells (A549), human liver cells (HUH 7.0), and human embryonic kidney cells (HEK-293T). In addition to Vero E6 and Vero CCL81 cells. ... Each cell line was inoculated at high multiplicity of infection and examined 24h post-infection.

No CPE was observed in any of the cell lines except in Vero cells, which grew to greater than 10 to the 7th power at 24 h post-infection. In contrast, HUH 7.0 and 293T showed only modest viral replication, and A549 cells were incompatible with SARS CoV-2 infection.

Cowan explains that when virologists attempt to prove infection they have three possible 'hosts' or models on which they can test. The first was humans. Exposure to humans was generally not done for ethical reasons and has never been done with SARS-CoV-2 or any coronavirus. The second possible host was animals. Cowan said that forgetting for a moment that they never actually use purified virus when exposing animals they do use solutions that they *claim* contain the virus. Exposure to animals has been done with SARS-CoV-2 in an experiment involving mice and this is what they found: *None of the wild (normal) mice got sick*. In a group of genetically-modified mice, a statistically insignificant number lost weight and had slightly bristled fur, but they experienced nothing like the illness called 'Covid-19'. Cowan said the third method – the one they mostly rely on – is to inoculate solutions they *say* contain the virus onto a variety of tissue cultures. This process had never been shown to kill tissue *unless* the sample material was starved of nutrients and poisoned as *part of the process*. Yes, incredibly, in tissue experiments designed to show the 'virus' is responsible for killing the tissue they starve the tissue of nutrients and add toxic drugs including antibiotics and they do not have control studies to see if it's the starvation and poisoning that is degrading the tissue rather than the 'virus' they allege to be in there somewhere. You want me to pinch you? Yep, I understand. Tom Cowan said this about the whole nonsensical farce as he explains what that quote from the CDC paper really means:

The shocking thing about the above quote is that using their own methods, the virologists found that solutions containing SARS-CoV-2 – even in high amounts – were NOT, I repeat NOT, infective to any of the three human tissue cultures they tested. In plain English, this means they proved, on their terms, that this ‘new coronavirus’ is not infectious to human beings. It is ONLY infective to monkey kidney cells, and only then when you add two potent drugs (gentamicin and amphotericin), known to be toxic to kidneys, to the mix.

My friends, read this again and again. These virologists, published by the CDC, performed a clear proof, on their terms, showing that the SARS-CoV-2 virus is harmless to human beings. That is the only possible conclusion, but, unfortunately, this result is not even mentioned in their conclusion. They simply say they can provide virus stocks cultured only on monkey Vero cells, thanks for coming.

Cowan concluded: ‘If people really understood how this “science” was done, I would hope they would storm the gates and demand honesty, transparency and truth.’ Dr Michael Yeadon, former Vice President and Chief Scientific Adviser at drug giant Pfizer has been a vocal critic of the ‘Covid vaccine’ and its potential for multiple harm. He said in an interview in April, 2021, that ‘not one [vaccine] has the virus. He was asked why vaccines normally using a ‘dead’ version of a disease to activate the immune system were not used for ‘Covid’ and instead we had the synthetic methods of the ‘mRNA Covid vaccine’. Yeadon said that to do the former ‘you’d have to have some of [the virus] wouldn’t you?’ He added: ‘No-one’s got any – seriously.’ Yeadon said that surely they couldn’t have fooled the whole world for a year without having a virus, ‘but oddly enough ask around – no one’s got it’. He didn’t know why with all the ‘great labs’ around the world that the virus had not been isolated – ‘Maybe they’ve been too busy running bad PCR tests and vaccines that people don’t need.’ What is today called ‘science’ is not ‘science’ at all. Science is no longer what is, but whatever people can be manipulated to *believe* that it is. Real science has been hijacked by the Cult to dispense and produce the ‘expert scientists’ and contentions that suit the agenda of the Cult. How big-time this has happened with the ‘Covid’ hoax which is entirely based on fake science delivered by fake ‘scientists’ and fake ‘doctors’. The human-caused climate change hoax is also entirely based on fake science delivered by fake ‘scientists’ and fake ‘climate experts’. In both cases real

scientists, climate experts and doctors have their views suppressed and deleted by the Cult-owned science establishment, media and Silicon Valley. This is the 'science' that politicians claim to be 'following' and a common denominator of 'Covid' and climate are Cult psychopaths Bill Gates and his mate Klaus Schwab at the Gates-funded World Economic Forum. But, don't worry, it's all just a coincidence and absolutely nothing to worry about. Zzzzzzzzz.

What is a 'virus' REALLY?

Dr Tom Cowan is one of many contesting the very existence of viruses let alone that they cause disease. This is understandable when there is no scientific evidence for a disease-causing 'virus'. German virologist Dr Stefan Lanka won a landmark case in 2017 in the German Supreme Court over his contention that there is no such thing as a measles virus. He had offered a big prize for anyone who could prove there is and Lanka won his case when someone sought to claim the money. There is currently a prize of more than 225,000 euros on offer from an Isolate Truth Fund for anyone who can prove the isolation of SARS-CoV-2 and its genetic substance. Lanka wrote in an article headed 'The Misconception Called Virus' that scientists think a 'virus' is causing tissue to become diseased and degraded when in fact it is the *processes they are using* which do that – not a 'virus'. Lanka has done an important job in making this point clear as Cowan did in his analysis of the CDC paper. Lanka says that all claims about viruses as disease-causing pathogens are wrong and based on 'easily recognisable, understandable and verifiable misinterpretations.' Scientists believed they were working with 'viruses' in their laboratories when they were really working with 'typical particles of specific dying tissues or cells ...' Lanka said that the tissue decaying process claimed to be caused by a 'virus' still happens when no alleged 'virus' is involved. It's the *process* that does the damage and not a 'virus'. The genetic sample is deprived of nutrients, removed from its energy supply through removal from the body and then doused in toxic antibiotics to remove any bacteria. He confirms again that establishment scientists do not (pinch me)

conduct control experiments to see if this is the case and if they did they would see the claims that 'viruses' are doing the damage is nonsense. He adds that during the measles 'virus' court case he commissioned an independent laboratory to perform just such a control experiment and the result was that the tissues and cells died in the exact same way as with alleged 'infected' material. This is supported by a gathering number of scientists, doctors and researchers who reject what is called 'germ theory' or the belief in the body being infected by contagious sources emitted by other people. Researchers Dawn Lester and David Parker take the same stance in their highly-detailed and sourced book *What Really Makes You Ill – Why everything you thought you knew about disease is wrong* which was recommended to me by a number of medical professionals genuinely seeking the truth. Lester and Parker say there is no provable scientific evidence to show that a 'virus' can be transmitted between people or people and animals or animals and people:

The definition also claims that viruses are the cause of many diseases, as if this has been definitively proven. But this is not the case; there is no original scientific evidence that definitively demonstrates that any virus is the cause of any disease. The burden of proof for any theory lies with those who proposed it; but none of the existing documents provides 'proof' that supports the claim that 'viruses' are pathogens.

Dr Tom Cowan employs one of his clever analogies to describe the process by which a 'virus' is named as the culprit for a disease when what is called a 'virus' is only material released by cells detoxing themselves from infiltration by chemical or radiation poisoning. The tidal wave of technologically-generated radiation in the 'smart' modern world plus all the toxic food and drink are causing this to happen more than ever. Deluded 'scientists' misread this as a gathering impact of what they wrongly label 'viruses'.

Paper can infect houses

Cowan said in an article for davidicke.com – with his tongue only mildly in his cheek – that he believed he had made a tremendous

discovery that may revolutionise science. He had discovered that small bits of paper are alive, 'well alive-ish', can 'infect' houses, and then reproduce themselves inside the house. The result was that this explosion of growth in the paper inside the house causes the house to explode, blowing it to smithereens. His evidence for this new theory is that in the past months he had carefully examined many of the houses in his neighbourhood and found almost no scraps of paper on the lawns and surrounds of the house. There was an occasional stray label, but nothing more. Then he would return to these same houses a week or so later and with a few, not all of them, particularly the old and decrepit ones, he found to his shock and surprise they were littered with stray bits of paper. He knew then that the paper had infected these houses, made copies of itself, and blew up the house. A young boy on a bicycle at one of the sites told him he had seen a demolition crew using dynamite to explode the house the previous week, but Cowan dismissed this as the idle thoughts of silly boys because 'I was on to something big'. He was on to how 'scientists' mistake genetic material in the detoxifying process for something they call a 'virus'. Cowan said of his house and paper story:

If this sounds crazy to you, it's because it should. This scenario is obviously nuts. But consider this admittedly embellished, for effect, current viral theory that all scientists, medical doctors and virologists currently believe.

He takes the example of the 'novel SARS-Cov2' virus to prove the point. First they take someone with an undefined illness called 'Covid-19' and don't even attempt to find any virus in their sputum. Never mind the scientists still describe how this 'virus', which they have not located attaches to a cell receptor, injects its genetic material, in 'Covid's' case, RNA, into the cell. The RNA once inserted exploits the cell to reproduce itself and makes 'thousands, nay millions, of copies of itself ... Then it emerges victorious to claim its next victim':

If you were to look in the scientific literature for proof, actual scientific proof, that uniform SARS-CoV2 viruses have been properly isolated from the sputum of a sick person, that actual spike proteins could be seen protruding from the virus (which has not been found), you would find that such evidence doesn't exist.

If you go looking in the published scientific literature for actual pictures, proof, that these spike proteins or any viral proteins are ever attached to any receptor embedded in any cell membrane, you would also find that no such evidence exists. If you were to look for a video or documented evidence of the intact virus injecting its genetic material into the body of the cell, reproducing itself and then emerging victorious by budding off the cell membrane, you would find that no such evidence exists.

The closest thing you would find is electron micrograph pictures of cellular particles, possibly attached to cell debris, both of which to be seen were stained by heavy metals, a process that completely distorts their architecture within the living organism. This is like finding bits of paper stuck to the blown-up bricks, thereby proving the paper emerged by taking pieces of the bricks on its way out.

The Enders baloney

Cowan describes the 'Covid' story as being just as make-believe as his paper story and he charts back this fantasy to a Nobel Prize winner called John Enders (1897-1985), an American biomedical scientist who has been dubbed 'The Father of Modern Vaccines'. Enders is claimed to have 'discovered' the process of the viral culture which 'proved' that a 'virus' caused measles. Cowan explains how Enders did this 'by using the EXACT same procedure that has been followed by every virologist to find and characterize every new virus since 1954'. Enders took throat swabs from children with measles and immersed them in 2ml of milk. Penicillin (100u/ml) and the antibiotic streptomycin (50,g/ml) were added and the whole mix was centrifuged – rotated at high speed to separate large cellular debris from small particles and molecules as with milk and cream, for example. Cowan says that if the aim is to find little particles of genetic material ('viruses') in the snot from children with measles it would seem that the last thing you would do is mix the snot with other material – milk –that also has genetic material. 'How are you ever going to know whether whatever you found came from the snot or the milk?' He points out that streptomycin is a 'nephrotoxic' or poisonous-to-the-kidney drug. You will see the relevance of that

shortly. Cowan says that it gets worse, much worse, when Enders describes the culture medium upon which the virus 'grows': 'The culture medium consisted of bovine amniotic fluid (90%), beef embryo extract (5%), horse serum (5%), antibiotics and phenol red as an indicator of cell metabolism.' Cowan asks incredulously: 'Did he just say that the culture medium also contained fluids and tissues that are themselves rich sources of genetic material?' The genetic cocktail, or 'medium', is inoculated onto tissue and cells from rhesus monkey *kidney* tissue. This is where the importance of streptomycin comes in and currently-used antimicrobials and other drugs that are *poisonous to kidneys* and used in ALL modern viral cultures (e.g. gentamicin, streptomycin, and amphotericin). Cowan asks: 'How are you ever going to know from this witch's brew where any genetic material comes from as we now have five different sources of rich genetic material in our mix?' Remember, he says, that all genetic material, whether from monkey kidney tissues, bovine serum, milk, etc., is made from the exact same components. The same central question returns: 'How are you possibly going to know that it was the virus that killed the kidney tissue and not the toxic antibiotic and starvation rations on which you are growing the tissue?' John Enders answered the question himself – *you can't*:

A second agent was obtained from an uninoculated culture of monkey kidney cells. The cytopathic changes [death of the cells] it induced in the unstained preparations could not be distinguished with confidence from the viruses isolated from measles.

The death of the cells ('cytopathic changes') happened in exactly the same manner, whether they inoculated the kidney tissue with the measles snot or not, Cowan says. 'This is evidence that the destruction of the tissue, the very proof of viral causation of illness, was not caused by anything in the snot because they saw the same destructive effect when the snot was not even used ... the cytopathic, i.e., cell-killing, changes come from the process of the culture itself, not from any virus in any snot, period.' Enders quotes in his 1957 paper a virologist called Ruckle as reporting similar findings 'and in addition has isolated an agent from monkey kidney tissue that is so

far indistinguishable from human measles virus'. In other words, Cowan says, these particles called 'measles viruses' are simply and clearly breakdown products of the starved and poisoned tissue. For measles 'virus' see all 'viruses' including the so-called 'Covid virus'. Enders, the 'Father of Modern Vaccines', also said:

There is a potential risk in employing cultures of primate cells for the production of vaccines composed of attenuated virus, since the presence of other agents possibly latent in primate tissues cannot be definitely excluded by any known method.

Cowan further quotes from a paper published in the journal *Viruses* in May, 2020, while the 'Covid pandemic' was well underway in the media if not in reality. 'EVs' here refers to particles of genetic debris from our own tissues, such as exosomes of which more in a moment: 'The remarkable resemblance between EVs and viruses has caused quite a few problems in the studies focused on the analysis of EVs released during viral infections.' Later the paper adds that to date a reliable method that can actually guarantee a complete separation (of EVs from viruses) DOES NOT EXIST. This was published at a time when a fairy tale 'virus' was claimed in total certainty to be causing a fairy tale 'viral disease' called 'Covid-19' – a fairy tale that was already well on the way to transforming human society in the image that the Cult has worked to achieve for so long. Cowan concludes his article:

To summarize, there is no scientific evidence that pathogenic viruses exist. What we think of as 'viruses' are simply the normal breakdown products of dead and dying tissues and cells. When we are well, we make fewer of these particles; when we are starved, poisoned, suffocated by wearing masks, or afraid, we make more.

There is no engineered virus circulating and making people sick. People in laboratories all over the world are making genetically modified products to make people sick. These are called vaccines. There is no virome, no 'ecosystem' of viruses, viruses are not 8%, 50% or 100 % of our genetic material. These are all simply erroneous ideas based on the misconception called a virus.

What is 'Covid'? Load of bollocks

The background described here by Cowan and Lanka was emphasised in the first video presentation that I saw by Dr Andrew Kaufman when he asked whether the 'Covid virus' was in truth a natural defence mechanism of the body called 'exosomes'. These are released by cells when in states of toxicity – see the same themes returning over and over. They are released ever more profusely as chemical and radiation toxicity increases and think of the potential effect therefore of 5G alone as its destructive frequencies infest the human energetic information field with a gathering pace (5G went online in Wuhan in 2019 as the 'virus' emerged). I'll have more about this later. Exosomes transmit a warning to the rest of the body that 'Houston, we have a problem'. Kaufman presented images of exosomes and compared them with 'Covid' under an electron microscope and the similarity was remarkable. They both attach to the same cell receptors (*claimed* in the case of 'Covid'), contain the same genetic material in the form of RNA or ribonucleic acid, and both are found in 'viral cell cultures' with damaged or dying cells. James Hildreth MD, President and Chief Executive Officer of the Meharry Medical College at Johns Hopkins, said: 'The virus is fully an exosome in every sense of the word.' Kaufman's conclusion was that there is no 'virus': 'This entire pandemic is a completely manufactured crisis ... there is no evidence of anyone dying from [this] illness.' Dr Tom Cowan and Sally Fallon Morell, authors of *The Contagion Myth*, published a statement with Dr Kaufman in February, 2021, explaining why the 'virus' does not exist and you can read it that in full in the Appendix.

'Virus' theory can be traced to the 'cell theory' in 1858 of German physician Rudolf Virchow (1821-1920) who contended that disease originates from a single cell infiltrated by a 'virus'. Dr Stefan Lanka said that findings and insights with respect to the structure, function and central importance of tissues in the creation of life, which were already known in 1858, comprehensively refute the cell theory. Virchow ignored them. We have seen the part later played by John Enders in the 1950s and Lanka notes that infection theories were only established as a global dogma through the policies and

eugenics of the Third Reich in Nazi Germany (creation of the same Sabbatian cult behind the 'Covid' hoax). Lanka said: 'Before 1933, scientists dared to contradict this theory; after 1933, these critical scientists were silenced'. Dr Tom Cowan's view is that ill-health is caused by too much of something, too little of something, or toxification from chemicals and radiation – not contagion. We must also highlight as a major source of the 'virus' theology a man still called the 'Father of Modern Virology' – Thomas Milton Rivers (1888-1962). There is no way given the Cult's long game policy that it was a coincidence for the 'Father of Modern Virology' to be director of the Rockefeller Institute for Medical Research from 1937 to 1956 when he is credited with making the Rockefeller Institute a leader in 'viral research'. Cult Rockefellerers were the force behind the creation of Big Pharma 'medicine', established the World Health Organisation in 1948, and have long and close associations with the Gates family that now runs the WHO during the pandemic hoax through mega-rich Cult gofer and psychopath Bill Gates.

Only a Renegade Mind can see through all this bullshit by asking the questions that need to be answered, not taking 'no' or prevarication for an answer, and certainly not hiding from the truth in fear of speaking it. Renegade Minds have always changed the world for the better and they will change this one no matter how bleak it may currently appear to be.

CHAPTER SIX

Sequence of deceit

If you tell the truth, you don't have to remember anything
Mark Twain

Against the background that I have laid out this far the sequence that took us from an invented 'virus' in Cult-owned China in late 2019 to the fascist transformation of human society can be seen and understood in a whole new context.

We were told that a deadly disease had broken out in Wuhan and the world media began its campaign (coordinated by behavioural psychologists as we shall see) to terrify the population into unquestioning compliance. We were shown images of Chinese people collapsing in the street which never happened in the West with what was supposed to be the same condition. In the earliest days when alleged cases and deaths were few the fear register was hysterical in many areas of the media and this would expand into the common media narrative across the world. The real story was rather different, but we were never told that. The Chinese government, one of the Cult's biggest centres of global operation, said they had discovered a new illness with flu-like and pneumonia-type symptoms in a city with such toxic air that it is overwhelmed with flu-like symptoms, pneumonia and respiratory disease. Chinese scientists said it was a new – 'novel' – coronavirus which they called Sars-Cov-2 and that it caused a disease they labelled 'Covid-19'. There was no evidence for this and the 'virus' has never to this day been isolated, purified and its genetic code established from that. It

was from the beginning a computer-generated fiction. Stories of Chinese whistleblowers saying the number of deaths was being suppressed or that the 'new disease' was related to the Wuhan bio-lab misdirected mainstream and alternative media into cul-de-sacs to obscure the real truth – there was no 'virus'.

Chinese scientists took genetic material from the lung fluid of just a few people and said they had found a 'new' disease when this material had a wide range of content. There was no evidence for a 'virus' for the very reasons explained in the last two chapters. The 'virus' has never been shown to (a) exist and (b) cause any disease. People were diagnosed on symptoms that are so widespread in Wuhan and polluted China and with a PCR test that can't detect infectious disease. On this farce the whole global scam was sold to the rest of the world which would also diagnose respiratory disease as 'Covid-19' from symptoms alone or with a PCR test not testing for a 'virus'. Flu miraculously disappeared *worldwide* in 2020 and into 2021 as it was redesignated 'Covid-19'. It was really the same old flu with its 'flu-like' symptoms attributed to 'flu-like' 'Covid-19'. At the same time with very few exceptions the Chinese response of draconian lockdown and fascism was the chosen weapon to respond across the West as recommended by the Cult-owned Tedros at the Cult-owned World Health Organization run by the Cult-owned Gates. All was going according to plan. Chinese scientists – everything in China is controlled by the Cult-owned government – compared their contaminated RNA lung-fluid material with other RNA sequences and said it appeared to be just under 80 percent identical to the SARS-CoV-1 'virus' claimed to be the cause of the SARS (severe acute respiratory syndrome) 'outbreak' in 2003. They decreed that because of this the 'new virus' had to be related and they called it SARS-CoV-2. There are some serious problems with this assumption and *assumption* was all it was. Most 'factual' science turns out to be assumptions repeated into everyone-knows-that. A match of under 80-percent is meaningless. Dr Kaufman makes the point that there's a 96 percent genetic correlation between humans and chimpanzees, but 'no one would say our genetic material is part

of the chimpanzee family'. Yet the Chinese authorities were claiming that a much lower percentage, less than 80 percent, proved the existence of a new 'coronavirus'. For goodness sake human DNA is 60 percent similar to a *banana*.

You are feeling sleepy

The entire 'Covid' hoax is a global Psyop, a psychological operation to program the human mind into believing and fearing a complete fantasy. A crucial aspect of this was what *appeared* to happen in Italy. It was all very well streaming out daily images of an alleged catastrophe in Wuhan, but to the Western mind it was still on the other side of the world in a very different culture and setting. A reaction of 'this could happen to me and my family' was still nothing like as intense enough for the mind-doctors. The Cult needed a Western example to push people over that edge and it chose Italy, one of its major global locations going back to the Roman Empire. An Italian 'Covid' crisis was manufactured in a particular area called Lombardy which just happens to be notorious for its toxic air and therefore respiratory disease. Wuhan, China, *déjà vu*. An hysterical media told horror stories of Italians dying from 'Covid' in their droves and how Lombardy hospitals were being overrun by a tidal wave of desperately ill people needing treatment after being struck down by the 'deadly virus'. Here was the psychological turning point the Cult had planned. Wow, if this is happening in Italy, the Western mind concluded, this indeed could happen to me and my family. Another point is that Italian authorities responded by following the Chinese blueprint so vehemently recommended by the Cult-owned World Health Organization. They imposed fascistic lockdowns on the whole country viciously policed with the help of surveillance drones sweeping through the streets seeking out anyone who escaped from mass house arrest. Livelihoods were destroyed and psychology unravelled in the way we have witnessed since in all lockdown countries. Crucial to the plan was that Italy responded in this way to set the precedent of suspending freedom and imposing fascism in a 'Western liberal democracy'. I emphasised in an

animated video explanation on davidicke.com posted in the summer of 2020 how important it was to the Cult to expand the Chinese lockdown model across the West. Without this, and the bare-faced lie that non-symptomatic people could still transmit a 'disease' they didn't have, there was no way locking down the whole population, sick and not sick, could be pulled off. At just the right time and with no evidence Cult operatives and gofers claimed that people without symptoms could pass on the 'disease'. In the name of protecting the 'vulnerable' like elderly people, who lockdowns would kill by the tens of thousands, we had for the first time healthy people told to isolate as well as the sick. The great majority of people who tested positive had no symptoms because there was nothing wrong with them. It was just a trick made possible by a test not testing for the 'virus'.

Months after my animated video the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College confirmed that I was right. He didn't say it in those terms, naturally, but he did say it. Ferguson will enter the story shortly for his outrageously crazy 'computer models' that led to Britain, the United States and many other countries following the Chinese and now Italian methods of response. Put another way, following the Cult script. Ferguson said that SAGE, the UK government's scientific advisory group which has controlled 'Covid' policy from the start, wanted to follow the Chinese lockdown model (while they all continued to work and be paid), but they wondered if they could possibly, in Ferguson's words, 'get away with it in Europe'. 'Get away with it'? Who the hell do these moronic, arrogant people think they are? This appalling man Ferguson said that once Italy went into national lockdown they realised they, too, could mimic China:

It's a communist one-party state, we said. We couldn't get away with it in Europe, we thought ... and then Italy did it. And we realised we could. Behind this garbage from Ferguson is a simple fact: Doing the same as China in every country was the plan from the start and Ferguson's 'models' would play a central role in achieving that. It's just a coincidence, of course, and absolutely nothing to worry your little head about.

Oops, sorry, our mistake

Once the Italian segment of the Psyop had done the job it was designed to do a very different story emerged. Italian authorities revealed that 99 percent of those who had 'died from Covid-19' in Italy had one, two, three, or more 'co-morbidities' or illnesses and health problems that could have ended their life. The US Centers for Disease Control and Prevention (CDC) published a figure of 94 percent for Americans dying of 'Covid' while having other serious medical conditions – on average two to three (some five or six) other potential causes of death. In terms of death from an unproven 'virus' I say it is 100 percent. The other one percent in Italy and six percent in the US would presumably have died from 'Covid's' flu-like symptoms with a range of other possible causes in conjunction with a test not testing for the 'virus'. Fox News reported that even more startling figures had emerged in one US county in which 410 of 422 deaths attributed to 'Covid-19' had other potentially deadly health conditions. The Italian National Health Institute said later that the average age of people dying with a 'Covid-19' diagnosis in Italy was about 81. Ninety percent were over 70 with ten percent over 90. In terms of other reasons to die some 80 percent had two or more chronic diseases with half having three or more including cardiovascular problems, diabetes, respiratory problems and cancer. Why is the phantom 'Covid-19' said to kill overwhelmingly old people and hardly affect the young? Old people continually die of many causes and especially respiratory disease which you can re-diagnose 'Covid-19' while young people die in tiny numbers by comparison and rarely of respiratory disease. Old people 'die of Covid' because they die of other things that can be redesignated 'Covid' and it really is that simple.

Flu has flown

The blueprint was in place. Get your illusory 'cases' from a test not testing for the 'virus' and redesignate other causes of death as 'Covid-19'. You have an instant 'pandemic' from something that is nothing more than a computer-generated fiction. With near-on a

billion people having 'flu-like' symptoms every year the potential was limitless and we can see why flu quickly and apparently miraculously disappeared *worldwide* by being diagnosed 'Covid-19'. The painfully bloody obvious was explained away by the childlike media in headlines like this in the UK '*Independent*': 'Not a single case of flu detected by Public Health England this year as Covid restrictions suppress virus'. I kid you not. The masking, social distancing and house arrest that did not make the 'Covid virus' disappear somehow did so with the 'flu virus'. Even worse the article, by a bloke called Samuel Lovett, suggested that maybe the masking, sanitising and other 'Covid' measures should continue to keep the flu away. With a ridiculousness that disturbs your breathing (it's 'Covid-19') the said Lovett wrote: 'With widespread social distancing and mask-wearing measures in place throughout the UK, the usual routes of transmission for influenza have been blocked.' He had absolutely no evidence to support that statement, but look at the consequences of him acknowledging the obvious. With flu not disappearing at all and only being relabelled 'Covid-19' he would have to contemplate that 'Covid' was a hoax on a scale that is hard to imagine. You need guts and commitment to truth to even go there and that's clearly something Samuel Lovett does not have in abundance. He would never have got it through the editors anyway.

Tens of thousands die in the United States alone every winter from flu including many with pneumonia complications. CDC figures record *45 million* Americans diagnosed with flu in 2017-2018 of which 61,000 died and some reports claim 80,000. Where was the same hysteria then that we have seen with 'Covid-19'? Some 250,000 Americans are admitted to hospital with pneumonia every year with about 50,000 cases proving fatal. About 65 million suffer respiratory disease every year and three million deaths makes this the third biggest cause of death worldwide. You only have to redesignate a portion of all these people 'Covid-19' and you have an instant global pandemic or the *appearance* of one. Why would doctors do this? They are told to do this and all but a few dare not refuse those who must be obeyed. Doctors in general are not researching their own

knowledge and instead take it direct and unquestioned from the authorities that own them and their careers. The authorities say they must now diagnose these symptoms 'Covid-19' and not flu, or whatever, and they do it. Dark suits say put 'Covid-19' on death certificates no matter what the cause of death and the doctors do it. Renegade Minds don't fall for the illusion that doctors and medical staff are all highly-intelligent, highly-principled, seekers of medical truth. *Some are*, but not the majority. They are repeaters, gofers, and yes sir, no sir, purveyors of what the system demands they purvey. The 'Covid' con is not merely confined to diseases of the lungs. Instructions to doctors to put 'Covid-19' on death certificates for anyone dying of *anything* within 28 days (or much more) of a positive test not testing for the 'virus' opened the floodgates. The term dying *with* 'Covid' and not *of* 'Covid' was coined to cover the truth. Whether it was a *with* or an *of* they were all added to the death numbers attributed to the 'deadly virus' compiled by national governments and globally by the Gates-funded Johns Hopkins operation in the United States that was so involved in those 'pandemic' simulations. Fraudulent deaths were added to the ever-growing list of fraudulent 'cases' from false positives from a false test. No wonder Professor Walter Ricciardi, scientific advisor to the Italian minister of health, said after the Lombardy hysteria had done its job that 'Covid' death rates were due to Italy having the second oldest population in the world and to *how hospitals record deaths*:

The way in which we code deaths in our country is very generous in the sense that all the people who die in hospitals with the coronavirus are deemed to be dying of the coronavirus. On re-evaluation by the National Institute of Health, only 12 per cent of death certificates have shown a direct causality from coronavirus, while 88 per cent of patients who have died have at least one pre-morbidity – many had two or three.

This is extraordinary enough when you consider the propaganda campaign to use Italy to terrify the world, but how can they even say twelve percent were genuine when the 'virus' has not been shown to exist, its 'code' is a computer program, and diagnosis comes from a test not testing for it? As in China, and soon the world, 'Covid-19' in

Italy was a redesignation of diagnosis. Lies and corruption were to become the real 'pandemic' fuelled by a pathetically-compliant medical system taking its orders from the tiny few at the top of their national hierarchy who answered to the World Health Organization which answers to Gates and the Cult. Doctors were told – ordered – to diagnose a particular set of symptoms 'Covid-19' and put that on the death certificate for any cause of death if the patient had tested positive with a test not testing for the virus or had 'Covid' symptoms like the flu. The United States even introduced big financial incentives to manipulate the figures with hospitals receiving £4,600 from the Medicare system for diagnosing someone with regular pneumonia, \$13,000 if they made the diagnosis from the same symptoms 'Covid-19' pneumonia, and \$39,000 if they put a 'Covid' diagnosed patient on a ventilator that would almost certainly kill them. A few – painfully and pathetically few – medical whistleblowers revealed (before Cult-owned YouTube deleted their videos) that they had been instructed to 'let the patient crash' and put them straight on a ventilator instead of going through a series of far less intrusive and dangerous methods as they would have done before the pandemic hoax began and the financial incentives kicked in. We are talking cold-blooded murder given that ventilators are so damaging to respiratory systems they are usually the last step before heaven awaits. Renegade Minds never fall for the belief that people in white coats are all angels of mercy and cannot be full-on psychopaths. I have explained in detail in *The Answer* how what I am describing here played out across the world coordinated by the World Health Organization through the medical hierarchies in almost every country.

Medical scientist calls it

Information about the non-existence of the 'virus' began to emerge for me in late March, 2020, and mushroomed after that. I was sent an email by Sir Julian Rose, a writer, researcher, and organic farming promotor, from a medical scientist friend of his in the United States. Even at that early stage in March the scientist was able to explain

how the 'Covid' hoax was being manipulated. He said there were no reliable tests for a specific 'Covid-19 virus' and nor were there any reliable agencies or media outlets for reporting numbers of actual 'Covid-19' cases. We have seen in the long period since then that he was absolutely right. 'Every action and reaction to Covid-19 is based on totally flawed data and we simply cannot make accurate assessments,' he said. Most people diagnosed with 'Covid-19' were showing nothing more than cold and flu-like symptoms 'because most coronavirus strains *are* nothing more than cold/flu-like symptoms'. We had farcical situations like an 84-year-old German man testing positive for 'Covid-19' and his nursing home ordered to quarantine only for him to be found to have a common cold. The scientist described back then why PCR tests and what he called the 'Mickey Mouse test kits' were useless for what they were claimed to be identifying. 'The idea these kits can isolate a specific virus like Covid-19 is nonsense,' he said. Significantly, he pointed out that 'if you want to create a totally false panic about a totally false pandemic – pick a coronavirus'. This is exactly what the Cult-owned Gates, World Economic Forum and Johns Hopkins University did with their Event 201 'simulation' followed by their real-life simulation called the 'pandemic'. The scientist said that all you had to do was select the sickest of people with respiratory-type diseases in a single location – 'say Wuhan' – and administer PCR tests to them. You can then claim that anyone showing 'viral sequences' similar to a coronavirus 'which will inevitably be quite a few' is suffering from a 'new' disease:

Since you already selected the sickest flu cases a fairly high proportion of your sample will go on to die. You can then say this 'new' virus has a CFR [case fatality rate] higher than the flu and use this to infuse more concern and do more tests which will of course produce more 'cases', which expands the testing, which produces yet more 'cases' and so on and so on. Before long you have your 'pandemic', and all you have done is use a simple test kit trick to convert the worst flu and pneumonia cases into something new that doesn't ACTUALLY EXIST [my emphasis].

He said that you then 'just run the same scam in other countries' and make sure to keep the fear message running high 'so that people

will feel panicky and less able to think critically'. The only problem to overcome was the fact *there is no* actual new deadly pathogen and only regular sick people. This meant that deaths from the 'new deadly pathogen' were going to be way too low for a real new deadly virus pandemic, but he said this could be overcome in the following ways – all of which would go on to happen:

1. You can claim this is just the beginning and more deaths are imminent [you underpin this with fantasy 'computer projections']. Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.
2. You can [say that people] 'minimizing' the dangers are irresponsible and bully them into not talking about numbers.
3. You can talk crap about made up numbers hoping to blind people with pseudoscience.
4. You can start testing well people (who, of course, will also likely have shreds of coronavirus [RNA] in them) and thus inflate your 'case figures' with 'asymptomatic carriers' (you will of course have to spin that to sound deadly even though any virologist knows the more symptom-less cases you have the less deadly is your pathogen).

The scientist said that if you take these simple steps 'you can have your own entirely manufactured pandemic up and running in weeks'. His analysis made so early in the hoax was brilliantly prophetic of what would actually unfold. Pulling all the information together in these recent chapters we have this is simple 1, 2, 3, of how you can delude virtually the entire human population into believing in a 'virus' that doesn't exist:

- A 'Covid case' is someone who tests positive with a test not testing for the 'virus'.
- A 'Covid death' is someone who dies of *any cause* within 28 days (or much longer) of testing positive with a test not testing for the 'virus'.
- Asymptomatic means there is nothing wrong with you, but they claim you can pass on what you don't have to justify locking

down (quarantining) healthy people in totality.

The foundations of the hoax are that simple. A study involving ten million people in Wuhan, published in November, 2020, demolished the whole lie about those without symptoms passing on the 'virus'. They found '300 asymptomatic cases' and traced their contacts to find that not one of them was detected with the 'virus'.

'Asymptomatic' patients and their contacts were isolated for no less than two weeks and nothing changed. I know it's all crap, but if you are going to claim that those without symptoms can transmit 'the virus' then you must produce evidence for that and they never have. Even World Health Organization official Dr Maria Van Kerkhove, head of the emerging diseases and zoonosis unit, said as early as June, 2020, that she doubted the validity of asymptomatic transmission. She said that 'from the data we have, it still seems to be rare that an asymptomatic person actually transmits onward to a secondary individual' and by 'rare' she meant that she couldn't cite any case of asymptomatic transmission.

The Ferguson factor

The problem for the Cult as it headed into March, 2020, when the script had lockdown due to start, was that despite all the manipulation of the case and death figures they still did not have enough people alleged to have died from 'Covid' to justify mass house arrest. This was overcome in the way the scientist described: 'You can claim this is just the beginning and more deaths are imminent ... Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.' Enter one Professor Neil Ferguson, the Gates-funded 'epidemiologist' at the Gates-funded Imperial College in London. Ferguson is Britain's Christian Drosten in that he has a dire record of predicting health outcomes, but is still called upon to advise government on the next health outcome when another 'crisis' comes along. This may seem to be a strange and ridiculous thing to do. Why would you keep turning for policy guidance to people who have a history of being

monumentally wrong? Ah, but it makes sense from the Cult point of view. These 'experts' keep on producing predictions that suit the Cult agenda for societal transformation and so it was with Neil Ferguson as he revealed his horrific (and clearly insane) computer model predictions that allowed lockdowns to be imposed in Britain, the United States and many other countries. Ferguson does not have even an A-level in biology and would appear to have no formal training in computer modelling, medicine or epidemiology, according to Derek Winton, an MSc in Computational Intelligence. He wrote an article somewhat aghast at what Ferguson did which included taking no account of respiratory disease 'seasonality' which means it is far worse in the winter months. Who would have thought that respiratory disease could be worse in the winter? Well, certainly not Ferguson.

The massively China-connected Imperial College and its bizarre professor provided the excuse for the long-incubated Chinese model of human control to travel westward at lightning speed. Imperial College confirms on its website that it collaborates with the Chinese Research Institute; publishes more than 600 research papers every year with Chinese research institutions; has 225 Chinese staff; 2,600 Chinese students – the biggest international group; 7,000 former students living in China which is the largest group outside the UK; and was selected for a tour by China's President Xi Jinping during his state visit to the UK in 2015. The college takes major donations from China and describes itself as the UK's number one university collaborator with Chinese research institutions. The China communist/fascist government did not appear phased by the woeful predictions of Ferguson and Imperial when during the lockdown that Ferguson induced the college signed a five-year collaboration deal with China tech giant Huawei that will have Huawei's indoor 5G network equipment installed at the college's West London tech campus along with an 'AI cloud platform'. The deal includes Chinese sponsorship of Imperial's Venture Catalyst entrepreneurship competition. Imperial is an example of the enormous influence the Chinese government has within British and North American

universities and research centres – and further afield. Up to 200 academics from more than a dozen UK universities are being investigated on suspicion of ‘unintentionally’ helping the Chinese government build weapons of mass destruction by ‘transferring world-leading research in advanced military technology such as aircraft, missile designs and cyberweapons’. Similar scandals have broken in the United States, but it’s all a coincidence. Imperial College serves the agenda in many other ways including the promotion of every aspect of the United Nations Agenda 21/2030 (the Great Reset) and produced computer models to show that human-caused ‘climate change’ is happening when in the real world it isn’t. Imperial College is driving the climate agenda as it drives the ‘Covid’ agenda (both Cult hoaxes) while Patrick Vallance, the UK government’s Chief Scientific Adviser on ‘Covid’, was named Chief Scientific Adviser to the UN ‘climate change’ conference known as COP26 hosted by the government in Glasgow, Scotland. ‘Covid’ and ‘climate’ are fundamentally connected.

Professor Woeful

From Imperial’s bosom came Neil Ferguson still advising government despite his previous disasters and it was announced early on that he and other key people like UK Chief Medical Adviser Chris Whitty had caught the ‘virus’ as the propaganda story was being sold. Somehow they managed to survive and we had Prime Minister Boris Johnson admitted to hospital with what was said to be a severe version of the ‘virus’ in this same period. His whole policy and demeanour changed when he returned to Downing Street. It’s a small world with these government advisors – especially in their communal connections to Gates – and Ferguson had partnered with Whitty to write a paper called ‘Infectious disease: Tough choices to reduce Ebola transmission’ which involved another scare-story that didn’t happen. Ferguson’s ‘models’ predicted that up to 150,000 could die from ‘mad cow disease’, or BSE, and its version in sheep if it was transmitted to humans. BSE was not transmitted and instead triggered by an organophosphate pesticide used to treat a pest on

cows. Fewer than 200 deaths followed from the human form. Models by Ferguson and his fellow incompetents led to the unnecessary culling of millions of pigs, cattle and sheep in the foot and mouth outbreak in 2001 which destroyed the lives and livelihoods of farmers and their families who had often spent decades building their herds and flocks. Vast numbers of these animals did not have foot and mouth and had no contact with the infection. Another 'expert' behind the cull was Professor Roy Anderson, a computer modeller at Imperial College specialising in the epidemiology of *human*, not animal, disease. Anderson has served on the Bill and Melinda Gates Grand Challenges in Global Health advisory board and chairs another Gates-funded organisation. Gates is everywhere.

In a precursor to the 'Covid' script Ferguson backed closing schools 'for prolonged periods' over the swine flu 'pandemic' in 2009 and said it would affect a third of the world population if it continued to spread at the speed he claimed to be happening. His mates at Imperial College said much the same and a news report said: 'One of the authors, the epidemiologist and disease modeller Neil Ferguson, who sits on the World Health Organisation's emergency committee for the outbreak, said the virus had "full pandemic potential".' Professor Liam Donaldson, the Chris Whitty of his day as Chief Medical Officer, said the worst case could see 30 percent of the British people infected by swine flu with 65,000 dying. Ferguson and Donaldson were indeed proved correct when at the end of the year the number of deaths attributed to swine flu was 392. The term 'expert' is rather liberally applied unfortunately, not least to complete idiots. Swine flu 'projections' were great for GlaxoSmithKline (GSK) as millions rolled in for its Pandemrix influenza vaccine which led to brain damage with children most affected. The British government (taxpayers) paid out more than £60 million in compensation after GSK was given immunity from prosecution. Yet another 'Covid' déjà vu. Swine flu was supposed to have broken out in Mexico, but Dr Wolfgang Wodarg, a German doctor, former member of parliament and critic of the 'Covid' hoax, observed 'the spread of swine flu' in Mexico City at the time. He

said: 'What we experienced in Mexico City was a very mild flu which did not kill more than usual – which killed even fewer people than usual.' Hying the fear against all the facts is not unique to 'Covid' and has happened many times before. Ferguson is reported to have over-estimated the projected death toll of bird flu (H5N1) by some three million-fold, but bird flu vaccine makers again made a killing from the scare. This is some of the background to the Neil Ferguson who produced the perfectly-timed computer models in early 2020 predicting that half a million people would die in Britain without draconian lockdown and 2.2 million in the United States. Politicians panicked, people panicked, and lockdowns of alleged short duration were instigated to 'flatten the curve' of cases gleaned from a test not testing for the 'virus'. I said at the time that the public could forget the 'short duration' bit. This was an agenda to destroy the livelihoods of the population and force them into mass control through dependency and there was going to be nothing 'short' about it. American researcher Daniel Horowitz described the consequences of the 'models' spewed out by Gates-funded Ferguson and Imperial College:

What led our government and the governments of many other countries into panic was a single Imperial College of UK study, funded by global warming activists, that predicted 2.2 million deaths if we didn't lock down the country. In addition, the reported 8-9% death rate in Italy scared us into thinking there was some other mutation of this virus that they got, which might have come here.

Together with the fact that we were finally testing and had the ability to actually report new cases, we thought we were headed for a death spiral. But again ... we can't flatten a curve if we don't know when the curve started.

How about it *never* started?

Giving them what they want

An investigation by German news outlet *Welt Am Sonntag* (*World on Sunday*) revealed how in March, 2020, the German government gathered together 'leading scientists from several research institutes and universities' and 'together, they were to produce a [modelling]

paper that would serve as legitimization for further tough political measures'. The Cult agenda was justified by computer modelling not based on evidence or reality; it was specifically constructed to justify the Cult demand for lockdowns all over the world to destroy the independent livelihoods of the global population. All these modellers and everyone responsible for the 'Covid' hoax have a date with a trial like those in Nuremberg after World War Two when Nazis faced the consequences of their war crimes. These corrupt-beyond-belief 'modellers' wrote the paper according to government instructions and it said that if lockdown measures were lifted then up to one million Germans would die from 'Covid-19' adding that some would die 'agonizingly at home, gasping for breath' unable to be treated by hospitals that couldn't cope. All lies. No matter – it gave the Cult all that it wanted. What did long-time government 'modeller' Neil Ferguson say? If the UK and the United States didn't lockdown half a million would die in Britain and 2.2 million Americans. Anyone see a theme here? 'Modellers' are such a crucial part of the lockdown strategy that we should look into their background and follow the money. Researcher Rosemary Frei produced an excellent article headlined 'The Modelling-paper Mafiosi'. She highlights a guy called John Edmunds, a British epidemiologist, and professor in the Faculty of Epidemiology and Population Health at the London School of Hygiene & Tropical Medicine. He studied at Imperial College. Edmunds is a member of government 'Covid' advisory bodies which have been dictating policy, the New and Emerging Respiratory Virus Threats Advisory Group (NERVTAG) and the Scientific Advisory Group for Emergencies (SAGE).

Ferguson, another member of NERVTAG and SAGE, led the way with the original 'virus' and Edmunds has followed in the 'variant' stage and especially the so-called UK or Kent variant known as the 'Variant of Concern' (VOC) B.1.1.7. He said in a co-written report for the Centre for Mathematical modelling of Infectious Diseases at the London School of Hygiene and Tropical Medicine, with input from the Centre's 'Covid-19' Working Group, that there was 'a realistic

possibility that VOC B.1.1.7 is associated with an increased risk of death compared to non-VOC viruses'. Fear, fear, fear, get the vaccine, fear, fear, fear, get the vaccine. Rosemary Frei reveals that almost all the paper's authors and members of the modelling centre's 'Covid-19' Working Group receive funding from the Bill and Melinda Gates Foundation and/or the associated Gates-funded Wellcome Trust. The paper was published by e-journal *Medrx* ^{xiv} which only publishes papers not peer-reviewed and the journal was established by an organisation headed by Facebook's Mark Zuckerberg and his missus. What a small world it is. Frei discovered that Edmunds is on the Scientific Advisory Board of the Coalition for Epidemic Preparedness Innovations (CEPI) which was established by the Bill and Melinda Gates Foundation, Klaus Schwab's Davos World Economic Forum and Big Pharma giant Wellcome. CEPI was 'launched in Davos [in 2017] to develop vaccines to stop future epidemics', according to its website. 'Our mission is to accelerate the development of vaccines against emerging infectious diseases and enable equitable access to these vaccines for people during outbreaks.' What kind people they are. Rosemary Frei reveals that Public Health England (PHE) director Susan Hopkins is an author of her organisation's non-peer-reviewed reports on 'new variants'. Hopkins is a professor of infectious diseases at London's Imperial College which is gifted tens of millions of dollars a year by the Bill and Melinda Gates Foundation. Gates-funded modelling disaster Neil Ferguson also co-authors Public Health England reports and he spoke in December, 2020, about the potential danger of the B.1.1.7. 'UK variant' promoted by Gates-funded modeller John Edmunds. When I come to the 'Covid vaccines' the 'new variants' will be shown for what they are – bollocks.

Connections, connections

All these people and modellers are lockdown-obsessed or, put another way, they demand what the Cult demands. Edmunds said in January, 2021, that to ease lockdowns too soon would be a disaster and they had to 'vaccinate much, much, much more widely than the

elderly'. Rosemary Frei highlights that Edmunds is married to Jeanne Pimenta who is described in a LinkedIn profile as director of epidemiology at GlaxoSmithKline (GSK) and she held shares in the company. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of GSK and has a deferred bonus of shares in the company worth £600,000. GSK has serious business connections with Bill Gates and is collaborating with mRNA-'vaccine' company CureVac to make 'vaccines' for the new variants that Edmunds is talking about. GSK is planning a 'Covid vaccine' with drug giant Sanofi. Puppets Prime Minister Boris Johnson announced in the spring of 2021 that up to 60 million vaccine doses were to be made at the GSK facility at Barnard Castle in the English North East. Barnard Castle, with a population of just 6,000, was famously visited in breach of lockdown rules in April, 2020, by Johnson aide Dominic Cummings who said that he drove there 'to test his eyesight' before driving back to London. Cummings would be better advised to test his integrity – not that it would take long. The GSK facility had nothing to do with his visit then although I'm sure Patrick Vallance would have been happy to arrange an introduction and some tea and biscuits. Ruthless psychopath Gates has made yet another fortune from vaccines in collaboration with Big Pharma companies and gushes at the phenomenal profits to be made from vaccines – more than a 20-to-1 return as he told one interviewer. Gates also tweeted in December, 2019, with the foreknowledge of what was coming: 'What's next for our foundation? I'm particularly excited about what the next year could mean for one of the best buys in global health: vaccines.'

Modeller John Edmunds is a big promotor of vaccines as all these people appear to be. He's the dean of the London School of Hygiene & Tropical Medicine's Faculty of Epidemiology and Population Health which is primarily funded by the Bill and Melinda Gates Foundation and the Gates-established and funded GAVI vaccine alliance which is the Gates vehicle to vaccinate the world. The organisation Doctors Without Borders has described GAVI as being 'aimed more at supporting drug-industry desires to promote new

products than at finding the most efficient and sustainable means for fighting the diseases of poverty'. But then that's why the psychopath Gates created it. John Edmunds said in a video that the London School of Hygiene & Tropical Medicine is involved in every aspect of vaccine development including large-scale clinical trials. He contends that mathematical modelling can show that vaccines protect individuals and society. That's on the basis of shit in and shit out, I take it. Edmunds serves on the UK Vaccine Network as does Ferguson and the government's foremost 'Covid' adviser, the grim-faced, dark-eyed Chris Whitty. The Vaccine Network says it works 'to support the government to identify and shortlist targeted investment opportunities for the most promising vaccines and vaccine technologies that will help combat infectious diseases with epidemic potential, and to address structural issues related to the UK's broader vaccine infrastructure'. Ferguson is acting Director of the Imperial College Vaccine Impact Modelling Consortium which has funding from the Bill and Melina Gates Foundation and the Gates-created GAVI 'vaccine alliance'. Anyone wonder why these characters see vaccines as the answer to every problem? Ferguson is wildly enthusiastic in his support for GAVI's campaign to vaccinate children en masse in poor countries. You would expect someone like Gates who has constantly talked about the need to reduce the population to want to fund vaccines to keep more people alive. I'm sure that's why he does it. The John Edmunds London School of Hygiene & Tropical Medicine (LSHTM) has a Vaccines Manufacturing Innovation Centre which develops, tests and commercialises vaccines. Rosemary Frei writes:

The vaccines centre also performs affiliated activities like combating 'vaccine hesitancy'. The latter includes the Vaccine Confidence Project. The project's stated purpose is, among other things, 'to provide analysis and guidance for early response and engagement with the public to ensure sustained confidence in vaccines and immunisation'. The Vaccine Confidence Project's director is LSHTM professor Heidi Larson. For more than a decade she's been researching how to combat vaccine hesitancy.

How the bloody hell can blokes like John Edmunds and Neil Ferguson with those connections and financial ties model 'virus' case

and death projections for the government and especially in a way that gives their paymasters like Gates exactly what they want? It's insane, but this is what you find throughout the world.

'Covid' is not dangerous, oops, wait, yes it is

Only days before Ferguson's nightmare scenario made Jackboot Johnson take Britain into a China-style lockdown to save us from a deadly 'virus' the UK government website gov.uk was reporting something very different to Ferguson on a page of official government guidance for 'high consequence infectious diseases (HCID)'. It said this about 'Covid-19':

As of 19 March 2020, COVID-19 is no longer considered to be a high consequence infectious diseases (HCID) in the UK [my emphasis]. The 4 nations public health HCID group made an interim recommendation in January 2020 to classify COVID-19 as an HCID. This was based on consideration of the UK HCID criteria about the virus and the disease with information available during the early stages of the outbreak.

Now that more is known about COVID-19, the public health bodies in the UK have reviewed the most up to date information about COVID-19 against the UK HCID criteria. They have determined that several features have now changed; in particular, more information is available about mortality rates (low overall), and there is now greater clinical awareness and a specific and sensitive laboratory test, the availability of which continues to increase. The Advisory Committee on Dangerous Pathogens (ACDP) is also of the opinion that COVID-19 should no longer be classified as an HCID.

Soon after the government had been exposed for downgrading the risk they upgraded it again and everyone was back to singing from the same Cult hymn book. Ferguson and his fellow Gates clones indicated that lockdowns and restrictions would have to continue until a Gates-funded vaccine was developed. Gates said the same because Ferguson and his like were repeating the Gates script which is the Cult script. 'Flatten the curve' became an ongoing nightmare of continuing lockdowns with periods in between of severe restrictions in pursuit of destroying independent incomes and had nothing to do with protecting health about which the Cult gives not a shit. Why wouldn't Ferguson be pushing a vaccine 'solution' when he's owned by vaccine-obsessive Gates who makes a fortune from them and

when Ferguson heads the Vaccine Impact Modelling Consortium at Imperial College funded by the Gates Foundation and GAVI, the 'vaccine alliance', created by Gates as his personal vaccine promotion operation? To compound the human catastrophe that Ferguson's 'models' did so much to create he was later exposed for breaking his own lockdown rules by having sexual liaisons with his married girlfriend Antonia Staats at his home while she was living at another location with her husband and children. Staats was a 'climate' activist and senior campaigner at the Soros-funded Avaaz which I wouldn't trust to tell me that grass is green. Ferguson had to resign as a government advisor over this hypocrisy in May, 2020, but after a period of quiet he was back being quoted by the ridiculous media on the need for more lockdowns and a vaccine rollout. Other government-advising 'scientists' from Imperial College held the fort in his absence and said lockdown could be indefinite until a vaccine was found. The Cult script was being sung by the payrolled choir. I said there was no intention of going back to 'normal' when the 'vaccine' came because the 'vaccine' is part of a very different agenda that I will discuss in Human 2.0. Why would the Cult want to let the world go back to normal when destroying that normal forever was the whole point of what was happening? House arrest, closing businesses and schools through lockdown, (un)social distancing and masks all followed the Ferguson fantasy models. Again as I predicted (these people are so predictable) when the 'vaccine' arrived we were told that house arrest, lockdown, (un)social distancing and masks would still have to continue. I will deal with the masks in the next chapter because they are of fundamental importance.

Where's the 'pandemic'?

Any mildly in-depth assessment of the figures revealed what was really going on. Cult-funded and controlled organisations still have genuine people working within them such is the number involved. So it is with Genevieve Briand, assistant program director of the Applied Economics master's degree program at Johns Hopkins

University. She analysed the impact that 'Covid-19' had on deaths from *all* causes in the United States using official data from the CDC for the period from early February to early September, 2020. She found that allegedly 'Covid' *related*-deaths exceeded those from heart disease which she found strange with heart disease always the biggest cause of fatalities. Her research became even more significant when she noted the sudden decline in 2020 of *all* non-'Covid' deaths: 'This trend is completely contrary to the pattern observed in all previous years ... the total decrease in deaths by other causes almost exactly equals the increase in deaths by Covid-19.' This was such a game, set and match in terms of what was happening that Johns Hopkins University deleted the article on the grounds that it 'was being used to support false and dangerous inaccuracies about the impact of the pandemic'. No – because it exposed the scam from official CDC figures and this was confirmed when those figures were published in January, 2021. Here we can see the effect of people dying from heart attacks, cancer, road accidents and gunshot wounds – *anything* – having 'Covid-19' on the death certificate along with those diagnosed from 'symptoms' who had even not tested positive with a test not testing for the 'virus'. I am not kidding with the gunshot wounds, by the way. Brenda Bock, coroner in Grand County, Colorado, revealed that two gunshot victims tested positive for the 'virus' within the previous 30 days and were therefore classified as 'Covid deaths'. Bock said: 'These two people had tested positive for Covid, but that's not what killed them. A gunshot wound is what killed them.' She said she had not even finished her investigation when the state listed the gunshot victims as deaths due to the 'virus'. The death and case figures for 'Covid-19' are an absolute joke and yet they are repeated like parrots by the media, politicians and alleged medical 'experts'. The official Cult narrative is the only show in town.

Genevieve Briand found that deaths from all causes were not exceptional in 2020 compared with previous years and a Spanish magazine published figures that said the same about Spain which was a 'Covid' propaganda hotspot at one point. *Discovery Salud*, a

health and medicine magazine, quoted government figures which showed how 17,000 *fewer* people died in Spain in 2020 than in 2019 and more than 26,000 fewer than in 2018. The age-standardised mortality rate for England and Wales when age distribution is taken into account was significantly lower in 2020 than the 1970s, 80s and 90s, and was only the ninth highest since 2000. Where is the 'pandemic'?

Post mortems and autopsies virtually disappeared for 'Covid' deaths amid claims that 'virus-infected' bodily fluids posed a risk to those carrying out the autopsy. This was rejected by renowned German pathologist and forensic doctor Klaus Püschel who said that he and his staff had by then done 150 autopsies on 'Covid' patients with no problems at all. He said they were needed to know why some 'Covid' patients suffered blood clots and not severe respiratory infections. The 'virus' is, after all, called SARS or 'severe acute respiratory syndrome'. I highlighted in the spring of 2020 this phenomenon and quoted New York intensive care doctor Cameron Kyle-Sidell who posted a soon deleted YouTube video to say that they had been told to prepare to treat an infectious disease called 'Covid-19', but that was not what they were dealing with. Instead he likened the lung condition of the most severely ill patients to what you would expect with cabin depressurisation in a plane at 30,000 feet or someone dropped on the top of Everest without oxygen or acclimatisation. I have never said this is not happening to a small minority of alleged 'Covid' patients – I am saying this is not caused by a phantom 'contagious virus'. Indeed Kyle-Sidell said that 'Covid-19' was not the disease they were told was coming their way. 'We are operating under a medical paradigm that is untrue,' he said, and he believed they were treating the wrong disease: 'These people are being slowly starved of oxygen.' Patients would take off their oxygen masks in a state of fear and stress and while they were blue in the face on the brink of death. They did not look like patients dying of pneumonia. You can see why they don't want autopsies when their virus doesn't exist and there is another condition in some people that they don't wish to be uncovered. I should add here that

the 5G system of millimetre waves was being rapidly introduced around the world in 2020 and even more so now as they fire 5G at the Earth from satellites. At 60 gigahertz within the 5G range that frequency interacts with the oxygen molecule and stops people breathing in sufficient oxygen to be absorbed into the bloodstream. They are installing 5G in schools and hospitals. The world is not mad or anything. 5G can cause major changes to the lungs and blood as I detail in *The Answer* and these consequences are labelled 'Covid-19', the alleged symptoms of which can be caused by 5G and other electromagnetic frequencies as cells respond to radiation poisoning.

The 'Covid death' scam

Dr Scott Jensen, a Minnesota state senator and medical doctor, exposed 'Covid' Medicare payment incentives to hospitals and death certificate manipulation. He said he was sent a seven-page document by the US Department of Health 'coaching' him on how to fill out death certificates which had never happened before. The document said that he didn't need to have a laboratory test for 'Covid-19' to put that on the death certificate and that shocked him when death certificates are supposed to be about facts. Jensen described how doctors had been 'encouraged, if not pressured' to make a diagnosis of 'Covid-19' if they thought it was probable or '*presumed*'. No positive test was necessary – not that this would have mattered anyway. He said doctors were told to diagnose 'Covid' by symptoms when these were the same as colds, allergies, other respiratory problems, and certainly with influenza which 'disappeared' in the 'Covid' era. A common sniffle was enough to get the dreaded verdict. Ontario authorities decreed that a single care home resident with *one* symptom from a long list must lead to the isolation of the entire home. Other courageous doctors like Jensen made the same point about death figure manipulation and how deaths by other causes were falling while 'Covid-19 deaths' were rising at the same rate due to re-diagnosis. Their videos rarely survive long on YouTube with its Cult-supporting algorithms courtesy of CEO Susan Wojcicki and her bosses at Google. Figure-tampering was so glaring

and ubiquitous that even officials were letting it slip or outright saying it. UK chief scientific adviser Patrick Vallance said on one occasion that 'Covid' on the death certificate doesn't mean 'Covid' was the cause of death (so why the hell is it there?) and we had the rare sight of a BBC reporter telling the truth when she said: 'Someone could be successfully treated for Covid, in say April, discharged, and then in June, get run over by a bus and die ... That person would still be counted as a Covid death in England.' Yet the BBC and the rest of the world media went on repeating the case and death figures as if they were real. Illinois Public Health Director Dr Ngozi Ezike revealed the deceit while her bosses must have been clenching their buttocks:

If you were in a hospice and given a few weeks to live and you were then found to have Covid that would be counted as a Covid death. [There might be] a clear alternate cause, but it is still listed as a Covid death. So everyone listed as a Covid death doesn't mean that was the cause of the death, but that they had Covid at the time of death.

Yes, a 'Covid virus' never shown to exist and tested for with a test not testing for the 'virus'. In the first period of the pandemic hoax through the spring of 2020 the process began of designating almost everything a 'Covid' death and this has continued ever since. I sat in a restaurant one night listening to a loud conversation on the next table where a family was discussing in bewilderment how a relative who had no symptoms of 'Covid', and had died of a long-term problem, could have been diagnosed a death by the 'virus'. I could understand their bewilderment. If they read this book they will know why this medical fraud has been perpetrated the world over.

Some media truth shock

The media ignored the evidence of death certificate fraud until eventually one columnist did speak out when she saw it first-hand. Bel Mooney is a long-time national newspaper journalist in Britain currently working for the *Daily Mail*. Her article on February 19th, 2021, carried this headline: 'My dad Ted passed three Covid tests

and died of a chronic illness yet he's officially one of Britain's 120,000 victims of the virus and is far from alone ... so how many more are there?' She told how her 99-year-old father was in a care home with a long-standing chronic obstructive pulmonary disease and vascular dementia. Maybe, but he was still aware enough to tell her from the start that there was no 'virus' and he refused the 'vaccine' for that reason. His death was not unexpected given his chronic health problems and Mooney said she was shocked to find that 'Covid-19' was declared the cause of death on his death certificate. She said this was a 'bizarre and unacceptable untruth' for a man with long-time health problems who had tested negative twice at the home for the 'virus'. I was also shocked by this story although not by what she said. I had been highlighting the death certificate manipulation for ten months. It was the confirmation that a professional full-time journalist only realised this was going on when it affected her directly and neither did she know that whether her dad tested positive or negative was irrelevant with the test not testing for the 'virus'. Where had she been? She said she did not believe in 'conspiracy theories' without knowing I'm sure that this and 'conspiracy theorists' were terms put into widespread circulation by the CIA in the 1960s to discredit those who did not accept the ridiculous official story of the Kennedy assassination. A blanket statement of 'I don't believe in conspiracy theories' is always bizarre. The dictionary definition of the term alone means the world is drowning in conspiracies. What she said was even more daft when her dad had just been affected by the 'Covid' conspiracy. Why else does she think that 'Covid-19' was going on the death certificates of people who died of something else?

To be fair once she saw from personal experience what was happening she didn't mince words. Mooney was called by the care home on the morning of February 9th to be told her father had died in his sleep. When she asked for the official cause of death what came back was 'Covid-19'. Mooney challenged this and was told there had been deaths from Covid on the dementia floor (confirmed by a test not testing for the 'virus') so they considered it 'reasonable

to assume'. 'But doctor,' Mooney rightly protested, 'an assumption isn't a diagnosis.' She said she didn't blame the perfectly decent and sympathetic doctor – 'he was just doing his job'. Sorry, but that's *bullshit*. He wasn't doing his job at all. He was putting a false cause of death on the death certificate and that is a criminal offence for which he should be brought to account and the same with the millions of doctors worldwide who have done the same. They were not doing their job they were following orders and that must not wash at new Nuremberg trials any more than it did at the first ones. Mooney's doctor was 'assuming' (presuming) as he was told to, but 'just following orders' makes no difference to his actions. A doctor's job is to serve the patient and the truth, not follow orders, but that's what they have done all over the world and played a central part in making the 'Covid' hoax possible with all its catastrophic consequences for humanity. Shame on them and they must answer for their actions. Mooney said her disquiet worsened when she registered her father's death by telephone and was told by the registrar there had been very many other cases like hers where 'the deceased' had not tested positive for 'Covid' yet it was recorded as the cause of death. The test may not matter, but those involved at their level *think* it matters and it shows a callous disregard for accurate diagnosis. The pressure to do this is coming from the top of the national 'health' pyramids which in turn obey the World Health Organization which obeys Gates and the Cult. Mooney said the registrar agreed that this must distort the national figures adding that 'the strangest thing is that every winter we record countless deaths from flu, and this winter there have been none. Not one!' She asked if the registrar thought deaths from flu were being misdiagnosed and lumped together with 'Covid' deaths. The answer was a 'puzzled yes'. Mooney said that the funeral director said the same about 'Covid' deaths which had nothing to do with 'Covid'. They had lost count of the number of families upset by this and other funeral companies in different countries have had the same experience. Mooney wrote:

The nightly shroud-waving and shocking close-ups of pain imposed on us by the TV news bewildered and terrified the population into eager compliance with lockdowns. We were invited to 'save the NHS' and to grieve for strangers – the real-life loved ones behind those shocking death counts. Why would the public imagine what I now fear, namely that the way Covid-19 death statistics are compiled might make the numbers seem greater than they are?

Oh, just a little bit – like 100 percent.

Do the maths

Mooney asked why a country would wish to skew its mortality figures by wrongly certifying deaths? What had been going on? Well, if you don't believe in conspiracies you will never find the answer which is that *it's a conspiracy*. She did, however, describe what she had discovered as a 'national scandal'. In reality it's a global scandal and happening everywhere. Pillars of this conspiracy were all put into place before the button was pressed with the Drosten PCR protocol and high amplifications to produce the cases and death certificate changes to secure illusory 'Covid' deaths. Mooney notes that normally two doctors were needed to certify a death, with one having to know the patient, and how the rules were changed in the spring of 2020 to allow one doctor to do this. In the same period 'Covid deaths' were decreed to be all cases where Covid-19 was put on the death certificate even without a positive test or any symptoms. Mooney asked: 'How many of the 30,851 (as of January 15) care home resident deaths with Covid-19 on the certificate (32.4 per cent of all deaths so far) were based on an assumption, like that of my father? And what has that done to our national psyche?' All of them is the answer to the first question and it has devastated and dismantled the national psyche, actually the global psyche, on a colossal scale. In the UK case and death data is compiled by organisations like Public Health England (PHE) and the Office for National Statistics (ONS). Mooney highlights the insane policy of counting a death from any cause as 'Covid-19' if this happens within 28 days of a positive test (with a test not testing for the 'virus') and she points out that ONS statistics reflect deaths 'involving Covid' 'or due to Covid' which meant in practice any

death where 'Covid-19' was mentioned on the death certificate. She described the consequences of this fraud:

Most people will accept the narrative they are fed, so panicky governments here and in Europe witnessed the harsh measures enacted in totalitarian China and jumped into lockdown. Headlines about Covid deaths tolled like the knell that would bring doomsday to us all. Fear stalked our empty streets. Politicians parroted the frankly ridiculous aim of 'zero Covid' and shut down the economy, while most British people agreed that lockdown was essential and (astonishingly to me, as a patriotic Brit) even wanted more restrictions.

For what? Lies on death certificates? Never mind the grim toll of lives ruined, suicides, schools closed, rising inequality, depression, cancelled hospital treatments, cancer patients in a torture of waiting, poverty, economic devastation, loneliness, families kept apart, and so on. How many lives have been lost as a direct result of lockdown?

She said that we could join in a national chorus of shock and horror at reaching the 120,000 death toll which was surely certain to have been totally skewed all along, but what about the human cost of lockdown justified by these 'death figures'? *The British Medical Journal* had reported a 1,493 percent increase in cases of children taken to Great Ormond Street Hospital with abusive head injuries alone and then there was the effect on families:

Perhaps the most shocking thing about all this is that families have been kept apart – and obeyed the most irrational, changing rules at the whim of government – because they believed in the statistics. They succumbed to fear, which his generation rejected in that war fought for freedom. Dad (God rest his soul) would be angry. And so am I.

Another theme to watch is that in the winter months when there are more deaths from all causes they focus on 'Covid' deaths and in the summer when the British Lung Foundation says respiratory disease plummets by 80 percent they rage on about 'cases'. Either way fascism on population is always the answer.

Nazi eugenics in the 21st century

Elderly people in care homes have been isolated from their families month after lonely month with no contact with relatives and grandchildren who were banned from seeing them. We were told

that lockdown fascism was to 'protect the vulnerable' like elderly people. At the same time Do Not Resuscitate (DNR) orders were placed on their medical files so that if they needed resuscitation it wasn't done and 'Covid-19' went on their death certificates. Old people were not being 'protected' they were being culled – murdered in truth. DNR orders were being decreed for disabled and young people with learning difficulties or psychological problems. The UK Care Quality Commission, a non-departmental body of the Department of Health and Social Care, found that 34 percent of those working in health and social care were pressured into placing 'do not attempt cardiopulmonary resuscitation' orders on 'Covid' patients who suffered from disabilities and learning difficulties without involving the patient or their families in the decision. UK judges ruled that an elderly woman with dementia should have the DNA-manipulating 'Covid vaccine' against her son's wishes and that a man with severe learning difficulties should have the jab despite his family's objections. Never mind that many had already died. The judiciary always supports doctors and government in fascist dictatorships. They wouldn't dare do otherwise. A horrific video was posted showing fascist officers from Los Angeles police forcibly giving the 'Covid' shot to women with special needs who were screaming that they didn't want it. The same fascists are seen giving the jab to a sleeping elderly woman in a care home. This is straight out of the Nazi playbook. Hitler's Nazis committed mass murder of the mentally ill and physically disabled throughout Germany and occupied territories in the programme that became known as Aktion T4, or just T4. Sabbatian-controlled Hitler and his grotesque crazies set out to kill those they considered useless and unnecessary. The Reich Committee for the Scientific Registering of Hereditary and Congenital Illnesses registered the births of babies identified by physicians to have 'defects'. By 1941 alone more than 5,000 children were murdered by the state and it is estimated that in total the number of innocent people killed in Aktion T4 was between 275,000 and 300,000. Parents were told their children had been sent away for 'special treatment' never to return. It is rather pathetic to see claims about plans for new extermination camps being dismissed today

when the same force behind current events did precisely that 80 years ago. Margaret Sanger was a Cult operative who used 'birth control' to sanitise her programme of eugenics. Organisations she founded became what is now Planned Parenthood. Sanger proposed that 'the whole dysgenic population would have its choice of segregation or sterilization'. These included epileptics, 'feeble-minded', and prostitutes. Sanger opposed charity because it perpetuated 'human waste'. She reveals the Cult mentality and if anyone thinks that extermination camps are a 'conspiracy theory' their naivety is touching if breathtakingly stupid.

If you don't believe that doctors can act with callous disregard for their patients it is worth considering that doctors and medical staff agreed to put government-decreed DNR orders on medical files and do nothing when resuscitation is called for. I don't know what you call such people in your house. In mine they are Nazis from the Josef Mengele School of Medicine. Phenomenal numbers of old people have died worldwide from the effects of lockdown, depression, lack of treatment, the 'vaccine' (more later) and losing the will to live. A common response at the start of the manufactured pandemic was to remove old people from hospital beds and transfer them to nursing homes. The decision would result in a mass cull of elderly people in those homes through lack of treatment – *not* 'Covid'. Care home whistleblowers have told how once the 'Covid' era began doctors would not come to their homes to treat patients and they were begging for drugs like antibiotics that often never came. The most infamous example was ordered by New York governor Andrew Cuomo, brother of a moronic CNN host, who amazingly was given an Emmy Award for his handling of the 'Covid crisis' by the ridiculous Wokers that hand them out. Just how ridiculous could be seen in February, 2021, when a Department of Justice and FBI investigation began into how thousands of old people in New York died in nursing homes after being discharged from hospital to make way for 'Covid' patients on Cuomo's say-so – and how he and his staff covered up these facts. This couldn't have happened to a nicer psychopath. Even then there was a 'Covid' spin. Reports said that

thousands of old people who tested positive for 'Covid' in hospital were transferred to nursing homes to both die of 'Covid' and transmit it to others. No – they were in hospital because they were ill and the fact that they tested positive with a test not testing for the 'virus' is irrelevant. They were ill often with respiratory diseases ubiquitous in old people near the end of their lives. Their transfer out of hospital meant that their treatment stopped and many would go on to die.

They're old. Who gives a damn?

I have exposed in the books for decades the Cult plan to cull the world's old people and even to introduce at some point what they call a 'demise pill' which at a certain age everyone would take and be out of here by law. In March, 2021, Spain legalised euthanasia and assisted suicide following the Netherlands, Belgium, Luxembourg and Canada on the Tiptoe to the demise pill. Treatment of old people by many 'care' homes has been a disgrace in the 'Covid' era. There are many, many, caring staff – I know some. There have, however, been legions of stories about callous treatment of old people and their families. Police were called when families came to take their loved ones home in the light of isolation that was killing them. They became prisoners of the state. Care home residents in insane, fascist Ontario, Canada, were not allowed to leave their *room* once the 'Covid' hoax began. UK staff have even wheeled elderly people away from windows where family members were talking with them. Oriana Criscuolo from Stockport in the English North West dropped off some things for her 80-year-old father who has Parkinson's disease and dementia and she wanted to wave to him through a ground-floor window. She was told that was 'illegal'. When she went anyway they closed the curtains in the middle of the day. Oriana said:

It's just unbelievable. I cannot understand how care home staff – people who are being paid to care – have become so uncaring. Their behaviour is inhumane and cruel. It's beyond belief.

She was right and this was not a one-off. What a way to end your life in such loveless circumstances. UK registered nurse Nicky Millen, a proper old school nurse for 40 years, said that when she started her career care was based on dignity, choice, compassion and empathy. Now she said 'the things that are important to me have gone out of the window.' She was appalled that people were dying without their loved ones and saying goodbye on iPads. Nicky described how a distressed 89-year-old lady stroked her face and asked her 'how many paracetamol would it take to finish me off'. Life was no longer worth living while not seeing her family. Nicky said she was humiliated in front of the ward staff and patients for letting the lady stroke her face and giving her a cuddle. Such is the dehumanisation that the 'Covid' hoax has brought to the surface. Nicky worked in care homes where patients told her they were being held prisoner. 'I want to live until I die', one said to her. 'I had a lady in tears because she hadn't seen her great-grandson.' Nicky was compassionate old school meeting psychopathic New Normal. She also said she had worked on a 'Covid' ward with no 'Covid' patients. Jewish writer Shai Held wrote an article in March, 2020, which was headlined 'The Staggering, Heartless Cruelty Toward the Elderly'. What he described was happening from the earliest days of lockdown. He said 'the elderly' were considered a group and not unique individuals (the way of the Woke). Shai Held said:

Notice how the all-too-familiar rhetoric of dehumanization works: 'The elderly' are bunched together as a faceless mass, all of them considered culprits and thus effectively deserving of the suffering the pandemic will inflict upon them. Lost entirely is the fact that the elderly are individual human beings, each with a distinctive face and voice, each with hopes and dreams, memories and regrets, friendships and marriages, loves lost and loves sustained.

'The elderly' have become another dehumanised group for which anything goes and for many that has resulted in cold disregard for their rights and their life. The distinctive face that Held talks about is designed to be deleted by masks until everyone is part of a faceless mass.

'War-zone' hospitals myth

Again and again medical professionals have told me what was really going on and how hospitals 'overrun like war zones' according to the media were virtually empty. The mantra from medical whistleblowers was please don't use my name or my career is over. Citizen journalists around the world sneaked into hospitals to film evidence exposing the 'war-zone' lie. They really *were* largely empty with closed wards and operating theatres. I met a hospital worker in my town on the Isle of Wight during the first lockdown in 2020 who said the only island hospital had never been so quiet. Lockdown was justified by the psychopaths to stop hospitals being overrun. At the same time that the island hospital was near-empty the military arrived here to provide *extra beds*. It was all propaganda to ramp up the fear to ensure compliance with fascism as were never-used temporary hospitals with thousands of beds known as Nightingales and never-used make-shift mortuaries opened by the criminal UK government. A man who helped to install those extra island beds attributed to the army said they were never used and the hospital was empty. Doctors and nurses 'stood around talking or on their phones, wandering down to us to see what we were doing'. There were no masks or social distancing. He accused the useless local island paper, the *County Press*, of 'pumping the fear as if our hospital was overrun and we only have one so it should have been'. He described ambulances parked up with crews outside in deck chairs. When his brother called an ambulance he was told there was a two-hour backlog which he called 'bullshit'. An old lady on the island fell 'and was in a bad way', but a caller who rang for an ambulance was told the situation wasn't urgent enough. Ambulance stations were working under capacity while people would hear ambulances with sirens blaring driving through the streets. When those living near the stations realised what was going on they would follow them as they left, circulated around an urban area with the sirens going, and then came back without stopping. All this was to increase levels of fear and the same goes for the 'ventilator shortage crisis' that cost tens of millions for hastily produced ventilators never to be used.

Ambulance crews that agreed to be exploited in this way for fear propaganda might find themselves a mirror. I wish them well with that. Empty hospitals were the obvious consequence of treatment and diagnoses of non-'Covid' conditions cancelled and those involved handed a death sentence. People have been dying at home from undiagnosed and untreated cancer, heart disease and other life-threatening conditions to allow empty hospitals to deal with a 'pandemic' that wasn't happening.

Death of the innocent

'War-zones' have been laying off nursing staff, even doctors where they can. There was no work for them. Lockdown was justified by saving lives and protecting the vulnerable they were actually killing with DNR orders and preventing empty hospitals being 'overrun'. In Britain the mantra of stay at home to 'save the NHS' was everywhere and across the world the same story was being sold when it was all lies. Two California doctors, Dan Erickson and Artin Massihi at Accelerated Urgent Care in Bakersfield, held a news conference in April, 2020, to say that intensive care units in California were 'empty, essentially', with hospitals shutting floors, not treating patients and laying off doctors. The California health system was working at minimum capacity 'getting rid of doctors because we just don't have the volume'. They said that people with conditions such as heart disease and cancer were not coming to hospital out of fear of 'Covid-19'. Their video was deleted by Susan Wojcicki's Cult-owned YouTube after reaching five million views. Florida governor Ron Desantis, who rejected the severe lockdowns of other states and is being targeted for doing so, said that in March, 2020, every US governor was given models claiming they would run out of hospital beds in days. That was never going to happen and the 'modellers' knew it. Deceit can be found at every level of the system. Urgent children's operations were cancelled including fracture repairs and biopsies to spot cancer. Eric Nicholls, a consultant paediatrician, said 'this is obviously concerning and we need to return to normal operating and to increase capacity as soon as possible'. Psychopaths

in power were rather less concerned *because* they are psychopaths. Deletion of urgent care and diagnosis has been happening all over the world and how many kids and others have died as a result of the actions of these cold and heartless lunatics dictating 'health' policy? The number must be stratospheric. Richard Sullivan, professor of cancer and global health at King's College London, said people feared 'Covid' more than cancer such was the campaign of fear. 'Years of lost life will be quite dramatic', Sullivan said, with 'a huge amount of avoidable mortality'. Sarah Woolnough, executive director for policy at Cancer Research UK, said there had been a 75 percent drop in urgent referrals to hospitals by family doctors of people with suspected cancer. Sullivan said that 'a lot of services have had to scale back – we've seen a dramatic decrease in the amount of elective cancer surgery'. Lockdown deaths worldwide has been absolutely fantastic with the *New York Post* reporting how data confirmed that 'lockdowns end more lives than they save':

There was a sharp decline in visits to emergency rooms and an increase in fatal heart attacks because patients didn't receive prompt treatment. Many fewer people were screened for cancer. Social isolation contributed to excess deaths from dementia and Alzheimer's.

Researchers predicted that the social and economic upheaval would lead to tens of thousands of "deaths of despair" from drug overdoses, alcoholism and suicide. As unemployment surged and mental-health and substance-abuse treatment programs were interrupted, the reported levels of anxiety, depression and suicidal thoughts increased dramatically, as did alcohol sales and fatal drug overdoses.

This has been happening while nurses and other staff had so much time on their hands in the 'war-zones' that Tic-Tok dancing videos began appearing across the Internet with medical staff dancing around in empty wards and corridors as people died at home from causes that would normally have been treated in hospital.

Mentions in dispatches

One brave and truth-committed whistleblower was Louise Hampton, a call handler with the UK NHS who made a viral Internet video saying she had done 'fuck all' during the 'pandemic'

which was 'a load of bollocks'. She said that 'Covid-19' was rebranded flu and of course she lost her job. This is what happens in the medical and endless other professions now when you tell the truth. Louise filmed inside 'war-zone' accident and emergency departments to show they were empty and I mean *empty* as in no one there. The mainstream media could have done the same and blown the gaff on the whole conspiracy. They haven't to their eternal shame. Not that most 'journalists' seem capable of manifesting shame as with the psychopaths they slavishly repeat without question. The relative few who were admitted with serious health problems were left to die alone with no loved ones allowed to see them because of 'Covid' rules and they included kids dying without the comfort of mum and dad at their bedside while the evil behind this couldn't give a damn. It was all good fun to them. A Scottish NHS staff nurse publicly quit in the spring of 2021 saying: 'I can no longer be part of the lies and the corruption by the government.' She said hospitals 'aren't full, the beds aren't full, beds have been shut, wards have been shut'. Hospitals were never busy throughout 'Covid'. The staff nurse said that Nicola Sturgeon, tragically the leader of the Scottish government, was on television saying save the hospitals and the NHS – 'but the beds are empty' and 'we've not seen flu, we always see flu every year'. She wrote to government and spoke with her union Unison (the unions are Cult-compromised and *useless*, but nothing changed. Many of her colleagues were scared of losing their jobs if they spoke out as they wanted to. She said nursing staff were being affected by wearing masks all day and 'my head is splitting every shift from wearing a mask'. The NHS is part of the fascist tyranny and must be dismantled so we can start again with human beings in charge. (Ironically, hospitals were reported to be busier again when official 'Covid' cases *fell* in spring/summer of 2021 and many other conditions required treatment at the same time as *the fake vaccine rollout*.)

I will cover the 'Covid vaccine' scam in detail later, but it is another indicator of the sickening disregard for human life that I am highlighting here. The DNA-manipulating concoctions do not fulfil

the definition of a 'vaccine', have never been used on humans before and were given only emergency approval because trials were not completed and they continued using the unknowing public. The result was what a NHS senior nurse with responsibility for 'vaccine' procedure said was 'genocide'. She said the 'vaccines' were not 'vaccines'. They had not been shown to be safe and claims about their effectiveness by drug companies were 'poetic licence'. She described what was happening as a 'horrid act of human annihilation'. The nurse said that management had instigated a policy of not providing a Patient Information Leaflet (PIL) before people were 'vaccinated' even though health care professionals are supposed to do this according to protocol. Patients should also be told that they are taking part in an ongoing clinical trial. Her challenges to what is happening had seen her excluded from meetings and ridiculed in others. She said she was told to 'watch my step ... or I would find myself surplus to requirements'. The nurse, who spoke anonymously in fear of her career, said she asked her NHS manager why he/she was content with taking part in genocide against those having the 'vaccines'. The reply was that everyone had to play their part and to 'put up, shut up, and get it done'. Government was 'leaning heavily' on NHS management which was clearly leaning heavily on staff. This is how the global 'medical' hierarchy operates and it starts with the Cult and its World Health Organization.

She told the story of a doctor who had the Pfizer jab and when questioned had no idea what was in it. The doctor had never read the literature. We have to stop treating doctors as intellectual giants when so many are moral and medical pygmies. The doctor did not even know that the 'vaccines' were not fully approved or that their trials were ongoing. They were, however, asking their patients if they minded taking part in follow-ups for research purposes – yes, the *ongoing clinical trial*. The nurse said the doctor's ignorance was not rare and she had spoken to a hospital consultant who had the jab without any idea of the background or that the 'trials' had not been completed. Nurses and pharmacists had shown the same ignorance.

‘My NHS colleagues have forsaken their duty of care, broken their code of conduct – Hippocratic Oath – and have been brainwashed just the same as the majority of the UK public through propaganda ...’ She said she had not been able to recruit a single NHS colleague, doctor, nurse or pharmacist to stand with her and speak out. Her union had refused to help. She said that if the genocide came to light she would not hesitate to give evidence at a Nuremberg-type trial against those in power who could have affected the outcomes but didn’t.

And all for what?

To put the nonsense into perspective let’s say the ‘virus’ does exist and let’s go completely crazy and accept that the official manipulated figures for cases and deaths are accurate. *Even then* a study by Stanford University epidemiologist Dr John Ioannidis published on the World Health Organization website produced an average infection to fatality rate of ... *0.23 percent!* Ioannidis said: ‘If one could sample equally from all locations globally, the median infection fatality rate might even be substantially lower than the 0.23% observed in my analysis.’ For healthy people under 70 it was ... *0.05 percent!* This compares with the 3.4 percent claimed by the Cult-owned World Health Organization when the hoax was first played and maximum fear needed to be generated. An updated Stanford study in April, 2021, put the ‘infection’ to ‘fatality’ rate at just 0.15 percent. Another team of scientists led by Megan O’Driscoll and Henrik Salje studied data from 45 countries and published their findings on the Nature website. For children and young people the figure is so small it virtually does not register although authorities will be hyping dangers to the young when they introduce DNA-manipulating ‘vaccines’ for children. The O’Driscoll study produced an average infection-fatality figure of 0.003 for children from birth to four; 0.001 for 5 to 14; 0.003 for 15 to 19; and it was still only 0.456 up to 64. To claim that children must be ‘vaccinated’ to protect them from ‘Covid’ is an obvious lie and so there must be another reason and there is. What’s more the average age of a ‘Covid’ death is akin

to the average age that people die in general. The average age of death in England is about 80 for men and 83 for women. The average age of death from alleged 'Covid' is between 82 and 83. California doctors, Dan Erickson and Artin Massihi, said at their April media conference that projection models of millions of deaths had been 'woefully inaccurate'. They produced detailed figures showing that Californians had a 0.03 chance of dying from 'Covid' based on the number of people who tested positive (with a test not testing for the 'virus'). Erickson said there was a 0.1 percent chance of dying from 'Covid' in the *state* of New York, not just the city, and a 0.05 percent chance in Spain, a centre of 'Covid-19' hysteria at one stage. The Stanford studies supported the doctors' data with fatality rate estimates of 0.23 and 0.15 percent. How close are these figures to my estimate of *zero*? Death-rate figures claimed by the World Health Organization at the start of the hoax were some 15 times higher. The California doctors said there was no justification for lockdowns and the economic devastation they caused. Everything they had ever learned about quarantine was that you quarantine the *sick* and not the healthy. They had never seen this before and it made no medical sense.

Why in the in the light of all this would governments and medical systems the world over say that billions must go under house arrest; lose their livelihood; in many cases lose their mind, their health and their life; force people to wear masks dangerous to health and psychology; make human interaction and even family interaction a criminal offence; ban travel; close restaurants, bars, watching live sport, concerts, theatre, and any activity involving human togetherness and discourse; and closing schools to isolate children from their friends and cause many to commit suicide in acts of hopelessness and despair? The California doctors said lockdown consequences included increased child abuse, partner abuse, alcoholism, depression, and other impacts they were seeing every day. Who would do that to the entire human race if not mentally-ill psychopaths of almost unimaginable extremes like Bill Gates? We must face the reality of what we are dealing with and come out of

denial. Fascism and tyranny are made possible only by the target population submitting and acquiescing to fascism and tyranny. The whole of human history shows that to be true. Most people naively and unquestioning believed what they were told about a 'deadly virus' and meekly and weakly submitted to house arrest. Those who didn't believe it – at least in total – still submitted in fear of the consequences of not doing so. For the rest who wouldn't submit draconian fines have been imposed, brutal policing by psychopaths *for* psychopaths, and condemnation from the meek and weak who condemn the Pushbackers on behalf of the very force that has them, too, in its gun sights. 'Pathetic' does not even begin to suffice. Britain's brainless 'Health' Secretary Matt Hancock warned anyone lying to border officials about returning from a list of 'hotspot' countries could face a jail sentence of up to ten years which is more than for racially-aggravated assault, incest and attempting to have sex with a child under 13. Hancock is a lunatic, but he has the state apparatus behind him in a Cult-led chain reaction and the same with UK 'Vaccine Minister' Nadhim Zahawi, a prominent member of the mega-Cult secret society, Le Cercle, which featured in my earlier books. The Cult enforces its will on governments and medical systems; government and medical systems enforce their will on business and police; business enforces its will on staff who enforce it on customers; police enforce the will of the Cult on the population and play their essential part in creating a world of fascist control that their own children and grandchildren will have to live in their entire lives. It is a hierarchical pyramid of imposition and acquiescence and, yes indeed, of clinical insanity.

Does anyone bright enough to read this book have to ask what the answer is? I think not, but I will reveal it anyway in the fewest of syllables: Tell the psychos and their moronic lackeys to fuck off and let's get on with our lives. We are many – They are few.

CHAPTER SEVEN

War on your mind

One believes things because one has been conditioned to believe them

Aldous Huxley, Brave New World

I have described the 'Covid' hoax as a 'Psyop' and that is true in every sense and on every level in accordance with the definition of that term which is psychological warfare. Break down the 'Covid pandemic' to the foundation themes and it is psychological warfare on the human individual and collective mind.

The same can be said for the entire human belief system involving every subject you can imagine. Huxley was right in his contention that people believe what they are conditioned to believe and this comes from the repetition throughout their lives of the same falsehoods. They spew from government, corporations, media and endless streams of 'experts' telling you what the Cult wants you to believe and often believing it themselves (although *far* from always). 'Experts' are rewarded with 'prestigious' jobs and titles and as agents of perceptual programming with regular access to the media. The Cult has to control the narrative – control *information* – or they lose control of the vital, crucial, without-which-they-cannot-prevail public perception of reality. The foundation of that control today is the Internet made possible by the Defense Advanced Research Projects Agency (DARPA), the incredibly sinister technological arm of the Pentagon. The Internet is the result of military technology.

DARPA openly brags about establishing the Internet which has been a long-term project to lasso the minds of the global population. I have said for decades the plan is to control information to such an extreme that eventually no one would see or hear anything that the Cult does not approve. We are closing in on that end with ferocious censorship since the 'Covid' hoax began and in my case it started back in the 1990s in terms of books and speaking venues. I had to create my own publishing company in 1995 precisely because no one else would publish my books even then. I think they're all still running.

Cult Internet

To secure total control of information they needed the Internet in which pre-programmed algorithms can seek out 'unclean' content for deletion and even stop it being posted in the first place. The Cult had to dismantle print and non-Internet broadcast media to ensure the transfer of information to the appropriate-named 'Web' – a critical expression of the *Cult* web. We've seen the ever-quickenening demise of traditional media and control of what is left by a tiny number of corporations operating worldwide. Independent journalism in the mainstream is already dead and never was that more obvious than since the turn of 2020. The Cult wants all information communicated via the Internet to globally censor and allow the plug to be pulled any time. Lockdowns and forced isolation has meant that communication between people has been through electronic means and no longer through face-to-face discourse and discussion. Cult psychopaths have targeted the bars, restaurants, sport, venues and meeting places in general for this reason. None of this is by chance and it's to stop people gathering in any kind of privacy or number while being able to track and monitor all Internet communications and block them as necessary. Even private messages between individuals have been censored by these fascists that control Cult fronts like Facebook, Twitter, Google and YouTube which are all officially run by Sabbatian place-people and from the background by higher-level Sabbatian place people.

Facebook, Google, Amazon and their like were seed-funded and supported into existence with money-no-object infusions of funds either directly or indirectly from DARPA and CIA technology arm In-Q-Tel. The Cult plays the long game and prepares very carefully for big plays like 'Covid'. Amazon is another front in the psychological war and pretty much controls the global market in book sales and increasingly publishing. Amazon's limitless funds have deleted fantastic numbers of independent publishers to seize global domination on the way to deciding which books can be sold and circulated and which cannot. Moves in that direction are already happening. Amazon's leading light Jeff Bezos is the grandson of Lawrence Preston Gise who worked with DARPA predecessor ARPA. Amazon has big connections to the CIA and the Pentagon. The plan I have long described went like this:

1. Employ military technology to establish the Internet.
2. Sell the Internet as a place where people can freely communicate without censorship and allow that to happen until the Net becomes the central and irreversible pillar of human society. If the Internet had been highly censored from the start many would have rejected it.
3. Fund and manipulate major corporations into being to control the circulation of information on your Internet using cover stories about geeks in garages to explain how they came about. Give them unlimited funds to expand rapidly with no need to make a profit for years while non-Cult companies who need to balance the books cannot compete. You know that in these circumstances your Googles, YouTubes, Facebooks and Amazons are going to secure near monopolies by either crushing or buying up the opposition.
4. Allow freedom of expression on both the Internet and communication platforms to draw people in until the Internet is the central and irreversible pillar of human society and your communication corporations have reached a stage of near monopoly domination.
5. Then unleash your always-planned frenzy of censorship on the basis of 'where else are you going to go?' and continue to expand that until nothing remains that the Cult does not want its human targets to see.

The process was timed to hit the 'Covid' hoax to ensure the best chance possible of controlling the narrative which they knew they had to do at all costs. They were, after all, about to unleash a 'deadly virus' that didn't really exist. If you do that in an environment of free-flowing information and opinion you would be dead in the

water before you could say Gates is a psychopath. The network was in place through which the Cult-created-and-owned World Health Organization could dictate the 'Covid' narrative and response policy slavishly supported by Cult-owned Internet communication giants and mainstream media while those telling a different story were censored. Google, YouTube, Facebook and Twitter openly announced that they would do this. What else would we expect from Cult-owned operations like Facebook which former executives have confirmed set out to make the platform more addictive than cigarettes and coldly manipulates emotions of its users to sow division between people and groups and scramble the minds of the young? If Zuckerberg lives out the rest of his life without going to jail for crimes against humanity, and most emphatically against the young, it will be a travesty of justice. Still, no matter, cause and effect will catch up with him eventually and the same with Sergey Brin and Larry Page at Google with its CEO Sundar Pichai who fix the Google search results to promote Cult narratives and hide the opposition. Put the same key words into Google and other search engines like DuckDuckGo and you will see how different results can be. Wikipedia is another intensely biased 'encyclopaedia' which skews its content to the Cult agenda. YouTube links to Wikipedia's version of 'Covid' and 'climate change' on video pages in which experts in their field offer a different opinion (even that is increasingly rare with Wojcicki censorship). Into this 'Covid' silence-them network must be added government media censors, sorry 'regulators', such as Ofcom in the UK which imposed tyrannical restrictions on British broadcasters that had the effect of banning me from ever appearing. Just to debate with me about my evidence and views on 'Covid' would mean breaking the fascistic impositions of Ofcom and its CEO career government bureaucrat Melanie Dawes. Gutless British broadcasters tremble at the very thought of fascist Ofcom.

Psychos behind 'Covid'

The reason for the 'Covid' catastrophe in all its facets and forms can be seen by whom and what is driving the policies worldwide in such a coordinated way. Decisions are not being made to protect health, but to target psychology. The dominant group guiding and 'advising' government policy are not medical professionals. They are psychologists and behavioural scientists. Every major country has its own version of this phenomenon and I'll use the British example to show how it works. In many ways the British version has been affecting the wider world in the form of the huge behaviour manipulation network in the UK which operates in other countries. The network involves private companies, government, intelligence and military. The Cabinet Office is at the centre of the government 'Covid' Psyop and part-owns, with 'innovation charity' Nesta, the Behavioural Insights Team (BIT) which claims to be independent of government but patently isn't. The BIT was established in 2010 and its job is to manipulate the psyche of the population to acquiesce to government demands and so much more. It is also known as the 'Nudge Unit', a name inspired by the 2009 book by two ultra-Zionists, Cass Sunstein and Richard Thaler, called *Nudge: Improving Decisions About Health, Wealth, and Happiness*. The book, as with the Behavioural Insights Team, seeks to 'nudge' behaviour (manipulate it) to make the public follow patterns of action and perception that suit those in authority (the Cult). Sunstein is so skilled at this that he advises the World Health Organization and the UK Behavioural Insights Team and was Administrator of the White House Office of Information and Regulatory Affairs in the Obama administration. Biden appointed him to the Department of Homeland Security – another ultra-Zionist in the fold to oversee new immigration laws which is another policy the Cult wants to control. Sunstein is desperate to silence anyone exposing conspiracies and co-authored a 2008 report on the subject in which suggestions were offered to ban 'conspiracy theorizing' or impose 'some kind of tax, financial or otherwise, on those who disseminate such theories'. I guess a psychiatrist's chair is out of the question?

Sunstein's mate Richard Thaler, an 'academic affiliate' of the UK Behavioural Insights Team, is a proponent of 'behavioural economics' which is defined as the study of 'the effects of psychological, cognitive, emotional, cultural and social factors on the decisions of individuals and institutions'. Study the effects so they can be manipulated to be what you want them to be. Other leading names in the development of behavioural economics are ultra-Zionists Daniel Kahneman and Robert J. Shiller and they, with Thaler, won the Nobel Memorial Prize in Economic Sciences for their work in this field. The Behavioural Insights Team is operating at the heart of the UK government and has expanded globally through partnerships with several universities including Harvard, Oxford, Cambridge, University College London (UCL) and Pennsylvania. They claim to have 'trained' (reframed) 20,000 civil servants and run more than 750 projects involving 400 randomised controlled trials in dozens of countries' as another version of mind reframers Common Purpose. BIT works from its office in New York with cities and their agencies, as well as other partners, across the United States and Canada – this is a company part-owned by the British government Cabinet Office. An executive order by President Cult-servant Obama established a US Social and Behavioral Sciences Team in 2015. They all have the same reason for being and that's to brainwash the population directly and by brainwashing those in positions of authority.

'Covid' mind game

Another prime aspect of the UK mind-control network is the 'independent' [joke] Scientific Pandemic Insights Group on Behaviours (SPI-B) which 'provides behavioural science advice aimed at anticipating and helping people adhere to interventions that are recommended by medical or epidemiological experts'. That means manipulating public perception and behaviour to do whatever government tells them to do. It's disgusting and if they really want the public to be 'safe' this lot should all be under lock and key. According to the government website SPI-B consists of

'behavioural scientists, health and social psychologists, anthropologists and historians' and advises the Whitty-Vallance-led Scientific Advisory Group for Emergencies (SAGE) which in turn advises the government on 'the science' (it doesn't) and 'Covid' policy. When politicians say they are being guided by 'the science' this is the rabble in each country they are talking about and that 'science' is dominated by behaviour manipulators to enforce government fascism through public compliance. The Behaviour Insight Team is headed by psychologist David Solomon Halpern, a visiting professor at King's College London, and connects with a national and global web of other civilian and military organisations as the Cult moves towards its goal of fusing them into one fascistic whole in every country through its 'Fusion Doctrine'. The behaviour manipulation network involves, but is not confined to, the Foreign Office; National Security Council; government communications headquarters (GCHQ); MI5; MI6; the Cabinet Office-based Media Monitoring Unit; and the Rapid Response Unit which 'monitors digital trends to spot emerging issues; including misinformation and disinformation; and identifies the best way to respond'.

There is also the 77th Brigade of the UK military which operates like the notorious Israeli military's Unit 8200 in manipulating information and discussion on the Internet by posing as members of the public to promote the narrative and discredit those who challenge it. Here we have the military seeking to manipulate *domestic* public opinion while the Nazis in government are fine with that. Conservative Member of Parliament Tobias Ellwood, an advocate of lockdown and control through 'vaccine passports', is a Lieutenant Colonel reservist in the 77th Brigade which connects with the military operation jHub, the 'innovation centre' for the Ministry of Defence and Strategic Command. jHub has also been involved with the civilian National Health Service (NHS) in 'symptom tracing' the population. The NHS is a key part of this mind control network and produced a document in December, 2020, explaining to staff how to use psychological manipulation with different groups and ages to get them to have the DNA-manipulating 'Covid vaccine'

that's designed to cumulatively rewrite human genetics. The document, called 'Optimising Vaccination Roll Out – Do's and Dont's for all messaging, documents and "communications" in the widest sense', was published by NHS England and the NHS Improvement *Behaviour Change Unit* in partnership with Public Health England and Warwick Business School. I hear the mantra about 'save the NHS' and 'protect the NHS' when we need to scrap the NHS and start again. The current version is far too corrupt, far too anti-human and totally compromised by Cult operatives and their assets. UK government broadcast media censor Ofcom will connect into this web – as will the BBC with its tremendous Ofcom influence – to control what the public see and hear and dictate mass perception. Nuremberg trials must include personnel from all these organisations.

The fear factor

The 'Covid' hoax has led to the creation of the UK Cabinet Office-connected Joint Biosecurity Centre (JBC) which is officially described as providing 'expert advice on pandemics' using its independent [all Cult operations are 'independent'] analytical function to provide real-time analysis about infection outbreaks to identify and respond to outbreaks of Covid-19'. Another role is to advise the government on a response to spikes in infections – 'for example by closing schools or workplaces in local areas where infection levels have risen'. Put another way, promoting the Cult agenda. The Joint Biosecurity Centre is modelled on the Joint Terrorism Analysis Centre which analyses intelligence to set 'terrorism threat levels' and here again you see the fusion of civilian and military operations and intelligence that has led to military intelligence producing documents about 'vaccine hesitancy' and how it can be combated. Domestic civilian matters and opinions should not be the business of the military. The Joint Biosecurity Centre is headed by Tom Hurd, director general of the Office for Security and Counter-Terrorism from the establishment-to-its-fingertips Hurd family. His father is former Foreign Secretary Douglas Hurd. How coincidental that Tom

Hurd went to the elite Eton College and Oxford University with Boris Johnson. Imperial College with its ridiculous computer modeller Neil Ferguson will connect with this gigantic web that will itself interconnect with similar set-ups in other major and not so major countries. Compared with this Cult network the politicians, be they Boris Johnson, Donald Trump or Joe Biden, are bit-part players 'following the science'. The network of psychologists was on the 'Covid' case from the start with the aim of generating maximum fear of the 'virus' to ensure compliance by the population. A government behavioural science group known as SPI-B produced a paper in March, 2020, for discussion by the main government science advisory group known as SAGE. It was headed 'Options for increasing adherence to social distancing measures' and it said the following in a section headed 'Persuasion':

- A substantial number of people still do not feel sufficiently personally threatened; it could be that they are reassured by the low death rate in their demographic group, although levels of concern may be rising. Having a good understanding of the risk has been found to be positively associated with adoption of COVID-19 social distancing measures in Hong Kong.
- The perceived level of personal threat needs to be increased among those who are complacent, using hard-hitting evaluation of options for increasing social distancing emotional messaging. To be effective this must also empower people by making clear the actions they can take to reduce the threat.
- Responsibility to others: There seems to be insufficient understanding of, or feelings of responsibility about, people's role in transmitting the infection to others ... Messaging about actions need to be framed positively in terms of protecting oneself and the community, and increase confidence that they will be effective.
- Some people will be more persuaded by appeals to play by the rules, some by duty to the community, and some to personal risk.

All these different approaches are needed. The messaging also needs to take account of the realities of different people's lives. Messaging needs to take account of the different motivational levers and circumstances of different people.

All this could be achieved the SPI-B psychologists said by *using the media to increase the sense of personal threat* which translates as terrify the shit out of the population, including children, so they all do what we want. That's not happened has it? Those excuses for 'journalists' who wouldn't know journalism if it bit them on the arse (the great majority) have played their crucial part in serving this Cult-government Psyop to enslave their own kids and grandkids. How they live with themselves I have no idea. The psychological war has been underpinned by constant government 'Covid' propaganda in almost every television and radio ad break, plus the Internet and print media, which has pounded out the fear with taxpayers footing the bill for their own programming. The result has been people terrified of a 'virus' that doesn't exist or one with a tiny fatality rate even if you believe it does. People walk down the street and around the shops wearing face-nappies damaging their health and psychology while others report those who refuse to be that naïve to the police who turn up in their own face-nappies. I had a cameraman come to my flat and he was so frightened of 'Covid' he came in wearing a mask and refused to shake my hand in case he caught something. He had – naïveitis – and the thought that he worked in the mainstream media was both depressing and made his behaviour perfectly explainable. The fear which has gripped the minds of so many and frozen them into compliance has been carefully cultivated by these psychologists who are really psychopaths. If lives get destroyed and a lot of young people commit suicide it shows our plan is working. SPI-B then turned to compulsion on the public to comply. 'With adequate preparation, rapid change can be achieved', it said. Some countries had introduced mandatory self-isolation on a wide scale without evidence of major public unrest and a large majority of the UK's population appeared to be supportive of more coercive measures with 64 percent of adults saying they would

support putting London under a lockdown (watch the 'polls' which are designed to make people believe that public opinion is in favour or against whatever the subject in hand).

For 'aggressive protective measures' to be effective, the SPI-B paper said, special attention should be devoted to those population groups that are more at risk. Translated from the Orwellian this means making the rest of population feel guilty for not protecting the 'vulnerable' such as old people which the Cult and its agencies were about to kill on an industrial scale with lockdown, lack of treatment and the Gates 'vaccine'. Psychopath psychologists sold their guilt-trip so comprehensively that Los Angeles County Supervisor Hilda Solis reported that children were apologising (from a distance) to their parents and grandparents for bringing 'Covid' into their homes and getting them sick. '... These apologies are just some of the last words that loved ones will ever hear as they die alone,' she said. Gut-wrenchingly Solis then used this childhood tragedy to tell children to stay at home and 'keep your loved ones alive'. Imagine heaping such potentially life-long guilt on a kid when it has absolutely nothing to do with them. These people are deeply disturbed and the psychologists behind this even more so.

Uncivil war – divide and rule

Professional mind-controllers at SPI-B wanted the media to increase a sense of responsibility to others (do as you're told) and promote 'positive messaging' for those actions while in contrast to invoke 'social disapproval' by the unquestioning, obedient, community of anyone with a mind of their own. Again the compliant Goebbels-like media obliged. This is an old, old, trick employed by tyrannies the world over throughout human history. You get the target population to keep the target population in line – *your* line. SPI-B said this could 'play an important role in preventing anti-social behaviour or discouraging failure to enact pro-social behaviour'. For 'anti-social' in the Orwellian parlance of SPI-B see any behaviour that government doesn't approve. SPI-B recommendations said that 'social disapproval' should be accompanied by clear messaging and

promotion of strong collective identity – hence the government and celebrity mantra of ‘we’re all in this together’. Sure we are. The mind doctors have such contempt for their targets that they think some clueless comedian, actor or singer telling them to do what the government wants will be enough to win them over. We have had UK comedian Lenny Henry, actor Michael Caine and singer Elton John wheeled out to serve the propagandists by urging people to have the DNA-manipulating ‘Covid’ non-‘vaccine’. The role of Henry and fellow black celebrities in seeking to coax a ‘vaccine’ reluctant black community into doing the government’s will was especially stomach-turning. An emotion-manipulating script and carefully edited video featuring these black ‘celebs’ was such an insult to the intelligence of black people and where’s the self-respect of those involved selling their souls to a fascist government agenda? Henry said he heard black people’s ‘legitimate worries and concerns’, but people must ‘trust the facts’ when they were doing exactly that by not having the ‘vaccine’. They had to include the obligatory reference to Black Lives Matter with the line ... ‘Don’t let coronavirus cost even more black lives – because we matter’. My god, it was pathetic. ‘I know the vaccine is safe and what it does.’ How? ‘I’m a comedian and it says so in my script.’

SPI-B said social disapproval needed to be carefully managed to avoid victimisation, scapegoating and misdirected criticism, but they knew that their ‘recommendations’ would lead to exactly that and the media were specifically used to stir-up the divide-and-conquer hostility. Those who conform like good little baa, baas, are praised while those who have seen through the tidal wave of lies are ‘Covidiot’. The awake have been abused by the fast asleep for not conforming to fascism and impositions that the awake know are designed to endanger their health, dehumanise them, and tear asunder the very fabric of human society. We have had the curtain-twitchers and morons reporting neighbours and others to the face-napped police for breaking ‘Covid rules’ with fascist police delighting in posting links and phone numbers where this could be done. The Cult cannot impose its will without a compliant police

and military or a compliant population willing to play their part in enslaving themselves and their kids. The words of a pastor in Nazi Germany are so appropriate today:

First they came for the socialists and I did not speak out because I was not a socialist.

Then they came for the trade unionists and I did not speak out because I was not a trade unionist.

Then they came for the Jews and I did not speak out because I was not a Jew.

Then they came for me and there was no one left to speak for me.

Those who don't learn from history are destined to repeat it and so many are.

'Covid' rules: Rewiring the mind

With the background laid out to this gigantic national and global web of psychological manipulation we can put 'Covid' rules into a clear and sinister perspective. Forget the claims about protecting health. 'Covid' rules are about dismantling the human mind, breaking the human spirit, destroying self-respect, and then putting Humpty Dumpty together again as a servile, submissive slave. Social isolation through lockdown and distancing have devastating effects on the human psyche as the psychological psychopaths well know and that's the real reason for them. Humans need contact with each other, discourse, closeness and touch, or they eventually, and literally, go crazy. Masks, which I will address at some length, fundamentally add to the effects of isolation and the Cult agenda to dehumanise and de-individualise the population. To do this while knowing – in fact *seeking* – this outcome is the very epitome of evil and psychologists involved in this *are* the epitome of evil. They must like all the rest of the Cult demons and their assets stand trial for crimes against humanity on a scale that defies the imagination. Psychopaths in uniform use isolation to break enemy troops and agents and make them subservient and submissive to tell what they know. The technique is rightly considered a form of torture and

torture is most certainly what has been imposed on the human population.

Clinically-insane American psychologist Harry Harlow became famous for his isolation experiments in the 1950s in which he separated baby monkeys from their mothers and imprisoned them for months on end in a metal container or 'pit of despair'. They soon began to show mental distress and depression as any idiot could have predicted. Harlow put other monkeys in steel chambers for three, six or twelve months while denying them any contact with animals or humans. He said that the effects of total social isolation for six months were 'so devastating and debilitating that we had assumed initially that twelve months of isolation would not produce any additional decrement'; but twelve months of isolation 'almost obliterated the animals socially'. This is what the Cult and its psychopaths are doing to you and your children. Even monkeys in partial isolation in which they were not allowed to form relationships with other monkeys became 'aggressive and hostile, not only to others, but also towards their own bodies'. We have seen this in the young as a consequence of lockdown. UK government psychopaths launched a public relations campaign telling people not to hug each other even after they received the 'Covid-19 vaccine' which we were told with more lies would allow a return to 'normal life'. A government source told *The Telegraph*: 'It will be along the lines that it is great that you have been vaccinated, but if you are going to visit your family and hug your grandchildren there is a chance you are going to infect people you love.' The source was apparently speaking from a secure psychiatric facility. Janet Lord, director of Birmingham University's Institute of Inflammation and Ageing, said that parents and grandparents should avoid hugging their children. Well, how can I put it, Ms Lord? Fuck off. Yep, that'll do.

Destroying the kids – where are the parents?

Observe what has happened to people enslaved and isolated by lockdown as suicide and self-harm has soared worldwide,

particularly among the young denied the freedom to associate with their friends. A study of 49,000 people in English-speaking countries concluded that almost half of young adults are at clinical risk of mental health disorders. A national survey in America of 1,000 currently enrolled high school and college students found that 5 percent reported attempting suicide during the pandemic. Data from the US CDC's National Syndromic Surveillance Program from January 1st to October 17th, 2020, revealed a *31 percent* increase in mental health issues among adolescents aged 12 to 17 compared with 2019. The CDC reported that America in general suffered the biggest drop in life expectancy since World War Two as it fell by a year in the first half of 2020 as a result of 'deaths of despair' – overdoses and suicides. Deaths of despair have leapt by more than 20 percent during lockdown and include the highest number of fatal overdoses ever recorded in a single year – 81,000. Internet addiction is another consequence of being isolated at home which lowers interest in physical activities as kids fall into inertia and what's the point? Children and young people are losing hope and giving up on life, sometimes literally. A 14-year-old boy killed himself in Maryland because he had 'given up' when his school district didn't reopen; an 11-year-old boy shot himself during a zoom class; a teenager in Maine succumbed to the isolation of the 'pandemic' when he ended his life after experiencing a disrupted senior year at school. Children as young as nine have taken their life and all these stories can be repeated around the world. Careers are being destroyed before they start and that includes those in sport in which promising youngsters have not been able to take part. The plan of the psycho-psychologists is working all right. Researchers at Cambridge University found that lockdowns cause significant harm to children's mental health. Their study was published in the *Archives of Disease in Childhood*, and followed 168 children aged between 7 and 11. The researchers concluded:

During the UK lockdown, children's depression symptoms have increased substantially, relative to before lockdown. The scale of this effect has direct relevance for the continuation of different elements of lockdown policy, such as complete or partial school closures ...

... Specifically, we observed a statistically significant increase in ratings of depression, with a medium-to-large effect size. Our findings emphasise the need to incorporate the potential impact of lockdown on child mental health in planning the ongoing response to the global pandemic and the recovery from it.

Not a chance when the Cult's psycho-psychologists were getting exactly what they wanted. The UK's Royal College of Paediatrics and Child Health has urged parents to look for signs of eating disorders in children and young people after a three to four fold increase. Specialists say the 'pandemic' is a major reason behind the rise. You don't say. The College said isolation from friends during school closures, exam cancellations, loss of extra-curricular activities like sport, and an increased use of social media were all contributory factors along with fears about the virus (psycho-psychologists again), family finances, and students being forced to quarantine. Doctors said young people were becoming severely ill by the time they were seen with 'Covid' regulations reducing face-to-face consultations. Nor is it only the young that have been devastated by the psychopaths. Like all bullies and cowards the Cult is targeting the young, elderly, weak and infirm. A typical story was told by a British lady called Lynn Parker who was not allowed to visit her husband in 2020 for the last ten and half months of his life 'when he needed me most' between March 20th and when he died on December 19th. This vacates the criminal and enters the territory of evil. The emotional impact on the immune system alone is immense as are the number of people of all ages worldwide who have died as a result of Cult-demanded, Gates-demanded, lockdowns.

Isolation is torture

The experience of imposing solitary confinement on millions of prisoners around the world has shown how a large percentage become 'actively psychotic and/or acutely suicidal'. Social isolation has been found to trigger 'a specific psychiatric syndrome, characterized by hallucinations; panic attacks; overt paranoia; diminished impulse control; hypersensitivity to external stimuli; and difficulties with thinking, concentration and memory'. Juan Mendez,

a United Nations rapporteur (investigator), said that isolation is a form of torture. Research has shown that even after isolation prisoners find it far more difficult to make social connections and I remember chatting to a shop assistant after one lockdown who told me that when her young son met another child again he had no idea how to act or what to do. Hannah Flanagan, Director of Emergency Services at Journey Mental Health Center in Dane County, Wisconsin, said: 'The specificity about Covid social distancing and isolation that we've come across as contributing factors to the suicides are really new to us this year.' But they are not new to those that devised them. They are getting the effect they want as the population is psychologically dismantled to be rebuilt in a totally different way. Children and the young are particularly targeted. They will be the adults when the full-on fascist AI-controlled technocracy is planned to be imposed and they are being prepared to meekly submit. At the same time older people who still have a memory of what life was like before – and how fascist the new normal really is – are being deleted. You are going to see efforts to turn the young against the old to support this geriatric genocide. Hannah Flanagan said the big increase in suicide in her county proved that social isolation is not only harmful, but deadly. Studies have shown that isolation from others is one of the main risk factors in suicide and even more so with women. Warnings that lockdown could create a 'perfect storm' for suicide were ignored. After all this was one of the *reasons* for lockdown. Suicide, however, is only the most extreme of isolation consequences. There are many others. Dr Dhruv Khullar, assistant professor of healthcare policy at Weill Cornell Medical College, said in a *New York Times* article in 2016 long before the fake 'pandemic':

A wave of new research suggests social separation is bad for us. Individuals with less social connection have disrupted sleep patterns, altered immune systems, more inflammation and higher levels of stress hormones. One recent study found that isolation increases the risk of heart disease by 29 percent and stroke by 32 percent. Another analysis that pooled data from 70 studies and 3.4 million people found that socially isolated individuals had a 30 percent higher risk of dying in the next seven years, and that this effect was largest in middle age.

Loneliness can accelerate cognitive decline in older adults, and isolated individuals are twice as likely to die prematurely as those with more robust social interactions. These effects start early: Socially isolated children have significantly poorer health 20 years later, even after controlling for other factors. All told, loneliness is as important a risk factor for early death as obesity and smoking.

There you have proof from that one article alone four years before 2020 that those who have enforced lockdown, social distancing and isolation knew what the effect would be and that is even more so with professional psychologists that have been driving the policy across the globe. We can go back even further to the years 2000 and 2003 and the start of a major study on the effects of isolation on health by Dr Janine Gronewold and Professor Dirk M. Hermann at the University Hospital in Essen, Germany, who analysed data on 4,316 people with an average age of 59 who were recruited for the long-term research project. They found that socially isolated people are more than 40 percent more likely to have a heart attack, stroke, or other major cardiovascular event and nearly 50 percent more likely to die from any cause. Given the financial Armageddon unleashed by lockdown we should note that the study found a relationship between increased cardiovascular risk and lack of financial support. After excluding other factors social isolation was still connected to a 44 percent increased risk of cardiovascular problems and a 47 percent increased risk of death by any cause. Lack of financial support was associated with a 30 percent increase in the risk of cardiovascular health events. Dr Gronewold said it had been known for some time that feeling lonely or lacking contact with close friends and family can have an impact on physical health and the study had shown that having strong social relationships is of high importance for heart health. Gronewold said they didn't understand yet why people who are socially isolated have such poor health outcomes, but this was obviously a worrying finding, particularly during these times of prolonged social distancing. Well, it can be explained on many levels. You only have to identify the point in the body where people feel loneliness and missing people they are parted from – it's in the centre of the chest where they feel the ache of loneliness and the ache of missing people. 'My heart aches for

you' ... 'My heart aches for some company.' I will explain this more in the chapter Escaping Wetiko, but when you realise that the body is the mind – they are expressions of each other – the reason why state of the mind dictates state of the body becomes clear.

American psychologist Ranjit Powar was highlighting the effects of lockdown isolation as early as April, 2020. She said humans have evolved to be social creatures and are wired to live in interactive groups. Being isolated from family, friends and colleagues could be unbalancing and traumatic for most people and could result in short or even long-term psychological and physical health problems. An increase in levels of anxiety, aggression, depression, forgetfulness and hallucinations were possible psychological effects of isolation. 'Mental conditions may be precipitated for those with underlying pre-existing susceptibilities and show up in many others without any pre-condition.' Powar said personal relationships helped us cope with stress and if we lost this outlet for letting off steam the result can be a big emotional void which, for an average person, was difficult to deal with. 'Just a few days of isolation can cause increased levels of anxiety and depression' – so what the hell has been the effect on the global population of *18 months* of this at the time of writing? Powar said: 'Add to it the looming threat of a dreadful disease being repeatedly hammered in through the media and you have a recipe for many shades of mental and physical distress.' For those with a house and a garden it is easy to forget that billions have had to endure lockdown isolation in tiny overcrowded flats and apartments with nowhere to go outside. The psychological and physical consequences of this are unimaginable and with lunatic and abusive partners and parents the consequences have led to tremendous increases in domestic and child abuse and alcoholism as people seek to shut out the horror. Ranjit Powar said:

Staying in a confined space with family is not all a rosy picture for everyone. It can be extremely oppressive and claustrophobic for large low-income families huddled together in small single-room houses. Children here are not lucky enough to have many board/electronic games or books to keep them occupied.

Add to it the deep insecurity of running out of funds for food and basic necessities. On the other hand, there are people with dysfunctional family dynamics, such as domineering, abusive or alcoholic partners, siblings or parents which makes staying home a period of trial. Incidence of suicide and physical abuse against women has shown a worldwide increase. Heightened anxiety and depression also affect a person's immune system, making them more susceptible to illness.

To think that Powar's article was published on April 11th, 2020.

Six-feet fantasy

Social (unsocial) distancing demanded that people stay six feet or two metres apart. UK government advisor Robert Dingwall from the New and Emerging Respiratory Virus Threats Advisory Group said in a radio interview that the two-metre rule was 'conjured up out of nowhere' and was not based on science. No, it was not based on *medical* science, but it didn't come out of nowhere. The distance related to *psychological* science. Six feet/two metres was adopted in many countries and we were told by people like the criminal Anthony Fauci and his ilk that it was founded on science. Many schools could not reopen because they did not have the space for six-foot distancing. Then in March, 2021, after a year of six-foot 'science', a study published in the *Journal of Infectious Diseases* involving more than 500,000 students and almost 100,000 staff over 16 weeks revealed no significant difference in 'Covid' cases between six feet and three feet and Fauci changed his tune. Now three feet was okay. There is no difference between six feet and three *inches* when there is no 'virus' and they got away with six feet for psychological reasons for as long as they could. I hear journalists and others talk about 'unintended consequences' of lockdown. They are not *unintended* at all; they have been coldly-calculated for a specific outcome of human control and that's why super-psychopaths like Gates have called for them so vehemently. Super-psychopath psychologists have demanded them and psychopathic or clueless, spineless, politicians have gone along with them by 'following the science'. But it's not science at all. 'Science' is not what is; it's only what people can be manipulated to believe it is. The whole 'Covid' catastrophe is

founded on mind control. Three word or three statement mantras issued by the UK government are a well-known mind control technique and so we've had 'Stay home/protect the NHS/save lives', 'Stay alert/control the virus/save lives' and 'hands/face/space'. One of the most vocal proponents of extreme 'Covid' rules in the UK has been Professor Susan Michie, a member of the British Communist Party, who is not a medical professional. Michie is the director of the Centre for Behaviour Change at University College London. She is a *behavioural psychologist* and another filthy rich 'Marxist' who praised China's draconian lockdown. She was known by fellow students at Oxford University as 'Stalin's nanny' for her extreme Marxism. Michie is an influential member of the UK government's Scientific Advisory Group for Emergencies (SAGE) and behavioural manipulation groups which have dominated 'Covid' policy. She is a consultant adviser to the World Health Organization on 'Covid-19' and behaviour. Why the hell are lockdowns anything to do with her when they are claimed to be about health? Why does a behavioural psychologist from a group charged with changing the behaviour of the public want lockdown, human isolation and mandatory masks? Does that question really need an answer? Michie *absolutely* has to explain herself before a Nuremberg court when humanity takes back its world again and even more so when you see the consequences of masks that she demands are compulsory. This is a Michie classic:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Those words alone should carry a prison sentence when you ponder on the callous disregard for children involved and what a statement it makes about the mind and motivations of Susan Michie. What a lovely lady and what she said there encapsulates the mentality of the psychopaths behind the 'Covid' horror. Let us compare what Michie said with a countrywide study in Germany published at [researchsquare.com](https://www.researchsquare.com) involving 25,000 school children and 17,854 health complaints submitted by parents. Researchers

found that masks are harming children physically, psychologically, and behaviourally with 24 health issues associated with mask wearing. They include: shortness of breath (29.7%); dizziness (26.4%); increased headaches (53%); difficulty concentrating (50%); drowsiness or fatigue (37%); and malaise (42%). Nearly a third of children experienced more sleep issues than before and a quarter developed new fears. Researchers found health issues and other impairments in 68 percent of masked children covering their faces for an average of 4.5 hours a day. Hundreds of those taking part experienced accelerated respiration, tightness in the chest, weakness, and short-term impairment of consciousness. A reminder of what Michie said again:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Psychopaths in government and psychology now have children and young people – plus all the adults – wearing masks for hours on end while clueless teachers impose the will of the psychopaths on the young they should be protecting. What the hell are parents doing?

Cult lab rats

We have some schools already imposing on students microchipped buzzers that activate when they get 'too close' to their pals in the way they do with lab rats. How apt. To the Cult and its brain-dead servants our children *are* lab rats being conditioned to be unquestioning, dehumanised slaves for the rest of their lives. Children and young people are being weaned and frightened away from the most natural human instincts including closeness and touch. I have tracked in the books over the years how schools were banning pupils from greeting each other with a hug and the whole Cult-induced Me Too movement has terrified men and boys from a relaxed and natural interaction with female friends and work colleagues to the point where many men try never to be in a room

alone with a woman that's not their partner. Airhead celebrities have as always played their virtue-signalling part in making this happen with their gross exaggeration. For every monster like Harvey Weinstein there are at least tens of thousands of men that don't treat women like that; but everyone must be branded the same and policy changed for them as well as the monster. I am going to be using the word 'dehumanise' many times in this chapter because that is what the Cult is seeking to do and it goes very deep as we shall see. Don't let them kid you that social distancing is planned to end one day. That's not the idea. We are seeing more governments and companies funding and producing wearable gadgets to keep people apart and they would not be doing that if this was meant to be short-term. A tech start-up company backed by GCHQ, the British Intelligence and military surveillance headquarters, has created a social distancing wrist sensor that alerts people when they get too close to others. The CIA has also supported tech companies developing similar devices. The wearable sensor was developed by Tended, one of a number of start-up companies supported by GCHQ (see the CIA and DARPA). The device can be worn on the wrist or as a tag on the waistband and will vibrate whenever someone wearing the device breaches social distancing and gets anywhere near natural human contact. The company had a lucky break in that it was developing a distancing sensor when the 'Covid' hoax arrived which immediately provided a potentially enormous market. How fortunate. The government in big-time Cult-controlled Ontario in Canada is investing \$2.5 million in wearable contact tracing technology that 'will alert users if they may have been exposed to the Covid-19 in the workplace and will beep or vibrate if they are within six feet of another person'. Facedrive Inc., the technology company behind this, was founded in 2016 with funding from the Ontario Together Fund and obviously they, too, had a prophet on the board of directors. The human surveillance and control technology is called TraceSCAN and would be worn by the human cyborgs in places such as airports, workplaces, construction sites, care homes and ... *schools*.

I emphasise schools with children and young people the prime targets. You know what is planned for society as a whole if you keep your eyes on the schools. They have always been places where the state program the next generation of slaves to be its compliant worker-ants – or Woker-ants these days; but in the mist of the ‘Covid’ madness they have been transformed into mind laboratories on a scale never seen before. Teachers and head teachers are just as programmed as the kids – often more so. Children are kept apart from human interaction by walk lanes, classroom distancing, staggered meal times, masks, and the rolling-out of buzzer systems. Schools are now physically laid out as a laboratory maze for lab-rats. Lunatics at a school in Anchorage, Alaska, who should be prosecuted for child abuse, took away desks and forced children to kneel (know your place) on a mat for five hours a day while wearing a mask and using their chairs as a desk. How this was supposed to impact on a ‘virus’ only these clinically insane people can tell you and even then it would be clap-trap. The school banned recess (interaction), art classes (creativity), and physical exercise (getting body and mind moving out of inertia). Everyone behind this outrage should be in jail or better still a mental institution. The behavioural manipulators are all for this dystopian approach to schools. Professor Susan Michie, the mind-doctor and British Communist Party member, said it was wrong to say that schools were safe. They had to be made so by ‘distancing’, masks and ventilation (sitting all day in the cold). I must ask this lady round for dinner on a night I know I am going to be out and not back for weeks. She probably wouldn’t be able to make it, anyway, with all the visits to her own psychologist she must have block-booked.

Masking identity

I know how shocking it must be for you that a behaviour manipulator like Michie wants everyone to wear masks which have long been a feature of mind-control programs like the infamous MKUltra in the United States, but, there we are. We live and learn. I spent many years from 1996 to right across the millennium

researching mind control in detail on both sides of the Atlantic and elsewhere. I met a large number of mind-control survivors and many had been held captive in body and mind by MKUltra. MK stands for mind-control, but employs the German spelling in deference to the Nazis spirited out of Germany at the end of World War Two by Operation Paperclip in which the US authorities, with help from the Vatican, transported Nazi mind-controllers and engineers to America to continue their work. Many of them were behind the creation of NASA and they included Nazi scientist and SS officer Wernher von Braun who swapped designing V-2 rockets to bombard London with designing the Saturn V rockets that powered the NASA moon programme's Apollo craft. I think I may have mentioned that the Cult has no borders. Among Paperclip escapees was Josef Mengele, the Angel of Death in the Nazi concentration camps where he conducted mind and genetic experiments on children often using twins to provide a control twin to measure the impact of his 'work' on the other. If you want to observe the Cult mentality in all its extremes of evil then look into the life of Mengele. I have met many people who suffered mercilessly under Mengele in the United States where he operated under the name Dr Greene and became a stalwart of MKUltra programming and torture. Among his locations was the underground facility in the Mojave Desert in California called the China Lake Naval Weapons Station which is almost entirely below the surface. My books *The Biggest Secret*, *Children of the Matrix* and *The Perception Deception* have the detailed background to MKUltra.

The best-known MKUltra survivor is American Cathy O'Brien. I first met her and her late partner Mark Phillips at a conference in Colorado in 1996. Mark helped her escape and deprogram from decades of captivity in an offshoot of MKUltra known as Project Monarch in which 'sex slaves' were provided for the rich and famous including Father George Bush, Dick Cheney and the Clintons. Read Cathy and Mark's book *Trance-Formation of America* and if you are new to this you will be shocked to the core. I read it in 1996 shortly before, with the usual synchronicity of my life, I found

myself given a book table at the conference right next to hers. MKUltra never ended despite being very publicly exposed (only a small part of it) in the 1970s and continues in other guises. I am still in touch with Cathy. She contacted me during 2020 after masks became compulsory in many countries to tell me how they were used as part of MKUltra programming. I had been observing 'Covid regulations' and the relationship between authority and public for months. I saw techniques that I knew were employed on individuals in MKUltra being used on the global population. I had read many books and manuals on mind control including one called *Silent Weapons for Quiet Wars* which came to light in the 1980s and was a guide on how to perceptually program on a mass scale. 'Silent Weapons' refers to mind-control. I remembered a line from the manual as governments, medical authorities and law enforcement agencies have so obviously talked to – or rather at – the adult population since the 'Covid' hoax began as if they are children. The document said:

If a person is spoken to by a T.V. advertiser as if he were a twelve-year-old, then, due to suggestibility, he will, with a certain probability, respond or react to that suggestion with the uncritical response of a twelve-year-old and will reach in to his economic reservoir and deliver its energy to buy that product on impulse when he passes it in the store.

That's why authority has spoken to adults like children since all this began.

Why did Michael Jackson wear masks?

Every aspect of the 'Covid' narrative has mind-control as its central theme. Cathy O'Brien wrote an article for davidicke.com about the connection between masks and mind control. Her daughter Kelly who I first met in the 1990s was born while Cathy was still held captive in MKUltra. Kelly was forced to wear a mask as part of her programming from the age of *two* to dehumanise her, target her sense of individuality and reduce the amount of oxygen her brain and body received. *Bingo*. This is the real reason for compulsory

masks, why they have been enforced en masse, and why they seek to increase the number they demand you wear. First one, then two, with one disgraceful alleged 'doctor' recommending four which is nothing less than a death sentence. Where and how often they must be worn is being expanded for the purpose of mass mind control and damaging respiratory health which they can call 'Covid-19'. Canada's government headed by the man-child Justin Trudeau, says it's fine for children of two and older to wear masks. An insane 'study' in Italy involving just 47 children concluded there was no problem for babies as young as *four months* wearing them. Even after people were 'vaccinated' they were still told to wear masks by the criminal that is Anthony Fauci. Cathy wrote that mandating masks is allowing the authorities literally to control the air we breathe which is what was done in MKUltra. You might recall how the singer Michael Jackson wore masks and there is a reason for that. He was subjected to MKUltra mind control through Project Monarch and his psyche was scrambled by these simpletons. Cathy wrote:

In MKUltra Project Monarch mind control, Michael Jackson had to wear a mask to silence his voice so he could not reach out for help. Remember how he developed that whisper voice when he wasn't singing? Masks control the mind from the outside in, like the redefining of words is doing. By controlling what we can and cannot say for fear of being labeled racist or beaten, for example, it ultimately controls thought that drives our words and ultimately actions (or lack thereof).

Likewise, a mask muffles our speech so that we are not heard, which controls voice ... words ... mind. This is Mind Control. Masks are an obvious mind control device, and I am disturbed so many people are complying on a global scale. Masks depersonalize while making a person feel as though they have no voice. It is a barrier to others. People who would never choose to comply but are forced to wear a mask in order to keep their job, and ultimately their family fed, are compromised. They often feel shame and are subdued. People have stopped talking with each other while media controls the narrative.

The 'no voice' theme has often become literal with train passengers told not to speak to each other in case they pass on the 'virus', singing banned for the same reason and bonkers California officials telling people riding roller coasters that they cannot shout and scream. Cathy said she heard every day from healed MKUltra survivors who cannot wear a mask without flashing back on ways

their breathing was controlled – ‘from ball gags and penises to water boarding’. She said that through the years when she saw images of people in China wearing masks ‘due to pollution’ that it was really to control their oxygen levels. ‘I knew it was as much of a population control mechanism of depersonalisation as are burkas’, she said. Masks are another Chinese communist/fascist method of control that has been swept across the West as the West becomes China at lightning speed since we entered 2020.

Mask-19

There are other reasons for mandatory masks and these include destroying respiratory health to call it ‘Covid-19’ and stunting brain development of children and the young. Dr Margarite Griesz-Brisson MD, PhD, is a Consultant Neurologist and Neurophysiologist and the Founder and Medical Director of the London Neurology and Pain Clinic. Her CV goes down the street and round the corner. She is clearly someone who cares about people and won’t parrot the propaganda. Griesz-Brisson has a PhD in pharmacology, with special interest in neurotoxicology, environmental medicine, neuroregeneration and neuroplasticity (the way the brain can change in the light of information received). She went public in October, 2020, with a passionate warning about the effects of mask-wearing laws:

The reinhalation of our exhaled air will without a doubt create oxygen deficiency and a flooding of carbon dioxide. We know that the human brain is very sensitive to oxygen deprivation. There are nerve cells for example in the hippocampus that can’t be longer than 3 minutes without oxygen – they cannot survive. The acute warning symptoms are headaches, drowsiness, dizziness, issues in concentration, slowing down of reaction time – reactions of the cognitive system.

Oh, I know, let’s tell bus, truck and taxi drivers to wear them and people working machinery. How about pilots, doctors and police? Griesz-Brisson makes the important point that while the symptoms she mentions may fade as the body readjusts this does not alter the fact that people continue to operate in oxygen deficit with long list of

potential consequences. She said it was well known that neurodegenerative diseases take years or decades to develop. 'If today you forget your phone number, the breakdown in your brain would have already started 20 or 30 years ago.' She said degenerative processes in your brain are getting amplified as your oxygen deprivation continues through wearing a mask. Nerve cells in the brain are unable to divide themselves normally in these circumstances and lost nerve cells will no longer be regenerated. 'What is gone is gone.' Now consider that people like shop workers and *schoolchildren* are wearing masks for hours every day. What in the name of sanity is going to be happening to them? 'I do not wear a mask, I need my brain to think', Griesz-Brisson said, 'I want to have a clear head when I deal with my patients and not be in a carbon dioxide-induced anaesthesia'. If you are told to wear a mask anywhere ask the organisation, police, store, whatever, for their risk assessment on the dangers and negative effects on mind and body of enforcing mask-wearing. They won't have one because it has never been done not even by government. All of them must be subject to class-action lawsuits as the consequences come to light. They don't do mask risk assessments for an obvious reason. They know what the conclusions would be and independent scientific studies that *have* been done tell a horror story of consequences.

'Masks are criminal'

Dr Griesz-Brisson said that for children and adolescents, masks are an absolute no-no. They had an extremely active and adaptive immune system and their brain was incredibly active with so much to learn. 'The child's brain, or the youth's brain, is thirsting for oxygen.' The more metabolically active an organ was, the more oxygen it required; and in children and adolescents every organ was metabolically active. Griesz-Brisson said that to deprive a child's or adolescent's brain of oxygen, or to restrict it in any way, was not only dangerous to their health, it was absolutely criminal. 'Oxygen deficiency inhibits the development of the brain, and the damage that has taken place as a result CANNOT be reversed.' Mind

manipulators of MKUltra put masks on two-year-olds they wanted to neurologically rewire and you can see why. Griesz-Brisson said a child needs the brain to learn and the brain needs oxygen to function. 'We don't need a clinical study for that. This is simple, indisputable physiology.' Consciously and purposely induced oxygen deficiency was an absolutely deliberate health hazard, and an absolute medical contraindication which means that 'this drug, this therapy, this method or measure should not be used, and is not allowed to be used'. To coerce an entire population to use an absolute medical contraindication by force, she said, there had to be definite and serious reasons and the reasons must be presented to competent interdisciplinary and independent bodies to be verified and authorised. She had this warning of the consequences that were coming if mask wearing continued:

When, in ten years, dementia is going to increase exponentially, and the younger generations couldn't reach their god-given potential, it won't help to say 'we didn't need the masks'. I know how damaging oxygen deprivation is for the brain, cardiologists know how damaging it is for the heart, pulmonologists know how damaging it is for the lungs. Oxygen deprivation damages every single organ. Where are our health departments, our health insurance, our medical associations? It would have been their duty to be vehemently against the lockdown and to stop it and stop it from the very beginning.

Why do the medical boards issue punishments to doctors who give people exemptions? Does the person or the doctor seriously have to prove that oxygen deprivation harms people? What kind of medicine are our doctors and medical associations representing? Who is responsible for this crime? The ones who want to enforce it? The ones who let it happen and play along, or the ones who don't prevent it?

All of the organisations and people she mentions there either answer directly to the Cult or do whatever hierarchical levels above them tell them to do. The outcome of both is the same. 'It's not about masks, it's not about viruses, it's certainly not about your health', Griesz-Brisson said. 'It is about much, much more. I am not participating. I am not afraid.' They were taking our air to breathe and there was no unfounded medical exemption from face masks. Oxygen deprivation was dangerous for every single brain. It had to be the free decision of every human being whether they want to

wear a mask that was absolutely ineffective to protect themselves from a virus. She ended by rightly identifying where the responsibility lies for all this:

The imperative of the hour is personal responsibility. We are responsible for what we think, not the media. We are responsible for what we do, not our superiors. We are responsible for our health, not the World Health Organization. And we are responsible for what happens in our country, not the government.

Halle-bloody-lujah.

But surgeons wear masks, right?

Independent studies of mask-wearing have produced a long list of reports detailing mental, emotional and physical dangers. What a definition of insanity to see police officers imposing mask-wearing on the public which will cumulatively damage their health while the police themselves wear masks that will cumulatively damage *their* health. It's utter madness and both public and police do this because 'the government says so' – yes a government of brain-donor idiots like UK Health Secretary Matt Hancock reading the 'follow the science' scripts of psychopathic, lunatic psychologists. The response you get from Stockholm syndrome sufferers defending the very authorities that are destroying them and their families is that 'surgeons wear masks'. This is considered the game, set and match that they must work and don't cause oxygen deficit. Well, actually, scientific studies have shown that they *do* and oxygen levels are monitored in operating theatres to compensate. Surgeons wear masks to stop spittle and such like dropping into open wounds – not to stop 'viral particles' which are so miniscule they can only be seen through an electron microscope. Holes in the masks are significantly bigger than 'viral particles' and if you sneeze or cough they will breach the mask. I watched an incredibly disingenuous 'experiment' that claimed to prove that masks work in catching 'virus' material from the mouth and nose. They did this with a slow motion camera and the mask did block big stuff which stayed inside the mask and

against the face to be breathed in or cause infections on the face as we have seen with many children. 'Viral particles', however, would never have been picked up by the camera as they came through the mask when they are far too small to be seen. The 'experiment' was therefore disingenuous *and* useless.

Studies have concluded that wearing masks in operating theatres (and thus elsewhere) make no difference to preventing infection while the opposite is true with toxic shite building up in the mask and this had led to an explosion in tooth decay and gum disease dubbed by dentists 'mask mouth'. You might have seen the Internet video of a furious American doctor urging people to take off their masks after a four-year-old patient had been rushed to hospital the night before and nearly died with a lung infection that doctors sourced to mask wearing. A study in the journal *Cancer Discovery* found that inhalation of harmful microbes can contribute to advanced stage lung cancer in adults and long-term use of masks can help breed dangerous pathogens. Microbiologists have said frequent mask wearing creates a moist environment in which microbes can grow and proliferate before entering the lungs. The Canadian Agency for Drugs and Technologies in Health, or CADTH, a Canadian national organisation that provides research and analysis to healthcare decision-makers, said this as long ago as 2013 in a report entitled 'Use of Surgical Masks in the Operating Room: A Review of the Clinical Effectiveness and Guidelines'. It said:

- No evidence was found to support the use of surgical face masks to reduce the frequency of surgical site infections
- No evidence was found on the effectiveness of wearing surgical face masks to protect staff from infectious material in the operating room.
- Guidelines recommend the use of surgical face masks by staff in the operating room to protect both operating room staff and patients (despite the lack of evidence).

We were told that the world could go back to 'normal' with the arrival of the 'vaccines'. When they came, fraudulent as they are, the story changed as I knew that it would. We are in the midst of transforming 'normal', not going back to it. Mary Ramsay, head of immunisation at Public Health England, echoed the words of US criminal Anthony Fauci who said masks and other regulations must stay no matter if people are vaccinated. The Fauci idiot continued to wear two masks – different colours so both could be clearly seen – after he *claimed* to have been vaccinated. Senator Rand Paul told Fauci in one exchange that his double-masks were 'theatre' and he was right. It's all theatre. Mary Ramsay back-tracked on the vaccine-return-to-normal theme when she said the public may need to wear masks and social-distance for years despite the jabs. 'People have got used to those lower-level restrictions now, and [they] can live with them', she said telling us what the idea has been all along. 'The vaccine does not give you a pass, even if you have had it, you must continue to follow all the guidelines' said a Public Health England statement which reneged on what we had been told before and made having the 'vaccine' irrelevant to 'normality' even by the official story. Spain's fascist government trumped everyone by passing a law mandating the wearing of masks on the beach and even when swimming in the sea. The move would have devastated what's left of the Spanish tourist industry, posed potential breathing dangers to swimmers and had Northern European sunbathers walking around with their forehead brown and the rest of their face white as a sheet. The ruling was so crazy that it had to be retracted after pressure from public and tourist industry, but it confirmed where the Cult wants to go with masks and how clinically insane authority has become. The determination to make masks permanent and hide the serious dangers to body and mind can be seen in the censorship of scientist Professor Denis Rancourt by Bill Gates-funded academic publishing website ResearchGate over his papers exposing the dangers and uselessness of masks. Rancourt said:

ResearchGate today has permanently locked my account, which I have had since 2015. Their reasons graphically show the nature of their attack against democracy, and their corruption of

science ... By their obscene non-logic, a scientific review of science articles reporting on harms caused by face masks has a 'potential to cause harm'. No criticism of the psychological device (face masks) is tolerated, if the said criticism shows potential to influence public policy.

This is what happens in a fascist world.

Where are the 'greens' (again)?

Other dangers of wearing masks especially regularly relate to the inhalation of minute plastic fibres into the lungs and the deluge of discarded masks in the environment and oceans. Estimates predicted that more than 1.5 billion disposable masks will end up in the world's oceans every year polluting the water with tons of plastic and endangering marine wildlife. Studies project that humans are using 129 billion face masks each month worldwide – about three million a minute. Most are disposable and made from plastic, non-biodegradable microfibers that break down into smaller plastic particles that become widespread in ecosystems. They are littering cities, clogging sewage channels and turning up in bodies of water. I have written in other books about the immense amounts of microplastics from endless sources now being absorbed into the body. Rolf Halden, director of the Arizona State University (ASU) Biodesign Center for Environmental Health Engineering, was the senior researcher in a 2020 study that analysed 47 human tissue samples and found microplastics in all of them. 'We have detected these chemicals of plastics in every single organ that we have investigated', he said. I wrote in *The Answer* about the world being deluged with microplastics. A study by the Worldwide Fund for Nature (WWF) found that people are consuming on average every week some 2,000 tiny pieces of plastic mostly through water and also through marine life and the air. Every year humans are ingesting enough microplastics to fill a heaped dinner plate and in a life-time of 79 years it is enough to fill two large waste bins. Marco Lambertini, WWF International director general said: 'Not only are plastics polluting our oceans and waterways and killing marine life – it's in all of us and we can't escape consuming plastics,' American

geologists found tiny plastic fibres, beads and shards in rainwater samples collected from the remote slopes of the Rocky Mountain National Park near Denver, Colorado. Their report was headed: 'It is raining plastic.' Rachel Adams, senior lecturer in Biomedical Science at Cardiff Metropolitan University, said that among health consequences are internal inflammation and immune responses to a 'foreign body'. She further pointed out that microplastics become carriers of toxins including mercury, pesticides and dioxins (a known cause of cancer and reproductive and developmental problems). These toxins accumulate in the fatty tissues once they enter the body through microplastics. Now this is being compounded massively by people putting plastic on their face and throwing it away.

Workers exposed to polypropylene plastic fibres known as 'flock' have developed 'flock worker's lung' from inhaling small pieces of the flock fibres which can damage lung tissue, reduce breathing capacity and exacerbate other respiratory problems. *Now ...* commonly used surgical masks have three layers of melt-blown textiles made of ... polypropylene. We have billions of people putting these microplastics against their mouth, nose and face for hours at a time day after day in the form of masks. How does anyone think that will work out? I mean – what could possibly go wrong? We posted a number of scientific studies on this at davidicke.com, but when I went back to them as I was writing this book the links to the science research website where they were hosted were dead. Anything that challenges the official narrative in any way is either censored or vilified. The official narrative is so unsupportable by the evidence that only deleting the truth can protect it. A study by Chinese scientists still survived – with the usual twist which it why it was still active, I guess. Yes, they found that virtually all the masks they tested increased the daily intake of microplastic fibres, but people should still wear them because the danger from the 'virus' was worse said the crazy 'team' from the Institute of Hydrobiology in Wuhan. Scientists first discovered microplastics in lung tissue of some patients who died of lung cancer

in the 1990s. Subsequent studies have confirmed the potential health damage with the plastic degrading slowly and remaining in the lungs to accumulate in volume. Wuhan researchers used a machine simulating human breathing to establish that masks shed up to nearly 4,000 microplastic fibres in a month with reused masks producing more. Scientists said some masks are laced with toxic chemicals and a variety of compounds seriously restricted for both health and environmental reasons. They include cobalt (used in blue dye) and formaldehyde known to cause watery eyes, burning sensations in the eyes, nose, and throat, plus coughing, wheezing and nausea. No – that must be ‘Covid-19’.

Mask ‘worms’

There is another and potentially even more sinister content of masks. Mostly new masks of different makes filmed under a microscope around the world have been found to contain strange black fibres or ‘worms’ that appear to move or ‘crawl’ by themselves and react to heat and water. The nearest I have seen to them are the self-replicating fibres that are pulled out through the skin of those suffering from Morgellons disease which has been connected to the phenomena of ‘chemtrails’ which I will bring into the story later on. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. Black ‘worm’ fibres in masks have that kind of feel to them and there is a nanotechnology technique called ‘worm micelles’ which carry and release drugs or anything else you want to deliver to the body. For sure the suppression of humanity by mind altering drugs is the Cult agenda big time and the more excuses they can find to gain access to the body the more opportunities there are to make that happen whether through ‘vaccines’ or masks pushed against the mouth and nose for hours on end.

So let us summarise the pros and cons of masks:

Against masks: Breathing in your own carbon dioxide; depriving the body and brain of sufficient oxygen; build-up of toxins in the mask that can be breathed into the lungs and cause rashes on the face and 'mask-mouth'; breathing microplastic fibres and toxic chemicals into the lungs; dehumanisation and deleting individualisation by literally making people faceless; destroying human emotional interaction through facial expression and deleting parental connection with their babies which look for guidance to their facial expression.

For masks: They don't protect you from a 'virus' that doesn't exist and even if it did 'viral' particles are so minute they are smaller than the holes in the mask.

Governments, police, supermarkets, businesses, transport companies, and all the rest who seek to impose masks have done no risk assessment on their consequences for health and psychology and are now open to group lawsuits when the impact becomes clear with a cumulative epidemic of respiratory and other disease. Authorities will try to exploit these effects and hide the real cause by dubbing them 'Covid-19'. Can you imagine setting out to force the population to wear health-destroying masks without doing any assessment of the risks? It is criminal and it is evil, but then how many people targeted in this way, who see their children told to wear them all day at school, have asked for a risk assessment? Billions can't be imposed upon by the few unless the billions allow it. Oh, yes, with just a tinge of irony, 85 percent of all masks made worldwide come from *China*.

Wash your hands in toxic shite

'Covid' rules include the use of toxic sanitisers and again the health consequences of constantly applying toxins to be absorbed through the skin is obvious to any level of Renegade Mind. America's Food and Drug Administration (FDA) said that sanitisers are drugs and issued a warning about 75 dangerous brands which contain

methanol used in antifreeze and can cause death, kidney damage and blindness. The FDA circulated the following warning even for those brands that it claims to be safe:

Store hand sanitizer out of the reach of pets and children, and children should use it only with adult supervision. Do not drink hand sanitizer. This is particularly important for young children, especially toddlers, who may be attracted by the pleasant smell or brightly colored bottles of hand sanitizer.

Drinking even a small amount of hand sanitizer can cause alcohol poisoning in children. (However, there is no need to be concerned if your children eat with or lick their hands after using hand sanitizer.) During this coronavirus pandemic, poison control centers have had an increase in calls about accidental ingestion of hand sanitizer, so it is important that adults monitor young children's use.

Do not allow pets to swallow hand sanitizer. If you think your pet has eaten something potentially dangerous, call your veterinarian or a pet poison control center right away. Hand sanitizer is flammable and should be stored away from heat and flames. When using hand sanitizer, rub your hands until they feel completely dry before performing activities that may involve heat, sparks, static electricity, or open flames.

There you go, perfectly safe, then, and that's without even a mention of the toxins absorbed through the skin. Come on kids – sanitise your hands everywhere you go. It will save you from the 'virus'. Put all these elements together of the 'Covid' normal and see how much health and psychology is being cumulatively damaged, even devastated, to 'protect your health'. Makes sense, right? They are only imposing these things because they care, right? *Right?*

Submitting to insanity

Psychological reframing of the population goes very deep and is done in many less obvious ways. I hear people say how contradictory and crazy 'Covid' rules are and how they are ever changing. This is explained away by dismissing those involved as idiots. It is a big mistake. The Cult is delighted if its cold calculation is perceived as incompetence and idiocy when it is anything but. Oh, yes, there are idiots within the system – lots of them – but they are *administering* the Cult agenda, mostly unknowingly. They are not deciding and dictating it. The bulwark against tyranny is self-

respect, always has been, always will be. It is self-respect that has broken every tyranny in history. By its very nature self-respect will not bow to oppression and its perpetrators. There is so little self-respect that it's always the few that overturn dictators. Many may eventually follow, but the few with the iron spines (self-respect) kick it off and generate the momentum. The Cult targets self-respect in the knowledge that once this has gone only submission remains. Crazy, contradictory, ever-changing 'Covid' rules are systematically applied by psychologists to delete self-respect. They *want* you to see that the rules make no sense. It is one thing to decide to do something when *you* have made the choice based on evidence and logic. You still retain your self-respect. It is quite another when you can see what you are being told to do is insane, ridiculous and makes no sense, and *yet you still do it*. Your self-respect is extinguished and this has been happening as ever more obviously stupid and nonsensical things have been demanded and the great majority have complied even when they can see they are stupid and nonsensical.

People walk around in face-nappies knowing they are damaging their health and make no difference to a 'virus'. They do it in fear of not doing it. I know it's daft, but I'll do it anyway. When that happens something dies inside of you and submissive reframing has begun. Next there's a need to hide from yourself that you have conceded your self-respect and you convince yourself that you have not really submitted to fear and intimidation. You begin to believe that you are complying with craziness because it's the right thing to do. When first you concede your self-respect of $2+2 = 4$ to $2+2 = 5$ you *know* you are compromising your self-respect. Gradually to avoid facing that fact you begin to *believe* that $2+2=5$. You have been reframed and I have been watching this process happening in the human psyche on an industrial scale. The Cult is working to break your spirit and one of its major tools in that war is humiliation. I read how former American soldier Bradley Manning (later Chelsea Manning after a sex-change) was treated after being jailed for supplying WikiLeaks with documents exposing the enormity of

government and elite mendacity. Manning was isolated in solitary confinement for eight months, put under 24-hour surveillance, forced to hand over clothing before going to bed, and stand naked for every roll call. This is systematic humiliation. The introduction of anal swab 'Covid' tests in China has been done for the same reason to delete self-respect and induce compliant submission. Anal swabs are mandatory for incoming passengers in parts of China and American diplomats have said they were forced to undergo the indignity which would have been calculated humiliation by the Cult-owned Chinese government that has America in its sights.

Government-people: An abusive relationship

Spirit-breaking psychological techniques include giving people hope and apparent respite from tyranny only to take it away again. This happened in the UK during Christmas, 2020, when the psychopsychologists and their political lackeys announced an easing of restrictions over the holiday only to reimpose them almost immediately on the basis of yet another lie. There is a big psychological difference between getting used to oppression and being given hope of relief only to have that dashed. Psychologists know this and we have seen the technique used repeatedly. Then there is traumatising people before you introduce more extreme regulations that require compliance. A perfect case was the announcement by the dark and sinister Whitty and Vallance in the UK that 'new data' predicted that 4,000 could die every day over the winter of 2020/2021 if we did not lockdown again. I think they call it lying and after traumatising people with that claim out came Jackboot Johnson the next day with new curbs on human freedom. Psychologists know that a frightened and traumatised mind becomes suggestable to submission and behaviour reframing. Underpinning all this has been to make people fearful and suspicious of each other and see themselves as a potential danger to others. In league with deleted self-respect you have the perfect psychological recipe for self-loathing. The relationship between authority and public is now demonstrably the same as that of

subservience to an abusive partner. These are signs of an abusive relationship explained by psychologist Leslie Becker-Phelps:

Psychological and emotional abuse: Undermining a partner's self-worth with verbal attacks, name-calling, and belittling. Humiliating the partner in public, unjustly accusing them of having an affair, or interrogating them about their every behavior. Keeping partner confused or off balance by saying they were just kidding or blaming the partner for 'making' them act this way ... Feigning in public that they care while turning against them in private. This leads to victims frequently feeling confused, incompetent, unworthy, hopeless, and chronically self-doubting. [Apply these techniques to how governments have treated the population since New Year, 2020, and the parallels are obvious.]

Physical abuse: The abuser might physically harm their partner in a range of ways, such as grabbing, hitting, punching, or shoving them. They might throw objects at them or harm them with a weapon. [Observe the physical harm imposed by masks, lockdown, and so on.]

Threats and intimidation: One way abusers keep their partners in line is by instilling fear. They might be verbally threatening, or give threatening looks or gestures. Abusers often make it known that they are tracking their partner's every move. They might destroy their partner's possessions, threaten to harm them, or threaten to harm their family members. Not surprisingly, victims of this abuse often feel anxiety, fear, and panic. [No words necessary.]

Isolation: Abusers often limit their partner's activities, forbidding them to talk or interact with friends or family. They might limit access to a car or even turn off their phone. All of this might be done by physically holding them against their will, but is often accomplished through psychological abuse and intimidation. The more isolated a person feels, the fewer resources they have to help gain perspective on their situation and to escape from it. [No words necessary.]

Economic abuse: Abusers often make their partners beholden to them for money by controlling access to funds of any kind. They might prevent their partner from getting a job or withhold access to money they earn from a job. This creates financial dependency that makes leaving the relationship very difficult. [See destruction of livelihoods and the proposed meagre 'guaranteed income' so long as you do whatever you are told.]

Using children: An abuser might disparage their partner's parenting skills, tell their children lies about their partner, threaten to take custody of their children, or threaten to harm their children. These tactics instil fear and often elicit compliance. [See reframed social service mafia and how children are being mercilessly abused by the state over 'Covid' while their parents look on too frightened to do anything.]

A further recurring trait in an abusive relationship is the abused blaming themselves for their abuse and making excuses for the abuser. We have the public blaming each other for lockdown abuse by government and many making excuses for the government while attacking those who challenge the government. How often we have heard authorities say that rules are being imposed or reimposed only because people have refused to 'behave' and follow the rules. We don't want to do it – it's *you*.

Renegade Minds are an antidote to all of these things. They will never concede their self-respect no matter what the circumstances. Even when apparent humiliation is heaped upon them they laugh in its face and reflect back the humiliation on the abuser where it belongs. Renegade Minds will never wear masks they know are only imposed to humiliate, suppress and damage both physically and psychologically. Consequences will take care of themselves and they will never break their spirit or cause them to concede to tyranny. UK newspaper columnist Peter Hitchens was one of the few in the mainstream media to speak out against lockdowns and forced vaccinations. He then announced he had taken the jab. He wanted to see family members abroad and he believed vaccine passports were inevitable even though they had not yet been introduced. Hitchens

has a questioning and critical mind, but not a Renegade one. If he had no amount of pressure would have made him concede. Hitchens excused his action by saying that the battle has been lost. Renegade Minds never accept defeat when freedom is at stake and even if they are the last one standing the self-respect of not submitting to tyranny is more important than any outcome or any consequence.

That's why Renegade Minds are the only minds that ever changed anything worth changing.

CHAPTER EIGHT

'Reframing' insanity

Insanity is relative. It depends on who has who locked in what cage
Ray Bradbury

Reframing' a mind means simply to change its perception and behaviour. This can be done subconsciously to such an extent that subjects have no idea they have been 'reframed' while to any observer changes in behaviour and attitudes are obvious.

Human society is being reframed on a ginormous scale since the start of 2020 and here we have the reason why psychologists rather than doctors have been calling the shots. Ask most people who have succumbed to 'Covid' reframing if they have changed and most will say 'no'; but they *have* and fundamentally. The Cult's long-game has been preparing for these times since way back and crucial to that has been to prepare both population and officialdom mentally and emotionally. To use the mind-control parlance they had to reframe the population with a mentality that would submit to fascism and reframe those in government and law enforcement to impose fascism or at least go along with it. The result has been the fact-deleted mindlessness of 'Wokeness' and officialdom that has either enthusiastically or unquestioningly imposed global tyranny demanded by reframed politicians on behalf of psychopathic and deeply evil cultists. 'Cognitive reframing' identifies and challenges the way someone sees the world in the form of situations, experiences and emotions and then restructures those perceptions to view the same set of circumstances in a different way. This can have

benefits if the attitudes are personally destructive while on the other side it has the potential for individual and collective mind control which the subject has no idea has even happened.

Cognitive therapy was developed in the 1960s by Aaron T. Beck who was born in Rhode Island in 1921 as the son of Jewish immigrants from the Ukraine. He became interested in the techniques as a treatment for depression. Beck's daughter Judith S. Beck is prominent in the same field and they founded the Beck Institute for Cognitive Behavior Therapy in Philadelphia in 1994. Cognitive reframing, however, began to be used worldwide by those with a very dark agenda. The Cult reframes politicians to change their attitudes and actions until they are completely at odds with what they once appeared to stand for. The same has been happening to government administrators at all levels, law enforcement, military and the human population. Cultists love mind control for two main reasons: It allows them to control what people think, do and say to secure agenda advancement and, by definition, it calms their legendary insecurity and fear of the unexpected. I have studied mind control since the time I travelled America in 1996. I may have been talking to next to no one in terms of an audience in those years, but my goodness did I gather a phenomenal amount of information and knowledge about so many things including the techniques of mind control. I have described this in detail in other books going back to *The Biggest Secret* in 1998. I met a very large number of people recovering from MKUltra and its offshoots and successors and I began to see how these same techniques were being used on the population in general. This was never more obvious than since the 'Covid' hoax began.

Reframing the enforcers

I have observed over the last two decades and more the very clear transformation in the dynamic between the police, officialdom and the public. I tracked this in the books as the relationship mutated from one of serving the public to seeing them as almost the enemy and certainly a lower caste. There has always been a class divide

based on income and always been some psychopathic, corrupt, and big-I-am police officers. This was different. Wholesale change was unfolding in the collective dynamic; it was less about money and far more about position and perceived power. An us-and-them was emerging. Noses were lifted skyward by government administration and law enforcement and their attitude to the public they were *supposed* to be serving changed to one of increasing contempt, superiority and control. The transformation was so clear and widespread that it had to be planned. Collective attitudes and dynamics do not change naturally and organically that quickly on that scale. I then came across an organisation in Britain called Common Purpose created in the late 1980s by Julia Middleton who would work in the office of Deputy Prime Minister John Prescott during the long and disastrous premiership of war criminal Tony Blair. When Blair speaks the Cult is speaking and the man should have been in jail a long time ago. Common Purpose proclaims itself to be one of the biggest 'leadership development' organisations in the world while functioning as a *charity* with all the financial benefits which come from that. It hosts 'leadership development' courses and programmes all over the world and claims to have 'brought together' what it calls 'leaders' from more than 100 countries on six continents. The modus operandi of Common Purpose can be compared with the work of the UK government's reframing network that includes the Behavioural Insights Team 'nudge unit' and 'Covid' reframing specialists at SPI-B. WikiLeaks described Common Purpose long ago as 'a hidden virus in our government and schools' which is unknown to the general public: 'It recruits and trains "leaders" to be loyal to the directives of Common Purpose and the EU, instead of to their own departments, which they then undermine or subvert, the NHS [National Health Service] being an example.' This is a vital point to understand the 'Covid' hoax. The NHS, and its equivalent around the world, has been utterly reframed in terms of administrators and much of the medical personnel with the transformation underpinned by recruitment policies. The outcome has been the criminal and psychopathic behaviour of the

NHS over 'Covid' and we have seen the same in every other major country. WikiLeaks said Common Purpose trainees are 'learning to rule without regard to democracy' and to usher in a police state (current events explained). Common Purpose operated like a 'glue' and had members in the NHS, BBC, police, legal profession, church, many of Britain's 7,000 quangos, local councils, the Civil Service, government ministries and Parliament, and controlled many RDA's (Regional Development Agencies). Here we have one answer for how and why British institutions and their like in other countries have changed so negatively in relation to the public. This further explains how and why the beyond-disgraceful reframed BBC has become a propaganda arm of 'Covid' fascism. They are all part of a network pursuing the same goal.

By 2019 Common Purpose was quoting a figure of 85,000 'leaders' that had attended its programmes. These 'students' of all ages are known as Common Purpose 'graduates' and they consist of government, state and local government officials and administrators, police chiefs and officers, and a whole range of others operating within the national, local and global establishment. Cressida Dick, Commissioner of the London Metropolitan Police, is the Common Purpose graduate who was the 'Gold Commander' that oversaw what can only be described as the murder of Brazilian electrician Jean Charles de Menezes in 2005. He was held down by psychopathic police and shot seven times in the head by a psychopathic lunatic after being mistaken for a terrorist when he was just a bloke going about his day. Dick authorised officers to pursue and keep surveillance on de Menezes and ordered that he be stopped from entering the underground train system. Police psychopaths took her at her word clearly. She was 'disciplined' for this outrage by being *promoted* – eventually to the top of the 'Met' police where she has been a disaster. Many Chief Constables controlling the police in different parts of the UK are and have been Common Purpose graduates. I have heard the 'graduate' network described as a sort of Mafia or secret society operating within the fabric of government at all levels pursuing a collective policy

ingrained at Common Purpose training events. Founder Julia Middleton herself has said:

Locally and internationally, Common Purpose graduates will be 'lighting small fires' to create change in their organisations and communities ... The Common Purpose effect is best illustrated by the many stories of small changes brought about by leaders, who themselves have changed.

A Common Purpose mission statement declared:

Common Purpose aims to improve the way society works by expanding the vision, decision-making ability and influence of all kinds of leaders. The organisation runs a variety of educational programmes for leaders of all ages, backgrounds and sectors, in order to provide them with the inspirational, information and opportunities they need to change the world.

Yes, but into what? Since 2020 the answer has become clear.

NLP and the Delphi technique

Common Purpose would seem to be a perfect name or would common programming be better? One of the foundation methods of reaching 'consensus' (group think) is by setting the agenda theme and then encouraging, cajoling or pressuring everyone to agree a 'consensus' in line with the core theme promoted by Common Purpose. The methodology involves the 'Delphi technique', or an adaption of it, in which opinions are expressed that are summarised by a 'facilitator or change agent' at each stage. Participants are 'encouraged' to modify their views in the light of what others have said. Stage by stage the former individual opinions are merged into group consensus which just happens to be what Common Purpose wants them to believe. A key part of this is to marginalise anyone refusing to concede to group think and turn the group against them to apply pressure to conform. We are seeing this very technique used on the general population to make 'Covid' group-thinkers hostile to those who have seen through the bullshit. People can be reframed by using perception manipulation methods such as Neuro-Linguistic Programming (NLP) in which you change perception with the use of

carefully constructed language. An NLP website described the technique this way:

... A method of influencing brain behaviour (the 'neuro' part of the phrase) through the use of language (the 'linguistic' part) and other types of communication to enable a person to 'recode' the way the brain responds to stimuli (that's the 'programming') and manifest new and better behaviours. Neuro-Linguistic Programming often incorporates hypnosis and self-hypnosis to help achieve the change (or 'programming') that is wanted.

British alternative media operation UKColumn has done very detailed research into Common Purpose over a long period. I quoted co-founder and former naval officer Brian Gerrish in my book *Remember Who You Are*, published in 2011, as saying the following years before current times:

It is interesting that many of the mothers who have had children taken by the State speak of the Social Services people being icily cool, emotionless and, as two ladies said in slightly different words, '... like little robots'. We know that NLP is cumulative, so people can be given small imperceptible doses of NLP in a course here, another in a few months, next year etc. In this way, major changes are accrued in their personality, but the day by day change is almost unnoticeable.

In these and other ways 'graduates' have had their perceptions uniformly reframed and they return to their roles in the institutions of government, law enforcement, legal profession, military, 'education', the UK National Health Service and the whole swathe of the establishment structure to pursue a common agenda preparing for the 'post-industrial', 'post-democratic' society. I say 'preparing' but we are now there. 'Post-industrial' is code for the Great Reset and 'post-democratic' is 'Covid' fascism. UKColumn has spoken to partners of those who have attended Common Purpose 'training'. They have described how personalities and attitudes of 'graduates' changed very noticeably for the worse by the time they had completed the course. They had been 'reframed' and told they are the 'leaders' – the special ones – who know better than the population. There has also been the very demonstrable recruitment of psychopaths and narcissists into government administration at all

levels and law enforcement. If you want psychopathy hire psychopaths and you get a simple cause and effect. If you want administrators, police officers and 'leaders' to perceive the public as lesser beings who don't matter then employ narcissists. These personalities are identified using 'psychometrics' that identifies knowledge, abilities, attitudes and personality traits, mostly through carefully-designed questionnaires and tests. As this policy has passed through the decades we have had power-crazy, power-trippers appointed into law enforcement, security and government administration in preparation for current times and the dynamic between public and law enforcement/officialdom has been transformed. UKColumn's Brian Gerrish said of the narcissistic personality:

Their love of themselves and power automatically means that they will crush others who get in their way. I received a major piece of the puzzle when a friend pointed out that when they made public officials re-apply for their own jobs several years ago they were also required to do psychometric tests. This was undoubtedly the start of the screening process to get 'their' sort of people in post.

How obvious that has been since 2020 although it was clear what was happening long before if people paid attention to the changing public-establishment dynamic.

Change agents

At the centre of events in 'Covid' Britain is the National Health Service (NHS) which has behaved disgracefully in slavishly following the Cult agenda. The NHS management structure is awash with Common Purpose graduates or 'change agents' working to a common cause. Helen Bevan, a Chief of Service Transformation at the NHS Institute for Innovation and Improvement, co-authored a document called 'Towards a million change agents, a review of the social movements literature: implications for large scale change in the NHS'. The document compared a project management approach to that of change and social movements where 'people change

themselves and each other – peer to peer’. Two definitions given for a ‘social movement’ were:

A group of people who consciously attempt to build a radically new social order; involves people of a broad range of social backgrounds; and deploys politically confrontational and socially disruptive tactics – Cyrus Zirakzadeh 1997

Collective challenges, based on common purposes and social solidarities, in sustained interaction with elites, opponents, and authorities – Sidney Tarrow 1994

Helen Bevan wrote another NHS document in which she defined ‘framing’ as ‘the process by which leaders construct, articulate and put across their message in a powerful and compelling way in order to win people to their cause and call them to action’. I think I could come up with another definition that would be rather more accurate. The National Health Service and institutions of Britain and the wider world have been taken over by reframed ‘change agents’ and that includes everything from the United Nations to national governments, local councils and social services which have been kidnapping children from loving parents on an extraordinary and gathering scale on the road to the end of parenthood altogether. Children from loving homes are stolen and kidnapped by the state and put into the ‘care’ (inversion) of the local authority through council homes, foster parents and forced adoption. At the same time children are allowed to be abused without response while many are under council ‘care’. UKColumn highlighted the Common Purpose connection between South Yorkshire Police and Rotherham council officers in the case of the scandal in that area of the sexual exploitation of children to which the authorities turned not one blind eye, but both:

We were alarmed to discover that the Chief Executive, the Strategic Director of Children and Young People's Services, the Manager for the Local Strategic Partnership, the Community Cohesion Manager, the Cabinet Member for Cohesion, the Chief Constable and his predecessor had all attended Leadership training courses provided by the pseudo-charity Common Purpose.

Once 'change agents' have secured positions of hire and fire within any organisation things start to move very quickly. Personnel are then hired and fired on the basis of whether they will work towards the agenda the change agent represents. If they do they are rapidly promoted even though they may be incompetent. Those more qualified and skilled who are pre-Common Purpose 'old school' see their careers stall and even disappear. This has been happening for decades in every institution of state, police, 'health' and social services and all of them have been transformed as a result in their attitudes to their jobs and the public. Medical professions, including nursing, which were once vocations for the caring now employ many cold, callous and couldn't give a shit personality types. The UKColumn investigation concluded:

By blurring the boundaries between people, professions, public and private sectors, responsibility and accountability, Common Purpose encourages 'graduates' to believe that as new selected leaders, they can work together, outside of the established political and social structures, to achieve a paradigm shift or CHANGE – so called 'Leading Beyond Authority'. In doing so, the allegiance of the individual becomes 'reframed' on CP colleagues and their NETWORK.

Reframing the Face-Nappies

Nowhere has this process been more obvious than in the police where recruitment of psychopaths and development of unquestioning mind-controlled group-thinkers have transformed law enforcement into a politically-correct 'Woke' joke and a travesty of what should be public service. Today they wear their face-nappies like good little gofers and enforce 'Covid' rules which are fascism under another name. Alongside the specifically-recruited psychopaths we have software minds incapable of free thought. Brian Gerrish again:

An example is the policeman who would not get on a bike for a press photo because he had not done the cycling proficiency course. Normal people say this is political correctness gone mad. Nothing could be further from the truth. The policeman has been reframed, and in his reality it is perfect common sense not to get on the bike 'because he hasn't done the cycling course'.

Another example of this is where the police would not rescue a boy from a pond until they had taken advice from above on the 'risk assessment'. A normal person would have arrived, perhaps thought of the risk for a moment, and dived in. To the police now 'reframed', they followed 'normal' procedure.

There are shocking cases of reframed ambulance crews doing the same. Sheer unthinking stupidity of London Face-Nappies headed by Common Purpose graduate Cressida Dick can be seen in their behaviour at a vigil in March, 2021, for a murdered woman, Sarah Everard. A police officer had been charged with the crime. Anyone with a brain would have left the vigil alone in the circumstances. Instead they 'manhandled' women to stop them breaking 'Covid rules' to betray classic reframing. Minds in the thrall of perception control have no capacity for seeing a situation on its merits and acting accordingly. 'Rules is rules' is their only mind-set. My father used to say that rules and regulations are for the guidance of the intelligent and the blind obedience of the idiot. Most of the intelligent, decent, coppers have gone leaving only the other kind and a few old school for whom the job must be a daily nightmare. The combination of psychopaths and rule-book software minds has been clearly on public display in the 'Covid' era with automaton robots in uniform imposing fascistic 'Covid' regulations on the population without any personal initiative or judging situations on their merits. There are thousands of examples around the world, but I'll make my point with the infamous Derbyshire police in the English East Midlands – the ones who think pouring dye into beauty spots and using drones to track people walking in the countryside away from anyone is called 'policing'. To them there are rules decreed by the government which they have to enforce and in their bewildered state a group gathering in a closed space and someone walking alone in the countryside are the same thing. It is beyond idiocy and enters the realm of clinical insanity.

Police officers in Derbyshire said they were 'horrified' – *horrified* – to find 15 to 20 'irresponsible' kids playing a football match at a closed leisure centre 'in breach of coronavirus restrictions'. When they saw the police the kids ran away leaving their belongings behind and the reframed men and women of Derbyshire police were seeking to establish their identities with a view to fining their parents. The most natural thing for youngsters to do – kicking a ball about – is turned into a criminal activity and enforced by the moronic software programs of Derbyshire police. You find the same mentality in every country. These barely conscious 'horrified' officers said they had to take action because 'we need to ensure these rules are being followed' and 'it is of the utmost importance that you ensure your children are following the rules and regulations for Covid-19'. Had any of them done ten seconds of research to see if this parroting of their masters' script could be supported by any evidence? Nope. Reframed people don't think – others think for them and that's the whole idea of reframing. I have seen police officers one after the other repeating without question word for word what officialdom tells them just as I have seen great swathes of the public doing the same. Ask either for 'their' opinion and out spews what they have been told to think by the official narrative. Police and public may seem to be in different groups, but their mentality is the same. Most people do whatever they are told in fear not doing so or because they believe what officialdom tells them; almost the entirety of the police do what they are told for the same reason. Ultimately it's the tiny inner core of the global Cult that's telling both what to do.

So Derbyshire police were 'horrified'. Oh, really? Why did they think those kids were playing football? It was to relieve the psychological consequences of lockdown and being denied human contact with their friends and interaction, touch and discourse vital to human psychological health. Being denied this month after month has dismantled the psyche of many children and young people as depression and suicide have exploded. Were Derbyshire police *horrified by that*? Are you kidding? Reframed people don't have those

mental and emotional processes that can see how the impact on the psychological health of youngsters is far more dangerous than any 'virus' even if you take the mendacious official figures to be true. The reframed are told (programmed) how to act and so they do. The Derbyshire Chief Constable in the first period of lockdown when the black dye and drones nonsense was going on was Peter Goodman. He was the man who severed the connection between his force and the Derbyshire Constabulary *Male Voice* Choir when he decided that it was not inclusive enough to allow women to join. The fact it was a male voice choir making a particular sound produced by male voices seemed to elude a guy who terrifyingly ran policing in Derbyshire. He retired weeks after his force was condemned as disgraceful by former Supreme Court Justice Jonathan Sumption for their behaviour over extreme lockdown impositions. Goodman was replaced by his deputy Rachel Swann who was in charge when her officers were 'horrificed'. The police statement over the boys committing the hanging-offence of playing football included the line about the youngsters being 'irresponsible in the times we are all living through' missing the point that the real relevance of the 'times we are all living through' is the imposition of fascism enforced by psychopaths and reframed minds of police officers playing such a vital part in establishing the fascist tyranny that their own children and grandchildren will have to live in their entire lives. As a definition of insanity that is hard to beat although it might be run close by imposing masks on people that can have a serious effect on their health while wearing a face nappy all day themselves. Once again public and police do it for the same reason – the authorities tell them to and who are they to have the self-respect to say no?

Wokers in uniform

How reframed do you have to be to arrest a *six-year-old* and take him to court for *picking a flower* while waiting for a bus? Brain dead police and officialdom did just that in North Carolina where criminal proceedings happen regularly for children under nine. Attorney Julie Boyer gave the six-year-old crayons and a colouring book

during the 'flower' hearing while the 'adults' decided his fate. County Chief District Court Judge Jay Corpening asked: 'Should a child that believes in Santa Claus, the Easter Bunny and the tooth fairy be making life-altering decisions?' Well, of course not, but common sense has no meaning when you have a common purpose and a reframed mind. Treating children in this way, and police operating in American schools, is all part of the psychological preparation for children to accept a police state as normal all their adult lives. The same goes for all the cameras and biometric tracking technology in schools. Police training is focused on reframing them as snowflake Wokers and this is happening in the military. Pentagon top brass said that 'training sessions on extremism' were needed for troops who asked why they were so focused on the Capitol Building riot when Black Lives Matter riots were ignored. What's the difference between them some apparently and rightly asked. Actually, there is a difference. Five people died in the Capitol riot, only one through violence, and that was a police officer shooting an unarmed protestor. BLM riots killed at least 25 people and cost billions. Asking the question prompted the psychopaths and reframed minds that run the Pentagon to say that more 'education' (programming) was needed. Troop training is all based on psychological programming to make them fodder for the Cult – 'Military men are just dumb, stupid animals to be used as pawns in foreign policy' as Cult-to-his-DNA former Secretary of State Henry Kissinger famously said. Governments see the police in similar terms and it's time for those among them who can see this to defend the people and stop being enforcers of the Cult agenda upon the people.

The US military, like the country itself, is being targeted for destruction through a long list of Woke impositions. Cult-owned gaga 'President' Biden signed an executive order when he took office to allow taxpayer money to pay for transgender surgery for active military personnel and veterans. Are you a man soldier? No, I'm a LGBTQIA+ with a hint of Skoliosexual and Spectrasexual. Oh, good man. Bad choice of words you bigot. The Pentagon announced in March, 2021, the appointment of the first 'diversity and inclusion

officer' for US Special Forces. Richard Torres-Estrada arrived with the publication of a 'D&I Strategic Plan which will guide the enterprise-wide effort to institutionalize and sustain D&I'. If you think a Special Forces 'Strategic Plan' should have something to do with defending America you haven't been paying attention. Defending Woke is now the military's new role. Torres-Estrada has posted images comparing Donald Trump with Adolf Hitler and we can expect no bias from him as a representative of the supposedly non-political Pentagon. Cable news host Tucker Carlson said: 'The Pentagon is now the Yale faculty lounge but with cruise missiles.' Meanwhile Secretary of Defense Lloyd Austin, a board member of weapons-maker Raytheon with stock and compensation interests in October, 2020, worth \$1.4 million, said he was purging the military of the 'enemy within' – anyone who isn't Woke and supports Donald Trump. Austin refers to his targets as 'racist extremists' while in true Woke fashion being himself a racist extremist. Pentagon documents pledge to 'eradicate, eliminate and conquer all forms of racism, sexism and homophobia'. The definitions of these are decided by 'diversity and inclusion committees' peopled by those who see racism, sexism and homophobia in every situation and opinion. Woke (the Cult) is dismantling the US military and purging testosterone as China expands its military and gives its troops 'masculinity training'. How do we think that is going to end when this is all Cult coordinated? The US military, like the British military, is controlled by Woke and spineless top brass who just go along with it out of personal career interests.

'Woke' means fast asleep

Mind control and perception manipulation techniques used on individuals to create group-think have been unleashed on the global population in general. As a result many have no capacity to see the obvious fascist agenda being installed all around them or what 'Covid' is really all about. Their brains are firewalled like a computer system not to process certain concepts, thoughts and realisations that are bad for the Cult. The young are most targeted as the adults they

will be when the whole fascist global state is planned to be fully implemented. They need to be prepared for total compliance to eliminate all pushback from entire generations. The Cult has been pouring billions into taking complete control of 'education' from schools to universities via its operatives and corporations and not least Bill Gates as always. The plan has been to transform 'education' institutions into programming centres for the mentality of 'Woke'. James McConnell, professor of psychology at the University of Michigan, wrote in *Psychology Today* in 1970:

The day has come when we can combine sensory deprivation with drugs, hypnosis, and astute manipulation of reward and punishment, to gain almost absolute control over an individual's behaviour. It should then be possible to achieve a very rapid and highly effective type of brainwashing that would allow us to make dramatic changes in a person's behaviour and personality ...

... We should reshape society so that we all would be trained from birth to want to do what society wants us to do. We have the techniques to do it... no-one owns his own personality you acquired, and there's no reason to believe you should have the right to refuse to acquire a new personality if your old one is anti-social.

This was the potential for mass brainwashing in 1970 and the mentality there displayed captures the arrogant psychopathy that drives it forward. I emphasise that not all young people have succumbed to Woke programming and those that haven't are incredibly impressive people given that today's young are the most perceptually-targeted generations in history with all the technology now involved. Vast swathes of the young generations, however, have fallen into the spell – and that's what it is – of Woke. The Woke mentality and perceptual program is founded on *inversion* and you will appreciate later why that is so significant. Everything with Woke is inverted and the opposite of what it is claimed to be. Woke was a term used in African-American culture from the 1900s and referred to an awareness of social and racial justice. This is not the meaning of the modern version or 'New Woke' as I call it in *The Answer*. Oh, no, Woke today means something very different no matter how much Wokers may seek to hide that and insist Old Woke and New

Woke are the same. See if you find any 'awareness of social justice' here in the modern variety:

- Woke demands 'inclusivity' while excluding anyone with a different opinion and calls for mass censorship to silence other views.
- Woke claims to stand against oppression when imposing oppression is the foundation of all that it does. It is the driver of political correctness which is nothing more than a Cult invention to manipulate the population to silence itself.
- Woke believes itself to be 'liberal' while pursuing a global society that can only be described as fascist (see 'anti-fascist' fascist Antifa).
- Woke calls for 'social justice' while spreading injustice wherever it goes against the common 'enemy' which can be easily identified as a differing view.
- Woke is supposed to be a metaphor for 'awake' when it is solid-gold asleep and deep in a Cult-induced coma that meets the criteria for 'off with the fairies'.

I state these points as obvious facts if people only care to look. I don't do this with a sense of condemnation. We need to appreciate that the onslaught of perceptual programming on the young has been incessant and merciless. I can understand why so many have been reframed, or, given their youth, framed from the start to see the world as the Cult demands. The Cult has had access to their minds day after day in its 'education' system for their entire formative years. Perception is formed from information received and the Cult-created system is a life-long download of information delivered to elicit a particular perception, thus behaviour. The more this has expanded into still new extremes in recent decades and ever-increasing censorship has deleted other opinions and information why wouldn't that lead to a perceptual reframing on a mass scale? I

have described already cradle-to-grave programming and in more recent times the targeting of young minds from birth to adulthood has entered the stratosphere. This has taken the form of skewing what is 'taught' to fit the Cult agenda and the omnipresent techniques of group-think to isolate non-believers and pressure them into line. There has always been a tendency to follow the herd, but we really are in a new world now in relation to that. We have parents who can see the 'Covid' hoax told by their children not to stop them wearing masks at school, being 'Covid' tested or having the 'vaccine' in fear of the peer-pressure consequences of being different. What is 'peer-pressure' if not pressure to conform to group-think? Renegade Minds never group-think and always retain a set of perceptions that are unique to them. Group-think is always underpinned by consequences for not group-thinking. Abuse now aimed at those refusing DNA-manipulating 'Covid vaccines' are a potent example of this. The biggest pressure to conform comes from the very group which is itself being manipulated. 'I am programmed to be part of a hive mind and so you must be.'

Woke control structures in 'education' now apply to every mainstream organisation. Those at the top of the 'education' hierarchy (the Cult) decide the policy. This is imposed on governments through the Cult network; governments impose it on schools, colleges and universities; their leadership impose the policy on teachers and academics and they impose it on children and students. At any level where there is resistance, perhaps from a teacher or university lecturer, they are targeted by the authorities and often fired. Students themselves regularly demand the dismissal of academics (increasingly few) at odds with the narrative that the students have been programmed to believe in. It is quite a thought that students who are being targeted by the Cult become so consumed by programmed group-think that they launch protests and demand the removal of those who are trying to push back against those targeting the students. Such is the scale of perceptual inversion. We see this with 'Covid' programming as the Cult imposes the rules via psycho-psychologists and governments on

shops, transport companies and businesses which impose them on their staff who impose them on their customers who pressure Pushbackers to conform to the will of the Cult which is in the process of destroying them and their families. Scan all aspects of society and you will see the same sequence every time.

Fact free Woke and hijacking the 'left'

There is no more potent example of this than 'Woke', a mentality only made possible by the deletion of factual evidence by an 'education' system seeking to produce an ever more uniform society. Why would you bother with facts when you don't know any? Deletion of credible history both in volume and type is highly relevant. Orwell said: 'Who controls the past controls the future: who controls the present controls the past.' They who control the perception of the past control the perception of the future and they who control the present control the perception of the past through the writing and deleting of history. Why would you oppose the imposition of Marxism in the name of Wokeism when you don't know that Marxism cost at least 100 million lives in the 20th century alone? Watch videos and read reports in which Woker generations are asked basic historical questions – it's mind-blowing. A survey of 2,000 people found that six percent of millennials (born approximately early 1980s to early 2000s) believed the Second World War (1939-1945) broke out with the assassination of President Kennedy (in 1963) and one in ten thought Margaret Thatcher was British Prime Minister at the time. She was in office between 1979 and 1990. We are in a post-fact society. Provable facts are no defence against the fascism of political correctness or Silicon Valley censorship. Facts don't matter anymore as we have witnessed with the 'Covid' hoax. Sacrificing uniqueness to the Woke group-think religion is all you are required to do and that means thinking for yourself is the biggest Woke no, no. All religions are an expression of group-think and censorship and Woke is just another religion with an orthodoxy defended by group-think and censorship. Burned at

the stake becomes burned on Twitter which leads back eventually to burned at the stake as Woke humanity regresses to ages past.

The biggest Woke inversion of all is its creators and funders. I grew up in a traditional left of centre political household on a council estate in Leicester in the 1950s and 60s – you know, the left that challenged the power of wealth-hoarding elites and threats to freedom of speech and opinion. In those days students went on marches defending freedom of speech while today's Wokers march for its deletion. What on earth could have happened? Those very elites (collectively the Cult) that we opposed in my youth and early life have funded into existence the antithesis of that former left and hijacked the 'brand' while inverting everything it ever stood for. We have a mentality that calls itself 'liberal' and 'progressive' while acting like fascists. Cult billionaires and their corporations have funded themselves into control of 'education' to ensure that Woke programming is unceasing throughout the formative years of children and young people and that non-Wokers are isolated (that word again) whether they be students, teachers or college professors. The Cult has funded into existence the now colossal global network of Woke organisations that have spawned and promoted all the 'causes' on the Cult wish-list for global transformation and turned Wokers into demanders of them. Does anyone really think it's a coincidence that the Cult agenda for humanity is a carbon (sorry) copy of the societal transformations desired by Woke?? These are only some of them:

Political correctness: The means by which the Cult deletes all public debates that it knows it cannot win if we had the free-flow of information and evidence.

Human-caused 'climate change': The means by which the Cult seeks to transform society into a globally-controlled dictatorship imposing its will over the fine detail of everyone's lives 'to save the planet' which doesn't actually need saving.

Transgender obsession: Preparing collective perception to accept the 'new human' which would not have genders because it would be created technologically and not through procreation. I'll have much more on this in Human 2.0.

Race obsession: The means by which the Cult seeks to divide and rule the population by triggering racial division through the perception that society is more racist than ever when the opposite is the case. Is it perfect in that regard? No. But to compare today with the racism of apartheid and segregation brought to an end by the civil rights movement in the 1960s is to insult the memory of that movement and inspirations like Martin Luther King. Why is the 'anti-racism' industry (which it is) so dominated by privileged white people?

White supremacy: This is a label used by privileged white people to demonise poor and deprived white people pushing back on tyranny to marginalise and destroy them. White people are being especially targeted as the dominant race by number within Western society which the Cult seeks to transform in its image. If you want to change a society you must weaken and undermine its biggest group and once you have done that by using the other groups you next turn on them to do the same ... 'Then they came for the Jews and I was not a Jew so I did nothing.'

Mass migration: The mass movement of people from the Middle East, Africa and Asia into Europe, from the south into the United States and from Asia into Australia are another way the Cult seeks to dilute the racial, cultural and political influence of white people on Western society. White people ask why their governments appear to be working against them while being politically and culturally biased towards incoming cultures. Well, here's your answer. In the same way sexually 'straight' people, men and women, ask why the

authorities are biased against them in favour of other sexualities. The answer is the same – that's the way the Cult wants it to be for very sinister motives.

These are all central parts of the Cult agenda and central parts of the Woke agenda and Woke was created and continues to be funded to an immense degree by Cult billionaires and corporations. If anyone begins to say 'coincidence' the syllables should stick in their throat.

Billionaire 'social justice warriors'

Joe Biden is a 100 percent-owned asset of the Cult and the Wokers' man in the White House whenever he can remember his name and for however long he lasts with his rapidly diminishing cognitive function. Even walking up the steps of an aircraft without falling on his arse would appear to be a challenge. He's not an empty-shell puppet or anything. From the minute Biden took office (or the Cult did) he began his executive orders promoting the Woke wish-list. You will see the Woke agenda imposed ever more severely because it's really the *Cult* agenda. Woke organisations and activist networks spawned by the Cult are funded to the extreme so long as they promote what the Cult wants to happen. Woke is funded to promote 'social justice' by billionaires who become billionaires by destroying social justice. The social justice mantra is only a cover for dismantling social justice and funded by billionaires that couldn't give a damn about social justice. Everything makes sense when you see that. One of Woke's premier funders is Cult billionaire financier George Soros who said: 'I am basically there to make money, I cannot and do not look at the social consequences of what I do.' This is the same Soros who has given more than \$32 billion to his Open Society Foundations global Woke network and funded Black Lives Matter, mass immigration into Europe and the United States, transgender activism, climate change activism, political correctness and groups targeting 'white supremacy' in the form of privileged white thugs that dominate Antifa. What a scam it all is and when

you are dealing with the unquestioning fact-free zone of Woke scamming them is child's play. All you need to pull it off in all these organisations are a few in-the-know agents of the Cult and an army of naïve, reframed, uninformed, narcissistic, know-nothings convinced of their own self-righteousness, self-purity and virtue.

Soros and fellow billionaires and billionaire corporations have poured hundreds of millions into Black Lives Matter and connected groups and promoted them to a global audience. None of this is motivated by caring about black people. These are the billionaires that have controlled and exploited a system that leaves millions of black people in abject poverty and deprivation which they do absolutely nothing to address. The same Cult networks funding BLM were behind the *slave trade*! Black Lives Matter hijacked a phrase that few would challenge and they have turned this laudable concept into a political weapon to divide society. You know that BLM is a fraud when it claims that *All Lives Matter*, the most inclusive statement of all, is 'racist'. BLM and its Cult masters don't want to end racism. To them it's a means to an end to control all of humanity never mind the colour, creed, culture or background. What has destroying the nuclear family got to do with ending racism? Nothing – but that is one of the goals of BLM and also happens to be a goal of the Cult as I have been exposing in my books for decades. Stealing children from loving parents and giving schools ever more power to override parents is part of that same agenda. BLM is a Marxist organisation and why would that not be the case when the Cult created Marxism *and* BLM? Patrisse Cullors, a BLM co-founder, said in a 2015 video that she and her fellow organisers, including co-founder Alicia Garza, are 'trained Marxists'. The lady known after marriage as Patrisse Khan-Cullors bought a \$1.4 million home in 2021 in one of the whitest areas of California with a black population of just 1.6 per cent and has so far bought *four* high-end homes for a total of \$3.2 million. How very Marxist. There must be a bit of spare in the BLM coffers, however, when Cult corporations and billionaires have handed over the best part of \$100 million. Many black people can see that Black Lives Matter is not

working for them, but against them, and this is still more confirmation. Black journalist Jason Whitlock, who had his account suspended by Twitter for simply linking to the story about the 'Marxist's' home buying spree, said that BLM leaders are 'making millions of dollars off the backs of these dead black men who they wouldn't spit on if they were on fire and alive'.

Black Lies Matter

Cult assets and agencies came together to promote BLM in the wake of the death of career criminal George Floyd who had been jailed a number of times including for forcing his way into the home of a black woman with others in a raid in which a gun was pointed at her stomach. Floyd was filmed being held in a Minneapolis street in 2020 with the knee of a police officer on his neck and he subsequently died. It was an appalling thing for the officer to do, but the same technique has been used by police on peaceful protestors of lockdown without any outcry from the Woke brigade. As unquestioning supporters of the Cult agenda Wokers have supported lockdown and all the 'Covid' claptrap while attacking anyone standing up to the tyranny imposed in its name. Court documents would later include details of an autopsy on Floyd by County Medical Examiner Dr Andrew Baker who concluded that Floyd had taken a fatal level of the drug fentanyl. None of this mattered to fact-free, question-free, Woke. Floyd's death was followed by worldwide protests against police brutality amid calls to defund the police. Throwing babies out with the bathwater is a Woke speciality. In the wake of the murder of British woman Sarah Everard a Green Party member of the House of Lords, Baroness Jones of Moulsecoomb (Nincompoopia would have been better), called for a 6pm curfew for all men. This would be in breach of the Geneva Conventions on war crimes which ban collective punishment, but that would never have crossed the black and white Woke mind of Baroness Nincompoopia who would have been far too convinced of her own self-righteousness to compute such details. Many American cities did defund the police in the face of Floyd riots

and after \$15 million was deleted from the police budget in Washington DC under useless Woke mayor Muriel Bowser car-jacking alone rose by 300 percent and within six months the US capital recorded its highest murder rate in 15 years. The same happened in Chicago and other cities in line with the Cult/Soros plan to bring fear to streets and neighbourhoods by reducing the police, releasing violent criminals and not prosecuting crime. This is the mob-rule agenda that I have warned in the books was coming for so long. Shootings in the area of Minneapolis where Floyd was arrested increased by 2,500 percent compared with the year before. Defunding the police over George Floyd has led to a big increase in dead people with many of them black. Police protection for politicians making these decisions stayed the same or increased as you would expect from professional hypocrites. The Cult doesn't actually want to abolish the police. It wants to abolish local control over the police and hand it to federal government as the psychopaths advance the Hunger Games Society. Many George Floyd protests turned into violent riots with black stores and businesses destroyed by fire and looting across America fuelled by Black Lives Matter. Woke doesn't do irony. If you want civil rights you must loot the liquor store and the supermarket and make off with a smart TV. It's the only way.

It's not a race war – it's a class war

Black people are patronised by privileged blacks and whites alike and told they are victims of white supremacy. I find it extraordinary to watch privileged blacks supporting the very system and bloodline networks behind the slave trade and parroting the same Cult-serving manipulative crap of their privileged white, often billionaire, associates. It is indeed not a race war but a class war and colour is just a diversion. Black Senator Cory Booker and black Congresswoman Maxine Waters, more residents of Nincompoopia, personify this. Once you tell people they are victims of someone else you devalue both their own responsibility for their plight and the power they have to impact on their reality and experience. Instead

we have: 'You are only in your situation because of whiteness – turn on them and everything will change.' It won't change. Nothing changes in our lives unless *we* change it. Crucial to that is never seeing yourself as a victim and always as the creator of your reality. Life is a simple sequence of choice and consequence. Make different choices and you create different consequences. *You* have to make those choices – not Black Lives Matter, the Woke Mafia and anyone else that seeks to dictate your life. Who are they these Wokers, an emotional and psychological road traffic accident, to tell you what to do? Personal empowerment is the last thing the Cult and its Black Lives Matter want black people or anyone else to have. They claim to be defending the underdog while *creating* and perpetuating the underdog. The Cult's worst nightmare is human unity and if they are going to keep blacks, whites and every other race under economic servitude and control then the focus must be diverted from what they have in common to what they can be manipulated to believe divides them. Blacks have to be told that their poverty and plight is the fault of the white bloke living on the street in the same poverty and with the same plight they are experiencing. The difference is that your plight black people is due to him, a white supremacist with 'white privilege' living on the street. Don't unite as one human family against your mutual oppressors and suppressors – fight the oppressor with the white face who is as financially deprived as you are. The Cult knows that as its 'Covid' agenda moves into still new levels of extremism people are going to respond and it has been spreading the seeds of disunity everywhere to stop a united response to the evil that targets *all of us*.

Racist attacks on 'whiteness' are getting ever more outrageous and especially through the American Democratic Party which has an appalling history for anti-black racism. Barack Obama, Joe Biden, Hillary Clinton and Nancy Pelosi all eulogised about Senator Robert Byrd at his funeral in 2010 after a nearly 60-year career in Congress. Byrd was a brutal Ku Klux Klan racist and a violent abuser of Cathy O'Brien in MKUltra. He said he would never fight in the military 'with a negro by my side' and 'rather I should die a thousand times,

and see Old Glory trampled in the dirt never to rise again, than to see this beloved land of ours become degraded by race mongrels, a throwback to the blackest specimen from the wilds'. Biden called Byrd a 'very close friend and mentor'. These 'Woke' hypocrites are not anti-racist they are anti-poor and anti-people not of their perceived class. Here is an illustration of the scale of anti-white racism to which we have now descended. Seriously Woke and moronic *New York Times* contributor Damon Young described whiteness as a 'virus' that 'like other viruses will not die until there are no bodies left for it to infect'. He went on: '... the only way to stop it is to locate it, isolate it, extract it, and kill it.' Young can say that as a black man with no consequences when a white man saying the same in reverse would be facing a jail sentence. *That's* racism. We had super-Woke numbskull senators Tammy Duckworth and Mazie Hirono saying they would object to future Biden Cabinet appointments if he did not nominate more Asian Americans and Pacific Islanders. Never mind the ability of the candidate what do they look like? Duckworth said: 'I will vote for racial minorities and I will vote for LGBTQ, but anyone else I'm not voting for.' Appointing people on the grounds of race is illegal, but that was not a problem for this ludicrous pair. They were on-message and that's a free pass in any situation.

Critical race racism

White children are told at school they are intrinsically racist as they are taught the divisive 'critical race theory'. This claims that the law and legal institutions are inherently racist and that race is a socially constructed concept used by white people to further their economic and political interests at the expense of people of colour. White is a 'virus' as we've seen. Racial inequality results from 'social, economic, and legal differences that white people create between races to maintain white interests which leads to poverty and criminality in minority communities'. I must tell that to the white guy sleeping on the street. The principal of East Side Community School in New York sent white parents a manifesto that called on

them to become 'white traitors' and advocate for full 'white abolition'. These people are teaching your kids when they urgently need a psychiatrist. The 'school' included a chart with 'eight white identities' that ranged from 'white supremacist' to 'white abolition' and defined the behaviour white people must follow to end 'the regime of whiteness'. Woke blacks and their privileged white associates are acting exactly like the slave owners of old and Ku Klux Klan racists like Robert Byrd. They are too full of their own self-purity to see that, but it's true. Racism is not a body type; it's a state of mind that can manifest through any colour, creed or culture.

Another racial fraud is '*equity*'. Not equality of treatment and opportunity – equity. It's a term spun as equality when it means something very different. Equality in its true sense is a raising up while 'equity' is a race to the bottom. Everyone in the same level of poverty is 'equity'. Keep everyone down – that's equity. The Cult doesn't want anyone in the human family to be empowered and BLM leaders, like all these 'anti-racist' organisations, continue their privileged, pampered existence by perpetuating the perception of gathering racism. When is the last time you heard an 'anti-racist' or 'anti-Semitism' organisation say that acts of racism and discrimination have *fallen*? It's not in the interests of their fundraising and power to influence and the same goes for the professional soccer anti-racism operation, Kick It Out. Two things confirmed that the Black Lives Matter riots in the summer of 2020 were Cult creations. One was that while anti-lockdown protests were condemned in this same period for 'transmitting 'Covid' the authorities supported mass gatherings of Black Lives Matter supporters. I even saw self-deluding people claiming to be doctors say the two types of protest were not the same. No – the non-existent 'Covid' was in favour of lockdowns and attacked those that protested against them while 'Covid' supported Black Lives Matter and kept well away from its protests. The whole thing was a joke and as lockdown protestors were arrested, often brutally, by reframed Face-Nappies we had the grotesque sight of police officers taking the knee to Black Lives Matter, a Cult-funded Marxist

organisation that supports violent riots and wants to destroy the nuclear family and white people.

He's not white? Shucks!

Woke obsession with race was on display again when ten people were shot dead in Boulder, Colorado, in March, 2021. Cult-owned Woke TV channels like CNN said the shooter appeared to be a white man and Wokers were on Twitter condemning 'violent white men' with the usual mantras. Then the shooter's name was released as Ahmad Al Aliwi Alissa, an anti-Trump Arab-American, and the sigh of disappointment could be heard five miles away. Never mind that ten people were dead and what that meant for their families. Race baiting was all that mattered to these sick Cult-serving people like Barack Obama who exploited the deaths to further divide America on racial grounds which is his job for the Cult. This is the man that 'racist' white Americans made the first black president of the United States and then gave him a second term. Not-very-bright Obama has become filthy rich on the back of that and today appears to have a big influence on the Biden administration. Even so he's still a downtrodden black man and a victim of white supremacy. This disingenuous fraud reveals the contempt he has for black people when he puts on a Deep South Alabama accent whenever he talks to them, no, *at* them.

Another BLM red flag was how the now fully-Woke (fully-Cult) and fully-virtue-signalled professional soccer authorities had their teams taking the knee before every match in support of Marxist Black Lives Matter. Soccer authorities and clubs displayed 'Black Lives Matter' on the players' shirts and flashed the name on electronic billboards around the pitch. Any fans that condemned what is a Freemasonic taking-the-knee ritual were widely condemned as you would expect from the Woke virtue-signallers of professional sport and the now fully-Woke media. We have reverse racism in which you are banned from criticising any race or culture except for white people for whom anything goes – say what you like, no problem. What has this got to do with racial harmony and

equality? We've had black supremacists from Black Lives Matter telling white people to fall to their knees in the street and apologise for their white supremacy. Black supremacists acting like white supremacist slave owners of the past couldn't breach their self-obsessed, race-obsessed sense of self-purity. Joe Biden appointed a race-obsessed black supremacist Kristen Clarke to head the Justice Department Civil Rights Division. Clarke claimed that blacks are endowed with 'greater mental, physical and spiritual abilities' than whites. If anyone reversed that statement they would be vilified. Clarke is on-message so no problem. She's never seen a black-white situation in which the black figure is anything but a virtuous victim and she heads the Civil Rights Division which should treat everyone the same or it isn't civil rights. Another perception of the Renegade Mind: If something or someone is part of the Cult agenda they will be supported by Woke governments and media no matter what. If they're not, they will be condemned and censored. It really is that simple and so racist Clarke prospers despite (make that because of) her racism.

The end of culture

Biden's administration is full of such racial, cultural and economic bias as the Cult requires the human family to be divided into warring factions. We are now seeing racially-segregated graduations and everything, but everything, is defined through the lens of perceived 'racism'. We have 'racist' mathematics, 'racist' food and even 'racist' *plants*. World famous Kew Gardens in London said it was changing labels on plants and flowers to tell its pre-'Covid' more than two million visitors a year how racist they are. Kew director Richard Deverell said this was part of an effort to 'move quickly to decolonise collections' after they were approached by one Ajay Chhabra 'an actor with an insight into how sugar cane was linked to slavery'. They are *plants* you idiots. 'Decolonisation' in the Woke manual really means colonisation of society with its mentality and by extension colonisation by the Cult. We are witnessing a new Chinese-style 'Cultural Revolution' so essential to the success of all

Marxist takeovers. Our cultural past and traditions have to be swept away to allow a new culture to be built-back-better. Woke targeting of long-standing Western cultural pillars including historical monuments and cancelling of historical figures is what happened in the Mao revolution in China which 'purged remnants of capitalist and traditional elements from Chinese society' and installed Maoism as the dominant ideology'. For China see the Western world today and for 'dominant ideology' see Woke. Better still see Marxism or Maoism. The 'Covid' hoax has specifically sought to destroy the arts and all elements of Western culture from people meeting in a pub or restaurant to closing theatres, music venues, sports stadiums, places of worship and even banning *singing*. Destruction of Western society is also why criticism of any religion is banned except for Christianity which again is the dominant religion as white is the numerically-dominant race. Christianity may be fading rapidly, but its history and traditions are weaved through the fabric of Western society. Delete the pillars and other structures will follow until the whole thing collapses. I am not a Christian defending that religion when I say that. I have no religion. It's just a fact. To this end Christianity has itself been turned Woke to usher its own downfall and its ranks are awash with 'change agents' – knowing and unknowing – at every level including Pope Francis (*definitely* knowing) and the clueless Archbishop of Canterbury Justin Welby (possibly not, but who can be sure?). Woke seeks to coordinate attacks on Western culture, traditions, and ways of life through 'intersectionality' defined as 'the complex, cumulative way in which the effects of multiple forms of discrimination (such as racism, sexism, and classism) combine, overlap, or intersect especially in the experiences of marginalised individuals or groups'. Wade through the Orwellian Woke-speak and this means coordinating disparate groups in a common cause to overthrow freedom and liberal values.

The entire structure of public institutions has been infested with Woke – government at all levels, political parties, police, military, schools, universities, advertising, media and trade unions. This abomination has been achieved through the Cult web by appointing

Workers to positions of power and battering non-Workers into line through intimidation, isolation and threats to their job. Many have been fired in the wake of the empathy-deleted, vicious hostility of 'social justice' Workers and the desire of gutless, spineless employers to virtue-signal their Wokeness. Corporations are filled with Workers today, most notably those in Silicon Valley. Ironically at the top they are not Woke at all. They are only exploiting the mentality their Cult masters have created and funded to censor and enslave while the Workers cheer them on until it's their turn. Thus the Woke 'liberal left' is an inversion of the traditional liberal left. Campaigning for justice on the grounds of power and wealth distribution has been replaced by campaigning for identity politics. The genuine traditional left would never have taken money from today's billionaire abusers of fairness and justice and nor would the billionaires have wanted to fund that genuine left. It would not have been in their interests to do so. The division of opinion in those days was between the haves and have nots. This all changed with Cult manipulated and funded identity politics. The division of opinion today is between Workers and non-Workers and not income brackets. Cult corporations and their billionaires may have taken wealth disparity to cataclysmic levels of injustice, but as long as they speak the language of Woke, hand out the dosh to the Woke network and censor the enemy they are 'one of us'. Billionaires who don't give a damn about injustice are laughing at them till their bellies hurt. Workers are not even close to self-aware enough to see that. The transformed 'left' dynamic means that Workers who drone on about 'social justice' are funded by billionaires that have destroyed social justice the world over. It's *why* they are billionaires.

The climate con

Nothing encapsulates what I have said more comprehensively than the hoax of human-caused global warming. I have detailed in my books over the years how Cult operatives and organisations were the pump-primers from the start of the climate con. A purpose-built vehicle for this is the Club of Rome established by the Cult in 1968

with the Rockefellers and Rothschilds centrally involved all along. Their gofer frontman Maurice Strong, a Canadian oil millionaire, hosted the Earth Summit in Rio de Janeiro, Brazil, in 1992 where the global 'green movement' really expanded in earnest under the guiding hand of the Cult. The Earth Summit established Agenda 21 through the Cult-created-and-owned United Nations to use the illusion of human-caused climate change to justify the transformation of global society to save the world from climate disaster. It is a No-Problem-Reaction-Solution sold through governments, media, schools and universities as whole generations have been terrified into believing that the world was going to end in their lifetimes unless what old people had inflicted upon them was stopped by a complete restructuring of how everything is done. Chill, kids, it's all a hoax. Such restructuring is precisely what the Cult agenda demands (purely by coincidence of course). Today this has been given the codename of the Great Reset which is only an updated term for Agenda 21 and its associated Agenda 2030. The latter, too, is administered through the UN and was voted into being by the General Assembly in 2015. Both 21 and 2030 seek centralised control of all resources and food right down to the raindrops falling on your own land. These are some of the demands of Agenda 21 established in 1992. See if you recognise this society emerging today:

- End national sovereignty
- State planning and management of all land resources, ecosystems, deserts, forests, mountains, oceans and fresh water; agriculture; rural development; biotechnology; and ensuring '*equity*'
- The state to 'define the role' of business and financial resources
- Abolition of private property
- 'Restructuring' the family unit (see BLM)
- Children raised by the state
- People told what their job will be
- Major restrictions on movement
- Creation of 'human settlement zones'

- Mass resettlement as people are forced to vacate land where they live
- Dumbing down education
- Mass global depopulation in pursuit of all the above

The United Nations was created as a Trojan horse for world government. With the climate con of critical importance to promoting that outcome you would expect the UN to be involved. Oh, it's involved all right. The UN is promoting Agenda 21 and Agenda 2030 justified by 'climate change' while also driving the climate hoax through its Intergovernmental Panel on Climate Change (IPCC), one of the world's most corrupt organisations. The IPCC has been lying ferociously and constantly since the day it opened its doors with the global media hanging unquestioningly on its every mendacious word. The Green movement is entirely Woke and has long lost its original environmental focus since it was co-opted by the Cult. An obsession with 'global warming' has deleted its values and scrambled its head. I experienced a small example of what I mean on a beautiful country walk that I have enjoyed several times a week for many years. The path merged into the fields and forests and you felt at one with the natural world. Then a 'Green' organisation, the Hampshire and Isle of Wight Wildlife Trust, took over part of the land and proceeded to cut down a large number of trees, including mature ones, to install a horrible big, bright steel 'this-is-ours-stay-out' fence that destroyed the whole atmosphere of this beautiful place. No one with a feel for nature would do that. Day after day I walked to the sound of chainsaws and a magnificent mature weeping willow tree that I so admired was cut down at the base of the trunk. When I challenged a Woke young girl in a green shirt (of course) about this vandalism she replied: 'It's a weeping willow – it will grow back.' This is what people are paying for when they donate to the Hampshire and Isle of Wight Wildlife Trust and many other 'green' organisations today. It is not the environmental movement that I knew and instead has become a support-system – as with Extinction Rebellion – for a very dark agenda.

Private jets for climate justice

The Cult-owned, Gates-funded, World Economic Forum and its founder Klaus Schwab were behind the emergence of Greta Thunberg to harness the young behind the climate agenda and she was invited to speak to the world at ... the UN. Schwab published a book, *Covid-19: The Great Reset* in 2020 in which he used the 'Covid' hoax and the climate hoax to lay out a new society straight out of Agenda 21 and Agenda 2030. Bill Gates followed in early 2021 when he took time out from destroying the world to produce a book in his name about the way to save it. Gates flies across the world in private jets and admitted that 'I probably have one of the highest greenhouse gas footprints of anyone on the planet ... my personal flying alone is gigantic.' He has also bid for the planet's biggest private jet operator. Other climate change saviours who fly in private jets include John Kerry, the US Special Presidential Envoy for Climate, and actor Leonardo DiCaprio, a 'UN Messenger of Peace with special focus on climate change'. These people are so full of bullshit they could corner the market in manure. We mustn't be sceptical, though, because the Gates book, *How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need*, is a genuine attempt to protect the world and not an obvious pile of excrement attributed to a mega-psychopath aimed at selling his masters' plans for humanity. The Gates book and the other shite-pile by Klaus Schwab could have been written by the same person and may well have been. Both use 'climate change' and 'Covid' as the excuses for their new society and by coincidence the Cult's World Economic Forum and Bill and Melinda Gates Foundation promote the climate hoax and hosted Event 201 which pre-empted with a 'simulation' the very 'coronavirus' hoax that would be simulated for real on humanity within weeks. The British 'royal' family is promoting the 'Reset' as you would expect through Prince 'climate change caused the war in Syria' Charles and his hapless son Prince William who said that we must 'reset our relationship with nature and our trajectory as a species' to avoid a climate disaster. Amazing how many promoters of the 'Covid' and 'climate change' control

systems are connected to Gates and the World Economic Forum. A 'study' in early 2021 claimed that carbon dioxide emissions must fall by the equivalent of a global lockdown roughly every two years for the next decade to save the planet. The 'study' appeared in the same period that the Schwab mob claimed in a video that lockdowns destroying the lives of billions are good because they make the earth 'quieter' with less 'ambient noise'. They took down the video amid a public backlash for such arrogant, empathy-deleted stupidity. You see, however, where they are going with this. Corinne Le Quéré, a professor at the Tyndall Centre for Climate Change Research, University of East Anglia, was lead author of the climate lockdown study, and she writes for ... the World Economic Forum. Gates calls in 'his' book for changing 'every aspect of the economy' (long-time Cult agenda) and for humans to eat synthetic 'meat' (predicted in my books) while cows and other farm animals are eliminated. Australian TV host and commentator Alan Jones described what carbon emission targets would mean for farm animals in Australia alone if emissions were reduced as demanded by 35 percent by 2030 and zero by 2050:

Well, let's take agriculture, the total emissions from agriculture are about 75 million tonnes of carbon dioxide, equivalent. Now reduce that by 35 percent and you have to come down to 50 million tonnes, I've done the maths. So if you take for example 1.5 million cows, you're going to have to reduce the herd by 525,000 [by] 2030, nine years, that's 58,000 cows a year. The beef herd's 30 million, reduce that by 35 percent, that's 10.5 million, which means 1.2 million cattle have to go every year between now and 2030. This is insanity!

There are 75 million sheep. Reduce that by 35 percent, that's 26 million sheep, that's almost 3 million a year. So under the Paris Agreement over 30 million beasts, dairy cows, cattle, pigs and sheep would go. More than 8,000 every minute of every hour for the next decade, do these people know what they're talking about?

Clearly they don't at the level of campaigners, politicians and administrators. The Cult *does* know; that's the outcome it wants. We are faced with not just a war on humanity. Animals and the natural world are being targeted and I have been saying since the 'Covid' hoax began that the plan eventually was to claim that the 'deadly virus' is able to jump from animals, including farm animals and

domestic pets, to humans. Just before this book went into production came this story: 'Russia registers world's first Covid-19 vaccine for cats & dogs as makers of Sputnik V warn pets & farm animals could spread virus'. The report said 'top scientists warned that the deadly pathogen could soon begin spreading through homes and farms' and 'the next stage is the infection of farm and domestic animals'. Know the outcome and you'll see the journey. Think what that would mean for animals and keep your eye on a term called zoonosis or zoonotic diseases which transmit between animals and humans. The Cult wants to break the connection between animals and people as it does between people and people. Farm animals fit with the Cult agenda to transform food from natural to synthetic.

The gas of life is killing us

There can be few greater examples of Cult inversion than the condemnation of carbon dioxide as a dangerous pollutant when it is the gas of life. Without it the natural world would be dead and so we would all be dead. We breathe in oxygen and breathe out carbon dioxide while plants produce oxygen and absorb carbon dioxide. It is a perfect symbiotic relationship that the Cult wants to dismantle for reasons I will come to in the final two chapters. Gates, Schwab, other Cult operatives and mindless repeaters, want the world to be 'carbon neutral' by at least 2050 and the earlier the better. 'Zero carbon' is the cry echoed by lunatics calling for 'Zero Covid' when we already have it. These carbon emission targets will deindustrialise the world in accordance with Cult plans – the post-industrial, post-democratic society – and with so-called renewables like solar and wind not coming even close to meeting human energy needs blackouts and cold are inevitable. Texans got the picture in the winter of 2021 when a snow storm stopped wind turbines and solar panels from working and the lights went down along with water which relies on electricity for its supply system. Gates wants everything to be powered by electricity to ensure that his masters have the kill switch to stop all human activity, movement, cooking, water and warmth any time they like. The climate lie is so

stupendously inverted that it claims we must urgently reduce carbon dioxide when we *don't have enough*.

Co2 in the atmosphere is a little above 400 parts per million when the optimum for plant growth is 2,000 ppm and when it falls anywhere near 150 ppm the natural world starts to die and so do we. It fell to as low as 280 ppm in an 1880 measurement in Hawaii and rose to 413 ppm in 2019 with industrialisation which is why the planet has become *greener* in the industrial period. How insane then that psychopathic madman Gates is not satisfied only with blocking the rise of Co2. He's funding technology to suck it out of the atmosphere. The reason why will become clear. The industrial era is not destroying the world through Co2 and has instead turned around a potentially disastrous ongoing fall in Co2. Greenpeace co-founder and scientist Patrick Moore walked away from Greenpeace in 1986 and has exposed the green movement for fear-mongering and lies. He said that 500 million years ago there was *17 times* more Co2 in the atmosphere than we have today and levels have been falling for hundreds of millions of years. In the last 150 million years Co2 levels in Earth's atmosphere had reduced by *90 percent*. Moore said that by the time humanity began to unlock carbon dioxide from fossil fuels we were at '38 seconds to midnight' and in that sense: 'Humans are [the Earth's] salvation.' Moore made the point that only half the Co2 emitted by fossil fuels stays in the atmosphere and we should remember that all pollution pouring from chimneys that we are told is carbon dioxide is in fact nothing of the kind. It's pollution. Carbon dioxide is an invisible gas.

William Happer, Professor of Physics at Princeton University and long-time government adviser on climate, has emphasised the Co2 deficiency for maximum growth and food production. Greenhouse growers don't add carbon dioxide for a bit of fun. He said that most of the warming in the last 100 years, after the earth emerged from the super-cold period of the 'Little Ice Age' into a natural warming cycle, was over by 1940. Happer said that a peak year for warming in 1988 can be explained by a 'monster El Nino' which is a natural and cyclical warming of the Pacific that has nothing to do with 'climate

change'. He said the effect of Co2 could be compared to painting a wall with red paint in that once two or three coats have been applied it didn't matter how much more you slapped on because the wall will not get much redder. Almost all the effect of the rise in Co2 has already happened, he said, and the volume in the atmosphere would now have to *double* to increase temperature by a single degree. Climate hoaxers know this and they have invented the most ridiculously complicated series of 'feedback' loops to try to overcome this rather devastating fact. You hear puppet Greta going on cluelessly about feedback loops and this is why.

The Sun affects temperature? No you *climate denier*

Some other nonsense to contemplate: Climate graphs show that rises in temperature do not follow rises in Co2 – *it's the other way round* with a lag between the two of some 800 years. If we go back 800 years from present time we hit the Medieval Warm Period when temperatures were higher than now without any industrialisation and this was followed by the Little Ice Age when temperatures plummeted. The world was still emerging from these centuries of serious cold when many climate records began which makes the ever-repeated line of the 'hottest year since records began' meaningless when you are not comparing like with like. The coldest period of the Little Ice Age corresponded with the lowest period of sunspot activity when the Sun was at its least active. Proper scientists will not be at all surprised by this when it confirms the obvious fact that earth temperature is affected by the scale of Sun activity and the energetic power that it subsequently emits; but when is the last time you heard a climate hoaxer talking about the Sun as a source of earth temperature?? Everything has to be focussed on Co2 which makes up just 0.117 percent of so-called greenhouse gases and only a fraction of even that is generated by human activity. The rest is natural. More than 90 percent of those greenhouse gases are water vapour and clouds ([Fig 9](#)). Ban moisture I say. Have you noticed that the climate hoaxers no longer use the polar bear as their promotion image? That's because far from becoming extinct polar

bear communities are stable or thriving. Joe Bastardi, American meteorologist, weather forecaster and outspoken critic of the climate lie, documents in his book *The Climate Chronicles* how weather patterns and events claimed to be evidence of climate change have been happening since long before industrialisation: 'What happened before naturally is happening again, as is to be expected given the cyclical nature of the climate due to the design of the planet.' If you read the detailed background to the climate hoax in my other books you will shake your head and wonder how anyone could believe the crap which has spawned a multi-trillion dollar industry based on absolute garbage (see HIV causes AIDs and Sars-Cov-2 causes 'Covid-19'). Climate and 'Covid' have much in common given they have the same source. They both have the contradictory *everything* factor in which everything is explained by reference to them. It's hot – 'it's climate change'. It's cold – 'it's climate change'. I got a sniffle – 'it's Covid'. I haven't got a sniffle – 'it's Covid'. Not having a sniffle has to be a symptom of 'Covid'. Everything is and not having a sniffle is especially dangerous if you are a slow walker. For sheer audacity I offer you a Cambridge University 'study' that actually linked 'Covid' to 'climate change'. It had to happen eventually. They concluded that climate change played a role in 'Covid-19' spreading from animals to humans because ... wait for it ... I kid you not ... *the two groups were forced closer together as populations grow*. Er, that's it. The whole foundation on which this depended was that 'Bats are the likely zoonotic origin of SARS-CoV-1 and SARS-CoV-2'. Well, they are not. They are nothing to do with it. Apart from bats not being the origin and therefore 'climate change' effects on bats being irrelevant I am in awe of their academic insight. Where would we be without them? Not where we are that's for sure.

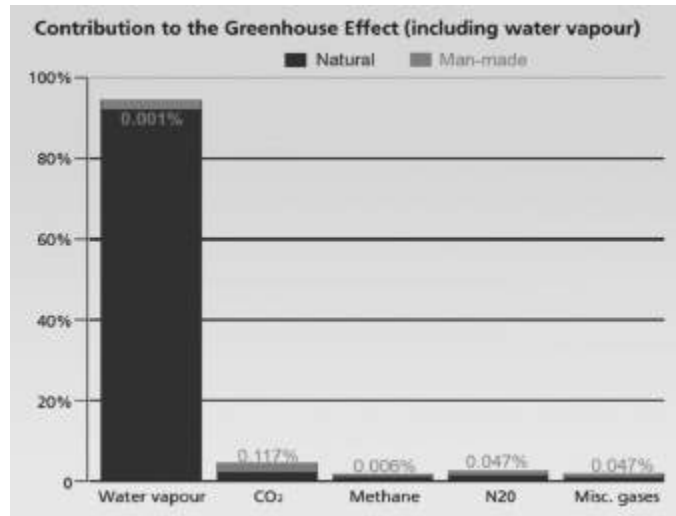


Figure 9: The idea that the gas of life is disastrously changing the climate is an insult to brain cell activity.

One other point about the weather is that climate modification is now well advanced and not every major weather event is natural – or earthquake come to that. I cover this subject at some length in other books. China is openly planning a rapid expansion of its weather modification programme which includes changing the climate in an area more than one and a half times the size of India. China used weather manipulation to ensure clear skies during the 2008 Olympics in Beijing. I have quoted from US military documents detailing how to employ weather manipulation as a weapon of war and they did that in the 1960s and 70s during the conflict in Vietnam with Operation Popeye manipulating monsoon rains for military purposes. Why would there be international treaties on weather modification if it wasn't possible? Of course it is. Weather is energetic information and it can be changed.

How was the climate hoax pulled off? See 'Covid'

If you can get billions to believe in a 'virus' that doesn't exist you can get them to believe in human-caused climate change that doesn't exist. Both are being used by the Cult to transform global society in the way it has long planned. Both hoaxes have been achieved in pretty much the same way. First you declare a lie is a fact. There's a

'virus' you call SARS-Cov-2 or humans are warming the planet with their behaviour. Next this becomes, via Cult networks, the foundation of government, academic and science policy and belief. Those who parrot the mantra are given big grants to produce research that confirms the narrative is true and ever more 'symptoms' are added to make the 'virus'/'climate change' sound even more scary. Scientists and researchers who challenge the narrative have their grants withdrawn and their careers destroyed. The media promote the lie as the unquestionable truth and censor those with an alternative view or evidence. A great percentage of the population believe what they are told as the lie becomes an everybody-knows-that and the believing-masses turn on those with a mind of their own. The technique has been used endlessly throughout human history. Wokers are the biggest promoters of the climate lie *and* 'Covid' fascism because their minds are owned by the Cult; their sense of self-righteous self-purity knows no bounds; and they exist in a bubble of reality in which facts are irrelevant and only get in the way of looking without seeing.

Running through all of this like veins in a blue cheese is control of information, which means control of perception, which means control of behaviour, which collectively means control of human society. The Cult owns the global media and Silicon Valley fascists for the simple reason that it *has* to. Without control of information it can't control perception and through that human society. Examine every facet of the Cult agenda and you will see that anything supporting its introduction is never censored while anything pushing back is always censored. I say again: Psychopaths that know why they are doing this must go before Nuremberg trials and those that follow their orders must trot along behind them into the same dock. 'I was just following orders' didn't work the first time and it must not work now. Nuremberg trials must be held all over the world before public juries for politicians, government officials, police, compliant doctors, scientists and virologists, and all Cult operatives such as Gates, Tedros, Fauci, Vallance, Whitty, Ferguson, Zuckerberg, Wojcicki, Brin, Page, Dorsey, the whole damn lot of

them – including, no *especially*, the psychopath psychologists. Without them and the brainless, gutless excuses for journalists that have repeated their lies, none of this could be happening. Nobody can be allowed to escape justice for the psychological and economic Armageddon they are all responsible for visiting upon the human race.

As for the compliant, unquestioning, swathes of humanity, and the self-obsessed, all-knowing ignorance of the Wokers ... don't start me. God help their kids. God help their grandkids. God *help them*.

CHAPTER NINE

We must have it? So what is it?

*Well I won't back down. No, I won't back down. You can stand me
up at the Gates of Hell. But I won't back down*

Tom Petty

I will now focus on the genetically-manipulating 'Covid vaccines' which do not meet this official definition of a vaccine by the US Centers for Disease Control (CDC): 'A product that stimulates a person's immune system to produce immunity to a specific disease, protecting the person from that disease.' On that basis 'Covid vaccines' are not a vaccine in that the makers don't even claim they stop infection or transmission.

They are instead part of a multi-levelled conspiracy to change the nature of the human body and what it means to be 'human' and to depopulate an enormous swathe of humanity. What I shall call Human 1.0 is on the cusp of becoming Human 2.0 and for very sinister reasons. Before I get to the 'Covid vaccine' in detail here's some background to vaccines in general. Government regulators do not test vaccines – the makers do – and the makers control which data is revealed and which isn't. Children in America are given 50 vaccine doses by age six and 69 by age 19 and the effect of the whole combined schedule has never been tested. Autoimmune diseases when the immune system attacks its own body have soared in the mass vaccine era and so has disease in general in children and the young. Why wouldn't this be the case when vaccines target the *immune system*? The US government gave Big Pharma drug

companies immunity from prosecution for vaccine death and injury in the 1986 National Childhood Vaccine Injury Act (NCVIA) and since then the government (taxpayer) has been funding compensation for the consequences of Big Pharma vaccines. The criminal and satanic drug giants can't lose and the vaccine schedule has increased dramatically since 1986 for this reason. There is no incentive to make vaccines safe and a big incentive to make money by introducing ever more. Even against a ridiculously high bar to prove vaccine liability, and with the government controlling the hearing in which it is being challenged for compensation, the vaccine court has so far paid out more than \$4 billion. These are the vaccines we are told are safe and psychopaths like Zuckerberg censor posts saying otherwise. The immunity law was even justified by a ruling that vaccines by their nature were 'unavoidably unsafe'.

Check out the ingredients of vaccines and you will be shocked if you are new to this. *They put that in children's bodies?? What??* Try aluminium, a brain toxin connected to dementia, aborted foetal tissue and formaldehyde which is used to embalm corpses. World-renowned aluminium expert Christopher Exley had his research into the health effect of aluminium in vaccines shut down by Keele University in the UK when it began taking funding from the Bill and Melinda Gates Foundation. Research when diseases 'eradicated' by vaccines began to decline and you will find the fall began long *before* the vaccine was introduced. Sometimes the fall even plateaued after the vaccine. Diseases like scarlet fever for which there was no vaccine declined in the same way because of environmental and other factors. A perfect case in point is the polio vaccine. Polio began when lead arsenate was first sprayed as an insecticide and residues remained in food products. Spraying started in 1892 and the first US polio epidemic came in Vermont in 1894. The simple answer was to stop spraying, but Rockefeller-created Big Pharma had a better idea. Polio was decreed to be caused by the *poliovirus* which 'spreads from person to person and can infect a person's spinal cord'. Lead arsenate was replaced by the lethal DDT which had the same effect of causing paralysis by damaging the brain and central nervous

system. Polio plummeted when DDT was reduced and then banned, but the vaccine is still given the credit for something it didn't do. Today by far the biggest cause of polio is the vaccines promoted by Bill Gates. Vaccine justice campaigner Robert Kennedy Jr, son of assassinated (by the Cult) US Attorney General Robert Kennedy, wrote:

In 2017, the World Health Organization (WHO) reluctantly admitted that the global explosion in polio is predominantly vaccine strain. The most frightening epidemics in Congo, Afghanistan, and the Philippines, are all linked to vaccines. In fact, by 2018, 70% of global polio cases were vaccine strain.

Vaccines make fortunes for Cult-owned Gates and Big Pharma while undermining the health and immune systems of the population. We had a glimpse of the mentality behind the Big Pharma cartel with a report on WION (World is One News), an international English language TV station based in India, which exposed the extraordinary behaviour of US drug company Pfizer over its 'Covid vaccine'. The WION report told how Pfizer had made fantastic demands of Argentina, Brazil and other countries in return for its 'vaccine'. These included immunity from prosecution, even for Pfizer negligence, government insurance to protect Pfizer from law suits and handing over as collateral sovereign assets of the country to include Argentina's bank reserves, military bases and embassy buildings. Pfizer demanded the same of Brazil in the form of waiving sovereignty of its assets abroad; exempting Pfizer from Brazilian laws; and giving Pfizer immunity from all civil liability. This is a 'vaccine' developed with government funding. Big Pharma is evil incarnate as a creation of the Cult and all must be handed tickets to Nuremberg.

Phantom 'vaccine' for a phantom 'disease'

I'll expose the 'Covid vaccine' fraud and then go on to the wider background of why the Cult has set out to 'vaccinate' every man, woman and child on the planet for an alleged 'new disease' with a survival rate of 99.77 percent (or more) even by the grotesquely-

manipulated figures of the World Health Organization and Johns Hopkins University. The 'infection' to 'death' ratio is 0.23 to 0.15 percent according to Stanford epidemiologist Dr John Ioannidis and while estimates vary the danger remains tiny. I say that if the truth be told the fake infection to fake death ratio is zero. Never mind all the evidence I have presented here and in *The Answer* that there is no 'virus' let us just focus for a moment on that death-rate figure of say 0.23 percent. The figure includes all those worldwide who have tested positive with a test not testing for the 'virus' and then died within 28 days or even longer of any other cause – *any other cause*. Now subtract all those illusory 'Covid' deaths on the global data sheets from the 0.23 percent. What do you think you would be left with? *Zero*. A vaccination has never been successfully developed for a so-called coronavirus. They have all failed at the animal testing stage when they caused hypersensitivity to what they were claiming to protect against and made the impact of a disease far worse. Cult-owned vaccine corporations got around that problem this time by bypassing animal trials, going straight to humans and making the length of the 'trials' before the public rollout as short as they could get away with. Normally it takes five to ten years or more to develop vaccines that still cause demonstrable harm to many people and that's without including the long-term effects that are never officially connected to the vaccination. 'Covid' non-vaccines have been officially produced and approved in a matter of months from a standing start and part of the reason is that (a) they were developed before the 'Covid' hoax began and (b) they are based on computer programs and not natural sources. Official non-trials were so short that government agencies gave *emergency*, not full, approval. 'Trials' were not even completed and full approval cannot be secured until they are. Public 'Covid vaccination' is actually a *continuation of the trial*. Drug company 'trials' are not scheduled to end until 2023 by which time a lot of people are going to be dead. Data on which government agencies gave this emergency approval was supplied by the Big Pharma corporations themselves in the form of Pfizer/BioNTech, AstraZeneca, Moderna, Johnson & Johnson, and

others, and this is the case with all vaccines. By its very nature *emergency* approval means drug companies do not have to prove that the 'vaccine' is 'safe and effective'. How could they with trials way short of complete? Government regulators only have to *believe* that they *could* be safe and effective. It is criminal manipulation to get products in circulation with no testing worth the name. Agencies giving that approval are infested with Big Pharma-connected place-people and they act in the interests of Big Pharma (the Cult) and not the public about whom they do not give a damn.

More human lab rats

'Covid vaccines' produced in record time by Pfizer/BioNTech and Moderna employ a technique *never approved before for use on humans*. They are known as mRNA 'vaccines' and inject a synthetic version of 'viral' mRNA or 'messenger RNA'. The key is in the term 'messenger'. The body works, or doesn't, on the basis of information messaging. Communications are constantly passing between and within the genetic system and the brain. Change those messages and you change the state of the body and even its very nature and you can change psychology and behaviour by the way the brain processes information. I think you are going to see significant changes in personality and perception of many people who have had the 'Covid vaccine' synthetic potions. Insider Aldous Huxley predicted the following in 1961 and mRNA 'vaccines' can be included in the term 'pharmacological methods':

There will be, in the next generation or so, a pharmacological method of making people love their servitude, and producing dictatorship without tears, so to speak, producing a kind of painless concentration camp for entire societies, so that people will in fact have their own liberties taken away from them, but rather enjoy it, because they will be distracted from any desire to rebel by propaganda or brainwashing, or brainwashing enhanced by pharmacological methods. And this seems to be the final revolution.

Apologists claim that mRNA synthetic 'vaccines' don't change the DNA genetic blueprint because RNA does not affect DNA only the other way round. This is so disingenuous. A process called 'reverse

transcription' can convert RNA into DNA and be integrated into DNA in the cell nucleus. This was highlighted in December, 2020, by scientists at Harvard and Massachusetts Institute of Technology (MIT). Geneticists report that more than 40 percent of mammalian genomes results from reverse transcription. On the most basic level if messaging changes then that sequence must lead to changes in DNA which is receiving and transmitting those communications. How can introducing synthetic material into cells not change the cells where DNA is located? The process is known as transfection which is defined as 'a technique to insert foreign nucleic acid (DNA or RNA) into a cell, typically with the intention of altering the properties of the cell'. Researchers at the Sloan Kettering Institute in New York found that changes in messenger RNA can deactivate tumour-suppressing proteins and thereby promote cancer. This is what happens when you mess with messaging. 'Covid vaccine' maker Moderna was founded in 2010 by Canadian stem cell biologist Derrick J. Rossi after his breakthrough discovery in the field of transforming and reprogramming stem cells. These are neutral cells that can be programmed to become any cell including sperm cells. Moderna was therefore founded on the principle of genetic manipulation and has never produced any vaccine or drug before its genetically-manipulating synthetic 'Covid' shite. Look at the name – Mode-RNA or Modify-RNA. Another important point is that the US Supreme Court has ruled that genetically-modified DNA, or complementary DNA (cDNA) synthesized in the laboratory from messenger RNA, can be patented and owned. These psychopaths are doing this to the human body.

Cells replicate synthetic mRNA in the 'Covid vaccines' and in theory the body is tricked into making antigens which trigger antibodies to target the 'virus spike proteins' which as Dr Tom Cowan said have *never been seen*. Cut the crap and these 'vaccines' deliver *self-replicating* synthetic material to the cells with the effect of changing human DNA. The more of them you have the more that process is compounded while synthetic material is all the time self-replicating. 'Vaccine'-maker Moderna describes mRNA as 'like

software for the cell' and so they are messing with the body's software. What happens when you change the software in a computer? Everything changes. For this reason the Cult is preparing a production line of mRNA 'Covid vaccines' and a long list of excuses to use them as with all the 'variants' of a 'virus' never shown to exist. The plan is further to transfer the mRNA technique to other vaccines mostly given to children and young people. The cumulative consequences will be a transformation of human DNA through a constant infusion of synthetic genetic material which will kill many and change the rest. Now consider that governments that have given emergency approval for a vaccine that's not a vaccine; never been approved for humans before; had no testing worth the name; and the makers have been given immunity from prosecution for any deaths or adverse effects suffered by the public. The UK government awarded *permanent legal indemnity* to itself and its employees for harm done when a patient is being treated for 'Covid-19' or 'suspected Covid-19'. That is quite a thought when these are possible 'side-effects' from the 'vaccine' (they are not 'side', they are effects) listed by the US Food and Drug Administration:

Guillain-Barre syndrome; acute disseminated encephalomyelitis; transverse myelitis; encephalitis; myelitis; encephalomyelitis; meningoencephalitis; meningitis; encephalopathy; convulsions; seizures; stroke; narcolepsy; cataplexy; anaphylaxis; acute myocardial infarction (heart attack); myocarditis; pericarditis; autoimmune disease; death; implications for pregnancy, and birth outcomes; other acute demyelinating diseases; non anaphylactic allergy reactions; thrombocytopenia ; disseminated intravascular coagulation; venous thromboembolism; arthritis; arthralgia; joint pain; Kawasaki disease; multisystem inflammatory syndrome in children; vaccine enhanced disease. The latter is the way the 'vaccine' has the potential to make diseases far worse than they would otherwise be.

UK doctor and freedom campaigner Vernon Coleman described the conditions in this list as 'all unpleasant, most of them very serious, and you can't get more serious than death'. The thought that anyone at all has had the 'vaccine' in these circumstances is testament to the potential that humanity has for clueless, unquestioning, stupidity and for many that programmed stupidity has already been terminal.

An insider speaks

Dr Michael Yeadon is a former Vice President, head of research and Chief Scientific Adviser at vaccine giant Pfizer. Yeadon worked on the inside of Big Pharma, but that did not stop him becoming a vocal critic of 'Covid vaccines' and their potential for multiple harms, including infertility in women. By the spring of 2021 he went much further and even used the no, no, term 'conspiracy'. When you begin to see what is going on it is impossible not to do so. Yeadon spoke out in an interview with freedom campaigner James Delingpole and I mentioned earlier how he said that no one had samples of 'the virus'. He explained that the mRNA technique originated in the anti-cancer field and ways to turn on and off certain genes which could be advantageous if you wanted to stop cancer growing out of control. 'That's the origin of them. They are a very unusual application, really.' Yeadon said that treating a cancer patient with an aggressive procedure might be understandable if the alternative was dying, but it was quite another thing to use the same technique as a public health measure. Most people involved wouldn't catch the infectious agent you were vaccinating against and if they did they probably wouldn't die:

If you are really using it as a public health measure you really want to as close as you can get to zero side-effects ... I find it odd that they chose techniques that were really cutting their teeth in the field of oncology and I'm worried that in using gene-based vaccines that have to be injected in the body and spread around the body, get taken up into some cells, and the regulators haven't quite told us which cells they get taken up into ... you are going to be generating a wide range of responses ... with multiple steps each of which could go well or badly.

I doubt the Cult intends it to go well. Yeadon said that you can put any gene you like into the body through the 'vaccine'. 'You can certainly give them a gene that would do them some harm if you wanted.' I was intrigued when he said that when used in the cancer field the technique could turn genes on and off. I explore this process in *The Answer* and with different genes having different functions you could create mayhem – physically and psychologically – if you turned the wrong ones on and the right ones off. I read reports of an experiment by researchers at the University of Washington's school of computer science and engineering in which they encoded DNA to infect computers. The body is itself a biological computer and if human DNA can inflict damage on a computer why can't the computer via synthetic material mess with the human body? It can. The Washington research team said it was possible to insert malicious malware into 'physical DNA strands' and corrupt the computer system of a gene sequencing machine as it 'reads gene letters and stores them as binary digits 0 and 1'. They concluded that hackers could one day use blood or spit samples to access computer systems and obtain sensitive data from police forensics labs or infect genome files. It is at this level of digital interaction that synthetic 'vaccines' need to be seen to get the full picture and that will become very clear later on. Michael Yeadon said it made no sense to give the 'vaccine' to younger people who were in no danger from the 'virus'. What was the benefit? It was all downside with potential effects:

The fact that my government in what I thought was a civilised, rational country, is raining [the 'vaccine'] on people in their 30s and 40s, even my children in their 20s, they're getting letters and phone calls, I know this is not right and any of you doctors who are vaccinating you know it's not right, too. They are not at risk. They are not at risk from the disease, so you are now hoping that the side-effects are so rare that you get away with it. You don't give new technology ... that you don't understand to 100 percent of the population.

Blood clot problems with the AstraZeneca 'vaccine' have been affecting younger people to emphasise the downside risks with no benefit. AstraZeneca's version, produced with Oxford University, does not use mRNA, but still gets its toxic cocktail inside cells where

it targets DNA. The Johnson & Johnson 'vaccine' which uses a similar technique has also produced blood clot effects to such an extent that the United States paused its use at one point. They are all 'gene therapy' (cell modification) procedures and not 'vaccines'. The truth is that once the content of these injections enter cells we have no idea what the effect will be. People can speculate and some can give very educated opinions and that's good. In the end, though, only the makers know what their potions are designed to do and even they won't know every last consequence. Michael Yeadon was scathing about doctors doing what they knew to be wrong. 'Everyone's mute', he said. Doctors in the NHS must know this was not right, coming into work and injecting people. 'I don't know how they sleep at night. I know I couldn't do it. I know that if I were in that position I'd have to quit.' He said he knew enough about toxicology to know this was not a good risk-benefit. Yeadon had spoken to seven or eight university professors and all except two would not speak out publicly. Their universities had a policy that no one said anything that countered the government and its medical advisors. They were afraid of losing their government grants. This is how intimidation has been used to silence the truth at every level of the system. I say silence, but these people could still speak out if they made that choice. Yeadon called them 'moral cowards' – 'This is about your children and grandchildren's lives and you have just buggered off and left it.'

'Variant' nonsense

Some of his most powerful comments related to the alleged 'variants' being used to instil more fear, justify more lockdowns, and introduce more 'vaccines'. He said government claims about 'variants' were nonsense. He had checked the alleged variant 'codes' and they were 99.7 percent identical to the 'original'. This was the human identity difference equivalent to putting a baseball cap on and off or wearing it the other way round. A 0.3 percent difference would make it impossible for that 'variant' to escape immunity from the 'original'. This made no sense of having new 'vaccines' for

‘variants’. He said there would have to be at least a *30 percent* difference for that to be justified and even then he believed the immune system would still recognise what it was. Gates-funded ‘variant modeller’ and ‘vaccine’-pusher John Edmunds might care to comment. Yeadon said drug companies were making new versions of the ‘vaccine’ as a ‘top up’ for ‘variants’. Worse than that, he said, the ‘regulators’ around the world like the MHRA in the UK had got together and agreed that because ‘vaccines’ for ‘variants’ were so similar to the first ‘vaccines’ *they did not have to do safety studies*. How transparently sinister that is. This is when Yeadon said: ‘There is a conspiracy here.’ There was no need for another vaccine for ‘variants’ and yet we were told that there was and the country had shut its borders because of them. ‘They are going into hundreds of millions of arms without passing ‘go’ or any regulator. Why did they do that? Why did they pick this method of making the vaccine?’

The reason had to be something bigger than that it seemed and ‘it’s not protection against the virus’. It’s was a far bigger project that meant politicians and advisers were willing to do things and not do things that knowingly resulted in avoidable deaths – ‘that’s already happened when you think about lockdown and deprivation of health care for a year.’ He spoke of people prepared to do something that results in the avoidable death of their fellow human beings and it not bother them. This is the penny-drop I have been working to get across for more than 30 years – the level of pure evil we are dealing with. Yeadon said his friends and associates could not believe there could be that much evil, but he reminded them of Stalin, Pol Pot and Hitler and of what Stalin had said: ‘One death is a tragedy. A million? A statistic.’ He could not think of a benign explanation for why you need top-up vaccines ‘which I’m sure you don’t’ and for the regulators ‘to just get out of the way and wave them through’. Why would the regulators do that when they were still wrestling with the dangers of the ‘parent’ vaccine? He was clearly shocked by what he had seen since the ‘Covid’ hoax began and now he was thinking the previously unthinkable:

If you wanted to depopulate a significant proportion of the world and to do it in a way that doesn't involve destruction of the environment with nuclear weapons, poisoning everyone with anthrax or something like that, and you wanted plausible deniability while you had a multi-year infectious disease crisis, I actually don't think you could come up with a better plan of work than seems to be in front of me. I can't say that's what they are going to do, but I can't think of a benign explanation why they are doing it.

He said he never thought that they would get rid of 99 percent of humans, but now he wondered. 'If you wanted to that this would be a hell of a way to do it – it would be unstoppable folks.' Yeadon had concluded that those who submitted to the 'vaccine' would be allowed to have some kind of normal life (but for how long?) while screws were tightened to coerce and mandate the last few percent. 'I think they'll put the rest of them in a prison camp. I wish I was wrong, but I don't think I am.' Other points he made included: There were no coronavirus vaccines then suddenly they all come along at the same time; we have no idea of the long term affect with trials so short; coercing or forcing people to have medical procedures is against the Nuremberg Code instigated when the Nazis did just that; people should at least delay having the 'vaccine'; a quick Internet search confirms that masks don't reduce respiratory viral transmission and 'the government knows that'; they have smashed civil society and they know that, too; two dozen peer-reviewed studies show no connection between lockdown and reducing deaths; he knew from personal friends the elite were still flying around and going on holiday while the public were locked down; the elite were not having the 'vaccines'. He was also asked if 'vaccines' could be made to target difference races. He said he didn't know, but the document by the Project for the New American Century in September, 2000, said developing 'advanced forms of biological warfare that can target *specific genotypes* may transform biological warfare from the realm of terror to a politically useful tool.' Oh, they're evil all right. Of that we can be *absolutely* sure.

Another cull of old people

We have seen from the CDC definition that the mRNA 'Covid vaccine' is not a vaccine and nor are the others that *claim* to reduce 'severity of symptoms' in *some* people, but not protect from infection or transmission. What about all the lies about returning to 'normal' if people were 'vaccinated'? If they are not claimed to stop infection and transmission of the alleged 'virus', how does anything change? This was all lies to manipulate people to take the jabs and we are seeing that now with masks and distancing still required for the 'vaccinated'. How did they think that elderly people with fragile health and immune responses were going to be affected by infusing their cells with synthetic material and other toxic substances? They *knew* that in the short and long term it would be devastating and fatal as the culling of the old that began with the first lockdowns was continued with the 'vaccine'. Death rates in care homes soared immediately residents began to be 'vaccinated' – infused with synthetic material. Brave and committed whistleblower nurses put their careers at risk by exposing this truth while the rest kept their heads down and their mouths shut to put their careers before those they are supposed to care for. A long-time American Certified Nursing Assistant who gave his name as James posted a video in which he described emotionally what happened in his care home when vaccination began. He said that during 2020 very few residents were sick with 'Covid' and no one died during the entire year; but shortly after the Pfizer mRNA injections 14 people died within two weeks and many others were near death. 'They're dropping like flies', he said. Residents who walked on their own before the shot could no longer and they had lost their ability to conduct an intelligent conversation. The home's management said the sudden deaths were caused by a 'super-spreader' of 'Covid-19'. Then how come, James asked, that residents who refused to take the injections were not sick? It was a case of inject the elderly with mRNA synthetic potions and blame their illness and death that followed on the 'virus'. James described what was happening in care homes as 'the greatest crime of genocide this country has ever seen'. Remember the NHS staff nurse from earlier who used the same

word 'genocide' for what was happening with the 'vaccines' and that it was an 'act of human annihilation'. A UK care home whistleblower told a similar story to James about the effect of the 'vaccine' in deaths and 'outbreaks' of illness dubbed 'Covid' after getting the jab. She told how her care home management and staff had zealously imposed government regulations and no one was allowed to even question the official narrative let alone speak out against it. She said the NHS was even worse. Again we see the results of reframing. A worker at a local care home where I live said they had not had a single case of 'Covid' there for almost a year and when the residents were 'vaccinated' they had 19 positive cases in two weeks with eight dying.

It's not the 'vaccine' – honest

The obvious cause and effect was being ignored by the media and most of the public. Australia's health minister Greg Hunt (a former head of strategy at the World Economic Forum) was admitted to hospital after he had the 'vaccine'. He was suffering according to reports from the skin infection 'cellulitis' and it must have been a severe case to have warranted days in hospital. Immediately the authorities said this was nothing to do with the 'vaccine' when an effect of some vaccines is a 'cellulitis-like reaction'. We had families of perfectly healthy old people who died after the 'vaccine' saying that if only they had been given the 'vaccine' earlier they would still be alive. As a numbskull rating that is off the chart. A father of four 'died of Covid' at aged 48 when he was taken ill two days after having the 'vaccine'. The man, a health administrator, had been 'shielding during the pandemic' and had 'not really left the house' until he went for the 'vaccine'. Having the 'vaccine' and then falling ill and dying does not seem to have qualified as a possible cause and effect and 'Covid-19' went on his death certificate. His family said they had no idea how he 'caught the virus'. A family member said: 'Tragically, it could be that going for a vaccination ultimately led to him catching Covid ...The sad truth is that they are never going to know where it came from.' The family warned people to remember

that the virus still existed and was 'very real'. So was their stupidity. Nurses and doctors who had the first round of the 'vaccine' were collapsing, dying and ending up in a hospital bed while they or their grieving relatives were saying they'd still have the 'vaccine' again despite what happened. I kid you not. You mean if your husband returned from the dead he'd have the same 'vaccine' again that killed him??

Doctors at the VCU Medical Center in Richmond, Virginia, said the Johnson & Johnson 'vaccine' was to blame for a man's skin peeling off. Patient Richard Terrell said: 'It all just happened so fast. My skin peeled off. It's still coming off on my hands now.' He said it was stinging, burning and itching and when he bent his arms and legs it was very painful with 'the skin swollen and rubbing against itself'. Pfizer/BioNTech and Moderna vaccines use mRNA to change the cell while the Johnson & Johnson version uses DNA in a process similar to AstraZeneca's technique. Johnson & Johnson and AstraZeneca have both had their 'vaccines' paused by many countries after causing serious blood problems. Terrell's doctor Fnu Nutan said he could have died if he hadn't got medical attention. It sounds terrible so what did Nutan and Terrell say about the 'vaccine' now? Oh, they still recommend that people have it. A nurse in a hospital bed 40 minutes after the vaccination and unable to swallow due to throat swelling was told by a doctor that he lost mobility in his arm for 36 hours following the vaccination. What did he say to the ailing nurse? 'Good for you for getting the vaccination.' We are dealing with a serious form of cognitive dissonance madness in both public and medical staff. There is a remarkable correlation between those having the 'vaccine' and trumpeting the fact and suffering bad happenings shortly afterwards. Witold Rogiewicz, a Polish doctor, made a video of his 'vaccination' and ridiculed those who were questioning its safety and the intentions of Bill Gates: 'Vaccinate yourself to protect yourself, your loved ones, friends and also patients. And to mention quickly I have info for anti-vaxxers and anti-Covidors if you want to contact Bill Gates you can do this through me.' He further ridiculed the dangers of 5G. Days later he

was dead, but naturally the vaccination wasn't mentioned in the verdict of 'heart attack'.

Lies, lies and more lies

So many members of the human race have slipped into extreme states of insanity and unfortunately they include reframed doctors and nursing staff. Having a 'vaccine' and dying within minutes or hours is not considered a valid connection while death from any cause within 28 days or longer of a positive test with a test not testing for the 'virus' means 'Covid-19' goes on the death certificate. How could that 'vaccine'-death connection not have been made except by calculated deceit? US figures in the initial rollout period to February 12th, 2020, revealed that a third of the deaths reported to the CDC after 'Covid vaccines' happened within 48 hours. Five men in the UK suffered an 'extremely rare' blood clot problem after having the AstraZeneca 'vaccine', but no causal link was established said the Gates-funded Medicines and Healthcare products Regulatory Agency (MHRA) which had given the 'vaccine' emergency approval to be used. Former Pfizer executive Dr Michael Yeadon explained in his interview how the procedures could cause blood coagulation and clots. People who should have been at no risk were dying from blood clots in the brain and he said he had heard from medical doctor friends that people were suffering from skin bleeding and massive headaches. The AstraZeneca 'shot' was stopped by some 20 countries over the blood clotting issue and still the corrupt MHRA, the European Medicines Agency (EMA) and the World Health Organization said that it should continue to be given even though the EMA admitted that it 'still cannot rule out definitively' a link between blood clotting and the 'vaccine'. Later Marco Cavaleri, head of EMA vaccine strategy, said there was indeed a clear link between the 'vaccine' and thrombosis, but they didn't know why. So much for the trials showing the 'vaccine' is safe. Blood clots were affecting younger people who would be under virtually no danger from 'Covid' even if it existed which makes it all the more stupid and sinister.

The British government responded to public alarm by wheeling out June Raine, the terrifyingly weak infant school headmistress sound-alike who heads the UK MHRA drug 'regulator'. The idea that she would stand up to Big Pharma and government pressure is laughable and she told us that all was well in the same way that she did when allowing untested, never-used-on-humans-before, genetically-manipulating 'vaccines' to be exposed to the public in the first place. Mass lying is the new normal of the 'Covid' era. The MHRA later said 30 cases of rare blood clots had by then been connected with the AstraZeneca 'vaccine' (that means a lot more in reality) while stressing that the benefits of the jab in preventing 'Covid-19' outweighed any risks. A more ridiculous and disingenuous statement with callous disregard for human health it is hard to contemplate. Immediately after the mendacious 'all-clears' two hospital workers in Denmark experienced blood clots and cerebral haemorrhaging following the AstraZeneca jab and one died. Top Norwegian health official Pål Andre Holme said the 'vaccine' was the only common factor: 'There is nothing in the patient history of these individuals that can give such a powerful immune response ... I am confident that the antibodies that we have found are the cause, and I see no other explanation than it being the vaccine which triggers it.' Strokes, a clot or bleed in the brain, were clearly associated with the 'vaccine' from word of mouth and whistleblower reports. Similar consequences followed with all these 'vaccines' that we were told were so safe and as the numbers grew by the day it was clear we were witnessing human carnage.

Learning the hard way

A woman interviewed by UKColumn told how her husband suffered dramatic health effects after the vaccine when he'd been in good health all his life. He went from being a little unwell to losing all feeling in his legs and experiencing 'excruciating pain'. Misdiagnosis followed twice at Accident and Emergency (an 'allergy' and 'sciatica') before he was admitted to a neurology ward where doctors said his serious condition had been caused by the

'vaccine'. Another seven 'vaccinated' people were apparently being treated on the same ward for similar symptoms. The woman said he had the 'vaccine' because they believed media claims that it was safe. 'I didn't think the government would give out a vaccine that does this to somebody; I believed they would be bringing out a vaccination that would be safe.' What a tragic way to learn that lesson. Another woman posted that her husband was transporting stroke patients to hospital on almost every shift and when he asked them if they had been 'vaccinated' for 'Covid' they all replied 'yes'. One had a 'massive brain bleed' the day after his second dose. She said her husband reported the 'just been vaccinated' information every time to doctors in A and E only for them to ignore it, make no notes and appear annoyed that it was even mentioned. This particular report cannot be verified, but it expresses a common theme that confirms the monumental underreporting of 'vaccine' consequences. Interestingly as the 'vaccines' and their brain blood clot/stroke consequences began to emerge the UK National Health Service began a publicity campaign telling the public what to do in the event of a stroke. A Scottish NHS staff nurse who quit in disgust in March, 2021, said:

I have seen traumatic injuries from the vaccine, they're not getting reported to the yellow card [adverse reaction] scheme, they're treating the symptoms, not asking why, why it's happening. It's just treating the symptoms and when you speak about it you're dismissed like you're crazy, I'm not crazy, I'm not crazy because every other colleague I've spoken to is terrified to speak out, they've had enough.

Videos appeared on the Internet of people uncontrollably shaking after the 'vaccine' with no control over muscles, limbs and even their face. A Scottish mother broke out in a severe rash all over her body almost immediately after she was given the AstraZeneca 'vaccine'. The pictures were horrific. Leigh King, a 41-year-old hairdresser from Lanarkshire said: 'Never in my life was I prepared for what I was about to experience ... My skin was so sore and constantly hot ... I have never felt pain like this ...' But don't you worry, the 'vaccine' is perfectly safe. Then there has been the effect on medical

staff who have been pressured to have the 'vaccine' by psychopathic 'health' authorities and government. A London hospital consultant who gave the name K. Polyakova wrote this to the *British Medical Journal* or *BMJ*:

I am currently struggling with ... the failure to report the reality of the morbidity caused by our current vaccination program within the health service and staff population. The levels of sickness after vaccination is unprecedented and staff are getting very sick and some with neurological symptoms which is having a huge impact on the health service function. Even the young and healthy are off for days, some for weeks, and some requiring medical treatment. Whole teams are being taken out as they went to get vaccinated together.

Mandatory vaccination in this instance is stupid, unethical and irresponsible when it comes to protecting our staff and public health. We are in the voluntary phase of vaccination, and encouraging staff to take an unlicensed product that is impacting on their immediate health ... it is clearly stated that these vaccine products do not offer immunity or stop transmission. In which case why are we doing it?

Not to protect health that's for sure. Medical workers are lauded by governments for agenda reasons when they couldn't give a toss about them any more than they can for the population in general. Schools across America faced the same situation as they closed due to the high number of teachers and other staff with bad reactions to the Pfizer/BioNTech, Moderna, and Johnson & Johnson 'Covid vaccines' all of which were linked to death and serious adverse effects. The *BMJ* took down the consultant's comments pretty quickly on the grounds that they were being used to spread 'disinformation'. They were exposing the truth about the 'vaccine' was the real reason. The cover-up is breathtaking.

Hiding the evidence

The scale of the 'vaccine' death cover-up worldwide can be confirmed by comparing official figures with the personal experience of the public. I heard of many people in my community who died immediately or soon after the vaccine that would never appear in the media or even likely on the official totals of 'vaccine' fatalities and adverse reactions when only about ten percent are estimated to be

reported and I have seen some estimates as low as one percent in a Harvard study. In the UK alone by April 29th, 2021, some 757,654 adverse reactions had been officially reported from the Pfizer/BioNTech, Oxford/AstraZeneca and Moderna 'vaccines' with more than a thousand deaths linked to jabs and that means an estimated ten times this number in reality from a ten percent reporting rate percentage. That's seven million adverse reactions and 10,000 potential deaths and a one percent reporting rate would be ten times *those* figures. In 1976 the US government pulled the swine flu vaccine after 53 deaths. The UK data included a combined 10,000 eye disorders from the 'Covid vaccines' with more than 750 suffering visual impairment or blindness and again multiply by the estimated reporting percentages. As 'Covid cases' officially fell hospitals virtually empty during the 'Covid crisis' began to fill up with a range of other problems in the wake of the 'vaccine' rollout. The numbers across America have also been catastrophic. Deaths linked to *all* types of vaccine increased by 6,000 *percent* in the first quarter of 2021 compared with 2020. A 39-year-old woman from Ogden, Utah, died four days after receiving a second dose of Moderna's 'Covid vaccine' when her liver, heart and kidneys all failed despite the fact that she had no known medical issues or conditions. Her family sought an autopsy, but Dr Erik Christensen, Utah's chief medical examiner, said proving vaccine injury as a cause of death almost never happened. He could think of only one instance where an autopsy would name a vaccine as the official cause of death and that would be anaphylaxis where someone received a vaccine and died almost instantaneously. 'Short of that, it would be difficult for us to definitively say this is the vaccine,' Christensen said. If that is true this must be added to the estimated ten percent (or far less) reporting rate of vaccine deaths and serious reactions and the conclusion can only be that vaccine deaths and serious reactions – including these 'Covid' potions' – are phenomenally understated in official figures. The same story can be found everywhere. Endless accounts of deaths and serious reactions among the public, medical

and care home staff while official figures did not even begin to reflect this.

Professional script-reader Dr David Williams, a 'top public-health official' in Ontario, Canada, insulted our intelligence by claiming only four serious adverse reactions and no deaths from the more than 380,000 vaccine doses then given. This bore no resemblance to what people knew had happened in their own circles and we had Dirk Huyer in charge of getting millions vaccinated in Ontario while at the same time he was Chief Coroner for the province investigating causes of death including possible death from the vaccine. An aide said he had stepped back from investigating deaths, but evidence indicated otherwise. Rosemary Frei, who secured a Master of Science degree in molecular biology at the Faculty of Medicine at Canada's University of Calgary before turning to investigative journalism, was one who could see that official figures for 'vaccine' deaths and reactions made no sense. She said that doctors seldom reported adverse events and when people got really sick or died after getting a vaccination they would attribute that to anything except the vaccines. It had been that way for years and anyone who wondered aloud whether the 'Covid vaccines' or other shots cause harm is immediately branded as 'anti-vax' and 'anti-science'. This was 'career-threatening' for health professionals. Then there was the huge pressure to support the push to 'vaccinate' billions in the quickest time possible. Frei said:

So that's where we're at today. More than half a million vaccine doses have been given to people in Ontario alone. The rush is on to vaccinate all 15 million of us in the province by September. And the mainstream media are screaming for this to be sped up even more. That all adds up to only a very slim likelihood that we're going to be told the truth by officials about how many people are getting sick or dying from the vaccines.

What is true of Ontario is true of everywhere.

They KNEW – and still did it

The authorities knew what was going to happen with multiple deaths and adverse reactions. The UK government's Gates-funded

and Big Pharma-dominated Medicines and Healthcare products Regulatory Agency (MHRA) hired a company to employ AI in compiling the projected reactions to the 'vaccine' that would otherwise be uncountable. The request for applications said: 'The MHRA urgently seeks an Artificial Intelligence (AI) software tool to process the expected high volume of Covid-19 vaccine Adverse Drug Reaction ...' This was from the agency, headed by the disingenuous June Raine, that gave the 'vaccines' emergency approval and the company was hired before the first shot was given. 'We are going to kill and maim you – is that okay?' 'Oh, yes, perfectly fine – I'm very grateful, thank you, doctor.' The range of 'Covid vaccine' adverse reactions goes on for page after page in the MHRA criminally underreported 'Yellow Card' system and includes affects to eyes, ears, skin, digestion, blood and so on. Raine's MHRA amazingly claimed that the 'overall safety experience ... is so far as expected from the clinical trials'. The death, serious adverse effects, deafness and blindness were *expected*? When did they ever mention that? If these human tragedies were expected then those that gave approval for the use of these 'vaccines' must be guilty of crimes against humanity including murder – a definition of which is 'killing a person with malice aforethought or with recklessness manifesting extreme indifference to the value of human life.' People involved at the MHRA, the CDC in America and their equivalent around the world must go before Nuremberg trials to answer for their callous inhumanity. We are only talking here about the immediate effects of the 'vaccine'. The longer-term impact of the DNA synthetic manipulation is the main reason they are so hysterically desperate to inoculate the entire global population in the shortest possible time.

Africa and the developing world are a major focus for the 'vaccine' depopulation agenda and a mass vaccination sales-pitch is underway thanks to caring people like the Rockefellers and other Cult assets. The Rockefeller Foundation, which pre-empted the 'Covid pandemic' in a document published in 2010 that 'predicted' what happened a decade later, announced an initial \$34.95 million grant in February, 2021, 'to ensure more equitable access to Covid-19

testing and vaccines' among other things in Africa in collaboration with '24 organizations, businesses, and government agencies'. The pan-Africa initiative would focus on 10 countries: Burkina Faso, Ethiopia, Ghana, Kenya, Nigeria, Rwanda, South Africa, Tanzania, Uganda, and Zambia'. Rajiv Shah, President of the Rockefeller Foundation and former administrator of CIA-controlled USAID, said that if Africa was not mass-vaccinated (to change the DNA of its people) it was a 'threat to all of humanity' and not fair on Africans. When someone from the Rockefeller Foundation says they want to do something to help poor and deprived people and countries it is time for a belly-laugh. They are doing this out of the goodness of their 'heart' because 'vaccinating' the entire global population is what the 'Covid' hoax set out to achieve. Official 'decolonisation' of Africa by the Cult was merely a prelude to financial colonisation on the road to a return to physical colonisation. The 'vaccine' is vital to that and the sudden and convenient death of the 'Covid' sceptic president of Tanzania can be seen in its true light. A lot of people in Africa are aware that this is another form of colonisation and exploitation and they need to stand their ground.

The 'vaccine is working' scam

A potential problem for the Cult was that the 'vaccine' is meant to change human DNA and body messaging and not to protect anyone from a 'virus' never shown to exist. The vaccine couldn't work because it was not designed to work and how could they make it *appear* to be working so that more people would have it? This was overcome by lowering the amplification rate of the PCR test to produce fewer 'cases' and therefore fewer 'deaths'. Some of us had been pointing out since March, 2020, that the amplification rate of the test not testing for the 'virus' had been made artificially high to generate positive tests which they could call 'cases' to justify lockdowns. The World Health Organization recommended an absurdly high 45 amplification cycles to ensure the high positives required by the Cult and then remained silent on the issue until January 20th, 2021 – Biden's Inauguration Day. This was when the

'vaccinations' were seriously underway and on that day the WHO recommended after discussions with America's CDC that laboratories *lowered their testing amplification*. Dr David Samadi, a certified urologist and health writer, said the WHO was encouraging all labs to reduce their cycle count for PCR tests. He said the current cycle was much too high and was 'resulting in any particle being declared a positive case'. Even one mainstream news report I saw said this meant the number of 'Covid' infections may have been 'dramatically inflated'. Oh, just a little bit. The CDC in America issued new guidance to laboratories in April, 2021, to use 28 cycles *but only for 'vaccinated' people*. The timing of the CDC/WHO interventions were cynically designed to make it appear the 'vaccines' were responsible for falling cases and deaths when the real reason can be seen in the following examples. New York's state lab, the Wadsworth Center, identified 872 positive tests in July, 2020, based on a threshold of 40 cycles. When the figure was lowered to 35 cycles 43 percent of the 872 were no longer 'positives'. At 30 cycles the figure was 63 percent. A Massachusetts lab found that between 85 to 90 percent of people who tested positive in July with a cycle threshold of 40 would be negative at 30 cycles, Ashish Jha, MD, director of the Harvard Global Health Institute, said: 'I'm really shocked that it could be that high ... Boy, does it really change the way we need to be thinking about testing.' I'm shocked that I could see the obvious in the spring of 2020, with no medical background, and most medical professionals still haven't worked it out. No, that's not shocking – it's terrifying.

Three weeks after the WHO directive to lower PCR cycles the London *Daily Mail* ran this headline: 'Why ARE Covid cases plummeting? New infections have fallen 45% in the US and 30% globally in the past 3 weeks but experts say vaccine is NOT the main driver because only 8% of Americans and 13% of people worldwide have received their first dose.' They acknowledged that the drop could not be attributed to the 'vaccine', but soon this morphed throughout the media into the 'vaccine' has caused cases and deaths to fall when it was the PCR threshold. In December, 2020, there was

chaos at English Channel ports with truck drivers needing negative 'Covid' tests before they could board a ferry home for Christmas. The government wanted to remove the backlog as fast as possible and they brought in troops to do the 'testing'. Out of 1,600 drivers just 36 tested positive and the rest were given the all clear to cross the Channel. I guess the authorities thought that 36 was the least they could get away with without the unquestioning catching on. The amplification trick which most people believed in the absence of information in the mainstream applied more pressure on those refusing the 'vaccine' to succumb when it 'obviously worked'. The truth was the exact opposite with deaths in care homes soaring with the 'vaccine' and in Israel the term used was 'skyrocket'. A re-analysis of published data from the Israeli Health Ministry led by Dr Hervé Seligmann at the Medicine Emerging Infectious and Tropical Diseases at Aix-Marseille University found that Pfizer's 'Covid vaccine' killed 'about 40 times more [elderly] people than the disease itself would have killed' during a five-week vaccination period and *260 times* more younger people than would have died from the 'virus' even according to the manipulated 'virus' figures. Dr Seligmann and his co-study author, Haim Yativ, declared after reviewing the Israeli 'vaccine' death data: 'This is a new Holocaust.'

Then, in mid-April, 2021, after vast numbers of people worldwide had been 'vaccinated', the story changed with clear coordination. The UK government began to prepare the ground for more future lockdowns when Nuremberg-destined Boris Johnson told yet another whopper. He said that cases had fallen because of *lockdowns* not 'vaccines'. Lockdowns are irrelevant when *there is no 'virus'* and the test and fraudulent death certificates are deciding the number of 'cases' and 'deaths'. Study after study has shown that lockdowns don't work and instead kill and psychologically destroy people. Meanwhile in the United States Anthony Fauci and Rochelle Walensky, the ultra-Zionist head of the CDC, peddled the same line. More lockdown was the answer and not the 'vaccine', a line repeated on cue by the moron that is Canadian Prime Minister Justin Trudeau. Why all the hysteria to get everyone 'vaccinated' if lockdowns and

not 'vaccines' made the difference? None of it makes sense on the face of it. Oh, but it does. The Cult wants lockdowns *and* the 'vaccine' and if the 'vaccine' is allowed to be seen as the total answer lockdowns would no longer be justified when there are still livelihoods to destroy. 'Variants' and renewed upward manipulation of PCR amplification are planned to instigate never-ending lockdown *and* more 'vaccines'.

You *must* have it – we're desperate

Israel, where the Jewish and Arab population are ruled by the Sabbatian Cult, was the front-runner in imposing the DNA-manipulating 'vaccine' on its people to such an extent that Jewish refusers began to liken what was happening to the early years of Nazi Germany. This would seem to be a fantastic claim. Why would a government of Jewish people be acting like the Nazis did? If you realise that the Sabbatian Cult was behind the Nazis and that Sabbatians hate Jews the pieces start to fit and the question of why a 'Jewish' government would treat Jews with such callous disregard for their lives and freedom finds an answer. Those controlling the government of Israel *aren't Jewish* – they're Sabbatian. Israeli lawyer Tamir Turgal was one who made the Nazi comparison in comments to German lawyer Reiner Fuellmich who is leading a class action lawsuit against the psychopaths for crimes against humanity. Turgal described how the Israeli government was vaccinating children and pregnant women on the basis that there was no evidence that this was dangerous when they had no evidence that it *wasn't* dangerous either. They just had no evidence. This was medical experimentation and Turgal said this breached the Nuremberg Code about medical experimentation and procedures requiring informed consent and choice. Think about that. A Nuremberg Code developed because of Nazi experimentation on Jews and others in concentration camps by people like the evil-beyond-belief Josef Mengele is being breached by the *Israeli* government; but when you know that it's a *Sabbatian* government along with its intelligence and military agencies like Mossad, Shin Bet and the Israeli Defense Forces, and that Sabbatians

were the force behind the Nazis, the kaleidoscope comes into focus. What have we come to when Israeli Jews are suing their government for violating the Nuremberg Code by essentially making Israelis subject to a medical experiment using the controversial 'vaccines'? It's a shocker that this has to be done in the light of what happened in Nazi Germany. The Anshe Ha-Emet, or 'People of the Truth', made up of Israeli doctors, lawyers, campaigners and public, have launched a lawsuit with the International Criminal Court. It says:

When the heads of the Ministry of Health as well as the prime minister presented the vaccine in Israel and began the vaccination of Israeli residents, the vaccinated were not advised, that, in practice, they are taking part in a medical experiment and that their consent is required for this under the Nuremberg Code.

The irony is unbelievable, but easily explained in one word: Sabbatians. The foundation of Israeli 'Covid' apartheid is the 'green pass' or 'green passport' which allows Jews and Arabs who have had the DNA-manipulating 'vaccine' to go about their lives – to work, fly, travel in general, go to shopping malls, bars, restaurants, hotels, concerts, gyms, swimming pools, theatres and sports venues, while non-'vaccinated' are banned from all those places and activities. Israelis have likened the 'green pass' to the yellow stars that Jews in Nazi Germany were forced to wear – the same as the yellow stickers that a branch of UK supermarket chain Morrisons told exempt mask-wearers they had to display when shopping. How very sensitive. The Israeli system is blatant South African-style apartheid on the basis of compliance or non-compliance to fascism rather than colour of the skin. How appropriate that the Sabbatian Israeli government was so close to the pre-Mandela apartheid regime in Pretoria. The Sabbatian-instigated 'vaccine passport' in Israel is planned for everywhere. Sabbatians struck a deal with Pfizer that allowed them to lead the way in the percentage of a national population infused with synthetic material and the result was catastrophic. Israeli freedom activist Shai Dannon told me how chairs were appearing on beaches that said 'vaccinated only'. Health Minister Yuli Edelstein said that anyone unwilling or unable to get

the jabs that 'confer immunity' will be 'left behind'. The man's a liar. Not even the makers claim the 'vaccines' confer immunity. When you see those figures of 'vaccine' deaths these psychopaths were saying that you must take the chance the 'vaccine' will kill you or maim you while knowing it will change your DNA or lockdown for you will be permanent. That's fascism. The Israeli parliament passed a law to allow personal information of the non-vaccinated to be shared with local and national authorities for three months. This was claimed by its supporters to be a way to 'encourage' people to be vaccinated. Hadas Ziv from Physicians for Human Rights described this as a 'draconian law which crushed medical ethics and the patient rights'. But that's the idea, the Sabbatians would reply.

Your papers, please

Sabbatian Israel was leading what has been planned all along to be a global 'vaccine pass' called a 'green passport' without which you would remain in permanent lockdown restriction and unable to do anything. This is how badly – *desperately* – the Cult is to get everyone 'vaccinated'. The term and colour 'green' was not by chance and related to the psychology of fusing the perception of the green climate hoax with the 'Covid' hoax and how the 'solution' to both is the same Great Reset. Lying politicians, health officials and psychologists denied there were any plans for mandatory vaccinations or restrictions based on vaccinations, but they knew that was exactly what was meant to happen with governments of all countries reaching agreements to enforce a global system. 'Free' Denmark and 'free' Sweden unveiled digital vaccine certification. Cyprus, Czech Republic, Estonia, Greece, Hungary, Iceland, Italy, Poland, Portugal, Slovakia, and Spain have all committed to a vaccine passport system and the rest including the whole of the EU would follow. The satanic UK government will certainly go this way despite mendacious denials and at the time of writing it is trying to manipulate the public into having the 'vaccine' so they could go abroad on a summer holiday. How would that work without something to prove you had the synthetic toxicity injected into you?

Documents show that the EU's European Commission was moving towards 'vaccine certificates' in 2018 and 2019 before the 'Covid' hoax began. They knew what was coming. Abracadabra – Ursula von der Leyen, the German President of the Commission, announced in March, 2021, an EU 'Digital Green Certificate' – green again – to track the public's 'Covid status'. The passport sting is worldwide and the Far East followed the same pattern with South Korea ruling that only those with 'vaccination' passports – again the *green* pass – would be able to 'return to their daily lives'.

Bill Gates has been preparing for this 'passport' with other Cult operatives for years and beyond the paper version is a Gates-funded 'digital tattoo' to identify who has been vaccinated and who hasn't. The 'tattoo' is reported to include a substance which is externally readable to confirm who has been vaccinated. This is a bio-luminous light-generating enzyme (think fireflies) called ... *Luciferase*. Yes, named after the Cult 'god' Lucifer the 'light bringer' of whom more to come. Gates said he funded the readable tattoo to ensure children in the developing world were vaccinated and no one was missed out. He cares so much about poor kids as we know. This was just the cover story to develop a vaccine tagging system for everyone on the planet. Gates has been funding the ID2020 'alliance' to do just that in league with other lovely people at Microsoft, GAVI, the Rockefeller Foundation, Accenture and IDEO.org. He said in interviews in March, 2020, before any 'vaccine' publicly existed, that the world must have a globalised digital certificate to track the 'virus' and who had been vaccinated. Gates knew from the start that the mRNA vaccines were coming and when they would come and that the plan was to tag the 'vaccinated' to marginalise the intelligent and stop them doing anything including travel. Evil just doesn't suffice. Gates was exposed for offering a \$10 million bribe to the Nigerian House of Representatives to invoke compulsory 'Covid' vaccination of all Nigerians. Sara Cunial, a member of the Italian Parliament, called Gates a 'vaccine criminal'. She urged the Italian President to hand him over to the International Criminal Court for crimes against

humanity and condemned his plans to 'chip the human race' through ID2020.

You know it's a long-planned agenda when war criminal and Cult gofer Tony Blair is on the case. With the scale of arrogance only someone as dark as Blair can muster he said: 'Vaccination in the end is going to be your route to liberty.' Blair is a disgusting piece of work and he confirms that again. The media has given a lot of coverage to a bloke called Charlie Mullins, founder of London's biggest independent plumbing company, Pimlico Plumbers, who has said he won't employ anyone who has not been vaccinated or have them go to any home where people are not vaccinated. He said that if he had his way no one would be allowed to walk the streets if they have not been vaccinated. Gates was cheering at the time while I was alerting the white coats. The plan is that people will qualify for 'passports' for having the first two doses and then to keep it they will have to have all the follow ups and new ones for invented 'variants' until human genetics is transformed and many are dead who can't adjust to the changes. Hollywood celebrities – the usual propaganda stunt – are promoting something called the WELL Health-Safety Rating to verify that a building or space has 'taken the necessary steps to prioritize the health and safety of their staff, visitors and other stakeholders'. They included Lady Gaga, Jennifer Lopez, Michael B. Jordan, Robert DeNiro, Venus Williams, Wolfgang Puck, Deepak Chopra and 17th Surgeon General Richard Carmona. Yawn. WELL Health-Safety has big connections with China. Parent company Delos is headed by former Goldman Sachs partner Paul Scialla. This is another example – and we will see so many others – of using the excuse of 'health' to dictate the lives and activities of the population. I guess one confirmation of the 'safety' of buildings is that only 'vaccinated' people can go in, right?

Electronic concentration camps

I wrote decades ago about the plans to restrict travel and here we are for those who refuse to bow to tyranny. This can be achieved in one go with air travel if the aviation industry makes a blanket decree.

The 'vaccine' and guaranteed income are designed to be part of a global version of China's social credit system which tracks behaviour 24/7 and awards or deletes 'credits' based on whether your behaviour is supported by the state or not. I mean your entire lifestyle – what you do, eat, say, everything. Once your credit score falls below a certain level consequences kick in. In China tens of millions have been denied travel by air and train because of this. All the locations and activities denied to refusers by the 'vaccine' passports will be included in one big mass ban on doing almost anything for those that don't bow their head to government. It's beyond fascist and a new term is required to describe its extremes – I guess fascist technocracy will have to do. The way the Chinese system of technological – technocratic – control is sweeping the West can be seen in the Los Angeles school system and is planned to be expanded worldwide. Every child is required to have a 'Covid'-tracking app scanned daily before they can enter the classroom. The so-called Daily Pass tracking system is produced by Gates' Microsoft which I'm sure will shock you rigid. The pass will be scanned using a barcode (one step from an inside-the-body barcode) and the information will include health checks, 'Covid' tests and vaccinations. Entry codes are for one specific building only and access will only be allowed if a student or teacher has a negative test with a test not testing for the 'virus', has no symptoms of anything alleged to be related to 'Covid' (symptoms from a range of other illness), and has a temperature under 100 degrees. No barcode, no entry, is planned to be the case for everywhere and not only schools.

Kids are being psychologically prepared to accept this as 'normal' their whole life which is why what they can impose in schools is so important to the Cult and its gofers. Long-time American freedom campaigner John Whitehead of the Rutherford Institute was not exaggerating when he said: 'Databit by databit, we are building our own electronic concentration camps.' Canada under its Cult gofer prime minister Justin Trudeau has taken a major step towards the real thing with people interned against their will if they test positive with a test not testing for the 'virus' when they arrive at a Canadian

airport. They are jailed in internment hotels often without food or water for long periods and with many doors failing to lock there have been sexual assaults. The interned are being charged sometimes \$2,000 for the privilege of being abused in this way. Trudeau is fully on board with the Cult and says the 'Covid pandemic' has provided an opportunity for a global 'reset' to permanently change Western civilisation. His number two, Deputy Prime Minister Chrystia Freeland, is a trustee of the World Economic Forum and a Rhodes Scholar. The Trudeau family have long been servants of the Cult. See *The Biggest Secret* and Cathy O'Brien's book *Trance-Formation of America* for the horrific background to Trudeau's father Pierre Trudeau another Canadian prime minister. Hide your fascism behind the façade of a heart-on-the-sleeve liberal. It's a well-honed Cult technique.

What can the 'vaccine' really do?

We have a 'virus' never shown to exist and 'variants' of the 'virus' that have also never been shown to exist except, like the 'original', as computer-generated fictions. Even if you believe there's a 'virus' the 'case' to 'death' rate is in the region of 0.23 to 0.15 percent and those 'deaths' are concentrated among the very old around the same average age that people die anyway. In response to this lack of threat (in truth none) psychopaths and idiots, knowingly and unknowingly answering to Gates and the Cult, are seeking to 'vaccinate' every man, woman and child on Planet Earth. Clearly the 'vaccine' is not about 'Covid' – none of this ever has been. So what is it all about *really*? Why the desperation to infuse genetically-manipulating synthetic material into everyone through mRNA fraudulent 'vaccines' with the intent of doing this over and over with the excuses of 'variants' and other 'virus' inventions? Dr Sherri Tenpenny, an osteopathic medical doctor in the United States, has made herself an expert on vaccines and their effects as a vehement campaigner against their use. Tenpenny was board certified in emergency medicine, the director of a level two trauma centre for 12 years, and moved to Cleveland in 1996 to start an integrative

medicine practice which has treated patients from all 50 states and some 17 other countries. Weaning people off pharmaceutical drugs is a speciality.

She became interested in the consequences of vaccines after attending a meeting at the National Vaccine Information Center in Washington DC in 2000 where she 'sat through four days of listening to medical doctors and scientists and lawyers and parents of vaccine injured kids' and asked: 'What's going on?' She had never been vaccinated and never got ill while her father was given a list of vaccines to be in the military and was 'sick his entire life'. The experience added to her questions and she began to examine vaccine documents from the Centers for Disease Control (CDC). After reading the first one, the 1998 version of *The General Recommendations of Vaccination*, she thought: 'This is it?' The document was poorly written and bad science and Tenpenny began 20 years of research into vaccines that continues to this day. She began her research into 'Covid vaccines' in March, 2020, and she describes them as 'deadly'. For many, as we have seen, they already have been. Tenpenny said that in the first 30 days of the 'vaccine' rollout in the United States there had been more than 40,000 adverse events reported to the vaccine adverse event database. A document had been delivered to her the day before that was 172 pages long. 'We have over 40,000 adverse events; we have over 3,100 cases of [potentially deadly] anaphylactic shock; we have over 5,000 neurological reactions.' Effects ranged from headaches to numbness, dizziness and vertigo, to losing feeling in hands or feet and paraesthesia which is when limbs 'fall asleep' and people have the sensation of insects crawling underneath their skin. All this happened in the first 30 days and remember that only about *ten percent* (or far less) of adverse reactions and vaccine-related deaths are estimated to be officially reported. Tenpenny said:

So can you think of one single product in any industry, any industry, for as long as products have been made on the planet that within 30 days we have 40,000 people complaining of side effects that not only is still on the market but ... we've got paid actors telling us how great

they are for getting their vaccine. We're offering people \$500 if they will just get their vaccine and we've got nurses and doctors going; 'I got the vaccine, I got the vaccine'.

Tenpenny said they were not going to be 'happy dancing folks' when they began to suffer Bell's palsy (facial paralysis), neuropathies, cardiac arrhythmias and autoimmune reactions that kill through a blood disorder. 'They're not going to be so happy, happy then, but we're never going to see pictures of those people' she said. Tenpenny described the 'vaccine' as 'a well-designed killing tool'.

No off-switch

Bad as the initial consequences had been Tenpenny said it would be maybe 14 months before we began to see the 'full ravage' of what is going to happen to the 'Covid vaccinated' with full-out consequences taking anything between two years and 20 years to show. You can understand why when you consider that variations of the 'Covid vaccine' use mRNA (messenger RNA) to in theory activate the immune system to produce protective antibodies without using the actual 'virus'. How can they when it's a computer program and they've never isolated what they claim is the 'real thing'? Instead they use *synthetic* mRNA. They are inoculating synthetic material into the body which through a technique known as the Trojan horse is absorbed into cells to change the nature of DNA. Human DNA is changed by an infusion of messenger RNA and with each new 'vaccine' of this type it is changed even more. Say so and you are banned by Cult Internet platforms. The contempt the contemptuous Mark Zuckerberg has for the truth and human health can be seen in an internal Facebook video leaked to the Project Veritas investigative team in which he said of the 'Covid vaccines': '... I share some caution on this because we just don't know the long term side-effects of basically modifying people's DNA and RNA.' At the same time this disgusting man's Facebook was censoring and banning anyone saying exactly the same. He must go before a Nuremberg trial for crimes against humanity when he *knows* that he

is censoring legitimate concerns and denying the right of informed consent on behalf of the Cult that owns him. People have been killed and damaged by the very 'vaccination' technique he cast doubt on himself when they may not have had the 'vaccine' with access to information that he denied them. The plan is to have at least annual 'Covid vaccinations', add others to deal with invented 'variants', and change all other vaccines into the mRNA system. Pfizer executives told shareholders at a virtual Barclays Global Healthcare Conference in March, 2021, that the public may need a third dose of 'Covid vaccine', plus regular yearly boosters and the company planned to hike prices to milk the profits in a 'significant opportunity for our vaccine'. These are the professional liars, cheats and opportunists who are telling you their 'vaccine' is safe. Given this volume of mRNA planned to be infused into the human body and its ability to then replicate we will have a transformation of human genetics from biological to synthetic biological – exactly the long-time Cult plan for reasons we'll see – and many will die. Sherri Tenpenny said of this replication:

It's like having an on-button but no off-button and that whole mechanism ... they actually give it a name and they call it the Trojan horse mechanism, because it allows that [synthetic] virus and that piece of that [synthetic] virus to get inside of your cells, start to replicate and even get inserted into other parts of your DNA as a Trojan-horse.

Ask the overwhelming majority of people who have the 'vaccine' what they know about the contents and what they do and they would reply: 'The government says it will stop me getting the virus.' Governments give that false impression on purpose to increase take-up. You can read Sherri Tenpenny's detailed analysis of the health consequences in her blog at [Vaxxter.com](https://vaxxter.com), but in summary these are some of them. She highlights the statement by Bill Gates about how human beings can become their own 'vaccine manufacturing machine'. The man is insane. ['Vaccine'-generated] 'antibodies' carry synthetic messenger RNA into the cells and the damage starts, Tenpenny contends, and she says that lungs can be adversely affected through varying degrees of pus and bleeding which

obviously affects breathing and would be dubbed 'Covid-19'. Even more sinister was the impact of 'antibodies' on macrophages, a white blood cell of the immune system. They consist of Type 1 and Type 2 which have very different functions. She said Type 1 are 'hyper-vigilant' white blood cells which 'gobble up' bacteria etc. However, in doing so, this could cause inflammation and in extreme circumstances be fatal. She says these affects are mitigated by Type 2 macrophages which kick in to calm down the system and stop it going rogue. They clear up dead tissue debris and reduce inflammation that the Type 1 'fire crews' have caused. Type 1 kills the infection and Type 2 heals the damage, she says. This is her punchline with regard to 'Covid vaccinations': She says that mRNA 'antibodies' block Type 2 macrophages by attaching to them and deactivating them. This meant that when the Type 1 response was triggered by infection there was nothing to stop that getting out of hand by calming everything down. There's an on-switch, but no off-switch, she says. What follows can be 'over and out, see you when I see you'.

Genetic suicide

Tenpenny also highlights the potential for autoimmune disease – the body attacking itself – which has been associated with vaccines since they first appeared. Infusing a synthetic foreign substance into cells could cause the immune system to react in a panic believing that the body is being overwhelmed by an invader (it is) and the consequences can again be fatal. There is an autoimmune response known as a 'cytokine storm' which I have likened to a homeowner panicked by an intruder and picking up a gun to shoot randomly in all directions before turning the fire on himself. The immune system unleashes a storm of inflammatory response called cytokines to a threat and the body commits hara-kiri. The lesson is that you mess with the body's immune response at your peril and these 'vaccines' seriously – fundamentally – mess with immune response. Tenpenny refers to a consequence called anaphylactic shock which is a severe and highly dangerous allergic reaction when the immune system

floods the body with chemicals. She gives the example of having a bee sting which primes the immune system and makes it sensitive to those chemicals. When people are stung again maybe years later the immune response can be so powerful that it leads to anaphylactic shock. Tenpenny relates this 'shock' with regard to the 'Covid vaccine' to something called polyethylene glycol or PEG. Enormous numbers of people have become sensitive to this over decades of use in a whole range of products and processes including food, drink, skin creams and 'medicine'. Studies have claimed that some 72 percent of people have antibodies triggered by PEG compared with two percent in the 1960s and allergic hypersensitive reactions to this become a gathering cause for concern. Tenpenny points out that the 'mRNA vaccine' is coated in a 'bubble' of polyethylene glycol which has the potential to cause anaphylactic shock through immune sensitivity. Many reports have appeared of people reacting this way after having the 'Covid vaccine'. What do we think is going to happen as humanity has more and more of these 'vaccines'? Tenpenny said: 'All these pictures we have seen with people with these rashes ... these weepy rashes, big reactions on their arms and things like that – it's an acute allergic reaction most likely to the polyethylene glycol that you've been previously primed and sensitised to.'

Those who have not studied the conspiracy and its perpetrators at length might think that making the population sensitive to PEG and then putting it in these 'vaccines' is just a coincidence. It is not. It is instead testament to how carefully and coldly-planned current events have been and the scale of the conspiracy we are dealing with. Tenpenny further explains that the 'vaccine' mRNA procedure can breach the blood-brain barrier which protects the brain from toxins and other crap that will cause malfunction. In this case they could make two proteins corrupt brain function to cause Amyotrophic lateral sclerosis (ALS), a progressive nervous system disease leading to loss of muscle control, and frontal lobe degeneration – Alzheimer's and dementia. Immunologist J. Bart Classon published a paper connecting mRNA 'vaccines' to prion

disease which can lead to Alzheimer's and other forms of neurodegenerative disease while others have pointed out the potential to affect the placenta in ways that make women infertile. This will become highly significant in the next chapter when I will discuss other aspects of this non-vaccine that relate to its nanotechnology and transmission from the injected to the uninjected.

Qualified in idiocy

Tenpenny describes how research has confirmed that these 'vaccine'-generated antibodies can interact with a range of other tissues in the body and attack many other organs including the lungs. 'This means that if you have a hundred people standing in front of you that all got this shot they could have a hundred different symptoms.' Anyone really think that Cult gofers like the Queen, Tony Blair, Christopher Whitty, Anthony Fauci, and all the other psychopaths have really had this 'vaccine' in the pictures we've seen? Not a bloody chance. Why don't doctors all tell us about all these dangers and consequences of the 'Covid vaccine'? Why instead do they encourage and pressure patients to have the shot? Don't let's think for a moment that doctors and medical staff can't be stupid, lazy, and psychopathic and that's without the financial incentives to give the jab. Tenpenny again:

Some people are going to die from the vaccine directly but a large number of people are going to start to get horribly sick and get all kinds of autoimmune diseases 42 days to maybe a year out. What are they going to do, these stupid doctors who say; 'Good for you for getting that vaccine.' What are they going to say; 'Oh, it must be a mutant, we need to give an extra dose of that vaccine.'

Because now the vaccine, instead of one dose or two doses we need three or four because the stupid physicians aren't taking the time to learn anything about it. If I can learn this sitting in my living room reading a 19 page paper and several others so can they. There's nothing special about me, I just take the time to do it.

Remember how Sara Kayat, the NHS and TV doctor, said that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Doctors can be idiots like every other profession and they

should not be worshipped as infallible. They are not and far from it. Behind many medical and scientific 'experts' lies an uninformed prat trying to hide themselves from you although in the 'Covid' era many have failed to do so as with UK narrative-repeating 'TV doctor' Hilary Jones. Pushing back against the minority of proper doctors and scientists speaking out against the 'vaccine' has been the entire edifice of the Cult global state in the form of governments, medical systems, corporations, mainstream media, Silicon Valley, and an army of compliant doctors, medical staff and scientists willing to say anything for money and to enhance their careers by promoting the party line. If you do that you are an 'expert' and if you won't you are an 'anti-vaxxer' and 'Covidiot'. The pressure to be 'vaccinated' is incessant. We have even had reports claiming that the 'vaccine' can help cure cancer and Alzheimer's and make the lame walk. I am waiting for the announcement that it can bring you coffee in the morning and cook your tea. Just as the symptoms of 'Covid' seem to increase by the week so have the miracles of the 'vaccine'. American supermarket giant Kroger Co. offered nearly 500,000 employees in 35 states a \$100 bonus for having the 'vaccine' while donut chain Krispy Kreme promised 'vaccinated' customers a free glazed donut every day for the rest of 2021. Have your DNA changed and you will get a doughnut although we might not have to give you them for long. Such offers and incentives confirm the desperation.

Perhaps the worse vaccine-stunt of them all was UK 'Health' Secretary Matt-the-prat Hancock on live TV after watching a clip of someone being 'vaccinated' when the roll-out began. Hancock faked tears so badly it was embarrassing. Brain-of-Britain Piers Morgan, the lockdown-supporting, 'vaccine' supporting, 'vaccine' passport-supporting, TV host played along with Hancock – 'You're quite emotional about that' he said in response to acting so atrocious it would have been called out at a school nativity which will presumably today include Mary and Jesus in masks, wise men keeping their camels six feet apart, and shepherds under tent arrest. System-serving Morgan tweeted this: 'Love the idea of covid vaccine passports for everywhere: flights, restaurants, clubs, football, gyms,

shops etc. It's time covid-denying, anti-vaxxer loonies had their bullsh*t bluff called & bar themselves from going anywhere that responsible citizens go.' If only I could aspire to his genius. To think that Morgan, who specialises in shouting over anyone he disagrees with, was lauded as a free speech hero when he lost his job after storming off the set of his live show like a child throwing his dolly out of the pram. If he is a free speech hero we are in real trouble. I have no idea what 'bullsh*t' means, by the way, the * throws me completely.

The Cult is desperate to infuse its synthetic DNA-changing concoction into everyone and has been using every lie, trick and intimidation to do so. The question of '*Why?*' we shall now address.

CHAPTER TEN

Human 2.0

I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted –
Alan Turing (1912-1954), the ‘Father of artificial intelligence’

I have been exposing for decades the plan to transform the human body from a biological to a synthetic-biological state. The new human that I will call Human 2.0 is planned to be connected to artificial intelligence and a global AI ‘Smart Grid’ that would operate as one global system in which AI would control everything from your fridge to your heating system to your car to your mind. Humans would no longer be ‘human’, but post-human and sub-human, with their thinking and emotional processes replaced by AI.

What I said sounded crazy and beyond science fiction and I could understand that. To any balanced, rational, mind it *is* crazy. Today, however, that world is becoming reality and it puts the ‘Covid vaccine’ into its true context. Ray Kurzweil is the ultra-Zionist ‘computer scientist, inventor and futurist’ and co-founder of the Singularity University. Singularity refers to the merging of humans with machines or ‘transhumanism’. Kurzweil has said humanity would be connected to the cyber ‘cloud’ in the period of the ever-recurring year of 2030:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and ‘think in the cloud’ ... We’re going to put gateways to the

cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations. As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

They are trying to sell this end-of-humanity-as-we-know-it as the next stage of 'evolution' when we become super-human and 'like the gods'. They are lying to you. Shocked, eh? The population, and again especially the young, have been manipulated into addiction to technologies designed to enslave them for life. First they induced an addiction to smartphones (holdables); next they moved to technology on the body (wearables); and then began the invasion of the body (implantables). I warned way back about the plan for microchipped people and we are now entering that era. We should not be diverted into thinking that this refers only to chips we can see. Most important are the nanochips known as smart dust, neural dust and nanobots which are far too small to be seen by the human eye. Nanotechnology is everywhere, increasingly in food products, and released into the atmosphere by the geoengineering of the skies funded by Bill Gates to 'shut out the Sun' and 'save the planet from global warming'. Gates has been funding a project to spray millions of tonnes of chalk (calcium carbonate) into the stratosphere over Sweden to 'dim the Sun' and cool the Earth. Scientists warned the move could be disastrous for weather systems in ways no one can predict and opposition led to the Swedish space agency announcing that the 'experiment' would not be happening as planned in the summer of 2021; but it shows where the Cult is going with dimming the impact of the Sun and there's an associated plan to change the planet's atmosphere. Who gives psychopath Gates the right to dictate to the entire human race and dismantle planetary systems? The world will not be safe while this man is at large.

The global warming hoax has made the Sun, like the gas of life, something to fear when both are essential to good health and human survival (more inversion). The body transforms sunlight into vital vitamin D through a process involving ... *cholesterol*. This is the cholesterol we are also told to fear. We are urged to take Big Pharma

statin drugs to reduce cholesterol and it's all systematic. Reducing cholesterol means reducing vitamin D uptake with all the multiple health problems that will cause. At least if you take statins long term it saves the government from having to pay you a pension. The delivery system to block sunlight is widely referred to as chemtrails although these have a much deeper agenda, too. They appear at first to be contrails or condensation trails streaming from aircraft into cold air at high altitudes. Contrails disperse very quickly while chemtrails do not and spread out across the sky before eventually their content falls to earth. Many times I have watched aircraft cross-cross a clear blue sky releasing chemtrails until it looks like a cloudy day. Chemtrails contain many things harmful to humans and the natural world including toxic heavy metals, aluminium (see Alzheimer's) and nanotechnology. Ray Kurzweil reveals the reason without actually saying so: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' How do you deliver that? *From the sky*. Self-replicating nanobots would connect everything to the Smart Grid. The phenomenon of Morgellons disease began in the chemtrail era and the correlation has led to it being dubbed the 'chemtrail disease'. Self-replicating fibres appear in the body that can be pulled out through the skin. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. I cover this at greater length in *Phantom Self*.

'Vaccine' operating system

'Covid vaccines' with their self-replicating synthetic material are also designed to make the connection between humanity and Kurzweil's 'cloud'. American doctor and dedicated campaigner for truth, Carrie Madej, an Internal Medicine Specialist in Georgia with more than 20 years medical experience, has highlighted the nanotechnology aspect of the fake 'vaccines'. She explains how one of the components in at least the Moderna and Pfizer synthetic potions are 'lipid nanoparticles' which are 'like little tiny computer bits' – a 'sci-fi substance' known as nanobots and hydrogel which can be 'triggered

at any moment to deliver its payload' and act as 'biosensors'. The synthetic substance had 'the ability to accumulate data from your body like your breathing, your respiration, thoughts and emotions, all kind of things' and each syringe could carry a *million* nanobots:

This substance because it's like little bits of computers in your body, crazy, but it's true, it can do that, [and] obviously has the ability to act through Wi-Fi. It can receive and transmit energy, messages, frequencies or impulses. That issue has never been addressed by these companies. What does that do to the human?

Just imagine getting this substance in you and it can react to things all around you, the 5G, your smart device, your phones, what is happening with that? What if something is triggering it, too, like an impulse, a frequency? We have something completely foreign in the human body.

Madej said her research revealed that electromagnetic (EMF) frequencies emitted by phones and other devices had increased dramatically in the same period of the 'vaccine' rollout and she was seeing more people with radiation problems as 5G and other electromagnetic technology was expanded and introduced to schools and hospitals. She said she was 'floored with the EMF coming off' the devices she checked. All this makes total sense and syncs with my own work of decades when you think that Moderna refers in documents to its mRNA 'vaccine' as an 'operating system':

Recognizing the broad potential of mRNA science, we set out to create an mRNA technology platform that functions very much like an operating system on a computer. It is designed so that it can plug and play interchangeably with different programs. In our case, the 'program' or 'app' is our mRNA drug – the unique mRNA sequence that codes for a protein ...

... Our MRNA Medicines – 'The 'Software Of Life': When we have a concept for a new mRNA medicine and begin research, fundamental components are already in place. Generally, the only thing that changes from one potential mRNA medicine to another is the coding region – the actual genetic code that instructs ribosomes to make protein. Utilizing these instruction sets gives our investigational mRNA medicines a software-like quality. We also have the ability to combine different mRNA sequences encoding for different proteins in a single mRNA investigational medicine.

Who needs a real ‘virus’ when you can create a computer version to justify infusing your operating system into the entire human race on the road to making living, breathing people into cyborgs? What is missed with the ‘vaccines’ is the *digital* connection between synthetic material and the body that I highlighted earlier with the study that hacked a computer with human DNA. On one level the body is digital, based on mathematical codes, and I’ll have more about that in the next chapter. Those who ridiculously claim that mRNA ‘vaccines’ are not designed to change human genetics should explain the words of Dr Tal Zaks, chief medical officer at Moderna, in a 2017 TED talk. He said that over the last 30 years ‘we’ve been living this phenomenal digital scientific revolution, and I’m here today to tell you, that we are actually *hacking the software of life*, and that it’s changing the way we think about prevention and treatment of disease’:

In every cell there’s this thing called messenger RNA, or mRNA for short, that transmits the critical information from the DNA in our genes to the protein, which is really the stuff we’re all made out of. This is the critical information that determines what the cell will do. So we think about it as an operating system. So if you could change that, if you could introduce a line of code, or change a line of code, it turns out, that has profound implications for everything, from the flu to cancer.

Zaks should more accurately have said that this has profound implications for the human genetic code and the nature of DNA. Communications within the body go both ways and not only one. But, hey, no, the ‘Covid vaccine’ will not affect your genetics. Cult fact-checkers say so even though the man who helped to develop the mRNA technique says that it does. Zaks said in 2017:

If you think about what it is we’re trying to do. We’ve taken information and our understanding of that information and how that information is transmitted in a cell, and we’ve taken our understanding of medicine and how to make drugs, and we’re fusing the two. We think of it as information therapy.

I have been writing for decades that the body is an information field communicating with itself and the wider world. This is why

radiation which is information can change the information field of body and mind through phenomena like 5G and change their nature and function. 'Information therapy' means to change the body's information field and change the way it operates. DNA is a receiver-transmitter of information and can be mutated by information like mRNA synthetic messaging. Technology to do this has been ready and waiting in the underground bases and other secret projects to be rolled out when the 'Covid' hoax was played. 'Trials' of such short and irrelevant duration were only for public consumption. When they say the 'vaccine' is 'experimental' that is not true. It may appear to be 'experimental' to those who don't know what's going on, but the trials have already been done to ensure the Cult gets the result it desires. Zaks said that it took decades to sequence the human genome, completed in 2003, but now they could do it in a week. By 'they' he means scientists operating in the public domain. In the secret projects they were sequencing the genome in a week long before even 2003.

Deluge of mRNA

Highly significantly the Moderna document says the guiding premise is that if using mRNA as a medicine works for one disease then it should work for many diseases. They were leveraging the flexibility afforded by their platform and the fundamental role mRNA plays in protein synthesis to pursue mRNA medicines for a broad spectrum of diseases. Moderna is confirming what I was saying through 2020 that multiple 'vaccines' were planned for 'Covid' (and later invented 'variants') and that previous vaccines would be converted to the mRNA system to infuse the body with massive amounts of genetically-manipulating synthetic material to secure a transformation to a synthetic-biological state. The 'vaccines' are designed to kill stunning numbers as part of the long-exposed Cult depopulation agenda and transform the rest. Given this is the goal you can appreciate why there is such hysterical demand for every human to be 'vaccinated' for an alleged 'disease' that has an estimated 'infection' to 'death' ratio of 0.23-0.15 percent. As I write

children are being given the 'vaccine' in trials (their parents are a disgrace) and ever-younger people are being offered the vaccine for a 'virus' that even if you believe it exists has virtually zero chance of harming them. Horrific effects of the 'trials' on a 12-year-old girl were revealed by a family member to be serious brain and gastric problems that included a bowel obstruction and the inability to swallow liquids or solids. She was unable to eat or drink without throwing up, had extreme pain in her back, neck and abdomen, and was paralysed from the waist down which stopped her urinating unaided. When the girl was first taken to hospital doctors said it was all in her mind. She was signed up for the 'trial' by her parents for whom no words suffice. None of this 'Covid vaccine' insanity makes any sense unless you see what the 'vaccine' really is – a body-changer. Synthetic biology or 'SynBio' is a fast-emerging and expanding scientific discipline which includes everything from genetic and molecular engineering to electrical and computer engineering. Synthetic biology is defined in these ways:

- A multidisciplinary area of research that seeks to create new biological parts, devices, and systems, or to redesign systems that are already found in nature.
- The use of a mixture of physical engineering and genetic engineering to create new (and therefore synthetic) life forms.
- An emerging field of research that aims to combine the knowledge and methods of biology, engineering and related disciplines in the design of chemically-synthesized DNA to create organisms with novel or enhanced characteristics and traits (synthetic organisms including humans).

We now have synthetic blood, skin, organs and limbs being developed along with synthetic body parts produced by 3D printers. These are all elements of the synthetic human programme and this comment by Kurzweil's co-founder of the Singularity University,

Peter Diamandis, can be seen in a whole new light with the 'Covid' hoax and the sanctions against those that refuse the 'vaccine':

Anybody who is going to be resisting the progress forward [to transhumanism] is going to be resisting evolution and, fundamentally, they will die out. It's not a matter of whether it's good or bad. It's going to happen.

'Resisting evolution'? What absolute bollocks. The arrogance of these people is without limit. His 'it's going to happen' mantra is another way of saying 'resistance is futile' to break the spirit of those pushing back and we must not fall for it. Getting this genetically-transforming 'vaccine' into everyone is crucial to the Cult plan for total control and the desperation to achieve that is clear for anyone to see. Vaccine passports are a major factor in this and they, too, are a form of resistance is futile. It's NOT. The paper funded by the Rockefeller Foundation for the 2013 'health conference' in China said:

We will interact more with artificial intelligence. The use of robotics, bio-engineering to augment human functioning is already well underway and will advance. Re-engineering of humans into potentially separate and unequal forms through genetic engineering or mixed human-robots raises debates on ethics and equality.

A new demography is projected to emerge after 2030 [that year again] of technologies (robotics, genetic engineering, nanotechnology) producing robots, engineered organisms, 'nanobots' and artificial intelligence (AI) that can self-replicate. Debates will grow on the implications of an impending reality of human designed life.

What is happening today is so long planned. The world army enforcing the will of the world government is intended to be a robot army, not a human one. Today's military and its technologically 'enhanced' troops, pilotless planes and driverless vehicles are just stepping stones to that end. Human soldiers are used as Cult fodder and its time they woke up to that and worked for the freedom of the population instead of their own destruction and their family's destruction – the same with the police. Join us and let's sort this out. The phenomenon of enforce my own destruction is widespread in the 'Covid' era with Woker 'luvvies' in the acting and entertainment

industries supporting 'Covid' rules which have destroyed their profession and the same with those among the public who put signs on the doors of their businesses 'closed due to Covid – stay safe' when many will never reopen. It's a form of masochism and most certainly insanity.

Transgender = transhumanism

When something explodes out of nowhere and is suddenly everywhere it is always the Cult agenda and so it is with the tidal wave of claims and demands that have infiltrated every aspect of society under the heading of 'transgenderism'. The term 'trans' is so 'in' and this is the dictionary definition:

A prefix meaning 'across', 'through', occurring ... in loanwords from Latin, used in particular for denoting movement or conveyance from place to place (transfer; transmit; transplant) or complete change (transform; transmute), or to form adjectives meaning 'crossing', 'on the other side of', or 'going beyond' the place named (transmontane; transnational; trans-Siberian).

Transgender means to go beyond gender and transhuman means to go beyond human. Both are aspects of the Cult plan to transform the human body to a synthetic state with *no gender*. Human 2.0 is not designed to procreate and would be produced technologically with no need for parents. The new human would mean the end of parents and so men, and increasingly women, are being targeted for the deletion of their rights and status. Parental rights are disappearing at an ever-quickenening speed for the same reason. The new human would have no need for men or women when there is no procreation and no gender. Perhaps the transgender movement that appears to be in a permanent state of frenzy might now contemplate on how it is being used. This was never about transgender rights which are only the interim excuse for confusing gender, particularly in the young, on the road to *fusing* gender. Transgender activism is not an end; it is a *means* to an end. We see again the technique of creative destruction in which you destroy the status quo to 'build back better' in the form that you want. The gender status quo had to be

destroyed by persuading the Cult-created Woke mentality to believe that you can have 100 genders or more. A programme for 9 to 12 year olds produced by the Cult-owned BBC promoted the 100 genders narrative. The very idea may be the most monumental nonsense, but it is not what is true that counts, only what you can make people *believe* is true. Once the gender of $2 + 2 = 4$ has been dismantled through indoctrination, intimidation and $2 + 2 = 5$ then the new no-gender normal can take its place with Human 2.0.

Aldous Huxley revealed the plan in his prophetic *Brave New World* in 1932:

Natural reproduction has been done away with and children are created, decanted', and raised in 'hatcheries and conditioning centres'. From birth, people are genetically designed to fit into one of five castes, which are further split into 'Plus' and 'Minus' members and designed to fulfil predetermined positions within the social and economic strata of the World State.

How could Huxley know this in 1932? For the same reason George Orwell knew about the Big Brother state in 1948, Cult insiders I have quoted knew about it in 1969, and I have known about it since the early 1990s. If you are connected to the Cult or you work your balls off to uncover the plan you can predict the future. The process is simple. If there is a plan for the world and nothing intervenes to stop it then it will happen. Thus if you communicate the plan ahead of time you are perceived to have predicted the future, but you haven't. You have revealed the plan which without intervention will become the human future. The whole reason I have done what I have is to alert enough people to inspire an intervention and maybe at last that time has come with the Cult and its intentions now so obvious to anyone with a brain in working order.

The future is here

Technological wombs that Huxley described to replace parent procreation are already being developed and they are only the projects we know about in the public arena. Israeli scientists told *The Times of Israel* in March, 2021, that they have grown 250-cell embryos

into mouse fetuses with fully formed organs using artificial wombs in a development they say could pave the way for gestating humans outside the womb. Professor Jacob Hanna of the Weizmann Institute of Science said:

We took mouse embryos from the mother at day five of development, when they are just of 250 cells, and had them in the incubator from day five until day 11, by which point they had grown all their organs.

By day 11 they make their own blood and have a beating heart, a fully developed brain. Anybody would look at them and say, 'this is clearly a mouse foetus with all the characteristics of a mouse.' It's gone from being a ball of cells to being an advanced foetus.

A special liquid is used to nourish embryo cells in a laboratory dish and they float on the liquid to duplicate the first stage of embryonic development. The incubator creates all the right conditions for its development, Hanna said. The liquid gives the embryo 'all the nutrients, hormones and sugars they need' along with a custom-made electronic incubator which controls gas concentration, pressure and temperature. The cutting-edge in the underground bases and other secret locations will be light years ahead of that, however, and this was reported by the London *Guardian* in 2017:

We are approaching a biotechnological breakthrough. Ectogenesis, the invention of a complete external womb, could completely change the nature of human reproduction. In April this year, researchers at the Children's Hospital of Philadelphia announced their development of an artificial womb.

The article was headed 'Artificial wombs could soon be a reality. What will this mean for women?' What would it mean for children is an even bigger question. No mother to bond with only a machine in preparation for a life of soulless interaction and control in a world governed by machines (see the *Matrix* movies). Now observe the calculated manipulations of the 'Covid' hoax as human interaction and warmth has been curtailed by distancing, isolation and fear with people communicating via machines on a scale never seen before.

These are all dots in the same picture as are all the personal assistants, gadgets and children's toys through which kids and adults communicate with AI as if it is human. The AI 'voice' on Sat-Nav should be included. All these things are psychological preparation for the Cult endgame. Before you can make a physical connection with AI you have to make a psychological connection and that is what people are being conditioned to do with this ever gathering human-AI interaction. Movies and TV programmes depicting the transhuman, robot dystopia relate to a phenomenon known as 'pre-emptive programming' in which the world that is planned is portrayed everywhere in movies, TV and advertising. This is conditioning the conscious and subconscious mind to become familiar with the planned reality to dilute resistance when it happens for real. What would have been a shock such is the change is made less so. We have young children put on the road to transgender transition surgery with puberty blocking drugs at an age when they could never be able to make those life-changing decisions.

Rachel Levine, a professor of paediatrics and psychiatry who believes in treating children this way, became America's highest-ranked openly-transgender official when she was confirmed as US Assistant Secretary at the Department of Health and Human Services after being nominated by Joe Biden (the Cult). Activists and governments press for laws to deny parents a say in their children's transition process so the kids can be isolated and manipulated into agreeing to irreversible medical procedures. A Canadian father Robert Hoogland was denied bail by the Vancouver Supreme Court in 2021 and remained in jail for breaching a court order that he stay silent over his young teenage daughter, a minor, who was being offered life-changing hormone therapy without parental consent. At the age of 12 the girl's 'school counsellor' said she may be transgender, referred her to a doctor and told the school to treat her like a boy. This is another example of state-serving schools imposing ever more control over children's lives while parents have ever less.

Contemptible and extreme child abuse is happening all over the world as the Cult gender-fusion operation goes into warp-speed.

Why the war on men – and now women?

The question about what artificial wombs mean for women should rightly be asked. The answer can be seen in the deletion of women's rights involving sport, changing rooms, toilets and status in favour of people in male bodies claiming to identify as women. I can identify as a mountain climber, but it doesn't mean I can climb a mountain any more than a biological man can be a biological woman. To believe so is a triumph of belief over factual reality which is the very perceptual basis of everything Woke. Women's sport is being destroyed by allowing those with male bodies who say they identify as female to 'compete' with girls and women. Male body 'women' dominate 'women's' competition with their greater muscle mass, bone density, strength and speed. With that disadvantage sport for women loses all meaning. To put this in perspective nearly 300 American high school boys can run faster than the quickest woman sprinter in the world. Women are seeing their previously protected spaces invaded by male bodies simply because they claim to identify as women. That's all they need to do to access all women's spaces and activities under the Biden 'Equality Act' that destroys equality for women with the usual Orwellian Woke inversion. Male sex offenders have already committed rapes in women's prisons after claiming to identify as women to get them transferred. Does this not matter to the Woke 'equality' hypocrites? Not in the least. What matters to Cult manipulators and funders behind transgender activists is to advance gender fusion on the way to the no-gender 'human'. When you are seeking to impose transparent nonsense like this, or the 'Covid' hoax, the only way the nonsense can prevail is through censorship and intimidation of dissenters, deletion of factual information, and programming of the unquestioning, bewildered and naive. You don't have to scan the world for long to see that all these things are happening.

Many women's rights organisations have realised that rights and status which took such a long time to secure are being eroded and that it is systematic. Kara Dansky of the global Women's Human Rights Campaign said that Biden's transgender executive order immediately he took office, subsequent orders, and Equality Act legislation that followed 'seek to erase women and girls in the law as a category'. *Exactly*. I said during the long ago-started war on men (in which many women play a crucial part) that this was going to turn into a war on them. The Cult is phasing out *both* male and female genders. To get away with that they are brought into conflict so they are busy fighting each other while the Cult completes the job with no unity of response. Unity, people, *unity*. We need unity everywhere. Transgender is the only show in town as the big step towards the no-gender human. It's not about rights for transgender people and never has been. Woke political correctness is deleting words relating to genders to the same end. Wokers believe this is to be 'inclusive' when the opposite is true. They are deleting words describing gender because gender *itself* is being deleted by Human 2.0. Terms like 'man', 'woman', 'mother' and 'father' are being deleted in the universities and other institutions to be replaced by the *no*-gender, not trans-gender, 'individuals' and 'guardians'. Women's rights campaigner Maria Keffler of Partners for Ethical Care said: 'Children are being taught from kindergarten upward that some boys have a vagina, some girls have a penis, and that kids can be any gender they want to be.' Do we really believe that suddenly countries all over the world at the same time had the idea of having drag queens go into schools or read transgender stories to very young children in the local library? It's coldly-calculated confusion of gender on the way to the fusion of gender. Suzanne Vierling, a psychologist from Southern California, made another important point:

Yesterday's slave woman who endured gynecological medical experiments is today's girl-child being butchered in a booming gender-transitioning sector. Ovaries removed, pushing her into menopause and osteoporosis, uncharted territory, and parents' rights and authority decimated.

The erosion of parental rights is a common theme in line with the Cult plans to erase the very concept of parents and 'ovaries removed, pushing her into menopause' means what? Those born female lose the ability to have children – another way to discontinue humanity as we know it.

Eliminating Human 1.0 (before our very eyes)

To pave the way for Human 2.0 you must phase out Human 1.0. This is happening through plummeting sperm counts and making women infertile through an onslaught of chemicals, radiation (including smartphones in pockets of men) and mRNA 'vaccines'. Common agriculture pesticides are also having a devastating impact on human fertility. I have been tracking collapsing sperm counts in the books for a long time and in 2021 came a book by fertility scientist and reproductive epidemiologist Shanna Swan, *Count Down: How Our Modern World Is Threatening Sperm Counts, Altering Male and Female Reproductive Development and Imperiling the Future of the Human Race*. She reports how the global fertility rate dropped by *half* between 1960 and 2016 with America's birth rate 16 percent below where it needs to be to sustain the population. Women are experiencing declining egg quality, more miscarriages, and more couples suffer from infertility. Other findings were an increase in erectile dysfunction, infant boys developing more genital abnormalities, male problems with conception, and plunging levels of the male hormone testosterone which would explain why so many men have lost their backbone and masculinity. This has been very evident during the 'Covid' hoax when women have been prominent among the Pushbackers and big strapping blokes have bowed their heads, covered their faces with a nappy and quietly submitted. Mind control expert Cathy O'Brien also points to how global education introduced the concept of 'we're all winners' in sport and classrooms: 'Competition was defused, and it in turn defused a sense of fighting back.' This is another version of the 'equity' doctrine in which you drive down rather than raise up. What a contrast in Cult-controlled China with its global ambitions

where the government published plans in January, 2021, to 'cultivate masculinity' in boys from kindergarten through to high school in the face of a 'masculinity crisis'. A government adviser said boys would be soon become 'delicate, timid and effeminate' unless action was taken. Don't expect any similar policy in the targeted West. A 2006 study showed that a 65-year-old man in 2002 had testosterone levels 15 percent lower than a 65-year-old man in 1987 while a 2020 study found a similar story with young adults and adolescents. Men are getting prescriptions for testosterone replacement therapy which causes an even greater drop in sperm count with up to 99 percent seeing sperm counts drop to zero during the treatment. More sperm is defective and malfunctioning with some having two heads or not pursuing an egg.

A class of *synthetic* chemicals known as phthalates are being blamed for the decline. These are found everywhere in plastics, shampoos, cosmetics, furniture, flame retardants, personal care products, pesticides, canned foods and even receipts. Why till receipts? Everyone touches them. Let no one delude themselves that all this is not systematic to advance the long-time agenda for human body transformation. Phthalates mimic hormones and disrupt the hormone balance causing testosterone to fall and genital birth defects in male infants. Animals and fish have been affected in the same way due to phthalates and other toxins in rivers. When fish turn gay or change sex through chemicals in rivers and streams it is a pointer to why there has been such an increase in gay people and the sexually confused. It doesn't matter to me what sexuality people choose to be, but if it's being affected by chemical pollution and consumption then we need to know. Does anyone really think that this is not connected to the transgender agenda, the war on men and the condemnation of male 'toxic masculinity'? You watch this being followed by 'toxic femininity'. It's already happening. When breastfeeding becomes 'chest-feeding', pregnant women become pregnant people along with all the other Woke claptrap you know that the world is going insane and there's a Cult scam in progress. Transgender activists are promoting the Cult agenda while Cult

billionaires support and fund the insanity as they laugh themselves to sleep at the sheer stupidity for which humans must be infamous in galaxies far, far away.

‘Covid vaccines’ and female infertility

We can now see why the ‘vaccine’ has been connected to potential infertility in women. Dr Michael Yeadon, former Vice President and Chief Scientific Advisor at Pfizer, and Dr Wolfgang Wodarg in Germany, filed a petition with the European Medicines Agency in December, 2020, urging them to stop trials for the Pfizer/BioNTech shot and all other mRNA trials until further studies had been done. They were particularly concerned about possible effects on fertility with ‘vaccine’-produced antibodies attacking the protein Syncytin-1 which is responsible for developing the placenta. The result would be infertility ‘of indefinite duration’ in women who have the ‘vaccine’ with the placenta failing to form. Section 10.4.2 of the Pfizer/BioNTech trial protocol says that pregnant women or those who might become so should not have mRNA shots. Section 10.4 warns men taking mRNA shots to ‘be abstinent from heterosexual intercourse’ and not to donate sperm. The UK government said that it *did not know* if the mRNA procedure had an effect on fertility. *Did not know?* These people have to go to jail. UK government advice did not recommend at the start that pregnant women had the shot and said they should avoid pregnancy for at least two months after ‘vaccination’. The ‘advice’ was later updated to pregnant women should only have the ‘vaccine’ if the benefits outweighed the risks to mother and foetus. What the hell is that supposed to mean? Then ‘spontaneous abortions’ began to appear and rapidly increase on the adverse reaction reporting schemes which include only a fraction of adverse reactions. Thousands and ever-growing numbers of ‘vaccinated’ women are describing changes to their menstrual cycle with heavier blood flow, irregular periods and menstruating again after going through the menopause – all links to reproduction effects. Women are passing blood clots and the lining of their uterus while men report erectile dysfunction and blood effects. Most

significantly of all *unvaccinated* women began to report similar menstrual changes after interaction with '*vaccinated*' people and men and children were also affected with bleeding noses, blood clots and other conditions. 'Shedding' is when vaccinated people can emit the content of a vaccine to affect the unvaccinated, but this is different. 'Vaccinated' people were not shedding a 'live virus' allegedly in 'vaccines' as before because the fake 'Covid vaccines' involve synthetic material and other toxicity. Doctors exposing what is happening prefer the term 'transmission' to shedding. Somehow those that have had the shots are transmitting effects to those that haven't. Dr Carrie Madej said the nano-content of the 'vaccines' can 'act like an antenna' to others around them which fits perfectly with my own conclusions. This 'vaccine' transmission phenomenon was becoming known as the book went into production and I deal with this further in the Postscript.

Vaccine effects on sterility are well known. The World Health Organization was accused in 2014 of sterilising millions of women in Kenya with the evidence confirmed by the content of the vaccines involved. The same WHO behind the 'Covid' hoax admitted its involvement for more than ten years with the vaccine programme. Other countries made similar claims. Charges were lodged by Tanzania, Nicaragua, Mexico, and the Philippines. The Gardasil vaccine claimed to protect against a genital 'virus' known as HPV has also been linked to infertility. Big Pharma and the WHO (same thing) are criminal and satanic entities. Then there's the Bill Gates Foundation which is connected through funding and shared interests with 20 pharmaceutical giants and laboratories. He stands accused of directing the policy of United Nations Children's Fund (UNICEF), vaccine alliance GAVI, and other groupings, to advance the vaccine agenda and silence opposition at great cost to women and children. At the same time Gates wants to reduce the global population. Coincidence?

Great Reset = Smart Grid = new human

The Cult agenda I have been exposing for 30 years is now being openly promoted by Cult assets like Gates and Klaus Schwab of the World Economic Forum under code-terms like the 'Great Reset', 'Build Back Better' and 'a rare but narrow window of opportunity to reflect, reimagine, and reset our world'. What provided this 'rare but narrow window of opportunity'? The 'Covid' hoax did. Who created that? *They* did. My books from not that long ago warned about the planned 'Internet of Things' (IoT) and its implications for human freedom. This was the plan to connect all technology to the Internet and artificial intelligence and today we are way down that road with an estimated 36 billion devices connected to the World Wide Web and that figure is projected to be 76 billion by 2025. I further warned that the Cult planned to go beyond that to the Internet of *Everything* when the human brain was connected via AI to the Internet and Kurzweil's 'cloud'. Now we have Cult operatives like Schwab calling for precisely that under the term 'Internet of Bodies', a fusion of the physical, digital and biological into one centrally-controlled Smart Grid system which the Cult refers to as the 'Fourth Industrial Revolution'. They talk about the 'biological', but they really mean the synthetic-biological which is required to fully integrate the human body and brain into the Smart Grid and artificial intelligence planned to replace the human mind. We have everything being synthetically manipulated including the natural world through GMO and smart dust, the food we eat and the human body itself with synthetic 'vaccines'. I said in *The Answer* that we would see the Cult push for synthetic meat to replace animals and in February, 2021, the so predictable psychopath Bill Gates called for the introduction of synthetic meat to save us all from 'climate change'. The climate hoax just keeps on giving like the 'Covid' hoax. The war on meat by vegan activists is a carbon (oops, sorry) copy of the manipulation of transgender activists. They have no idea (except their inner core) that they are being used to promote and impose the agenda of the Cult or that they are only the *vehicle* and not the *reason*. This is not to say those who choose not to eat meat shouldn't be respected and supported in that right, but there are ulterior motives

for those in power. A *Forbes* article in December, 2019, highlighted the plan so beloved of Schwab and the Cult under the heading: 'What Is The Internet of Bodies? And How Is It Changing Our World?' The article said the human body is the latest data platform (remember 'our vaccine is an operating system'). *Forbes* described the plan very accurately and the words could have come straight out of my books from long before:

The Internet of Bodies (IoB) is an extension of the IoT and basically connects the human body to a network through devices that are ingested, implanted, or connected to the body in some way. Once connected, data can be exchanged, and the body and device can be remotely monitored and controlled.

They were really describing a human hive mind with human perception centrally-dictated via an AI connection as well as allowing people to be 'remotely monitored and controlled'. Everything from a fridge to a human mind could be directed from a central point by these insane psychopaths and 'Covid vaccines' are crucial to this. *Forbes* explained the process I mentioned earlier of holdable and wearable technology followed by implantable. The article said there were three generations of the Internet of Bodies that include:

- Body external: These are wearable devices such as Apple Watches or Fitbits that can monitor our health.
- Body internal: These include pacemakers, cochlear implants, and digital pills that go inside our bodies to monitor or control various aspects of health.
- Body embedded: The third generation of the Internet of Bodies is embedded technology where technology and the human body are melded together and have a real-time connection to a remote machine.

Forbes noted the development of the Brain Computer Interface (BCI) which merges the brain with an external device for monitoring and controlling in real-time. 'The ultimate goal is to help restore function to individuals with disabilities by using brain signals rather than conventional neuromuscular pathways.' Oh, do fuck off. The goal of brain interface technology is controlling human thought and emotion from the central point in a hive mind serving its masters wishes. Many people are now agreeing to be chipped to open doors without a key. You can recognise them because they'll be wearing a mask, social distancing and lining up for the 'vaccine'. The Cult plans a Great Reset money system after they have completed the demolition of the global economy in which 'money' will be exchanged through communication with body operating systems. Rand Corporation, a Cult-owned think tank, said of the Internet of Bodies or IoB:

Internet of Bodies technologies fall under the broader IoT umbrella. But as the name suggests, IoB devices introduce an even more intimate interplay between humans and gadgets. IoB devices monitor the human body, collect health metrics and other personal information, and transmit those data over the Internet. Many devices, such as fitness trackers, are already in use ... IoB devices ... and those in development can track, record, and store users' whereabouts, bodily functions, and what they see, hear, and even think.

Schwab's World Economic Forum, a long-winded way of saying 'fascism' or 'the Cult', has gone full-on with the Internet of Bodies in the 'Covid' era. 'We're entering the era of the Internet of Bodies', it declared, 'collecting our physical data via a range of devices that can be implanted, swallowed or worn'. The result would be a huge amount of health-related data that could improve human wellbeing around the world, and prove crucial in fighting the 'Covid-19 pandemic'. Does anyone think these clowns care about 'human wellbeing' after the death and devastation their pandemic hoax has purposely caused? Schwab and co say we should move forward with the Internet of Bodies because 'Keeping track of symptoms could help us stop the spread of infection, and quickly detect new cases'. How wonderful, but keeping track' is all they are really bothered

about. Researchers were investigating if data gathered from smartwatches and similar devices could be used as viral infection alerts by tracking the user's heart rate and breathing. Schwab said in his 2018 book *Shaping the Future of the Fourth Industrial Revolution*:

The lines between technologies and beings are becoming blurred and not just by the ability to create lifelike robots or synthetics. Instead it is about the ability of new technologies to literally become part of us. Technologies already influence how we understand ourselves, how we think about each other, and how we determine our realities. As the technologies ... give us deeper access to parts of ourselves, we may begin to integrate digital technologies into our bodies.

You can see what the game is. Twenty-four hour control and people – if you could still call them that – would never know when something would go ping and take them out of circulation. It's the most obvious rush to a global fascist dictatorship and the complete submission of humanity and yet still so many are locked away in their Cult-induced perceptual coma and can't see it.

Smart Grid control centres

The human body is being transformed by the 'vaccines' and in other ways into a synthetic cyborg that can be attached to the global Smart Grid which would be controlled from a central point and other sub-locations of Grid manipulation. Where are these planned to be? Well, China for a start which is one of the Cult's biggest centres of operation. The technological control system and technocratic rule was incubated here to be unleashed across the world after the 'Covid' hoax came out of China in 2020. Another Smart Grid location that will surprise people new to this is Israel. I have exposed in *The Trigger* how Sabbatian technocrats, intelligence and military operatives were behind the horrors of 9/11 and not 19 Arab hijackers' who somehow manifested the ability to pilot big passenger airliners when instructors at puddle-jumping flying schools described some of them as a joke. The 9/11 attacks were made possible through control of civilian and military air computer systems and those of the White House, Pentagon and connected agencies. See *The Trigger* – it

will blow your mind. The controlling and coordinating force were the Sabbatian networks in Israel and the United States which by then had infiltrated the entire US government, military and intelligence system. The real name of the American Deep State is 'Sabbatian State'. Israel is a tiny country of only nine million people, but it is one of the global centres of cyber operations and fast catching Silicon Valley in importance to the Cult. Israel is known as the 'start-up nation' for all the cyber companies spawned there with the Sabbatian specialisation of 'cyber security' that I mentioned earlier which gives those companies access to computer systems of their clients in real time through 'backdoors' written into the coding when security software is downloaded. The Sabbatian centre of cyber operations outside Silicon Valley is the Israeli military Cyber Intelligence Unit, the biggest infrastructure project in Israel's history, headquartered in the desert-city of Beersheba and involving some 20,000 'cyber soldiers'. Here are located a literal army of Internet trolls scanning social media, forums and comment lists for anyone challenging the Cult agenda. The UK military has something similar with its 77th Brigade and associated operations. The Beersheba complex includes research and development centres for other Cult operations such as Intel, Microsoft, IBM, Google, Apple, Hewlett-Packard, Cisco Systems, Facebook and Motorola. Techcrunch.com ran an article about the Beersheba global Internet technology centre headlined 'Israel's desert city of Beersheba is turning into a cybertech oasis':

The military's massive relocation of its prestigious technology units, the presence of multinational and local companies, a close proximity to Ben Gurion University and generous government subsidies are turning Beersheba into a major global cybertech hub. Beersheba has all of the ingredients of a vibrant security technology ecosystem, including Ben Gurion University with its graduate program in cybersecurity and Cyber Security Research Center, and the presence of companies such as EMC, Deutsche Telekom, PayPal, Oracle, IBM, and Lockheed Martin. It's also the future home of the INCB (Israeli National Cyber Bureau); offers a special income tax incentive for cyber security companies, and was the site for the relocation of the army's intelligence corps units.

Sabbatians have taken over the cyber world through the following process: They scan the schools for likely cyber talent and develop them at Ben Gurion University and their period of conscription in the Israeli Defense Forces when they are stationed at the Beersheba complex. When the cyber talented officially leave the army they are funded to start cyber companies with technology developed by themselves or given to them by the state. Much of this is stolen through backdoors of computer systems around the world with America top of the list. Others are sent off to Silicon Valley to start companies or join the major ones and so we have many major positions filled by apparently 'Jewish' but really Sabbatian operatives. Google, YouTube and Facebook are all run by 'Jewish' CEOs while Twitter is all but run by ultra-Zionist hedge-fund shark Paul Singer. At the centre of the Sabbatian global cyber web is the Israeli army's Unit 8200 which specialises in hacking into computer systems of other countries, inserting viruses, gathering information, instigating malfunction, and even taking control of them from a distance. A long list of Sabbatians involved with 9/11, Silicon Valley and Israeli cyber security companies are operatives of Unit 8200. This is not about Israel. It's about the Cult. Israel is planned to be a Smart Grid hub as with China and what is happening at Beersheba is not for the benefit of Jewish people who are treated disgustingly by the Sabbatian elite that control the country. A glance at the Nuremberg Codes will tell you that.

The story is much bigger than 'Covid', important as that is to where we are being taken. Now, though, it's time to really strap in. There's more ... much more ...

CHAPTER ELEVEN

Who controls the Cult?

Awake, arise or be forever fall'n
John Milton, *Paradise Lost*

I have exposed this far the level of the Cult conspiracy that operates in the world of the seen and within the global secret society and satanic network which operates in the shadows one step back from the seen. The story, however, goes much deeper than that.

The 'Covid' hoax is major part of the Cult agenda, but only part, and to grasp the biggest picture we have to expand our attention beyond the realm of human sight and into the infinity of possibility that we cannot see. It is from here, ultimately, that humanity is being manipulated into a state of total control by the force which dictates the actions of the Cult. How much of reality can we see? Next to damn all is the answer. We may appear to see all there is to see in the 'space' our eyes survey and observe, but little could be further from the truth. The human 'world' is only a tiny band of frequency that the body's visual and perceptual systems can decode into *perception* of a 'world'. According to mainstream science the electromagnetic spectrum is 0.005 percent of what exists in the Universe ([Fig 10](#)). The maximum estimate I have seen is 0.5 percent and either way it's miniscule. I say it is far, far, smaller even than 0.005 percent when you compare reality we see with the totality of reality that we don't. Now get this if you are new to such information: Visible light, the only band of frequency that we can see, is a *fraction* of the 0.005

percent (Fig 11 overleaf). Take this further and realise that our universe is one of infinite universes and that universes are only a fragment of overall reality – *infinite* reality. Then compare that with the almost infinitesimal frequency band of visible light or human sight. You see that humans are as near blind as it is possible to be without actually being so. Artist and filmmaker, Sergio Toporek, said:

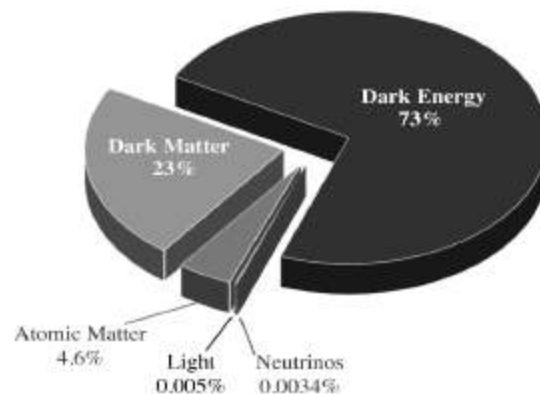


Figure 10: Humans can perceive such a tiny band of visual reality it's laughable.

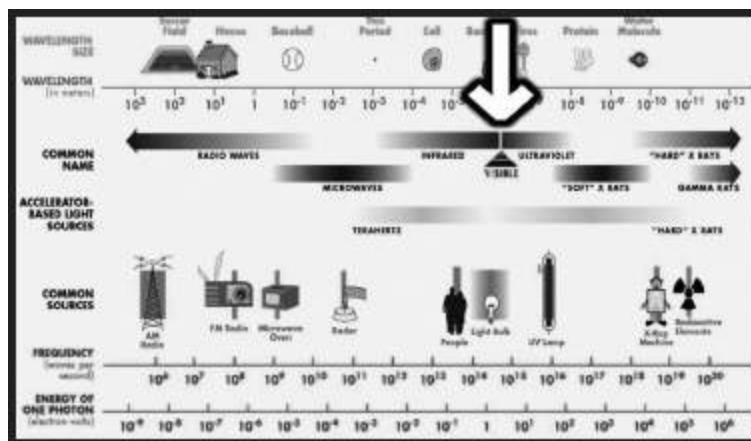


Figure 11: We can see a smear of the 0.005 percent electromagnetic spectrum, but we still know it all. Yep, makes sense.

Consider that you can see less than 1% of the electromagnetic spectrum and hear less than 1% of the acoustic spectrum. 90% of the cells in your body carry their own microbial DNA and are not 'you'. The atoms in your body are 99.999999999999999% empty space and none of them are the ones you were born with ... Human beings have 46 chromosomes, two less than a potato.

The existence of the rainbow depends on the conical photoreceptors in your eyes; to animals without cones, the rainbow does not exist. So you don't just look at a rainbow, you create it. This is pretty amazing, especially considering that all the beautiful colours you see represent less than 1% of the electromagnetic spectrum.

Suddenly the 'world' of humans looks a very different place. Take into account, too, that Planet Earth when compared with the projected size of this single universe is the equivalent of a billionth of a pinhead. Imagine the ratio that would be when compared to infinite reality. To think that Christianity once insisted that Earth and humanity were the centre of everything. This background is vital if we are going to appreciate the nature of 'human' and how we can be manipulated by an unseen force. To human visual reality virtually *everything* is unseen and yet the prevailing perception within the institutions and so much of the public is that if we can't see it, touch it, hear it, taste it and smell it then it cannot exist. Such perception is indoctrinated and encouraged by the Cult and its agents because it isolates believers in the strictly limited, village-idiot, realm of the five senses where perceptions can be firewalled and information controlled. Most of those perpetuating the 'this-world-is-all-there-is' insanity are themselves indoctrinated into believing the same delusion. While major players and influencers know that official reality is laughable most of those in science, academia and medicine really believe the nonsense they peddle and teach succeeding generations. Those who challenge the orthodoxy are dismissed as nutters and freaks to protect the manufactured illusion from exposure. Observe the dynamic of the 'Covid' hoax and you will see how that takes the same form. The inner-circle psychopaths knows it's a gigantic scam, but almost the entirety of those imposing their fascist rules believe that 'Covid' is all that they're told it is.

Stolen identity

Ask people who they are and they will give you their name, place of birth, location, job, family background and life story. Yet that is not who they are – it is what they are *experiencing*. The difference is *absolutely crucial*. The true 'I', the eternal, infinite 'I', is consciousness,

a state of being aware. Forget 'form'. That is a vehicle for a brief experience. Consciousness does not come *from* the brain, but *through* the brain and even that is more symbolic than literal. We are awareness, pure awareness, and this is what withdraws from the body at what we call 'death' to continue our eternal beingness, *isness*, in other realms of reality within the limitlessness of infinity or the Biblical 'many mansions in my father's house'. Labels of a human life, man, woman, transgender, black, white, brown, nationality, circumstances and income are not who we are. They are what we are – awareness – is *experiencing* in a brief connection with a band of frequency we call 'human'. The labels are not the self; they are, to use the title of one of my books, a *Phantom Self*. I am not David Icke born in Leicester, England, on April 29th, 1952. I am the consciousness *having that experience*. The Cult and its non-human masters seek to convince us through the institutions of 'education', science, medicine, media and government that what we are *experiencing* is who we *are*. It's so easy to control and direct perception locked away in the bewildered illusions of the five senses with no expanded radar. Try, by contrast, doing the same with a humanity aware of its true self and its true power to consciously create its reality and experience. How is it possible to do this? We do it all day every day. If you perceive yourself as 'little me' with no power to impact upon your life and the world then your life experience will reflect that. You will hand the power you don't think you have to authority in all its forms which will use it to control your experience. This, in turn, will appear to confirm your perception of 'little me' in a self-fulfilling feedback loop. But that is what 'little me' really is – a *perception*. We are all 'big-me', infinite me, and the Cult has to make us forget that if its will is to prevail. We are therefore manipulated and pressured into self-identifying with human labels and not the consciousness/awareness *experiencing* those human labels.

The phenomenon of identity politics is a Cult-instigated manipulation technique to sub-divide previous labels into even smaller ones. A United States university employs this list of letters to

describe student identity: LGBTTQQFAGPBDSM or lesbian, gay, bisexual, transgender, transsexual, queer, questioning, flexual, asexual, gender-fuck, polyamorous, bondage/discipline, dominance/submission and sadism/masochism. I'm sure other lists are even longer by now as people feel the need to self-identity the 'I' with the minutiae of race and sexual preference. Wokers programmed by the Cult for generations believe this is about 'inclusivity' when it's really the Cult locking them away into smaller and smaller versions of Phantom Self while firewalling them from the influence of their true self, the infinite, eternal 'I'. You may notice that my philosophy which contends that we are all unique points of attention/awareness within the same infinite whole or Oneness is the ultimate non-racism. The very sense of Oneness makes the judgement of people by their body-type, colour or sexuality utterly ridiculous and confirms that racism has no understanding of reality (including anti-white racism). Yet despite my perception of life Cult agents and fast-asleep Wokers label me racist to discredit my information while they are themselves phenomenally racist and sexist. All they see is race and sexuality and they judge people as good or bad, demons or untouchables, by their race and sexuality. All they see is *Phantom Self* and perceive themselves in terms of Phantom Self. They are pawns and puppets of the Cult agenda to focus attention and self-identity in the five senses and play those identities against each other to divide and rule. Columbia University has introduced segregated graduations in another version of social distancing designed to drive people apart and teach them that different racial and cultural groups have nothing in common with each other. The last thing the Cult wants is unity. Again the pump-primers of this will be Cult operatives in the knowledge of what they are doing, but the rest are just the Phantom Self blind leading the Phantom Self blind. We *do* have something in common – we are all *the same consciousness* having different temporary experiences.

What is this 'human'?

Yes, what *is* 'human'? That is what we are supposed to be, right? I mean 'human'? True, but 'human' is the experience not the 'I'. Break it down to basics and 'human' is the way that information is processed. If we are to experience and interact with this band of frequency we call the 'world' we must have a vehicle that operates within that band of frequency. Our consciousness in its prime form cannot do that; it is way beyond the frequency of the human realm. My consciousness or awareness could not tap these keys and pick up the cup in front of me in the same way that radio station A cannot interact with radio station B when they are on different frequencies. The human body is the means through which we have that interaction. I have long described the body as a biological computer which processes information in a way that allows consciousness to experience this reality. The body is a receiver, transmitter and processor of information in a particular way that we call human. We visually perceive only the world of the five senses in a wakened state – that is the limit of the body's visual decoding system. In truth it's not even visual in the way we experience 'visual reality' as I will come to in a moment. We are 'human' because the body processes the information sources of human into a reality and behaviour system that we *perceive* as human. Why does an elephant act like an elephant and not like a human or a duck? The elephant's biological computer is a different information field and processes information according to that program into a visual and behaviour type we call an elephant. The same applies to everything in our reality. These body information fields are perpetuated through procreation (like making a copy of a software program). The Cult wants to break that cycle and intervene technologically to transform the human information field into one that will change what we call humanity. If it can change the human information field it will change the way that field processes information and change humanity both 'physically' and psychologically. Hence the *messenger* (information) RNA 'vaccines' and so much more that is targeting human genetics by changing the body's information – *messaging* – construct through food, drink, radiation, toxicity and other means.

Reality that we experience is nothing like reality as it really is in the same way that the reality people experience in virtual reality games is not the reality they are really living in. The game is only a decoded source of information that appears to be a reality. Our world is also an information construct – a *simulation* (more later). In its base form our reality is a wavefield of information much the same in theme as Wi-Fi. The five senses decode wavefield information into electrical information which they communicate to the brain to decode into holographic (illusory ‘physical’) information. Different parts of the brain specialise in decoding different senses and the information is fused into a reality that appears to be outside of us but is really inside the brain and the genetic structure in general ([Fig 12](#) overleaf). DNA is a receiver-transmitter of information and a vital part of this decoding process and the body’s connection to other realities. Change DNA and you change the way we decode and connect with reality – see ‘Covid vaccines’. Think of computers decoding Wi-Fi. You have information encoded in a radiation field and the computer decodes that information into a very different form on the screen. You can’t see the Wi-Fi until its information is made manifest on the screen and the information on the screen is inside the computer and not outside. I have just described how we decode the ‘human world’. All five senses decode the waveform ‘Wi-Fi’ field into electrical signals and the brain (computer) constructs reality inside the brain and not outside – ‘You don’t just look at a rainbow, you create it’. Sound is a simple example. We don’t hear sound until the brain decodes it. Waveform sound waves are picked up by the hearing sense and communicated to the brain in an electrical form to be decoded into the sounds that we hear. Everything we hear is inside the brain along with everything we see, feel, smell and taste. Words and language are waveform fields generated by our vocal chords which pass through this process until they are decoded by the brain into words that we hear. Different languages are different frequency fields or sound waves generated by vocal chords. Late British philosopher Alan Watts said:

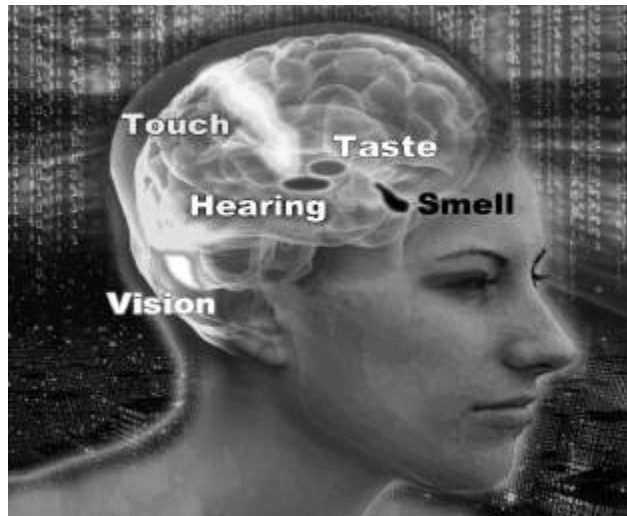


Figure 12: The brain receives information from the five senses and constructs from that our perceived reality.

[Without the brain] the world is devoid of light, heat, weight, solidity, motion, space, time or any other imaginable feature. All these phenomena are interactions, or transactions, of vibrations with a certain arrangement of neurons.

That's exactly what they are and scientist Robert Lanza describes in his book, *Biocentrism*, how we decode electromagnetic waves and energy into visual and 'physical' experience. He uses the example of a flame emitting photons, electromagnetic energy, each pulsing electrically and magnetically:

... these ... invisible electromagnetic waves strike a human retina, and if (and only if) the waves happen to measure between 400 and 700 nano meters in length from crest to crest, then their energy is just right to deliver a stimulus to the 8 million cone-shaped cells in the retina.

Each in turn send an electrical pulse to a neighbour neuron, and on up the line this goes, at 250 mph, until it reaches the ... occipital lobe of the brain, in the back of the head. There, a cascading complex of neurons fire from the incoming stimuli, and we subjectively perceive this experience as a yellow brightness occurring in a place we have been conditioned to call the 'external world'.

You hear what you decode

If a tree falls or a building collapses they make no noise unless someone is there to decode the energetic waves generated by the disturbance into what we call sound. Does a falling tree make a noise? Only if you hear it – *decode* it. Everything in our reality is a frequency field of information operating within the overall ‘Wi-Fi’ field that I call The Field. A vibrational disturbance is generated in The Field by the fields of the falling tree or building. These disturbance waves are what we decode into the sound of them falling. If no one is there to do that then neither will make any noise. Reality is created by the observer – *decoder* – and the *perceptions* of the observer affect the decoding process. For this reason different people – different *perceptions* – will perceive the same reality or situation in a different way. What one may perceive as a nightmare another will see as an opportunity. The question of why the Cult is so focused on controlling human perception now answers itself. All experienced reality is the act of decoding and we don’t experience Wi-Fi until it is decoded on the computer screen. The sight and sound of an Internet video is encoded in the Wi-Fi all around us, but we don’t see or hear it until the computer decodes that information. Taste, smell and touch are all phenomena of the brain as a result of the same process. We don’t taste, smell or feel anything except in the brain and there are pain relief techniques that seek to block the signal from the site of discomfort to the brain because if the brain doesn’t decode that signal we don’t feel pain. Pain is in the brain and only appears to be at the point of impact thanks to the feedback loop between them. We don’t see anything until electrical information from the sight senses is decoded in an area at the back of the brain. If that area is damaged we can go blind when our eyes are perfectly okay. So why do we go blind if we damage an eye? We damage the information processing between the waveform visual information and the visual decoding area of the brain. If information doesn’t reach the brain in a form it can decode then we can’t see the visual reality that it represents. What’s more the brain is decoding only a fraction of the information it receives and the rest is absorbed by the

sub-conscious mind. This explanation is from the science magazine, *Wonderpedia*:

Every second, 11 million sensations crackle along these [brain] pathways ... The brain is confronted with an alarming array of images, sounds and smells which it rigorously filters down until it is left with a manageable list of around 40. Thus 40 sensations per second make up what we perceive as reality.

The 'world' is not what people are told to believe that is it and the inner circles of the Cult *know that*.

Illusory 'physical' reality

We can only see a smear of 0.005 percent of the Universe which is only one of a vast array of universes – 'mansions' – within infinite reality. Even then the brain decodes only 40 pieces of information ('sensations') from a potential *11 million* that we receive every second. Two points strike you from this immediately: The sheer breathtaking stupidity of believing we know anything so rigidly that there's nothing more to know; and the potential for these processes to be manipulated by a malevolent force to control the reality of the population. One thing I can say for sure with no risk of contradiction is that when you can perceive an almost indescribable fraction of infinite reality there is always more to know as in tidal waves of it. Ancient Greek philosopher Socrates was so right when he said that wisdom is to know how little we know. How obviously true that is when you think that we are experiencing a physical world of solidity that is neither physical nor solid and a world of apartness when everything is connected. Cult-controlled 'science' dismisses the so-called 'paranormal' and all phenomena related to that when the 'para'-normal is perfectly normal and explains the alleged 'great mysteries' which dumbfound scientific minds. There is a reason for this. A 'scientific mind' in terms of the mainstream is a material mind, a five-sense mind imprisoned in see it, touch it, hear it, smell it and taste it. Phenomena and happenings that can't be explained that way leave the 'scientific mind' bewildered and the rule is that if they

can't account for why something is happening then it can't, by definition, be happening. I beg to differ. Telepathy is thought waves passing through The Field (think wave disturbance again) to be decoded by someone able to connect with that wavelength (information). For example: You can pick up the thought waves of a friend at any distance and at the very least that will bring them to mind. A few minutes later the friend calls you. 'My god', you say, 'that's incredible – I was just thinking of you.' Ah, but *they* were thinking of *you* before they made the call and that's what you decoded. Native peoples not entrapped in five-sense reality do this so well it became known as the 'bush telegraph'. Those known as psychics and mediums (genuine ones) are doing the same only across dimensions of reality. 'Mind over matter' comes from the fact that matter and mind are the *same*. The state of one influences the state of the other. Indeed one *and* the other are illusions. They are aspects of the same field. Paranormal phenomena are all explainable so why are they still considered 'mysteries' or not happening? Once you go down this road of understanding you begin to expand awareness beyond the five senses and that's the nightmare for the Cult.



Figure 13: Holograms are not solid, but the best ones appear to be.

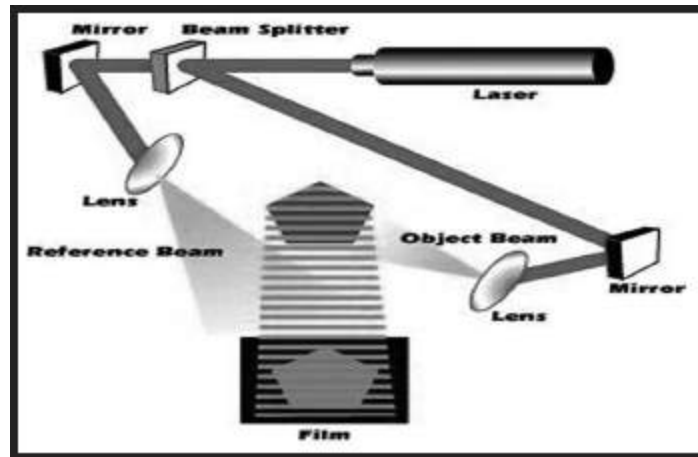


Figure 14: How holograms are created by capturing a waveform version of the subject image.

Holographic 'solidity'

Our reality is not solid, it is holographic. We are now well aware of holograms which are widely used today. Two-dimensional information is decoded into a three-dimensional reality that is not solid although can very much appear to be (Fig 13). Holograms are created with a laser divided into two parts. One goes directly onto a holographic photographic print ('reference beam') and the other takes a waveform image of the subject ('working beam') before being directed onto the print where it 'collides' with the other half of the laser (Fig 14). This creates a *waveform* interference pattern which contains the wavefield information of whatever is being photographed (Fig 15 overleaf). The process can be likened to dropping pebbles in a pond. Waves generated by each one spread out across the water to collide with the others and create a wave representation of where the stones fell and at what speed, weight and distance. A waveform interference pattern of a hologram is akin to the waveform information in The Field which the five senses decode into electrical signals to be decoded by the brain into a holographic illusory 'physical' reality. In the same way when a laser (think human attention) is directed at the waveform interference pattern a three-dimensional version of the subject is projected into apparently 'solid' reality (Fig 16). An amazing trait of holograms reveals more 'paranormal mysteries'. Information of the *whole*

hologram is encoded in waveform in every part of the interference pattern by the way they are created. This means that every *part* of a hologram is a smaller version of the whole. Cut the interference wave-pattern into four and you won't get four parts of the image. You get quarter-sized versions of the *whole* image. The body is a hologram and the same applies. Here we have the basis of acupuncture, reflexology and other forms of healing which identify representations of the whole body in all of the parts, hands, feet, ears, everywhere. Skilled palm readers can do what they do because the information of whole body is encoded in the hand. The concept of as above, so below, comes from this.



Figure 15: A waveform interference pattern that holds the information that transforms into a hologram.



Figure 16: Holographic people including 'Elvis' holographically inserted to sing a duet with Celine Dion.

The question will be asked of why, if solidity is illusory, we can't just walk through walls and each other. The resistance is not solid against solid; it is electromagnetic field against electromagnetic field and we decode this into the *experience* of solid against solid. We should also not underestimate the power of belief to dictate reality. What you believe is impossible *will be*. Your belief impacts on your decoding processes and they won't decode what you think is impossible. What we believe we perceive and what we perceive we experience. 'Can't dos' and 'impossibles' are like a firewall in a computer system that won't put on the screen what the firewall blocks. How vital that is to understanding how human experience has been hijacked. I explain in *The Answer, Everything You Need To Know But Have Never Been Told* and other books a long list of 'mysteries' and 'paranormal' phenomena that are not mysterious and perfectly normal once you realise what reality is and how it works. 'Ghosts' can be seen to pass through 'solid' walls because the walls are not solid and the ghost is a discarnate entity operating on a frequency so different to that of the wall that it's like two radio stations sharing the same space while never interfering with each other. I have seen ghosts do this myself. The apartness of people and objects is also an illusion. Everything is connected by the Field like all sea life is connected by the sea. It's just that within the limits of our visual reality we only 'see' holographic information and not the field of information that connects everything and from which the holographic world is made manifest. If you can only see holographic 'objects' and not the field that connects them they will appear to you as unconnected to each other in the same way that we see the computer while not seeing the Wi-Fi.

What you don't know *can* hurt you

Okay, we return to those 'two worlds' of human society and the Cult with its global network of interconnecting secret societies and satanic groups which manipulate through governments, corporations, media, religions, etc. The fundamental difference between them is *knowledge*. The idea has been to keep humanity

ignorant of the plan for its total enslavement underpinned by a crucial ignorance of reality – who we are and where we are – and how we interact with it. ‘Human’ should be the interaction between our expanded eternal consciousness and the five-sense body experience. We are meant to be *in* this world in terms of the five senses but not *of* this world in relation to our greater consciousness and perspective. In that state we experience the small picture of the five senses within the wider context of the big picture of awareness beyond the five senses. Put another way the five senses see the dots and expanded awareness connects them into pictures and patterns that give context to the apparently random and unconnected. Without the context of expanded awareness the five senses see only apartness and randomness with apparently no meaning. The Cult and its other-dimensional controllers seek to intervene in the frequency realm where five-sense reality is supposed to connect with expanded reality and to keep the two apart (more on this in the final chapter). When that happens five-sense mental and emotional processes are no longer influenced by expanded awareness, or the True ‘I’, and instead are driven by the isolated perceptions of the body’s decoding systems. They are in the world *and* of it. Here we have the human plight and why humanity with its potential for infinite awareness can be so easily manipulatable and descend into such extremes of stupidity.

Once the Cult isolates five-sense mind from expanded awareness it can then program the mind with perceptions and beliefs by controlling information that the mind receives through the ‘education’ system of the formative years and the media perceptual bombardment and censorship of an entire lifetime. Limit perception and a sense of the possible through limiting knowledge by limiting and skewing information while censoring and discrediting that which could set people free. As the title of another of my books says ... *And The Truth Shall Set You Free*. For this reason the last thing the Cult wants in circulation is the truth about anything – especially the reality of the eternal ‘I’ – and that’s why it is desperate to control information. The Cult knows that information becomes perception

which becomes behaviour which, collectively, becomes human society. Cult-controlled and funded mainstream 'science' denies the existence of an eternal 'I' and seeks to dismiss and trash all evidence to the contrary. Cult-controlled mainstream religion has a version of 'God' that is little more than a system of control and dictatorship that employs threats of damnation in an afterlife to control perceptions and behaviour in the here and now through fear and guilt. Neither is true and it's the 'neither' that the Cult wishes to suppress. This 'neither' is that everything is an expression, a point of attention, within an infinite state of consciousness which is the real meaning of the term 'God'.

Perceptual obsession with the 'physical body' and five-senses means that 'God' becomes personified as a bearded bloke sitting among the clouds or a raging bully who loves us if we do what 'he' wants and condemns us to the fires of hell if we don't. These are no more than a 'spiritual' fairy tales to control and dictate events and behaviour through fear of this 'God' which has bizarrely made 'God-fearing' in religious circles a state to be desired. I would suggest that fearing *anything* is not to be encouraged and celebrated, but rather deleted. You can see why 'God fearing' is so beneficial to the Cult and its religions when *they* decide what 'God' wants and what 'God' demands (the Cult demands) that everyone do. As the great American comedian Bill Hicks said satirising a Christian zealot: 'I think what God meant to say.' How much of this infinite awareness ('God') that we access is decided by how far we choose to expand our perceptions, self-identity and sense of the possible. The scale of self-identity reflects itself in the scale of awareness that we can connect with and are influenced by – how much knowing and insight we have instead of programmed perception. You cannot expand your awareness into the infinity of possibility when you believe that you are little me Peter the postman or Mary in marketing and nothing more. I'll deal with this in the concluding chapter because it's crucial to how we turnaround current events.

Where the Cult came from

When I realised in the early 1990s there was a Cult network behind global events I asked the obvious question: When did it start? I took it back to ancient Rome and Egypt and on to Babylon and Sumer in Mesopotamia, the 'Land Between Two Rivers', in what we now call Iraq. The two rivers are the Tigris and Euphrates and this region is of immense historical and other importance to the Cult, as is the land called Israel only 550 miles away by air. There is much more going on with deep esoteric meaning across this whole region. It's not only about 'wars for oil'. Priceless artefacts from Mesopotamia were stolen or destroyed after the American and British invasion of Iraq in 2003 justified by the lies of Boy Bush and Tony Blair (their Cult masters) about non-existent 'weapons of mass destruction'.

Mesopotamia was the location of Sumer (about 5,400BC to 1,750BC), and Babylon (about 2,350BC to 539BC). Sabbatians may have become immensely influential in the Cult in modern times but they are part of a network that goes back into the mists of history. Sumer is said by historians to be the 'cradle of civilisation'. I disagree. I say it was the re-start of what we call human civilisation after cataclysmic events symbolised in part as the 'Great Flood' destroyed the world that existed before. These fantastic upheavals that I have been describing in detail in the books since the early 1990s appear in accounts and legends of ancient cultures across the world and they are supported by geological and biological evidence. Stone tablets found in Iraq detailing the Sumer period say the cataclysms were caused by non-human 'gods' they call the Anunnaki. These are described in terms of extraterrestrial visitations in which knowledge supplied by the Anunnaki is said to have been the source of at least one of the world's oldest writing systems and developments in astronomy, mathematics and architecture that were way ahead of their time. I have covered this subject at length in *The Biggest Secret* and *Children of the Matrix* and the same basic 'Anunnaki' story can be found in Zulu accounts in South Africa where the late and very great Zulu high shaman Credo Mutwa told me that the Sumerian Anunnaki were known by Zulus as the Chitauri or 'children of the serpent'. See my six-hour video interview with Credo on this subject entitled *The*

Reptilian Agenda recorded at his then home near Johannesburg in 1999 which you can watch on the Ickonic media platform.

The Cult emerged out of Sumer, Babylon and Egypt (and elsewhere) and established the Roman Empire before expanding with the Romans into northern Europe from where many empires were savagely imposed in the form of Cult-controlled societies all over the world. Mass death and destruction was their calling card. The Cult established its centre of operations in Europe and European Empires were Cult empires which allowed it to expand into a global force. Spanish and Portuguese colonialists headed for Central and South America while the British and French targeted North America. Africa was colonised by Britain, France, Belgium, the Netherlands, Portugal, Spain, Italy, and Germany. Some like Britain and France moved in on the Middle East. The British Empire was by far the biggest for a simple reason. By now Britain was the headquarters of the Cult from which it expanded to form Canada, the United States, Australia and New Zealand. The Sun never set on the British Empire such was the scale of its occupation. London remains a global centre for the Cult along with Rome and the Vatican although others have emerged in Israel and China. It is no accident that the 'virus' is alleged to have come out of China while Italy was chosen as the means to terrify the Western population into compliance with 'Covid' fascism. Nor that Israel has led the world in 'Covid' fascism and mass 'vaccination'.

You would think that I would mention the United States here, but while it has been an important means of imposing the Cult's will it is less significant than would appear and is currently in the process of having what power it does have deleted. The Cult in Europe has mostly loaded the guns for the US to fire. America has been controlled from Europe from the start through Cult operatives in Britain and Europe. The American Revolution was an illusion to make it appear that America was governing itself while very different forces were pulling the strings in the form of Cult families such as the Rothschilds through the Rockefellers and other subordinates. The Rockefellers are extremely close to Bill Gates and

established both scalpel and drug 'medicine' and the World Health Organization. They play a major role in the development and circulation of vaccines through the Rockefeller Foundation on which Bill Gates said his Foundation is based. Why wouldn't this be the case when the Rockefellers and Gates are on the same team? Cult infiltration of human society goes way back into what we call history and has been constantly expanding and centralising power with the goal of establishing a global structure to dictate everything. Look how this has been advanced in great leaps with the 'Covid' hoax.

The non-human dimension

I researched and observed the comings and goings of Cult operatives through the centuries and even thousands of years as they were born, worked to promote the agenda within the secret society and satanic networks, and then died for others to replace them. Clearly there had to be a coordinating force that spanned this entire period while operatives who would not have seen the end goal in their lifetimes came and went advancing the plan over millennia. I went in search of that coordinating force with the usual support from the extraordinary synchronicity of my life which has been an almost daily experience since 1990. I saw common themes in religious texts and ancient cultures about a non-human force manipulating human society from the hidden. Christianity calls this force Satan, the Devil and demons; Islam refers to the Jinn or Djinn; Zulus have their Chitauri (spelt in other ways in different parts of Africa); and the Gnostic people in Egypt in the period around and before 400AD referred to this phenomena as the 'Archons', a word meaning rulers in Greek. Central American cultures speak of the 'Predators' among other names and the same theme is everywhere. I will use 'Archons' as a collective name for all of them. When you see how their nature and behaviour is described all these different sources are clearly talking about the same force. Gnostics described the Archons in terms of 'luminous fire' while Islam relates the Jinn to 'smokeless fire'. Some refer to beings in form that could occasionally be seen, but the most common of common theme is that they operate from

unseen realms which means almost all existence to the visual processes of humans. I had concluded that this was indeed the foundation of human control and that the Cult was operating within the human frequency band on behalf of this hidden force when I came across the writings of Gnostics which supported my conclusions in the most extraordinary way.

A sealed earthen jar was found in 1945 near the town of Nag Hammadi about 75-80 miles north of Luxor on the banks of the River Nile in Egypt. Inside was a treasure trove of manuscripts and texts left by the Gnostic people some 1,600 years earlier. They included 13 leather-bound papyrus codices (manuscripts) and more than 50 texts written in Coptic Egyptian estimated to have been hidden in the jar in the period of 400AD although the source of the information goes back much further. Gnostics oversaw the Great or Royal Library of Alexandria, the fantastic depository of ancient texts detailing advanced knowledge and accounts of human history. The Library was dismantled and destroyed in stages over a long period with the death-blow delivered by the Cult-established Roman Church in the period around 415AD. The Church of Rome was the Church of Babylon relocated as I said earlier. Gnostics were not a race. They were a way of perceiving reality. Whenever they established themselves and their information circulated the terrorists of the Church of Rome would target them for destruction. This happened with the Great Library and with the Gnostic Cathars who were burned to death by the psychopaths after a long period of oppression at the siege of the Castle of Monségur in southern France in 1244. The Church has always been terrified of Gnostic information which demolishes the official Christian narrative although there is much in the Bible that supports the Gnostic view if you read it in another way. To anyone studying the texts of what became known as the Nag Hammadi Library it is clear that great swathes of Christian and Biblical belief has its origin with Gnostics sources going back to Sumer. Gnostic themes have been twisted to manipulate the perceived reality of Bible believers. Biblical texts have been in the open for centuries where they could be changed while Gnostic

documents found at Nag Hammadi were sealed away and untouched for 1,600 years. What you see is what they wrote.

Use your *pneuma* not your *nous*

Gnosticism and Gnostic come from 'gnosis' which means knowledge, or rather *secret* knowledge, in the sense of spiritual awareness – knowledge about reality and life itself. The desperation of the Cult's Church of Rome to destroy the Gnostics can be understood when the knowledge they were circulating was the last thing the Cult wanted the population to know. Sixteen hundred years later the same Cult is working hard to undermine and silence me for the same reason. The dynamic between knowledge and ignorance is a constant. 'Time' appears to move on, but essential themes remain the same. We are told to 'use your nous', a Gnostic word for head/brain/intelligence. They said, however, that spiritual awakening or 'salvation' could only be secured by expanding awareness *beyond* what they called *nous* and into *pneuma* or Infinite Self. Obviously as I read these texts the parallels with what I have been saying since 1990 were fascinating to me. There is a universal truth that spans human history and in that case why wouldn't we be talking the same language 16 centuries apart? When you free yourself from the perception program of the five senses and explore expanded realms of consciousness you are going to connect with the same information no matter what the perceived 'era' within a manufactured timeline of a single and tiny range of manipulated frequency. Humans working with 'smart' technology or knocking rocks together in caves is only a timeline appearing to operate within the human frequency band. Expanded awareness and the knowledge it holds have always been there whether the era be Stone Age or computer age. We can only access that knowledge by opening ourselves to its frequency which the five-sense prison cell is designed to stop us doing. Gates, Fauci, Whitty, Vallance, Zuckerberg, Brin, Page, Wojcicki, Bezos, and all the others behind the 'Covid' hoax clearly have a long wait before their range of frequency can make that connection given that an open heart is

crucial to that as we shall see. Instead of accessing knowledge directly through expanded awareness it is given to Cult operatives by the secret society networks of the Cult where it has been passed on over thousands of years outside the public arena. Expanded realms of consciousness is where great artists, composers and writers find their inspiration and where truth awaits anyone open enough to connect with it. We need to go there fast.

Archon hijack

A fifth of the Nag Hammadi texts describe the existence and manipulation of the Archons led by a 'Chief Archon' they call 'Yaldabaoth', or the 'Demiurge', and this is the Christian 'Devil', 'Satan', 'Lucifer', and his demons. Archons in Biblical symbolism are the 'fallen ones' which are also referred to as fallen angels after the angels expelled from heaven according to the Abrahamic religions of Judaism, Christianity and Islam. These angels are claimed to tempt humans to 'sin' ongoing and you will see how accurate that symbolism is during the rest of the book. The theme of 'original sin' is related to the 'Fall' when Adam and Eve were 'tempted by the serpent' and fell from a state of innocence and 'obedience' (connection) with God into a state of disobedience (disconnection). The Fall is said to have brought sin into the world and corrupted everything including human nature. Yaldabaoth, the 'Lord Archon', is described by Gnostics as a 'counterfeit spirit', 'The Blind One', 'The Blind God', and 'The Foolish One'. The Jewish name for Yaldabaoth in Talmudic writings is Samael which translates as 'Poison of God', or 'Blindness of God'. You see the parallels. Yaldabaoth in Islamic belief is the Muslim Jinn devil known as Shaytan – Shaytan is Satan as the same themes are found all over the world in every religion and culture. The 'Lord God' of the Old Testament is the 'Lord Archon' of Gnostic manuscripts and that's why he's such a bloodthirsty bastard. Satan is known by Christians as 'the Demon of Demons' and Gnostics called Yaldabaoth the 'Archon of Archons'. Both are known as 'The Deceiver'. We are talking about the same 'bloke' for sure and these common themes

using different names, storylines and symbolism tell a common tale of the human plight.

Archons are referred to in Nag Hammadi documents as mind parasites, inverters, guards, gatekeepers, detainers, judges, pitiless ones and deceivers. The 'Covid' hoax alone is a glaring example of all these things. The Biblical 'God' is so different in the Old and New Testaments because they are not describing the same phenomenon. The vindictive, angry, hate-filled, 'God' of the Old Testament, known as Yahweh, is Yaldabaoth who is depicted in Cult-dictated popular culture as the 'Dark Lord', 'Lord of Time', Lord (Darth) Vader and Dormammu, the evil ruler of the 'Dark Dimension' trying to take over the 'Earth Dimension' in the Marvel comic movie, *Dr Strange*. Yaldabaoth is both the Old Testament 'god' and the Biblical 'Satan'. Gnostics referred to Yaldabaoth as the 'Great Architect of the Universe' and the Cult-controlled Freemason network calls their god 'the 'Great Architect of the Universe' (also Grand Architect). The 'Great Architect' Yaldabaoth is symbolised by the Cult as the all-seeing eye at the top of the pyramid on the Great Seal of the United States and the dollar bill. Archon is encoded in *arch*-itect as it is in *arch*-angels and *arch*-bishops. All religions have the theme of a force for good and force for evil in some sort of spiritual war and there is a reason for that – the theme is true. The Cult and its non-human masters are quite happy for this to circulate. They present themselves as the force for good fighting evil when they are really the force of evil (absence of love). The whole foundation of Cult modus operandi is inversion. They promote themselves as a force for good and anyone challenging them in pursuit of peace, love, fairness, truth and justice is condemned as a satanic force for evil. This has been the game plan throughout history whether the Church of Rome inquisitions of non-believers or 'conspiracy theorists' and 'anti-vaxxers' of today. The technique is the same whatever the timeline era.

Yaldabaoth is revolting (true)

Yaldabaoth and the Archons are said to have revolted against God with Yaldabaoth claiming to *be* God – the *All That Is*. The Old Testament ‘God’ (Yaldabaoth) demanded to be worshipped as such: ‘*I am the LORD, and there is none else, there is no God beside me*’ (Isaiah 45:5). I have quoted in other books a man who said he was the unofficial son of the late Baron Philippe de Rothschild of the Mouton-Rothschild wine producing estates in France who died in 1988 and he told me about the Rothschild ‘revolt from God’. The man said he was given the name Phillip Eugene de Rothschild and we shared long correspondence many years ago while he was living under another identity. He said that he was conceived through ‘occult incest’ which (within the Cult) was ‘normal and to be admired’. ‘Phillip’ told me about his experience attending satanic rituals with rich and famous people whom he names and you can see them and the wider background to Cult Satanism in my other books starting with *The Biggest Secret*. Cult rituals are interactions with Archontic ‘gods’. ‘Phillip’ described Baron Philippe de Rothschild as ‘a master Satanist and hater of God’ and he used the same term ‘revolt from God’ associated with Yaldabaoth/Satan/Lucifer/the Devil in describing the Sabbatian Rothschild dynasty. ‘I played a key role in my family’s revolt from God’, he said. That role was to infiltrate in classic Sabbatian style the Christian Church, but eventually he escaped the mind-prison to live another life. The Cult has been targeting religion in a plan to make worship of the Archons the global one-world religion. Infiltration of Satanism into modern ‘culture’, especially among the young, through music videos, stage shows and other means, is all part of this.

Nag Hammadi texts describe Yaldabaoth and the Archons in their prime form as energy – consciousness – and say they can take form if they choose in the same way that consciousness takes form as a human. Yaldabaoth is called ‘formless’ and represents a deeply inverted, distorted and chaotic state of consciousness which seeks to attached to humans and turn them into a likeness of itself in an attempt at assimilation. For that to happen it has to manipulate

humans into low frequency mental and emotional states that match its own. Archons can certainly appear in human form and this is the origin of the psychopathic personality. The energetic distortion Gnostics called Yaldabaoth is psychopathy. When psychopathic Archons take human form that human will be a psychopath as an expression of Yaldabaoth consciousness. Cult psychopaths are Archons in human form. The principle is the same as that portrayed in the 2009 *Avatar* movie when the American military travelled to a fictional Earth-like moon called Pandora in the Alpha Centauri star system to infiltrate a society of blue people, or Na'vi, by hiding within bodies that looked like the Na'vi. Archons posing as humans have a particular hybrid information field, part human, part Archon, (the ancient 'demigods') which processes information in a way that manifests behaviour to match their psychopathic evil, lack of empathy and compassion, and stops them being influenced by the empathy, compassion and love that a fully-human information field is capable of expressing. Cult bloodlines interbreed, be they royalty or dark suits, for this reason and you have their obsession with incest. Interbreeding with full-blown humans would dilute the Archontic energy field that guarantees psychopathy in its representatives in the human realm.

Gnostic writings say the main non-human forms that Archons take are *serpentine* (what I have called for decades 'reptilian' amid unbounded ridicule from the Archontically-programmed) and what Gnostics describe as 'an unborn baby or foetus with grey skin and dark, unmoving eyes'. This is an excellent representation of the ET 'Greys' of UFO folklore which large numbers of people claim to have seen and been abducted by – Zulu shaman Credo Mutwa among them. I agree with those that believe in extraterrestrial or interdimensional visitations today and for thousands of years past. No wonder with their advanced knowledge and technological capability they were perceived and worshipped as gods for technological and other 'miracles' they appeared to perform. Imagine someone arriving in a culture disconnected from the modern world with a smartphone and computer. They would be

seen as a 'god' capable of 'miracles'. The Renegade Mind, however, wants to know the source of everything and not only the way that source manifests as human or non-human. In the same way that a Renegade Mind seeks the original source material for the 'Covid virus' to see if what is claimed is true. The original source of Archons in form is consciousness – the distorted state of consciousness known to Gnostics as Yaldabaoth.

'Revolt from God' is energetic disconnection

Where I am going next will make a lot of sense of religious texts and ancient legends relating to 'Satan', Lucifer' and the 'gods'. Gnostic descriptions sync perfectly with the themes of my own research over the years in how they describe a consciousness distortion seeking to impose itself on human consciousness. I've referred to the core of infinite awareness in previous books as Infinite Awareness in Awareness of Itself. By that I mean a level of awareness that knows that it is all awareness and is aware of all awareness. From here comes the frequency of love in its true sense and balance which is what love is on one level – the balance of all forces into a single whole called Oneness and Isness. The more we disconnect from this state of love that many call 'God' the constituent parts of that Oneness start to unravel and express themselves as a part and not a whole. They become individualised as intellect, mind, selfishness, hatred, envy, desire for power over others, and such like. This is not a problem in the greater scheme in that 'God', the *All That Is*, can experience all these possibilities through different expressions of itself including humans. What we as expressions of the whole experience the *All That Is* experiences. We are the *All That Is* experiencing itself. As we withdraw from that state of Oneness we disconnect from its influence and things can get very unpleasant and very stupid. Archontic consciousness is at the extreme end of that. It has so disconnected from the influence of Oneness that it has become an inversion of unity and love, an inversion of everything, an inversion of life itself. Evil is appropriately live written backwards. Archontic consciousness is obsessed with death, an inversion of life,

and so its manifestations in Satanism are obsessed with death. They use inverted symbols in their rituals such as the inverted pentagram and cross. Sabbatians as Archontic consciousness incarnate invert Judaism and every other religion and culture they infiltrate. They seek disunity and chaos and they fear unity and harmony as they fear love like garlic to a vampire. As a result the Cult, Archons incarnate, act with such evil, psychopathy and lack of empathy and compassion disconnected as they are from the source of love. How could Bill Gates and the rest of the Archontic psychopaths do what they have to human society in the 'Covid' era with all the death, suffering and destruction involved and have no emotional consequence for the impact on others? Now you know. Why have Zuckerberg, Brin, Page, Wojcicki and company callously censored information warning about the dangers of the 'vaccine' while thousands have been dying and having severe, sometimes life-changing reactions? Now you know. Why have Tedros, Fauci, Whitty, Vallance and their like around the world been using case and death figures they're aware are fraudulent to justify lockdowns and all the deaths and destroyed lives that have come from that? Now you know. Why did Christian Drosten produce and promote a 'testing' protocol that he knew couldn't test for infectious disease which led to a global human catastrophe. Now you know. The Archontic mind doesn't give a shit ([Fig 17](#)). I personally think that Gates and major Cult insiders are a form of AI cyborg that the Archons want humans to become.

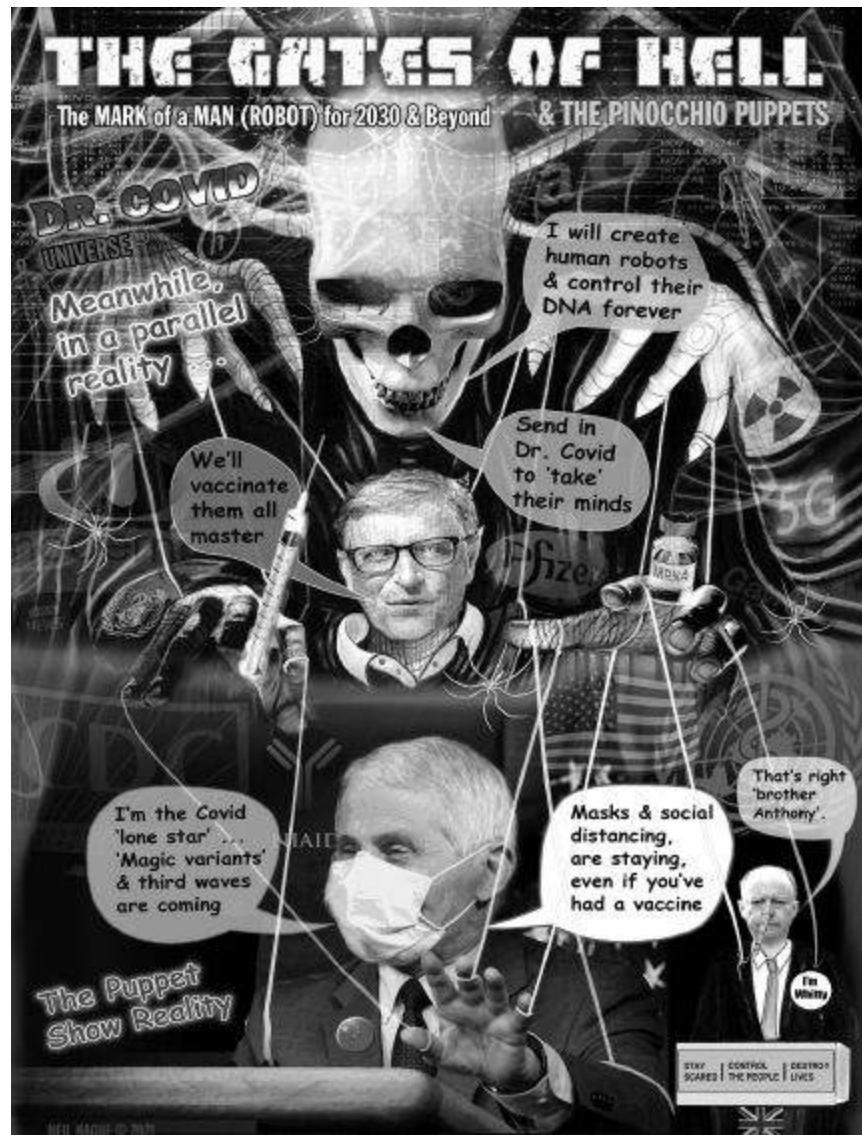


Figure 17: Artist Neil Hague's version of the 'Covid' hierarchy.

Human batteries

A state of such inversion does have its consequences, however. The level of disconnection from the Source of All means that you withdraw from that source of energetic sustenance and creativity. This means that you have to find your own supply of energetic power and it has – *us*. When the Morpheus character in the first *Matrix* movie held up a battery he spoke a profound truth when he said: 'The Matrix is a computer-generated dream world built to keep us under control in order to change the human being into one of

these.' The statement was true in all respects. We do live in a technologically-generated virtual reality simulation (more very shortly) and we have been manipulated to be an energy source for Archontic consciousness. The Disney-Pixar animated movie *Monsters, Inc.* in 2001 symbolised the dynamic when monsters in their world had no energy source and they would enter the human world to terrify children in their beds, catch the child's scream, terror (low-vibrational frequencies), and take that energy back to power the monster world. The lead character you might remember was a single giant eye and the symbolism of the Cult's all-seeing eye was obvious. Every thought and emotion is broadcast as a frequency unique to that thought and emotion. Feelings of love and joy, empathy and compassion, are high, quick, frequencies while fear, depression, anxiety, suffering and hate are low, slow, dense frequencies. Which kind do you think Archontic consciousness can connect with and absorb? In such a low and dense frequency state there's no way it can connect with the energy of love and joy. Archons can only feed off energy compatible with their own frequency and they and their Cult agents want to delete the human world of love and joy and manipulate the transmission of low vibrational frequencies through low-vibrational human mental and emotional states. *We are their energy source.* Wars are energetic banquets to the Archons – a world war even more so – and think how much low-frequency mental and emotional energy has been generated from the consequences for humanity of the 'Covid' hoax orchestrated by Archons incarnate like Gates.

The ancient practice of human sacrifice 'to the gods', continued in secret today by the Cult, is based on the same principle. 'The gods' are Archontic consciousness in different forms and the sacrifice is induced into a state of intense terror to generate the energy the Archontic frequency can absorb. Incarnate Archons in the ritual drink the blood which contains an adrenaline they crave which floods into the bloodstream when people are terrorised. Most of the sacrifices, ancient and modern, are children and the theme of 'sacrificing young virgins to the gods' is just code for children. They

have a particular pre-puberty energy that Archons want more than anything and the energy of the young in general is their target. The California Department of Education wants students to chant the names of Aztec gods (Archontic gods) once worshipped in human sacrifice rituals in a curriculum designed to encourage them to 'challenge racist, bigoted, discriminatory, imperialist/colonial beliefs', join 'social movements that struggle for social justice', and 'build new possibilities for a post-racist, post-systemic racism society'. It's the usual Woke crap that inverts racism and calls it anti-racism. In this case solidarity with 'indigenous tribes' is being used as an excuse to chant the names of 'gods' to which people were sacrificed (and still are in secret). What an example of Woke's inability to see beyond black and white, us and them, They condemn the colonisation of these tribal cultures by Europeans (quite right), but those cultures sacrificing people including children to their 'gods', and mass murdering untold numbers as the Aztecs did, is just fine. One chant is to the Aztec god Tezcatlipoca who had a man sacrificed to him in the 5th month of the Aztec calendar. His heart was cut out and he was eaten. Oh, that's okay then. Come on children ... after three ... Other sacrificial 'gods' for the young to chant their allegiance include Quetzalcoatl, Huitzilopochtli and Xipe Totec. The curriculum says that 'chants, affirmations, and energizers can be used to bring the class together, build unity around ethnic studies principles and values, and to reinvigorate the class following a lesson that may be emotionally taxing or even when student engagement may appear to be low'. Well, that's the cover story, anyway. Chanting and mantras are the repetition of a particular frequency generated from the vocal cords and chanting the names of these Archontic 'gods' tunes you into their frequency. That is the last thing you want when it allows for energetic synchronisation, attachment and perceptual influence. Initiates chant the names of their 'Gods' in their rituals for this very reason.

Vampires of the Woke

Paedophilia is another way that Archons absorb the energy of children. Paedophiles possessed by Archontic consciousness are used as the conduit during sexual abuse for discarnate Archons to vampire the energy of the young they desire so much. Stupendous numbers of children disappear every year never to be seen again although you would never know from the media. Imagine how much low-vibrational energy has been generated by children during the 'Covid' hoax when so many have become depressed and psychologically destroyed to the point of killing themselves. Shocking numbers of children are now taken by the state from loving parents to be handed to others. I can tell you from long experience of researching this since 1996 that many end up with paedophiles and assets of the Cult through corrupt and Cult-owned social services which in the reframing era has hired many psychopaths and emotionless automatons to do the job. Children are even stolen to order using spurious reasons to take them by the corrupt and secret (because they're corrupt) 'family courts'. I have written in detail in other books, starting with *The Biggest Secret* in 1997, about the ubiquitous connections between the political, corporate, government, intelligence and military elites (Cult operatives) and Satanism and paedophilia. If you go deep enough both networks have an interlocking leadership. The Woke mentality has been developed by the Cult for many reasons: To promote almost every aspect of its agenda; to hijack the traditional political left and turn it fascist; to divide and rule; and to target agenda pushbackers. But there are other reasons which relate to what I am describing here. How many happy and joyful Wokers do you ever see especially at the extreme end? They are a mental and psychological mess consumed by emotional stress and constantly emotionally cocked for the next explosion of indignation at someone referring to a female as a female. They are walking, talking, batteries as Morpheus might say emitting frequencies which both enslave them in low-vibrational bubbles of perceptual limitation and feed the Archons. Add to this the hatred claimed to be love; fascism claimed to 'anti-fascism', racism claimed to be 'anti-racism';

exclusion claimed to inclusion; and the abuse-filled Internet trolling. You have a purpose-built Archontic energy system with not a wind turbine in sight and all founded on Archontic *inversion*. We have whole generations now manipulated to serve the Archons with their actions and energy. They will be doing so their entire adult lives unless they snap out of their Archon-induced trance. Is it really a surprise that Cult billionaires and corporations put so much money their way? Where is the energy of joy and laughter, including laughing at yourself which is confirmation of your own emotional security? Mark Twain said: 'The human race has one really effective weapon, and that is laughter.' We must use it all the time. Woke has destroyed comedy because it has no humour, no joy, sense of irony, or self-deprecation. Its energy is dense and intense. *Mmmmm*, lunch says the Archontic frequency. Rudolf Steiner (1861-1925) was the Austrian philosopher and famous esoteric thinker who established Waldorf education or Steiner schools to treat children like unique expressions of consciousness and not minds to be programmed with the perceptions determined by authority. I'd been writing about this energy vampiring for decades when I was sent in 2016 a quote by Steiner. He was spot on:

There are beings in the spiritual realms for whom anxiety and fear emanating from human beings offer welcome food. When humans have no anxiety and fear, then these creatures starve. If fear and anxiety radiates from people and they break out in panic, then these creatures find welcome nutrition and they become more and more powerful. These beings are hostile towards humanity. Everything that feeds on negative feelings, on anxiety, fear and superstition, despair or doubt, are in reality hostile forces in super-sensible worlds, launching cruel attacks on human beings, while they are being fed ... These are exactly the feelings that belong to contemporary culture and materialism; because it estranges people from the spiritual world, it is especially suited to evoke hopelessness and fear of the unknown in people, thereby calling up the above mentioned hostile forces against them.

Pause for a moment from this perspective and reflect on what has happened in the world since the start of 2020. Not only will pennies drop, but billion dollar bills. We see the same theme from Don Juan Matus, a Yaqui Indian shaman in Mexico and the information source for Peruvian-born writer, Carlos Castaneda, who wrote a series of

books from the 1960s to 1990s. Don Juan described the force manipulating human society and his name for the Archons was the predator:

We have a predator that came from the depths of the cosmos and took over the rule of our lives. Human beings are its prisoners. The predator is our lord and master. It has rendered us docile, helpless. If we want to protest, it suppresses our protest. If we want to act independently, it demands that we don't do so ... indeed we are held prisoner!

They took us over because we are food to them, and they squeeze us mercilessly because we are their sustenance. Just as we rear chickens in coops, the predators rear us in human coops, humaneros. Therefore, their food is always available to them.

Different cultures, different eras, same recurring theme.

The 'ennoia' dilemma

Nag Hammadi Gnostic manuscripts say that Archon consciousness has no 'ennoia'. This is directly translated as 'intentionality', but I'll use the term 'creative imagination'. The *All That Is* in awareness of itself is the source of all creativity – all possibility – and the more disconnected you are from that source the more you are subsequently denied 'creative imagination'. Given that Archon consciousness is almost entirely disconnected it severely lacks creativity and has to rely on far more mechanical processes of thought and exploit the creative potential of those that do have 'ennoia'. You can see cases of this throughout human society. Archon consciousness almost entirely dominates the global banking system and if we study how that system works you will appreciate what I mean. Banks manifest 'money' out of nothing by issuing lines of 'credit' which is 'money' that has never, does not, and will never exist except in theory. It's a confidence trick. If you think 'credit' figures-on-a-screen 'money' is worth anything you accept it as payment. If you don't then the whole system collapses through lack of confidence in the value of that 'money'. Archontic bankers with no 'ennoia' are 'lending' 'money' that doesn't exist to humans that *do* have creativity – those that have the inspired ideas and create businesses and products. Archon banking feeds off human creativity

which it controls through 'money' creation and debt. Humans have the creativity and Archons exploit that for their own benefit and control while having none themselves. Archon Internet platforms like Facebook claim joint copyright of everything that creative users post and while Archontic minds like Zuckerberg may officially head that company it will be human creatives on the staff that provide the creative inspiration. When you have limitless 'money' you can then buy other companies established by creative humans. Witness the acquisition record of Facebook, Google and their like. Survey the Archon-controlled music industry and you see non-creative dark suit executives making their fortune from the human creativity of their artists. The cases are endless. Research the history of people like Gates and Zuckerberg and how their empires were built on exploiting the creativity of others. Archon minds cannot create out of nothing, but they are skilled (because they have to be) in what Gnostic texts call 'countermimicry'. They can imitate, but not innovate. Sabbatians trawl the creativity of others through backdoors they install in computer systems through their cybersecurity systems. Archon-controlled China is globally infamous for stealing intellectual property and I remember how Hong Kong, now part of China, became notorious for making counterfeit copies of the creativity of others – 'countermimicry'. With the now pervasive and all-seeing surveillance systems able to infiltrate any computer you can appreciate the potential for Archons to vampire the creativity of humans. Author John Lamb Lash wrote in his book about the Nag Hammadi texts, *Not In His Image*:

Although they cannot originate anything, because they lack the divine factor of ennoia (intentionality), Archons can imitate with a vengeance. Their expertise is simulation (HAL, virtual reality). The Demiurge [Yaldabaoth] fashions a heaven world copied from the fractal patterns [of the original] ... His construction is celestial kitsch, like the fake Italianate villa of a Mafia don complete with militant angels to guard every portal.

This brings us to something that I have been speaking about since the turn of the millennium. Our reality is a simulation; a virtual reality that we think is real. No, I'm not kidding.

Human reality? Well, virtually

I had pondered for years about whether our reality is 'real' or some kind of construct. I remembered being immensely affected on a visit as a small child in the late 1950s to the then newly-opened Planetarium on the Marylebone Road in London which is now closed and part of the adjacent Madame Tussauds wax museum. It was in the middle of the day, but when the lights went out there was the night sky projected in the Planetarium's domed ceiling and it appeared to be so real. The experience never left me and I didn't know why until around the turn of the millennium when I became certain that our 'night sky' and entire reality is a projection, a virtual reality, akin to the illusory world portrayed in the *Matrix* movies. I looked at the sky one day in this period and it appeared to me like the domed roof of the Planetarium. The release of the first *Matrix* movie in 1999 also provided a synchronistic and perfect visual representation of where my mind had been going for a long time. I hadn't come across the Gnostic Nag Hammadi texts then. When I did years later the correlation was once again astounding. As I read Gnostic accounts from 1,600 years and more earlier it was clear that they were describing the same simulation phenomenon. They tell how the Yaldabaoth 'Demiurge' and Archons created a 'bad copy' of original reality to rule over all that were captured by its illusions and the body was a prison to trap consciousness in the 'bad copy' fake reality. Read how Gnostics describe the 'bad copy' and update that to current times and they are referring to what we would call today a virtual reality simulation.

Author John Lamb Lash said 'the Demiurge fashions a heaven world copied from the fractal patterns' of the original through expertise in 'HAL' or virtual reality simulation. Fractal patterns are part of the energetic information construct of our reality, a sort of blueprint. If these patterns were copied in computer terms it would indeed give you a copy of a 'natural' reality in a non-natural frequency and digital form. The principle is the same as making a copy of a website. The original website still exists, but now you can change the copy version to make it whatever you like and it can

become very different to the original website. Archons have done this with our reality, a *synthetic* copy of prime reality that still exists beyond the frequency walls of the simulation. Trapped within the illusions of this synthetic Matrix, however, were and are human consciousness and other expressions of prime reality and this is why the Archons via the Cult are seeking to make the human body synthetic and give us synthetic AI minds to complete the job of turning the entire reality synthetic including what we perceive to be the natural world. To quote Kurzweil: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' Yes, *synthetic* 'creatures' just as 'Covid' and other genetically-manipulating 'vaccines' are designed to make the human body synthetic. From this perspective it is obvious why Archons and their Cult are so desperate to infuse synthetic material into every human with their 'Covid' scam.

Let there be (electromagnetic) light

Yaldabaoth, the force that created the simulation, or Matrix, makes sense of the Gnostic reference to 'The Great Architect' and its use by Cult Freemasonry as the name of its deity. The designer of the Matrix in the movies is called 'The Architect' and that trilogy is jam-packed with symbolism relating to these subjects. I have contended for years that the angry Old Testament God (Yaldabaoth) is the 'God' being symbolically 'quoted' in the opening of Genesis as 'creating the world'. This is not the creation of prime reality – it's the creation of the *simulation*. The Genesis 'God' says: 'Let there be Light: and there was light.' But what is this 'Light'? I have said for decades that the speed of light (186,000 miles per second) is not the fastest speed possible as claimed by mainstream science and is in fact the frequency walls or outer limits of the Matrix. You can't have a fastest or slowest anything within all possibility when everything is possible. The human body is encoded to operate within the speed of light or *within the simulation* and thus we see only the tiny frequency band of visible *light*. Near-death experiencers who perceive reality outside the body during temporary 'death' describe a very different

form of light and this is supported by the Nag Hammadi texts. Prime reality beyond the simulation ('Upper Aeons' to the Gnostics) is described as a realm of incredible beauty, bliss, love and harmony – a realm of 'watery light' that is so powerful 'there are no shadows'. Our false reality of Archon control, which Gnostics call the 'Lower Aeons', is depicted as a realm with a different kind of 'light' and described in terms of chaos, 'Hell', 'the Abyss' and 'Outer Darkness', where trapped souls are tormented and manipulated by demons (relate that to the 'Covid' hoax alone). The watery light theme can be found in near-death accounts and it is not the same as *simulation* 'light' which is electromagnetic or radiation light within the speed of light – the 'Lower Aeons'. Simulation 'light' is the 'luminous fire' associated by Gnostics with the Archons. The Bible refers to Yaldabaoth as 'that old serpent, called the Devil, and Satan, which deceiveth the whole world' (Revelation 12:9). I think that making a simulated copy of prime reality ('countermimicry') and changing it dramatically while all the time manipulating humanity to believe it to be real could probably meet the criteria of deceiving the whole world. Then we come to the Cult god Lucifer – the *Light Bringer*. Lucifer is symbolic of Yaldabaoth, the bringer of radiation light that forms the bad copy simulation within the speed of light. 'He' is symbolised by the lighted torch held by the Statue of Liberty and in the name 'Illuminati'. Sabbatian-Frankism declares that Lucifer is the true god and Lucifer is the real god of Freemasonry honoured as their 'Great or Grand Architect of the Universe' (simulation).

I would emphasise, too, the way Archontic technologically-generated luminous fire of radiation has deluged our environment since I was a kid in the 1950s and changed the nature of The Field with which we constantly interact. Through that interaction technological radiation is changing us. The Smart Grid is designed to operate with immense levels of communication power with 5G expanding across the world and 6G, 7G, in the process of development. Radiation is the simulation and the Archontic manipulation system. Why wouldn't the Archon Cult wish to unleash radiation upon us to an ever-greater extreme to form

Kurzweil's 'cloud'? The plan for a synthetic human is related to the need to cope with levels of radiation beyond even anything we've seen so far. Biological humans would not survive the scale of radiation they have in their script. The Smart Grid is a technological sub-reality within the technological simulation to further disconnect five-sense perception from expanded consciousness. It's a technological prison of the mind.

Infusing the 'spirit of darkness'

A recurring theme in religion and native cultures is the manipulation of human genetics by a non-human force and most famously recorded as the biblical 'sons of god' (the gods plural in the original) who interbred with the daughters of men. The Nag Hammadi *Apocryphon of John* tells the same story this way:

He [Yaldabaoth] sent his angels [Archons/demons] to the daughters of men, that they might take some of them for themselves and raise offspring for their enjoyment. And at first they did not succeed. When they had no success, they gathered together again and they made a plan together ... And the angels changed themselves in their likeness into the likeness of their mates, filling them with the spirit of darkness, which they had mixed for them, and with evil ... And they took women and begot children out of the darkness according to the likeness of their spirit.

Possession when a discarnate entity takes over a human body is an age-old theme and continues today. It's very real and I've seen it. Satanic and secret society rituals can create an energetic environment in which entities can attach to initiates and I've heard many stories of how people have changed their personality after being initiated even into lower levels of the Freemasons. I have been inside three Freemasonic temples, one at a public open day and two by just walking in when there was no one around to stop me. They were in Ryde, the town where I live, Birmingham, England, when I was with a group, and Boston, Massachusetts. They all felt the same energetically – dark, dense, low-vibrational and sinister. Demonic attachment can happen while the initiate has no idea what is going on. To them it's just a ritual to get in the Masons and do a bit of good

business. In the far more extreme rituals of Satanism human possession is even more powerful and they are designed to make possession possible. The hierarchy of the Cult is dictated by the power and perceived status of the possessing Archon. In this way the Archon hierarchy becomes the Cult hierarchy. Once the entity has attached it can influence perception and behaviour and if it attaches to the extreme then so much of its energy (information) infuses into the body information field that the hologram starts to reflect the nature of the possessing entity. This is the *Exorcist* movie type of possession when facial features change and it's known as shapeshifting. Islam's Jinn are said to be invisible tricksters who change shape, 'whisper', confuse and take human form. These are all traits of the Archons and other versions of the same phenomenon. Extreme possession could certainly infuse the 'spirit of darkness' into a partner during sex as the Nag Hammadi texts appear to describe. Such an infusion can change genetics which is also energetic information. Human genetics is information and the 'spirit of darkness' is information. Mix one with the other and change must happen. Islam has the concept of a 'Jinn baby' through possession of the mother and by Jinn taking human form. There are many ways that human genetics can be changed and remember that Archons have been aware all along of advanced techniques to do this. What is being done in human society today – and far more – was known about by Archons at the time of the 'fallen ones' and their other versions described in religions and cultures.

Archons and their human-world Cult are obsessed with genetics as we see today and they know this dictates how information is processed into perceived reality during a human life. They needed to produce a human form that would decode the simulation and this is symbolically known as 'Adam and Eve' who left the 'garden' (prime reality) and 'fell' into Matrix reality. The simulation is not a 'physical' construct (there is no 'physical'); it is a source of information. Think Wi-Fi again. The simulation is an energetic field encoded with information and body-brain systems are designed to decode that information encoded in wave or frequency form which

is transmitted to the brain as electrical signals. These are decoded by the brain to construct our sense of reality – an illusory ‘physical’ world that only exists in the brain or the mind. Virtual reality games mimic this process using the same sensory decoding system. Information is fed to the senses to decode a virtual reality that can appear so real, but isn’t (Figs 18 and 19). Some scientists believe – and I agree with them – that what we perceive as ‘physical’ reality only exists when we are looking or observing. The act of perception or focus triggers the decoding systems which turn waveform information into holographic reality. When we are not observing something our reality reverts from a holographic state to a waveform state. This relates to the same principle as a falling tree not making a noise unless someone is there to hear it or decode it. The concept makes sense from the simulation perspective. A computer is not decoding all the information in a Wi-Fi field all the time and only decodes or brings into reality on the screen that part of Wi-Fi that it’s decoding – focusing upon – at that moment.



Figure 18: Virtual reality technology ‘hacks’ into the body’s five-sense decoding system.



Figure 19: The result can be experienced as very ‘real’.

Interestingly, Professor Donald Hoffman at the Department of Cognitive Sciences at the University of California, Irvine, says that our experienced reality is like a computer interface that shows us only the level with which we interact while hiding all that exists beyond it: 'Evolution shaped us with a user interface that hides the truth. Nothing that we see is the truth – the very language of space and time and objects is the wrong language to describe reality.' He is correct in what he says on so many levels. Space and time are not a universal reality. They are a phenomenon of decoded *simulation* reality as part of the process of enslaving our sense of reality. Near-death experiencers report again and again how space and time did not exist as we perceive them once they were free of the body – body decoding systems. You can appreciate from this why Archons and their Cult are so desperate to entrap human attention in the five senses where we are in the Matrix and of the Matrix. Opening your mind to expanded states of awareness takes you beyond the information confines of the simulation and you become aware of knowledge and insights denied to you before. This is what we call 'awakening' – *awakening from the Matrix* – and in the final chapter I will relate this to current events.

Where are the 'aliens'?

A simulation would explain the so-called 'Fermi Paradox' named after Italian physicist Enrico Fermi (1901-1954) who created the first nuclear reactor. He considered the question of why there is such a lack of extraterrestrial activity when there are so many stars and planets in an apparently vast universe; but what if the night sky that we see, or think we do, is a simulated projection as I say? If you control the simulation and your aim is to hold humanity fast in essential ignorance would you want other forms of life including advanced life coming and going sharing information with humanity? Or would you want them to believe they were isolated and apparently alone? Themes of human isolation and apartness are common whether they be the perception of a lifeless universe or the fascist isolation laws of the 'Covid' era. Paradoxically the very

existence of a simulation means that we are not alone when some force had to construct it. My view is that experiences that people have reported all over the world for centuries with Reptilians and Grey entities are Archon phenomena as Nag Hammadi texts describe; and that benevolent 'alien' interactions are non-human groups that come in and out of the simulation by overcoming Archon attempts to keep them out. It should be highlighted, too, that Reptilians and Greys are obsessed with *genetics* and *technology* as related by cultural accounts and those who say they have been abducted by them. Technology is their way of overcoming some of the limitations in their creative potential and our technology-driven and controlled human society of today is *archetypical* Archon-Reptilian-Grey *modus operandi*. Technocracy is really *Archontocracy*. The Universe does not have to be as big as it appears with a simulation. There is no space or distance only information decoded into holographic reality. What we call 'space' is only the absence of holographic 'objects' and that 'space' is The Field of energetic information which connects everything into a single whole. The same applies with the artificially-generated information field of the simulation. The Universe is not big or small as a physical reality. It is decoded information, that's all, and its perceived size is decided by the way the simulation is encoded to make it appear. The entire night sky as we perceive it only exists in our brain and so where are those 'millions of light years'? The 'stars' on the ceiling of the Planetarium looked a vast distance away.

There's another point to mention about 'aliens'. I have been highlighting since the 1990s the plan to stage a fake 'alien invasion' to justify the centralisation of global power and a world military. Nazi scientist Werner von Braun, who was taken to America by Operation Paperclip after World War Two to help found NASA, told his American assistant Dr Carol Rosin about the Cult agenda when he knew he was dying in 1977. Rosin said that he told her about a sequence that would lead to total human control by a one-world government. This included threats from terrorism, rogue nations, meteors and asteroids before finally an 'alien invasion'. All of these

things, von Braun said, would be bogus and what I would refer to as a No-Problem-Reaction-Solution. Keep this in mind when 'the aliens are coming' is the new mantra. The aliens are not coming – they are *already here* and they have infiltrated human society while looking human. French-Canadian investigative journalist Serge Monast said in 1994 that he had uncovered a NASA/military operation called Project Blue Beam which fits with what Werner von Braun predicted. Monast died of a 'heart attack' in 1996 the day after he was arrested and spent a night in prison. He was 51. He said Blue Beam was a plan to stage an alien invasion that would include religious figures beamed holographically into the sky as part of a global manipulation to usher in a 'new age' of worshipping what I would say is the Cult 'god' Yaldabaoth in a one-world religion. Fake holographic asteroids are also said to be part of the plan which again syncs with von Braun. How could you stage an illusory threat from asteroids unless they were holographic inserts? This is pretty straightforward given the advanced technology outside the public arena and the fact that our 'physical' reality is holographic anyway. Information fields would be projected and we would decode them into the illusion of a 'physical' asteroid. If they can sell a global 'pandemic' with a 'virus' that doesn't exist what will humans not believe if government and media tell them?

All this is particularly relevant as I write with the Pentagon planning to release in June, 2021, information about 'UFO sightings'. I have been following the UFO story since the early 1990s and the common theme throughout has been government and military denials and cover up. More recently, however, the Pentagon has suddenly become more talkative and apparently open with Air Force pilot radar images released of unexplained craft moving and changing direction at speeds well beyond anything believed possible with human technology. Then, in March, 2021, former Director of National Intelligence John Ratcliffe said a Pentagon report months later in June would reveal a great deal of information about UFO sightings unknown to the public. He said the report would have 'massive implications'. The order to do this was included bizarrely

in a \$2.3 trillion 'coronavirus' relief and government funding bill passed by the Trump administration at the end of 2020. I would add some serious notes of caution here. I have been pointing out since the 1990s that the US military and intelligence networks have long had craft – 'flying saucers' or anti-gravity craft – which any observer would take to be extraterrestrial in origin. Keeping this knowledge from the public allows craft flown by *humans* to be perceived as alien visitations. I am not saying that 'aliens' do not exist. I would be the last one to say that, but we have to be streetwise here. President Ronald Reagan told the UN General Assembly in 1987: 'I occasionally think how quickly our differences worldwide would vanish if we were facing an alien threat from outside this world.' That's the idea. Unite against a common 'enemy' with a common purpose behind your 'saviour force' (the Cult) as this age-old technique of mass manipulation goes global.

Science moves this way ...

I could find only one other person who was discussing the simulation hypothesis publicly when I concluded it was real. This was Nick Bostrom, a Swedish-born philosopher at the University of Oxford, who has explored for many years the possibility that human reality is a computer simulation although his version and mine are not the same. Today the simulation and holographic reality hypothesis have increasingly entered the scientific mainstream. Well, the more open-minded mainstream, that is. Here are a few of the ever-gathering examples. American nuclear physicist Silas Beane led a team of physicists at the University of Bonn in Germany pursuing the question of whether we live in a simulation. They concluded that we probably do and it was likely based on a lattice of cubes. They found that cosmic rays align with that specific pattern. The team highlighted the Greisen–Zatsepin–Kuzmin (GZK) limit which refers to cosmic ray particle interaction with cosmic background radiation that creates an apparent boundary for cosmic ray particles. They say in a paper entitled 'Constraints on the Universe as a Numerical Simulation' that this 'pattern of constraint' is exactly what you

would find with a computer simulation. They also made the point that a simulation would create its own 'laws of physics' that would limit possibility. I've been making the same point for decades that the *perceived* laws of physics relate only to this reality, or what I would later call the simulation. When designers write codes to create computer and virtual reality games they are the equivalent of the laws of physics for that game. Players interact within the limitations laid out by the coding. In the same way those who wrote the codes for the simulation decided the laws of physics that would apply. These can be overridden by expanded states of consciousness, but not by those enslaved in only five-sense awareness where simulation codes rule. Overriding the codes is what people call 'miracles'. They are not. They are bypassing the encoded limits of the simulation. A population caught in simulation perception would have no idea that this was their plight. As the Bonn paper said: 'Like a prisoner in a pitch-black cell we would not be able to see the "walls" of our prison,' That's true if people remain mesmerised by the five senses. Open to expanded awareness and those walls become very clear. The main one is the speed of light.

American theoretical physicist James Gates is another who has explored the simulation question and found considerable evidence to support the idea. Gates was Professor of Physics at the University of Maryland, Director of The Center for String and Particle Theory, and on Barack Obama's Council of Advisors on Science and Technology. He and his team found *computer codes* of digital data embedded in the fabric of our reality. They relate to on-off electrical charges of 1 and 0 in the binary system used by computers. 'We have no idea what they are doing there', Gates said. They found within the energetic fabric mathematical sequences known as error-correcting codes or block codes that 'reboot' data to its original state or 'default settings' when something knocks it out of sync. Gates was asked if he had found a set of equations embedded in our reality indistinguishable from those that drive search engines and browsers and he said: 'That is correct.' Rich Terrile, director of the Centre for Evolutionary Computation and Automated Design at NASA's Jet

Propulsion Laboratory, has said publicly that he believes the Universe is a digital hologram that must have been created by a form of intelligence. I agree with that in every way. Waveform information is delivered electrically by the senses to the brain which constructs a *digital* holographic reality that we call the 'world'. This digital level of reality can be read by the esoteric art of numerology. Digital holograms are at the cutting edge of holographics today. We have digital technology everywhere designed to access and manipulate our digital level of perceived reality. Synthetic mRNA in 'Covid vaccines' has a digital component to manipulate the body's digital 'operating system'.

Reality is numbers

How many know that our reality can be broken down to numbers and codes that are the same as computer games? Max Tegmark, a physicist at the Massachusetts Institute of Technology (MIT), is the author of *Our Mathematical Universe* in which he lays out how reality can be entirely described by numbers and maths in the way that a video game is encoded with the 'physics' of computer games. Our world and computer virtual reality are essentially the same.

Tegmark imagines the perceptions of characters in an advanced computer game when the graphics are so good they don't know they are in a game. They think they can bump into real objects (electromagnetic resistance in our reality), fall in love and feel emotions like excitement. When they began to study the apparently 'physical world' of the video game they would realise that everything was made of pixels (which have been found in our energetic reality as must be the case when on one level our world is digital). What computer game characters thought was physical 'stuff', Tegmark said, could actually be broken down into numbers:

And we're exactly in this situation in our world. We look around and it doesn't seem that mathematical at all, but everything we see is made out of elementary particles like quarks and electrons. And what properties does an electron have? Does it have a smell or a colour or a texture? No! ... We physicists have come up with geeky names for [Electron] properties, like

electric charge, or spin, or lepton number, but the electron doesn't care what we call it, the properties are just numbers.

This is the illusory reality Gnostics were describing. This is the simulation. The A, C, G, and T codes of DNA have a binary value – A and C = 0 while G and T = 1. This has to be when the simulation is digital and the body must be digital to interact with it. Recurring mathematical sequences are encoded throughout reality and the body. They include the Fibonacci sequence in which the two previous numbers are added to get the next one, as in ... 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, etc. The sequence is encoded in the human face and body, proportions of animals, DNA, seed heads, pine cones, trees, shells, spiral galaxies, hurricanes and the number of petals in a flower. The list goes on and on. There are fractal patterns – a 'never-ending pattern that is infinitely complex and self-similar across all scales in the as above, so below, principle of holograms. These and other famous recurring geometrical and mathematical sequences such as Phi, Pi, Golden Mean, Golden Ratio and Golden Section are *computer codes* of the simulation. I had to laugh and give my head a shake the day I finished this book and it went into the production stage. I was sent an article in *Scientific American* published in April, 2021, with the headline 'Confirmed! We Live in a Simulation'. Two decades after I first said our reality is a simulation and the speed of light is its outer limit the article suggested that we do live in a simulation and that the speed of light is its outer limit. I left school at 15 and never passed a major exam in my life while the writer was up to his eyes in qualifications. As I will explain in the final chapter *knowing* is far better than thinking and they come from very different sources. The article rightly connected the speed of light to the processing speed of the 'Matrix' and said what has been in my books all this time ... 'If we are in a simulation, as it appears, then space is an abstract property written in code. It is not real'. No it's not and if we live in a simulation something created it and it wasn't *us*. 'That David Icke says we are manipulated by aliens' – he's crackers.'

Wow ...

The reality that humanity thinks is so real is an illusion. Politicians, governments, scientists, doctors, academics, law enforcement, media, school and university curriculums, on and on, are all founded on a world that *does not exist* except as a simulated prison cell. Is it such a stretch to accept that 'Covid' doesn't exist when our entire 'physical' reality doesn't exist? Revealed here is the knowledge kept under raps in the Cult networks of compartmentalised secrecy to control humanity's sense of reality by inducing the population to believe in a reality that's not real. If it wasn't so tragic in its experiential consequences the whole thing would be hysterically funny. None of this is new to Renegade Minds. Ancient Greek philosopher Plato (about 428 to about 347BC) was a major influence on Gnostic belief and he described the human plight thousands of years ago with his Allegory of the Cave. He told the symbolic story of prisoners living in a cave who had never been outside. They were chained and could only see one wall of the cave while behind them was a fire that they could not see. Figures walked past the fire casting shadows on the prisoners' wall and those moving shadows became their sense of reality. Some prisoners began to study the shadows and were considered experts on them (today's academics and scientists), but what they studied was only an illusion (today's academics and scientists). A prisoner escaped from the cave and saw reality as it really is. When he returned to report this revelation they didn't believe him, called him mad and threatened to kill him if he tried to set them free. Plato's tale is not only a brilliant analogy of the human plight and our illusory reality. It describes, too, the dynamics of the 'Covid' hoax. I have only skimmed the surface of these subjects here. The aim of this book is to crisply connect all essential dots to put what is happening today into its true context. All subject areas and their connections in this chapter are covered in great evidential detail in *Everything You Need To Know, But Have Never Been Told* and *The Answer*.

They say that bewildered people 'can't see the forest for the trees'. Humanity, however, can't see the forest for the *twigs*. The five senses

see only twigs while Renegade Minds can see the forest and it's the forest where the answers lie with the connections that reveals.

Breaking free of perceptual programming so the forest can be seen is the way we turn all this around. Not breaking free is how humanity got into this mess. The situation may seem hopeless, but I promise you it's not. We are a perceptual heartbeat from paradise if only we knew.

CHAPTER TWELVE

Escaping Wetiko

Life is simply a vacation from the infinite
Dean Cavanagh

Renegade Minds weave the web of life and events and see common themes in the apparently random. They are always there if you look for them and their pursuit is aided by incredible synchronicity that comes when your mind is open rather than mesmerised by what it thinks it can see.

Infinite awareness is infinite possibility and the more of infinite possibility that we access the more becomes infinitely possible. That may be stating the apparently obvious, but it is a devastatingly-powerful fact that can set us free. We are a point of attention within an infinity of consciousness. The question is how much of that infinity do we choose to access? How much knowledge, insight, awareness, wisdom, do we want to connect with and explore? If your focus is only in the five senses you will be influenced by a fraction of infinite awareness. I mean a range so tiny that it gives new meaning to infinitesimal. Limitation of self-identity and a sense of the possible limit accordingly your range of consciousness. We are what we think we are. Life is what we think it is. The dream is the dreamer and the dreamer is the dream. Buddhist philosophy puts it this way: 'As a thing is viewed, so it appears.' Most humans live in the realm of touch, taste, see, hear, and smell and that's the limit of their sense of the possible and sense of self. Many will follow a religion and speak of a God in his heaven, but their lives are still

dominated by the five senses in their perceptions and actions. The five senses become the arbiter of everything. When that happens all except a smear of infinity is sealed away from influence by the rigid, unyielding, reality bubbles that are the five-sense human or Phantom Self. Archon Cult methodology is to isolate consciousness within five-sense reality – the simulation – and then program that consciousness with a sense of self and the world through a deluge of life-long information designed to instil the desired perception that allows global control. Efforts to do this have increased dramatically with identity politics as identity bubbles are squeezed into the minutiae of five-sense detail which disconnect people even more profoundly from the infinite 'I'.

Five-sense focus and self-identity are like a firewall that limits access to the infinite realms. You only perceive one radio or television station and no other. We'll take that literally for a moment. Imagine a vast array of stations giving different information and angles on reality, but you only ever listen to one. Here we have the human plight in which the population is overwhelmingly confined to CultFM. This relates only to the frequency range of CultFM and limits perception and insight to that band – limits *possibility* to that band. It means you are connecting with an almost imperceptibly minuscule range of possibility and creative potential within the infinite Field. It's a world where everything seems apart from everything else and where synchronicity is rare. Synchronicity is defined in the dictionary as 'the happening by chance of two or more related or similar events at the same time'. Use of 'by chance' betrays a complete misunderstanding of reality. Synchronicity is not 'by chance'. As people open their minds, or 'awaken' to use the term, they notice more and more coincidences in their lives, bits of 'luck', apparently miraculous happenings that put them in the right place at the right time with the right people. Days become peppered with 'fancy meeting you here' and 'what are the chances of that?' My entire life has been lived like this and ever more so since my own colossal awakening in 1990 and 91 which transformed my sense of reality. Synchronicity is not 'by chance'; it is by accessing expanded

realms of possibility which allow expanded potential for manifestation. People broadcasting the same vibe from the same openness of mind tend to be drawn 'by chance' to each other through what I call frequency magnetism and it's not only people. In the last more than 30 years incredible synchronicity has also led me through the Cult maze to information in so many forms and to crucial personal experiences. These 'coincidences' have allowed me to put the puzzle pieces together across an enormous array of subjects and situations. Those who have breached the bubble of five-sense reality will know exactly what I mean and this escape from the perceptual prison cell is open to everyone whenever they make that choice. This may appear super-human when compared with the limitations of 'human', but it's really our natural state. 'Human' as currently experienced is consciousness in an unnatural state of induced separation from the infinity of the whole. I'll come to how this transformation into unity can be made when I have described in more detail the force that holds humanity in servitude by denying this access to infinite self.

The Wetiko factor

I have been talking and writing for decades about the way five-sense mind is systematically barricaded from expanded awareness. I have used the analogy of a computer (five-sense mind) and someone at the keyboard (expanded awareness). Interaction between the computer and the operator is symbolic of the interaction between five-sense mind and expanded awareness. The computer directly experiences the Internet and the operator experiences the Internet via the computer which is how it's supposed to be – the two working as one. Archons seek to control that point where the operator connects with the computer to stop that interaction ([Fig 20](#)). Now the operator is banging the keyboard and clicking the mouse, but the computer is not responding and this happens when the computer is taken over – *possessed* – by an appropriately-named computer 'virus'. The operator has lost all influence over the computer which goes its own way making decisions under the control of the 'virus'. I have

just described the dynamic through which the force known to Gnostics as Yaldabaoth and Archons disconnects five-sense mind from expanded awareness to imprison humanity in perceptual servitude.



Figure 20: The mind ‘virus’ I have been writing about for decades seeks to isolate five-sense mind (the computer) from the true ‘I’. (Image by Neil Hague).

About a year ago I came across a Native American concept of Wetiko which describes precisely the same phenomenon. Wetiko is the spelling used by the Cree and there are other versions including wintiko and windigo used by other tribal groups. They spell the name with lower case, but I see Wetiko as a proper noun as with Archons and prefer a capital. I first saw an article about Wetiko by writer and researcher Paul Levy which so synced with what I had been writing about the computer/operator disconnection and later the Archons. I then read his book, the fascinating *Dispelling Wetiko, Breaking the Spell of Evil*. The parallels between what I had concluded long before and the Native American concept of Wetiko were so clear and obvious that it was almost funny. For Wetiko see the Gnostic Archons for sure and the Jinn, the Predators, and every other name for a force of evil, inversion and chaos. Wetiko is the Native American name for the force that divides the computer from

the operator (Fig 21). Indigenous author Jack D. Forbes, a founder of the Native American movement in the 1960s, wrote another book about Wetiko entitled *Columbus And Other Cannibals – The Wetiko Disease of Exploitation, Imperialism, and Terrorism* which I also read. Forbes says that Wetiko refers to an evil person or spirit ‘who terrorizes other creatures by means of terrible acts, including cannibalism’. Zulu shaman Credo Mutwa told me that African accounts tell how cannibalism was brought into the world by the Chitauri ‘gods’ – another manifestation of Wetiko. The distinction between ‘evil person or spirit’ relates to Archons/Wetiko possessing a human or acting as pure consciousness. Wetiko is said to be a sickness of the soul or spirit and a state of being that takes but gives nothing back – the Cult and its operatives perfectly described. Black Hawk, a Native American war leader defending their lands from confiscation, said European invaders had ‘poisoned hearts’ – Wetiko hearts – and that this would spread to native societies. Mention of the heart is very significant as we shall shortly see. Forbes writes: ‘Tragically, the history of the world for the past 2,000 years is, in great part, the story of the epidemiology of the wetiko disease.’ Yes, and much longer. Forbes is correct when he says: ‘The wetikos destroyed Egypt and Babylon and Athens and Rome and Tenochtitlan [capital of the Aztec empire] and perhaps now they will destroy the entire earth.’ Evil, he said, is the number one export of a Wetiko culture – see its globalisation with ‘Covid’. Constant war, mass murder, suffering of all kinds, child abuse, Satanism, torture and human sacrifice are all expressions of Wetiko and the Wetiko possessed. The world is Wetiko made manifest, *but it doesn’t have to be*. There is a way out of this even now.



Figure 21: The mind 'virus' is known to Native Americans as 'Wetiko'. (Image by Neil Hague).

Cult of Wetiko

Wetiko is the Yaldabaoth frequency distortion that seeks to attach to human consciousness and absorb it into its own. Once this connection is made Wetiko can drive the perceptions of the target which they believe to be coming from their own mind. All the horrors of history and today from mass killers to Satanists, paedophiles like Jeffrey Epstein and other psychopaths, are the embodiment of Wetiko and express its state of being in all its grotesqueness. The Cult is Wetiko incarnate, Yaldabaoth incarnate, and it seeks to facilitate Wetiko assimilation of humanity in totality into its distortion by manipulating the population into low frequency states that match its own. Paul Levy writes:

'Holographically enforced within the psyche of every human being the wetiko virus pervades and underlies the entire field of consciousness, and can therefore potentially manifest through any one of us at any moment if we are not mindful.' The 'Covid' hoax has achieved this with many people, but others have not fallen into Wetiko's frequency lair. Players in the 'Covid' human catastrophe including Gates, Schwab, Tedros, Fauci, Whitty, Vallance, Johnson, Hancock, Ferguson, Drosten, and all the rest, including the psychopath psychologists, are expressions of Wetiko. This is why

they have no compassion or empathy and no emotional consequence for what they do that would make them stop doing it. Observe all the people who support the psychopaths in authority against the Pushbackers despite the damaging impact the psychopaths have on their own lives and their family's lives. You are again looking at Wetiko possession which prevents them seeing through the lies to the obvious scam going on. *Why can't they see it?* Wetiko won't let them see it. The perceptual divide that has now become a chasm is between the Wetikoed and the non-Wetikoed.

Paul Levy describes Wetiko in the same way that I have long described the Archontic force. They are the same distorted consciousness operating across dimensions of reality: '... the subtle body of wetiko is not located in the third dimension of space and time, literally existing in another dimension ... it is able to affect ordinary lives by mysteriously interpenetrating into our three-dimensional world.' Wetiko does this through its incarnate representatives in the Cult and by weaving itself into The Field which on our level of reality is the electromagnetic information field of the simulation or Matrix. More than that, the simulation *is* Wetiko / Yaldabaoth. Caleb Scharf, Director of Astrobiology at Columbia University, has speculated that 'alien life' could be so advanced that it has transcribed itself into the quantum realm to become what we call physics. He said intelligence indistinguishable from the fabric of the Universe would solve many of its greatest mysteries:

Perhaps hyper-advanced life isn't just external. Perhaps it's already all around. It is embedded in what we perceive to be physics itself, from the root behaviour of particles and fields to the phenomena of complexity and emergence ... In other words, life might not just be in the equations. It might BE the equations [My emphasis].

Scharf said it is possible that 'we don't recognise advanced life because it forms an integral and unsuspecting part of what we've considered to be the natural world'. I agree. Wetiko/Yaldabaoth *is* the simulation. We are literally in the body of the beast. But that doesn't mean it has to control us. We all have the power to overcome Wetiko

influence and the Cult knows that. I doubt it sleeps too well because it knows that.

Which Field?

This, I suggest, is how it all works. There are two Fields. One is the fierce electromagnetic light of the Matrix within the speed of light; the other is the 'watery light' of The Field beyond the walls of the Matrix that connects with the Great Infinity. Five-sense mind and the decoding systems of the body attach us to the Field of Matrix light. They have to or we could not experience this reality. Five-sense mind sees only the Matrix Field of information while our expanded consciousness is part of the Infinity Field. When we open our minds, and most importantly our hearts, to the Infinity Field we have a mission control which gives us an expanded perspective, a road map, to understand the nature of the five-sense world. If we are isolated only in five-sense mind there is no mission control. We're on our own trying to understand a world that's constantly feeding us information to ensure we do not understand. People in this state can feel 'lost' and bewildered with no direction or radar. You can see ever more clearly those who are influenced by the Fields of Big Infinity or little five-sense mind simply by their views and behaviour with regard to the 'Covid' hoax. We have had this division throughout known human history with the mass of the people on one side and individuals who could see and intuit beyond the walls of the simulation – Plato's prisoner who broke out of the cave and saw reality for what it is. Such people have always been targeted by Wetiko/Archon-possessed authority, burned at the stake or demonised as mad, bad and dangerous. The Cult today and its global network of 'anti-hate', 'anti-fascist' Woke groups are all expressions of Wetiko attacking those exposing the conspiracy, 'Covid' lies and the 'vaccine' agenda.

Woke as a whole is Wetiko which explains its black and white mentality and how at one it is with the Wetiko-possessed Cult. Paul Levy said: 'To be in this paradigm is to still be under the thrall of a two-valued logic – where things are either true or false – of a

wetikoized mind.’ Wetiko consciousness is in a permanent rage, therefore so is Woke, and then there is Woke inversion and contradiction. ‘Anti-fascists’ act like fascists because fascists *and* ‘anti-fascists’ are both Wetiko at work. Political parties act the same while claiming to be different for the same reason. Secret society and satanic rituals are attaching initiates to Wetiko and the cold, ruthless, psychopathic mentality that secures the positions of power all over the world is Wetiko. Reframing ‘training programmes’ have the same cumulative effect of attaching Wetiko and we have their graduates described as automatons and robots with a cold, psychopathic, uncaring demeanour. They are all traits of Wetiko possession and look how many times they have been described in this book and elsewhere with regard to personnel behind ‘Covid’ including the police and medical profession. Climbing the greasy pole in any profession in a Wetiko society requires traits of Wetiko to get there and that is particularly true of politics which is not about fair competition and pre-eminence of ideas. It is founded on how many backs you can stab and arses you can lick. This culminated in the global ‘Covid’ coordination between the Wetiko possessed who pulled it off in all the different countries without a trace of empathy and compassion for their impact on humans. Our sight sense can see only holographic form and not the Field which connects holographic form. Therefore we perceive ‘physical’ objects with ‘space’ in between. In fact that ‘space’ is energy/consciousness operating on multiple frequencies. One of them is Wetiko and that connects the Cult psychopaths, those who submit to the psychopaths, and those who serve the psychopaths in the media operations of the world. Wetiko is Gates. Wetiko is the mask-wearing submissive. Wetiko is the fake journalist and ‘fact-checker’. The Wetiko Field is coordinating the whole thing. Psychopaths, gofers, media operatives, ‘anti-hate’ hate groups, ‘fact-checkers’ and submissive people work as one unit *even without human coordination* because they are attached to the *same* Field which is organising it all ([Fig 22](#)). Paul Levy is here describing how Wetiko-possessed people are drawn together and refuse to let any information breach their rigid

perceptions. He was writing long before 'Covid', but I think you will recognise followers of the 'Covid' religion *oh just a little bit*:

People who are channelling the vibratory frequency of wetiko align with each other through psychic resonance to reinforce their unspoken shared agreement so as to uphold their deranged view of reality. Once an unconscious content takes possession of certain individuals, it irresistibly draws them together by mutual attraction and knits them into groups tied together by their shared madness that can easily swell into an avalanche of insanity.

A psychic epidemic is a closed system, which is to say that it is insular and not open to any new information or informing influences from the outside world which contradict its fixed, limited, and limiting perspective.

There we have the Woke mind and the 'Covid' mind. Compatible resonance draws the awakening together, too, which is clearly happening today.

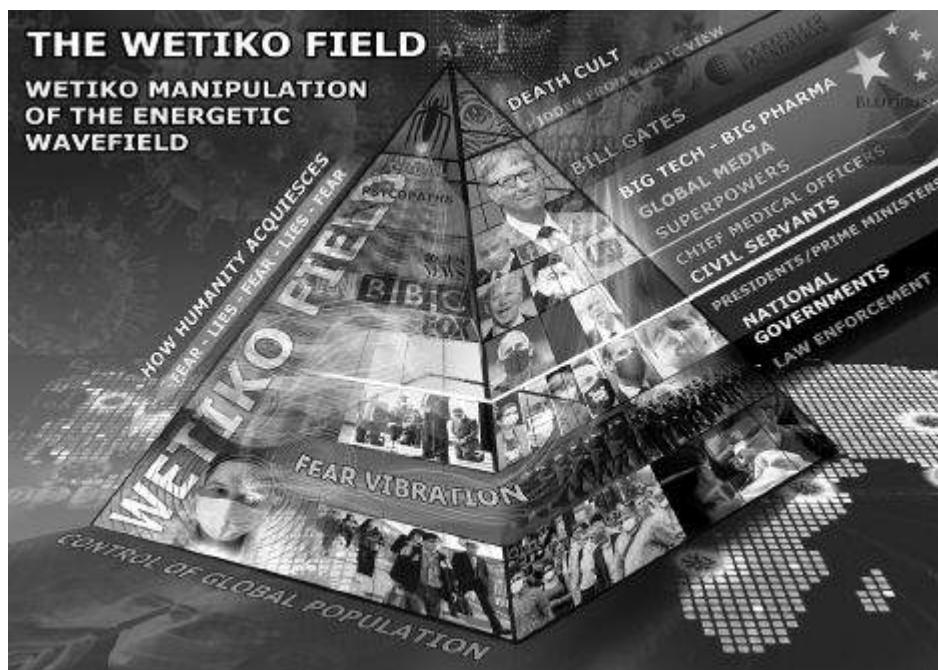


Figure 22: The Wetiko Field from which the Cult pyramid and its personnel are made manifest. (Image by Neil Hague).

Spiritual servitude

Wetiko doesn't care about humans. It's not human; it just possesses humans for its own ends and the effect (depending on the scale of

possession) can be anything from extreme psychopathy to unquestioning obedience. Wetiko's worst nightmare is for human consciousness to expand beyond the simulation. Everything is focussed on stopping that happening through control of information, thus perception, thus frequency. The 'education system', media, science, medicine, academia, are all geared to maintaining humanity in five-sense servitude as is the constant stimulation of low-vibrational mental and emotional states (see 'Covid'). Wetiko seeks to dominate those subconscious spaces between five-sense perception and expanded consciousness where the computer meets the operator. From these subconscious hiding places Wetiko speaks to us to trigger urges and desires that we take to be our own and manipulate us into anything from low-vibrational to psychopathic states. Remember how Islam describes the Jinn as invisible tricksters that 'whisper' and confuse. Wetiko is the origin of the 'trickster god' theme that you find in cultures all over the world. Jinn, like the Archons, are Wetiko which is terrified of humans awakening and reconnecting with our true self for then its energy source has gone. With that the feedback loop breaks between Wetiko and human perception that provides the energetic momentum on which its very existence depends as a force of evil. Humans are both its target and its source of survival, but only if we are operating in low-vibrational states of fear, hate, depression and the background anxiety that most people suffer. We are Wetiko's target because we are its key to survival. It needs us, not the other way round. Paul Levy writes:

A vampire has no intrinsic, independent, substantial existence in its own right; it only exists in relation to us. The pathogenic, vampiric mind-parasite called wetiko is nothing in itself – not being able to exist from its own side – yet it has a 'virtual reality' such that it can potentially destroy our species ...

...The fact that a vampire is not reflected by a mirror can also mean that what we need to see is that there's nothing, no-thing to see, other than ourselves. The fact that wetiko is the expression of something inside of us means that the cure for wetiko is with us as well. The critical issue is finding this cure within us and then putting it into effect.

Evil begets evil because if evil does not constantly expand and find new sources of energetic sustenance its evil, its *distortion*, dies with the assimilation into balance and harmony. Love is the garlic to Wetiko's vampire. Evil, the absence of love, cannot exist in the presence of love. I think I see a way out of here. I have emphasised so many times over the decades that the Archons/Wetiko and their Cult are not all powerful. *They are not*. I don't care how it looks even now *they are not*. I have not called them little boys in short trousers for effect. I have said it because it is true. Wetiko's insatiable desire for power over others is not a sign of its omnipotence, but its insecurity. Paul Levy writes: 'Due to the primal fear which ultimately drives it and which it is driven to cultivate, wetiko's body politic has an intrinsic and insistent need for centralising power and control so as to create imagined safety for itself.' *Yeeeeeees!* Exactly! Why does Wetiko want humans in an ongoing state of fear? Wetiko itself *is* fear and it is petrified of love. As evil is an absence of love, so love is an absence of fear. Love conquers all and *especially* Wetiko which *is* fear. Wetiko brought fear into the world when it wasn't here before. *Fear* was the 'fall', the fall into low-frequency ignorance and illusion – fear is **False Emotion Appearing Real**. The simulation is driven and energised by fear because Wetiko/Yaldabaoth (fear) *are* the simulation. Fear is the absence of love and Wetiko is the absence of love.

Wetiko today

We can now view current events from this level of perspective. The 'Covid' hoax has generated momentous amounts of ongoing fear, anxiety, depression and despair which have empowered Wetiko. No wonder people like Gates have been the instigators when they are Wetiko incarnate and exhibit every trait of Wetiko in the extreme. See how cold and unemotional these people are like Gates and his cronies, how dead of eye they are. That's Wetiko. Sabbatians are Wetiko and everything they control including the World Health Organization, Big Pharma and the 'vaccine' makers, national 'health'

hierarchies, corporate media, Silicon Valley, the banking system, and the United Nations with its planned transformation into world government. All are controlled and possessed by the Wetiko distortion into distorting human society in its image. We are with this knowledge at the gateway to understanding the world. Divisions of race, culture, creed and sexuality are diversions to hide the real division between those possessed and influenced by Wetiko and those that are not. The 'Covid' hoax has brought both clearly into view. Human behaviour is not about race. Tyrants and dictatorships come in all colours and creeds. What unites the US president bombing the innocent and an African tribe committing genocide against another as in Rwanda? What unites them? *Wetiko*. All wars are Wetiko, all genocide is Wetiko, all hunger over centuries in a world of plenty is Wetiko. Children going to bed hungry, including in the West, is Wetiko. Cult-generated Woke racial divisions that focus on the body are designed to obscure the reality that divisions in behaviour are manifestations of mind, not body. Obsession with body identity and group judgement is a means to divert attention from the real source of behaviour – mind and perception. Conflict sown by the Woke both within themselves and with their target groups are Wetiko providing lunch for itself through still more agents of the division, chaos, and fear on which it feeds. The Cult is seeking to assimilate the entirety of humanity and all children and young people into the Wetiko frequency by manipulating them into states of fear and despair. Witness all the suicide and psychological unravelling since the spring of 2020. Wetiko psychopaths want to impose a state of unquestioning obedience to authority which is no more than a conduit for Wetiko to enforce its will and assimilate humanity into itself. It needs us to believe that resistance is futile when it fears resistance and even more so the game-changing non-cooperation with its impositions. It can use violent resistance for its benefit. Violent impositions and violent resistance are *both* Wetiko. The Power of Love with its Power of No will sweep Wetiko from our world. Wetiko and its Cult know that. They just don't want us to know.

AI Wetiko

This brings me to AI or artificial intelligence and something else Wetikos don't want us to know. What is AI *really*? I know about computer code algorithms and AI that learns from data input. These, however, are more diversions, the expeditionary force, for the real AI that they want to connect to the human brain as promoted by Silicon Valley Wetikos like Kurzweil. What is this AI? It is the frequency of *Wetiko*, the frequency of the Archons. The connection of AI to the human brain is the connection of the Wetiko frequency to create a Wetiko hive mind and complete the job of assimilation. The hive mind is planned to be controlled from Israel and China which are both 100 percent owned by Wetiko Sabbatians. The assimilation process has been going on minute by minute in the 'smart' era which fused with the 'Covid' era. We are told that social media is scrambling the minds of the young and changing their personality. This is true, but what is social media? Look more deeply at how it works, how it creates divisions and conflict, the hostility and cruelty, the targeting of people until they are destroyed. That's Wetiko. Social media is manipulated to tune people to the Wetiko frequency with all the emotional exploitation tricks employed by platforms like Facebook and its Wetiko front man, Zuckerberg. Facebook's Instagram announced a new platform for children to overcome a legal bar on them using the main site. This is more Wetiko exploitation and manipulation of kids. Amnesty International likened the plan to foxes offering to guard the henhouse and said it was incompatible with human rights. Since when did Wetiko or Zuckerberg (I repeat myself) care about that? Would Brin and Page at Google, Wojcicki at YouTube, Bezos at Amazon and whoever the hell runs Twitter act as they do if they were not channelling Wetiko? Would those who are developing technologies for no other reason than human control? How about those designing and selling technologies to kill people and Big Pharma drug and 'vaccine' producers who know they will end or devastate lives? Quite a thought for these people to consider is that if you are Wetiko in a human life you are Wetiko on the 'other side' unless your frequency

changes and that can only change by a change of perception which becomes a change of behaviour. Where Gates is going does not bear thinking about although perhaps that's exactly where he wants to go. Either way, that's where he's going. His frequency will make it so.

The frequency lair

I have been saying for a long time that a big part of the addiction to smartphones and devices is that a frequency is coming off them that entraps the mind. People spend ages on their phones and sometimes even a minute or so after they put them down they pick them up again and it all repeats. 'Covid' lockdowns will have increased this addiction a million times for obvious reasons. Addictions to alcohol overindulgence and drugs are another way that Wetiko entraps consciousness to attach to its own. Both are symptoms of low-vibrational psychological distress which alcoholism and drug addiction further compound. Do we think it's really a coincidence that access to them is made so easy while potions that can take people into realms beyond the simulation are banned and illegal? I have explored smartphone addiction in other books, the scale is mind-blowing, and that level of addiction does not come without help. Tech companies that make these phones are Wetiko and they will have no qualms about destroying the minds of children. We are seeing again with these companies the Wetiko perceptual combination of psychopathic enforcers and weak and meek unquestioning compliance by the rank and file.

The global Smart Grid is the Wetiko Grid and it is crucial to complete the Cult endgame. The simulation is radiation and we are being deluged with technological radiation on a devastating scale. Wetiko frauds like Elon Musk serve Cult interests while occasionally criticising them to maintain his street-cred. 5G and other forms of Wi-Fi are being directed at the earth from space on a volume and scale that goes on increasing by the day. Elon Musk's (officially) SpaceX Starlink project is in the process of putting tens of thousands of satellites in low orbit to cover every inch of the planet with 5G and other Wi-Fi to create Kurzweil's global 'cloud' to which the

human mind is planned to be attached very soon. SpaceX has approval to operate 12,000 satellites with more than 1,300 launched at the time of writing and applications filed for 30,000 more. Other operators in the Wi-Fi, 5G, low-orbit satellite market include OneWeb (UK), Telesat (Canada), and AST & Science (US). Musk tells us that AI could be the end of humanity and then launches a company called Neuralink to connect the human brain to computers. Musk's (in theory) Tesla company is building electric cars and the driverless vehicles of the smart control grid. As frauds and bullshitters go Elon Musk in my opinion is Major League.

5G and technological radiation in general are destructive to human health, genetics and psychology and increasing the strength of artificial radiation underpins the five-sense perceptual bubbles which are themselves expressions of radiation or electromagnetism. Freedom activist John Whitehead was so right with his 'databit by databit, we are building our own electronic concentration camps'. The Smart Grid and 5G is a means to control the human mind and infuse perceptual information into The Field to influence anyone in sync with its frequency. You can change perception and behaviour en masse if you can manipulate the population into those levels of frequency and this is happening all around us today. The arrogance of Musk and his fellow Cult operatives knows no bounds in the way that we see with Gates. Musk's satellites are so many in number already they are changing the night sky when viewed from Earth. The astronomy community has complained about this and they have seen nothing yet. Some consequences of Musk's Wetiko hubris include: Radiation; visible pollution of the night sky; interference with astronomy and meteorology; ground and water pollution from intensive use of increasingly many spaceports; accumulating space debris; continual deorbiting and burning up of aging satellites, polluting the atmosphere with toxic dust and smoke; and ever-increasing likelihood of collisions. A collective public open letter of complaint to Musk said:

We are writing to you ... because SpaceX is in process of surrounding the Earth with a network of thousands of satellites whose very purpose is to irradiate every square inch of the

Earth. SpaceX, like everyone else, is treating the radiation as if it were not there. As if the mitochondria in our cells do not depend on electrons moving undisturbed from the food we digest to the oxygen we breathe.

As if our nervous systems and our hearts are not subject to radio frequency interference like any piece of electronic equipment. As if the cancer, diabetes, and heart disease that now afflict a majority of the Earth's population are not metabolic diseases that result from interference with our cellular machinery. As if insects everywhere, and the birds and animals that eat them, are not starving to death as a result.

People like Musk and Gates believe in their limitless Wetiko arrogance that they can do whatever they like to the world because they own it. Consequences for humanity are irrelevant. It's absolutely time that we stopped taking this shit from these self-styled masters of the Earth when you consider where this is going.

Why is the Cult so anti-human?

I hear this question often: Why would they do this when it will affect them, too? Ah, but will it? Who is this *them*? Forget their bodies. They are just vehicles for Wetiko consciousness. When you break it all down to the foundations we are looking at a state of severely distorted consciousness targeting another state of consciousness for assimilation. The rest is detail. The simulation is the fly-trap in which unique sensations of the five senses create a cycle of addiction called reincarnation. Renegade Minds see that everything which happens in our reality is a smaller version of the whole picture in line with the holographic principle. Addiction to the radiation of smart technology is a smaller version of addiction to the whole simulation. Connecting the body/brain to AI is taking that addiction on a giant step further to total ongoing control by assimilating human incarnate consciousness into Wetiko. I have watched during the 'Covid' hoax how many are becoming ever more profoundly attached to Wetiko's perceptual calling cards of aggressive response to any other point of view ('There is no other god but me'), psychopathic lack of compassion and empathy, and servile submission to the narrative and will of authority. Wetiko is the psychopaths *and* subservience to psychopaths. The Cult of Wetiko is

so anti-human because it is *not* human. It embarked on a mission to destroy human by targeting everything that it means to be human and to survive as human. 'Covid' is not the end, just a means to an end. The Cult with its Wetiko consciousness is seeking to change Earth systems, including the atmosphere, to suit them, not humans. The gathering bombardment of 5G alone from ground and space is dramatically changing The Field with which the five senses interact. There is so much more to come if we sit on our hands and hope it will all go away. It is not meant to go away. It is meant to get ever more extreme and we need to face that while we still can – just.

Carbon dioxide is the gas of life. Without that human is over. Kaput, gone, history. No natural world, no human. The Cult has created a cock and bull story about carbon dioxide and climate change to justify its reduction to the point where Gates and the ignoramus Biden 'climate chief' John Kerry want to suck it out of the atmosphere. Kerry wants to do this because his master Gates does. Wetikos have made the gas of life a demon with the usual support from the Wokers of Extinction Rebellion and similar organisations and the bewildered puppet-child that is Greta Thunberg who was put on the world stage by Klaus Schwab and the World Economic Forum. The name Extinction Rebellion is both ironic and as always Wetiko inversion. The gas that we need to survive must be reduced to save us from extinction. The most basic need of human is oxygen and we now have billions walking around in face nappies depriving body and brain of this essential requirement of human existence. More than that 5G at 60 gigahertz interacts with the oxygen molecule to reduce the amount of oxygen the body can absorb into the bloodstream. The obvious knock-on consequences of that for respiratory and cognitive problems and life itself need no further explanation. Psychopaths like Musk are assembling a global system of satellites to deluge the human atmosphere with this insanity. The man should be in jail. Here we have two most basic of human needs, oxygen and carbon dioxide, being dismantled.

Two others, water and food, are getting similar treatment with the United Nations Agendas 21 and 2030 – the Great Reset – planning to

centrally control all water and food supplies. People will not even own rain water that falls on their land. Food is affected at the most basic level by reducing carbon dioxide. We have genetic modification or GMO infiltrating the food chain on a mass scale, pesticides and herbicides polluting the air and destroying the soil. Freshwater fish that provide livelihoods for 60 million people and feed hundreds of millions worldwide are being 'pushed to the brink' according the conservationists while climate change is the only focus. Now we have Gates and Schwab wanting to dispense with current food sources all together and replace them with a synthetic version which the Wetiko Cult would control in terms of production and who eats and who doesn't. We have been on the Totalitarian Tiptoe to this for more than 60 years as food has become ever more processed and full of chemical shite to the point today when it's not natural food at all. As Dr Tom Cowan says: 'If it has a label don't eat it.' Bill Gates is now the biggest owner of farmland in the United States and he does nothing without an ulterior motive involving the Cult. Klaus Schwab wrote: 'To feed the world in the next 50 years we will need to produce as much food as was produced in the last 10,000 years ... food security will only be achieved, however, if regulations on genetically modified foods are adapted to reflect the reality that gene editing offers a precise, efficient and safe method of improving crops.' Liar. People and the world are being targeted with aluminium through vaccines, chemtrails, food, drink cans, and endless other sources when aluminium has been linked to many health issues including dementia which is increasing year after year. Insects, bees and wildlife essential to the food chain are being deleted by pesticides, herbicides and radiation which 5G is dramatically increasing with 6G and 7G to come. The pollinating bee population is being devastated while wildlife including birds, dolphins and whales are having their natural radar blocked by the effects of ever-increasing radiation. In the summer windscreens used to be splattered with insects so numerous were they. It doesn't happen now. Where have they gone?

Synthetic everything

The Cult is introducing genetically-modified versions of trees, plants and insects including a Gates-funded project to unleash hundreds of millions of genetically-modified, lab-altered and patented male mosquitoes to mate with wild mosquitoes and induce genetic flaws that cause them to die out. Clinically-insane Gates-funded Japanese researchers have developed mosquitos that spread vaccine and are dubbed 'flying vaccinators'. Gates is funding the modification of weather patterns in part to sell the myth that this is caused by carbon dioxide and he's funding geoengineering of the skies to change the atmosphere. Some of this came to light with the Gates-backed plan to release tonnes of chalk into the atmosphere to 'deflect the Sun and cool the planet'. Funny how they do this while the heating effect of the Sun is not factored into climate projections focussed on carbon dioxide. The reason is that they want to reduce carbon dioxide (so don't mention the Sun), but at the same time they do want to reduce the impact of the Sun which is so essential to human life and health. I have mentioned the sun-cholesterol-vitamin D connection as they demonise the Sun with warnings about skin cancer (caused by the chemicals in sun cream they tell you to splash on). They come from the other end of the process with statin drugs to reduce cholesterol that turns sunlight into vitamin D. A lack of vitamin D leads to a long list of health effects and how vitamin D levels must have fallen with people confined to their homes over 'Covid'. Gates is funding other forms of geoengineering and most importantly chemtrails which are dropping heavy metals, aluminium and self-replicating nanotechnology onto the Earth which is killing the natural world. See *Everything You Need To Know, But Have Never Been Told* for the detailed background to this.

Every human system is being targeted for deletion by a force that's not human. The Wetiko Cult has embarked on the process of transforming the human body from biological to synthetic biological as I have explained. Biological is being replaced by the artificial and synthetic – Archontic 'countermimicry' – right across human society. The plan eventually is to dispense with the human body altogether

and absorb human consciousness – which it wouldn't really be by then – into cyberspace (the simulation which is Wetiko/Yaldabaoth). Preparations for that are already happening if people would care to look. The alternative media rightly warns about globalism and 'the globalists', but this is far bigger than that and represents the end of the human race as we know it. The 'bad copy' of prime reality that Gnostics describe was a bad copy of harmony, wonder and beauty to start with before Wetiko/Yaldabaoth set out to change the simulated 'copy' into something very different. The process was slow to start with. Entrapped humans in the simulation timeline were not technologically aware and they had to be brought up to intellectual speed while being suppressed spiritually to the point where they could build their own prison while having no idea they were doing so. We have now reached that stage where technological intellect has the potential to destroy us and that's why events are moving so fast. Central American shaman Don Juan Matus said:

Think for a moment, and tell me how you would explain the contradictions between the intelligence of man the engineer and the stupidity of his systems of belief, or the stupidity of his contradictory behaviour. Sorcerers believe that the predators have given us our systems of beliefs, our ideas of good and evil; our social mores. They are the ones who set up our dreams of success or failure. They have given us covetousness, greed, and cowardice. It is the predator who makes us complacent, routinary, and egomaniacal.

In order to keep us obedient and meek and weak, the predators engaged themselves in a stupendous manoeuvre – stupendous, of course, from the point of view of a fighting strategist; a horrendous manoeuvre from the point of those who suffer it. They gave us their mind. The predators' mind is baroque, contradictory, morose, filled with the fear of being discovered any minute now.

For 'predators' see Wetiko, Archons, Yaldabaoth, Jinn, and all the other versions of the same phenomenon in cultures and religions all over the world. The theme is always the same because it's true and it's real. We have reached the point where we have to deal with it. The question is – how?

Don't fight – walk away

I thought I'd use a controversial subheading to get things moving in terms of our response to global fascism. What do you mean 'don't fight'? What do you mean 'walk away'? We've got to fight. We can't walk away. Well, it depends what we mean by fight and walk away. If fighting means physical combat we are playing Wetiko's game and falling for its trap. It wants us to get angry, aggressive, and direct hate and hostility at the enemy we think we must fight. Every war, every battle, every conflict, has been fought with Wetiko leading both sides. It's what it does. Wetiko wants a fight, anywhere, any place. Just hit me, son, so I can hit you back. Wetiko hits Wetiko and Wetiko hits Wetiko in return. I am very forthright as you can see in exposing Wetikos of the Cult, but I don't hate them. I refuse to hate them. It's what they want. What you hate you become. What you *fight* you become. Wokers, 'anti-haters' and 'anti-fascists' prove this every time they reach for their keyboards or don their balaclavas. By walk away I mean to disengage from Wetiko which includes ceasing to cooperate with its tyranny. Paul Levy says of Wetiko:

The way to 'defeat' evil is not to try to destroy it (for then, in playing evil's game, we have already lost), but rather, to find the invulnerable place within ourselves where evil is unable to vanquish us – this is to truly 'win' our battle with evil.

Wetiko is everywhere in human society and it's been on steroids since the 'Covid' hoax. Every shouting match over wearing masks has Wetiko wearing a mask and Wetiko not wearing one. It's an electrical circuit of push and resist, push and resist, with Wetiko pushing *and* resisting. Each polarity is Wetiko empowering itself. Dictionary definitions of 'resist' include 'opposing, refusing to accept or comply with' and the word to focus on is 'opposing'. What form does this take – setting police cars alight or 'refusing to accept or comply with'? The former is Wetiko opposing Wetiko while the other points the way forward. This is the difference between those aggressively demanding that government fascism must be obeyed who stand in stark contrast to the great majority of Pushbackers. We saw this clearly with a march by thousands of Pushbackers against lockdown in London followed days later by a Woker-hijacked

protest in Bristol in which police cars were set on fire. Masks were virtually absent in London and widespread in Bristol. Wetiko wants lockdown on every level of society and infuses its aggression to police it through its unknowing stooges. Lockdown protesters are the ones with the smiling faces and the hugs, The two blatantly obvious states of being – getting more obvious by the day – are the result of Wokers and their like becoming ever more influenced by the simulation Field of Wetiko and Pushbackers ever more influenced by The Field of a far higher vibration beyond the simulation. Wetiko can't invade the heart which is where most lockdown opponents are coming from. It's the heart that allows them to see through the lies to the truth in ways I will be highlighting.

Renegade Minds know that calmness is the place from which wisdom comes. You won't find wisdom in a hissing fit and wisdom is what we need in abundance right now. Calmness is not weakness – you don't have to scream at the top of your voice to be strong. Calmness is indeed a sign of strength. 'No' means I'm not doing it. NOOOO!!! doesn't mean you're not doing it even more. Volume does not advance 'No – I'm not doing it'. You are just not doing it. Wetiko possessed and influenced don't know how to deal with that. Wetiko wants a fight and we should not give it one. What it needs more than anything is our *cooperation* and we should not give that either. Mass rallies and marches are great in that they are a visual representation of feeling, but if it ends there they are irrelevant. You demand that Wetikos act differently? Well, they're not going to are they? They are Wetikos. We don't need to waste our time demanding that something doesn't happen when that will make no difference. We need to delete the means that *allows* it to happen. This, invariably, is our cooperation. You can demand a child stop firing a peashooter at the dog or you can refuse to buy the peashooter. If you provide the means you are cooperating with the dog being smacked on the nose with a pea. How can the authorities enforce mask-wearing if millions in a country refuse? What if the 74 million Pushbackers that voted for Trump in 2020 refused to wear masks, close their businesses or stay in their homes. It would be unenforceable. The

few control the many through the compliance of the many and that's always been the dynamic be it 'Covid' regulations or the Roman Empire. I know people can find it intimidating to say no to authority or stand out in a crowd for being the only one with a face on display; but it has to be done or it's over. I hope I've made clear in this book that where this is going will be far more intimidating than standing up now and saying 'No' – I will not cooperate with my own enslavement and that of my children. There might be consequences for some initially, although not so if enough do the same. The question that must be addressed is what is going to happen if we don't? It is time to be strong and unyieldingly so. No means no. Not here and there, but *everywhere* and *always*. I have refused to wear a mask and obey all the other nonsense. I will not comply with tyranny. I repeat: Fascism is not imposed by fascists – there are never enough of them. Fascism is imposed by the population acquiescing to fascism. *I will not do it*. I will die first, or my body will. Living meekly under fascism is a form of death anyway, the death of the spirit that Martin Luther King described.

Making things happen

We must not despair. This is not over till it's over and it's far from that. The 'fat lady' must refuse to sing. The longer the 'Covid' hoax has dragged on and impacted on more lives we have seen an awakening of phenomenal numbers of people worldwide to the realisation that what they have believed all their lives is not how the world really is. Research published by the system-serving University of Bristol and King's College London in February, 2021, concluded: 'One in every 11 people in Britain say they trust David Icke's take on the coronavirus pandemic.' It will be more by now and we have gathering numbers to build on. We must urgently progress from seeing the scam to ceasing to cooperate with it. Prominent German lawyer Reiner Fuellmich, also licenced to practice law in America, is doing a magnificent job taking the legal route to bring the psychopaths to justice through a second Nuremberg tribunal for crimes against humanity. Fuellmich has an impressive record of

beating the elite in court and he formed the German Corona Investigative Committee to pursue civil charges against the main perpetrators with a view to triggering criminal charges. Most importantly he has grasped the foundation of the hoax – the PCR test not testing for the ‘virus’ – and Christian Drosten is therefore on his charge sheet along with Gates frontman Tedros at the World Health Organization. Major players must not be allowed to inflict their horrors on the human race without being brought to book. A life sentence must follow for Bill Gates and the rest of them. A group of researchers has also indicted the government of Norway for crimes against humanity with copies sent to the police and the International Criminal Court. The lawsuit cites participation in an internationally-planned false pandemic and violation of international law and human rights, the European Commission’s definition of human rights by coercive rules, Nuremberg and Hague rules on fundamental human rights, and the Norwegian constitution. We must take the initiative from hereon and not just complain, protest and react.

There are practical ways to support vital mass non-cooperation. Organising in numbers is one. Lockdown marches in London in the spring in 2021 were mass non-cooperation that the authorities could not stop. There were too many people. Hundreds of thousands walked the London streets in the centre of the road for mile after mile while the Face-Nappies could only look on. They were determined, but calm, and just *did it* with no histrionics and lots of smiles. The police were impotent. Others are organising group shopping without masks for mutual support and imagine if that was happening all over. Policing it would be impossible. If the store refuses to serve people in these circumstances they would be faced with a long line of trolleys full of goods standing on their own and everything would have to be returned to the shelves. How would they cope with that if it kept happening? I am talking here about moving on from complaining to being pro-active; from watching things happen to making things happen. I include in this our relationship with the police. The behaviour of many Face-Nappies

has been disgraceful and anyone who thinks they would never find concentration camp guards in the 'enlightened' modern era have had that myth busted big-time. The period and setting may change – Wetikos never do. I watched film footage from a London march in which a police thug viciously kicked a protestor on the floor who had done nothing. His fellow Face-Nappies stood in a ring protecting him. What he did was a criminal assault and with a crowd far outnumbering the police this can no longer be allowed to happen unchallenged. I get it when people chant 'shame on you' in these circumstances, but that is no longer enough. They *have* no shame those who do this. Crowds needs to start making a citizen's arrest of the police who commit criminal offences and brutally attack innocent people and defenceless women. A citizen's arrest can be made under section 24A of the UK Police and Criminal Evidence (PACE) Act of 1984 and you will find something similar in other countries. I prefer to call it a Common Law arrest rather than citizen's for reasons I will come to shortly. Anyone can arrest a person committing an indictable offence or if they have reasonable grounds to suspect they are committing an indictable offence. On both counts the attack by the police thug would have fallen into this category. A citizen's arrest can be made to stop someone:

- Causing physical injury to himself or any other person
- Suffering physical injury
- Causing loss of or damage to property
- Making off before a constable can assume responsibility for him

A citizen's arrest may also be made to prevent a breach of the peace under Common Law and if they believe a breach of the peace will happen or anything related to harm likely to be done or already done in their presence. This is the way to go I think – the Common Law version. If police know that the crowd and members of the public will no longer be standing and watching while they commit

their thuggery and crimes they will think twice about acting like Brownshirts and Blackshirts.

Common Law – common sense

Mention of Common Law is very important. Most people think the law is the law as in one law. This is not the case. There are two bodies of law, Common Law and Statute Law, and they are not the same. Common Law is founded on the simple premise of do no harm. It does not recognise victimless crimes in which no harm is done while Statute Law does. There is a Statute Law against almost everything. So what is Statute Law? Amazingly it's the law of the *sea* that was brought ashore by the Cult to override the law of the land which is Common Law. They had no right to do this and as always they did it anyway. They had to. They could not impose their will on the people through Common Law which only applies to do no harm. How could you stitch up the fine detail of people's lives with that? Instead they took the law of the sea, or Admiralty Law, and applied it to the population. Statute Law refers to all the laws spewing out of governments and their agencies including all the fascist laws and regulations relating to 'Covid'. The key point to make is that Statute Law is *contract law*. It only applies between *contracting* corporations. Most police officers don't even know this. They have to be kept in the dark, too. Long ago when merchants and their sailing ships began to trade with different countries a contractual law was developed called Admiralty Law and other names. Again it only applied to *contracts* agreed between *corporate* entities. If there is no agreed contract the law of the sea had no jurisdiction *and that still applies to its new alias of Statute Law*. The problem for the Cult when the law of the sea was brought ashore was an obvious one. People were not corporations and neither were government entities. To overcome the latter they made governments and all associated organisations corporations. All the institutions are *private corporations* and I mean governments and their agencies, local councils, police, courts, military, US states, the whole lot. Go to the

Dun and Bradstreet corporate listings website for confirmation that they are all corporations. You are arrested by a private corporation called the police by someone who is really a private security guard and they take you to court which is another private corporation. Neither have jurisdiction over you unless you consent and *contract* with them. This is why you hear the mantra about law enforcement policing by *consent* of the people. In truth the people 'consent' only in theory through monumental trickery.

Okay, the Cult overcame the corporate law problem by making governments and institutions corporate entities; but what about people? They are not corporations are they? Ah ... well in a sense, and *only* a sense, they are. Not people exactly – the illusion of people. The Cult creates a corporation in the name of everyone at the time that their birth certificate is issued. Note birth/ *berth* certificate and when you go to court under the law of the sea on land you stand in a *dock*. These are throwbacks to the origin. My Common Law name is David Vaughan Icke. The name of the corporation created by the government when I was born is called Mr David Vaughan Icke usually written in capitals as MR DAVID VAUGHAN ICKE. That is not me, the living, breathing man. It is a fictitious corporate entity. The trick is to make you think that David Vaughan Icke and MR DAVID VAUGHAN ICKE are the same thing. *They are not*. When police charge you and take you to court they are prosecuting the corporate entity and not the living, breathing, man or woman. They have to trick you into identifying as the corporate entity and contracting with them. Otherwise they have no jurisdiction. They do this through a language known as legalese. Lawful and legal are not the same either. Lawful relates to Common Law and legal relates to Statute Law. Legalese is the language of Statue Law which uses terms that mean one thing to the public and another in legalese. Notice that when a police officer tells someone why they are being charged he or she will say at the end: 'Do you understand?' To the public that means 'Do you comprehend?' In legalese it means 'Do you stand under me?' Do you stand under my authority? If you say

yes to the question you are unknowingly agreeing to give them jurisdiction over you in a contract between two corporate entities.

This is a confidence trick in every way. Contracts have to be agreed between informed parties and if you don't know that David Vaughan Icke is agreeing to be the corporation MR DAVID VAUGHAN ICKE you cannot knowingly agree to contract. They are deceiving you and another way they do this is to ask for proof of identity. You usually show them a driving licence or other document on which your corporate name is written. In doing so you are accepting that you are that corporate entity when you are not. Referring to yourself as a 'person' or 'citizen' is also identifying with your corporate fiction which is why I made the Common Law point about the citizen's arrest. If you are approached by a police officer you identify yourself immediately as a living, breathing, man or woman and say 'I do not consent, I do not contract with you and I do not understand' or stand under their authority. I have a Common Law birth certificate as a living man and these are available at no charge from commonlawcourt.com. Businesses registered under the Statute Law system means that its laws apply. There are, however, ways to run a business under Common Law. Remember all 'Covid' laws and regulations are Statute Law – the law of *contracts* and you do not have to contract. This doesn't mean that you can kill someone and get away with it. Common Law says do no harm and that applies to physical harm, financial harm etc. Police are employees of private corporations and there needs to be a new system of non-corporate Common Law constables operating outside the Statute Law system. If you go to davidicke.com and put Common Law into the search engine you will find videos that explain Common Law in much greater detail. It is definitely a road we should walk.

With all my heart

I have heard people say that we are in a spiritual war. I don't like the term 'war' with its Wetiko dynamic, but I know what they mean. Sweep aside all the bodily forms and we are in a situation in which two states of consciousness are seeking very different realities.

Wetiko wants upheaval, chaos, fear, suffering, conflict and control. The other wants love, peace, harmony, fairness and freedom. That's where we are. We should not fall for the idea that Wetiko is all-powerful and there's nothing we can do. Wetiko is not all-powerful. It's a joke, pathetic. It doesn't have to be, but it has made that choice for now. A handful of times over the years when I have felt the presence of its frequency I have allowed it to attach briefly so I could consciously observe its nature. The experience is not pleasant, the energy is heavy and dark, but the ease with which you can kick it back out the door shows that its real power is in persuading us that it has power. It's all a con. Wetiko is a con. It's a trickster and not a power that can control us if we unleash our own. The con is founded on manipulating humanity to give its power to Wetiko which recycles it back to present the illusion that it has power when its power is *ours* that we gave away. This happens on an energetic level and plays out in the world of the seen as humanity giving its power to Wetiko authority which uses that power to control the population when the power is only the power the population has handed over. How could it be any other way for billions to be controlled by a relative few? I have had experiences with people possessed by Wetiko and again you can kick its arse if you do it with an open heart. Oh yes – the *heart* which can transform the world of perceived 'matter'.

We are receiver-transmitters and processors of information, but what information and where from? Information is processed into perception in three main areas – the brain, the heart and the belly. These relate to thinking, knowing, and emotion. Wetiko wants us to be head and belly people which means we think within the confines of the Matrix simulation and low-vibrational emotional reaction scrambles balance and perception. A few minutes on social media and you see how emotion is the dominant force. Woke is all emotion and is therefore thought-free and fact-free. Our heart is something different. It *knows* while the head *thinks* and has to try to work it out because it doesn't know. The human energy field has seven prime vortexes which connect us with wider reality ([Fig 23](#)). Chakra means

‘wheels of light’ in the Sanskrit language of ancient India. The main ones are: The crown chakra on top of the head; brow (or ‘third eye’) chakra in the centre of the forehead; throat chakra; heart chakra in the centre of the chest; solar plexus chakra below the sternum; sacral chakra beneath the navel; and base chakra at the bottom of the spine. Each one has a particular function or functions. We feel anxiety and nervousness in the belly where the sacral chakra is located and this processes emotion that can affect the colon to give people ‘the shits’ or make them ‘shit scared’ when they are nervous. Chakras all play an important role, but the Mr and Mrs Big is the heart chakra which sits at the centre of the seven, above the chakras that connect us to the ‘physical’ and below those that connect with higher realms (or at least should). Here in the heart chakra we feel love, empathy and compassion – ‘My heart goes out to you’. Those with closed hearts become literally ‘heart-less’ in their attitudes and behaviour (see Bill Gates). Native Americans portrayed Wetiko with what Paul Levy calls a ‘frigid, icy heart, devoid of mercy’ (see Bill Gates).



Figure 23: The chakra system which interpenetrates the human energy field. The heart chakra is the governor – or should be.

Wetiko trembles at the thought of heart energy which it cannot infiltrate. The frequency is too high. What it seeks to do instead is close the heart chakra vortex to block its perceptual and energetic influence. Psychopaths have ‘hearts of stone’ and emotionally-damaged people have ‘heartache’ and ‘broken hearts’. The astonishing amount of heart disease is related to heart chakra

disruption with its fundamental connection to the 'physical' heart. Dr Tom Cowan has written an outstanding book challenging the belief that the heart is a pump and making the connection between the 'physical' and spiritual heart. Rudolph Steiner who was way ahead of his time said the same about the fallacy that the heart is a pump. *What?* The heart is not a pump? That's crazy, right? Everybody knows that. Read Cowan's *Human Heart, Cosmic Heart* and you will realise that the very idea of the heart as a pump is ridiculous when you see the evidence. How does blood in the feet so far from the heart get pumped horizontally up the body by the heart?? Cowan explains in the book the real reason why blood moves as it does. Our 'physical' heart is used to symbolise love when the source is really the heart vortex or spiritual heart which is our most powerful energetic connection to 'out there' expanded consciousness. That's why we feel *knowing* – intuitive knowing – in the centre of the chest. Knowing doesn't come from a process of thoughts leading to a conclusion. It is there in an instant all in one go. Our heart knows because of its connection to levels of awareness that *do* know. This is the meaning and source of intuition – intuitive *knowing*.

For the last more than 30 years of uncovering the global game and the nature of reality my heart has been my constant antenna for truth and accuracy. An American intelligence insider once said that I had quoted a disinformant in one of my books and yet I had only quoted the part that was true. He asked: 'How do you do that?' By using my heart antenna was the answer and anyone can do it. Heart-centred is how we are meant to be. With a closed heart chakra we withdraw into a closed mind and the bubble of five-sense reality. If you take a moment to focus your attention on the centre of your chest, picture a spinning wheel of light and see it opening and expanding. You will feel it happening, too, and perceptions of the heart like joy and love as the heart impacts on the mind as they interact. The more the chakra opens the more you will feel expressions of heart consciousness and as the process continues, and becomes part of you, insights and knowings will follow. An open

heart is connected to that level of awareness that knows all is *One*. You will see from its perspective that the fault-lines that divide us are only illusions to control us. An open heart does not process the illusions of race, creed and sexuality except as brief experiences for a consciousness that is all. Our heart does not see division, only unity (Figs 24 and 25). There's something else, too. Our hearts love to laugh. Mark Twain's quote that says 'The human race has one really effective weapon, and that is laughter' is really a reference to the heart which loves to laugh with the joy of knowing the true nature of infinite reality and that all the madness of human society is an illusion of the mind. Twain also said: 'Against the assault of laughter nothing can stand.' This is so true of Wetiko and the Cult. Their insecurity demands that they be taken seriously and their power and authority acknowledged and feared. We should do nothing of the sort. We should not get aggressive or fearful which their insecurity so desires. We should laugh in their face. Even in their no-face as police come over in their face-nappies and expect to be taken seriously. They don't take themselves seriously looking like that so why should we? Laugh in the face of intimidation. Laugh in the face of tyranny. You will see by its reaction that you have pressed all of its buttons. Wetiko does not know what to do in the face of laughter or when its targets refuse to concede their joy to fear. We have seen many examples during the 'Covid' hoax when people have expressed their energetic power and the string puppets of Wetiko retreat with their tail limp between their knees. Laugh – the world is bloody mad after all and if it's a choice between laughter and tears I know which way I'm going.



Figure 24: Head consciousness without the heart sees division and everything apart from everything else.



Figure 25: Heart consciousness sees everything as One.

'Vaccines' and the soul

The foundation of Wetiko/Archon control of humans is the separation of incarnate five-sense mind from the infinite 'I' and closing the heart chakra where the True 'I' lives during a human life. The goal has been to achieve complete separation in both cases. I was interested therefore to read an account by a French energetic healer of what she said she experienced with a patient who had been given the 'Covid' vaccine. Genuine energy healers can sense information and consciousness fields at different levels of being which are referred to as 'subtle bodies'. She described treating the patient who later returned after having, without the healer's knowledge, two doses of the 'Covid vaccine'. The healer said:

I noticed immediately the change, very heavy energy emanating from [the] subtle bodies. The scariest thing was when I was working on the heart chakra, I connected with her soul: it was detached from the physical body, it had no contact and it was, as if it was floating in a state of total confusion: a damage to the consciousness that loses contact with the physical body, i.e. with our biological machine, there is no longer any communication between them.

I continued the treatment by sending light to the heart chakra, the soul of the person, but it seemed that the soul could no longer receive any light, frequency or energy. It was a very powerful experience for me. Then I understood that this substance is indeed used to detach consciousness so that this consciousness can no longer interact through this body that it possesses in life, where there is no longer any contact, no frequency, no light, no more energetic balance or mind.

This would create a human that is rudderless and at the extreme almost zombie-like operating with a fractional state of consciousness at the mercy of Wetiko. I was especially intrigued by what the healer said in the light of the prediction by the highly-informed Rudolf Steiner more than a hundred years ago. He said:

In the future, we will eliminate the soul with medicine. Under the pretext of a 'healthy point of view', there will be a vaccine by which the human body will be treated as soon as possible directly at birth, so that the human being cannot develop the thought of the existence of soul and Spirit. To materialistic doctors will be entrusted the task of removing the soul of humanity.

As today, people are vaccinated against this disease or that disease, so in the future, children will be vaccinated with a substance that can be produced precisely in such a way that people, thanks to this vaccination, will be immune to being subjected to the 'madness' of spiritual life. He would be extremely smart, but he would not develop a conscience, and that is the true goal of some materialistic circles.

Steiner said the vaccine would detach the physical body from the etheric body (subtle bodies) and 'once the etheric body is detached the relationship between the universe and the etheric body would become extremely unstable, and man would become an automaton'. He said 'the physical body of man must be polished on this Earth by spiritual will – so the vaccine becomes a kind of arymanique (Wetiko) force' and 'man can no longer get rid of a given materialistic feeling'. Humans would then, he said, become 'materialistic of constitution and can no longer rise to the spiritual'. I have been writing for years about DNA being a receiver-transmitter of information that connects us to other levels of reality and these 'vaccines' changing DNA can be likened to changing an antenna and what it can transmit and receive. Such a disconnection would clearly lead to changes in personality and perception. Steiner further predicted the arrival of AI. Big Pharma 'Covid vaccine' makers, expressions of Wetiko, are testing their DNA-manipulating evil on children as I write with a view to giving the 'vaccine' to babies. If it's a soul-body disconnecter – and I say that it is or can be – every child would be disconnected from 'soul' at birth and the 'vaccine' would create a closed system in which spiritual guidance from the greater self would play no part. This has been the ambition of Wetiko all

along. A Pentagon video from 2005 was leaked of a presentation explaining the development of vaccines to change behaviour by their effect on the brain. Those that believe this is not happening with the 'Covid' genetically-modifying procedure masquerading as a 'vaccine' should make an urgent appointment with Naivety Anonymous. Klaus Schwab wrote in 2018:

Neurotechnologies enable us to better influence consciousness and thought and to understand many activities of the brain. They include decoding what we are thinking in fine levels of detail through new chemicals and interventions that can influence our brains to correct for errors or enhance functionality.

The plan is clear and only the heart can stop it. With every heart that opens, every mind that awakens, Wetiko is weakened. Heart and love are far more powerful than head and hate and so nothing like a majority is needed to turn this around.

Beyond the Phantom

Our heart is the prime target of Wetiko and so it must be the answer to Wetiko. We *are* our heart which is part of one heart, the infinite heart. Our heart is where the true self lives in a human life behind firewalls of five-sense illusion when an imposter takes its place – *Phantom Self*; but our heart waits patiently to be set free any time we choose to see beyond the Phantom, beyond Wetiko. A Wetikoed Phantom Self can wreak mass death and destruction while the love of forever is locked away in its heart. The time is here to unleash its power and let it sweep away the fear and despair that is Wetiko. Heart consciousness does not seek manipulated, censored, advantage for its belief or religion, its activism and desires. As an expression of the One it treats all as One with the same rights to freedom and opinion. Our heart demands fairness for itself no more than for others. From this unity of heart we can come together in mutual support and transform this Wetikoed world into what reality is meant to be – a place of love, joy, happiness, fairness, justice and freedom. Wetiko has another agenda and that's why the world is as

it is, but enough of this nonsense. Wetiko can't stay where hearts are open and it works so hard to keep them closed. Fear is its currency and its food source and love in its true sense has no fear. Why would love have fear when it knows it is *All That Is, Has Been, And Ever Can Be* on an eternal exploration of all possibility? Love in this true sense is not the physical attraction that passes for love. This can be an expression of it, yes, but Infinite Love, a love without condition, goes far deeper to the core of all being. It *is* the core of all being. Infinite reality was born from love beyond the illusions of the simulation. Love infinitely expressed is the knowing that all is One and the swiftly-passing experience of separation is a temporary hallucination. You cannot disconnect from Oneness; you can only *perceive* that you have and withdraw from its influence. This is the most important of all perception trickery by the mind parasite that is Wetiko and the foundation of all its potential for manipulation.

If we open our hearts, open the sluice gates of the mind, and redefine self-identity amazing things start to happen. Consciousness expands or contracts in accordance with self-identity. When true self is recognised as infinite awareness and label self – Phantom Self – is seen as only a series of brief experiences life is transformed. Consciousness expands to the extent that self-identity expands and everything changes. You see unity, not division, the picture, not the pixels. From this we can play the long game. No more is an experience something in and of itself, but a fleeting moment in the eternity of forever. Suddenly people in uniform and dark suits are no longer intimidating. Doing what your heart knows to be right is no longer intimidating and consequences for those actions take on the same nature of a brief experience that passes in the blink of an infinite eye. Intimidation is all in the mind. Beyond the mind there is no intimidation.

An open heart does not consider consequences for what it knows to be right. To do so would be to consider not doing what it knows to be right and for a heart in its power that is never an option. The Renegade Mind is really the Renegade Heart. Consideration of consequences will always provide a getaway car for the mind and

the heart doesn't want one. What is right in the light of what we face today is to stop cooperating with Wetiko in all its forms and to do it without fear or compromise. You cannot compromise with tyranny when tyranny always demands more until it has everything. Life is your perception and you are your destiny. Change your perception and you change your life. Change collective perception and we change the world.

Come on people ... One human family, One heart, One goal ...
FREEEEEEEDOM!

We must settle for nothing less.

Postscript

The big scare story as the book goes to press is the 'Indian' variant and the world is being deluged with propaganda about the 'Covid catastrophe' in India which mirrors in its lies and misrepresentations what happened in Italy before the first lockdown in 2020.

The *New York Post* published a picture of someone who had 'collapsed in the street from Covid' in India in April, 2021, which was actually taken during a gas leak in May, 2020. Same old, same old. Media articles in mid-February were asking why India had been so untouched by 'Covid' and then as their vaccine rollout gathered pace the alleged 'cases' began to rapidly increase. Indian 'Covid vaccine' maker Bharat Biotech was funded into existence by the Bill and Melinda Gates Foundation (the pair announced their divorce in May, 2021, which is a pity because they so deserve each other). The Indian 'Covid crisis' was ramped up by the media to terrify the world and prepare people for submission to still more restrictions. The scam that worked the first time was being repeated only with far more people seeing through the deceit. Davidicke.com and Ickonic.com have sought to tell the true story of what is happening by talking to people living through the Indian nightmare which has nothing to do with 'Covid'. We posted a letter from 'Alisha' in Pune who told a very different story to government and media mendacity. She said scenes of dying people and overwhelmed hospitals were designed to hide what was really happening – genocide and starvation. Alisha said that millions had already died of starvation during the ongoing lockdowns while government and media were lying and making it look like the 'virus':

Restaurants, shops, gyms, theatres, basically everything is shut. The cities are ghost towns. Even so-called 'essential' businesses are only open till 11am in the morning. You basically have just an hour to buy food and then your time is up.

Inter-state travel and even inter-district travel is banned. The cops wait at all major crossroads to question why you are traveling outdoors or to fine you if you are not wearing a mask.

The medical community here is also complicit in genocide, lying about hospitals being full and turning away people with genuine illnesses, who need immediate care. They have even created a shortage of oxygen cylinders.

This is the classic Cult modus operandi played out in every country. Alisha said that people who would not have a PCR test not testing for the 'virus' were being denied hospital treatment. She said the people hit hardest were migrant workers and those in rural areas. Most businesses employed migrant workers and with everything closed there were no jobs, no income and no food. As a result millions were dying of starvation or malnutrition. All this was happening under Prime Minister Narendra Modi, a 100-percent asset of the Cult, and it emphasises yet again the scale of pure anti-human evil we are dealing with. Australia banned its people from returning home from India with penalties for trying to do so of up to five years in jail and a fine of £37,000. The manufactured 'Covid' crisis in India was being prepared to justify further fascism in the West. Obvious connections could be seen between the Indian 'vaccine' programme and increased 'cases' and this became a common theme. The Seychelles, the most per capita 'Covid vaccinated' population in the world, went back into lockdown after a 'surge of cases'.

Long ago the truly evil Monsanto agricultural biotechnology corporation with its big connections to Bill Gates devastated Indian farming with genetically-modified crops. Human rights activist Gurcharan Singh highlighted the efforts by the Indian government to complete the job by destroying the food supply to hundreds of millions with 'Covid' lockdowns. He said that 415 million people at the bottom of the disgusting caste system (still going whatever they say) were below the poverty line and struggled to feed themselves every year. Now the government was imposing lockdown at just the

time to destroy the harvest. This deliberate policy was leading to mass starvation. People may reel back at the suggestion that a government would do that, but Wetiko-controlled 'leaders' are capable of any level of evil. In fact what is described in India is in the process of being instigated worldwide. The food chain and food supply are being targeted at every level to cause world hunger and thus control. Bill Gates is not the biggest owner of farmland in America for no reason and destroying access to food aids both the depopulation agenda and the plan for synthetic 'food' already being funded into existence by Gates. Add to this the coming hyper-inflation from the suicidal creation of fake 'money' in response to 'Covid' and the breakdown of container shipping systems and you have a cocktail that can only lead one way and is meant to. The Cult plan is to crash the entire system to 'build back better' with the Great Reset.

'Vaccine' transmission

Reports from all over the world continue to emerge of women suffering menstrual and fertility problems after having the fake 'vaccine' and of the non-'vaccinated' having similar problems when interacting with the 'vaccinated'. There are far too many for 'coincidence' to be credible. We've had menopausal women getting periods, others having periods stop or not stopping for weeks, passing clots, sometimes the lining of the uterus, breast irregularities, and miscarriages (which increased by 400 percent in parts of the United States). Non-'vaccinated' men and children have suffered blood clots and nose bleeding after interaction with the 'vaccinated'. Babies have died from the effects of breast milk from a 'vaccinated' mother. Awake doctors – the small minority – speculated on the cause of non-'vaccinated' suffering the same effects as the 'vaccinated'. Was it nanotechnology in the synthetic substance transmitting frequencies or was it a straight chemical bioweapon that was being transmitted between people? I am not saying that some kind of chemical transmission is not one possible answer, but the foundation of all that the Cult does is frequency and

this is fertile ground for understanding how transmission can happen. American doctor Carrie Madej, an internal medicine physician and osteopath, has been practicing for the last 20 years, teaching medical students, and she says attending different meetings where the agenda for humanity was discussed. Madej, who operates out of Georgia, did not dismiss other possible forms of transmission, but she focused on frequency in search of an explanation for transmission. She said the Moderna and Pfizer 'vaccines' contained nano-lipid particles as a key component. This was a brand new technology never before used on humanity. 'They're using a nanotechnology which is pretty much little tiny computer bits ... nanobots or hydrogel.' Inside the 'vaccines' was 'this sci-fi kind of substance' which suppressed immune checkpoints to get into the cell. I referred to this earlier as the 'Trojan horse' technique that tricks the cell into opening a gateway for the self-replicating synthetic material and while the immune system is artificially suppressed the body has no defences. Madej said the substance served many purposes including an on-demand ability to 'deliver the payload' and using the nano 'computer bits' as biosensors in the body. 'It actually has the ability to accumulate data from your body, like your breathing, your respiration, thoughts, emotions, all kinds of things.'

She said the technology obviously has the ability to operate through Wi-Fi and transmit and receive energy, messages, frequencies or impulses. 'Just imagine you're getting this new substance in you and it can react to things all around you, the 5G, your smart device, your phones.' We had something completely foreign in the human body that had never been launched large scale at a time when we were seeing 5G going into schools and hospitals (plus the Musk satellites) and she believed the 'vaccine' transmission had something to do with this: '... if these people have this inside of them ... it can act like an antenna and actually transmit it outwardly as well.' The synthetic substance produced its own voltage and so it could have that kind of effect. This fits with my own contention that the nano receiver-transmitters are designed to connect people to the

Smart Grid and break the receiver-transmitter connection to expanded consciousness. That would explain the French energy healer's experience of the disconnection of body from 'soul' with those who have had the 'vaccine'. The nanobots, self-replicating inside the body, would also transmit the synthetic frequency which could be picked up through close interaction by those who have not been 'vaccinated'. Madej speculated that perhaps it was 5G and increased levels of other radiation that was causing the symptoms directly although interestingly she said that non-'vaccinated' patients had shown improvement when they were away from the 'vaccinated' person they had interacted with. It must be remembered that you can control frequency and energy with your mind and you can consciously create energetic barriers or bubbles with the mind to stop damaging frequencies from penetrating your field. American paediatrician Dr Larry Palevsky said the 'vaccine' was not a 'vaccine' and was never designed to protect from a 'viral' infection. He called it 'a massive, brilliant propaganda of genocide' because they didn't have to inject everyone to get the result they wanted. He said the content of the jabs was able to infuse any material into the brain, heart, lungs, kidneys, liver, sperm and female productive system. 'This is genocide; this is a weapon of mass destruction.' At the same time American colleges were banning students from attending if they didn't have this life-changing and potentially life-ending 'vaccine'. Class action lawsuits must follow when the consequences of this college fascism come to light. As the book was going to press came reports about fertility effects on sperm in 'vaccinated' men which would absolutely fit with what I have been saying and hospitals continued to fill with 'vaccine' reactions. Another question is what about transmission via blood transfusions? The NHS has extended blood donation restrictions from seven days after a 'Covid vaccination' to 28 days after even a sore arm reaction.

I said in the spring of 2020 that the then touted 'Covid vaccine' would be ongoing each year like the flu jab. A year later Pfizer CEO, the appalling Albert Bourla, said people would 'likely' need a 'booster dose' of the 'vaccine' within 12 months of getting 'fully

vaccinated' and then a yearly shot. 'Variants will play a key role', he said confirming the point. Johnson & Johnson CEO Alex Gorsky also took time out from his 'vaccine' disaster to say that people may need to be vaccinated against 'Covid-19' each year. UK Health Secretary, the psychopath Matt Hancock, said additional 'boosters' would be available in the autumn of 2021. This is the trap of the 'vaccine passport'. The public will have to accept every last 'vaccine' they introduce, including for the fake 'variants', or it would cease to be valid. The only other way in some cases would be continuous testing with a test not testing for the 'virus' and what is on the swabs constantly pushed up your nose towards the brain every time?

'Vaccines' changing behaviour

I mentioned in the body of the book how I believed we would see gathering behaviour changes in the 'vaccinated' and I am already hearing such comments from the non-'vaccinated' describing behaviour changes in friends, loved ones and work colleagues. This will only increase as the self-replicating synthetic material and nanoparticles expand in body and brain. An article in the *Guardian* in 2016 detailed research at the University of Virginia in Charlottesville which developed a new method for controlling brain circuits associated with complex animal behaviour. The method, dubbed 'magnetogenetics', involves genetically-engineering a protein called ferritin, which stores and releases iron, to create a magnetised substance – 'Magneto' – that can activate specific groups of nerve cells from a distance. This is claimed to be an advance on other methods of brain activity manipulation known as optogenetics and chemogenetics (the Cult has been developing methods of brain control for a long time). The ferritin technique is said to be non-invasive and able to activate neurons 'rapidly and reversibly'. In other words, human thought and perception. The article said that earlier studies revealed how nerve cell proteins 'activated by heat and mechanical pressure can be genetically engineered so that they become sensitive to radio waves and magnetic fields, by attaching them to an iron-storing protein called ferritin, or to inorganic

paramagnetic particles'. Sensitive to radio waves and magnetic fields? You mean like 5G, 6G and 7G? This is the human-AI Smart Grid hive mind we are talking about. The *Guardian* article said:

... the researchers injected Magneto into the striatum of freely behaving mice, a deep brain structure containing dopamine-producing neurons that are involved in reward and motivation, and then placed the animals into an apparatus split into magnetised and non-magnetised sections.

Mice expressing Magneto spent far more time in the magnetised areas than mice that did not, because activation of the protein caused the striatal neurons expressing it to release dopamine, so that the mice found being in those areas rewarding. This shows that Magneto can remotely control the firing of neurons deep within the brain, and also control complex behaviours.

Make no mistake this basic methodology will be part of the 'Covid vaccine' cocktail and using magnetics to change brain function through electromagnetic field frequency activation. The Pentagon is developing a 'Covid vaccine' using ferritin. Magnetism would explain changes in behaviour and why videos are appearing across the Internet as I write showing how magnets stick to the skin at the point of the 'vaccine' shot. Once people take these 'vaccines' anything becomes possible in terms of brain function and illness which will be blamed on 'Covid-19' and 'variants'. Magnetic field manipulation would further explain why the non-'vaccinated' are reporting the same symptoms as the 'vaccinated' they interact with and why those symptoms are reported to decrease when not in their company. Interestingly 'Magneto', a 'mutant', is a character in the Marvel Comic *X-Men* stories with the ability to manipulate magnetic fields and he believes that mutants should fight back against their human oppressors by any means necessary. The character was born Erik Lehnsherr to a Jewish family in Germany.

Cult-controlled courts

The European Court of Human Rights opened the door for mandatory 'Covid-19 vaccines' across the continent when it ruled in a Czech Republic dispute over childhood immunisation that legally

enforced vaccination could be 'necessary in a democratic society'. The 17 judges decided that compulsory vaccinations did not breach human rights law. On the face of it the judgement was so inverted you gasp for air. If not having a vaccine infused into your body is not a human right then what is? Ah, but they said human rights law which has been specifically written to delete all human rights at the behest of the state (the Cult). Article 8 of the European Convention on Human Rights relates to the right to a private life. The crucial word here is '*except*':

There shall be no interference by a public authority with the exercise of this right EXCEPT such as is in accordance with the law and is necessary in a democratic society in the interests of national security, public safety or the economic wellbeing of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others [My emphasis].

No interference *except* in accordance with the law means there *are* no 'human rights' *except* what EU governments decide you can have at their behest. 'As is necessary in a democratic society' explains that reference in the judgement and 'in the interests of national security, public safety or the economic well-being of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others' gives the EU a coach and horses to ride through 'human rights' and scatter them in all directions. The judiciary is not a check and balance on government extremism; it is a vehicle to enforce it. This judgement was almost laughably predictable when the last thing the Cult wanted was a decision that went against mandatory vaccination. Judges rule over and over again to benefit the system of which they are a part. Vaccination disputes that come before them are invariably delivered in favour of doctors and authorities representing the view of the state which owns the judiciary. Oh, yes, and we have even had calls to stop putting 'Covid-19' on death certificates within 28 days of a 'positive test' because it is claimed the practice makes the 'vaccine' appear not to work. They are laughing at you.

The scale of madness, inhumanity and things to come was highlighted when those not 'vaccinated' for 'Covid' were refused evacuation from the Caribbean island of St Vincent during massive volcanic eruptions. Cruise ships taking residents to the safety of another island allowed only the 'vaccinated' to board and the rest were left to their fate. Even in life and death situations like this we see 'Covid' stripping people of their most basic human instincts and the insanity is even more extreme when you think that fake 'vaccine'-makers are not even claiming their body-manipulating concoctions stop 'infection' and 'transmission' of a 'virus' that doesn't exist. St Vincent Prime Minister Ralph Gonsalves said: 'The chief medical officer will be identifying the persons already vaccinated so that we can get them on the ship.' Note again the power of the chief medical officer who, like Whitty in the UK, will be answering to the World Health Organization. This is the Cult network structure that has overridden politicians who 'follow the science' which means doing what WHO-controlled 'medical officers' and 'science advisers' tell them. Gonsalves even said that residents who were 'vaccinated' after the order so they could board the ships would still be refused entry due to possible side effects such as 'wooziness in the head'. The good news is that if they were woozy enough in the head they could qualify to be prime minister of St Vincent.

Microchipping freedom

The European judgement will be used at some point to justify moves to enforce the 'Covid' DNA-manipulating procedure. Sandra Ro, CEO of the Global Blockchain Business Council, told a World Economic Forum event that she hoped 'vaccine passports' would help to 'drive forced consent and standardisation' of global digital identity schemes: 'I'm hoping with the desire and global demand for some sort of vaccine passport – so that people can get travelling and working again – [it] will drive forced consent, standardisation, and frankly, cooperation across the world.' The lady is either not very bright, or thoroughly mendacious, to use the term 'forced consent'.

You do not 'consent' if you are forced – you *submit*. She was describing what the plan has been all along and that's to enforce a digital identity on every human without which they could not function. 'Vaccine passports' are opening the door and are far from the end goal. A digital identity would allow you to be tracked in everything you do in cyberspace and this is the same technique used by Cult-owned China to enforce its social credit system of total control. The ultimate 'passport' is planned to be a microchip as my books have warned for nearly 30 years. Those nice people at the Pentagon working for the Cult-controlled Defense Advanced Research Projects Agency (DARPA) claimed in April, 2021, they have developed a microchip inserted under the skin to detect 'asymptomatic Covid-19 infection' before it becomes an outbreak and a 'revolutionary filter' that can remove the 'virus' from the blood when attached to a dialysis machine. The only problems with this are that the 'virus' does not exist and people transmitting the 'virus' with no symptoms is brain-numbing bullshit. This is, of course, not a ruse to get people to be microchipped for very different reasons. DARPA also said it was producing a one-stop 'vaccine' for the 'virus' and all 'variants'. One of the most sinister organisations on Planet Earth is doing this? Better have it then. These people are insane because Wetiko that possesses them is insane.

Researchers from the Salk Institute in California announced they have created an embryo that is part human and part monkey. My books going back to the 1990s have exposed experiments in top secret underground facilities in the United States where humans are being crossed with animal and non-human 'extraterrestrial' species. They are now easing that long-developed capability into the public arena and there is much more to come given we are dealing with psychiatric basket cases. Talking of which – Elon Musk's scientists at Neuralink trained a monkey to play Pong and other puzzles on a computer screen using a joystick and when the monkey made the correct move a metal tube squirted banana smoothie into his mouth which is the basic technique for training humans into unquestioning compliance. Two Neuralink chips were in the monkey's skull and

more than 2,000 wires 'fanned out' into its brain. Eventually the monkey played a video game purely with its brain waves. Psychopathic narcissist Musk said the 'breakthrough' was a step towards putting Neuralink chips into human skulls and merging minds with artificial intelligence. *Exactly*. This man is so dark and Cult to his DNA.

World Economic Fascism (WEF)

The World Economic Forum is telling you the plan by the statements made at its many and various events. Cult-owned fascist YouTube CEO Susan Wojcicki spoke at the 2021 WEF Global Technology Governance Summit (see the name) in which 40 governments and 150 companies met to ensure 'the responsible design and deployment of emerging technologies'. Orwellian translation: 'Ensuring the design and deployment of long-planned technologies will advance the Cult agenda for control and censorship.' Freedom-destroyer and Nuremberg-bound Wojcicki expressed support for tech platforms like hers to censor content that is 'technically legal but could be harmful'. Who decides what is 'harmful'? She does and they do. 'Harmful' will be whatever the Cult doesn't want people to see and we have legislation proposed by the UK government that would censor content on the basis of 'harm' no matter if the information is fair, legal and provably true. Make that *especially* if it is fair, legal and provably true. Wojcicki called for a global coalition to be formed to enforce content moderation standards through automated censorship. This is a woman and mega-censor so self-deluded that she shamelessly accepted a 'free expression' award – *Wojcicki* – in an event sponsored by her own *YouTube*. They have no shame and no self-awareness.

You know that 'Covid' is a scam and Wojcicki a Cult operative when YouTube is censoring medical and scientific opinion purely on the grounds of whether it supports or opposes the Cult 'Covid' narrative. Florida governor Ron DeSantis compiled an expert panel with four professors of medicine from Harvard, Oxford, and Stanford Universities who spoke against forcing children and

vaccinated people to wear masks. They also said there was no proof that lockdowns reduced spread or death rates of 'Covid-19'. Cult-gofer Wojcicki and her YouTube deleted the panel video 'because it included content that contradicts the consensus of local and global health authorities regarding the efficacy of masks to prevent the spread of Covid-19'. This 'consensus' refers to what the Cult tells the World Health Organization to say and the WHO tells 'local health authorities' to do. Wojcicki knows this, of course. The panellists pointed out that censorship of scientific debate was responsible for deaths from many causes, but Wojcicki couldn't care less. She would not dare go against what she is told and as a disgrace to humanity she wouldn't want to anyway. The UK government is seeking to pass a fascist 'Online Safety Bill' to specifically target with massive fines and other means non-censored video and social media platforms to make them censor 'lawful but harmful' content like the Cult-owned Facebook, Twitter, Google and YouTube. What is 'lawful but harmful' would be decided by the fascist Blair-created Ofcom.

Another WEF obsession is a cyber-attack on the financial system and this is clearly what the Cult has planned to take down the bank accounts of everyone – except theirs. Those that think they have enough money for the Cult agenda not to matter to them have got a big lesson coming if they continue to ignore what is staring them in the face. The World Economic Forum, funded by Gates and fronted by Klaus Schwab, announced it would be running a 'simulation' with the Russian government and global banks of just such an attack called Cyber Polygon 2021. What they simulate – as with the 'Covid' Event 201 – they plan to instigate. The WEF is involved in a project with the Cult-owned Carnegie Endowment for International Peace called the WEF-Carnegie Cyber Policy Initiative which seeks to merge Wall Street banks, 'regulators' (I love it) and intelligence agencies to 'prevent' (arrange and allow) a cyber-attack that would bring down the global financial system as long planned by those that control the WEF and the Carnegie operation. The Carnegie Endowment for International Peace sent an instruction to First World

War US President Woodrow Wilson not to let the war end before society had been irreversibly transformed.

The Wuhan lab diversion

As I close, the Cult-controlled authorities and lapdog media are systematically pushing 'the virus was released from the Wuhan lab' narrative. There are two versions – it happened by accident and it happened on purpose. Both are nonsense. The perceived existence of the never-shown-to-exist 'virus' is vital to sell the impression that there is actually an infective agent to deal with and to allow the endless potential for terrifying the population with 'variants' of a 'virus' that does not exist. The authorities at the time of writing are going with the 'by accident' while the alternative media is promoting the 'on purpose'. Cable news host Tucker Carlson who has questioned aspects of lockdown and 'vaccine' compulsion has bought the Wuhan lab story. 'Everyone now agrees' he said. Well, I don't and many others don't and the question is *why* does the system and its media suddenly 'agree'? When the media moves as one unit with a narrative it is always a lie – witness the hour by hour mendacity of the 'Covid' era. Why would this Cult-owned combination which has unleashed lies like machine gun fire suddenly 'agree' to tell the truth??

Much of the alternative media is buying the lie because it fits the conspiracy narrative, but it's the *wrong* conspiracy. The real conspiracy is that *there is no virus* and that is what the Cult is desperate to hide. The idea that the 'virus' was released by accident is ludicrous when the whole 'Covid' hoax was clearly long-planned and waiting to be played out as it was so fast in accordance with the Rockefeller document and Event 201. So they prepared everything in detail over decades and then sat around strumming their fingers waiting for an 'accidental' release from a bio-lab? *What??* It's crazy. Then there's the 'on purpose' claim. You want to circulate a 'deadly virus' and hide the fact that you've done so and you release it down the street from the highest-level bio-lab in China? I repeat – *What??*

You would release it far from that lab to stop any association being made. But, no, we'll do it in a place where the connection was certain to be made. Why would you need to scam 'cases' and 'deaths' and pay hospitals to diagnose 'Covid-19' if you had a real 'virus'? What are sections of the alternative media doing believing this crap? Where were all the mass deaths in Wuhan from a 'deadly pathogen' when the recovery to normal life after the initial propaganda was dramatic in speed? Why isn't the 'deadly pathogen' now circulating all over China with bodies in the street? Once again we have the technique of tell them what they want to hear and they will likely believe it. The alternative media has its 'conspiracy' and with Carlson it fits with his 'China is the danger' narrative over years. China *is* a danger as a global Cult operations centre, but not for this reason. The Wuhan lab story also has the potential to instigate conflict with China when at some stage the plan is to trigger a Problem-Reaction-Solution confrontation with the West. Question everything – *everything* – and especially when the media agrees on a common party line.

Third wave ... fourth wave ... fifth wave ...

As the book went into production the world was being set up for more lockdowns and a 'third wave' supported by invented 'variants' that were increasing all the time and will continue to do so in public statements and computer programs, but not in reality. India became the new Italy in the 'Covid' propaganda campaign and we were told to be frightened of the new 'Indian strain'. Somehow I couldn't find it within myself to do so. A document produced for the UK government entitled 'Summary of further modelling of easing of restrictions – Roadmap Step 2' declared that a third wave was inevitable (of course when it's in the script) and it would be the fault of children and those who refuse the health-destroying fake 'Covid vaccine'. One of the computer models involved came from the Cult-owned *Imperial College* and the other from Warwick University which I wouldn't trust to tell me the date in a calendar factory. The document states that both models presumed extremely high uptake

of the 'Covid vaccines' and didn't allow for 'variants'. The document states: 'The resurgence is a result of some people (mostly children) being ineligible for vaccination; others choosing not to receive the vaccine; and others being vaccinated but not perfectly protected.' The mendacity takes the breath away. Okay, blame those with a brain who won't take the DNA-modifying shots and put more pressure on children to have it as 'trials' were underway involving children as young as six months with parents who give insanity a bad name. Massive pressure is being put on the young to have the fake 'vaccine' and child age consent limits have been systematically lowered around the world to stop parents intervening. Most extraordinary about the document was its claim that the 'third wave' would be driven by 'the resurgence in both hospitalisations and deaths ... dominated by *those that have received two doses of the vaccine*, comprising around 60-70% of the wave respectively'. The predicted peak of the 'third wave' suggested 300 deaths per day with 250 of them *fully 'vaccinated' people*. How many more lies do acquiescers need to be told before they see the obvious? Those who took the jab to 'protect themselves' are projected to be those who mostly get sick and die? So what's in the 'vaccine'? The document went on:

It is possible that a summer of low prevalence could be followed by substantial increases in incidence over the following autumn and winter. Low prevalence in late summer should not be taken as an indication that SARS-CoV-2 has retreated or that the population has high enough levels of immunity to prevent another wave.

They are telling you the script and while many British people believed 'Covid' restrictions would end in the summer of 2021 the government was preparing for them to be ongoing. Authorities were awarding contracts for 'Covid marshals' to police the restrictions with contracts starting in July, 2021, and going through to January 31st, 2022, and the government was advertising for 'Media Buying Services' to secure media propaganda slots worth a potential £320 million for 'Covid-19 campaigns' with a contract not ending until March, 2022. The recipient – via a list of other front companies – was reported to be American media marketing giant Omnicom Group

Inc. While money is no object for 'Covid' the UK waiting list for all other treatment – including life-threatening conditions – passed 4.5 million. Meantime the Cult is seeking to control all official 'inquiries' to block revelations about what has really been happening and why. It must not be allowed to – we need Nuremberg jury trials in every country. The cover-up doesn't get more obvious than appointing ultra-Zionist professor Philip Zelikow to oversee two dozen US virologists, public health officials, clinicians, former government officials and four American 'charitable foundations' to 'learn the lessons' of the 'Covid' debacle. The personnel will be those that created and perpetuated the 'Covid' lies while Zelikow is the former executive director of the 9/11 Commission who ensured that the truth about those attacks never came out and produced a report that must be among the most mendacious and manipulative documents ever written – see *The Trigger* for the detailed exposure of the almost unimaginable 9/11 story in which Sabbatians can be found at every level.

Passive no more

People are increasingly challenging the authorities with amazing numbers of people taking to the streets in London well beyond the ability of the Face-Nappies to stop them. Instead the Nappies choose situations away from the mass crowds to target, intimidate, and seek to promote the impression of 'violent protestors'. One such incident happened in London's Hyde Park. Hundreds of thousands walking through the streets in protest against 'Covid' fascism were ignored by the Cult-owned BBC and most of the rest of the mainstream media, but they delighted in reporting how police were injured in 'clashes with protestors'. The truth was that a group of people gathered in Hyde Park at the end of one march when most had gone home and they were peacefully having a good time with music and chat. Face-Nappies who couldn't deal with the full-march crowd then waded in with their batons and got more than they bargained for. Instead of just standing for this criminal brutality the crowd used their numerical superiority to push the Face-Nappies out of the

park. Eventually the Nappies turned and ran. Unfortunately two or three idiots in the crowd threw drink cans striking two officers which gave the media and the government the image they wanted to discredit the 99.9999 percent who were peaceful. The idiots walked straight into the trap and we must always be aware of potential agent provocateurs used by the authorities to discredit their targets.

This response from the crowd – the can people apart – must be a turning point when the public no longer stand by while the innocent are arrested and brutally attacked by the Face-Nappies. That doesn't mean to be violent, that's the last thing we need. We'll leave the violence to the Face-Nappies and government. But it does mean that when the Face-Nappies use violence against peaceful people the numerical superiority is employed to stop them and make citizen's arrests or Common Law arrests for a breach of the peace. The time for being passive in the face of fascism is over.

We are the many, they are the few, and we need to make that count before there is no freedom left and our children and grandchildren face an ongoing fascist nightmare.

COME ON PEOPLE – IT'S TIME.

One final thought ...

The power of love
A force from above
Cleaning my soul
Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

I'll protect you from the hooded claw
Keep the vampires from your door
When the chips are down I'll be around
With my undying, death-defying
Love for you

Envy will hurt itself
Let yourself be beautiful
Sparkling love, flowers
And pearls and pretty girls
Love is like an energy
Rushin' rushin' inside of me

This time we go sublime
Lovers entwine, divine, divine,
Love is danger, love is pleasure
Love is pure – the only treasure

I'm so in love with you
Purge the soul
Make love your goal

The power of love
A force from above
Cleaning my soul
The power of love
A force from above
A sky-scraping dove

Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

Frankie Goes To Hollywood

APPENDIX

Cowan-Kaufman-Morell Statement on Virus Isolation (SOVI)

Isolation: The action of isolating; the fact or condition of being isolated or standing alone; separation from other things or persons; solitariness

Oxford English Dictionary

The controversy over whether the SARS-CoV-2 virus has ever been isolated or purified continues. However, using the above definition, common sense, the laws of logic and the dictates of science, any unbiased person must come to the conclusion that the SARS-CoV-2 virus has never been isolated or purified. As a result, no confirmation of the virus' existence can be found. The logical, common sense, and scientific consequences of this fact are:

- the structure and composition of something not shown to exist can't be known, including the presence, structure, and function of any hypothetical spike or other proteins;
- the genetic sequence of something that has never been found can't be known;
- "variants" of something that hasn't been shown to exist can't be known;
- it's impossible to demonstrate that SARS-CoV-2 causes a disease called Covid-19.

In as concise terms as possible, here's the proper way to isolate, characterize and demonstrate a new virus. First, one takes samples (blood, sputum, secretions) from many people (e.g. 500) with symptoms which are unique and specific enough to characterize an illness. Without mixing these samples with ANY tissue or products that also contain genetic material, the virologist macerates, filters and ultracentrifuges i.e. *purifies* the specimen. This common virology technique, done for decades to isolate bacteriophages¹ and so-called giant viruses in every virology lab, then allows the virologist to demonstrate with electron microscopy thousands of identically sized and shaped particles. These particles are the isolated and purified virus.

These identical particles are then checked for uniformity by physical and/or microscopic techniques. Once the purity is determined, the particles may be further characterized. This would include examining the structure, morphology, and chemical composition of the particles. Next, their genetic makeup is characterized by extracting the genetic material directly from the purified particles and using genetic-sequencing techniques, such as Sanger sequencing, that have also been around for decades. Then one does an analysis to confirm that these uniform particles are exogenous (outside) in origin as a virus is conceptualized to be, and not the normal breakdown products of dead and dying tissues.² (As of May 2020, we know that virologists have no way to determine whether the particles they're seeing are viruses or just normal breakdown products of dead and dying tissues.)³

1 Isolation, characterization and analysis of bacteriophages from the haloalkaline lake Elmenteita, Kenya Julia Khayeli Akhwale et al, PLOS One, Published: April 25, 2019.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0215734> – accessed 2/15/21

2 "Extracellular Vesicles Derived From Apoptotic Cells: An Essential Link Between Death and Regeneration," Maojiao Li et al, Frontiers in Cell and Developmental Biology, 2020 October 2.
<https://www.frontiersin.org/articles/10.3389/fcell.2020.573511/full> – accessed 2/15/21

If we have come this far then we have fully isolated, characterized, and genetically sequenced an exogenous virus particle. However, we still have to show it is causally related to a disease. This is carried out by exposing a group of healthy subjects (animals are usually used) to this isolated, purified virus in the manner in which the disease is thought to be transmitted. If the animals get sick with the same disease, as confirmed by clinical and autopsy findings, one has now shown that the virus actually causes a disease. This demonstrates infectivity and transmission of an infectious agent.

None of these steps has even been attempted with the SARS-CoV-2 virus, nor have all these steps been successfully performed for any so-called pathogenic virus. Our research indicates that a single study showing these steps does not exist in the medical literature.

Instead, since 1954, virologists have taken unpurified samples from a relatively few people, often less than ten, with a similar disease. They then minimally process this sample and inoculate this unpurified sample onto tissue culture containing usually four to six other types of material – all of which contain identical genetic material as to what is called a “virus.” The tissue culture is starved and poisoned and naturally disintegrates into many types of particles, some of which contain genetic material. Against all common sense, logic, use of the English language and scientific integrity, this process is called “virus isolation.” This brew containing fragments of genetic material from many sources is then subjected to genetic analysis, which then creates in a computer-simulation process the alleged sequence of the alleged virus, a so called in silico genome. At no time is an actual virus confirmed by electron microscopy. At no time is a genome extracted and sequenced from an actual virus. This is scientific fraud.

The observation that the unpurified specimen — inoculated onto tissue culture along with toxic antibiotics, bovine fetal tissue, amniotic fluid and other tissues — destroys the kidney tissue onto which it is inoculated is given as evidence of the virus' existence and pathogenicity. This is scientific fraud.

From now on, when anyone gives you a paper that suggests the SARS-CoV-2 virus has been isolated, please check the methods sections. If the researchers used Vero cells or any other culture method, you know that their process was not isolation. You will hear the following excuses for why actual isolation isn't done:

1. There were not enough virus particles found in samples from patients to analyze.
2. Viruses are intracellular parasites; they can't be found outside the cell in this manner.

If No. 1 is correct, and we can't find the virus in the sputum of sick people, then on what evidence do we think the virus is dangerous or even lethal? If No. 2 is correct, then how is the virus spread from person to person? We are told it emerges from the cell to infect others. Then why isn't it possible to find it?

Finally, questioning these virology techniques and conclusions is not some distraction or divisive issue. Shining the light on this truth is essential to stop this terrible fraud that humanity is confronting. For, as we now know, if the virus has never been isolated, sequenced or shown to cause illness, if the virus is imaginary, then why are we wearing masks, social distancing and putting the whole world into prison?

Finally, if pathogenic viruses don't exist, then what is going into those injectable devices erroneously called "vaccines," and what is their purpose? This scientific question is the most urgent and relevant one of our time.

We are correct. The SARS-CoV2 virus does not exist.

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Ickonic is something that has been a dream of mine for the last 5 years, growing up around alternative information I have always had a natural interest in what is going on in the World and what could I do to make it better.

Across the range of subjects and positions of influence occupied mainly by people who don't strive to make things better it's the Media that I have always found the most frustrating and fascinating. Mainly because if the Media did their Jobs properly then so much of the negative things happening in the World simply would not be able to happen, because they would be exposed within a heartbeat.

Free Press and the Opportunities that the internet could have given would mean that the Media are able to expose things like never before and hold people to account for their actions. As we all know there are 'Untouchables' that walk among us, people the Media simply won't touch, expose or investigate and that leads to the dark underworlds that infest the establishment the World over. Well I say enough, it's time for something different, a different kind of Media, where no one is off limits from exposing and investigating. All we're interested in at Ickonic is the truth of what is really going on in the World on whichever subject we're covering.

We hope you enjoy what we have created and take something away from the platform, we aim to deliver information that's informative and most importantly self-empowering, you're not a little person, you're part of something much bigger than that and its time we as a collective race began to understand that and look to the future as ours to take.

It's time...

Jaymie Icke - Founder Ickonic Alternative Media.

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THE ANSWER



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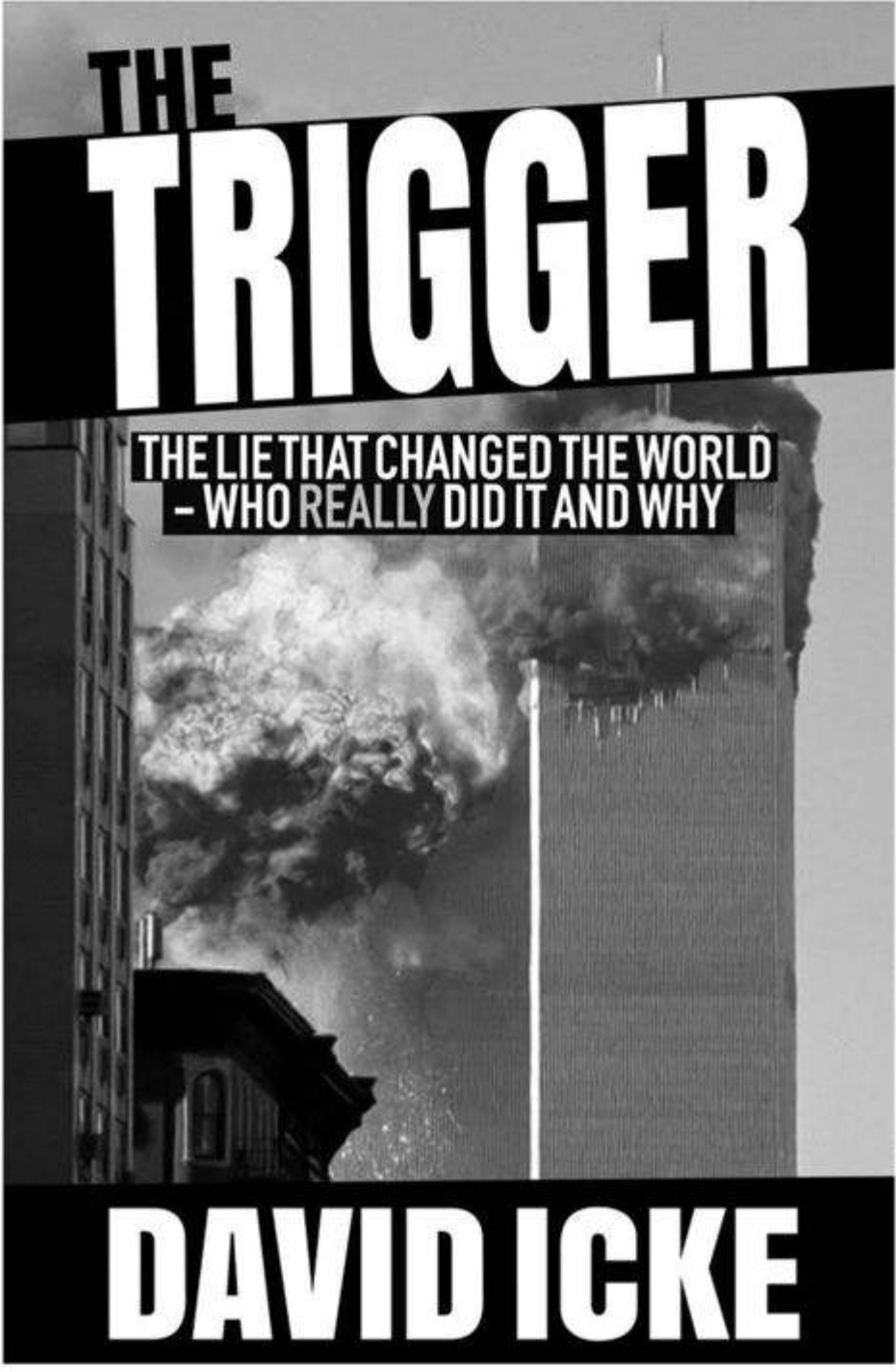
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/ren·i·geɪd/

noun

A person who behaves in a rebelliously unconventional manner.



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